

Men's 110m Hurdles Touchdown Times - by athletics meeting

LAST UPDATE: 12-Nov-23

2023 Prefontaine Classic (Eugene, OR)

FINAL

date 17-Sep-23 wind 0.9 m/s

Omega Timing (2023) - diamond league race analysis

			H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10	
Parchment, Hansle (JAM) (1998)	time		2.47	3.50	4.48	5.47	6.45	7.42	8.40	9.40	10.40	11.41		12.93	5 / 1				
	reaction time	0.168	interval	7 steps	1.03	0.98	0.99	0.98	0.97	0.98	1.00	1.00	1.01	1.52	PB		3.00	2.93	3.01
			velocity		5.55	8.87	9.33	9.23	9.33	9.42	9.33	9.14	9.14	9.05	9.22	8.51	9.14	9.36	9.11
Holloway, Grant (USA) (1997)	time		2.33	3.35	4.34	5.34	6.34	7.34	8.35	9.38	10.41	11.47		13.06	6 / 2				
	reaction time	0.153	interval	7 steps	1.02	0.99	1.00	1.00	1.00	1.01	1.03	1.03	1.06	1.59			3.01	3.01	3.12
			velocity		5.89	8.96	9.23	9.14	9.14	9.14	9.05	8.87	8.87	8.62	8.82	8.42	9.11	9.11	8.79
Roberts, Daniel (USA) (1998)	time		2.34	3.37	4.37	5.38	6.39	7.40	8.42	9.46	10.50	11.55		13.07	3 / 3				
	reaction time	0.149	interval		1.03	1.00	1.01	1.01	1.01	1.02	1.04	1.04	1.05	1.52			3.04	3.04	3.13
			velocity		5.86	8.87	9.14	9.05	9.05	9.05	8.96	8.79	8.79	8.70	9.22	8.42	9.02	9.02	8.76
Izumiya, Shunsuke (JPN) (2000)	time		2.34	3.38	4.38	5.40	6.41	7.41	8.44	9.48	10.51	11.55		13.10	7 / 4				
	reaction time	0.136	interval	7 steps	1.04	1.00	1.02	1.01	1.00	1.03	1.04	1.03	1.04	1.55			3.06	3.04	3.11
			velocity		5.86	8.79	9.14	8.96	9.05	9.14	8.87	8.79	8.87	8.79	9.05	8.40	8.96	9.02	8.82
Joseph, Jason (SUI) (1998)	time		2.43	3.49	4.52	5.52	6.51	7.50	8.51	9.52	10.54	11.57		13.12	8 / 5				
	reaction time	0.156	interval	7 steps	1.06	1.03	1.00	0.99	0.99	1.01	1.01	1.02	1.03	1.55			3.09	2.99	3.06
			velocity		5.65	8.62	8.87	9.14	9.23	9.23	9.05	9.05	8.96	8.87	9.05	8.38	8.87	9.17	8.96
Crittenden, Freddie (USA) (1998)	time		2.39	3.45	4.46	5.46	6.47	7.47	8.47	9.49	10.54	11.60		13.15	4 / 6				
	reaction time	0.146	interval		1.06	1.01	1.00	1.01	1.00	1.00	1.02	1.05	1.06	1.55			3.07	3.01	3.13
			velocity		5.74	8.62	9.05	9.14	9.05	9.14	9.14	8.96	8.70	8.62	9.05	8.37	8.93	9.11	8.76
Tinch, Cordell (USA) (2000)	time		2.36	3.41	4.41	5.41	6.41	7.42	8.43	9.47	10.52	11.60		13.21	2 / 7				
	reaction time	0.148	interval	7 steps	1.05	1.00	1.00	1.00	1.01	1.01	1.04	1.05	1.08	1.61			3.05	3.02	3.17
			velocity		5.81	8.70	9.14	9.14	9.14	9.05	9.05	8.79	8.70	8.46	8.71	8.33	8.99	9.08	8.65
Britt, Jamal (USA) (1998)	time		2.41	3.49	4.52	5.55	6.58	7.59	8.62	9.66	10.71	11.78		13.36	9 / 8				
	reaction time	0.140	interval	7 steps	1.08	1.03	1.03	1.03	1.01	1.03	1.04	1.05	1.07	1.58			3.14	3.07	3.16
			velocity		5.69	8.46	8.87	8.87	8.87	9.05	8.87	8.79	8.70	8.54	8.87	8.23	8.73	8.93	8.68
Kwaou-Mathey, Just (FRA) (1998)	time		2.45	3.53	4.56	5.57	6.59	7.60	8.63	9.69	10.75	11.85		13.46	1 / 9				
	reaction time	0.159	interval	7 steps	1.08	1.03	1.01	1.02	1.01	1.03	1.06	1.06	1.10	1.61			3.12	3.06	3.22
			velocity		5.60	8.46	8.87	9.05	8.96	9.05	8.87	8.62	8.62	8.31	8.71	8.17	8.79	8.96	8.52

2023 Xiamen Diamond League (Xiamen, CHN)

FINAL

date 02-Sep-23 wind 0.0 m/s

Omega Timing (2023) - diamond league race analysis

			H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
Parchment, Hansle (JAM) (1998)	time		2.40	3.40	4.40	5.40	6.40	7.40	8.40	9.40	10.40	11.40		12.96	5 / 1			
	reaction time	0.163	7 steps	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.56			3.00	3.00	3.00
	interval		velocity	5.72	9.14	9.14	9.14	9.14	9.14	9.14	9.14	9.14	8.99	8.49		9.14	9.14	9.14
Roberts, Daniel (USA) (1998)	time		2.33	3.37	4.37	5.35	6.35	7.35	8.36	9.39	10.43	11.48		13.03	4 / 2			
	reaction time	0.140		1.04	1.00	0.98	1.00	1.00	1.01	1.03	1.04	1.05	1.55			3.02	3.01	3.12
	interval		velocity	5.89	8.79	9.14	9.33	9.14	9.14	9.05	8.87	8.79	8.70	9.05	8.44		9.08	9.11
Holloway, Grant (USA) (1997)	time		2.32	3.34	4.32	5.30	6.29	7.28	8.28	9.31	10.36	11.43		13.12	6 / 3			
	reaction time	0.141	7 steps	1.02	0.98	0.98	0.99	0.99	1.00	1.03	1.05	1.07	1.69			2.98	2.98	3.15
	interval		velocity	5.91	8.96	9.33	9.33	9.23	9.23	9.14	8.87	8.70	8.54	8.30	8.38		9.20	9.20
Belocian, Wilhem (FRA) (1995)	time		2.38	3.43	4.44	5.44	6.45	7.45	8.47	9.51	10.56	11.62		13.17	3 / 4			
	reaction time	0.130		1.05	1.01	1.00	1.01	1.00	1.02	1.04	1.05	1.06	1.55			3.06	3.03	3.15
	interval		velocity	5.76	8.70	9.05	9.14	9.05	9.14	8.96	8.79	8.70	8.62	9.05	8.35		8.96	9.05
Muratake, Rachiddo (JPN) (2000)	time		2.37	3.42	4.43	5.44	6.46	7.47	8.51	9.54	10.59	11.65		13.19	2 / 5			
	reaction time	0.153	7 steps	1.05	1.01	1.01	1.02	1.01	1.04	1.03	1.05	1.06	1.54			3.07	3.07	3.14
	interval		velocity	5.79	8.70	9.05	9.05	8.96	9.05	8.79	8.87	8.70	8.62	9.10	8.34		8.93	8.93
Crittenden, Freddie (USA) (1998)	time		2.42	3.48	4.49	5.49	6.50	7.51	8.53	9.56	10.61	11.66		13.26	7 / 6			
	reaction time	0.148		1.06	1.01	1.00	1.01	1.01	1.02	1.03	1.05	1.05	1.60			3.07	3.04	3.13
	interval		velocity	5.67	8.62	9.05	9.14	9.05	9.05	8.96	8.87	8.70	8.70	8.76	8.30		8.93	9.02
Tinch, Cordell (USA) (2000)	time		2.41	3.43	4.45	5.47	6.48	7.51	8.54	9.59	10.66	11.74		13.38	8 / 7			
	reaction time	0.170	7 steps	1.02	1.02	1.02	1.01	1.03	1.03	1.05	1.07	1.08	1.64			3.06	3.07	3.20
	interval		velocity	5.69	8.96	8.96	8.96	9.05	8.87	8.87	8.70	8.54	8.46	8.55	8.22		8.96	8.93
Pereira, Rafael (BRA) (1998)	0.130	no information available											13.42	10 / 8				
Zeller, Joshua (GBR) (2000)	time		2.47	3.56	4.59	5.62	6.64	7.68	8.72	9.75	10.81	11.88		13.45	1 / 9			
	reaction time	0.147	interval	1.09	1.03	1.03	1.02	1.04	1.04	1.03	1.06	1.07	1.57			3.15	3.10	3.16

WR=World Rec. AR=Area Rec. OR=Olympic Rec. CR=Championship Rec. NR=National Rec. PB=Personal Best DQ=Disqualified DV=Drug Violation

velocity	5.55	8.39	8.87	8.87	8.96	8.79	8.79	8.87	8.62	8.54	8.93	8.18	8.70	8.85	8.68
time	2.49	3.56	4.63	5.68	6.72	7.75	8.80	9.85	10.92	11.98		13.56	9 / 10		
interval	7 steps	1.07	1.07	1.05	1.04	1.03	1.05	1.05	1.07	1.06	1.58		3.19	3.12	3.18
velocity	5.51	8.54	8.54	8.70	8.79	8.87	8.70	8.70	8.54	8.62	8.87	8.11	8.60	8.79	8.62

2023 Weltklasse (Zürich, SUI)

FINAL

INAL		date 31-Aug-23		wind -0.1 m/s										Omega Timing (2023) - diamond league race analysis						
				H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10	
Joseph, Jason (SUI) (1998)	time			2.40	3.46	4.46	5.45	6.45	7.45	8.46	9.46	10.49	11.54		13.08	5 / 1				
	reaction time	0.137	interval	7 steps	1.06	1.00	0.99	1.00	1.00	1.01	1.00	1.03	1.05	1.54	NR PB		3.05	3.01	3.08	
			velocity		5.72	8.62	9.14	9.23	9.14	9.14	9.05	9.14	8.87	8.70	9.10	8.41		8.99	9.11	8.90
Llopis, Enrique (ESP) (2000)	time			2.43	3.51	4.53	5.55	6.55	7.57	8.58	9.63	10.67	11.74		13.31	3 / 2				
	reaction time	0.117	interval	7 steps	1.08	1.02	1.02	1.00	1.02	1.01	1.05	1.04	1.07	1.57			3.12	3.03	3.16	
			velocity		5.65	8.46	8.96	8.96	9.14	8.96	9.05	8.70	8.79	8.54	8.93	8.26		8.79	9.05	8.68
Edwards, Eric (USA) (2000)	time			2.36	3.43	4.47	5.49	6.50	7.54	8.57	9.61	10.70	11.83		13.45	6 / 3				
	reaction time	0.125	interval		1.07	1.04	1.02	1.01	1.04	1.03	1.04	1.09	1.13	1.62			3.13	3.08	3.26	
			velocity		5.81	8.54	8.79	8.96	9.05	8.79	8.87	8.79	8.39	8.09	8.65	8.18		8.76	8.90	8.41
King, David (GBR) (1994)	time			2.44	3.51	4.55	5.57	6.61	7.66	8.72	9.79	10.87	11.98		13.60	8 / 4				
	reaction time	0.122	interval	7 steps	1.07	1.04	1.02	1.04	1.05	1.06	1.07	1.08	1.11	1.62			3.13	3.15	3.26	
			velocity		5.62	8.54	8.79	8.96	8.79	8.70	8.62	8.54	8.46	8.23	8.65	8.09		8.76	8.70	8.41
Simonelli, Lorenzo Ndele (ITA)	time			2.43	3.50	4.50	5.50	6.50	7.56	8.60	9.60	10.70	11.80		13.60	4 / 5				
	reaction time	0.142	interval	7 steps	1.07	1.00	1.00	1.00	1.06	1.04	1.00	1.10	1.10	1.80			3.07	3.10	3.20	
			velocity		5.65	8.54	9.14	9.14	9.14	8.62	8.79	9.14	8.31	8.31	7.79	8.09		8.93	8.85	8.57
Czykier, Damian (POL) (1992)	time			2.42	3.52	4.56	5.59	6.62	7.66	8.73	9.81	10.90	12.01		13.64	7 / 6				
	reaction time	0.144	interval	7 steps	1.10	1.04	1.03	1.03	1.04	1.07	1.08	1.09	1.11	1.63			3.17	3.14	3.28	
			velocity		5.67	8.31	8.79	8.87	8.87	8.79	8.54	8.46	8.39	8.23	8.60	8.06		8.65	8.73	8.36
Thorne, Craig (CAN) (2001)	time			2.47	3.55	4.60	5.64	6.68	7.74	8.79	9.87	10.96	12.08		13.70	2 / 7				
	reaction time	0.148	interval		1.08	1.05	1.04	1.04	1.06	1.05	1.08	1.09	1.12	1.62			3.17	3.15	3.29	
			velocity		5.55	8.46	8.70	8.79	8.79	8.62	8.70	8.46	8.39	8.16	8.65	8.03		8.65	8.70	8.33
Gaio, Finley (SUI) (1999)	time			2.45	3.55	4.63	5.69	6.74	7.81	8.92	10.01	11.10	12.18		13.77	9 / 8				
	reaction time	0.154	interval	7 steps	1.10	1.08	1.06	1.05	1.07	1.11	1.09	1.09	1.08	1.59			3.24	3.23	3.26	
			velocity		5.60	8.31	8.46	8.62	8.70	8.54	8.23	8.39	8.39	8.46	8.82	7.99		8.46	8.49	8.41

2023 World Athletics Championships (Budapest, HUN) (TV Analysis)

FINAL

INAL		date 21-Aug-23		wind 0.0 m/s		Henson (2023) - Athlete First: 2023 year end hurdle report													
				H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
Holloway, Grant (USA) (1997)	time			2.44	3.44	4.40	5.36	6.36	7.36	8.36	9.38	10.42	11.52		12.96	5 / 1			
	reaction time	0.135	interval	7 steps	1.00	0.96	0.96	1.00	1.00	1.00	1.02	1.04	1.10	1.44			2.92	3.00	3.16
			velocity		5.62	9.14	9.52	9.52	9.14	9.14	9.14	8.96	8.79	8.31	9.74	8.49		9.39	9.14
Parchment, Hansle (JAM) (1996)	time			2.56	3.62	4.62	5.60	6.60	7.62	8.62	9.62	10.64	11.66		13.07	8 / 2			
	reaction time	0.169	interval	7 steps	1.06	1.00	0.98	1.00	1.02	1.00	1.00	1.02	1.02	1.41			3.04	3.02	3.04
			velocity		5.36	8.62	9.14	9.33	9.14	8.96	9.14	9.14	8.96	8.96	9.94	8.42		9.02	9.08
Roberts, Daniel (USA) (1998)	time			2.48	3.50	4.50	5.50	6.50	7.54	8.54	9.60	10.64	11.68		13.09	3 / 3			
	reaction time	0.143	interval		1.02	1.00	1.00	1.00	1.04	1.00	1.06	1.04	1.04	1.41			3.02	3.04	3.14
			velocity		5.53	8.96	9.14	9.14	9.14	8.79	9.14	8.62	8.79	8.79	9.94	8.40		9.08	9.02
Crittenden, Freddie (USA) (1998)	time			2.52	3.56	4.58	5.58	6.60	7.60	8.62	9.66	10.70	11.74		13.16	4 / 4			
	reaction time	0.140	interval		1.04	1.02	1.00	1.02	1.00	1.02	1.04	1.04	1.04	1.42			3.06	3.04	3.12
			velocity		5.44	8.79	8.96	9.14	8.96	9.14	8.96	8.79	8.79	8.79	9.87	8.36		8.96	9.02
Izumiya, Shunsuke (JPN) (2000)	time			2.52	3.56	4.60	5.64	6.66	7.66	8.68	9.72	10.76	11.84		13.19	6 / 5			
	reaction time	0.143	interval	7 steps	1.04	1.04	1.04	1.02	1.00	1.02	1.04	1.04	1.08	1.35			3.12	3.04	3.16
			velocity		5.44	8.79	8.79	8.79	8.96	9.14	8.96	8.79	8.79	8.46	10.39	8.34		8.79	9.02
Zhoya, Sasha (FRA) (2002)	time			2.58	3.62	4.64	5.68	6.68	7.70	8.72	9.74	10.76	11.86		13.26	7 / 6			
	reaction time	0.166	interval	7 steps	1.04	1.02	1.04	1.00	1.02	1.02	1.02	1.02	1.10	1.40			3.10	3.04	3.14
			velocity		5.32	8.79	8.96	8.79	9.14	8.96	8.96	8.96	8.96	8.31	10.01	8.30		8.85	9.02
Joseph, Jason (SUI) (1998)	time			2.54	3.56	4.56	5.58	6.60	7.64	8.68	9.72	10.76	11.84		13.28	9 / 7			
	reaction time	0.143	interval	7 steps	1.02	1.00	1.02	1.02	1.04	1.04	1.04	1.04	1.08	1.44			3.04	3.10	3.16
			velocity		5.40	8.96	9.14	8.96	8.96	8.79	8.79	8.79	8.79	8.46	9.74	8.28		9.02	8.85
Belocian, Wilhem (FRA) (1995)	time			2.56	3.60	4.60	5.60	6.62	7.68	8.72	9.74	10.78	11.86		13.32	2 / 8			
	reaction time	0.158	interval		1.04	1.00	1.00	1.02	1.06	1.04	1.02	1.04	1.08	1.46			3.04	3.12	3.14
			velocity		5.36	8.79	9.14	9.14	8.96	8.62	8.79	8.96	8.79	8.46	9.60	8.26		9.02	8.79

Semi-Final 3

date 21-Aug-23 wind -0.1 m/s

Henson (2023) - Athlete First: 2023 year end hurdle report

			H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
Crittenden, Freddie (USA) (1991)	time		2.54	3.58	4.58	5.58	6.61	7.60	8.63	9.68	10.68	11.73		13.17	6 / 1			
	reaction time	0.156		1.04	1.00	1.00	1.03	0.99	1.03	1.05	1.00	1.05	1.44			3.04	3.05	3.10
	interval															9.02	8.99	8.85
	velocity		5.40	8.79	9.14	9.14	8.87	9.23	8.87	8.70	9.14	8.70	9.74	8.35				
Parchment, Hansle (JAM) (1991)	time		2.52	3.55	4.55	5.53	6.56	7.56	8.60	9.63	10.66	11.73		13.18	5 / 2			
	reaction time	0.138		7 steps	1.03	1.00	0.98	1.03	1.00	1.04	1.03	1.03	1.07	1.45		3.01	3.07	3.13
	interval															9.11	8.93	8.76
	velocity		5.44	8.87	9.14	9.33	8.87	9.14	8.79	8.87	8.87	8.54	9.67	8.35				
Belocian, Wilhem (FRA) (1995)	time		2.50	3.52	4.53	5.53	6.55	7.56	8.60	9.63	10.72	11.77		13.23	4 / 3			
	reaction time	0.140		1.02	1.01	1.00	1.02	1.01	1.04	1.03	1.09	1.05	1.46			3.03	3.07	3.17
	interval															9.05	8.93	8.65
	velocity		5.49	8.96	9.05	9.14	8.96	9.05	8.79	8.87	8.39	8.70	9.60	8.31				
Joseph, Jason (SUI) (1998)	time		2.56	3.66	4.68	5.70	6.72	7.73	8.73	9.73	10.77	11.80		13.25	8 / 4			
	reaction time	0.163		7 steps	1.10	1.02	1.02	1.02	1.01	1.00	1.04	1.03	1.45			3.14	3.03	3.07
	interval															8.73	9.05	8.93
	velocity		5.36	8.31	8.96	8.96	8.96	9.05	9.14	9.14	8.79	8.87	9.67	8.30				
Takayama, Shunya (JPN) (1994)	time		2.52	3.54	4.56	5.56	6.58	7.62	8.68	9.73	10.77	11.87		13.34	7 / 5			
	reaction time	0.135		1.02	1.02	1.00	1.02	1.04	1.06	1.05	1.04	1.10	1.47			3.04	3.12	3.19
	interval															9.02	8.79	8.60
	velocity		5.44	8.96	8.96	9.14	8.96	8.79	8.62	8.70	8.79	8.31	9.54	8.25				
Al-Yoha, Yaqoub (KUW) (1993)	time		2.58	3.63	4.66	5.68	6.70	7.70	8.73	9.80	10.86	11.95		13.44	2 / 6			
	reaction time	0.157		7 steps	1.05	1.03	1.02	1.02	1.00	1.03	1.07	1.06	1.09	1.49		3.10	3.05	3.22
	interval															8.85	8.99	8.52
	velocity		5.32	8.70	8.87	8.96	8.96	9.14	8.87	8.54	8.62	8.39	9.41	8.18				
Fofana, Hassane (ITA) (1992)	time		2.56	3.66	4.73	5.75	6.78	7.82	8.88	9.92	10.96	12.03		13.50	9 / 7			
	reaction time	0.147		7 steps	1.10	1.07	1.02	1.03	1.04	1.06	1.04	1.07	1.47			3.19	3.13	3.15
	interval															8.60	8.76	8.70
	velocity		5.36	8.31	8.54	8.96	8.87	8.79	8.62	8.79	8.54	9.54	8.15					
Pereira, Rafael (BRA) (1997)	time		2.56	3.60	4.70	5.70	6.74	7.76	8.80	9.86	10.93	12.00		13.52	3 / 8			
	reaction time	0.160		7 steps	1.04	1.10	1.00	1.04	1.02	1.04	1.06	1.07	1.52			3.14	3.10	3.20
	interval															8.73	8.85	8.57
	velocity		5.36	8.79	8.31	9.14	8.79	8.96	8.79	8.62	8.54	8.54	9.22	8.14				

Semi-Final 2

date 21-Aug-23 wind -0.2 m/s

Henson (2023) - Athlete First: 2023 year end hurdle report

			H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
Holloway, Grant (USA) (1997)	time		2.44	3.44	4.42	5.40	6.38	7.38	8.40	9.42	10.46	11.53		13.02	5 / 1			
	reaction time	0.137		7 steps	1.00	0.98	0.98	0.98	1.00	1.02	1.02	1.04	1.07	1.49		2.96	3.00	3.13
	interval															9.26	9.14	8.76
	velocity		5.62	9.14	9.33	9.33	9.33	9.14	8.96	8.96	8.79	8.54	9.41	8.45				
Zhoya, Sasha (FRA) (2002)	time		2.50	3.53	4.56	5.56	6.58	7.58	8.62	9.64	10.66	11.73		13.15	7 / 2			
	reaction time	0.129		7 steps	1.03	1.03	1.00	1.02	1.00	1.04	1.02	1.02	1.07	1.42	PB	3.06	3.06	3.11
	interval															8.96	8.96	8.82
	velocity		5.49	8.87	8.87	9.14	8.96	9.14	8.79	8.96	8.96	8.54	9.87	8.37				
Llopis, Enrique (ESP) (2000)	time		2.56	3.62	4.64	5.62	6.63	7.65	8.67	9.70	10.76	11.86		13.30	6 / 3			
	reaction time	0.146		7 steps	1.06	1.02	0.98	1.01	1.02	1.02	1.03	1.06	1.10	1.44	PB	3.06	3.05	3.19
	interval															8.96	8.99	8.60
	velocity		5.36	8.62	8.96	9.33	9.05	8.96	8.96	8.87	8.62	8.31	9.74	8.27				
Tinch, Cordell (USA) (2000)	time		2.50	3.58	4.60	5.60	6.62	7.65	8.67	9.70	10.76	11.83		13.31	3 / 4			
	reaction time	0.168		7 steps	1.08	1.02	1.00	1.02	1.03	1.02	1.03	1.06	1.07	1.48		3.10	3.07	3.16
	interval															8.85	8.93	8.68
	velocity		5.49	8.46	8.96	9.14	8.96	8.87	8.96	8.87	8.62	8.54	9.47	8.26				
Trajkovic, Milan (CYP) (1992)	time		2.50	3.56	4.58	5.56	6.58	7.60	8.64	9.68	10.78	11.85		13.33	4 / 5			
	reaction time	0.137		7 steps	1.06	1.02	0.98	1.02	1.02	1.04	1.04	1.10	1.07	1.48		3.06	3.08	3.21
	interval															8.96	8.90	8.54
	velocity		5.49	8.62	8.96	9.33	8.96	8.96	8.79	8.79	8.31	8.54	9.47	8.25				
Rodrigues, Eduardo (BRA) (1991)	time		2.66	3.75	4.80	5.80	6.82	7.83	8.83	9.86	10.93	12.03		13.52	8 / 6			
	reaction time	0.142		7 steps	1.09	1.05	1.00	1.02	1.01	1.00	1.03	1.07	1.10	1.49		3.14	3.03	3.20
	interval															8.73	9.05	8.57
	velocity		5.16	8.39	8.70	9.14	8.96	9.05	9.14	8.87	8.54	8.31	9.41	8.14				
Cherubin, Yves (HAI) (1999)	time		2.63	3.70	4.73	5.80	6.83	7.86	8.93	9.98	11.05	12.14		13.66	9 / 7			
	reaction time	0.178		1.07	1.03	1.07	1.03	1.03	1.07	1.05	1.07	1.09	1.52			3.17	3.13	3.21
	interval															8.65	8.76	8.54
	velocity		5.22	8.54	8.87	8.54	8.87	8.87	8.54	8.70	8.54	8.39	9.22	8.05				
Simonelli, Lorenzo Ndele (ITA) (1991)	time		2.53	3.60	4.62	5.65	6.70	7.78	8.85	9.93	11.02	12.13		13.69	2 / 8			
	reaction time	0.151		7 steps	1.07	1.02	1.03	1.05	1.08	1.07	1.08	1.09	1.11	1.56		3.12	3.20	3.28
	interval															8.79	8.57	8.36
	velocity		5.42	8.54	8.96	8.87	8.70	8.46	8.54	8.46	8.39	8.23	8.99	8.04				
Yokochi, Taiga (JPN) (2000)	time		2.60	3.70	4.80	5.93	7.05	8.18	9.36	10.50	11.80	13.06		14.93	1 / 9			
	reaction time	0.173		7 steps	1.10	1.10	1.13	1.12	1.13	1.18	1.14	1.30	1.26	1.87		3.33	3.43	3.70
	interval															8.23	7.99	7.41
	velocity		5.28	8.31	8.31	8.09	8.16	8.09	7.75	8.02	7.03	7.25	7.50	7.37				

Semi-Final 1

date 21-Aug-23 wind -0.2 m/s

Henson (2023) - Athlete First: 2023 year end hurdle report

			H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
Izumiya, Shunsuke (JPN) (2000)	time		2.52	3.60	4.60	5.62	6.64	7.66	8.66	9.69	10.72	11.80		13.16	6 / 1			
	reaction time	0.129		7 steps	1.08	1.00	1.02	1.02	1.00	1.03	1.03	1.08	1.36			3.10	3.04	3.14
	interval																	

	velocity	5.44	8.46	9.14	8.96	8.96	8.96	9.14	8.87	8.87	8.46	10.31	8.36		8.85	9.02	8.73
Roberts, Daniel (USA) (1998)	time	2.46	3.48	4.52	5.54	6.56	7.60	8.62	9.63	10.66	11.70		13.19	5 / 2			
	reaction time	0.144		1.02	1.04	1.02	1.04	1.02	1.01	1.03	1.04	1.49			3.08	3.08	3.08
	interval																
	velocity	5.58	8.96	8.79	8.96	8.96	8.79	8.96	9.05	8.87	8.79	9.41	8.34		8.90	8.90	8.90
Kwaou-Mathey, Just (FRA) (1999)	time	2.52	3.56	4.60	5.62	6.62	7.65	8.65	9.69	10.73	11.84		13.31	2 / 3			
	reaction time	0.165		1.04	1.04	1.02	1.00	1.03	1.00	1.04	1.11	1.47			3.10	3.03	3.19
	interval																
	velocity	5.44	8.79	8.79	8.96	9.14	8.87	9.14	8.79	8.79	8.23	9.54	8.26		8.85	9.05	8.60
Bennett, Orlando (JAM) (1999)	time	2.54	3.62	4.65	5.67	6.70	7.70	8.73	9.78	10.83	11.90		13.34	9 / 4			
	reaction time	0.144		1.08	1.03	1.02	1.03	1.00	1.03	1.05	1.05	1.07	1.44		3.13	3.06	3.17
	interval																
	velocity	5.40	8.46	8.87	8.96	8.87	9.14	8.87	8.70	8.70	8.54	9.74	8.25		8.76	8.96	8.65
Ojora, Tade (GBR) (1999)	time	2.58	3.63	4.67	5.70	6.72	7.78	8.80	9.83	10.87	11.93		13.43	7 / 5			
	reaction time	0.144		1.05	1.04	1.03	1.02	1.06	1.02	1.03	1.04	1.06	1.50		3.12	3.10	3.13
	interval																
	velocity	5.32	8.70	8.79	8.87	8.96	8.62	8.96	8.87	8.79	8.62	9.35	8.19		8.79	8.85	8.76
Hrelja, Max (SWE) (1998)	time	2.63	3.70	4.80	5.85	6.88	7.93	9.00	10.03	11.07	12.15		13.60	8 / 6			
	reaction time	0.162		1.07	1.10	1.05	1.03	1.05	1.07	1.03	1.04	1.08	1.45		3.22	3.15	3.15
	interval																
	velocity	5.22	8.54	8.31	8.70	8.87	8.70	8.54	8.87	8.79	8.46	9.67	8.09		8.52	8.70	8.70
Czykier, Damian (POL) (1992)	time	2.58	3.62	4.65	5.66	6.68	7.68	8.80	9.92	11.13	12.26		13.97	3 / 7			
	reaction time	0.162		1.04	1.03	1.01	1.02	1.00	1.12	1.12	1.21	1.13	1.71		3.08	3.14	3.46
	interval																
	velocity	5.32	8.79	8.87	9.05	8.96	9.14	8.16	8.16	7.55	8.09	8.20	7.87		8.90	8.73	7.92

Heat 5

Heat 5		date		20-Aug-23		wind		-0.9 m/s		Henson (2023) - Athlete First: 2023 year end hurdle report												
				H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10			
Ojora, Tade (GBR) (1999)				time	2.58	3.62	4.68	5.70	6.72	7.74	8.77	9.82	10.88	11.92		13.32	2 / 1					
				reaction time	0.148	interval	7 steps	1.04	1.06	1.02	1.02	1.02	1.03	1.05	1.06	1.04	1.40			3.12	3.07	3.15
				velocity		5.32	8.79	8.62	8.96	8.96	8.96	8.87	8.70	8.62	8.79	10.01	8.26			8.79	8.93	8.70
Takayama, Shunya (JPN) (1994)				time	2.54	3.58	4.60	5.62	6.66	7.70	8.73	9.77	10.86	11.92		13.35	9 / 2					
				reaction time	0.146	interval		1.04	1.02	1.02	1.04	1.04	1.03	1.04	1.09	1.06	1.43			3.08	3.11	3.19
				velocity		5.40	8.79	8.96	8.96	8.79	8.79	8.87	8.79	8.39	8.62	9.80	8.24			8.90	8.82	8.60
Roberts, Daniel (USA) (1998)				time	2.53	3.55	4.57	5.60	6.62	7.70	8.74	9.79	10.88	11.94		13.36	5 / 3					
				reaction time	0.155	interval		1.02	1.02	1.03	1.02	1.08	1.04	1.05	1.09	1.06	1.42			3.07	3.14	3.20
				velocity		5.42	8.96	8.96	8.87	8.96	8.46	8.79	8.70	8.39	8.62	9.87	8.23			8.93	8.73	8.57
Hrelja, Max (SWE) (1998)				time	2.63	3.71	4.74	5.76	6.78	7.81	8.86	9.91	10.95	12.05		13.42	7 / 4					
				reaction time	0.138	interval	7 steps	1.08	1.03	1.02	1.02	1.03	1.05	1.05	1.04	1.10	1.37	PB		3.13	3.10	3.19
				velocity		5.22	8.46	8.87	8.96	8.96	8.87	8.70	8.70	8.79	8.31	10.23	8.20			8.76	8.85	8.60
Kwaou-Mathey, Just (FRA) (1999)				time	2.55	3.57	4.60	5.60	6.66	7.71	8.75	9.78	10.90	11.94		13.42	3 / 5					
				reaction time	0.160	interval	7 steps	1.02	1.03	1.00	1.06	1.05	1.04	1.03	1.12	1.04	1.48			3.05	3.15	3.19
				velocity		5.38	8.96	8.87	9.14	8.62	8.70	8.79	8.87	8.16	8.79	9.47	8.20			8.99	8.70	8.60

Heat 4

Heat 4		date 20-Aug-23		wind -0.6 m/s										Henson (2023) - Athlete First: 2023 year end hurdle report						
				H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10	
Holloway, Grant (USA) (1997)	time			2.45	3.47	4.47	5.47	6.47	7.47	8.50	9.57	10.63	11.68		13.18	5 / 1				
	reaction time	0.153	interval	7 steps	1.02	1.00	1.00	1.00	1.00	1.03	1.07	1.06	1.05	1.50			3.02	3.03	3.18	
		velocity		5.60	8.96	9.14	9.14	9.14	9.14	8.87	8.54	8.62	8.70	9.35	8.35		9.08	9.05	8.62	
Trajkovic, Milan (CYP) (1992)	time			2.50	3.55	4.58	5.60	6.62	7.65	8.67	9.70	10.78	11.85		13.33	4 / 2				
	reaction time	0.164	interval	7 steps	1.05	1.03	1.02	1.02	1.03	1.02	1.03	1.08	1.07	1.48			3.10	3.07	3.18	
		velocity		5.49	8.70	8.87	8.96	8.96	8.87	8.96	8.87	8.46	8.54	9.47	8.25		8.85	8.93	8.62	
Rodrigues, Eduardo (BRA) (1999)	time			2.63	3.68	4.70	5.70	6.70	7.72	8.74	9.77	10.83	11.88		13.37	1 / 3				
	reaction time	0.150	interval	7 steps	1.05	1.02	1.00	1.00	1.02	1.02	1.03	1.06	1.05	1.49			3.07	3.04	3.14	
		velocity		5.22	8.70	8.96	9.14	9.14	8.96	8.96	8.87	8.62	8.70	9.41	8.23		8.93	9.02	8.73	
Joseph, Jason (SUI) (1998)	time			2.53	3.60	4.62	5.64	6.68	7.72	8.78	9.82	10.86	11.90		13.38	2 / 4				
	reaction time	0.159	interval	7 steps	1.07	1.02	1.02	1.04	1.04	1.06	1.04	1.04	1.04	1.48			3.11	3.14	3.12	
		velocity		5.42	8.54	8.96	8.96	8.79	8.79	8.62	8.79	8.79	8.79	9.47	8.22		8.82	8.73	8.79	
Bennett, Orlando (JAM) (1999)	time			2.53	3.62	4.66	5.70	6.73	7.76	8.80	9.83	10.88	11.93		13.39	9 / 5				
	reaction time	0.140	interval		1.09	1.04	1.04	1.03	1.03	1.04	1.03	1.05	1.05	1.46			3.17	3.10	3.13	
		velocity		5.42	8.39	8.79	8.79	8.87	8.87	8.79	8.87	8.70	8.70	9.60	8.22		8.65	8.85	8.76	
Cherubin, Yves (HAI) (1999)	time			2.52	3.60	4.64	5.67	6.70	7.72	8.80	9.83	10.93	12.05		13.56	6 / 6				
	reaction time	0.180	interval		1.08	1.04	1.03	1.03	1.02	1.08	1.03	1.10	1.12	1.51	PB		3.15	3.13	3.25	
		velocity		5.44	8.46	8.79	8.87	8.87	8.96	8.46	8.87	8.31	8.16	9.28	8.11		8.70	8.76	8.44	

Heat 3

date	20-Aug-23	wind	0.0 m/s	Henson (2023) - Athlete First: 2023 year end hurdle report															
				H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10

Mendy, Louis François (SEN) (time	2.56	3.60	4.60	5.61	6.61	7.64	8.66	9.72	10.76	11.83		13.24	6 / 1			
reaction time	0.176	interval	1.04	1.00	1.01	1.00	1.03	1.02	1.06	1.04	1.07	1.41		3.05	3.05	3.17
velocity	5.36	8.79	9.14	9.05	9.14	8.87	8.96	8.62	8.79	8.54	9.94	8.31		8.99	8.99	8.65
Crittenden, Freddie (USA) (199 time	2.56	3.60	4.63	5.65	6.66	7.72	8.75	9.80	10.86	11.95		13.40	3 / 2			
reaction time	0.148	interval	1.04	1.03	1.02	1.01	1.06	1.03	1.05	1.06	1.09	1.45		3.09	3.10	3.20
velocity	5.36	8.79	8.87	8.96	9.05	8.62	8.87	8.70	8.62	8.39	9.67	8.21		8.87	8.85	8.57
Czykier, Damian (POL) (1992) time	2.56	3.60	4.62	5.63	6.65	7.70	8.73	9.82	10.92	12.00		13.49	7 / 3			
reaction time	0.152	interval	7 steps	1.04	1.02	1.01	1.02	1.05	1.03	1.09	1.10	1.08	1.49	3.07	3.10	3.27
velocity	5.36	8.79	8.96	9.05	8.96	8.70	8.87	8.39	8.31	8.46	9.41	8.15		8.93	8.85	8.39
Simonelli, Lorenzo Ndele (ITA) time	2.53	3.58	4.63	5.65	6.72	7.76	8.80	9.86	10.95	12.03		13.50	4 / 4			
reaction time	0.152	interval	7 steps	1.05	1.05	1.02	1.07	1.04	1.06	1.09	1.08	1.47		3.12	3.15	3.23
velocity	5.42	8.70	8.70	8.96	8.54	8.79	8.79	8.62	8.39	8.46	9.54	8.15		8.79	8.70	8.49

Heat 2

date 20-Aug-23 wind 0.5 m/s

Henson (2023) - Athlete First: 2023 year end hurdle report

	H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
Belocian, Wilhem (FRA) (1995) time	2.55	3.58	4.63	5.63	6.64	7.65	8.69	9.75	10.81	11.87		13.31	9 / 1			
reaction time	0.156	interval	1.03	1.05	1.00	1.01	1.04	1.06	1.06	1.06	1.44			3.08	3.06	3.18
velocity	5.38	8.87	8.70	9.14	9.05	9.05	8.79	8.62	8.62	8.62	9.74	8.26		8.90	8.96	8.62
Izumiya, Shunsuke (JPN) (2000) time	2.57	3.62	4.64	5.66	6.68	7.76	8.80	9.86	10.90	11.98		13.33	6 / 2			
reaction time	0.146	interval	7 steps	1.05	1.02	1.02	1.08	1.04	1.06	1.04	1.08	1.35		3.09	3.14	3.18
velocity	5.34	8.70	8.96	8.96	8.96	8.46	8.79	8.62	8.79	8.46	10.39	8.25		8.87	8.73	8.62
Pereira, Rafael (BRA) (1997) time	2.58	3.63	4.66	5.66	6.68	7.74	8.78	9.82	10.92	12.02		13.52	4 / 3			
reaction time	0.146	interval	1.05	1.03	1.00	1.02	1.06	1.04	1.10	1.10	1.50			3.08	3.12	3.24
velocity	5.32	8.70	8.87	9.14	8.96	8.62	8.79	8.79	8.31	8.31	9.35	8.14		8.90	8.79	8.46
Fofana, Hassane (ITA) (1992) time	2.63	3.68	4.78	5.83	6.86	7.90	8.95	10.00	11.04	12.10		13.53	5 / 4			
reaction time	0.141	interval	7 steps	1.05	1.10	1.05	1.03	1.04	1.05	1.04	1.06	1.43		3.20	3.12	3.15
velocity	5.22	8.70	8.31	8.70	8.87	8.79	8.70	8.70	8.79	8.62	9.80	8.13		8.57	8.79	8.70

Heat 1

date 20-Aug-23 wind -0.3 m/s

Henson (2023) - Athlete First: 2023 year end hurdle report

	H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
Parchment, Hansle (JAM) (199 time	2.58	3.63	4.63	5.65	6.63	7.66	8.70	9.73	10.77	11.80		13.30	3 / 1			
reaction time	0.195	interval	7 steps	1.05	1.00	1.02	0.98	1.03	1.04	1.03	1.50			3.07	3.05	3.10
velocity	5.32	8.70	9.14	8.96	9.33	8.87	8.79	8.87	8.79	8.87	9.35	8.27		8.93	8.99	8.85
Llopis, Enrique (ESP) (2000) time	2.56	3.60	4.60	5.62	6.63	7.68	8.70	9.73	10.80	11.88		13.33	2 / 2			
reaction time	0.136	interval	7 steps	1.04	1.00	1.02	1.01	1.05	1.02	1.03	1.07	1.08	1.45	3.06	3.08	3.18
velocity	5.36	8.79	9.14	8.96	9.05	8.70	8.96	8.87	8.54	8.46	9.67	8.25		8.96	8.90	8.62
Zhoya, Sasha (FRA) (2002) time	2.56	3.63	4.70	5.72	6.73	7.75	8.78	9.80	10.88	11.93		13.35	7 / 3			
reaction time	0.157	interval	7 steps	1.07	1.07	1.02	1.01	1.02	1.03	1.02	1.08	1.05	1.42	3.16	3.06	3.15
velocity	5.36	8.54	8.54	8.96	9.05	8.96	8.87	8.96	8.46	8.70	9.87	8.24		8.68	8.96	8.70
Tinch, Cordell (USA) (2000) time	2.50	3.55	4.56	5.58	6.60	7.63	8.72	9.78	10.88	11.96		13.49	9 / 4			
reaction time	0.170	interval	7 steps	1.05	1.01	1.02	1.02	1.03	1.09	1.06	1.10	1.08	1.53	3.08	3.14	3.24
velocity	5.49	8.70	9.05	8.96	8.96	8.87	8.39	8.62	8.31	8.46	9.16	8.15		8.90	8.73	8.46
Al-Yoha, Yaqoub (KUW) (1993) time	2.60	3.68	4.72	5.75	6.77	7.80	8.86	9.92	10.96	12.06		13.56	1 / 5			
reaction time	0.140	interval	7 steps	1.08	1.04	1.03	1.02	1.03	1.06	1.06	1.04	1.10	1.50	3.15	3.11	3.20
velocity	5.28	8.46	8.79	8.87	8.96	8.87	8.62	8.62	8.79	8.31	9.35	8.11		8.70	8.82	8.57

2023 French National Championships (Albi, FRA) (TV Analysis)

FINAL

date 29-Jul-23 wind 2.3 m/s

Henson (2023) - Athlete First: 2023 year end hurdle report

	H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
Zhoya, Sasha (FRA) (2002) time	2.50	3.54	4.54	5.52	6.50	7.50	8.52	9.52	10.54	11.58		13.01	6 / 1			
reaction time	0.187	interval	7 steps	1.04	1.00	0.98	0.98	1.00	1.02	1.04	1.43			3.02	3.00	3.06
velocity	5.49	8.79	9.14	9.33	9.33	9.14	8.96	9.14	8.96	8.79	9.80	8.46		9.08	9.14	8.96
Belocian, Wilhem (FRA) (1995) time	2.46	3.50	4.50	5.50	6.48	7.50	8.52	9.52	10.56	11.60		13.07	5 / 2			
reaction time	0.168	interval	1.04	1.00	1.00	0.98	1.02	1.02	1.00	1.04	1.04	1.47		3.04	3.02	3.08
velocity	5.58	8.79	9.14	9.14	9.33	8.96	8.96	9.14	8.79	8.79	9.54	8.42		9.02	9.08	8.90
Kwaou-Mathey, Just (FRA) (19 time	2.54	3.56	4.56	5.56	6.58	7.60	8.60	9.64	10.70	11.74		13.19	3 / 3			
reaction time	0.206	interval	7 steps	1.02	1.00	1.00	1.02	1.00	1.04	1.06	1.04	1.45		3.02	3.04	3.14
velocity	5.40	8.96	9.14	9.14	8.96	8.96	9.14	8.79	8.62	8.79	9.67	8.34		9.08	9.02	8.73
Martinot-Lagarde, Pascal (FRA) time	2.52	3.56	4.58	5.60	6.60	7.62	8.64	9.66	10.70	11.76		13.25	2 / 4			
reaction time	0.180	interval	7 steps	1.04	1.02	1.00	1.02	1.02	1.04	1.06	1.49			3.08	3.04	3.12
velocity	5.44	8.79	8.96	8.96	9.14	8.96	8.96	8.96	8.79	8.62	9.41	8.30		8.90	9.02	8.79
Manga, Aurel (FRA) (1992) time	2.52	3.56	4.62	5.62	6.64	7.64	8.68	9.72	10.78	11.84		13.39	7 / 5			

reaction time	0.166	interval	7 steps	1.04	1.06	1.00	1.02	1.00	1.04	1.04	1.06	1.06	1.55			3.10	3.06	3.16
velocity				5.44	8.79	8.62	9.14	8.96	9.14	8.79	8.79	8.62	8.62	9.05	8.22	8.85	8.96	8.68
Lecoeur, Romain (FRA) (1997)	time	2.56	3.60	4.64	5.68	6.70	7.74	8.78	9.82	10.90	12.00			13.51	1 / 6			
reaction time	0.203	interval	7 steps	1.04	1.04	1.04	1.02	1.04	1.04	1.04	1.08	1.10	1.51			3.12	3.10	3.22
velocity				5.36	8.79	8.79	8.79	8.96	8.79	8.79	8.79	8.46	8.31	9.28	8.14	8.79	8.85	8.52

Heat 2

Heat 2

date	29-Jul-23	wind	5.3 m/s	Henson (2023) - Athlete First: 2023 year end hurdle report															
				H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
Zhoya, Sasha (FRA) (2002)	time			2.52	3.56	4.58	5.56	6.52	7.52	8.52	9.52	10.54	11.58		13.01	6 / 1			
reaction time	0.173	interval	7 steps	1.04	1.02	0.98	0.96	1.00	1.00	1.00	1.02	1.04	1.43				3.04	2.96	3.06
		velocity		5.44	8.79	8.96	9.33	9.52	9.14	9.14	9.14	8.96	8.79	9.80	8.46		9.02	9.26	8.96
Bascou, Dimitri (FRA) (1987)	time			2.54	3.58	4.58	5.56	6.54	7.54	8.54	9.58	10.66	11.70		13.14	3 / 2			
reaction time	0.162	interval	7 steps	1.04	1.00	0.98	0.98	1.00	1.00	1.04	1.08	1.04	1.44				3.02	2.98	3.16
		velocity		5.40	8.79	9.14	9.33	9.33	9.14	9.14	8.79	8.46	8.79	9.74	8.37		9.08	9.20	8.68

2023 London Athletics Meet (London, GBR)

FINAL

FINAL

	date	23-Jul-23	wind	1.3 m/s	Omega Timing (2023) - diamond league race analysis														
			H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10	
Holloway, Grant (USA) (1997)	time		2.36	3.35	4.33	5.30	6.29	7.28	8.29	9.32	10.37	11.43		13.01	5 / 1				
	reaction time	0.165	interval	7 steps	0.99	0.98	0.97	0.99	0.99	1.01	1.03	1.05	1.06	1.58			2.94	2.99	3.14
			velocity		5.81	9.23	9.33	9.42	9.23	9.23	9.05	8.87	8.70	8.62	8.87	8.46		9.33	9.17
Izumiya, Shunsuke (JPN) (2000)	time		2.35	3.39	4.41	5.42	6.41	7.41	8.42	9.45	10.49	11.54		13.06	3 / 2				
	reaction time	0.138	interval	7 steps	1.04	1.02	1.01	0.99	1.00	1.01	1.03	1.04	1.05	1.52			3.07	3.00	3.12
			velocity		5.84	8.79	8.96	9.05	9.23	9.14	9.05	8.87	8.79	8.70	9.22	8.42		8.93	9.14
Britt, Jamal (USA) (1998)	time		2.39	3.46	4.50	5.50	6.51	7.52	8.54	9.57	10.62	11.68		13.25	6 / 3				
	reaction time	0.145	interval	7 steps	1.07	1.04	1.00	1.01	1.01	1.02	1.03	1.05	1.06	1.57			3.11	3.04	3.14
			velocity		5.74	8.54	8.79	9.14	9.05	9.05	8.96	8.87	8.70	8.62	8.93	8.30		8.82	9.02
Parchment, Hansle (JAM) (1990)	time		2.41	3.45	4.44	5.43	6.44	7.45	8.46	9.49	10.56	11.63		13.26	4 / 4				
	reaction time	0.135	interval	7 steps	1.04	0.99	0.99	1.01	1.01	1.01	1.03	1.07	1.07	1.63			3.02	3.03	3.17
			velocity		5.69	8.79	9.23	9.23	9.05	9.05	9.05	8.87	8.54	8.54	8.60	8.30		9.08	9.05
Ojora, Tade (GBR) (1999)	time		2.42	3.51	4.54	5.54	6.54	7.55	8.57	9.61	10.66	11.70		13.27	8 / 5				
	reaction time	0.148	interval	7 steps	1.09	1.03	1.00	1.00	1.01	1.02	1.04	1.05	1.04	1.57			3.12	3.03	3.13
			velocity		5.67	8.39	8.87	9.14	9.14	9.05	8.96	8.79	8.70	8.79	8.93	8.29		8.79	9.05
Zeller, Joshua (GBR) (2000)	time		2.49	3.57	4.60	5.61	6.67	7.75	8.81	9.89	10.99	12.10		13.82	1 / 6				
	reaction time	0.144	interval		1.08	1.03	1.01	1.06	1.08	1.06	1.08	1.10	1.11	1.72			3.12	3.20	3.29
			velocity		5.51	8.46	8.87	9.05	8.62	8.46	8.62	8.46	8.31	8.23	8.15	7.96		8.79	8.57
Crittenden, Freddie (USA) (1998)	time		2.39	3.44	4.47	5.50	6.53	7.61	8.78	10.04	11.35	12.67		14.83	2 / 7				
	reaction time	0.132	interval		1.05	1.03	1.03	1.03	1.08	1.17	1.26	1.31	1.32	2.16			3.11	3.28	3.89
			velocity		5.74	8.70	8.87	8.87	8.87	8.46	7.81	7.25	6.98	6.92	6.49	7.42		8.82	8.36

2023 Kamila Skolimowska Memorial (Silesia, POL)

FINAL

FINAL		date	16-Jul-23	wind 0.3 m/s										Omega Timing (2023) - diamond league race analysis						
				H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10	
Iribarne, Roger (CUB) (1996)	time			2.37	3.42	4.45	5.46	6.47	7.49	8.52	9.56	10.61	11.68		13.25	4 / 1				
	reaction time	0.147	interval	7 steps	1.05	1.03	1.01	1.01	1.02	1.03	1.04	1.05	1.07	1.57			3.09	3.06	3.16	
			velocity		5.79	8.70	8.87	9.05	9.05	8.96	8.87	8.79	8.70	8.54	8.93	8.30		8.87	8.96	8.68
Kwaou-Mathey, Just (FRA) (1990)	time			2.39	3.46	4.50	5.53	6.55	7.57	8.59	9.62	10.65	11.71		13.26	7 / 2				
	reaction time	0.157	interval	7 steps	1.07	1.04	1.03	1.02	1.02	1.02	1.03	1.03	1.06	1.55			3.14	3.06	3.12	
			velocity		5.74	8.54	8.79	8.87	8.96	8.96	8.96	8.87	8.87	8.62	9.05	8.30		8.73	8.96	8.79
Mason, Tyler (JAM) (1995)	time			2.38	3.45	4.50	5.53	6.55	7.57	8.59	9.63	10.67	11.72		13.29	5 / 3				
	reaction time	0.138	interval	7 steps	1.07	1.05	1.03	1.02	1.02	1.02	1.04	1.04	1.05	1.57			3.15	3.06	3.13	
			velocity		5.76	8.54	8.70	8.87	8.96	8.96	8.96	8.79	8.79	8.70	8.93	8.28		8.70	8.96	8.76
Cunningham, Trey (USA) (1998)	time			2.41	3.46	4.48	5.49	6.50	7.52	8.54	9.58	10.63	11.70		13.36	3 / 4				
	reaction time	0.147	interval	7 steps	1.05	1.02	1.01	1.01	1.02	1.02	1.04	1.05	1.07	1.66			3.08	3.05	3.16	
			velocity		5.69	8.70	8.96	9.05	9.05	8.96	8.96	8.79	8.70	8.54	8.45	8.23		8.90	8.99	8.68
Zeller, Joshua (GBR) (2000)	time			2.49	3.56	4.59	5.62	6.64	7.65	8.68	9.72	10.77	11.84		13.43	1 / 5				
	reaction time	0.158	interval		1.07	1.03	1.03	1.02	1.01	1.03	1.04	1.05	1.07	1.59			3.13	3.06	3.16	
			velocity		5.51	8.54	8.87	8.87	8.96	9.05	8.87	8.79	8.70	8.54	8.82	8.19		8.76	8.96	8.68
Edwards, Eric (USA) (2000)	time			2.38	3.46	4.51	5.57	6.61	7.65	8.69	9.75	10.82	11.92		13.54	9 / 6				
	reaction time	0.130	interval		1.08	1.05	1.06	1.04	1.04	1.04	1.06	1.07	1.10	1.62			3.19	3.12	3.23	
			velocity		5.76	8.46	8.70	8.62	8.79	8.79	8.79	8.62	8.54	8.31	8.65	8.12		8.60	8.79	8.49
Roberts, Daniel (USA) (1998)	time			2.33	3.39	4.43	5.45	6.48	7.50	8.57	9.73	10.92	12.10		13.90	6 / 7				

reaction time	0.159	interval		1.06	1.04	1.02	1.03	1.02	1.07	1.16	1.19	1.18	1.80			3.12	3.12	3.53
		velocity		5.89	8.62	8.79	8.96	8.87	8.96	8.54	7.88	7.68	7.75	7.79	7.91	8.79	8.79	7.77
Trajkovic, Milan (CYP) (1992)	time	2.40	3.48	4.53	5.58	6.62	7.67	8.74	10.06	12.38	20.77				8 / --			
reaction time	0.159	interval	7 steps	1.08	1.05	1.05	1.04	1.05	1.07	1.32	2.32	8.39		DQ		3.18	3.16	12.03
		velocity		5.72	8.46	8.70	8.70	8.79	8.70	8.54	6.92	3.94	1.09			8.62	8.68	2.28

Heat 2

Heat 2

		date	16-Jul-23	wind	-0.2 m/s															Omega Timing (2023) - diamond league race analysis						
				H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10							
Roberts, Daniel (USA) (1998)	time			2.33	3.39	4.42	5.44	6.46	7.49	8.51	9.55	10.59	11.64		13.24	5 / 1										
	reaction time	0.146	interval		1.06	1.03	1.02	1.02	1.03	1.02	1.04	1.04	1.05	1.60			3.11	3.07	3.13							
			velocity	5.89	8.62	8.87	8.96	8.96	8.87	8.96	8.79	8.79	8.70	8.76	8.31		8.82	8.93	8.76							
Mason, Tyler (JAM) (1995)	time			2.40	3.46	4.49	5.52	6.53	7.57	8.60	9.64	10.67	11.71		13.28	4 / 2										
	reaction time	0.152	interval	7 steps	1.06	1.03	1.03	1.01	1.04	1.03	1.04	1.03	1.04	1.57			3.12	3.08	3.11							
			velocity	5.72	8.62	8.87	8.87	9.05	8.79	8.87	8.79	8.87	8.79	8.93	8.28		8.79	8.90	8.82							
Trajkovic, Milan (CYP) (1992)	time			2.42	3.47	4.50	5.52	6.53	7.55	8.59	9.63	10.68	11.76		13.34	9 / 3										
	reaction time	0.160	interval	7 steps	1.05	1.03	1.02	1.01	1.02	1.04	1.04	1.05	1.08	1.58			3.10	3.07	3.17							
			velocity	5.67	8.70	8.87	8.96	9.05	8.96	8.79	8.79	8.70	8.46	8.87	8.25		8.85	8.93	8.65							
Manga, Aurel (FRA) (1992)	time			2.45	3.55	4.59	5.63	6.66	7.69	8.72	9.76	10.82	11.88		13.45	1 / 4										
	reaction time	0.156	interval	7 steps	1.10	1.04	1.04	1.03	1.03	1.03	1.04	1.06	1.06	1.57			3.18	3.09	3.16							
			velocity	5.60	8.31	8.79	8.79	8.87	8.87	8.87	8.79	8.62	8.62	8.93	8.18		8.62	8.87	8.68							
Pereira, Rafael (BRA) (1997)	time			2.43	3.52	4.57	5.61	6.66	7.70	8.74	9.78	10.83	11.89		13.50	7 / 5										
	reaction time	0.136	interval		1.09	1.05	1.04	1.05	1.04	1.04	1.04	1.05	1.06	1.61			3.18	3.13	3.15							
			velocity	5.65	8.39	8.70	8.79	8.70	8.79	8.79	8.79	8.70	8.62	8.71	8.15		8.62	8.76	8.70							
Dickson, Michael (USA) (1997)	time			2.46	3.55	4.60	5.65	6.69	7.74	8.81	9.87	10.93	12.01		13.57	8 / 6										
	reaction time	0.183	interval	7 steps	1.09	1.05	1.05	1.04	1.05	1.07	1.06	1.06	1.08	1.56			3.19	3.16	3.20							
			velocity	5.58	8.39	8.70	8.70	8.79	8.70	8.54	8.62	8.62	8.46	8.99	8.11		8.60	8.68	8.57							
Martinot-Lagarde, Pascal (FRA)	time			2.45	3.55	4.60	5.64	6.69	7.74	8.79	9.85	10.91	11.99		13.59	3 / 7										
	reaction time	0.153	interval	7 steps	1.10	1.05	1.04	1.05	1.05	1.05	1.06	1.06	1.08	1.60			3.19	3.15	3.20							
			velocity	5.60	8.31	8.70	8.79	8.70	8.70	8.70	8.62	8.62	8.46	8.76	8.09		8.60	8.70	8.57							
Ishikawa, Shuhei (JPN) (1995)	time			2.41	3.48	4.53	5.58	6.62	7.67	8.74	9.82	10.91	12.00		13.64	2 / 8										
	reaction time	0.133	interval	7 steps	1.07	1.05	1.05	1.04	1.05	1.07	1.08	1.09	1.09	1.64			3.17	3.16	3.26							
			velocity	5.69	8.54	8.70	8.70	8.79	8.70	8.54	8.46	8.39	8.39	8.55	8.06		8.65	8.68	8.41							

Heat 1

Heat 1		date 16-Jul-23		wind 0.3 m/s										Omega Timing (2023) - diamond league race analysis						
				H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10	
Iribarne, Roger (CUB) (1996)	time			2.40	3.45	4.47	5.48	6.48	7.49	8.52	9.55	10.59	11.66		13.21	2 / 1				
	reaction time	0.157	interval	7 steps	1.05	1.02	1.01	1.00	1.01	1.03	1.03	1.04	1.07	1.55	PB		3.08	3.04	3.14	
			velocity		5.72	8.70	8.96	9.05	9.14	9.05	8.87	8.87	8.79	8.54	9.05	8.33		8.90	9.02	8.73
Kwaou-Mathey, Just (FRA) (1999)	time			2.47	3.56	4.59	5.60	6.61	7.62	8.63	9.65	10.68	11.72		13.27	6 / 2				
	reaction time	0.169	interval	7 steps	1.09	1.03	1.01	1.01	1.01	1.01	1.02	1.03	1.04	1.55			3.13	3.03	3.09	
			velocity		5.55	8.39	8.87	9.05	9.05	9.05	9.05	8.96	8.87	8.79	9.05	8.29		8.76	9.05	8.87
Cunningham, Trey (USA) (1998)	time			2.41	3.46	4.47	5.48	6.49	7.51	8.54	9.59	10.66	11.73		13.31	5 / 3				
	reaction time	0.153	interval	7 steps	1.05	1.01	1.01	1.01	1.02	1.03	1.05	1.07	1.07	1.58			3.07	3.06	3.19	
			velocity		5.69	8.70	9.05	9.05	9.05	8.96	8.87	8.70	8.54	8.54	8.87	8.26		8.93	8.96	8.60
Edwards, Eric (USA) (2000)	time				3.46	4.50	5.53	6.57	7.62	8.64	9.68	10.73	11.80		13.36	3 / 4				
	reaction time	0.137	interval			1.04	1.03	1.04	1.05	1.02	1.04	1.05	1.07	1.56				3.11	3.16	
			velocity		6.61	8.79	8.87	8.79	8.70	8.96	8.79	8.70	8.54	8.99	8.23			8.82	8.68	
Dunning, Robert (USA) (1997)	time			2.51	3.59	4.63	5.65	6.66	7.67	8.69	9.72	10.76	11.81		13.38	4 / 5				
	reaction time	0.140	interval	7 steps	1.08	1.04	1.02	1.01	1.01	1.02	1.03	1.04	1.05	1.57			3.14	3.04	3.12	
			velocity		5.47	8.46	8.79	8.96	9.05	9.05	8.96	8.87	8.79	8.70	8.93	8.22		8.73	9.02	8.79
Zeller, Joshua (GBR) (2000)	time			2.49	3.57	4.61	5.64	6.65	7.67	8.70	9.74	10.78	11.83		13.40	8 / 6				
	reaction time	0.156	interval		1.08	1.04	1.03	1.01	1.02	1.03	1.04	1.04	1.05	1.57			3.15	3.06	3.13	
			velocity		5.51	8.46	8.79	8.87	9.05	8.96	8.87	8.79	8.79	8.70	8.93	8.21		8.70	8.96	8.76
Alkana, Antonio (RSA) (1990)	time			2.44	3.51	4.55	5.58	6.60	7.63	8.67	9.71	10.77	11.84		13.44	1 / 7				
	reaction time	0.132	interval		1.07	1.04	1.03	1.02	1.03	1.04	1.04	1.06	1.07	1.60			3.14	3.09	3.17	
			velocity		5.62	8.54	8.79	8.87	8.96	8.87	8.79	8.79	8.62	8.54	8.76	8.18		8.73	8.87	8.65
Thomas, Damion (JAM) (1999)	time			2.47	3.55	4.59	5.62	6.65	7.69	8.73	9.78	10.85	11.93		13.53	9 / 8				
	reaction time	0.147	interval	7 steps	1.08	1.04	1.03	1.03	1.04	1.04	1.05	1.07	1.08	1.60			3.15	3.11	3.20	
			velocity		5.55	8.46	8.79	8.87	8.87	8.79	8.79	8.70	8.54	8.46	8.76	8.13		8.70	8.82	8.57
Czykier, Damian (POL) (1992)	time			2.48	3.57	4.62	5.65	6.70	7.75	8.80	9.85	10.93	12.03		13.65	7 / 9				
	reaction time	0.154	interval	7 steps	1.09	1.05	1.03	1.05	1.05	1.05	1.05	1.08	1.10	1.62			3.17	3.15	3.23	

velocity 5.53 8.39 8.70 8.87 8.70 8.70 8.70 8.70 8.46 8.31 8.65 8.06 8.65 8.70 8.49

2023 USATF National Championships (Eugene, OR) (TV Analysis)

FINAL

		date	09-Jul-23	wind	-0.2 m/s											Henson (2023) - Athlete First: 2023 year end hurdle report			
				H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
Roberts, Daniel (USA) (1998)	time			2.46	3.47	4.47	5.47	6.47	7.47	8.48	9.51	10.55	11.63		13.05	6 / 1			
	reaction time			interval		1.01	1.00	1.00	1.00	1.01	1.03	1.04	1.08	1.42			3.01	3.01	3.15
	velocity			5.58	9.05	9.14	9.14	9.14	9.14	9.05	8.87	8.79	8.46	9.87	8.43		9.11	9.11	8.70
Tinch, Cordell (USA) (2000)	time			2.47	3.49	4.50	5.50	6.48	7.48	8.50	9.52	10.55	11.62		13.08	3 / 2			
	reaction time			interval	7 steps	1.02	1.01	1.00	0.98	1.00	1.02	1.02	1.03	1.07	1.46		3.03	3.00	3.12
	velocity			5.55	8.96	9.05	9.14	9.33	9.14	8.96	8.96	8.87	8.54	9.60	8.41		9.05	9.14	8.79
Crittenden, Freddie (USA) (1999)	time			2.53	3.57	4.60	5.60	6.60	7.61	8.63	9.66	10.73	11.77		13.23	7 / 3			
	reaction time			interval		1.04	1.03	1.00	1.00	1.01	1.02	1.03	1.07	1.04	1.46		3.07	3.03	3.14
	velocity			5.42	8.79	8.87	9.14	9.14	9.05	8.96	8.87	8.54	8.79	9.60	8.31		8.93	9.05	8.73
Edwards, Eric (USA) (2000)	time			2.53	3.58	4.60	5.60	6.62	7.65	8.68	9.71	10.75	11.81		13.26	1 / 4			
	reaction time			interval		1.05	1.02	1.00	1.02	1.03	1.03	1.04	1.06	1.45			3.07	3.08	3.13
	velocity			5.42	8.70	8.96	9.14	8.96	8.87	8.87	8.87	8.79	8.62	9.67	8.30		8.93	8.90	8.76
Cunningham, Trey (USA) (1998)	time			2.51	3.56	4.57	5.58	6.60	7.62	8.66	9.70	10.76	11.82		13.29	4 / 5			
	reaction time			interval	7 steps	1.05	1.01	1.01	1.02	1.02	1.04	1.04	1.06	1.06	1.47		3.07	3.08	3.16
	velocity			5.47	8.70	9.05	9.05	8.96	8.96	8.79	8.79	8.62	8.62	9.54	8.28		8.93	8.90	8.68
Britt, Jamal (USA) (1998)	time			2.52	3.57	4.60	5.62	6.64	7.66	8.69	9.72	10.76	11.84		13.31	8 / 6			
	reaction time			interval	7 steps	1.05	1.03	1.02	1.02	1.03	1.03	1.04	1.08	1.47			3.10	3.07	3.15
	velocity			5.44	8.70	8.87	8.96	8.96	8.96	8.87	8.87	8.79	8.46	9.54	8.26		8.85	8.93	8.70
Dunning, Robert (USA) (1997)	time			2.53	3.59	4.60	5.62	6.64	7.67	8.71	9.75	10.81	11.89		13.40	2 / 7			
	reaction time			interval	7 steps	1.06	1.01	1.02	1.02	1.03	1.04	1.04	1.06	1.08	1.51		3.09	3.09	3.18
	velocity			5.42	8.62	9.05	8.96	8.96	8.87	8.79	8.79	8.62	8.46	9.28	8.21		8.87	8.87	8.62

2023 Jamaican National Championships (Kingston, JAM) (TV Analysis)

FINAL

		date	09-Jul-23	wind	0.7 m/s											Henson (2023) - Athlete First: 2023 year end hurdle report			
				H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
Broadbell, Rasheed (JAM) (2000)	time			2.50	3.55	4.55	5.53	6.53	7.53	8.50	9.50	10.48	11.48		12.94	3 / 1			
	reaction time			interval	7 steps	1.05	1.00	0.98	1.00	1.00	0.97	1.00	0.98	1.00	1.46	PB	3.03	2.97	2.98
	velocity			5.49	8.70	9.14	9.33	9.14	9.14	9.42	9.14	9.33	9.14	9.60	8.50		9.05	9.23	9.20
Parchment, Hansle (JAM) (1999)	time			2.53	3.58	4.60	5.58	6.56	7.56	8.54	9.56	10.58	11.62		13.12	4 / 2			
	reaction time			interval	7 steps	1.05	1.02	0.98	0.98	1.00	0.98	1.02	1.02	1.04	1.50		3.05	2.96	3.08
	velocity			5.42	8.70	8.96	9.33	9.33	9.14	9.33	8.96	8.96	8.79	9.35	8.38		8.99	9.26	8.90
Bennett, Orlando (JAM) (1999)	time			2.50	3.56	4.58	5.60	6.58	7.58	8.60	9.60	10.63	11.70		13.19	6 / 3			
	reaction time			interval		1.06	1.02	1.02	0.98	1.00	1.02	1.00	1.03	1.07	1.49		3.10	3.00	3.10
	velocity			5.49	8.62	8.96	8.96	9.33	9.14	8.96	9.14	8.87	8.54	9.41	8.34		8.85	9.14	8.85
Mason, Tyler (JAM) (1995)	time			2.53	3.58	4.62	5.62	6.60	7.60	8.62	9.65	10.67	11.72		13.22	5 / 4			
	reaction time			interval	7 steps	1.05	1.04	1.00	0.98	1.00	1.02	1.03	1.02	1.05	1.50		3.09	3.00	3.10
	velocity			5.42	8.70	8.79	9.14	9.33	9.14	8.96	8.87	8.96	8.70	9.35	8.32		8.87	9.14	8.85

2023 Athletissima (Lausanne, SUI)

FINAL

		date	30-Jun-23	wind	-1.0 m/s											Omega Timing (2023) - diamond league race analysis			
				H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
Izumiya, Shunsuke (JPN) (2000)	time			2.33	3.40	4.42	5.45	6.46	7.47	8.50	9.55	10.60	11.67		13.22	4 / 1			
	reaction time	0.101		interval	7 steps	1.07	1.02	1.03	1.01	1.01	1.03	1.05	1.05	1.07	1.55		3.12	3.05	3.17
	velocity			5.89	8.54	8.96	8.87	9.05	9.05	8.87	8.70	8.70	8.54	9.05	8.32		8.79	8.99	8.65
Joseph, Jason (SUI) (1998)	time			2.40	3.47	4.49	5.50	6.50	7.50	8.52	9.58	10.63	11.68		13.23	5 / 2			
	reaction time	0.153		interval	7 steps	1.07	1.02	1.01	1.00	1.00	1.02	1.06	1.05	1.05	1.55		3.10	3.02	3.16
	velocity			5.72	8.54	8.96	9.05	9.14	9.14	8.96	8.62	8.70	8.70	9.05	8.31		8.85	9.08	8.68
Kwaou-Mathey, Just (FRA) (1999)	time			2.43	3.50	4.53	5.53	6.55	7.56	8.60	9.65	10.70	11.79		13.37	3 / 3			
	reaction time	0.134		interval	7 steps	1.07	1.03	1.00	1.02	1.01	1.04	1.05	1.05	1.09	1.58		3.10	3.07	3.19
	velocity			5.65	8.54	8.87	9.14	8.96	9.05	8.79	8.70	8.70	8.39	8.87	8.23		8.85	8.93	8.60
Simonelli, Lorenzo Ndele (ITA)	time			2.39	3.46	4.50	5.53	6.57	7.60	8.64	9.68	10.74	11.82		13.41	1 / 4			
	reaction time	0.136		interval	7 steps	1.07	1.04	1.03	1.04	1.03	1.04	1.06	1.08	1.59			3.14	3.11	3.18
	velocity			5.74	8.54	8.79	8.87	8.79	8.87	8.79	8.79	8.62	8.46	8.82	8.20		8.73	8.82	8.62
Llopis, Enrique (ESP) (2000)	time			2.44	3.52	4.54	5.56	6.57	7.59	8.62	9.67	10.74	11.82		13.43	7 / 5			
	reaction time	0.135		interval	7 steps	1.08	1.02	1.02	1.01	1.02	1.03	1.05	1.07	1.08	1.61		3.12	3.06	3.20
	velocity			5.62	8.46	8.96	8.96	9.05	8.96	8.87	8.70	8.54	8.46	8.71	8.19		8.79	8.96	8.57
Pereira, Rafael (BRA) (1997)	time			2.43	3.50	4.55	5.58	6.61	7.64	8.69	9.75	10.83	11.92		13.56	2 / 6			

WR=World Rec. AR=Area Rec. OR=Olympic Rec. CR=Championship Rec. NR=National Rec. PB=Personal Best DQ=Disqualified DV=Drug Violation

FINAL

INAL		date		24-Jun-23		wind		1.6 m/s		Henson (2023) - Athlete First: 2023 year end hurdle report									
				H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
Roberts, Daniel (USA) (1998)		time		2.47	3.50	4.50	5.50	6.54	7.54	8.56	9.58	10.60	11.61		13.01	3 / 1			
reaction time		0.159	interval		1.03	1.00	1.00	1.04	1.00	1.02	1.02	1.02	1.01	1.40			3.03	3.06	3.05
			velocity	5.55	8.87	9.14	9.14	8.79	9.14	8.96	8.96	8.96	9.05	10.01	8.46		9.05	8.96	8.99
Allen, Devon (USA) (1994)		time		2.57	3.62	4.64	5.62	6.61	7.61	8.61	9.62	10.64	11.66		13.04	5 / 2			
reaction time		0.173	interval	7 steps	1.05	1.02	0.98	0.99	1.00	1.00	1.01	1.02	1.02	1.38			3.05	2.99	3.05
			velocity	5.34	8.70	8.96	9.33	9.23	9.14	9.14	9.05	8.96	8.96	10.16	8.44		8.99	9.17	8.99
Britt, Jamal (USA) (1998)		time		2.52	3.57	4.60	5.61	6.61	7.65	8.67	9.71	10.74	11.79		13.19	7 / 3			
reaction time		0.169	interval	7 steps	1.05	1.03	1.01	1.00	1.04	1.02	1.04	1.03	1.05	1.40			3.09	3.06	3.12
			velocity	5.44	8.70	8.87	9.05	9.14	8.79	8.96	8.79	8.87	8.70	10.01	8.34		8.87	8.96	8.79
Edwards, Eric (USA) (2000)		time		2.50	3.54	4.59	5.61	6.62	7.64	8.67	9.71	10.76	11.83		13.23	8 / 4			
reaction time		0.127	interval		1.04	1.05	1.02	1.01	1.02	1.03	1.04	1.05	1.07	1.40			3.11	3.06	3.16
			velocity	5.49	8.79	8.70	8.96	9.05	8.96	8.87	8.79	8.70	8.54	10.01	8.31		8.82	8.96	8.68
Dickson, Michael (USA) (1997)		time		2.53	3.59	4.61	5.63	6.64	7.67	8.71	9.74	10.78	11.85		13.26	1 / 5			
reaction time		0.212	interval	7 steps	1.06	1.02	1.02	1.01	1.03	1.04	1.03	1.04	1.07	1.41			3.10	3.08	3.14
			velocity	5.42	8.62	8.96	8.96	9.05	8.87	8.79	8.87	8.79	8.54	9.94	8.30		8.85	8.90	8.73
Dunning, Robert (USA) (1997)		time		2.53	3.57	4.61	5.61	6.62	7.64	8.66	9.68	10.74	11.83		13.29	6 / 6			
reaction time		0.181	interval	7 steps	1.04	1.04	1.00	1.01	1.02	1.02	1.02	1.06	1.09	1.46			3.08	3.05	3.17
			velocity	5.42	8.79	8.79	9.14	9.05	8.96	8.96	8.96	8.62	8.39	9.60	8.28		8.90	8.99	8.65
Bennett, Orlando (JAM) (1999)		time		2.58	3.64	4.67	5.70	6.72	7.74	8.76	9.79	10.84	11.91		13.34	2 / 7			
reaction time		0.150	interval		1.06	1.03	1.03	1.02	1.02	1.02	1.03	1.05	1.07	1.43			3.12	3.06	3.15
			velocity	5.32	8.62	8.87	8.87	8.96	8.96	8.96	8.87	8.70	8.54	9.80	8.25		8.79	8.96	8.70
Thomas, Damion (JAM) (1999)		time		2.57	3.60	4.65	5.68	6.71	7.73	8.77	9.83	10.92	12.02		13.53	9 / 8			
reaction time		0.163	interval	7 steps	1.03	1.05	1.03	1.03	1.02	1.04	1.06	1.09	1.10	1.51			3.11	3.09	3.25
			velocity	5.34	8.87	8.70	8.87	8.87	8.96	8.79	8.62	8.39	8.31	9.28	8.13		8.82	8.87	8.44
Cunningham, Trey (USA) (1998)		time		2.53	3.59	4.61	5.64	6.67	7.71	8.77	9.85	10.94	12.05		13.57	4 / 9			
reaction time		0.157	interval	7 steps	1.06	1.02	1.03	1.03	1.04	1.06	1.08	1.09	1.11	1.52			3.11	3.13	3.28
			velocity	5.42	8.62	8.96	8.87	8.87	8.79	8.62	8.46	8.39	8.23	9.22	8.11		8.82	8.76	8.36

FINAL

INAL	date		09-Jun-23		wind		1.8 m/s		Henson (2023) - Athlete First: 2023 year end hurdle report									
			H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
Lemonious, Phillip (JAM) (1998)	time		2.50	3.54	4.57	5.57	6.60	7.61	8.65	9.69	10.72	11.76		13.24	5 / 1			
	reaction time	interval	7 steps	1.04	1.03	1.00	1.03	1.01	1.04	1.04	1.03	1.04	1.48			3.07	3.08	3.11
		velocity		5.49	8.79	8.87	9.14	8.87	9.05	8.79	8.79	8.87	8.79	9.47	8.31		8.93	8.90
Wilson, De'vion (USA) (2000)	time		2.50	3.57	4.60	5.64	6.64	7.64	8.65	9.69	10.72	11.78		13.26	2 / 2			
	reaction time	interval	7 steps	1.07	1.03	1.04	1.00	1.00	1.01	1.04	1.03	1.06	1.48	PB		3.14	3.01	3.13
		velocity		5.49	8.54	8.87	8.79	9.14	9.14	9.05	8.79	8.87	8.62	9.47	8.30		8.73	9.11
Hayles, Jaheem (JAM) (2001)	time		2.57	3.62	4.66	5.70	6.71	7.72	8.74	9.77	10.79	11.84		13.28	9 / 3			
	reaction time	interval		1.05	1.04	1.04	1.01	1.01	1.02	1.03	1.02	1.05	1.44	PB		3.13	3.04	3.10
		velocity		5.34	8.70	8.79	8.79	9.05	9.05	8.96	8.87	8.96	8.70	9.74	8.28		8.76	9.02
Ojora, Tade (GBR) (1999)	time		2.57	3.63	4.65	5.67	6.69	7.71	8.74	9.77	10.80	11.86		13.29	3 / 4			
	reaction time	interval	7 steps	1.06	1.02	1.02	1.02	1.02	1.03	1.03	1.03	1.06	1.43	PB		3.10	3.07	3.12
		velocity		5.34	8.62	8.96	8.96	8.96	8.96	8.87	8.87	8.87	8.62	9.80	8.28		8.85	8.93
Roberts, Giano (JAM) (2000)	time		2.50	3.57	4.60	5.63	6.64	7.68	8.72	9.77	10.80	11.83		13.31	7 / 5			
	reaction time	interval		1.07	1.03	1.03	1.01	1.04	1.04	1.05	1.03	1.03	1.48	PB		3.13	3.09	3.11
		velocity		5.49	8.54	8.87	8.87	9.05	8.79	8.79	8.70	8.87	8.87	9.47	8.26		8.76	8.87
Luff, Darius (USA) (2001)	time		2.53	3.57	4.60	5.63	6.64	7.66	8.68	9.72	10.76	11.83		13.38	8 / 6			
	reaction time	interval		1.04	1.03	1.03	1.01	1.02	1.02	1.04	1.04	1.07	1.55			3.10	3.05	3.15
		velocity		5.42	8.79	8.87	8.87	9.05	8.96	8.96	8.79	8.79	8.54	9.05	8.22		8.85	8.99
Shulman, Connor (USA) (2001)	time		2.53	3.57	4.62	5.64	6.66	7.72	8.79	9.84	10.89	11.97		13.47	6 / 7			
	reaction time	interval	7 steps	1.04	1.05	1.02	1.02	1.06	1.07	1.05	1.05	1.08	1.50			3.11	3.15	3.18
		velocity		5.42	8.79	8.70	8.96	8.96	8.62	8.54	8.70	8.70	8.46	9.35	8.17		8.82	8.70

WR=World Rec. AR=Area Rec. OR=Olympic Rec. CR=Championship Rec. NR=National Rec. PB=Personal Best DQ=Disqualified DV=Drug Violation

Brown, Rasheem (CAY) (2000)	time	2.57	3.62	4.67	5.71	6.74	7.78	8.84	9.90	10.96	12.01		13.50	1 / 8			
	reaction time	interval	1.05	1.05	1.04	1.03	1.04	1.06	1.06	1.06	1.05	1.49	PB		3.14	3.13	3.17
	velocity		5.34	8.70	8.70	8.79	8.87	8.79	8.62	8.62	8.62	8.70	9.41	8.15	8.73	8.76	8.65
Murray, Cameron (USA) (1999)	time	2.67											dnf	4 / --			
	reaction time	interval	7 steps														
	velocity		5.14														

2023 Meeting de Paris (Paris, FRA)

FINAL

date 09-Jun-23		wind -0.5 m/s										Omega Timing (2023) - diamond league race analysis					
		H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
Holloway, Grant (USA) (1997)	time	2.30	3.30	4.28	5.25	6.24	7.23	8.24	9.26	10.30	11.36		12.98	6 / 1			
	reaction time 0.157	interval	7 steps	1.00	0.98	0.97	0.99	1.01	1.02	1.04	1.06	1.62			2.95	2.99	3.12
	velocity		5.97	9.14	9.33	9.42	9.23	9.05	8.96	8.79	8.62	8.65	8.47		9.29	9.17	8.79
Kwaou-Mathey, Just (FRA) (1999)	time	2.40	3.45	4.45	5.43	6.43	7.43	8.44	9.46	10.50	11.55		13.09	7 / 2			
	reaction time 0.171	interval	7 steps	1.05	1.00	0.98	1.00	1.01	1.02	1.04	1.05	1.54	PB		3.03	3.01	3.11
	velocity		5.72	8.70	9.14	9.33	9.14	9.05	8.96	8.79	8.70	9.10	8.40		9.05	9.11	8.82
Britt, Jamal (USA) (1998)	time	2.37	3.43	4.46	5.46	6.47	7.47	8.48	9.50	10.52	11.57		13.14	5 / 3			
	reaction time 0.141	interval	7 steps	1.06	1.03	1.00	1.01	1.00	1.02	1.02	1.05	1.57			3.09	3.02	3.09
	velocity		5.79	8.62	8.87	9.14	9.05	9.14	8.96	8.96	8.70	8.93	8.37		8.87	9.08	8.87
Roberts, Daniel (USA) (1998)	time	2.30	3.34	4.34	5.34	6.35	7.39	8.42	9.46	10.49	11.54		13.14	4 / 3			
	reaction time 0.154	interval		1.04	1.00	1.00	1.01	1.04	1.03	1.04	1.05	1.60			3.04	3.08	3.12
	velocity		5.97	8.79	9.14	9.14	9.05	8.79	8.87	8.79	8.87	8.70	8.37		9.02	8.90	8.79
Belocian, Wilhem (FRA) (1995)	time	2.35	3.40	4.42	5.43	6.45	7.47	8.50	9.53	10.57	11.62		13.20	8 / 5			
	reaction time 0.142	interval		1.05	1.02	1.01	1.02	1.03	1.03	1.04	1.05	1.58			3.08	3.07	3.12
	velocity		5.84	8.70	8.96	9.05	8.96	8.96	8.87	8.87	8.79	8.87	8.33		8.90	8.93	8.79
Crittenden, Freddie (USA) (1999)	time	2.37	3.43	4.46	5.48	6.49	7.52	8.55	9.58	10.62	11.68		13.26	2 / 6			
	reaction time 0.136	interval		1.06	1.03	1.02	1.01	1.03	1.03	1.04	1.06	1.58			3.11	3.07	3.13
	velocity		5.79	8.62	8.87	8.96	9.05	8.87	8.87	8.87	8.79	8.62	8.87		8.82	8.93	8.76
Joseph, Jason (SUI) (1998)	time	2.43	3.47	4.49	5.49	6.50	7.51	8.53	9.56	10.61	11.67		13.29	3 / 7			
	reaction time 0.195	interval	7 steps	1.04	1.02	1.00	1.01	1.02	1.03	1.05	1.06	1.62			3.06	3.04	3.14
	velocity		5.65	8.79	8.96	9.14	9.05	8.96	8.87	8.70	8.62	8.65	8.28		8.96	9.02	8.73
Edwards, Eric (USA) (2000)	time	2.32	3.39	4.42	5.45	6.48	7.51	8.55	9.58	10.65	11.72		13.32	1 / 8			
	reaction time 0.130	interval		1.07	1.03	1.03	1.03	1.04	1.03	1.07	1.07	1.60			3.13	3.10	3.17
	velocity		5.91	8.54	8.87	8.87	8.87	8.79	8.87	8.54	8.54	8.76	8.26		8.76	8.85	8.65

Heat 2

date 09-Jun-23		wind -0.6 m/s										Omega Timing (2023) - diamond league race analysis					
		H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
Holloway, Grant (USA) (1997)	time	2.30	3.30	4.30	5.30	6.30	7.30	8.30	9.40	10.40	11.50		13.20	6 / 1			
	reaction time 0.147	interval	7 steps	1.00	1.00	1.00	1.00	1.00	1.10	1.00	1.10	1.70			3.00	3.00	3.20
	velocity		5.97	9.14	9.14	9.14	9.14	9.14	8.31	9.14	8.31	8.25	8.33		9.14	9.14	8.57
Britt, Jamal (USA) (1998)	time	2.37	3.42	4.45	5.46	6.47	7.49	8.53	9.56	10.61	11.66		13.21	4 / 2			
	reaction time 0.136	interval	7 steps	1.05	1.03	1.01	1.01	1.02	1.04	1.03	1.05	1.55			3.09	3.07	3.13
	velocity		5.79	8.70	8.87	9.05	9.05	8.96	8.79	8.87	8.70	9.05	8.33		8.87	8.93	8.76
Kwaou-Mathey, Just (FRA) (1999)	time	2.40	3.46	4.47	5.47	6.47	7.48	8.51	9.55	10.61	11.67		13.25	5 / 3			
	reaction time 0.171	interval	7 steps	1.06	1.01	1.00	1.01	1.03	1.04	1.06	1.06	1.58	PB		3.07	3.04	3.16
	velocity		5.72	8.62	9.05	9.14	9.14	9.05	8.87	8.79	8.62	8.62	8.87		8.93	9.02	8.68
Crittenden, Freddie (USA) (1999)	time	2.36	3.43	4.46	5.47	6.48	7.50	8.54	9.59	10.66	11.73		13.34	7 / 4			
	reaction time 0.136	interval		1.07	1.03	1.01	1.01	1.02	1.04	1.05	1.07	1.61			3.11	3.07	3.19
	velocity		5.81	8.54	8.87	9.05	9.05	8.96	8.79	8.70	8.54	8.54	8.71		8.82	8.93	8.60
Martinot-Lagarde, Pascal (FRA) (1999)	time	2.42	3.49	4.52	5.56	6.59	7.63	8.68	9.73	10.79	11.85		13.45	3 / 5			
	reaction time 0.156	interval	7 steps	1.07	1.03	1.04	1.03	1.04	1.05	1.05	1.06	1.60			3.14	3.12	3.17
	velocity		5.67	8.54	8.87	8.79	8.87	8.79	8.70	8.70	8.62	8.62	8.76		8.73	8.79	8.65
Alkana, Antonio (RSA) (1990)	time	2.39	3.46	4.50	5.52	6.54	7.57	8.62	9.69	10.78	11.88		13.52	2 / 6			
	reaction time 0.114	interval		1.07	1.04	1.02	1.02	1.03	1.05	1.07	1.09	1.64			3.13	3.10	3.26
	velocity		5.74	8.54	8.79	8.96	8.96	8.87	8.70	8.54	8.39	8.31	8.55		8.76	8.85	8.41
Manga, Aurel (FRA) (1992)	time	2.41	3.48	4.52	5.55	6.58	7.61	8.67	9.76	10.84	11.92		13.55	1 / 7			
	reaction time 0.163	interval	7 steps	1.07	1.04	1.03	1.03	1.06	1.09	1.08	1.08	1.63			3.14	3.12	3.25
	velocity		5.69	8.54	8.79	8.87	8.87	8.62	8.39	8.46	8.46	8.60	8.12		8.73	8.79	8.44
Pereira, Rafael (BRA) (1997)	time	2.42	3.48	4.52	5.55	6.56	7.61	8.65	9.72	10.81	11.90		13.57	8 / 8			
	reaction time 0.149	interval		1.06	1.04	1.03	1.01	1.05	1.04	1.07	1.09	1.67			3.13	3.10	3.25

velocity	5.67	8.62	8.79	8.87	9.05	8.70	8.79	8.54	8.39	8.39	8.40	8.11	8.76	8.85	8.44
----------	------	------	------	------	------	------	------	------	------	------	------	------	------	------	------

Heat 1

date 09-Jun-23 wind -2.0 m/s

Omega Timing (2023) - diamond league race analysis

		H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
Roberts, Daniel (USA) (1998)	time	2.32	3.39	4.42	5.45	6.48	7.51	8.54	9.58	10.63	11.71		13.32	3 / 1			
	reaction time	0.153	interval	1.07	1.03	1.03	1.03	1.03	1.04	1.05	1.08	1.61			3.13	3.09	3.17
	velocity	5.91	8.54	8.87	8.87	8.87	8.87	8.87	8.79	8.70	8.46	8.71	8.26		8.76	8.87	8.65
Joseph, Jason (SUI) (1998)	time	2.47	3.54	4.57	5.59	6.60	7.62	8.65	9.68	10.71	11.78		13.36	4 / 2			
	reaction time	0.201	interval	7 steps	1.07	1.03	1.02	1.01	1.02	1.03	1.03	1.07	1.58		3.12	3.06	3.13
	velocity	5.55	8.54	8.87	8.96	9.05	8.96	8.87	8.87	8.87	8.54	8.87	8.23		8.79	8.96	8.76
Belocian, Wilhem (FRA) (1995)	time	2.36	3.43	4.46	5.49	6.52	7.56	8.60	9.64	10.70	11.78		13.37	6 / 3			
	reaction time	0.129	interval	1.07	1.03	1.03	1.04	1.04	1.04	1.06	1.08	1.59			3.13	3.11	3.18
	velocity	5.81	8.54	8.87	8.87	8.87	8.79	8.79	8.79	8.62	8.46	8.82	8.23		8.76	8.82	8.62
Edwards, Eric (USA) (2000)	time	2.39	3.46	4.50	5.53	6.56	7.60	8.64	9.69	10.75	11.81		13.39	2 / 4			
	reaction time	0.149	interval	1.07	1.04	1.03	1.04	1.04	1.05	1.06	1.06	1.58			3.14	3.11	3.17
	velocity	5.74	8.54	8.79	8.87	8.87	8.79	8.79	8.70	8.62	8.62	8.87	8.22		8.73	8.82	8.65
Iribarne, Roger (CUB) (1996)	time	2.37	3.44	4.47	5.50	6.53	7.58	8.62	9.68	10.75	11.84		13.44	7 / 5			
	reaction time	0.152	interval	7 steps	1.07	1.03	1.03	1.03	1.05	1.04	1.06	1.07	1.09	1.60	3.13	3.12	3.22
	velocity	5.79	8.54	8.87	8.87	8.87	8.70	8.79	8.62	8.54	8.39	8.76	8.18		8.76	8.79	8.52
Mohamed, Raphael (FRA) (199)	time	2.48	3.56	4.62	5.66	6.70	7.75	8.82	9.89	10.97	12.05		13.65	1 / 6			
	reaction time	0.165	interval	7 steps	1.08	1.06	1.04	1.04	1.05	1.07	1.07	1.08	1.08	1.60	3.18	3.16	3.23
	velocity	5.53	8.46	8.62	8.79	8.79	8.70	8.54	8.54	8.46	8.46	8.76	8.06		8.62	8.68	8.49
Bascou, Dimitri (FRA) (1987)	time	2.43	3.49	4.54	5.58	6.61	7.65	8.69	9.74	10.85	11.98		13.66	8 / 7			
	reaction time	0.162	interval	7 steps	1.06	1.05	1.04	1.03	1.04	1.04	1.05	1.11	1.13	1.68	3.15	3.11	3.29
	velocity	5.65	8.62	8.70	8.79	8.87	8.79	8.79	8.70	8.23	8.09	8.35	8.05		8.70	8.82	8.33
Allen, Devon (USA) (1994)	time	2.44	4.48										dnf	5 / --			
	reaction time	0.140	interval	7 steps	2.04												
	velocity	5.62	4.48														

2023 FBK Games (Hengelo, NED) (TV Analysis)

FINAL

date 04-Jun-23 wind 0.4 m/s

Henson (2023) - Athlete First: 2023 year end hurdle report

		H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
Holloway, Grant (USA) (1997)	time	2.53	3.53	4.51	5.49	6.45	7.43	8.45	9.47	10.53	11.60		13.03	6 / 1			
	reaction time	0.187	interval	7 steps	1.00	0.98	0.98	0.96	0.98	1.02	1.02	1.06	1.07	1.43	2.96	2.96	3.15
	velocity	5.42	9.14	9.33	9.33	9.52	9.33	8.96	8.96	8.62	8.54	9.80	8.44		9.26	9.26	8.70
Allen, Devon (USA) (1994)	time	2.56	3.61	4.62	5.63	6.62	7.62	8.64	9.68	10.72	11.78		13.12	5 / 2			
	reaction time	0.137	interval	7 steps	1.05	1.01	1.01	0.99	1.00	1.02	1.04	1.06	1.34		3.07	3.01	3.14
	velocity	5.36	8.70	9.05	9.05	9.23	9.14	8.96	8.79	8.79	8.62	10.46	8.38		8.93	9.11	8.73
Edwards, Eric (USA) (2000)	time	2.56	3.60	4.64	5.67	6.70	7.73	8.76	9.80	10.83	11.93		13.29	3 / 3			
	reaction time	0.129	interval	1.04	1.04	1.03	1.03	1.03	1.04	1.03	1.10	1.36			3.11	3.09	3.17
	velocity	5.36	8.79	8.79	8.87	8.87	8.87	8.87	8.79	8.87	8.31	10.31	8.28		8.82	8.87	8.65
Crittenden, Freddie (USA) (199)	time	2.60	3.65	4.70	5.72	6.74	7.76	8.78	9.80	10.86	11.95		13.33	4 / 4			
	reaction time	0.145	interval	1.05	1.05	1.02	1.02	1.02	1.02	1.06	1.09	1.38			3.12	3.06	3.17
	velocity	5.28	8.70	8.70	8.96	8.96	8.96	8.96	8.96	8.62	8.39	10.16	8.25		8.79	8.96	8.65
Trajkovic, Milan (CYP) (1992)	time	2.58	3.63	4.68	5.70	6.72	7.75	8.78	9.83	10.93	12.03		13.44	2 / 5			
	reaction time	0.117	interval	7 steps	1.05	1.05	1.02	1.02	1.03	1.03	1.05	1.10	1.10	1.41	3.12	3.08	3.25
	velocity	5.32	8.70	8.70	8.96	8.96	8.87	8.87	8.70	8.31	8.31	9.94	8.18		8.79	8.90	8.44
Rodrigues, Eduardo (BRA) (19)	time	2.65	3.74	4.77	5.78	6.79	7.81	8.83	9.93	10.98	12.10		13.51	7 / 6			
	reaction time	0.149	interval	7 steps	1.09	1.03	1.01	1.01	1.02	1.02	1.10	1.05	1.12	1.41	3.13	3.05	3.27
	velocity	5.18	8.39	8.87	9.05	9.05	8.96	8.96	8.31	8.70	8.16	9.94	8.14		8.76	8.99	8.39
Manga, Aurel (FRA) (1992)	time	2.58	3.66	4.72	5.75	6.77	7.82	8.86	9.95	11.03	12.14		13.58	8 / 7			
	reaction time	0.151	interval	7 steps	1.08	1.06	1.03	1.02	1.05	1.04	1.09	1.08	1.11	1.44	3.17	3.11	3.28
	velocity	5.32	8.46	8.62	8.87	8.96	8.70	8.79	8.39	8.46	8.23	9.74	8.10		8.65	8.82	8.36
Koster, Timme (NED) (2002)	time	2.73	3.83	4.93	5.97	7.00	8.06	9.13	10.20	11.27	12.36		13.78	1 / 8			
	reaction time	0.157	interval	1.10	1.10	1.04	1.03	1.06	1.07	1.07	1.07	1.09	1.42		3.24	3.16	3.23
	velocity	5.03	8.31	8.31	8.79	8.87	8.62	8.54	8.54	8.54	8.39	9.87	7.98		8.46	8.68	8.49
Smet, Koen (NED) (1992)	time	2.65	3.73	4.80	5.87	6.96	8.03	9.13	10.20	11.33	12.40		13.85	9 / 9			
	reaction time	0.147	interval	1.08	1.07	1.07	1.09	1.07	1.10	1.07	1.13	1.07	1.45		3.22	3.26	3.27
	velocity	5.18	8.46	8.54	8.54	8.39	8.54	8.31	8.54	8.09	8.54	9.67	7.94		8.52	8.41	8.39

2023 Golden Gala Pietro Mennea (Firenze, ITA) (TV Analysis)

WR=World Rec. AR=Area Rec. OR=Olympic Rec. CR=Championship Rec. NR=National Rec. PB=Personal Best DQ=Disqualified DV=Drug Violation

FINAL

date 02-Jun-23 wind -0.2 m/s

Henson (2023) - Athlete First: 2023 year end hurdle report

			H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
Holloway, Grant (USA) (1997)	time		2.45	3.46	4.44	5.42	6.40	7.41	8.42	9.45	10.51	11.59		13.04	6 / 1			
	reaction time	0.152	interval	7 steps	1.01	0.98	0.98	0.98	1.01	1.01	1.03	1.06	1.08	1.45		2.97	3.00	3.17
	velocity		5.60	9.05	9.33	9.33	9.33	9.05	9.05	8.87	8.62	8.46	9.67	8.44		9.23	9.14	8.65
Joseph, Jason (SUI) (1998)	time		2.53	3.56	4.54	5.52	6.54	7.54	8.54	9.57	10.60	11.66		13.10	3 / 2			
	reaction time	0.164	interval	7 steps	1.03	0.98	0.98	1.02	1.00	1.00	1.03	1.03	1.06	1.44	NR	2.99	3.02	3.12
	velocity		5.42	8.87	9.33	9.33	8.96	9.14	9.14	8.87	8.87	8.62	9.74	8.40		9.17	9.08	8.79
Allen, Devon (USA) (1994)	time		2.56	3.61	4.65	5.65	6.65	7.65	8.66	9.68	10.72	11.76		13.19	5 / 3			
	reaction time	0.136	interval	7 steps	1.05	1.04	1.00	1.00	1.01	1.02	1.04	1.04	1.43			3.09	3.01	3.10
	velocity		5.36	8.70	8.79	9.14	9.14	9.14	9.05	8.96	8.79	8.79	9.80	8.34		8.87	9.11	8.85
Britt, Jamal (USA) (1998)	time		2.48	3.56	4.54	5.56	6.60	7.62	8.65	9.72	10.77	11.83		13.26	7 / 4			
	reaction time	0.134	interval	7 steps	1.08	0.98	1.02	1.04	1.02	1.03	1.07	1.05	1.06	1.43		3.08	3.09	3.18
	velocity		5.53	8.46	9.33	8.96	8.79	8.96	8.87	8.54	8.70	8.62	9.80	8.30		8.90	8.87	8.62
Kwaou-Mathey, Just (FRA) (1999)	time		2.54	3.57	4.60	5.63	6.63	7.66	8.68	9.74	10.78	11.84		13.35	1 / 5			
	reaction time	0.161	interval	7 steps	1.03	1.03	1.03	1.00	1.03	1.02	1.06	1.04	1.06	1.51		3.09	3.05	3.16
	velocity		5.40	8.87	8.87	8.87	9.14	8.87	8.96	8.62	8.79	8.62	9.28	8.24		8.87	8.99	8.68
Crittenden, Freddie (USA) (1999)	time		2.54	3.57	4.58	5.60	6.62	7.66	8.68	9.74	10.80	11.90		13.38	4 / 6			
	reaction time	0.132	interval		1.03	1.01	1.02	1.02	1.04	1.02	1.06	1.06	1.10	1.48		3.06	3.08	3.22
	velocity		5.40	8.87	9.05	8.96	8.96	8.79	8.96	8.62	8.62	8.31	9.47	8.22		8.96	8.90	8.52
Pereira, Rafael (BRA) (1997)	time		2.54	3.58	4.64	5.64	6.66	7.70	8.74	9.82	10.88	11.97		13.47	2 / 7			
	reaction time	0.145	interval		1.04	1.06	1.00	1.02	1.04	1.04	1.06	1.09	1.50			3.10	3.10	3.23
	velocity		5.40	8.79	8.62	9.14	8.96	8.79	8.79	8.46	8.62	8.39	9.35	8.17		8.85	8.85	8.49
Simonelli, Lorenzo Ndele (ITA)	time		2.54	3.58	4.66	5.74	6.78	7.84	8.86	9.92	11.00	12.10		13.57	8 / 8			
	reaction time	0.153	interval	7 steps	1.04	1.08	1.08	1.04	1.06	1.02	1.06	1.08	1.10	1.47	PB	3.20	3.12	3.24
	velocity		5.40	8.79	8.46	8.46	8.79	8.62	8.96	8.62	8.46	8.31	9.54	8.11		8.57	8.79	8.46

2023 Meeting International Mohammed VI d'Athlétisme (Rabat, MAR)

FINAL

date 28-May-23 wind -1.3 m/s

Omega Timing (2023) - diamond league race analysis

			H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
Broadbell, Rasheed (JAM) (200)	time		2.41	3.46	4.47	5.48	6.49	7.50	8.51	9.51	10.53	11.55		13.08	5 / 1			
	reaction time	0.161	interval	7 steps	1.05	1.01	1.01	1.01	1.01	1.00	1.02	1.02	1.53			3.07	3.03	3.04
	velocity		5.69	8.70	9.05	9.05	9.05	9.05	9.05	9.14	8.96	8.96	9.16	8.41		8.93	9.05	9.02
Holloway, Grant (USA) (1997)	time		2.35	3.36	4.34	5.33	6.32	7.34	8.36	9.40	10.45	11.52		13.12	6 / 2			
	reaction time	0.177	interval	7 steps	1.01	0.98	0.99	0.99	1.02	1.02	1.04	1.05	1.07	1.60		2.98	3.03	3.16
	velocity		5.84	9.05	9.33	9.23	9.23	8.96	8.96	8.79	8.70	8.54	8.76	8.38		9.20	9.05	8.68
Parchment, Hansle (JAM) (1999)	time		2.48	3.52	4.52	5.53	6.52	7.52	8.53	9.55	10.58	11.65		13.24	4 / 3			
	reaction time	0.157	interval	7 steps	1.04	1.00	1.01	0.99	1.00	1.01	1.02	1.03	1.07	1.59		3.05	3.00	3.12
	velocity		5.53	8.79	9.14	9.05	9.23	9.14	9.05	8.96	8.87	8.54	8.82	8.31		8.99	9.14	8.79
Allen, Devon (USA) (1994)	time		2.38	3.45	4.47	5.48	6.49	7.51	8.54	9.57	10.62	11.68		13.25	3 / 4			
	reaction time	0.152	interval	7 steps	1.07	1.02	1.01	1.01	1.02	1.03	1.03	1.05	1.06	1.57		3.10	3.06	3.14
	velocity		5.76	8.54	8.96	9.05	9.05	8.96	8.87	8.87	8.70	8.62	8.93	8.30		8.85	8.96	8.73
Crittenden, Freddie (USA) (1999)	time		2.39	3.44	4.49	5.52	6.55	7.57	8.62	9.67	10.73	11.80		13.43	7 / 5			
	reaction time	0.166	interval		1.05	1.05	1.03	1.03	1.02	1.05	1.05	1.06	1.07	1.63		3.13	3.10	3.18
	velocity		5.74	8.70	8.70	8.87	8.87	8.96	8.70	8.70	8.62	8.54	8.60	8.19		8.76	8.85	8.62
Pereira, Rafael (BRA) (1997)	time		2.43	3.52	4.57	5.62	6.66	7.70	8.77	9.82	10.90	12.01		13.68	8 / 6			
	reaction time	0.165	interval		1.09	1.05	1.05	1.04	1.04	1.07	1.05	1.08	1.11	1.67		3.19	3.15	3.24
	velocity		5.65	8.39	8.70	8.70	8.79	8.79	8.54	8.70	8.46	8.23	8.40	8.04		8.60	8.70	8.46
Martinot-Lagarde, Pascal (FRA)	time		2.45	3.53	4.57	5.64	6.68	7.75	8.82	9.89	10.97	12.07		13.69	2 / 7			
	reaction time	0.183	interval	7 steps	1.08	1.04	1.07	1.04	1.07	1.07	1.08	1.10	1.62			3.19	3.18	3.25
	velocity		5.60	8.46	8.79	8.54	8.79	8.54	8.54	8.54	8.46	8.31	8.65	8.04		8.60	8.62	8.44
Koussi, Mohamed (MAR) (1999)	time		2.42	3.49	4.53	5.59	6.64	7.71	8.78	9.85	10.92	12.02		13.70	1 / 8			
	reaction time	0.141	interval		1.07	1.04	1.06	1.05	1.07	1.07	1.07	1.10	1.68			3.17	3.19	3.24
	velocity		5.67	8.54	8.79	8.62	8.70	8.54	8.54	8.54	8.54	8.31	8.35	8.03		8.65	8.60	8.46
Czykier, Damian (POL) (1992)	time		2.48	3.56	4.62	5.67	6.73	7.79	8.86	9.94	11.02	12.14		13.83	9 / 9			
	reaction time	0.193	interval	7 steps	1.08	1.06	1.05	1.06	1.06	1.07	1.08	1.08	1.12	1.69		3.19	3.19	3.28
	velocity		5.53	8.46	8.62	8.70	8.62	8.62	8.54	8.46	8.46	8.16	8.30	7.95		8.60	8.60	8.36

2023 NCAA Division II National Championships (Pueblo, CO) (Altitude) (TV Analysis)

FINAL

date 27-May-23 wind 6.0 m/s

Henson (2023) - Athlete First: 2023 year end hurdle report

Tinch, Cordell (USA) (2000)	time	2.50	3.50	4.50	5.48	6.46	7.44	8.41	9.43	10.43	11.46		12.87	5 / 1			
	reaction time	interval	7 steps	1.00	1.00	0.98	0.98	0.98	0.97	1.02	1.00	1.03	1.41		2.98	2.93	3.05
	velocity			5.49	9.14	9.14	9.33	9.33	9.33	9.42	8.96	9.14	8.87	9.94	8.55	9.20	9.36

2023 Trackwired Arkansas Grand Prix (Fayetteville, AR) (TV Analysis)

FINAL

date 23-May-23 wind 1.3 m/s

Henson (2023) - Athlete First: 2023 year end hurdle report

		H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
Tinch, Cordell (USA) (2000)	time	2.46	3.50	4.50	5.50	6.48	7.46	8.43	9.47	10.47	11.49		12.96	5 / 1			
	reaction time	interval	7 steps	1.04	1.00	1.00	0.98	0.98	0.97	1.04	1.00	1.02	1.47	PB	3.04	2.93	3.06
	velocity	5.58	8.79	9.14	9.14	9.33	9.33	9.42	8.79	9.14	8.96	9.54	8.49		9.02	9.36	8.96
Ojora, Tade (GBR) (1999)	time	2.58	3.65	4.67	5.69	6.69	7.69	8.67	9.70	10.76	11.82		13.26	6 / 2			
	reaction time	interval	7 steps	1.07	1.02	1.02	1.00	1.00	0.98	1.03	1.06	1.06	1.44	PB	3.11	2.98	3.15
	velocity	5.32	8.54	8.96	8.96	9.14	9.14	9.33	8.87	8.62	8.62	9.74	8.30		8.82	9.20	8.70

2023 USATF Bermuda Grand Prix (Devonshire, BER) (TV Analysis)

FINAL

date 21-May-23 wind 4.0 m/s

Henson (2023) - Athlete First: 2023 year end hurdle report

		H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10	
Britt, Jamal (USA) (1998)	time	2.50	3.52	4.53	5.53	6.52	7.52	8.53	9.55	10.60	11.65		12.99	6 / 1				
	reaction time	0.124	interval	7 steps	1.02	1.01	1.00	0.99	1.00	1.01	1.02	1.05	1.05	1.34		3.03	3.00	3.12
			velocity		5.49	8.96	9.05	9.14	9.23	9.14	9.05	8.96	8.70	8.70	10.46	8.47	9.05	9.14
Edwards, Eric (USA) (2000)	time	2.53	3.56	4.56	5.56	6.56	7.56	8.56	9.60	10.64	11.70		13.07	2 / 2				
	reaction time	0.122	interval		1.03	1.00	1.00	1.00	1.00	1.04	1.04	1.06	1.37			3.03	3.00	3.14
			velocity		5.42	8.87	9.14	9.14	9.14	9.14	8.79	8.79	8.62	10.23	8.42	9.05	9.14	8.73
Crittenden, Freddie (USA) (199)	time	2.60	3.65	4.67	5.66	6.66	7.66	8.67	9.68	10.71	11.74		13.13	4 / 3				
	reaction time	0.162	interval		1.05	1.02	0.99	1.00	1.00	1.01	1.01	1.03	1.39			3.06	3.01	3.07
			velocity		5.28	8.70	8.96	9.23	9.14	9.14	9.05	9.05	8.87	8.87	10.09	8.38	8.96	9.11
Mason, Tyler (JAM) (1995)	time	2.56	3.66	4.67	5.66	6.66	7.66	8.68	9.72	10.75	11.86		13.30	3 / 4				
	reaction time	0.155	interval	7 steps	1.10	1.01	0.99	1.00	1.00	1.02	1.04	1.03	1.11	1.44		3.10	3.02	3.18
			velocity		5.36	8.31	9.05	9.23	9.14	9.14	8.96	8.79	8.87	8.23	9.74	8.27	8.85	9.08
Thomas, Damion (JAM) (1999)	time	2.63	3.67	4.70	5.70	6.72	7.73	8.75	9.82	10.87	11.93		13.38	8 / 5				
	reaction time	0.166	interval	7 steps	1.04	1.03	1.00	1.02	1.01	1.02	1.07	1.05	1.06	1.45		3.07	3.05	3.18
			velocity		5.22	8.79	8.87	9.14	8.96	9.05	8.96	8.54	8.70	8.62	9.67	8.22	8.93	8.99
Dickson, Michael (USA) (1997)	time	2.56	3.69	4.73	5.76	6.82	7.87	8.90	9.93	11.02	12.06		13.48	7 / 6				
	reaction time	0.190	interval	7 steps	1.13	1.04	1.03	1.06	1.05	1.03	1.03	1.09	1.04	1.42		3.20	3.14	3.16
			velocity		5.36	8.09	8.79	8.87	8.62	8.70	8.87	8.87	8.39	8.79	9.87	8.16	8.57	8.73
Rollins, Louis (USA) (1996)	time	2.60	3.70	4.75	5.83	6.87	7.92	8.97	10.04	11.13	12.25		13.67	1 / 7				
	reaction time	0.179	interval		1.10	1.05	1.08	1.04	1.05	1.05	1.07	1.09	1.12	1.42		3.23	3.14	3.28
			velocity		5.28	8.31	8.70	8.46	8.79	8.70	8.70	8.54	8.39	8.16	9.87	8.05	8.49	8.73
Dunning, Robert (USA) (1997)	time	2.56	3.62	4.63	5.63	6.66	7.66	8.68	9.76	10.83	12.00		22.28	5 / 8				
	reaction time	0.152	interval	7 steps	1.06	1.01	1.00	1.03	1.00	1.02	1.08	1.07	1.17	10.28		3.07	3.05	3.32
			velocity		5.36	8.62	9.05	9.14	8.87	9.14	8.96	8.46	8.54	7.81	1.36	4.94	8.93	8.99

2023 MIAA Championships (Jefferson City, MO) (TV Analysis)

FINAL

date 07-May-23 wind 3.0 m/s

Henson (2023) - Athlete First: 2023 year end hurdle report

		H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
Tinch, Cordell (USA) (2000)	time	2.40	3.40	4.40	5.37	6.34	7.31	8.34	9.34	10.36	11.40		12.97	4 / 1			
	reaction time	interval	7 steps	1.00	1.00	0.97	0.97	0.97	1.03	1.00	1.02	1.04	1.57		2.97	2.97	3.06
	velocity			5.72	9.14	9.14	9.42	9.42	9.42	8.87	9.14	8.96	8.79	8.93	8.48	9.23	9.23

2023 Adidas Atlanta City Games (Atlanta, GA) (street) (TV Analysis)

FINAL

date 05-May-23 wind 0.0 m/s

Henson (2023) - Athlete First: 2023 year end hurdle report

		H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
Holloway, Grant (USA) (1997)	time	2.50	3.48	4.48	5.48	6.45	7.45	8.47	9.50	10.53	11.60		13.01	3 / 1			
	reaction time 0.181	interval	7 steps	0.98	1.00	1.00	0.97	1.00	1.02	1.03	1.03	1.07	1.41		2.98	2.99	3.13
	velocity			5.49	9.33	9.14	9.14	9.42	9.14	8.96	8.87	8.87	8.54	9.94	8.46	9.20	9.17
Dunning, Robert (USA) (1997)	time	2.57	3.63	4.65	5.65	6.65	7.65	8.65	9.65	10.68	11.73		13.09	1 / 2			
	reaction time 0.187	interval	7 steps	1.06	1.02	1.00	1.00	1.00	1.00	1.00	1.03	1.05	1.36	PB	3.08	3.00	3.08
	velocity			5.34	8.62	8.96	9.14	9.14	9.14	9.14	8.87	8.70	10.31	8.40	8.90	9.14	8.90
Britt, Jamal (USA) (1998)	time	2.52	3.57	4.60	5.60	6.60	7.60	8.62	9.65	10.68	11.73		13.14	4 / 3			
	reaction time 0.188	interval	7 steps	1.05	1.03	1.00	1.00	1.02	1.03	1.03	1.05	1.41			3.08	3.02	3.11
	velocity			5.44	8.70	8.87	9.14	9.14	9.14	8.96	8.87	8.87	8.70	9.94	8.37	8.90	9.08
Cunningham, Trey (USA) (1998)	time	2.58	3.62	4.64	5.66	6.66	7.68	8.73	9.80	10.87	11.98		13.41	2 / 4			
	reaction time 0.174	interval	7 steps	1.04	1.02	1.00	1.02	1.05	1.07	1.07	1.11	1.43			3.08	3.07	3.25
	velocity			5.32	8.79	8.96	8.96	9.14	8.96	8.70	8.54	8.54	8.23	9.80	8.20	8.90	8.93

Crittenden, Freddie (USA) (199)	time	2.62	3.68	4.72	5.75	6.75	7.78	8.80	9.85	10.93	12.01		13.43	5 / 5					
reaction time	0.159	interval		1.06	1.04	1.03	1.00	1.03	1.02	1.05	1.08	1.08	1.42			3.13	3.05	3.21	
		velocity		5.24	8.62	8.79	8.87	9.14	8.87	8.96	8.70	8.46	8.46	9.87	8.19		8.76	8.99	8.54

2022 Weltklasse (Zürich, SUI) (TV Analysis)**FINAL**

date 08-Sep-22 wind -1.0 m/s

Henson (2022) - Athlete First: 2022 year end hurdle report

		H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10	
Holloway, Grant (USA) (1997)	time	2.44	3.44	4.41	5.37	6.34	7.34	8.34	9.38	10.44	11.51		13.02	6 / 1				
	reaction time	0.140	interval	7 steps	1.00	0.97	0.96	0.97	1.00	1.00	1.04	1.06	1.07	1.51		2.93	2.97	3.17
			velocity		5.62	9.14	9.42	9.52	9.42	9.14	9.14	8.79	8.62	8.54	9.28	8.45	9.36	9.23
Broadbell, Rasheed (JAM) (200)	time	2.44	3.47	4.48	5.47	6.45	7.48	8.50	9.52	10.61	11.64		13.06	7 / 2				
	reaction time	0.149	interval	7 steps	1.03	1.01	0.99	0.98	1.03	1.02	1.02	1.09	1.03	1.42		3.03	3.03	3.14
			velocity		5.62	8.87	9.05	9.23	9.33	8.87	8.96	8.96	8.39	8.87	9.87	8.42	9.05	9.05
Parchment, Hansle (JAM) (199)	time	2.53	3.60	4.64	5.64	6.64	7.64	8.65	9.67	10.71	11.75		13.26	5 / 3				
	reaction time	0.153	interval	7 steps	1.07	1.04	1.00	1.00	1.00	1.01	1.02	1.04	1.04	1.51		3.11	3.01	3.10
			velocity		5.42	8.54	8.79	9.14	9.14	9.14	9.05	8.96	8.79	8.79	9.28	8.30	8.82	9.11
Martínez, Asier (ESP) (2000)	time	2.57	3.65	4.68	5.70	6.71	7.74	8.74	9.81	10.84	11.88		13.29	3 / 4				
	reaction time	0.144	interval	7 steps	1.08	1.03	1.02	1.01	1.03	1.00	1.07	1.03	1.04	1.41		3.13	3.04	3.14
			velocity		5.34	8.46	8.87	8.96	9.05	8.87	9.14	8.54	8.87	8.79	9.94	8.28	8.76	9.02
Cunningham, Trey (USA) (199)	time	2.50	3.50	4.50	5.50	6.54	7.54	8.64	9.66	10.74	11.81		13.30	4 / 5				
	reaction time	0.153	interval	7 steps	1.00	1.00	1.00	1.04	1.00	1.10	1.02	1.08	1.07	1.49		3.00	3.14	3.17
			velocity		5.49	9.14	9.14	9.14	8.79	9.14	8.31	8.96	8.46	8.54	9.41	8.27	9.14	8.73
Joseph, Jason (SUI) (1998)	time	2.60	3.64	4.68	5.70	6.74	7.76	8.81	9.88	10.96	12.04		13.54	9 / 6				
	reaction time	0.178	interval	7 steps	1.04	1.04	1.02	1.04	1.02	1.05	1.07	1.08	1.08	1.50		3.10	3.11	3.23
			velocity		5.28	8.79	8.79	8.96	8.79	8.96	8.70	8.54	8.46	8.46	9.35	8.12	8.85	8.82
Czykier, Damian (POL) (1992)	time	2.60	3.67	4.70	5.74	6.77	7.81	8.84	9.88	11.01	12.11		13.65	1 / 7				
	reaction time	0.142	interval	7 steps	1.07	1.03	1.04	1.03	1.04	1.03	1.04	1.13	1.10	1.54		3.14	3.10	3.27
			velocity		5.28	8.54	8.87	8.79	8.87	8.79	8.87	8.79	8.09	8.31	9.10	8.06	8.73	8.85
Kwaou-Mathey, Just (FRA) (19)	time	2.58	3.64	4.70	5.74	6.80	7.84	8.88	9.98	11.01	12.13		13.73	2 / 8				
	reaction time	0.150	interval	7 steps	1.06	1.06	1.04	1.06	1.04	1.10	1.03	1.12	1.60		3.16	3.14	3.25	
			velocity		5.32	8.62	8.62	8.79	8.62	8.79	8.79	8.31	8.87	8.16	8.76	8.01	8.68	8.73
Pereira, Rafael (BRA) (1997)	time	2.50	3.55	4.57	5.60	6.61	7.64	8.65	9.81	11.01	12.12		13.73	8 / 9				
	reaction time	0.130	interval		1.05	1.02	1.03	1.01	1.03	1.01	1.16	1.20	1.11	1.61		3.10	3.05	3.47
			velocity		5.49	8.70	8.96	8.87	9.05	8.87	9.05	7.88	7.62	8.23	8.71	8.01	8.85	8.99

2022 ISTAF (Berlin, GER) (TV Analysis)**FINAL**

date 04-Sep-22 wind 0.6 m/s

Henson (2022) - Athlete First: 2022 year end hurdle report

				H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
Holloway, Grant (USA) (1997)	time			2.42	3.44	4.40	5.36	6.34	7.34	8.36	9.40	10.44	11.56		13.05	4 / 1			
	reaction time	0.140	interval	7 steps	1.02	0.96	0.96	0.98	1.00	1.02	1.04	1.04	1.12	1.49			2.94	3.00	3.20
			velocity		5.67	8.96	9.52	9.52	9.33	9.14	8.96	8.79	8.79	9.41	8.43		9.33	9.14	8.57
Crittenden, Freddie (USA) (199)	time			2.48	3.50	4.50	5.50	6.50	7.52	8.52	9.54	10.58	11.62		13.07	5 / 2			
	reaction time	0.142	interval		1.02	1.00	1.00	1.00	1.02	1.00	1.02	1.04	1.04	1.45			3.02	3.02	3.10
			velocity		5.53	8.96	9.14	9.14	9.14	8.96	9.14	8.96	8.79	9.67	8.42		9.08	9.08	8.85
Roberts, Daniel (USA) (1998)	time			2.50	3.52	4.54	5.56	6.56	7.60	8.62	9.66	10.70	11.78		13.25	6 / 3			
	reaction time	0.145	interval		1.02	1.02	1.02	1.00	1.04	1.02	1.04	1.04	1.08	1.47			3.06	3.06	3.16
			velocity		5.49	8.96	8.96	8.96	9.14	8.79	8.96	8.79	8.79	8.46	9.54	8.30		8.96	8.96
Izumiya, Shunsuke (JPN) (2000)	time			2.50	3.54	4.58	5.60	6.62	7.66	8.68	9.74	10.78	11.88		13.26	8 / 4			
	reaction time	0.129	interval	7 steps	1.04	1.04	1.02	1.02	1.04	1.02	1.06	1.04	1.10	1.38			3.10	3.08	3.20
			velocity		5.49	8.79	8.79	8.96	8.96	8.79	8.96	8.62	8.79	8.31	10.16	8.30		8.85	8.90

2022 Athletissima (Lausanne, SUI) (TV Analysis)**FINAL**

date 26-Aug-22 wind 0.0 m/s

Henson (2022) - Athlete First: 2022 year end hurdle report

				H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10	
Broadbell, Rasheed (JAM) (200)	time			2.47	3.50	4.50	5.50	6.50	7.52	8.51	9.52	10.54	11.58		12.99	5 / 1				
	reaction time	0.142	interval	7 steps	1.03	1.00	1.00	1.00	1.02	0.99	1.01	1.02	1.04	1.41	PB			3.03	3.01	3.07
			velocity		5.55	8.87	9.14	9.14	9.14	8.96	9.23	9.05	8.96	8.79	9.94	8.47		9.05	9.11	8.93
Cunningham, Trey (USA) (199)	time			2.47	3.44	4.44	5.42	6.38	7.38	8.40	9.46	10.52	11.62		13.10	3 / 2				
	reaction time	0.140	interval	7 steps	0.97	1.00	0.98	0.96	1.00	1.02	1.06	1.06	1.10	1.48				2.95	2.98	3.22
			velocity		5.55	9.42	9.14	9.33	9.52	9.14	8.96	8.62	8.62	8.31	9.47	8.40		9.29	9.20	8.52
Holloway, Grant (USA) (1997)	time			2.50	3.54	4.54	5.54	6.52	7.54	8.58	9.61	10.64	11.69		13.11	6 / 3				
	reaction time	0.122	interval	7 steps	1.04	1.00	1.00	0.98	1.02	1.04	1.03	1.03	1.05	1.42				3.04	3.04	3.11
			velocity		5.49	8.79	9.14	9.14	9.33	8.96	8.79	8.87	8.87	8.70	9.87	8.39		9.02	9.02	8.82

Parchment, Hansle (JAM) (1996)	time	2.60	3.64	4.64	5.64	6.64	7.64	8.64	9.64	10.67	11.71		13.13	4 / 4			
	reaction time	0.151	interval	7 steps	1.04	1.00	1.00	1.00	1.00	1.00	1.03	1.04	1.42		3.04	3.00	3.07
	velocity	5.28	8.79	9.14	9.14	9.14	9.14	9.14	9.14	8.87	8.79	9.87	8.38		9.02	9.14	8.93
Pereira, Rafael (BRA) (1997)	time	2.57	3.60	4.67	5.67	6.71	7.74	8.81	9.88	10.95	12.03		13.49	1 / 5			
	reaction time	0.140	interval		1.03	1.07	1.00	1.04	1.03	1.07	1.07	1.08	1.46		3.10	3.14	3.22
	velocity	5.34	8.87	8.54	9.14	8.79	8.87	8.54	8.54	8.54	8.46	9.60	8.15		8.85	8.73	8.52
Czykier, Damian (POL) (1992)	time	2.64	3.70	4.74	5.74	6.77	7.81	8.87	9.94	11.01	12.11		13.58	8 / 6			
	reaction time	0.166	interval	7 steps	1.06	1.04	1.00	1.03	1.04	1.06	1.07	1.10	1.47		3.10	3.13	3.24
	velocity	5.20	8.62	8.79	9.14	8.87	8.79	8.62	8.54	8.54	8.31	9.54	8.10		8.85	8.76	8.46
Martinot-Lagarde, Pascal (FRA) (1999)	time	2.60	3.65	4.70	5.74	6.76	7.81	8.84	9.91	10.99	12.09		13.58	7 / 7			
	reaction time	0.155	interval	7 steps	1.05	1.05	1.04	1.02	1.05	1.03	1.07	1.08	1.10	1.49	3.14	3.10	3.25
	velocity	5.28	8.70	8.70	8.79	8.96	8.70	8.87	8.54	8.46	8.31	9.41	8.10		8.73	8.85	8.44
Joseph, Jason (SUI) (1998)	time	2.58	3.65	4.72	5.77	6.81	7.84	8.91	9.98	11.08	12.18		13.66	2 / 8			
	reaction time	0.158	interval	7 steps	1.07	1.07	1.05	1.04	1.03	1.07	1.10	1.10	1.48		3.19	3.14	3.27
	velocity	5.32	8.54	8.54	8.70	8.79	8.87	8.54	8.54	8.31	8.31	9.47	8.05		8.60	8.73	8.39

2022 NACAC Championships (Freeport, BAH) (TV Analysis)

FINAL

	date	20-Aug-22	wind	0.3 m/s	Henson (2022) - Athlete First: 2022 year end hurdle report																
					H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10	
Crittenden, Freddie (USA) (1996)	time		2.43	3.45	4.44	5.43	6.40	7.40	8.40	9.43	10.47	11.52		13.00	4 / 1						
	reaction time			interval		1.02	0.99	0.99	0.97	1.00	1.00	1.03	1.04	1.05	1.48	PB			3.00	2.97	3.12
	velocity		5.65	8.96	9.23	9.23	9.42	9.14	9.14	8.87	8.79	8.70	9.47	8.46					9.14	9.23	8.79
Britt, Jamal (USA) (1998)	time		2.48	3.53	4.53	5.55	6.55	7.55	8.57	9.60	10.62	11.67		13.08	7 / 2						
	reaction time			interval	7 steps	1.05	1.00	1.02	1.00	1.00	1.02	1.03	1.02	1.05	1.41	PB			3.07	3.02	3.10
	velocity		5.53	8.70	9.14	8.96	9.14	9.14	8.96	8.87	8.96	8.70	9.94	8.41					8.93	9.08	8.85
Bennett, Orlando (JAM) (1999)	time		2.47	3.53	4.56	5.55	6.55	7.55	8.57	9.62	10.64	11.69		13.18	6 / 3						
	reaction time			interval		1.06	1.03	0.99	1.00	1.00	1.02	1.05	1.02	1.05	1.49	PB			3.08	3.02	3.12
	velocity		5.55	8.62	8.87	9.23	9.14	9.14	8.96	8.70	8.96	8.70	9.41	8.35					8.90	9.08	8.79
Brathwaite, Shane (BAR) (1990)	time		2.50	3.56	4.56	5.57	6.60	7.60	8.63	9.70	10.76	11.83		13.42	5 / 4						
	reaction time			interval		1.06	1.00	1.01	1.03	1.00	1.03	1.07	1.06	1.07	1.59				3.07	3.06	3.20
	velocity		5.49	8.62	9.14	9.05	8.87	9.14	8.87	8.54	8.62	8.54	8.82	8.20					8.93	8.96	8.57

2022 European Athletics Championships (Munich, GER) (TV Analysis)

FINAL

INAL	date		17-Aug-22		wind -0.2 m/s										Henson (2022) - Athlete First: 2022 year end hurdle report						
			H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10			
Martínez, Asier (ESP) (2000)	time		2.56	3.60	4.60	5.60	6.62	7.62	8.64	9.66	10.68	11.74		13.14	4 / 1						
	reaction time	0.149	interval	7 steps	1.04	1.00	1.00	1.02	1.00	1.02	1.02	1.02	1.06	1.40	PB		3.04	3.04	3.10		
			velocity	5.36	8.79	9.14	9.14	8.96	9.14	8.96	8.96	8.96	8.62	10.01	8.37		9.02	9.02	8.85		
Martinot-Lagarde, Pascal (FRA)	time		2.52	3.54	4.58	5.58	6.56	7.58	8.60	9.62	10.64	11.70		13.14	5 / 2						
	reaction time	0.167	interval	7 steps	1.02	1.04	1.00	0.98	1.02	1.02	1.02	1.02	1.06	1.44		3.06	3.02	3.10			
			velocity	5.44	8.96	8.79	9.14	9.33	8.96	8.96	8.96	8.96	8.62	9.74	8.37		8.96	9.08	8.85		
Kwaou-Mathey, Just (FRA) (1999)	time		2.56	3.58	4.58	5.58	6.58	7.58	8.64	9.70	10.76	11.86		13.33	3 / 3						
	reaction time	0.178	interval	7 steps	1.02	1.00	1.00	1.00	1.00	1.06	1.06	1.06	1.10	1.47		3.02	3.06	3.22			
			velocity	5.36	8.96	9.14	9.14	9.14	9.14	8.62	8.62	8.62	8.31	9.54	8.25		9.08	8.96	8.52		
Joseph, Jason (SUI) (1998)	time		2.52	3.58	4.62	5.66	6.68	7.72	8.76	9.80	10.84	11.88		13.35	8 / 4						
	reaction time	0.169	interval	7 steps	1.06	1.04	1.04	1.02	1.04	1.04	1.04	1.04	1.04	1.47		3.14	3.10	3.12			
			velocity	5.44	8.62	8.79	8.79	8.96	8.79	8.79	8.79	8.79	8.79	9.54	8.24		8.73	8.85	8.79		
Gaio, Finley (SUI) (1999)	time		2.58	3.64	4.70	5.72	6.74	7.78	8.82	9.86	10.94	12.02		13.50	2 / 5						
	reaction time	0.172	interval	7 steps	1.06	1.06	1.02	1.02	1.04	1.04	1.04	1.08	1.08	1.48		3.14	3.10	3.20			
			velocity	5.32	8.62	8.62	8.96	8.96	8.79	8.79	8.79	8.46	8.46	9.47	8.15		8.73	8.85	8.57		
Pozzi, Andy (GBR) (1992)	time		2.52	3.58	4.64	5.68	6.72	7.76	8.82	9.92	11.02	12.14		13.66	1 / 6						
	reaction time	0.166	interval	7 steps	1.06	1.06	1.04	1.04	1.04	1.06	1.10	1.10	1.12	1.52		3.16	3.14	3.32			
			velocity	5.44	8.62	8.62	8.79	8.79	8.79	8.62	8.31	8.31	8.16	9.22	8.05		8.68	8.73	8.26		
Llopi, Enrique (ESP) (2000)	time		2.56	3.64	4.84	6.00	7.20	8.36	9.52	10.68	11.84	13.06		14.81	6 / 7						
	reaction time	0.145	interval	7 steps	1.08	1.20	1.16	1.20	1.16	1.16	1.16	1.22	1.75			3.44	3.52	3.54			
			velocity	5.36	8.46	7.62	7.88	7.62	7.88	7.88	7.88	7.88	7.49	8.01	7.43		7.97	7.79	7.75		
Zhoya, Sasha (FRA) (2002)	time		2.52	3.58	4.62	5.64	6.66	7.68	8.72	9.78	10.86	11.92		16.51	7 / 8						
	reaction time	0.153	interval	7 steps	1.06	1.04	1.02	1.02	1.04	1.06	1.08	1.06	4.59			3.12	3.08	3.20			
			velocity	5.44	8.62	8.79	8.96	8.96	8.96	8.79	8.62	8.46	8.62	3.05	6.66		8.79	8.90	8.57		

2022 Herculis Meeting International d'Athlétisme (Monaco, MON) (TV Analysis)

FINAL

date 10-Aug-22 wind 0.6 m/s

Henson (2022) - Athlete First: 2022 year end hurdle report

			H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
Holloway, Grant (USA) (1997)	time		2.40	3.40	4.36	5.33	6.30	7.30	8.30	9.30	10.36	11.43		12.99	6 / 1			
	reaction time	0.148	interval	7 steps	1.00	0.96	0.97	0.97	1.00	1.00	1.00	1.06	1.07	1.56		2.93	2.97	3.13
	velocity		5.72	9.14	9.52	9.42	9.42	9.14	9.14	9.14	8.62	8.54	8.99	8.47		9.36	9.23	8.76
Cunningham, Trey (USA) (1996)	time		2.43	3.43	4.40	5.40	6.40	7.43	8.46	9.46	10.53	11.60		13.03	4 / 2			
	reaction time	0.141	interval	7 steps	1.00	0.97	1.00	1.00	1.03	1.03	1.00	1.07	1.07	1.43		2.97	3.06	3.14
	velocity		5.65	9.14	9.42	9.14	9.14	8.87	8.87	9.14	8.54	8.54	9.80	8.44		9.23	8.96	8.73
Parchment, Hansle (JAM) (1996)	time		2.50	3.53	4.53	5.53	6.53	7.53	8.53	9.56	10.56	11.60		13.08	5 / 3			
	reaction time	0.128	interval	7 steps	1.03	1.00	1.00	1.00	1.00	1.03	1.00	1.04	1.48			3.03	3.00	3.07
	velocity		5.49	8.87	9.14	9.14	9.14	9.14	8.87	9.14	8.79	8.79	9.47	8.41		9.05	9.14	8.93
Roberts, Daniel (USA) (1998)	time		2.43	3.46	4.50	5.50	6.53	7.53	8.56	9.56	10.63	11.73		13.20	3 / 4			
	reaction time	0.157	interval		1.03	1.04	1.00	1.03	1.00	1.03	1.00	1.07	1.10	1.47		3.07	3.06	3.17
	velocity		5.65	8.87	8.79	9.14	8.87	9.14	8.87	9.14	8.54	8.31	9.54	8.33		8.93	8.96	8.65
Zhoya, Sasha (FRA) (2002)	time		2.50	3.53	4.56	5.56	6.56	7.56	8.60	9.63	10.66	11.73		13.21	2 / 5			
	reaction time	0.138	interval	7 steps	1.03	1.03	1.00	1.00	1.04	1.03	1.03	1.07	1.48			3.06	3.04	3.13
	velocity		5.49	8.87	8.87	9.14	9.14	9.14	8.79	8.87	8.87	8.54	9.47	8.33		8.96	9.02	8.76
Martinot-Lagarde, Pascal (FRA)	time		2.46	3.53	4.53	5.53	6.53	7.56	8.60	9.63	10.66	11.76		13.26	8 / 6			
	reaction time	0.137	interval	7 steps	1.07	1.00	1.00	1.00	1.03	1.04	1.03	1.10	1.50			3.07	3.07	3.16
	velocity		5.58	8.54	9.14	9.14	9.14	8.87	8.79	8.87	8.87	8.31	9.35	8.30		8.93	8.93	8.68
Czykier, Damian (POL) (1992)	time		2.53	3.56	4.56	5.60	6.63	7.66	8.73	9.76	10.83	11.93		13.46	7 / 7			
	reaction time	0.156	interval	7 steps	1.03	1.00	1.04	1.03	1.03	1.07	1.03	1.07	1.10	1.53		3.07	3.13	3.20
	velocity		5.42	8.87	9.14	8.79	8.87	8.87	8.54	8.87	8.54	8.31	9.16	8.17		8.93	8.76	8.57
Kwaou-Mathey, Just (FRA) (1996)	time		2.53	3.63	4.73	5.73	6.76	7.83	8.90	10.00	11.06	12.16		13.78	1 / 8			
	reaction time	0.144	interval	7 steps	1.10	1.10	1.00	1.03	1.07	1.07	1.10	1.06	1.10	1.62		3.20	3.17	3.26
	velocity		5.42	8.31	8.31	9.14	8.87	8.54	8.54	8.31	8.62	8.31	8.65	7.98		8.57	8.65	8.41

2022 Japanese National High School Championships (Naruto, JPN)

FINAL

date 07-Aug-22 wind -3.8 m/s

Kishima (2022) - national high school championships biomechanics data collection

			H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
Konno, Ryoma (JPN) (2004)	time		2.69	3.82	4.96	6.08	7.20	8.32	9.47	10.66	11.82	13.06		14.57	4 / 1			
	reaction time	0.147	interval		1.13	1.14	1.12	1.12	1.15	1.19	1.16	1.24	1.51			3.39	3.39	3.59
	velocity		5.10	8.09	8.02	8.16	8.16	8.16	7.95	7.68	7.88	7.37	9.28	7.55		8.09	8.09	7.64
Sotono, Haru (JPN) (2004)	time		2.66	3.80	4.94	6.08	7.19	8.35	9.51	10.68	11.89	13.09		14.61	6 / 2			
	reaction time	0.150	interval		1.14	1.14	1.14	1.11	1.16	1.17	1.21	1.20	1.52	PB		3.42	3.43	3.58
	velocity		5.16	8.02	8.02	8.02	8.23	7.88	7.88	7.81	7.55	7.62	9.22	7.53		8.02	7.99	7.66
Imanishi, Ryota (JPN) (2004)	time		2.74	3.91	5.08	6.21	7.39	8.57	9.77	10.98	12.17	13.36		14.86	9 / 3			
	reaction time	0.144	interval		1.17	1.17	1.13	1.18	1.18	1.20	1.21	1.19	1.19	1.50		3.47	3.56	3.59
	velocity		5.01	7.81	7.81	8.09	7.75	7.75	7.62	7.55	7.68	7.68	9.35	7.40		7.90	7.70	7.64
Uchida, Kaisei (JPN) (2004)	time		2.72	3.90	5.07	6.24	7.42	8.58	9.78	10.98	12.18	13.39		14.87	5 / 4			
	reaction time	0.159	interval		1.18	1.17	1.17	1.18	1.16	1.20	1.20	1.21	1.48			3.52	3.54	3.61
	velocity		5.04	7.75	7.81	7.81	7.75	7.88	7.62	7.62	7.62	7.55	9.47	7.40		7.79	7.75	7.60
Miyata, Ryusei (JPN) (2004)	time		2.70	3.85	5.02	6.16	7.30	8.47	9.66	10.92	12.12	13.36		14.92	8 / 5			
	reaction time	0.159	interval		1.15	1.17	1.14	1.14	1.17	1.19	1.26	1.20	1.24	1.56		3.46	3.50	3.70
	velocity		5.08	7.95	7.81	8.02	8.02	7.81	7.68	7.25	7.62	7.37	8.99	7.37		7.92	7.83	7.41
Takeuchi, Tomoki (JPN) (2004)	time		2.70	3.86	5.02	6.16	7.34	8.52	9.70	10.92	12.17	13.45		14.96	2 / 6			
	reaction time	0.157	interval		1.16	1.16	1.14	1.18	1.18	1.22	1.25	1.28	1.51			3.46	3.54	3.75
	velocity		5.08	7.88	7.88	8.02	7.75	7.75	7.75	7.49	7.31	7.14	9.28	7.35		7.92	7.75	7.31
Asai, Seira (JPN) (2006)	time		2.74	3.88	5.03	6.19	7.35	8.50	9.70	10.92	12.17	13.40		14.97	3 / 7			
	reaction time	0.145	interval		1.14	1.15	1.16	1.16	1.15	1.20	1.22	1.23	1.57			3.45	3.51	3.70
	velocity		5.01	8.02	7.95	7.88	7.88	7.95	7.62	7.49	7.31	7.43	8.93	7.35		7.95	7.81	7.41
Yoneyama, Aiki (JPN) (2004)	time		2.69	3.84	4.99	6.12	7.26	8.38	9.65	10.92	12.22	13.47		15.00	7 / 8			
	reaction time	0.124	interval		1.15	1.15	1.13	1.14	1.12	1.27	1.30	1.25	1.53			3.43	3.53	3.82
	velocity		5.10	7.95	7.95	8.09	8.02	8.16	7.20	7.20	7.03	7.31	9.16	7.33		7.99	7.77	7.18

2022 World Athletics Championships (Eugene, OR)

FINAL

date 17-Jul-22 wind 1.2 m/s

Shibayama (2023) - race analysis of the men's 110m hurdles in the 2022 season

Henson (2022) - Athlete First: 2022 year end hurdle report (TV Analysis)

			H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
Holloway, Grant (USA) (1997)	time		2.44	3.43	4.41	5.38	6.38	7.38	8.38	9.38	10.44	11.54		13.03	4 / 1			
	reaction time	0.124	interval	7 steps	0.99	0.98	0.97	1.00	1.00	1.00	1.06	1.10	1.49			2.94	3.00	3.16
	velocity		5.62	9.23	9.33	9.42	9.14	9.14	9.14	9.14	8.62	8.31	9.41	8.44		9.33	9.14	8.68

Cunningham, Trey (USA) (1998)	time	2.52	3.53	4.53	5.54	6.53	7.58	8.60	9.63	10.65	11.72		13.08	6 / 2	Shibayama (2023)			
	reaction time	0.109	interval	7 steps	1.01	1.00	1.01	0.99	1.05	1.02	1.03	1.02	1.07	1.36		3.02	3.06	3.12
	velocity	5.44	9.05	9.14	9.05	9.23	8.70	8.96	8.87	8.96	8.54	10.31	8.41		9.08	8.96	8.79	
Martínez, Asier (ESP) (2000)	time	2.62	3.70	4.72	5.73	6.73	7.74	8.75	9.75	10.78	11.83		13.17	8 / 3	Shibayama (2023)			
	reaction time	0.126	interval	7 steps	1.08	1.02	1.01	1.00	1.01	1.01	1.00	1.03	1.05	1.34	PB	3.11	3.02	3.08
	velocity	5.24	8.46	8.96	9.05	9.14	9.05	9.05	9.14	8.87	8.70	10.46	8.35		8.82	9.08	8.90	
Czykier, Damian (POL) (1992)	time	2.61	3.64	4.64	5.64	6.68	7.68	8.71	9.75	10.81	11.89		13.32	2 / 4	Shibayama (2023)			
	reaction time	0.140	interval	7 steps	1.03	1.00	1.00	1.04	1.00	1.03	1.04	1.06	1.08	1.43		3.03	3.07	3.18
	velocity	5.26	8.87	9.14	9.14	8.79	9.14	8.87	8.79	8.62	8.46	9.80	8.26		9.05	8.93	8.62	
Zeller, Joshua (GBR) (2000)	time	2.61	3.67	4.69	5.71	6.72	7.73	8.75	9.78	10.82	11.89		13.33	7 / 5	Shibayama (2023)			
	reaction time	0.145	interval		1.06	1.02	1.02	1.01	1.02	1.03	1.04	1.07	1.44		3.10	3.04	3.14	
	velocity	5.26	8.62	8.96	8.96	9.05	9.05	8.96	8.87	8.79	8.54	9.74	8.25		8.85	9.02	8.73	
Brathwaite, Shane (BAR) (1990)	time	2.48	3.56	4.60	5.60	6.63	7.65						dnf	1 / --	Henson (2022)			
	reaction time	0.108	interval	7 steps	1.08	1.04	1.00	1.03	1.02				DQ		3.12			
	velocity	5.53	8.46	8.79	9.14	8.87	8.96								8.79			
Allen, Devon (USA) (1994)	reaction time		0.099										DQ	3 / --				
Parchment, Hansle (JAM) (1990)													dns	5 / --				

Shibayama (2023) - race analysis of the men's 110m hurdles in the 2022 season

Semi-Final 3

date 17-Jul-22 wind 2.5 m/s

Henson (2022) - Athlete First: 2022 year end hurdle report (TV Analysis)

		date 17-Jul-22		wind 2.0 m/s		Henson (2022) - All race first 2022 year end hurdle report (TV Analysis)												
		H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10	
Parchment, Hansle (JAM) (1990)	time	2.57	3.60	4.60	5.57	6.53	7.55	8.53	9.53	10.55	11.58		13.02	3 / 1	Shibayama (2023)			
	reaction time	0.128	interval	7 steps	1.03	1.00	0.97	0.96	1.02	0.98	1.00	1.02	1.03	1.44		3.00	2.96	3.05
	velocity	5.34	8.87	9.14	9.42	9.52	8.96	9.33	9.14	8.96	8.87	9.74	8.45		9.14	9.26	8.99	
Allen, Devon (USA) (1994)	time	2.60	3.64	4.65	5.65	6.63	7.62	8.63	9.63	10.67	11.72		13.09	5 / 2	Shibayama (2023)			
	reaction time	0.101	interval	7 steps	1.04	1.01	1.00	0.98	0.99	1.01	1.00	1.04	1.05	1.37		3.05	2.98	3.09
	velocity	5.28	8.79	9.05	9.14	9.33	9.23	9.05	9.14	8.79	8.70	10.23	8.40		8.99	9.20	8.87	
Brathwaite, Shane (BAR) (1990)	time	2.55	3.58	4.60	5.60	6.59	7.60	8.63	9.65	10.71	11.76		13.21	8 / 3	Shibayama (2023)			
	reaction time	0.115	interval	7 steps	1.03	1.02	1.00	0.99	1.01	1.03	1.02	1.06	1.05	1.45		3.05	3.03	3.13
	velocity	5.38	8.87	8.96	9.14	9.23	9.05	8.87	8.96	8.62	8.70	9.67	8.33		8.99	9.05	8.76	
Czykier, Damian (POL) (1992)	time	2.56	3.58	4.60	5.60	6.60	7.58	8.60	9.63	10.70	11.78		13.22	6 / 4	Henson (2022)			
	reaction time	0.135	interval	7 steps	1.02	1.02	1.00	0.98	1.02	1.03	1.07	1.08	1.44		3.04	3.00	3.18	
	velocity	5.36	8.96	8.96	9.14	9.14	9.33	8.96	8.87	8.54	8.46	9.74	8.32		9.02	9.14	8.62	
Kwaou-Mathey, Just (FRA) (1990)	time	2.60	3.63	4.65	5.65	6.67	7.67	8.67	9.70	10.76	11.83		13.25	4 / 5	Henson (2022)			
	reaction time	0.148	interval	7 steps	1.03	1.02	1.00	1.02	1.00	1.03	1.06	1.07	1.42		3.05	3.02	3.16	
	velocity	5.28	8.87	8.96	9.14	8.96	9.14	9.14	8.87	8.62	8.54	9.87	8.30		8.99	9.08	8.68	
Pozzi, Andy (GBR) (1992)	time	2.56	3.56	4.60	5.60	6.60	7.60	8.63	9.66	10.76	11.83		13.35	7 / 6	Henson (2022)			
	reaction time	0.141	interval	7 steps	1.00	1.04	1.00	1.00	1.03	1.03	1.10	1.07	1.52		3.04	3.03	3.20	
	velocity	5.36	9.14	8.79	9.14	9.14	9.14	8.87	8.87	8.31	8.54	9.22	8.24		9.02	9.05	8.57	
Hough, Nicholas (AUS) (1993)	time	2.63	3.70	4.73	5.76	6.76	7.80	8.83	9.86	10.90	11.96		13.42	1 / 7	Henson (2022)			
	reaction time	0.131	interval	7 steps	1.07	1.03	1.03	1.00	1.04	1.03	1.03	1.04	1.06	1.46		3.13	3.07	3.13
	velocity	5.22	8.54	8.87	8.87	9.14	8.79	8.87	8.87	8.79	8.62	9.60	8.20		8.76	8.93	8.76	
Llopis, Enrique (ESP) (2000)	time	2.56	3.60	4.60	5.63	6.63	7.66	8.70	9.80	10.86	11.96		13.44	2 / 8	Henson (2022)			
	reaction time	0.113	interval	7 steps	1.04	1.00	1.03	1.00	1.03	1.04	1.10	1.06	1.10	1.48		3.07	3.07	3.26
	velocity	5.36	8.79	9.14	8.87	9.14	8.87	8.79	8.31	8.62	8.31	9.47	8.18		8.93	8.93	8.41	

Shibayama (2023) - race analysis of the men's 110m hurdles in the 2022 season

Semi-Final 2

date 17-Jul-22 wind 0.3 m/s

Henson (2022) - Athlete First: 2022 year end hurdle report (TV Analysis)

			H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
Cunningham, Trey (USA) (1998)	time		2.48	3.50	4.50	5.50	6.50	7.53	8.53	9.56	10.60	11.66		13.07	3 / 1		Henson (2022)	
	reaction time	0.120	interval	7 steps	1.02	1.00	1.00	1.00	1.03	1.00	1.03	1.04	1.06	1.41		3.02	3.03	3.13
			velocity	5.53	8.96	9.14	9.14	9.14	8.87	9.14	8.87	8.79	8.62	9.94	8.42		9.08	9.05
Martínez, Asier (ESP) (2000)	time		2.66	3.70	4.74	5.76	6.80	7.80	8.83	9.83	10.83	11.87		13.26	6 / 2		Henson (2022)	
	reaction time	0.141	interval	7 steps	1.04	1.04	1.02	1.04	1.00	1.03	1.00	1.00	1.04	1.39		3.10	3.07	3.04
			velocity	5.16	8.79	8.79	8.96	8.79	9.14	8.87	9.14	9.14	8.79	10.09	8.30		8.85	8.93
Broadbell, Rasheed (JAM) (2000)	time		2.50	3.53	4.60	5.66	6.66	7.66	8.70	9.73	10.83	11.87		13.27	4 / 3		Henson (2022)	
	reaction time	0.129	interval	7 steps	1.03	1.07	1.06	1.00	1.00	1.04	1.03	1.10	1.04	1.40		3.16	3.04	3.17
			velocity	5.49	8.87	8.54	8.62	9.14	9.14	8.79	8.87	8.31	8.79	10.01	8.29		8.68	9.02
Martinot-Lagarde, Pascal (FRA) (1990)	time		2.50	3.56	4.60	5.63	6.63	7.66	8.70	9.76	10.83	11.90		13.40	1 / 4		Henson (2022)	
	reaction time	0.129	interval	7 steps	1.06	1.04	1.03	1.00	1.03	1.04	1.06	1.07	1.07	1.50		3.13	3.07	3.20
			velocity	5.49	8.62	8.79	8.87	9.14	8.87	8.79	8.62	8.54	8.54	9.35	8.21		8.76	8.93
Izumiya, Shunsuke (JPN) (2000)	time		2.59	3.65	4.68	5.72	6.77	7.81	8.85	9.91	10.97	12.04		13.42	8 / 5		Shibayama (2023)	

	reaction time	0.112	interval	7 steps	1.06	1.03	1.04	1.05	1.04	1.04	1.06	1.06	1.07	1.38		3.13	3.13	3.19
			velocity	5.30	8.62	8.87	8.79	8.70	8.79	8.79	8.62	8.62	8.54	10.16	8.20	8.76	8.76	8.60
Trajkovic, Milan (CYP) (1992)	time			2.56	3.63	4.70	5.70	6.73	7.76	8.83	9.88	10.93	12.03		13.49	5 / 6		Henson (2022)
	reaction time	0.142	interval	7 steps	1.07	1.07	1.00	1.03	1.03	1.07	1.05	1.05	1.10	1.46		3.14	3.13	3.20
			velocity	5.36	8.54	8.54	9.14	8.87	8.87	8.54	8.70	8.70	8.31	9.60	8.15	8.73	8.76	8.57
King, David (GBR) (1994)	time			2.56	3.63	4.66	5.66	6.73	7.76	8.83	9.88	10.93	12.03		13.51	2 / 7		Henson (2022)
	reaction time	0.120	interval	7 steps	1.07	1.03	1.00	1.07	1.03	1.07	1.05	1.05	1.10	1.48		3.10	3.17	3.20
			velocity	5.36	8.54	8.87	9.14	8.54	8.87	8.54	8.70	8.70	8.31	9.47	8.14	8.85	8.65	8.57
Rodrigues, Eduardo (BRA) (1995)	time			2.63	3.70	4.73	5.76	6.86	7.90	8.93	10.00	11.03	12.13		13.62	7 / 8		Henson (2022)
	reaction time	0.130	interval	7 steps	1.07	1.03	1.03	1.10	1.04	1.03	1.07	1.03	1.10	1.49		3.13	3.17	3.20
			velocity	5.22	8.54	8.87	8.87	8.31	8.79	8.87	8.54	8.87	8.31	9.41	8.08	8.76	8.65	8.57

Shibayama (2023) - race analysis of the men's 110m hurdles in the 2022 season

Semi-Final 1

Henson (2022) - Athlete First: 2022 year end hurdle report (TV Analysis)

Semi-Final 1		date	17-Jul-22	wind	Henson (2022) - Athlete First: 2022 year end hurdle report (TV Analysis)															
				H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10	
Holloway, Grant (USA) (1997)	time			2.43	3.40	4.40	5.37	6.37	7.37	8.37	9.40	10.43	11.48		13.01	6 / 1		Henson (2022)		
	reaction time	0.128	interval	7 steps	0.97	1.00	0.97	1.00	1.00	1.00	1.03	1.03	1.05	1.53			2.94	3.00	3.11	
			velocity		5.65	9.42	9.14	9.42	9.14	9.14	9.14	8.87	8.87	8.70	9.16	8.46		9.33	9.14	8.82
Zeller, Joshua (GBR) (2000)	time			2.56	3.60	4.63	5.63	6.65	7.67	8.70	9.73	10.80	11.86		13.31	5 / 2		Henson (2022)		
	reaction time	0.124	interval		1.04	1.03	1.00	1.02	1.02	1.03	1.03	1.07	1.06	1.45			3.07	3.07	3.16	
			velocity		5.36	8.79	8.87	9.14	8.96	8.96	8.87	8.87	8.54	8.62	9.67	8.26		8.93	8.93	8.68
Bouanani, Amine (ALG) (1997)	time			2.60	3.66	4.70	5.70	6.70	7.76	8.80	9.83	10.87	11.96		13.37	3 / 3		Henson (2022)		
	reaction time	0.130	interval	7 steps	1.06	1.04	1.00	1.00	1.06	1.04	1.03	1.04	1.09	1.41	NR PB		3.10	3.10	3.16	
			velocity		5.28	8.62	8.79	9.14	9.14	8.62	8.79	8.87	8.79	8.39	9.94	8.23		8.85	8.85	8.68
Pereira, Rafael (BRA) (1997)	time			2.50	3.63	4.66	5.70	6.70	7.76	8.80	9.83	10.87	11.96		13.46	4 / 4		Henson (2022)		
	reaction time	0.125	interval		1.13	1.03	1.04	1.00	1.06	1.04	1.03	1.04	1.09	1.50			3.20	3.10	3.16	
			velocity		5.49	8.09	8.87	8.79	9.14	8.62	8.79	8.87	8.79	8.39	9.35	8.17		8.57	8.85	8.68
Zhoya, Sasha (FRA) (2002)	time			2.50	3.63	4.70	5.76	6.80	7.83	8.83	9.88	10.96	12.00		13.47	8 / 5		Henson (2022)		
	reaction time	0.154	interval	7 steps	1.13	1.07	1.06	1.04	1.03	1.00	1.05	1.08	1.04	1.47			3.26	3.07	3.17	
			velocity		5.49	8.09	8.54	8.62	8.79	8.87	9.14	8.70	8.46	8.79	9.54	8.17		8.41	8.93	8.65
Bennett, Orlando (JAM) (1999)	time			2.56	3.60	4.70	5.76	6.80	7.85	8.90	9.96	11.06	12.17		13.67	2 / 6		Henson (2022)		
	reaction time	0.139	interval		1.04	1.10	1.06	1.04	1.05	1.05	1.06	1.10	1.11	1.50			3.20	3.14	3.27	
			velocity		5.36	8.79	8.31	8.62	8.79	8.70	8.62	8.31	8.23	9.35	8.05		8.57	8.73	8.39	
Joseph, Jason (SUI) (1998)	time			2.58	3.63	4.66	5.70	6.75	7.80	8.86	9.96	11.03	12.17		13.67	7 / 7		Henson (2022)		
	reaction time	0.149	interval	7 steps	1.05	1.03	1.04	1.05	1.05	1.06	1.10	1.07	1.14	1.50			3.12	3.16	3.31	
			velocity		5.32	8.70	8.87	8.79	8.70	8.70	8.62	8.31	8.54	8.02	9.35	8.05		8.79	8.68	8.28
Ishikawa, Shuhei (JPN) (1995)	time			2.63	3.70	4.78	5.85	6.90	7.95	9.02	10.07	11.14	12.25		13.68	1 / 8		Shibayama (2023)		
	reaction time	0.144	interval	7 steps	1.07	1.08	1.07	1.05	1.05	1.07	1.05	1.07	1.11	1.43			3.22	3.17	3.23	
			velocity		5.22	8.54	8.46	8.54	8.70	8.70	8.54	8.70	8.54	8.23	9.80	8.04		8.52	8.65	8.49

Shibayama (2023) - race analysis of the men's 110m hurdles in the 2022 season

Heat 5

Henson (2022) - Athlete First: 2022 year end hurdle report (TV Analysis)

Heat 5

	date	16-Jul-22	wind	0.4 m/s	Henson (2022) - Athlete First: 2022 year end hurdle report (TV Analysis)															
					H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
Allen, Devon (USA) (1994)	time				2.56	3.58	4.63	5.63	6.66	7.70	8.76	9.83	10.93	12.00		13.47	3 / 1		Henson (2022)	
	reaction time	0.123	interval	7 steps	1.02	1.05	1.00	1.03	1.04	1.06	1.07	1.10	1.07	1.47				3.07	3.13	3.24
			velocity		5.36	8.96	8.70	9.14	8.87	8.79	8.62	8.54	8.31	8.54	9.54	8.17		8.93	8.76	8.46
Trajkovic, Milan (CYP) (1992)	time				2.53	3.60	4.60	5.66	6.70	7.73	8.76	9.83	10.90	11.98		13.52	4 / 2		Henson (2022)	
	reaction time	0.133	interval	7 steps	1.07	1.00	1.06	1.04	1.03	1.07	1.07	1.07	1.08	1.54				3.13	3.10	3.22
			velocity		5.42	8.54	9.14	8.62	8.79	8.87	8.87	8.54	8.54	8.46	9.10	8.14		8.76	8.85	8.52
Izumiya, Shunsuke (JPN) (2000)	time				2.62	3.67	4.73	5.77	6.81	7.85	8.93	9.98	11.07	12.17		13.56	1 / 3		Shibayama (2023)	
	reaction time	0.128	interval	7 steps	1.05	1.06	1.04	1.04	1.04	1.08	1.05	1.09	1.10	1.39				3.15	3.16	3.24
			velocity		5.24	8.70	8.62	8.79	8.79	8.79	8.46	8.70	8.39	8.31	10.09	8.11		8.70	8.68	8.46
King, David (GBR) (1994)	time				2.58	3.66	4.73	5.73	6.76	7.83	8.90	9.93	11.00	12.10		13.57	5 / 4		Henson (2022)	
	reaction time	0.117	interval	7 steps	1.08	1.07	1.00	1.03	1.07	1.07	1.07	1.03	1.07	1.10	1.47			3.15	3.17	3.20
			velocity		5.32	8.46	8.54	9.14	8.87	8.54	8.54	8.87	8.54	8.31	9.54	8.11		8.70	8.65	8.57

Shibayama (2023) - race analysis of the men's 110m hurdles in the 2022 season

Heat 4

Henson (2022) - Athlete First: 2022 year end hurdle report (TV Analysis)

Heat 4

	date	16-Jul-22	wind	0.2 m/s	Henson (2022) - Athlete First: 2022 year end hurdle report (TV Analysis)															
					H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
Parchment, Hansle (JAM) (1996)	time				2.53	3.56	4.63	5.63	6.66	7.66	8.66	9.66	10.66	11.70		13.17	4 / 1		Henson (2022)	
	reaction time	0.153	interval	7 steps	1.03	1.07	1.00	1.03	1.00	1.00	1.00	1.00	1.00	1.04	1.47			3.10	3.03	3.04
			velocity		5.42	8.87	8.54	9.14	8.87	9.14	9.14	9.14	9.14	8.79	9.54	8.35		8.85	9.05	9.02
Pereira, Rafael (BRA) (1997)	time				2.50	3.56	4.63	5.63	6.63	7.63	8.66	9.66	10.70	11.76		13.23	6 / 2		Henson (2022)	
	reaction time	0.140	interval			1.06	1.07	1.00	1.00	1.00	1.03	1.00	1.04	1.06	1.47			3.13	3.03	3.10
			velocity		5.49	8.62	8.54	9.14	9.14	9.14	8.87	9.14	8.79	8.62	9.54	8.31		8.76	9.05	8.85

Kwaou-Mathey, Just (FRA) (19		time	2.56	3.64	4.66	5.66	6.66	7.70	8.70	9.73	10.76	11.86		13.32	2 / 3		<i>Henson (2022)</i>	
reaction time	0.150	interval	7 steps	1.08	1.02	1.00	1.00	1.04	1.00	1.03	1.03	1.10	1.46			3.10	3.04	3.16
		velocity	5.36	8.46	8.96	9.14	9.14	8.79	9.14	8.87	8.87	8.31	9.60	8.26		8.85	9.02	8.68

Brathwaite, Shane (BAR) (1990)	time	2.56	3.66	4.70	5.73	6.73	7.76	8.80	9.83	10.90	11.96		13.47	3 / 4		<i>Henson (2022)</i>		
reaction time	0.134	interval	7 steps	1.10	1.04	1.03	1.00	1.03	1.04	1.03	1.07	1.06	1.51		3.17	3.07	3.16	
		velocity		5.36	8.31	8.79	8.87	9.14	8.87	8.79	8.87	8.54	8.62	9.28	8.17	8.65	8.93	8.68

Muratake, Rashiddo (JPN) (2001)		time	2.67	3.78	4.83	5.86	6.92	7.98	9.03	10.09	11.16	12.30		13.73	8 / 6	<i>Shibayama (2023)</i>			
reaction time	0.182	interval	7 steps	1.11	1.05	1.03	1.06	1.06	1.05	1.06	1.07	1.14	1.43			3.19	3.17	3.27	
		velocity		5.14	8.23	8.70	8.87	8.62	8.62	8.70	8.62	8.54	8.02	9.80	8.01		8.60	8.65	8.39

Shibayama (2023) - race analysis of the men's 110m hurdles in the 2022 season

Heat 3

date 16-Jul-22 wind -0.3 m/s

Henson (2022) - Athlete First: 2022 year end hurdle report (TV Analysis)

		date	10 Oct 22	wind	0.0 m/s	Henson (2022)										Atlanta first 2022 year and made report (TV Analysis)			
				H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
Martínez, Asier (ESP) (2000)	time			2.56	3.60	4.60	5.60	6.63	7.66	8.70	9.73	10.80	11.83		13.37	2 / 1			
	reaction time	0.137	interval	7 steps	1.04	1.00	1.00	1.03	1.03	1.04	1.03	1.07	1.03	1.54			3.04	3.10	3.13
			velocity		5.36	8.79	9.14	9.14	8.87	8.87	8.79	8.87	8.54	8.87	9.10	8.23		9.02	8.85

Bouanani, Amine (ALG) (1997)	time	2.56	3.63	4.63	5.66	6.66	7.70	8.70	9.80	10.86	11.93		13.44	1 / 2		<i>Henson (2022)</i>		
reaction time	0.134	interval	7 steps	1.07	1.00	1.03	1.00	1.04	1.00	1.10	1.06	1.07	1.51		3.10	3.04	3.23	
		velocity		5.36	8.54	9.14	8.87	9.14	8.79	9.14	8.31	8.62	8.54	9.28	8.18	8.85	9.02	8.49

Zhoya, Sasha (FRA) (2002)	time	2.46	3.53	4.60	5.60	6.63	7.66	8.70	9.80	10.86	12.00		13.48	6 / 3		<i>Henson (2022)</i>		
reaction time	0.132	interval	7 steps	1.07	1.07	1.00	1.03	1.03	1.04	1.10	1.06	1.14	1.48		3.14	3.10	3.30	
		velocity		5.58	8.54	8.54	9.14	8.87	8.87	8.79	8.31	8.62	8.02	9.47	8.16	8.73	8.85	8.31

Ishikawa, Shuhei (JPN) (1995)	time	2.58	3.65	4.69	5.73	6.76	7.80	8.86	9.93	11.00	12.10		13.53	3 / 4	Shibayama (2023)			
	reaction time	0.144	interval	7 steps	1.07	1.04	1.04	1.03	1.04	1.06	1.07	1.07	1.10	1.43		3.15	3.13	3.24
			velocity	5.32	8.54	8.79	8.79	8.87	8.79	8.62	8.54	8.54	8.31	9.80	8.13		8.70	8.76

Roberts, Daniel (USA) (1998)	time	2.50	3.50	4.50	5.50	6.53	7.56	8.60			dnf	4 / --		Henson (2022)
	reaction time	0.179	interval	1.00	1.00	1.03	1.03	1.04			DQ		3.00	3.10
		velocity	5.49	9.14	9.14	9.14	8.87	8.87	8.79				9.14	8.85

Heat 2 (TV Analysis)

date 16-Jul-22 wind 0.4 m/s

Henson (2022) - Athlete First: 2022 year end hurdle report

		H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
Holloway, Grant (USA) (1997)	time	2.43	3.46	4.46	5.43	6.43	7.43	8.46	9.50	10.53	11.63		13.14	1 / 1			
reaction time	0.147	interval	7 steps	1.03	1.00	0.97	1.00	1.03	1.04	1.03	1.10	1.51			3.00	3.03	3.17
velocity	5.65	8.87	9.14	9.42	9.14	9.14	8.87	8.79	8.87	8.31	9.28	8.37			9.14	9.05	8.65

Czykier, Damian (POL) (1992)	time	2.56	3.58	4.60	5.60	6.63	7.66	8.70	9.73	10.76	11.86		13.37	3 / 2			
reaction time	0.146	interval	7 steps	1.02	1.02	1.00	1.03	1.04	1.03	1.03	1.10	1.51			3.04	3.10	3.16
velocity	5.36	8.96	8.96	9.14	8.87	8.87	8.79	8.87	8.87	8.87	8.31	9.28	8.23		9.02	8.85	8.68

Zeller, Joshua (GBR) (2000)	time	2.56	3.63	4.68	5.70	6.73	7.73	8.76	9.80	10.86	11.93		13.41	5 / 3			
reaction time	0.126	interval	7 steps	1.07	1.05	1.02	1.03	1.00	1.03	1.04	1.06	1.07	1.48		3.14	3.06	3.17
velocity	5.36	8.54	8.70	8.96	8.87	9.14	8.87	8.79	8.62	8.54	9.47	8.20			8.73	8.96	8.65

Rodrigues, Eduardo (BRA) (1991)	time	2.56	3.63	4.70	5.73	6.73	7.73	8.76	9.80	10.90	11.96		13.46	8 / 4			
reaction time	0.134	interval	7 steps	1.07	1.07	1.03	1.00	1.00	1.03	1.04	1.10	1.06	1.50		3.17	3.03	3.20
velocity	5.36	8.54	8.54	8.87	9.14	9.14	8.87	8.79	8.31	8.62	9.35	8.17			8.65	9.05	8.57

Heat 1 (TV Analysis)

date 16-Jul-22 wind -0.5 m/s

Henson (2022) - Athlete First: 2022 year end hurdle report

		H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
Cunningham, Trey (USA) (1998)	time	2.50	3.53	4.56	5.56	6.58	7.60	8.63	9.66	10.76	11.80		13.28	4 / 1			
reaction time	0.115	interval	7 steps	1.03	1.03	1.00	1.02	1.02	1.03	1.10	1.04	1.48			3.06	3.07	3.17
velocity	5.49	8.87	8.87	9.14	8.96	8.96	8.87	8.87	8.87	8.31	8.79	9.47	8.28		8.96	8.93	8.65

Broadbell, Rasheed (JAM) (2001)	time	2.46	3.53	4.56	5.56	6.60	7.63	8.73	9.80	10.83	11.86		13.36	8 / 2			
reaction time	0.139	interval	7 steps	1.07	1.03	1.00	1.04	1.03	1.10	1.07	1.03	1.03	1.50		3.10	3.17	3.13
velocity	5.58	8.54	8.87	9.14	8.79	8.87	8.31	8.54	8.87	8.87	8.87	9.35	8.23		8.85	8.65	8.76

Pozzi, Andy (GBR) (1992)	time	2.46	3.53	4.53	5.56	6.58	7.60	8.63	9.70	10.80	11.86		13.45	2 / 3			
reaction time	0.120	interval	7 steps	1.07	1.00	1.03	1.02	1.02	1.03	1.07	1.10	1.06	1.59		3.10	3.07	3.23
velocity	5.58	8.54	9.14	8.87	8.96	8.96	8.87	8.54	8.31	8.62	8.82	8.18			8.85	8.93	8.49

Joseph, Jason (SUI) (1998)	time	2.53	3.56	4.60	5.60	6.63	7.66	8.76	9.80	10.86	11.96		13.49	7 / 4			
reaction time	0.140	interval	7 steps	1.03	1.04	1.00	1.03	1.03	1.10	1.04	1.06	1.10	1.53		3.07	3.16	3.20
velocity	5.42	8.87	8.79	9.14	8.87	8.87	8.31	8.79	8.62	8.31	9.16	8.15			8.93	8.68	8.57

Martinot-Lagarde, Pascal (FRA) (1991)	time	2.53	3.56	4.60	5.63	6.66	7.70	8.76	9.80	10.86	11.96		13.49	1 / 5			
reaction time	0.134	interval	7 steps	1.03	1.04	1.03	1.03	1.04	1.06	1.04	1.06	1.10	1.53		3.10	3.13	3.20
velocity	5.42	8.87	8.79	8.87	8.87	8.79	8.62	8.79	8.62	8.31	9.16	8.15			8.85	8.76	8.57

2022 USATF National Championships (Eugene, OR)

FINAL (TV Analysis)

date 26-Jun-22 wind 1.2 m/s

Henson (2022) - Athlete First: 2022 year end hurdle report

		H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
Roberts, Daniel (USA) (1998)	time	2.43	3.47	4.47	5.47	6.47	7.47	8.50	9.52	10.54	11.61		13.03	7 / 1			
	reaction time																
	interval		1.04	1.00	1.00	1.00	1.00	1.03	1.02	1.02	1.07	1.42			3.04	3.03	3.11
	velocity	5.65	8.79	9.14	9.14	9.14	9.14	8.87	8.96	8.96	8.54	9.87	8.44		9.02	9.05	8.82
Cunningham, Trey (USA) (1998)	time	2.47	3.47	4.52	5.51	6.52	7.54	8.57	9.57	10.61	11.68		13.08	5 / 2			
	reaction time																
	interval	7 steps	1.00	1.05	0.99	1.01	1.02	1.03	1.00	1.04	1.07	1.40			3.04	3.06	3.11
	velocity	5.55	9.14	8.70	9.23	9.05	8.96	8.87	9.14	8.79	8.54	10.01	8.41		9.02	8.96	8.82
Allen, Devon (USA) (1994)	time	2.53	3.56	4.55	5.58	6.57	7.57	8.59	9.61	10.64	11.70		13.09	4 / 3			
	reaction time																
	interval	7 steps	1.03	0.99	1.03	0.99	1.00	1.02	1.02	1.03	1.06	1.39			3.05	3.01	3.11
	velocity	5.42	8.87	9.23	8.87	9.23	9.14	8.96	8.96	8.87	8.62	10.09	8.40		8.99	9.11	8.82
Britt, Jamal (USA) (1998)	time	2.50	3.55	4.55	5.55	6.55	7.55	8.56	9.57	10.62	11.66		13.09	3 / 4			
	reaction time																
	interval	7 steps	1.05	1.00	1.00	1.00	1.00	1.01	1.01	1.05	1.04	1.43	PB		3.05	3.01	3.10
	velocity	5.49	8.70	9.14	9.14	9.14	9.14	9.05	9.05	8.70	8.79	9.80	8.40		8.99	9.11	8.85
Crittenden, Freddie (USA) (1998)	time	2.53	3.53	4.55	5.57	6.57	7.57	8.57	9.61	10.64	11.68		13.14	2 / 5			
	reaction time																
	interval		1.00	1.02	1.02	1.00	1.00	1.00	1.04	1.03	1.04	1.46	PB		3.04	3.00	3.11
	velocity	5.42	9.14	8.96	8.96	9.14	9.14	9.14	8.79	8.87	8.79	9.60	8.37		9.02	9.14	8.82
Dunning, Robert (USA) (1997)	time	2.53	3.55	4.57	5.59	6.61	7.61	8.61	9.64	10.67	11.74		13.27	9 / 6			
	reaction time																
	interval	7 steps	1.02	1.02	1.02	1.02	1.00	1.00	1.03	1.03	1.07	1.53			3.06	3.02	3.13
	velocity	5.42	8.96	8.96	8.96	8.96	9.14	9.14	8.87	8.87	8.54	9.16	8.29		8.96	9.08	8.76
Edwards Jr., Eric (USA) (2000)	time	2.57	3.63	4.70	5.74	6.78	7.82	8.87	9.94	11.02	12.11		13.76	8 / 7			
	reaction time																
	interval		1.06	1.07	1.04	1.04	1.04	1.05	1.07	1.08	1.09	1.65			3.17	3.13	3.24
	velocity	5.34	8.62	8.54	8.79	8.79	8.79	8.70	8.54	8.46	8.39	8.50	7.99		8.65	8.76	8.46

Semi-Final 2

date 26-Jun-22 wind 0.1 m/s

USATF (2022) - Results powered by Karmarush.com

		H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
Cunningham, Trey (USA) (1998)	time	2.43	3.46	4.56	5.45	6.46	7.52	8.53	9.55	10.57	11.63		13.09	6 / 1			
	reaction time																
	interval	7 steps	1.03	1.10	0.89	1.01	1.06	1.01	1.02	1.02	1.06	1.46			3.02	3.08	3.10
	velocity	5.65	8.87	8.31	10.27	9.05	8.62	9.05	8.96	8.96	8.62	9.60	8.40		9.08	8.90	8.85
Allen, Devon (USA) (1994)	time	2.45	3.52	4.55	5.54	6.55	7.61	8.55	9.61	10.63	11.64		13.14	5 / 2			
	reaction time																
	interval	7 steps	1.07	1.03	0.99	1.01	1.06	0.94	1.06	1.02	1.01	1.50			3.09	3.01	3.09
	velocity	5.60	8.54	8.87	9.23	9.05	8.62	9.72	8.62	8.96	9.05	9.35	8.37		8.87	9.11	8.87
Britt, Jamal (USA) (1998)	time	2.51	3.53	4.57	5.60	6.61	7.64	8.65	9.72	10.67	11.77		13.25	7 / 3			
	reaction time																
	interval	7 steps	1.02	1.04	1.03	1.01	1.03	1.01	1.07	0.95	1.10	1.48			3.09	3.05	3.12
	velocity	5.47	8.96	8.79	8.87	9.05	8.87	9.05	8.54	9.62	8.31	9.47	8.30		8.87	8.99	8.79
Crittenden, Freddie (USA) (1998)	time	2.50	3.57	4.60	5.64	6.70	7.69	8.70	9.77	10.78	11.86		13.35	4 / 4			
	reaction time																
	interval		1.07	1.03	1.04	1.06	0.99	1.01	1.07	1.01	1.08	1.49			3.14	3.06	3.16
	velocity	5.49	8.54	8.87	8.79	8.62	9.23	9.05	8.54	9.05	8.46	9.41	8.24		8.73	8.96	8.68
Gilbert, Tre'Bien (USA) (1999)	time	2.54	3.58	4.62	5.65	6.66	7.68	8.71	9.77	10.85	11.95		13.51	8 / 5			
	reaction time																
	interval		1.04	1.04	1.03	1.01	1.02	1.03	1.06	1.08	1.10	1.56			3.11	3.06	3.24
	velocity	5.40	8.79	8.79	8.87	9.05	8.96	8.87	8.62	8.46	8.31	8.99	8.14		8.82	8.96	8.46
Brockman, Joshua (USA) (2001)	time	2.47	3.56	4.60	5.60	6.62	7.69	8.76	9.85	10.92	12.00		13.55	2 / 6			
	reaction time																
	interval		1.09	1.04	1.00	1.02	1.07	1.07	1.09	1.07	1.08	1.55	PB		3.13	3.16	3.24
	velocity	5.55	8.39	8.79	9.14	8.96	8.54	8.54	8.39	8.54	8.46	9.05	8.12		8.76	8.68	8.46
Anderson, Nick (USA) (1995)	time	2.50	3.56	4.61	5.66	6.66	7.78	8.82	9.90	10.94	12.04		13.69	3 / 7			
	reaction time																
	interval		1.06	1.05	1.05	1.00	1.12	1.04	1.08	1.04	1.10	1.65			3.16	3.16	3.22
	velocity	5.49	8.62	8.70	8.70	9.14	8.16	8.79	8.46	8.79	8.31	8.50	8.04		8.68	8.68	8.52
Wilson, De'vion (USA) (2000)	time	2.58	3.70	4.77	5.80	6.87	7.94	9.01	10.10	11.17	12.28		13.86	9 / 8			
	reaction time																
	interval		1.12	1.07	1.03	1.07	1.07	1.07	1.09	1.07	1.11	1.58			3.22	3.21	3.27
	velocity	5.32	8.16	8.54	8.87	8.54	8.54	8.54	8.39	8.54	8.23	8.87	7.94		8.52	8.54	8.39

Semi-Final 1

date 26-Jun-22 wind 0.4 m/s

USATF (2022) - Results powered by Karmarush.com

		H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
Holloway, Grant (USA) (1997)	time	2.41	3.41	4.38	5.35	6.34	7.36	8.36	9.40	10.45	11.53		13.03	4 / 1			
	reaction time																
	interval	7 steps	1.00	0.97	0.97	0.99	1.02	1.00	1.04	1.05	1.08	1.50			2.94	3.01	3.17
	velocity	5.69	9.14	9.42	9.42	9.23	8.96	9.14	8.79	8.70	8.46	9.35	8.44		9.33	9.11	8.65
Roberts, Daniel (USA) (1998)	time	2.40	3.45	4.55	5.43	6.46	7.52	8.54	9.57	10.60	11.65		13.17	6 / 2			
	reaction time																
	interval		1.05	1.10	0.88	1.03	1.06	1.02	1.03	1.03	1.05	1.52			3.03	3.11	3.11
	velocity	5.72	8.70	8.31	10.39	8.87	8.62	8.96	8.87	8.87	8.70	9.22	8.35		9.05	8.82	8.82
Edwards Jr., Eric (USA) (2000)	time	2.42	3.45	4.53	5.51	6.54	7.60	8.61	9.64	10.56	11.66		13.29	5 / 3			
	reaction time																
	interval		1.03	1.08	0.98	1.03	1.06	1.01	1.03	0.92	1.10	1.63			3.09	3.10	3.05

FINAL

INAL		date		18-Jun-22		wind		-0.2 m/s		Henson (2022) - Athlete First: 2022 year end hurdle report									
				H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
Allen, Devon (USA) (1994)		time		2.50	3.54	4.56	5.56	6.56	7.60	8.60	9.64	10.70	11.76		13.16	5 / 1			
reaction time		0.138	interval	7 steps	1.04	1.02	1.00	1.00	1.04	1.00	1.04	1.06	1.06	1.40			3.06	3.04	3.16
			velocity	5.49	8.79	8.96	9.14	9.14	8.79	9.14	8.79	8.62	8.62	10.01	8.36		8.96	9.02	8.68
Pereira, Rafael (BRA) (1997)		time		2.50	3.54	4.56	5.56	6.58	7.58	8.60	9.62	10.68	11.76		13.25	4 / 2			
reaction time		0.141	interval		1.04	1.02	1.00	1.02	1.00	1.02	1.02	1.06	1.08	1.49	=PB		3.06	3.04	3.16
			velocity	5.49	8.79	8.96	9.14	8.96	9.14	8.96	8.96	8.62	8.46	9.41	8.30		8.96	9.02	8.68
Kwaou-Mathey, Just (FRA) (1992)		time		2.56	3.60	4.62	5.62	6.62	7.66	8.66	9.70	10.74	11.80		13.27	2 / 3			
reaction time		0.169	interval	7 steps	1.04	1.02	1.00	1.00	1.04	1.00	1.04	1.04	1.06	1.47	PB		3.06	3.04	3.14
			velocity	5.36	8.79	8.96	9.14	9.14	8.79	9.14	8.79	8.79	8.62	9.54	8.29		8.96	9.02	8.73
Czykier, Damian (POL) (1992)		time		2.52	3.54	4.58	5.60	6.62	7.64	8.66	9.70	10.74	11.80		13.31	7 / 4			
reaction time		0.144	interval	7 steps	1.02	1.04	1.02	1.02	1.02	1.02	1.04	1.04	1.06	1.51			3.08	3.06	3.14
			velocity	5.44	8.96	8.79	8.96	8.96	8.96	8.96	8.79	8.79	8.62	9.28	8.26		8.90	8.96	8.73
Trajkovic, Milan (CYP) (1992)		time		2.50	3.56	4.58	5.58	6.60	7.64	8.68	9.72	10.80	11.86		13.39	8 / 5			
reaction time		0.154	interval	7 steps	1.06	1.02	1.00	1.02	1.04	1.04	1.04	1.08	1.06	1.53			3.08	3.10	3.18
			velocity	5.49	8.62	8.96	9.14	8.96	8.79	8.79	8.79	8.46	8.62	9.16	8.22		8.90	8.85	8.62
Pozzi, Andy (GBR) (1992)		time		2.50	3.54	4.58	5.60	6.62	7.66	8.70	9.76	10.84	11.94		13.49	1 / 6			
reaction time		0.126	interval	7 steps	1.04	1.04	1.02	1.02	1.04	1.04	1.06	1.08	1.10	1.55			3.10	3.10	3.24
			velocity	5.49	8.79	8.79	8.96	8.96	8.79	8.79	8.62	8.46	8.31	9.05	8.15		8.85	8.85	8.46
Martínez, Asier (ESP) (2000)		time		2.58	3.64	4.66	5.68	6.68	7.70	8.76	9.84	10.92	12.02		13.69	6 / 7			
reaction time		0.161	interval	7 steps	1.06	1.02	1.02	1.00	1.02	1.06	1.08	1.08	1.10	1.67			3.10	3.08	3.26
			velocity	5.32	8.62	8.96	8.96	9.14	8.96	8.62	8.46	8.46	8.31	8.40	8.04		8.85	8.90	8.41
Mallett, Aaron (USA) (1994)		time		2.50	3.56	4.58	5.62	6.62	7.62	8.66	9.68					3 / --			
reaction time		0.150	interval	7 steps	1.06	1.02	1.04	1.00	1.00	1.04	1.02				DQ		3.12	3.04	
			velocity	5.49	8.62	8.96	8.79	9.14	9.14	8.79	8.96						8.79	9.02	

FINAL

INAL		date	16-Jun-22	wind -1.2 m/s										Henson (2022) - Athlete First: 2022 year end hurdle report						
				H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10	
Allen, Devon (USA) (1994)	time			2.52	3.56	4.57	5.58	6.56	7.58	8.60	9.62	10.70	11.80		13.22	5 / 1				
	reaction time	0.140	interval	7 steps	1.04	1.01	1.01	0.98	1.02	1.02	1.02	1.08	1.10	1.42			3.06	3.02	3.20	
			velocity	5.44	8.79	9.05	9.05	9.33	8.96	8.96	8.96	8.46	8.31	9.87	8.32		8.96	9.08	8.57	
Martínez, Asier (ESP) (2000)	time			2.56	3.62	4.64	5.66	6.68	7.70	8.72	9.76	10.80	11.84		13.30	6 / 2				
	reaction time	0.137	interval	7 steps	1.06	1.02	1.02	1.02	1.02	1.04	1.04	1.04	1.04	1.46			3.10	3.06	3.12	
			velocity	5.36	8.62	8.96	8.96	8.96	8.96	8.96	8.79	8.79	8.79	9.60	8.27		8.85	8.96	8.79	
Pereira, Rafael (BRA) (1997)	time			2.48	3.54	4.58	5.62	6.62	7.66	8.68	9.74	10.78	11.86		13.37	3 / 3				
	reaction time	0.137	interval		1.06	1.04	1.04	1.00	1.04	1.02	1.06	1.04	1.08	1.51			3.14	3.06	3.18	
			velocity	5.53	8.62	8.79	8.79	9.14	8.79	8.96	8.62	8.79	8.46	9.28	8.23		8.73	8.96	8.62	
Mallett, Aaron (USA) (1994)	time			2.52	3.58	4.62	5.62	6.62	7.66	8.68	9.74	10.80	11.88		13.40	1 / 4				
	reaction time	0.173	interval	7 steps	1.06	1.04	1.00	1.00	1.04	1.02	1.06	1.06	1.08	1.52			3.10	3.06	3.20	
			velocity	5.44	8.62	8.79	9.14	9.14	8.79	8.96	8.62	8.62	8.46	9.22	8.21		8.85	8.96	8.57	

Joseph, Jason (SUI) (1998)	time	2.56	3.62	4.64	5.68	6.68	7.72	8.76	9.84	10.92	12.02				13.55	7 / 5					
	reaction time	0.159	interval	7 steps	1.06	1.02	1.04	1.00	1.04	1.04	1.08	1.08	1.10	1.53				3.12	3.08	3.26	
		velocity	5.36	8.62	8.96	8.79	9.14	8.79	8.79	8.46	8.46	8.31	9.16				8.12		8.79	8.90	8.41
Dal Molin, Paolo (ITA) (1987)	time	2.52	3.60	4.66	5.72	6.76	7.84	8.94	10.02	11.12	12.26				13.76	8 / 6					
	reaction time	0.122	interval	7 steps	1.08	1.06	1.06	1.04	1.08	1.10	1.08	1.10	1.14	1.50				3.20	3.22	3.32	
		velocity	5.44	8.46	8.62	8.62	8.79	8.46	8.31	8.46	8.31	8.02	9.35				7.99		8.57	8.52	8.26
Belocian, Wilhem (FRA) (1995)	time	2.48	3.52	4.58	5.64											dnf	1 / --				
	reaction time	0.111	interval	7 steps	1.04	1.06													3.16		
		velocity	5.53	8.79	8.62	8.62													8.68		

2022 New York Grand Prix (New York, NY) (TV Analysis)

FINAL

NAME	date	12-Jun-22	wind 1.6 m/s											Henson (2022) - Athlete First: 2022 year end hurdle report						
				H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10	
Allen, Devon (USA) (1994)	time			2.42	3.44	4.43	5.42	6.38	7.38	8.36	9.38	10.38	11.42		12.84	5 / 1				
	reaction time	0.128	interval	7 steps	1.02	0.99	0.99	0.96	1.00	0.98	1.02	1.00	1.04	1.42	PB		3.00	2.94	3.06	
	velocity			5.67	8.96	9.23	9.23	9.52	9.14	9.33	8.96	9.14	8.79	9.87	8.57		9.14	9.33	8.96	
Holloway, Grant (USA) (1997)	time			2.35	3.35	4.33	5.33	6.30	7.28	8.30	9.35	10.38	11.52		13.06	6 / 2				
	reaction time	0.116	interval	7 steps	1.00	0.98	1.00	0.97	0.98	1.02	1.05	1.03	1.14	1.54			2.98	2.97	3.22	
	velocity			5.84	9.14	9.33	9.14	9.42	9.33	8.96	8.70	8.87	8.02	9.10	8.42		9.20	9.23	8.52	
Roberts, Daniel (USA) (1997)	time			2.40	3.43	4.45	5.47	6.50	7.52	8.52	9.56	10.60	11.67		13.17	4 / 3				
	reaction time	0.156	interval		1.03	1.02	1.02	1.03	1.02	1.00	1.04	1.04	1.07	1.50	PB		3.07	3.05	3.15	
	velocity			5.72	8.87	8.96	8.96	8.87	8.96	9.14	8.79	8.79	8.54	9.35	8.35		8.93	8.99	8.70	
Britt, Jamal (USA) (1998)	time			2.43	3.52	4.53	5.55	6.55	7.58	8.62	9.65	10.72	11.76		13.22	3 / 4				
	reaction time	0.139	interval	7 steps	1.09	1.01	1.02	1.00	1.03	1.04	1.03	1.07	1.04	1.46	PB		3.12	3.07	3.14	
	velocity			5.65	8.39	9.05	8.96	9.14	8.87	8.79	8.87	8.54	8.79	9.60	8.32		8.79	8.93	8.73	
Brathwaite, Shane (BAR) (1990)	time			2.52	3.57	4.60	5.62	6.63	7.63	8.68	9.72	10.78	11.84		13.40	8 / 5				
	reaction time	0.133	interval		1.05	1.03	1.02	1.01	1.00	1.05	1.04	1.06	1.06	1.56			3.10	3.06	3.16	
	velocity			5.44	8.70	8.87	8.96	9.05	9.14	8.70	8.79	8.62	8.62	8.99	8.21		8.85	8.96	8.68	
Thomas, Damion (JAM) (1999)	time			2.52	3.56	4.57	5.58	6.61	7.63	8.70	9.75	10.82	11.90		13.40	7 / 6				
	reaction time	0.149	interval	7 steps	1.04	1.01	1.01	1.03	1.02	1.07	1.05	1.07	1.08	1.50			3.06	3.12	3.20	
	velocity			5.44	8.79	9.05	9.05	8.87	8.96	8.54	8.70	8.54	8.46	9.35	8.21		8.96	8.79	8.57	
Walters, Ruebin (TTO) (1995)	time			2.53	3.60	4.66	5.73	6.77	7.83	8.90	9.95	11.03	12.12		13.67	2 / 7				
	reaction time	0.139	interval		1.07	1.06	1.07	1.04	1.06	1.07	1.05	1.08	1.09	1.55			3.20	3.17	3.22	
	velocity			5.42	8.54	8.62	8.54	8.79	8.62	8.54	8.70	8.46	8.39	9.05	8.05		8.57	8.65	8.52	

2022 Japanese National Championships (Osaka, JPN)

FINAL

NAME	date 12-Jun-22			wind -1.2 m/s			Shibayama (2023) - race analysis of the men's 110m hurdles in the 2022 season												
				H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
Izumiya, Shunzuke (JPN) (2000)	time			2.52	3.55	4.58	5.60	6.63	7.54	8.66	9.71	10.77	11.83		13.21	5 / 1			
	reaction time	0.142	interval	7 steps	1.03	1.03	1.02	1.03	0.91	1.12	1.05	1.06	1.06	1.38			3.08	3.06	3.17
	velocity			5.44	8.87	8.87	8.96	8.87	10.04	8.16	8.70	8.62	8.62	10.16	8.33		8.90	8.96	8.65
Muratake, Rashiddo (JPN) (200)	time			2.62	3.65	4.67	5.68	6.70	7.72	8.73	9.78	10.83	11.89		13.31	7 / 2			
	reaction time	0.174	interval	7 steps	1.03	1.02	1.01	1.02	1.02	1.01	1.05	1.05	1.06	1.42			3.06	3.05	3.16
	velocity			5.24	8.87	8.96	9.05	8.96	8.96	9.05	8.70	8.70	8.62	9.87	8.26		8.96	8.99	8.68
Ishikawa, Shuhei (JPN) (1995)	time			2.60	3.67	4.70	5.72	6.75	7.77	8.83	9.88	10.93	12.03		13.48	4 / 3			
	reaction time	0.155	interval	7 steps	1.07	1.03	1.02	1.03	1.02	1.06	1.05	1.05	1.10	1.45			3.12	3.11	3.20
	velocity			5.28	8.54	8.87	8.96	8.87	8.96	8.62	8.70	8.70	8.31	9.67	8.16		8.79	8.82	8.57
Takahashi, Yusuke (JPN) (1996)	time			2.63	3.70	4.74	5.78	6.83	7.87	8.94	9.99	11.08	12.17		13.57	6 / 4			
	reaction time	0.152	interval		1.07	1.04	1.04	1.05	1.04	1.07	1.05	1.09	1.09	1.40			3.15	3.16	3.23
	velocity			5.22	8.54	8.79	8.79	8.70	8.79	8.54	8.70	8.39	8.39	10.01	8.11		8.70	8.68	8.49
Takayama, Shunya (JPN) (1994)	time			2.56	3.58	4.64	5.69	6.73	7.79	8.85	9.93	11.03	12.13		13.60	2 / 5			
	reaction time	0.136	interval		1.02	1.06	1.05	1.04	1.06	1.06	1.08	1.10	1.10	1.47			3.13	3.16	3.28
	velocity			5.36	8.96	8.62	8.70	8.79	8.62	8.62	8.46	8.31	8.31	9.54	8.09		8.76	8.68	8.36
Fujii, Ryota (JPN) (1996)	time			2.58	3.65	4.70	5.73	6.76	7.82	8.89	9.97	11.05	12.15		13.63	9 / 6			
	reaction time	0.145	interval		1.07	1.05	1.03	1.03	1.06	1.07	1.08	1.08	1.10	1.48			3.15	3.16	3.26
	velocity			5.32	8.54	8.70	8.87	8.87	8.62	8.54	8.46	8.46	8.31	9.47	8.07		8.70	8.68	8.41
Ishida Thomas, Higashi (JPN)	time			2.55	3.65	4.68	5.71	6.75	7.81	8.89	9.97	11.09	12.20		13.68	8 / 7			
	reaction time	0.147	interval	7 steps	1.10	1.03	1.03	1.04	1.06	1.08	1.08	1.12	1.11	1.48			3.16	3.18	3.31
	velocity			5.38	8.31	8.87	8.87	8.79	8.62	8.46	8.46	8.16	8.23	9.47	8.04		8.68	8.62	8.28
Yokochi, Taiga (JPN) (2000)	time			2.64	3.73	4.77	5.82	6.85	7.92	8.97	10.05	11.13	12.25		13.72	3 / 8			
	reaction time	0.161	interval	7 steps	1.09	1.04	1.05	1.03	1.07	1.05	1.08	1.08	1.12	1.47			3.18	3.15	3.28
	velocity																		

velocity 5.20 8.39 8.79 8.70 8.87 8.54 8.70 8.46 8.46 8.16 9.54 8.02 8.62 8.70 8.36

2022 Japanese U20 National Championships (Osaka, JPN) (99cm)

FINAL

	date	12-Jun-22	wind 1.1 m/s	Shibayama (2023) - race analysis of the men's 110m hurdles in the 2022 season												Lane / Place	H1-H4	H4-H7	H7-H10
				H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time				
Abe, Tatsuki (JPN) (2003)	time			2.65	3.73	4.77	5.80	6.82	7.85	8.91	9.96	11.01	12.09		13.49	4 / 1			
	reaction time	0.157	interval	7 steps	1.08	1.04	1.03	1.02	1.03	1.06	1.05	1.05	1.08	1.40	PB		3.15	3.11	3.18
	velocity			5.18	8.46	8.79	8.87	8.96	8.87	8.62	8.70	8.70	8.46	10.01	8.15		8.70	8.82	8.62
Manago, Ryona (JPN) (2003)	time			2.58	3.67	4.70	5.73	6.76	7.80	8.88	9.94	11.02	12.08		13.55	5 / 2			
	reaction time	0.151	interval	7 steps	1.09	1.03	1.03	1.03	1.04	1.08	1.06	1.08	1.06	1.47			3.15	3.15	3.20
	velocity			5.32	8.39	8.87	8.87	8.87	8.79	8.46	8.62	8.46	8.62	9.54	8.12		8.70	8.70	8.57
Koike, Ryo (JPN) (2003)	time			2.58	3.69	4.74	5.79	6.80	7.83	8.87	9.93	11.00	12.10		13.57	7 / 3			
	reaction time	0.155	interval	7 steps	1.11	1.05	1.05	1.01	1.03	1.04	1.06	1.07	1.10	1.47	PB		3.21	3.08	3.23
	velocity			5.32	8.23	8.70	8.70	9.05	8.87	8.79	8.62	8.54	8.31	9.54	8.11		8.54	8.90	8.49
Shikata, Makoto (JPN) (2003)	time			2.65	3.72	4.78	5.83	6.87	7.90	9.00	10.08	11.14	12.23		13.63	6 / 4			
	reaction time	0.145	interval		1.07	1.06	1.05	1.04	1.03	1.10	1.08	1.06	1.09	1.40	PB		3.18	3.17	3.23
	velocity			5.18	8.54	8.62	8.70	8.79	8.87	8.31	8.46	8.62	8.39	10.01	8.07		8.62	8.65	8.49
Yamada, Masahiro (JPN) (2003)	time			2.67	3.77	4.87	5.93	6.98	8.04	9.10	10.17	11.25	12.35		13.82	8 / 5			
	reaction time	0.155	interval		1.10	1.10	1.06	1.05	1.06	1.06	1.07	1.08	1.10	1.47	PB		3.26	3.17	3.25
	velocity			5.14	8.31	8.31	8.62	8.70	8.62	8.62	8.54	8.46	8.31	9.54	7.96		8.41	8.65	8.44
Matsumoto, Nozomu (JPN) (2003)	time			2.68	3.77	4.81	5.86	6.91	7.95	9.01	10.11	11.20	12.31		13.83	9 / 6			
	reaction time	0.171	interval	7 steps	1.09	1.04	1.05	1.05	1.04	1.06	1.10	1.09	1.11	1.52			3.18	3.15	3.30
	velocity			5.12	8.39	8.79	8.70	8.70	8.79	8.62	8.31	8.39	8.23	9.22	7.95		8.62	8.70	8.31
Tahara, Keigo (JPN) (2003)	time			2.63	3.73	4.83	5.86	6.92	7.96	9.05	10.13	11.23	12.34		13.87	2 / 7			
	reaction time	0.155	interval		1.10	1.10	1.03	1.06	1.04	1.09	1.08	1.10	1.11	1.53	PB		3.23	3.19	3.29
	velocity			5.22	8.31	8.31	8.87	8.62	8.79	8.39	8.46	8.31	8.23	9.16	7.93		8.49	8.60	8.33
Sasaki, Keito (JPN) (2003)	time			2.75	3.86	4.96	6.05	7.13	8.20	9.30	10.39	11.51	12.64		14.18	3 / 8			
	reaction time	0.175	interval		1.11	1.10	1.09	1.08	1.07	1.10	1.09	1.12	1.13	1.54			3.30	3.25	3.34
	velocity			4.99	8.23	8.31	8.39	8.46	8.54	8.31	8.39	8.16	8.09	9.10	7.76		8.31	8.44	8.21

2022 NCAA Championships (Eugene, OR) (TV Analysis)

FINAL

	date	10-Jun-22	wind 0.0 m/s	Henson (2022) - Athlete First: 2022 year end hurdle report												Lane / Place	H1-H4	H4-H7	H7-H10
				H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time				
Cunningham, Trey (USA) (1998)	time			2.44	3.45	4.44	5.42	6.40	7.44	8.44	9.46	10.51	11.56		13.00	5 / 1			
	reaction time		interval	7 steps	1.01	0.99	0.98	0.98	1.04	1.00	1.02	1.05	1.05	1.44	PB		2.98	3.02	3.12
	velocity			5.62	9.05	9.23	9.33	9.33	8.79	9.14	8.96	8.70	8.70	9.74	8.46		9.20	9.08	8.79
Edwards Jr., Eric (USA) (2000)	time			2.44	3.47	4.47	5.46	6.46	7.46	8.47	9.49	10.53	11.61		13.15	6 / 2			
	reaction time		interval		1.03	1.00	0.99	1.00	1.00	1.01	1.02	1.04	1.08	1.54	PB		3.02	3.01	3.14
	velocity			5.62	8.87	9.14	9.23	9.14	9.14	9.05	8.96	8.79	8.46	9.10	8.37		9.08	9.11	8.73
Zeller, Joshua (GBR) (2000)	time			2.50	3.57	4.59	5.61	6.61	7.62	8.64	9.66	10.69	11.73		13.26	7 / 3			
	reaction time		interval		1.07	1.02	1.02	1.00	1.01	1.02	1.02	1.03	1.04	1.53			3.11	3.03	3.09
	velocity			5.49	8.54	8.96	8.96	9.14	9.05	8.96	8.96	8.87	8.79	9.16	8.30		8.82	9.05	8.87
Roberts, Giano (JAM) (2000)	time			2.50	3.59	4.64	5.67	6.69	7.74	8.77	9.82	10.86	11.91		13.45	1 / 4			
	reaction time		interval		1.09	1.05	1.03	1.02	1.05	1.03	1.05	1.04	1.05	1.54			3.17	3.10	3.14
	velocity			5.49	8.39	8.70	8.87	8.96	8.70	8.87	8.70	8.79	8.70	9.10	8.18		8.65	8.85	8.73
Campbell, Lafranz (JAM) (1997)	time			2.51	3.55	4.59	5.60	6.64	7.66	8.69	9.74	10.79	11.91		13.45	8 / 5			
	reaction time		interval	7 steps	1.04	1.04	1.01	1.04	1.02	1.03	1.05	1.05	1.12	1.54			3.09	3.09	3.22
	velocity			5.47	8.79	8.79	9.05	8.79	8.96	8.87	8.70	8.70	8.16	9.10	8.18		8.87	8.87	8.52
Patterson, Kentre (USA) (1999)	time			2.50	3.55	4.60	5.60	6.64	7.67	8.72	9.78	10.84	11.91		13.46	9 / 6			
	reaction time		interval		1.05	1.05	1.00	1.04	1.03	1.05	1.06	1.06	1.07	1.55			3.10	3.12	3.19
	velocity			5.49	8.70	8.70	9.14	8.79	8.87	8.70	8.62	8.62	8.54	9.05	8.17		8.85	8.79	8.60
Leaston, Justin (USA) (2001)	time			2.52	3.58	4.63	5.66	6.70	7.74	8.77	9.83	10.89	11.98		13.50	3 / 7			
	reaction time		interval		1.06	1.05	1.03	1.04	1.04	1.03	1.06	1.06	1.09	1.52			3.14	3.11	3.21
	velocity			5.44	8.62	8.70	8.87	8.79	8.79	8.87	8.62	8.62	8.39	9.22	8.15		8.73	8.82	8.54
Ojora, Tade (GBR) (1999)	time			2.53	3.60	4.67	5.70	6.74	7.81	8.86	9.93	10.98	12.05		13.58	2 / 8			
	reaction time		interval	7 steps	1.07	1.07	1.03	1.04	1.07	1.05	1.07	1.05	1.07	1.53			3.17	3.16	3.19
	velocity			5.42	8.54	8.54	8.87	8.79	8.54	8.70	8.54	8.70	8.54	9.16	8.10		8.65	8.68	8.60
Hayles, Jaheem (JAM) (2001)	time			2.55	3.65	4.72	5.77	6.82	7.86	8.89	9.96	11.06	12.19		13.73	4 / 9			
	reaction time		interval		1.10	1.07	1.05	1.05	1.04	1.03	1.07	1.10	1.13	1.54			3.22	3.12	3.30
	velocity			5.38	8.31	8.54	8.70	8.70	8.79	8.87	8.54	8.31	8.09	9.10	8.01		8.52	8.79	8.31

2022 Japanese Multi-Event National Championships (Akita, JPN)

Decathlon - FINAL

date 05-Jun-22 wind -2.3 m/s

Takefu (2023) - performance analysis of decathlon athletes in the 2022 season

		H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
Taue, Shun (JPN) (1997)	time	2.72	3.87	4.99	6.09	7.23	8.35	9.49	10.65	11.81	12.99		14.48	7 / 1			
	reaction time		interval	1.15	1.12	1.10	1.14	1.12	1.14	1.16	1.16	1.18	1.49		3.37	3.40	3.50
	velocity	5.04	7.95	8.16	8.31	8.02	8.16	8.02	7.88	7.88	7.75	9.41	7.60		8.14	8.06	7.83
Nakamura, Akihiko (JPN) (1990)	time	2.76	3.92	5.06	6.21	7.35	8.49	9.64	10.81	11.98	13.15		14.65	5 / 3			
	reaction time		interval	1.16	1.14	1.15	1.14	1.15	1.17	1.17	1.17	1.50			3.45	3.43	3.51
	velocity	4.97	7.88	8.02	7.95	8.02	8.02	7.95	7.81	7.81	7.81	9.35	7.51		7.95	7.99	7.81
Okuda, Keisuke (JPN) (1996)	time	2.75	3.90	5.05	6.20	7.35	8.50	9.66	10.82	12.00	13.21		14.71	9 / 4			
	reaction time		interval	1.15	1.15	1.15	1.15	1.16	1.16	1.18	1.21	1.50			3.45	3.46	3.55
	velocity	4.99	7.95	7.95	7.95	7.95	7.95	7.88	7.88	7.75	7.55	9.35	7.48		7.95	7.92	7.72
Ushiro, Keisuke (JPN) (1986)	time	2.79	3.98	5.19	6.41	7.62	8.86	10.12	11.41	12.67	13.95		15.67	8 / 8			
	reaction time		interval	1.19	1.21	1.22	1.21	1.24	1.26	1.29	1.26	1.28	1.72		3.62	3.71	3.83
	velocity	4.92	7.68	7.55	7.49	7.55	7.37	7.25	7.09	7.25	7.14	8.15	7.02		7.57	7.39	7.16

2022 NCAA East Region Preliminary (Bloomington, IN) (TV Analysis)

FINAL

date 27-May-22 wind -1.5 m/s

Henson (2022) - Athlete First: 2022 year end hurdle report

		H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
Cunningham, Trey (USA) (1998)	time	2.50	3.53	4.53	5.53	6.53	7.53	8.54	9.56	10.61	11.66		13.07	5 / 1			
	reaction time		interval	7 steps	1.03	1.00	1.00	1.00	1.01	1.02	1.05	1.05	1.41		3.03	3.01	3.12
	velocity	5.49	8.87	9.14	9.14	9.14	9.14	9.05	8.96	8.70	8.70	9.94	8.42		9.05	9.11	8.79

2022 Müller Birmingham Diamond League (Birmingham, GBR) (TV Analysis)

FINAL

date 21-May-22 wind 0.7 m/s

Henson (2022) - Athlete First: 2022 year end hurdle report

		Horse Performance Data (2022)										Horse / Rider			2022 Year Summary Report			
		H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10	
Parchment, Hansle (JAM) (1998)	time	2.47	3.50	4.50	5.49	6.47	7.45	8.45	9.45	10.49	11.53		13.09	4 / 1				
	reaction time	0.143	interval	7 steps	1.03	1.00	0.99	0.98	0.98	1.00	1.00	1.04	1.04	1.56		3.02	2.96	3.08
	velocity	5.55	8.87	9.14	9.23	9.33	9.33	9.14	9.14	8.79	8.79	8.79	8.99	8.40		9.08	9.26	8.90
McLeod, Omar (JAM) (1994)	time	2.40	3.47	4.47	5.47	6.47	7.45	8.49	9.53	10.58	11.66		13.17	5 / 2				
	reaction time	0.132	interval		1.07	1.00	1.00	1.00	0.98	1.04	1.04	1.05	1.08	1.51		3.07	3.02	3.17
	velocity	5.72	8.54	9.14	9.14	9.14	9.33	8.79	8.79	8.70	8.46	9.28	8.35		8.93	9.08	8.65	
Martinez, Asier (ESP) (2000)	time	2.50	3.57	4.60	5.63	6.68	7.68	8.73	9.77	10.80	11.87		13.32	6 / 3				
	reaction time	0.150	interval	7 steps	1.07	1.03	1.03	1.05	1.00	1.05	1.04	1.03	1.07	1.45		3.13	3.10	3.14
	velocity	5.49	8.54	8.87	8.87	8.70	9.14	8.70	8.79	8.87	8.54	9.67	8.26		8.76	8.85	8.73	
Czykier, Damian (POL) (1992)	time	2.49	3.52	4.53	5.53	6.55	7.57	8.59	9.65	10.72	11.77		13.32	1 / 4				
	reaction time	0.158	interval	7 steps	1.03	1.01	1.00	1.02	1.02	1.06	1.07	1.05	1.55		3.04	3.06	3.18	
	velocity	5.51	8.87	9.05	9.14	8.96	8.96	8.96	8.62	8.54	8.70	9.05	8.26		9.02	8.96	8.62	
Pozzi, Andy (GBR) (1992)	time	2.47	3.50	4.53	5.53	6.55	7.57	8.61	9.69	10.77	11.87		13.39	3 / 5				
	reaction time	0.102	interval	7 steps	1.03	1.03	1.00	1.02	1.02	1.04	1.08	1.08	1.10	1.52		3.06	3.08	3.26
	velocity	5.55	8.87	8.87	9.14	8.96	8.96	8.79	8.46	8.46	8.31	9.22	8.22		8.96	8.90	8.41	
Manga, Aurel (FRA) (1992)	time	2.47	3.53	4.59	5.67	6.70	7.75	8.80	9.87	10.93	12.03		13.61	7 / 6				
	reaction time	0.153	interval	7 steps	1.06	1.06	1.08	1.03	1.05	1.05	1.07	1.06	1.10	1.58		3.20	3.13	3.23
	velocity	5.55	8.62	8.62	8.46	8.87	8.70	8.70	8.54	8.62	8.31	8.87	8.08		8.57	8.76	8.49	
King, David (GBR) (1994)	time	2.53	3.60	4.70	5.72	6.74	7.80	8.87	9.93	11.00	12.08		13.64	2 / 7				
	reaction time	0.137	interval	7 steps	1.07	1.10	1.02	1.02	1.06	1.07	1.06	1.07	1.08	1.56		3.19	3.15	3.21
	velocity	5.42	8.54	8.31	8.96	8.96	8.62	8.54	8.62	8.54	8.46	8.99	8.06		8.60	8.70	8.54	
Zaza, Wellington (LBR) (1995)	time	2.47	3.57	4.67	5.73	6.77	7.90	8.95	10.08	11.17	12.28		13.81	8 / 8				
	reaction time	0.141	interval		1.10	1.10	1.06	1.04	1.13	1.05	1.13	1.09	1.11	1.53		3.26	3.22	3.33
	velocity	5.55	8.31	8.31	8.62	8.79	8.09	8.70	8.09	8.39	8.23	9.16	7.97		8.41	8.52	8.23	
Fillery, Cameron (GBR) (1998)	time	2.57	3.67	4.73	5.79	6.88	7.93	8.98	10.08	11.17	12.28		13.90	9 / 9				
	reaction time	0.126	interval	7 steps	1.10	1.06	1.06	1.09	1.05	1.05	1.10	1.09	1.11	1.62		3.22	3.19	3.30
	velocity	5.34	8.31	8.62	8.62	8.39	8.70	8.70	8.31	8.39	8.23	8.65	7.91		8.52	8.60	8.31	

2022 Seiko Golden Grand Prix (Tokyo, JPN)

FINAL

date 08-May-22 wind 0.1 m/s

Shibayama (2023) - race analysis of the men's 110m hurdles in the 2022 season

		H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10	
Muratake, Rashiddo (JPN) (2000)	time	2.60	3.66	4.69	5.71	6.73	7.75	8.80	9.84	10.89	11.95		13.34	5 / 1				
	reaction time	0.159	interval	7 steps	1.06	1.03	1.02	1.02	1.02	1.05	1.04	1.05	1.06	1.39		3.11	3.09	3.15
	velocity	5.28	8.62	8.87	8.96	8.96	8.96	8.70	8.79	8.70	8.62	10.09	8.25		8.82	8.87	8.70	
Chen Kuei-Ru (TPE) (1993)	time	2.58	3.62	4.68	5.70	6.74	7.77	8.82	9.88	10.95	12.02		13.49	2 / 2				
	reaction time	0.166	interval	7 steps	1.04	1.06	1.02	1.04	1.03	1.05	1.06	1.07	1.07	1.47		3.12	3.12	3.20
	velocity	5.32	8.79	8.62	8.96	8.79	8.87	8.70	8.62	8.54	8.54	9.54	8.15		8.79	8.79	8.57	

Hough, Nicholas (AUS) (1993)	time	2.66	3.77	4.82	5.85	6.87	7.89	8.90	9.94	10.98	12.05		13.49	9 / 3				
	reaction time	0.155	interval	7 steps	1.11	1.05	1.03	1.02	1.02	1.01	1.04	1.04	1.07	1.44		3.19	3.05	3.15
	velocity	5.16	8.23	8.70	8.87	8.96	8.96	9.05	8.79	8.79	8.54	9.74	8.15		8.60	8.99	8.70	
Nomoto, Syuusei (JPN) (1995)	time	2.57	3.70	4.79	5.83	6.88	7.91	8.97	10.05	11.14	12.23		13.68	3 / 4				
	reaction time	0.156	interval	7 steps	1.13	1.09	1.04	1.05	1.03	1.06	1.08	1.09	1.09	1.45		3.26	3.14	3.26
	velocity	5.34	8.09	8.39	8.79	8.70	8.87	8.62	8.46	8.39	8.39	9.67	8.04		8.41	8.73	8.41	
Fujii, Ryota (JPN) (1996)	time	2.62	3.70	4.75	5.80	6.85	7.91	8.98	10.06	11.15	12.25		13.70	7 / 5				
	reaction time	0.153	interval		1.08	1.05	1.05	1.06	1.07	1.08	1.09	1.10	1.45		3.18	3.18	3.27	
	velocity	5.24	8.46	8.70	8.70	8.70	8.62	8.54	8.46	8.39	8.31	9.67	8.03		8.62	8.62	8.39	
Yokochi, Taiga (JPN) (2000)	time	2.65	3.75	4.79	5.83	6.88	7.94	9.00	10.07	11.17	12.27		13.73	8 / 6				
	reaction time	0.155	interval	7 steps	1.10	1.04	1.04	1.05	1.06	1.06	1.07	1.10	1.10	1.46		3.18	3.17	3.27
	velocity	5.18	8.31	8.79	8.79	8.70	8.62	8.62	8.54	8.31	8.31	9.60	8.01		8.62	8.65	8.39	
Takayama, Shunya (JPN) (1994)	time	2.63	3.69	4.73	5.80	6.86	7.93	9.00	10.09	11.18	12.31		13.75	4 / 7				
	reaction time	0.156	interval		1.06	1.04	1.07	1.06	1.07	1.09	1.09	1.13	1.44		3.17	3.20	3.31	
	velocity	5.22	8.62	8.79	8.54	8.62	8.54	8.54	8.39	8.39	8.09	9.74	8.00		8.65	8.57	8.28	
Ishikawa, Shuhei (JPN) (1995)	time	2.62	3.70	4.76	5.81	6.89	7.95	9.03	10.12	11.23	12.35		13.85	6 / 8				
	reaction time	0.162	interval	7 steps	1.08	1.06	1.05	1.08	1.06	1.08	1.09	1.11	1.12	1.50		3.19	3.22	3.32
	velocity	5.24	8.46	8.62	8.70	8.46	8.62	8.46	8.39	8.23	8.16	9.35	7.94		8.60	8.52	8.26	

2022 Michitaka Kinami Memorial (Osaka, JPN)**FINAL**

INAL	date 01-May-22			wind 0.8 m/s					Shibayama (2023) - race analysis of the men's 110m hurdles in the 2022 season												
				H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10		
Hough, Nicholas (AUS) (1993)	time			2.70	3.76	4.80	5.82	6.86	7.90	8.92	9.95	11.00	12.05		13.50	3 / 1					
	reaction time	0.168	interval	7 steps	1.06	1.04	1.02	1.04	1.04	1.02	1.03	1.05	1.05	1.45				3.12	3.10	3.13	
			velocity		5.08	8.62	8.79	8.96	8.79	8.79	8.96	8.87	8.70	8.70	9.67	8.15		8.79	8.85	8.76	
Chen Kuei-Ru (TPE) 1993)	time			2.59	3.63	4.68	5.71	6.74	7.78	8.84	9.89	10.98	12.07		13.51	7 / 2					
	reaction time	0.158	interval	7 steps	1.04	1.05	1.03	1.03	1.04	1.06	1.05	1.09	1.09	1.44				3.12	3.13	3.23	
			velocity		5.30	8.79	8.70	8.87	8.87	8.79	8.62	8.70	8.39	8.39	9.74	8.14		8.79	8.76	8.49	
Ishikawa, Shuhei (JPN) (1995)	time			2.60	3.65	4.70	5.75	6.80	7.83	8.89	9.95	11.02	12.12		13.58	6 / 3					
	reaction time	0.157	interval	7 steps	1.05	1.05	1.05	1.05	1.03	1.06	1.06	1.07	1.10	1.46				3.15	3.14	3.23	
			velocity		5.28	8.70	8.70	8.70	8.70	8.87	8.62	8.62	8.54	8.31	9.60	8.10		8.70	8.73	8.49	
Fujii, Ryota (JPN) (1996)	time			2.59	3.65	4.69	5.73	6.77	7.83	8.88	9.95	11.02	12.14		13.59	4 / 4					
	reaction time	0.147	interval		1.06	1.04	1.04	1.04	1.06	1.05	1.07	1.07	1.12	1.45				3.14	3.15	3.26	
			velocity		5.30	8.62	8.79	8.79	8.79	8.62	8.70	8.54	8.54	8.16	9.67	8.09		8.73	8.70	8.41	
Andrews, Nicholas (AUS) (1997)	time			2.60	3.66	4.72	5.76	6.81	7.86	8.93	10.04	11.17	12.27		13.72	8 / 5					
	reaction time	0.119	interval		1.06	1.06	1.04	1.05	1.05	1.07	1.11	1.13	1.10	1.45				3.16	3.17	3.34	
			velocity		5.28	8.62	8.62	8.79	8.70	8.70	8.54	8.23	8.09	8.31	9.67	8.02		8.68	8.65	8.21	
Higuchi, Rikuto (JPN) (1999)	time			2.59	3.70	4.76	5.85	6.91	7.98	9.04	10.13	11.23	12.35		13.81	5 / 6					
	reaction time	0.153	interval	7 steps	1.11	1.06	1.09	1.06	1.07	1.06	1.09	1.10	1.12	1.46				3.26	3.19	3.31	
			velocity		5.30	8.23	8.62	8.39	8.62	8.54	8.62	8.39	8.31	8.16	9.60	7.97		8.41	8.60	8.28	
Kubota, Taichi (JPN) (2000)	time			2.69	3.79	4.86	5.94	7.02	8.08	9.17	10.25	11.35	12.48		13.95	2 / 7					
	reaction time	0.165	interval		1.10	1.07	1.08	1.08	1.06	1.09	1.08	1.10	1.13	1.47	PB			3.25	3.23	3.31	
			velocity		5.10	8.31	8.54	8.46	8.46	8.62	8.39	8.46	8.31	8.09	9.54	7.89		8.44	8.49	8.28	
Miyazaki, Takumi (JPN) (2001)	time			2.64	3.71	4.79	5.87	6.96	8.05	9.15	10.26	11.39	12.53		13.99	9 / 8					
	reaction time	0.152	interval		1.07	1.08	1.08	1.09	1.09	1.10	1.11	1.13	1.14	1.46				3.23	3.28	3.38	
			velocity		5.20	8.54	8.46	8.46	8.39	8.39	8.31	8.23	8.09	8.02	9.60	7.86		8.49	8.36	8.11	

Decathlon - Heat 2

Decathlon - Heat 2		date 01-May-22		wind 1.2 m/s		Takefu (2023) - performance analysis of decathlon athletes in the 2022 season														
				H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10	
Maruyama, Yuma (JPN) (1998)	time			2.69	3.82	4.91	5.99	7.07	8.14	9.23	10.36	11.49	12.61		14.13	6 / 1				
	reaction time	0.146	interval		1.13	1.09	1.08	1.08	1.07	1.09	1.13	1.13	1.12	1.52				3.30	3.24	3.38
			velocity		5.10	8.09	8.39	8.46	8.46	8.54	8.39	8.09	8.09	8.16	9.22	7.78		8.31	8.46	8.11
Okuda, Keisuke (JPN) (1996)	time			2.79	3.96	5.13	6.30	7.47	8.65	9.84	11.03	12.23	13.44		14.95	5 / 3				
	reaction time	0.132	interval		1.17	1.17	1.17	1.17	1.18	1.19	1.19	1.20	1.21	1.51				3.51	3.54	3.60
			velocity		4.92	7.81	7.81	7.81	7.81	7.75	7.68	7.68	7.62	7.55	9.28	7.36		7.81	7.75	7.62

Decathlon - Heat 1

Decathlon - Heat 1		date 01-May-22		wind 0.0 m/s		Takefu (2023) - performance analysis of decathlon athletes in the 2022 season														
				H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10	
Taue, Shun (JPN) (1997)	time			2.75	3.90	5.03	6.16	7.28	8.40	9.53	10.69	11.84	13.02	14.54		4 / 1				
	reaction time	0.139	interval			1.15	1.13	1.13	1.12	1.13	1.16	1.15	1.18	1.52			3.41	3.37	3.49	
			velocity			4.99	7.95	8.09	8.09	8.16	8.16	8.09	7.88	7.95	7.75	9.22	7.57	8.04	8.14	7.86
Nakamura, Akihiko (JPN) (1990)		time			2.79	3.95	5.09	6.22	7.36	8.50	9.63	10.78	11.94	13.11	14.64		9 / 2			

reaction time	0.147	interval	1.16	1.14	1.13	1.14	1.14	1.13	1.15	1.16	1.17	1.53			3.43	3.41	3.48
velocity	4.92		7.88	8.02	8.09	8.02	8.02	8.09	7.95	7.88	7.81	9.16	7.51		7.99	8.04	7.88
Ushiro, Keisuke (JPN) (1986)	time	2.77	3.96	5.15	6.35	7.57	8.81	10.03	11.28	12.54	13.81		15.50	3 / 4			
reaction time	0.146	interval	1.19	1.19	1.20	1.22	1.24	1.22	1.25	1.26	1.27	1.69			3.58	3.68	3.78
velocity		4.95	7.68	7.68	7.62	7.49	7.37	7.49	7.31	7.25	7.20	8.30	7.10		7.66	7.45	7.25

2022 Mikio Oda Memorial (Hiroshima, JPN)

FINAL

date 29-Apr-22 wind -1.5 m/s

Shibayama (2023) - race analysis of the men's 110m hurdles in the 2022 season

			H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
Muratake, Rashiddo (JPN) (200)	time	2.66	3.73	4.78	5.80	6.84	7.88	8.92	9.97	11.03	12.12			13.55	6 / 1			
reaction time	0.159	interval	7 steps	1.07	1.05	1.02	1.04	1.04	1.05	1.06	1.09	1.43				3.14	3.12	3.20
velocity		5.16	8.54	8.70	8.96	8.79	8.79	8.79	8.70	8.62	8.39	9.80	8.12			8.73	8.79	8.57
Hough, Nicholas (AUS) (1993)	time	2.69	3.79	4.85	5.91	6.95	8.00	9.05	10.08	11.14	12.23			13.73	1 / 2			
reaction time	0.154	interval	7 steps	1.10	1.06	1.06	1.04	1.05	1.03	1.06	1.09	1.50				3.22	3.14	3.18
velocity		5.10	8.31	8.62	8.62	8.79	8.70	8.70	8.87	8.62	8.39	9.35	8.01			8.52	8.73	8.62
Nomoto, Syuusei (JPN) (1995)	time	2.59	3.66	4.73	5.79	6.86	7.92	8.98	10.08	11.18	12.31			13.83	3 / 3			
reaction time	0.157	interval	7 steps	1.07	1.07	1.06	1.07	1.06	1.10	1.10	1.13	1.52				3.20	3.19	3.33
velocity		5.30	8.54	8.54	8.62	8.54	8.62	8.62	8.31	8.31	8.09	9.22	7.95			8.57	8.60	8.23
Takayama, Shunya (JPN) (1994)	time	2.59	3.67	4.75	5.82	6.90	7.98	9.07	10.18	11.31	12.43			13.92	7 / 4			
reaction time	0.165	interval		1.08	1.08	1.07	1.08	1.08	1.09	1.11	1.13	1.12	1.49			3.23	3.25	3.36
velocity		5.30	8.46	8.46	8.54	8.46	8.46	8.39	8.23	8.09	8.16	9.41	7.90			8.49	8.44	8.16
Ishikawa, Shuhei (JPN) (1995)	time	2.63	3.73	4.80	5.89	6.95	8.06	9.15	10.25	11.35	12.46			13.94	5 / 5			
reaction time	0.140	interval	7 steps	1.10	1.07	1.09	1.06	1.11	1.09	1.10	1.11	1.48				3.26	3.26	3.31
velocity		5.22	8.31	8.54	8.39	8.62	8.23	8.39	8.31	8.31	8.23	9.47	7.89			8.41	8.41	8.28
Fujii, Ryota (JPN) (1996)	time	2.64	3.73	4.81	5.89	6.96	8.05	9.14	10.23	11.33	12.46			13.94	8 / 6			
reaction time	0.155	interval		1.09	1.08	1.08	1.07	1.09	1.09	1.10	1.13	1.48				3.25	3.25	3.32
velocity		5.20	8.39	8.46	8.46	8.54	8.39	8.39	8.39	8.31	8.09	9.47	7.89			8.44	8.44	8.26
Yokochi, Taiga (JPN) (2000)	time	2.67	3.77	4.83	5.90	6.95	8.00	9.05	10.20	11.33	12.48			13.98	2 / 7			
reaction time	0.157	interval	7 steps	1.10	1.06	1.07	1.05	1.05	1.15	1.13	1.15	1.50				3.23	3.15	3.43
velocity		5.14	8.31	8.62	8.54	8.70	8.70	8.70	7.95	8.09	7.95	9.35	7.87			8.49	8.70	7.99

B FINAL

date 29-Apr-22 wind -0.7 m/s

Shibayama (2023) - race analysis of the men's 110m hurdles in the 2022 season

			H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
Tokuoka, Ryo (JPN) (1999)	time	2.67	3.75	4.80	5.85	6.91	7.99	9.07	10.18	11.28	12.39			13.86	3 / 1			
reaction time	0.162	interval	7 steps	1.08	1.05	1.05	1.06	1.08	1.08	1.11	1.10	1.11	1.47			3.18	3.22	3.32
velocity		5.14	8.46	8.70	8.70	8.62	8.46	8.46	8.23	8.31	8.23	9.54	7.94			8.62	8.52	8.26
Ishida Thomas, Higashi (JPN)	time	2.65	3.73	4.79	5.85	6.90	7.98	9.06	10.16	11.26	12.39			13.87	5 / 2			
reaction time	0.153	interval	7 steps	1.08	1.06	1.06	1.05	1.08	1.08	1.10	1.10	1.13	1.48			3.20	3.21	3.33
velocity		5.18	8.46	8.62	8.62	8.70	8.46	8.46	8.31	8.31	8.09	9.47	7.93			8.57	8.54	8.23
Andrews, Nicholas (AUS) (1997)	time	2.64	3.76	4.86	5.95	7.01	8.11	9.20	10.29	11.40	12.53			13.98	4 / 3			
reaction time	0.150	interval		1.12	1.10	1.09	1.06	1.10	1.09	1.09	1.11	1.13	1.45			3.31	3.25	3.33
velocity		5.20	8.16	8.31	8.39	8.62	8.31	8.39	8.39	8.23	8.09	9.67	7.87			8.28	8.44	8.23
Fujihara, Koki (JPN) (2002)	time	2.63	3.74	4.80	5.85	6.90	7.99	9.10	10.23	11.38	12.50			14.03	8 / 4			
reaction time	0.177	interval	7 steps	1.11	1.06	1.05	1.05	1.09	1.11	1.13	1.15	1.12	1.53			3.22	3.25	3.40
velocity		5.22	8.23	8.62	8.70	8.70	8.39	8.23	8.09	7.95	8.16	9.16	7.84			8.52	8.44	8.06
Omuro, Hideki (JPN) (1990)	time	2.68	3.78	4.87	5.95	7.02	8.12	9.23	10.35	11.46	12.58			14.04	7 / 5			
reaction time	0.124	interval		1.10	1.09	1.08	1.07	1.10	1.11	1.12	1.11	1.12	1.46			3.27	3.28	3.35
velocity		5.12	8.31	8.39	8.46	8.54	8.31	8.23	8.16	8.23	8.16	9.60	7.83			8.39	8.36	8.19
Yoshima, Kaito (JPN) (1998)	time	2.70	3.81	4.90	5.99	7.06	8.13	9.24	10.35	11.48	12.64			14.16	1 / 6			
reaction time	0.175	interval	7 steps	1.11	1.09	1.09	1.07	1.07	1.11	1.11	1.13	1.16	1.52			3.29	3.25	3.40
velocity		5.08	8.23	8.39	8.39	8.54	8.54	8.23	8.23	8.09	7.88	9.22	7.77			8.33	8.44	8.06
Miyazaki, Takumi (JPN) (2001)	time	2.65	3.76	4.84	5.95	7.06	8.17	9.29	10.42	11.58	12.74			14.24	2 / 7			
reaction time	0.161	interval		1.11	1.08	1.11	1.11	1.12	1.13	1.16	1.16	1.50				3.30	3.34	3.45
velocity		5.18	8.23	8.46	8.23	8.23	8.23	8.16	8.09	7.88	7.88	9.35	7.72			8.31	8.21	7.95

2022 World Athletics Indoor Championships (Belgrade, SRB) (TV Analysis)

FINAL

date 20-Mar-22

Henson (2022) - Athlete First: Grant Holloway 2022 Men's 60m hurdles world indoor record

			H1	H2	H3	H4	H5											
Holloway, Grant (USA) (1997)	time	2.36	3.40	4.38	5.38	6.38								7.39	4 / 1			
reaction time	0.151	interval	7 steps	1.04	0.98	1.00	1.00					1.01				3.02		
velocity		5.81	8.79	9.33	9.14	9.14						9.62	8.12			9.08		
Martinot-Lagarde, Pascal (FRA)	time	2.46	3.50	4.52	5.52	6.54								7.50	5 / 2			

reaction time	0.154	interval	7 steps	1.04	1.02	1.00	1.02	0.96		3.06
		velocity	5.58	8.79	8.96	9.14	8.96	10.13	8.00	8.96
Eaton, Jarrett (USA) (1989)	time	2.46	3.48	4.52	5.52	6.54		7.53	6 / 3	
reaction time	0.166	interval	7 steps	1.02	1.04	1.00	1.02	0.99		3.06
		velocity	5.58	8.96	8.79	9.14	8.96	9.82	7.97	8.96
Martinez, Asier (ESP) (2000)	time	2.50	3.54	4.58	5.58	6.58		7.57	2 / 4	
reaction time	0.150	interval	7 steps	1.04	1.04	1.00	1.00	0.99		3.08
		velocity	5.49	8.79	8.79	9.14	9.14	9.82	7.93	8.90
Douglas, Chris (AUS) (1997)	time	2.46	3.50	4.54	5.56	6.56		7.60	8 / 5	
reaction time	0.159	interval	7 steps	1.04	1.04	1.02	1.00	1.04		3.10
		velocity	5.58	8.79	8.79	8.96	9.14	9.35	7.89	8.85
King, David (GBR) (1994)	time	2.48	3.54	4.58	5.60	6.60		7.62	1 / 6	
reaction time	0.142	interval	7 steps	1.06	1.04	1.02	1.00	1.02		3.12
		velocity	5.53	8.62	8.79	8.96	9.14	9.53	7.87	8.79
Trajkovic, Milan (CYP) (1992)	time	2.52	3.58	4.62	5.62	6.62		7.62	7 / 7	
reaction time	0.207	interval	7 steps	1.06	1.04	1.00	1.00	1.00		3.10
		velocity	5.44	8.62	8.79	9.14	9.14	9.72	7.87	8.85
Belocian, Wilhem (FRA) (1995)	time	2.50	3.54	4.58	5.62	6.66		7.67	3 / 8	
reaction time	0.177	interval		1.04	1.04	1.04	1.04	1.01		3.12
		velocity	5.49	8.79	8.79	8.79	8.79	9.62	7.82	8.79

Semi-Final 3

date 20-Mar-22

Henson (2022) - Athlete First: 2022 year end hurdle report

			H1	H2	H3	H4	H5	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
Belocian, Wilhem (FRA) (1995)	time	2.53	3.56	4.56	5.56	6.56		7.53		4 / 1			
reaction time	0.129	interval		1.03	1.00	1.00	1.00	0.97			3.03		
		velocity	5.42	8.87	9.14	9.14	9.14	10.02	7.97		9.05		
Douglas, Chris (AUS) (1997)	time	2.46	3.53	4.56	5.56	6.58		7.56		8 / 2			
reaction time	0.145	interval	7 steps	1.07	1.03	1.00	1.02	0.98	NR PB		3.10		
		velocity	5.58	8.54	8.87	9.14	8.96	9.92	7.94		8.85		
King, David (GBR) (1994)	time	2.53	3.56	4.60	5.62	6.63		7.57		3 / 3			
reaction time	0.121	interval	7 steps	1.03	1.04	1.02	1.01	0.94	=PB		3.09		
		velocity	5.42	8.87	8.79	8.96	9.05	10.34	7.93		8.87		
Obasuyi, Michael (BEL) (1999)	time	2.56	3.60	4.63	5.63	6.66		7.58		5 / 4			
reaction time	0.130	interval	7 steps	1.04	1.03	1.00	1.03	0.92	PB		3.07		
		velocity	5.36	8.79	8.87	9.14	8.87	10.57	7.92		8.93		
Szymański, Jakub (POL) (2002)	time	2.53	3.56	4.58	5.60	6.60		7.59		7 / 5			
reaction time	0.146	interval	7 steps	1.03	1.02	1.02	1.00	0.99	PB		3.07		
		velocity	5.42	8.87	8.96	8.96	9.14	9.82	7.91		8.93		
Mallett, Aaron (USA) (1994)	time	2.53	3.60	4.63	5.66	6.70		7.67		6 / 6			
reaction time	0.152	interval	7 steps	1.07	1.03	1.03	1.04	0.97			3.13		
		velocity	5.42	8.54	8.87	8.87	8.79	10.02	7.82		8.76		
Van Der Schaaf, Liam (NED) (1991)	time	2.53	3.60	4.63	5.70	6.73		7.69		1 / 7			
reaction time	0.129	interval	7 steps	1.07	1.03	1.07	1.03	0.96	PB		3.17		
		velocity	5.42	8.54	8.87	8.54	8.87	10.13	7.80		8.65		
Larrinaga, Abdel-Kader (POL) (2000)	time	2.56	3.63	4.66	5.70	6.76		7.70		2 / 8			
reaction time	0.137	interval	7 steps	1.07	1.03	1.04	1.06	0.94			3.14		
		velocity	5.36	8.54	8.87	8.79	8.62	10.34	7.79		8.73		

Semi-Final 2

date 20-Mar-22

Henson (2022) - Athlete First: Grant Holloway 2022 Men's 60m hurdles world indoor record

			H1	H2	H3	H4	H5	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
Holloway, Grant (USA) (1997)	time	2.36	3.36	4.34	5.32	6.32		7.29		6 / 1			
reaction time	0.130	interval	7 steps	1.00	0.98	0.98	1.00	0.97	=WR		2.96		
		velocity	5.81	9.14	9.33	9.33	9.14	10.02	8.23		9.26		
Martinot-Lagarde, Pascal (FRA) (1991)	time	2.50	3.52	4.54	5.54	6.56		7.53		4 / 2			
reaction time	0.136	interval	7 steps	1.02	1.02	1.00	1.02	0.97			3.04		
		velocity	5.49	8.96	8.96	9.14	8.96	10.02	7.97		9.02		
Martinez, Asier (ESP) (2000)	time	2.50	3.56	4.58	5.58	6.62		7.55		2 / 3			
reaction time	0.138	interval	7 steps	1.06	1.02	1.00	1.04	0.93	=PB		3.08		
		velocity	5.49	8.62	8.96	9.14	8.79	10.45	7.95		8.90		
Pozzi, Andy (GBR) (1992)	time	2.48	3.52	4.56	5.56	6.58		7.60		3 / 4			
reaction time	0.122	interval	7 steps	1.04	1.04	1.00	1.02	1.02			3.08		

WR=World Rec. AR=Area Rec. OR=Olympic Rec. CR=Championship Rec. NR=National Rec. PB=Personal Best DQ=Disqualified DV=Drug Violation

	velocity	5.53	8.79	8.79	9.14	8.96	9.53	7.89	8.90
Czykier, Damian (POL) (1992)	time	2.50	3.54	4.58	5.60	6.64		7.61	5 / 5
reaction time	0.157	interval	7 steps	1.04	1.04	1.02	1.04	0.97	3.10
	velocity	5.49	8.79	8.79	8.96	8.79	10.02	7.88	8.85
Fofana, Hassane (ITA) (1992)	time	2.48	3.54	4.58	5.60	6.64		7.65	1 / 6
reaction time	0.122	interval		1.06	1.04	1.02	1.04	1.01	3.12
	velocity	5.53	8.62	8.79	8.96	8.79	9.62	7.84	8.79
Traber, Gregor (GER) (1992)	time	2.54	3.58	4.62	5.66	6.70		7.67	8 / 7
reaction time	0.140	interval	7 steps	1.04	1.04	1.04	1.04	0.97	3.12
	velocity	5.40	8.79	8.79	8.79	8.79	10.02	7.82	8.79
Chen Kuei-Ju (TPE) (1993)	time	2.50	3.58	4.62	5.64	6.66		7.67	7 / 8
reaction time	0.168	interval	7 steps	1.08	1.04	1.02	1.02	1.01	3.14
	velocity	5.49	8.46	8.79	8.96	8.96	9.62	7.82	8.73

Semi-Final 1

date 20-Mar-22

Henson (2022) - Athlete First: 2022 year end hurdle report

		H1	H2	H3	H4	H5	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
Eaton, Jarrett (USA) (1989)	time	2.43	3.46	4.46	5.48	6.50		7.52	4 / 1			
reaction time	0.134	interval	7 steps	1.03	1.00	1.02	1.02	1.02		3.05		
	velocity	5.65	8.87	9.14	8.96	8.96	9.53	7.98		8.99		
Trajkovic, Milan (CYP) (1992)	time	2.47	3.53	4.53	5.55	6.57		7.53	1 / 2			
reaction time	0.145	interval	7 steps	1.06	1.00	1.02	1.02	0.96		3.08		
	velocity	5.55	8.62	9.14	8.96	8.96	10.13	7.97		8.90		
Nomoto, Syuusei (JPN) (1995)	time	2.40	3.46	4.50	5.55	6.60		7.57	8 / 3			
reaction time	0.127	interval	7 steps	1.06	1.04	1.05	1.05	0.97		3.15		
	velocity	5.72	8.62	8.79	8.70	8.70	10.02	7.93		8.70		
Pereira, Rafeal (BRA) (1997)	time	2.50	3.53	4.56	5.60	6.63		7.58	3 / 4			
reaction time	0.141	interval		1.03	1.03	1.04	1.03	0.95		3.10		
	velocity	5.49	8.87	8.87	8.79	8.87	10.23	7.92		8.85		
Al-Yoha, Yaqoub (KUW) (1993)	time	2.50	3.53	4.56	5.56	6.60		7.59	6 / 5			
reaction time	0.156	interval		1.03	1.03	1.00	1.04	0.99		3.06		
	velocity	5.49	8.87	8.87	9.14	8.79	9.82	7.91		8.96		
Walters, Ruebin (TTO) (1995)	time	2.50	3.56	4.63	5.67	6.70		7.68	7 / 6			
reaction time	0.135	interval		1.06	1.07	1.04	1.03	0.98		3.17		
	velocity	5.49	8.62	8.54	8.79	8.87	9.92	7.81		8.65		
Svoboda, Petr (CZE) (1984)	time	2.50	3.56	4.58	5.58	6.63		7.71	5 / 7			
reaction time	0.136	interval		1.06	1.02	1.00	1.05	1.08		3.08		
	velocity	5.49	8.62	8.96	9.14	8.70	9.00	7.78		8.90		
Sevier, Mikdat (TUR) (1998)	time	2.53	3.60	4.66	5.66	6.73		7.75	2 / 8			
reaction time	0.129	interval		1.07	1.06	1.00	1.07	1.02		3.13		
	velocity	5.42	8.54	8.62	9.14	8.54	9.53	7.74		8.76		

Heat 6

date 20-Mar-22

Henson (2022) - Athlete First: Grant Holloway 2022 Men's 60m hurdles world indoor record

		H1	H2	H3	H4	H5	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
Belocian, Wilhem (FRA) (1995)	time	2.50	3.55	4.57	5.57	6.60		7.55	1 / 1			
reaction time	0.109	interval		1.05	1.02	1.00	1.03	0.95		3.07		
	velocity	5.49	8.70	8.96	9.14	8.87	10.23	7.95		8.93		

Heat 5

date 20-Mar-22

Henson (2022) - Athlete First: Grant Holloway 2022 Men's 60m hurdles world indoor record

		H1	H2	H3	H4	H5	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
Holloway, Grant (USA) (1997)	time	2.50	3.50	4.48	5.46	6.48		7.40	3 / 1			
reaction time	0.140	interval	7 steps	1.00	0.98	0.98	1.02	0.92		2.96		
	velocity	5.49	9.14	9.33	9.33	8.96	10.57	8.11		9.26		
Pozzi, Andy (GBR) (1992)	time	2.56	3.64	4.65	5.65	6.67		7.60	4 / 2			
reaction time	0.132	interval	7 steps	1.08	1.01	1.00	1.02	0.93		3.09		
	velocity	5.36	8.46	9.05	9.14	8.96	10.45	7.89		8.87		

Heat 4

date 20-Mar-22

Henson (2022) - Athlete First: Grant Holloway 2022 Men's 60m hurdles world indoor record

		H1	H2	H3	H4	H5	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
Svoboda, Petr (CZE) (1984)	time	2.53	3.58	4.60	5.62	6.66		7.59	4 / 1			
reaction time	0.132	interval		1.05	1.02	1.02	1.04	0.93		3.09		
	velocity	5.42	8.70	8.96	8.96	8.79	10.45	7.91		8.87		

Heat 2

date 20-Mar-22

Henson (2022) - Athlete First: Grant Holloway 2022 Men's 60m hurdles world indoor record

		H1	H2	H3	H4	H5	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
--	--	----	----	----	----	----	--------	---------------	--------------	-------	-------	--------

Pereira, Rafael (BRA) (1997)	time	2.49	3.53	4.56	5.58	6.65
reaction time	0.130	interval	1.04	1.03	1.02	1.07
velocity	5.51	8.79	8.87	8.96	8.54	

7.60	5 / 1	
0.95		3.09
10.23	7.89	8.87

Martinot-Lagarde, Pascal (FRA)	time	2.51	3.56	4.60	5.65	6.70
reaction time	0.137	interval	7 steps 1.05	1.04	1.05	1.05
velocity	5.47	8.70	8.79	8.70	8.70	

7.69	3 / 2	
0.99		3.14
9.82	7.80	8.73

Heat 1

date 20-Mar-22

Henson (2022) - Athlete First: Grant Holloway 2022 Men's 60m hurdles world indoor record

		H1	H2	H3	H4	H5	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
King, David (GBR) (1994)	time	2.53	3.60	4.63	5.65	6.68		7.65	3 / 1			
reaction time	0.118	interval	7 steps 1.07	1.03	1.02	1.03	0.97					3.12
velocity	5.42	8.54	8.87	8.96	8.87		10.02	7.84				8.79
Eaton, Jarrett (USA) (1989)	time	2.56	3.63	4.63	5.65	6.66		7.66	8 / 2			
reaction time	0.128	interval	7 steps 1.07	1.00	1.02	1.01	1.00					3.09
velocity	5.36	8.54	9.14	8.96	9.05		9.72	7.83				8.87

2022 Müller Indoor Grand Prix (Birmingham, GBR) (TV Analysis)

FINAL

date 19-Feb-22

Henson (2022) - Athlete First: Grant Holloway 2022 Men's 60m hurdles world indoor record

		H1	H2	H3	H4	H5	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
Holloway, Grant (USA) (1997)	time	2.37	3.38	4.39	5.39	6.39		7.41	3 / 1			
reaction time	0.149	interval	7 steps 1.01	1.01	1.00	1.00	1.02					3.02
velocity	5.79	9.05	9.05	9.14	9.14		9.53	8.10				9.08
Pozzi, Andy (GBR) (1992)	time	2.47	3.53	4.55	5.56	6.57		7.59	4 / 2			
reaction time	0.141	interval	7 steps 1.06	1.02	1.01	1.01	1.02					3.09
velocity	5.55	8.62	8.96	9.05	9.05		9.53	7.91				8.87
Eaton, Jarrett (USA) (1989)	time	2.46	3.49	4.51	5.52	6.53		7.60	6 / 3			
reaction time	0.154	interval	7 steps 1.03	1.02	1.01	1.01	1.07					3.06
velocity	5.58	8.87	8.96	9.05	9.05		9.08	7.89				8.96
Szymański, Jakub (POL) (2002)	time	2.46	3.50	4.52	5.55	6.57		7.62	8 / 4			
reaction time	0.165	interval	7 steps 1.04	1.02	1.03	1.02	1.05					3.09
velocity	5.58	8.79	8.96	8.87	8.96		9.26	7.87				8.87
King, David (GBR) (1994)	time	2.46	3.53	4.55	5.57	6.60		7.63	7 / 5			
reaction time	0.132	interval	7 steps 1.07	1.02	1.02	1.03	1.03	=PB				3.11
velocity	5.58	8.54	8.96	8.96	8.87		9.44	7.86				8.82
Al-Yoha, Yaqoub (KUW) (1993)	time	2.49	3.53	4.58	5.61	6.65		7.65	5 / 6			
reaction time	0.146	interval	1.04	1.05	1.03	1.04	1.00					3.12
velocity	5.51	8.79	8.70	8.87	8.79		9.72	7.84				8.79
Ortega, Orlando (ESP) (1991)	time	2.50	3.58	4.62	5.67	6.72		7.79	2 / 7			
reaction time	0.165	interval	1.08	1.04	1.05	1.05	1.07					3.17
velocity	5.49	8.46	8.79	8.70	8.70		9.08	7.70				8.65
Levy, Ronald (JAM) (1992)	time	2.52	3.60	4.68	5.75	6.81		7.87	1 / 8			
reaction time	0.146	interval	7 steps 1.08	1.08	1.07	1.06	1.06					3.23
velocity	5.44	8.46	8.46	8.54	8.62		9.17	7.62				8.49

2022 Meeting Hauts-de-France Pas-de-Calais (Liévin, FRA) (TV Analysis)

FINAL

date 17-Feb-22

Henson (2022) - Athlete First: Grant Holloway 2022 Men's 60m hurdles world indoor record

		H1	H2	H3	H4	H5	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
Holloway, Grant (USA) (1997)	time	2.39	3.40	4.38	5.36	6.36		7.35	4 / 1			
reaction time	0.167	interval	7 steps 1.01	0.98	0.98	1.00	0.99					2.97
velocity	5.74	9.05	9.33	9.33	9.14		9.82	8.16				9.23
Martinot-Lagarde, Pascal (FRA)	time	2.43	3.46	4.46	5.46	6.46		7.46	6 / 2			
reaction time	0.144	interval	7 steps 1.03	1.00	1.00	1.00	1.00					3.03
velocity	5.65	8.87	9.14	9.14	9.14		9.72	8.04				9.05
Eaton, Jarrett (USA) (1989)	time	2.43	3.44	4.44	5.44	6.46		7.51	5 / 3			
reaction time	0.146	interval	7 steps 1.01	1.00	1.00	1.02	1.05					3.01
velocity	5.65	9.05	9.14	9.14	8.96		9.26	7.99				9.11
Llopis, Enrique (ESP) (2000)	time	2.55	3.58	4.60	5.60	6.60		7.62	2 / 4			
reaction time	0.163	interval	7 steps 1.03	1.02	1.00	1.00	1.02					3.05
velocity	5.38	8.87	8.96	9.14	9.14		9.53	7.87				8.99
Pozzi, Andy (GBR) (1992)	time	2.55	3.58	4.58	5.60	6.62		7.62	1 / 5			
reaction time	0.163	interval	7 steps 1.03	1.00	1.02	1.02	1.00					3.05
velocity	5.38	8.87	9.14	8.96	8.96		9.72	7.87				8.99

Zaza, Wellington (LBR) (1995)	time	2.45	3.52	4.55	5.57	6.63		7.63	8 / 6	
	reaction time	0.149		1.07	1.03	1.02	1.06	1.00		3.12
	velocity	5.60	8.54	8.87	8.96	8.62		9.72	7.86	8.79

Roberts, Daniel (USA) (1998)	time	2.43	3.46	4.53	5.55	6.60		7.67	7 / 7	
	reaction time	0.172		1.03	1.07	1.02	1.05	1.07		3.12
	velocity	5.65	8.87	8.54	8.96	8.70		9.08	7.82	8.79

Belocian, Wilhem (FRA) (1995)	time	2.46	3.52	4.55	5.57	6.66		7.72	3 / 8	
	reaction time	0.130		1.06	1.03	1.02	1.09	1.06		3.11
	velocity	5.58	8.62	8.87	8.96	8.39		9.17	7.77	8.82

Heat 2

date 17-Feb-22

Henson (2022) - Athlete First: Grant Holloway 2022 Men's 60m hurdles world indoor record

			H1	H2	H3	H4	H5	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
Holloway, Grant (USA) (1997)	time	2.35	3.38	4.37	5.37	6.38			7.40	4 / 1			
	reaction time	0.153		1.03	0.99	1.00	1.01	1.02			3.02		
	velocity	5.84	8.87	9.23	9.14	9.05		9.53	8.11		9.08		
Belocian, Wilhem (FRA) (1995)	time	2.43	3.45	4.50	5.54	6.54			7.57	5 / 2			
	reaction time	0.139		1.02	1.05	1.04	1.00	1.03			3.11		
	velocity	5.65	8.96	8.70	8.79	9.14		9.44	7.93		8.82		
Roberts, Daniel (USA) (1998)	time	2.37	3.40	4.44	5.48	6.54			7.58	3 / 3			
	reaction time	0.149		1.03	1.04	1.04	1.06	1.04			3.11		
	velocity	5.79	8.87	8.79	8.79	8.62		9.35	7.92		8.82		
Dickson, Michael (USA) (1997)	time	2.46	3.55	4.58	5.60	6.63			7.64	6 / 4			
	reaction time	0.183		1.09	1.03	1.02	1.03	1.01			3.14		
	velocity	5.58	8.39	8.87	8.96	8.87		9.62	7.85		8.73		
Obasuyi, Michael (BEL) (1999)	time	2.48	3.57	4.60	5.62	6.60			7.65	2 / 5			
	reaction time	0.150		1.09	1.03	1.02	0.98	1.05			3.14		
	velocity	5.53	8.39	8.87	8.96	9.33		9.26	7.84		8.73		
Al-Yoha, Yaqoub (KUW) (1993)	time	2.53	3.58	4.62	5.66	6.70			7.73	7 / 6			
	reaction time	0.166		1.05	1.04	1.04	1.04	1.03			3.13		
	velocity	5.42	8.70	8.79	8.79	8.79		9.44	7.76		8.76		
Lakka, Elmo (FIN) (1993)	time	2.53	3.62	4.72	5.80	6.85			7.98	8 / 7			
	reaction time	0.152		1.09	1.10	1.08	1.05	1.13			3.27		
	velocity	5.42	8.39	8.31	8.46	8.70		8.60	7.52		8.39		

2022 New Balance Indoor Grand Prix (Staten Island, NY) (TV Analysis)

FINAL

date 06-Feb-22

Henson (2022) - Athlete First: Grant Holloway 2022 Men's 60m hurdles world indoor record

			H1	H2	H3	H4	H5	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
Holloway, Grant (USA) (1997)	time	2.45	3.44	4.45	5.44	6.42			7.37	3 / 1			
	reaction time	0.181		1.03	0.99	0.98	0.98	0.95			2.99		
	velocity	5.60	9.23	9.05	9.23	9.33		10.23	8.14		9.17		
Cunningham, Trey (USA) (1998)	time	2.48	3.50	4.51	5.51	6.55			7.47	4 / 2			
	reaction time	0.189		1.02	1.01	1.00	1.04	0.92			3.03		
	velocity	5.53	8.96	9.05	9.14	8.79		10.57	8.03		9.05		
Allen, Devon (USA) (1994)	time	2.51	3.55	4.58	5.57	6.58			7.51	5 / 3			
	reaction time	0.143		1.04	1.03	0.99	1.01	0.93			3.06		
	velocity	5.47	8.79	8.87	9.23	9.05		10.45	7.99		8.96		
Roberts, Daniel (USA) (1998)	time	2.47	3.54	4.60	5.61	6.65			7.60	6 / 4			
	reaction time	0.175		1.07	1.06	1.01	1.04	0.95			3.14		
	velocity	5.55	8.54	8.62	9.05	8.79		10.23	7.89		8.73		
Crittenden, Freddie (USA) (1998)	time	2.57	3.62	4.67	5.69	6.71			7.71	2 / 5			
	reaction time	0.169		1.05	1.05	1.02	1.02	1.00			3.12		
	velocity	5.34	8.70	8.70	8.96	8.96		9.72	7.78		8.79		
Szücs, Valdó (HUN) (1995)	time	2.57	3.64	4.70	5.72	6.72			7.73	1 / 6			
	reaction time	0.160		1.07	1.06	1.02	1.00	1.01			3.15		
	velocity	5.34	8.54	8.62	8.96	9.14		9.62	7.76		8.70		
Walters, Ruebin (TTO) (1995)	time	2.61	3.70	4.76	5.79	6.81			7.76	8 / 7			
	reaction time	0.176		1.09	1.06	1.03	1.02	0.95			3.18		
	velocity	5.26	8.39	8.62	8.87	8.96		10.23	7.73		8.62		
Dunning, Robert (USA) (1997)	time	2.55	3.64	4.70	5.74	6.85			7.82	7 / 8			
	reaction time	0.179		1.09	1.06	1.04	1.11	0.97			3.19		
	velocity	5.38	8.39	8.62	8.79	8.23		10.02	7.67		8.60		

2021 Japanese U16 National Championships (Matsuyama, JPN) (99cm)

FINAL

FINAL	date	24-Oct-21	wind 1.0 m/s			Shibayama (2021) - race analysis of men's 110m hurdles in the 2021 season												
			H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
Hashimoto, Yu (JPN)	time		2.67	3.77	4.88	6.00	7.12	8.23	9.36	10.47	11.55	12.68		14.19	/ 1			
	reaction time			1.10	1.11	1.12	1.12	1.11	1.13	1.11	1.08	1.13	1.51			3.33	3.36	3.32
	velocity		5.14	8.31	8.23	8.16	8.16	8.23	8.09	8.23	8.46	8.09	9.28	7.75		8.23	8.16	8.26
Kawahara, Ren (JPN) (2006)	time		2.74	3.89	5.05	6.21	7.39	8.54	9.69	10.81	11.94	13.11		14.68	/ 2			
	reaction time			1.15	1.16	1.16	1.18	1.15	1.15	1.12	1.13	1.17	1.57			3.47	3.48	3.42
	velocity		5.01	7.95	7.88	7.88	7.75	7.95	7.95	8.16	8.09	7.81	8.93	7.49		7.90	7.88	8.02
Asai, Seira (JPN (2006)	time		2.83	3.96	5.10	6.27	7.44	8.60	9.75	10.87	12.02	13.18		14.77	/ 3			
	reaction time			1.13	1.14	1.17	1.17	1.16	1.15	1.12	1.15	1.16	1.59			3.44	3.48	3.43
	velocity		4.85	8.09	8.02	7.81	7.81	7.88	7.95	8.16	7.95	7.88	8.82	7.45		7.97	7.88	7.99
Nakamura, Shyuunta (JPN)	time		2.70	3.84	5.00	6.16	7.32	8.51	9.68	10.84	12.01	13.23		14.81	/ 4			
	reaction time			1.14	1.16	1.16	1.16	1.19	1.17	1.16	1.17	1.22	1.58			3.46	3.52	3.55
	velocity		5.08	8.02	7.88	7.88	7.88	7.68	7.81	7.88	7.81	7.49	8.87	7.43		7.92	7.79	7.72
Yamauti, Masanari (JPN)	time		2.74	3.90	5.06	6.21	7.37	8.50	9.69	10.85	12.03	13.22		14.83	/ 5			
	reaction time			1.16	1.16	1.15	1.16	1.13	1.19	1.16	1.18	1.19	1.61			3.47	3.48	3.53
	velocity		5.01	7.88	7.88	7.95	7.88	8.09	7.68	7.88	7.75	7.68	8.71	7.42		7.90	7.88	7.77
Susami, Youhei (JPN) (2006)	time		2.73	3.90	5.05	6.23	7.37	8.56	9.72	10.92	12.08	13.30		14.89	/ 6			
	reaction time			1.17	1.15	1.18	1.14	1.19	1.16	1.20	1.16	1.22	1.59			3.50	3.49	3.58
	velocity		5.03	7.81	7.95	7.75	8.02	7.68	7.88	7.62	7.88	7.49	8.82	7.39		7.83	7.86	7.66
Adachi, Eiji (JPN) (2006)	time		2.73	3.92	5.07	6.25	7.42	8.61	9.79	10.97	12.13	13.32		14.89	/ 7			
	reaction time			1.19	1.15	1.18	1.17	1.19	1.18	1.18	1.16	1.19	1.57			3.52	3.54	3.53
	velocity		5.03	7.68	7.95	7.75	7.81	7.68	7.75	7.75	7.88	7.68	8.93	7.39		7.79	7.75	7.77
Takahashi, Hayato (JPN)	time		2.78	3.96	5.13	6.32	7.52	8.73	9.93	11.13	12.32	13.57		15.22	/ 8			
	reaction time			1.18	1.17	1.19	1.20	1.21	1.20	1.20	1.19	1.25	1.65			3.54	3.61	3.64
	velocity		4.94	7.75	7.81	7.68	7.62	7.55	7.62	7.62	7.68	7.31	8.50	7.23		7.75	7.60	7.53

2021 Japanese U18 National Championships (Matsuyama, JPN) (99cm)

FINAL

FINAL		date 22-Oct-21		wind -0.6 m/s		Shibayama (2021) - race analysis of men's 110m hurdles in the 2021 season													
						H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4
Nishi, Tetsuro (JPN) (2004)	time		2.61	3.68	4.72	5.75	6.77	7.80	8.83	9.87	10.92	11.98		13.48	5 / 1				
	reaction time	0.131	interval	7 steps	1.07	1.04	1.03	1.02	1.03	1.03	1.04	1.05	1.06	1.50			3.14	3.08	3.15
		velocity		5.26	8.54	8.79	8.87	8.96	8.87	8.87	8.79	8.70	8.62	9.35	8.16		8.73	8.90	8.70
Tahara, Ayuma (JPN) (2004)	time		2.63	3.74	4.82	5.88	6.95	8.03	9.11	10.20	11.31	12.43		13.94	6 / 2				
	reaction time	0.150	interval	7 steps	1.11	1.08	1.06	1.07	1.08	1.08	1.09	1.11	1.12	1.51			3.25	3.23	3.32
		velocity		5.22	8.23	8.46	8.62	8.54	8.46	8.46	8.39	8.23	8.16	9.28	7.89		8.44	8.49	8.26
Hirotsu, Ato (JPN) (2004)	time		2.73	3.83	4.91	5.99	7.07	8.14	9.25	10.33	11.45	12.56		14.06	4 / 3				
	reaction time	0.160	interval		1.10	1.08	1.08	1.08	1.07	0.11	2.08	1.12	1.11	1.50			3.26	2.26	4.31
		velocity		5.03	8.31	8.46	8.46	8.46	8.54	83.09	4.39	8.16	8.23	9.35	7.82		8.41	12.13	6.36
Uchida, Kaisei (JPN) (2004)	time		2.64	3.76	4.86	5.96	7.07	8.16	9.27	10.39	11.53	12.65		14.12	8 / 4				
	reaction time	0.139	interval		1.12	1.10	1.10	1.11	1.09	1.11	1.12	1.14	1.12	1.47			3.32	3.31	3.38
		velocity		5.20	8.16	8.31	8.31	8.23	8.39	8.23	8.16	8.02	8.16	9.54	7.79		8.26	8.28	8.11
Takahashi, Ryo (JPN) (2005)	time		2.71	3.83	4.93	6.02	7.10	8.19	9.31	10.40	11.52	12.64		14.15	9 / 5				
	reaction time	0.163	interval		1.12	1.10	1.09	1.08	1.09	1.12	1.09	1.12	1.12	1.51			3.31	3.29	3.33
		velocity		5.06	8.16	8.31	8.39	8.46	8.39	8.16	8.39	8.16	8.16	9.28	7.77		8.28	8.33	8.23
Tomita, Kanto (JPN) (2004)	time		2.71	3.82	4.91	6.00	7.10	8.19	9.30	10.41	11.54	12.68		14.20	3 / 6				
	reaction time	0.191	interval		1.11	1.09	1.09	1.10	1.09	1.11	1.11	1.13	1.14	1.52			3.29	3.30	3.38
		velocity		5.06	8.23	8.39	8.39	8.31	8.39	8.23	8.23	8.09	8.02	9.22	7.75		8.33	8.31	8.11
Nakai, Shouta (JPN) (2004)	time		2.74	3.90	5.01	6.10	7.21	8.30	9.39	10.48	11.62	12.75		14.28	7 / 7				
	reaction time	0.188	interval		1.16	1.11	1.09	1.11	1.09	1.09	1.14	1.13	1.53				3.36	3.29	3.36
		velocity		5.01	7.88	8.23	8.39	8.23	8.39	8.39	8.39	8.02	8.09	9.16	7.70		8.16	8.33	8.16

2021 Boris Hanžeković Memorial (Zagreb, CRO) (TV Analysis)

FINAL

date		14-Sep-21		wind 0.7 m/s		Henson (2021) - Athlete First: 2021 year end hurdle report													
						H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4
Allen, Devon (USA) (1994)	time			2.46	3.50	4.50	5.47	6.47	7.47	8.47	9.47	10.50	11.53		12.99	5 / 1			
	reaction time	0.126	interval		1.04	1.00	0.97	1.00	1.00	1.00	1.00	1.03	1.03	1.46	PB		3.01	3.00	3.06
		velocity			5.58	8.79	9.14	9.42	9.14	9.14	9.14	9.14	8.87	8.87	9.60	8.47		9.11	9.14
Levy, Ronald (JAM) (1992)	time			2.48	3.52	4.52	5.52	6.50	7.50	8.50	9.53	10.56	11.63		13.11	4 / 2			
	reaction time	0.156	interval		7 steps	1.04	1.00	1.00	0.98	1.00	1.00	1.03	1.03	1.07	1.48		3.04	2.98	3.13
		velocity																	

	velocity	5.53	8.79	9.14	9.14	9.33	9.14	9.14	8.87	8.87	8.54	9.47	8.39		9.02	9.20	8.76
Parchment, Hansle (JAM) (1991)	time	2.50	3.55	4.55	5.54	6.53	7.52	8.53	9.54	10.56	11.63		13.12	6 / 3			
	reaction time	0.141	interval	7 steps	1.05	1.00	0.99	0.99	1.01	1.01	1.02	1.07	1.49		3.04	2.99	3.10
	velocity	5.49	8.70	9.14	9.23	9.23	9.23	9.05	9.05	8.96	8.54	9.41	8.38		9.02	9.17	8.85
Svoboda, Petr (CZE) (1984)	time	2.50	3.60	4.63	5.63	6.66	7.70	8.73	9.76	10.86	11.93		13.51	7 / 4			
	reaction time	0.167	interval		1.10	1.03	1.00	1.03	1.04	1.03	1.10	1.07	1.58		3.13	3.10	3.20
	velocity	5.49	8.31	8.87	9.14	8.87	8.79	8.87	8.87	8.31	8.54	8.87	8.14		8.76	8.85	8.57
King, David (GBR) (1994)	time	2.50	3.60	4.63	5.66	6.66	7.70	8.73	9.76	10.86	12.03		13.56	3 / 5			
	reaction time	0.140	interval	7 steps	1.10	1.03	1.03	1.00	1.04	1.03	1.10	1.17	1.53		3.16	3.07	3.30
	velocity	5.49	8.31	8.87	8.87	9.14	8.79	8.87	8.87	8.31	7.81	9.16	8.11		8.68	8.93	8.31
Pedriks, Keiso (EST) (1994)	time	2.63	3.68	4.73	5.78	6.81	7.83	8.86	9.93				13.63	8 / 6			
	reaction time	0.182	interval	7 steps	1.05	1.05	1.05	1.03	1.02	1.03	1.07		PB		3.15	3.08	
	velocity	5.22	8.70	8.70	8.70	8.87	8.96	8.87	8.54				8.07		8.70	8.90	
Vukicevic, Vladimir (NOR) (1991)	time	2.56	3.63	4.66	5.70	6.73	7.76	8.83	9.90	11.00	12.10		13.65	2 / 7			
	reaction time	0.136	interval		1.07	1.03	1.04	1.03	1.03	1.07	1.10	1.10	1.55		3.14	3.13	3.27
	velocity	5.36	8.54	8.87	8.79	8.87	8.87	8.54	8.54	8.31	8.31	9.05	8.06		8.73	8.76	8.39

2021 Weltklasse (Zürich, SUI) (TV Analysis)

FINAL

date 09-Sep-21 wind 0.6 m/s Henson (2021) - Athlete First: 2021 year end hurdle report

		H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
Allen, Devon (USA) (1994)	time	2.48	3.52	4.52	5.52	6.54	7.56	8.58	9.60	10.60	11.64		13.06	4 / 1			
	reaction time	0.115	interval	1.04	1.00	1.00	1.02	1.02	1.02	1.00	1.04	1.42			3.04	3.06	3.06
	velocity	5.53	8.79	9.14	9.14	8.96	8.96	8.96	8.96	9.14	8.79	9.87	8.42		9.02	8.96	8.96
Levy, Ronald (JAM) (1992)	time	2.46	3.50	4.54	5.54	6.56	7.58	8.58	9.58	10.62	11.66		13.06	7 / 2			
	reaction time	0.128	interval	7 steps	1.04	1.04	1.00	1.02	1.02	1.00	1.04	1.40			3.08	3.04	3.08
	velocity	5.58	8.79	8.79	9.14	8.96	8.96	9.14	9.14	8.79	8.79	10.01	8.42		8.90	9.02	8.90
Parchment, Hansle (JAM) (1991)	time	2.50	3.52	4.54	5.52	6.54	7.54	8.54	9.56	10.60	11.68		13.17	5 / 3			
	reaction time	0.132	interval	7 steps	1.02	1.02	0.98	1.02	1.00	1.02	1.04	1.08	1.49		3.02	3.02	3.14
	velocity	5.49	8.96	8.96	9.33	8.96	9.14	9.14	8.96	8.79	8.46	9.41	8.35		9.08	9.08	8.73
Roberts, Daniel (USA) (1997)	time	2.46	3.52	4.54	5.56	6.58	7.60	8.64	9.68	10.74	11.80		13.31	3 / 4			
	reaction time	0.173	interval	1.06	1.02	1.02	1.02	1.04	1.04	1.06	1.06	1.51			3.10	3.08	3.16
	velocity	5.58	8.62	8.96	8.96	8.96	8.96	8.79	8.79	8.62	8.62	9.28	8.26		8.85	8.90	8.68
Dal Molin, Paolo (ITA) (1987)	time	2.54	3.54	4.56	5.60	6.64	7.68	8.76	9.80	10.86	11.96		13.43	2 / 5			
	reaction time	0.135	interval	7 steps	1.00	1.02	1.04	1.04	1.08	1.04	1.06	1.10	1.47		3.06	3.16	3.20
	velocity	5.40	9.14	8.96	8.79	8.79	8.79	8.46	8.79	8.62	8.31	9.54	8.19		8.96	8.68	8.57
Gaio, Finley (SUI) (1999)	time	2.56	3.64	4.70	5.74	6.82	7.88	8.96	10.02	11.12	12.22		13.72	8 / 6			
	reaction time	0.135	interval	7 steps	1.08	1.06	1.04	1.08	1.06	1.08	1.10	1.10	1.50		3.18	3.22	3.26
	velocity	5.36	8.46	8.62	8.79	8.46	8.62	8.46	8.62	8.31	8.31	9.35	8.02		8.62	8.52	8.41
Smet, Koen (NED) (1992)	time	2.56	3.62	4.66	5.70	6.78	7.84	8.94	10.02	11.10	12.22		13.77	1 / 7			
	reaction time	0.132	interval		1.06	1.04	1.04	1.08	1.06	1.10	1.08	1.12	1.55		3.14	3.24	3.28
	velocity	5.36	8.62	8.79	8.79	8.46	8.62	8.31	8.46	8.46	8.16	9.05	7.99		8.73	8.46	8.36

2021 Meeting de Paris (Paris, FRA) (TV Analysis)

FINAL

date 28-Aug-21 wind 0.7 m/s Henson (2021) - Athlete First: 2021 year end hurdle report

		H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
Parchment, Hansle (JAM) (1991)	time	2.46	3.48	4.48	5.48	6.46	7.46	8.44	9.46	10.50	11.54		13.03	6 / 1			
	reaction time	0.138	interval	7 steps	1.02	1.00	1.00	0.98	1.00	0.98	1.02	1.04	1.49		3.02	2.96	3.10
	velocity	5.58	8.96	9.14	9.14	9.33	9.14	9.33	8.96	8.79	8.79	9.41	8.44		9.08	9.26	8.85
Allen, Devon (USA) (1994)	time	2.52	3.54	4.56	5.56	6.56	7.56	8.58	9.58	10.60	11.62		13.08	5 / 2			
	reaction time	0.141	interval		1.02	1.02	1.00	1.00	1.02	1.00	1.02	1.46			3.04	3.02	3.04
	velocity	5.44	8.96	8.96	9.14	9.14	9.14	8.96	9.14	8.96	8.96	9.60	8.41		9.02	9.08	9.02
Roberts, Daniel (USA) (1997)	time	2.46	3.50	4.52	5.54	6.52	7.56	8.58	9.58	10.62	11.68		13.16	2 / 3			
	reaction time	0.169	interval	1.04	1.02	1.02	0.98	1.04	1.02	1.00	1.04	1.06	1.48		3.08	3.04	3.10
	velocity	5.58	8.79	8.96	8.96	9.33	8.79	8.96	9.14	8.79	8.62	9.47	8.36		8.90	9.02	8.85
Levy, Ronald (JAM) (1992)	time	2.48	3.54	4.58	5.60	6.60	7.62	8.64	9.68	10.72	11.78		13.24	4 / 4			
	reaction time	0.144	interval	7 steps	1.06	1.04	1.02	1.00	1.02	1.02	1.04	1.06	1.46		3.12	3.04	3.14
	velocity	5.53	8.62	8.79	8.96	9.14	8.96	8.96	8.79	8.79	8.62	9.60	8.31		8.79	9.02	8.73
Manga, Aurel (FRA) (1992)	time	2.44	3.50	4.54	5.58	6.60	7.62	8.64	9.70	10.76	11.84		13.40	7 / 5			
	reaction time	0.154	interval	7 steps	1.06	1.04	1.04	1.02	1.02	1.06	1.06	1.08	1.56		3.14	3.06	3.20
	velocity	5.62	8.62	8.79	8.79	8.96	8.96	8.96	8.62	8.62	8.46	8.99	8.21		8.73	8.96	8.57

Pozzi, Andy (GBR) (1992)	time	2.52	3.56	4.60	5.62	6.64	7.68	8.72	9.76	10.86	11.94		13.49	8 / 6			
	reaction time	0.126	interval	7 steps	1.04	1.04	1.02	1.02	1.04	1.04	1.10	1.08	1.55		3.10	3.10	3.22
	velocity	5.44	8.79	8.79	8.96	8.96	8.79	8.79	8.79	8.31	8.46	9.05	8.15		8.85	8.85	8.52
Kwaou-Mathey, Just (FRA) (1992)	time	2.60	3.64	4.68	5.72	6.74	7.76	8.78	9.84	10.92	12.00		13.50	1 / 7			
	reaction time	0.161	interval	7 steps	1.04	1.04	1.04	1.02	1.02	1.06	1.08	1.08	1.50		3.12	3.06	3.22
	velocity	5.28	8.79	8.79	8.79	8.96	8.96	8.96	8.62	8.46	8.46	9.35	8.15		8.79	8.96	8.52
Martinot-Lagarde, Pascal (FRA) (1992)	time	2.46	3.48	4.50	5.52	6.54	7.58	8.66						3 / --			
	reaction time	0.150	interval	7 steps	1.02	1.02	1.02	1.04	1.08						3.06	3.14	
	velocity	5.58	8.96	8.96	8.96	8.96	8.79	8.46							8.96	8.73	

2021 Athletissima (Lausanne, SUI) (TV Analysis)**FINAL**

		date	26-Aug-21	wind	2.5 m/s	Henson (2021) - Athlete First: 2021 year end hurdle report													
				H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
Allen, Devon (USA) (1994)	time			2.48	3.48	4.52	5.56	6.56	7.56	8.56	9.56	10.56	11.62		13.07	5 / 1			
	reaction time	0.140	interval		1.00	1.04	1.04	1.00	1.00	1.00	1.00	1.00	1.06	1.45			3.08	3.00	3.06
			velocity	5.53	9.14	8.79	8.79	9.14	9.14	9.14	9.14	9.14	9.14	8.62	9.67	8.42		8.90	9.14
Joseph, Jason (SUI) (1998)	time			2.56	3.60	4.64	5.62	6.60	7.60	8.60	9.60	10.62	11.62		13.11	3 / 2			
	reaction time	0.149	interval	7 steps	1.04	1.04	0.98	0.98	1.00	1.00	1.00	1.02	1.00	1.49			3.06	2.98	3.02
			velocity	5.36	8.79	8.79	9.33	9.33	9.14	9.14	9.14	8.96	9.14	9.41	8.39		8.96	9.20	9.08
Martinot-Lagarde, Pascal (FRA)	time			2.52	3.54	4.56	5.56	6.56	7.56	8.60	9.62	10.68	11.72		13.17	2 / 3			
	reaction time	0.149	interval	7 steps	1.02	1.02	1.00	1.00	1.00	1.04	1.02	1.06	1.04	1.45			3.04	3.04	3.12
			velocity	5.44	8.96	8.96	9.14	9.14	9.14	8.79	8.96	8.62	8.79	9.67	8.35		9.02	9.02	8.79
Roberts, Daniel (USA) (1997)	time			2.48	3.52	4.56	5.56	6.58	7.60	8.62	9.66	10.72	11.78		13.23	7 / 4			
	reaction time	0.178	interval		1.04	1.04	1.00	1.02	1.02	1.02	1.04	1.06	1.06	1.45			3.08	3.06	3.16
			velocity	5.53	8.79	8.79	9.14	8.96	8.96	8.96	8.79	8.62	8.62	9.67	8.31		8.90	8.96	8.68
Levy, Ronald (JAM) (1992)	time			2.48	3.56	4.60	5.64	6.68	7.70	8.72	9.76	10.80	11.86		13.40	4 / 5			
	reaction time	0.113	interval	7 steps	1.08	1.04	1.04	1.04	1.02	1.02	1.04	1.04	1.06	1.54			3.16	3.08	3.14
			velocity	5.53	8.46	8.79	8.79	8.79	8.96	8.96	8.79	8.79	8.62	9.10	8.21		8.68	8.90	8.73
Smet, Koen (NED) (1992)	time			2.56	3.60	4.62	5.64	6.68	7.72	8.78	9.78	10.88	11.94		13.42	1 / 6			
	reaction time	0.144	interval		1.04	1.02	1.02	1.04	1.04	1.06	1.00	1.10	1.06	1.48			3.08	3.14	3.16
			velocity	5.36	8.79	8.96	8.96	8.79	8.79	8.62	9.14	8.31	8.62	9.47	8.20		8.90	8.73	8.68
Pozzi, Andy (GBR) (1992)	time			2.48	3.52	4.56	5.56	6.56	7.58	8.62	9.68	10.80	11.92		13.45	8 / 7			
	reaction time	0.149	interval	7 steps	1.04	1.04	1.00	1.00	1.02	1.04	1.06	1.12	1.12	1.53			3.08	3.06	3.30
			velocity	5.53	8.79	8.79	9.14	9.14	8.96	8.79	8.62	8.16	8.16	9.16	8.18		8.90	8.96	8.31
Parchment, Hansle (JAM) (1992)	time			2.52	3.52	4.54	5.52	6.54	7.54	8.56	9.56	10.66	11.76		13.58	6 / 8			
	reaction time	0.133	interval	7 steps	1.00	1.02	0.98	1.02	1.00	1.02	1.00	1.10	1.10	1.82			3.00	3.04	3.20
			velocity	5.44	9.14	8.96	9.33	8.96	9.14	8.96	9.14	8.31	8.31	7.70	8.10		9.14	9.02	8.57

2021 World Athletics U20 Championships (Nairobi, KEN) (99.1cm) (Altitude) (TV Analysis)**FINAL**

INAL	date	21-Aug-21	wind	1.0 m/s	Henson (2021) - Athlete First: 2021 Sasha Zhoya 110m Hurdles World U20 Record														
			H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10	
Zhoya, Sasha (FRA) (2002)	time		2.32	3.36	4.32	5.28	6.24	7.20	8.20	9.20	10.16	11.16		12.72	4 / 1				
	reaction time	0.138	interval	7 steps	1.04	0.96	0.96	0.96	0.96	1.00	1.00	0.96	1.00	1.56	WU20R		2.96	2.92	2.96
			velocity	5.91	8.79	9.52	9.52	9.52	9.52	9.14	9.14	9.52	9.14	8.99	8.65		9.26	9.39	9.26

Semi-Final 2

emi-Final 2		date	20-Aug-21	wind	-0.4 m/s																Henson (2021) - Athlete First: 2021 Sasha Zhoya 110m Hurdles World U20 Record											
					H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10												
Zhoya, Sasha (FRA) (2002)	time				2.36	3.36	4.36	5.34	6.32	7.28	8.26	9.26	10.28	11.32		12.93	6 / 1															
	reaction time	0.160	interval	7 steps	1.00	1.00	0.98	0.98	0.96	0.98	1.00	1.02	1.04	1.61	WU20R			2.98	2.92	3.06												
			velocity		5.81	9.14	9.14	9.33	9.33	9.52	9.33	9.14	8.96	8.79	8.71	8.51			9.20	9.39	8.96											

Heat 3

Heat 3	date 20-Aug-21		wind 0.3 m/s				Henson (2021) - Athlete First: 2021 Sasha Zhoya 110m Hurdles World U20 Record												
			H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10	
	Zhoya, Sasha (FRA) (2002)	time	2.43	3.50	4.50	5.50	6.50	7.50	8.50	9.52	10.54	11.58		13.12	4 / 1				
	reaction time 0.175	interval	7 steps	1.07	1.00	1.00	1.00	1.00	1.02	1.02	1.04	1.54				3.07	3.00	3.08	
		velocity	5.65	8.54	9.14	9.14	9.14	9.14	9.14	8.96	8.96	8.79	9.10	8.38		8.93	9.14	8.90	

2020 Olympic Games (Tokyo, JPN) (TV Analysis)**FINAL**

FINAL	date		06-Aug-21		wind -0.5 m/s										Henson (2021) - Athlete First: 2021 year end hurdle report						
					H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10	
Parchment, Hansle (JAM) (1992)	time				2.50	3.54	4.56	5.54	6.52	7.50	8.50	9.54	10.56	11.58		13.04	7 / 1				
	reaction time	0.130	interval	7 steps	1.04	1.02	0.98	0.98	0.98	1.00	1.04	1.02	1.02	1.46				3.04	2.96	3.08	
			velocity		5.49	8.79	8.96	9.33	9.33	9.33	9.14	8.79	8.96	8.96	9.60	8.44			9.02	9.26	8.90
Holloway, Grant (USA) (1997)	time				2.40	3.40	4.40	5.40	6.40	7.38	8.40	9.44	10.50	11.58		13.09	4 / 2				
	reaction time	0.136	interval	7 steps	1.00	1.00	1.00	1.00	0.98	1.02	1.04	1.06	1.08	1.51				3.00	3.00	3.18	
			velocity		5.72	9.14	9.14	9.14	9.14	9.33	8.96	8.79	8.62	8.46	9.28	8.40			9.14	9.14	8.62

Levy, Ronald (JAM) (1992)	time	2.46	3.50	4.54	5.56	6.58	7.60	8.60	9.62	10.66	11.72		13.10	5 / 3				
	reaction time	0.146	interval	7 steps	1.04	1.04	1.02	1.02	1.02	1.00	1.02	1.04	1.06	1.38		3.10	3.04	3.12
	velocity		5.58	8.79	8.79	8.96	8.96	8.96	9.14	8.96	8.79	8.62	10.16	8.40		8.85	9.02	8.79
Allen, Devon (USA) (1994)	time	2.52	3.56	4.58	5.60	6.58	7.62	8.62	9.66	10.70	11.72		13.14	6 / 4				
	reaction time	0.133	interval		1.04	1.02	1.02	0.98	1.04	1.00	1.04	1.04	1.02	1.42		3.08	3.02	3.10
	velocity		5.44	8.79	8.96	8.96	9.33	8.79	9.14	8.79	8.79	8.96	9.87	8.37		8.90	9.08	8.85
Martinot-Lagarde, Pascal (FRA)	time	2.48	3.52	4.54	5.54	6.54	7.56	8.56	9.58	10.62	11.70		13.16	8 / 5				
	reaction time	0.120	interval	7 steps	1.04	1.02	1.00	1.00	1.02	1.00	1.02	1.04	1.08	1.46		3.06	3.02	3.14
	velocity		5.53	8.79	8.96	9.14	9.14	8.96	9.14	8.96	8.79	8.46	9.60	8.36		8.96	9.08	8.73
Martínez, Asier (ESP) (2000)	time	2.56	3.60	4.64	5.66	6.66	7.68	8.70	9.72	10.76	11.80		13.22	2 / 6				
	reaction time	0.155	interval	7 steps	1.04	1.04	1.02	1.00	1.02	1.02	1.02	1.04	1.04	1.42	PB	3.10	3.04	3.10
	velocity		5.36	8.79	8.79	8.96	9.14	8.96	8.96	8.96	8.79	8.79	9.87	8.32		8.85	9.02	8.85
Pozzi, Andy (GBR) (1992)	time	2.46	3.50	4.52	5.52	6.52	7.54	8.58	9.62	10.68	11.76		13.30	3 / 7				
	reaction time	0.140	interval	7 steps	1.04	1.02	1.00	1.00	1.02	1.04	1.06	1.08	1.54			3.06	3.06	3.18
	velocity		5.58	8.79	8.96	9.14	9.14	8.96	8.79	8.79	8.62	8.46	9.10	8.27		8.96	8.96	8.62
Manga, Aurel (FRA) (1992)	time	2.50	3.58	4.62	5.62	6.64	7.66	8.68	9.70	10.78	11.86		13.38	9 / 8				
	reaction time	0.151	interval	7 steps	1.08	1.04	1.00	1.02	1.02	1.02	1.08	1.08	1.52			3.12	3.06	3.18
	velocity		5.49	8.46	8.79	9.14	8.96	8.96	8.96	8.96	8.46	8.46	9.22	8.22		8.79	8.96	8.62

Semi-Final 3

Semi-Final 3		date 04-Aug-21		wind -0.1 m/s		Henson (2021) - Athlete First: 2021 year end hurdle report													
				H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
Holloway, Grant (USA) (1997)	time			2.43	3.43	4.43	5.43	6.44	7.44	8.47	9.49	10.52	11.60		13.13	4 / 1			
	reaction time	0.130	interval	7 steps	1.00	1.00	1.00	1.01	1.00	1.03	1.02	1.03	1.08	1.53			3.00	3.04	3.13
		velocity		5.65	9.14	9.14	9.14	9.05	9.14	8.87	8.96	8.87	8.46	9.16	8.38		9.14	9.02	8.76
Parchment, Hansle (JAM) (1999)	time			2.53	3.60	4.61	5.61	6.61	7.57	8.57	9.65	10.65	11.72		13.23	6 / 2			
	reaction time	0.145	interval	7 steps	1.07	1.01	1.00	1.00	0.96	1.00	1.08	1.00	1.07	1.51			3.08	2.96	3.15
		velocity		5.42	8.54	9.05	9.14	9.14	9.52	9.14	8.46	9.14	8.54	9.28	8.31		8.90	9.26	8.70

Semi-Final 2

emi-Final 2		date	04-Aug-21	wind 0.1 m/s										Henson (2021) - Athlete First: 2021 year end hurdle report						
				H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10	
Allen, Devon (USA) (1994)	time			2.52	3.55	4.57	5.60	6.60	7.60	8.64	9.67	10.71	11.76		13.18	5 / 1				
	reaction time	0.121	interval		1.03	1.02	1.03	1.00	1.00	1.04	1.03	1.04	1.05	1.42				3.08	3.04	3.12
			velocity	5.44	8.87	8.96	8.87	9.14	9.14	8.79	8.87	8.79	8.70	9.87	8.35			8.90	9.02	8.79

Heat 3

Heat 3

date		02-Aug-21		wind		0.1 m/s		Henson (2021) - Athlete First: 2021 year end hurdle report										
Holloway, Grant (USA) (1997)			H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
	time		2.37	3.37	4.35	5.34	6.31	7.31	8.31	9.34	10.41	11.47		13.02	9 / 1			
	reaction time	0.127	7 steps	1.00	0.98	0.99	0.97	1.00	1.00	1.03	1.07	1.06	1.55			2.97	2.97	3.16
	interval																	
	velocity		5.79	9.14	9.33	9.23	9.42	9.14	9.14	8.87	8.54	8.62	9.05	8.45		9.23	9.23	8.68
Parchment, Hansle (JAM) (1999)																		
	time		2.51	3.56	4.59	5.59	6.59	7.59	8.61	9.63	10.64	11.70		13.23	7 / 2			
	reaction time	0.155	7 steps	1.05	1.03	1.00	1.00	1.00	1.02	1.02	1.01	1.06	1.53			3.08	3.02	3.09
	interval																	
	velocity		5.47	8.70	8.87	9.14	9.14	9.14	8.96	8.96	9.05	8.62	9.16	8.31		8.90	9.08	8.87

2021 Japanese High School National Championships (Fukui, JPN)

FINAL

INAL	date 01-Aug-21		wind 1.4 m/s		Shibayama (2021) - national high school and U20 national championships														
			H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10	
Nishi, Tetsuro (JPN) (2004)	time		2.65	3.74	4.79	5.85	6.89	7.95	9.00	10.07	11.14	12.23		13.69	3 / 1				
	reaction time	0.128	interval	7 steps	1.09	1.05	1.06	1.04	1.06	1.05	1.07	1.07	1.09	1.46	PB		3.20	3.15	3.23
			velocity		5.18	8.39	8.70	8.62	8.79	8.62	8.70	8.54	8.54	8.39	9.60	8.04	8.57	8.70	8.49
Koike, Rio (JPN) (2003)	time		2.65	3.78	4.88	5.93	7.00	8.08	9.15	10.24	11.35	12.45		13.93	5 / 2				
	reaction time	0.147	interval	7 steps	1.13	1.10	1.05	1.07	1.08	1.07	1.09	1.11	1.10	1.48	=PB		3.28	3.22	3.30
			velocity		5.18	8.09	8.31	8.70	8.54	8.46	8.54	8.39	8.23	8.31	9.47	7.90	8.36	8.52	8.31
Higuchi, Hayato (JPN) (2003)	time		2.62	3.70	4.80	5.88	6.97	8.06	9.18	10.30	11.40	12.53		13.95	6 / 3				
	reaction time	0.121	interval		1.08	1.10	1.08	1.09	1.09	1.12	1.12	1.10	1.13	1.42	PB		3.26	3.30	3.35
			velocity		5.24	8.46	8.31	8.46	8.39	8.39	8.16	8.16	8.31	8.09	9.87	7.89	8.41	8.31	8.19
Matsumoto, Nozomu (JPN) (2004)	time		2.63	3.74	4.81	5.91	6.98	8.07	9.18	10.28	11.41	12.54		14.05	4 / 4				
	reaction time	0.158	interval		1.11	1.07	1.10	1.07	1.09	1.11	1.10	1.13	1.13	1.51	PB		3.28	3.27	3.36
			velocity		5.22	8.23	8.54	8.31	8.54	8.39	8.23	8.31	8.09	8.09	9.28	7.83	8.36	8.39	8.16
Hiroto, Ato (JPN) (2004)	time		2.73	3.80	4.92	5.99	7.10	8.22	9.33	10.45	11.58	12.70		14.17	2 / 5				
	reaction time	0.136	interval		1.07	1.12	1.07	1.11	1.12	1.11	1.12	1.13	1.12	1.47	PB		3.26	3.34	3.37
			velocity		5.03	8.54	8.16	8.54	8.23	8.16	8.23	8.16	8.09	8.16	9.54	7.76	8.41	8.21	8.14
Imanishi, Ryota (JPN) (2004)	time		2.74	3.87	4.97	6.07	7.16	8.28	9.39	10.50	11.63	12.76		14.22	8 / 6				
	reaction time	0.187	interval		1.13	1.10	1.10	1.09	1.12	1.11	1.11	1.13	1.13	1.46	PB		3.33	3.32	3.37

velocity	5.01	8.09	8.31	8.31	8.39	8.16	8.23	8.23	8.09	8.09	9.60	7.74	8.23	8.26	8.14
Kawano, Hayato (JPN) (2003) time	2.64	3.75	4.85	5.97	7.07	8.18	9.33	10.50	11.62	12.79		14.36	7 / 7		
reaction time 0.143 interval		1.11	1.10	1.12	1.10	1.11	1.15	1.17	1.12	1.17	1.57	PB		3.33	3.36
velocity	5.20	8.23	8.31	8.16	8.31	8.23	7.95	7.81	8.16	7.81	8.93	7.66		8.23	8.16
Uchino, Rio (JPN) (2003) time	2.73	3.85	4.95	6.06	7.17	8.28	9.40	10.51	11.66	12.84		14.37	1 / 8		
reaction time 0.139 interval		1.12	1.10	1.11	1.11	1.11	1.12	1.11	1.15	1.18	1.53	PB		3.33	3.34
velocity	5.03	8.16	8.31	8.23	8.23	8.23	8.16	8.23	7.95	7.75	9.16	7.65		8.23	8.21

2021 European U20 Championships (Tallinn, EST) (99.1cm) (TV Analysis)

FINAL

INAL	date	17-Jul-21	wind	0.2 m/s	Henson (2021) - Athlete First: 2021 Sasha Zhoya 110m Hurdles World U20 Record													
			H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
Zhoya, Sasha (FRA) (2002)	time		2.48	3.48	4.48	5.48	6.48	7.48	8.48	9.48	10.52	11.52		13.05	4 / 1			
reaction time	0.196	interval	7 steps	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.04	1.00	1.53	CR		3.00	3.00	3.04
		velocity	5.53	9.14	9.14	9.14	9.14	9.14	9.14	9.14	8.79	9.14	9.16	8.43		9.14	9.14	9.02

Semi-Final 3

emi-Final 3		date	16-Jul-21		wind	2.4 m/s		Henson (2021) - Athlete First: 2021 Sasha Zhoya 110m Hurdles World U20 Record										
			H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
Zhoya, Sasha (FRA) (2002)		time	2.48	3.48	4.48	5.46	6.42	7.40	8.38	9.38	10.40	11.44		12.98	5 / 1			
reaction time 0.219		interval	7 steps	1.00	1.00	0.98	0.96	0.98	0.98	1.00	1.02	1.04	1.54			2.98	2.92	3.06
		velocity	5.53	9.14	9.14	9.33	9.52	9.33	9.33	9.14	8.96	8.79	9.10	8.47		9.20	9.39	8.96

2021 Müller British Grand Prix (Gateshead, GBR) (TV Analysis)

FINAL

INAL

		date	13-Jul-21	wind 0.8 m/s			Henson (2021) - Athlete First: 2021 year end hurdle report												
				H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
Levy, Ronald (JAM) (1992)	time			2.54	3.60	4.64	5.66	6.66	7.68	8.70	9.70	10.76	11.82		13.22	4 / 1			
	reaction time	0.150	interval	7 steps	1.06	1.04	1.02	1.00	1.02	1.02	1.00	1.06	1.06	1.40			3.12	3.04	3.12
			velocity		5.40	8.62	8.79	8.96	9.14	8.96	8.96	9.14	8.62	8.62	10.01	8.32		8.79	9.02
McLeod, Omar (JAM) (1994)	time			2.56	3.62	4.62	5.66	6.68	7.68	8.68	9.74	10.78	11.94		13.42	5 / 2			
	reaction time	0.140	interval		1.06	1.00	1.04	1.02	1.00	1.00	1.06	1.04	1.16	1.48			3.10	3.02	3.26
			velocity		5.36	8.62	9.14	8.79	8.96	9.14	9.14	8.62	8.79	7.88	9.47	8.20		8.85	9.08
Pozzi, Andy (GBR) (1992)	time			2.56	3.62	4.64	5.64	6.66	7.72	8.76	9.80	10.88	11.98		13.45	3 / 3			
	reaction time	0.114	interval	7 steps	1.06	1.02	1.00	1.02	1.06	1.04	1.04	1.08	1.10	1.47			3.08	3.12	3.22
			velocity		5.36	8.62	8.96	9.14	8.96	8.62	8.79	8.79	8.46	8.31	9.54	8.18		8.90	8.79
Dunning, Robert (USA) (1997)	time			2.60	3.62	4.72	5.76	6.82	7.88	8.92	10.00	11.08	12.18		13.71	6 / 4			
	reaction time	0.130	interval	7 steps	1.02	1.10	1.04	1.06	1.06	1.04	1.08	1.08	1.10	1.53			3.16	3.16	3.26
			velocity		5.28	8.96	8.31	8.79	8.62	8.62	8.79	8.46	8.46	8.31	9.16	8.02		8.68	8.68
Broadbell, Rasheed (JAM) (200)	time			2.60	3.72	4.76	5.82	6.88	7.98	9.04	10.10	11.22	12.32		13.84	2 / 5			
	reaction time	0.165	interval	7 steps	1.12	1.04	1.06	1.06	1.10	1.06	1.06	1.12	1.10	1.52			3.22	3.22	3.28
			velocity		5.28	8.16	8.79	8.62	8.62	8.31	8.62	8.62	8.16	8.31	9.22	7.95		8.52	8.52
Ojora, Tade (GBR) (1999)	time			2.64	3.72	4.80	5.84	6.92	8.00	9.04	10.18	11.28	12.40		13.87	7 / 6			
	reaction time	0.151	interval		1.08	1.08	1.04	1.08	1.08	1.04	1.14	1.10	1.12	1.47			3.20	3.20	3.36
			velocity		5.20	8.46	8.46	8.79	8.46	8.46	8.79	8.02	8.31	8.16	9.54	7.93		8.57	8.57
Dal Molin, Paolo (ITA) (1987)	time			2.52	3.60	4.64	5.74	6.86	7.96	9.04	10.16	11.24	12.36		13.97	8 / 7			
	reaction time	0.150	interval	7 steps	1.08	1.04	1.10	1.12	1.10	1.08	1.12	1.08	1.12	1.61			3.22	3.30	3.32
			velocity		5.44	8.46	8.79	8.31	8.16	8.31	8.46	8.16	8.46	8.16	8.71	7.87		8.52	8.31

2021 French National U20 Championships (Bondoufle, FRA) (99.1cm) (TV Analysis)

FINAL

INAL	date	10-Jul-21	wind	2.2 m/s															Henson (2021) - Athlete First: 2021 Sasha Zhoya 110m Hurdles World U20 Record														
				H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10														
Zhoya, Sasha (FRA) (2002)	time			2.53	3.56	4.56	5.58	6.56	7.58	8.60	9.60	10.62	11.66		13.06	6 / 1																	
reaction time	0.150			7 steps	1.03	1.00	1.02	0.98	1.02	1.00	1.02	1.04	1.40				3.05	3.02	3.06														
	interval																																
	velocity			5.42	8.87	9.14	8.96	9.33	8.96	8.96	9.14	8.96	8.79	10.01	8.42		8.99	9.08	8.96														

2021 Flavour Hangzhou Athletics Meeting (Hangzhou, CHN)

FINAL

INAL	date	09-Jul-21	wind	0.4 m/s	CAA Hurdle Development (2021)													
			H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
Xie Wenjun (CHN) (1990)	time		2.64	3.70	4.74	5.77	6.79	7.82	8.86	9.91	10.98	12.06		13.49	4 / 1			
reaction time	0.154	interval		1.06	1.04	1.03	1.02	1.03	1.04	1.05	1.07	1.08	1.43			3.13	3.09	3.20
velocity			5.20	8.62	8.79	8.87	8.96	8.87	8.79	8.70	8.54	8.46	9.80	8.15		8.76	8.87	8.57

2021 István Gyulai Memorial (Székesfehérvár, HUN) (TV Analysis)

FINAL

Holloway, Grant (USA) (1997)		date	06-Jul-21	wind -0.1 m/s										Henson (2021) - Athlete First: 2021 year end hurdle report							
				H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10		
Holloway, Grant (USA) (1997)		time		2.36	3.37	4.37	5.35	6.34	7.34	8.36	9.40	10.47	11.56		13.08	4 / 1					
reaction time 0.145		interval		7 steps	1.01	1.00	0.98	0.99	1.00	1.02	1.04	1.07	1.09	1.52			2.99	3.01	3.20		
		velocity		5.81	9.05	9.14	9.33	9.23	9.14	8.96	8.79	8.54	8.39	9.22	8.41		9.17	9.11	8.57		

Ortega, Orlando (ESP) (1991)	time	2.47	3.50	4.54	5.54	6.54	7.54	8.56	9.57	10.58	11.63		13.15	5 / 2				
	reaction time	0.134	interval	7 steps	1.03	1.04	1.00	1.00	1.00	1.02	1.01	1.01	1.05	1.52		3.07	3.02	3.07
		velocity	5.55	8.87	8.79	9.14	9.14	9.14	8.96	9.05	9.05	8.70	9.22	8.37		8.93	9.08	8.93
Shubenkov, Sergey (ANA) (1991)	time	2.50	3.53	4.54	5.56	6.55	7.54	8.54	9.57	10.61	11.64		13.19	6 / 3				
	reaction time	0.150	interval	7 steps	1.03	1.01	1.02	0.99	0.99	1.00	1.03	1.04	1.03	1.55		3.06	2.98	3.10
		velocity	5.49	8.87	9.05	8.96	9.23	9.23	9.14	8.87	8.79	8.87	9.05	8.34		8.96	9.20	8.85
Levy, Ronald (JAM) (1992)	time	2.43	3.46	4.52	5.57	6.59	7.61	8.65	9.68	10.70	11.74		13.25	3 / 4				
	reaction time	0.140	interval	7 steps	1.03	1.06	1.05	1.02	1.02	1.04	1.03	1.02	1.04	1.51		3.14	3.08	3.09
		velocity	5.65	8.87	8.62	8.70	8.96	8.96	8.79	8.87	8.96	8.79	9.28	8.30		8.73	8.90	8.87
Szűcs, Valdó (HUN) (1995)	time	2.45	3.50	4.52	5.57	6.61	7.61	8.67	9.72	10.78	11.86		13.47	2 / 5				
	reaction time	0.147	interval	7 steps	1.05	1.02	1.05	1.04	1.00	1.06	1.05	1.06	1.08	1.61		3.12	3.10	3.19
		velocity	5.60	8.70	8.96	8.70	8.79	9.14	8.62	8.70	8.62	8.46	8.71	8.17		8.79	8.85	8.60

2021 Japanese National Championships (Osaka, JPN)

FINAL

INAL	date		27-Jun-21		wind		1.2 m/s		Shibayama (2021) - race analysis of men's 110m hurdles in the 2021 season													
					H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10		
Izumiya, Shunsuke (JPN) (2000)	time				2.55	3.57	4.61	5.63	6.64	7.65	8.67	9.68	10.69	11.75		13.06	6 / 1					
	reaction time	0.155	interval	7 steps		1.02	1.04	1.02	1.01	1.01	1.02	1.01	1.01	1.06	1.31	NR PB		3.08	3.04	3.08		
			velocity		5.38	8.96	8.79	8.96	9.05	9.05	8.96	9.05	9.05	8.62	10.70	8.42		8.90	9.02	8.90		
Kanai, Taio (JPN) (1995)	time				2.52	3.55	4.56	5.59	6.61	7.63	8.66	9.72	10.76	11.82		13.22	9 / 2					
	reaction time	0.141	interval	7 steps		1.03	1.01	1.03	1.02	1.02	1.03	1.06	1.04	1.06	1.40			3.07	3.07	3.16		
			velocity		5.44	8.87	9.05	8.87	8.96	8.96	8.87	8.62	8.79	8.62	10.01	8.32		8.93	8.93	8.68		
Takayama, Shunya (JPN) (1994)	time				2.57	3.60	4.64	5.68	6.71	7.74	8.78	9.83	10.88	11.95		13.37	3 / 3					
	reaction time	0.139	interval			1.03	1.04	1.04	1.03	1.03	1.04	1.05	1.05	1.07	1.42			3.11	3.10	3.17		
			velocity		5.34	8.87	8.79	8.79	8.87	8.87	8.79	8.70	8.70	8.54	9.87	8.23		8.82	8.85	8.65		
Nomoto, Syuusei (JPN) (1995)	time				2.54	3.60	4.65	5.68	6.71	7.75	8.79	9.85	10.90	11.97		13.38	4 / 4					
	reaction time	0.154	interval	7 steps		1.06	1.05	1.03	1.03	1.04	1.04	1.06	1.05	1.07	1.41	PB		3.14	3.11	3.18		
			velocity		5.40	8.62	8.70	8.87	8.87	8.79	8.79	8.62	8.70	8.54	9.94	8.22		8.73	8.82	8.62		
Yokochi, Taiga (JPN) (2000)	time				2.62	3.70	4.72	5.75	6.76	7.77	8.84	9.89	10.96	12.03		13.47	8 / 5					
	reaction time	0.167	interval	7 steps		1.08	1.02	1.03	1.01	1.01	1.07	1.05	1.07	1.07	1.44	PB		3.13	3.09	3.19		
			velocity		5.24	8.46	8.96	8.87	9.05	9.05	8.54	8.70	8.54	8.54	9.74	8.17		8.76	8.87	8.60		
Fujii, Ryota (JPN) (1996)	time				2.59	3.65	4.69	5.73	6.77	7.80	8.86	9.92	11.00	12.08		13.54	2 / 6					
	reaction time	0.149	interval			1.06	1.04	1.04	1.04	1.03	1.06	1.06	1.08	1.08	1.46	PB		3.14	3.13	3.22		
			velocity		5.30	8.62	8.79	8.79	8.79	8.87	8.62	8.62	8.46	8.46	9.60	8.12		8.73	8.76	8.52		

2020 USA Olympic Trials (Eugene, OR)

FINAL (TV Analysis)

INAL (TV Analysis)		date	26-Jun-21	wind	0.4 m/s										Henson (2021) - Athlete First: 2021 year end hurdle report						
				H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10		
Holloway, Grant (USA) (1997)	time			2.40	3.37	4.34	5.30	6.27	7.24	8.26	9.27	10.33	11.42		12.96	4 / 1					
	reaction time	0.146	interval	7 steps	0.97	0.97	0.96	0.97	0.97	1.02	1.01	1.06	1.09	1.54			2.90	2.96	3.16		
			velocity		5.72	9.42	9.42	9.52	9.42	9.42	8.96	9.05	8.62	8.39	9.10	8.49		9.46	9.26	8.68	
Allen, Devon (USA) (1994)	time			2.50	3.53	4.53	5.53	6.54	7.57	8.56	9.58	10.60	11.64		13.10	5 / 2					
	reaction time	0.152	interval		1.03	1.00	1.00	1.01	1.03	0.99	1.02	1.02	1.04	1.46			3.03	3.03	3.08		
			velocity		5.49	8.87	9.14	9.14	9.05	8.87	9.23	8.96	8.96	8.79	9.60	8.40		9.05	9.05	8.90	
Roberts, Daniel (USA) (1998)	time			2.43	3.47	4.47	5.47	6.47	7.49	8.51	9.53	10.56	11.61		13.11	6 / 3					
	reaction time	0.201	interval		1.04	1.00	1.00	1.00	1.02	1.02	1.02	1.03	1.05	1.50			3.04	3.04	3.10		
			velocity		5.65	8.79	9.14	9.14	9.14	8.96	8.96	8.96	8.87	8.70	9.35	8.39		9.02	9.02	8.85	
Cunningham, Trey (USA) (1998)	time			2.43	3.47	4.47	5.47	6.47	7.49	8.52	9.56	10.62	11.70		13.21	3 / 4					
	reaction time	0.165	interval	7 steps	1.04	1.00	1.00	1.00	1.02	1.03	1.04	1.06	1.08	1.51	=PB		3.04	3.05	3.18		
			velocity		5.65	8.79	9.14	9.14	9.14	8.96	8.87	8.79	8.62	8.46	9.28	8.33		9.02	8.99	8.62	
Brown, Tai (USA) (2000)	time			2.60	3.63	4.65	5.67	6.67	7.67	8.69	9.72	10.76	11.84		13.32	8 / 5					
	reaction time	0.174	interval		1.03	1.02	1.02	1.00	1.00	1.02	1.03	1.04	1.08	1.48	PB		3.07	3.02	3.15		
			velocity		5.28	8.87	8.96	8.96	9.14	9.14	8.96	8.87	8.79	8.46	9.47	8.26		8.93	9.08	8.70	
Dickson, Michael (USA) (1997)	time			2.50	3.55	4.60	5.62	6.64	7.67	8.71	9.74	10.78	11.85		13.34	2 / 6					
	reaction time	0.201	interval	7 steps	1.05	1.05	1.02	1.02	1.03	1.04	1.03	1.04	1.07	1.49			3.12	3.09	3.14		
			velocity		5.49	8.70	8.70	8.96	8.96	8.87	8.79	8.87	8.79	8.54	9.41	8.25		8.79	8.87	8.73	
Britt, Jamal (USA) (1998)	time			2.47	3.53	4.57	5.60	6.64	7.67	8.72	9.77	10.85	11.95		13.47	7 / 7					
	reaction time	0.158	interval	7 steps	1.06	1.04	1.03	1.04	1.03	1.05	1.05	1.08	1.10	1.52			3.13	3.12	3.23		
			velocity		5.55	8.62	8.79	8.87	8.79	8.87	8.70	8.70	8.46	8.31	9.22	8.17		8.76	8.79	8.49	
Eaton, Jarrett (USA) (1989)	time			2.47	3.50	4.50	5.54	6.54	7.57	8.64	9.71	10.84	11.98		13.64	1 / 8					
	reaction time	0.137	interval	7 steps	1.03	1.00	1.04	1.00	1.03	1.07	1.07	1.13	1.14	1.66			3.07	3.10	3.34		

velocity 5.55 8.87 9.14 8.79 9.14 8.87 8.54 8.54 8.09 8.02 8.45 8.06 8.93 8.85 8.21

Semi-Final 2 (TV Analysis)

		date	26-Jun-21	wind	1.2 m/s											Henson (2021) - Athlete First: 2021 year end hurdle report			
				H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
Allen, Devon (USA) (1994)	time			2.53	3.56	4.56	5.54	6.55	7.56	8.55	9.58	10.61	11.64		13.10	5 / 1			
	reaction time	0.124	interval		1.03	1.00	0.98	1.01	1.01	0.99	1.03	1.03	1.03	1.46			3.01	3.01	3.09
	velocity			5.42	8.87	9.14	9.33	9.05	9.05	9.23	8.87	8.87	8.87	9.60	8.40		9.11	9.11	8.87
Roberts, Daniel (USA) (1998)	time			2.46	3.53	4.53	5.53	6.55	7.56	8.55	9.61	10.63	11.72		13.25	4 / 2			
	reaction time	0.179	interval		1.07	1.00	1.00	1.02	1.01	0.99	1.06	1.02	1.09	1.53			3.07	3.02	3.17
	velocity			5.58	8.54	9.14	9.14	8.96	9.05	9.23	8.62	8.96	8.39	9.16	8.30		8.93	9.08	8.65
Dickson, Michael (USA) (1997)	time			2.50	3.60	4.57	5.60	6.63	7.67	8.67	9.71	10.74	11.86		13.29	6 / 3			
	reaction time	0.173	interval		7 steps	1.10	0.97	1.03	1.03	1.04	1.00	1.04	1.03	1.12	1.43		3.10	3.07	3.19
	velocity			5.49	8.31	9.42	8.87	8.87	8.79	9.14	8.79	8.87	8.16	9.80	8.28		8.85	8.93	8.60
Eaton, Jarrett (USA) (1989)	time			2.50	3.56	4.53	5.54	6.55	7.60	8.60	9.63	10.70	11.80		13.32	3 / 4			
	reaction time	0.157	interval		7 steps	1.06	0.97	1.01	1.01	1.05	1.00	1.03	1.07	1.10	1.52		3.04	3.06	3.20
	velocity			5.49	8.62	9.42	9.05	9.05	8.70	9.14	8.87	8.54	8.31	9.22	8.26		9.02	8.96	8.57
Mallett, Aaron (USA) (1994)	time			2.56	3.60	4.63	5.63	6.63	7.67	8.67	9.71	10.76	11.86		13.33	7 / 5			
	reaction time	0.198	interval			1.04	1.03	1.00	1.00	1.04	1.00	1.04	1.05	1.10	1.47		3.07	3.04	3.19
	velocity			5.36	8.79	8.87	9.14	9.14	8.79	9.14	8.79	8.70	8.31	9.54	8.25		8.93	9.02	8.60
Marshall, Jamar (USA) (2001)	time			2.50	3.60	4.60	5.60	6.65	7.69	8.70	9.73	10.80	11.88		13.38	1 / 6			
	reaction time	0.137	interval		7 steps	1.10	1.00	1.00	1.05	1.04	1.01	1.03	1.07	1.08	1.50	DQ	3.10	3.10	3.18
	velocity			5.49	8.31	9.14	9.14	8.70	8.79	9.05	8.87	8.54	8.46	9.35	8.22		8.85	8.85	8.62
Crittenden, Freddie (USA) (1999)	time			2.56	3.60	4.63	5.70	6.74	7.77	8.80	9.90	10.99	12.07		13.63	2 / 7 6			
	reaction time	0.175	interval			1.04	1.03	1.07	1.04	1.03	1.03	1.10	1.09	1.08	1.56		3.14	3.10	3.27
	velocity			5.36	8.79	8.87	8.54	8.79	8.87	8.87	8.31	8.39	8.46	8.99	8.07		8.73	8.85	8.39
Ames, Brendan (USA) (1988)	time			2.56	3.63	4.65	5.67	6.70	7.73	8.88	10.00	11.13	12.30		13.96	8 / 8 7			
	reaction time	0.208	interval		7 steps	1.07	1.02	1.03	1.03	1.15	1.12	1.13	1.17	1.66			3.11	3.21	3.42
	velocity			5.36	8.54	8.96	8.96	8.87	8.87	7.95	8.16	8.09	7.81	8.45	7.88		8.82	8.54	8.02

Henson (2021) - Athlete First: 2021 year end hurdle report (TV Analysis)

Semi-Final 1

		date	26-Jun-21	wind	1.8 m/s											Vazel (2021) - https://twitter.com/pjvazel/status/1409323602432999426?s=21			
				H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
Holloway, Grant (USA) (1997)	time			2.49	3.49	4.45	5.40	6.36	7.36	8.35	9.36	10.38	11.42		12.81	5 / 1			
	reaction time	0.169	interval		7 steps	1.00	0.96	0.95	0.96	1.00	0.99	1.01	1.02	1.04	1.39	PB	2.91	2.95	3.07
	velocity			5.51	9.14	9.52	9.62	9.52	9.14	9.23	9.05	8.96	8.79	10.09	8.59		9.42	9.29	8.93
Cunningham, Trey (USA) (1998)	time			2.46	3.48	4.49	5.47	6.48	7.52	8.56	9.59	10.66	11.72		13.21	6 / 2			Henson (2021)
	reaction time	0.163	interval		7 steps	1.02	1.01	0.98	1.01	1.04	1.04	1.03	1.07	1.06	1.49	PB	3.01	3.09	3.16
	velocity			5.58	8.96	9.05	9.33	9.05	8.79	8.79	8.87	8.54	8.62	9.41	8.33		9.11	8.87	8.68
Britt, Jamal (USA) (1998)	time			2.47	3.53	4.55	5.55	6.57	7.58	8.62	9.67	10.73	11.81		13.27	3 / 3			Henson (2021)
	reaction time	0.147	interval		7 steps	1.06	1.02	1.00	1.02	1.01	1.05	1.06	1.08	1.46	PB		3.08	3.07	3.19
	velocity			5.55	8.62	8.96	9.14	8.96	9.05	8.79	8.70	8.62	8.46	9.60	8.29		8.90	8.93	8.60
Brown, Tai (USA) (2000)	time			2.62	3.65	4.67	5.69	6.70	7.69	8.71	9.74	10.78	11.86		13.33	2 / 4			Henson (2021)
	reaction time	0.173	interval			1.03	1.02	1.01	0.99	1.02	1.03	1.04	1.08	1.47	PB		3.07	3.02	3.15
	velocity			5.24	8.87	8.96	8.96	9.05	9.23	8.96	8.87	8.79	8.46	9.54	8.25		8.93	9.08	8.70
Gilbert, Tre'Bien (USA) (1999)	time			2.50	3.58	4.60	5.61	6.63	7.64	8.71	9.76	10.80	11.92		13.41	8 / 5			Henson (2021)
	reaction time	0.191	interval		7 steps	1.08	1.02	1.01	1.02	1.01	1.07	1.05	1.04	1.12	1.49	PB	3.11	3.10	3.21
	velocity			5.49	8.46	8.96	9.05	8.96	9.05	8.54	8.70	8.79	8.16	9.41	8.20		8.82	8.85	8.54
McConico, Jaylan (USA) (1998)	time			2.55	3.62	4.64	5.68	6.70	7.72	8.74	9.81	10.86	11.94		13.43	4 / 6			Henson (2021)
	reaction time	0.209	interval			1.07	1.02	1.04	1.02	1.02	1.02	1.07	1.05	1.08	1.49		3.13	3.06	3.20
	velocity			5.38	8.54	8.96	8.79	8.96	8.96	8.96	8.54	8.70	8.46	9.41	8.19		8.76	8.96	8.57
Brooks, Devon (USA) (1998)	time			2.62	3.66	4.68	5.70	6.72	7.74	8.77	9.81	10.87	12.01		13.55	1 / 7			Henson (2021)
	reaction time	0.264	interval		7 steps	1.04	1.02	1.02	1.02	1.02	1.03	1.04	1.06	1.14	1.54		3.08	3.07	3.24
	velocity			5.24	8.79	8.96	8.96	8.96	8.96	8.87	8.79	8.62	8.02	9.10	8.12		8.90	8.93	8.46
Moore, Isaiah (USA) (1996)	time			2.54	3.63	4.69	5.72	6.77	7.81	8.86	9.91	10.96	12.06		13.59	7 / 8			Henson (2021)
	reaction time	0.147	interval			1.09	1.06	1.03	1.05	1.04	1.05	1.05	1.10	1.53			3.18	3.14	3.20
	velocity			5.40	8.39	8.62	8.87	8.70	8.79	8.70	8.70	8.70	8.31	9.16	8.09		8.62	8.73	8.57
Brixey, Sam (USA) (1998)	time			2.50	3.58	4.67	5.70	6.77	7.86	8.99	10.11	11.21	12.38		13.99	9 / 9			Henson (2021)
	reaction time	0.148	interval		7 steps	1.08	1.09	1.03	1.07	1.09	1.13	1.12	1.10	1.17	1.61		3.20	3.29	3.39
	velocity			5.49	8.46	8.39	8.87	8.54	8.39	8.09	8.16	8.31	7.81	8.71	7.86		8.57	8.33	8.09

Heat 3 (TV Analysis)

		date	25-Jun-21	wind	0.5 m/s											Henson (2021) - Athlete First: 2021 year end hurdle report			
				H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10

Allen, Devon (USA) (1994)	time	2.50	3.53	4.56	5.63	6.63	7.63	8.66	9.73	10.73	11.76		13.26	3 / 1			
	reaction time	0.139	interval	1.03	1.03	1.07	1.00	1.00	1.03	1.07	1.00	1.03	1.50		3.13	3.03	3.10
	velocity	5.49	8.87	8.87	8.54	9.14	9.14	8.87	8.54	9.14	8.87	9.35	8.30		8.76	9.05	8.85
Dickson, Michael (USA) (1997)	time	2.53	3.57	4.60	5.66	6.66	7.66	8.70	9.73	10.73	11.78		13.26	2 / 2			
	reaction time	0.208	interval	7 steps	1.04	1.03	1.06	1.00	1.00	1.04	1.03	1.00	1.48	PB	3.13	3.04	3.08
	velocity	5.42	8.79	8.87	8.62	9.14	9.14	8.79	8.87	9.14	8.70	9.47	8.30		8.76	9.02	8.90

Heat 2 (TV Analysis)

date	25-Jun-21	wind	1.5 m/s	Henson (2021) - Athlete First: 2021 year end hurdle report													
		H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
Roberts, Daniel (USA) (1998)	time	2.46	3.50	4.53	5.53	6.53	7.53	8.56	9.60	10.63	11.70		13.22	1 / 1			
	reaction time	0.182	interval	1.04	1.03	1.00	1.00	1.03	1.04	1.03	1.07	1.52			3.07	3.03	3.14
	velocity	5.58	8.79	8.87	9.14	9.14	9.14	8.87	8.79	8.87	8.54	9.22	8.32		8.93	9.05	8.73
Cunningham, Trey (USA) (1998)	time	2.43	3.46	4.50	5.50	6.50	7.50	8.56	9.60	10.66	11.73		13.23	3 / 2			
	reaction time	0.162	interval	7 steps	1.03	1.04	1.00	1.00	1.06	1.04	1.06	1.07	1.50	PB	3.07	3.06	3.17
	velocity	5.65	8.87	8.79	9.14	9.14	9.14	8.62	8.79	8.62	8.54	9.35	8.31		8.93	8.96	8.65

Heat 1 (TV Analysis)

date		25-Jun-21		wind		0.7 m/s		Henson (2021) - Athlete First: 2021 year end hurdle report									
		H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
997	time	2.43	3.43	4.43	5.40	6.42	7.43	8.46	9.49	10.51	11.55		13.11	4 / 1			
175	interval	7 steps	1.00	1.00	0.97	1.02	1.01	1.03	1.03	1.02	1.04	1.56			2.97	3.06	3.09
	velocity	5.65	9.14	9.14	9.42	8.96	9.05	8.87	8.87	8.96	8.79	8.99	8.39		9.23	8.96	8.87

2021 Jamaican National Championships (Kingston, JAM) (TV Analysis)**Heat 1**

date	26-Jun-21	wind	1.3 m/s	Henson (2021) - Athlete First: 2021 year end hurdle report													
		H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
McLeod, Omar (JAM) (1994)	time	2.47	3.53	4.53	5.56	6.53	7.57	8.57	9.57	10.58	11.59		13.04	4 / 1			
	reaction time		interval	1.06	1.00	1.03	0.97	1.04	1.00	1.00	1.01	1.01	1.45		3.09	3.01	3.02
	velocity	5.55	8.62	9.14	8.87	9.42	8.79	9.14	9.14	9.05	9.05	9.67	8.44		8.87	9.11	9.08
Levy, Ronald (JAM) (1992)	time	2.53	3.53	4.56	5.56	6.53	7.53	8.53	9.53	10.56	11.60		13.08	5 / 2			
	reaction time		interval	7 steps	1.00	1.03	1.00	0.97	1.00	1.00	1.03	1.04	1.48		3.03	2.97	3.07
	velocity	5.42	9.14	8.87	9.14	9.42	9.14	9.14	9.14	8.87	8.79	9.47	8.41		9.05	9.23	8.93

2021 Chinese National Championships (Chongqing, CHN)**Decathlon - Heat 3**

date		wind -0.2 m/s										CAA Hurdle Development (2021)						
		H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10	
177	time	2.71	3.89	5.05	6.19	7.32	8.46	9.62	10.81	12.01	13.21		14.76	2 / 1				
	interval		1.18	1.16	1.14	1.13	1.14	1.16	1.19	1.20	1.20	1.55			3.48	3.43	3.59	
	velocity	5.06	7.75	7.88	8.02	8.09	8.02	7.88	7.68	7.62	7.62	9.05	7.45		7.88	7.99	7.64	
172	time	2.85	4.07	5.24	6.42	7.57	8.74	9.91	11.11	12.29	13.53		15.07	7 / 2				
	interval		1.22	1.17	1.18	1.15	1.17	1.17	1.20	1.18	1.24	1.54			3.57	3.49	3.62	
	velocity	4.81	7.49	7.81	7.75	7.95	7.81	7.81	7.62	7.75	7.37	9.10	7.30		7.68	7.86	7.57	
163	time	2.80	3.97	5.14	6.31	7.47	8.64	9.81	10.99	12.19	13.45		15.09	6 / 3				
	interval		1.17	1.17	1.17	1.16	1.17	1.17	1.18	1.20	1.26	1.64			3.51	3.50	3.64	
	velocity	4.90	7.81	7.81	7.81	7.88	7.81	7.81	7.75	7.62	7.25	8.55	7.29		7.81	7.83	7.53	
273	time	2.85	4.02	5.19	6.35	7.52	8.69	9.88	11.04	12.24	13.50		15.15	8 / 4				
	interval		1.17	1.17	1.16	1.17	1.17	1.19	1.16	1.20	1.26	1.65			3.50	3.53	3.62	
	velocity	4.81	7.81	7.81	7.88	7.81	7.81	7.68	7.88	7.62	7.25	8.50	7.26		7.83	7.77	7.57	
186	time	2.82	4.02	5.20	6.39	7.57	8.79	9.99	11.19	12.43	13.66		15.28	4 / 5				
	interval		1.20	1.18	1.19	1.18	1.22	1.20	1.20	1.24	1.23	1.62			3.57	3.60	3.67	
	velocity	4.87	7.62	7.75	7.68	7.75	7.49	7.62	7.62	7.37	7.43	8.65	7.20		7.68	7.62	7.47	
188	time	2.77	3.94	5.12	6.30	7.50	8.71	9.94	11.19	12.46	13.75		15.41	3 / 6				
	interval	7 steps	1.17	1.18	1.18	1.20	1.21	1.23	1.25	1.27	1.29	1.66			3.53	3.64	3.81	
	velocity	4.95	7.81	7.75	7.75	7.62	7.55	7.43	7.31	7.20	7.09	8.45	7.14		7.77	7.53	7.20	
179	time	2.83	4.04	5.19	6.42	7.66	8.89	10.14	11.40	12.68	14.01		15.72	5 / 7				
	interval		1.21	1.15	1.23	1.24	1.23	1.25	1.26	1.28	1.33	1.71			3.59	3.72	3.87	
	velocity	4.85	7.55	7.95	7.43	7.37	7.43	7.31	7.25	7.14	6.87	8.20	7.00		7.64	7.37	7.09	

Decathlon - Heat 2

date	27-Jun-21	wind	-0.2 m/s	CAA Hurdle Development (2021)													
		H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
Cao Jin (CHN) (1996)	time	2.73	3.89	4.99	6.12	7.24	8.37	9.52	10.68	11.86	13.06		14.60	5 / 1			
	reaction time	0.185	interval	7 steps	1.10	1.13	1.12	1.13	1.15	1.16	1.18	1.20	1.54		3.39	3.40	3.54
	velocity	5.03	7.88	8.31	8.09	8.16	8.09	7.95	7.88	7.75	7.62	9.10	7.53		8.09	8.06	7.75
Chen Xinghua (CHN) (2002)	time	2.77	3.94	5.07	6.17	7.31	8.46	9.59	10.73	11.94	13.16		14.79	3 / 2			
	reaction time		interval	1.17	1.13	1.10	1.14	1.15	1.13	1.14	1.21	1.22	1.63		3.40	3.42	3.57
	velocity	4.95	7.81	8.09	8.31	8.02	7.95	8.09	8.02	7.55	7.49	8.60	7.44		8.06	8.02	7.68

Hu Yufei (CHN) (1993)	time	2.75	3.94	5.09	6.23	7.39	8.52	9.67	10.84	12.00	13.18		14.80	7 / 3				
	reaction time	0.159	interval	7 steps	1.19	1.15	1.14	1.16	1.13	1.15	1.17	1.16	1.18	1.62		3.48	3.44	3.51
	velocity	4.99	7.68	7.95	8.02	7.88	8.09	7.95	7.81	7.88	7.75	8.65	7.43		7.88	7.97	7.81	
Hua Zihui (CHN) (2000)	time	2.78	3.97	5.17	6.37	7.59	8.82	10.04	11.29	12.53	13.82		15.45	8 / 4				
	reaction time	0.164	interval		1.19	1.20	1.20	1.22	1.23	1.22	1.25	1.24	1.29	1.63		3.59	3.67	3.78
	velocity	4.94	7.68	7.62	7.62	7.49	7.43	7.49	7.31	7.37	7.09	8.60	7.12		7.64	7.47	7.25	
Gu Yihe (CHN) (2000)	time	2.87	4.07	5.34	6.57	7.82	9.07	10.33	11.58	12.86	14.16		15.89	2 / 5				
	reaction time	0.208	interval		1.20	1.27	1.23	1.25	1.25	1.26	1.25	1.28	1.30	1.73		3.70	3.76	3.83
	velocity	4.78	7.62	7.20	7.43	7.31	7.31	7.25	7.31	7.14	7.03	8.10	6.92		7.41	7.29	7.16	
Qin Guoyuan (CHN) (1996)	time	2.95	4.19	5.47	6.71	8.01	9.27	10.54	11.81	13.11	14.40		16.03	6 / 6				
	reaction time	0.235	interval		1.24	1.28	1.24	1.30	1.26	1.27	1.30	1.29	1.63		3.76	3.83	3.86	
	velocity	4.65	7.37	7.14	7.37	7.03	7.25	7.20	7.20	7.03	7.09	8.60	6.86		7.29	7.16	7.10	

Decathlon - Heat 1

date 27-Jun-21 wind 0.5 m/s

CAA Hurdle Development (2021)

			H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
Li Xiaodong (CHN) (1995)	time		2.73	3.90	5.02	6.14	7.27	8.39	9.52	10.68	11.84	13.01		14.55	7 / 1			
	reaction time	0.154	interval		1.17	1.12	1.12	1.13	1.12	1.13	1.16	1.16	1.17	1.54		3.41	3.38	3.49
			velocity	5.03	7.81	8.16	8.16	8.09	8.16	8.09	7.88	7.88	7.81	9.10	7.56		8.04	8.11
Yang Zihao (CHN) (1998)	time		2.72	3.89	5.04	6.17	7.32	8.44	9.56	10.73	11.91	13.08		14.63	8 / 2			
	reaction time	0.172	interval		1.17	1.15	1.13	1.15	1.12	1.12	1.17	1.18	1.17	1.55		3.45	3.39	3.52
			velocity	5.04	7.81	7.95	8.09	7.95	8.16	8.16	7.81	7.75	7.81	9.05	7.52		7.95	8.09
Yue Xuesong (CHN) (1997)	time		2.80	3.95	5.10	6.25	7.42	8.59	9.74	10.91	12.08	13.28		14.82	4 / 3			
	reaction time	0.193	interval		1.15	1.15	1.15	1.17	1.17	1.15	1.17	1.17	1.20	1.54		3.45	3.49	3.54
			velocity	4.90	7.95	7.95	7.95	7.81	7.81	7.95	7.81	7.81	7.62	9.10	7.42		7.95	7.86
Guo Qi (CHN) (1990)	time		2.78	3.99	5.17	6.35	7.52	8.71	9.89	11.08	12.28	13.53		15.18	5 / 4			
	reaction time	0.188	interval	7 steps	1.21	1.18	1.18	1.17	1.19	1.18	1.19	1.20	1.25	1.65		3.57	3.54	3.64
			velocity	4.94	7.55	7.75	7.75	7.81	7.68	7.75	7.68	7.62	7.31	8.50	7.25		7.68	7.75
Li Ning (CHN) (2000)	time		2.83	4.04	5.25	6.45	7.67	8.92	10.19	11.46	12.78	14.10		15.84	2 / 5			
	reaction time	0.133	interval	7 steps	1.21	1.21	1.20	1.22	1.25	1.27	1.27	1.32	1.32	1.74		3.62	3.74	3.91
			velocity	4.85	7.55	7.55	7.62	7.49	7.31	7.20	7.20	6.92	6.92	8.06	6.94		7.57	7.33
Xiang Qingqiang (CHN) (1999)	time		2.87	4.15	5.40	6.66	7.89	9.14	10.38	11.66	12.96	14.26		15.98	6 / 6			
	reaction time	0.170	interval		1.28	1.25	1.26	1.23	1.25	1.24	1.28	1.30	1.30	1.72		3.79	3.72	3.88
			velocity	4.78	7.14	7.31	7.25	7.43	7.31	7.37	7.14	7.03	7.03	8.15	6.88		7.23	7.37
Chen Junyi (CHN) (1999)	time		2.87	4.19	5.45	6.72	8.00	9.29	10.56	11.81	13.11	14.43		16.05	3 / 7			
	reaction time	0.187	interval	7 steps	1.32	1.26	1.27	1.28	1.29	1.27	1.25	1.30	1.32	1.62		3.85	3.84	3.87
			velocity	4.78	6.92	7.25	7.20	7.14	7.09	7.20	7.31	7.03	6.92	8.65	6.85		7.12	7.14

FINAL

date 25-Jun-21 wind 1.6 m/s

CAA Hurdle Development (2021)

		H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10	
Sun Zhenjiang (CHN) (1999)	time	2.60	3.70	4.73	5.77	6.80	7.86	8.88	9.96	11.02	12.09		13.56	7 / 1				
	reaction time	0.157	interval	7 steps	1.10	1.03	1.04	1.03	1.06	1.02	1.08	1.06	1.07	1.47	PB	3.17	3.11	3.21
	velocity	5.28	8.31	8.87	8.79	8.87	8.62	8.96	8.46	8.62	8.54	9.54	8.11		8.65	8.82	8.54	
Zhang Honglin (CHN) (1994)	time	2.60	3.67	4.73	5.76	6.82	7.87	8.93	9.99	11.12	12.21		13.70	5 / 2				
	reaction time	0.158	interval		1.07	1.06	1.03	1.06	1.05	1.06	1.06	1.13	1.09	1.49		3.16	3.17	3.28
	velocity	5.28	8.54	8.62	8.87	8.62	8.70	8.62	8.62	8.09	8.39	9.41	8.03		8.68	8.65	8.36	
Ning Xiaohan (CHN) (2000)	time	2.57	3.64	4.67	5.70	6.74	7.77	8.86	9.94	11.05	12.18		13.75	4 / 3				
	reaction time	0.134	interval		1.07	1.03	1.03	1.04	1.03	1.09	1.08	1.11	1.13	1.57		3.13	3.16	3.32
	velocity	5.34	8.54	8.87	8.87	8.79	8.87	8.39	8.46	8.23	8.09	8.93	8.00		8.76	8.68	8.26	
Zhu Shenglong (CHN) (2000)	time	2.66	3.73	4.78	5.84	6.91	7.99	9.05	10.15	11.27	12.39		13.87	8 / 4				
	reaction time	0.176	interval		1.07	1.05	1.06	1.07	1.08	1.06	1.10	1.12	1.12	1.48		3.18	3.21	3.34
	velocity	5.16	8.54	8.70	8.62	8.54	8.46	8.62	8.31	8.16	8.16	9.47	7.93		8.62	8.54	8.21	
Guo Zhongjie (CHN) (2001)	time	2.70	3.78	4.85	5.89	6.97	8.04	9.12	10.21	11.31	12.44		13.91	6 / 5				
	reaction time	0.182	interval		1.08	1.07	1.04	1.08	1.07	1.08	1.09	1.10	1.13	1.47	PB	3.19	3.23	3.32
	velocity	5.08	8.46	8.54	8.79	8.46	8.54	8.46	8.39	8.31	8.09	9.54	7.91		8.60	8.49	8.26	
Shi Guoxing (CHN) (2000)	time	2.69	3.77	4.84	5.91	6.99	8.06	9.16	10.24	11.39	12.51		13.93	3 / 6				
	reaction time	0.196	interval		1.08	1.07	1.07	1.08	1.07	1.10	1.08	1.15	1.12	1.42	PB	3.22	3.25	3.35
	velocity	5.10	8.46	8.54	8.54	8.46	8.54	8.31	8.46	7.95	8.16	9.87	7.90		8.52	8.44	8.19	
Wu Zhenhua (CHN) (1998)	time	2.65	3.75	4.82	5.90	6.99	8.07	9.16	10.26	11.38	12.53		13.98	2 / 7				
	reaction time	0.164	interval		1.10	1.07	1.08	1.09	1.08	1.09	1.10	1.12	1.15	1.45	PB	3.25	3.26	3.37
	velocity	5.18	8.31	8.54	8.46	8.39	8.46	8.39	8.31	8.16	7.95	9.67	7.87		8.44	8.41	8.14	
Pan Zijie (CHN) (1995)	time	2.64	3.75	4.81	5.88	6.95	8.02	9.16	10.29	11.43	12.55		14.10	9 / 8				

reaction time	0.185	interval	1.11	1.06	1.07	1.07	1.07	1.14	1.13	1.14	1.12	1.55			3.24	3.28	3.39
velocity	5.20	8.23	8.62	8.54	8.54	8.54	8.02	8.09	8.02	8.16	9.05	7.80			8.46	8.36	8.09

Heat 4

date 25-Jun-21 wind -0.2 m/s

CAA Hurdle Development (2021)

		H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
Ning Xiaohan (CHN) (2000)	time	2.65	3.72	4.80	5.87	6.92	7.99	9.08	10.16	11.28	12.41		13.96	3 / 1			
reaction time	0.149		1.07	1.08	1.07	1.05	1.07	1.09	1.08	1.12	1.13	1.55			3.22	3.21	3.33
interval		5.18	8.54	8.46	8.54	8.70	8.54	8.39	8.46	8.16	8.09	9.05	7.88		8.52	8.54	8.23
velocity																	
Li Huaihai (CHN) (1999)	time	2.68	3.77	4.85	5.94	7.02	8.11	9.21	10.31	11.41	12.56		14.06	8 / 2			
reaction time	0.169		7 steps 1.09	1.08	1.09	1.08	1.09	1.10	1.10	1.10	1.15	1.50			3.26	3.27	3.35
interval		5.12	8.39	8.46	8.39	8.46	8.39	8.31	8.31	8.31	7.95	9.35	7.82		8.41	8.39	8.19
velocity																	
Liu Bowen (CHN) (2001)	time	2.60	3.70	4.80	5.91	6.99	8.11	9.22	10.33	11.48	12.65		14.19	7 / 3			
reaction time	0.201		1.10	1.10	1.11	1.08	1.12	1.11	1.11	1.15	1.17	1.54			3.31	3.31	3.43
interval		5.28	8.31	8.31	8.23	8.46	8.16	8.23	8.23	7.95	7.81	9.10	7.75		8.28	8.28	7.99
velocity																	
Zhang Tao (CHN) (1997)	time	2.70	3.80	4.92	6.04	7.14	8.24	9.38	10.51	11.66	12.85		14.42	5 / 4			
reaction time	0.189		1.10	1.12	1.12	1.10	1.10	1.14	1.13	1.15	1.19	1.57			3.34	3.34	3.47
interval		5.08	8.31	8.16	8.16	8.31	8.31	8.02	8.09	7.95	7.68	8.93	7.63		8.21	8.21	7.90
velocity																	
Jiang Wei (CHN) (1994)	time	2.69	3.82	4.97	6.09	7.21	8.31	9.43	10.56	11.73	12.90		14.49	4 / 5			
reaction time	0.162		1.13	1.15	1.12	1.12	1.10	1.12	1.13	1.17	1.17	1.59			3.40	3.34	3.47
interval		5.10	8.09	7.95	8.16	8.16	8.31	8.16	8.09	7.81	7.81	8.82	7.59		8.06	8.21	7.90
velocity																	
Lai Junjie (CHN) (2003)	time	2.75	3.90	5.04	6.16	7.29	8.42	9.56	10.74	11.94	13.16		14.78	9 / 6			
reaction time	0.181		1.15	1.14	1.12	1.13	1.13	1.14	1.18	1.20	1.22	1.62			3.41	3.40	3.60
interval		4.99	7.95	8.02	8.16	8.09	8.09	8.02	7.75	7.62	7.49	8.65	7.44		8.04	8.06	7.62
velocity																	

Heat 3

date 25-Jun-21 wind 0.5 m/s

CAA Hurdle Development (2021)

		H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
Zhang Honglin (CHN) (1994)	time	2.60	3.68	4.75	5.81	6.87	7.94	8.99	10.09	11.21	12.36		13.91	5 / 1			
reaction time	0.156		1.08	1.07	1.06	1.06	1.07	1.05	1.10	1.12	1.15	1.55			3.21	3.18	3.37
interval		5.28	8.46	8.54	8.62	8.62	8.54	8.70	8.31	8.16	7.95	9.05	7.91		8.54	8.62	8.14
velocity																	
Guo Zhongjie (CHN) (2001)	time	2.66	3.72	4.79	5.84	6.90	7.96	9.06	10.15	11.29	12.45		13.94	4 / 2			
reaction time	0.178		1.06	1.07	1.05	1.06	1.06	1.10	1.09	1.14	1.16	1.49	PB		3.18	3.22	3.39
interval		5.16	8.62	8.54	8.70	8.62	8.62	8.31	8.39	8.02	7.88	9.41	7.89		8.62	8.52	8.09
velocity																	
Pan Zijie (CHN) (1995)	time	2.62	3.70	4.76	5.87	6.96	8.04	9.11	10.20	11.32	12.47		14.00	6 / 3			
reaction time	0.167		1.08	1.06	1.11	1.09	1.08	1.07	1.09	1.12	1.15	1.53			3.25	3.24	3.36
interval		5.24	8.46	8.62	8.23	8.39	8.46	8.54	8.39	8.16	7.95	9.16	7.86		8.44	8.46	8.16
velocity																	
Wu Zhenhua (CHN) (1998)	time	2.69	3.77	4.87	5.94	7.02	8.11	9.19	10.29	11.38	12.51		14.01	3 / 4			
reaction time	0.201		1.08	1.10	1.07	1.08	1.09	1.08	1.10	1.09	1.13	1.50	PB		3.25	3.25	3.32
interval		5.10	8.46	8.31	8.54	8.46	8.39	8.46	8.31	8.39	8.09	9.35	7.85		8.44	8.44	8.26
velocity																	
Shi Guoxing (CHN) (2000)	time	2.70	3.79	4.85	5.94	7.02	8.09	9.19	10.31	11.43	12.56		14.03	8 / 5			
reaction time	0.172		1.09	1.06	1.09	1.08	1.07	1.10	1.12	1.12	1.13	1.47	PB		3.24	3.25	3.37
interval		5.08	8.39	8.62	8.39	8.46	8.54	8.31	8.16	8.16	8.09	9.54	7.84		8.46	8.44	8.14
velocity																	
Long Wei (CHN) (1997)	time	2.62	3.71	4.81	5.91	7.01	8.11	9.24	10.34	11.50	12.65		14.17	2 / 6			
reaction time	0.162		1.09	1.10	1.10	1.10	1.10	1.13	1.10	1.16	1.15	1.52	PB		3.29	3.33	3.41
interval		5.24	8.39	8.31	8.31	8.31	8.31	8.09	8.31	7.88	7.95	9.22	7.76		8.33	8.23	8.04
velocity																	
Ji Haohao (CHN) (2001)	time	2.72	3.84	4.94	6.07	7.18	8.29	9.43	10.56	11.71	12.88		14.44	7 / 7			
reaction time	0.177		1.12	1.10	1.13	1.11	1.11	1.14	1.13	1.15	1.17	1.56			3.35	3.36	3.45
interval		5.04	8.16	8.31	8.09	8.23	8.23	8.02	8.09	7.95	7.81	8.99	7.62		8.19	8.16	7.95
velocity																	
Huan Jiaying (CHN) (2001)	time	2.69	3.82	4.95	6.09	7.22	8.34	9.48	10.64	11.80	12.96		14.54	9 / 8			
reaction time	0.172		1.13	1.13	1.14	1.13	1.12	1.14	1.16	1.16	1.16	1.58			3.40	3.39	3.48
interval		5.10	8.09	8.09	8.02	8.09	8.16	8.02	7.88	7.88	7.88	8.87	7.57		8.06	8.09	7.88
velocity																	

Heat 2

date 25-Jun-21 wind -1.2 m/s

CAA Hurdle Development (2021)

		H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
Zhu Shenglong (CHN) (2000)	time	2.65	3.77	4.82	5.89	6.97	8.04	9.13	10.23	11.34	12.48		13.99	6 / 1			
reaction time	0.163		1.12	1.05	1.07	1.08	1.07	1.09	1.10	1.11	1.14	1.51			3.24	3.24	3.35
interval		5.18	8.16	8.70	8.54	8.46	8.54	8.39	8.31	8.23	8.02	9.28	7.86		8.46	8.46	8.19
velocity																	
Qin Weibo (CHN) (2002)	time	2.64	3.75	4.83	5.92	6.99	8.06	9.16	10.25	11.37	12.52		14.05	3 / 2			
reaction time	0.198		7 steps 1.11	1.08	1.09	1.07	1.07	1.10	1.09	1.12	1.15	1.53	PB		3.28	3.24	3.36
interval		5.20	8.23	8.46	8.39	8.54	8.54	8.31	8.39	8.16	7.95	9.16	7.83		8.36	8.46	8.16
velocity																	
Zhang Jinzhao (CHN) (1999)	time	2.70	3.82	4.88	5.95	7.04	8.12	9.23	10.34	11.46	12.59		14.09	4 / 3			
reaction time	0.189		1.12	1.06	1.07	1.09	1.08	1.11	1.11	1.12	1.13	1.50			3.25	3.28	3.36
interval		5.08	8.16	8.62	8.54	8.39	8.46	8.23	8.23	8.16	8.09	9.35	7.81		8.44	8.36	8.16
velocity																	

Lu Yang (CHN) (1996)	time	2.70	3.82	4.90	5.99	7.07	8.17	9.27	10.38	11.53	12.70		14.28	5 / 4			
	reaction time	0.163		1.12	1.08	1.09	1.08	1.10	1.10	1.11	1.15	1.17	1.58		3.29	3.28	3.43
	velocity		5.08	8.16	8.46	8.39	8.46	8.31	8.31	8.23	7.95	7.81	8.87	7.70	8.33	8.36	7.99
Sun Haoru (CHN) (1998)	time	2.75	3.89	5.00	6.11	7.25	8.38	9.51	10.64	11.88	13.09		14.69	8 / 5			
	reaction time	0.166		1.14	1.11	1.11	1.14	1.13	1.13	1.13	1.24	1.21	1.60		3.36	3.40	3.58
	velocity		4.99	8.02	8.23	8.23	8.02	8.09	8.09	8.09	7.37	7.55	8.76	7.49	8.16	8.06	7.66
Yan Chenyu (CHN) (2003)	time	2.72	3.85	4.97	6.11	7.27	8.38	9.58	10.77	11.96	13.17		14.74	2 / 6			
	reaction time	0.182		1.13	1.12	1.14	1.16	1.11	1.20	1.19	1.19	1.21	1.57		3.39	3.47	3.59
	velocity		5.04	8.09	8.16	8.02	7.88	8.23	7.62	7.68	7.68	7.55	8.93	7.46	8.09	7.90	7.64
Zhao Jiefu (CHN) (2001)	time	2.79	3.94	5.05	6.18	7.31	8.43	9.58	10.71	11.91	13.11		14.77	9 / 7			
	reaction time	0.214		1.15	1.11	1.13	1.13	1.12	1.15	1.13	1.20	1.20	1.66		3.39	3.40	3.53
	velocity		4.92	7.95	8.23	8.09	8.09	8.16	7.95	8.09	7.62	7.62	8.45	7.45	8.09	8.06	7.77

Heat 1

Heat 1		date	25-Jun-21	wind -1.0 m/s										CAA Hurdle Development (2021)						
				H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10	
Sun Zhenjiang (CHN) (1999)	time			2.58	3.66	4.72	5.79	6.85	7.91	9.00	10.09	11.21	12.33		13.91	4 / 1				
	reaction time	0.150	interval	7 steps	1.08	1.06	1.07	1.06	1.06	1.09	1.09	1.12	1.12	1.58			3.21	3.21	3.33	
			velocity		5.32	8.46	8.62	8.54	8.62	8.62	8.39	8.39	8.16	8.16	8.87	7.91		8.54	8.54	8.23
Yang Quanlei (CHN) (1999)	time			2.68	3.78	4.88	5.96	7.05	8.14	9.22	10.33	11.45	12.57		14.05	5 / 2				
	reaction time	0.164	interval		1.10	1.10	1.08	1.09	1.09	1.08	1.11	1.12	1.12	1.48			3.28	3.26	3.35	
			velocity		5.12	8.31	8.31	8.46	8.39	8.39	8.46	8.23	8.16	8.16	9.47	7.83		8.36	8.41	8.19
Xu Zhuoyi (CHN) (2003)	time			2.68	3.80	4.89	5.97	7.06	8.15	9.26	10.38	11.54	12.71		14.23	6 / 3				
	reaction time	0.156	interval	7 steps	1.12	1.09	1.08	1.09	1.09	1.11	1.12	1.16	1.17	1.52			3.29	3.29	3.45	
			velocity		5.12	8.16	8.39	8.46	8.39	8.39	8.23	8.16	7.88	7.81	9.22	7.73		8.33	8.33	7.95
Yang Yuxuan (CHN) (2002)	time			2.72	3.83	4.94	6.05	7.14	8.24	9.36	10.49	11.64	12.80		14.29	3 / 4				
	reaction time	0.146	interval		1.11	1.11	1.11	1.09	1.10	1.12	1.13	1.15	1.16	1.49			3.33	3.31	3.44	
			velocity		5.04	8.23	8.23	8.23	8.39	8.31	8.16	8.09	7.95	7.88	9.41	7.70		8.23	8.28	7.97
Chen Jie (CHN) (2002)	time			2.72	3.81	4.93	6.04	7.15	8.27	9.40	10.53	11.68	12.81		14.31	8 / 5				
	reaction time	0.151	interval		1.09	1.12	1.11	1.11	1.12	1.13	1.13	1.15	1.13	1.50			3.32	3.36	3.41	
			velocity		5.04	8.39	8.16	8.23	8.23	8.16	8.09	8.09	7.95	8.09	9.35	7.69		8.26	8.16	8.04
Yang Jiajun (CHN) (2002)	time			2.71	3.85	4.97	6.06	7.16	8.26	9.37	10.52	11.67	12.84		14.40	9 / 6				
	reaction time	0.170	interval		1.14	1.12	1.09	1.10	1.10	1.11	1.15	1.15	1.17	1.56	=PB		3.35	3.31	3.47	
			velocity		5.06	8.02	8.16	8.39	8.31	8.31	8.23	7.95	7.95	7.81	8.99	7.64		8.19	8.28	7.90
Lin Ken (CHN) (1999)	time			2.69	3.78	4.88	6.02	7.10	8.22	9.37	10.50	11.67	12.84		14.41	2 / 7				
	reaction time	0.190	interval		1.09	1.10	1.14	1.08	1.12	1.15	1.13	1.17	1.17	1.57	PB		3.33	3.35	3.47	
			velocity		5.10	8.39	8.31	8.02	8.46	8.16	7.95	8.09	7.81	7.81	8.93	7.63		8.23	8.19	7.90
Yang Haoran (CHN) (2001)	time			2.71	3.83	4.94	6.06	7.20	8.35	9.50	10.67	11.83	12.99		14.54	7 / 8				
	reaction time	0.142	interval		1.12	1.11	1.12	1.14	1.15	1.15	1.17	1.16	1.16	1.55			3.35	3.44	3.49	
			velocity		5.06	8.16	8.23	8.16	8.02	7.95	7.95	7.81	7.88	7.88	9.05	7.57		8.19	7.97	7.86

2021 Japanese Multi-Event National Championships (Nagano, JPN)

Decathlon - Heat 3

Decathlon - Heat 3		date	13-Jun-21	wind	Matsubayashi (2021) - 110m hurdling race analysis of decathlon athletes in the 2021 season														
				H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
Nakamura, Akihiko (JPN) (1990)	time			2.73	3.85	4.97	6.09	7.21	8.32	9.45	10.58	11.72	12.85		14.34	2 / 2			
	reaction time	0.152	interval		1.12	1.12	1.12	1.12	1.11	1.13	1.13	1.14	1.13	1.49			3.36	3.36	3.40
			velocity	5.03	8.16	8.16	8.16	8.16	8.23	8.09	8.09	8.02	8.09	9.41	7.67		8.16	8.16	8.06
Maruyama, Yuma (JPN) (1998)	time			2.74	3.91	5.05	6.17	7.27	8.40	9.51	10.66	11.80	12.95		14.50	6 / 3			
	reaction time	0.170	interval		1.17	1.14	1.12	1.10	1.13	1.11	1.15	1.14	1.15	1.55			3.43	3.34	3.44
			velocity	5.01	7.81	8.02	8.16	8.31	8.09	8.23	7.95	8.02	7.95	9.05	7.59		7.99	8.21	7.97

Decathlon - Heat 2

Decathlon - Heat 2		date	13-Jun-21		wind	-0.9 m/s		Matsubayashi (2021) - 110m hurdling race analysis of decathlon athletes in the 2021 season												
				H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10	
Okuda, Keisuke (JPN) (1996)	time			2.74	3.88	5.02	6.15	7.26	8.41	9.56	10.71	11.86	13.06		14.54	2 / 1				
	reaction time	0.151	interval		1.14	1.14	1.13	1.11	1.15	1.15	1.15	1.15	1.20	1.48			3.41	3.41	3.50	
			velocity		5.01	8.02	8.02	8.09	8.23	7.95	7.95	7.95	7.95	7.62	9.47	7.57		8.04	8.04	7.83

2020 Chinese Olympic Trials (Shaoxing, CHN)

FINAL

INAL	date	12-Jun-21	wind	CAA Hurdle Development (2021)															
				H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
Ning Xiaohan (CHN) (2000)	time			2.62	3.68	4.72	5.76	6.81	7.86	8.91	9.98	11.09	12.19		13.69	3 / 1			
	reaction time	0.143			1.06	1.04	1.04	1.05	1.05	1.05	1.07	1.11	1.10	1.50			3.14	3.15	3.28
	velocity			5.24	8.62	8.79	8.79	8.70	8.70	8.70	8.54	8.23	8.31	9.35	8.04		8.73	8.70	8.36
Zhang Honglin (CHN) (1994)	time			2.65	3.75	4.79	5.82	6.87	7.94	9.01	10.07	11.18	12.29		13.75	5 / 2			

reaction time	0.161	interval		1.10	1.04	1.03	1.05	1.07	1.07	1.06	1.11	1.11	1.46		3.17	3.19	3.28
		velocity	5.18	8.31	8.79	8.87	8.70	8.54	8.54	8.62	8.23	8.23	9.60	8.00	8.65	8.60	8.36
Zhang Jinzhao (CHN) (1999)	time	2.77	3.87	4.95	6.01	7.07	8.14	9.19	10.28	11.38	12.47		13.94	4 / 3			
reaction time	0.190	interval		1.10	1.08	1.06	1.06	1.07	1.05	1.09	1.10	1.09	1.47	PB	3.24	3.18	3.28
		velocity	4.95	8.31	8.46	8.62	8.62	8.54	8.70	8.39	8.31	8.39	9.54	7.89	8.46	8.62	8.36
Li Huaihai (CHN) (1999)	time	2.65	3.75	4.84	5.93	7.02	8.11	9.19	10.29	11.39	12.51		14.00	7 / 4			
reaction time	0.139	interval	7 steps	1.10	1.09	1.09	1.09	1.09	1.08	1.10	1.10	1.12	1.49	PB	3.28	3.26	3.32
		velocity	5.18	8.31	8.39	8.39	8.39	8.39	8.46	8.31	8.31	8.16	9.41	7.86	8.36	8.41	8.26
Pan Zijie (CHN) (1995)	time	2.63	3.74	4.81	5.93	7.03	8.13	9.22	10.30	11.43	12.55		14.02	1 / 5			
reaction time	0.158	interval		1.11	1.07	1.12	1.10	1.10	1.09	1.08	1.13	1.12	1.47		3.30	3.29	3.33
		velocity	5.22	8.23	8.54	8.16	8.31	8.31	8.39	8.46	8.09	8.16	9.54	7.85	8.31	8.33	8.23
Xie Wenjun (CHN) (1990)	time	2.68	3.77	4.84	5.89	6.96	8.04	9.14	10.27	11.39	12.53		14.06	2 / 6			
reaction time	0.152	interval		1.09	1.07	1.05	1.07	1.08	1.10	1.13	1.12	1.14	1.53		3.21	3.25	3.39
		velocity	5.12	8.39	8.54	8.70	8.54	8.46	8.31	8.09	8.16	8.02	9.16	7.82	8.54	8.44	8.09
Yang Quanlei (CHN) (1999)	time	2.72	3.79	4.88	5.97	7.05	8.17	9.24	10.36	11.51	12.65		14.16	6 / 7			
reaction time	0.172	interval		1.07	1.09	1.09	1.08	1.12	1.07	1.12	1.15	1.14	1.51		3.25	3.27	3.41
		velocity	5.04	8.54	8.39	8.39	8.46	8.16	8.54	8.16	7.95	8.02	9.28	7.77	8.44	8.39	8.04
Guo Zhongjie (CHN) (2001)	time	2.71	3.77	4.85	5.95	7.10							dnf	8 / --			
reaction time		interval		1.06	1.08	1.10	1.15								3.24		
		velocity	5.06	8.62	8.46	8.31	7.95								8.46		

Heat 2

date	12-Jun-21	wind	-0.8 m/s	CAA Hurdle Development (2021)													
		H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
Yang Quanlei (CHN) (1999)	time	2.55	3.63	4.71	5.77	6.85	7.92	9.01	10.09	11.19	12.29		13.87	5 / 1			
reaction time	0.161	interval		1.08	1.08	1.06	1.08	1.07	1.09	1.08	1.10	1.10	1.58	PB	3.22	3.24	3.28
		velocity	5.38	8.46	8.46	8.62	8.46	8.54	8.39	8.46	8.31	8.31	8.87	7.93	8.52	8.46	8.36
Zhang Jinzhao (CHN) (1999)	time	2.64	3.76	4.84	5.92	7.00	8.09	9.17	10.28	11.39	12.51		14.10	3 / 2			
reaction time	0.172	interval		1.12	1.08	1.08	1.08	1.09	1.08	1.11	1.11	1.12	1.59		3.28	3.25	3.34
		velocity	5.20	8.16	8.46	8.46	8.46	8.39	8.46	8.23	8.23	8.16	8.82	7.80	8.36	8.44	8.21
Guo Zhongjie (CHN) (2001)	time	2.61	3.70	4.77	5.85	6.93	8.02	9.14	10.26	11.36	12.54		14.15	6 / 3			
reaction time	0.177	interval		1.09	1.07	1.08	1.08	1.09	1.12	1.12	1.10	1.18	1.61		3.24	3.29	3.40
		velocity	5.26	8.39	8.54	8.46	8.46	8.39	8.16	8.16	8.31	7.75	8.71	7.77	8.46	8.33	8.06
Xie Wenjun (CHN) (1990)	time	2.59	3.69	4.76	5.83	6.90	8.01	9.13	10.28	11.42	12.59		14.25	8 / 4			
reaction time	0.162	interval		1.10	1.07	1.07	1.07	1.11	1.12	1.15	1.14	1.17	1.66		3.24	3.30	3.46
		velocity	5.30	8.31	8.54	8.54	8.54	8.23	8.16	7.95	8.02	7.81	8.45	7.72	8.46	8.31	7.92
Qin Weibo (CHN) (2002)	time	2.58	3.72	4.81	5.91	7.00	8.10	9.23	10.36	11.49	12.64		14.26	4 / 5			
reaction time	0.174	interval	7 steps	1.14	1.09	1.10	1.09	1.10	1.13	1.13	1.13	1.15	1.62		3.33	3.32	3.41
		velocity	5.32	8.02	8.39	8.31	8.39	8.31	8.09	8.09	8.09	7.95	8.65	7.71	8.23	8.26	8.04
Shi Guoxing (CHN) (2000)	time	2.65	3.76	4.87	5.97	7.11	8.24	9.36	10.49	11.63	12.77		14.39	7 / 6			
reaction time	0.156	interval		1.11	1.11	1.10	1.14	1.13	1.12	1.13	1.14	1.14	1.62		3.32	3.39	3.41
		velocity	5.18	8.23	8.23	8.31	8.02	8.09	8.16	8.09	8.02	8.02	8.65	7.64	8.26	8.09	8.04

Heat 1

date	12-Jun-21	wind	-0.4 m/s	CAA Hurdle Development (2021)													
		H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
Zhang Honglin (CHN) (1994)	time	2.52	3.60	4.69	5.77	6.84	7.90	8.96	10.06	11.19	12.33		13.94	6 / 1			
reaction time	0.166	interval		1.08	1.09	1.08	1.07	1.06	1.10	1.13	1.14	1.61			3.25	3.19	3.37
		velocity	5.44	8.46	8.39	8.46	8.54	8.62	8.62	8.31	8.09	8.02	8.71	7.89	8.44	8.60	8.14
Ning Xiaohan (CHN) (2000)	time	2.50	3.58	4.65	5.72	6.80	7.87	8.95	10.05	11.17	12.31		13.97	4 / 2			
reaction time	0.141	interval		1.08	1.07	1.07	1.08	1.07	1.08	1.10	1.12	1.14	1.66		3.22	3.23	3.36
		velocity	5.49	8.46	8.54	8.54	8.46	8.54	8.46	8.31	8.16	8.02	8.45	7.87	8.52	8.49	8.16
Li Huaihai (CHN) (1999)	time	2.55	3.64	4.72	5.79	6.87	7.96	9.04	10.20	11.34	12.50		14.09	8 / 3			
reaction time	0.154	interval	7 steps	1.09	1.08	1.07	1.08	1.09	1.08	1.16	1.14	1.16	1.59		3.24	3.25	3.46
		velocity	5.38	8.39	8.46	8.54	8.46	8.39	8.46	7.88	8.02	7.88	8.82	7.81	8.46	8.44	7.92
Pan Zijie (CHN) (1995)	time	2.51	3.61	4.72	5.80	6.91	8.03	9.14	10.25	11.40	12.53		14.16	7 / 4			
reaction time	0.111	interval		1.10	1.11	1.08	1.11	1.12	1.11	1.11	1.15	1.13	1.63		3.29	3.34	3.39
		velocity	5.47	8.31	8.23	8.46	8.23	8.16	8.23	8.23	7.95	8.09	8.60	7.77	8.33	8.21	8.09
Liu Bowen (CHN) (2001)	time	2.51	3.60	4.72	5.84	6.94	8.07	9.18	10.32	11.46	12.65		14.30	3 / 5			
reaction time	0.148	interval		1.09	1.12	1.12	1.10	1.13	1.11	1.14	1.14	1.19	1.65		3.33	3.34	3.47
		velocity	5.47	8.39	8.16	8.16	8.31	8.09	8.23	8.02	8.02	7.68	8.50	7.69	8.23	8.21	7.90
Xu Zhuoyi (CHN) (2003)	time	2.64	3.73	4.85	5.94	7.04	8.14	9.26	10.38	11.52	12.70		14.33	5 / 6			
reaction time	0.173	interval	7 steps	1.09	1.12	1.09	1.10	1.10	1.12	1.12	1.14	1.18	1.63		3.30	3.32	3.44

velocity 5.20 8.39 8.16 8.39 8.31 8.31 8.16 8.16 8.02 7.75 8.60 7.68 8.31 8.26 7.97

2021 NCAA Championships (Eugene, OR) (TV Analysis)

FINAL

		date	11-Jun-21	wind	-0.6 m/s											Henson (2021) - Athlete First: 2021 year end hurdle report			
				H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
Dunning, Robert (USA) (1997)	time			2.53	3.57	4.60	5.62	6.60	7.63	8.63	9.65	10.69	11.75		13.25	2 / 1			
	reaction time	0.161	interval	7 steps	1.04	1.03	1.02	0.98	1.03	1.00	1.02	1.04	1.06	1.50			3.09	3.01	3.12
	velocity			5.42	8.79	8.87	8.96	9.33	8.87	9.14	8.96	8.79	8.62	9.35	8.30		8.87	9.11	8.79
McConico, Jaylan (USA) (1998)	time			2.55	3.57	4.60	5.63	6.63	7.63	8.73	9.80	10.83	11.93		13.38	4 / 2			
	reaction time	0.194	interval		1.02	1.03	1.03	1.00	1.00	1.10	1.07	1.03	1.10	1.45			3.08	3.10	3.20
	velocity			5.38	8.96	8.87	8.87	9.14	9.14	8.31	8.54	8.87	8.31	9.67	8.22		8.90	8.85	8.57
Lemonious, Phillip (JAM) (1998)	time			2.57	3.60	4.70	5.70	6.73	7.73	8.77	9.80	10.83	11.90		13.39	5 / 3			
	reaction time	0.150	interval	7 steps	1.03	1.10	1.00	1.03	1.00	1.04	1.03	1.03	1.07	1.49	PB		3.13	3.07	3.13
	velocity			5.34	8.87	8.31	9.14	8.87	9.14	8.79	8.87	8.87	8.54	9.41	8.22		8.76	8.93	8.76
Britt, Jamal (USA) (1998)	time			2.57	3.63	4.67	5.70	6.70	7.73	8.77	9.83	10.87	11.97		13.45	8 / 4			
	reaction time	0.203	interval	7 steps	1.06	1.04	1.03	1.00	1.03	1.04	1.06	1.04	1.10	1.48	PB		3.13	3.07	3.20
	velocity			5.34	8.62	8.79	8.87	9.14	8.87	8.79	8.62	8.79	8.31	9.47	8.18		8.76	8.93	8.57
Marshall, Jamar (USA) (2001)	time			2.50	3.53	4.57	5.60	6.60	7.60	8.63	9.70	10.80	11.93		13.53	7 / 5			
	reaction time	0.161	interval	7 steps	1.03	1.04	1.03	1.00	1.00	1.03	1.07	1.10	1.13	1.60			3.10	3.03	3.30
	velocity			5.49	8.87	8.79	8.87	9.14	9.14	8.87	8.54	8.31	8.09	8.76	8.13		8.85	9.05	8.31
Ojora, Tade (GBR) (1999)	time			2.63	3.70	4.77	5.83	6.85	7.87	8.93	9.97	11.07	12.13		13.57	9 / 6			
	reaction time	0.224	interval		1.07	1.07	1.06	1.02	1.02	1.06	1.04	1.10	1.06	1.44	PB		3.20	3.10	3.20
	velocity			5.22	8.54	8.54	8.62	8.96	8.96	8.62	8.79	8.31	8.62	9.74	8.11		8.57	8.85	8.57
Roberts, Giano (JAM) (2000)	time			2.57	3.63	4.70	5.73	6.75	7.79	8.83	9.95	11.07	12.13		13.65	3 / 7			
	reaction time	0.152	interval		1.06	1.07	1.03	1.02	1.04	1.04	1.12	1.12	1.06	1.52			3.16	3.10	3.30
	velocity			5.34	8.62	8.54	8.87	8.96	8.79	8.79	8.16	8.16	8.62	9.22	8.06		8.68	8.85	8.31
Thomas, Damion (JAM) (1999)	time			2.60	3.67	4.70	5.75	6.77	7.80	8.90	10.00	11.07	12.23		13.76	6 / 8			
	reaction time	0.200	interval	7 steps	1.07	1.03	1.05	1.02	1.03	1.10	1.10	1.07	1.16	1.53			3.15	3.15	3.33
	velocity			5.28	8.54	8.87	8.70	8.96	8.87	8.31	8.31	8.54	7.88	9.16	7.99		8.70	8.70	8.23
Brown, Tai (USA) (2000)	time			2.70	3.77	4.85	5.90	6.93	7.97	9.03	10.07	11.13	12.23		13.89	1 / 9			
	reaction time	0.161	interval		1.07	1.08	1.05	1.03	1.04	1.06	1.04	1.06	1.10	1.66			3.20	3.13	3.20
	velocity			5.08	8.54	8.46	8.70	8.87	8.79	8.62	8.79	8.62	8.31	8.45	7.92		8.57	8.76	8.57

2021 Golden Gala Pietro Mennea (Florence, ITA) (TV Analysis)

FINAL

		date	10-Jun-21	wind	-0.1 m/s											Henson (2021) - Athlete First: 2021 year end hurdle report			
				H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
McLeod, Omar (JAM) (1994)	time			2.44	3.50	4.48	5.48	6.48	7.48	8.48	9.50	10.50	11.56		13.01	4 / 1			
	reaction time	0.147	interval		1.06	0.98	1.00	1.00	1.00	1.00	1.02	1.00	1.06	1.45			3.04	3.00	3.08
	velocity			5.62	8.62	9.33	9.14	9.14	9.14	9.14	8.96	9.14	8.62	9.67	8.46		9.02	9.14	8.90
Pozzi, Andy (GBR) (1992)	time			2.48	3.52	4.50	5.50	6.52	7.54	8.56	9.60	10.64	11.74		13.25	3 / 2			
	reaction time	0.113	interval	7 steps	1.04	0.98	1.00	1.02	1.02	1.02	1.04	1.04	1.10	1.51			3.02	3.06	3.18
	velocity			5.53	8.79	9.33	9.14	8.96	8.96	8.96	8.79	8.79	8.31	9.28	8.30		9.08	8.96	8.62
Belocian, Wilhem (FRA) (1995)	time			2.48	3.52	4.56	5.58	6.60	7.64	8.66	9.68	10.76	11.82		13.31	6 / 3			
	reaction time	0.138	interval		1.04	1.04	1.02	1.02	1.04	1.02	1.02	1.08	1.06	1.49			3.10	3.08	3.16
	velocity			5.53	8.79	8.79	8.96	8.96	8.79	8.96	8.96	8.46	8.62	9.41	8.26		8.85	8.90	8.68
Allen, Devon (USA) (1994)	time			2.52	3.56	4.58	5.60	6.62	7.64	8.68	9.72	10.76	11.82		13.32	5 / 4			
	reaction time	0.136	interval		1.04	1.02	1.02	1.02	1.02	1.04	1.04	1.04	1.06	1.50			3.08	3.08	3.14
	velocity			5.44	8.79	8.96	8.96	8.96	8.96	8.79	8.79	8.79	8.62	9.35	8.26		8.90	8.90	8.73
Brathwaite, Shane (BAR) (1990)	time			2.50	3.56	4.60	5.64	6.66	7.68	8.74	9.80	10.86	11.94		13.46	1 / 5			
	reaction time	0.122	interval		1.06	1.04	1.04	1.02	1.02	1.06	1.06	1.06	1.08	1.52			3.14	3.10	3.20
	velocity			5.49	8.62	8.79	8.79	8.96	8.96	8.62	8.62	8.62	8.46	9.22	8.17		8.73	8.85	8.57
Perini, Lorenzo (ITA) (1994)	time			2.54	3.60	4.68	5.72	6.76	7.80	8.84	9.90	10.98	12.06		13.63	8 / 6			
	reaction time	0.140	interval		1.06	1.08	1.04	1.04	1.04	1.04	1.06	1.08	1.08	1.57			3.18	3.12	3.22
	velocity			5.40	8.62	8.46	8.79	8.79	8.79	8.79	8.62	8.46	8.46	8.93	8.07		8.62	8.79	8.52
Dal Molin, Paolo (ITA) (1987)	time			2.48	3.54	4.60	5.68	6.72	7.80	8.86	9.94	11.00	12.12		13.64	2 / 7			
	reaction time	0.142	interval	7 steps	1.06	1.06	1.08	1.04	1.08	1.06	1.08	1.06	1.12	1.52			3.20	3.18	3.26
	velocity			5.53	8.62	8.62	8.46	8.79	8.46	8.62	8.46	8.62	8.16	9.22	8.06		8.57	8.62	8.41
Martinot-Lagarde, Pascal (FRA)	time			2.56	3.64	4.74	5.80	6.88	7.98	9.08	10.24	11.42	12.58		14.26	7 / 8			
	reaction time	0.162	interval	7 steps	1.08	1.10	1.06	1.08	1.10	1.10	1.16	1.18	1.16	1.68			3.24	3.28	3.50
	velocity			5.36	8.46	8.31	8.62	8.46	8.31	8.31	7.88	7.75	7.88	8.35	7.71		8.46	8.36	7.83

2021 Fanny Blankers Koen Games (Hengelo, NED) (TV Analysis)

FINAL

INAL	date	06-Jun-21	wind	Henson (2021) - Athlete First: 2021 year end hurdle report															
				H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
McLeod, Omar (JAM) (1994)	time			2.37	3.43	4.48	5.50	6.47	7.47	8.54	9.56	10.58	11.66		13.08	5 / 1			
	reaction time				1.06	1.05	1.02	0.97	1.00	1.07	1.02	1.02	1.08	1.42			3.13	3.04	3.12
	velocity			5.79	8.62	8.70	8.96	9.42	9.14	8.54	8.96	8.96	8.46	9.87	8.41		8.76	9.02	8.79
Allen, Devon (USA) (1994)	time			2.47	3.54	4.54	5.60	6.64	7.67	8.73	9.74	10.77	11.86		13.32	4 / 2			
	reaction time				1.07	1.00	1.06	1.04	1.03	1.06	1.01	1.03	1.09	1.46			3.13	3.13	3.13
	velocity			5.55	8.54	9.14	8.62	8.79	8.87	8.62	9.05	8.87	8.39	9.60	8.26		8.76	8.76	8.76
Belocian, Wilhem (FRA) (1995)	time			2.47	3.54	4.54	5.57	6.60	7.62	8.64	9.67	10.74	11.84		13.34	6 / 3			
	reaction time				1.07	1.00	1.03	1.03	1.02	1.02	1.03	1.07	1.10	1.50			3.10	3.07	3.20
	velocity			5.55	8.54	9.14	8.87	8.87	8.96	8.96	8.87	8.54	8.31	9.35	8.25		8.85	8.93	8.57

2021 Fuse Sprint (Tottori, JPN)

FINAL

		date	06-Jun-21	wind	1.8 m/s	Shibayama (2021) - race analysis of men's 110m hurdles in the 2021 season													
				H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
Kanai, Taio (JPN) (1995)	time			2.55	3.58	4.62	5.65	6.68	7.71	8.76	9.81	10.88	11.96		13.40	4 / 1			
	reaction time	0.128	interval	7 steps	1.03	1.04	1.03	1.03	1.03	1.05	1.05	1.07	1.08	1.44			3.10	3.11	3.20
		velocity			5.38	8.87	8.79	8.87	8.87	8.87	8.70	8.70	8.54	8.46	9.74	8.21		8.85	8.82
Ishikawa, Shuhei (JPN) (1995)	time			2.60	3.64	4.68	5.71	6.73	7.75	8.82	9.86	10.93	12.03		13.48	7 / 2			
	reaction time	0.152	interval	7 steps	1.04	1.04	1.03	1.02	1.02	1.07	1.04	1.07	1.10	1.45			3.11	3.11	3.21
		velocity			5.28	8.79	8.79	8.87	8.96	8.96	8.54	8.79	8.54	8.31	9.67	8.16		8.82	8.82
Takayama, Shunya (JPN) (1994)	time			2.56	3.61	4.64	5.68	6.71	7.75	8.79	9.85	10.91	12.02		13.48	6 / 3			
	reaction time	0.145	interval		1.05	1.03	1.04	1.03	1.04	1.04	1.06	1.06	1.11	1.46			3.12	3.11	3.23
		velocity			5.36	8.70	8.87	8.79	8.87	8.79	8.79	8.62	8.62	8.23	9.60	8.16		8.79	8.82
Nomoto, Syuusei (JPN) (1995)	time			2.57	3.62	4.68	5.75	6.80	7.85	8.93	10.00	11.11	12.22		13.66	8 / 4			
	reaction time	0.144	interval	7 steps	1.05	1.06	1.07	1.05	1.05	1.08	1.07	1.11	1.11	1.44			3.18	3.18	3.29
		velocity			5.34	8.70	8.62	8.54	8.70	8.70	8.46	8.54	8.23	8.23	9.74	8.05		8.62	8.62
Kawashima, Ryota (JPN) (1999)	time			2.59	3.68	4.75	5.79	6.85	7.92	8.98	10.06	11.14	12.27		13.77	3 / 5			
	reaction time	0.139	interval		1.09	1.07	1.04	1.06	1.07	1.06	1.08	1.08	1.13	1.50	PB		3.20	3.19	3.29
		velocity			5.30	8.39	8.54	8.79	8.62	8.54	8.62	8.46	8.46	8.09	9.35	7.99		8.57	8.60
Fujii, Ryota (JPN) (1996)	time			2.62	3.71	4.76	5.81	6.90	7.96	9.02	10.13	11.21	12.31		13.79	9 / 6			
	reaction time	0.152	interval		1.09	1.05	1.05	1.09	1.06	1.06	1.11	1.08	1.10	1.48			3.19	3.21	3.29
		velocity			5.24	8.39	8.70	8.70	8.39	8.62	8.62	8.23	8.46	8.31	9.47	7.98		8.60	8.54
Omuro, Hideki (JPN) (1990)	time			2.60	3.68	4.74	5.81	6.87	7.95	9.02	10.10	11.19	12.30		13.80	2 / 7			
	reaction time	0.150	interval		1.08	1.06	1.07	1.06	1.08	1.07	1.08	1.09	1.11	1.50			3.21	3.21	3.28
		velocity			5.28	8.46	8.62	8.54	8.62	8.46	8.54	8.46	8.39	8.23	9.35	7.97		8.54	8.54

2021 Michitaka Kinami Memorial (Osaka, JPN)

FINAL

INAL		date		01-Jun-21		wind		0.3 m/s		Shibayama (2021) - race analysis of men's 110m hurdles in the 2021 season												
		H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10					
Muratake, Rashiddo (JPN) (200)	time			2.60	3.65	4.68	5.70	6.72	7.76	8.81	9.86	10.91	11.99		13.35	7 / 1						
	reaction time	0.150	interval	7 steps	1.05	1.03	1.02	1.02	1.04	1.05	1.05	1.05	1.08	1.36	NJR PB		3.10	3.11	3.18			
			velocity		5.28	8.70	8.87	8.96	8.96	8.79	8.70	8.70	8.70	8.46	10.31	8.24		8.85	8.82	8.62		
Ishikawa, Shuhei (JPN) (1995)	time			2.58	3.62	4.65	5.68	6.70	7.75	8.79	9.85	10.90	11.98		13.42	6 / 2						
	reaction time	0.156	interval	7 steps	1.04	1.03	1.03	1.02	1.05	1.04	1.06	1.05	1.08	1.44			3.10	3.11	3.19			
			velocity		5.32	8.79	8.87	8.87	8.96	8.70	8.79	8.62	8.70	8.46	9.74	8.20		8.85	8.82	8.60		
Tokuoka, Ryo (JPN) (1999)	time			2.68	3.76	4.80	5.84	6.89	7.95	9.03	10.08	11.16	12.25		13.69	8 / 3						
	reaction time	0.185	interval		1.08	1.04	1.04	1.05	1.06	1.08	1.05	1.08	1.09	1.44			3.16	3.19	3.22			
			velocity		5.12	8.46	8.79	8.79	8.70	8.62	8.46	8.70	8.46	8.39	9.74	8.04		8.68	8.60	8.52		
Shimizu, Koichiro (JPN) (2000)	time			2.69	3.75	4.85	5.91	6.96	8.03	9.10	10.18	11.28	12.39		13.85	9 / 4						
	reaction time	0.169	interval		1.06	1.10	1.06	1.05	1.07	1.07	1.08	1.10	1.11	1.46			3.22	3.19	3.29			
			velocity		5.10	8.62	8.31	8.62	8.70	8.54	8.54	8.46	8.31	8.23	9.60	7.94		8.52	8.60	8.33		
Ishida-Thomas, Higashi (JPN)	time			2.70	3.78	4.85	5.92	6.98	8.05	9.13	10.22	11.32	12.43		13.87	2 / 5						
	reaction time	0.208	interval	7 steps	1.08	1.07	1.07	1.06	1.07	1.08	1.09	1.10	1.11	1.44			3.22	3.21	3.30			
			velocity		5.08	8.46	8.54	8.54	8.62	8.54	8.46	8.39	8.31	8.23	9.74	7.93		8.52	8.54	8.31		
Ikeda, Kai (JPN) (2002)	time			2.76	3.88	4.95	6.03	7.10	8.16	9.24	10.32	11.39	12.50		13.97	4 / 6						
	reaction time	0.185	interval	7 steps	1.12	1.07	1.08	1.07	1.06	1.08	1.08	1.07	1.11	1.47			3.27	3.21	3.26			
			velocity		4.97	8.16	8.54	8.46	8.54	8.62	8.46	8.46	8.54	8.23	9.54	7.87		8.39	8.54	8.41		
Yazawa, Wataru (JPN) (1991)	time			2.68	3.80	4.89	5.96	7.03	8.11	9.19	10.29	11.40	12.55		14.09	3 / 7						
	reaction time	0.176	interval	7 steps	1.12	1.09	1.07	1.07	1.08	1.08	1.10	1.11	1.15	1.54			3.28	3.23	3.36			
			velocity		5.12	8.16	8.39	8.54	8.54	8.46	8.46	8.31	8.23	7.95	9.10	7.81		8.36	8.49	8.16		

Omuro, Hideki (JPN) (1990)	time	2.65	3.78	4.88	6.00	7.13	8.25	9.42	10.56	11.75	12.95		14.65	5 / 8			
	reaction time	0.165		1.13	1.10	1.12	1.13	1.12	1.17	1.14	1.19	1.20	1.70		3.35	3.42	3.53
	velocity		5.18	8.09	8.31	8.16	8.09	8.16	7.81	8.02	7.68	7.62	8.25	7.51	8.19	8.02	7.77

2021 Shimane High School Championships (Honshu, JPN)

FINAL

date 30-May-21 wind 1.7 m/s

Shimizu (2021) - shimane high school championships 2021 - biomechanics data analysis

		H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
山本 俊介	time	2.75	3.95	5.12	6.30	7.45	8.60	9.80	10.97	12.17	13.37		14.95	6 / 1			
	reaction time			1.20	1.17	1.18	1.15	1.15	1.20	1.17	1.20	1.58			3.55	3.50	3.57
	velocity	4.99	7.62	7.81	7.75	7.95	7.95	7.62	7.81	7.62	7.62	8.87	7.36		7.72	7.83	7.68
外野 晴琉	time	2.78	3.95	5.17	6.35	7.55	8.75	9.93	11.20	12.43	13.70		15.32	4 / 2			
	reaction time			1.17	1.22	1.18	1.20	1.18	1.27	1.23	1.27	1.62			3.57	3.58	3.77
	velocity	4.94	7.81	7.49	7.75	7.62	7.62	7.75	7.20	7.43	7.20	8.65	7.18		7.68	7.66	7.27
園山 天志	time	2.82	4.02	5.22	6.40	7.60	8.80	10.02	11.27	12.50	13.82		15.45	3 / 3			
	reaction time			1.20	1.20	1.18	1.20	1.22	1.25	1.23	1.32	1.63			3.58	3.62	3.80
	velocity	4.87	7.62	7.62	7.75	7.62	7.62	7.49	7.31	7.43	6.92	8.60	7.12		7.66	7.57	7.22
森木 大翔	time	2.78	4.02	5.28	6.50	7.72	9.00	10.25	11.52	12.82	14.12		15.80	5 / 4			
	reaction time			1.24	1.26	1.22	1.22	1.28	1.25	1.27	1.30	1.68			3.72	3.75	3.87
	velocity	4.94	7.37	7.25	7.49	7.49	7.14	7.31	7.20	7.03	7.03	8.35	6.96		7.37	7.31	7.09
東本 匠	time	2.83	4.08	5.40	6.68	8.03	9.37	10.75	12.07	13.48	14.85		16.57	2 / 5			
	reaction time			1.25	1.32	1.28	1.35	1.34	1.38	1.32	1.41	1.37	1.72		3.85	4.07	4.10
	velocity	4.85	7.31	6.92	7.14	6.77	6.82	6.62	6.92	6.48	6.67	8.15	6.64		7.12	6.74	6.69
品川 侑斗	time	3.02	4.35	5.70	7.05	8.37	9.72	11.07	12.42	13.80	15.22		17.07	1 / 6			
	reaction time			1.33	1.35	1.35	1.32	1.35	1.35	1.38	1.42	1.85			4.03	4.02	4.15
	velocity	4.54	6.87	6.77	6.77	6.92	6.77	6.77	6.77	6.62	6.44	7.58	6.44		6.80	6.82	6.61

2021 Philippe Cup Athletics Invitational (Wuhan, CHN)

FINAL

date 19-May-21 wind 1.9 m/s

CAA Hurdle Development (2021)

		H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
Ning Xiaohan (CHN) (2000)	time	2.60	3.67	4.70	5.74	6.79	7.82	8.89	9.98	11.08	12.20		13.70	3 / 1			
	reaction time	0.158		1.07	1.03	1.04	1.05	1.03	1.07	1.09	1.10	1.50			3.14	3.15	3.31
	velocity	5.28	8.54	8.87	8.79	8.70	8.87	8.54	8.39	8.31	8.16	9.35	8.03		8.73	8.70	8.28
Zhu Shenglong (CHN) (2000)	time	2.63	3.70	4.75	5.79	6.84	7.92	8.97	10.09	11.18	12.28		13.78	6 / 2			
	reaction time	0.169		1.07	1.05	1.04	1.05	1.08	1.05	1.12	1.09	1.10	1.50	PB	3.16	3.18	3.31
	velocity	5.22	8.54	8.70	8.79	8.70	8.46	8.70	8.16	8.39	8.31	9.35	7.98		8.68	8.62	8.28
Liu Bowen (CHN) (2001)	time	2.67	3.77	4.87	5.93	7.02	8.09	9.21	10.33	11.46	12.61		14.14	5 / 3			
	reaction time	0.163		1.10	1.10	1.06	1.09	1.07	1.12	1.12	1.13	1.15	1.53	PB	3.26	3.28	3.40
	velocity	5.14	8.31	8.31	8.62	8.39	8.54	8.16	8.16	8.09	7.95	9.16	7.78		8.41	8.36	8.06
Yang Jiajun (CHN) (2002)	time	2.72	3.83	4.94	6.06	7.16	8.26	9.39	10.56	11.71	12.88		14.40	8 / 4			
	reaction time	0.126		1.11	1.11	1.12	1.10	1.13	1.17	1.15	1.17	1.52	PB		3.34	3.33	3.49
	velocity	5.04	8.23	8.23	8.16	8.31	8.31	8.09	7.81	7.95	7.81	9.22	7.64		8.21	8.23	7.86
Jiang Wei (CHN) (1994)	time	2.74	3.89	5.00	6.10	7.22	8.36	9.44	10.59	11.73	12.90		14.43	2 / 5			
	reaction time	0.194		1.15	1.11	1.10	1.12	1.14	1.08	1.15	1.14	1.17	1.53		3.36	3.34	3.46
	velocity	5.01	7.95	8.23	8.31	8.16	8.02	8.46	7.95	8.02	7.81	9.16	7.62		8.16	8.21	7.92
Sun Haoru (CHN) (1998)	time	2.74	3.85	4.97	6.19								dnf	7 / --			
	reaction time	0.156		1.11	1.12	1.22									3.45		
	velocity	5.01	8.23	8.16	7.49										7.95		

2021 Ready Steady Tokyo (Tokyo, JPN)

FINAL

date 09-May-21 wind -0.8 m/s

Shibayama (2021) - race analysis of men's 110m hurdles in the 2021 season

		H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
Kanai, Taio (JPN) (1995)	time	2.53	3.57	4.59	5.65	6.68	7.73	8.78	9.84	10.91	11.98		13.38	4 / 1			
	reaction time	0.133		7 steps	1.04	1.02	1.06	1.03	1.05	1.05	1.06	1.07	1.07		3.12	3.13	3.20
	velocity	5.42	8.79	8.96	8.62	8.87	8.70	8.70	8.62	8.54	8.54	10.01	8.22		8.79	8.76	8.57
Izumiya, Shunsuke (JPN) (2000)	time	2.51	3.60	4.63	5.71	6.76	7.83	8.90	9.94	11.01	12.06		13.43	7 / 2			
	reaction time	0.132		7 steps	1.09	1.03	1.08	1.05	1.07	1.07	1.04	1.07	1.05		3.20	3.19	3.16
	velocity	5.47	8.39	8.87	8.46	8.70	8.54	8.54	8.79	8.54	8.70	10.23	8.19		8.57	8.60	8.68
Takayama, Shunya (JPN) (1994)	time	2.58	3.64	4.65	5.67	6.70	7.74	8.77	9.84	10.92	12.01		13.45	5 / 3			
	reaction time	0.163		1.06	1.01	1.02	1.03	1.04	1.03	1.07	1.08	1.09	1.44		3.09	3.10	3.24
	velocity	5.32	8.62	9.05	8.96	8.87	8.79	8.87	8.54	8.46	8.39	9.74	8.18		8.87	8.85	8.46
Muratake, Rashiddo (JPN) (2000)	time	2.56	3.63	4.66	5.72	6.77	7.81	8.85	9.93	11.01	12.11		13.51	9 / 4			
	reaction time	0.119		7 steps	1.07	1.03	1.06	1.05	1.04	1.04	1.08	1.10	1.40	PB	3.16	3.13	3.26
	velocity																

	velocity	5.36	8.54	8.87	8.62	8.70	8.79	8.79	8.46	8.46	8.31	10.01	8.14		8.68	8.76	8.41
Chen Kuei-Ru (TPE) (1993)	time	2.58	3.61	4.65	5.69	6.73	7.77	8.84	9.92	11.02	12.13		13.62	2 / 5			
	reaction time	0.161	interval	7 steps	1.03	1.04	1.04	1.04	1.07	1.08	1.10	1.11	1.49		3.11	3.15	3.29
	velocity	5.32	8.87	8.79	8.79	8.79	8.79	8.54	8.46	8.31	8.23	9.41	8.08		8.82	8.70	8.33
Ishikawa, Shuhei (JPN) (1995)	time	2.65	3.74	4.77	5.80	6.86	7.93	8.99	10.07	11.15	12.25		13.67	8 / 6			
	reaction time	0.141	interval	7 steps	1.09	1.03	1.03	1.06	1.07	1.06	1.08	1.10	1.42		3.15	3.19	3.26
	velocity	5.18	8.39	8.87	8.87	8.62	8.54	8.62	8.46	8.46	8.31	9.87	8.05		8.70	8.60	8.41
Yokochi, Taiga (JPN) (2000)	time	2.58	3.64	4.69	5.73	6.78	7.83	8.88	9.98	11.09	12.20		13.67	6 / 7			
	reaction time	0.139	interval	7 steps	1.06	1.05	1.04	1.05	1.05	1.10	1.11	1.11	1.47		3.15	3.15	3.32
	velocity	5.32	8.62	8.70	8.79	8.70	8.70	8.70	8.31	8.23	8.23	9.54	8.05		8.70	8.70	8.26
Tokuoka, Ryo (JPN) (1999)	time	2.63	3.69	4.74	5.82	6.87	7.92	9.05	10.15	11.25	12.35		13.77	3 / 8			
	reaction time	0.166	interval		1.06	1.05	1.08	1.05	1.05	1.13	1.10	1.10	1.42	PB	3.19	3.23	3.30
	velocity	5.22	8.62	8.70	8.46	8.70	8.70	8.09	8.31	8.31	8.31	9.87	7.99		8.60	8.49	8.31
Omuro, Hideki (JPN) (1990)	time	2.59	3.66	4.74	5.81	6.90	7.99	9.09	10.21	11.33	12.73		14.46	1 / 9			
	reaction time	0.153	interval		1.07	1.08	1.07	1.09	1.09	1.10	1.12	1.12	1.40	1.73	3.22	3.28	3.64
	velocity	5.30	8.54	8.46	8.54	8.39	8.39	8.31	8.16	8.16	6.53	8.10	7.61		8.52	8.36	7.53

2021 Mikio Oda Memorial (Hiroshima, JPN)

FINAL

date 29-Apr-21 wind 1.7 m/s Shibayama (2021) - race analysis of men's 110m hurdles in the 2021 season

		date	25-Apr-21	wind	1.7 m/s	Shibayama (2021) - Race analysis of men's 110m hurdles in the 2021 season													
				H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
Kanai, Taio (JPN) (1995)	time		2.51	3.54	4.55	5.56	6.58	7.60	8.62	9.67	10.70	11.75			13.16	4 / 1			
	reaction time	interval	7 steps	1.03	1.01	1.01	1.02	1.02	1.02	1.05	1.03	1.05	1.41	NR PB			3.05	3.06	3.13
		velocity		5.47	8.87	9.05	9.05	8.96	8.96	8.96	8.70	8.87	8.70	9.94	8.36			8.99	8.96
Izumiya, Shunsuke (JPN) (2000)	time		2.57	3.65	4.66	5.69	6.70	7.74	8.79	9.83	10.88	11.93			13.33	3 / 2			
	reaction time	interval	7 steps	1.08	1.01	1.03	1.01	1.04	1.05	1.04	1.05	1.05	1.40	PB			3.12	3.10	3.14
		velocity		5.34	8.46	9.05	8.87	9.05	8.79	8.70	8.79	8.70	8.70	10.01	8.25			8.79	8.85
Ishikawa, Shuhei (JPN) (1995)	time		2.58	3.65	4.66	5.69	6.73	7.77	8.80	9.86	10.91	11.98			13.44	5 / 3			
	reaction time	interval	7 steps	1.07	1.01	1.03	1.04	1.04	1.03	1.06	1.05	1.07	1.46				3.11	3.11	3.18
		velocity		5.32	8.54	9.05	8.87	8.79	8.79	8.87	8.62	8.70	8.54	9.60	8.18			8.82	8.82
Muratake, Rashiddo (JPN) (2000)	time		2.63	3.68	4.72	5.75	6.79	7.84	8.91	9.98	11.04	12.11			13.53	6 / 4			
	reaction time	interval	7 steps	1.05	1.04	1.03	1.04	1.05	1.07	1.07	1.06	1.07	1.42	PB			3.12	3.16	3.20
		velocity		5.22	8.70	8.79	8.87	8.79	8.70	8.54	8.54	8.62	8.54	9.87	8.13			8.79	8.68
Tokuoka, Ryo (JPN) (1999)	time		2.64	3.70	4.75	5.80	6.85	7.88	8.95	10.00	11.09	12.20			13.67	8 / 5			
	reaction time	interval		1.06	1.05	1.05	1.05	1.03	1.07	1.05	1.09	1.11	1.47				3.16	3.15	3.25
		velocity		5.20	8.62	8.70	8.70	8.70	8.87	8.54	8.70	8.39	8.23	9.54	8.05			8.68	8.70
Yokochi, Taiga (JPN) (2000)	time		2.65	3.75	4.81	5.85	6.89	7.93	8.99	10.06	11.14	12.24			13.68	1 / 6			
	reaction time	interval	7 steps	1.10	1.06	1.04	1.04	1.04	1.06	1.07	1.08	1.10	1.44				3.20	3.14	3.25
		velocity		5.18	8.31	8.62	8.79	8.79	8.79	8.62	8.54	8.46	8.31	9.74	8.04			8.57	8.73
Omuro, Hideki (JPN) (1990)	time		2.60	3.66	4.72	5.76	6.82	7.88	8.96	10.03	11.14	12.25			13.75	7 / 7			
	reaction time	interval		1.06	1.06	1.04	1.06	1.06	1.08	1.07	1.11	1.11	1.50				3.16	3.20	3.29
		velocity		5.28	8.62	8.62	8.79	8.62	8.62	8.46	8.54	8.23	8.23	9.35	8.00			8.68	8.57
Fujii, Ryota (JPN) (1996)	time		2.64	3.73	4.78	5.85	6.91	7.98	9.05	10.13	11.22	12.34			13.84	2 / 8			
	reaction time	interval		1.09	1.05	1.07	1.06	1.07	1.07	1.08	1.09	1.12	1.50				3.21	3.20	3.29
		velocity		5.20	8.39	8.70	8.54	8.62	8.54	8.54	8.46	8.39	8.16	9.35	7.95			8.54	8.57

2021 East China District Meeting (Zhaoqing, CHN)

FINAL

date 24-Apr-21 wind -0.3 m/s CAA Hurdle Development (2021)

			H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
Yang Quanlei (CHN) (1999)	time		2.64	3.74	4.80	5.88	6.95	8.02	9.11	10.20	11.31	12.43		13.89	6 / 1			
	reaction time	0.185	interval	1.10	1.06	1.08	1.07	1.07	1.09	1.09	1.11	1.12	1.46	PB		3.24	3.23	3.32
	velocity		5.20	8.31	8.62	8.46	8.54	8.54	8.39	8.39	8.23	8.16	9.60	7.92		8.46	8.49	8.26
Zhu Shenglong (CHN) (2000)	time		2.64	3.70	4.76	5.82	6.88	7.96	9.03	10.12	11.24	12.38		13.91	3 / 2			
	reaction time	0.165	interval	1.06	1.06	1.06	1.06	1.08	1.07	1.09	1.12	1.14	1.53			3.18	3.21	3.35
	velocity		5.20	8.62	8.62	8.62	8.62	8.46	8.54	8.39	8.16	8.02	9.16	7.91		8.62	8.54	8.19
Zhang Jinzhao (CHN) (1999)	time		2.74	3.84	4.94	5.99	7.06	8.14	9.22	10.29	11.40	12.52		14.00	5 / 3			
	reaction time	0.208	interval	1.10	1.10	1.05	1.07	1.08	1.08	1.07	1.11	1.12	1.48	PB		3.25	3.23	3.30
	velocity		5.01	8.31	8.31	8.70	8.54	8.46	8.46	8.54	8.23	8.16	9.47	7.86		8.44	8.49	8.31
Wu Zhenhua (CHN) (1998)	time		2.70	3.79	4.90	5.98	7.06	8.16	9.26	10.35	11.46	12.59		14.04	9 / 4			
	reaction time	0.174	interval	1.09	1.11	1.08	1.08	1.10	1.10	1.09	1.11	1.13	1.45	PB		3.28	3.28	3.33
	velocity		5.08	8.39	8.23	8.46	8.46	8.31	8.31	8.39	8.23	8.09	9.67	7.83		8.36	8.36	8.23

Pan Zijie (CHN) (1995)	time	2.68	3.79	4.91	5.99	7.08	8.19	9.28	10.39	11.54	12.68		14.19	2 / 5			
	reaction time	0.186	interval	1.11	1.12	1.08	1.09	1.11	1.09	1.11	1.15	1.14	1.51		3.31	3.29	3.40
	velocity	5.12	8.23	8.16	8.46	8.39	8.23	8.39	8.23	7.95	8.02	9.28	7.75		8.28	8.33	8.06
Chen Jie (CHN) (2002)	time	2.68	3.76	4.89	5.97	7.10	8.19	9.38	10.56	11.76	12.96		14.57	8 / 6			
	reaction time	0.171	interval	1.08	1.13	1.08	1.13	1.09	1.19	1.18	1.20	1.20	1.61		3.29	3.41	3.58
	velocity	5.12	8.46	8.09	8.46	8.09	8.39	7.68	7.75	7.62	7.62	8.71	7.55		8.33	8.04	7.66

Heat 2

date 24-Apr-21

wind -1.2 m/s

CAA Hurdle Development (2021)

		H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
Zeng Jianhang (CHN) (1998)	time	2.56	3.62	4.66	5.69	6.72	7.77	8.84	9.91	11.00	12.10		13.62	5 / 1			
	reaction time	0.151	interval	1.06	1.04	1.03	1.05	1.07	1.07	1.09	1.10	1.52			3.13	3.15	3.26
	velocity	5.36	8.62	8.79	8.87	8.87	8.70	8.54	8.54	8.39	8.31	9.22	8.08		8.76	8.70	8.41
Li Huaihai (CHN) (1999)	time	2.66	3.76	4.86	5.94	7.03	8.13	9.25	10.36	11.51	12.66		14.20	4 / 2			
	reaction time	0.162	interval	7 steps	1.10	1.10	1.08	1.09	1.10	1.12	1.11	1.15	1.54		3.28	3.31	3.41
	velocity	5.16	8.31	8.31	8.46	8.39	8.31	8.16	8.23	7.95	7.95	9.10	7.75		8.36	8.28	8.04
Zhu Shenglong (CHN) (2000)	time	2.64	3.73	4.78	5.87	6.97	8.05	9.23	10.38	11.57	12.74		14.29	3 / 3			
	reaction time	0.164	interval	1.09	1.05	1.09	1.10	1.08	1.18	1.15	1.19	1.17	1.55		3.23	3.36	3.51
	velocity	5.20	8.39	8.70	8.39	8.31	8.46	7.75	7.95	7.68	7.81	9.05	7.70		8.49	8.16	7.81
Pan Zijie (CHN) (1995)	time	2.68	3.80	4.90	5.98	7.10	8.25	9.36	10.50	11.63	12.78		14.31	9 / 4			
	reaction time	0.178	interval	1.12	1.10	1.08	1.12	1.15	1.11	1.14	1.13	1.15	1.53		3.30	3.38	3.42
	velocity	5.12	8.16	8.31	8.46	8.16	7.95	8.23	8.02	8.09	7.95	9.16	7.69		8.31	8.11	8.02
Zhao Jiefu (CHN) (2001)	time	2.80	3.96	5.08	6.23	7.35	8.46	9.60	10.74	11.88	13.08		14.69	2 / 5			
	reaction time	0.213	interval	1.16	1.12	1.15	1.12	1.11	1.14	1.14	1.20	1.61			3.43	3.37	3.48
	velocity	4.90	7.88	8.16	7.95	8.16	8.23	8.02	8.02	8.02	7.62	8.71	7.49		7.99	8.14	7.88
Yang Jiajun (CHN) (2002)	time	2.76	3.93	5.04	6.17	7.30	8.46	9.61	10.78	11.96	13.15		14.72	6 / 6			
	reaction time	0.171	interval	1.17	1.11	1.13	1.13	1.16	1.15	1.17	1.18	1.19	1.57		3.41	3.44	3.54
	velocity	4.97	7.81	8.23	8.09	8.09	7.88	7.95	7.81	7.75	7.68	8.93	7.47		8.04	7.97	7.75
Zhang Tao (CHN) (1997)	time	2.72	3.91	5.04	6.21	7.38	8.54	9.73	10.93	12.16	13.37		15.13	7 / 7			
	reaction time	0.186	interval	1.19	1.13	1.17	1.17	1.16	1.19	1.20	1.23	1.21	1.76		3.49	3.52	3.64
	velocity	5.04	7.68	8.09	7.81	7.81	7.88	7.68	7.62	7.43	7.55	7.97	7.27		7.86	7.79	7.53
Wang Yanbo (CHN) (2002)	time	2.77	3.97	5.13	6.33	7.57	8.84	10.08	11.38	12.66	13.94		15.70	8 / 8			
	reaction time	0.164	interval	1.20	1.16	1.20	1.24	1.27	1.24	1.30	1.28	1.28	1.76		3.56	3.75	3.86
	velocity	4.95	7.62	7.88	7.62	7.37	7.20	7.37	7.03	7.14	7.14	7.97	7.01		7.70	7.31	7.10

Heat 1

date 24-Apr-21

wind -1.1 m/s

CAA Hurdle Development (2021)

		H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
Yang Quanlei (CHN) (1999)	time	2.66	3.76	4.85	5.94	7.02	8.10	9.20	10.28	11.39	12.51		14.00	4 / 1			
	reaction time	0.164	interval	1.10	1.09	1.09	1.08	1.08	1.10	1.08	1.11	1.12	1.49		3.28	3.26	3.31
	velocity	5.16	8.31	8.39	8.39	8.46	8.46	8.31	8.46	8.23	8.16	9.41	7.86		8.36	8.41	8.28
Zhang Jinzhao (CHN) (1999)	time	2.74	3.84	4.91	5.99	7.07	8.14	9.24	10.34	11.45	12.59		14.09	3 / 2			
	reaction time	0.192	interval	1.10	1.07	1.08	1.08	1.07	1.10	1.10	1.11	1.14	1.50		3.25	3.25	3.35
	velocity	5.01	8.31	8.54	8.46	8.46	8.54	8.31	8.31	8.23	8.02	9.35	7.81		8.44	8.44	8.19
Wu Zhenhua (CHN) (1998)	time	2.68	3.80	4.88	5.98	7.08	8.20	9.32	10.44	11.59	12.72		14.21	9 / 3			
	reaction time	0.156	interval	1.12	1.08	1.10	1.10	1.12	1.12	1.15	1.13	1.49			3.30	3.34	3.40
	velocity	5.12	8.16	8.46	8.31	8.31	8.16	8.16	8.16	7.95	8.09	9.41	7.74		8.31	8.21	8.06
Chen Jie (CHN) (2002)	time	2.73	3.85	4.94	6.04	7.16	8.26	9.38	10.50	11.60	12.75		14.22	7 / 4			
	reaction time	0.168	interval	1.12	1.09	1.10	1.12	1.10	1.12	1.10	1.15	1.47			3.31	3.34	3.37
	velocity	5.03	8.16	8.39	8.31	8.16	8.31	8.16	8.16	8.31	7.95	9.54	7.74		8.28	8.21	8.14
Xu Zhuoyi (CHN) (2003)	time	2.74	3.88	5.02	6.15	7.25	8.38	9.50	10.62	11.75	12.91		14.42	8 / 5			
	reaction time	0.179	interval	7 steps	1.14	1.14	1.13	1.10	1.13	1.12	1.13	1.16	1.51		3.41	3.35	3.41
	velocity	5.01	8.02	8.02	8.09	8.31	8.09	8.16	8.16	8.09	7.88	9.28	7.63		8.04	8.19	8.04
Jian Jinlong (CHN) (2000)	time	2.75	3.93	5.06	6.20	7.32	8.46	9.59	10.74	11.90	13.10		14.66	2 / 6			
	reaction time	0.166	interval	1.18	1.13	1.14	1.12	1.14	1.13	1.15	1.16	1.20	1.56		3.45	3.39	3.51
	velocity	4.99	7.75	8.09	8.02	8.16	8.02	8.09	7.95	7.88	7.62	8.99	7.50		7.95	8.09	7.81
Liu Bowen (CHN) (2001)	time	2.66	3.77	4.92	6.02	7.14	8.29	9.49	10.70	11.90	13.14		14.78	5 / 7			
	reaction time	0.164	interval	1.11	1.15	1.10	1.12	1.15	1.20	1.21	1.20	1.24	1.64		3.36	3.47	3.65
	velocity	5.16	8.23	7.95	8.31	8.16	7.95	7.62	7.55	7.62	7.37	8.55	7.44		8.16	7.90	7.51

2021 Tom Jones Invitational (Gainesville, FL) (TV Analysis)

FINAL

date 17-Apr-21

wind 1.3 m/s

Henson (2021) - Athlete First: 2021 year end hurdle report

		H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
Holloway, Grant (USA) (1997)	time	2.50	3.52	4.49	5.47	6.45	7.44	8.42	9.47	10.53	11.59		13.07	6 / 1			

reaction time	interval	7 steps	1.02	0.97	0.98	0.98	0.99	0.98	1.05	1.06	1.06	1.48		2.97	2.95	3.17	
	velocity		5.49	8.96	9.42	9.33	9.33	9.23	9.33	8.70	8.62	8.62	9.47	8.42	9.23	9.29	8.65

2021 Yangtze River Delta Athletics Meeting (Shaoxing, CHN)**FINAL**

date 08-Apr-21 wind -0.1 m/s

CAA Hurdle Development (2021)

		H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
Zeng Jianhang (CHN) (1998)	time	2.56	3.64	4.66	5.68	6.68	7.72	8.76	9.80	10.88	11.96		13.43	6 / 1			
reaction time	interval		1.08	1.02	1.02	1.00	1.04	1.04	1.04	1.08	1.08	1.47	PB		3.12	3.08	3.20
	velocity	5.36	8.46	8.96	8.96	9.14	8.79	8.79	8.79	8.46	8.46	9.54	8.19		8.79	8.90	8.57
Yang Quanlei (CHN) (1999)	time	2.64	3.76	4.84	5.92	7.00	8.08	9.18	10.26	11.36	12.44		13.90	4 / 2			
reaction time	interval		1.12	1.08	1.08	1.08	1.08	1.10	1.08	1.10	1.08	1.46	PB		3.28	3.26	3.26
	velocity	5.20	8.16	8.46	8.46	8.46	8.46	8.31	8.46	8.31	8.46	9.60	7.91		8.36	8.41	8.41
Zhang Jinzhao (CHN) (1999)	time	2.76	3.86	4.96	6.04	7.12	8.22	9.32	10.42	11.52	12.64		14.12	9 / 3			
reaction time	interval		1.10	1.10	1.08	1.08	1.10	1.10	1.10	1.10	1.12	1.48	PB		3.28	3.28	3.32
	velocity	4.97	8.31	8.31	8.46	8.46	8.31	8.31	8.31	8.31	8.16	9.47	7.79		8.36	8.36	8.26
Xu Zhuoyi (CHN) (2003)	time	2.72	3.84	4.96	6.04	7.12	8.24	9.32	10.44	11.56	12.68		14.17	5 / 4			
reaction time	interval		7 steps	1.12	1.12	1.08	1.12	1.08	1.12	1.12	1.12	1.49	PB		3.32	3.28	3.36
	velocity	5.04	8.16	8.16	8.46	8.46	8.16	8.46	8.16	8.16	8.16	9.41	7.76		8.26	8.36	8.16
Lu Yang (CHN) (1996)	time	2.76	3.88	4.96	6.08	7.16	8.24	9.34	10.48	11.60	12.76		14.30	7 / 5			
reaction time	interval		1.12	1.08	1.12	1.08	1.08	1.10	1.14	1.12	1.16	1.54			3.32	3.26	3.42
	velocity	4.97	8.16	8.46	8.16	8.46	8.46	8.31	8.02	8.16	7.88	9.10	7.69		8.26	8.41	8.02
Pan Zijie (CHN) (1995)	time	2.72	3.84	4.96	6.04	7.16	8.28	9.40	10.52	11.66	12.78		14.33	3 / 6			
reaction time	interval		1.12	1.12	1.08	1.12	1.12	1.12	1.12	1.14	1.12	1.55			3.32	3.36	3.38
	velocity	5.04	8.16	8.16	8.46	8.16	8.16	8.16	8.16	8.02	8.16	9.05	7.68		8.26	8.16	8.11
Liu Junqian (CHN) (2003)	time	2.68	3.80	4.92	6.04	7.16	8.28	9.40	10.52	11.72	12.90		14.51	8 / 7			
reaction time	interval		1.12	1.12	1.12	1.12	1.12	1.12	1.12	1.20	1.18	1.61			3.36	3.36	3.50
	velocity	5.12	8.16	8.16	8.16	8.16	8.16	8.16	8.16	7.62	7.75	8.71	7.58		8.16	8.16	7.83
Huan Jiaxing (CHN) (2001)	time	2.80	3.92	5.08	6.22	7.34	8.48	9.66	10.84	12.00	13.20		14.74	2 / 8			
reaction time	interval		1.12	1.16	1.14	1.12	1.14	1.18	1.18	1.16	1.20	1.54			3.42	3.44	3.54
	velocity	4.90	8.16	7.88	8.02	8.16	8.02	7.75	7.75	7.88	7.62	9.10	7.46		8.02	7.97	7.75

2021 National Grand Prix (Chengdu, CHN)**FINAL**

date 02-Apr-21 wind -0.1 m/s

CAA Hurdle Development (2021)

		H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
Zhang Honglin (CHN) (1994)	time	2.66	3.76	4.80	5.87	6.91	7.94	9.02	10.08	11.20	12.30		13.76	4 / 1			
reaction time	0.179 interval		1.10	1.04	1.07	1.04	1.03	1.08	1.06	1.12	1.10	1.46			3.21	3.15	3.28
	velocity	5.16	8.31	8.79	8.54	8.79	8.87	8.46	8.62	8.16	8.31	9.60	7.99		8.54	8.70	8.36
Sun Zhenjiang (CHN) (1999)	time	2.62	3.72	4.78	5.82	6.88	7.93	9.01	10.10	11.24	12.36		13.98	5 / 2			
reaction time	0.168 interval		7 steps	1.10	1.06	1.04	1.06	1.05	1.08	1.09	1.14	1.12	1.62		3.20	3.19	3.35
	velocity	5.24	8.31	8.62	8.79	8.62	8.70	8.46	8.39	8.02	8.16	8.65	7.87		8.57	8.60	8.19
Lai Junjie (CHN) (2003)	time	2.78	3.95	5.06	6.18	7.27	8.38	9.48	10.60	11.75	12.90		14.43	3 / 3			
reaction time	0.176 interval		1.17	1.11	1.12	1.09	1.11	1.10	1.12	1.15	1.15	1.53			3.40	3.30	3.42
	velocity	4.94	7.81	8.23	8.16	8.39	8.23	8.31	8.16	7.95	7.95	9.16	7.62		8.06	8.31	8.02
Yang Haoran (CHN) (2001)	time	2.69	3.82	4.97	6.10	7.25	8.38	9.52	10.66	11.82	12.96		14.44	6 / 4			
reaction time	0.145 interval		1.13	1.15	1.13	1.15	1.13	1.14	1.14	1.16	1.14	1.48	PB		3.41	3.42	3.44
	velocity	5.10	8.09	7.95	8.09	7.95	8.09	8.02	8.02	7.88	8.02	9.47	7.62		8.04	8.02	7.97
Yan Chenyu (CHN) (2003)	time	2.74	3.88	5.00	6.14	7.27	8.40	9.54	10.68	11.81	12.94		14.50	8 / 5			
reaction time	0.192 interval		1.14	1.12	1.14	1.13	1.13	1.14	1.14	1.13	1.13	1.56			3.40	3.40	3.40
	velocity	5.01	8.02	8.16	8.02	8.09	8.09	8.02	8.02	8.09	8.09	8.99	7.59		8.06	8.06	8.06
Wang Jiawei (CHN) (2004)	time	2.90	4.12	5.34	6.52	7.72	8.90	10.10	11.32	12.52	13.78		15.43	2 / 6			
reaction time	0.210 interval		1.22	1.22	1.18	1.20	1.18	1.20	1.22	1.20	1.26	1.65			3.62	3.58	3.68
	velocity	4.73	7.49	7.49	7.75	7.62	7.75	7.62	7.49	7.62	7.25	8.50	7.13		7.57	7.66	7.45
Tan Junyu (CHN) (2004)	time	2.76	3.92	5.14	6.36	7.56	8.74	10.06	11.36	12.68	13.98		15.64	7 / 7			
reaction time	0.194 interval		1.16	1.22	1.22	1.20	1.18	1.32	1.30	1.32	1.30	1.66			3.60	3.70	3.92
	velocity	4.97	7.88	7.49	7.49	7.62	7.75	6.92	7.03	6.92	7.03	8.45	7.03		7.62	7.41	6.99

Heat 2

date 02-Apr-21 wind 0.3 m/s

CAA Hurdle Development (2021)

		H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
Zhang Honglin (CHN) (1994)	time	2.62	3.72	4.78	5.88	6.94	8.02	9.10	10.18	11.28	12.38		13.90	6 / 1			
reaction time	0.172 interval		1.10	1.06	1.10	1.06	1.08	1.08	1.08	1.10	1.10	1.52			3.26	3.22	3.28
	velocity	5.24	8.31	8.62	8.31	8.62	8.46	8.46	8.46	8.31	8.31	9.22	7.91		8.41	8.52	8.36
Lai Junjie (CHN) (2003)	time	2.72	3.88	5.02	6.10	7.20	8.30	9.42	10.56	11.68	12.84		14.38	3 / 2			

	reaction time	0.189	interval		1.16	1.14	1.08	1.10	1.10	1.12	1.14	1.12	1.16	1.54	PB		3.38	3.32	3.42
			velocity	5.04	7.88	8.02	8.46	8.31	8.31	8.16	8.02	8.16	7.88	9.10	7.94		8.11	8.26	8.02
Yang Haoran (CHN) (2001)	time			2.70	3.82	4.94	6.10	7.22	8.32	9.48	10.60	11.78	12.96		14.54	5 / 3			
	reaction time	0.162	interval		1.12	1.12	1.16	1.12	1.10	1.16	1.12	1.18	1.18	1.58	PB		3.40	3.38	3.48
			velocity	5.08	8.16	8.16	7.88	8.16	8.31	7.88	8.16	7.75	7.75	8.87	7.81		8.06	8.11	7.88
Tan Junyu (CHN) (2004)	time			2.80	3.96	5.16	6.32	7.52	8.72	9.92	11.10	12.32	13.54		15.14	4 / 4			
	reaction time	0.218	interval		1.16	1.20	1.16	1.20	1.20	1.20	1.18	1.22	1.22	1.60			3.52	3.60	3.62
			velocity	4.90	7.88	7.62	7.88	7.62	7.62	7.62	7.75	7.49	7.49	8.76	7.76		7.79	7.62	7.57

Heat 1

eat 1

date	02-Apr-21	wind	0.5 m/s	CAA Hurdle Development (2021)													
		H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
Sun Zhenjiang (CHN) (1999)	time	2.66	3.74	4.80	5.88	6.92	7.96	9.04	10.14	11.24	12.36		13.88	4 / 1			
	reaction time	0.189	interval	7 steps	1.08	1.06	1.08	1.04	1.08	1.10	1.10	1.12	1.52		3.22	3.16	3.32
	velocity		5.16	8.46	8.62	8.46	8.79	8.79	8.46	8.31	8.31	8.16	9.22	7.93	8.52	8.68	8.26
Yan Chenyu (CHN) (2003)	time	2.74	3.94	5.10	6.26	7.40	8.56	9.70	10.84	12.02	13.20		14.79	5 / 2			
	reaction time	0.216	interval	1.20	1.16	1.16	1.14	1.16	1.14	1.14	1.18	1.18	1.59		3.52	3.44	3.50
	velocity		5.01	7.62	7.88	7.88	8.02	7.88	8.02	8.02	7.75	7.75	8.82	7.44	7.79	7.97	7.83
Wang Jiawei (CHN) (2004)	time	2.88	4.12	5.34	6.56	7.78	9.02	10.30	11.56	12.80	14.06		15.75	3 / 3			
	reaction time	0.195	interval	1.24	1.22	1.22	1.24	1.28	1.26	1.24	1.26	1.69		3.68	3.74	3.76	
	velocity		4.76	7.37	7.49	7.49	7.49	7.37	7.14	7.25	7.37	7.25	8.30	6.98	7.45	7.33	7.29

2021 National Indoor Grand Prix (Jinan, CHN)

FINAL

date	18-Mar-21	CAA Hurdle Development (2021)															
		H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
Zeng Jianhang (CHN) (1998)	time	2.54	3.59	4.63	5.65	6.67							7.57	4 / 1			
reaction time	0.161	interval	1.05	1.04	1.02	1.02						0.90			3.11		
velocity		5.40	8.70	8.79	8.96	8.96						10.80	7.93		8.82		
Zhang Honglin (CHN) (1994)	time	2.66	3.74	4.81	5.86	6.91							7.81	3 / 2			
reaction time	0.204	interval	7 steps	1.08	1.07	1.05	1.05					0.90			3.20		
velocity		5.16	8.46	8.54	8.70	8.70						10.80	7.68		8.57		
Yang Quanlei (CHN) (1999)	time	2.63	3.74	4.82	5.90	6.96							7.86	5 / 3			
reaction time	0.173	interval	1.11	1.08	1.08	1.06						0.90	PB		3.27		
velocity		5.22	8.23	8.46	8.46	8.62						10.80	7.63		8.39		
Ning Xiaohan (CHN) (2000)	time	2.65	3.75	4.82	5.90	7.01							7.97	6 / 4			
reaction time	0.166	interval	7 steps	1.10	1.07	1.08	1.11					0.96			3.25		
velocity		5.18	8.31	8.54	8.46	8.23						10.13	7.53		8.44		
Sun Haoru (CHN) (1998)	time	2.77	3.89	4.98	6.09	7.17							8.15	1 / 5			
reaction time	0.188	interval	1.12	1.09	1.11	1.08						0.98			3.32		
velocity		4.95	8.16	8.39	8.23	8.46						9.92	7.36		8.26		
Xu Zhuoyi (CHN) (2003)	time	2.76	3.90	5.03	6.15	7.29							8.30	2 / 6			
reaction time	0.171	interval	7 steps	1.14	1.13	1.12	1.14					1.01			3.39		
velocity		4.97	8.02	8.09	8.16	8.02						9.62	7.23		8.09		

2021 National Indoor Grand Prix (Chengdu, CHN)

A FINAL

date	13-Mar-21	CAA Hurdle Development (2021)															
		H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
Zeng Jianhang (CHN) (1998)	time	2.54	3.58	4.62	5.62	6.64							7.55	3 / 1			
reaction time	0.163	interval	1.04	1.04	1.00	1.02						0.91	PB		3.08		
velocity		5.40	8.79	8.79	9.14	8.96						10.68	7.95		8.90		
Sun Zhenjiang (CHN) (1999)	time	2.62	3.70	4.78	5.83	6.88							7.81	5 / 2			
reaction time	0.187	interval	7 steps	1.08	1.08	1.05	1.05					0.93			3.21		
velocity		5.24	8.46	8.46	8.70	8.70						10.45	7.68		8.54		
Li Guancheng (CHN) (2002)	time	2.70	3.92	5.08	6.22	7.35							8.34	2 / 3			
reaction time	0.201	interval	1.22	1.16	1.14	1.13						0.99	PB		3.52		
velocity		5.08	7.49	7.88	8.02	8.09						9.82	7.19		7.79		

B FINAL

date	13-Mar-21	CAA Hurdle Development (2021)															
		H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
Yang Quanlei (CHN) (1999)	time	2.66	3.77	4.89	5.98	7.06							7.97	2 / 1			
reaction time	0.186	interval	1.11	1.12	1.09	1.08						0.91	PB		3.32		
velocity		5.16	8.23	8.16	8.39	8.46						10.68	7.53		8.26		
Xu Zhuoyi (CHN) (2003)	time	2.66	3.77	4.88	5.98	7.06							7.97	3 / 2			
reaction time	0.173	interval	7 steps	1.11	1.11	1.10	1.08					0.91	PB		3.32		
velocity		5.16	8.23	8.23	8.31	8.46						10.68	7.53		8.26		

Yang Haoran (CHN) (2001)	time	2.68	3.80	4.94	6.04	7.14		8.05	6 / 3
	reaction time	0.154	interval	1.12	1.14	1.10	0.91	PB	3.36
	velocity	5.12	8.16	8.02	8.31	8.31	10.68	7.45	8.16
Lai Junjie (CHN) (2003)	time	2.80	3.98	5.14	6.28	7.44		8.48	5 / 4
	reaction time	0.197	interval	1.18	1.16	1.14	1.04		3.48
	velocity	4.90	7.75	7.88	8.02	7.88	9.35	7.08	7.88
Luo Xinyu (CHN) (2002)	time	2.85	4.14	5.38	6.66	7.92		9.00	4 / 5
	reaction time	0.199	interval	1.29	1.24	1.28	1.08	PB	3.81
	velocity	4.81	7.09	7.37	7.14	7.25	9.00	6.67	7.20

2021 European Indoor Championships (Toruń, POL) (TV Analysis)

FINAL

date 07-Mar-21

Henson (2021) - Athlete First: 2021 year end hurdle report

		H1	H2	H3	H4	H5	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
Belocian, Wilhem (FRA) (1995)	time	2.40	3.46	4.44	5.44	6.42		7.42	5 / 1			
	reaction time	0.129	interval	1.06	0.98	1.00	1.00	PB		3.04		
	velocity	5.72	8.62	9.33	9.14	9.33	9.72	8.09		9.02		
Pozzi, Andy (GBR) (1992)	time	2.42	3.44	4.44	5.42	6.44		7.43	6 / 2			
	reaction time	0.151	interval	7 steps	1.02	1.00	0.99	=PB		3.00		
	velocity	5.67	8.96	9.14	9.33	8.96	9.82	8.08		9.14		
Dal Molin, Paolo (ITA) (1987)	time	2.40	3.46	4.50	5.54	6.58		7.56	7 / 3			
	reaction time	0.133	interval	7 steps	1.06	1.04	0.98			3.14		
	velocity	5.72	8.62	8.79	8.79	8.79	9.92	7.94		8.73		
Martinez, Asier (ESP) (2000)	time	2.50	3.54	4.56	5.58	6.62		7.60	8 / 4			
	reaction time	0.150	interval	7 steps	1.04	1.02	0.98	PB		3.08		
	velocity	5.49	8.79	8.96	8.96	8.79	9.92	7.89		8.90		
Manga, Aurel (FRA) (1992)	time	2.42	3.48	4.50	5.52	6.58		7.63	3 / 5			
	reaction time	0.152	interval	7 steps	1.06	1.02	1.05			3.10		
	velocity	5.67	8.62	8.96	8.96	8.62	9.26	7.86		8.85		
Czykier, Damian (POL) (1992)	time	2.46	3.48	4.48	5.46	6.60		7.63	4 / 6			
	reaction time	0.148	interval	7 steps	1.02	1.00	1.03			3.00		
	velocity	5.58	8.96	9.14	9.33	8.02	9.44	7.86		9.14		
Smet, Koen (BEL) (1992)	time	2.54	3.60	4.64	5.68	6.74		7.73	1 / 7			
	reaction time	0.144	interval	1.06	1.04	1.06	0.99			3.14		
	velocity	5.40	8.62	8.79	8.79	8.62	9.82	7.76		8.73		
Koua, Franck Brice (ITA) (2001)	time	2.52	3.60	4.66	5.70	6.76		7.76	2 / 8			
	reaction time	0.165	interval	1.08	1.06	1.04	1.00			3.18		
	velocity	5.44	8.46	8.62	8.79	8.62	9.72	7.73		8.62		

2021 Villa de Madrid (Madrid, ESP)

FINAL

date 24-Feb-21

Vazel (2021) - <https://twitter.com/pjvazel/status/1364677209525485569>

Henson (2021) - men's 60m hurdles world record season - Grant Holloway (TV Analysis)

		H1	H2	H3	H4	H5	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
Holloway, Grant (USA) (1997)	time	2.48	3.47	4.46	5.45	6.44		7.29	6 / 1			Vazel (2021)
	reaction time		interval	7 steps	0.99	0.99	0.85	WR		2.97		
	velocity	5.53	9.23	9.23	9.23	9.23	11.44	8.23		9.23		
Pozzi, Andy (GBR) (1992)	time	2.40	3.44	4.45	5.45	6.47		7.51	3 / 2			
	reaction time		interval	7 steps	1.04	1.01	1.04			3.05		
	velocity	5.72	8.79	9.05	9.14	8.96	9.35	7.99		8.99		
Dal Molin, Paolo (ITA) (1987)	time	2.37	3.43	4.47	5.50	6.54		7.57	5 / 3			
	reaction time		interval	7 steps	1.06	1.04	1.03			3.13		
	velocity	5.79	8.62	8.79	8.87	8.79	9.44	7.93		8.76		
Martinez, Asier (ESP) (2000)	time	2.48	3.54	4.56	5.58	6.59		7.61	4 / 4			
	reaction time		interval	7 steps	1.06	1.02	1.02	=PB		3.10		
	velocity	5.53	8.62	8.96	8.96	9.05	9.53	7.88		8.85		
Obasuyi, Michael (BEL) (1999)	time	2.45	3.50	4.54	5.58	6.61		7.69	8 / 5			
	reaction time		interval	7 steps	1.05	1.04	1.08			3.13		
	velocity	5.60	8.70	8.79	8.79	8.87	9.00	7.80		8.76		
Llopis, Enrique (ESP) (2000)	time	2.45	3.52	4.58	5.62	6.65		7.69	1 / 6			
	reaction time		interval	7 steps	1.07	1.06	1.04	PB		3.17		
	velocity	5.60	8.54	8.62	8.79	8.87	9.35	7.80		8.65		
Smet, Koen (BEL) (1992)	time	2.50	3.59	4.63	5.67	6.71		7.73	2 / 7			

reaction time	interval	1.09	1.04	1.04	1.04		
	velocity	5.49	8.39	8.79	8.79		
Lakka, Elmo (FIN) (1993)	time	2.45	3.50	4.58	5.65	6.69	
	reaction time	interval	7 steps	1.05	1.08	1.07	1.04
		velocity	5.60	8.70	8.46	8.54	8.79

1.02		3.17
9.53	7.76	8.65
	7.77	7 / 8
1.08		3.20
9.00	7.72	8.57

Heat 2 (TV Analysis)

date 24-Feb-21

Henson (2021) - men's 60m hurdles world record season - Grant Holloway

				H1	H2	H3	H4	H5
Pozzi, Andy (GBR) (1992)	time			2.57	3.63	4.63	5.65	6.66
	reaction time	0.132	interval	7 steps	1.06	1.00	1.02	1.01
			velocity	5.34	8.62	9.14	8.96	9.05
Dal Molin, Paolo (ITA) (1987)	time			2.47	3.53	4.63	5.72	6.78
	reaction time	0.134	interval	7 steps	1.06	1.10	1.09	1.06
			velocity	5.55	8.62	8.31	8.39	8.62
Obasuyi, Michael (BEL) (1999)	time			2.60	3.66	4.73	5.75	6.78
	reaction time	0.172	interval	7 steps	1.06	1.07	1.02	1.03
			velocity	5.28	8.62	8.54	8.96	8.87

Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
	7.53	4 / 1			
0.87			3.08		
11.17	7.97		8.90		
	7.71	5 / 2			
0.93			3.25		
10.45	7.78		8.44		
	7.74	3 / 3			
0.96	PB		3.15		
10.13	7.75		8.70		

Heat 1 (TV Analysis)

date 24-Feb-21

Henson (2021) - men's 60m hurdles world record season - Grant Holloway

				H1	H2	H3	H4	H5
Holloway, Grant (USA) (1997)	time			2.36	3.36	4.34	5.34	6.33
	reaction time	0.164	interval	7 steps	1.00	0.98	1.00	0.99
			velocity	5.81	9.14	9.33	9.14	9.23
Martinez, Asier (ESP) (2000)	time			2.52	3.56	4.57	5.58	6.60
	reaction time	0.131	interval	7 steps	1.04	1.01	1.01	1.02
			velocity	5.44	8.79	9.05	9.05	8.96
Lakka, Elmo (FIN) (1993)	time			2.50	3.56	4.60	5.67	6.70
	reaction time	0.142	interval	7 steps	1.06	1.04	1.07	1.03
			velocity	5.49	8.62	8.79	8.54	8.87

Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
	7.32	5 / 1			
0.99	=AR		2.98		
9.82	8.20		9.20		
	7.61	4 / 2			
1.01	PB		3.06		
9.62	7.88		8.96		
	7.75	6 / 3			
1.05			3.17		
9.26	7.74		8.65		

2021 Copernicus Cup (Toruń, POL) (TV Analysis)

FINAL

date 17-Feb-21

Henson (2021) - men's 60m hurdles world record season - Grant Holloway

				H1	H2	H3	H4	H5
Holloway, Grant (USA) (1997)	time			2.44	3.45	4.43	5.42	6.45
	reaction time	0.122	interval	7 steps	1.01	0.98	0.99	1.03
			velocity	5.62	9.05	9.33	9.23	8.87
Belocian, Wilhem (FRA) (1995)	time			2.54	3.56	4.58	5.60	6.61
	reaction time	0.143	interval		1.02	1.02	1.02	1.01
			velocity	5.40	8.96	8.96	8.96	9.05
Mallett, Aaron (USA) (1994)	time			2.54	3.58	4.63	5.67	6.67
	reaction time	0.170	interval		1.04	1.05	1.04	1.00
			velocity	5.40	8.79	8.70	8.79	9.14
Czykier, Damian (POL) (1992)	time			2.54	3.59	4.60	5.60	6.67
	reaction time	0.149	interval	7 steps	1.05	1.01	1.00	1.07
			velocity	5.40	8.70	9.05	9.14	8.54
Al-Yoha, Yaqoub (KUW) (1993)	time			2.59	3.65	4.70	5.72	6.75
	reaction time	0.166	interval		1.06	1.05	1.02	1.03
			velocity	5.30	8.62	8.70	8.96	8.87
Kiljan, Krzysztof (POL) (1999)	time			2.58	3.67	4.71	5.77	6.84
	reaction time	0.155	interval	7 steps	1.09	1.04	1.06	1.07
			velocity	5.32	8.39	8.79	8.62	8.54
Żevrowski, Dawid (POL) (1997)	time			2.63	3.71	4.76	5.80	6.86
	reaction time	0.167	interval		1.08	1.05	1.04	1.06
			velocity	5.22	8.46	8.70	8.79	8.62

Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
	7.38	4 / 1			
0.93			2.98		
10.45	8.13		9.20		
	7.57	3 / 2			
0.96			3.06		
10.13	7.93		8.96		
	7.60	6 / 3			
0.93			3.13		
10.45	7.89		8.76		
	7.63	7 / 4			
0.96			3.06		
10.13	7.86		8.96		
	7.75	5 / 5			
1.00			3.13		
9.72	7.74		8.76		
	7.80	2 / 6			
0.96			3.19		
10.13	7.69		8.60		
	7.84	1 / 7			
0.98			3.17		
9.92	7.65		8.65		

2021 Meeting Elite en Salle de L'Eure (Val-de-Reuil, FRA) (TV Analysis)

FINAL

date 14-Feb-21

Henson (2021) - men's 60m hurdles world record season - Grant Holloway

		H1	H2	H3	H4	H5		
Holloway, Grant (USA) (1997)	time	2.40	3.40	4.40	5.42	6.42		
	reaction time	0.161	interval	7 steps	1.00	1.00	1.02	1.00
			velocity	5.72	9.14	9.14	8.96	9.14
Constantino, Gabriel (BRA) (21)	time	2.50	3.56	4.60	5.62	6.66		
	reaction time	0.166	interval	7 steps	1.06	1.04	1.02	1.04
			velocity	5.49	8.62	8.79	8.96	8.79

Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
	7.41	5 / 1			
0.99			3.02		
9.82	8.10		9.08		
	7.65	4 / 2			
0.99			3.12		
9.82	7.84		8.79		

Mallett, Aaron (USA) (1995)	time	2.50	3.58	4.62	5.64	6.66		7.66	3 / 3	
	reaction time	0.137	interval	1.08	1.04	1.02	1.02	1.00		3.14
	velocity	5.49	8.46	8.79	8.96	8.96		9.72	7.83	8.73
Manga, Aurel (FRA) (1992)	time	2.50	3.58	4.66	5.70	6.74		7.77	6 / 4	
	reaction time	0.160	interval	7 steps	1.08	1.08	1.04	1.04	1.03	3.20
	velocity	5.49	8.46	8.46	8.79	8.79		9.44	7.72	8.57
Kwaou-Mathey, Just (FRA) (1992)	time	2.60	3.68	4.72	5.74	6.82		7.83	1 / 5	
	reaction time	0.168	interval	7 steps	1.08	1.04	1.02	1.08	1.01	3.14
	velocity	5.28	8.46	8.79	8.96	8.46		9.62	7.66	8.73
Laviolette, Jeanice (FRA) (2000)	time	2.60	3.68	4.74	5.84	6.90		8.01	2 / 6	
	reaction time	0.158	interval	1.08	1.06	1.10	1.06	1.11		3.24
	velocity	5.28	8.46	8.62	8.31	8.62		8.76	7.49	8.46

Heat 2

date 14-Feb-21

Henson (2021) - men's 60m hurdles world record season - Grant Holloway

		H1	H2	H3	H4	H5	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
Mallett, Aaron (USA) (1995)	time	2.58	3.64	4.68	5.70	6.72		7.68	3 / 1			
	reaction time	0.166	interval	1.06	1.04	1.02	1.02	0.96		3.12		
	velocity	5.32	8.62	8.79	8.96	8.96	10.13	7.81		8.79		
Manga, Aurel (FRA) (1992)	time	2.58	3.64	4.68	5.72	6.74		7.69	4 / 2			
	reaction time	0.176	interval	7 steps	1.06	1.04	1.04	1.02	0.95	3.14		
	velocity	5.32	8.62	8.79	8.79	8.96	10.23	7.80		8.73		
Kwaou-Mathey, Just (FRA) (1992)	time	2.68	3.76	4.80	5.86	6.90		7.83	2 / 3			
	reaction time	0.158	interval	7 steps	1.08	1.04	1.06	1.04	0.93	3.18		
	velocity	5.12	8.46	8.79	8.62	8.79	10.45	7.66		8.62		

Heat 1

date 14-Feb-21

Henson (2021) - men's 60m hurdles world record season - Grant Holloway

		H1	H2	H3	H4	H5	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
Holloway, Grant (USA) (1997)	time	2.42	3.42	4.42	5.42	6.42		7.47	4 / 1			
	reaction time	0.142	interval	7 steps	1.00	1.00	1.00	1.05		3.00		
	velocity	5.67	9.14	9.14	9.14	9.14	9.26	8.03		9.14		
Constantino, Gabriel (BRA) (2019)	time	2.54	3.58	4.62	5.66	6.68		7.68	5 / 2			
	reaction time	0.185	interval	7 steps	1.04	1.04	1.04	1.02	1.00	3.12		
	velocity	5.40	8.79	8.79	8.79	8.96	9.72	7.81		8.79		
Lecoeur, Romain (FRA) (1997)	time	2.60	3.74	4.82	5.92	6.98		8.01	2 / 3			
	reaction time	0.116	interval	7 steps	1.14	1.08	1.10	1.06	1.03	3.32		
	velocity	5.28	8.02	8.46	8.31	8.62	9.44	7.49		8.26		

2021 Meeting Hauts-de-France Pas-de-Calais (Liévin, FRA)

Vazel (2021) - <https://twitter.com/pjvazel/status/1359286403834724356>

FINAL

date 09-Feb-21

Henson (2021) - men's 60m hurdles world record season - Grant Holloway (TV Analysis)

		H1	H2	H3	H4	H5	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
Holloway, Grant (USA) (1997)	time	2.50	3.49	4.48	5.47	6.45		7.32	5 / 1			Vazel (2021)
	reaction time	0.150	interval	7 steps	0.99	0.99	0.98	0.87	AR	2.97		
	velocity	5.49	9.23	9.23	9.23	9.33	11.17	8.20		9.23		
Belocian, Wilhem (FRA) (1995)	time	2.52	3.54	4.56	5.55	6.56		7.52	4 / 2			
	reaction time	0.160	interval	1.02	1.02	0.99	1.01	0.96		3.03		
	velocity	5.44	8.96	8.96	9.23	9.05	10.13	7.98		9.05		
Dal Molin, Paolo (ITA) (1987)	time	2.43	3.49	4.55	5.57	6.60		7.55	2 / 3			
	reaction time	0.126	interval	7 steps	1.06	1.06	1.02	1.03	0.95	3.14		
	velocity	5.65	8.62	8.62	8.96	8.87	10.23	7.95		8.73		
Mallett, Aaron (USA) (1994)	time	2.54	3.57	4.60	5.60	6.62		7.57	7 / 4			
	reaction time	0.150	interval	1.03	1.03	1.00	1.02	0.95		3.06		
	velocity	5.40	8.87	8.87	9.14	8.96	10.23	7.93		8.96		
Czykier, Damian (POL) (1992)	time	2.55	3.58	4.60	5.60	6.62		7.59	3 / 5			
	reaction time	0.161	interval	7 steps	1.03	1.02	1.00	1.02	0.97	3.05		
	velocity	5.38	8.87	8.96	9.14	8.96	10.02	7.91		8.99		
Eaton, Jarrett (USA) (1989)	time	2.50	3.52	4.55	5.59	6.62		7.60	6 / 6			
	reaction time	0.140	interval	7 steps	1.02	1.03	1.04	1.03	0.98	3.09		
	velocity	5.49	8.96	8.87	8.79	8.87	9.92	7.89		8.87		
Obasuyi, Michael (BEL) (1999)	time	2.57	3.60	4.62	5.62	6.64		7.67	8 / 7			
	reaction time	0.158	interval	7 steps	1.03	1.02	1.00	1.02	1.03	3.05		
	velocity	5.34	8.87	8.96	9.14	8.96	9.44	7.82		8.99		

Mayer, Kevin (FRA) (1992)	time	2.58	3.68	4.76	5.77	6.81	
reaction time	0.179	interval	7 steps	1.10	1.08	1.01	1.04
		velocity	5.32	8.31	8.46	9.05	8.79

	7.79	1 / 8	
0.98			3.19
9.92	7.70		8.60

Heat 2 (TV Analysis) date 09-Feb-21

Henson (2021) - men's 60m hurdles world record season - Grant Holloway

		H1	H2	H3	H4	H5		
Holloway, Grant (USA) (1997)	time	2.38	3.38	4.38	5.36	6.34		
	reaction time	0.194	interval	7 steps	1.00	1.00	0.98	0.98
	velocity	5.76	9.14	9.14	9.33	9.33		

Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
	7.38	4 / 1			
1.04			2.98		
9.35	8.13		9.20		

Czykier, Damian (POL) (1992)	time	2.45	3.50	4.53	5.53	6.53
reaction time	0.155	interval	7 steps	1.05	1.03	1.00
		velocity	5.60	8.70	8.87	9.14

	7.63	6 / 2	
1.10			3.08
8.84	7.86		8.90

Obasuyi, Michael (BEL) (1999)	time	2.45	3.50	4.53	5.56	6.60		
	reaction time	0.152	interval	7 steps	1.05	1.03	1.03	1.04
		velocity	5.60	8.70	8.87	8.87	8.79	

	7.67	7 / 3	
1.07	PB		3.11
9.08	7.82		8.82

Heat 1 (TV Analysis) date 09-Feb-21

Henson (2021) - men's 60m hurdles world record season - Grant Holloway

		H1	H2	H3	H4	H5
Belocian, Wilhem (FRA) (1995)		2.40	3.43	4.46	5.46	6.46
reaction time	0.141 interval		1.03	1.03	1.00	1.00
	velocity	5.72	8.87	8.87	9.14	9.14

Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
	7.50	5 / 1			
1.04			3.06		
9.35	8.00		8.96		

Eaton, Jarrett (USA) (1989)	time	2.37	3.40	4.43	5.46	6.49
reaction time	0.161	interval	7 steps	1.03	1.03	1.03
		velocity	5.79	8.87	8.87	8.87

	7.55	4 / 2	
1.06			3.09
9.17	7.95		8.87

Mallett, Aaron (USA) (1994)		time	2.46	3.51	4.53	5.53	6.56
reaction time	0.152	interval		1.05	1.02	1.00	1.03
		velocity	5.58	8.70	8.96	9.14	8.87

7.61	3 / 3	
1.05		3.07
9.26	7.88	8.93

Dal Molin, Paolo (ITA) (1987)	time	2.40	3.48	4.53	5.55	6.58	
reaction time	0.134	interval	7 steps	1.08	1.05	1.02	1.03
		velocity	5.72	8.46	8.70	8.96	8.87

	7.65	7 / 4	
1.07			3.15
9.08	7.84		8.70

Mayer, Kevin (FRA) (1992)	time	2.46	3.51	4.53	5.56	6.60	
reaction time	0.174	interval	7 steps	1.05	1.02	1.03	1.04
		velocity	5.58	8.70	8.96	8.87	8.79

	7.68	6 / 5	
1.08	PB		3.10
9.00	7.81		8.85

2021 Meeting Elite en Salle de Metz (Metz, FRA)**FINAL**

date 06-Feb-21

Vazel (2021) - <https://twitter.com/pjvazel/status/1358428540241399810>

		H1	H2	H3	H4	H5
Belocian, Wilhem (FRA) (1995)	time	2.53	3.55	4.54	5.61	
reaction time		interval	1.02	0.99	1.07	
velocity		5.42	8.96	9.23	8.54	

Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
dnf		3 / --			
			3.08		
			8.90		

Heat 1

date 06-Feb-21

Vazel (2021) - <https://twitter.com/pjvazel/status/1358428540241399810>

		H1	H2	H3	H4	H5
Belocian, Wilhem (FRA) (1995)	time	2.54	3.57	4.56	5.56	6.56
reaction time	0.148 interval		1.03	0.99	1.00	1.00
	velocity	5.40	8.87	9.23	9.14	9.14

Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
	7.45	/ 1			
0.89	PB		3.02		
10.92	14.77		9.08		

2021 American Track League 1 (Fayetteville, AR) (TV Analysis)**FINAL**

date 24-Jan-21

Henson (2021) - men's 60m hurdles world record season - Grant Holloway

		H1	H2	H3	H4	H5		
Holloway, Grant (USA) (1997)	time	2.42	3.43	4.43	5.42	6.39		
	reaction time	0.174	interval	7 steps	1.01	1.00	0.99	0.97
		velocity	5.67	9.05	9.14	9.23	9.42	

Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
	7.35	4 / 1			
0.96	=AR =PB		3.00		
10.13	8.16		9.14		

Dickson, Michael (USA) (1997)	time	2.50	3.55	4.59	5.62	6.62	
reaction time	0.186	interval	7 steps	1.05	1.04	1.03	1.00
		velocity	5.49	8.70	8.79	8.87	9.14

	7.54	3 / 2	
0.92	PB		3.12
10.57	7.96		8.79

Walters, Ruebin (TTO) (1995)	time	2.55	3.63	4.68	5.70	6.72
reaction time	0.142	interval	1.08	1.05	1.02	1.02
		velocity	5.38	8.46	8.70	8.96

	7.65	5 / 3	
0.93			3.15
10.45	7.84		8.70

Zallow, Chad (USA) (1997)	time	2.55	3.60	4.64	5.70	6.72
reaction time	0.189	interval	1.05	1.04	1.06	1.02
	velocity	5.38	8.70	8.79	8.62	8.96

	7.71	6 / 4	
0.99			3.15
9.82	7.78		8.70

Lovett, Eddie (ISV) (1992)	time	2.58	3.67	4.70	5.78	6.85	
reaction time	0.182	interval	1.09	1.03	1.08	1.07	
		velocity	5.32	8.39	8.87	8.46	8.54

	7.84	7 / 5	
0.99			3.20
9.82	7.65		8.57

Zaza, Wellington (LBR) (1995)	time	2.50	3.60	4.70	5.78	6.88							7.87	2 / 6	
	reaction time	0.153		1.10	1.10	1.08	1.10					0.99			3.28
	velocity		5.49	8.31	8.31	8.46	8.31					9.82	7.62		8.36
Holloway, Trey (USA) (1994)	time	2.57	3.76		5.97	7.03							8.10	1 / 7	
	reaction time	0.166		7 steps	1.19	2.21	1.06					1.07			3.40
	velocity		5.34	7.68		8.27	8.62					9.08	7.41		8.06

2020 Michitaka Kinami Memorial (Osaka, JPN)**FINAL**

date 24-Oct-20

wind 0.6 m/s

Shibayama (2020) - race analysis of men's 110m hurdles in the 2020 season

		H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
Takayama, Shunya (JPN) (1994)	time	2.53	3.58	4.61	5.65	6.68	7.73	8.79	9.84	10.93	12.01		13.49	7 / 1			
	reaction time			1.05	1.03	1.04	1.03	1.05	1.06	1.05	1.09	1.08	1.48		3.12	3.14	3.22
	velocity		5.42	8.70	8.87	8.79	8.87	8.70	8.62	8.70	8.39	8.46	9.47	8.15	8.79	8.73	8.52
Nomoto, Syuusei (JPN) (1995)	time	2.53	3.59	4.65	5.69	6.75	7.81	8.88	9.95	11.03	12.12		13.55	4 / 2			
	reaction time			7 steps	1.06	1.06	1.04	1.06	1.07	1.07	1.08	1.09	1.43		3.16	3.19	3.24
	velocity		5.42	8.62	8.62	8.79	8.62	8.62	8.54	8.54	8.46	8.39	9.80	8.12	8.68	8.60	8.46
Yokochi, Taiga (JPN) (2000)	time	2.67	3.77	4.80	5.87	6.91	7.95	9.04	10.07	11.15	12.24		13.67	5 / 3			
	reaction time			7 steps	1.10	1.03	1.07	1.04	1.04	1.09	1.03	1.08	1.09	1.43	3.20	3.17	3.20
	velocity		5.14	8.31	8.87	8.54	8.79	8.79	8.39	8.87	8.46	8.39	9.80	8.05	8.57	8.65	8.57
Ishikawa, Shuhei (JPN) (1995)	time	2.60	3.66	4.71	5.75	6.82	7.87	8.94	10.00	11.10	12.20		13.69	6 / 4			
	reaction time			7 steps	1.06	1.05	1.04	1.07	1.05	1.07	1.06	1.10	1.10	1.49	3.15	3.19	3.26
	velocity		5.28	8.62	8.70	8.79	8.54	8.70	8.54	8.62	8.31	8.31	9.41	8.04	8.70	8.60	8.41
Masuno, Genta (JPN) (1993)	time	2.61	3.68	4.75	5.80	6.87	7.92	8.98	10.06	11.18	12.27		13.77	2 / 5			
	reaction time			1.07	1.07	1.05	1.07	1.05	1.06	1.08	1.12	1.09	1.50		3.19	3.18	3.29
	velocity		5.26	8.54	8.54	8.70	8.54	8.70	8.62	8.46	8.16	8.39	9.35	7.99	8.60	8.62	8.33
Tokuoka, Ryo (JPN) (1999)	time	2.65	3.73	4.80	5.85	6.93	7.99	9.08	10.17	11.26	12.39		13.89	3 / 6			
	reaction time			1.08	1.07	1.05	1.08	1.06	1.09	1.09	1.09	1.13	1.50		3.20	3.23	3.31
	velocity		5.18	8.46	8.54	8.70	8.46	8.62	8.39	8.39	8.39	8.09	9.35	7.92	8.57	8.49	8.28
Ishida Thomas, Higashi (JPN)	time	2.62	3.71	4.76	5.83	6.89	7.95	9.01	10.14	11.24	12.38		13.90	9 / 7			
	reaction time			7 steps	1.09	1.05	1.07	1.06	1.06	1.13	1.10	1.14	1.52		3.21	3.18	3.37
	velocity		5.24	8.39	8.70	8.54	8.62	8.62	8.09	8.31	8.02	9.22	7.91		8.54	8.62	8.14
Omuro, Hideki (JPN) (1990)	time	2.61	3.69	4.78	5.88	6.97	8.04	9.14	10.24	11.33	12.44		13.94	8 / 8			
	reaction time			1.08	1.09	1.10	1.09	1.07	1.10	1.10	1.09	1.11	1.50		3.27	3.26	3.30
	velocity		5.26	8.46	8.39	8.31	8.39	8.54	8.31	8.31	8.39	8.23	9.35	7.89	8.39	8.41	8.31

2020 Japanese National Championships (Niigata, JPN)**FINAL**

date 03-Oct-20

wind -0.1 m/s

Shibayama (2020) - race analysis of men's 110m hurdles in the 2020 season

				H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
Kanai, Taioh (JPN) (1995)	time			2.56	3.60	4.63	5.66	6.68	7.73	8.77	9.82	10.89	11.95		13.36	6 / 1			
	reaction time	0.137	interval	7 steps	1.04	1.03	1.03	1.02	1.05	1.04	1.05	1.07	1.06	1.41			3.10	3.11	3.18
			velocity		5.36	8.79	8.87	8.87	8.96	8.70	8.79	8.70	8.54	8.62	9.94	8.23		8.85	8.82
Takayama, Shunya (JPN) (1994)	time			2.57	3.60	4.63	5.66	6.70	7.73	8.78	9.84	10.92	12.02		13.47	7 / 2			
	reaction time	0.146	interval		1.03	1.03	1.03	1.04	1.03	1.05	1.06	1.08	1.10	1.45			3.09	3.12	3.24
			velocity		5.34	8.87	8.87	8.87	8.79	8.87	8.70	8.62	8.46	8.31	9.67	8.17		8.87	8.79
Izumiya, Shunsuke (JPN) (2000)	time			2.59	3.65	4.69	5.71	6.74	7.81	8.85	9.92	10.97	12.07		13.48	5 / 3			
	reaction time	0.130	interval		1.06	1.04	1.02	1.03	1.07	1.04	1.07	1.05	1.10	1.41			3.12	3.14	3.22
			velocity		5.30	8.62	8.79	8.96	8.87	8.54	8.79	8.54	8.70	8.31	9.94	8.16		8.79	8.73
Nomoto, Syuusei (JPN) (1995)	time			2.54	3.63	4.68	5.73	6.78	7.83	8.90	9.97	11.05	12.13		13.55	8 / 4			
	reaction time	0.145	interval	7 steps	1.09	1.05	1.05	1.05	1.05	1.07	1.07	1.08	1.08	1.42			3.19	3.17	3.23
			velocity		5.40	8.39	8.70	8.70	8.70	8.70	8.54	8.54	8.46	8.46	9.87	8.12		8.60	8.65
Masuno, Genta (JPN) (1993)	time			2.59	3.65	4.71	5.76	6.81	7.87	8.94	10.00	11.09	12.18		13.61	4 / 5			
	reaction time	0.136	interval		1.06	1.06	1.05	1.05	1.06	1.07	1.06	1.09	1.09	1.43			3.17	3.18	3.24
			velocity		5.30	8.62	8.62	8.70	8.70	8.62	8.54	8.62	8.39	8.39	9.80	8.08		8.65	8.62
Ishida Thomas, Higashi (JPN)	time			2.64	3.71	4.75	5.79	6.83	7.88	8.94	10.00	11.08	12.21		13.67	9 / 6			
	reaction time	0.153	interval	7 steps	1.07	1.04	1.04	1.04	1.05	1.06	1.06	1.08	1.13	1.46			3.15	3.15	3.27
			velocity		5.20	8.54	8.79	8.79	8.79	8.70	8.62	8.62	8.46	8.09	9.60	8.05		8.70	8.70
Ishikawa, Shuhei (JPN) (1995)	time			2.61	3.68	4.75	5.81	6.87	7.94	9.00	10.07	11.15	12.25		13.71	2 / 7			
	reaction time	0.159	interval	7 steps	1.07	1.07	1.06	1.06	1.07	1.06	1.07	1.08	1.10	1.46			3.20	3.19	3.25
			velocity		5.26	8.54	8.54	8.62	8.62	8.54	8.62	8.54	8.46	8.31	9.60	8.02		8.57	8.60
Yokochi, Taiga (JPN) (2000)	time			2.67	3.75	4.80	5.83	6.88	7.93	8.97	10.05	11.12	12.23		13.72	3 / 8			
	reaction time	0.158	interval	7 steps	1.08	1.05	1.03	1.05	1.05	1.04	1.08	1.07	1.11	1.49			3.16	3.14	3.26
			velocity																

velocity 5.14 8.46 8.70 8.87 8.70 8.70 8.79 8.46 8.54 8.23 9.41 8.02 8.68 8.73 8.41

Heat 2

date 02-Oct-20 wind 0.4 m/s

Shibayama (2020) - race analysis of men's 110m hurdles in the 2020 season

		H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
Masuno, Genta (JPN) (1993)	time	2.63	3.69	4.75	5.79	6.83	7.88	8.93	10.00	11.08	12.16		13.59	8 / 2			
	reaction time	0.148	interval	1.06	1.06	1.04	1.04	1.05	1.05	1.07	1.08	1.08	1.43		3.16	3.14	3.23
	velocity	5.22	8.62	8.62	8.79	8.79	8.70	8.70	8.54	8.46	8.46	9.80	8.09		8.68	8.73	8.49
Ishida Thomas, Higashi (JPN) (2000)	time	2.62	3.68	4.72	5.75	6.81	7.86	8.92	10.00	11.06	12.17		13.61	7 / 3			
	reaction time	0.151	interval	7 steps	1.06	1.04	1.03	1.06	1.05	1.06	1.08	1.06	1.11	1.44	3.13	3.17	3.25
	velocity	5.24	8.62	8.79	8.87	8.62	8.70	8.62	8.46	8.62	8.23	9.74	8.08		8.76	8.65	8.44
Muratake, Rashiddo (JPN) (2000)	time	2.63	3.69	4.74	5.80	6.85	7.90	8.98	10.05	11.12	12.20		13.61	5 / 4			
	reaction time	0.168	interval	1.06	1.05	1.06	1.05	1.05	1.08	1.07	1.07	1.08	1.41	PB	3.17	3.18	3.22
	velocity	5.22	8.62	8.70	8.62	8.70	8.70	8.46	8.54	8.54	8.46	9.94	8.08		8.65	8.62	8.52

Heat 1

date 02-Oct-20 wind 0.8 m/s

Shibayama (2020) - race analysis of men's 110m hurdles in the 2020 season

		H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
Izumiya, Shunsuke (JPN) (2000)	time	2.58	3.62	4.67	5.70	6.73	7.79	8.81	9.87	10.92	11.99		13.45	4 / 2			
	reaction time	0.128	interval	1.04	1.05	1.03	1.06	1.02	1.06	1.05	1.07	1.46			3.12	3.11	3.18
	velocity	5.32	8.79	8.70	8.87	8.87	8.62	8.96	8.62	8.70	8.54	9.60	8.18		8.79	8.82	8.62
Yokochi, Taiga (JPN) (2000)	time	2.69	3.76	4.80	5.83	6.85	7.90	8.93	10.00	11.06	12.15		13.60	7 / 5			
	reaction time	0.160	interval	7 steps	1.07	1.04	1.03	1.02	1.03	1.07	1.06	1.09	1.45	PB	3.14	3.10	3.22
	velocity	5.10	8.54	8.79	8.87	8.96	8.70	8.87	8.54	8.62	8.39	9.67	8.09		8.73	8.85	8.52

2020 Japanese Multi-Event National Championships (Nagano, JPN)**Decathlon - Heat 3**

date 27-Sep-20 wind 0.7 m/s

Matsubayashi (2020) - research on athlete performance and technique- 2020 data book

		H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
Taue, Shun (JPN) (1997)	time	2.77	3.88	4.97	6.06	7.16	8.24	9.34	10.44	11.56	12.68		14.15	7 / 1			
	reaction time	interval	1.11	1.09	1.09	1.10	1.08	1.10	1.10	1.12	1.12	1.47			3.29	3.28	3.34
	velocity	4.95	8.23	8.39	8.39	8.31	8.46	8.31	8.31	8.16	8.16	9.54	7.77		8.33	8.36	8.21
Nakamura, Akihiko (JPN) (1990)	time	2.71	3.85	4.94	6.05	7.15	8.26	9.37	10.49	11.60	12.73		14.21	4 / 2			
	reaction time	interval	1.14	1.09	1.11	1.10	1.11	1.11	1.12	1.11	1.13	1.48			3.34	3.32	3.36
	velocity	5.06	8.02	8.39	8.23	8.31	8.23	8.23	8.16	8.23	8.09	9.47	7.74		8.21	8.26	8.16

Decathlon - Heat 2

date 27-Sep-20 wind 1.1 m/s

Matsubayashi (2020) - research on athlete performance and technique- 2020 data book

		H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
Ushiro, Keisuke (JPN) (1986)	time	2.80	3.97	5.12	6.30	7.47	8.63	9.81	11.03	12.25	13.47		15.14	4 / 7			
	reaction time	interval	1.17	1.15	1.18	1.17	1.16	1.18	1.22	1.22	1.22	1.67			3.50	3.51	3.66
	velocity	4.90	7.81	7.95	7.75	7.81	7.88	7.75	7.49	7.49	7.49	8.40	7.27		7.83	7.81	7.49

2020 Doha Diamond League (Doha, QAT) (TV Analysis)**FINAL**

date 25-Sep-20 wind 0.3 m/s

Henson (2020) - Athlete First: 2020 year end hurdle report

		H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
Mallett, Aaron (USA) (1994)	time	2.46	3.50	4.52	5.52	6.52	7.54	8.56	9.58	10.60	11.64		13.15	4 / 1			
	reaction time	0.149	interval	1.04	1.02	1.00	1.02	1.02	1.02	1.02	1.04	1.51	PB		3.06	3.04	3.08
	velocity	5.58	8.79	8.96	9.14	9.14	8.96	8.96	8.96	8.96	8.79	9.28	8.37		8.96	9.02	8.90
Joseph, Jason (SUI) (1998)	time	2.56	3.64	4.68	5.70	6.72	7.76	8.80	9.82	10.82	11.88		13.40	3 / 2			
	reaction time	0.196	interval	7 steps	1.08	1.04	1.02	1.04	1.02	1.00	1.06	1.52			3.14	3.10	3.08
	velocity	5.36	8.46	8.79	8.96	8.96	8.79	8.79	8.96	9.14	8.62	9.22	8.21		8.73	8.85	8.90
King, David (GBR) (1994)	time	2.48	3.56	4.60	5.64	6.68	7.70	8.76	9.82	10.90	11.98		13.54	1 / 3			
	reaction time	0.109	interval	7 steps	1.08	1.04	1.04	1.02	1.06	1.08	1.08	1.56			3.16	3.12	3.22
	velocity	5.53	8.46	8.79	8.79	8.79	8.96	8.62	8.62	8.46	8.46	8.99	8.12		8.68	8.79	8.52
Constantino, Gabriel (BRA) (1991)	time	2.44	3.48	4.52	5.54	6.56	7.60	8.64	9.68	10.74	11.94		13.60	7 / 4			
	reaction time	0.122	interval	7 steps	1.04	1.04	1.02	1.04	1.04	1.06	1.20	1.66			3.10	3.10	3.30
	velocity	5.62	8.79	8.79	8.96	8.96	8.79	8.79	8.79	8.62	7.62	8.45	8.09		8.85	8.85	8.31
Fillery, Cameron (GBR) (1998)	time	2.52	3.64	4.72	5.76	6.80	7.84	8.92	10.00	11.08	12.16		13.76	6 / 5			
	reaction time	0.128	interval	7 steps	1.12	1.08	1.04	1.04	1.08	1.08	1.08	1.60			3.24	3.16	3.24
	velocity	5.44	8.16	8.46	8.79	8.79	8.79	8.46	8.46	8.46	8.46	8.76	7.99		8.46	8.68	8.46

2020 All Japan Corporate Championships (Kumagaya, JPN)**FINAL**

date 20-Sep-20 wind 0.1 m/s

Shibayama (2020) - race analysis of men's 110m hurdles in the 2020 season

		H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
Takayama, Shunya (JPN) (1994)	time	2.58	3.64	4.67	5.69	6.73	7.78	8.83	9.88	10.95	12.05		13.51	5 / 1			
	reaction time	0.153	interval	1.06	1.03	1.02	1.04	1.05	1.05	1.07	1.10	1.46			3.11	3.14	3.22
	velocity	5.32	8.62	8.87	8.96	8.79	8.70	8.70	8.70	8.54	8.31	9.60	8.14		8.82	8.73	8.52
Ishikawa, Shuhei (JPN) (1995)	time	2.58	3.63	4.68	5.73	6.78	7.83	8.88	9.97	11.06	12.19		13.69	4 / 2			
	reaction time	0.151	interval	1.05	1.05	1.05	1.05	1.05	1.09	1.09	1.13	1.50			3.15	3.15	3.31

	velocity	5.32	8.70	8.70	8.70	8.70	8.70	8.70	8.39	8.39	8.09	9.35	8.04		8.70	8.70	8.28
Fujii, Ryota (JPN) (1996)	time	2.60	3.68	4.73	5.79	6.83	7.90	8.98	10.06	11.16	12.28		13.78	7 / 3			
reaction time	0.151		1.08	1.05	1.06	1.04	1.07	1.08	1.08	1.10	1.12	1.50			3.19	3.19	3.30
	velocity	5.28	8.46	8.70	8.62	8.79	8.54	8.46	8.46	8.31	8.16	9.35	7.98		8.60	8.60	8.31
Omuro, Hideki (JPN) (1990)	time	2.60	3.68	4.74	5.82	6.89	7.98	9.07	10.18	11.30	12.44		13.92	8 / 4			
reaction time	0.142		1.08	1.06	1.08	1.07	1.09	1.09	1.11	1.12	1.14	1.48			3.22	3.25	3.37
	velocity	5.28	8.46	8.62	8.46	8.54	8.39	8.39	8.23	8.16	8.02	9.47	7.90		8.52	8.44	8.14
Kuriki, Anthony Tyrrell (JPN) (1990)	time	2.68	3.77	4.85	5.92	6.97	8.05	9.11	10.20	11.30	12.43		13.92	9 / 4			
reaction time	0.130		1.09	1.08	1.07	1.05	1.08	1.06	1.09	1.10	1.13	1.49			3.24	3.19	3.32
	velocity	5.12	8.39	8.46	8.54	8.70	8.46	8.62	8.39	8.31	8.09	9.41	7.90		8.46	8.60	8.26
Wado, Tatsuya (JPN) (1990)	time	2.62	3.72	4.81	5.89	6.97	8.05	9.15	10.28	11.40	12.54		14.09	2 / 6			
reaction time	0.134		1.10	1.09	1.08	1.08	1.10	1.13	1.12	1.14	1.55				3.27	3.26	3.39
	velocity	5.24	8.31	8.39	8.46	8.46	8.46	8.31	8.09	8.16	8.02	9.05	7.81		8.39	8.41	8.09

Heat 2

Heat 2

	date	20-Sep-20	wind	1.7 m/s	Shibayama (2020) - race analysis of men's 110m hurdles in the 2020 season													
			H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
Kanai, Taioh (JPN) (1995)	time		2.55	3.60	4.65	5.66	6.69	7.72	8.75	9.82	10.88	11.95		13.38	5 / 1			
	reaction time	0.138		1.05	1.05	1.01	1.03	1.03	1.03	1.07	1.06	1.07	1.43			3.11	3.09	3.20
	velocity		5.38	8.70	8.70	9.05	8.87	8.87	8.87	8.54	8.62	8.54	9.80	8.22		8.82	8.87	8.57
Tanaka, Hideaki (JPN) (1995)	time		2.68	3.78	4.86	5.95	7.04	8.13	9.22	10.33	11.47	12.61		14.09	3 / 4			
	reaction time	0.131		1.10	1.08	1.09	1.09	1.09	1.09	1.11	1.14	1.14	1.48			3.27	3.27	3.39
	velocity		5.12	8.31	8.46	8.39	8.39	8.39	8.39	8.23	8.02	8.02	9.47	7.81		8.39	8.39	8.09

2020 Golden Gala Pietro Mennea (Rome, ITA) (TV Analysis)

FINAL

		date	17-Sep-20	wind 0.1 m/s			Henson (2020) - Athlete First: 2020 year end hurdle report													
				H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10	
Pozzi, Andy (GBR) (1992)	time			2.50	3.54	4.52	5.52	6.52	7.52	8.54	9.58	10.62	11.68		13.15	4 / 1				
	reaction time	0.128	interval	7 steps	1.04	0.98	1.00	1.00	1.00	1.02	1.04	1.04	1.06	1.47				3.02	3.02	3.14
		velocity		5.49	8.79	9.33	9.14	9.14	9.14	8.96	8.79	8.79	8.62	9.54	8.37			9.08	9.08	8.73
Mallett, Aaron (USA) (1994)	time			2.52	3.58	4.60	5.62	6.66	7.68	8.68	9.70	10.72	11.78		13.23	3 / 2				
	reaction time	0.149	interval		1.06	1.02	1.02	1.04	1.02	1.00	1.02	1.02	1.06	1.45	PB			3.10	3.06	3.10
		velocity		5.44	8.62	8.96	8.96	8.79	8.96	9.14	8.96	8.96	8.62	9.67	8.31			8.85	8.96	8.85
Crittendon, Freddie (USA) (1995)	time			2.48	3.52	4.56	5.58	6.62	7.64	8.68	9.72	10.76	11.82		13.31	6 / 3				
	reaction time	0.139	interval		1.04	1.04	1.02	1.04	1.02	1.04	1.04	1.04	1.06	1.49				3.10	3.10	3.14
		velocity		5.53	8.79	8.79	8.96	8.79	8.96	8.79	8.79	8.79	8.62	9.41	8.26			8.85	8.85	8.73
Belocian, Wilhem (FRA) (1995)	time			2.50	3.52	4.56	5.60	6.64	7.68	8.72	9.78	10.86	11.94		13.49	5 / 4				
	reaction time	0.134	interval		1.02	1.04	1.04	1.04	1.04	1.04	1.06	1.08	1.08	1.55				3.10	3.12	3.22
		velocity		5.49	8.96	8.79	8.79	8.79	8.79	8.79	8.62	8.46	8.46	9.05	8.15			8.85	8.79	8.52
Al Yoha, Jaqoub (KUW) (1993)	time			2.56	3.64	4.68	5.70	6.74	7.78	8.82	9.86	10.94	12.08		13.60	9 / 5				
	reaction time	0.139	interval		1.08	1.04	1.02	1.04	1.04	1.04	1.04	1.08	1.14	1.52				3.14	3.12	3.26
		velocity		5.36	8.46	8.79	8.96	8.79	8.79	8.79	8.79	8.46	8.02	9.22	8.09			8.73	8.79	8.41
Perini, Lorenzo (ITA) (1994)	time			2.54	3.62	4.68	5.72	6.78	7.82	8.88	9.94	11.02	12.08		13.61	7 / 6				
	reaction time	0.132	interval		1.08	1.06	1.04	1.06	1.04	1.06	1.06	1.08	1.06	1.53				3.18	3.16	3.20
		velocity		5.40	8.46	8.62	8.79	8.62	8.79	8.62	8.62	8.46	8.62	9.16	8.08			8.62	8.68	8.57
Alkana, Antonio (RSA) (1990)	time			2.52	3.60	4.68	5.76	6.80	7.84	8.88	9.96	11.04	12.14		13.66	8 / 7				
	reaction time	0.130	interval		1.08	1.08	1.08	1.04	1.04	1.04	1.08	1.08	1.10	1.52				3.24	3.12	3.26
		velocity		5.44	8.46	8.46	8.46	8.79	8.79	8.79	8.46	8.46	8.31	9.22	8.05			8.46	8.79	8.41
Constantino, Gabriel (BRA) (1991)	time			2.54	3.58	4.62	5.66	6.70	7.76	8.82	9.92	11.04	12.16		13.67	2 / 8				
	reaction time	0.161	interval	7 steps	1.04	1.04	1.04	1.04	1.06	1.06	1.10	1.12	1.12	1.51				3.12	3.16	3.34
		velocity		5.40	8.79	8.79	8.79	8.79	8.62	8.62	8.31	8.16	8.16	9.28	8.05			8.79	8.68	8.21
Dal Molin, Paolo (ITA) (1987)	time			2.50	3.52	4.60	5.66	6.70	7.78	8.88	9.96	11.06	12.18		13.70	1 / 9				
	reaction time	0.134	interval	7 steps	1.02	1.08	1.06	1.04	1.08	1.10	1.08	1.10	1.12	1.52				3.16	3.22	3.30
		velocity		5.49	8.96	8.46	8.62	8.79	8.46	8.31	8.46	8.31	8.16	9.22	8.03			8.68	8.52	8.31

2020 Chinese National Championships (Shaoxing, CHN)

FINAL

INAL	date	16-Sep-20	wind	0.4 m/s	CAA Hurdle Development (2020)													
			H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
Xie Wenjun (CHN) (1990)	time		2.57	3.62	4.65	5.68	6.69	7.71	8.73	9.75	10.78	11.83		13.24	5 / 1			
	reaction time	0.162	interval	7 steps	1.05	1.03	1.03	1.01	1.02	1.02	1.03	1.05	1.41			3.11	3.05	3.10
		velocity	5.34	8.70	8.87	8.87	9.05	8.96	8.96	8.96	8.87	8.70	9.94	8.31		8.82	8.99	8.85
Zeng Jianhang (CHN) (1998)	time		2.58	3.62	4.66	5.69	6.72	7.76	8.82	9.90	10.98	12.09		13.54	6 / 2			
	reaction time	0.156	interval		1.04	1.04	1.03	1.03	1.04	1.06	1.08	1.08	1.11	1.45		3.11	3.13	3.27

	velocity	5.32	8.79	8.79	8.87	8.87	8.79	8.62	8.46	8.46	8.23	9.67	8.12		8.82	8.76	8.39
Ning Xiaohan (CHN) (2000)	time	2.60	3.69	4.76	5.81	6.86	7.90	8.95	10.01	11.07	12.15		13.60	7 / 3			
	reaction time	0.144	interval	7 steps	1.09	1.07	1.05	1.05	1.04	1.06	1.06	1.08	1.45	PB	3.21	3.14	3.20
	velocity	5.28	8.39	8.54	8.70	8.70	8.79	8.70	8.62	8.62	8.46	9.67	8.09		8.54	8.73	8.57
Zhang Honglin (CHN) (1994)	time	2.63	3.74	4.80	5.86	6.90	7.94	9.01	10.08	11.16	12.26		13.66	4 / 4			
	reaction time	0.167	interval		1.11	1.06	1.06	1.04	1.04	1.07	1.08	1.10	1.40		3.23	3.15	3.25
	velocity	5.22	8.23	8.62	8.62	8.79	8.79	8.54	8.54	8.46	8.31	10.01	8.05		8.49	8.70	8.44
Sun Zhenjiang (CHN) (1999)	time	2.60	3.70	4.77	5.83	6.90	7.95	9.03	10.13	11.24	12.36		13.88	9 / 5			
	reaction time	0.176	interval	7 steps	1.10	1.07	1.06	1.07	1.05	1.08	1.10	1.11	1.12	1.52	3.23	3.20	3.33
	velocity	5.28	8.31	8.54	8.62	8.54	8.70	8.46	8.31	8.23	8.16	9.22	7.93		8.49	8.57	8.23
Li Huaihai (CHN) (1999)	time	2.66	3.77	4.86	5.94	7.00	8.07	9.15	10.26	11.40	12.55		14.07	3 / 6			
	reaction time	0.171	interval		1.11	1.09	1.08	1.06	1.07	1.08	1.11	1.14	1.15	1.52	3.28	3.21	3.40
	velocity	5.16	8.23	8.39	8.46	8.62	8.54	8.46	8.23	8.02	7.95	9.22	7.82		8.36	8.54	8.06
Qin Weibo (CHN) (2002)	time	2.69	3.80	4.89	5.96	7.02	8.14	9.25	10.38	11.52	12.68		14.17	8 / 7			
	reaction time	0.169	interval		1.11	1.09	1.07	1.06	1.12	1.11	1.13	1.14	1.16	1.49	3.27	3.29	3.43
	velocity	5.10	8.23	8.39	8.54	8.62	8.16	8.23	8.09	8.02	7.88	9.41	7.76		8.39	8.33	7.99
Guo Zhongjie (CHN) (2001)	time	2.74	3.81	4.89	5.96	7.05	8.14	9.24	10.36	11.51	12.67		14.22	2 / 8			
	reaction time	0.185	interval		1.07	1.08	1.07	1.09	1.09	1.10	1.12	1.15	1.16	1.55	3.22	3.28	3.43
	velocity	5.01	8.54	8.46	8.54	8.39	8.39	8.31	8.16	7.95	7.88	9.05	7.74		8.52	8.36	7.99

Heat 2

Heat 2

		date	16-Sep-20	wind -0.5 m/s										CAA Hurdle Development (2020)						
				H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10	
Xie Wenjun (CHN) (1990)	time			2.60	3.67	4.72	5.77	6.82	7.87	8.94	10.01	11.08	12.18		13.63	2 / 1				
	reaction time	0.175	interval	7 steps	1.07	1.05	1.05	1.05	1.05	1.07	1.07	1.07	1.10	1.45			3.17	3.17	3.24	
			velocity	5.28	8.54	8.70	8.70	8.70	8.70	8.54	8.54	8.54	8.31	9.67	8.07		8.65	8.65	8.46	
Sun Zhenjiang (CHN) (1999)	time			2.59	3.71	4.76	5.82	6.88	7.95	9.03	10.11	11.23	12.34		13.86	3 / 2				
	reaction time	0.152	interval	7 steps	1.12	1.05	1.06	1.06	1.07	1.08	1.08	1.12	1.11	1.52			3.23	3.21	3.31	
			velocity	5.30	8.16	8.70	8.62	8.62	8.54	8.46	8.46	8.16	8.23	9.22	7.94		8.49	8.54	8.28	
Qin Weibo (CHN) (2002)	time			2.67	3.80	4.86	5.93	7.01	8.09	9.18	10.30	11.44	12.58		14.08	2 / 3				
	reaction time	0.187	interval		1.13	1.06	1.07	1.08	1.08	1.09	1.12	1.14	1.14	1.50			3.26	3.25	3.40	
			velocity	5.14	8.09	8.62	8.54	8.46	8.46	8.39	8.16	8.02	8.02	9.35	7.81		8.41	8.44	8.06	
Lu Yang (CHN) (1996)	time			2.72	3.84	4.95	6.02	7.11	8.19	9.27	10.38	11.50	12.68		14.17	8 / 4				
	reaction time	0.145	interval		1.12	1.11	1.07	1.09	1.08	1.08	1.11	1.12	1.18	1.49			3.30	3.25	3.41	
			velocity	5.04	8.16	8.23	8.54	8.39	8.46	8.46	8.23	8.16	7.75	9.41	7.76		8.31	8.44	8.04	
Jiang Fan (CHN) (1989)	time			2.69	3.82	4.92	6.01	7.12	8.21	9.30	10.45	11.59	12.74		14.27	4 / 5				
	reaction time	0.140	interval		1.13	1.10	1.09	1.11	1.09	1.09	1.15	1.14	1.15	1.53			3.32	3.29	3.44	
			velocity	5.10	8.09	8.31	8.39	8.23	8.39	8.39	7.95	8.02	7.95	9.16	7.71		8.26	8.33	7.97	
Sun Haoru (CHN) (1998)	time			2.68	3.79	4.89	5.98	7.05	8.17	9.28	10.41	11.57	12.74		14.30	7 / 6				
	reaction time	0.153	interval		1.11	1.10	1.09	1.07	1.12	1.11	1.13	1.16	1.17	1.56			3.30	3.30	3.46	
			velocity	5.12	8.23	8.31	8.39	8.54	8.16	8.23	8.09	7.88	7.81	8.99	7.69		8.31	8.31	7.92	
Pan Zijie (CHN) (1995)	time			2.64	3.78	4.90	6.00	7.13	8.26	9.39	10.53	11.68	12.84		14.40	9 / 7				
	reaction time	0.154	interval		1.14	1.12	1.10	1.13	1.13	1.13	1.14	1.15	1.16	1.56			3.36	3.39	3.45	
			velocity	5.20	8.02	8.16	8.31	8.09	8.09	8.09	8.02	7.95	7.88	8.99	7.64		8.16	8.09	7.95	
Wu Zhenhua (CHN) (1998)	time			2.73	3.89	5.03	6.14	7.27	8.39	9.59	10.79	11.97	13.18		14.73	5 / 8				
	reaction time	0.150	interval		1.16	1.14	1.11	1.13	1.12	1.20	1.20	1.18	1.21	1.55			3.41	3.45	3.59	
			velocity	5.03	7.88	8.02	8.23	8.09	8.16	7.62	7.62	7.75	7.55	9.05	7.47		8.04	7.95	7.64	

Heat 1

Heat 1	date		16-Sep-20		wind		0.3 m/s		CAA Hurdle Development (2020)												
					H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10	
Ning Xiaohan (CHN) (2000)	time				2.60	3.67	4.72	5.77	6.82	7.87	8.94	10.01	11.08	12.18		13.63	2 / 1				
	reaction time	0.142	interval	7 steps	1.07	1.05	1.05	1.05	1.05	1.05	1.07	1.07	1.07	1.10	1.45	PB		3.17	3.17	3.24	
			velocity		5.28	8.54	8.70	8.70	8.70	8.70	8.54	8.54	8.54	8.31	9.67	8.07		8.65	8.65	8.46	
Zeng Jianhang (CHN) (1998)	time				2.61	3.69	4.72	5.75	6.81	7.87	8.94	10.02	11.12	12.22		13.69	6 / 2				
	reaction time	0.159	interval			1.08	1.03	1.03	1.06	1.06	1.07	1.08	1.10	1.10	1.47			3.14	3.19	3.28	
			velocity		5.26	8.46	8.87	8.87	8.62	8.62	8.54	8.46	8.31	8.31	9.54	8.04		8.73	8.60	8.36	
Zhang Honglin (CHN) (1994)	time				2.63	3.73	4.78	5.84	6.90	7.96	9.04	10.12	11.22	12.35		13.79	8 / 3				
	reaction time	0.167	interval			1.10	1.05	1.06	1.06	1.06	1.08	1.08	1.10	1.13	1.44			3.21	3.20	3.31	
			velocity		5.22	8.31	8.70	8.62	8.62	8.62	8.46	8.46	8.31	8.09	9.74	7.98		8.54	8.57	8.28	
Yang Quanlei (CHN) (1999)	time				2.70	3.79	4.89	6.00	7.08	8.18	9.29	10.43	11.56	12.68		14.17	5 / 4				
	reaction time	0.175	interval			1.09	1.10	1.11	1.08	1.10	1.11	1.14	1.13	1.12	1.49			3.30	3.29	3.39	
			velocity		5.08	8.39	8.31	8.23	8.46	8.31	8.23	8.02	8.09	8.16	9.41	7.76		8.31	8.33	8.09	

Guo Zhongjie (CHN) (2001)	time	2.71	3.79	4.87	5.95	7.05	8.16	9.26	10.37	11.52	12.68		14.22	7 / 5			
	reaction time	0.179		1.08	1.08	1.08	1.10	1.11	1.10	1.11	1.15	1.16	1.54		3.24	3.31	3.42
	velocity		5.06	8.46	8.46	8.46	8.31	8.23	8.31	8.23	7.95	7.88	9.10	7.74	8.46	8.28	8.02
Li Huaihai (CHN) (1999)	time	2.69	3.77	4.93	6.03	7.13	8.23	9.32	10.45	11.57	12.72		14.25	3 / 6			
	reaction time	0.185		1.08	1.16	1.10	1.10	1.10	1.09	1.13	1.12	1.15	1.53		3.34	3.29	3.40
	velocity		5.10	8.46	7.88	8.31	8.31	8.31	8.39	8.09	8.16	7.95	9.16	7.72	8.21	8.33	8.06
Shi Guoxing (CHN) (2000)	time	2.72	3.86	5.00	6.11	7.24	8.37	9.50	10.63	11.76	12.90		14.34	9 / 7			
	reaction time	0.169		1.14	1.14	1.11	1.13	1.13	1.13	1.13	1.14	1.44	PB		3.39	3.39	3.40
	velocity		5.04	8.02	8.02	8.23	8.09	8.09	8.09	8.09	8.02	9.74	7.67		8.09	8.09	8.06
Zhu Shenglong (CHN) (2000)	time	2.73	3.84	4.94	6.02	7.11	8.22	9.34	10.46	11.62	12.80		14.36	1 / 8			
	reaction time	0.157		1.11	1.10	1.08	1.09	1.11	1.12	1.12	1.16	1.18	1.56		3.29	3.32	3.46
	velocity		5.03	8.23	8.31	8.46	8.39	8.23	8.16	8.16	7.88	7.75	8.99	7.66	8.33	8.26	7.92
Yang Yuxuan (CHN) (2002)	time	2.72	3.88	4.98	6.10	7.20	8.32	9.45	10.59	11.75	12.91		14.43	4 / 9			
	reaction time	0.138		1.16	1.10	1.12	1.10	1.12	1.13	1.14	1.16	1.16	1.52		3.38	3.35	3.46
	velocity		5.04	7.88	8.31	8.16	8.31	8.16	8.09	8.02	7.88	7.88	9.22	7.62	8.11	8.19	7.92

2020 French National Championships (Albi, FRA) (TV Analysis)**FINAL**

INAL	date 13-Sep-20			wind 0.0 m/s										Henson (2020) - Athlete First: 2020 year end hurdle report							
														H1	H2	H3	H4	H5	H6	H7	H8
Belocian, Wilhem (FRA) (1995)	time			2.50	3.51	4.49	5.50	6.48	7.53	8.57	9.59	10.63	11.67		13.20	4 / 1					
	reaction time	0.156	interval		1.01	0.98	1.01	0.98	1.05	1.04	1.02	1.04	1.04	1.53			3.00	3.07	3.10		
	velocity			5.49	9.05	9.33	9.05	9.33	8.70	8.79	8.96	8.79	8.79	9.16	8.33		9.14	8.93	8.85		
Laviolette, Jeanice (FRA) (2000)	time			2.56	3.64	4.69	5.74	6.71	7.76	8.81	9.83	10.87	11.95		13.52	5 / 2					
	reaction time	0.222	interval	7 steps	1.08	1.05	1.05	0.97	1.05	1.05	1.02	1.04	1.08	1.57	PB		3.18	3.07	3.14		
	velocity			5.36	8.46	8.70	8.70	9.42	8.70	8.70	8.96	8.79	8.46	8.93	8.14		8.62	8.93	8.73		
Kwaou-Mathey, Just (FRA) (1999)	time			2.63	3.71	4.69	5.77	6.75	7.87	8.90	9.95	11.00	12.08		13.61	3 / 3					
	reaction time	0.199	interval	7 steps	1.08	0.98	1.08	0.98	1.12	1.03	1.05	1.05	1.08	1.53	PB		3.14	3.13	3.18		
	velocity			5.22	8.46	9.33	8.46	9.33	8.16	8.87	8.70	8.70	8.46	9.16	8.08		8.73	8.76	8.62		
Ngo, Matteo (FRA) (1998)	time			2.63	3.71	4.73	5.84	6.89	8.00	9.05	10.13	11.24	12.39		14.02	2 / 4					
	reaction time	0.218	interval		1.08	1.02	1.11	1.05	1.11	1.05	1.08	1.11	1.15	1.63			3.21	3.21	3.34		
	velocity			5.22	8.46	8.96	8.23	8.70	8.23	8.70	8.46	8.23	7.95	8.60	7.85		8.54	8.54	8.21		
Pistol, Marvin (FRA) (1996)	time			2.67	3.75	4.86	5.94	6.96	8.10	9.15	10.23	11.34	12.49		14.20	1 / 5					
	reaction time	0.202	interval		1.08	1.11	1.08	1.02	1.14	1.05	1.08	1.11	1.15	1.71			3.27	3.21	3.34		
	velocity			5.14	8.46	8.23	8.46	8.96	8.02	8.70	8.46	8.23	7.95	8.20	7.75		8.39	8.54	8.21		
Mohamed, Raphael (FRA) (1999)	time			2.70	3.75	4.79	5.84	6.89	8.00	9.05	10.13	11.14		dnf	7 / --						
	reaction time	0.270	interval	7 steps	1.05	1.04	1.05	1.05	1.11	1.05	1.08	1.01					3.14	3.21			
	velocity			5.08	8.70	8.79	8.70	8.70	8.23	8.70	8.46	9.05					8.73	8.54			

2020 World Athletics Trials (Fuji, JPN)**Race 2**

Race 2	date	06-Sep-20	wind 1.3 m/s										Shibayama (2020) - race analysis of men's 110m hurdles in the 2020 season							
			H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10		
Masuno, Genta (JPN) (1993)	time		2.60	3.67	4.73	5.76	6.81	7.88	8.95	10.03	11.12	12.23		13.69	3 / 1					
	reaction time			1.07	1.06	1.03	1.05	1.07	1.07	1.08	1.09	1.11	1.46			3.16	3.19	3.28		
	velocity		5.28	8.54	8.62	8.87	8.70	8.54	8.54	8.46	8.39	8.23	9.60	8.04		8.68	8.60	8.36		
Fujii, Ryota (JPN) (1996)	time		2.60	3.67	4.73	5.78	6.83	7.89	8.95	10.03	11.15	12.26		13.78	4 / 2					
	reaction time			1.07	1.06	1.05	1.05	1.06	1.06	1.08	1.12	1.11	1.52			3.18	3.17	3.31		
	velocity		5.28	8.54	8.62	8.70	8.70	8.62	8.62	8.46	8.16	8.23	9.22	7.98		8.62	8.65	8.28		
Tanaka, Hideaki (JPN) (1995)	time		2.71	3.83	4.92	6.00	7.10	8.19	9.28	10.40	11.51	12.65		14.13	6 / 3					
	reaction time		7 steps	1.12	1.09	1.08	1.10	1.09	1.09	1.12	1.11	1.14	1.48			3.29	3.28	3.37		
	velocity		5.06	8.16	8.39	8.46	8.31	8.39	8.39	8.16	8.23	8.02	9.47	7.78		8.33	8.36	8.14		
Furuya, Shinta (JPN) (2000)	time		2.65	3.76	4.83	5.92	7.00	8.10	9.22	10.36	11.49	12.64		14.16	7 / 4					
	reaction time			1.11	1.07	1.09	1.08	1.10	1.12	1.14	1.13	1.15	1.52	PB		3.27	3.30	3.42		
	velocity		5.18	8.23	8.54	8.39	8.46	8.31	8.16	8.02	8.09	7.95	9.22	7.77		8.39	8.31	8.02		
Tomiyama, Hiroki (JPN) (1998)	time		2.68	3.81	4.93	6.04	7.13	8.24	9.35	10.49	11.62	12.78		14.30	5 / 5					
	reaction time		7 steps	1.13	1.12	1.11	1.09	1.11	1.11	1.14	1.13	1.16	1.52			3.36	3.31	3.43		
	velocity		5.12	8.09	8.16	8.23	8.39	8.23	8.23	8.02	8.09	7.88	9.22	7.69		8.16	8.28	7.99		
[Redacted]	time		2.83	4.00	5.18	6.38	7.56	8.74	9.93	11.12	12.36	13.61		15.21	2 / 6					
	reaction time			1.17	1.18	1.20	1.18	1.18	1.19	1.19	1.24	1.25	1.60			3.55	3.55	3.68		
	velocity		4.85	7.81	7.75	7.62	7.75	7.75	7.68	7.68	7.37	7.31	8.76	7.23		7.72	7.72	7.45		

2020 Meeting Pro Athlé Tour de Marseille (Marseille, FRA) (TV Analysis)

FINAL

date 03-Sep-20 wind -0.3 m/s

Henson (2020) - Athlete First: 2020 year end hurdle report

			H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
Ortega, Orlando (ESP) (1991)	time		2.49	3.55	4.61	5.59	6.60	7.57	8.55	9.62	10.67	11.70		13.15	5 / 1			
	reaction time	0.214	interval	7 steps	1.06	1.06	0.98	1.01	0.97	0.98	1.07	1.05	1.03	1.45		3.10	2.96	3.15
	velocity		5.51	8.62	8.62	9.33	9.05	9.42	9.33	8.54	8.70	8.87	9.67	8.37		8.85	9.26	8.70
Belocian, Wilhem (FRA) (1995)	time		2.52	3.57	4.57	5.55	6.60	7.61	8.62	9.66	10.70	11.75		13.27	6 / 2			
	reaction time	0.211	interval		1.05	1.00	0.98	1.05	1.01	1.01	1.04	1.04	1.05	1.52		3.03	3.07	3.13
	velocity		5.44	8.70	9.14	9.33	8.70	9.05	9.05	8.79	8.79	8.70	9.22	8.29		9.05	8.93	8.76
Martinot-Lagarde, Pascal (FRA)	time		2.56	3.60	4.67	5.65	6.70	7.74	8.78	9.86	11.00	12.06		13.63	4 / 3			
	reaction time		interval	7 steps	1.04	1.07	0.98	1.05	1.04	1.08	1.14	1.06	1.57			3.09	3.13	3.28
	velocity		5.36	8.79	8.54	9.33	8.70	8.79	8.79	8.46	8.02	8.62	8.93	8.07		8.87	8.76	8.36
Alkana, Antonio (RSA) (1990)	time		2.56	3.60	4.71	5.75	6.81	7.84	8.89	9.96	11.11	12.18		13.81	3 / 4			
	reaction time	0.201	interval		1.04	1.11	1.04	1.06	1.03	1.05	1.07	1.15	1.07	1.63		3.19	3.14	3.29
	velocity		5.36	8.79	8.23	8.79	8.62	8.87	8.70	8.54	7.95	8.54	8.60	7.97		8.60	8.73	8.33
Obasuyi, Michael (BEL) (1999)	time		2.66	3.74	4.81	5.86	6.93	8.01	9.05	10.16	11.24	12.32		13.88	7 / 5			
	reaction time	0.208	interval	7 steps	1.08	1.07	1.05	1.07	1.08	1.04	1.11	1.08	1.08	1.56		3.20	3.19	3.27
	velocity		5.16	8.46	8.54	8.70	8.54	8.46	8.79	8.23	8.46	8.46	8.99	7.93		8.57	8.60	8.39
Kwaou-Mathey, Just (FRA) (19)	time		2.70	3.80	4.91	5.89	7.00	8.04	9.05	10.20	11.31	12.39		13.99	2 / 6			
	reaction time		interval	7 steps	1.10	1.11	0.98	1.11	1.04	1.01	1.15	1.11	1.08	1.60		3.19	3.16	3.34
	velocity		5.08	8.31	8.23	9.33	8.23	8.79	9.05	7.95	8.23	8.46	8.76	7.86		8.60	8.68	8.21
Chaverra, Joan Camilo (COL) (	time		2.62	3.67	4.78	5.79	6.86	7.92	9.00	10.17	11.29	12.42		14.08	8 / 7			
	reaction time		interval	7 steps	1.05	1.11	1.01	1.07	1.06	1.08	1.17	1.12	1.13	1.66		3.17	3.21	3.42
	velocity		5.24	8.70	8.23	9.05	8.54	8.62	8.46	7.81	8.16	8.09	8.45	7.81		8.65	8.54	8.02
Greff, Ronan (FRA) (1998)	time		2.69	3.80	4.88	5.89	7.03	8.08	9.15	10.30	11.41	12.49		14.16	1 / 8			
	reaction time	0.247	interval		1.11	1.08	1.01	1.14	1.05	1.07	1.15	1.11	1.08	1.67		3.20	3.26	3.34
	velocity		5.10	8.23	8.46	9.05	8.02	8.70	8.54	7.95	8.23	8.46	8.40	7.77		8.57	8.41	8.21

2020 Athlete Night Games (Fukui, JPN)

FINAL

date 29-Aug-20 wind 1.4 m/s

Shibayama (2020) - race analysis of men's 110m hurdles in the 2020 season

			H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
Kanai, Taioh (JPN) (1995)	time		2.55	3.60	4.63	5.65	6.66	7.70	8.73	9.78	10.83	11.90		13.27	5 / 1			
	reaction time		interval	7 steps	1.05	1.03	1.02	1.01	1.04	1.03	1.05	1.05	1.07	PB		3.10	3.08	3.17
	velocity		5.38	8.70	8.87	8.96	9.05	8.79	8.87	8.70	8.70	8.54	10.23	8.29		8.85	8.90	8.65
Takayama, Shunya (JPN) (1994)	time		2.57	3.60	4.62	5.63	6.64	7.68	8.72	9.75	10.82	11.90		13.34	3 / 2			
	reaction time		interval		1.03	1.02	1.01	1.01	1.04	1.04	1.03	1.07	1.08	1.44		3.06	3.09	3.18
	velocity		5.34	8.87	8.96	9.05	9.05	8.79	8.87	8.54	8.46	8.74		8.25		8.96	8.87	8.62
Ishikawa, Shuhei (JPN) (1995)	time		2.56	3.62	4.65	5.67	6.70	7.73	8.75	9.80	10.85	11.92		13.39	6 / 3			
	reaction time		interval	7 steps	1.06	1.03	1.02	1.03	1.03	1.02	1.05	1.05	1.07	1.47	PB	3.11	3.08	3.17
	velocity		5.36	8.62	8.87	8.96	8.87	8.87	8.96	8.70	8.70	8.54	9.54	8.22		8.82	8.90	8.65
Nomoto, Syuusei (JPN) (1995)	time		2.55	3.62	4.68	5.71	6.75	7.80	8.84	9.91	10.99	12.06		13.45	7 / 4			
	reaction time		interval	7 steps	1.07	1.06	1.03	1.04	1.05	1.04	1.07	1.08	1.07	1.39	PB	3.16	3.13	3.22
	velocity		5.38	8.54	8.62	8.87	8.79	8.70	8.79	8.54	8.46	8.54	10.09	8.18		8.68	8.76	8.52
Fujii, Ryota (JPN) (1996)	time		2.59	3.65	4.68	5.70	6.76	7.82	8.87	9.94	11.02	12.14		13.58	8 / 5			
	reaction time		interval		1.06	1.03	1.02	1.06	1.06	1.05	1.07	1.08	1.12	1.44		3.11	3.17	3.27
	velocity		5.30	8.62	8.87	8.96	8.62	8.62	8.70	8.54	8.46	8.16	9.74	8.10		8.82	8.65	8.39
Takahashi, Yusuke (JPN) (1996)	time		2.62	3.72	4.76	5.81	6.87	7.93	9.01	10.07	11.15	12.23		13.60	4 / 6			
	reaction time		interval		1.10	1.04	1.05	1.06	1.06	1.08	1.06	1.08	1.08	1.37	PB	3.19	3.20	3.22
	velocity		5.24	8.31	8.79	8.70	8.62	8.62	8.46	8.62	8.46	8.46	10.23	8.09		8.60	8.57	8.52
Masuno, Genta (JPN) (1993)	time		2.61	3.67	4.72	5.75	6.81	7.88	8.94	10.00	11.09	12.19		13.62	1 / 7			
	reaction time		interval		1.06	1.05	1.03	1.06	1.07	1.06	1.09	1.10	1.43			3.14	3.19	3.25
	velocity		5.26	8.62	8.70	8.87	8.62	8.54	8.62	8.62	8.39	8.31	9.80	8.08		8.73	8.60	8.44
Kuriki, Anthony Tyrrell (JPN) (	time		2.65	3.72	4.77	5.83	6.85	7.90	8.95	10.04	11.10	12.19		13.64	2 / 8			
	reaction time		interval	7 steps	1.07	1.05	1.06	1.02	1.05	1.05	1.09	1.06	1.09	1.45		3.18	3.12	3.24
	velocity		5.18	8.54	8.70	8.62	8.96	8.70	8.70	8.39	8.62	8.39	9.67	8.06		8.62	8.79	8.46

Heat 2

date 29-Aug-20 wind 1.5 m/s

Shibayama (2020) - race analysis of men's 110m hurdles in the 2020 season

			H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
Fujii, Ryota (JPN) (1996)	time		2.57	3.61	4.64	5.68	6.72	7.78	8.87	9.93	11.03	12.10		13.57	1 / 1			
	reaction time		interval		1.04	1.03	1.04	1.06	1.09	1.06	1.10	1.07	1.47	PB		3.11	3.19	3.23
	velocity		5.34	8.79	8.87	8.79	8.79	8.62	8.39	8.62	8.31	8.54	9.54	8.11		8.82	8.60	8.49

2020 Seiko Golden Grand Prix (Tokyo, JPN)

FINAL

date 23-Aug-20 wind -0.4 m/s

Shibayama (2020) - race analysis of men's 110m hurdles in the 2020 season

			H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
Kanai, Taioh (JPN) (1995)	time		2.55	3.60	4.65	5.68	6.73	7.77	8.81	9.88	10.93	12.03		13.45	6 / 1			
	reaction time	0.121	7 steps	1.05	1.05	1.03	1.05	1.04	1.04	1.07	1.05	1.10	1.42			3.13	3.13	3.22
	velocity		5.38	8.70	8.70	8.87	8.70	8.79	8.79	8.54	8.70	8.31	9.87	8.18		8.76	8.76	8.52
Muratake, Rashiddo (JPN) (2001)	time		2.58	3.67	4.72	5.79	6.84	7.90	8.96	10.04	11.13	12.22		13.65	2 / 2			
	reaction time	0.138		1.09	1.05	1.07	1.05	1.06	1.06	1.08	1.09	1.09	1.43	PB		3.21	3.17	3.26
	velocity		5.32	8.39	8.70	8.54	8.70	8.62	8.62	8.46	8.39	8.39	9.80	8.06		8.54	8.65	8.41
Takayama, Shunya (JPN) (1994)	time		2.58	3.65	4.70	5.75	6.81	7.87	8.95	10.06	11.16	12.27		13.74	5 / 3			
	reaction time	0.135		1.07	1.05	1.05	1.06	1.06	1.08	1.11	1.10	1.11	1.47			3.17	3.20	3.32
	velocity		5.32	8.54	8.70	8.70	8.62	8.62	8.46	8.23	8.31	8.23	9.54	8.01		8.65	8.57	8.26
Kuriki, Anthony Tyrrell (JPN) (2019)	time		2.63	3.72	4.79	5.85	6.91	7.97	9.05	10.13	11.21	12.31		13.79	3 / 4			
	reaction time	0.108	7 steps	1.09	1.07	1.06	1.06	1.06	1.08	1.08	1.08	1.10	1.48			3.22	3.20	3.26
	velocity		5.22	8.39	8.54	8.62	8.62	8.62	8.46	8.46	8.46	8.31	9.47	7.98		8.52	8.57	8.41
Fujii, Ryota (JPN) (1996)	time		2.60	3.67	4.73	5.77	6.82	7.88	8.97	10.08	11.19	12.31		13.79	9 / 5			
	reaction time	0.144		1.07	1.06	1.04	1.05	1.06	1.09	1.11	1.11	1.12	1.48			3.17	3.20	3.34
	velocity		5.28	8.54	8.62	8.79	8.70	8.62	8.39	8.23	8.23	8.16	9.47	7.98		8.65	8.57	8.21
Ishida Thomas, Higashi (JPN) (2019)	time		2.70	3.79	4.86	5.93	7.01	8.11	9.23	10.34	11.46	12.61		14.09	7 / 6			
	reaction time	0.146	7 steps	1.09	1.07	1.07	1.08	1.10	1.12	1.11	1.12	1.15	1.48			3.23	3.30	3.38
	velocity		5.08	8.39	8.54	8.54	8.46	8.31	8.16	8.23	8.16	7.95	9.47	7.81		8.49	8.31	8.11
Toyoda, Ken (JPN) (2002)	time		2.60	3.68	4.76	5.85	6.98	8.08	9.19	10.33	11.48	12.62		14.15	1 / 7			
	reaction time	0.136	7 steps	1.08	1.08	1.09	1.13	1.10	1.11	1.14	1.15	1.14	1.53	PB		3.25	3.34	3.43
	velocity		5.28	8.46	8.46	8.39	8.09	8.31	8.23	8.02	7.95	8.02	9.16	7.77		8.44	8.21	7.99
Yazawa, Wataru (JPN) (1991)	time		2.67	3.80	4.90	6.00	7.10	8.19	9.29	10.42	11.56	12.72		14.29	8 / 8			
	reaction time	0.158	7 steps	1.13	1.10	1.10	1.10	1.09	1.10	1.13	1.14	1.16	1.57			3.33	3.29	3.43
	velocity		5.14	8.09	8.31	8.31	8.31	8.39	8.31	8.09	8.02	7.88	8.93	7.70		8.23	8.33	7.99

2020 István Gyulai Memorial (Székesfehérvár, HUN) (TV Analysis)

FINAL

date 19-Aug-20 wind 0.3 m/s

Henson (2020) - Athlete First: 2020 year end hurdle report

			H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
Ortega, Orlando (ESP) (1991)	time		2.50	3.54	4.57	5.60	6.64	7.64	8.66	9.68	10.71	11.74		13.21	4 / 1			
	reaction time	0.146	7 steps	1.04	1.03	1.03	1.04	1.00	1.02	1.02	1.03	1.03	1.47			3.10	3.06	3.08
	velocity		5.49	8.79	8.87	8.87	8.79	9.14	8.96	8.96	8.87	8.87	9.54	8.33		8.85	8.96	8.90
Holloway, Grant (USA) (1997)	time		2.36	3.37	4.34	5.34	6.31	7.34	8.34	9.44	10.51	11.64		13.22	3 / 2			
	reaction time	0.141	7 steps	1.01	0.97	1.00	0.97	1.03	1.00	1.10	1.07	1.13	1.58			2.98	3.00	3.30
	velocity		5.81	9.05	9.42	9.14	9.42	8.87	9.14	8.31	8.54	8.09	8.87	8.32		9.20	9.14	8.31
Crittendon, Freddie (USA) (1995)	time		2.47	3.50	4.52	5.54	6.57		8.64	9.68	10.74	11.78		13.30	8 / 3			
	reaction time	0.146		1.03	1.02	1.02	1.03		2.07	1.04	1.06	1.04	1.52			3.07	3.10	3.14
	velocity		5.55	8.87	8.96	8.96	8.87		8.83	8.79	8.62	8.79	9.22	8.27		8.93	8.85	8.73
Belocian, Wilhem (FRA) (1995)	time		2.47	3.50	4.54	5.57	6.57	7.60	8.64	9.68	10.74	11.84		13.34	6 / 4			
	reaction time	0.134		1.03	1.04	1.03	1.00	1.03	1.04	1.04	1.06	1.10	1.50			3.10	3.07	3.20
	velocity		5.55	8.87	8.79	8.87	9.14	8.87	8.79	8.79	8.62	8.31	9.35	8.25		8.85	8.93	8.57
Mallett, Aaron (USA) (1994)	time		2.49	3.54	4.57	5.61	6.64	7.69	8.71	9.75	10.82	11.91		13.45	1 / 5			
	reaction time	0.168		1.05	1.03	1.04	1.03	1.05	1.02	1.04	1.07	1.09	1.54			3.12	3.10	3.20
	velocity		5.51	8.70	8.87	8.79	8.87	8.70	8.96	8.79	8.54	8.39	9.10	8.18		8.79	8.85	8.57
Pozzi, Andy (GBR) (1992)	time		2.50	3.54	4.54	5.57	6.57	7.61	8.64	9.68	10.74	11.91		13.60	5 / 6			
	reaction time	0.127	7 steps	1.04	1.00	1.03	1.00	1.04	1.03	1.04	1.06	1.17	1.69			3.07	3.07	3.27
	velocity		5.49	8.79	9.14	8.87	9.14	8.79	8.87	8.79	8.62	7.81	8.30	8.09		8.93	8.93	8.39
Szűcs, Valdó (HUN) (1995)	time		2.50	3.67	4.74	5.80	6.85	7.91	8.94	9.98	11.08	12.18		13.74	7 / 7			
	reaction time	0.141	7 steps	1.17	1.07	1.06	1.05	1.06	1.03	1.04	1.10	1.10	1.56			3.30	3.14	3.24
	velocity		5.49	7.81	8.54	8.62	8.70	8.62	8.87	8.79	8.31	8.31	8.99	8.01		8.31	8.73	8.46
Baji, Balázs (HUN) (1989)	time		2.54	3.67	4.74	5.80	6.84	7.91	8.95	10.04	11.14	12.28		13.86	2 / 8			
	reaction time	0.137	7 steps	1.13	1.07	1.06	1.04	1.07	1.04	1.09	1.10	1.14	1.58			3.26	3.15	3.33
	velocity		5.40	8.09	8.54	8.62	8.79	8.54	8.79	8.39	8.31	8.02	8.87	7.94		8.41	8.70	8.23

2020 Herculis Meeting International d'Athletisme (Monaco, MON) (TV Analysis)

FINAL

date 14-Aug-20 wind 0.8 m/s

Henson (2020) - Athlete First: 2020 year end hurdle report

			H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
Ortega, Orlando (ESP) (1991)	time		2.54	3.57	4.60	5.60	6.57	7.57	8.61	9.61	10.67	11.71		13.11	4 / 1			
	reaction time	0.125	7 steps	1.03	1.03	1.00	0.97	1.00	1.04	1.00	1.06	1.04	1.40			3.06	3.01	3.10
	velocity		5.40	8.87	8.87	9.14	9.42	9.14	8.79	9.14	8.62	8.79	10.01	8.39		8.96	9.11	8.85

Pozzi, Andy (GBR) (1992)	time	2.52	3.54	4.54	5.54	6.54	7.54	8.57	9.57	10.61	11.68		13.14	5 / 2				
	reaction time	0.125	interval	7 steps	1.02	1.00	1.00	1.00	1.00	1.03	1.00	1.04	1.07	1.46	=PB	3.02	3.03	3.11
	velocity	5.44	8.96	9.14	9.14	9.14	9.14	8.87	9.14	8.79	8.54	9.60	8.37		9.08	9.05	8.82	
Belocian, Wilhem (FRA) (1995)	time	2.54	3.57	4.60	5.60	6.60	7.61	8.64	9.67	10.71	11.77		13.18	6 / 3				
	reaction time	0.134	interval		1.03	1.03	1.00	1.00	1.01	1.03	1.03	1.04	1.06	1.41	PB	3.06	3.04	3.13
	velocity	5.40	8.87	8.87	9.14	9.14	9.05	8.87	8.87	8.79	8.62	9.94	8.35		8.96	9.02	8.76	
Holloway, Grant (USA) (1997)	time	2.50	3.50	4.50	5.50	6.50	7.50	8.54	9.57	10.64	11.74		13.19	3 / 4				
	reaction time	0.163	interval	7 steps	1.00	1.00	1.00	1.00	1.04	1.03	1.07	1.10	1.45		3.00	3.04	3.20	
	velocity	5.49	9.14	9.14	9.14	9.14	9.14	8.79	8.87	8.54	8.31	9.67	8.34		9.14	9.02	8.57	
Dal Molin, Paolo (ITA) (1987)	time	2.50	3.57	4.64	5.70	6.77	7.80	8.87	9.97	11.04	12.14		13.61	8 / 5				
	reaction time	0.123	interval	7 steps	1.07	1.07	1.06	1.07	1.03	1.07	1.10	1.07	1.10	1.47		3.20	3.17	3.27
	velocity	5.49	8.54	8.54	8.62	8.54	8.87	8.54	8.31	8.54	8.31	9.54	8.08		8.57	8.65	8.39	
Vukicevic, Vladimir (NOR) (1991)	time	2.63	3.70	4.77	5.80	6.87	7.91	9.00	10.07	11.14	12.24		13.75	1 / 6				
	reaction time	0.130	interval		1.07	1.07	1.03	1.07	1.04	1.09	1.07	1.07	1.10	1.51		3.17	3.20	3.24
	velocity	5.22	8.54	8.54	8.87	8.54	8.79	8.39	8.54	8.54	8.31	9.28	8.00		8.65	8.57	8.46	
Joseph, Jason (SUI) (1998)	time	2.73	3.80	4.87	5.90	6.92	7.97	9.00	10.07	11.18	12.31		13.84	2 / 7				
	reaction time	0.233	interval	7 steps	1.07	1.07	1.03	1.02	1.05	1.03	1.07	1.11	1.13	1.53		3.17	3.10	3.31
	velocity	5.03	8.54	8.54	8.87	8.96	8.70	8.87	8.54	8.23	8.09	9.16	7.95		8.65	8.85	8.28	
Alkana, Antonio (RSA) (1990)	time	2.57	3.63	4.70	5.77	6.84	7.91	8.98	10.07	11.18	12.31		13.87	7 / 8				
	reaction time	0.134	interval		1.06	1.07	1.07	1.07	1.07	1.09	1.11	1.13	1.56		3.20	3.21	3.33	
	velocity	5.34	8.62	8.54	8.54	8.54	8.54	8.54	8.39	8.23	8.09	8.99	7.93		8.57	8.54	8.23	

2020 Paavo Nurmi Games (Turku, FIN) (TV Analysis)

FINAL

INAL		date	11-Aug-20	wind 1.1 m/s			Henson (2020) - Athlete First: 2020 year end hurdle report												
				H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
Pozzi, Andy (GBR) (1992)	time			2.57	3.57	4.57	5.57	6.57	7.57	8.59	9.61	10.65	11.71		13.17	6 / 1			
	reaction time	0.149	interval	7 steps	1.00	1.00	1.00	1.00	1.00	1.02	1.02	1.04	1.06	1.46			3.00	3.02	3.12
	velocity			5.34	9.14	9.14	9.14	9.14	9.14	8.96	8.96	8.79	8.62	9.60	8.35		9.14	9.08	8.79
Belocian, Wilhem (FRA) (1995)	time			2.57	3.60	4.60	5.64	6.64	7.64	8.67	9.74	10.81	11.88		13.38	3 / 2			
	reaction time	0.147	interval		1.03	1.00	1.04	1.00	1.00	1.03	1.07	1.07	1.07	1.50			3.07	3.03	3.21
	velocity			5.34	8.87	9.14	8.79	9.14	9.14	8.87	8.54	8.54	8.54	9.35	8.22		8.93	9.05	8.54
King, David (GBR) (1994)	time			2.60	3.67	4.70	5.75	6.77	7.84	8.88	9.91	11.01	12.08		13.57	5 / 3			
	reaction time	0.148	interval	7 steps	1.07	1.03	1.05	1.02	1.07	1.04	1.03	1.10	1.07	1.49			3.15	3.13	3.20
	velocity			5.28	8.54	8.87	8.70	8.96	8.54	8.79	8.87	8.31	8.54	9.41	8.11		8.70	8.76	8.57
Alkana, Antonio (RSA) (1990)	time			2.57	3.60	4.64	5.70	6.77	7.84	8.88	9.91	11.01	12.08		13.58	8 / 4			
	reaction time	0.158	interval		1.03	1.04	1.06	1.07	1.07	1.04	1.03	1.10	1.07	1.50			3.13	3.18	3.20
	velocity			5.34	8.87	8.79	8.62	8.54	8.54	8.79	8.87	8.31	8.54	9.35	8.10		8.76	8.62	8.57
Lakka, Elmo (FIN) (1993)	time			2.62	3.70	4.77	5.84	6.87	7.91	9.01	10.08	11.18	12.28		13.79	4 / 5			
	reaction time	0.186	interval		1.08	1.07	1.07	1.03	1.04	1.10	1.07	1.10	1.10	1.51			3.22	3.17	3.27
	velocity			5.24	8.46	8.54	8.54	8.87	8.79	8.31	8.54	8.31	8.31	9.28	7.98		8.52	8.65	8.39
Fofana, Hassane (ITA) (1992)	time			2.60	3.70	4.80	5.85	6.91	8.00	9.08	10.14	11.22	12.30		13.88	7 / 6			
	reaction time	0.170	interval	7 steps	1.10	1.10	1.05	1.06	1.09	1.08	1.06	1.08	1.08	1.58			3.25	3.23	3.22
	velocity			5.28	8.31	8.31	8.70	8.62	8.39	8.46	8.62	8.46	8.46	8.87	7.93		8.44	8.49	8.52
Manninen, Ilari (FIN) (1997)	time			2.60	3.70	4.80	5.90	7.00	8.08	9.21	10.31	11.44	12.58		14.11	1 / 7			
	reaction time	0.201	interval	7 steps	1.10	1.10	1.10	1.10	1.08	1.13	1.10	1.13	1.14	1.53			3.30	3.31	3.37
	velocity			5.28	8.31	8.31	8.31	8.31	8.46	8.09	8.31	8.09	8.02	9.16	7.80		8.31	8.28	8.14
Kraus, Simon (FRA) (1992)	time			2.64	3.70	4.84	5.91	7.00	8.08	9.21					dnf	2 / --			
	reaction time	0.154	interval	7 steps	1.06	1.14	1.07	1.09	1.08	1.13							3.27	3.30	
	velocity			5.20	8.62	8.02	8.54	8.39	8.46	8.09							8.39	8.31	

2020 CAA Shanghai Training Base Permit Meeting (Shanghai, CHN)

FINAL

INAL	date	19-Jul-20	wind	Shanghai Sport Institute (2020) - competition analysis															
				H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
Xie Wenjun (CHN) (1990)	time			2.69	3.70	4.70	5.71	6.70	7.70	8.72	9.75	10.77	11.82		13.20	/ 1			
	reaction time			7 steps	1.01	1.00	1.01	0.99	1.00	1.02	1.03	1.02	1.05	1.38			3.02	3.01	3.10
	velocity			5.10	9.05	9.14	9.05	9.23	9.14	8.96	8.87	8.96	8.70	10.16	8.33		9.08	9.11	8.85

2020 Shimane High School Championships (Honshu, JPN)

FINAL

INAL	date	11-Jul-20	wind	-0.1 m/s	Shimizu (2020) - shimane high school championships 2020 - biomechanics data analysis														
			H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10	
堀 汐音	time		2.73	3.92	5.10	6.27	7.47	8.68	9.92	11.18	12.50	13.80		15.47	4 / 1				
	reaction time			1.19	1.18	1.17	1.20	1.21	1.24	1.26	1.32	1.30	1.67				3.54	3.65	3.88
	velocity		5.03	7.68	7.75	7.81	7.62	7.55	7.37	7.25	6.92	7.03	8.40	7.11			7.75	7.51	7.07

山本 俊介	time	2.83	4.03	5.23	6.45	7.70	8.93	10.18	11.47	12.75	14.02		15.66	1 / 2			
	reaction time	interval	1.20	1.20	1.22	1.25	1.23	1.25	1.29	1.28	1.27	1.64			3.62	3.73	3.84
	velocity	4.85	7.62	7.62	7.49	7.31	7.43	7.31	7.09	7.14	7.20	8.55	7.02		7.57	7.35	7.14
外野 晴琉	time	2.85	4.12	5.38	6.62	7.88	9.17	10.45	11.75	13.05	14.37		16.11	9 / 3			
	reaction time	interval	1.27	1.26	1.24	1.26	1.29	1.28	1.30	1.30	1.32	1.74			3.77	3.83	3.92
	velocity	4.81	7.20	7.25	7.37	7.25	7.09	7.14	7.03	7.03	6.92	8.06	6.83		7.27	7.16	6.99
秦 伸成	time	2.88	4.13	5.38	6.62	7.88	9.17	10.45	11.75	13.05	14.37		16.22	7 / 4			
	reaction time	interval	1.25	1.25	1.24	1.26	1.29	1.28	1.30	1.30	1.32	1.85			3.74	3.83	3.92
	velocity	4.76	7.31	7.31	7.37	7.25	7.09	7.14	7.03	7.03	6.92	7.58	6.78		7.33	7.16	6.99
圓山 天志	time	2.83	4.08	5.32	6.53	7.82	9.15	10.52	11.83	13.18	14.58		16.36	5 / 5			
	reaction time	interval	1.25	1.24	1.21	1.29	1.33	1.37	1.31	1.35	1.40	1.78			3.70	3.99	4.06
	velocity	4.85	7.31	7.37	7.55	7.09	6.87	6.67	6.98	6.77	6.53	7.88	6.72		7.41	6.87	6.75

2020 USATF National Indoor Championships (Albuquerque, NM) (Altitude)

FINAL

date 15-Feb-20

CAA Hurdle Development (2020)

		H1	H2	H3	H4	H5	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
Mallett, Aaron (USA) (1994)	time	2.55	3.58	4.62	5.61	6.62		7.54	3 / 1			
	reaction time 0.160	interval	1.03	1.04	0.99	1.01	0.92	PB		3.06		
	velocity	5.38	8.87	8.79	9.23	9.05	10.57	7.96		8.96		
Eaton, Jarret (USA) (1989)	time	2.55	3.57	4.60	5.62	6.65		7.57	4 / 2			
	reaction time 0.155	interval	7 Steps	1.02	1.03	1.02	0.92			3.07		
	velocity	5.38	8.96	8.87	8.96	8.87	10.57	7.93		8.93		
Ames, Brendan (USA) (1988)	time	2.58	3.63	4.67	5.72	6.74		7.65	5 / 3			
	reaction time 0.145	interval	7 Steps	1.05	1.04	1.05	0.91			3.14		
	velocity	5.32	8.70	8.79	8.70	8.96	10.68	7.84		8.73		
Zallow, Chad (USA) (1997)	time	2.53	3.59	4.65	5.70	6.75		7.65	2 / 4			
	reaction time 0.151	interval	1.06	1.06	1.05	1.05	0.90			3.17		
	velocity	5.42	8.62	8.62	8.70	8.70	10.80	7.84		8.65		
Harris, Aleec (USA) (1990)	time	2.60	3.63	4.70	5.71	6.75		7.67	7 / 5			
	reaction time 0.128	interval	1.03	1.07	1.01	1.04	0.92			3.11		
	velocity	5.28	8.87	8.54	9.05	8.79	10.57	7.82		8.82		
Dickson, Michael (USA) (1997)	time	2.58	3.65	4.72	5.74	6.79		7.69	6 / 6			
	reaction time 0.171	interval	1.07	1.07	1.02	1.05	0.90			3.16		
	velocity	5.32	8.54	8.54	8.96	8.70	10.80	7.80		8.68		
Fontenot, Ryan (USA) (1986)	time	2.67	3.73	4.80	5.84	6.87		7.84	8 / 7			
	reaction time 0.170	interval	1.06	1.07	1.04	1.03	0.97			3.17		
	velocity	5.14	8.62	8.54	8.79	8.87	10.02	7.65		8.65		
Moore, Isaiah (USA) (1996)	time	2.65	3.77	4.84	5.91	6.97		7.93	1 / 8			
	reaction time 0.139	interval	1.12	1.07	1.07	1.06	0.96			3.26		
	velocity	5.18	8.16	8.54	8.54	8.62	10.13	7.57		8.41		

Semi-Final 2

date 15-Feb-20

CAA Hurdle Development (2020)

		H1	H2	H3	H4	H5	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
Ames, Brendan (USA) (1988)	time	2.58	3.65	4.69	5.70	6.75		7.65	3 / 1			
	reaction time 0.174	interval	7 Steps	1.07	1.04	1.01	0.90			3.12		
	velocity	5.32	8.54	8.79	9.05	8.70	10.80	7.84		8.79		
Dickson, Michael (USA) (1997)	time	2.57	3.62	4.69	5.74	6.79		7.69	5 / 2			
	reaction time 0.126	interval	1.05	1.07	1.05	1.05	0.90			3.17		
	velocity	5.34	8.70	8.54	8.70	8.70	10.80	7.80		8.65		
Zallow, Chad (USA) (1997)	time	2.55	3.60	4.67	5.72	6.80		7.72	4 / 3			
	reaction time 0.156	interval	1.05	1.07	1.05	1.08	0.92			3.17		
	velocity	5.38	8.70	8.54	8.70	8.46	10.57	7.77		8.65		
Moore, Isaiah (USA) (1996)	time	2.63	3.72	4.79	5.84	6.89		7.83	6 / 4			
	reaction time 0.161	interval	1.09	1.07	1.05	1.05	0.94			3.21		
	velocity	5.22	8.39	8.54	8.70	8.70	10.34	7.66		8.54		
Graham, Javon (USA)	time	2.60	3.70	4.77	5.83	6.91		7.84	2 / 5			
	reaction time 0.158	interval	1.10	1.07	1.06	1.08	0.93	PB		3.23		
	velocity	5.28	8.31	8.54	8.62	8.46	10.45	7.65		8.49		
Wilt, Caleb (USA) (1997)	time	2.58	3.67	4.75	5.82	6.89		7.85	7 / 6			
	reaction time 0.112	interval	7 Steps	1.09	1.08	1.07	0.96			3.24		

	velocity	5.32	8.39	8.46	8.54	8.54	10.13	7.64	8.46	
Fareed, Naim (USA) (1996)	time	2.63	3.72	4.80	5.86	6.96		7.95	1 / 7	
reaction time	0.136	interval	1.09	1.08	1.06	1.10	0.99			3.23
	velocity	5.22	8.39	8.46	8.62	8.31	9.82	7.55		8.49
Tawiah, Casimir (USA) (1997)	time	2.68	3.80	4.92	6.00	7.15		8.09	8 / 8	
reaction time	0.162	interval	7 Steps	1.12	1.12	1.08	0.94			3.32
	velocity	5.12	8.16	8.16	8.46	7.95	10.34	7.42		8.26

Semi-Final 1

date 15-Feb-20

CAA Hurdle Development (2020)

		H1	H2	H3	H4	H5	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
Eaton, Jarret (USA) (1989)	time	2.58	3.62	4.63	5.65	6.70		7.63	4 / 1			
reaction time	0.148	interval	7 Steps	1.04	1.01	1.02	0.93			3.07		
	velocity	5.32	8.79	9.05	8.96	8.70	10.45	7.86		8.93		
Mallett, Aaron (USA) (1994)	time	2.57	3.62	4.67	5.68	6.72		7.65	5 / 2			
reaction time	0.149	interval		1.05	1.05	1.01	0.93			3.11		
	velocity	5.34	8.70	8.70	9.05	8.79	10.45	7.84		8.82		
Harris, Aleec (USA) (1990)	time	2.63	3.70	4.74	5.77	6.84		7.75	6 / 3			
reaction time	0.145	interval		1.07	1.04	1.03	0.91			3.14		
	velocity	5.22	8.54	8.79	8.87	8.54	10.68	7.74		8.73		
Fontenot, Ryan (USA) (1986)	time	2.67	3.73	4.79	5.82	6.85		7.79	3 / 4			
reaction time	0.151	interval		1.06	1.06	1.03	0.94			3.15		
	velocity	5.14	8.62	8.62	8.87	8.87	10.34	7.70		8.70		
Moore, Titus (USA) (1999)	time	2.66	3.77	4.84	5.90	6.97		7.87	7 / 5			
reaction time	0.179	interval		1.11	1.07	1.06	0.90			3.24		
	velocity	5.16	8.23	8.54	8.62	8.54	10.80	7.62		8.46		
Tait, Emmanuel (USA)	time	2.63	3.75	4.85	5.92	7.04		7.99	2 / 6			
reaction time	0.158	interval		1.12	1.10	1.07	0.95			3.29		
	velocity	5.22	8.16	8.31	8.54	8.16	10.23	7.51		8.33		
Anderson, Joseph (USA) (1999)	time	2.68	3.78	4.88	5.97	7.07		8.03	1 / 7			
reaction time	0.171	interval		1.10	1.10	1.09	0.96			3.29		
	velocity	5.12	8.31	8.31	8.39	8.31	10.13	7.47		8.33		
Prescott, Matt (USA) (1984)	time	2.72	3.83	4.92	6.05	7.17		8.19	8 / 8			
reaction time	0.159	interval		1.11	1.09	1.13	1.02			3.33		
	velocity	5.04	8.23	8.39	8.09	8.16	9.53	7.33		8.23		

Heat 3

date 14-Feb-20

CAA Hurdle Development (2020)

		H1	H2	H3	H4	H5	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
Mallett, Aaron (USA) (1994)	time	2.62	3.68	4.74	5.75	6.79		7.70	4 / 1			
reaction time	0.196	interval	1.06	1.06	1.01	1.04	0.91			3.13		
	velocity	5.24	8.62	8.62	9.05	8.79	10.68	7.79		8.76		
Ames, Brendan (USA) (1988)	time	2.60	3.69	4.74	5.75	6.81		7.71	3 / 2			
reaction time	0.183	interval	7 Steps	1.09	1.05	1.01	0.90			3.15		
	velocity	5.28	8.39	8.70	9.05	8.62	10.80	7.78		8.70		
Harris, Aleec (USA) (1990)	time	2.62	3.67	4.72	5.77	6.84		7.77	5 / 3			
reaction time	0.157	interval		1.05	1.05	1.05	0.93			3.15		
	velocity	5.24	8.70	8.70	8.70	8.54	10.45	7.72		8.70		
Graham, Javon (USA)	time	2.65	3.77	4.82	5.92	6.97		7.92	8 / 4			
reaction time	0.166	interval		1.12	1.05	1.10	0.95			3.27		
	velocity	5.18	8.16	8.70	8.31	8.70	10.23	7.58		8.39		
Anderson, Joseph (USA) (1999)	time	2.67	3.79	4.89	5.96	7.06		7.97	2 / 5			
reaction time	0.160	interval		1.12	1.10	1.07	0.91			3.29		
	velocity	5.14	8.16	8.31	8.54	8.31	10.68	7.53		8.33		
Tawiah, Casimir (USA) (1997)	time	2.70	3.80	4.90	5.98	7.07		8.03	7 / 6			
reaction time	0.168	interval	7 Steps	1.10	1.10	1.08	0.96			3.28		
	velocity	5.08	8.31	8.31	8.46	8.39	10.13	7.47		8.36		
Snell, Errik (USA)	time	2.65	3.77	4.90	5.98	7.11		8.08	6 / 7			
reaction time	0.159	interval		1.12	1.13	1.08	0.97			3.33		
	velocity	5.18	8.16	8.09	8.46	8.09	10.02	7.43		8.23		

Heat 2

date 14-Feb-20

CAA Hurdle Development (2020)

H1 H2 H3 H4 H5 Run-In Official Time Lane / Place H1-H4 H4-H7 H7-H10

Eaton, Jarret (USA) (1989)	time	2.62	3.67	4.69	5.74	6.81		7.74	3 / 1		
	reaction time	0.154	interval	7 Steps	1.05	1.02	1.05	1.07		3.12	
	velocity	5.24	8.70	8.96	8.70	8.54		10.45	7.75	8.79	
Fontenot, Ryan (USA) (1986)	time	2.66	3.74	4.79	5.84	6.89		7.81	4 / 2		
	reaction time	0.160	interval		1.08	1.05	1.05	1.05		3.18	
	velocity	5.16	8.46	8.70	8.70	8.70		10.57	7.68	8.62	
Moore, Isaiah (USA) (1996)	time	2.64	3.74	4.79	5.82	6.89		7.82	1 / 3		
	reaction time	0.154	interval		1.10	1.05	1.03	1.07		3.18	
	velocity	5.20	8.31	8.70	8.87	8.54		10.45	7.67	8.62	
Moore, Titus (USA) (1999)	time	2.64	3.73	4.79	5.85	6.92		7.84	6 / 4		
	reaction time	0.159	interval		1.09	1.06	1.06	1.07	0.92	PB	3.21
	velocity	5.20	8.39	8.62	8.62	8.54		10.57	7.65	8.54	
Rooney, Taylor (USA) (1999)	time	2.70	3.82	4.89	5.99	7.09		8.06	5 / 5		
	reaction time	0.219	interval	7 Steps	1.12	1.07	1.10	1.10	0.97		3.29
	velocity	5.08	8.16	8.54	8.31	8.31		10.02	7.44	8.33	
Kimball, Patrick (USA) (1998)	time	2.68	3.80	4.92	5.99	7.09		8.09	7 / 6		
	reaction time	0.156	interval		1.12	1.12	1.07	1.10	1.00		3.31
	velocity	5.12	8.16	8.16	8.54	8.31		9.72	7.42	8.28	

Heat 1

Heat 1		date 14-Feb-20		CAA Hurdle Development (2020)									
		H1	H2	H3	H4	H5	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10	
Zallow, Chad (USA) (1997)	time	2.57	3.62	4.67	5.72	6.81		7.67	8 / 1				
	reaction time	0.168	interval	1.05	1.05	1.05	1.09	0.86		3.15			
	velocity	5.34	8.70	8.70	8.70	8.39		11.30	7.82		8.70		
Dickson, Michael (USA) (1997)	time	2.60	3.67	4.72	5.75	6.79		7.68	3 / 2				
	reaction time	0.159	interval	1.07	1.05	1.03	1.04	0.89		3.15			
	velocity	5.28	8.54	8.70	8.87	8.79		10.92	7.81		8.70		
Wilt, Caleb (USA) (1997)	time	2.59	3.65	4.74	5.80	6.87		7.82	2 / 3				
	reaction time	0.144	interval	7 Steps	1.06	1.09	1.06	1.07	0.95	=PB	3.21		
	velocity	5.30	8.62	8.39	8.62	8.54		10.23	7.67		8.54		
Tait, Emmanuel (USA)	time	2.64	3.72	4.84	5.87	6.96		7.89	1 / 4				
	reaction time	0.146	interval	1.08	1.12	1.03	1.09	0.93	PB	3.23			
	velocity	5.20	8.46	8.16	8.87	8.39		10.45	7.60		8.49		
Fareed, Naim (USA) (1996)	time	2.64	3.70	4.79	5.86	6.99		7.94	7 / 5				
	reaction time	0.153	interval	1.06	1.09	1.07	1.13	0.95		3.22			
	velocity	5.20	8.62	8.39	8.54	8.09		10.23	7.56		8.52		
Prescott, Matt (USA) (1984)	time	2.68	3.79	4.89	5.96	7.06		8.01	4 / 6				
	reaction time	0.161	interval	1.11	1.10	1.07	1.10	0.95		3.28			
	velocity	5.12	8.23	8.31	8.54	8.31		10.23	7.49		8.36		
Golaube, Kamal (USA)	time	2.63	3.75	4.89	6.00	7.12		8.16	5 / 7				
	reaction time	0.130	interval	1.12	1.14	1.11	1.12	1.04		3.37			
	velocity	5.22	8.16	8.02	8.23	8.16		9.35	7.35		8.14		

2020 Don Kirby Elite Invitational (Albuquerque, NM) (Altitude)

FINAL

		date 14-Feb-20					CAA Hurdle Development (2020)					
		H1	H2	H3	H4	H5	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
Brixey, Sam (USA) (1998)	time	2.55	3.59	4.64	5.71	6.76		7.72	4 / 1			
	reaction time	interval	7 steps	1.04	1.05	1.07	1.05	0.96		3.16		
	velocity	5.38	8.79	8.70	8.54	8.70		10.13	7.77		8.68	
Zhu Shenglong (CHN) (2000)	time	2.69	3.77	4.86	5.89	6.96		7.88	5 / 2			
	reaction time	interval		1.08	1.09	1.03	1.07	0.92		3.20		
	velocity	5.10	8.46	8.39	8.87	8.54		10.57	7.61		8.57	
Smith, Shakeem (BAH) (1997)	time	2.65	3.75	4.84	5.91	6.99		7.91	6 / 3			
	reaction time	interval		1.10	1.09	1.07	1.08	0.92		3.26		
	velocity	5.18	8.31	8.39	8.54	8.46		10.57	7.59		8.41	
Foster, Jasher (USA) (1998)	time	2.64	3.74	4.82	5.89	6.97		7.95	7 / 4			
	reaction time	interval		1.10	1.08	1.07	1.08	0.98		3.25		
	velocity	5.20	8.31	8.46	8.54	8.46		9.92	7.55		8.44	
Jiang Fan (CHN) (1989)	time	2.70	3.82	4.90	6.01	7.11		8.14	2 / 6			
	reaction time	interval		1.12	1.08	1.11	1.10	1.03		3.31		
	velocity	5.08	8.16	8.46	8.23	8.31		9.44	7.37		8.28	

Semi-Final 3

date 14-Feb-20

CAA Hurdle Development (2020)

		H1	H2	H3	H4	H5	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
Jiang Fan (CHN) (1989)	time	2.68	3.80	4.85	5.94	7.00		7.96	1 / 2			
	reaction time		1.12	1.05	1.09	1.06	0.96			3.26		
	velocity	5.12	8.16	8.70	8.39	8.62	10.13	7.54		8.41		

Semi-Final 2

date 14-Feb-20

CAA Hurdle Development (2020)

		H1	H2	H3	H4	H5	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
Zhu Shenglong (CHN) (2000)	time	2.70	3.80	4.87	5.90	6.97		7.89	2 / 1			
	reaction time		1.10	1.07	1.03	1.07	0.92			3.20		
	velocity	5.08	8.31	8.54	8.87	8.54	10.57	7.60		8.57		

Heat 3

date 14-Feb-20

CAA Hurdle Development (2020)

		H1	H2	H3	H4	H5	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
Zhu Shenglong (CHN) (2000)	time	2.59	3.64	4.74	5.79	6.84		7.92	2 / 1			
	reaction time		1.05	1.10	1.05	1.05	1.08			3.20		
	velocity	5.30	8.70	8.31	8.70	8.70	9.00	7.58		8.57		

Heat 2

date 14-Feb-20

CAA Hurdle Development (2020)

		H1	H2	H3	H4	H5	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
Jiang Fan (CHN) (1989)	time	2.55	3.65	4.72	5.79	6.87		7.94	2 / 1			
	reaction time		1.10	1.07	1.07	1.08	1.07			3.24		
	velocity	5.38	8.31	8.54	8.54	8.46	9.08	7.56		8.46		

2020 Millrose Games (New York, NY) (TV Analysis)**FINAL**

date 08-Feb-20

Henson (2020) - Athlete First: 2020 year end hurdle report

		H1	H2	H3	H4	H5	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
Roberts, Daniel (USA) (1998)	time	2.47	3.54	4.57	5.61	6.67		7.64	5 / 1			
	reaction time 0.194		7 steps	1.07	1.03	1.04	0.97			3.14		
	velocity	5.55	8.54	8.87	8.79	8.62	10.02	7.85		8.73		
Xie Wenjun (CHN) (1990)	time	2.53	3.60	4.64	5.67	6.70		7.69	6 / 2			
	reaction time 0.156		7 steps	1.07	1.04	1.03	0.99			3.14		
	velocity	5.42	8.54	8.79	8.87	8.87	9.82	7.80		8.73		
Zallow, Chad (USA) (1997)	time	2.50	3.55	4.63	5.67	6.72		7.69	8 / 3			
	reaction time 0.172		1.05	1.08	1.04	1.05	0.97			3.17		
	velocity	5.49	8.70	8.46	8.79	8.70	10.02	7.80		8.65		
Harris, Aleec (USA) (1990)	time	2.56	3.64	4.70	5.73	6.80		7.77	4 / 4			
	reaction time 0.167		1.08	1.06	1.03	1.07	0.97			3.17		
	velocity	5.36	8.46	8.62	8.87	8.54	10.02	7.72		8.65		
Dickson, Michael (USA) (1997)	time	2.60	3.67	4.74	5.80	6.84		7.83	7 / 5			
	reaction time 0.231		1.07	1.07	1.06	1.04	0.99			3.20		
	velocity	5.28	8.54	8.54	8.62	8.79	9.82	7.66		8.57		
Lovett, Eddie (ISV) (1994)	time	2.57	3.67	4.74	5.77	6.87		7.85	3 / 6			
	reaction time 0.180		7 steps	1.10	1.07	1.03	0.98			3.20		
	velocity	5.34	8.31	8.54	8.87	8.31	9.92	7.64		8.57		
Hairston, Max (USA) (1994)	time	2.57	3.67	4.77	5.81	6.87		7.87	2 / 7			
	reaction time 0.206		7 steps	1.10	1.10	1.04	1.00			3.24		
	velocity	5.34	8.31	8.31	8.79	8.62	9.72	7.62		8.46		

2020 Mountain T's Invitational (Flagstaff AZ) (Altitude)**FINAL**

date 01-Feb-20

CAA Hurdle Development (2020)

		H1	H2	H3	H4	H5	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
Zhu Shenglong (CHN) (2000)	time	2.64	3.72	4.76	5.79	6.82		7.74	6 / 1			
	reaction time		1.08	1.04	1.03	1.03	0.92	PB		3.15		
	velocity	5.20	8.46	8.79	8.87	8.87	10.57	7.75		8.70		
Xie Wenjun (CHN) (1990)	time	2.65	3.70	4.75	5.78	6.82		7.75	4 / 2			
	reaction time		7 steps	1.05	1.05	1.03	0.93			3.13		
	velocity	5.18	8.70	8.70	8.87	8.79	10.45	7.74		8.76		
Charles, Jordan (USA) (1995)	time	2.60	3.70	4.76	5.80	6.85		7.78	5 / 3			
	reaction time		7 steps	1.10	1.06	1.04	0.93	PB		3.20		
	velocity	5.28	8.31	8.62	8.79	8.70	10.45	7.71		8.57		
Hairston, Max (USA) (1994)	time	2.65	3.75	4.82	5.92	6.99		7.91	3 / 4			
	reaction time		7 steps	1.10	1.07	1.10	0.92			3.27		
	velocity	5.18	8.31	8.54	8.31	8.54	10.57	7.59		8.39		
Jiang Fan (CHN) (1989)	time	2.72	3.81	4.89	5.98	7.04		7.97	2 / 5			

reaction time	interval	1.09	1.08	1.09	1.06	0.93	3.26					
	velocity	5.04	8.39	8.46	8.39	8.62	10.45	7.53	8.41			
Heat 3												
	date	01-Feb-20					CAA Hurdle Development (2020)					
		H1	H2	H3	H4	H5	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
Xie Wenjun (CHN) (1990)	time	2.64	3.74	4.79	5.83	6.87		7.80	4 / 1			
	reaction time	interval	7 steps	1.10	1.05	1.04	1.04	0.93		3.19		
		velocity	5.20	8.31	8.70	8.79	8.79	10.45	7.69	8.60		

Heat 2		date 01-Feb-20					CAA Hurdle Development (2020)						
		H1	H2	H3	H4	H5		Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
Hairston, Max (USA) (1994)	time	2.66	3.75	4.86	5.94	7.03			7.96	4 / 1			
	reaction time	7 steps	1.09	1.11	1.08	1.09		0.93			3.28		
	velocity	5.16	8.39	8.23	8.46	8.39		10.45	7.54			8.36	
Jiang Fan (CHN) (1989)	time	2.72	3.82	4.91	5.97	7.02			7.97	5 / 2			
	reaction time		1.10	1.09	1.06	1.05		0.95			3.25		
	velocity	5.04	8.31	8.39	8.62	8.70		10.23	7.53			8.44	

Heat 1		date 01-Feb-20					CAA Hurdle Development (2020)					
		H1	H2	H3	H4	H5	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
Zhu Shenglong (CHN) (2000)	time	2.67	3.75	4.82	5.86	6.90		7.83	5 / 2			
	reaction time		1.08	1.07	1.04	1.04	0.93	PB		3.19		
	velocity	5.14	8.46	8.54	8.79	8.79	10.45	7.66		8.60		

2020 Leonard Hilton Memorial Invitational (Houston, TX)

FINAL		date 17-Jan-20					CAA Hurdle Development (2020)						
		H1	H2	H3	H4	H5	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10	
Akindede, Oladayo (USA) (1996)	time	2.53	3.60	4.64	5.67	6.71		7.78	5 / 1				
	reaction time	interval	7 steps	1.07	1.04	1.03	1.04	1.07			3.14		
	velocity	5.42	8.54	8.79	8.87	8.79	9.08	PB			7.71	8.73	
Burt, John (USA) (1997)	time	2.49	3.58	4.65	5.71	6.79		7.88	6 / 2				
	reaction time	interval		1.09	1.07	1.06	1.08	1.09			3.22		
	velocity	5.51	8.39	8.54	8.62	8.46	8.92	7.61			8.52		
Zhu Shenglong (CHN) (2000)	time	2.60	3.67	4.76	5.81	6.87		7.91	3 / 3				
	reaction time	interval		1.07	1.09	1.05	1.06	1.04			PB	3.21	
	velocity	5.28	8.54	8.39	8.70	8.62	9.35	7.59			8.54		

Heast 3		date 17-Jan-20					CAA Hurdle Development (2020)						
		H1	H2	H3	H4	H5	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10	
Zhu Shenglong (CHN) (2000)	time	2.62	3.71	4.79	5.84	6.92		8.00	5 / 1				
	reaction time		1.09	1.08	1.05	1.08	1.08			3.22			
	interval												
	velocity	5.24	8.39	8.46	8.70	8.46	9.00	7.50		8.52			

2019 Japanese National U18/U20 Championships (Hiroshima, JPN)

U20 FINAL (99cm)		date	18-Oct-19	wind	-0.6 m/s			Kijima (2019) - research on athlete performance and technique- 2019 data book												
				H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10	
Muratake, Rashiddo (JPN) (2000)				time	2.60	3.64	4.70	5.74	6.78	7.83	8.87	9.94	11.01	12.08		13.54	5 / 1			
reaction time				0.144	interval	1.04	1.06	1.04	1.04	1.05	1.04	1.07	1.07	1.46				3.14	3.13	3.21
velocity					5.28	8.79	8.62	8.79	8.79	8.70	8.79	8.54	8.54	8.54	9.60	8.12		8.73	8.76	8.54
Kurokawa, Kazuki (JPN) (2001)				time	2.59	3.68	4.74	5.79	6.85	7.90	8.99	10.08	11.17	12.25		13.73	6 / 2			
reaction time				0.131	interval	7 steps	1.09	1.06	1.05	1.06	1.09	1.09	1.09	1.08	1.48			3.20	3.20	3.26
velocity					5.30	8.39	8.62	8.70	8.62	8.70	8.39	8.39	8.39	8.46	9.47	8.01		8.57	8.57	8.41
Furuya, Shinta (JPN) (2000)				time	2.60	3.69	4.75	5.79	6.85	7.91	8.99	10.06	11.17	12.28		13.79	7 / 3			
reaction time				0.122	interval	1.09	1.06	1.04	1.06	1.06	1.08	1.07	1.11	1.11	1.51			3.19	3.20	3.29
velocity					5.28	8.39	8.62	8.79	8.62	8.62	8.46	8.54	8.23	8.23	9.28	7.98		8.60	8.57	8.33

U18 FINAL (99cm)		date	18-Oct-19	wind	-0.4 m/s										Kijima (2019) - research on athlete performance and technique- 2019 data book						
				H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10		
Takahashi, Harumasa (JPN) (2000)		time		2.62	3.73	4.80	5.85	6.90	7.92	8.97	10.01	11.08	12.15		13.60	6 / 1					
reaction time		0.131	interval		1.11	1.07	1.05	1.05	1.02	1.05	1.04	1.07	1.07	1.45			3.23	3.12	3.18		
			velocity	5.24	8.23	8.54	8.70	8.70	8.96	8.70	8.79	8.54	8.54	9.67	8.09		8.49	8.79	8.62		
Kondo, Mizuki (JPN) (2003)		time		2.64	3.70	4.76	5.80	6.83	7.88	8.92	9.97	11.03	12.13		13.60	3 / 2					
reaction time		0.152	interval		1.06	1.06	1.04	1.03	1.05	1.04	1.05	1.06	1.10	1.47			3.16	3.12	3.21		
			velocity	5.20	8.62	8.62	8.79	8.87	8.70	8.79	8.70	8.62	8.31	9.54	8.09		8.68	8.79	8.54		
片岡 巧 (JPN)		time		2.64	3.74	4.82	5.93	6.99	8.07	9.16	10.25	11.34	12.46		13.94	8 / 3					
reaction time		0.115	interval		1.10	1.08	1.11	1.06	1.08	1.09	1.09	1.09	1.12	1.48			3.29	3.23	3.30		
			velocity	5.20	8.31	8.46	8.23	8.62	8.46	8.39	8.39	8.39	8.16	9.47	7.89		8.33	8.49	8.31		

2019 National Sports Festival (Hitachinaka, JPN) (99cm)

FINAL

date 06-Oct-19 wind 2.7 m/s

Enomoto (2019) - research on athlete performance and technique- 2019 data book

			H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
Muratake, Rashiddo (JPN) (200)	time		2.60	3.66	4.69	5.73	6.75	7.79	8.83	9.88	10.92	11.97		13.37	4 / 1			
	reaction time	0.160		1.06	1.03	1.04	1.02	1.04	1.04	1.05	1.04	1.05	1.40			3.13	3.10	3.14
	interval															8.76	8.85	8.73
Kondo, Mizuki (JPN) (2003)	time		2.63	3.68	4.72	5.73	6.76	7.78	8.82	9.87	10.92	11.98		13.45	6 / 2			
	reaction time	0.142		1.05	1.04	1.01	1.03	1.02	1.04	1.05	1.05	1.06	1.47			3.10	3.09	3.16
	interval															8.85	8.87	8.68
Kurokawa, Kazuki (JPN) (2001)	time		2.61	3.69	4.75	5.78	6.80	7.83	8.88	9.92	10.97	12.02		13.49	9 / 3			
	reaction time	0.157		1.08	1.06	1.03	1.02	1.03	1.05	1.04	1.05	1.05	1.47			3.17	3.10	3.14
	interval															8.65	8.85	8.73
Kikuchi, Rintaro (JPN) (2001)	time		2.57	3.62	4.70	5.76	6.83	7.85	8.92	9.94	11.02	12.10		13.54	2 / 4			
	reaction time	0.132		1.05	1.08	1.06	1.07	1.02	1.07	1.02	1.08	1.08	1.44			3.19	3.16	3.18
	interval															8.60	8.68	8.62
Miyazaki, Takumi (JPN) (2001)	time		2.64	3.72	4.79	5.86	6.89	7.94	8.97	10.03	11.08	12.15		13.60	5 / 5			
	reaction time	0.150		1.08	1.07	1.07	1.03	1.05	1.03	1.06	1.05	1.07	1.45			3.22	3.11	3.18
	interval															8.52	8.82	8.62
Machi, Ryota (JPN) (2001)	time		2.64	3.74	4.80	5.86	6.93	7.95	9.01	10.08	11.16	12.26		13.72	7 / 6			
	reaction time	0.145		1.10	1.06	1.06	1.07	1.02	1.06	1.07	1.08	1.10	1.46			3.22	3.15	3.25
	interval															8.52	8.70	8.44
Ono, Kazuto (JPN) (2001)	time		2.68	3.76	4.84	5.91	6.97	8.03	9.10	10.16	11.24	12.32		13.79	3 / 7			
	reaction time	0.157		1.08	1.08	1.07	1.06	1.06	1.07	1.06	1.08	1.08	1.47			3.23	3.19	3.22
	interval															8.49	8.60	8.52
Sugaya, Takumi (JPN) (2001)	time		2.68	3.76	4.82	5.88	6.93	7.98	9.04	10.13	11.20	12.28		13.79	8 / 8			
	reaction time	0.150		1.08	1.06	1.06	1.05	1.05	1.06	1.09	1.07	1.08	1.51			3.20	3.16	3.24
	interval															8.57	8.68	8.46

2019 IAAF World Championships (Doha, QAT)

Decathlon - Heat 3

date 03-Oct-19 wind 0.8 m/s

Henson (2021) - Athlete First: 2019 year end hurdle report

			H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
Warner, Damion (CAN) (1989)	time		2.48	3.56	4.58	5.62	6.66	7.70	8.76	9.82	10.88	11.96		13.56	5 / 1			
	reaction time	0.143		7 steps	1.08	1.02	1.04	1.04	1.06	1.06	1.06	1.08	1.60			3.14	3.14	3.20
	interval															8.73	8.73	8.57
	velocity		5.53	8.46	8.96	8.79	8.79	8.79	8.62	8.62	8.62	8.46	8.76	8.11				

FINAL

date 02-Oct-19 wind 0.6 m/s

CAA Hurdle Development (2019)

			H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
Holloway, Grant (USA) (1997)	time		2.46	3.46	4.44	5.41	6.41	7.40	8.42	9.44	10.50	11.60		13.10	6 / 1			
	reaction time	0.158		7 steps	1.00	0.98	0.97	1.00	0.99	1.02	1.06	1.10	1.50			2.95	3.01	3.18
	interval															9.29	9.11	8.62
Shubenkov, Sergey (ANA) (199)	time		2.62	3.66	4.66	5.66	6.65	7.66	8.67	9.68	10.74	11.79		13.15	9 / 2			
	reaction time	0.150		7 steps	1.04	1.00	1.00	0.99	1.01	1.01	1.06	1.05	1.36			3.04	3.01	3.12
	interval															9.02	9.11	8.79
Martinot-Lagarde, Pascal (FRA)	time		2.55	3.59	4.61	5.61	6.61	7.63	8.64	9.65	10.69	11.75		13.18	7 / 3			
	reaction time	0.161		7 steps	1.04	1.02	1.00	1.00	1.02	1.01	1.04	1.06	1.43			3.06	3.03	3.11
	interval															8.96	9.05	8.82
Ortega, Orlando (ESP) (1991)	time		2.58	3.60	4.65	5.63	6.66	7.65	8.63	9.66	10.69	11.79		13.30	5 / 3			
	reaction time	0.130		7 steps	1.02	1.05	0.98	1.03	0.99	1.03	1.03	1.10	1.51			3.05	3.00	3.16
	interval															8.99	9.14	8.68
Xie Wenjun (CHN) (1990)	time		2.59	3.64	4.66	5.68	6.69	7.71	8.73	9.76	10.80	11.86		13.29	2 / 5			
	reaction time	0.140		7 steps	1.05	1.02	1.02	1.01	1.02	1.03	1.04	1.06	1.43			3.09	3.05	3.13
	interval															8.87	8.99	8.76
Brathwaite, Shane (BAR) (1990)	time		2.62	3.68	4.78	5.85	6.91	7.96	8.99	10.04	11.09	12.15		13.61	1 / 6			
	reaction time	0.149		1.06	1.10	1.07	1.06	1.05	1.03	1.05	1.05	1.06	1.46			3.23	3.14	3.16
	interval															8.49	8.73	8.68
Allen, Devon (USA) (1994)	time		2.58	3.67	4.73	5.77	6.80	7.82	8.88	9.94	11.05	12.19		13.70	3 / 7			
	reaction time	0.124		1.09	1.06	1.04	1.03	1.02	1.06	1.06	1.11	1.14	1.51			3.19	3.11	3.31
	interval															8.60	8.82	8.28
Trajkovic, Milan (CYP) (1992)	time		2.57	3.62	4.62	5.64	6.63	7.65	8.81	10.07	11.25	12.39		13.87	8 / 8			
	reaction time	0.150		7 steps	1.05	1.00	1.02	0.99	1.02	1.16	1.26	1.18	1.14	1.48		3.07	3.17	3.58
	interval															8.93	8.65	7.66

McLeod, Omar (JAM) (1994)	time	2.53	3.53	4.54	5.57	6.56	7.53	8.54	9.54	10.70	12.00		14.10	4 / --			
	reaction time	0.128	interval	1.00	1.01	1.03	0.99	0.97	1.01	1.00	1.16	1.30	2.10	DQ	3.04	2.97	3.46
	velocity		5.42	9.14	9.05	8.87	9.23	9.42	9.05	9.14	7.88	7.03	6.68	7.80	9.02	9.23	7.92

Semi-Final 3

date 02-Oct-19 wind 0.6 m/s

CAA Hurdle Development (2019)

		H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
Ortega, Orlando (ESP) (1991)	time	2.58	3.63	4.63	5.67	6.66	7.67	8.68	9.69	10.74	11.77		13.16	4 / 1			
	reaction time	0.131	interval	7 steps	1.05	1.00	1.04	0.99	1.01	1.01	1.05	1.03	1.39		3.09	3.01	3.09
	velocity		5.32	8.70	9.14	8.79	9.23	9.05	9.05	9.05	8.70	8.87	10.09	8.36	8.87	9.11	8.87
Trajkovic, Milan (CYP) (1992)	time	2.58	3.61	4.63	5.62	6.62	7.63	8.65	9.69	10.77	11.85		13.29	7 / 2			
	reaction time	0.162	interval	7 steps	1.03	1.02	0.99	1.00	1.01	1.02	1.04	1.08	1.44		3.04	3.03	3.20
	velocity		5.32	8.87	8.96	9.23	9.14	9.05	8.96	8.79	8.46	8.46	9.74	8.28	9.02	9.05	8.57
Alkana, Antonio (RSA) (1990)	time	2.58	3.61	4.67	5.69	6.73	7.77	8.81	9.87	10.94	12.03		13.47	5 / 3			
	reaction time	0.123	interval		1.03	1.06	1.02	1.04	1.04	1.06	1.07	1.09	1.44		3.11	3.12	3.22
	velocity		5.32	8.87	8.62	8.96	8.79	8.79	8.62	8.54	8.39	9.74	8.17		8.82	8.79	8.52
Douvalidis, Konstadinos (GRE)	time	2.65	3.71	4.75	5.77	6.81	7.86	8.92	9.99	11.05	12.13		13.54	9 / 4			
	reaction time	0.198	interval	7 steps	1.06	1.04	1.02	1.04	1.05	1.06	1.07	1.06	1.41		3.12	3.15	3.21
	velocity		5.18	8.62	8.79	8.96	8.79	8.70	8.62	8.54	8.62	8.46	9.94	8.12	8.79	8.70	8.54
Riley, Andrew (JAM) (1988)	time	2.57	3.60	4.66	5.68	6.76	7.82	8.88	9.94	11.02	12.13		13.57	8 / 5			
	reaction time	0.136	interval		1.03	1.06	1.02	1.08	1.06	1.06	1.08	1.11	1.44		3.11	3.20	3.25
	velocity		5.34	8.87	8.62	8.96	8.46	8.62	8.62	8.62	8.46	8.23	9.74	8.11	8.82	8.57	8.44
Takayama, Shunya (JPN) (1994)	time	2.54	3.58	4.58	5.60	6.61	7.67	8.86	9.99	11.07	12.16		13.58	6 / 6			
	reaction time	0.144	interval		1.04	1.00	1.02	1.01	1.06	1.19	1.13	1.08	1.09	1.42	3.06	3.26	3.30
	velocity		5.40	8.79	9.14	8.96	9.05	8.62	7.68	8.09	8.46	8.39	9.87	8.10	8.96	8.41	8.31
Belocian, Wilhem (FRA) (1995)	time	2.58	3.64	4.66	5.67	6.72	7.74	8.80	9.90	11.00	12.12		13.60	3 / 7			
	reaction time	0.133	interval	7 steps	1.06	1.02	1.01	1.05	1.02	1.06	1.10	1.10	1.12	1.48	3.09	3.13	3.32
	velocity		5.32	8.62	8.96	9.05	8.70	8.96	8.62	8.31	8.31	8.16	9.47	8.09	8.87	8.76	8.26
Hough, Nicholas (AUS) (1993)	time	2.66	3.72	4.77	5.78	6.84	7.90	8.93	10.00	11.06	12.15		13.61	2 / 8			
	reaction time	0.176	interval	7 steps	1.06	1.05	1.01	1.06	1.06	1.03	1.07	1.06	1.09	1.46	3.12	3.15	3.22
	velocity		5.16	8.62	8.70	9.05	8.62	8.62	8.87	8.54	8.62	8.39	9.60	8.08	8.79	8.70	8.52

Semi-Final 2

date 02-Oct-19 wind 0.9 m/s

CAA Hurdle Development (2019)

		H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
McLeod, Omar (JAM) (1994)	time	2.51	3.54	4.54	5.54	6.58	7.59	8.61	9.65	10.70	11.76		13.08	5 / 1			
	reaction time	0.141	interval		1.03	1.00	1.00	1.04	1.01	1.02	1.04	1.05	1.32		3.03	3.07	3.15
	velocity		5.47	8.87	9.14	9.14	8.79	9.05	8.96	8.79	8.70	8.62	10.62	8.41	9.05	8.93	8.70
Martinot-Lagarde, Pascal (FRA)	time	2.55	3.60	4.61	5.61	6.62	7.62	8.62	9.65	10.68	11.73		13.12	6 / 2			
	reaction time	0.163	interval	7 steps	1.05	1.01	1.00	1.01	1.00	1.03	1.03	1.05	1.39		3.06	3.01	3.11
	velocity		5.38	8.70	9.05	9.14	9.05	9.14	8.87	8.87	8.70	10.09	8.38		8.96	9.11	8.82
Xie Wenjun (CHN) (1990)	time	2.56	3.60	4.63	5.64	6.66	7.67	8.68	9.69	10.73	11.78		13.22	7 / 3			
	reaction time	0.137	interval	7 steps	1.04	1.03	1.01	1.02	1.01	1.01	1.04	1.05	1.44		3.08	3.04	3.10
	velocity		5.36	8.79	8.87	9.05	8.96	9.05	9.05	9.05	8.79	8.70	9.74	8.32	8.90	9.02	8.85
Allen, Devon (USA) (1994)	time	2.60	3.64	4.67	5.69	6.76	7.75	8.80	9.83	10.87	11.95		13.36	9 / 4			
	reaction time	0.119	interval		1.04	1.03	1.02	1.07	0.99	1.05	1.03	1.04	1.08	1.41	3.09	3.11	3.15
	velocity		5.28	8.79	8.87	8.96	8.54	9.23	8.70	8.87	8.79	8.46	9.94	8.23	8.87	8.82	8.70
Chen Keui-Ru (TPE) (1993)	time	2.53	3.59	4.63	5.66	6.70	7.74	8.82	9.89	10.96	12.05		13.52	8 / 5			
	reaction time	0.132	interval	7 steps	1.06	1.04	1.03	1.04	1.04	1.08	1.07	1.09	1.47		3.13	3.16	3.23
	velocity		5.42	8.62	8.79	8.87	8.79	8.79	8.46	8.54	8.54	8.39	9.54	8.14	8.76	8.68	8.49
Fofana, Hassane (ITA) (1992)	time	2.60	3.67	4.72	5.75	6.79	7.83	8.87	9.92	10.99	12.08		13.52	2 / 6			
	reaction time	0.155	interval	7 steps	1.07	1.05	1.03	1.04	1.04	1.05	1.07	1.09	1.44		3.15	3.12	3.21
	velocity		5.28	8.54	8.70	8.87	8.79	8.79	8.70	8.54	8.39	9.74	8.14		8.70	8.79	8.54
Joseph, Jason (SUI) (1998)	time	2.62	3.67	4.71	5.75	6.77	7.79	8.82	9.90	10.97	12.08		13.53	4 / 7			
	reaction time	0.158	interval	7 steps	1.05	1.04	1.04	1.02	1.02	1.03	1.08	1.07	1.11	1.45	3.13	3.07	3.26
	velocity		5.24	8.70	8.79	8.79	8.96	8.96	8.87	8.46	8.54	8.23	9.67	8.13	8.76	8.93	8.41
Chaverra, Joan Camilo (PUR) (1997)	time	2.60	3.66	4.71	5.74	6.77	7.81	8.85	9.93	11.06	12.19		13.76	3 / 8			
	reaction time	0.164	interval	7 steps	1.06	1.05	1.03	1.03	1.04	1.04	1.08	1.13	1.57		3.14	3.11	3.34
	velocity		5.28	8.62	8.70	8.87	8.87	8.79	8.79	8.46	8.09	8.09	8.93	7.99	8.73	8.82	8.21

Semi-Final 1

date 02-Oct-19 wind 1.1 m/s

CAA Hurdle Development (2019)

		H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
Holloway, Grant (USA) (1997)	time	2.50	3.50	4.46	5.43	6.40	7.39	8.40	9.42	10.48	11.62		13.10	5 / 1			
	reaction time	0.166	interval	7 steps	1.00	0.96	0.97	0.97	0.99	1.01	1.02	1.06	1.14	1.48	2.93	2.97	3.22
	velocity																

velocity	5.49	9.14	9.52	9.42	9.42	9.23	9.05	8.96	8.62	8.02	9.47	8.40			9.36	9.23	8.52
time	2.62	3.66	4.68	5.68	6.68	7.68	8.68	9.70	10.74	11.78		13.18	6 / 2				
interval	7 steps	1.04	1.02	1.00	1.00	1.00	1.00	1.02	1.04	1.04	1.40				3.06	3.00	3.10
velocity	5.24	8.79	8.96	9.14	9.14	9.14	9.14	8.96	8.79	8.79	10.01	8.35			8.96	9.14	8.85
time	2.58	3.62	4.66	5.68	6.70	7.74	8.80	9.88	10.94	12.04		13.48	2 / 3				
interval	7 steps	1.04	1.04	1.02	1.02	1.04	1.06	1.08	1.06	1.10	1.44				3.10	3.12	3.24
velocity	5.32	8.79	8.79	8.96	8.96	8.79	8.62	8.46	8.62	8.31	9.74	8.16			8.85	8.79	8.46
time	2.62	3.66	4.70	5.72	6.74	7.76	8.82	9.90	10.98	12.10		13.57	9 / 4				
interval		1.04	1.04	1.02	1.02	1.02	1.06	1.08	1.08	1.12	1.47				3.10	3.10	3.28
velocity	5.24	8.79	8.79	8.96	8.96	8.96	8.62	8.46	8.46	8.16	9.54	8.11			8.85	8.85	8.36
time	2.58	3.61	4.64	5.66	6.68	7.74	8.86	9.94	11.04	12.14		13.60	4 / 5				
interval	7 steps	1.03	1.03	1.02	1.02	1.06	1.12	1.08	1.10	1.10	1.46				3.08	3.20	3.28
velocity	5.32	8.87	8.87	8.96	8.96	8.62	8.16	8.46	8.31	8.31	9.60	8.09			8.90	8.57	8.36
time	2.62	3.72	4.76	5.80	6.84	7.90	8.92	10.00	11.08	12.18		13.60	3 / 6				
interval		1.10	1.04	1.04	1.04	1.06	1.02	1.08	1.08	1.10	1.42				3.18	3.12	3.26
velocity	5.24	8.31	8.79	8.79	8.79	8.62	8.96	8.46	8.46	8.31	9.87	8.09			8.62	8.79	8.41
time	2.62	3.70	4.74	5.78	6.80	7.82	8.86	9.90	10.96	12.04		13.60	7 / 7				
interval	7 steps	1.08	1.04	1.04	1.02	1.02	1.04	1.04	1.06	1.08	1.56	DQ			3.16	3.08	3.18
velocity	5.24	8.46	8.79	8.79	8.96	8.96	8.79	8.79	8.62	8.46	8.99	8.09			8.68	8.90	8.62
time	2.62	3.68	4.70	5.72	6.74	7.76	8.79	9.86	10.92	12.12		14.29	8 / 8-7				
interval		1.06	1.02	1.02	1.02	1.02	1.03	1.07	1.06	1.20	2.17				3.10	3.07	3.33
velocity	5.24	8.62	8.96	8.96	8.96	8.96	8.87	8.54	8.62	7.62	6.46	7.70			8.85	8.93	8.23

Heat 5

date		30-Sep-19		wind -0.5 m/s										Shibayama (2020) - race analysis of men's 110m hurdles in the 2020 season							
		H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10				
991)	time	2.58	3.63	4.63	5.63	6.63	7.64	8.63	9.66	10.68	11.74		13.15	9 / 1							
140	interval	7 steps	1.05	1.00	1.00	1.00	1.01	0.99	1.03	1.02	1.06	1.41			3.05	3.00	3.11				
	velocity	5.32	8.70	9.14	9.14	9.14	9.05	9.23	8.87	8.96	8.62	9.94	8.37		8.99	9.14	8.82				

Henson (2021) - Athlete First: 2019 year end hurdle report (TV Analysis)

Heat 4

date		30-Sep-19		wind		0.4 m/s		Shibayama (2019) - race analysis of men's 110m hurdles in the 2019 season												
		H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10			
1997	time	2.38	3.38	4.34	5.32	6.32	7.30	8.32	9.36	10.42	11.56		13.22	8 / 1			Henson (2021)			
160	interval	7 steps	1.00	0.96	0.98	1.00	0.98	1.02	1.04	1.06	1.14	1.66			2.94	3.00	3.24			
	velocity	5.76	9.14	9.52	9.33	9.14	9.33	8.96	8.79	8.62	8.02	8.45	8.32		9.33	9.14	8.46			
1994	time	2.56	3.58	4.62	5.64	6.66	7.69	8.72	9.77	10.83	11.89		13.32	7 / 2			Shibayama (2019)			
146	interval		1.02	1.04	1.02	1.02	1.03	1.03	1.05	1.06	1.06	1.43			3.08	3.08	3.17			
	velocity	5.36	8.96	8.79	8.96	8.96	8.87	8.87	8.70	8.62	8.62	9.80	8.26		8.90	8.90	8.65			

Heat 3 (TV Analysis)

date		30-Sep-19		wind		-0.4 m/s		Henson (2021) - Athlete First: 2019 year end hurdle report												
		H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10			
1998)	time	2.42	3.46	4.48	5.48	6.48	7.52	8.54	9.58	10.62	11.76		13.38	4 / 4						
186	interval		1.04	1.02	1.00	1.00	1.04	1.02	1.04	1.04	1.14	1.62	DQ		3.06	3.06	3.22			
	velocity	5.67	8.79	8.96	9.14	9.14	8.79	8.96	8.79	8.79	8.02	8.65	8.22		8.96	8.96	8.52			

Martinot-Lagarde, Pascal (FRA)

time	2.48	3.52	4.56	5.58	6.60	7.62	8.68	9.72	10.80	11.90		13.45	5 / 2-1				
reaction time	0.160	interval	7 steps	1.04	1.04	1.02	1.02	1.06	1.04	1.08	1.10	1.55			3.10	3.10	3.22
velocity	5.53	8.79	8.79	8.96	8.96	8.96	8.62	8.79	8.46	8.31	9.05	8.18			8.85	8.85	8.52

Shibayama (2020) - race analysis of men's 110m hurdles in the 2020 season

Heat 2

	date	30-Sep-19	wind 0.5 m/s										Shibayama (2019) - race analysis of men's 110m hurdles in the 2019 season							
			H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10		
Shubenkov, Sergey (ANA) (1999)	time		2.52	3.52	4.54	5.54	6.54	7.54	8.58	9.62	10.68	11.74		13.27	6 / 1					
	reaction time		0.138																	
	interval		7 steps	1.00	1.02	1.00	1.00	1.00	1.04	1.04	1.06	1.06	1.53				3.02	3.04	3.16	
	velocity		5.44	9.14	8.96	9.14	9.14	9.14	8.79	8.79	8.62	8.62	9.16	8.29			9.08	9.02	8.68	

Brathwaite, Shane (BAR) (1990)

time	2.62	3.65	4.72	5.73	6.76	7.81	8.84	9.88	10.96	12.04		13.51	3 / 4	Shibayama (2020)		
interval		1.03	1.07	1.01	1.03	1.05	1.03	1.04	1.08	1.08	1.47			3.11	3.11	3.20
velocity	5.24	8.87	8.54	9.05	8.87	8.70	8.87	8.79	8.46	8.46	9.54	8.14		8.82	8.82	8.57

Shibayama (2020)

Kanai, Taio (JPN) (1995)

time	2.58	3.64	4.70	5.76	6.84	7.90	8.99	10.09	11.20	12.31		13.74	7 / 7	Shibayama (2019)		
interval	7 steps	1.06	1.06	1.06	1.08	1.06	1.09	1.10	1.11	1.11	1.43			3.18	3.23	3.32
velocity	5.32	8.62	8.62	8.62	8.46	8.62	8.39	8.31	8.23	8.23	9.80	8.01		8.62	8.49	8.26

Shibayama (2019)

Heat 1 (TV Analysis)

date		30-Sep-19		wind		0.2 m/s		Henson (2021) - Athlete First: 2019 year end hurdle report												
		H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10			
94)	time	2.52	3.54	4.56	5.56	6.58	7.60	8.62	9.66	10.70	11.76		13.17	9 / 1						
132	interval		1.02	1.02	1.00	1.02	1.02	1.02	1.04	1.04	1.06	1.41			3.04	3.06	3.14			
	velocity	5.44	8.96	8.96	9.14	8.96	8.96	8.96	8.79	8.79	8.62	9.94	8.35		9.02	8.96	8.73			

2019 The Match - Europe v USA (Minsk, BLR) (TV Analysis)

FINAL

		date	10-Sep-19	wind	0.1 m/s													Henson (2020) - Athlete First: 2019 year end hurdle report			
						H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
Ortega, Orlando (ESP) (1991)	time					2.58	3.64	4.66	5.71	6.71	7.73	8.73	9.73	10.75	11.77		13.21	5 / 1			
	reaction time	0.158	interval	7 steps		1.06	1.02	1.05	1.00	1.02	1.00	1.00	1.00	1.02	1.02	1.44			3.13	3.02	3.04
	velocity					5.32	8.62	8.96	8.70	9.14	8.96	9.14	9.14	8.96	8.96	9.74	8.33		8.76	9.08	9.02
Shubenkov, Sergey (ANA) (1991)	time					2.60	3.66	4.73	5.76	6.80	7.83	8.83	9.86	10.90	11.96		13.39	7 / 2			
	reaction time	0.186	interval	7 steps		1.06	1.07	1.03	1.04	1.03	1.00	1.03	1.03	1.04	1.06	1.43			3.16	3.07	3.13
	velocity					5.28	8.62	8.54	8.87	8.79	8.87	9.14	8.87	8.79	8.62	9.80	8.22		8.68	8.93	8.76
Crittenden, Freddie (USA) (1991)	time					2.60	3.63	4.66	5.71	6.73	7.77	8.77	9.77	10.88	11.90		13.43	4 / 3			
	reaction time	0.161	interval			1.03	1.03	1.05	1.02	1.04	1.00	1.00	1.00	1.11	1.02	1.53			3.11	3.06	3.13
	velocity					5.28	8.87	8.87	8.70	8.96	8.79	9.14	9.14	8.23	8.96	9.16	8.19		8.82	8.96	8.76
Pozzi, Andy (GBR) (1992)	time					2.53	3.60	4.63	5.63	6.65	7.71	8.75	9.82	10.90	12.06		13.59	3 / 4			
	reaction time	0.155	interval	7 steps		1.07	1.03	1.00	1.02	1.06	1.04	1.07	1.08	1.16	1.53				3.10	3.12	3.31
	velocity					5.42	8.54	8.87	9.14	8.96	8.62	8.79	8.54	8.46	7.88	9.16	8.09		8.85	8.79	8.28
Dickson, Michael (USA) (1997)	time					2.60	3.68	4.72	5.78	6.86	7.93	8.96	10.10	11.16	12.23		13.70	6 / 5			
	reaction time	0.189	interval			1.08	1.04	1.06	1.08	1.07	1.03	1.14	1.06	1.07	1.47				3.18	3.18	3.27
	velocity					5.28	8.46	8.79	8.62	8.46	8.54	8.87	8.02	8.62	8.54	9.54	8.03		8.62	8.62	8.39
Parakhonka, Vitali (BLR) (1993)	time					2.63	3.73	4.78	5.80	6.83	7.92	8.96	10.10	11.16	12.28		13.80	9 / 6			
	reaction time	0.162	interval			1.10	1.05	1.02	1.03	1.09	1.04	1.14	1.06	1.12	1.52				3.17	3.16	3.32
	velocity					5.22	8.31	8.70	8.96	8.87	8.39	8.79	8.02	8.62	8.16	9.22	7.97		8.65	8.68	8.26
Dunning, Robert (USA) (1997)	time					2.60	3.68	4.73	5.80	6.85	7.91	8.95	10.03	11.13	12.28		13.86	2 / 7			
	reaction time	0.183	interval	7 steps		1.08	1.05	1.07	1.05	1.06	1.04	1.08	1.10	1.15	1.58				3.20	3.15	3.33
	velocity					5.28	8.46	8.70	8.54	8.70	8.62	8.79	8.46	8.31	7.95	8.87	7.94		8.57	8.70	8.23
Eaton, Jarret (USA) (1989)	time					2.73	3.93	5.13	6.33	7.60	8.76	10.00	11.22	12.43	13.73		15.54	8 / 8			
	reaction time	0.180	interval	7 steps		1.20	1.20	1.20	1.27	1.16	1.24	1.22	1.21	1.30	1.81				3.60	3.67	3.73
	velocity					5.03	7.62	7.62	7.62	7.20	7.88	7.37	7.49	7.55	7.03	7.75	7.08		7.62	7.47	7.35

2019 AG Memorial van Damme (Brussels, BEL) (TV Analysis)

FINAL

		date	06-Sep-19	wind	0.0 m/s													Henson (2020) - Athlete First: 2019 year end hurdle report			
						H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
Ortega, Orlando (ESP) (1991)	time					2.52	3.56	4.58	5.60	6.58	7.58	8.60	9.64	10.68	11.72		13.22	5 / 1			
	reaction time	0.126	interval	7 steps		1.04	1.02	1.02	0.98	1.00	1.02	1.04	1.04	1.04	1.50				3.08	3.00	3.12
	velocity					5.44	8.79	8.96	8.96	9.33	9.14	8.96	8.79	8.79	8.79	9.35	8.32		8.90	9.14	8.79
Levy, Ronald (JAM) (1992)	time					2.50	3.54	4.60	5.64	6.66	7.66	8.68	9.72	10.78	11.82		13.31	3 / 2			
	reaction time	0.134	interval	7 steps		1.04	1.06	1.04	1.02	1.00	1.02	1.04	1.06	1.04	1.49				3.14	3.04	3.14
	velocity					5.49	8.79	8.62	8.79	8.96	9.14	8.96	8.79	8.62	8.79	9.41	8.26		8.73	9.02	8.73
Shubenkov, Sergey (ANA) (1991)	time					2.54	3.60	4.64	5.64	6.64	7.64	8.68	9.72	10.76	11.82		13.33	7 / 3			
	reaction time	0.153	interval	7 steps		1.06	1.04	1.00	1.00	1.00	1.04	1.04	1.04	1.04	1.06	1.51			3.10	3.04	3.14
	velocity					5.40	8.62	8.79	9.14	9.14	9.14	8.79	8.79	8.79	8.62	9.28	8.25		8.85	9.02	8.73
Crittenden, Freddie (USA) (1991)	time					2.52	3.56	4.60	5.62	6.62	7.64	8.68	9.70	10.74	11.80		13.35	6 / 4			
	reaction time	0.144	interval			1.04	1.04	1.02	1.00	1.02	1.04	1.02	1.04	1.04	1.06	1.55			3.10	3.06	3.12
	velocity					5.44	8.79	8.79	8.96	9.14	8.96	8.79	8.96	8.79	8.62	9.05	8.24		8.85	8.96	8.79
Xie Wenjun (CHN) (1990)	time					2.54	3.60	4.66	5.70	6.72	7.74	8.80	9.84	10.88	11.98		13.49	4 / 5			
	reaction time	0.150	interval	7 steps		1.06	1.06	1.04	1.02	1.02	1.06	1.04	1.04	1.04	1.10	1.51			3.16	3.10	3.18
	velocity					5.40	8.62	8.62	8.79	8.96	8.96	8.62	8.79	8.79	8.31	9.28	8.15		8.68	8.85	8.62
Pozzi, Andy (GBR) (1992)	time					2.52	3.58	4.60	5.60	6.62	7.66	8.70	9.72	10.82	11.90		13.50	2 / 6			
	reaction time	0.141	interval	7 steps		1.06	1.02	1.00	1.02	1.04	1.04	1.04	1.02	1.10	1.08	1.60			3.08	3.10	3.20
	velocity					5.44	8.62	8.96	9.14	8.96	8.79	8.79	8.96	8.31	8.46	8.76	8.15		8.90	8.85	8.57
Alkana, Antonio (RSA) (1990)	time					2.48	3.56	4.60	5.64	6.70	7.76	8.82	9.88	10.96	12.04		13.61	8 / 7			
	reaction time	0.136	interval			1.08	1.04	1.04	1.06	1.06	1.06	1.06	1.06	1.08	1.08	1.57			3.16	3.18	3.22
	velocity					5.53	8.46	8.79	8.79	8.62	8.62	8.62	8.62	8.46	8.46	8.93	8.08		8.68	8.62	8.52
Obasuyi, Michael (BEL) (1999)	time					2.52	3.64	4.68	5.72	6.78	7.82	8.88	9.92	11.00	12.08			1 / -			
	reaction time	0.149	interval	7 steps		1.12	1.04	1.04	1.06	1.04	1.06	1.04	1.04	1.08	1.08		DQ		3.20	3.16	3.20
	velocity					5.44	8.16	8.79	8.79	8.62	8.79	8.62	8.79	8.46	8.46				8.57	8.68	8.57

2019 ISTAF (Berlin, GER) (TV Analysis)

FINAL

		date	01-Sep-19	wind	0.8 m/s													Henson (2020) - Athlete First: 2019 year end hurdle report			
						H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
McLeod, Omar (JAM) (1994)	time					2.46	3.46	4.46	5.50	6.50	7.50	8.52	9.58	10.62	11.68		13.07	5 / 1			
	reaction time	0.130	interval			1.00	1.00	1.04	1.00	1.00	1.00	1.02	1.06	1.04	1.06	1.39			3.04	3.02	3.16
	velocity					5.58	9.14	9.14	8.79	9.14	9.14	8.96	8.62	8.79	8.62	10.09	8.42		9.02	9.08	8.68

Martinot-Lagarde, Pascal (FRA)	time	2.53	3.56	4.58	5.58	6.58	7.62	8.63	9.65	10.70	11.78		13.25	7 / 2				
	reaction time	0.158	interval	7 steps	1.03	1.02	1.00	1.00	1.04	1.01	1.02	1.05	1.08	1.47		3.05	3.05	3.15
	velocity	5.42	8.87	8.96	9.14	9.14	8.79	9.05	8.96	8.70	8.46	9.54	8.30		8.99	8.99	8.70	
Brathwaite, Shane (BAR) (1990)	time	2.53	3.58	4.61	5.63	6.65	7.65	8.68	9.75	10.80	11.86		13.36	1 / 3				
	reaction time	0.125	interval		1.05	1.03	1.02	1.02	1.00	1.03	1.07	1.05	1.06	1.50		3.10	3.05	3.18
	velocity	5.42	8.70	8.87	8.96	8.96	9.14	8.87	8.54	8.70	8.62	9.35	8.23		8.85	8.99	8.62	
Crittenden, Freddie (USA) (199)	time	2.53	3.53	4.63	5.65	6.67	7.68	8.73	9.80	10.84	11.88		13.38	6 / 4				
	reaction time	0.137	interval		1.00	1.10	1.02	1.02	1.01	1.05	1.07	1.04	1.04	1.50		3.12	3.08	3.15
	velocity	5.42	9.14	8.31	8.96	8.96	9.05	8.70	8.54	8.79	8.79	9.35	8.22		8.79	8.90	8.70	

2019 Japanese World Championships Trials (Fujiyoshida, JPN)

FINAL

date 01-Sep-19 wind 1.4 m/s

Shibayama (2019) - race analysis of men's 110m hurdles in the 2019 season

		date	07-Sep-19	wind	1.4 m/s	Shibayama (2019) - Race analysis of men's 110m hurdles in the 2019 season													
				H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
Takayama, Shunya (JPN) (1994)	time			2.53	3.61	4.63	5.68	6.70	7.73	8.75	9.80	10.83	11.88		13.29	5 / 1			
	reaction time					1.08	1.02	1.05	1.02	1.03	1.02	1.05	1.03	1.05	1.41		3.15	3.07	3.13
	velocity			5.42	8.46	8.96	8.70	8.96	8.87	8.96	8.70	8.87	8.70	9.94	8.28		8.70	8.93	8.76
Chen Kuei-Ru (TPE) (1993)	time			2.56	3.61	4.65	5.69	6.72	7.75	8.78	9.80	10.85	11.91		13.36	6 / 2			
	reaction time			7 steps		1.05	1.04	1.04	1.03	1.03	1.03	1.02	1.05	1.06	1.45		3.13	3.09	3.13
	velocity			5.36	8.70	8.79	8.79	8.87	8.87	8.87	8.96	8.70	8.62	9.67	8.23		8.76	8.87	8.76
Kanai, Taio (JPN) (1995)	time			2.58	3.63	4.68	5.72	6.77	7.83	8.88	9.95	11.02	12.10		13.51	4 / 3			
	reaction time					1.05	1.05	1.04	1.05	1.06	1.05	1.07	1.07	1.08	1.41		3.14	3.16	3.22
	velocity			5.32	8.70	8.70	8.79	8.70	8.62	8.70	8.54	8.54	8.46	9.94	8.14		8.73	8.68	8.52
Yazawa, Wataru (JPN) (1991)	time			2.62	3.70	4.73	5.76	6.81	7.85	8.90	9.96	11.04	12.12		13.51	7 / 4			
	reaction time			7 steps		1.08	1.03	1.03	1.05	1.04	1.05	1.06	1.08	1.08	1.39		3.14	3.14	3.22
	velocity			5.24	8.46	8.87	8.87	8.70	8.79	8.70	8.62	8.46	8.46	10.09	8.14		8.73	8.73	8.52
Kuriki, Anthony Tyrell (JPN) (1991)	time			2.60	3.69	4.73	5.77	6.82	7.86	8.91	9.97	11.03	12.10		13.59	8 / 5			
	reaction time			7 steps		1.09	1.04	1.04	1.05	1.04	1.05	1.06	1.06	1.07	1.49	PB	3.17	3.14	3.19
	velocity			5.28	8.39	8.79	8.79	8.70	8.79	8.70	8.62	8.62	8.54	9.41	8.09		8.65	8.73	8.60
Fujii, Ryota (JPN) (1996)	time			2.59	3.65	4.70	5.74	6.80	7.86	8.92	10.00	11.10	12.21		13.65	2 / 6			
	reaction time					1.06	1.05	1.04	1.06	1.06	1.08	1.10	1.11	1.44	PB		3.15	3.18	3.29
	velocity			5.30	8.62	8.70	8.79	8.62	8.62	8.62	8.46	8.31	8.23	9.74	8.06		8.70	8.62	8.33
Ishikawa, Shuhei (JPN) (1995)	time			2.56	3.64	4.69	5.75	6.82	7.89	8.95	10.02	11.12	12.23		13.67	3 / 7			
	reaction time			7 steps		1.08	1.05	1.06	1.07	1.07	1.06	1.07	1.10	1.11	1.44		3.19	3.20	3.28
	velocity			5.36	8.46	8.70	8.62	8.54	8.54	8.62	8.54	8.31	8.23	9.74	8.05		8.60	8.57	8.36

2019 Meeting de Paris (Paris, FRA) (TV Analysis)

FINAL

date 24-Aug-19 wind 0.8 m/s

Henson (2020) - Athlete First: 2019 year end hurdle report

		H1-H10										Run-In		Official Time	Lane / Place	H1-H4			H4-H7	H7-H10
		H1	H2	H3	H4	H5	H6	H7	H8	H9	H10									
Roberts, Daniel (USA) (1998)	time	2.46	3.48	4.50	5.52	6.52	7.52	8.54	9.56	10.58	11.66		13.08	3 / 1						
	reaction time	0.161	interval	1.02	1.02	1.02	1.00	1.00	1.02	1.02	1.02	1.08	1.42			3.06	3.02	3.12		
	velocity	5.58	8.96	8.96	8.96	9.14	9.14	8.96	8.96	8.96	8.96	8.46	9.87	8.41		8.96	9.08	8.79		
Ortega, Orlando (ESP) (1991)	time	2.50	3.52	4.54	5.54	6.52	7.56	8.58	9.62	10.66	11.70		13.14	6 / 2						
	reaction time	0.132	interval	7 steps	1.02	1.02	1.00	0.98	1.04	1.02	1.04	1.04	1.44			3.04	3.04	3.12		
	velocity	5.49	8.96	8.96	9.14	9.33	8.79	8.96	8.79	8.79	8.79	8.79	9.74	8.37		9.02	9.02	8.79		
Crittenden, Freddie (USA) (199)	time	2.52	3.54	4.56	5.56	6.56	7.58	8.58	9.62	10.66	11.70		13.17	1 / 3						
	reaction time	0.150	interval	1.02	1.02	1.00	1.00	1.02	1.00	1.04	1.04	1.04	1.47	PB		3.04	3.02	3.12		
	velocity	5.44	8.96	8.96	9.14	9.14	8.96	9.14	8.79	8.79	8.79	8.79	9.54	8.35		9.02	9.08	8.79		
Levy, Ronald (JAM) (1992)	time	2.52	3.56	4.60	5.62	6.64	7.66	8.70	9.70	10.74	11.78		13.22	8 / 4						
	reaction time	0.151	interval	7 steps	1.04	1.04	1.02	1.02	1.04	1.00	1.04	1.04	1.44			3.10	3.08	3.08		
	velocity	5.44	8.79	8.79	8.96	8.96	8.96	8.79	9.14	8.79	8.79	8.79	9.74	8.32		8.85	8.90	8.90		
Martinot-Lagarde, Pascal (FRA)	time	2.50	3.54	4.56	5.56	6.56	7.58	8.58	9.62	10.66	11.76		13.24	2 / 5						
	reaction time	0.165	interval	7 steps	1.04	1.02	1.00	1.00	1.02	1.00	1.04	1.04	1.10	1.48		3.06	3.02	3.18		
	velocity	5.49	8.79	8.96	9.14	9.14	8.96	9.14	8.79	8.79	8.31	9.47	8.31		8.96	9.08	8.62			
Holloway, Grant (USA) (1997)	time	2.44	3.44	4.44	5.44	6.42	7.44	8.44	9.50	10.58	11.66		13.25	5 / 6						
	reaction time	0.162	interval	7 steps	1.00	1.00	1.00	0.98	1.02	1.00	1.06	1.08	1.08	1.59		3.00	3.00	3.22		
	velocity	5.62	9.14	9.14	9.14	9.33	8.96	9.14	8.62	8.46	8.46	8.82	8.30		9.14	9.14	8.52			
Xie Wenjun (CHN) (1990)	time	2.56	3.60	4.66	5.68	6.70	7.74	8.76	9.80	10.86	11.98		13.46	7 / 7						
	reaction time	0.162	interval	7 steps	1.04	1.06	1.02	1.02	1.04	1.02	1.04	1.06	1.12	1.48		3.12	3.08	3.22		
	velocity	5.36	8.79	8.62	8.96	8.96	8.79	8.96	8.79	8.62	8.16	9.47	8.17		8.79	8.90	8.52			
Shubenkov, Sergey (ANA) (199)	time	2.56	3.64	4.72	5.76	6.80	7.88	8.94	10.00	11.12	12.26		13.88	4 / 8						

reaction time	0.144	interval	7 steps	1.08	1.08	1.04	1.04	1.08	1.06	1.06	1.12	1.14	1.62		3.20	3.18	3.32	
		velocity		5.36	8.46	8.46	8.79	8.79	8.46	8.62	8.62	8.16	8.02	8.65	7.93	8.57	8.62	8.26

2019 Chinese National Grand Prix Final (Daqing, CHN)

FINAL

date 24-Aug-19 wind -0.1 m/s

CAA Hurdle Development (2019)

			H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
Zeng Jianhang (CHN) (1998)	time		2.62	3.67	4.72	5.75	6.79	7.84	8.89	9.98	11.07	12.17		13.65	5 / 1			
reaction time	0.177	interval		1.05	1.05	1.03	1.04	1.05	1.05	1.09	1.09	1.10	1.48			3.13	3.14	3.28
		velocity	5.24	8.70	8.70	8.87	8.79	8.70	8.70	8.39	8.39	8.31	9.47	8.06		8.76	8.73	8.36
Sun Zhenjiang (CHN) (1999)	time		2.60	3.68	4.76	5.80	6.89	8.00	9.09	10.19	11.31	12.46		13.96	7 / 2			
reaction time	0.181	interval		7 steps	1.08	1.08	1.04	1.09	1.11	1.09	1.10	1.12	1.15	1.50		3.20	3.29	3.37
		velocity	5.28	8.46	8.46	8.79	8.39	8.23	8.39	8.31	8.16	7.95	9.35	7.88		8.57	8.33	8.14
Li Jiming (CHN) (1996)	time		2.71	3.80	4.89	5.94	6.99	8.06	9.15	10.24	11.35	12.46		13.97	6 / 3			
reaction time	0.213	interval		1.09	1.09	1.05	1.05	1.07	1.09	1.09	1.11	1.11	1.51			3.23	3.21	3.31
		velocity	5.06	8.39	8.39	8.70	8.70	8.54	8.39	8.39	8.23	8.23	9.28	7.87		8.49	8.54	8.28
Pan Zijie (CHN) (1995)	time		2.68	3.80	4.90	6.00	7.09	8.17	9.27	10.38	11.49	12.61		14.08	2 / 4			
reaction time	0.204	interval		1.12	1.10	1.10	1.09	1.08	1.10	1.11	1.11	1.12	1.47			3.32	3.27	3.34
		velocity	5.12	8.16	8.31	8.31	8.39	8.46	8.31	8.23	8.23	8.16	9.54	7.81		8.26	8.39	8.21
Zhang Tao (CHN) (1997)	time		2.72	3.85	4.93	6.03	7.11	8.21	9.31	10.42	11.55	12.67		14.18	4 / 5			
reaction time	0.181	interval		1.13	1.08	1.10	1.08	1.10	1.10	1.11	1.13	1.12	1.51			3.31	3.28	3.36
		velocity	5.04	8.09	8.46	8.31	8.46	8.31	8.31	8.23	8.09	8.16	9.28	7.76		8.28	8.36	8.16
Shi Guoxing (CHN) (2000)	time		2.78	3.89	4.99	6.10	7.19	8.32	9.44	10.56	11.72	12.88		14.36	3 / 6			
reaction time	0.201	interval		1.11	1.10	1.11	1.09	1.13	1.12	1.12	1.16	1.16	1.48	PB		3.32	3.34	3.44
		velocity	4.94	8.23	8.31	8.23	8.39	8.09	8.16	8.16	7.88	7.88	9.47	7.66		8.26	8.21	7.97
Guo Zhongjie (CHN) (2001)	time		2.73	3.80	4.90	6.00	7.11	8.24	9.37	10.52	11.68	12.88		14.46	8 / 7			
reaction time	0.226	interval		1.07	1.10	1.10	1.11	1.13	1.13	1.15	1.16	1.20	1.58			3.27	3.37	3.51
		velocity	5.03	8.54	8.31	8.31	8.23	8.09	8.09	7.95	7.88	7.62	8.87	7.61		8.39	8.14	7.81
Wu Zhenhua (CHN) (1998)	time		2.74	3.87	5.00	6.12	7.23	8.36	9.50	10.73	11.92	13.16		14.83	9 / 8			
reaction time	0.184	interval		1.13	1.13	1.12	1.11	1.13	1.14	1.23	1.19	1.24	1.67			3.38	3.38	3.66
		velocity	5.01	8.09	8.09	8.16	8.23	8.09	8.02	7.43	7.68	7.37	8.40	7.42		8.11	8.11	7.49

Heat 2

date 23-Aug-19 wind -0.2 m/s

CAA Hurdle Development (2019)

			H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
Zhang Tao (CHN) (1997)	time		2.74	3.85	4.95	6.05	7.14	8.22	9.33	10.46	11.58	12.75		14.30	4 / 1			
reaction time	0.180	interval		1.11	1.10	1.10	1.09	1.08	1.11	1.13	1.12	1.17	1.55			3.31	3.28	3.42
		velocity	5.01	8.23	8.31	8.31	8.39	8.46	8.23	8.09	8.16	7.81	9.05	7.69		8.28	8.36	8.02
Pan Zijie (CHN) (1995)	time		2.67	3.80	4.90	6.01	7.12	8.24	9.36	10.52	11.67	12.86		14.48	7 / 2			
reaction time	0.195	interval		1.13	1.10	1.11	1.11	1.12	1.12	1.16	1.15	1.19	1.62			3.34	3.35	3.50
		velocity	5.14	8.09	8.31	8.23	8.23	8.16	8.16	7.88	7.95	7.68	8.65	7.60		8.21	8.19	7.83
Shi Guoxing (CHN) (2000)	time		2.81	3.94	5.06	6.17	7.34	8.48	9.62	10.76	11.92	13.08		14.58	5 / 3			
reaction time	0.195	interval		1.13	1.12	1.11	1.17	1.14	1.14	1.14	1.16	1.16	1.50			3.36	3.45	3.46
		velocity	4.88	8.09	8.16	8.23	7.81	8.02	8.02	8.02	7.88	7.88	9.35	7.54		8.16	7.95	7.92
Jiang Wei (CHN) (1994)	time		2.72	3.87	4.99	6.11	7.26	8.39	9.51	10.68	11.84	13.04		14.60	3 / 4			
reaction time	0.200	interval		7 steps	1.15	1.12	1.12	1.15	1.13	1.12	1.17	1.16	1.20	1.56		3.39	3.40	3.53
		velocity	5.04	7.95	8.16	8.16	7.95	8.09	8.16	7.81	7.88	7.62	8.99	7.53		8.09	8.06	7.77
Lu Wei (CHN) (2000)	time		2.83	3.98	5.09	6.26	7.41	8.54	9.67	10.80	11.98	13.18		14.80	6 / 5			
reaction time	0.208	interval		1.15	1.11	1.17	1.15	1.13	1.13	1.13	1.18	1.20	1.62			3.43	3.41	3.51
		velocity	4.85	7.95	8.23	7.81	7.95	8.09	8.09	8.09	7.75	7.62	8.65	7.43		7.99	8.04	7.81
Jiang Fan (CHN) (1989)	time		2.70	3.84	4.95	6.06	7.17	8.29	9.41	10.59	11.83	13.09		14.80	9 / 6			
reaction time	0.159	interval		1.14	1.11	1.11	1.11	1.12	1.12	1.18	1.24	1.26	1.71			3.36	3.35	3.68
		velocity	5.08	8.02	8.23	8.23	8.23	8.16	8.16	7.75	7.37	7.25	8.20	7.43		8.16	8.19	7.45
Yang Quanlei (CHN) (1999)	time		2.74	3.89	5.03	6.19	7.35	8.52	9.72	10.93	12.16	13.38		14.97	8 / 7			
reaction time	0.193	interval		1.15	1.14	1.16	1.16	1.17	1.20	1.21	1.23	1.22	1.59			3.45	3.53	3.66
		velocity	5.01	7.95	8.02	7.88	7.88	7.81	7.62	7.55	7.43	7.49	8.82	7.35		7.95	7.77	7.49
Long Wei (CHN) (1997)	time		2.77	3.94	5.11	6.27	7.46	8.67	9.86	11.08	12.29	13.51		15.11	2 / 8			
reaction time	0.199	interval		1.17	1.17	1.16	1.19	1.21	1.19	1.22	1.21	1.22	1.60			3.50	3.59	3.65
		velocity	4.95	7.81	7.81	7.88	7.68	7.55	7.68	7.49	7.55	7.49	8.76	7.28		7.83	7.64	7.51

Heat 1

date 23-Aug-19 wind -0.2 m/s

CAA Hurdle Development (2019)

			H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
Zeng Jianhang (CHN) (1998)	time		2.65	3.71	4.77	5.85	6.94	8.02	9.11	10.24	11.38	12.53		14.11	4 / 1			
reaction time	0.176	interval		1.06	1.06	1.08	1.09	1.08	1.09	1.13	1.14	1.15	1.58			3.20	3.26	3.42

		velocity	5.18	8.62	8.62	8.46	8.39	8.46	8.39	8.09	8.02	7.95	8.87	7.80		8.57	8.41	8.02
Sun Zhenjiang (CHN) (1999)	time		2.62	3.72	4.80	5.90	6.99	8.09	9.21	10.33	11.46	12.64		14.20	7 / 2			
	reaction time	0.177	interval	7 steps	1.10	1.08	1.10	1.09	1.10	1.12	1.12	1.13	1.18	1.56		3.28	3.31	3.43
		velocity	5.24	8.31	8.46	8.31	8.39	8.31	8.16	8.16	8.09	7.75	8.99	7.75		8.36	8.28	7.99
Li Jiming (CHN) (1996)	time		2.70	3.80	4.89	5.97	7.06	8.16	9.26	10.36	11.48	12.64		14.23	8 / 3			
	reaction time	0.192	interval		1.10	1.09	1.08	1.09	1.10	1.10	1.12	1.16	1.59			3.27	3.29	3.38
		velocity	5.08	8.31	8.39	8.46	8.39	8.31	8.31	8.31	8.16	7.88	8.82	7.73		8.39	8.33	8.11
Wu Zhenhua (CHN) (1998)	time		2.74	3.84	4.95	6.06	7.20	8.31	9.44	10.59	11.73	12.88		14.41	9 / 4			
	reaction time	0.175	interval		1.10	1.11	1.11	1.14	1.11	1.13	1.15	1.14	1.15	1.53		3.32	3.38	3.44
		velocity	5.01	8.31	8.23	8.23	8.02	8.23	8.09	7.95	8.02	7.95	9.16	7.63		8.26	8.11	7.97
Guo Zhongjie (CHN) (2001)	time		2.73	3.84	4.92	6.01	7.12	8.22	9.36	10.51	11.66	12.85		14.42	5 / 5			
	reaction time	0.193	interval		1.11	1.08	1.09	1.11	1.10	1.14	1.15	1.15	1.19	1.57		3.28	3.35	3.49
		velocity	5.03	8.23	8.46	8.39	8.23	8.31	8.02	7.95	7.95	7.68	8.93	7.63		8.36	8.19	7.86
Yang Jianxing (CHN) (1996)	time		2.70	3.85	4.97	6.10	7.22	8.37	9.53	10.69	11.89	13.15		14.79	3 / 6			
	reaction time	0.174	interval		1.15	1.12	1.13	1.12	1.15	1.16	1.16	1.20	1.26	1.64		3.40	3.43	3.62
		velocity	5.08	7.95	8.16	8.09	8.16	7.95	7.88	7.88	7.62	7.25	8.55	7.44		8.06	7.99	7.57
Zhu Shenglong (CHN) (2000)	time		2.80	3.92	4.99	6.09	7.19	8.30	9.46	10.59	11.85	13.11		14.85	6 / 7			
	reaction time	0.184	interval		1.12	1.07	1.10	1.10	1.11	1.16	1.13	1.26	1.26	1.74		3.29	3.37	3.65
		velocity	4.90	8.16	8.54	8.31	8.31	8.23	7.88	8.09	7.25	7.25	8.06	7.41		8.33	8.14	7.51
Lu Yang (CHN) (1996)	time		2.84	4.00	5.15									dnf	2 / --			
	reaction time	0.179	interval		1.16	1.15												
		velocity	4.83	7.88	7.95													

Decathlon - Heat 2

date 23-Aug-19		wind ?? m/s										CAA Hurdle Development (2019)						
		H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10	
165	time	2.79	3.94	5.09	6.26	7.41	8.58	9.78	10.94	12.11	13.30		14.86	4 / 1				
	interval		1.15	1.15	1.17	1.15	1.17	1.20	1.16	1.17	1.19	1.56			3.47	3.52	3.52	
	velocity	4.92	7.95	7.95	7.81	7.95	7.81	7.62	7.88	7.81	7.68	8.99		7.40	7.90	7.79	7.79	
197	time	2.87	4.10	5.30	6.54	7.74	8.96	10.19	11.41	12.64	13.88		15.46	5 / 2				
	interval		1.23	1.20	1.24	1.20	1.22	1.23	1.22	1.23	1.24	1.58			3.67	3.65	3.69	
	velocity	4.78	7.43	7.62	7.37	7.62	7.49	7.43	7.49	7.43	7.37	8.87		7.12	7.47	7.51	7.43	
202)	time	2.85	4.09	5.29	6.49	7.71	8.91	10.14	11.33	12.58	13.83		15.51	9 / 3				
	interval		1.24	1.20	1.20	1.22	1.20	1.23	1.19	1.25	1.25	1.68			3.64	3.65	3.69	
	velocity	4.81	7.37	7.62	7.62	7.49	7.62	7.43	7.68	7.31	7.31	8.35		7.09	7.53	7.51	7.43	
190	time	2.84	4.05	5.29	6.56	7.79	9.02	10.29	11.59	12.86	14.25		15.94	8 / 4				
	interval		1.21	1.24	1.27	1.23	1.23	1.27	1.30	1.27	1.39	1.69			3.72	3.73	3.96	
	velocity	4.83	7.55	7.37	7.20	7.43	7.43	7.20	7.03	7.20	6.58	8.30		6.90	7.37	7.35	6.92	

Decathlon - Heat 1

date		23-Aug-19	wind ?? m/s										CAA Hurdle Development (2019)						
			H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10	
209	time		2.78	3.90	5.04	6.14	7.29	8.41	9.53	10.71	11.86	13.05		14.61	4 / 1				
	interval	7 steps	1.12	1.14	1.10	1.15	1.12	1.12	1.18	1.15	1.19	1.56				3.36	3.39	3.52	
	velocity		4.94	8.16	8.02	8.31	7.95	8.16	8.16	7.75	7.95	7.68	8.99	7.53		8.16	8.09	7.79	
187	time		2.82	4.00	5.16	6.32	7.49	8.69	9.89	11.13	12.33	13.61		15.25	7 / 2				
	interval		1.18	1.16	1.16	1.17	1.20	1.20	1.24	1.20	1.28	1.64				3.50	3.57	3.72	
	velocity		4.87	7.75	7.88	7.88	7.81	7.62	7.62	7.37	7.62	7.14	8.55	7.21		7.83	7.68	7.37	
5)	time		2.84	4.00	5.17	6.38	7.56	8.77	9.99	11.23	12.46	13.78		15.40	5 / 3				
	interval		1.16	1.17	1.21	1.18	1.21	1.22	1.24	1.23	1.32	1.62				3.54	3.61	3.79	
	velocity		4.83	7.88	7.81	7.55	7.75	7.55	7.49	7.37	7.43	6.92	8.65	7.14		7.75	7.60	7.23	
194	time		2.87	4.02	5.30	6.54	7.76	9.01	10.24	11.51	12.80	14.11		15.76	6 / 4				
	interval		1.15	1.28	1.24	1.22	1.25	1.23	1.27	1.29	1.31	1.65				3.67	3.70	3.87	
	velocity		4.78	7.95	7.14	7.37	7.49	7.31	7.43	7.20	7.09	6.98	8.50	6.98		7.47	7.41	7.09	
)	time		2.84	4.07	5.29	6.52	7.81	9.06	10.31	11.63	12.90	14.25		16.02	8 / 5				
	interval		1.23	1.22	1.23	1.29	1.25	1.25	1.32	1.27	1.35	1.77				3.68	3.79	3.94	
	velocity		4.83	7.43	7.49	7.43	7.09	7.31	7.31	6.92	7.20	6.77	7.92	6.87		7.45	7.23	6.96	
1999)	time		2.90	4.12	5.37	6.65	7.96	9.28	10.59	11.89	13.28	14.65		16.43	3 / 6				
	interval		1.22	1.25	1.28	1.31	1.32	1.31	1.30	1.39	1.37	1.78				3.75	3.94	4.06	
	velocity		4.73	7.49	7.31	7.14	6.98	6.92	6.98	7.03	6.58	6.67	7.88	6.70		7.31	6.96	6.75	

2019 Müller Grand Prix (Birmingham, GBR) (TV Analysis)

FINAL

date 18-Aug-19 wind -0.2 m/s

Henson (2020) - Athlete First: 2019 year end hurdle report

		H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
McLeod, Omar (JAM) (1994)	time	2.54	3.56	4.57	5.58	6.57	7.61	8.59	9.63	10.65	11.73		13.21	5 / 1			
	reaction time	0.139	interval	1.02	1.01	0.99	1.04	0.98	1.04	1.02	1.08	1.48			3.04	3.01	3.14
	velocity	5.40	8.96	9.05	9.05	9.23	8.79	9.33	8.79	8.96	8.46	9.47	8.33		9.02	9.11	8.73
Crittenden, Freddie (USA) (1991)	time	2.54	3.58	4.62	5.65	6.63	7.65	8.67	9.69	10.72	11.78		13.31	4 / 2			
	reaction time	0.116	interval	1.04	1.04	1.03	0.98	1.02	1.02	1.02	1.03	1.06	1.53		3.11	3.02	3.11
	velocity	5.40	8.79	8.79	8.87	9.33	8.96	8.96	8.96	8.87	8.62	9.16	8.26		8.82	9.08	8.82
Xie Wenjun (CHN) (1990)	time	2.58	3.63	4.66	5.68	6.70	7.72	8.73	9.77	10.81	11.90		13.43	7 / 3			
	reaction time	0.140	interval	7 steps 1.05	1.03	1.02	1.02	1.02	1.01	1.04	1.04	1.09	1.53		3.10	3.05	3.17
	velocity	5.32	8.70	8.87	8.96	8.96	8.96	9.05	8.79	8.79	8.39	9.16	8.19		8.85	8.99	8.65
Roberts, Daniel (USA) (1998)	time	2.54	3.56	4.61	5.64	6.62	7.64	8.66	9.69	10.77	11.87		13.48	6 / 4			
	reaction time	0.200	interval	1.02	1.05	1.03	0.98	1.02	1.02	1.03	1.08	1.10	1.61		3.10	3.02	3.21
	velocity	5.40	8.96	8.70	8.87	9.33	8.96	8.96	8.87	8.46	8.31	8.71	8.16		8.85	9.08	8.54
Fillery, Cameron (GBR) (1998)	time	2.58	3.65	4.74	5.76	6.77	7.78	8.83	9.88	10.92	12.02		13.54	9 / 5			
	reaction time	0.109	interval	7 steps 1.07	1.09	1.02	1.01	1.05	1.05	1.04	1.10	1.52	PB		3.18	3.07	3.19
	velocity	5.32	8.54	8.39	8.96	9.05	9.05	8.70	8.70	8.79	8.31	9.22	8.12		8.62	8.93	8.60
Hough, Nicholas (AUS) (1993)	time	2.62	3.69	4.75	5.77	6.77	7.78	8.84	9.90	10.94	12.10		13.66	8 / 6			
	reaction time	0.149	interval	7 steps 1.07	1.06	1.02	1.00	1.06	1.06	1.04	1.16	1.56			3.15	3.07	3.26
	velocity	5.24	8.54	8.62	8.96	9.14	9.05	8.62	8.62	8.79	7.88	8.99	8.05		8.70	8.93	8.41
King, David (GBR) (1994)	time	2.62	3.73	4.80	5.83	6.86	7.89	8.91	9.96	11.05	12.15		13.69	3 / 7			
	reaction time	0.136	interval	7 steps 1.11	1.07	1.03	1.03	1.02	1.05	1.09	1.10	1.54			3.21	3.08	3.24
	velocity	5.24	8.23	8.54	8.87	8.87	8.87	8.96	8.70	8.39	8.31	9.10	8.04		8.54	8.90	8.46
Moore, Isaiah (USA) (1996)	time	2.64	3.73	4.80	5.87	6.91	7.93	9.00	10.09	11.16	12.26		13.81	2 / 8			
	reaction time	0.144	interval	1.09	1.07	1.04	1.02	1.07	1.09	1.07	1.10	1.55			3.23	3.13	3.26
	velocity	5.20	8.39	8.54	8.54	8.79	8.96	8.54	8.39	8.54	8.31	9.05	7.97		8.49	8.76	8.41

2019 Athlete Night Games (Fukui, JPN)

FINAL

date 17-Aug-19 wind 1.1 m/s

(2019) - <http://tfd-data-store.com/2019/09/03/post-1615/>

		H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
Takayama, Shunya (JPN) (1994)	time	2.54	3.57	4.59	5.62	6.63	7.65	8.68	9.72	10.77	11.82		13.25	4 / 1			
	reaction time		interval	1.03	1.02	1.03	1.01	1.02	1.03	1.04	1.05	1.43	NR		3.08	3.06	3.14
	velocity	5.40	8.87	8.96	8.87	9.05	8.96	8.87	8.79	8.70	8.70	9.80	8.30		8.90	8.96	8.73

2019 Chinese National Youth Games (Taiyuan, CHN)

U18 FINAL (91cm)

date 14-Aug-19 wind -0.1 m/s

CAA Hurdle Development (2019)

		H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
Yang Jiajun (CHN) (2002)	time	2.62	3.69	4.69	5.70	6.71	7.69	8.71	9.73	10.78	11.81		13.30	6 / 1			
	reaction time	0.152	interval	1.07	1.00	1.01	0.98	1.02	1.02	1.05	1.03	1.49			3.08	3.01	3.10
	velocity	5.24	8.54	9.14	9.05	9.05	9.33	8.96	8.96	8.70	8.87	9.41	8.27		8.90	9.11	8.85
Zheng Wang (CHN) (2003)	time	2.64	3.74	4.80	5.87	6.93	8.02	9.09	10.18	11.33	12.46		14.06	7 / 2			
	reaction time	0.167	interval	1.10	1.06	1.07	1.06	1.09	1.09	1.15	1.13	1.60			3.23	3.22	3.37
	velocity	5.20	8.31	8.62	8.54	8.62	8.39	8.54	8.39	7.95	8.09	8.76	7.82		8.49	8.52	8.14
Mao Chenchen (CHN) (2002)	time	2.67	3.82	4.93	6.02	7.09	8.19	9.29	10.44	11.59	12.73		14.35	5 / 3			
	reaction time	0.162	interval	1.15	1.11	1.09	1.07	1.10	1.10	1.15	1.14	1.62			3.35	3.27	3.44
	velocity	5.14	7.95	8.23	8.39	8.54	8.31	8.31	7.95	7.95	8.02	8.65	7.67		8.19	8.39	7.97
Zhong Shulun (CHN) (2003)	time	2.80	3.97	5.12	6.23	7.36	8.51	9.66	10.79	11.93	13.10		14.72	8 / 4			
	reaction time	0.225	interval	1.17	1.15	1.13	1.15	1.15	1.13	1.14	1.17	1.62			3.43	3.43	3.44
	velocity	4.90	7.81	7.95	8.23	8.09	7.95	7.95	8.09	8.02	7.81	8.65	7.47		7.99	7.99	7.97
Ni Mengfan (CHN) (2003)	time	2.70	3.87	5.02	6.17	7.31	8.44	9.64	10.81	11.98	13.16		14.73	2 / 5			
	reaction time	0.199	interval	1.17	1.15	1.15	1.14	1.13	1.20	1.17	1.17	1.57			3.47	3.47	3.52
	velocity	5.08	7.81	7.95	7.95	8.02	8.09	7.62	7.81	7.81	7.75	8.93	7.47		7.90	7.90	7.79
Zhou Jun (CHN) (2002)	time	2.83	3.99	5.11	6.25	7.36	8.51	9.66	10.79	11.98	13.16		14.80	9 / 6			
	reaction time	0.173	interval	1.16	1.12	1.14	1.11	1.15	1.15	1.13	1.19	1.18	1.64		3.42	3.41	3.50
	velocity	4.85	7.88	8.16	8.02	8.23	7.95	7.95	8.09	7.68	7.75	8.55	7.43		8.02	8.04	7.83
Yan Yuwei (CHN) (2002)	time	2.75	3.93	5.12	6.29	7.47	8.64	9.84	11.03	12.24	13.46		15.15	3 / 7			
	reaction time	0.172	interval	1.18	1.19	1.17	1.18	1.17	1.20	1.19	1.21	1.22	1.69		3.54	3.55	3.62
	velocity	4.99	7.75	7.68	7.81	7.75	7.81	7.62	7.68	7.55	7.49	8.30	7.26		7.75	7.72	7.57
Liu Junxi (CHN) (2003)	time	2.61	3.78										dnf	4 / --			
	reaction time		interval	1.17													
	velocity	5.26	7.81														

U20 FINAL (99cm)

date 14-Aug-19 wind -0.3 m/s

CAA Hurdle Development (2019)

			H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
Ning Xiaohan (CHN) (2000)	time		2.57	3.62	4.65	5.67	6.69	7.71	8.74	9.78	10.83	11.91		13.39	4 / 1			
	reaction time	0.134		1.05	1.03	1.02	1.02	1.02	1.03	1.04	1.05	1.08	1.48	PB		3.10	3.07	3.17
	velocity		5.34	8.70	8.87	8.96	8.96	8.96	8.87	8.79	8.70	8.46	9.47	8.22		8.85	8.93	8.65
Liu Bowen (CHN) (2001)	time		2.54	3.61	4.67	5.69	6.73	7.76	8.80	9.86	10.91	11.99		13.47	5 / 2			
	reaction time	0.144		1.07	1.06	1.02	1.04	1.03	1.04	1.06	1.05	1.08	1.48	PB		3.15	3.11	3.19
	velocity		5.40	8.54	8.62	8.96	8.79	8.87	8.79	8.62	8.70	8.46	9.47	8.17		8.70	8.82	8.60
Shi Guoxing (CHN) (2000)	time		2.67	3.76	4.81	5.85	6.87	7.91	8.96	10.00	11.07	12.13		13.52	8 / 3			
	reaction time	0.144		1.09	1.05	1.04	1.02	1.04	1.05	1.04	1.07	1.06	1.39	PB		3.18	3.11	3.17
	velocity		5.14	8.39	8.70	8.79	8.96	8.79	8.70	8.79	8.54	8.62	10.09	8.14		8.62	8.82	8.65
Lu Wei (CHN) (2000)	time		2.67	3.73	4.77	5.82	6.82	7.86	8.89	9.93	11.01	12.09		13.55	7 / 4			
	reaction time	0.170		1.06	1.04	1.05	1.00	1.04	1.03	1.04	1.08	1.08	1.46	PB		3.15	3.07	3.20
	velocity		5.14	8.62	8.79	8.70	9.14	8.79	8.87	8.79	8.46	8.46	9.60	8.12		8.70	8.93	8.57
Guo Zhongjie (CHN) (2001)	time		2.64	3.70	4.75	5.77	6.79	7.82	8.87	9.93	11.03	12.14		13.68	9 / 5			
	reaction time	0.162		1.06	1.05	1.02	1.02	1.03	1.05	1.06	1.10	1.11	1.54	PB		3.13	3.10	3.27
	velocity		5.20	8.62	8.70	8.96	8.96	8.87	8.70	8.62	8.31	8.23	9.10	8.04		8.76	8.85	8.39
Ji Haohao (CHN) (2001)	time		2.69	3.77	4.84	5.90	6.94	8.04	9.09	10.19	11.31	12.43		13.96	2 / 6			
	reaction time	0.187		1.08	1.07	1.06	1.04	1.10	1.05	1.10	1.12	1.12	1.53			3.21	3.19	3.34
	velocity		5.10	8.46	8.54	8.62	8.79	8.31	8.70	8.31	8.16	8.16	9.16	7.88		8.54	8.60	8.21
Fung Kin Hei (HKG) (2000)	time		2.68	3.78	4.90	5.99	7.07	8.17	9.24	10.37	11.48	12.61		14.10	3 / 7			
	reaction time	0.155		1.10	1.12	1.09	1.08	1.10	1.07	1.13	1.11	1.13	1.49			3.31	3.25	3.37
	velocity		5.12	8.31	8.16	8.39	8.46	8.31	8.54	8.09	8.23	8.09	9.41	7.80		8.28	8.44	8.14

U20 Heat 3 (99cm)

date 14-Aug-19 wind 0.6 m/s

CAA Hurdle Development (2019)

			H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
Ning Xiaohan (CHN) (2000)	time		2.58	3.62	4.64	5.67	6.69	7.71	8.74	9.80	10.88	12.04		13.59	6 / 1			
	reaction time	0.135		1.04	1.02	1.03	1.02	1.02	1.03	1.06	1.08	1.16	1.55			3.09	3.07	3.30
	velocity		5.32	8.79	8.96	8.87	8.96	8.96	8.87	8.62	8.46	7.88	9.05	8.09		8.87	8.93	8.31
Liu Bowen (CHN) (2001)	time		2.57	3.61	4.67	5.71	6.74	7.77	8.83	9.89	10.98	12.08		13.66	7 / 2			
	reaction time	0.152		1.04	1.06	1.04	1.03	1.03	1.06	1.06	1.09	1.10	1.58			3.14	3.12	3.25
	velocity		5.34	8.79	8.62	8.79	8.87	8.87	8.62	8.62	8.39	8.31	8.87	8.05		8.73	8.79	8.44
Yu Haonan (CHN) (2000)	time		2.64	3.75	4.83	5.94	7.02	8.12	9.21	10.31	11.41	12.53		14.04	4 / 3			
	reaction time	0.154		1.11	1.08	1.11	1.08	1.10	1.09	1.10	1.10	1.12	1.51			3.30	3.27	3.32
	velocity		5.20	8.23	8.46	8.23	8.46	8.31	8.39	8.31	8.31	8.16	9.28	7.83		8.31	8.39	8.26
Pan Yicheng (CHN) (2000)	time		2.70	3.79	4.87	5.95	7.04	8.11	9.21	10.31	11.44	12.58		14.05	5 / 4			
	reaction time	0.163		1.09	1.08	1.08	1.09	1.07	1.10	1.10	1.13	1.14	1.47			3.25	3.26	3.37
	velocity		5.08	8.39	8.46	8.46	8.39	8.54	8.31	8.31	8.09	8.02	9.54	7.83		8.44	8.41	8.14
Gao Shunyang (CHN) (2001)	time		2.75	3.85	4.95	6.02	7.11	8.19	9.28	10.38	11.49	12.64		14.21	2 / 5			
	reaction time	0.184		1.10	1.10	1.07	1.09	1.08	1.09	1.10	1.11	1.15	1.57			3.27	3.26	3.36
	velocity		4.99	8.31	8.31	8.54	8.39	8.46	8.39	8.31	8.23	7.95	8.93	7.74		8.39	8.41	8.16
Li Fuqiang (CHN) (2000)	time		2.65	3.77	4.90	6.01	7.15	8.26	9.39	10.55	11.68	12.83		14.35	8 / 6			
	reaction time	0.162		1.12	1.13	1.11	1.14	1.11	1.13	1.16	1.13	1.15	1.52			3.36	3.38	3.44
	velocity		5.18	8.16	8.09	8.23	8.02	8.23	8.09	7.88	8.09	7.95	9.22	7.67		8.16	8.11	7.97
Xu Xin (MAC) (2000)	time		2.69	3.87	5.02	6.22	7.44	8.61	9.83	11.06	12.31	13.56		15.26	3 / 7			
	reaction time	0.138		1.18	1.15	1.20	1.22	1.17	1.22	1.23	1.25	1.25	1.70			3.53	3.61	3.73
	velocity		5.10	7.75	7.95	7.62	7.49	7.81	7.49	7.43	7.31	7.31	8.25	7.21		7.77	7.60	7.35

U20 Heat 2 (99cm)

date 14-Aug-19 wind -0.3 m/s

CAA Hurdle Development (2019)

			H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
Lu Wei (CHN) (2000)	time		2.71	3.80	4.84	5.89	6.91	7.94	8.99	10.04	11.13	12.21		13.72	6 / 1			
	reaction time	0.205		1.09	1.04	1.05	1.02	1.03	1.05	1.05	1.09	1.08	1.51			3.18	3.10	3.22
	velocity		5.06	8.39	8.79	8.70	8.96	8.87	8.70	8.70	8.39	8.46	9.28	8.02		8.62	8.85	8.52
Shi Guoxing (CHN) (2000)	time		2.72	3.81	4.86	5.91	6.97	8.04	9.11	10.19	11.26	12.34		13.77	5 / 2			
	reaction time	0.179		1.09	1.05	1.05	1.06	1.07	1.07	1.08	1.07	1.08	1.43			3.19	3.20	3.23
	velocity		5.04	8.39	8.70	8.70	8.62	8.54	8.54	8.46	8.54	8.46	9.80	7.99		8.60	8.57	8.49
Fung Kin Hei (HKG) (2000)	time		2.65	3.72	4.81	5.91	6.97	8.07	9.18	10.28	11.39	12.53		14.04	7 / 3			
	reaction time	0.160		1.07	1.09	1.10	1.06	1.10	1.11	1.10	1.11	1.14	1.51			3.26	3.27	3.35
	velocity		5.18	8.54	8.39	8.31	8.62	8.31	8.23	8.31	8.23	8.02	9.28	7.83		8.41	8.39	8.19
Wang Weihao (CHN) (2001)	time		2.64	3.72	4.80	5.87	6.96	8.09	9.20	10.33	11.45	12.63		14.15	4 / 4			
	reaction time	0.174		1.08	1.08	1.07	1.09	1.13	1.11	1.13	1.12	1.18	1.52			3.23	3.33	3.43
	velocity		5.20	8.46	8.46	8.54	8.39	8.09	8.23	8.09	8.16	7.75	9.22	7.77		8.49	8.23	7.99

Jiang Jinlong (CHN) (2000)	time	2.70	3.80	4.90	5.99	7.09	8.18	9.26	10.37	11.48	12.61		14.16	3 / 5			
	reaction time	0.168		1.10	1.10	1.09	1.10	1.09	1.08	1.11	1.11	1.13	1.55		3.29	3.27	3.35
	velocity		5.08	8.31	8.31	8.39	8.31	8.39	8.46	8.23	8.23	8.09	9.05	7.77	8.33	8.39	8.19
Xu Jialin (CHN) (2001)	time	2.69	3.81	4.91	6.04	7.17	8.29	9.44	10.59	11.80	12.98		14.63	8 / 6			
	reaction time	0.166		1.12	1.10	1.13	1.13	1.12	1.15	1.15	1.21	1.18	1.65		3.35	3.40	3.54
	velocity		5.10	8.16	8.31	8.09	8.09	8.16	7.95	7.95	7.55	7.75	8.50	7.52	8.19	8.06	7.75

U20 Heat 1 (99cm)

date 14-Aug-19 wind 0.5 m/s

CAA Hurdle Development (2019)

		H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
Zhu Shenglong (CHN) (2000)	time	2.54	3.63	4.71	5.74	6.77	7.82	8.87	9.94	11.04	12.14		13.82	5 / 1			
	reaction time	0.145		1.09	1.08	1.03	1.05	1.05	1.07	1.10	1.10	1.68			3.20	3.13	3.27
	velocity		5.40	8.39	8.46	8.87	8.87	8.70	8.70	8.54	8.31	8.31	8.35	7.96	8.57	8.76	8.39
Guo Zhongjie (CHN) (2001)	time	2.60	3.68	4.75	5.77	6.81	7.86	8.92	10.03	11.11	12.26		13.90	4 / 2			
	reaction time	0.211		1.08	1.07	1.02	1.04	1.05	1.06	1.11	1.08	1.15	1.64		3.17	3.15	3.34
	velocity		5.28	8.46	8.54	8.96	8.79	8.70	8.62	8.23	8.46	7.95	8.55	7.91	8.65	8.70	8.21
Ji Haohao (CHN) (2001)	time	2.56	3.64	4.71	5.77	6.82	7.88	8.96	10.05	11.16	12.28		13.91	6 / 3			
	reaction time	0.168		1.08	1.07	1.06	1.05	1.06	1.08	1.09	1.11	1.12	1.63		3.21	3.19	3.32
	velocity		5.36	8.46	8.54	8.62	8.70	8.62	8.46	8.39	8.23	8.16	8.60	7.91	8.54	8.60	8.26
Wang Luoxi (CHN) (2001)	time	2.57	3.67	4.74	5.80	6.87	7.96	9.04	10.14	11.24	12.39		14.04	3 / 4			
	reaction time	0.178		1.10	1.07	1.06	1.07	1.09	1.08	1.10	1.10	1.15	1.65		3.23	3.24	3.35
	velocity		5.34	8.31	8.54	8.62	8.54	8.39	8.46	8.31	8.31	7.95	8.50	7.83	8.49	8.46	8.19
Jia Qihang (CHN) (2001)	time	2.59	3.72	4.81	5.91	6.99	8.14	9.26	10.41	11.51	12.70		14.30	7 / 5			
	reaction time	0.177		1.13	1.09	1.10	1.08	1.15	1.12	1.15	1.10	1.19	1.60		3.32	3.35	3.44
	velocity		5.30	8.09	8.39	8.31	8.46	7.95	8.16	7.95	8.31	7.68	8.76	7.69	8.26	8.19	7.97
Huan Jiaying (CHN) (2001)	time	2.59	3.69	4.81	5.89	6.99	8.11	9.23	10.36	11.49	12.66		14.32	8 / 6			
	reaction time	0.196		1.10	1.12	1.08	1.10	1.12	1.12	1.13	1.13	1.17	1.66		3.30	3.34	3.43
	velocity		5.30	8.31	8.16	8.46	8.31	8.16	8.16	8.09	8.09	7.81	8.45	7.68	8.31	8.21	7.99

2019 Pan American Games (Lima, PER) (TV Analysis)**FINAL**

date 10-Aug-19 wind 1.8 m/s

Henson (2020) - Athlete First: 2019 year end hurdle report

		H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
Brathwaite, Shane (BAR) (1990)	time	2.63	3.68	4.72	5.74	6.74	7.77	8.79	9.81	10.84	11.89		13.31	7 / 1			
	reaction time	0.153		1.05	1.04	1.02	1.00	1.03	1.02	1.03	1.05	1.42			3.11	3.05	3.10
	velocity		5.22	8.70	8.79	8.96	9.14	8.87	8.96	8.87	8.70	9.87	8.26		8.82	8.99	8.85
Crittenden, Freddie (USA) (1991)	time	2.63	3.67	4.68	5.70	6.70	7.72	8.76	9.79	10.84	11.89		13.32	6 / 2			
	reaction time	0.140		1.04	1.01	1.02	1.00	1.02	1.03	1.05	1.05	1.43			3.07	3.06	3.13
	velocity		5.22	8.79	9.05	8.96	9.14	8.96	8.79	8.87	8.70	9.80	8.26		8.93	8.96	8.76
Santos Rodriques, Eduardo (BRA) (2001)	time	2.70	3.74	4.77	5.77	6.80	7.80	8.84	9.88	10.92	12.03		13.48	8 / 3			
	reaction time	0.193		1.04	1.03	1.00	1.03	1.04	1.04	1.04	1.11	1.45			3.07	3.07	3.19
	velocity		5.08	8.79	8.87	9.14	8.87	9.14	8.79	8.79	8.23	9.67	8.16		8.93	8.93	8.60
Swift, Greggmar (BAR) (1991)	time	2.60	3.60	4.65	5.68	6.70	7.74	8.79	9.86	10.95	12.05		13.51	4 / 4			
	reaction time	0.140		1.00	1.05	1.03	1.02	1.04	1.05	1.07	1.09	1.10	1.46		3.08	3.11	3.26
	velocity		5.28	9.14	8.70	8.87	8.96	8.79	8.70	8.54	8.39	8.31	9.60	8.14	8.90	8.82	8.41

2019 Japanese National High School Championships (Okinawa, JPN) (99cm)**FINAL**

date 08-Aug-19 wind -3.0 m/s

Kota (2019) - 72nd high school championships: JAF scientific committee - biomechanics data collection

		H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
Muratake, Rashido (JPN) (2000)	time	2.64	3.73	4.83	5.92	7.02	8.13	9.25	10.37	11.49	12.63		14.04	6 / 1			
	reaction time			1.09	1.10	1.09	1.10	1.11	1.12	1.12	1.14	1.41	PB		3.28	3.33	3.38
	velocity		5.20	8.39	8.31	8.39	8.31	8.23	8.16	8.16	8.02	9.94	7.83		8.36	8.23	8.11
Kondo, Mizuki (JPN) (2003)	time	2.69	3.80	4.92	6.00	7.12	8.23	9.33	10.46	11.57	12.72		14.21	5 / 2			
	reaction time			1.11	1.12	1.08	1.12	1.11	1.10	1.13	1.11	1.15	1.49	PB	3.31	3.33	3.39
	velocity		5.10	8.23	8.16	8.46	8.16	8.23	8.31	8.09	8.23	7.95	9.41	7.74	8.28	8.23	8.09
Sugaya, Takumi (JPN) (2001)	time	2.74	3.86	4.99	6.12	7.24	8.37	9.50	10.63	11.75	12.91		14.40	7 / 3			
	reaction time			1.12	1.13	1.13	1.12	1.13	1.13	1.12	1.16	1.49	PB		3.38	3.38	3.41
	velocity		5.01	8.16	8.09	8.09	8.16	8.09	8.09	8.09	8.16	7.88	9.41	7.64	8.11	8.11	8.04
Fujiwara, Koki (JPN) (2002)	time	2.71	3.84	4.93	6.01	7.10	8.26	9.37	10.54	11.71	12.93		14.47	3 / 4			
	reaction time			1.13	1.09	1.08	1.09	1.16	1.11	1.17	1.17	1.22	1.54		3.30	3.36	3.56
	velocity		5.06	8.09	8.39	8.46	8.39	7.88	8.23	7.81	7.81	7.49	9.10	7.60	8.31	8.16	7.70
Kurokawa, Kazuki (JPN) (2001)	time	2.72	3.90	5.03	6.19	7.35	8.52	9.68	10.84	12.01	13.22		14.76	4 / 5			
	reaction time			1.18	1.13	1.16	1.16	1.17	1.16	1.16	1.17	1.21	1.54	PB	3.47	3.49	3.54
	velocity		5.04	7.75	8.09	7.88	7.88	7.81	7.88	7.88	7.81	7.55	9.10	7.45	7.90	7.86	7.75

Bondoreo, Sho (JPN)	time	2.78	3.95	5.12	6.27	7.41	8.60	9.78	10.95	12.14	13.33		14.91	2 / 6			
	reaction time	interval	1.17	1.17	1.15	1.14	1.19	1.18	1.17	1.19	1.19	1.58			3.49	3.51	3.55
	velocity	4.94	7.81	7.81	7.95	8.02	7.68	7.75	7.81	7.68	7.68	8.87	7.38		7.86	7.81	7.72
Tochioka, Takei (JPN) (2001)	time	2.76	3.96	5.13	6.29	7.47	8.67	9.93	11.18	12.45	13.78		15.49	1 / 7			
	reaction time	interval	1.20	1.17	1.16	1.18	1.20	1.26	1.25	1.27	1.33	1.71			3.53	3.64	3.85
	velocity	4.97	7.62	7.81	7.88	7.75	7.62	7.25	7.31	7.20	6.87	8.20	7.10		7.77	7.53	7.12

2019 Chinese World Championship Trials (Shenyang, CHN)

FINAL

date 02-Aug-19 wind 0.0 m/s

CAA Hurdle Development (2019)

			H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
Xie Wenjun (CHN) (1990)	time		2.61	3.64	4.67	5.70	6.72	7.74	8.77	9.80	10.87	11.97		13.43	4 / 1			
	reaction time	0.153	interval	7 steps	1.03	1.03	1.03	1.02	1.02	1.03	1.07	1.10	1.46			3.09	3.07	3.20
	velocity		5.26	8.87	8.87	8.87	8.96	8.96	8.87	8.87	8.54	8.31	9.60	8.19		8.87	8.93	8.57
Zeng Jianhang (CHN) (1998)	time		2.56	3.60	4.64	5.67	6.74	7.76	8.82	9.88	10.96	12.08		13.54	3 / 2			
	reaction time	0.150	interval		1.04	1.04	1.03	1.07	1.02	1.06	1.08	1.12	1.46			3.11	3.15	3.26
	velocity		5.36	8.79	8.79	8.87	8.54	8.96	8.62	8.62	8.46	8.16	9.60	8.12		8.82	8.70	8.41
Pan Zijie (CHN) (1995)	time		2.62	3.70	4.77	5.84	6.91	7.96	9.04	10.11	11.21	12.32		13.77	8 / 3			
	reaction time	0.171	interval		1.08	1.07	1.07	1.07	1.05	1.08	1.07	1.10	1.45	PB		3.22	3.20	3.28
	velocity		5.24	8.46	8.54	8.54	8.54	8.70	8.46	8.54	8.31	8.23	9.67	7.99		8.52	8.57	8.36
Jiang Fan (CHN) (1989)	time		2.65	3.74	4.80	5.87	6.92	7.99	9.07	10.16	11.26	12.38		13.89	5 / 4			
	reaction time	0.148	interval		1.09	1.06	1.07	1.05	1.07	1.08	1.09	1.10	1.51			3.22	3.20	3.31
	velocity		5.18	8.39	8.62	8.54	8.70	8.54	8.46	8.39	8.31	8.16	9.28	7.92		8.52	8.57	8.28
Zhang Tao (CHN) (1997)	time		2.69	3.78	4.87	5.94	7.01	8.07	9.14	10.24	11.34	12.48		13.98	6 / 5			
	reaction time	0.159	interval		1.09	1.09	1.07	1.06	1.07	1.10	1.10	1.14	1.50			3.25	3.20	3.34
	velocity		5.10	8.39	8.39	8.54	8.54	8.62	8.54	8.31	8.31	8.02	9.35	7.87		8.44	8.57	8.21
Li Jiming (CHN) (1996)	time		2.64	3.73	4.80	5.91	7.00	8.09	9.18	10.29	11.40	12.58		14.14	1 / 6			
	reaction time	0.176	interval		1.09	1.07	1.11	1.09	1.09	1.11	1.11	1.18	1.56			3.27	3.27	3.40
	velocity		5.20	8.39	8.54	8.23	8.39	8.39	8.39	8.23	8.23	7.75	8.99	7.78		8.39	8.39	8.06
Wu Zhenhua (CHN) (1998)	time		2.70	3.82	4.93	6.02	7.12	8.21	9.32	10.43	11.58	12.75		14.25	7 / 7			
	reaction time	0.151	interval		1.12	1.11	1.09	1.10	1.09	1.11	1.11	1.15	1.17	1.50		3.32	3.30	3.43
	velocity		5.08	8.16	8.23	8.39	8.31	8.39	8.23	8.23	7.95	7.81	9.35	7.72		8.26	8.31	7.99

Heat 2

date 02-Aug-19 wind -0.2 m/s

CAA Hurdle Development (2019)

			H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
Zeng Jianhang (CHN) (1998)	time		2.62	3.70	4.78	5.86	6.94	8.02	9.10	10.20	11.31	12.47		13.99	7 / 1			
	reaction time	0.153	interval		1.08	1.08	1.08	1.08	1.08	1.10	1.11	1.16	1.52			3.24	3.24	3.37
	velocity		5.24	8.46	8.46	8.46	8.46	8.46	8.46	8.31	8.23	7.88	9.22	7.86		8.46	8.46	8.14
Zhang Tao (CHN) (1997)	time		2.70	3.82	4.90	5.96	7.07	8.16	9.24	10.36	11.48	12.65		14.20	4 / 2			
	reaction time	0.174	interval		1.12	1.08	1.06	1.11	1.09	1.12	1.12	1.17	1.55			3.26	3.28	3.41
	velocity		5.08	8.16	8.46	8.62	8.23	8.39	8.46	8.16	8.16	7.81	9.05	7.75		8.41	8.36	8.04
Wu Zhenhua (CHN) (1998)	time		2.74	3.87	4.98	6.10	7.21	8.34	9.48	10.61	11.74	12.91		14.43	8 / 3			
	reaction time	0.174	interval		1.13	1.11	1.12	1.11	1.13	1.14	1.13	1.13	1.52			3.36	3.38	3.43
	velocity		5.01	8.09	8.23	8.16	8.23	8.09	8.02	8.09	8.09	7.81	9.22	7.62		8.16	8.11	7.99
Li Jiming (CHN) (1996)	time		2.71	3.85	4.96	6.06	7.17	8.31	9.46	10.63	11.81	13.01		14.69	3 / 4			
	reaction time	0.191	interval		1.14	1.11	1.10	1.11	1.14	1.15	1.17	1.18	1.20	1.68		3.35	3.40	3.55
	velocity		5.06	8.02	8.23	8.31	8.23	8.02	7.95	7.81	7.75	7.62	8.35	7.49		8.19	8.06	7.72
Lu Yang (CHN) (1996)	time		2.85	4.01	5.15	6.30	7.44	8.61	9.80	11.05	12.34	13.72		15.90	6 / 5			
	reaction time	0.196	interval		1.16	1.14	1.15	1.14	1.17	1.19	1.25	1.29	1.38	2.18		3.45	3.50	3.92
	velocity		4.81	7.88	8.02	7.95	8.02	7.81	7.68	7.31	7.09	6.62	6.43	6.92		7.95	7.83	6.99
Sun Zhenjiang (CHN) (1999)	time		2.72											dnf	5 / --			
	reaction time	0.166	interval	7 steps														
	velocity		5.04															

Heat 1

date 02-Aug-19 wind -0.3 m/s

CAA Hurdle Development (2019)

			H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
Xie Wenjun (CHN) (1990)	time		2.63	3.72	4.80	5.88	6.97	8.06	9.12	10.21	11.31	12.42		13.94	6 / 1			
	reaction time	0.175	interval	7 steps	1.09	1.08	1.08	1.09	1.09	1.06	1.09	1.10	1.52			3.25	3.24	3.30
	velocity		5.22	8.39	8.46	8.46	8.39	8.39	8.62	8.39	8.31	8.23	9.22	7.89		8.44	8.46	8.31
Jiang Fan (CHN) (1989)	time		2.68	3.78	4.88	5.97	7.05	8.12	9.24	10.36	11.48	12.61		14.13	7 / 2			
	reaction time	0.144	interval		1.10	1.10	1.09	1.08	1.07	1.12	1.12	1.12	1.52			3.29	3.27	3.37
	velocity		5.12	8.31	8.31	8.39	8.46	8.54	8.16	8.16	8.16	8.09	9.22	7.78		8.33	8.39	8.14
Pan Zijie (CHN) (1995)	time		2.65	3.77	4.87	6.00	7.14	8.24	9.35	10.46	11.57	12.68		14.17	3 / 3			

reaction time	0.173	interval		1.12	1.10	1.13	1.14	1.10	1.11	1.11	1.11	1.11	1.49		3.35	3.35	3.33	
		velocity		5.18	8.16	8.31	8.09	8.02	8.31	8.23	8.23	8.23	9.41	7.76	8.19	8.19	8.23	
Zhao Pengchuan (CHN) (1993)	time		2.69	3.82	4.93	6.01	7.12	8.21	9.32	10.43	11.56	12.69		14.25	4 / 4			
reaction time	0.168	interval		1.13	1.11	1.08	1.11	1.09	1.11	1.11	1.13	1.13	1.56		3.32	3.31	3.37	
		velocity		5.10	8.09	8.23	8.46	8.23	8.39	8.23	8.23	8.09	8.09	8.99	7.72	8.26	8.28	8.14
Zhu Shenglong (CHN) (2000)	time		2.74	3.88	5.02	6.14	7.28	8.42	9.55	10.73	11.89	13.11		14.82	5 / 5			
reaction time	0.154	interval		1.14	1.14	1.12	1.14	1.14	1.13	1.18	1.16	1.22	1.71		3.40	3.41	3.56	
		velocity		5.01	8.02	8.02	8.16	8.02	8.09	7.75	7.88	7.49	8.20	7.42	8.06	8.04	7.70	

2019 USATF National Championships (Des Moines, IA) (TV Analysis)

FINAL

FINAL

	date	28-Jul-19	wind -0.8 m/s													Henson (2020) - Athlete First: 2019 year end hurdle report						
			H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10				
Roberts, Daniel (USA) (1998)	time		2.46	3.50	4.54	5.57	6.57	7.60	8.61	9.64	10.67	11.73		13.23	4 / 1							
	reaction time			1.04	1.04	1.03	1.00	1.03	1.01	1.03	1.03	1.06	1.50				3.11	3.04	3.12			
	velocity		5.58	8.79	8.79	8.87	9.14	8.87	9.05	8.87	8.87	8.62	9.35	8.31			8.82	9.02	8.79			
Holloway, Grant (USA) (1997)	time		2.46	3.50	4.48	5.47	6.47	7.48	8.54	9.58	10.67	11.77		13.36	7 / 2							
	reaction time		7 steps	1.04	0.98	0.99	1.00	1.01	1.06	1.04	1.09	1.10	1.59				3.01	3.07	3.23			
	velocity		5.58	8.79	9.33	9.23	9.14	9.05	8.62	8.79	8.39	8.31	8.82	8.23			9.11	8.93	8.49			
Allen, Devon (USA) (1994)	time		2.50	3.60	4.63	5.67	6.69	7.70	8.74	9.77	10.83	11.90		13.38	8 / 3							
	reaction time			1.10	1.03	1.04	1.02	1.01	1.04	1.03	1.06	1.07	1.48				3.17	3.07	3.16			
	velocity		5.49	8.31	8.87	8.79	8.96	9.05	8.79	8.87	8.62	8.54	9.47	8.22			8.65	8.93	8.68			
Crittendon, Freddie (USA) (1995)	time		2.60	3.67	4.70	5.70	6.73	7.73	8.74	9.77	10.84	11.88		13.39	6 / 4							
	reaction time			1.07	1.03	1.00	1.03	1.00	1.01	1.03	1.07	1.04	1.51				3.10	3.04	3.14			
	velocity		5.28	8.54	8.87	9.14	8.87	9.14	9.05	8.87	8.54	8.79	9.28	8.22			8.85	9.02	8.73			
Dickson, Michael (USA) (1989)	time		2.56	3.60	4.67	5.70	6.74	7.77	8.80	9.85	10.94	12.00		13.45	3 / 5							
	reaction time			1.04	1.07	1.03	1.04	1.03	1.03	1.05	1.09	1.06	1.45	PB			3.14	3.10	3.20			
	velocity		5.36	8.79	8.54	8.87	8.79	8.87	8.87	8.70	8.39	8.62	9.67	8.18			8.73	8.85	8.57			
Eaton, Jarret (USA) (1989)	time		2.50	3.54	4.54	5.57	6.57	7.60	8.60	9.71	10.81	11.93		13.51	1 / 6							
	reaction time		7 steps	1.04	1.00	1.03	1.00	1.03	1.00	1.11	1.10	1.12	1.58				3.07	3.03	3.33			
	velocity		5.49	8.79	9.14	8.87	9.14	8.87	9.14	8.23	8.31	8.16	8.87	8.14			8.93	9.05	8.23			
Mallett, Aaron (USA) (1994)	time		2.53	3.60	4.63	5.67	6.67	7.73	8.78	9.84	10.91	11.97		13.51	5 / 7							
	reaction time		7 steps	1.07	1.03	1.04	1.00	1.06	1.05	1.06	1.07	1.06	1.54				3.14	3.11	3.19			
	velocity		5.42	8.54	8.87	8.79	9.14	8.62	8.70	8.62	8.54	8.62	9.10	8.14			8.73	8.82	8.60			
Cunningham, Trey (USA) (1995)	time		2.51	3.57	4.60	5.63	6.67	7.70	8.76	9.85	10.94	12.11		13.72	2 / 8							
	reaction time		7 steps	1.06	1.03	1.03	1.04	1.03	1.06	1.09	1.09	1.17	1.61				3.12	3.13	3.35			
	velocity		5.47	8.62	8.87	8.87	8.79	8.87	8.62	8.39	8.39	7.81	8.71	8.02			8.79	8.76	8.19			

2019 Corporate Team vs College Team (Hiratsukai, JPN)

FINAL

FINAL

	date	27-Jul-19	wind	1.9 m/s	(2019) - http://fddata-store.com/2019/09/03/post-1615/														
			H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10	
Takayama, Shunya (JPN) (1994)	time		2.55	3.58	4.62	5.63	6.66	7.71	8.75	9.79	10.84	11.91		13.30	4 / 1				
reaction time	0.129	interval		1.03	1.04	1.01	1.03	1.05	1.04	1.04	1.05	1.07	1.39	NR		3.08	3.12	3.16	
		velocity	5.38	8.87	8.79	9.05	8.87	8.70	8.79	8.79	8.70	8.54	10.09	8.27		8.90	8.79	8.68	

2019 Müller Anniversary Games (London, GBR) (TV Analysis)

FINAL

FINAL

	date	21-Jul-19	wind	0.4 m/s	Henson (2020) - Athlete First: 2019 year end hurdle report														
			H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10	
Xie Wenjun (CHN) (1990)	time		2.46	3.53	4.55	5.57	6.58	7.59	8.63	9.66	10.69	11.72		13.28	5 / 1				
	reaction time	0.135	interval	7 steps	1.07	1.02	1.02	1.01	1.01	1.04	1.03	1.03	1.03	1.56			3.11	3.06	3.09
			velocity		5.58	8.54	8.96	8.96	9.05	9.05	8.79	8.87	8.87	8.87	8.99	8.28		8.82	8.96
Belocian, Wilhem (FRA) (1995)	time		2.43	3.46	4.48	5.50	6.50	7.53	8.56	9.58	10.61	11.66		13.28	6 / 2				
	reaction time	0.125	interval		1.03	1.02	1.02	1.00	1.03	1.03	1.02	1.03	1.05	1.62			3.07	3.06	3.10
			velocity		5.65	8.87	8.96	8.96	9.14	8.87	8.87	8.96	8.87	8.70	8.65	8.28		8.93	8.96
McLeod, Omar (JAM) (1994)	time		2.39	3.45	4.48	5.53	6.55	7.56	8.58	9.64	10.72	11.77		13.32	4 / 3				
	reaction time	0.134	interval		1.06	1.03	1.05	1.02	1.01	1.02	1.06	1.08	1.05	1.55			3.14	3.05	3.19
			velocity		5.74	8.62	8.87	8.70	8.96	9.05	8.96	8.62	8.46	8.70	9.05	8.26		8.73	8.99
Martinot-Lagarde, Pascal (FRA)	time		2.45	3.51	4.55	5.58	6.60	7.63	8.66	9.73	10.80	11.90		13.49	8 / 4				
	reaction time	0.152	interval	7 steps	1.06	1.04	1.03	1.02	1.03	1.03	1.07	1.07	1.10	1.59			3.13	3.08	3.24
			velocity		5.60	8.62	8.79	8.87	8.96	8.87	8.87	8.54	8.54	8.31	8.82	8.15		8.76	8.90
Alkana, Antonio (RSA) (1990)	time		2.43	3.46	4.51	5.53	6.55	7.57	8.62	9.71	10.78	11.91		13.51	7 / 5				
	reaction time	0.111	interval		1.03	1.05	1.02	1.02	1.02	1.05	1.09	1.07	1.13	1.60			3.10	3.09	3.29
			velocity		5.65	8.87	8.70	8.96	8.96	8.96	8.70	8.39	8.54	8.09	8.76	8.14		8.85	8.87

Pozzi, Andy (GBR) (1992)	time	2.38	3.43	4.45	5.46	6.46	7.50	8.55	9.63	10.77	11.89		13.52	3 / 6					
	reaction time	0.131	interval	7 steps	1.05	1.02	1.01	1.00	1.04	1.05	1.08	1.14	1.12	1.63		3.08	3.09	3.34	
		velocity				5.76	8.70	8.96	9.05	9.14	8.79	8.70	8.46	8.02	8.16	8.60	8.14	8.90	8.87
Levy, Ronald (JAM) (1992)	time	2.43	3.51	4.56	5.61	6.63	7.70	8.73	9.82	10.92	11.99		13.60	2 / 7					
	reaction time	0.165	interval	7 steps	1.08	1.05	1.05	1.02	1.07	1.03	1.09	1.10	1.07	1.61		3.18	3.12	3.26	
		velocity				5.65	8.46	8.70	8.70	8.96	8.54	8.87	8.39	8.31	8.54	8.71	8.09	8.62	8.79
King, David (GBR) (1994)	time	2.45	3.51	4.56	5.61	6.65	7.72	8.77	9.85	10.92	12.05		13.66	1 / 8					
	reaction time	0.135	interval	7 steps	1.06	1.05	1.05	1.04	1.07	1.05	1.08	1.07	1.13	1.61		3.16	3.16	3.28	
		velocity				5.60	8.62	8.70	8.70	8.79	8.54	8.70	8.46	8.54	8.09	8.71	8.05	8.68	8.68

2019 Chinese National Championships (Shenyang, CHN)

Decathlon - Heat 3

Decathlon - Heat 3			date	11-Jul-19	wind	0.0 m/s										CAA Hurdle Development (2019)						
					H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10		
Yue Xuesong (CHN) (1997)	time			2.77	3.92	5.05	6.20	7.36	8.52	9.69	10.86	12.03	13.23			14.76	5 / 1					
	reaction time	0.203	interval		1.15	1.13	1.15	1.16	1.16	1.17	1.17	1.17	1.20	1.53		PB		3.43	3.49	3.54		
			velocity		4.95	7.95	8.09	7.95	7.88	7.88	7.81	7.81	7.81	7.62	9.16		7.45		7.99	7.86	7.75	
Li Xiaodong (CHN) (1995)	time			2.74	3.90	5.05	6.19	7.34	8.50	9.66	10.87	12.06	13.26			14.79	2 / 2					
	reaction time	0.153	interval		1.16	1.15	1.14	1.15	1.16	1.16	1.21	1.19	1.20	1.53				3.45	3.47	3.60		
			velocity		5.01	7.88	7.95	8.02	7.95	7.88	7.88	7.55	7.68	7.62	9.16		7.44		7.95	7.90	7.62	
Li Shuo (CHN) (1994)	time			2.87	4.07	5.25	6.42	7.59	8.76	9.94	11.13	12.34	13.56			15.19	3 / 3					
	reaction time	0.179	interval		1.20	1.18	1.17	1.17	1.17	1.18	1.19	1.21	1.22	1.63				3.55	3.52	3.62		
			velocity		4.78	7.62	7.75	7.81	7.81	7.81	7.75	7.68	7.55	7.49	8.60		7.24		7.72	7.79	7.57	
Wang Chi (CHN) (1997)	time			2.82	4.04	5.24	6.44	7.67	8.89	10.12	11.39	12.64	13.91			15.55	6 / 4					
	reaction time	0.177	interval		1.22	1.20	1.20	1.23	1.22	1.23	1.27	1.25	1.27	1.64		PB		3.62	3.68	3.79		
			velocity		4.87	7.49	7.62	7.62	7.43	7.49	7.43	7.20	7.31	7.20	8.55		7.07		7.57	7.45	7.23	
Wei Peng (CHN) (2002)	time			2.95	4.27	5.60	6.97	8.32	9.71	11.09	12.48	13.88	15.66			18.65	4 / 5					
	reaction time	0.189	interval		1.32	1.33	1.37	1.35	1.39	1.38	1.39	1.40	1.78	2.99		PB		4.02	4.12	4.57		
			velocity		4.65	6.92	6.87	6.67	6.77	6.58	6.62	6.58	6.53	5.13	4.69		5.90		6.82	6.66	6.00	

Decathlon - Heat 2

Decathlon - Heat 2		date	11-Jul-19	wind -0.1 m/s										CAA Hurdle Development (2019)						
				H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10	
Hu Yufei (CHN) (1993)	time			2.72	3.85	4.97	6.07	7.20	8.35	9.51	10.66	11.84	12.98		14.56	7 / 1				
	reaction time	0.162	interval	7 steps	1.13	1.12	1.10	1.13	1.15	1.16	1.15	1.18	1.14	1.58				3.35	3.44	3.47
			velocity		5.04	8.09	8.16	8.31	8.09	7.95	7.88	7.95	7.75	8.02	8.87	7.55		8.19	7.97	7.90
Yang Zihao (CHN) (1998)	time			2.75	3.90	5.05	6.19	7.32	8.49	9.69	10.84	12.04	13.21		14.78	5 / 2				
	reaction time	0.171	interval		1.15	1.15	1.14	1.13	1.17	1.20	1.15	1.20	1.17	1.57	PB			3.44	3.50	3.52
			velocity		4.99	7.95	7.95	8.02	8.09	7.81	7.62	7.95	7.62	7.81	8.93	7.44		7.97	7.83	7.79
Tian Yang (CHN) (1997)	time			2.85	4.00	5.15	6.29	7.44	8.61	9.77	10.96	12.14	13.40		15.02	2 / 3				
	reaction time	0.223	interval		1.15	1.15	1.14	1.15	1.17	1.16	1.19	1.18	1.26	1.62	PB			3.44	3.48	3.63
			velocity		4.81	7.95	7.95	8.02	7.95	7.81	7.88	7.68	7.75	7.25	8.65	7.32		7.97	7.88	7.55
Zhang Zhen (CHN) (1996)	time			2.83	4.00	5.13	6.29	7.43	8.59	9.77	10.96	12.18	13.42		15.10	3 / 4				
	reaction time	0.169	interval		1.17	1.13	1.16	1.14	1.16	1.18	1.19	1.22	1.24	1.68				3.46	3.48	3.65
			velocity		4.85	7.81	8.09	7.88	8.02	7.88	7.75	7.68	7.49	7.37	8.35	7.28		7.92	7.88	7.51
Zhao Guorui (CHN) (1996)	time			2.78	4.02	5.24	6.44	7.64	8.86	10.09	11.33	12.59	13.86		15.63	6 / 5				
	reaction time	0.198	interval		1.24	1.22	1.20	1.20	1.22	1.23	1.24	1.26	1.27	1.77	PB			3.66	3.65	3.77
			velocity		4.94	7.37	7.49	7.62	7.62	7.49	7.43	7.37	7.25	7.20	7.92	7.04		7.49	7.51	7.27
Jiang Yuhong (CHN) (1997)	time			2.98	4.25	5.51	6.79	8.10	9.56	11.04	12.50	13.98	15.48		17.39	4 / 6				
	reaction time	0.247	interval		1.27	1.26	1.28	1.31	1.46	1.48	1.46	1.48	1.50	1.91				3.81	4.25	4.44
			velocity		4.60	7.20	7.25	7.14	6.98	6.26	6.18	6.26	6.18	6.09	7.34	6.33		7.20	6.45	6.18

Decathlon - Heat 1

Decathlon - Heat 1			date	11-Jul-19	wind -0.6 m/s										CAA Hurdle Development (2019)						
					H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10	
Cao Jin (CHN) (1996)		time			2.72	3.84	4.93	6.02	7.17	8.29	9.46	10.62	11.78	13.00		14.57	5 / 1				
	reaction time	0.169	interval	7 steps	1.12	1.09	1.09	1.15	1.12	1.17	1.16	1.16	1.16	1.22	1.57			3.30	3.44	3.54	
			velocity		5.04	8.16	8.39	8.39	7.95	8.16	7.81	7.88	7.88	7.49	8.93	7.55		8.31	7.97	7.75	
Guo Qi (CHN) (1990)		time			2.75	3.91	5.05	6.19	7.32	8.47	9.62	10.77	11.98	13.16		14.75	6 / 2				
	reaction time	0.169	interval	7 steps	1.16	1.14	1.14	1.13	1.15	1.15	1.15	1.15	1.21	1.18	1.59			3.44	3.43	3.54	
			velocity		4.99	7.88	8.02	8.02	8.09	7.95	7.95	7.95	7.55	7.75	8.82	7.46		7.97	7.99	7.75	
Fei Xiang (CHN) (1998)		time			2.82	4.00	5.17	6.34	7.52	8.71	9.89	11.11	12.31	13.54		15.10	3 / 3				
	reaction time	0.165	interval			1.18	1.17	1.17	1.18	1.19	1.18	1.22	1.20	1.23	1.56	PB		3.52	3.55	3.65	
			velocity		4.87	7.75	7.81	7.81	7.75	7.68	7.75	7.49	7.62	7.43	8.99	7.28		7.79	7.72	7.51	
Qian Wenze (CHN) (2001)		time			2.87	4.10	5.29	6.50	7.72	8.96	10.17	11.43	12.71	13.96		15.59	4 / 4				

reaction time	0.211	interval	1.23	1.19	1.21	1.22	1.24	1.21	1.26	1.28	1.25	1.63	PB	3.63	3.67	3.79	
		velocity	4.78	7.43	7.68	7.55	7.49	7.37	7.55	7.25	7.14	7.31	8.60	7.06	7.55	7.47	7.23
Aihaiti Yisikandaer (CHN) (1998)	time	2.88	4.10	5.34	6.59	7.81	9.07	10.34	11.58	12.84	14.14		15.80	8 / 5			
reaction time	0.193	interval	1.22	1.24	1.25	1.22	1.26	1.27	1.24	1.26	1.30	1.66			3.71	3.75	3.80
		velocity	4.76	7.49	7.37	7.31	7.49	7.25	7.20	7.37	7.25	7.03	8.45	6.96	7.39	7.31	7.22
Chen Junyi (CHN) (1999)	time	2.95	4.30	5.64	6.94	8.27	9.61	10.96	12.28	13.61	14.95		16.62	2 / 6			
reaction time	0.190	interval	7 steps	1.35	1.34	1.30	1.33	1.34	1.35	1.32	1.33	1.34	1.67		3.99	4.02	3.99
		velocity	4.65	6.77	6.82	7.03	6.87	6.82	6.77	6.92	6.87	6.82	8.40	6.62	6.87	6.82	6.87

FINAL

FINAL

		date	08-Jul-19	wind	0.1 m/s														CAA Hurdle Development (2019)		
				H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10		
Zeng Jianhang (CHN) (1998)	time			2.58	3.65	4.70	5.75	6.77	7.86	8.92	9.99	11.09	12.18		13.59	8 / 1					
	reaction time	0.174			1.07	1.05	1.05	1.02	1.09	1.06	1.07	1.10	1.09	1.41			3.17	3.17	3.26		
	velocity			5.32	8.54	8.70	8.70	8.96	8.39	8.62	8.54	8.31	8.39	9.94	8.09		8.65	8.65	8.41		
Sun Zhenjiang (CHN) (1999)	time			2.58	3.64	4.69	5.72	6.77	7.82	8.88	9.96	11.04	12.13		13.60	5 / 2					
	reaction time	0.164			7 steps 1.06	1.05	1.03	1.05	1.05	1.06	1.08	1.08	1.09	1.47	PB		3.14	3.16	3.25		
	velocity			5.32	8.62	8.70	8.87	8.70	8.70	8.62	8.46	8.46	8.39	9.54	8.09		8.73	8.68	8.44		
Pan Zijie (CHN) (1995)	time			2.63	3.70	4.80	5.87	6.96	8.02	9.11	10.21	11.31	12.41		13.87	3 / 3					
	reaction time	0.187				1.07	1.10	1.07	1.09	1.06	1.09	1.10	1.10	1.46			3.24	3.24	3.30		
	velocity			5.22	8.54	8.31	8.54	8.39	8.62	8.39	8.31	8.31	8.31	9.60	7.93		8.46	8.46	8.31		
Zhu Shenglong (CHN) (2000)	time			2.72	3.80	4.87	5.92	6.99	8.04	9.09	10.18	11.29	12.40		13.90	2 / 4					
	reaction time	0.170				1.08	1.07	1.05	1.07	1.05	1.05	1.09	1.11	1.11	1.50	PB	3.20	3.17	3.31		
	velocity			5.04	8.46	8.54	8.70	8.54	8.70	8.70	8.39	8.23	8.23	9.35	7.91		8.57	8.65	8.28		
Jiang Fan (CHN) (1989)	time			2.63	3.73	4.80	5.89	6.97	8.06	9.15	10.24	11.34	12.48		13.96	7 / 5					
	reaction time	0.143				1.10	1.07	1.09	1.08	1.09	1.09	1.10	1.14	1.48			3.26	3.26	3.33		
	velocity			5.22	8.31	8.54	8.39	8.46	8.39	8.39	8.39	8.31	8.02	9.47	7.88		8.41	8.41	8.23		
Li Jiming (CHN) (1996)	time			2.65	3.74	4.80	5.87	6.95	8.02	9.12	10.21	11.31	12.46		13.96	1 / 6					
	reaction time	0.180				1.09	1.06	1.07	1.08	1.07	1.10	1.09	1.10	1.15	1.50		3.22	3.25	3.34		
	velocity			5.18	8.39	8.62	8.54	8.46	8.54	8.31	8.39	8.31	7.95	9.35	7.88		8.52	8.44	8.21		
Zhao Pengchuan (CHN) (1993)	time			2.67	3.77	4.84	5.90	6.96	8.02	9.11	10.22	11.35	12.46		14.04	4 / 7					
	reaction time	0.164				1.10	1.07	1.06	1.06	1.06	1.09	1.11	1.13	1.11	1.58		3.23	3.21	3.35		
	velocity			5.14	8.31	8.54	8.62	8.62	8.62	8.39	8.23	8.09	8.23	8.87	7.83		8.49	8.54	8.19		

Heat 3

Heat 3	date	08-Jul-19		wind -0.5 m/s										CAA Hurdle Development (2019)						
				H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10	
Zhang Tao (CHN) (1997)	time			2.65	3.75	4.82	5.91	6.96	7.99	9.07	10.16	11.26	12.41		13.89	2 / 1				
	reaction time	0.158			1.10	1.07	1.09	1.05	1.03	1.08	1.09	1.10	1.15	1.48			3.26	3.16	3.34	
	velocity		5.18	8.31	8.54	8.39	8.70	8.87	8.46	8.39	8.31	7.95	9.47	7.92			8.41	8.68	8.21	
Zeng Jianhang (CHN) (1998)	time			2.69	3.79	4.85	5.94	7.02	8.11	9.18	10.27	11.39	12.51		13.98	6 / 2				
	reaction time	0.204			1.10	1.06	1.09	1.08	1.09	1.07	1.09	1.12	1.12	1.47			3.25	3.24	3.33	
	velocity		5.10	8.31	8.62	8.39	8.46	8.39	8.54	8.39	8.16	8.16	9.54	7.87			8.44	8.46	8.23	
Zhu Shenglong (CHN) (2000)	time			2.71	3.80	4.89	5.95	7.02	8.09	9.20	10.31	11.43	12.57		14.08	5 / 3				
	reaction time	0.169			1.09	1.09	1.06	1.07	1.07	1.11	1.11	1.12	1.14	1.51			3.24	3.25	3.37	
	velocity		5.06	8.39	8.39	8.62	8.54	8.54	8.23	8.23	8.16	8.02	9.28	7.81			8.46	8.44	8.14	
Lu Yang (CHN) (1996)	time			2.70	3.79	4.89	5.94	7.01	8.07	9.15	10.26	11.34	12.55		14.10	3 / 4				
	reaction time	0.170			1.09	1.10	1.05	1.07	1.06	1.08	1.11	1.08	1.21	1.55			3.24	3.21	3.40	
	velocity		5.08	8.39	8.31	8.70	8.54	8.62	8.46	8.23	8.46	7.55	9.05	7.80			8.46	8.54	8.06	
Si Zhengdong (CHN) (1997)	time			2.72	3.84	4.92	6.01	7.09	8.17	9.27	10.38	11.57	12.80		14.43	7 / 5				
	reaction time	0.185			1.12	1.08	1.09	1.08	1.08	1.10	1.11	1.19	1.23	1.63	PB		3.29	3.26	3.53	
	velocity		5.04	8.16	8.46	8.39	8.46	8.46	8.31	8.23	7.68	7.43	8.60	7.62			8.33	8.41	7.77	
Zhang Jinzhao (CHN) (1999)	time			2.82	3.95	5.06	6.17	7.29	8.41	9.54	10.69	11.85	13.03		14.55	4 / 6				
	reaction time	0.233			1.13	1.11	1.11	1.12	1.12	1.13	1.15	1.16	1.18	1.52	PB		3.35	3.37	3.49	
	velocity		4.87	8.09	8.23	8.23	8.16	8.16	8.09	7.95	7.88	7.75	9.22	7.56			8.19	8.14	7.86	
Zhang Shuai (CHN) (1996)	time			2.69	3.84	5.01	6.14	7.26	8.37	9.53	10.69	11.86	13.05		14.67	8 / 7				
	reaction time	0.260			1.15	1.17	1.13	1.12	1.11	1.16	1.16	1.17	1.19	1.62			3.45	3.39	3.52	
	velocity		5.10	7.95	7.81	8.09	8.16	8.23	7.88	7.88	7.81	7.68	8.65	7.50			7.95	8.09	7.79	

Heat 2

Heat 2

date	08-Jul-19	wind	0.4 m/s	CAA Hurdle Development (2019)													
		H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
Zhao Pengchuan (CHN) (1993)	time	2.65	3.77	4.82	5.89	6.95	8.02	9.10	10.18	11.26	12.37		13.85	3 / 1			
reaction time	0.168	interval	1.12	1.05	1.07	1.06	1.07	1.08	1.08	1.08	1.11	1.48			3.24	3.21	3.27
velocity		5.18	8.16	8.70	8.54	8.62	8.54	8.46	8.46	8.46	8.23	9.47	7.94		8.46	8.54	8.39

Pan Zijie (CHN) (1995)	time	2.65	3.77	4.83	5.90	6.97	8.06	9.14	10.23	11.33	12.43		13.88	4 / 2			
	reaction time	0.191	interval	1.12	1.06	1.07	1.07	1.09	1.08	1.09	1.10	1.10	1.45		3.25	3.24	3.29
	velocity	5.18	8.16	8.62	8.54	8.54	8.39	8.46	8.39	8.31	8.31	9.67	7.93		8.44	8.46	8.33
Li Jiming (CHN) (1996)	time	2.65	3.75	4.80	5.87	6.94	8.01	9.09	10.21	11.34	12.48		14.03	5 / 3			
	reaction time	0.182	interval	1.10	1.05	1.07	1.07	1.08	1.12	1.13	1.14	1.55			3.22	3.22	3.39
	velocity	5.18	8.31	8.70	8.54	8.54	8.54	8.46	8.16	8.09	8.02	9.05	7.84		8.52	8.52	8.09
Sun Haoru (CHN) (1998)	time	2.65	3.77	4.87	5.98	7.06	8.14	9.25	10.36	11.48	12.63		14.14	6 / 4			
	reaction time	0.174	interval	1.12	1.10	1.11	1.08	1.08	1.11	1.11	1.12	1.15	PB		3.33	3.27	3.38
	velocity	5.18	8.16	8.31	8.23	8.46	8.46	8.23	8.23	8.16	7.95	9.28	7.78		8.23	8.39	8.11
Yang Yuxuan (CHN) (2002)	time	2.70	3.82	4.90	6.00	7.11	8.22	9.34	10.46	11.59	12.74		14.23	8 / 5			
	reaction time	0.158	interval	1.12	1.08	1.10	1.11	1.11	1.12	1.12	1.13	1.15	PB		3.30	3.34	3.40
	velocity	5.08	8.16	8.46	8.31	8.23	8.23	8.16	8.16	8.09	7.95	9.41	7.73		8.31	8.21	8.06
Wu Zhenhua (CHN) (1998)	time	2.70	3.81	4.94	6.04	7.20	8.34	9.51	10.68	11.86	13.05		14.69	7 / 6			
	reaction time	0.157	interval	1.11	1.13	1.10	1.16	1.14	1.17	1.17	1.18	1.19	1.64		3.34	3.47	3.54
	velocity	5.08	8.23	8.09	8.31	7.88	8.02	7.81	7.81	7.75	7.68	8.55	7.49		8.21	7.90	7.75
Zhao Jiefu (CHN) (2001)	time	2.83	3.99	5.12	6.26	7.37	8.51	9.66	10.79	11.97	13.16		14.78	2 / 7			
	reaction time	0.218	interval	1.16	1.13	1.14	1.11	1.14	1.15	1.13	1.18	1.19	1.62		3.43	3.40	3.50
	velocity	4.85	7.88	8.09	8.02	8.23	8.02	7.95	8.09	7.75	7.68	8.65	7.44		7.99	8.06	7.83

Heat 1

Heat 1

		date	08-Jul-19	wind 0.1 m/s										CAA Hurdle Development (2019)						
				H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10	
Sun Zhenjiang (CHN) (1999)	time			2.60	3.70	4.76	5.86	6.94	8.05	9.13	10.24	11.34	12.47		13.96	5 / 1				
	reaction time	0.163	interval	7 steps	1.10	1.06	1.10	1.08	1.11	1.08	1.11	1.10	1.13	1.49				3.26	3.27	3.34
			velocity	5.28	8.31	8.62	8.31	8.46	8.23	8.46	8.23	8.31	8.09	9.41	7.88			8.41	8.39	8.21
Jiang Fan (CHN) (1989)	time			2.69	3.80	4.87	5.95	7.05	8.12	9.22	10.34	11.44	12.59		14.12	6 / 2				
	reaction time	0.150	interval		1.11	1.07	1.08	1.10	1.07	1.10	1.12	1.10	1.15	1.53				3.26	3.27	3.37
			velocity	5.10	8.23	8.54	8.46	8.31	8.54	8.31	8.16	8.31	7.95	9.16	7.79			8.41	8.39	8.14
Ning Xiaohan (CHN) (2000)	time			2.68	3.77	4.85	5.94	7.02	8.12	9.22	10.34	11.46	12.60		14.17	4 / 3				
	reaction time	0.163	interval		1.09	1.08	1.09	1.08	1.10	1.10	1.12	1.12	1.14	1.57				3.26	3.28	3.38
			velocity	5.12	8.39	8.46	8.39	8.46	8.31	8.31	8.16	8.16	8.02	8.93	7.76			8.41	8.36	8.11
Ma Lei (CHN) (1989)	time			2.72	3.85	4.95	6.04	7.14	8.22	9.31	10.43	11.54	12.69		14.21	7 / 4				
	reaction time	0.155	interval	7 steps	1.13	1.10	1.09	1.10	1.08	1.09	1.12	1.11	1.15	1.52				3.32	3.27	3.38
			velocity	5.04	8.09	8.31	8.39	8.31	8.46	8.39	8.16	8.23	7.95	9.22	7.74			8.26	8.39	8.11
Guo Zhongjie (CHN) (2001)	time			2.70	3.80	4.92	6.00	7.11	8.19	9.32	10.43	11.58	12.73		14.30	8 / 5				
	reaction time	0.190	interval		1.10	1.12	1.08	1.11	1.08	1.13	1.11	1.15	1.15	1.57				3.30	3.32	3.41
			velocity	5.08	8.31	8.16	8.46	8.23	8.46	8.09	8.23	7.95	7.95	8.93	7.69			8.31	8.26	8.04
Yang Jianxing (CHN) (1996)	time			2.69	3.82	4.94	6.07	7.16	8.31	9.44	10.57	11.73	12.88		14.43	3 / 6				
	reaction time	0.141	interval		1.13	1.12	1.13	1.09	1.15	1.13	1.13	1.16	1.15	1.55				3.38	3.37	3.44
			velocity	5.10	8.09	8.16	8.09	8.39	7.95	8.09	8.09	7.88	7.95	9.05	7.62			8.11	8.14	7.97

2019 Athletissima (Lausanne, SUI) (TV Analysis)

FINAL

INAL	date		05-Jul-19	wind		1.0 m/s												Henson (2020) - Athlete First: 2019 year end hurdle report						
				H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10					
Ortega, Orlando (ESP) (1991)	time			2.52	3.52	4.54	5.54	6.54	7.54	8.54	9.56	10.58	11.62		13.05	4 / 1								
	reaction time	0.134	interval	7 steps	1.00	1.02	1.00	1.00	1.00	1.00	1.02	1.02	1.04	1.43			3.02	3.00	3.08					
			velocity	5.44	9.14	8.96	9.14	9.14	9.14	9.14	8.96	8.96	8.79	9.80	8.43		9.08	9.14	8.90					
Roberts, Daniel (USA) (1998)	time			2.50	3.54	4.56	5.58	6.58	7.58	8.60	9.62	10.64	11.66		13.11	5 / 2								
	reaction time	0.204	interval		1.04	1.02	1.02	1.00	1.00	1.02	1.02	1.02	1.02	1.45			3.08	3.02	3.06					
			velocity	5.49	8.79	8.96	8.96	9.14	9.14	8.96	8.96	8.96	8.96	9.67	8.39		8.90	9.08	8.96					
Levy, Ronald (JAM) (1992)	time			2.46	3.50	4.52	5.50	6.50	7.50	8.50	9.52	10.68	11.72		13.25	6 / 3								
	reaction time	0.138	interval	7 steps	1.04	1.02	0.98	1.00	1.00	1.00	1.02	1.16	1.04	1.53			3.04	3.00	3.22					
			velocity	5.58	8.79	8.96	9.33	9.14	9.14	9.14	8.96	7.88	8.79	9.16	8.30		9.02	9.14	8.52					
Pozzi, Andy (GBR) (1992)	time			2.50	3.54	4.58	5.58	6.60	7.62	8.64	9.68	10.74	11.80		13.28	8 / 4								
	reaction time	0.150	interval	7 steps	1.04	1.04	1.00	1.02	1.02	1.02	1.04	1.06	1.06	1.48			3.08	3.06	3.16					
			velocity	5.49	8.79	8.79	9.14	8.96	8.96	8.96	8.79	8.62	8.62	9.47	8.28		8.90	8.96	8.68					
Xie Wenjun (CHN) (1990)	time			2.52	3.56	4.60	5.60	6.60	7.64	8.68	9.72	10.76	11.84		13.29	3 / 5								
	reaction time	0.135	interval	7 steps	1.04	1.04	1.00	1.00	1.04	1.04	1.04	1.04	1.08	1.45			3.08	3.08	3.16					
			velocity	5.44	8.79	8.79	9.14	9.14	8.79	8.79	8.79	8.79	8.46	9.67	8.28		8.90	8.90	8.68					
Martinot-Lagarde, Pascal (FRA)	time			2.56	3.60	4.60	5.60	6.62	7.64	8.66	9.68	10.76	11.84		13.34	2 / 6								
	reaction time	0.157	interval	7 steps	1.04	1.00	1.00	1.02	1.02	1.02	1.02	1.08	1.08	1.50			3.04	3.06	3.18					

velocity	5.36	8.79	9.14	9.14	8.96	8.96	8.96	8.96	8.46	8.46	9.35	8.25		9.02	8.96	8.62
Allen, Devon (USA) (1994) time	2.56	3.58	4.58	5.60	6.60	7.64	8.68	9.72	10.78	11.86		13.35	1 / 7			
reaction time 0.133 interval		1.02	1.00	1.02	1.00	1.04	1.04	1.04	1.06	1.08	1.49			3.04	3.08	3.18
velocity	5.36	8.96	9.14	8.96	9.14	8.79	8.79	8.79	8.62	8.46	9.41	8.24		9.02	8.90	8.62

2019 Prefontaine Classic (Stanford, CA) (TV Analysis)

FINAL

date 30-Jun-19 wind 0.3 m/s

Henson (2020) - Athlete First: 2019 year end hurdle report

		H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
Ortega, Orlando (ESP) (1991) time		2.55	3.60	4.62	5.63	6.62	7.64	8.68	9.70	10.75	11.80		13.24	4 / 1			
reaction time 0.163 interval		7 steps	1.05	1.02	1.01	0.99	1.02	1.04	1.02	1.05	1.05	1.44			3.08	3.05	3.12
velocity		5.38	8.70	8.96	9.05	9.23	8.96	8.79	8.96	8.70	8.70	9.74	8.31		8.90	8.99	8.79
Belocian, Wilhem (FRA) (1995) time		2.56	3.60	4.63	5.63	6.63	7.65	8.68	9.70	10.76	11.83		13.29	2 / 2			
reaction time 0.151 interval			1.04	1.03	1.00	1.00	1.02	1.03	1.02	1.06	1.07	1.46			3.07	3.05	3.15
velocity		5.36	8.79	8.87	9.14	9.14	8.96	8.87	8.96	8.62	8.54	9.60	8.28		8.93	8.99	8.70
McLeod, Omar (JAM) (1994) time		2.53	3.60	4.62	5.64	6.66	7.70	8.75	9.80	10.83	11.88		13.29	5 / 3			
reaction time 0.164 interval			1.07	1.02	1.02	1.02	1.04	1.05	1.05	1.03	1.05	1.41			3.11	3.11	3.13
velocity		5.42	8.54	8.96	8.96	8.96	8.79	8.70	8.70	8.87	8.70	9.94	8.28		8.82	8.82	8.76
Levy, Ronald (JAM) (1992) time		2.60	3.63	4.66	5.73	6.73	7.73	8.75	9.78	10.83	11.86		13.30	6 / 4			
reaction time 0.224 interval		7 steps	1.03	1.03	1.07	1.00	1.00	1.02	1.03	1.05	1.03	1.44			3.13	3.02	3.11
velocity		5.28	8.87	8.87	8.54	9.14	9.14	8.96	8.87	8.70	8.87	9.74	8.27		8.76	9.08	8.82
Allen, Devon (USA) (1994) time		2.56	3.60	4.62	5.64	6.66	7.68	8.75	9.76	10.82	11.86		13.33	3 / 5			
reaction time 0.145 interval			1.04	1.02	1.02	1.02	1.02	1.07	1.01	1.06	1.04	1.47			3.08	3.11	3.11
velocity		5.36	8.79	8.96	8.96	8.96	8.96	8.54	9.05	8.62	8.79	9.54	8.25		8.90	8.82	8.82
Xie Wenjun (CHN) (1990) time		2.56	3.60	4.63	5.66	6.70	7.73	8.76	9.80	10.83	11.88		13.36	7 / 6			
reaction time 0.175 interval		7 steps	1.04	1.03	1.03	1.04	1.03	1.03	1.04	1.03	1.05	1.48			3.10	3.10	3.12
velocity		5.36	8.79	8.87	8.87	8.79	8.87	8.87	8.79	8.87	8.70	9.47	8.23		8.85	8.85	8.79
Crittendon, Freddie (USA) (1995) time		2.60	3.63	4.66	5.66	6.70	7.73	8.78	9.82	10.85	11.92		13.38	8 / 7			
reaction time 0.220 interval			1.03	1.03	1.00	1.04	1.03	1.05	1.04	1.03	1.07	1.46			3.06	3.12	3.14
velocity		5.28	8.87	8.87	9.14	8.79	8.87	8.70	8.79	8.87	8.54	9.60	8.22		8.96	8.79	8.73
Harris, Aleec (USA) (1990) time		2.60	3.63	4.66	5.72	6.74	7.76	8.78	9.82	10.88	11.94		13.39	1 / 8			
reaction time 0.161 interval			1.03	1.03	1.06	1.02	1.02	1.02	1.04	1.06	1.06	1.45			3.12	3.06	3.16
velocity		5.28	8.87	8.87	8.62	8.96	8.96	8.96	8.79	8.62	8.62	9.67	8.22		8.79	8.96	8.68
Swift, Greggmar (BAR) (1991) time		2.55	3.58	4.62	5.68	6.70	7.74	8.82	9.86	10.93	12.03		13.56	9 / 9			
reaction time 0.150 interval			1.03	1.04	1.06	1.02	1.04	1.08	1.04	1.07	1.10	1.53			3.13	3.14	3.21
velocity		5.38	8.87	8.79	8.62	8.96	8.79	8.46	8.79	8.54	8.31	9.16	8.11		8.76	8.73	8.54

2019 Japanese National Championships (Fukuoka, JPN)

FINAL

date 30-Jun-19 wind -0.6 m/s

Yamanaka (2019) - research on athlete performance and technique- 2019 data book

		H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
Takayama, Shunya (JPN) (1994) time		2.54	3.59	4.61	5.63	6.66	7.69	8.73	9.78	10.84	11.92		13.36	7 / 1			
reaction time 0.146 interval			1.05	1.02	1.02	1.03	1.03	1.04	1.05	1.06	1.08	1.44	=NR =PB		3.09	3.10	3.19
velocity		5.40	8.70	8.96	8.96	8.87	8.87	8.79	8.70	8.62	8.46	9.74	8.23		8.87	8.85	8.60
Izumiya, Shunsuke (JPN) (2000) time		2.55	3.59	4.63	5.65	6.68	7.72	8.75	9.80	10.89	11.97		13.36	6 / 2			
reaction time 0.144 interval			1.04	1.04	1.02	1.03	1.04	1.03	1.05	1.09	1.08	1.39	=NR NJR PB		3.10	3.10	3.22
velocity		5.38	8.79	8.79	8.96	8.87	8.79	8.87	8.70	8.39	8.46	10.09	8.23		8.85	8.85	8.52
Ishikawa, Shuhei (JPN) (1995) time		2.57	3.65	4.71	5.76	6.83	7.87	8.93	10.00	11.08	12.20		13.67	4 / 3			
reaction time 0.137 interval		7 steps	1.08	1.06	1.05	1.07	1.04	1.06	1.07	1.08	1.12	1.47			3.19	3.17	3.27
velocity		5.34	8.46	8.62	8.70	8.54	8.79	8.62	8.54	8.46	8.16	9.54	8.05		8.60	8.65	8.39
Yazawa, Wataru (JPN) (1991) time		2.64	3.71	4.77	5.83	6.87	7.92	9.00	10.07	11.18	12.29		13.76	9 / 4			
reaction time 0.138 interval		7 steps	1.07	1.06	1.06	1.04	1.05	1.08	1.07	1.11	1.11	1.47			3.19	3.17	3.29
velocity		5.20	8.54	8.62	8.62	8.79	8.70	8.46	8.54	8.23	8.23	9.54	7.99		8.60	8.65	8.33
Kuriki, Anthony Tyrell (JPN) (1991) time		2.64	3.74	4.79	5.87	6.94	8.01	9.10	10.20	11.30	12.42		13.92	3 / 5			
reaction time 0.136 interval		7 steps	1.10	1.05	1.08	1.07	1.07	1.09	1.10	1.10	1.12	1.50			3.23	3.23	3.32
velocity		5.20	8.31	8.70	8.46	8.54	8.54	8.39	8.31	8.31	8.16	9.35	7.90		8.49	8.49	8.26
Fujii, Ryota (JPN) (1996) time		2.64	3.72	4.79	5.87	6.95	8.05	9.15	10.26	11.39	12.52		13.99	2 / 6			
reaction time 0.148 interval			1.08	1.07	1.08	1.08	1.10	1.10	1.11	1.13	1.13	1.47			3.23	3.28	3.37
velocity		5.20	8.46	8.54	8.46	8.46	8.31	8.31	8.23	8.09	8.09	9.54	7.86		8.49	8.36	8.14
Fudaba, Hiroki (JPN) (1994) time		2.66	3.75	4.81	5.91	6.98	8.07	9.15	10.26	11.37	12.49		14.00	8 / 7			
reaction time 0.148 interval		7 steps	1.09	1.06	1.10	1.07	1.09	1.08	1.11	1.11	1.12	1.51			3.25	3.24	3.34
velocity		5.16	8.39	8.62	8.31	8.54	8.39	8.46	8.23	8.23	8.16	9.28	7.86		8.44	8.46	8.21

Sato, Hiroyuki (JPN) (1990)	time	2.58	3.68	4.75	5.84	6.92	8.00	9.10	10.21	11.44	12.63		14.16	5 / 8			
reaction time	0.157	interval		1.10	1.07	1.09	1.08	1.08	1.10	1.11	1.23	1.19	1.53		3.26	3.26	3.53
		velocity	5.32	8.31	8.54	8.39	8.46	8.46	8.31	8.23	7.43	7.68	9.16	7.77	8.41	8.41	7.77

2019 Meeting International Mohammed VI d'Athlétisme (Rabat, MAR) (TV Analysis)

FINAL

		date	16-Jun-19	wind	0.3 m/s	Henson (2020) - Athlete First: 2019 year end hurdle report													
				H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
Shubenkov, Sergey (ANA) (1991)	time			2.50	3.54	4.56	5.58	6.58	7.60	8.60	9.62	10.66	11.70		13.12	6 / 1			
	reaction time	0.135	interval	7 steps	1.04	1.02	1.02	1.00	1.02	1.00	1.02	1.04	1.04	1.42			3.08	3.02	3.10
			velocity		5.49	8.79	8.96	8.96	9.14	8.96	9.14	8.96	8.79	8.79	9.87	8.38		8.90	9.08
Pozzi, Andy (GBR) (1992)	time			2.48	3.52	4.52	5.52	6.54	7.58	8.62	9.66	10.72	11.78		13.30	4 / 2			
	reaction time	0.118	interval	7 steps	1.04	1.00	1.00	1.02	1.04	1.04	1.04	1.06	1.06	1.52			3.04	3.10	3.16
			velocity		5.53	8.79	9.14	9.14	8.96	8.79	8.79	8.79	8.62	8.62	9.22	8.27		9.02	8.85
Constantino, Gabriel (BRA) (1991)	time			2.54	3.58	4.60	5.64	6.68	7.70	8.72	9.76	10.82	11.90		13.41	2 / 3			
	reaction time	0.177	interval	7 steps	1.04	1.02	1.04	1.04	1.02	1.02	1.04	1.06	1.08	1.51			3.10	3.08	3.18
			velocity		5.40	8.79	8.96	8.79	8.79	8.96	8.96	8.79	8.62	8.46	9.28	8.20		8.85	8.90
Belocian, Wilhem (FRA) (1995)	time			2.54	3.58	4.60	5.60	6.64	7.66	8.70	9.74	10.80	11.90		13.43	7 / 4			
	reaction time	0.137	interval		1.04	1.02	1.00	1.04	1.02	1.04	1.04	1.06	1.10	1.53			3.06	3.10	3.20
			velocity		5.40	8.79	8.96	9.14	8.79	8.96	8.79	8.79	8.62	8.31	9.16	8.19		8.96	8.85
McLeod, Omar (JAM) (1994)	time			2.44	3.46	4.48	5.48	6.50	7.52	8.56	9.58	10.62	11.72		13.48	5 / 5			
	reaction time	0.145	interval		1.02	1.02	1.00	1.02	1.02	1.04	1.02	1.04	1.10	1.76			3.04	3.08	3.16
			velocity		5.62	8.96	8.96	9.14	8.96	8.96	8.79	8.96	8.79	8.31	7.97	8.16		9.02	8.90
Alkana, Antonio (RSA) (1990)	time			2.48	3.52	4.52	5.56	6.60	7.64	8.68	9.74	10.88	11.98		13.51	3 / 6			
	reaction time	0.168	interval		1.04	1.00	1.04	1.04	1.04	1.04	1.06	1.14	1.10	1.53			3.08	3.12	3.30
			velocity		5.53	8.79	9.14	8.79	8.79	8.79	8.62	8.02	8.31	9.16	8.14		8.90	8.79	8.31
Trajkovic, Milan (CYP) (1992)	time			2.54	3.62	4.64	5.68	6.70	7.74	8.80	9.86	10.90	12.00		13.53	8 / 7			
	reaction time	0.098	interval	7 steps	1.08	1.02	1.04	1.02	1.04	1.06	1.06	1.04	1.10	1.53			3.14	3.12	3.20
			velocity		5.40	8.46	8.96	8.79	8.96	8.79	8.62	8.62	8.79	8.31	9.16	8.13		8.73	8.79
Santos Rodriques, Eduardo (BRA) (1991)	time			2.58	3.66	4.70	5.74	6.78	7.80	8.86	9.90	10.98	12.06		13.58	9 / 8			
	reaction time	0.156	interval		1.08	1.04	1.04	1.04	1.02	1.06	1.04	1.08	1.08	1.52			3.16	3.12	3.20
			velocity		5.32	8.46	8.79	8.79	8.79	8.96	8.62	8.79	8.46	8.46	9.22	8.10		8.68	8.79
Manga, Aurel (FRA) (1992)	time			2.52	3.58	4.66	5.72	6.80	7.86	8.94	10.04	11.12	12.26		13.88	1 / 9			
	reaction time	0.163	interval	7 steps	1.06	1.08	1.06	1.08	1.06	1.08	1.10	1.08	1.14	1.62			3.20	3.22	3.32
			velocity		5.44	8.62	8.46	8.62	8.46	8.62	8.46	8.31	8.46	8.02	8.65	7.93		8.57	8.52

2019 Japanese National Multi-Event Championships (Nagano, JPN)

Decathlon - Heat 2

ecathlon - Heat 2		date	09-Jun-19		wind	2.1 m/s		Matsubayashi (2019) - research on athlete performance and technique- 2019 data book												
					H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
Nakamura, Akihiko (JPN) (1990)	time				2.69	3.80	4.88	5.98	7.07	8.17	9.26	10.37	11.50	12.62		14.06	4 / 2			
	reaction time		interval			1.11	1.08	1.10	1.09	1.10	1.09	1.11	1.13	1.12	1.44			3.29	3.28	3.36
			velocity		5.10	8.23	8.46	8.31	8.39	8.31	8.39	8.23	8.09	8.16	9.74	7.82		8.33	8.36	8.16
Maruyama, Yuma (JPN) (1998)	time				2.69	3.80	4.91	5.99	7.08	8.19	9.28	10.41	11.53	12.68		14.19	5 / 3			
	reaction time		interval			1.11	1.11	1.08	1.09	1.11	1.09	1.13	1.12	1.15	1.51			3.30	3.29	3.40
			velocity		5.10	8.23	8.23	8.46	8.39	8.23	8.39	8.09	8.16	7.95	9.28	7.75		8.31	8.33	8.06

Decathlon - Heat 1

scaethlon - Heat 1		date	09-Jun-19	wind	1.8 m/s	Matsubayashi (2019) - research on athlete performance and technique- 2019 data book														
				H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10	
Ushiro, Keisuke (JPN) (1986)	time		2.74	3.92	5.11	6.39	7.63	8.83	10.07	11.27	12.47	13.69			15.36	6 / 4				
	reaction time	interval		1.18	1.19	1.28	1.24	1.20	1.24	1.20	1.20	1.22	1.67				3.65	3.68	3.62	
		velocity		5.01	7.75	7.68	7.14	7.37	7.62	7.37	7.62	7.62	7.49	8.40	7.16		7.51	7.45	7.57	

2019 NCAA National Championships (Austin, TX) (TV Analysis)

FINAL

NAME	date		07-Jun-19	wind 0.8 m/s										Henson (2020) - Athlete First: 2019 year end hurdle report						
				H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10	
Holloway, Grant (USA) (1997)	time			2.47	3.47	4.47	5.44	6.41	7.40	8.40	9.43	10.47	11.54		12.98	5 / 1				
	reaction time	0.180	interval	7 steps	1.00	1.00	0.97	0.97	0.99	1.00	1.03	1.04	1.07	1.44	PB		2.97	2.96	3.14	
			velocity		5.55	9.14	9.14	9.42	9.42	9.23	9.14	8.87	8.79	8.54	9.74	8.47		9.23	9.26	8.73
Roberts, Daniel (USA) (1998)	time			2.50	3.53	4.53	5.53	6.51	7.51	8.51	9.54	10.55	11.60		13.00	4 / 2				
	reaction time	0.191	interval		1.03	1.00	1.00	0.98	1.00	1.00	1.03	1.01	1.05	1.40	PB		3.03	2.98	3.09	
			velocity		5.49	8.87	9.14	9.14	9.33	9.14	9.14	8.87	9.05	8.70	10.01	8.46		9.05	9.20	8.87
Moore, Isaiah (USA) (1996)	time			2.60	3.67	4.70	5.70	6.74	7.77	8.81	9.84	10.87	11.94		13.37	3 / 3				
	reaction time	0.176	interval		1.07	1.03	1.00	1.04	1.03	1.04	1.03	1.03	1.07	1.43	PB		3.10	3.11	3.13	
			velocity		5.28	8.54	8.87	9.14	8.79	8.87	8.79	8.87	8.87	8.54	9.80	8.23		8.85	8.82	8.76
Jackson, Dashaun (USA) (1997)	time			2.54	3.60	4.64	5.67	6.70	7.70	8.71	9.76	10.81	11.91		13.38	6 / 4				

reaction time	0.175	interval	1.06	1.04	1.03	1.03	1.00	1.01	1.05	1.05	1.10	1.47			3.13	3.04	3.20
velocity	5.40	8.62	8.79	8.87	8.87	9.14	9.05	8.70	8.70	8.31	9.54	8.22			8.76	9.02	8.57

Semi-Final 1

date 05-Jun-19 wind 0.9 m/s

Henson (2020) - Athlete First: 2019 year end hurdle report

		H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
Roberts, Daniel (USA) (1998)	time	2.33	3.35	4.35	5.35	6.35	7.38	8.41	9.43	10.46	11.50		13.06	6 / 1			
reaction time	0.203	interval	1.02	1.00	1.00	1.00	1.03	1.03	1.02	1.03	1.04	1.56	PB		3.02	3.06	3.09
	velocity	5.89	8.96	9.14	9.14	9.14	8.87	8.87	8.96	8.87	8.79	8.99	8.42		9.08	8.96	8.87
Jackson, Dashaun (USA) (1997)	time	2.38	3.43	4.46	5.48	6.50	7.50	8.60	9.63	10.68	11.74		13.37	7 / 2			
reaction time	0.213	interval	1.05	1.03	1.02	1.02	1.00	1.10	1.03	1.05	1.06	1.63	PB		3.10	3.12	3.14
	velocity	5.76	8.70	8.87	8.96	8.96	9.14	8.31	8.87	8.70	8.62	8.60	8.23		8.85	8.79	8.73

2019 Asian Grand Prix 2 (Chongqing, CHN)**FINAL**

date 07-Jun-19 wind 1.2 m/s

CAA Hurdle Development (2019)

		H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
Chen Kuei-Ru (TPE) (1993)	time	2.58	3.63	4.67	5.69	6.72	7.75	8.79	9.84	10.91	11.99		13.43	5 / 1			
reaction time	0.164	interval	7 steps	1.05	1.04	1.02	1.03	1.03	1.04	1.05	1.07	1.08	1.44		3.11	3.10	3.20
	velocity	5.32	8.70	8.79	8.96	8.87	8.87	8.79	8.70	8.54	8.46	9.74	8.19		8.82	8.85	8.57
Zeng Jianhang (CHN) (1998)	time	2.58	3.62	4.66	5.67	6.70	7.72	8.76	9.80	10.92	12.03		13.45	3 / 2			
reaction time	0.168	interval	1.04	1.04	1.01	1.03	1.02	1.04	1.04	1.12	1.11	1.42	PB		3.09	3.09	3.27
	velocity	5.32	8.79	8.79	9.05	8.87	8.96	8.79	8.79	8.16	8.23	9.87	8.18		8.87	8.87	8.39
Kuriki, Anthony Tyrell (JPN) (1991)	time	2.60	3.70	4.74	5.79	6.82	7.86	8.92	9.99	11.07	12.17		13.62	4 / 3			
reaction time	0.130	interval	7 steps	1.10	1.04	1.05	1.03	1.04	1.06	1.07	1.08	1.10	1.45		3.19	3.13	3.25
	velocity	5.28	8.31	8.79	8.70	8.87	8.79	8.62	8.54	8.46	8.31	9.67	8.08		8.60	8.76	8.44
Efremov, David (KAZ) (1999)	time	2.66	3.75	4.78	5.84	6.90	7.95	9.02	10.12	11.20	12.32		13.81	8 / 4			
reaction time	0.186	interval	7 steps	1.09	1.03	1.06	1.06	1.07	1.10	1.08	1.12	1.49	PB		3.18	3.18	3.30
	velocity	5.16	8.39	8.87	8.62	8.62	8.70	8.54	8.31	8.46	8.16	9.41	7.97		8.62	8.62	8.31
Ogata, Akihiro (JPN) (1993)	time	2.66	3.75	4.80	5.87	6.92	7.98	9.07	10.16	11.27	12.41		13.90	6 / 5			
reaction time	0.201	interval	7 steps	1.09	1.05	1.07	1.06	1.09	1.09	1.11	1.14	1.49			3.21	3.20	3.34
	velocity	5.16	8.39	8.70	8.54	8.70	8.62	8.39	8.39	8.23	8.02	9.41	7.91		8.54	8.57	8.21
Chan Chung Wang (HKG) (1995)	time	2.70	3.78	4.85	5.91	6.96	8.02	9.07	10.16	11.31	12.49		14.07	7 / 6			
reaction time	0.158	interval	1.08	1.07	1.06	1.05	1.06	1.05	1.09	1.15	1.18	1.58			3.21	3.16	3.42
	velocity	5.08	8.46	8.54	8.62	8.70	8.62	8.70	8.39	7.95	7.75	8.87	7.82		8.54	8.68	8.02
Bautista, Clinton Kingsley (PHI) (1992)	time	2.65	3.77	4.87	5.99	7.07	8.17	9.29	10.41	11.52	12.67		14.13	2 / 7			
reaction time	0.180	interval	7 steps	1.12	1.10	1.12	1.08	1.10	1.12	1.12	1.11	1.15	1.46		3.34	3.30	3.38
	velocity	5.18	8.16	8.31	8.16	8.46	8.31	8.16	8.16	8.23	7.95	9.60	7.78		8.21	8.31	8.11

2019 Golden Gala Pietro Mennea (Rome, ITA) (TV Analysis)**FINAL**

date 06-Jun-19 wind 0.4 m/s

Henson (2020) - Athlete First: 2019 year end hurdle report

		H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
Shubenkov, Sergey (ANA) (1991)	time	2.61	3.66	4.68	5.67	6.68	7.69	8.71	9.73	10.75	11.83		13.26	5 / 1			
reaction time	0.132	interval	7 steps	1.05	1.02	0.99	1.01	1.01	1.02	1.02	1.08	1.43			3.06	3.04	3.12
	velocity	5.26	8.70	8.96	9.23	9.05	9.05	8.96	8.96	8.96	8.46	9.80	8.30		8.96	9.02	8.79
Pozzi, Andy (GBR) (1992)	time	2.53	3.55	4.56	5.55	6.55	7.60	8.63	9.68	10.73	11.80		13.29	7 / 2			
reaction time	0.139	interval	7 steps	1.02	1.01	0.99	1.00	1.05	1.03	1.05	1.05	1.07	1.49		3.02	3.08	3.17
	velocity	5.42	8.96	9.05	9.23	9.14	8.70	8.87	8.70	8.70	8.54	9.41	8.28		9.08	8.90	8.65
Alkana, Antonio (RSA) (1990)	time	2.53	3.60	4.62	5.63	6.63	7.63	8.68	9.72	10.78	11.83		13.30	8 / 3			
reaction time	0.133	interval	1.07	1.02	1.01	1.00	1.00	1.05	1.04	1.06	1.05	1.47			3.10	3.05	3.15
	velocity	5.42	8.54	8.96	9.05	9.14	9.14	8.70	8.79	8.62	8.70	9.54	8.27		8.85	8.99	8.70
Constantino, Gabriel (BRA) (1991)	time	2.55	3.60	4.62	5.63	6.66	7.70	8.75	9.80	10.88	11.98		13.50	4 / 4			
reaction time	0.160	interval	7 steps	1.05	1.02	1.01	1.03	1.04	1.05	1.05	1.08	1.10	1.52		3.08	3.12	3.23
	velocity	5.38	8.70	8.96	9.05	8.87	8.79	8.70	8.70	8.46	8.31	9.22	8.15		8.90	8.79	8.49
Trajkovic, Milan (CYP) (1992)	time	2.55	3.63	4.68	5.70	6.73	7.82	8.82	9.92	10.95	12.06		13.50	3 / 5			
reaction time	0.136	interval	7 steps	1.08	1.05	1.02	1.03	1.09	1.00	1.10	1.03	1.11	1.44		3.15	3.12	3.24
	velocity	5.38	8.46	8.70	8.96	8.87	8.39	9.14	8.31	8.87	8.23	9.74	8.15		8.70	8.79	8.46
Manga, Aurel (FRA) (1992)	time	2.53	3.60	4.62	5.65	6.71	7.75	8.80	9.88	10.94	12.03		13.51	2 / 6			
reaction time	0.150	interval	7 steps	1.07	1.02	1.03	1.06	1.04	1.05	1.08	1.06	1.09	1.48		3.12	3.15	3.23
	velocity	5.42	8.54	8.96	8.87	8.62	8.79	8.70	8.46	8.62	8.39	9.47	8.14		8.79	8.70	8.49
Perini, Lorenzo (ITA) (1994)	time	2.60	3.66	4.72	5.75	6.78	7.83	8.88	9.94	11.00	12.08		13.58	9 / 7			
reaction time	0.161	interval	1.06	1.06	1.03	1.03	1.05	1.05	1.06	1.06	1.08	1.50			3.15	3.13	3.20
	velocity	5.28	8.62	8.62	8.87	8.87	8.70	8.70	8.62	8.62	8.46	9.35	8.10		8.70	8.76	8.57
Cabral, Johathan (CAN) (1992)	time	2.60	3.68	4.71	5.75	6.78	7.83	8.88	9.95	11.00	12.11		13.61	1 / 8			

reaction time	0.142	interval	7 steps	1.08	1.03	1.04	1.03	1.05	1.05	1.07	1.05	1.11	1.50			3.15	3.13	3.23
		velocity		5.28	8.46	8.87	8.79	8.87	8.70	8.70	8.54	8.70	8.23	9.35	8.08	8.70	8.76	8.49
Ortega, Orlando (ESP) (1991)	time	2.60	3.71	4.80	5.82	6.87	7.89	8.97	10.02	11.13	12.27			14.00	6 / 9			
reaction time	0.127	interval	7 steps	1.11	1.09	1.02	1.05	1.02	1.08	1.05	1.11	1.14	1.73			3.22	3.15	3.30
		velocity		5.28	8.23	8.39	8.96	8.70	8.96	8.46	8.70	8.23	8.02	8.10	7.86	8.52	8.70	8.31

2019 Asian Grand Prix 1 (Chongqing, CHN)

FINAL

		date	04-Jun-19	wind	-0.3 m/s														CAA Hurdle Development (2019)			
				H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10			
Chen Kuei-Ru (TPE) (1993)	time			2.61	3.67	4.70	5.74	6.79	7.84	8.90	9.96	11.02	12.13		13.58	4 / 1						
reaction time	0.187	interval	7 steps		1.06	1.03	1.04	1.05	1.05	1.06	1.06	1.06	1.11	1.45			3.13	3.16	3.23			
		velocity		5.26	8.62	8.87	8.79	8.70	8.70	8.62	8.62	8.62	8.23	9.67	8.10		8.76	8.68	8.49			
Zeng Jianhang (CHN) (1998)	time			2.58	3.65	4.70	5.75	6.83	7.90	8.95	10.05	11.16	12.26		13.72	6 / 2						
reaction time	0.159	interval			1.07	1.05	1.05	1.08	1.07	1.05	1.10	1.11	1.10	1.46			3.17	3.20	3.31			
		velocity		5.32	8.54	8.70	8.70	8.46	8.54	8.70	8.31	8.23	8.31	9.60	8.02		8.65	8.57	8.28			
Chan Chung Wang (HKG) (1995)	time			2.65	3.73	4.78	5.85	6.89	7.94	9.04	10.11	11.20	12.29		13.75	7 / 3						
reaction time	0.149	interval			1.08	1.05	1.07	1.04	1.05	1.10	1.07	1.09	1.09	1.46			3.20	3.19	3.25			
		velocity		5.18	8.46	8.70	8.54	8.79	8.70	8.31	8.54	8.39	8.39	9.60	8.00		8.57	8.60	8.44			
Efremov, David (KAZ) (1999)	time			2.65	3.70	4.77	5.79	6.87	7.95	9.05	10.16	11.26	12.39		13.91	9 / 4						
reaction time	0.174	interval	7 steps		1.05	1.07	1.02	1.08	1.08	1.10	1.11	1.10	1.13	1.52			3.14	3.26	3.34			
		velocity		5.18	8.70	8.54	8.96	8.46	8.46	8.31	8.23	8.31	8.09	9.22	7.91		8.73	8.41	8.21			
Ogata, Akihiro (JPN) (1993)	time			2.65	3.73	4.82	5.90	6.95	8.04	9.12	10.22	11.34	12.44		13.94	5 / 5						
reaction time	0.173	interval			1.08	1.09	1.08	1.05	1.09	1.08	1.10	1.12	1.10	1.50			3.25	3.22	3.32			
		velocity		5.18	8.46	8.39	8.46	8.70	8.39	8.46	8.31	8.16	8.31	9.35	7.89		8.44	8.52	8.26			
Bautista, Clinton Kingsley (PHI) (1993)	time			2.60	3.70	4.80	5.88	6.99	8.10	9.25	10.41	11.57	12.72		14.22	2 / 6						
reaction time	0.169	interval	7 steps		1.10	1.10	1.08	1.11	1.11	1.15	1.16	1.16	1.15	1.50			3.28	3.37	3.47			
		velocity		5.28	8.31	8.31	8.46	8.23	8.23	7.95	7.88	7.88	7.95	9.35	7.74		8.36	8.14	7.90			

2019 Fuse Sprint (Tottori, JPN)

FINAL

		date	02-Jun-19	wind	1.9 m/s														Kijima (2019) - research on athlete performance and technique- 2019 data book			
				H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10			
Takayama, Shunya (JPN) (1994)	time			2.54	3.58	4.62	5.66	6.68	7.70	8.75	9.80	10.87	11.92		13.36	7 / 1						
reaction time	0.128	interval			1.04	1.04	1.04	1.02	1.02	1.05	1.05	1.07	1.05	1.44	=NR PB		3.12	3.09	3.17			
		velocity		5.40	8.79	8.79	8.79	8.96	8.96	8.70	8.70	8.54	8.70	9.74	8.23		8.79	8.87	8.65			
Ishikawa, Shuhei (JPN) (1995)	time			2.57	3.62	4.66	5.70	6.73	7.77	8.83	9.88	10.94	12.04		13.49	5 / 2						
reaction time	0.149	interval	7 steps		1.05	1.04	1.04	1.03	1.04	1.06	1.05	1.06	1.10	1.45	PB		3.13	3.13	3.21			
		velocity		5.34	8.70	8.79	8.79	8.87	8.79	8.62	8.70	8.62	8.31	9.67	8.15		8.76	8.76	8.54			
Yazawa, Wataru (JPN) (1991)	time			2.61	3.69	4.78	5.83	6.86	7.91	8.95	10.01	11.08	12.16		13.60	4 / 3						
reaction time	0.128	interval	7 steps		1.08	1.09	1.05	1.03	1.05	1.04	1.06	1.07	1.08	1.44			3.22	3.12	3.21			
		velocity		5.26	8.46	8.39	8.70	8.87	8.70	8.79	8.62	8.54	8.46	9.74	8.09		8.52	8.79	8.54			
Sato, Hiroyuki (JPN) (1990)	time			2.53	3.61	4.67	5.73	6.77	7.85	8.93	10.05	11.16	12.30		13.77	6 / 4						
reaction time	0.126	interval			1.08	1.06	1.06	1.04	1.08	1.08	1.12	1.11	1.14	1.47			3.20	3.20	3.37			
		velocity		5.42	8.46	8.62	8.62	8.79	8.46	8.46	8.16	8.23	8.02	9.54	7.99		8.57	8.57	8.14			
Fujii, Ryota (JPN) (1996)	time			2.60	3.68	4.74	5.81	6.87	7.95	9.02	10.11	11.21	12.31		13.78	8 / 5						
reaction time	0.146	interval			1.08	1.06	1.07	1.06	1.08	1.07	1.09	1.10	1.10	1.47	PB		3.21	3.21	3.29			
		velocity		5.28	8.46	8.62	8.54	8.62	8.46	8.54	8.39	8.31	8.31	9.54	7.98		8.54	8.54	8.33			
Wado, Tatsuya (JPN) (1990)	time			2.58	3.67	4.75	5.82	6.87	7.95	9.03	10.11	11.22	12.33		13.82	9 / 6						
reaction time	0.131	interval	7 steps		1.09	1.08	1.07	1.05	1.08	1.08	1.08	1.11	1.11	1.49			3.24	3.21	3.30			
		velocity		5.32	8.39	8.46	8.54	8.70	8.46	8.46	8.46	8.23	8.23	9.41	7.96		8.46	8.54	8.31			
Fudaba, Hiroki (JPN) (1994)	time			2.66	3.75	4.84	5.89	6.96	8.04	9.11	10.20	11.29	12.39		13.89	3 / 7						
reaction time	0.150	interval	7 steps		1.09	1.09	1.05	1.07	1.08	1.07	1.09	1.09	1.10	1.50			3.23	3.22	3.28			
		velocity		5.16	8.39	8.39	8.70	8.54	8.46	8.54	8.39	8.39	8.31	9.35	7.92		8.49	8.52	8.36			
Shimizu, Koichiro (JPN) (2000)	time			2.69	3.77	4.87	5.92	7.00	8.09	9.20	10.29	11.39	12.52		13.96	2 / 8						
reaction time	0.156	interval			1.08	1.10	1.05	1.08	1.09	1.11	1.09	1.10	1.13	1.44	PB		3.23	3.28	3.32			
		velocity		5.10	8.46	8.31	8.70	8.46	8.39	8.23	8.39	8.31	8.09	9.74	7.88		8.49	8.36	8.26			

2019 Chinese National Grand Prix 4 (Luoyang, CHN)

FINAL

		date	30-May-19	wind	0.0 m/s														CAA Hurdle Development (2019)			
				H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10			
Zhao Pengchuan (CHN) (1993)	time			2.67	3.77	4.84	5.93	6.99	8.07	9.16	10.26	11.35	12.46		13.94	6 / 1						
reaction time	0.165	interval			1.10	1.07	1.09	1.06	1.08	1.09	1.10	1.09	1.11	1.48			3.26	3.23	3.30			
		velocity		5.14	8.31	8.54	8.39	8.62	8.46	8.39	8.31	8.39	8.23	9.47	7.89		8.41	8.49	8.31			

Li Jiming (CHN) (1996)	time	2.70	3.78	4.86	5.92	7.00	8.06	9.17	10.26	11.36	12.47		14.00	5 / 2				
	reaction time	0.183														3.22	3.25	3.30
	interval		1.08	1.08	1.06	1.08	1.06	1.11	1.09	1.10	1.11	1.53				8.52	8.44	8.31
	velocity		5.08	8.46	8.46	8.62	8.46	8.62	8.23	8.39	8.31	8.23	9.16	7.86				
Zhu Shenglong (CHN) (2000)	time	2.71	3.80	4.89	5.96	7.03	8.09	9.16	10.26	11.37	12.51		14.01	8 / 3				
	reaction time	0.171											PB			3.25	3.20	3.35
	interval		1.09	1.09	1.07	1.07	1.06	1.07	1.10	1.11	1.14	1.50				8.44	8.57	8.19
	velocity		5.06	8.39	8.39	8.54	8.54	8.62	8.54	8.31	8.23	8.02	9.35	7.85				
Pan Zijie (CHN) (1995)	time	2.64	3.72	4.82	5.89	6.97	8.05	9.14	10.24	11.34	12.51		14.03	2 / 4				
	reaction time	0.177														3.25	3.25	3.37
	interval		1.08	1.10	1.07	1.08	1.08	1.09	1.10	1.10	1.17	1.52				8.44	8.44	8.14
	velocity		5.20	8.46	8.31	8.54	8.46	8.46	8.39	8.31	8.31	7.81	9.22	7.84				
Wu Zhenhua (CHN) (1998)	time	2.67	3.78	4.87	5.97	7.06	8.17	9.27	10.36	11.46	12.63		14.09	7 / 5				
	reaction time	0.145											PB			3.30	3.30	3.36
	interval		1.11	1.09	1.10	1.09	1.11	1.10	1.09	1.10	1.17	1.46				8.31	8.31	8.16
	velocity		5.14	8.23	8.39	8.31	8.39	8.23	8.31	8.39	8.31	7.81	9.60	7.81				
Sun Zhenjiang (CHN) (1999)	time	2.62	3.70	4.74	5.80	6.86	7.93	9.02	10.11	11.22	12.38		14.15	3 / 6				
	reaction time	0.173														3.18	3.22	3.36
	interval		1.08	1.04	1.06	1.06	1.07	1.09	1.09	1.09	1.11	1.16	1.77			8.62	8.52	8.16
	velocity		5.24	8.46	8.79	8.62	8.62	8.54	8.39	8.39	8.23	7.88	7.92	7.77				
Guo Zhongjie (CHN) (2001)	time	2.68	3.77	4.87	5.96	7.05	8.12	9.24	10.36	11.53	12.68		14.22	1 / 7				
	reaction time	0.192														3.28	3.28	3.44
	interval		1.09	1.10	1.09	1.09	1.07	1.12	1.12	1.12	1.17	1.15	1.54			8.36	8.36	7.97
	velocity		5.12	8.39	8.31	8.39	8.39	8.54	8.16	8.16	7.81	7.95	9.10	7.74				
Zhang Shuai (CHN) (1996)	time	2.68	3.78	4.88	5.97	7.07	8.19	9.30	10.41	11.54	12.72		14.27	4 / 8				
	reaction time	0.174														3.29	3.33	3.42
	interval		1.10	1.10	1.09	1.10	1.12	1.11	1.11	1.11	1.13	1.18	1.55			8.33	8.23	8.02
	velocity		5.12	8.31	8.31	8.39	8.31	8.16	8.23	8.23	8.09	7.75	9.05	7.71				

2019 Shimane High School Championships (Honshu, JPN)

FINAL

date 26-May-19 wind 2.3 m/s

Shimizu (2019) - shimane high school championships 2019 - biomechanics data analysis

		H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
足立 駿人	time	2.82	3.99	5.14	6.29	7.47	8.62	9.79	11.01	12.24	13.42		14.99	4 / 1			
	reaction time																
	interval		1.17	1.15	1.15	1.18	1.15	1.17	1.22	1.23	1.18	1.57			3.47	3.50	3.63
荻田 陽向	time	2.85	4.09	5.29	6.54	7.77	9.00	10.22	11.46	12.71	13.99		15.53	6 / 2			
	reaction time																
	interval		1.24	1.20	1.25	1.23	1.23	1.22	1.24	1.25	1.28	1.54			3.69	3.68	3.77
原田 真紘	time	2.75	3.95	5.19	6.40	7.64	8.85	10.09	11.37	12.62	13.89		15.53	5 / 3			
	reaction time																
	interval		1.20	1.24	1.21	1.24	1.21	1.24	1.28	1.25	1.27	1.64			3.65	3.69	3.80
秦 伸成	time	2.87	4.12	5.32	6.60	7.82	9.04	10.31	11.56	12.82	14.11		15.74	7 / 4			
	reaction time																
	interval		1.25	1.20	1.28	1.22	1.22	1.27	1.25	1.26	1.29	1.63			3.73	3.71	3.80
堀 汐音	time	2.80	3.99	5.25	6.50	7.72	8.97	10.27	11.56	12.89	14.24		15.90	2 / 5			
	reaction time																
	interval		1.19	1.26	1.25	1.22	1.25	1.30	1.29	1.33	1.35	1.66			3.70	3.77	3.97
的場 真那斗	time	3.02	4.34	5.64	6.90	8.20	9.47	10.72	12.06	13.41	14.81		16.60	9 / 6			
	reaction time																
	interval		1.32	1.30	1.26	1.30	1.27	1.25	1.34	1.35	1.40	1.79			3.88	3.82	4.09

2019 NCAA Championships - East Preliminary Round (Jacksonville, FL) (TV Analysis)

Quarter-Final 3

date 25-May-19 wind 1.5 m/s

Henson (2020) - Athlete First: 2019 year end hurdle report

		H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
Roberts, Daniel (USA) (1998)	time	2.36	3.40	4.40	5.44	6.48	7.48	8.50	9.54	10.56	11.62		13.13	5 / 1			
	reaction time																
	interval		1.04	1.00	1.04	1.04	1.00	1.02	1.04	1.02	1.06	1.51			3.08	3.06	3.12

Quarter-Final 1

date 25-May-19 wind 1.3 m/s

Henson (2020) - Athlete First: 2019 year end hurdle report

		H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
Holloway, Grant (USA) (1997)	time	2.36	3.36	4.36	5.36	6.36	7.36	8.38	9.40	10.44	11.50		13.10	6 / 1			
	reaction time																
	interval		1.00	1.00	1.00	1.00	1.00	1.02	1.02	1.04	1.06	1.60			3.00	3.02	3.12

2019 Chinese National Grand Prix 3 (Jinhau, CHN)

FINAL

date 23-May-19 wind 2.5 m/s

CAA Hurdle Development (2019)

		H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
Sun Zhenjiang (CHN) (1999)	time	2.62	3.68	4.74	5.78	6.82	7.90	8.96	10.02	11.12	12.22		13.71	2 / 1			
	reaction time	0.179															
	interval		1.06	1.06	1.04	1.04	1.08	1.06	1.06	1.10	1.10	1.49			3.16	3.18	3.26
Pan Zijie (CHN) (1995)	time	2.68	3.76	4.84	5.92	7.00	8.08	9.14	10.20	11.30	12.38		13.83	5 / 2			
	reaction time																
	interval		1.08	1.08	1.06	1.06	1.06	1.06	1.06	1.10	1.10	1.49			8.68	8.62	8.41

reaction time	0.182	interval	1.08	1.08	1.08	1.08	1.08	1.06	1.06	1.10	1.08	1.45			3.24	3.22	3.24
		velocity	5.12	8.46	8.46	8.46	8.46	8.46	8.62	8.62	8.31	8.46	9.67	7.95	8.46	8.52	8.46
Li Jiming (CHN) (1996)	time	2.66	3.74	4.80	5.88	6.92	8.02	9.08	10.16	11.28	12.40			13.89	6 / 3		
reaction time	0.181	interval	1.08	1.06	1.08	1.04	1.10	1.06	1.08	1.12	1.12	1.49			3.22	3.20	3.32
		velocity	5.16	8.46	8.62	8.46	8.79	8.31	8.62	8.46	8.16	8.16	9.41	7.92	8.52	8.57	8.26
Ma Lei (CHN) (1989)	time	2.68	3.78	4.86	5.92	6.98	8.06	9.14	10.22	11.32	12.42			13.91	4 / 4		
reaction time	0.158	interval	7 steps	1.10	1.08	1.06	1.08	1.08	1.08	1.10	1.10	1.49			3.24	3.22	3.28
		velocity	5.12	8.31	8.46	8.62	8.62	8.46	8.46	8.46	8.31	8.31	9.41	7.91	8.46	8.52	8.36
Yang Yuxuan (CHN) (2002)	time	2.70	3.82	4.90	5.98	7.06	8.12	9.22	10.30	11.38	12.50			13.94	8 / 5		
reaction time	0.166	interval		1.12	1.08	1.08	1.08	1.06	1.10	1.08	1.08	1.12	1.44		3.28	3.24	3.28
		velocity	5.08	8.16	8.46	8.46	8.46	8.62	8.31	8.46	8.46	8.16	9.74	7.89	8.36	8.46	8.36
Zhang Tao (CHN) (1997)	time	2.70	3.82	4.92	6.00	7.08	8.16	9.22	10.32	11.42	12.52			13.99	7 / 6		
reaction time	0.174	interval		1.12	1.10	1.08	1.08	1.08	1.06	1.10	1.10	1.10	1.47		3.30	3.22	3.30
		velocity	5.08	8.16	8.31	8.46	8.46	8.46	8.62	8.31	8.31	8.31	9.54	7.86	8.31	8.52	8.31
Sun Haoru (CHN) (1998)	time	2.70	3.82	4.92	6.00	7.08	8.20	9.28	10.38	11.50	12.60			14.13	9 / 7		
reaction time	0.231	interval		1.12	1.10	1.08	1.08	1.12	1.08	1.10	1.12	1.10	1.53		3.30	3.28	3.32
		velocity	5.08	8.16	8.31	8.46	8.46	8.16	8.46	8.31	8.16	8.31	9.16	7.78	8.31	8.36	8.26
Guo Zhongjie (CHN) (2001)	time	2.74	3.82	4.88	5.94	7.02	8.12	9.22	10.32	11.46	12.66			14.25	3 / 8		
reaction time	0.219	interval		1.08	1.06	1.06	1.08	1.10	1.10	1.10	1.14	1.20	1.59		3.20	3.28	3.44
		velocity	5.01	8.46	8.62	8.62	8.46	8.31	8.31	8.31	8.02	7.62	8.82	7.72	8.57	8.36	7.97

2019 Nanjing World Challenge (Nanjing, CHN)

FINAL

INAL			date	21-May-19	wind 0.1 m/s										CAA Hurdle Development (2019)						
					H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10	
Ortega, Orlando (ESP) (1991)	time				2.57	3.63	4.65	5.68	6.68	7.72	8.74	9.78	10.81	11.87		13.27	5 / 1				
	reaction time	0.145	interval	7 steps	1.06	1.02	1.03	1.00	1.04	1.02	1.04	1.03	1.06	1.40				3.11	3.06	3.13	
			velocity		5.34	8.62	8.96	8.87	9.14	8.79	8.96	8.79	8.87	8.62	10.01	8.29		8.82	8.96	8.76	
McLeod, Omar (JAM) (1994)	time				2.55	3.57	4.58	5.60	6.64	7.66	8.71	9.75	10.79	11.89		13.28	6 / 2				
	reaction time	0.152	interval		1.02	1.01	1.02	1.04	1.02	1.05	1.04	1.04	1.10	1.39				3.05	3.11	3.18	
			velocity		5.38	8.96	9.05	8.96	8.79	8.96	8.70	8.79	8.79	8.31	10.09	8.28		8.99	8.82	8.62	
Crittenden, Freddie (USA) (1991)	time				2.55	3.59	4.64	5.69	6.71	7.74	8.78	9.83	10.88	11.96		13.37	9 / 3				
	reaction time	0.137	interval		1.04	1.05	1.05	1.02	1.03	1.04	1.05	1.05	1.08	1.41				3.14	3.09	3.18	
			velocity		5.38	8.79	8.70	8.70	8.96	8.87	8.79	8.70	8.70	8.46	9.94	8.23		8.73	8.87	8.62	
Shubenkov, Sergey (ANA) (1991)	time				2.62	3.66	4.68	5.68	6.71	7.73	8.76	9.79	10.85	11.94		13.37	7 / 4				
	reaction time	0.165	interval	7 steps	1.04	1.02	1.00	1.03	1.02	1.03	1.03	1.03	1.06	1.09	1.43			3.06	3.08	3.18	
			velocity		5.24	8.79	8.96	9.14	8.87	8.96	8.87	8.87	8.62	8.39	9.80	8.23		8.96	8.90	8.62	
Chen Kuei-Ru (TPE) (1993)	time				2.56	3.62	4.64	5.66	6.70	7.74	8.80	9.87	10.94	12.01		13.49	2 / 5				
	reaction time	0.169	interval	7 steps	1.06	1.02	1.02	1.04	1.04	1.06	1.07	1.07	1.07	1.48				3.10	3.14	3.21	
			velocity		5.36	8.62	8.96	8.96	8.79	8.79	8.62	8.54	8.54	8.54	9.47	8.15		8.85	8.73	8.54	
Trajkovic, Milan (CYP) (1992)	time				2.62	3.67	4.73	5.76	6.79	7.85	8.90	9.96	11.02	12.11		13.56	4 / 6				
	reaction time	0.192	interval	7 steps	1.05	1.06	1.03	1.03	1.06	1.05	1.06	1.06	1.06	1.09	1.45			3.14	3.14	3.21	
			velocity		5.24	8.70	8.62	8.87	8.87	8.62	8.70	8.62	8.62	8.39	9.67	8.11		8.73	8.73	8.54	
Alkana, Antonio (RSA) (1990)	time				2.55	3.60	4.66	5.72	6.76	7.81	8.87	9.95	11.03	12.14		13.61	3 / 7				
	reaction time	0.140	interval		1.05	1.06	1.06	1.04	1.05	1.06	1.08	1.08	1.11	1.47				3.17	3.15	3.27	
			velocity		5.38	8.70	8.62	8.62	8.79	8.70	8.62	8.46	8.46	8.23	9.54	8.08		8.65	8.70	8.39	
Zeng Jianhang (CHN) (1998)	time				2.59	3.64	4.70	5.73	6.76	7.82	8.87	9.95	11.04	12.16		13.62	10 / 8				
	reaction time	0.146	interval		1.05	1.06	1.03	1.03	1.06	1.05	1.08	1.09	1.12	1.46	PB			3.14	3.14	3.29	
			velocity		5.30	8.70	8.62	8.87	8.87	8.62	8.70	8.46	8.39	8.16	9.60	8.08		8.73	8.73	8.33	
Pozzi, Andy (GBR) (1992)	time				2.57	3.60	4.60	5.61	6.64	7.67	8.73	9.78	10.88	12.02		13.66	8 / 9				
	reaction time	0.146	interval	7 steps	1.03	1.00	1.01	1.03	1.03	1.06	1.05	1.10	1.14	1.64				3.04	3.12	3.29	
			velocity		5.34	8.87	9.14	9.05	8.87	8.87	8.62	8.70	8.31	8.02	8.55	8.05		9.02	8.79	8.33	

2019 Seiko Golden Grand Prix (Osaka, JPN)

FINAL

FINAL	date		19-May-19		wind		2.9 m/s		Shibayama (2019) - race analysis of men's 110m hurdles in the 2019 season												
					H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10	
Izumiya, Shunsuke (JPN) (2000)	time				2.53	3.60	4.62	5.66	6.71	7.72	8.74	9.79	10.84	11.93		13.26	3 / 1				
	reaction time	0.121	interval			1.07	1.02	1.04	1.05	1.01	1.02	1.05	1.05	1.09	1.33			3.13	3.08	3.19	
			velocity		5.42	8.54	8.96	8.79	8.70	9.05	8.96	8.70	8.70	8.70	8.39	10.54	8.30		8.76	8.90	8.60
Swift, Greggmar (BAR) (1991)	time				2.54	3.61	4.66	5.69	6.70	7.76	8.79	9.84	10.91	12.02		13.45	2 / 2				
	reaction time	0.150	interval			1.07	1.05	1.03	1.01	1.06	1.03	1.05	1.07	1.11	1.43			3.15	3.10	3.23	
			velocity		5.40	8.54	8.70	8.87	9.05	8.62	8.87	8.70	8.54	8.23	9.80	8.18		8.70	8.85	8.49	

Kanai, Taio (JPN) (1995)	time	2.54	3.61	4.67	5.71	6.75	7.79	8.82	9.90	10.98	12.06		13.47	5 / 3				
	reaction time	0.102	interval	7 steps	1.07	1.06	1.04	1.04	1.04	1.03	1.08	1.08	1.08	1.41		3.17	3.11	3.24
			velocity		5.40	8.54	8.62	8.79	8.79	8.79	8.87	8.46	8.46	8.46	9.94	8.17	8.65	8.82
Takayama, Shunya (JPN) (1994)	time	2.57	3.62	4.68	5.72	6.76	7.80	8.84	9.90	10.96	12.05		13.51	7 / 4				
	reaction time	0.143	interval		1.05	1.06	1.04	1.04	1.04	1.06	1.06	1.09	1.46			3.15	3.12	3.21
			velocity		5.34	8.70	8.62	8.79	8.79	8.79	8.62	8.62	8.39	9.60	8.14	8.70	8.79	8.54
Ishikawa, Shuhei (JPN) (1995)	time	2.60	3.68	4.76	5.83	6.88	7.92	8.99	10.03	11.10	12.20		13.63	9 / 5				
	reaction time	0.172	interval	7 steps	1.08	1.08	1.07	1.05	1.04	1.07	1.04	1.07	1.10	1.43		3.23	3.16	3.21
			velocity		5.28	8.46	8.46	8.54	8.70	8.79	8.54	8.79	8.54	8.31	9.80	8.07	8.49	8.68
Fontenot, Ryan (USA) (1986)	time	2.67	3.75	4.82	5.91	6.95	8.01	9.06	10.10	11.18	12.27		13.70	6 / 6				
	reaction time	0.157	interval		1.08	1.07	1.09	1.04	1.06	1.05	1.04	1.08	1.09	1.43		3.24	3.15	3.21
			velocity		5.14	8.46	8.54	8.39	8.79	8.62	8.70	8.79	8.46	8.39	9.80	8.03	8.46	8.70
Cabral, Johnathan (CAN) (1992)	time	2.72	3.83	4.88	5.92	6.95	7.99	9.04	10.10	11.16	12.26		13.74	8 / 7				
	reaction time	0.245	interval	7 steps	1.11	1.05	1.04	1.03	1.04	1.05	1.06	1.06	1.10	1.48		3.20	3.12	3.22
			velocity		5.04	8.23	8.70	8.79	8.87	8.79	8.70	8.62	8.62	8.31	9.47	8.01	8.57	8.79
Omuro, Hideki (JPN) (1990)	time	2.59	3.70	4.77	5.86	6.95	8.03	9.08	10.16	11.24	12.33		13.76	1 / 8				
	reaction time	0.143	interval		1.11	1.07	1.09	1.09	1.08	1.05	1.08	1.08	1.09	1.43		3.27	3.22	3.25
			velocity		5.30	8.23	8.54	8.39	8.39	8.46	8.70	8.46	8.46	8.39	9.80	7.99	8.39	8.52
Walters, Ruebin (TTO) (1995)	time	2.67	3.77	4.85	5.92	6.97	8.02	9.06	10.13	11.22	12.34		13.84	4 / 9				
	reaction time	0.147	interval		1.10	1.08	1.07	1.05	1.05	1.04	1.07	1.09	1.12	1.50		3.25	3.14	3.28
			velocity		5.14	8.31	8.46	8.54	8.70	8.70	8.79	8.54	8.39	8.16	9.35	7.95	8.44	8.73

2019 Shanghai Diamond League (Shanghai, CHN)

FINAL

		date	18-May-19	wind 0.7 m/s										CAA Hurdle Development (2019)						
				H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10	
McLeod, Omar (JAM) (1994)	time			2.52	3.56	4.58	5.56	6.56	7.58	8.60	9.62	10.68	11.76		13.12	4 / 1				
	reaction time	0.149	interval		1.04	1.02	0.98	1.00	1.02	1.02	1.02	1.06	1.08	1.36				3.04	3.04	3.16
			velocity	5.44	8.79	8.96	9.33	9.14	8.96	8.96	8.96	8.62	8.46	10.31	8.38			9.02	9.02	8.68
Xie Wenjun (CHN) (1990)	time			2.58	3.62	4.62	5.64	6.62	7.64	8.64	9.68	10.70	11.78		13.17	3 / 2				
	reaction time	0.145	interval	7 steps	1.04	1.00	1.02	0.98	1.02	1.00	1.04	1.02	1.08	1.39	PB			3.06	3.00	3.14
			velocity	5.32	8.79	9.14	8.96	9.33	8.96	9.14	8.79	8.96	8.46	10.09	8.35			8.96	9.14	8.73
Shubenkov, Sergey (ANA) (1991)	time			2.60	3.66	4.68	5.66	6.66	7.68	8.74	9.76	10.82	11.90		13.28	5 / 3				
	reaction time	0.126	interval	7 steps	1.06	1.02	0.98	1.00	1.02	1.06	1.02	1.06	1.08	1.38				3.06	3.08	3.16
			velocity	5.28	8.62	8.96	9.33	9.14	8.96	8.62	8.96	8.62	8.46	10.16	8.28			8.96	8.90	8.68
Ortega, Orlando (ESP) (1991)	time			2.58	3.62	4.64	5.64	6.66	7.68	8.70	9.74	10.80	11.86		13.28	6 / 4				
	reaction time	0.154	interval	7 steps	1.04	1.02	1.00	1.02	1.02	1.02	1.04	1.06	1.06	1.42				3.06	3.06	3.16
			velocity	5.32	8.79	8.96	9.14	8.96	8.96	8.96	8.79	8.62	8.62	9.87	8.28			8.96	8.96	8.68
Crittenden, Freddie (USA) (1991)	time			2.58	3.64	4.68	5.70	6.72	7.76	8.78	9.82	10.86	11.94		13.36	8 / 5				
	reaction time	0.151	interval		1.06	1.04	1.02	1.02	1.04	1.02	1.04	1.04	1.08	1.42				3.12	3.08	3.16
			velocity	5.32	8.62	8.79	8.96	8.96	8.79	8.96	8.79	8.79	8.46	9.87	8.23			8.79	8.90	8.68
Pozzi, Andy (GBR) (1992)	time			2.52	3.58	4.58	5.62	6.62	7.66	8.70	9.74	10.82	11.92		13.39	7 / 6				
	reaction time	0.133	interval	7 steps	1.06	1.00	1.04	1.00	1.04	1.04	1.04	1.08	1.10	1.47				3.10	3.08	3.22
			velocity	5.44	8.62	9.14	8.79	9.14	8.79	8.79	8.79	8.46	8.31	9.54	8.22			8.85	8.90	8.52
Trajkovic, Milan (CYP) (1992)	time			2.60	3.68	4.70	5.70	6.72	7.74	8.76	9.80	10.84	11.94		13.41	9 / 7				
	reaction time	0.145	interval	7 steps	1.08	1.02	1.00	1.02	1.02	1.02	1.04	1.04	1.10	1.47				3.10	3.06	3.18
			velocity	5.28	8.46	8.96	9.14	8.96	8.96	8.96	8.79	8.79	8.31	9.54	8.20			8.85	8.96	8.62
Alkana, Antonio (RSA) (1990)	time			2.54	3.60	4.68	5.72	6.76	7.78	8.82	9.86	10.92	12.02		13.47	2 / 8				
	reaction time	0.132	interval		1.06	1.08	1.04	1.04	1.02	1.04	1.04	1.06	1.10	1.45				3.18	3.10	3.20
			velocity	5.40	8.62	8.46	8.79	8.79	8.96	8.79	8.79	8.62	8.31	9.67	8.17			8.62	8.85	8.57
Zeng Jianhang (CHN) (1998)	time			2.58	3.68	4.72	5.78	6.82	7.88	8.96	10.02	11.12	12.24		13.71	1 / 9				
	reaction time	0.154	interval		1.10	1.04	1.06	1.04	1.06	1.08	1.06	1.10	1.12	1.47				3.20	3.18	3.28
			velocity	5.32	8.31	8.79	8.62	8.79	8.62	8.46	8.62	8.31	8.16	9.54	8.02			8.57	8.62	8.36

2019 South Eastern Conference Championships (Fayetteville, AR) (TV Analysis)

FINAL

INAL	date	11-May-19	wind	1.0 m/s	Henson (2020) - Athlete First: 2019 year end hurdle report														
			H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10	
Roberts, Daniel (USA) (1998)	time		2.53	3.56	4.58	5.60	6.60	7.58	8.60	9.62	10.66	11.72		13.07	4 / 1				
	reaction time	interval		1.03	1.02	1.02	1.00	0.98	1.02	1.02	1.04	1.06	1.35	PB		3.07	3.00	3.12	
		velocity	5.42	8.87	8.96	8.96	9.14	9.33	8.96	8.96	8.79	8.62	10.39	8.42		8.93	9.14	8.79	
Holloway, Grant (USA) (1997)	time		2.53	3.53	4.53	5.53	6.50	7.50	8.50	9.53	10.58	11.68		13.12	5 / 2				

reaction time	interval	7 steps	1.00	1.00	1.00	0.97	1.00	1.00	1.03	1.05	1.10	1.44			3.00	2.97	3.18
	velocity		5.42	9.14	9.14	9.14	9.42	9.14	9.14	8.87	8.70	8.31	9.74	8.38	9.14	9.23	8.62
Moore, Isaiah (USA) (1996)	time		2.66	3.75	4.80	5.84	6.86	7.86	8.90	9.93	10.96	12.03		13.42	6 / 3		
reaction time	interval			1.09	1.05	1.04	1.02	1.00	1.04	1.03	1.03	1.07	1.39			3.18	3.06
	velocity		5.16	8.39	8.70	8.79	8.96	9.14	8.79	8.87	8.87	8.54	10.09	8.20		8.62	8.96

2019 IAAF World Relays - Shuttle Hurdle Relay (Yokohama, JPN)

Heat 2 - 2nd Leg

date	11-May-19	wind	na	CAA Hurdle Development (2019)													
			H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7
Sun Zhenjiang (CHN) (1999)	time		2.45	3.53	4.62	5.72	6.79	7.87	8.97	10.07	11.21	12.34		13.88	4 /		
reaction time	interval	7 steps		1.08	1.09	1.10	1.07	1.08	1.10	1.10	1.14	1.13	1.54			3.27	3.25
	velocity		5.60	8.46	8.39	8.31	8.54	8.46	8.31	8.31	8.02	8.09	9.10	7.93	57.54	8.39	8.44

Heat 2 - 4th Leg

date	11-May-19	wind	na	CAA Hurdle Development (2019)													
			H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7
Zeng Jianhang (CHN) (1998)	time		2.50	3.58	4.68	5.78	6.88	7.97	9.04	10.17	11.27	12.41		13.90	4 / 3		
reaction time	interval			1.08	1.10	1.10	1.10	1.09	1.07	1.13	1.10	1.14	1.49			3.28	3.26
	velocity		5.49	8.46	8.31	8.31	8.31	8.39	8.54	8.09	8.31	8.02	9.41	7.91	57.54	8.36	8.41

2019 Kinami Memorial (Osaka, JPN)

FINAL

date	06-May-19	wind	0.3 m/s	Matsubayashi (2019) - research on athlete performance and technique- 2019 data book													
			H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7
Crittenden, Freddie (USA) (199)	time		2.58	3.62	4.65	5.67	6.70	7.72	8.75	9.81	10.86	11.95		13.38	4 / 1		
reaction time	interval			1.04	1.03	1.02	1.03	1.02	1.03	1.06	1.05	1.09	1.43			3.09	3.08
	velocity		5.32	8.79	8.87	8.96	8.87	8.96	8.87	8.62	8.70	8.39	9.80	8.22		8.87	8.90
Ishikawa, Shuhei (JPN) (1995)	time		2.57	3.61	4.66	5.72	6.77	7.81	8.85	9.91	10.96	12.06		13.50	6 / 2		
reaction time	interval	7 steps		1.04	1.05	1.06	1.05	1.04	1.04	1.06	1.05	1.10	1.44	PB		3.15	3.13
	velocity		5.34	8.79	8.70	8.62	8.70	8.79	8.79	8.62	8.70	8.31	9.74	8.15		8.70	8.76
Izumiya, Shunsuke (JPN) (2000)	time		2.59	3.63	4.71	5.76	6.78	7.87	8.96	10.01	11.07	12.20		13.58	7 / 3		
reaction time	interval			1.04	1.08	1.05	1.02	1.09	1.09	1.05	1.06	1.13	1.38			3.17	3.20
	velocity		5.30	8.79	8.46	8.70	8.96	8.39	8.39	8.70	8.62	8.09	10.16	8.10		8.65	8.57
Yazawa, Wataru (JPN) (1991)	time		2.64	3.73	4.80	5.86	6.92	7.97	9.04	10.11	11.19	12.27		13.71	8 / 4		
reaction time	interval	7 steps		1.09	1.07	1.06	1.06	1.05	1.07	1.07	1.08	1.08	1.44			3.22	3.18
	velocity		5.20	8.39	8.54	8.62	8.62	8.70	8.54	8.54	8.46	8.46	9.74	8.02		8.52	8.62
Wado, Tatsuya (JPN) (1990)	time		2.60	3.67	4.73	5.78	6.84	7.90	8.99	10.08	11.17	12.32		13.79	5 / 5		
reaction time	interval	7 steps		1.07	1.06	1.05	1.06	1.06	1.09	1.09	1.09	1.15	1.47			3.18	3.21
	velocity		5.28	8.54	8.62	8.70	8.62	8.62	8.39	8.39	8.39	7.95	9.54	7.98		8.62	8.54
Sato, Hiroyuki (JPN) (1990)	time		2.58	3.64	4.71	5.78	6.86	7.92	9.01	10.11	11.22	12.35		13.81	2 / 6		
reaction time	interval			1.06	1.07	1.07	1.08	1.06	1.09	1.10	1.11	1.13	1.46			3.20	3.23
	velocity		5.32	8.62	8.54	8.54	8.46	8.62	8.39	8.31	8.23	8.09	9.60	7.97		8.57	8.49
Fujii, Ryota (JPN) (1996)	time		2.60	3.70	4.78	5.87	6.97	8.05	9.15	10.24	11.35	12.49		13.94	3 / 7		
reaction time	interval			1.10	1.08	1.09	1.10	1.08	1.10	1.09	1.11	1.14	1.45			3.27	3.28
	velocity		5.28	8.31	8.46	8.39	8.31	8.46	8.31	8.39	8.23	8.02	9.67	7.89		8.39	8.36
Kagimoto, Masahiro (JPN) (199)	time		2.66	3.74	4.80	5.90	6.97	8.05	9.16	10.26	11.38	12.49		13.95	9 / 8		
reaction time	interval			1.08	1.06	1.10	1.07	1.08	1.11	1.10	1.12	1.11	1.46			3.24	3.26
	velocity		5.16	8.46	8.62	8.31	8.54	8.46	8.23	8.31	8.16	8.23	9.60	7.89		8.46	8.41

2019 Mikio Oda Memorial (Hiroshima, JPN)

A FINAL

date	28-Apr-19	wind	0.6 m/s	Kobayashi (2019) - research on athlete performance and technique- 2019 data book													
			H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7
Ishikawa, Shuhei (JPN) (1995)	time		2.59	3.65	4.68	5.70	6.74	7.77	8.81	9.86	10.97	12.09		13.54	6 / 1		
reaction time	interval	7 steps		1.06	1.03	1.02	1.04	1.03	1.04	1.05	1.11	1.12	1.45	PB		3.11	3.11
	velocity		5.30	8.62	8.87	8.96	8.79	8.87	8.79	8.70	8.23	8.16	9.67	8.12		8.82	8.82
Izumiya, Shunsuke (JPN) (2000)	time		2.56	3.62	4.68	5.72	6.77	7.82	8.88	9.95	11.04	12.13		13.56	4 / 2		
reaction time	interval			1.06	1.06	1.04	1.05	1.05	1.06	1.07	1.09	1.09	1.43	NJR PB		3.16	3.16
	velocity		5.36	8.62	8.62	8.79	8.70	8.70	8.62	8.54	8.39	8.39	9.80	8.11		8.68	8.68
Omuro, Hideki (JPN) (1990)	time		2.64	3.73	4.79	5.83	6.89	7.95	9.04	10.12	11.23	12.35		13.78	2 / 3		
reaction time	interval			1.09	1.06	1.04	1.06	1.06	1.09	1.08	1.11	1.12	1.43			3.19	3.21
	velocity		5.20	8.39	8.62	8.79	8.62	8.62	8.39	8.46	8.23	8.16	9.80	7.98		8.60	8.54
Yazawa, Wataru (JPN) (1991)	time		2.70	3.80	4.86	5.92	6.97	8.03	9.10	10.19	11.29	12.41		13.83	7 / 4		
reaction time	interval	7 steps		1.10	1.06	1.06	1.05	1.06	1.07	1.09	1.10	1.12	1.42			3.22	3.18
	velocity		5.08	8.31	8.62	8.62	8.70	8.62	8.54	8.39	8.31	8.16	9.87	7.95		8.52	8.62
Wado, Tatsuya (JPN) (1990)	time		2.60	3.69	4.75	5.80	6.86	7.93	9.00	10.10	11.22	12.35		13.88	8 / 5		
reaction time	interval	7 steps		1.09	1.06	1.05	1.06	1.07	1.07	1.10	1.12	1.13	1.53			3.20	3.20

	velocity	5.28	8.39	8.62	8.70	8.62	8.54	8.54	8.31	8.16	8.09	9.16	7.93		8.57	8.57	8.19
Ogata, Akihiro (JPN) (1993)	time	2.65	3.73	4.78	5.82	6.89	7.97	9.07	10.17	11.29	12.42		13.97	3 / 6			
	reaction time	interval	7 steps	1.08	1.05	1.04	1.07	1.08	1.10	1.10	1.12	1.13	1.55		3.17	3.25	3.35
	velocity	5.18	8.46	8.70	8.79	8.54	8.46	8.31	8.31	8.16	8.09	9.05	7.87		8.65	8.44	8.19
Fujii, Ryota (JPN) (1996)	time	2.74	3.84	4.89	5.96	7.03	8.12	9.22	10.32	11.44	12.58		14.07	1 / 7			
	reaction time	interval		1.10	1.05	1.07	1.07	1.09	1.10	1.10	1.12	1.14	1.49		3.22	3.26	3.36
	velocity	5.01	8.31	8.70	8.54	8.54	8.39	8.31	8.31	8.16	8.02	9.41	7.82		8.52	8.41	8.16

B FINAL

date 28-Apr-19 wind 1.4 m/s Kobayashi (2019) - research on athlete performance and technique- 2019 data book

		H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
Kagimoto, Masahiro (JPN) (1993)	time	2.67	3.76	4.83	5.90	6.98	8.07	9.18	10.29	11.40	12.51		13.97	5 / 1			
	reaction time	interval		1.09	1.07	1.08	1.09	1.11	1.11	1.11	1.11	1.46			3.23	3.28	3.33
	velocity	5.14	8.39	8.54	8.54	8.46	8.39	8.23	8.23	8.23	8.23	9.60	7.87		8.49	8.36	8.23
Kokubu, Toru (JPN) (1984)	time	2.69	3.79	4.88	5.96	7.05	8.15	9.25	10.38	11.52	12.67		14.17	6 / 2			
	reaction time	interval	7 steps	1.10	1.09	1.08	1.09	1.10	1.10	1.13	1.14	1.15	1.50		3.27	3.29	3.42
	velocity	5.10	8.31	8.39	8.46	8.39	8.31	8.31	8.09	8.02	7.95	9.35	7.76		8.39	8.33	8.02
McCorry, Jacob (AUS) (1997)	time	2.67	3.76	4.83	5.92	7.03	8.14	9.25	10.36	11.46	12.63		14.20	3 / 3			
	reaction time	interval		1.09	1.07	1.09	1.11	1.11	1.11	1.10	1.17	1.57			3.25	3.33	3.38
	velocity	5.14	8.39	8.54	8.39	8.23	8.23	8.23	8.23	8.31	7.81	8.93	7.75		8.44	8.23	8.11
Sato, Hiroyuki (JPN) (1990)	time	2.59	3.69	4.76	5.84	6.93	8.03	9.13	10.28	11.47	12.65		14.21	2 / 4			
	reaction time	interval		1.10	1.07	1.08	1.09	1.10	1.10	1.15	1.19	1.18	1.56		3.25	3.29	3.52
	velocity	5.30	8.31	8.54	8.46	8.39	8.31	8.31	7.95	7.68	7.75	8.99	7.74		8.44	8.33	7.79
Furuya, Takumu (JPN) (1997)	time	2.73	3.82	4.93	6.00	7.10	8.20	9.31	10.43	11.57	12.73		14.28	4 / 5			
	reaction time	interval		1.09	1.11	1.07	1.10	1.10	1.11	1.12	1.14	1.16	1.55		3.27	3.31	3.42
	velocity	5.03	8.39	8.23	8.54	8.31	8.31	8.23	8.16	8.02	7.88	9.05	7.70		8.39	8.28	8.02
Tokuoka, Ryo (JPN) (1999)	time	2.74	3.86	4.99	6.11	7.20	8.33	9.44	10.60	11.75	12.91		14.44	1 / 6			
	reaction time	interval		1.12	1.13	1.12	1.09	1.13	1.11	1.16	1.15	1.16	1.53		3.37	3.33	3.47
	velocity	5.01	8.16	8.09	8.16	8.39	8.09	8.23	7.88	7.95	7.88	9.16	7.62		8.14	8.23	7.90
Inui, Daisuke (JPN) (1995)	time	2.69	3.83	4.92	6.02	7.13	8.27	9.41	10.56	11.71	12.91		14.47	7 / 7			
	reaction time	interval		1.14	1.09	1.10	1.11	1.14	1.15	1.15	1.20	1.56			3.33	3.39	3.50
	velocity	5.10	8.02	8.39	8.31	8.23	8.02	8.02	7.95	7.95	7.62	8.99	7.60		8.23	8.09	7.83

2019 Asian Athletics Championships (Doha, QAT)**FINAL**

date 24-Apr-19 wind 1.7 m/s CAA Hurdle Development (2019)

		H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
Xie Wenjun (CHN) (1990)	time	2.55	3.58	4.63	5.63	6.64	7.65	8.65	9.67	10.72	11.79		13.21	7 / 1			
	reaction time	0.151	interval	7 steps	1.03	1.05	1.00	1.01	1.00	1.02	1.05	1.07	1.42	CR PB	3.08	3.02	3.14
	velocity	5.38	8.87	8.70	9.14	9.05	9.05	9.14	8.96	8.70	8.54	9.87	8.33		8.90	9.08	8.73
Al-Yoha, Yaqoub (KUW) (1993)	time	2.51	3.56	4.58	5.60	6.62	7.64	8.69	9.74	10.82	11.91		13.35	6 / 2			
	reaction time	0.120	interval		1.05	1.02	1.02	1.05	1.05	1.08	1.09	1.44	NR PB		3.09	3.09	3.22
	velocity	5.47	8.70	8.96	8.96	8.96	8.96	8.70	8.70	8.46	8.39	9.74	8.24		8.87	8.87	8.52
Chen Kuei-Ru (TPE) (1993)	time	2.53	3.58	4.60	5.63	6.65	7.69	8.74	9.79	10.84	11.94		13.39	5 / 3			
	reaction time	0.137	interval	7 steps	1.05	1.02	1.03	1.04	1.05	1.05	1.10	1.45	=NR =PB		3.10	3.11	3.20
	velocity	5.42	8.70	8.96	8.87	8.96	8.79	8.70	8.70	8.70	8.31	9.67	8.22		8.85	8.82	8.57
Takayama, Shunya (JPN) (1994)	time	2.57	3.59	4.65	5.69	6.75	7.82	8.89	9.98	11.08	12.16		13.59	4 / 4			
	reaction time	0.145	interval		1.02	1.06	1.04	1.06	1.07	1.09	1.10	1.08	1.43		3.12	3.20	3.27
	velocity	5.34	8.96	8.62	8.79	8.62	8.54	8.54	8.39	8.31	8.46	9.80	8.09		8.79	8.57	8.39
Kanai, Taio (JPN) (1995)	time	2.58	3.64	4.70	5.73	6.80	7.87	8.94	10.02	11.11	12.21		13.64	9 / 5			
	reaction time	0.138	interval	7 steps	1.06	1.06	1.03	1.07	1.07	1.08	1.09	1.10	1.43		3.15	3.21	3.27
	velocity	5.32	8.62	8.62	8.87	8.54	8.54	8.54	8.46	8.39	8.31	9.80	8.06		8.70	8.54	8.39
Efremov, David (KAZ) (1999)	time	2.65	3.72	4.76	5.81	6.82	7.89	8.96	10.03	11.13	12.25		13.83	3 / 6			
	reaction time	0.168	interval	7 steps	1.07	1.04	1.05	1.01	1.07	1.07	1.10	1.12	1.58	PB	3.16	3.15	3.29
	velocity	5.18	8.54	8.79	8.70	9.05	8.54	8.54	8.54	8.31	8.16	8.87	7.95		8.68	8.70	8.33
Zeng Jianhang (CHN) (1998)	time	2.59	3.63	4.67	5.72	6.75	7.81	8.85	9.92	11.03	12.03		13.85	2 / 7			
	reaction time	0.152	interval		1.04	1.04	1.05	1.03	1.06	1.04	1.07	1.11	1.00	1.82	3.13	3.13	3.18
	velocity	5.30	8.79	8.79	8.70	8.87	8.62	8.79	8.54	8.23	9.14	7.70	7.94		8.76	8.76	8.62
Mui Ching Yeung (HKG) (1993)	time	2.63	3.71	4.78	5.84	6.92	7.97	9.10	10.22	11.36	12.51		13.96	8 / 8			
	reaction time	0.151	interval		1.08	1.07	1.06	1.08	1.05	1.13	1.12	1.14	1.45		3.21	3.26	3.41
	velocity	5.22	8.46	8.54	8.62	8.46	8.70	8.09	8.16	8.02	7.95	9.67	7.88		8.54	8.41	8.04

Heat 3

date 23-Apr-19 wind 1.6 m/s

CAA Hurdle Development (2019)

		H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10	
Xie Wenjun (CHN) (1990)	time	2.61	3.67	4.71	5.77	6.79	7.82	8.86	9.91	10.95	12.01		13.47	3 / 1				
	reaction time	0.159	interval	7 steps	1.06	1.04	1.06	1.02	1.03	1.04	1.05	1.04	1.06	1.46		3.16	3.09	3.15
		velocity	5.26	8.62	8.79	8.62	8.96	8.87	8.79	8.70	8.79	8.62	9.60	8.17		8.68	8.87	8.70

Heat 1

date 23-Apr-19 wind 2.1 m/s

CAA Hurdle Development (2019)

		H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
Zeng Jianhang (CHN) (1998)	time	2.55	3.60	4.67	5.70	6.76	7.82	8.89	9.98	11.07	12.21		13.69	7 / 3			
	reaction time	0.153	interval	1.05	1.07	1.03	1.06	1.06	1.07	1.09	1.09	1.14	1.48		3.15	3.19	3.32
	velocity	5.38	8.70	8.54	8.87	8.62	8.62	8.54	8.39	8.39	8.02	9.47	8.04		8.70	8.60	8.26

2019 Chinese National Grand Prix 2 (Huangshi, CHN)**FINAL**

date 13-Apr-19 wind -0.3 m/s

CAA Hurdle Development (2019)

		H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
Zeng Jianhang (CHN) (1998)	time	2.62	3.70	4.74	5.78	6.82	7.86	8.94	10.02	11.12	12.22		13.67	4 / 1			
	reaction time	0.186	interval	1.08	1.04	1.04	1.04	1.08	1.08	1.10	1.10	1.45			3.16	3.16	3.28
	velocity	5.24	8.46	8.79	8.79	8.79	8.79	8.46	8.46	8.31	8.31	9.67	8.05		8.68	8.68	8.36
Sun Zhenjiang (CHN) (1999)	time	2.65	3.72	4.76	5.80	6.84	7.90	8.98	10.06	11.14	12.26		13.75	2 / 2			
	reaction time	0.181	interval	7 steps	1.07	1.04	1.04	1.06	1.08	1.08	1.12	1.49	PB		3.15	3.18	3.28
	velocity	5.18	8.54	8.79	8.79	8.79	8.62	8.46	8.46	8.46	8.16	9.41	8.00		8.70	8.62	8.36
Lu Yang (CHN) (1996)	time	2.74	3.86	4.92	6.00	7.06	8.14	9.22	10.32	11.42	12.56		14.03	6 / 3			
	reaction time	0.187	interval	1.12	1.06	1.06	1.08	1.08	1.10	1.10	1.14	1.47			3.26	3.22	3.34
	velocity	5.01	8.16	8.62	8.46	8.62	8.46	8.46	8.31	8.31	8.02	9.54	7.84		8.41	8.52	8.21
Zhang Tao (CHN) (1997)	time	2.76	3.86	4.96	6.06	7.14	8.23	9.32	10.42	11.52	12.66		14.11	9 / 4			
	reaction time	0.216	interval	1.10	1.10	1.10	1.08	1.09	1.09	1.10	1.10	1.14	1.45		3.30	3.26	3.34
	velocity	4.97	8.31	8.31	8.31	8.46	8.39	8.39	8.31	8.31	8.02	9.67	7.80		8.31	8.41	8.21
Ning Xiaohan (CHN) (2000)	time	2.68	3.80	4.86	5.96	7.02	8.14	9.24	10.38	11.52	12.67		14.19	5 / 5			
	reaction time	0.163	interval	1.12	1.06	1.10	1.06	1.12	1.10	1.14	1.14	1.15	1.52		3.28	3.28	3.43
	velocity	5.12	8.16	8.62	8.31	8.62	8.16	8.31	8.02	8.02	7.95	9.22	7.75		8.36	8.36	7.99
Wu Zhenhua (CHN) (1998)	time	2.72	3.81	4.92	6.03	7.10	8.19	9.34	10.45	11.57	12.73		14.19	8 / 6			
	reaction time	0.178	interval	1.09	1.11	1.11	1.07	1.09	1.15	1.11	1.12	1.16	1.46	PB	3.31	3.31	3.39
	velocity	5.04	8.39	8.23	8.23	8.54	8.39	7.95	8.23	8.16	7.88	9.60	7.75		8.28	8.28	8.09
Zhao Pengchuan (CHN) (1993)	time	2.68	3.81	4.90	5.96	7.06	8.18	9.28	10.42	11.54	12.68		14.24	7 / 7			
	reaction time	0.176	interval	1.13	1.09	1.06	1.10	1.12	1.10	1.14	1.12	1.14	1.56		3.28	3.32	3.40
	velocity	5.12	8.09	8.39	8.62	8.31	8.16	8.31	8.02	8.16	8.02	8.99	7.72		8.36	8.26	8.06
Sun Haoru (CHN) (1998)	time	2.70	3.81	4.92	6.02	7.12	8.20	9.29	10.42	11.54	12.70		14.25	3 / 8			
	reaction time	0.181	interval	1.11	1.11	1.10	1.10	1.08	1.09	1.13	1.12	1.16	1.55	PB	3.32	3.27	3.41
	velocity	5.08	8.23	8.23	8.31	8.31	8.46	8.39	8.09	8.16	7.88	9.05	7.72		8.26	8.39	8.04

Heat 5

date 13-Apr-19 wind 0.6 m/s

CAA Hurdle Development (2019)

		H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
Zhao Pengchuan (CHN) (1993)	time	2.62	3.73	4.80	5.85	6.92	8.00	9.08	10.16	11.30	12.44		14.00	6 / 1			
	reaction time	0.161	interval	1.11	1.07	1.05	1.07	1.08	1.08	1.14	1.14	1.56			3.23	3.23	3.36
	velocity	5.24	8.23	8.54	8.70	8.54	8.46	8.46	8.46	8.02	8.02	8.99	7.86		8.49	8.49	8.16
Si Zhengdong (CHN) (1997)	time	2.72	3.86	5.00	6.14	7.24	8.34	9.46	10.59	11.76	12.94		14.55	5 / 3			
	reaction time	0.197	interval	1.14	1.14	1.14	1.10	1.12	1.13	1.17	1.18	1.61			3.42	3.32	3.48
	velocity	5.04	8.02	8.02	8.02	8.31	8.31	8.16	8.09	7.81	7.75	8.71	7.56		8.02	8.26	7.88

Heat 4

date 13-Apr-19 wind -0.8 m/s

CAA Hurdle Development (2019)

		H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
Wu Zhenhua (CHN) (1998)	time	2.70	3.80	4.90	5.99	7.11	8.22	9.36	10.46	11.60	12.72		14.23	6 / 1			
	reaction time	0.174	interval	1.10	1.10	1.09	1.12	1.11	1.14	1.10	1.14	1.12	1.51	PB	3.29	3.37	3.36
	velocity	5.08	8.31	8.31	8.39	8.16	8.23	8.02	8.31	8.02	8.16	9.28	7.73		8.33	8.14	8.16
Ma Lei (CHN) (1989)	time	2.76	3.88	4.98	6.09	7.17	8.28	9.38	10.50	11.66	12.80		14.36	7 / 2			
	reaction time	0.164	interval	7 steps	1.12	1.10	1.11	1.08	1.11	1.12	1.16	1.14	1.56		3.33	3.29	3.42
	velocity	4.97	8.16	8.31	8.23	8.46	8.23	8.31	8.16	7.88	8.02	8.99	7.66		8.23	8.33	8.02
Jiang Fan (CHN) (1989)	time	2.69	3.84	4.96	6.08	7.22	8.40						dnf	8 / --			
	reaction time	0.153	interval	1.15	1.12	1.12	1.14	1.18							3.39		
	velocity	5.10	7.95	8.16	8.16	8.02	7.75								8.09		

Heat 3

date 13-Apr-19 wind 0.0 m/s

CAA Hurdle Development (2019)

		H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
Lu Yang (CHN) (1996)	time	2.68	3.78	4.84	5.92	6.98	8.03	9.11	10.22	11.34	12.52		14.07	9 / 1			
	reaction time	0.160	interval	1.10	1.06	1.08	1.06	1.05	1.08	1.11	1.12	1.18	1.55		3.24	3.19	3.41
	velocity	5.12	8.31	8.62	8.46	8.62	8.70	8.46	8.23	8.16	7.75	9.05	7.82		8.46	8.60	8.04

Ning Xiaohan (CHN) (2000)	time	2.68	3.78	4.86	5.94	7.02	8.10	9.22	10.32	11.45	12.60		14.15	5 / 2				
	reaction time	0.145														3.26	3.28	3.38
	interval		1.10	1.08	1.08	1.08	1.08	1.12	1.10	1.13	1.15	1.55				8.41	8.36	8.11
	velocity	5.12	8.31	8.46	8.46	8.46	8.46	8.16	8.31	8.09	7.95	9.05	7.77					
Zhang Tao (CHN) (1997)	time	2.72	3.82	4.90	5.98	7.06	8.16	9.26	10.40	11.52	12.68		14.16	8 / 3				
	reaction time	0.198														3.26	3.28	3.42
	interval		1.10	1.08	1.08	1.08	1.10	1.10	1.14	1.12	1.16	1.48				8.41	8.36	8.02
	velocity	5.04	8.31	8.46	8.46	8.46	8.31	8.31	8.02	8.16	7.88	9.47	7.77					

Heat 2

date 13-Apr-19		wind 0.1 m/s												CAA Hurdle Development (2019)				
Zeng Jianhang (CHN) (1998)	time	2.62	3.67	4.73	5.78	6.82	7.88	8.96	10.06	11.20	12.39		13.98	5 / 1				
	reaction time	0.176														3.16	3.18	3.43
	interval		1.05	1.06	1.05	1.04	1.06	1.08	1.10	1.14	1.19	1.59				8.68	8.62	7.99
Sun Zhenjiang (CHN) (1999)	time	2.64	3.72	4.79	5.86	6.94	8.04	9.18	10.31	11.45	12.68		14.33	3 / 2				
	reaction time	0.169														3.22	3.32	3.50
	interval		7 steps	1.08	1.07	1.07	1.08	1.10	1.14	1.13	1.14	1.23	1.65			8.52	8.26	7.83
Li Jiming (CHN) (1996)	time	2.70	3.80	4.90	6.00	7.10	8.22	9.34	10.47	11.64	12.83		14.48	1 / 5				
	reaction time	0.179														3.30	3.34	3.49
	interval		1.10	1.10	1.10	1.10	1.12	1.12	1.13	1.17	1.19	1.65				8.31	8.21	7.86
	velocity		5.08	8.31	8.31	8.31	8.31	8.16	8.16	8.09	7.81	7.68	8.50	7.60				

Heat 1

date 13-Apr-19		wind 0.0 m/s												CAA Hurdle Development (2019)				
Sun Haoru (CHN) (1998)	time	2.67	3.79	4.90	5.97	7.07	8.19	9.31	10.46	11.61	12.78		14.35	9 / 1				
	reaction time	0.185														3.30	3.34	3.47
	interval		1.12	1.11	1.07	1.10	1.12	1.12	1.15	1.15	1.17	1.57				8.31	8.21	7.90
Lu Wei (CHN) (2000)	time	2.77	3.90	5.02	6.11	7.22	8.36	9.50	10.67	11.82	12.98		14.52	7 / 2				
	reaction time	0.170														3.34	3.39	3.48
	interval		1.13	1.12	1.09	1.11	1.14	1.14	1.17	1.15	1.16	1.54	PB			8.21	8.09	7.88
	velocity		4.95	8.09	8.16	8.39	8.23	8.02	8.02	7.81	7.95	7.88	9.10	7.58				

2019 Chinese National Grand Prix 1 (Zhaoqing, CHN)

FINAL

date 09-Apr-19		wind 0.2 m/s												CAA Hurdle Development (2019)				
Sun Zhenjiang (CHN) (1999)	time	2.62	3.70	4.78	5.82	6.88	7.94	9.02	10.10	11.22	12.32		13.83	4 / 1				
	reaction time	0.177														3.20	3.20	3.30
	interval		7 steps	1.08	1.08	1.04	1.06	1.06	1.08	1.08	1.12	1.10	1.51	PB		8.57	8.57	8.31
Zeng Jianhang (CHN) (1998)	time	2.58	3.65	4.70	5.73	6.77	7.80	8.96	10.22	11.35	12.51		14.00	9 / 2				
	reaction time	0.180														3.15	3.23	3.55
	interval		1.07	1.05	1.03	1.04	1.03	1.16	1.26	1.13	1.16	1.49				8.70	8.49	7.72
Zhang Tao (CHN) (1997)	time	2.70	3.81	4.90	5.99	7.07	8.16	9.26	10.34	11.44	12.55		14.03	5 / 3				
	reaction time	0.195														3.29	3.27	3.29
	interval		1.11	1.09	1.09	1.08	1.09	1.10	1.08	1.10	1.11	1.48				8.33	8.39	8.33
Ning Xiaohan (CHN) (2000)	time	2.64	3.71	4.81	5.89	6.98	8.05	9.16	10.26	11.38	12.52		14.07	7 / 4				
	reaction time	0.165														3.25	3.27	3.36
	interval		1.07	1.10	1.08	1.09	1.07	1.11	1.10	1.12	1.14	1.55	PB			8.44	8.39	8.16
Lu Yang (CHN) (1996)	time	2.78	3.90	4.98	6.06	7.13	8.20	9.30	10.37	11.48	12.60		14.09	6 / 5				
	reaction time	0.192														3.28	3.24	3.30
	interval		1.12	1.08	1.08	1.07	1.07	1.10	1.07	1.11	1.12	1.49				8.36	8.46	8.31
Mui Ching Yeung (HKG) (1993)	time	2.68	3.79	4.88	6.00	7.10	8.20	9.27	10.41	11.52	12.66		14.11	1 / --				
	reaction time	0.184														3.32	3.27	3.39
	interval		1.11	1.09	1.12	1.10	1.10	1.07	1.14	1.11	1.14	1.45				8.26	8.39	8.09
Yang Yuxuan (CHN) (2002)	time	2.76	3.88	4.98	6.08	7.18	8.26	9.38	10.52	11.66	12.82		14.36	3 / 6				
	reaction time	0.180														3.32	3.30	3.44
	interval		1.12	1.10	1.10	1.10	1.08	1.12	1.14	1.14	1.16	1.54				8.26	8.31	7.97
Jiang Fan (CHN) (1989)	time	2.76	3.88	4.97	6.11	7.21	8.34	9.48	10.63	11.80	12.98		14.65	8 / 7				
	reaction time	0.209														3.35	3.37	3.50
	interval		1.12	1.09	1.14	1.10	1.13	1.14	1.15	1.17	1.18	1.67				8.19	8.14	7.83
Ma Lei (CHN) (1989)	time	2.94	4.10	5.22	6.34	7.46	8.59	9.71	10.87	12.04	13.24		14.87	2 / 8				
	reaction time	0.328														3.40	3.37	3.53
	interval		7 steps	1.16	1.12	1.12	1.12	1.13	1.12	1.16	1.17	1.20	1.63			8.06	8.14	7.77
	velocity		4.67	7.88	8.16	8.16	8.16	8.09	8.16	7.88	7.81	7.62	8.60	7.40				

Heat 5

date 08-Apr-19		wind 0.6 m/s												CAA Hurdle Development (2019)				
Lu Yang (CHN) (1996)	time	2.74	3.84	4.92	5.98	7.06	8.15	9.24	10.31	11.44	12.58		14.11	2 / 1				
	reaction time	0.201														3.24	3.26	3.34
	interval		1.10	1.08	1.06	1.08	1.09	1.09	1.07	1.13	1.14	1.53				8.46	8.41	8.21
	velocity		5.01	8.31	8.46	8.62	8.46	8.39	8.39	8.54	8.09	8.02	9.16	7.80				

Ma Lei (CHN) (1989)	time	2.75	3.89	5.00	6.09	7.18	8.30	9.42	10.54	11.68	12.82		14.34	8 / 2				
	reaction time	0.159	interval	7 steps	1.14	1.11	1.09	1.09	1.12	1.12	1.12	1.14	1.14	1.52		3.34	3.33	3.40
	velocity	4.99	8.02	8.23	8.39	8.39	8.16	8.16	8.16	8.02	8.02	9.22	7.67		8.21	8.23	8.06	

Heat 4

date 08-Apr-19 wind -0.8 m/s CAA Hurdle Development (2019)

		H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
Ning Xiaohan (CHN) (2000)	time	2.62	3.72	4.80	5.88	6.95	8.05	9.14	10.26	11.40	12.58		14.17	4 / 1			
	reaction time	0.158	interval	1.10	1.08	1.07	1.10	1.09	1.12	1.14	1.18	1.59	PB		3.26	3.26	3.44
	velocity	5.24	8.31	8.46	8.46	8.54	8.31	8.39	8.16	8.02	7.75	8.82	7.76		8.41	8.41	7.97
Li Jiming (CHN) (1996)	time	2.73	3.85	4.96	6.06	7.14	8.26	9.37	10.52	11.68	12.87		14.46	6 / 2			
	reaction time	0.190	interval	1.12	1.11	1.10	1.08	1.12	1.11	1.15	1.16	1.19	1.59		3.33	3.31	3.50
	velocity	5.03	8.16	8.23	8.31	8.46	8.16	8.23	7.95	7.88	7.68	8.82	7.61		8.23	8.28	7.83

Heat 3

date 08-Apr-19 wind 0.0 m/s CAA Hurdle Development (2019)

		H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
Sun Zhenjiang (CHN) (1999)	time	2.63	3.72	4.78	5.83	6.88	7.97	9.06	10.16	11.30	12.43		14.00	6 / 1			
	reaction time	0.177	interval	7 steps	1.09	1.06	1.05	1.09	1.10	1.14	1.13	1.57	PB		3.20	3.23	3.37
	velocity	5.22	8.39	8.62	8.70	8.70	8.39	8.39	8.31	8.02	8.09	8.93	7.86		8.57	8.49	8.14
Yang Yuxuan (CHN) (2002)	time	2.76	3.88	4.99	6.13	7.25	8.36	9.46	10.58	11.70	12.83		14.34	8 / 2			
	reaction time	0.165	interval	1.12	1.11	1.14	1.12	1.11	1.10	1.12	1.12	1.13	1.51	PB	3.37	3.33	3.37
	velocity	4.97	8.16	8.23	8.02	8.16	8.23	8.31	8.16	8.16	8.09	9.28	7.67		8.14	8.23	8.14

Heat 2

date 08-Apr-19 wind -0.6 m/s CAA Hurdle Development (2019)

		H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
Zhang Tao (CHN) (1997)	time	2.72	3.82	4.90	5.98	7.06	8.14	9.26	10.34	11.44	12.58		14.05	7 / 1			
	reaction time	0.200	interval	1.10	1.08	1.08	1.08	1.12	1.08	1.10	1.14	1.47			3.26	3.28	3.32
	velocity	5.04	8.31	8.46	8.46	8.46	8.46	8.16	8.46	8.31	8.02	9.54	7.83		8.41	8.36	8.26
Mui Ching Yeung (HKG) (1993)	time	2.68	3.77	4.85	5.91	7.01	8.09	9.21	10.31	11.47	12.62		14.09	6 / --			
	reaction time	0.172	interval	1.09	1.08	1.06	1.10	1.08	1.12	1.10	1.16	1.15	1.47		3.23	3.30	3.41
	velocity	5.12	8.39	8.46	8.62	8.31	8.46	8.16	8.31	7.88	7.95	9.54	7.81		8.49	8.31	8.04
Jiang Fan (CHN) (1989)	time	2.71	3.85	4.95	6.05	7.15	8.26	9.38	10.50	11.65	12.79		14.33	5 / 2			
	reaction time	0.176	interval	1.14	1.10	1.10	1.11	1.12	1.12	1.15	1.14	1.54			3.34	3.33	3.41
	velocity	5.06	8.02	8.31	8.31	8.31	8.23	8.16	8.16	7.95	8.02	9.10	7.68		8.21	8.23	8.04

Heat 1

date 08-Apr-19 wind -1.5 m/s CAA Hurdle Development (2019)

		H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
Zeng Jianhang (CHN) (1998)	time	2.69	3.81	4.87	5.96	7.08	8.20	9.33	10.44	11.58	12.73		14.31	2 / 1			
	reaction time	0.290	interval	1.12	1.06	1.09	1.12	1.12	1.13	1.11	1.14	1.15	1.58		3.27	3.37	3.40
	velocity	5.10	8.16	8.62	8.39	8.16	8.16	8.09	8.23	8.02	7.95	8.87	7.69		8.39	8.14	8.06
Zhao Pengchuan (CHN) (1993)	time	2.78	3.89	4.98	6.06	7.18	8.26	9.38	10.48	11.60	12.75		14.38	3 / 2			
	reaction time	0.288	interval	1.11	1.09	1.08	1.12	1.08	1.12	1.10	1.12	1.15	1.63		3.28	3.32	3.37
	velocity	4.94	8.23	8.39	8.46	8.16	8.46	8.16	8.31	8.16	7.95	8.60	7.65		8.36	8.26	8.14

2019 Chinese Indoor National Grand Prix Final (Hangzhou, CHN)**A-FINAL**

date 20-Mar-19 CAA Hurdle Development (2019)

		H1	H2	H3	H4	H5	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
Zeng Jianhang (CHN) (1998)	time	2.55	3.62	4.68	5.69	6.74		7.62	3 / 1			
	reaction time		interval	1.07	1.06	1.01	0.88	PB		3.14		
	velocity	5.38	8.54	8.62	9.05	8.70	11.05	7.87		8.73		
Sun Zhenjiang (CHN) (1999)	time	2.61	3.69	4.73	5.77	6.86		7.79	5 / 2			
	reaction time		interval	7 steps	1.08	1.04	0.93			3.16		
	velocity	5.26	8.46	8.79	8.79	8.39	10.45	7.70		8.68		
Ning Xiaohan (CHN) (2000)	time	2.64	3.71	4.79	5.82	6.91		7.85	4 / 3			
	reaction time		interval	1.07	1.08	1.03	0.94			3.18		
	velocity	5.20	8.54	8.46	8.87	8.39	10.34	7.64		8.62		
Zhao Pengchuan (CHN) (1993)	time	2.64	3.75	4.84	5.92	7.02		8.00	2 / 4			
	reaction time		interval	1.11	1.09	1.08	0.98			3.28		
	velocity	5.20	8.23	8.39	8.46	8.31	9.92	7.50		8.36		

B-FINAL

date 20-Mar-19 CAA Hurdle Development (2019)

		H1	H2	H3	H4	H5	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
Li Jiming (CHN) (1996)	time	2.68	3.80	4.88	5.97	7.04		7.96	3 / 1			
	reaction time		interval	1.12	1.08	1.09	0.92			3.29		
	velocity	5.12	8.16	8.46	8.39	8.54	10.57	7.54		8.33		
Zhu Shenglong (CHN) (2000)	time	2.71	3.79	4.87	5.97	7.04		7.97	2 / 2			
	reaction time		interval	1.08	1.08	1.10	0.93			3.26		

	velocity	5.06	8.46	8.46	8.31	8.54	10.45	7.53	8.41
Pan Zijie (CHN) (1995)	time	2.66	3.76	4.89	5.96	7.05		7.97	4 / 3
reaction time	interval		1.10	1.13	1.07	1.09	0.92		3.30
	velocity	5.16	8.31	8.09	8.54	8.39	10.57	7.53	8.31

2019 NCAA National Indoor Championships (Birmingham, AL) (TV Analysis)

FINAL

date 9-Mar-19

Henson (2020) - Athlete First: 2019 year end hurdle report

		H1	H2	H3	H4	H5	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
Holloway, Grant (USA) (1997)	time	2.51	3.52	4.53	5.51	6.48		7.35	4 / 1			
reaction time	interval	7 steps	1.01	1.01	0.98	0.97	0.87	AR PB		3.00		
	velocity	5.47	9.05	9.05	9.33	9.42	11.17	8.16		9.14		
Roberts, Daniel (USA) (1998)	time	2.53	3.54	4.54	5.54	6.57		7.41	3 / 2			
reaction time	interval	0.181	1.01	1.00	1.00	1.03	0.84	PB		3.01		
	velocity	5.42	9.05	9.14	9.14	8.87	11.57	8.10		9.11		
Cunningham, Trey (USA) (1998)	time	2.56	3.61	4.64	5.71	6.73		7.64	5 / 3			
reaction time	interval	0.179	7 steps	1.05	1.03	1.07	0.91			3.15		
	velocity	5.36	8.70	8.87	8.54	8.96	10.68	7.85		8.70		

2019 Chinese Indoor National Grand Prix 4 (Xi'an, CHN)

A-FINAL

date 4-Mar-19

CAA Hurdle Development (2019)

		H1	H2	H3	H4	H5	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
Zhao Pengchuan (CHN) (1993)	time	2.62	3.72	4.80	5.86	6.90		7.82	3 / 1			
reaction time	interval		1.10	1.08	1.06	1.04	0.92			3.24		
	velocity	5.24	8.31	8.46	8.62	8.79	10.57	7.67		8.46		
Ning Xiaohan (CHN) (2000)	time	2.64	3.72	4.78	5.82	6.88		7.82	4 / 2			
reaction time	interval		1.08	1.06	1.04	1.06	0.94	PB		3.18		
	velocity	5.20	8.46	8.62	8.79	8.62	10.34	7.67		8.62		
Zhang Honglin (CHN) (1994)	time	2.67	3.75	4.83	5.91	7.01		7.93	5 / 3			
reaction time	interval		1.08	1.08	1.08	1.10	0.92			3.24		
	velocity	5.14	8.46	8.46	8.46	8.31	10.57	7.57		8.46		
Zhu Shenglong (CHN) (2000)	time	2.68	3.76	4.83	5.90	6.99		7.96	2 / 4			
reaction time	interval		1.08	1.07	1.07	1.09	0.97	PB		3.22		
	velocity	5.12	8.46	8.54	8.54	8.39	10.02	7.54		8.52		

B-FINAL

date 4-Mar-19

CAA Hurdle Development (2019)

		H1	H2	H3	H4	H5	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
Sun Haoru (CHN) (1998)	time	2.65	3.75	4.85	5.93	7.02		7.97	2 / 1			
reaction time	interval		1.10	1.10	1.08	1.09	0.95	PB		3.28		
	velocity	5.18	8.31	8.31	8.46	8.39	10.23	7.53		8.36		
Jiang Wei (CHN) (1994)	time	2.69	3.79	4.87	5.93	7.01		7.98	3 / 2			
reaction time	interval	7 steps	1.10	1.08	1.06	1.08	0.97			3.24		
	velocity	5.10	8.31	8.46	8.62	8.46	10.02	7.52		8.46		
Wu Zhenhua (CHN) (1998)	time	2.66	3.80	4.91	6.03	7.11		8.07	5 / 3			
reaction time	interval		1.14	1.11	1.12	1.08	0.96	PB		3.37		
	velocity	5.16	8.02	8.23	8.16	8.46	10.13	7.43		8.14		
Zhang Shuai (CHN) (1996)	time	2.70	3.79	4.96	6.08	7.21		8.26	4 / 4			
reaction time	interval		1.09	1.17	1.12	1.13	1.05			3.38		
	velocity	5.08	8.39	7.81	8.16	8.09	9.26	7.26		8.11		

2019 Chinese Indoor National Grand Prix 3 (Xi'an, CHN)

A-FINAL

date 28-Feb-19

CAA Hurdle Development (2019)

		H1	H2	H3	H4	H5	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
Sun Zhenjiang (CHN) (1999)	time	2.58	3.63	4.69	5.75	6.81		7.80	3 / 1			
reaction time	interval	7 steps	1.05	1.06	1.06	1.06	0.99			3.17		
	velocity	5.32	8.70	8.62	8.62	8.62	9.82	7.69		8.65		
Zhao Pengchuan (CHN) (1993)	time	2.61	3.71	4.77	5.85	6.91		7.84	4 / 2			
reaction time	interval		1.10	1.06	1.08	1.06	0.93			3.24		
	velocity	5.26	8.31	8.62	8.46	8.62	10.45	7.65		8.46		
Lei Yiwen (CHN) (1993)	time	2.64	3.74	4.86	5.96	7.08		8.06	2 / 3			
reaction time	interval		1.10	1.12	1.10	1.12	0.98			3.32		
	velocity	5.20	8.31	8.16	8.31	8.16	9.92	7.44		8.26		
Si Zhengdong (CHN) (1997)	time	2.62						DQ	5 / --			
reaction time	interval											
	velocity	5.24										

B-FINAL

date 28-Feb-19

CAA Hurdle Development (2019)

		H1	H2	H3	H4	H5	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
Zhang Honglin (CHN) (1994)	time	2.69	3.81	4.89	5.96	7.06		7.94	3 / 1			
	reaction time		1.12	1.08	1.07	1.10	0.88			3.27		
	velocity	5.10	8.16	8.46	8.54	8.31	11.05	7.56		8.39		
Zhu Shenglong (CHN) (2000)	time	2.72	3.80	4.88	5.94	7.04		7.98	4 / 2			
	reaction time		1.08	1.08	1.06	1.10	0.94	PB		3.22		
	velocity	5.04	8.46	8.46	8.62	8.31	10.34	7.52		8.52		
Sun Haoru (CHN) (1998)	time	2.70	3.80	4.90	5.96	7.06		8.00	5 / 3			
	reaction time		1.10	1.10	1.06	1.10	0.94	PB		3.26		
	velocity	5.08	8.31	8.31	8.62	8.31	10.34	7.50		8.41		
Yang Quanlei (CHN) (1999)	time	2.67	3.81	4.91	6.01	7.15		8.10	2 / 4			
	reaction time		1.14	1.10	1.10	1.14	0.95			3.34		
	velocity	5.14	8.02	8.31	8.31	8.02	10.23	7.41		8.21		

2019 Chinese Indoor National Grand Prix 2 (Nanjing, CHN)**A-FINAL**

date 24-Feb-19

CAA Hurdle Development (2019)

		H1	H2	H3	H4	H5	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
Zeng Jianhang (CHN) (1998)	time	2.60	3.69	4.73	5.77	6.81		7.70	3 / 1			
	reaction time		1.09	1.04	1.04	1.04	0.89			3.17		
	velocity	5.28	8.39	8.79	8.79	8.79	10.92	7.79		8.65		
Pan Zijie (CHN) (1995)	time	2.60	3.75	4.85	5.94	7.04		7.97	4 / 2			
	reaction time		1.15	1.10	1.09	1.10	0.93			3.34		
	velocity	5.28	7.95	8.31	8.39	8.31	10.45	7.53		8.21		
Jiang Fan (CHN) (1989)	time	2.68	3.80	4.89	5.99	7.09		8.03	5 / 3			
	reaction time		1.12	1.09	1.10	1.10	0.94			3.31		
	velocity	5.12	8.16	8.39	8.31	8.31	10.34	7.47		8.28		
Yang Jianxing (CHN) (1996)	time	2.68	3.82	4.94	6.04	7.16		8.10	2 / 4	0.00		
	reaction time		1.14	1.12	1.10	1.12	0.94			3.36		
	velocity	5.12	8.02	8.16	8.31	8.16	10.34	7.41		8.16		

B-FINAL

date 24-Feb-19

CAA Hurdle Development (2019)

		H1	H2	H3	H4	H5	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
Guo Zhongjie (CHN) (2001)	time	2.70	3.82	4.90	5.97	7.10		8.04	4 / 1			
	reaction time		1.12	1.08	1.07	1.13	0.94			3.27		
	velocity	5.08	8.16	8.46	8.54	8.09	10.34	7.46		8.39		
Yang Yuxuan (CHN) (2002)	time	2.73	3.85	4.99	6.09	7.20		8.12	5 / 2			
	reaction time		1.12	1.14	1.10	1.11	0.92	PB		3.36		
	velocity	5.03	8.16	8.02	8.31	8.23	10.57	7.39		8.16		
Li Huaihai (CHN) (1999)	time	2.67	3.80	4.89	6.03	7.15		8.15	2 / 3			
	reaction time		1.13	1.09	1.14	1.12	1.00	PB		3.36		
	velocity	5.14	8.09	8.39	8.02	8.16	9.72	7.36		8.16		
Lu Wei (CHN) (2000)	time	2.77	3.89	4.99	6.09	7.19		8.17	3 / 4			
	reaction time		1.12	1.10	1.10	1.10	0.98			3.32		
	velocity	4.95	8.16	8.31	8.31	8.31	9.92	7.34		8.26		

2019 South Eastern Conference Indoor Championships (Fayetteville, AR) (TV Analysis)**FINAL**

date 23-Feb-19

Henson (2020) - Athlete First: 2019 year end hurdle report

		H1	H2	H3	H4	H5	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
Holloway, Grant (USA) (1997)	time	2.55	3.57	4.57	5.56	6.54		7.44	4 / 1			
	reaction time		7 steps	1.02	1.00	0.99	0.90			3.01		
	velocity	5.38	8.96	9.14	9.23	9.33	10.80	8.06		9.11		
Roberts, Daniel (USA) (1998)	time	2.55	3.60	4.61	5.60	6.62		7.48	5 / 2			
	reaction time		1.05	1.01	0.99	1.02	0.86			3.05		
	velocity	5.38	8.70	9.05	9.23	8.96	11.30	8.02		8.99		

2019 PSD Bank Meeting (Düsseldorf, GER) (TV Analysis)**FINAL**

date 20-Feb-19

Henson (2020) - Athlete First: 2019 year end hurdle report

		H1	H2	H3	H4	H5	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
Ortega, Orlando (ESP) (1991)	time	2.43	3.46	4.48	5.50	6.48		7.52	4 / 1			
	reaction time	0.120	7 steps	1.03	1.02	0.98	1.04			3.07		
	velocity	5.65	8.87	8.96	8.96	9.33	9.35	7.98		8.93		
Trajkovic, Milan (CYP) (1992)	time	2.40	3.42	4.44	5.44	6.44		7.52	7 / 2			
	reaction time	0.153	7 steps	1.02	1.00	1.00	1.08			3.04		
	velocity											

	velocity	5.72	8.96	8.96	9.14	9.14	9.00	7.98	9.02
Crittenden, Freddie (USA) (1999)	time	2.42	3.44	4.48	5.52	6.54		7.60	5 / 3
	reaction time	0.156	interval	1.02	1.04	1.04	1.06		3.10
	velocity	5.67	8.96	8.79	8.79	8.96	9.17	7.89	8.85
Mallett, Aaron (USA) (1994)	time	2.40	3.46	4.50	5.52	6.56		7.66	6 / 4
	reaction time	0.166	interval	7 steps 1.06	1.04	1.02	1.10		3.12
	velocity	5.72	8.62	8.79	8.96	8.79	8.84	7.83	8.79
Pozzi, Andy (GBR) (1992)	time	2.43	3.46	4.52	5.52	6.56		7.67	3 / 5
	reaction time	0.106	interval	7 steps 1.03	1.06	1.00	1.11		3.09
	velocity	5.65	8.87	8.62	9.14	8.79	8.76	7.82	8.87
Balnuweit, Erik (GER) (1988)	time	2.42	3.46	4.52	5.52	6.56		7.67	1 / 6
	reaction time	0.148	interval	1.04	1.06	1.00	1.11		3.10
	velocity	5.67	8.79	8.62	9.14	8.79	8.76	7.82	8.85

2019 Chinese Indoor National Grand Prix 1 (Nanjing, CHN)

A-FINAL

date 20-Feb-19

CAA Hurdle Development (2019)

		H1	H2	H3	H4	H5	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
Jiang Fan (CHN) (1989)	time	2.67	3.79	4.85	5.94	7.02		7.94	5 / 1			
	reaction time	interval	1.12	1.06	1.09	1.08	0.92			3.27		
	velocity	5.14	8.16	8.62	8.39	8.46	10.57	7.56		8.39		
Li Jiming (CHN) (1996)	time	2.65	3.79	4.83	5.94	7.02		7.95	4 / 2			
	reaction time	interval	1.14	1.04	1.11	1.08	0.93			3.29		
	velocity	5.18	8.02	8.79	8.23	8.46	10.45	7.55		8.33		
Guo Zhongjie (CHN) (2001)	time	2.73	3.82	4.90	6.00	7.09		8.03	2 / 3			
	reaction time	interval	1.09	1.08	1.10	1.09	0.94	PB		3.27		
	velocity	5.03	8.39	8.46	8.31	8.39	10.34	7.47		8.39		
Zhang Tao (CHN) (1997)	time	2.71	3.82	4.92	6.00	7.14		8.12	3 / 4			
	reaction time	interval	1.11	1.10	1.08	1.14	0.98			3.29		
	velocity	5.06	8.23	8.31	8.46	8.02	9.92	7.39		8.33		

B-FINAL

date 20-Feb-19

CAA Hurdle Development (2019)

		H1	H2	H3	H4	H5	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
Zeng Jianhang (CHN) (1998)	time	2.60	3.68	4.74	5.79	6.84		7.75	4 / 1			
	reaction time	interval	1.08	1.06	1.05	1.05	0.91			3.19		
	velocity	5.28	8.46	8.62	8.70	8.70	10.68	7.74		8.60		
Pan Zijie (CHN) (1995)	time	2.68	3.78	4.89	5.95	7.04		7.96	3 / 2			
	reaction time	interval	1.10	1.11	1.06	1.09	0.92			3.27		
	velocity	5.12	8.31	8.23	8.62	8.39	10.57	7.54		8.39		
Lu Yang (CHN) (1996)	time	2.73	3.85	4.95	6.02	7.12		8.04	5 / 3			
	reaction time	interval	1.12	1.10	1.07	1.10	0.92			3.29		
	velocity	5.03	8.16	8.31	8.54	8.31	10.57	7.46		8.33		
Yang Yuxuan (CHN) (2002)	time	2.75	3.85	4.95	6.02	7.12		8.19	2 / 4			
	reaction time	interval	1.10	1.10	1.07	1.10	1.07			3.27		
	velocity	4.99	8.31	8.31	8.54	8.31	9.08	7.33		8.39		

2019 Müller Indoor Grand Prix (Birmingham, GBR) (TV Analysis)

FINAL

date 16-Feb-19

Henson (2020) - Athlete First: 2019 year end hurdle report

		H1	H2	H3	H4	H5	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
Eaton, Jarret (USA) (1989)	time	2.50	3.54	4.54	5.54	6.56		7.51	4 / 1			
	reaction time	0.136	interval	7 steps 1.04	1.00	1.00	0.95			3.04		
	velocity	5.49	8.79	9.14	9.14	8.96	10.23	7.99		9.02		
Crittenden, Freddie (USA) (1999)	time	2.50	3.52	4.55	5.55	6.56		7.53	8 / 2			
	reaction time	0.146	interval	1.02	1.03	1.00	0.97			3.05		
	velocity	5.49	8.96	8.87	9.14	9.05	10.02	7.97		8.99		
Trajkovic, Milan (CYP) (1992)	time	2.52	3.55	4.55	5.55	6.56		7.54	6 / 3			
	reaction time	0.166	interval	7 steps 1.03	1.00	1.00	0.98			3.03		
	velocity	5.44	8.87	9.14	9.14	9.05	9.92	7.96		9.05		
Constantino, Gabriel (BRA) (1991)	time	2.54	3.54	4.58	5.62	6.66		7.62	5 / 4			
	reaction time	0.136	interval	7 steps 1.00	1.04	1.04	0.96			3.08		
	velocity	5.40	9.14	8.79	8.79	8.79	10.13	7.87		8.90		
Mallett, Aaron (USA) (1994)	time	2.54	3.58	4.65	5.65	6.68		7.63	7 / 5			
	reaction time	0.152	interval	7 steps 1.04	1.07	1.00	0.95			3.11		

velocity	5.40	8.79	8.54	9.14	8.87	10.23	7.86	8.82
King, David (GBR) (1994) time	2.54	3.58	4.65	5.68	6.72		7.68	3 / 6
reaction time 0.130 interval	7 steps	1.04	1.07	1.03	1.04	0.96		3.14
velocity	5.40	8.79	8.54	8.87	8.79	10.13	7.81	8.73

2019 French National Indoor Championships (Miramas, FRA) (TV Analysis)

FINAL

date 16-Feb-19

Henson (2020) - Athlete First: 2019 year end hurdle report

		H1	H2	H3	H4	H5	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
Martinot-Lagarde, Pascal (FRA) time		2.48	3.52	4.54	5.54	6.54		7.52	5 / 1			
reaction time 0.150 interval	7 steps	1.04	1.02	1.00	1.00		0.98			3.06		
velocity		5.53	8.79	8.96	9.14	9.14	9.92	7.98		8.96		
Manga, Aurel (FRA) (1992) time		2.52	3.58	4.58	5.58	6.58		7.57	4 / 2			
reaction time 0.155 interval	7 steps	1.06	1.00	1.00	1.00		0.99			3.06		
velocity		5.44	8.62	9.14	9.14	9.14	9.82	7.93		8.96		
Belocian, Wilhem (FRA) (1995) time		2.56	3.58	4.62	5.64	6.66		7.63	3 / 3			
reaction time 0.147 interval			1.02	1.04	1.02	1.02	0.97			3.08		
velocity		5.36	8.96	8.79	8.96	8.96	10.02	7.86		8.90		

2019 Tyson Indoor Invitational (Fayetteville, AR) (TV Analysis)

FINAL

date 8-Feb-19

Henson (2020) - Athlete First: 2019 year end hurdle report

		H1	H2	H3	H4	H5	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
Holloway, Grant (USA) (1997) time		2.43	3.44	4.43	5.42	6.44		7.43	4 / 1			
reaction time interval	7 steps	1.01	0.99	0.99	1.02		0.99			2.99		
velocity		5.65	9.05	9.23	9.23	8.96	9.82	8.08		9.17		

2018 Japanese U20 National Championships (Nagoya, JPN) (99cm)

FINAL

date 19-Oct-18

wind -0.6 m/s

Shibayama (2021) race analysis of men's 110m hurdles in the 2021 season

		H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
Izumiya, Shunsuke (JPN) (2000) time		2.53	3.57	4.61	5.64	6.63	7.66	8.68	9.71	10.75	11.81		13.19	6 / 1			
reaction time 0.123 interval			1.04	1.04	1.03	0.99	1.03	1.02	1.03	1.04	1.06	1.38	NJR PB		3.11	3.04	3.13
velocity		5.42	8.79	8.79	8.87	9.23	8.87	8.96	8.87	8.79	8.62	10.16	8.34		8.82	9.02	8.76

2018 National Sports Festival (Fukui, JPN)

A FINAL

date 08-Oct-18

wind 1.2 m/s

Shibayama (2018) - Race analysis of men's 110m hurdle run in 2018 season

		H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
Kanai, Taio (JPN) (1995) time		2.56	3.63	4.66	5.70	6.73	7.78	8.83	9.88	10.95	12.03		13.46	6 / 1			
reaction time 0.121 interval			1.07	1.03	1.04	1.03	1.05	1.05	1.05	1.07	1.08	1.43			3.14	3.13	3.20
velocity		5.36	8.54	8.87	8.79	8.87	8.70	8.70	8.70	8.54	8.46	9.80	8.17		8.73	8.76	8.57
Masuno, Genta (JPN) (1993) time		2.60	3.65	4.67	5.70	6.74	7.78	8.84	9.91	10.98	12.07		13.50	5 / 2			
reaction time 0.143 interval			1.05	1.02	1.03	1.04	1.04	1.06	1.07	1.07	1.09	1.43			3.10	3.14	3.23
velocity		5.28	8.70	8.96	8.87	8.79	8.79	8.62	8.54	8.54	8.39	9.80	8.15		8.85	8.73	8.49
Ogata, Akihiro (JPN) (1993) time		2.60	3.66	4.72	5.75	6.80	7.87	8.93	10.00	11.08	12.18		13.65	4 / 3			
reaction time 0.161 interval			1.06	1.06	1.03	1.05	1.07	1.06	1.07	1.08	1.10	1.47	=PB		3.15	3.18	3.25
velocity		5.28	8.62	8.62	8.87	8.70	8.54	8.62	8.54	8.46	8.31	9.54	8.06		8.70	8.62	8.44
Ishikawa, Shuhei (JPN) (1995) time		2.59	3.67	4.74	5.80	6.87	7.93	8.99	10.07	11.15	12.27		13.74	3 / 4			
reaction time 0.139 interval			1.08	1.07	1.06	1.07	1.06	1.06	1.08	1.08	1.12	1.47			3.21	3.19	3.28
velocity		5.30	8.46	8.54	8.62	8.54	8.62	8.62	8.46	8.46	8.16	9.54	8.01		8.54	8.60	8.36
Kuriki, Anthony Tyrell (JPN) (1991) time		2.64	3.70	4.78	5.87	6.92	8.00	9.06	10.15	11.23	12.30		13.77	8 / 5			
reaction time 0.129 interval			1.06	1.08	1.09	1.05	1.08	1.06	1.09	1.08	1.07	1.47			3.23	3.19	3.24
velocity		5.20	8.62	8.46	8.39	8.70	8.46	8.62	8.39	8.46	8.54	9.54	7.99		8.49	8.60	8.46
Fujii, Ryota (JPN) (1996) time		2.61	3.70	4.75	5.82	6.89	7.98	9.05	10.14	11.24	12.33		13.81	1 / 6			
reaction time 0.142 interval			1.09	1.05	1.07	1.07	1.09	1.07	1.09	1.10	1.09	1.48	PB		3.21	3.23	3.28
velocity		5.26	8.39	8.70	8.54	8.54	8.39	8.54	8.39	8.31	8.39	9.47	7.97		8.54	8.49	8.36
Furukawa, Yutaro (JPN) (1985) time		2.63	3.70	4.76	5.84	6.90	7.97	9.05	10.15	11.24	12.36		13.86	2 / 7			
reaction time 0.145 interval			1.07	1.06	1.08	1.06	1.07	1.08	1.10	1.09	1.12	1.50			3.21	3.21	3.31
velocity		5.22	8.54	8.62	8.46	8.62	8.54	8.46	8.31	8.39	8.16	9.35	7.94		8.54	8.54	8.28
Kagimoto, Masahiro (JPN) (1991) time		2.66	3.79	4.92	5.99	7.09	8.20	9.29	10.39	11.50	12.62		14.09	7 / 8			
reaction time 0.144 interval			1.13	1.13	1.07	1.10	1.11	1.09	1.10	1.11	1.12	1.47			3.33	3.30	3.33
velocity		5.16	8.09	8.09	8.54	8.31	8.23	8.39	8.31	8.23	8.16	9.54	7.81		8.23	8.31	8.23

U20 FINAL (99cm)

date 07-Oct-18

wind 1.5 m/s

Joshi (2018) - research on athlete performance and technique- 2018 data book

		H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
Tawada, Asahi (JPN) (2001) time		2.59	3.66	4.69	5.73	6.75	7.78	8.80	9.83	10.86	11.91		13.31	3 / 1			
reaction time 0.122 interval			1.07	1.03	1.04	1.02	1.03	1.02	1.03	1.03	1.05	1.40			3.14	3.07	3.11
velocity		5.30	8.54	8.87	8.79	8.96	8.87	8.96	8.87	8.87	8.70	10.01	8.26		8.73	8.93	8.82

Amaoka, Naoki (JPN) (2000)	time	2.64	3.71	4.77	5.83	6.86	7.91	8.96	10.02	11.08	12.17		13.57	5 / 2				
	reaction time	0.140														3.19	3.13	3.21
	interval		1.07	1.06	1.06	1.03	1.05	1.05	1.06	1.06	1.09	1.40				8.60	8.76	8.54
	velocity		5.20	8.54	8.62	8.62	8.87	8.70	8.70	8.62	8.62	8.39	10.01	8.11				
Morita, Kanon (JPN) (2000)	time	2.59	3.67	4.72	5.79	6.83	7.90	8.97	10.04	11.11	12.20		13.60	6 / 3				
	reaction time	0.126														3.20	3.18	3.23
	interval		1.08	1.05	1.07	1.04	1.07	1.07	1.07	1.07	1.09	1.40				8.57	8.62	8.49
	velocity		5.30	8.46	8.70	8.54	8.79	8.54	8.54	8.54	8.39	10.01	8.09					
Yamashita, Tomoki (JPN) (2000)	time	2.65	3.75	4.81	5.85	6.91	7.97	9.04	10.12	11.20	12.28		13.71	4 / 4				
	reaction time	0.134														3.20	3.19	3.24
	interval		1.10	1.06	1.04	1.06	1.06	1.07	1.08	1.08	1.08	1.43				8.57	8.60	8.46
	velocity		5.18	8.31	8.62	8.79	8.62	8.62	8.54	8.46	8.46	8.46	9.80	8.02				
Fukumoto, Ren (JPN) (2000)	time	2.62	3.70	4.76	5.82	6.88	7.96	9.03	10.11	11.20	12.31		13.72	1 / 5				
	reaction time	0.135														3.20	3.21	3.28
	interval		1.08	1.06	1.06	1.06	1.08	1.07	1.08	1.09	1.11	1.41				8.57	8.54	8.36
	velocity		5.24	8.46	8.62	8.62	8.62	8.46	8.54	8.46	8.39	8.23	9.94	8.02				
Soga, Kentaro (JPN) (2001)	time	2.69	3.78	4.85	5.92	6.96	8.04	9.09	10.14	11.25	12.34		13.79	2 / 6				
	reaction time	0.178														3.23	3.17	3.25
	interval		1.09	1.07	1.07	1.04	1.08	1.05	1.05	1.11	1.09	1.45				8.49	8.65	8.44
	velocity		5.10	8.39	8.54	8.54	8.79	8.46	8.70	8.70	8.23	8.39	9.67	7.98				
Kumagai, Kai (JPN)	time	2.62	3.73	4.80	5.86	6.93	8.02	9.10	10.19	11.29	12.40		13.84	7 / 7				
	reaction time	0.147														3.24	3.24	3.30
	interval		1.11	1.07	1.06	1.07	1.09	1.08	1.09	1.10	1.11	1.44				8.46	8.46	8.31
	velocity		5.24	8.23	8.54	8.62	8.54	8.39	8.46	8.39	8.31	8.23	9.74	7.95				

2018 Chinese National Championships (Taiyuan, CHN)

FINAL

		date	16-Sep-18	wind	-0.3 m/s											Henson (2018) - coaching observations			
				H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
Xie Wenjun (CHN) (1990)	time		2.59	3.64	4.67	5.71	6.72	7.76	8.77	9.83	10.88	11.96			13.40	6 / 1			
	reaction time			7 steps	1.05	1.03	1.04	1.01	1.04	1.01	1.06	1.05	1.08	1.44			3.12	3.06	3.19
	interval																8.79	8.96	8.60
Zeng Jianhang (CHN) (1998)	time		2.72	3.78	4.81	5.84	6.87	7.92	8.99	10.07	11.14	12.30			13.77	4 / 2			
	reaction time																3.12	3.15	3.31
	interval																8.79	8.70	8.28
Zhao Pengchuan (CHN) (1993)	time		2.67	3.79	4.87	5.94	7.06	8.09	9.20	10.29	11.39	12.51			13.99	7 / 3			
	reaction time																3.27	3.26	3.31
	interval																8.39	8.41	8.28
Zhu Shenglong (CHN) (2000)	time		2.72	3.82	4.89	5.96	7.04	8.12	9.22	10.33	11.46	12.58			14.09	2 / 4			
	reaction time																3.24	3.26	3.36
	interval																8.46	8.41	8.16
Pan Zijie (CHN) (1995)	time		2.64	3.75	4.84	5.96	7.04	8.13	9.23	10.33	11.46	12.60			14.12	8 / 5			
	reaction time																3.32	3.27	3.37
	interval																8.26	8.39	8.14
Li Jiming (CHN) (1996)	time		2.70	3.80	4.87	5.94	7.05	8.12	9.23	10.39	11.53	12.68			14.26	1 / 6			
	reaction time																3.24	3.29	3.45
	interval																8.46	8.33	7.95
Ma Lei (CHN) (1989)	time		2.74	3.87	4.96	6.07	7.21	8.36	9.53	10.69	11.89	13.11			14.80	5 / 7			
	reaction time																3.33	3.46	3.58
	interval																8.23	7.92	7.66
Zhang Tao (CHN) (1997)	time		2.80	3.89	4.96	6.02	7.12	8.19	9.29	10.39	11.50	12.53			16.34	3 / 8			
	reaction time																3.22	3.27	3.24
	interval																8.52	8.39	8.46

Heat 2

		date	15-Sep-18	wind	0.7 m/s											Henson (2018) - coaching observations			
				H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
Zeng Jianhang (CHN) (1998)	time		2.58	3.67	4.71	5.77	6.84	7.90	8.98	10.08	11.17	12.29			13.83	3 / 1			
	reaction time																3.19	3.21	3.31
	interval																8.60	8.54	8.28
Ma Lei (CHN) (1989)	time		2.72	3.82	4.90	5.99	7.07	8.14	9.24	10.34	11.46	12.61			14.10	5 / 2			
	reaction time																3.27	3.25	3.37
	interval																8.39	8.44	8.14
Pan Zijie (CHN) (1995)	time		2.70	3.80	4.87	5.98	7.07	8.15	9.26	10.38	11.48	12.64			14.11	7 / 3			
	reaction time																3.28	3.28	3.38
	interval																8.36	8.36	8.11
Zhu Shenglong (CHN) (2000)	time		2.74	3.85	4.94	6.00	7.09	8.19	9.26	10.38	11.49	12.61			14.13	2 / 4			
	reaction time																3.26	3.26	3.35
	interval																		

date		15-Sep-18										wind -0.2 m/s							Henson (2018) - coaching observations						
		H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10								
)	time	2.63	3.69	4.74	5.77	6.82	7.87	8.93	10.02	11.11	12.21		13.76	5 / 1											
	interval	7 steps	1.06	1.05	1.03	1.05	1.05	1.06	1.09	1.09	1.10	1.55			3.14	3.16	3.28								
	velocity	5.22	8.62	8.70	8.87	8.70	8.70	8.62	8.39	8.39	8.31	9.05	7.99		8.73	8.68	8.36								
	time	2.77	3.85	4.90	5.99	7.50	8.11	9.18	10.29	11.36	12.51		14.00	8 / 2											
	interval		1.08	1.05	1.09	1.51	0.61	1.07	1.11	1.07	1.15	1.49			3.22	3.19	3.33								
	velocity	4.95	8.46	8.70	8.39	6.05	14.98	8.54	8.23	8.54	7.95	9.41	7.86		8.52	8.60	8.23								
(1993)	time	2.67	3.77	4.84	5.92	6.99	8.09	9.17	10.28	11.40	12.50		14.03	2 / 3											
	interval		1.10	1.07	1.08	1.07	1.10	1.08	1.11	1.12	1.10	1.53			3.25	3.25	3.33								
	velocity	5.14	8.31	8.54	8.46	8.54	8.31	8.46	8.23	8.16	8.31	9.16	7.84		8.44	8.44	8.23								
	time	2.67	3.77	4.84	5.91	6.97	8.06	9.14	10.23	11.36	12.51		14.07	4 / 4											
	interval		1.10	1.07	1.07	1.06	1.09	1.08	1.09	1.13	1.15	1.56			3.24	3.23	3.37								
	velocity	5.14	8.31	8.54	8.54	8.62	8.39	8.46	8.39	8.09	7.95	8.99	7.82		8.46	8.49	8.14								
3)	time	2.78	3.92	5.00	6.09	7.17	8.27	9.36	10.46	11.61	12.76		14.27	7 / 5											
	interval		1.14	1.08	1.09	1.08	1.10	1.09	1.10	1.15	1.15	1.51			3.31	3.27	3.40								
	velocity	4.94	8.02	8.46	8.39	8.46	8.31	8.39	8.31	7.95	7.95	9.28	7.71		8.28	8.39	8.06								
	time	2.75	3.84	4.92	5.97	7.07	8.14	9.31	10.47	11.64	12.80		14.34	3 / 6											
	interval		1.09	1.08	1.05	1.10	1.07	1.17	1.16	1.17	1.16	1.54			3.22	3.34	3.49								
	velocity	4.99	8.39	8.46	8.70	8.31	8.54	7.81	7.88	7.81	7.88	9.10	7.67		8.52	8.21	7.86								
98)	time	2.72	3.84	4.96	6.06	7.17	8.31	9.43	10.56	11.68	12.84		14.36	6 / 7											
	interval		1.12	1.12	1.10	1.11	1.14	1.12	1.13	1.12	1.16	1.52			3.34	3.37	3.41								
	velocity	5.04	8.16	8.16	8.31	8.23	8.02	8.16	8.09	8.16	7.88	9.22	7.66		8.21	8.14	8.04								

date		09-Sep-18	wind		0.9 m/s		Henson (2021) - Athlete First: 2018 year end hurdle report												
			H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10	
A) (1990)	time		2.52	3.58	4.60	5.56	6.54	7.54	8.54	9.54	10.54	11.58		13.03	2 / 1				
	interval		7 steps	1.06	1.02	0.96	0.98	1.00	1.00	1.00	1.00	1.04	1.45				3.04	2.98	3.04
0.143	velocity		5.44	8.62	8.96	9.52	9.33	9.14	9.14	9.14	9.14	8.79	9.67	8.44			9.02	9.20	9.02
B) (2011)	time		2.46	3.50	4.52	5.54	6.54	7.56	8.58	9.60	10.62	11.68		13.12	3 / 2				
	interval		7 steps	1.04	1.02	1.02	1.00	1.02	1.02	1.02	1.02	1.06	1.44				3.08	3.04	3.10
0.115	velocity		5.58	8.79	8.96	8.96	9.14	8.96	8.96	8.96	8.96	8.62	9.74	8.38			8.90	9.02	8.85
C) (FRA)	time		2.48	3.56	4.58	5.60	6.60	7.60	8.60	9.64	10.70	11.78		13.31	6 / 3				
	interval		7 steps	1.08	1.02	1.02	1.00	1.00	1.00	1.04	1.06	1.08	1.53				3.12	3.00	3.18
0.146	velocity		5.53	8.46	8.96	8.96	9.14	9.14	9.14	8.79	8.62	8.46	9.16	8.26			8.79	9.14	8.62
D) (1990)	time		2.48	3.56	4.60	5.62	6.64	7.64	8.66	9.68	10.76	11.84		13.36	4 / 4				
	interval			1.08	1.04	1.02	1.02	1.00	1.02	1.02	1.08	1.08	1.52				3.14	3.04	3.18
0.117	velocity		5.53	8.46	8.79	8.96	8.96	9.14	8.96	8.96	8.46	8.46	9.22	8.23			8.73	9.02	8.62
E) (2014)	time		2.52	3.60	4.64	5.68	6.72	7.72	8.80	9.88	10.96	12.02		13.57	7 / 5				
	interval			1.08	1.04	1.04	1.04	1.00	1.08	1.08	1.08	1.06	1.55				3.16	3.12	3.22
0.118	velocity		5.44	8.46	8.79	8.79	8.79	9.14	8.46	8.46	8.46	8.62	9.05	8.11			8.68	8.79	8.52
F) (2011)	time		2.52	3.64	4.68	5.74	6.80	7.84	8.88	10.00	11.08	12.18		13.72	5 / 6				
	interval		7 steps	1.12	1.04	1.06	1.06	1.04	1.04	1.12	1.08	1.10	1.54				3.22	3.14	3.30
0.119	velocity		5.44	8.16	8.79	8.62	8.62	8.79	8.79	8.16	8.46	8.31	9.10	8.02			8.52	8.73	8.31
G) (A) (19)	time		2.56	3.62	4.68	5.72	6.76	7.86	8.92	10.04	11.16	12.28		13.83	1 / 7				
	interval			1.06	1.06	1.04	1.04	1.10	1.06	1.12	1.12	1.12	1.55				3.16	3.20	3.36
0.143	velocity		5.36	8.62	8.62	8.79	8.79	8.31	8.62	8.16	8.16	8.16	9.05	7.95			8.68	8.57	8.16
H) (1993)	time		2.60																

reaction time	0.225	interval	7 steps	1.16	1.06	1.06	1.08	1.04	1.08	1.06	1.06	1.10	1.55	PB	3.28	3.20	3.22	
		velocity		5.28	7.88	8.62	8.62	8.46	8.79	8.46	8.62	8.62	8.31	9.05	7.94	8.36	8.57	8.52

2018 Japanese University National Championships (Nagoya, JPN)**FINAL**

date 09-Sep-18 wind -1.1 m/s

Shibayama (2021) race analysis of men's 110m hurdles in the 2021 season

		H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
Izumiya, Shunsuke (JPN) (2000)	time	2.60	3.65	4.69	5.77	6.80	7.88	8.94	10.04	11.15	12.28		13.75	1 / 1			
reaction time	interval		1.05	1.04	1.08	1.03	1.08	1.06	1.10	1.11	1.13	1.47	PB		3.17	3.17	3.34
velocity		5.28	8.70	8.79	8.46	8.87	8.46	8.62	8.31	8.23	8.09	9.54	8.00		8.65	8.65	8.21

2018 Meeting Città di Padova (Padova, ITA) (TV Analysis)**FINAL**

date 02-Sep-18 wind 1.7 m/s

Henson (2020) - Athlete First: 2018 year end hurdle report

		H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
Shubenkov, Sergey (ANA) (199)	time	2.56	3.58	4.60	5.58	6.56	7.56	8.56	9.56	10.56	11.60		13.09	4 / 1			
reaction time	interval		7 steps	1.02	1.02	0.98	0.98	1.00	1.00	1.00	1.04	1.49			3.02	2.98	3.04
velocity		5.36	8.96	8.96	9.33	9.33	9.14	9.14	9.14	9.14	8.79	9.41	8.40		9.08	9.20	9.02

2018 AG Memorial van Damme (Brussels, BEL) (TV Analysis)**FINAL**

date 31-Aug-18 wind -0.1 m/s

Henson (2020) - Athlete First: 2018 year end hurdle report

		H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
Shubenkov, Sergey (ANA) (199)	time	2.54	3.56	4.58	5.58	6.56	7.54	8.52	9.52	10.54	11.56		12.97	5 / 1			
reaction time	0.168 interval		7 steps	1.02	1.02	1.00	0.98	0.98	1.00	1.02	1.02	1.41			3.04	2.94	3.04
velocity		5.40	8.96	8.96	9.14	9.33	9.33	9.33	9.14	8.96	8.96	9.94	8.48		9.02	9.33	9.02
Ortega, Orlando (ESP) (1991)	time	2.54	3.56	4.58	5.58	6.56	7.56	8.58	9.58	10.62	11.66		13.10	6 / 2			
reaction time	0.124 interval		7 steps	1.02	1.02	1.00	0.98	1.00	1.02	1.00	1.04	1.44			3.04	3.00	3.08
velocity		5.40	8.96	8.96	9.14	9.33	9.14	8.96	9.14	8.79	8.79	9.74	8.40		9.02	9.14	8.90
Parchment, Hansle (JAM) (199)	time	2.56	3.62	4.66	5.66	6.70	7.72	8.74	9.78	10.82	11.86		13.35	7 / 3			
reaction time	0.152 interval		7 steps	1.06	1.04	1.00	1.04	1.02	1.04	1.04	1.04	1.49			3.10	3.08	3.12
velocity		5.36	8.62	8.79	9.14	8.79	8.96	8.96	8.79	8.79	8.79	9.41	8.24		8.85	8.90	8.79
Martinot-Lagarde, Pascal (FRA)	time	2.50	3.54	4.56	5.58	6.58	7.64	8.68	9.74	10.80	11.88		13.36	3 / 4			
reaction time	0.153 interval		7 steps	1.04	1.02	1.02	1.00	1.06	1.04	1.06	1.06	1.48			3.08	3.10	3.20
velocity		5.49	8.79	8.96	8.96	9.14	8.62	8.79	8.62	8.62	8.46	9.47	8.23		8.90	8.85	8.57
Crittenden, Freddie (USA) (199)	time	2.52	3.56	4.60	5.62	6.66	7.68	8.72	9.76	10.82	11.90		13.39	8 / 5			
reaction time	0.151 interval		1.04	1.04	1.02	1.04	1.02	1.04	1.04	1.06	1.08	1.49			3.10	3.10	3.18
velocity		5.44	8.79	8.79	8.96	8.79	8.96	8.79	8.79	8.62	8.46	9.41	8.22		8.85	8.85	8.62
Allen, Devon (USA) (1994)	time	2.54	3.56	4.60	5.62	6.64	7.68	8.72	9.78	10.84	11.94		13.41	2 / 6			
reaction time	0.126 interval		1.02	1.04	1.02	1.02	1.04	1.04	1.06	1.06	1.10	1.47			3.08	3.10	3.22
velocity		5.40	8.96	8.79	8.96	8.96	8.79	8.79	8.62	8.62	8.31	9.54	8.20		8.90	8.85	8.52
Levy, Ronald (JAM) (1992)	time	2.50	3.56	4.60	5.62	6.62	7.64	8.68	9.74	10.86	11.96		13.47	4 / 7			
reaction time	0.148 interval		7 steps	1.06	1.04	1.02	1.00	1.02	1.04	1.06	1.12	1.10	1.51		3.12	3.06	3.28
velocity		5.49	8.62	8.79	8.96	9.14	8.96	8.79	8.62	8.16	8.31	9.28	8.17		8.79	8.96	8.36
Baji, Balázs (HUN) (1989)	time	2.56	3.62	4.64	5.68	6.70	7.72	8.82	9.88	10.96	12.08		13.63	1 / 8			
reaction time	0.140 interval		7 steps	1.06	1.02	1.04	1.02	1.10	1.06	1.08	1.12	1.55			3.12	3.14	3.26
velocity		5.36	8.62	8.96	8.79	8.96	8.96	8.31	8.62	8.46	8.16	9.05	8.07		8.79	8.73	8.41

2018 Asian Games (Jakarta, INA)**FINAL**

date 28-Aug-18 wind 0.0 m/s

Shibayama (2018) - Race analysis of men's 110m hurdle run in 2018 season

		H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
Xie Wenjun (CHN) (1990)	time	2.58	3.63	4.66	5.69	6.71	7.73	8.76	9.81	10.85	11.93		13.34	6 / 1			
reaction time	0.142 interval		7 steps	1.05	1.03	1.03	1.02	1.03	1.05	1.04	1.08	1.41			3.11	3.07	3.17
velocity		5.32	8.70	8.87	8.87	8.96	8.96	8.87	8.70	8.79	8.46	9.94	8.25		8.82	8.93	8.65
Chen Kuei-Ru (TPE) (1993)	time	2.53	3.60	4.62	5.64	6.67	7.69	8.71	9.78	10.83	11.94		13.39	5 / 2			
reaction time	0.165 interval		1.07	1.02	1.02	1.03	1.02	1.02	1.07	1.05	1.11	1.45	NR PB		3.11	3.07	3.23
velocity		5.42	8.54	8.96	8.96	8.87	8.96	8.96	8.54	8.70	8.23	9.67	8.22		8.82	8.93	8.49
Takayama, Shunya (JPN) (1994)	time	2.56	3.60	4.63	5.66	6.70	7.74	8.78	9.85	10.93	12.03		13.48	4 / 3			
reaction time	0.147 interval		1.04	1.03	1.03	1.04	1.04	1.04	1.07	1.08	1.10	1.45			3.10	3.12	3.25
velocity		5.36	8.79	8.87	8.87	8.79	8.79	8.79	8.54	8.46	8.31	9.67	8.16		8.85	8.79	8.44
Al-Moualeed, Ahmad (KSA) (19)	time	2.53	3.58	4.62	5.63	6.66	7.70	8.73	9.82	10.92	12.03		13.50	3 / 4			
reaction time	0.115 interval		1.05	1.04	1.01	1.03	1.04	1.03	1.09	1.10	1.11	1.47			3.10	3.10	3.30
velocity		5.42	8.70	8.79	9.05	8.87	8.79	8.87	8.39	8.31	8.23	9.54	8.15		8.85	8.85	8.31
Kim Byung-Jun (KOR) (1991)	time	2.62	3.69	4.72	5.75	6.79	7.82	8.86	9.93	11.00	12.09		13.57	2 / 5			
reaction time	0.188 interval		1.07	1.03	1.03	1.04	1.03	1.04	1.07	1.07	1.09	1.48			3.13	3.11	3.23
velocity		5.24	8.54	8.87	8.87	8.79	8.87	8.79	8.54	8.54	8.39	9.47	8.11		8.76	8.82	8.49
Zeng Jianhang (CHN) (1998)	time	2.63	3.66	4.70	5.76	6.79	7.85	8.90	9.98	11.06	12.19		13.65	1 / 6			

	reaction time	0.221	interval		1.03	1.04	1.06	1.03	1.06	1.05	1.08	1.08	1.13	1.46	=PB		3.13	3.14	3.29
			velocity	5.22	8.87	8.79	8.62	8.87	8.62	8.70	8.46	8.46	8.09	9.60	8.06		8.76	8.73	8.33
Kanai, Taio (JPN) (1995)	time			2.58	3.65	4.70	5.76	6.82	7.88	8.96	10.06	11.16	12.28		13.74		7 / 7		
	reaction time	0.125	interval		1.07	1.05	1.06	1.06	1.06	1.08	1.10	1.10	1.12	1.46			3.18	3.20	3.32
			velocity	5.32	8.54	8.70	8.62	8.62	8.62	8.46	8.31	8.31	8.16	9.60	8.01		8.62	8.57	8.26
Yang Wei-Ting (TPE) (1994)	time			2.62	3.68	4.72	5.76	6.82	7.87	8.95	10.05	11.13	12.29		13.75		8 / 8		
	reaction time	0.136	interval		1.06	1.04	1.04	1.06	1.05	1.08	1.10	1.08	1.16	1.46			3.14	3.19	3.34
			velocity	5.24	8.62	8.79	8.79	8.62	8.70	8.46	8.31	8.46	7.88	9.60	8.00		8.73	8.60	8.21

2018 Müller Grand Prix (Birmingham, GBR) (TV Analysis)

FINAL

INAL	date		18-Aug-18	wind	1.3 m/s										Henson (2020) - Athlete First: 2018 year end hurdle report						
				H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10		
Ortega, Orlando (ESP) (1991)	time			2.48	3.52	4.54	5.54	6.52	7.52	8.52	9.54	10.56	11.58		13.08	6 / 1					
	reaction time	0.139	interval	7 steps	1.04	1.02	1.00	0.98	1.00	1.00	1.02	1.02	1.02	1.50			3.06	2.98	3.06		
			velocity		5.53	8.79	8.96	9.14	9.33	9.14	9.14	8.96	8.96	8.96	9.35	8.41		8.96	9.20	8.96	
Levy, Ronald (JAM) (1992)	time			2.42	3.54	4.56	5.58	6.58	7.58	8.62	9.62	10.66	11.72		13.22	3 / 2					
	reaction time	0.132	interval	7 steps	1.12	1.02	1.02	1.00	1.00	1.04	1.00	1.04	1.06	1.50			3.16	3.04	3.10		
			velocity		5.67	8.16	8.96	8.96	9.14	9.14	8.79	9.14	8.79	8.62	9.35	8.32		8.68	9.02	8.85	
Martinot-Lagarde, Pascal (FRA)	time			2.46	3.52	4.56	5.56	6.54	7.58	8.58	9.60	10.62	11.70		13.27	7 / 3					
	reaction time	0.166	interval	7 steps	1.06	1.04	1.00	0.98	1.04	1.00	1.02	1.02	1.08	1.57			3.10	3.02	3.12		
			velocity		5.58	8.62	8.79	9.14	9.33	8.79	9.14	8.96	8.96	8.46	8.93	8.29		8.85	9.08	8.79	
Crittenden, Freddie (USA) (1991)	time			2.46	3.50	4.54	5.56	6.58	7.60	8.60	9.62	10.66	11.74		13.27	2 / 4					
	reaction time	0.128	interval		1.04	1.04	1.02	1.02	1.02	1.00	1.02	1.04	1.08	1.53	PB		3.10	3.04	3.14		
			velocity		5.58	8.79	8.79	8.96	8.96	8.96	9.14	8.96	8.79	8.46	9.16	8.29		8.85	9.02	8.73	
Pozzi, Andy (GBR) (1992)	time			2.42	3.46	4.48	5.48	6.50	7.54	8.58	9.62	10.70	11.78		13.35	5 / 5					
	reaction time	0.128	interval	7 steps	1.04	1.02	1.00	1.02	1.04	1.04	1.04	1.08	1.08	1.57			3.06	3.10	3.20		
			velocity		5.67	8.79	8.96	9.14	8.96	8.79	8.79	8.79	8.46	8.46	8.93	8.24		8.96	8.85	8.57	
Constantino, Gabriel (BRA) (1991)	time			2.48	3.52	4.56	5.58	6.60	7.62	8.66	9.70	10.78	11.88		13.41	8 / 6					
	reaction time	0.126	interval	7 steps	1.04	1.04	1.02	1.02	1.02	1.04	1.04	1.04	1.08	1.10	1.53			3.10	3.08	3.22	
			velocity		5.53	8.79	8.79	8.96	8.96	8.96	8.79	8.79	8.46	8.31	9.16	8.20		8.85	8.90	8.52	
Cabral, Johnathan (CAN) (1992)	time			2.52	3.60	4.64	5.68	6.72	7.76	8.78	9.78	10.88	11.90		13.46	4 / 7					
	reaction time	0.117	interval	7 steps	1.08	1.04	1.04	1.04	1.04	1.02	1.00	1.10	1.02	1.56			3.16	3.10	3.12		
			velocity		5.44	8.46	8.79	8.79	8.79	8.79	8.96	9.14	8.31	8.96	8.99	8.17		8.68	8.85	8.79	
King, David (GBR) (1994)	time			2.52	3.58	4.62	5.64	6.66	7.70	8.74	9.78	10.84	11.92		13.53	9 / 8					
	reaction time	0.124	interval	7 steps	1.06	1.04	1.02	1.02	1.04	1.04	1.04	1.06	1.08	1.61			3.12	3.10	3.18		
			velocity		5.44	8.62	8.79	8.96	8.96	8.79	8.79	8.79	8.62	8.46	8.71	8.13		8.79	8.85	8.62	

2018 Japanese National High School Championships (Nagoya, JPN) (99ci)

FINAL

		date	06-Aug-18	wind	1.6 m/s	Kota (2018) - 71st high school championships: JAF scientific committee - biomechanics data collection														
				H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10	
Abe, Ryuto (JPN) (2000)	time			2.62	3.69	4.79	5.86	6.96	8.06	9.17	10.28	11.41	12.58		14.09	8 / 1				
	reaction time		interval		1.07	1.10	1.07	1.10	1.10	1.11	1.11	1.13	1.17	1.51	PB			3.24	3.31	3.41
		velocity		5.24	8.54	8.31	8.54	8.31	8.31	8.23	8.23	8.09	7.81	9.28	7.81			8.46	8.28	8.04
Iizuka, Kaisei (JPN) (2001)	time			2.71	3.82	4.94	6.03	7.13	8.21	9.31	10.42	11.52	12.62		14.12	5 / 2				
	reaction time		interval		1.11	1.12	1.09	1.10	1.08	1.10	1.11	1.10	1.10	1.50	PB			3.32	3.28	3.31
		velocity		5.06	8.23	8.16	8.39	8.31	8.46	8.31	8.23	8.31	8.31	9.35	7.79			8.26	8.36	8.28
Kubota, Taichi (JPN) (2000)	time			2.70	3.80	4.90	6.00	7.10	8.21	9.29	10.41	11.52	12.64		14.12	3 / 3				
	reaction time		interval		1.10	1.10	1.10	1.10	1.11	1.08	1.12	1.11	1.12	1.48	PB			3.30	3.29	3.35
		velocity		5.08	8.31	8.31	8.31	8.31	8.23	8.46	8.16	8.23	8.16	9.47	7.79			8.31	8.33	8.19
Sasa, Atsuhiko (JPN) (2001)	time			2.76	3.87	4.98	6.08	7.20	8.31	9.44	10.56	11.69	12.83		14.29	2 / 4				
	reaction time		interval		1.11	1.11	1.10	1.12	1.11	1.13	1.12	1.13	1.14	1.46				3.32	3.36	3.39
		velocity		4.97	8.23	8.23	8.31	8.16	8.23	8.09	8.16	8.09	8.02	9.60	7.70			8.26	8.16	8.09
Tawada, Asahi (JPN) (2001)	time			2.65	3.76	4.88	6.00	7.09	8.19	9.30	10.45	11.59	12.75		14.31	6 / 5				
	reaction time		interval		1.11	1.12	1.12	1.09	1.10	1.11	1.15	1.14	1.16	1.56				3.35	3.30	3.45
		velocity		5.18	8.23	8.16	8.16	8.39	8.31	8.23	7.95	8.02	7.88	8.99	7.69			8.19	8.31	7.95
Yamamura, Keita (JPN) (1999)	time			2.69	3.82	4.95	6.05	7.15	8.26	9.38	10.50	11.65	12.80		14.32	9 / 6				
	reaction time		interval		1.13	1.13	1.10	1.10	1.11	1.12	1.12	1.15	1.15	1.52				3.36	3.33	3.42
		velocity		5.10	8.09	8.09	8.31	8.31	8.23	8.16	8.16	7.95	7.95	9.22	7.68			8.16	8.23	8.02
Eida, Ryusei (JPN) (2000)	time			2.70	3.83	4.94	6.06	7.18	8.32	9.47	10.60	11.76	12.89		14.35	7 / 7				
	reaction time		interval		1.13	1.11	1.12	1.12	1.14	1.15	1.13	1.16	1.13	1.46				3.36	3.41	3.42
		velocity		5.08	8.09	8.23	8.16	8.16	8.02	7.95	8.09	7.88	8.09	9.60	7.67			8.16	8.04	8.02

Muratake, Rashiddo (JPN) (200)	time	2.66	3.75	4.89	5.99	7.11	8.24	9.43	10.60	11.77	12.98		14.43	4 / 8			
reaction time	interval		1.09	1.14	1.10	1.12	1.13	1.19	1.17	1.17	1.21	1.45			3.33	3.44	3.55
	velocity	5.16	8.39	8.02	8.31	8.16	8.09	7.68	7.81	7.81	7.55	9.67	7.62		8.23	7.97	7.72

2018 Müller Anniversary Games (London, GBR) (TV Analysis)

FINAL

INAL		date	22-Jul-18	wind 0.2 m/s			Henson (2020) - Athlete First: 2018 year end hurdle report													
				H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10	
Levy, Ronald (JAM) (1992)	time			2.48	3.52	4.56	5.54	6.56	7.56	8.56	9.58	10.62	11.70		13.13	5 / 1				
	reaction time	0.134	interval	7 steps	1.04	1.04	0.98	1.02	1.00	1.00	1.02	1.04	1.08	1.43				3.06	3.02	3.14
			velocity		5.53	8.79	8.79	9.33	8.96	9.14	9.14	8.96	8.79	8.46	9.80	8.38		8.96	9.08	8.73
Allen, Devon (USA) (1994)	time			2.54	3.58	4.62	5.62	6.62	7.66	8.70	9.74	10.78	11.90		13.30	6 / 2				
	reaction time	0.111	interval		1.04	1.04	1.00	1.00	1.04	1.04	1.04	1.04	1.12	1.40				3.08	3.08	3.20
			velocity		5.40	8.79	8.79	9.14	9.14	8.79	8.79	8.79	8.79	8.16	10.01	8.27		8.90	8.90	8.57
Crittenden, Freddie (USA) (199)	time			2.50	3.54	4.58	5.62	6.66	7.70	8.72	9.82	10.84	11.92		13.33	2 / 3				
	reaction time	0.137	interval		1.04	1.04	1.04	1.04	1.04	1.02	1.10	1.02	1.08	1.41				3.12	3.10	3.20
			velocity		5.49	8.79	8.79	8.79	8.79	8.79	8.96	8.31	8.96	8.46	9.94	8.25		8.79	8.85	8.57
Pozzi, Andy (GBR) (1992)	time			2.50	3.54	4.56	5.56	6.58	7.60	8.66	9.70	10.76	11.84		13.36	3 / 4				
	reaction time	0.129	interval	7 steps	1.04	1.02	1.00	1.02	1.02	1.06	1.04	1.06	1.08	1.52				3.06	3.10	3.18
			velocity		5.49	8.79	8.96	9.14	8.96	8.96	8.62	8.79	8.62	8.46	9.22	8.23		8.96	8.85	8.62
Alkana, Antonio (RSA) (1990)	time			2.52	3.56	4.58	5.60	6.62	7.66	8.70	9.74	10.84	11.94		13.40	7 / 5				
	reaction time	0.140	interval		1.04	1.02	1.02	1.02	1.04	1.04	1.04	1.10	1.10	1.46				3.08	3.10	3.24
			velocity		5.44	8.79	8.96	8.96	8.96	8.79	8.79	8.79	8.31	8.31	9.60	8.21		8.90	8.85	8.46
Brathwaite, Shane (BAR) (1990)	time			2.56	3.62	4.66	5.68	6.70	7.74	8.78	9.84	10.88	11.96		13.41	9 / 6				
	reaction time	0.119	interval	7 steps	1.06	1.04	1.02	1.02	1.04	1.04	1.06	1.04	1.08	1.45				3.12	3.10	3.18
			velocity		5.36	8.62	8.79	8.96	8.96	8.79	8.79	8.62	8.79	8.46	9.67	8.20		8.79	8.85	8.62
Baji, Balázs (HUN) (1989)	time			2.54	3.60	4.64	5.66	6.68	7.70	8.74	9.78	10.86	11.94		13.42	8 / 7				
	reaction time	0.146	interval	7 steps	1.06	1.04	1.02	1.02	1.02	1.04	1.04	1.08	1.08	1.48				3.12	3.08	3.20
			velocity		5.40	8.62	8.79	8.96	8.96	8.96	8.79	8.79	8.46	8.46	9.47	8.20		8.79	8.90	8.57
Merritt, Aries (USA) (1985)	time			2.54	3.60	4.64	5.66	6.70	7.74	8.76	9.82	10.86	11.94		13.44	4 / 8				
	reaction time	0.132	interval	7 steps	1.06	1.04	1.02	1.04	1.04	1.02	1.06	1.04	1.08	1.50				3.12	3.10	3.18
			velocity		5.40	8.62	8.79	8.96	8.79	8.96	8.62	8.79	8.46	9.35	8.18			8.79	8.85	8.62

2018 Herculis Meeting International d'Athletisme (Monaco, MON) (TV Analysis)

FINAL

			date	20-Jul-18	wind -0.2 m/s										Henson (2020) - Athlete First: 2018 year end hurdle report						
					H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10	
Shubenkov, Sergey (ANA) (199)	time				2.52	3.56	4.56	5.56	6.56	7.56	8.56	9.56	10.58	11.60		13.07	4 / 1				
	reaction time	0.132	interval	7 steps	1.04	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.02	1.02	1.47			3.04	3.00	3.04	
			velocity		5.44	8.79	9.14	9.14	9.14	9.14	9.14	9.14	8.96	8.96	9.54	8.42		9.02	9.14	9.02	
Ortega, Orlando (ESP) (1991)	time				2.48	3.56	4.56	5.56	6.58	7.60	8.60	9.64	10.66	11.70		13.18	5 / 2				
	reaction time	0.129	interval	7 steps	1.08	1.00	1.00	1.02	1.02	1.00	1.04	1.02	1.04	1.48				3.08	3.04	3.10	
			velocity		5.53	8.46	9.14	9.14	8.96	8.96	9.14	8.79	8.96	8.79	9.47	8.35		8.90	9.02	8.85	
Martinot-Lagarde, Pascal (FRA)	time				2.46	3.50	4.52	5.52	6.52	7.52	8.56	9.60	10.64	11.70		13.20	6 / 3				
	reaction time	0.157	interval	7 steps	1.04	1.02	1.00	1.00	1.00	1.04	1.04	1.04	1.06	1.50				3.06	3.04	3.14	
			velocity		5.58	8.79	8.96	9.14	9.14	9.14	8.79	8.79	8.79	8.62	9.35	8.33		8.96	9.02	8.73	
Parchment, Hansle (JAM) (199)	time				2.52	3.56	4.58	5.58	6.58	7.58	8.60	9.62	10.66	11.70		13.21	3 / 4				
	reaction time	0.155	interval	7 steps	1.04	1.02	1.00	1.00	1.00	1.02	1.02	1.02	1.04	1.04	1.51			3.06	3.02	3.10	
			velocity		5.44	8.79	8.96	9.14	9.14	9.14	8.96	8.96	8.79	8.79	9.28	8.33		8.96	9.08	8.85	
Allen, Devon (USA) (1994)	time				2.50	3.56	4.60	5.62	6.66	7.70	8.74	9.78	10.82	11.88		13.38	2 / 5				
	reaction time	0.141	interval			1.06	1.04	1.02	1.04	1.04	1.04	1.04	1.04	1.06	1.50			3.12	3.12	3.14	
			velocity		5.49	8.62	8.79	8.96	8.79	8.79	8.79	8.79	8.79	8.62	9.35	8.22		8.79	8.79	8.73	
Baji, Balázs (HUN) (1989)	time				2.48	3.52	4.56	5.58	6.60	7.62	8.66	9.70	10.74	11.82		13.38	7 / 6				
	reaction time	0.139	interval	7 steps	1.04	1.04	1.02	1.02	1.02	1.04	1.04	1.04	1.04	1.08	1.56			3.10	3.08	3.16	
			velocity		5.53	8.79	8.79	8.96	8.96	8.96	8.79	8.79	8.79	8.46	8.99	8.22		8.85	8.90	8.68	
Merritt, Aries (USA) (1985)	time				2.46	3.52	4.58	5.60	6.62	7.66	8.70	9.74	10.82	11.90		13.49	8 / 7				
	reaction time	0.136	interval	7 steps	1.06	1.06	1.02	1.02	1.04	1.04	1.04	1.04	1.08	1.08	1.59			3.14	3.10	3.20	
			velocity		5.58	8.62	8.62	8.96	8.96	8.79	8.79	8.79	8.46	8.46	8.82	8.15		8.73	8.85	8.57	

2018 IAAF World Junior Championships (Tampere, FIN) (99cm)

FINAL

		date	12-Jul-18	wind	Koyama (2018) - research on athlete performance and technique- 2018 data book														
				H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
Thomas, Damion (JAM) (1999)	time			2.58	3.63	4.66	5.69	6.70	7.76	8.79	9.86	10.92	11.98		13.16	6 / 1			
	reaction time	0.186	interval		1.05	1.03	1.03	1.01	1.06	1.03	1.07	1.06	1.06	1.18			3.11	3.10	3.19

velocity	5.32	8.70	8.87	8.87	9.05	8.62	8.87	8.54	8.62	8.62	11.88	8.36		8.82	8.85	8.60
Bennett, Orlando (JAM) (1999) time	2.62	3.66	4.68	5.68	6.69	7.69	8.69	9.71	10.74	11.79		13.33	4 / 2			
reaction time	0.171	interval	1.04	1.02	1.00	1.01	1.00	1.02	1.03	1.05	1.54			3.06	3.01	3.10
velocity	5.24	8.79	8.96	9.14	9.05	9.14	9.14	8.96	8.87	8.70	9.10	8.25		8.96	9.11	8.85

Izumiya, Shunsuke (JPN) (2000)	time	2.59	3.63	4.70	5.69	6.70	7.73	8.78	9.81	10.87	11.92		13.38	2 / 3				
reaction time	0.170	interval	1.04	1.07	0.99	1.01	1.03	1.05	1.03	1.06	1.05	1.46		PB	3.10	3.09	3.14	
		velocity	5.30	8.79	8.54	9.23	9.05	8.87	8.70	8.87	8.62	8.70	9.60		8.22	8.85	8.87	8.73

Heat 3

date 11-Jul-18 wind 0.7 m/s Koyama (2018) - research on athlete performance and technique- 2018 data book

Hiraga, Kentaro (JPN) (1999) time	2.63	3.72	4.80	5.86	6.92	8.02	9.10	10.19	11.29	12.38		13.80	5 / 4			
reaction time	0.147	interval	1.09	1.08	1.06	1.10	1.08	1.09	1.10	1.09	1.42			3.23	3.24	3.28
velocity	5.22	8.39	8.46	8.62	8.62	8.31	8.46	8.39	8.31	8.39	9.87	7.97		8.49	8.46	8.36

2018 Athletissima (Lausanne, SUI) (TV Analysis)

FINAL

date 05-Jul-18 wind 0.6 m/s Henson (2020) - Athlete First: 2018 year end hurdle report

Shubenkov, Sergey (ANA) (199) time	2.59	3.61	4.62	5.60	6.58	7.56	8.54	9.54	10.58	11.65		12.95	6 / 1			
reaction time	0.147	interval	7 steps	1.02	1.01	0.98	0.98	1.00	1.04	1.07	1.30			3.01	2.94	3.11
velocity	5.30	8.96	9.05	9.33	9.33	9.33	9.33	9.14	8.79	8.54	10.78	8.49		9.11	9.33	8.82
Allen, Devon (USA) (1994) time	2.57	3.59	4.63	5.63	6.64	7.68	8.70	9.72	10.80	11.87		13.29	4 / 2			
reaction time	0.121	interval	1.02	1.04	1.00	1.01	1.04	1.02	1.02	1.08	1.07	1.42		3.06	3.07	3.17
velocity	5.34	8.96	8.79	9.14	9.05	8.79	8.96	8.96	8.46	8.54	9.87	8.28		8.96	8.93	8.65
Martinot-Lagarde, Pascal (FRA) time	2.57	3.59	4.63	5.63	6.64	7.68	8.70	9.72	10.80	11.86		13.30	3 / 3			
reaction time	0.147	interval	7 steps	1.02	1.04	1.00	1.01	1.04	1.02	1.02	1.08	1.06	1.44	3.06	3.07	3.16
velocity	5.34	8.96	8.79	9.14	9.05	8.79	8.96	8.96	8.46	8.62	9.74	8.27		8.96	8.93	8.68
Baji, Balázs (HUN) (1989) time	2.59	3.63	4.67	5.69	6.71	7.71	8.74	9.76	10.86	11.93		13.36	7 / 4			
reaction time	0.135	interval	7 steps	1.04	1.04	1.02	1.02	1.03	1.02	1.10	1.07	1.43		3.10	3.05	3.19
velocity	5.30	8.79	8.79	8.96	8.96	9.14	8.87	8.96	8.31	8.54	9.80	8.23		8.85	8.99	8.60
McLeod, Omar (JAM) (1994) time	2.57	3.59	4.61	5.60	6.61	7.62	8.63	9.66	10.81	11.95		13.41	5 / 5			
reaction time	0.134	interval	1.02	1.02	0.99	1.01	1.01	1.03	1.15	1.14	1.46			3.03	3.03	3.32
velocity	5.34	8.96	8.96	9.23	9.05	9.05	9.05	8.87	7.95	8.02	9.60	8.20		9.05	9.05	8.26
Merritt, Aries (USA) (1985) time	2.59	3.61	4.63	5.65	6.69	7.73	8.77	9.83	10.87	11.95		13.44	2 / 6			
reaction time	0.129	interval	7 steps	1.02	1.02	1.02	1.04	1.04	1.06	1.04	1.08	1.49		3.06	3.12	3.18
velocity	5.30	8.96	8.96	8.96	8.79	8.79	8.79	8.62	8.79	8.46	9.41	8.18		8.96	8.79	8.62

2018 István Gyulai Memorial (Székesfehérvár, HUN) (TV Analysis)

FINAL

date 02-Jul-18 wind 0.6 m/s Henson (2020) - Athlete First: 2018 year end hurdle report

Shubenkov, Sergey (ANA) (199) time	2.52	3.54	4.54	5.52	6.50	7.48	8.46	9.46	10.48	11.52		12.92	6 / 1			
reaction time	0.169	interval	7 steps	1.02	1.00	0.98	0.98	0.98	1.00	1.02	1.04	1.40	NR PB	3.00	2.94	3.06
velocity	5.44	8.96	9.14	9.33	9.33	9.33	9.33	9.14	8.96	8.79	10.01	8.51		9.14	9.33	8.96
Baji, Balázs (HUN) (1989) time	2.48	3.54	4.56	5.58	6.58	7.62	8.64	9.68	10.72	11.76		13.27	2 / 2			
reaction time	0.143	interval	7 steps	1.06	1.02	1.02	1.00	1.04	1.02	1.04	1.04	1.51		3.10	3.06	3.12
velocity	5.53	8.62	8.96	8.96	9.14	8.79	8.96	8.79	8.79	8.79	9.28	8.29		8.85	8.96	8.79
Allen, Devon (USA) (1994) time	2.52	3.58	4.62	5.64	6.66	7.70	8.74	9.78	10.84	11.90		13.36	5 / 3			
reaction time	0.130	interval	1.06	1.04	1.02	1.02	1.04	1.04	1.06	1.06	1.46			3.12	3.10	3.16
velocity	5.44	8.62	8.79	8.96	8.96	8.79	8.79	8.79	8.62	8.62	9.60	8.23		8.79	8.85	8.68
Merritt, Aries (USA) (1985) time	2.46	3.54	4.58	5.62	6.64	7.68	8.72	9.76	10.80	11.86		13.37	3 / 4			
reaction time	0.144	interval	7 steps	1.08	1.04	1.04	1.02	1.04	1.04	1.04	1.06	1.51		3.16	3.10	3.14
velocity	5.58	8.46	8.79	8.79	8.96	8.79	8.79	8.79	8.79	8.62	9.28	8.23		8.68	8.85	8.73
Harris, Aleec (USA) (1990) time	2.56	3.60	4.64	5.68	6.72	7.74	8.78	9.84	10.88	11.96		13.43	7 / 5			
reaction time	0.163	interval	1.04	1.04	1.04	1.02	1.04	1.06	1.04	1.08	1.47			3.12	3.10	3.18
velocity	5.36	8.79	8.79	8.79	8.79	8.96	8.79	8.62	8.79	8.46	9.54	8.19		8.79	8.85	8.62
McLeod, Omar (JAM) (1994) time	2.48	3.52	4.54	5.56	6.56	7.56	8.58	9.74	10.82	11.94		13.45	4 / 6			
reaction time	0.152	interval	1.04	1.02	1.00	1.00	1.02	1.16	1.08	1.12	1.51			3.08	3.02	3.36
velocity	5.53	8.79	8.96	8.96	9.14	9.14	8.96	7.88	8.46	8.16	9.28	8.18		8.90	9.08	8.16

2018 Meeting de Paris (Paris, FRA) (TV Analysis)

FINAL

date 30-Jun-18 wind 1.5 m/s Henson (2020) - Athlete First: 2018 year end hurdle report

Levy, Ronald (JAM) (1992) time	2.46	3.50	4.54	5.58	6.58	7.58	8.60	9.62	10.70	11.76		13.18	3 / 1			
reaction time	0.123	interval	7 steps	1.04	1.04	1.00	1.02	1.02	1.08	1.06	1.42			3.12	3.02	3.16
velocity	5.58	8.79	8.79	8.79	9.14	9.14	8.96	8.96	8.46	8.62	9.87	8.35		8.79	9.08	8.68

Parchment, Hansle (JAM) (1991)	time	2.54	3.58	4.62	5.62	6.62	7.62	8.62	9.66	10.66	11.74		13.22	6 / 2			
	reaction time	0.160	interval	7 steps	1.04	1.04	1.00	1.00	1.00	1.04	1.00	1.08	1.48		3.08	3.00	3.12
	velocity	5.40	8.79	8.79	9.14	9.14	9.14	9.14	8.79	9.14	8.46	9.47	8.32		8.90	9.14	8.79
Allen, Devon (USA) (1994)	time	2.52	3.56	4.60	5.64	6.64	7.68	8.70	9.70	10.72	11.76		13.23	1 / 3			
	reaction time	0.123	interval		1.04	1.04	1.04	1.00	1.04	1.02	1.00	1.02	1.04	1.47	3.12	3.06	3.06
	velocity	5.44	8.79	8.79	8.79	9.14	8.79	8.96	9.14	8.96	8.79	9.54	8.31		8.79	8.96	8.96
Alkana, Antonio (RSA) (1990)	time	2.50	3.54	4.58	5.58	6.60	7.64	8.66	9.72	10.74	11.82		13.32	7 / 4			
	reaction time	0.138	interval		1.04	1.04	1.00	1.02	1.04	1.02	1.06	1.02	1.08	1.50	3.08	3.08	3.16
	velocity	5.49	8.79	8.79	9.14	8.96	8.79	8.96	8.62	8.96	8.46	9.35	8.26		8.90	8.90	8.68
Eaton, Jarret (USA) (1989)	time	2.44	3.48	4.52	5.54	6.54	7.58	8.60	9.64	10.74	11.84		13.40	2 / 5			
	reaction time	0.129	interval	7 steps	1.04	1.04	1.02	1.00	1.04	1.02	1.04	1.10	1.10	1.56	3.10	3.06	3.24
	velocity	5.62	8.79	8.79	8.96	9.14	8.79	8.96	8.79	8.31	8.31	8.99	8.21		8.85	8.96	8.46
Ortega, Orlando (ESP) (1991)	time	2.52	3.60	4.64	5.68	6.68	7.70	8.72	9.74	10.80	11.88		13.44	5 / 6			
	reaction time	0.135	interval	7 steps	1.08	1.04	1.04	1.02	1.02	1.02	1.06	1.08	1.56		3.16	3.04	3.16
	velocity	5.44	8.46	8.79	8.79	9.14	8.96	8.96	8.96	8.62	8.46	8.99	8.18		8.68	9.02	8.68
Manga, Aurel (FRA) (1992)	time	2.50	3.56	4.60	5.64	6.66	7.70	8.74	9.78	10.86	11.94		13.48	8 / 7			
	reaction time	0.157	interval	7 steps	1.06	1.04	1.04	1.02	1.04	1.04	1.08	1.08	1.54		3.14	3.10	3.20
	velocity	5.49	8.62	8.79	8.79	8.96	8.79	8.79	8.79	8.46	8.46	9.10	8.16		8.73	8.85	8.57

2018 USATF Championships (Des Moines, IA) (TV Analysis)

FINAL

INAL	date	24-Jun-18	wind -1.8 m/s										Henson (2020) - Athlete First: 2018 year end hurdle report						
			H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10	
Allen, Devon (USA) (1994)	time		2.50	3.55	4.62	5.66	6.70	7.74	8.77	9.82	10.88	11.95		13.46	3 / 1				
	reaction time		interval	1.05	1.07	1.04	1.04	1.04	1.03	1.05	1.06	1.07	1.51				3.16	3.11	3.18
		velocity	5.49	8.70	8.54	8.79	8.79	8.79	8.87	8.70	8.62	8.54	9.28	8.17			8.68	8.82	8.62
Holloway, Grant (USA) (1997)	time		2.43	3.44	4.46	5.48	6.50	7.52	8.57	9.65	10.74	11.88		13.46	5 / 2				
	reaction time		interval	7 steps	1.01	1.02	1.02	1.02	1.02	1.05	1.08	1.09	1.14	1.58			3.05	3.09	3.31
		velocity	5.65	9.05	8.96	8.96	8.96	8.96	8.70	8.46	8.39	8.02	8.87	8.17			8.99	8.87	8.28
Eaton, Jarret (USA) (1989)	time		2.47	3.50	4.52	5.54	6.56	7.56	8.60	9.68	10.76	11.88		13.51	7 / 3				
	reaction time		interval	7 steps	1.03	1.02	1.02	1.02	1.00	1.04	1.08	1.08	1.12	1.63			3.07	3.06	3.28
		velocity	5.55	8.87	8.96	8.96	8.96	9.14	8.79	8.46	8.46	8.16	8.60	8.14			8.93	8.96	8.36
Merritt, Aries (USA) (1985)	time		2.47	3.54	4.60	5.65	6.68	7.72	8.76	9.81	10.88	11.96		13.52	4 / 4				
	reaction time		interval	7 steps	1.07	1.06	1.05	1.03	1.04	1.04	1.05	1.07	1.08	1.56			3.18	3.11	3.20
		velocity	5.55	8.54	8.62	8.70	8.87	8.79	8.79	8.70	8.54	8.46	8.99	8.14			8.62	8.82	8.57
Cunningham, Trey (USA) (1991)	time		2.50	3.57	4.63	5.67	6.72	7.77	8.84	9.94	11.04	12.14		13.71	1 / 5				
	reaction time		interval	1.07	1.06	1.04	1.05	1.05	1.07	1.10	1.10	1.10	1.57				3.17	3.17	3.30
		velocity	5.49	8.54	8.62	8.79	8.70	8.70	8.54	8.31	8.31	8.31	8.93	8.02			8.65	8.65	8.31
Lloyd, Antoine (USA) (1996)	time		2.47	3.57	4.63	5.70	6.74	7.77	8.84	9.94	11.04	12.14		13.72	2 / 6				
	reaction time		interval	7 steps	1.10	1.06	1.07	1.04	1.03	1.07	1.10	1.10	1.10	1.58			3.23	3.14	3.30
		velocity	5.55	8.31	8.62	8.54	8.79	8.87	8.54	8.31	8.31	8.31	8.87	8.02			8.49	8.73	8.31
Harris, Aleec (USA) (1990)	time		2.57	3.63	4.70	5.77	6.80	7.85	8.90	10.01	11.11	12.21		13.76	6 / 7				
	reaction time		interval	1.06	1.07	1.07	1.03	1.05	1.05	1.11	1.10	1.10	1.55				3.20	3.13	3.31
		velocity	5.34	8.62	8.54	8.54	8.87	8.70	8.70	8.23	8.31	8.31	9.05	7.99			8.57	8.76	8.28

2018 Japanese National Championships (Yamaguchi, JPN)

FINAL

INAL	date		24-Jun-18		wind		0.7 m/s		Shibayama (2018) - Race analysis of men's 110m hurdle run in 2018 season											
					H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
Kanai, Taio (JPN) (1995)	time				2.56	3.60	4.62	5.66	6.68	7.71	8.77	9.83	10.89	11.96		13.36	4 / 1			
	reaction time	0.144	interval	7 steps	1.04	1.02	1.04	1.02	1.03	1.06	1.06	1.06	1.06	1.07	1.40	NR PB		3.10	3.11	3.19
			velocity		5.36	8.79	8.96	8.79	8.96	8.87	8.62	8.62	8.62	8.54	10.01	8.23		8.85	8.82	8.60
Takayama, Shunya (JPN) (1994)	time				2.56	3.60	4.63	5.66	6.69	7.72	8.77	9.84	10.90	12.01		13.45	7 / 2			
	reaction time	0.145	interval			1.04	1.03	1.03	1.03	1.03	1.05	1.07	1.06	1.11	1.44			3.10	3.11	3.24
			velocity		5.36	8.79	8.87	8.87	8.87	8.87	8.70	8.54	8.62	8.23	9.74	8.18		8.85	8.82	8.46
Tanaka, Shinya (JPN) (1985)	time				2.60	3.66	4.70	5.73	6.83	7.90	8.98	10.06	11.15	12.23		13.64	5 / 3			
	reaction time	0.145	interval			1.06	1.04	1.03	1.10	1.07	1.08	1.08	1.09	1.08	1.41	=PB		3.13	3.25	3.25
			velocity		5.28	8.62	8.79	8.87	8.31	8.54	8.46	8.46	8.39	8.46	9.94	8.06		8.76	8.44	8.44
Fudaba, Hiroki (JPN) (1994)	time				2.68	3.75	4.82	5.85	6.92	7.97	9.04	10.11	11.20	12.32		13.76	3 / 4			
	reaction time	0.164	interval	7 steps	1.07	1.07	1.03	1.07	1.05	1.07	1.07	1.07	1.09	1.12	1.44			3.17	3.19	3.28
			velocity		5.12	8.54	8.54	8.87	8.54	8.70	8.54	8.54	8.39	8.16	9.74	7.99		8.65	8.60	8.36
Ogata, Akihiro (JPN) (1993)	time				2.62	3.69	4.75	5.78	6.86	7.85	9.03	10.11	11.20	12.32		13.80	9 / 5			

reaction time	0.183	interval	7 steps	1.07	1.06	1.03	1.08	0.99	1.18	1.08	1.09	1.12	1.48			3.16	3.25	3.29
		velocity		5.24	8.54	8.62	8.87	8.46	9.23	7.75	8.46	8.39	8.16	9.47	7.97	8.68	8.44	8.33
Ishikawa, Shuhei (JPN) (1995)	time	2.60	3.69	4.76	5.81	6.89	7.97	9.04	10.12	11.22	12.34			13.81	8 / 6			
reaction time	0.155	interval	7 steps	1.09	1.07	1.05	1.08	1.08	1.07	1.08	1.10	1.12	1.47			3.21	3.23	3.30
		velocity		5.28	8.39	8.54	8.70	8.46	8.46	8.54	8.46	8.31	8.16	9.54	7.97	8.54	8.49	8.31
Masuno, Genta (JPN) (1993)	time	2.61	3.68	4.72	5.76	6.81	7.87	8.95	10.08	11.20	12.30			13.82	6 / 7			
reaction time	0.152	interval		1.07	1.04	1.04	1.05	1.06	1.08	1.13	1.12	1.10	1.52			3.15	3.19	3.35
		velocity		5.26	8.54	8.79	8.79	8.70	8.62	8.46	8.09	8.16	8.31	9.22	7.96	8.70	8.60	8.19
Wado, Tatsuya (JPN) (1990)	time	2.55	3.64	4.70	5.74	6.83	7.92	9.01	10.13	11.26	12.40			13.91	2 / 8			
reaction time	0.117	interval		1.09	1.06	1.04	1.09	1.09	1.09	1.12	1.13	1.14	1.51			3.19	3.27	3.39
		velocity		5.38	8.39	8.62	8.79	8.39	8.39	8.16	8.09	8.02	9.28	7.91		8.60	8.39	8.09

2018 Meeting International de Montreuil (Montreuil, FRA) (TV Analysis)

FINAL

FINAL

date	19-Jun-18	wind	0.5 m/s	Henson (2020) - Athlete First: 2018 year end hurdle report																
				H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10	
Shubenkov, Sergey (ANA) (199)	time			2.62	3.64	4.64	5.62	6.60	7.58	8.56	9.56	10.58	11.62		12.99	5 / 1				
	reaction time	0.143	interval	7 steps	1.02	1.00	0.98	0.98	0.98	0.98	1.00	1.02	1.04	1.37				3.00	2.94	3.06
			velocity		5.24	8.96	9.14	9.33	9.33	9.33	9.33	9.14	8.96	8.79	10.23	8.47		9.14	9.33	8.96
Constantino, Gabriel (BRA) (19)	time			2.64	3.66	4.68	5.68	6.68	7.68	8.70	9.74	10.78	11.86		13.23	1 / 2				
	reaction time	0.165	interval	7 steps	1.02	1.02	1.00	1.00	1.00	1.02	1.04	1.04	1.08	1.37	AR PB			3.04	3.02	3.16
			velocity		5.20	8.96	8.96	9.14	9.14	8.96	8.79	8.79	8.46	10.23	8.31			9.02	9.08	8.68
Martinot-Lagarde, Pascal (FRA)	time			2.58	3.62	4.66	5.68	6.68	7.68	8.70	9.72	10.74	11.86		13.28	4 / 3				
	reaction time	0.156	interval	7 steps	1.04	1.04	1.02	1.00	1.00	1.02	1.02	1.02	1.12	1.42				3.10	3.02	3.16
			velocity		5.32	8.79	8.79	8.96	9.14	9.14	8.96	8.96	8.96	8.16	9.87	8.28		8.85	9.08	8.68

2018 National Grand Prix Final (Guiyang, CHN) (Altitude)

FINAL

FINAL

	date	17-Jun-18	wind	-0.3 m/s	Zhang (2019) - comparative analysis of characteristics of Chinese elite men's 110m hurdles in different age groups														
			H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10	
Zeng Jianhang (CHN) (1998)	time		2.60	3.64	4.69	5.72	6.75	7.78	8.85	9.91	11.00	12.11		13.65	/ 1				
	reaction time																		
	interval			1.04	1.05	1.03	1.03	1.03	1.07	1.06	1.09	1.11	1.54	PB			3.12	3.13	3.26
	velocity		5.28	8.79	8.70	8.87	8.87	8.87	8.54	8.62	8.39	8.23	9.10	8.06			8.79	8.76	8.41
Zhang Tao (CHN) (1997)	time		2.71	3.78	4.86	5.91	6.99	8.05	9.14	10.21	11.32	12.41		13.92	/ 2				
	reaction time																		
	interval			1.07	1.08	1.05	1.08	1.06	1.09	1.07	1.11	1.09	1.51				3.20	3.23	3.27
	velocity		5.06	8.54	8.46	8.70	8.46	8.62	8.39	8.54	8.23	8.39	9.28	7.90			8.57	8.49	8.39
Ma Lei (CHN) (1989)	time		2.69	3.78	4.84	5.91	6.98	8.04	9.13	10.21	11.32	12.45		13.93	/ 3				
	reaction time																		
	interval	7 steps		1.09	1.06	1.07	1.07	1.06	1.09	1.08	1.11	1.13	1.48				3.22	3.22	3.32
	velocity		5.10	8.39	8.62	8.54	8.54	8.62	8.39	8.46	8.23	8.09	9.47	7.90			8.52	8.52	8.26
Zhang Honglin (CHN) (1994)	time		2.69	3.77	4.83	5.92	6.99	8.06	9.15	10.24	11.35	12.47		13.98	/ 4				
	reaction time																		
	interval			1.08	1.06	1.09	1.07	1.07	1.09	1.09	1.11	1.12	1.51				3.23	3.23	3.32
	velocity		5.10	8.46	8.62	8.39	8.54	8.54	8.39	8.39	8.23	8.16	9.28	7.87			8.49	8.49	8.26
Chu Pengfei (CHN) (1993)	time		2.75	3.86	4.96	6.03	7.10	8.18	9.26	10.35	11.45	12.55		14.03	/ 5				
	reaction time																		
	interval			1.11	1.10	1.07	1.07	1.08	1.08	1.09	1.10	1.10	1.48				3.28	3.23	3.29
	velocity		4.99	8.23	8.31	8.54	8.54	8.46	8.46	8.39	8.31	8.31	9.47	7.84			8.36	8.49	8.33
Li Jiming (CHN) (1996)	time		2.70	3.81	4.89	5.97	7.03	8.10	9.18	10.28	11.40	12.58		14.12	/ 6				
	reaction time																		
	interval			1.11	1.08	1.08	1.06	1.07	1.08	1.10	1.12	1.18	1.54				3.27	3.21	3.40
	velocity		5.08	8.23	8.46	8.46	8.62	8.54	8.46	8.31	8.16	7.75	9.10	7.79			8.39	8.54	8.06
Zhang Xiangyu (CHN) (1991)	time		2.71	3.80	4.89	5.97	7.08	8.16	9.27	10.37	11.50	12.62		14.15	/ 7				
	reaction time																		
	interval			1.09	1.09	1.08	1.11	1.08	1.11	1.10	1.13	1.12	1.53				3.26	3.30	3.35
	velocity		5.06	8.39	8.39	8.46	8.23	8.46	8.23	8.31	8.09	8.16	9.16	7.77			8.41	8.31	8.19

2018 Japanese National Multi-Event Championships (Nagano, JPN)

Decathlon - Heat 3

Decathlon - Heat 3		date	17-Jun-18		wind <td colspan="2">0.2 m/s</td> <th colspan="13">Matsubayashi (2018) - research on athlete performance and technique- 2018 data book</th>	0.2 m/s		Matsubayashi (2018) - research on athlete performance and technique- 2018 data book												
			H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10		
Maruyama, Yuma (JPN) (1998)	time		2.70	3.81	4.91	5.99	7.07	8.14	9.22	10.34	11.45	12.61		14.08	6 / 1					
	reaction time	interval		1.11	1.10	1.08	1.08	1.07	1.08	1.12	1.11	1.16	1.47	PB		3.29	3.23	3.39		
		velocity	5.08	8.23	8.31	8.46	8.46	8.54	8.46	8.16	8.23	7.88	9.54	7.81		8.33	8.49	8.09		
Nakamura, Akihiko (JPN) (1990)	time		2.69	3.85	4.95	6.04	7.14	8.25	9.34	10.48	11.61	12.75		14.20	4 / 2					
	reaction time	interval		1.16	1.10	1.09	1.10	1.11	1.09	1.14	1.13	1.14	1.45			3.35	3.30	3.41		
		velocity	5.10	7.88	8.31	8.39	8.31	8.23	8.39	8.02	8.09	8.02	9.67	7.75		8.19	8.31	8.04		

Decathlon - Heat 2

Decathlon - Heat 2	date	17-Jun-18	wind	0.5 m/s			Matsubayashi (2018) - research on athlete performance and technique- 2018 data book														
				H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10		
	Morimoto, Kimihito (JPN) (199)	time		2.70	3.89	5.07	6.23	7.39	8.55	9.76	10.95	12.13	13.35		14.91	5 / 1					
	reaction time	interval			1.19	1.18	1.16	1.16	1.16	1.21	1.19	1.18	1.22	1.56				3.53	3.53	3.59	

velocity	5.08	7.68	7.75	7.88	7.88	7.88	7.55	7.68	7.75	7.49	8.99	7.38		7.77	7.77	7.64
Ushiro, Keisuke (JPN) (1986) time	2.71	3.87	5.03	6.21	7.37	8.55	9.74	10.94	12.17	13.42		15.05	4 / 5			
reaction time interval		1.16	1.16	1.18	1.16	1.18	1.19	1.20	1.23	1.25	1.63			3.50	3.53	3.68
velocity	5.06	7.88	7.88	7.75	7.88	7.75	7.68	7.62	7.43	7.31	8.60	7.31		7.83	7.77	7.45

2018 NCAA Championships (Eugene, OR) (TV Analysis)

FINAL

date 08-Jun-18 wind -1.1 m/s

Henson (2020) - Athlete First: 2018 year end hurdle report

		H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
Holloway, Grant (USA) (1997) time		2.45	3.48	4.52	5.53	6.56	7.58	8.65	9.72	10.78	11.92		13.42	4 / 1			
reaction time interval		7 steps	1.03	1.04	1.01	1.03	1.02	1.07	1.07	1.06	1.14	1.50			3.08	3.12	3.27
velocity		5.60	8.87	8.79	9.05	8.87	8.96	8.54	8.54	8.62	8.02	9.35	8.20		8.90	8.79	8.39
Kendziera, Dave (USA) (1994) time		2.54	3.63	4.67	5.70	6.71	7.74	8.78	9.82	10.87	11.92		13.43	6 / 2			
reaction time interval		7 steps	1.09	1.04	1.03	1.01	1.03	1.04	1.04	1.05	1.05	1.51			3.16	3.08	3.14
velocity		5.40	8.39	8.79	8.87	9.05	8.87	8.79	8.79	8.70	8.70	9.28	8.19		8.68	8.90	8.73
Thomas, Damion (JAM) (1999) time		2.60	3.67	4.67	5.70	6.71	7.74	8.78	9.82	10.89	11.95		13.45	3 / 3			
reaction time interval			1.07	1.00	1.03	1.01	1.03	1.04	1.04	1.07	1.06	1.50			3.10	3.08	3.17
velocity		5.28	8.54	9.14	8.87	9.05	8.87	8.79	8.79	8.54	8.62	9.35	8.18		8.85	8.90	8.65
Cunningham, Trey (USA) (1998) time		2.58	3.65	4.68	5.72	6.76	7.80	8.85	9.90	11.00	12.08		13.64	8 / 4			
reaction time interval			1.07	1.03	1.04	1.04	1.04	1.05	1.05	1.10	1.08	1.56			3.14	3.13	3.23
velocity		5.32	8.54	8.87	8.79	8.79	8.79	8.70	8.70	8.31	8.46	8.99	8.06		8.73	8.76	8.49

2018 Fuse Sprint (Tottori, JPN)

FINAL

date 03-Jun-18 wind 0.5 m/s

Shibayama (2018) - Race analysis of men's 110m hurdle run in 2018 season

		H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
Kanai, Taio (JPN) (1995) time		2.58	3.63	4.67	5.73	6.75	7.80	8.85	9.92	10.99	12.10		13.52	/ 1			
reaction time interval			1.05	1.04	1.06	1.02	1.05	1.05	1.07	1.07	1.11	1.42			3.15	3.12	3.25
velocity		5.32	8.70	8.79	8.62	8.96	8.70	8.70	8.54	8.54	8.23	9.87	8.14		8.70	8.79	8.44
Takayama, Shunya (JPN) (1994) time		2.63	3.69	4.74	5.79	6.83	7.88	8.96	10.03	11.11	12.20		13.67	/ 2			
reaction time interval			1.06	1.05	1.05	1.04	1.05	1.08	1.07	1.08	1.09	1.47			3.16	3.17	3.24
velocity		5.22	8.62	8.70	8.70	8.79	8.70	8.46	8.54	8.46	8.39	9.54	8.05		8.68	8.65	8.46
Ogata, Akihiro (JPN) (1993) time		2.58	3.64	4.69	5.73	6.77	7.82	8.89	9.98	11.07	12.18		13.71	/ 3			
reaction time interval			1.06	1.05	1.04	1.04	1.05	1.07	1.09	1.09	1.11	1.53			3.15	3.16	3.29
velocity		5.32	8.62	8.70	8.79	8.79	8.70	8.54	8.39	8.39	8.23	9.16	8.02		8.70	8.68	8.33
Ishikawa, Shuhei (JPN) (1995) time		2.61	3.68	4.77	5.83	6.90	7.96	9.05	10.14	11.24	12.35		13.86	/ 4			
reaction time interval			1.07	1.09	1.06	1.07	1.06	1.09	1.09	1.10	1.11	1.51			3.22	3.22	3.30
velocity		5.26	8.54	8.39	8.62	8.54	8.62	8.39	8.39	8.31	8.23	9.28	7.94		8.52	8.52	8.31
Wado, Tatsuya (JPN) (1990) time		2.60	3.70	4.77	5.83	6.88	7.95	9.04	10.14	11.24	12.37		13.88	/ 5			
reaction time interval			1.10	1.07	1.06	1.05	1.07	1.09	1.10	1.10	1.13	1.51			3.23	3.21	3.33
velocity		5.28	8.31	8.54	8.62	8.70	8.54	8.39	8.31	8.31	8.09	9.28	7.93		8.49	8.54	8.23
Kagimoto, Masahiro (JPN) (1995) time		2.65	3.74	4.80	5.86	6.93	8.02	9.11	10.23	11.33	12.46		13.93	/ 6			
reaction time interval			1.09	1.06	1.06	1.07	1.09	1.09	1.12	1.10	1.13	1.47			3.21	3.25	3.35
velocity		5.18	8.39	8.62	8.62	8.54	8.39	8.39	8.16	8.31	8.09	9.54	7.90		8.54	8.44	8.19
Tanaka, Shinya (JPN) (1985) time		2.65	3.75	4.83	5.92	7.00	8.11	9.25	10.38	11.53	12.70		14.25	/ 7			
reaction time interval			1.10	1.08	1.09	1.08	1.11	1.14	1.13	1.15	1.17	1.55			3.27	3.33	3.45
velocity		5.18	8.31	8.46	8.39	8.46	8.23	8.02	8.09	7.95	7.81	9.05	7.72		8.39	8.23	7.95

2018 Shimane High School Championships (Honshu, JPN)

FINAL

date 27-May-18 wind 1.2 m/s

Shimizu (2018) - shimane high school championships 2018 - biomechanics data analysis

		H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
京極 史弥 time		2.82	3.98	5.15	6.32	7.50	8.67	9.92	11.17	12.40	13.63		15.28	5 / 1			
reaction time interval			1.16	1.17	1.17	1.18	1.17	1.25	1.25	1.23	1.23	1.65			3.50	3.60	3.71
velocity		4.87	7.88	7.81	7.81	7.75	7.81	7.31	7.31	7.43	7.43	8.50	7.20		7.83	7.62	7.39
足立 駿人 time		2.85	4.03	5.23	6.43	7.65	8.90	10.17	11.45	12.73	14.03		15.65	4 / 2			
reaction time interval			1.18	1.20	1.20	1.22	1.25	1.27	1.28	1.28	1.30	1.62			3.58	3.74	3.86
velocity		4.81	7.75	7.62	7.62	7.49	7.31	7.20	7.14	7.14	7.03	8.65	7.03		7.66	7.33	7.10
原田 真紘 time		2.83	4.03	5.30	6.53	7.78	9.07	10.38	11.68	13.02	14.35		16.06	6 / 3			
reaction time interval			1.20	1.27	1.23	1.25	1.29	1.31	1.30	1.34	1.33	1.71			3.70	3.85	3.97
velocity		4.85	7.62	7.20	7.43	7.31	7.09	6.98	7.03	6.82	6.87	8.20	6.85		7.41	7.12	6.91
藤岡 哲也 time		2.92	4.15	5.42	6.68	7.98	9.27	10.55	11.83	13.13	14.48		16.10	3 / 4			
reaction time interval			1.23	1.27	1.26	1.30	1.29	1.28	1.28	1.30	1.35	1.62			3.76	3.87	3.93
velocity		4.70	7.43	7.20	7.25	7.03	7.09	7.14	7.14	7.03	6.77	8.65	6.83		7.29	7.09	6.98
の場 真那斗 time		2.92	4.20	5.50	6.75	8.05	9.35	10.65	11.97	13.32	14.63		16.28	/ 5			

reaction time	interval	1.28	1.30	1.25	1.30	1.30	1.30	1.30	1.32	1.35	1.31	1.65		3.83	3.90	3.98
	velocity	4.70	7.14	7.03	7.31	7.03	7.03	7.03	6.92	6.77	6.98	8.50	6.76	7.16	7.03	6.89
荏田 陽向	time	2.87	4.13	5.43	6.73	8.05	9.35	10.65	11.97	13.30	14.65		16.32	7 / 6		
reaction time	interval		1.26	1.30	1.30	1.32	1.30	1.30	1.32	1.33	1.35	1.67		3.86	3.92	4.00
	velocity	4.78	7.25	7.03	7.03	6.92	7.03	7.03	6.92	6.87	6.77	8.40	6.74	7.10	6.99	6.86

2018 Prefontaine Classic (Eugene, OR) (TV Analysis)

FINAL

date 26-May-18 wind 3.0 m/s

Henson (2020) - Athlete First: 2018 year end hurdle report

		H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
McLeod, Omar (JAM) (1994)	time	2.54	3.54	4.56	5.56	6.56	7.54	8.52	9.54	10.54	11.58		13.01	4 / 1			
reaction time	0.176	interval	1.00	1.02	1.00	1.00	0.98	0.98	1.02	1.00	1.04	1.43			3.02	2.96	3.06
		velocity	5.40	9.14	8.96	9.14	9.33	9.33	8.96	9.14	8.79	9.80	8.46		9.08	9.26	8.96
Shubenkov, Sergey (ANA) (1999)	time	2.56	3.58	4.58	5.56	6.56	7.56	8.56	9.58	10.62	11.66		13.08	6 / 2			
reaction time	0.173	interval	7 steps	1.02	1.00	0.98	1.00	1.00	1.02	1.04	1.04	1.42			3.00	3.00	3.10
		velocity	5.36	8.96	9.14	9.33	9.14	9.14	8.96	8.79	8.79	9.87	8.41		9.14	9.14	8.85
Allen, Devon (USA) (1994)	time	2.52	3.56	4.58	5.60	6.60	7.60	8.62	9.66	10.68	11.72		13.13	8 / 3			
reaction time	0.122	interval		1.04	1.02	1.02	1.00	1.00	1.02	1.04	1.04	1.41			3.08	3.02	3.10
		velocity	5.44	8.79	8.96	8.96	9.14	9.14	8.96	8.79	8.96	9.94	8.38		8.90	9.08	8.85
Ortega, Orlando (ESP) (1991)	time	2.54	3.60	4.62	5.62	6.62	7.64	8.64	9.66	10.66	11.72		13.17	5 / 4			
reaction time	0.158	interval	7 steps	1.06	1.02	1.00	1.00	1.02	1.00	1.06	1.06	1.45			3.08	3.02	3.08
		velocity	5.40	8.62	8.96	9.14	9.14	8.96	9.14	8.96	8.62	9.67	8.35		8.90	9.08	8.90
Levy, Ronald (JAM) (1992)	time	2.56	3.60	4.62	5.64	6.66	7.70	8.74	9.76	10.80	11.82		13.26	2 / 5			
reaction time	0.171	interval	7 steps	1.04	1.02	1.02	1.04	1.04	1.02	1.04	1.02	1.44			3.08	3.10	3.08
		velocity	5.36	8.79	8.96	8.96	8.79	8.79	8.96	8.79	8.96	9.74	8.30		8.90	8.85	8.90
Merritt, Aries (USA) (1985)	time	2.52	3.58	4.60	5.60	6.62	7.64	8.66	9.70	10.74	11.80		13.27	7 / 6			
reaction time	0.144	interval	7 steps	1.06	1.02	1.00	1.02	1.02	1.04	1.04	1.06	1.47			3.08	3.06	3.14
		velocity	5.44	8.62	8.96	9.14	8.96	8.96	8.79	8.79	8.62	9.54	8.29		8.90	8.96	8.73
Pozzi, Andy (GBR) (1992)	time	2.54	3.60	4.60	5.60	6.60	7.64	8.68	9.74	10.84	11.94		13.51	3 / 7			
reaction time	0.164	interval	7 steps	1.06	1.00	1.00	1.00	1.04	1.04	1.06	1.10	1.57			3.06	3.08	3.26
		velocity	5.40	8.62	9.14	9.14	9.14	8.79	8.79	8.62	8.31	8.31	8.93	8.14	8.96	8.90	8.41
Harris, Aleec (USA) (1990)	time	2.58	3.64	4.68	5.72	6.76	7.84	8.88	9.92	10.98	12.06		13.52	1 / 8			
reaction time	0.170	interval		1.06	1.04	1.04	1.04	1.08	1.04	1.04	1.06	1.08	1.46		3.14	3.16	3.18
		velocity	5.32	8.62	8.79	8.79	8.79	8.46	8.79	8.79	8.62	8.46	9.60	8.14	8.73	8.68	8.62

2018 Seiko Golden Grand Prix (Osaka, JPN)

FINAL

date 20-May-18 wind -0.4 m/s

Shibayama (2018) - Race analysis of men's 110m hurdle run in 2018 season

		H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
Chen Kuei-Ru (TPE) (1993)	time	2.60	3.66	4.71	5.75	6.78	7.83	8.90	9.96	11.05	12.13		13.49	5 / 1			
reaction time	0.153	interval	7 steps	1.06	1.05	1.04	1.03	1.05	1.06	1.09	1.08	1.36	NR PB		3.15	3.15	3.23
		velocity	5.28	8.62	8.70	8.79	8.87	8.70	8.54	8.62	8.39	8.46	10.31	8.15	8.70	8.70	8.49
Kanai, Taio (JPN) (1995)	time	2.67	3.74	4.77	5.81	6.87	7.91	8.96	10.02	11.10	12.18		13.53	8 / 2			
reaction time	0.159	interval	7 steps	1.07	1.03	1.04	1.06	1.04	1.05	1.06	1.08	1.08	1.35	=PB	3.14	3.15	3.22
		velocity	5.14	8.54	8.87	8.79	8.62	8.79	8.70	8.62	8.46	8.46	10.39	8.13	8.73	8.70	8.52
Takayama, Shunya (JPN) (1994)	time	2.61	3.69	4.74	5.78	6.83	7.89	8.94	10.00	11.07	12.17		13.55	6 / 3			
reaction time	0.150	interval		1.08	1.05	1.04	1.05	1.06	1.05	1.06	1.07	1.10	1.38		3.17	3.16	3.23
		velocity	5.26	8.46	8.70	8.79	8.70	8.62	8.70	8.62	8.54	8.31	10.16	8.12	8.65	8.68	8.49
Furuya, Takumu (JPN) (1994)	time	2.66	3.73	4.77	5.82	6.86	7.92	8.98	10.04	11.12	12.22		13.61	2 / 4			
reaction time	0.133	interval		1.07	1.04	1.05	1.04	1.06	1.06	1.08	1.10	1.39	PB		3.16	3.16	3.24
		velocity	5.16	8.54	8.79	8.70	8.79	8.62	8.62	8.62	8.46	8.31	10.09	8.08	8.68	8.68	8.46
Kuriki, Anthony Tyrell (JPN) (1991)	time	2.64	3.71	4.78	5.83	6.89	7.95	9.01	10.09	11.17	12.25		13.64	3 / 5			
reaction time	0.134	interval	7 steps	1.07	1.07	1.05	1.06	1.06	1.08	1.08	1.08	1.39			3.19	3.18	3.24
		velocity	5.20	8.54	8.54	8.70	8.62	8.62	8.46	8.46	8.46	10.09	8.06		8.60	8.62	8.46
Kim Byung-Jun (KOR) (1991)	time	2.64	3.73	4.80	5.85	6.91	7.97	9.04	10.15	11.23	12.34		13.85	4 / 6			
reaction time	0.162	interval		1.09	1.07	1.05	1.06	1.06	1.07	1.11	1.08	1.11	1.51		3.21	3.19	3.30
		velocity	5.20	8.39	8.54	8.70	8.62	8.62	8.54	8.23	8.46	8.23	9.28	7.94	8.54	8.60	8.31

2018 Chinese National U20 Championships (Nanchang, CHN) (99.1cm)

FINAL

date 13-May-18 wind 1.0 m/s

Zhang (2019) - comparative analysis of characteristics of Chinese elite men's 110m hurdles in different age groups

		H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
Ning Xiaohan (CHN) (2000)	time	2.63	3.69	4.74	5.79	6.84	7.89	8.93	10.00	11.07	12.17		13.40	/ 1			
reaction time		interval	1.06	1.05	1.05	1.05	1.05	1.04	1.07	1.07	1.10	1.23			3.16	3.14	3.24
		velocity	5.22	8.62	8.70	8.70	8.70	8.79	8.54	8.54	8.31	11.40	8.21		8.68	8.73	8.46

Shi Guoxing (CHN) (2000)	time	2.72	3.82	4.89	5.97	7.01	8.07	9.12	10.19	11.29	12.38		13.81	/ 2			
	reaction time	interval	1.10	1.07	1.08	1.04	1.06	1.05	1.07	1.10	1.09	1.43			3.25	3.15	3.26
	velocity	5.04	8.31	8.54	8.46	8.79	8.62	8.70	8.54	8.31	8.39	9.80	7.97		8.44	8.70	8.41
Zhu Shenglong (CHN) (2000)	time	2.69	3.79	4.85	5.90	6.94	8.00	9.06	10.15	11.24	12.36		13.83	/ 3			
	reaction time	interval	1.10	1.06	1.05	1.04	1.06	1.06	1.09	1.09	1.12	1.47			3.21	3.16	3.30
	velocity	5.10	8.31	8.62	8.70	8.79	8.62	8.62	8.39	8.39	8.16	9.54	7.95		8.54	8.68	8.31
Sun Zhenjiang (CHN) (1999)	time	2.61	3.70	4.76	5.83	6.92	7.99	9.08	10.18	11.30	12.42		13.96	/ 4			
	reaction time	interval	7 steps	1.09	1.06	1.07	1.09	1.07	1.09	1.10	1.12	1.12	1.54		3.22	3.25	3.34
	velocity	5.26	8.39	8.62	8.54	8.39	8.54	8.39	8.31	8.16	8.16	9.10	7.88		8.52	8.44	8.21
Chen Mingjie (CHN) (2000)	time	2.24	3.41	4.52	5.61	6.69	7.77	8.86	9.94	11.06	12.22		14.12	/ 5			
	reaction time	interval	1.17	1.11	1.09	1.08	1.08	1.09	1.08	1.12	1.16	1.90			3.37	3.25	3.36
	velocity	6.13	7.81	8.23	8.39	8.46	8.46	8.39	8.46	8.16	7.88	7.38	7.79		8.14	8.44	8.16
Yang Quanlei (CHN) (1999)	time	2.70	3.82	4.90	5.98	7.06	8.15	9.22	10.32	11.44	12.57		14.13	/ 6			
	reaction time	interval	1.12	1.08	1.08	1.08	1.09	1.07	1.10	1.12	1.13	1.56			3.28	3.24	3.35
	velocity	5.08	8.16	8.46	8.46	8.46	8.39	8.54	8.31	8.16	8.09	8.99	7.78		8.36	8.46	8.19

2018 Shanghai Diamond League (Shanghai, CHN) (TV Analysis)

FINAL

INAL			date	12-May-18	wind				0.2 m/s				Henson (2020) - Athlete First: 2018 year end hurdle report									
					H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10		
McLeod, Omar (JAM) (1994)	time				2.52	3.58	4.60	5.60	6.60	7.62	8.64	9.66	10.68	11.74		13.16	5 / 1					
	reaction time	0.119		interval		1.06	1.02	1.00	1.00	1.02	1.02	1.02	1.02	1.06	1.42			3.08	3.04	3.10		
			velocity		5.44	8.62	8.96	9.14	9.14	8.96	8.96	8.96	8.96	8.62	9.87	8.36		8.90	9.02	8.85		
Ortega, Orlando (ESP) (1991)	time				2.52	3.58	4.60	5.60	6.62	7.64	8.64	9.64	10.68	11.72		13.17	8 / 2					
	reaction time	0.135		interval	7 steps	1.06	1.02	1.00	1.02	1.02	1.00	1.00	1.04	1.04	1.45			3.08	3.04	3.08		
			velocity		5.44	8.62	8.96	9.14	8.96	8.96	9.14	9.14	8.79	8.79	9.67	8.35		8.90	9.02	8.90		
Shubenkov, Sergey (ANA) (1991)	time				2.52	3.58	4.62	5.66	6.68	7.70	8.72	9.76	10.78	11.82		13.27	6 / 3					
	reaction time	0.148		interval	7 steps	1.06	1.04	1.04	1.02	1.02	1.02	1.04	1.02	1.04	1.45			3.14	3.06	3.10		
			velocity		5.44	8.62	8.79	8.79	8.96	8.96	8.96	8.79	8.96	8.79	9.67	8.29		8.73	8.96	8.85		
Levy, Ronald (JAM) (1992)	time				2.52	3.56	4.60	5.62	6.66	7.70	8.74	9.78	10.82	11.88		13.33	4 / 4					
	reaction time	0.122		interval	7 steps	1.04	1.04	1.02	1.04	1.04	1.04	1.04	1.04	1.06	1.45			3.10	3.12	3.14		
			velocity		5.44	8.79	8.79	8.96	8.79	8.79	8.79	8.79	8.79	8.62	9.67	8.25		8.85	8.79	8.73		
Parchment, Hansle (JAM) (1991)	time				2.58	3.64	4.68	5.70	6.74	7.74	8.78	9.82	10.86	11.94		13.48	2 / 5					
	reaction time	0.137		interval	7 steps	1.06	1.04	1.02	1.04	1.00	1.04	1.04	1.04	1.08	1.54			3.12	3.08	3.16		
			velocity		5.32	8.62	8.79	8.96	8.79	9.14	8.79	8.79	8.79	8.46	9.10	8.16		8.79	8.90	8.68		
Merritt, Aries (USA) (1985)	time				2.48	3.54	4.60	5.64	6.68	7.72	8.84	9.92	11.00	12.10		13.65	7 / 6					
	reaction time	0.126		interval	7 steps	1.06	1.06	1.04	1.04	1.04	1.12	1.08	1.08	1.10	1.55			3.16	3.20	3.26		
			velocity		5.53	8.62	8.62	8.79	8.79	8.79	8.16	8.46	8.46	8.31	9.05	8.06		8.68	8.57	8.41		
Darien, Garfield (FRA) (1987)	time				2.56	3.62	4.68	5.74	6.76	7.84	8.92	10.00	11.08	12.20		13.74	3 / 7					
	reaction time	0.144		interval	7 steps	1.06	1.06	1.06	1.02	1.08	1.08	1.08	1.08	1.12	1.54			3.18	3.18	3.28		
			velocity		5.36	8.62	8.62	8.62	8.96	8.46	8.46	8.46	8.46	8.16	9.10	8.01		8.62	8.62	8.36		
Zeng Jianhang (CHN) (1998)	time				2.56	3.60	4.64	5.68	6.72	7.78	8.84	9.92	11.00	12.16		13.76	1 / 8					
	reaction time	0.148		interval		1.04	1.04	1.04	1.04	1.06	1.06	1.08	1.08	1.16	1.60	PB		3.12	3.16	3.32		
			velocity		5.36	8.79	8.79	8.79	8.79	8.62	8.62	8.46	8.46	7.88	8.76	7.99		8.79	8.68	8.26		
Manga, Aurel (FRA) (1992)	time				2.52	3.60	4.70	5.76	6.82	7.90	8.96	10.04	11.16	12.28		13.86	9 / 9					
	reaction time	0.147		interval	7 steps	1.08	1.10	1.06	1.06	1.08	1.06	1.08	1.12	1.12	1.58			3.24	3.20	3.32		
			velocity		5.44	8.46	8.31	8.62	8.62	8.46	8.62	8.46	8.16	8.16	8.87	7.94		8.46	8.57	8.26		

2018 Kinami Michitaka Memorial (Osaka, JPN)

FINAL

FINAL	date 06-May-18		wind 1.6 m/s		Shibayama (2018) - Race analysis of men's 110m hurdle run in 2018 season														
			H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10	
Masuno, Genta (JPN) (1993)	time		2.61	3.66	4.73	5.77	6.83	7.87	8.91	10.00	11.10	12.20		13.64	7 / 1				
	reaction time	interval		1.05	1.07	1.04	1.06	1.04	1.04	1.09	1.10	1.10	1.44			3.16	3.14	3.29	
		velocity	5.26	8.70	8.54	8.79	8.62	8.79	8.79	8.39	8.31	8.31	9.74	8.06		8.68	8.73	8.33	
Sato, Hiroyuki (JPN) (1990)	time		2.58	3.67	4.76	5.82	6.87	7.93	9.00	10.08	11.18	12.28		13.72	4 / 2				
	reaction time	interval		1.09	1.09	1.06	1.05	1.06	1.07	1.08	1.10	1.10	1.44			3.24	3.18	3.28	
		velocity	5.32	8.39	8.39	8.62	8.70	8.62	8.54	8.46	8.31	8.31	9.74	8.02		8.46	8.62	8.36	
Takeyoshi, Daiki (JPN) (1994)	time		2.65	3.72	4.79	5.85	6.90	7.95	9.03	10.12	11.22	12.34		13.83	5 / 3				
	reaction time	interval		1.07	1.07	1.06	1.05	1.05	1.08	1.09	1.10	1.12	1.49			3.20	3.18	3.31	
		velocity	5.18	8.54	8.54	8.62	8.70	8.70	8.46	8.39	8.31	8.16	9.41	7.95		8.57	8.62	8.28	
Kagimoto, Masahiro (JPN) (1991)	time		2.65	3.72	4.80	5.86	6.94	8.00	9.10	10.20	11.33	12.47		13.91	8 / 4				
	reaction time	interval		1.07	1.08	1.06	1.08	1.06	1.10	1.10	1.13	1.14	1.44			3.21	3.24	3.37	
		velocity																	

	velocity	5.18	8.54	8.46	8.62	8.46	8.62	8.31	8.31	8.09	8.02	9.74	7.91		8.54	8.46	8.14
Tanaka, Shinya (JPN) (1985)	time	2.63	3.78	4.87	5.95	7.03	8.10	9.16	10.28	11.38	12.50		13.96	6 / 5			
	reaction time		1.15	1.09	1.08	1.08	1.07	1.06	1.12	1.10	1.12	1.46			3.32	3.21	3.34
	interval														8.26	8.54	8.21
	velocity	5.22	7.95	8.39	8.46	8.46	8.54	8.62	8.16	8.31	8.16	9.60	7.88				
Hatanaka, Yushi (JPN) (1994)	time	2.65	3.76	4.88	5.98	7.07	8.16	9.28	10.40	11.53	12.65		14.11	3 / 6			
	reaction time		1.11	1.12	1.10	1.09	1.09	1.12	1.12	1.13	1.12	1.46			3.33	3.30	3.37
	interval														8.23	8.31	8.14
	velocity	5.18	8.23	8.16	8.31	8.39	8.39	8.16	8.16	8.09	8.16	9.60	7.80				
Maeda, Koki (JPN)	time	2.65	3.75	4.82	5.91	6.99	8.09	9.18	10.31	11.45	12.59		14.11	9 / 7			
	reaction time		1.10	1.07	1.09	1.08	1.10	1.09	1.13	1.14	1.14	1.52			3.26	3.27	3.41
	interval														8.41	8.39	8.04
	velocity	5.18	8.31	8.54	8.39	8.46	8.31	8.39	8.09	8.02	8.02	9.22	7.80				
Kawamura, Naoya (JPN) (1996)	time	2.73	3.84	4.94	6.05	7.13	8.23	9.32	10.43	11.53	12.67		14.14	2 / 8			
	reaction time		1.11	1.10	1.11	1.08	1.10	1.09	1.11	1.10	1.14	1.47			3.32	3.27	3.35
	interval														8.26	8.39	8.19
	velocity	5.03	8.23	8.31	8.23	8.46	8.31	8.39	8.23	8.31	8.02	9.54	7.78				

2018 Mikio Oda Memorial (Hiroshima, JPN)**FINAL**

date 29-Apr-18 wind 2.6 m/s Shibayama (2018) - Race analysis of men's 110m hurdle run in 2018 season

		H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
Kanai, Taio (JPN) (1995)	time	2.60	3.66	4.71	5.75	6.78	7.83	8.90	9.96	11.05	12.13		13.52	6 / 1			
	reaction time		1.06	1.05	1.04	1.03	1.05	1.07	1.06	1.09	1.08	1.39			3.15	3.15	3.23
	interval														8.70	8.70	8.49
	velocity	5.28	8.62	8.70	8.79	8.87	8.70	8.54	8.62	8.39	8.46	10.09	8.14				
Furuya, Takumu (JPN) (1997)	time	2.67	3.74	4.77	5.81	6.87	7.91	8.96	10.02	11.10	12.18		13.57	2 / 2			
	reaction time		1.07	1.03	1.04	1.06	1.04	1.05	1.06	1.08	1.08	1.39			3.14	3.15	3.22
	interval														8.73	8.70	8.52
	velocity	5.14	8.54	8.87	8.79	8.62	8.79	8.70	8.62	8.46	8.46	10.09	8.11				
Omuro, Hideki (JPN) (1990)	time	2.61	3.69	4.74	5.78	6.83	7.89	8.94	10.00	11.07	12.17		13.57	3 / 3			
	reaction time		1.08	1.05	1.04	1.05	1.06	1.05	1.06	1.07	1.10	1.40			3.17	3.16	3.23
	interval														8.65	8.68	8.49
	velocity	5.26	8.46	8.70	8.79	8.70	8.62	8.70	8.62	8.54	8.31	10.01	8.11				
Kuriki, Anthony Tyrell (JPN) (1991)	time	2.66	3.73	4.77	5.82	6.86	7.92	8.98	10.04	11.12	12.22		13.64	4 / 4			
	reaction time		1.07	1.04	1.05	1.04	1.06	1.06	1.06	1.08	1.10	1.42			3.16	3.16	3.24
	interval														8.68	8.68	8.46
	velocity	5.16	8.54	8.79	8.70	8.79	8.62	8.62	8.62	8.46	8.31	9.87	8.06				
Takayama, Shunya (JPN) (1994)	time	2.64	3.71	4.78	5.83	6.89	7.95	9.01	10.09	11.20	12.32		13.76	5 / 5			
	reaction time		1.07	1.07	1.05	1.06	1.06	1.06	1.08	1.11	1.12	1.44			3.19	3.18	3.31
	interval														8.60	8.62	8.28
	velocity	5.20	8.54	8.54	8.70	8.62	8.62	8.62	8.46	8.23	8.16	9.74	7.99				
Sato, Hiroyuki (JPN) (1990)	time	2.64	3.72	4.79	5.84	6.90	7.97	9.05	10.15	11.27	12.39		13.82	8 / 6			
	reaction time		1.08	1.07	1.05	1.06	1.07	1.08	1.10	1.12	1.12	1.43			3.20	3.21	3.34
	interval														8.57	8.54	8.21
	velocity	5.20	8.46	8.54	8.70	8.62	8.54	8.46	8.31	8.16	8.16	9.80	7.96				
Ishikawa, Shuhei (JPN) (1995)	time	2.69	3.77	4.81	5.87	6.93	8.00	9.08	10.18	11.29	12.41		13.85	1 / 7			
	reaction time		1.08	1.04	1.06	1.06	1.07	1.08	1.10	1.11	1.12	1.44			3.18	3.21	3.33
	interval														8.62	8.54	8.23
	velocity	5.10	8.46	8.79	8.62	8.62	8.54	8.46	8.31	8.23	8.16	9.74	7.94				
Masuno, Genta (JPN) (1993)	time	2.57	3.78	4.91	5.97	7.03	8.09	9.15	10.22	11.32	12.40		13.88	7 / 8			
	reaction time		1.21	1.13	1.06	1.06	1.06	1.06	1.07	1.10	1.08	1.48			3.40	3.18	3.25
	interval														8.06	8.62	8.44
	velocity	5.34	7.55	8.09	8.62	8.62	8.62	8.62	8.54	8.31	8.46	9.47	7.93				

Heat 3date 29-Apr-18 wind 2.2 m/s (2018.05.01) - https://twitter.com/touchdown_time/media?lang=en&lang=en&lang=en&lang=en

		H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
Furuya, Takumu (JPN) (1997)	time	2.65	3.71	4.77	5.83	6.88	7.95	9.00	10.07	11.15	12.24		13.70	2 / 3			
	reaction time		1.06	1.06	1.06	1.05	1.07	1.05	1.07	1.08	1.09	1.46			3.18	3.17	3.24
	interval														8.62	8.65	8.46
	velocity	5.18	8.62	8.62	8.62	8.70	8.54	8.70	8.54	8.46	8.39	9.60	8.03				

2018 Tokyo Combined Event Meeting (Tokyo, JPN)**Decathlon - Heat 2**

date 22-Apr-18 wind 1.1 m/s Matsubayashi (2018) - research on athlete performance and technique- 2018 data book

		H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
Nakamura, Akihiko (JPN) (1990)	time	2.73	3.85	4.96	6.06	7.16	8.26	9.36	10.49	11.62	12.76		14.22	5 / 1			
	reaction time		1.12	1.11	1.10	1.10	1.10	1.10	1.13	1.13	1.14	1.46			3.33	3.30	3.40
	interval														8.23	8.31	8.06
	velocity	5.03	8.16	8.23	8.31	8.31	8.31	8.31	8.09	8.09	8.02	9.60	7.74				
Shlmizu, Tsuyoshi (JPN) (1993)	time	2.75	3.92	5.09	6.27	7.45	8.63	9.83	11.07	12.32	13.60		15.27	2 / 6			
	reaction time		1.17	1.17	1.18	1.18	1.18	1.20	1.24	1.25	1.28	1.67			3.52	3.56	3.77
	interval														7.79	7.70	7.27
	velocity	4.99	7.81	7.81	7.75	7.75	7.75	7.62	7.37	7.31	7.14	8.40	7.20				

Decathlon - Heat 1

date 22-Apr-18 wind 1.8 m/s Matsubayashi (2018) - research on athlete performance and technique- 2018 data book

		H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
Ushiro, Keisuke (JPN) (1986)	time	2.73	3.90	5.05	6.24	7.39	8.58	9.76	10.94	12.12	13.31		14.92	6 / 3			
	reaction time		1.17	1.15	1.19	1.15	1.19	1.18	1.18	1.18	1.19	1.61			3.51	3.52	3.55
	interval														7.81	7.79	7.72
	velocity	5.03	7.81	7.95	7.68	7.95	7.68	7.75	7.75	7.75	7.68	8.71	7.37				

2018 IAAF World Indoor Championships (Birmingham, GBR)**FINAL**

date 04-Mar-18

Walker (2019) - Biomechanical Report for the IAAF World Indoor Championships 2018: 60 Metres Hurdles Men

		H1	H2	H3	H4	H5	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
Pozzi, Andy (GBR) (1992)	time	2.51	3.54	4.59	5.58	6.57		7.46	4 / 1			
	reaction time	0.143	interval	7 steps	1.03	1.05	0.99	0.99		3.07		
	velocity	5.47	8.87	8.70	9.23	9.23	0.89	8.04		8.93		
Eaton, Jarret (USA) (1989)	time	2.47	3.49	4.51	5.50	6.52		7.47	7 / 2			
	reaction time	0.129	interval	7 steps	1.02	1.02	0.99	1.02		3.03		
	velocity	5.55	8.96	8.96	9.23	8.96	0.95	8.03		9.05		
Manga, Aurel (FRA) (1992)	time	2.57	3.63	4.68	5.69	6.68		7.54	5 / 3			
	reaction time	0.169	interval	7 steps	1.06	1.05	1.01	0.99		3.12		
	velocity	5.34	8.62	8.70	9.05	9.23	0.86	7.96		8.79		
Merritt, Aries (USA) (1985)	time	2.53	3.58	4.62	5.63	6.64		7.56	1 / 4			
	reaction time	0.141	interval	7 steps	1.05	1.04	1.01	1.01		3.10		
	velocity	5.42	8.70	8.79	9.05	9.05	0.92	7.94		8.85		
Martinot-Lagarde, Pascal (FRA)	time	2.55	3.61	4.70	5.74	6.77		7.68	3 / 5			
	reaction time	0.154	interval	7 steps	1.06	1.09	1.04	1.03		3.19		
	velocity	5.38	8.62	8.39	8.79	8.87	0.91	7.81		8.60		
Constantino, Gabriel (BRA) (1991)	time	2.58	3.65	4.71	5.74	6.79		7.71	2 / 6			
	reaction time	0.176	interval	7 steps	1.07	1.06	1.03	1.05		3.16		
	velocity	5.32	8.54	8.62	8.87	8.70	0.92	7.78		8.68		
Iribarne, Roger (CUB) (1996)	time	2.52	3.58	4.68	5.77	6.83		7.77	8 / 7			
	reaction time	0.172	interval	7 steps	1.06	1.10	1.09	1.06		3.25		
	velocity	5.44	8.62	8.31	8.39	8.62	0.94	7.72		8.44		
Trajkovic, Milan (CYP) (1992)		0.084						DQ	6 / --			

2018 Müller Indoor Grand Prix (Glasgow, GBR)**FINAL**

date 25-Feb-18

(2018.03.04) - https://twitter.com/touchdown_time/media?lang=en&lang=en&lang=en&lang=en

		H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
Levy, Ronald (JAM) (1992)	time	2.54	3.59	4.61	5.63	6.63							7.49	1 / 1			
	reaction time	0.148	interval		1.05	1.02	1.02	1.00				0.86	PB		3.09		
	velocity	5.40	8.70	8.96	8.96	9.14						11.30	8.01		8.87		

2018 Japanese National Indoor U20 Championships (Osaka, JPN) (99cm)**FINAL**

date 03-Feb-18

(2018.03.04) - https://twitter.com/touchdown_time/media?lang=en&lang=en&lang=en&lang=en

		H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
Higuchi, Rikuto (JPN) (1999)	time	2.62	3.71	4.75	5.77	6.81							7.71	4 / 1			
	reaction time		interval		1.09	1.04	1.02	1.04				0.90	NJR		3.15		
	velocity	5.24	8.39	8.79	8.96	8.79						10.80	7.78		8.70		

2017 National Sports Festival (Matsuyama, JPN) (99cm)**U19 FINAL**

date 08-Oct-17

wind -0.1 m/s

(2017.10.10) - https://twitter.com/touchdown_time/media?lang=en&lang=en&lang=en&lang=en

		H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
Higuchi, Rikuto (JPN) (1999)	time	2.62	3.68	4.70	5.74	6.77	7.81	8.84	9.87	10.93	11.99		13.40	5 / 1			
	reaction time	0.172	interval		1.06	1.02	1.04	1.03	1.03	1.06	1.06	1.41	PB		3.12	3.10	3.15
	velocity	5.24	8.62	8.96	8.79	8.87	8.79	8.87	8.87	8.62	8.62	9.94	7.46		8.79	8.85	8.70

2017 Chinese National Games (Tianjin, CHN)**FINAL**

date 03-Sep-17

wind 0.3 m/s

Henson (2017) - coaching observations

		H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
Xie Wunjun (CHN) (1990)	time	2.68	3.75	4.79	5.80	6.82	7.86	8.89	9.92	10.99	12.07		13.52	6 / 1			
	reaction time		interval		1.07	1.04	1.01	1.02	1.04	1.03	1.03	1.07	1.08	1.45	3.12	3.09	3.18
	velocity	5.12	8.54	8.79	9.05	8.96	8.79	8.87	8.87	8.54	8.46	9.67	8.14		8.79	8.87	8.62
Jiang Fan (CHN) (1990)	time	2.65	3.72	4.79	5.84	6.89	7.94	9.02	10.11	11.20	12.31		13.78	2 / 2			
	reaction time		interval		1.07	1.07	1.05	1.05	1.08	1.09	1.09	1.11	1.47		3.19	3.18	3.29
	velocity	5.18	8.54	8.54	8.70	8.70	8.70	8.46	8.39	8.39	8.23	9.54	7.98		8.60	8.62	8.33
Zhang Honglin (CHN) (1994)	time	2.67	3.75	4.82	5.87	6.94	7.99	9.09	10.16	11.26	12.36		13.78	4 / 3			
	reaction time		interval		1.08	1.07	1.05	1.07	1.10	1.07	1.10	1.10	1.42		3.20	3.22	3.27
	velocity	5.14	8.46	8.54	8.70	8.54	8.70	8.31	8.54	8.31	8.31	9.87	7.98		8.57	8.52	8.39
Zhao Pengchuan (CHN) (1993)	time	2.63	3.74	4.82	5.89	6.97	8.04	9.12	10.22	11.32	12.46		13.94	3 / 4			
	reaction time		interval	7 steps	1.11	1.08	1.07	1.08	1.07	1.10	1.10	1.14	1.48		3.26	3.23	3.34
	velocity	5.22	8.23	8.46	8.54	8.46	8.54	8.46	8.31	8.31	8.02	9.47	7.89		8.41	8.49	8.21
Ma Lei (CHN) (1989)	time	2.72	3.81	4.87	5.94	6.99	8.07	9.13	10.23	11.36	12.48		13.96	7 / 5			
	reaction time		interval	7 steps	1.09	1.06	1.07	1.05	1.08	1.06	1.10	1.13	1.48		3.22	3.19	3.35

	velocity	5.04	8.39	8.62	8.54	8.70	8.46	8.62	8.31	8.09	8.16	9.47	7.88		8.52	8.60	8.19
Ji Wei (CHN) (1984)	time	2.73	3.84	4.92	5.99	7.03	8.08	9.16	10.26	11.36	12.50		14.01	9 / 6			
	reaction time																
	interval		1.11	1.08	1.07	1.04	1.05	1.08	1.10	1.10	1.14	1.51			3.26	3.17	3.34
	velocity	5.03	8.23	8.46	8.54	8.79	8.70	8.46	8.31	8.31	8.02	9.28	7.85		8.41	8.65	8.21
Chu Pengfei (CHN) (1993)	time	2.71	3.83	4.89	5.97	7.02	8.09	9.21	10.32	11.43	12.54		14.01	5 / 7			
	reaction time																
	interval		1.12	1.06	1.08	1.05	1.07	1.12	1.11	1.11	1.11	1.47			3.26	3.24	3.33
	velocity	5.06	8.16	8.62	8.46	8.70	8.54	8.16	8.23	8.23	8.23	9.54	7.85		8.41	8.46	8.23
Li Haifeng (CHN) (1989)	time	2.70	3.80	4.91	5.99	7.05	8.15	9.24	10.34	11.48	12.61		14.05	8 / 8			
	reaction time																
	interval		1.10	1.11	1.08	1.06	1.10	1.09	1.10	1.14	1.13	1.44			3.29	3.25	3.37
	velocity	5.08	8.31	8.23	8.46	8.62	8.31	8.39	8.31	8.02	8.09	9.74	7.83		8.33	8.44	8.14

2017 Memorial van Damme (Brussels, BEL) (TV Analysis)**FINAL**

date 01-Sep-17		wind 0.5 m/s												Henson (2020) - Athlete First: 2017 year end hurdle report			
		H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
Shubenkov, Sergey (ANA) (1991)	time	2.56	3.60	4.60	5.60	6.58	7.56	8.58	9.58	10.62	11.66		13.14	5 / 1			
	reaction time	0.146															
	interval		7 steps	1.04	1.00	0.98	0.98	1.02	1.00	1.04	1.04	1.48			3.04	2.98	3.08
	velocity	5.36	8.79	9.14	9.14	9.33	9.33	8.96	9.14	8.79	8.79	9.47	8.37		9.02	9.20	8.90
Ortega, Orlando (ESP) (1991)	time	2.54	3.62	4.62	5.62	6.62	7.64	8.64	9.66	10.68	11.72		13.17	7 / 2			
	reaction time	0.102															
	interval		7 steps	1.08	1.00	1.00	1.02	1.00	1.02	1.02	1.04	1.45			3.08	3.02	3.08
	velocity	5.40	8.46	9.14	9.14	9.14	8.96	9.14	8.96	8.96	8.79	9.67	8.35		8.90	9.08	8.90
Merritt, Aries (USA) (1985)	time	2.48	3.52	4.54	5.54	6.54	7.54	8.58	9.60	10.62	11.68		13.20	6 / 3			
	reaction time	0.133															
	interval		7 steps	1.04	1.02	1.00	1.00	1.04	1.02	1.02	1.06	1.52			3.06	3.04	3.10
	velocity	5.53	8.79	8.96	9.14	9.14	9.14	8.79	8.96	8.96	8.62	9.22	8.33		8.96	9.02	8.85
Allen, Devon (USA) (1994)	time	2.52	3.58	4.62	5.62	6.66	7.68	8.72	9.74	10.78	11.82		13.24	4 / 4			
	reaction time	0.122															
	interval			1.06	1.04	1.00	1.04	1.02	1.04	1.02	1.04	1.42			3.10	3.10	3.10
	velocity	5.44	8.62	8.79	9.14	8.79	8.96	8.79	8.96	8.79	8.79	9.87	8.31		8.85	8.85	8.85
Levy, Ronald (JAM) (1992)	time	2.54	3.58	4.62	5.62	6.64	7.68	8.72	9.78	10.86	11.92		13.41	8 / 5			
	reaction time	0.108															
	interval		7 steps	1.04	1.04	1.00	1.02	1.04	1.06	1.08	1.06	1.49			3.08	3.10	3.20
	velocity	5.40	8.79	8.79	9.14	8.96	8.79	8.79	8.62	8.46	8.62	9.41	8.20		8.90	8.85	8.57
Darien, Garfield (FRA) (1987)	time	2.54	3.62	4.66	5.68	6.70	7.72	8.76	9.80	10.86	11.92		13.42	9 / 6			
	reaction time	0.138															
	interval		7 steps	1.08	1.04	1.02	1.02	1.04	1.04	1.06	1.06	1.50			3.14	3.08	3.16
	velocity	5.40	8.46	8.79	8.96	8.96	8.96	8.79	8.79	8.62	8.62	9.35	8.20		8.73	8.90	8.68
Trajkovic, Milan (CYP) (1992)	time	2.52	3.58	4.62	5.64	6.68	7.72	8.76	9.82	10.88	11.96		13.47	2 / 7			
	reaction time	0.142															
	interval		7 steps	1.06	1.04	1.02	1.04	1.04	1.06	1.06	1.08	1.51			3.12	3.12	3.20
	velocity	5.44	8.62	8.79	8.96	8.79	8.79	8.79	8.62	8.62	8.46	9.28	8.17		8.79	8.79	8.57
Brathwaite, Shane (BAR) (1990)	time	2.54	3.60	4.64	5.66	6.70	7.74	8.78	9.84	10.92	12.00		13.49	3 / 8			
	reaction time	0.141															
	interval		7 steps	1.06	1.04	1.02	1.04	1.04	1.06	1.08	1.08	1.49			3.12	3.12	3.22
	velocity	5.40	8.62	8.79	8.96	8.79	8.79	8.79	8.62	8.46	8.46	9.41	8.15		8.79	8.79	8.52

2017 Müller Grand Prix (Birmingham, GBR) (TV Analysis)**FINAL**

date 20-Aug-17		wind -0.6 m/s												Henson (2020) - Athlete First: 2017 year end hurdle report			
		H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
Merritt, Aries (USA) (1985)	time	2.48	3.52	4.54	5.52	6.56	7.60	8.62	9.66	10.70	11.78		13.29	6 / 1			
	reaction time	0.127															
	interval		7 steps	1.04	1.02	0.98	1.04	1.02	1.04	1.04	1.08	1.51			3.04	3.10	3.16
	velocity	5.53	8.79	8.96	9.33	8.79	8.79	8.96	8.79	8.79	8.46	9.28	8.28		9.02	8.85	8.68
Shubenkov, Sergey (ANA) (1991)	time	2.54	3.58	4.58	5.60	6.62	7.64	8.66	9.72	10.78	11.82		13.31	7 / 2			
	reaction time	0.133															
	interval		7 steps	1.04	1.00	1.02	1.02	1.02	1.06	1.06	1.04	1.49			3.06	3.06	3.16
	velocity	5.40	8.79	9.14	8.96	8.96	8.96	8.96	8.62	8.62	8.79	9.41	8.26		8.96	8.96	8.68
Allen, Devon (USA) (1994)	time	2.52	3.56	4.58	5.58	6.62	7.62	8.68	9.74	10.78	11.86		13.40	9 / 3			
	reaction time	0.119															
	interval			1.04	1.02	1.00	1.04	1.00	1.06	1.06	1.04	1.08	1.54		3.06	3.10	3.18
	velocity	5.44	8.79	8.96	9.14	8.79	9.14	8.62	8.62	8.79	8.46	9.10	8.21		8.96	8.85	8.62
Baji, Balázs (HUN) (1989)	time	2.48	3.56	4.60	5.64	6.64	7.66	8.70	9.74	10.82	11.90		13.47	8 / 4			
	reaction time	0.132															
	interval		7 steps	1.08	1.04	1.04	1.00	1.02	1.04	1.04	1.08	1.08	1.57		3.16	3.06	3.20
	velocity	5.53	8.46	8.79	8.79	9.14	8.96	8.79	8.79	8.46	8.46	8.93	8.17		8.68	8.96	8.57
Ortega, Orlando (ESP) (1991)	time	2.56	3.62	4.68	5.72	6.74	7.78	8.82	9.86	10.92	12.02		13.48	4 / 5			
	reaction time	0.140															
	interval			1.06	1.06	1.04	1.02	1.04	1.04	1.06	1.10	1.46			3.16	3.10	3.20
	velocity	5.36	8.62	8.62	8.79	8.96	8.79	8.79	8.79	8.62	8.31	9.60	8.16		8.68	8.85	8.57
Pozzi, Andy (GBR) (1992)	time	2.50	3.54	4.56	5.58	6.62	7.70	8.74	9.80	10.88	11.98		13.53	3 / 6			
	reaction time	0.119															
	interval		7 steps	1.04	1.02	1.02	1.04	1.08	1.04	1.06	1.08	1.10	1.55		3.08	3.16	3.24
	velocity	5.49	8.79	8.96	8.96	8.79	8.46	8.79	8.62	8.46	8.31	9.05	8.13		8.90	8.68	8.46

King, David (GBR) (19940)	time	2.58	3.64	4.70	5.76	6.82	7.88	8.92	10.00	11.04	12.12		13.65	1 / 7			
reaction time	0.138	interval	7 steps	1.06	1.06	1.06	1.06	1.06	1.04	1.08	1.04	1.08	1.53		3.18	3.16	3.20
		velocity		5.32	8.62	8.62	8.62	8.62	8.79	8.46	8.79	8.46	9.16	8.06	8.62	8.68	8.57

2017 IAAF World Championships (London, GBR)

FINAL

date 07-Aug-17 wind 0.0 m/s

Pollitt (2018) - biomechanical report for the IAAF World Champs - London 2017 - Men's 110m Hurdles

		H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
McLeod, Omar (JAM) (1994)	time	2.36	3.42	4.42	5.43	6.42	7.44	8.45	9.46	10.48	11.51		13.04	4 / 1			
reaction time	0.123	interval		1.06	1.00	1.01	0.99	1.02	1.01	1.01	1.02	1.03	1.53		3.07	3.02	3.06
		velocity		5.81	8.62	9.14	9.05	9.23	8.96	9.05	9.05	8.96	8.87	8.44	8.93	9.08	8.96
Shubenkov, Sergey (ANA) (199)	time	2.41	3.45	4.45	5.46	6.45	7.46	8.47	9.48	10.52	11.57		13.14	2 / 2			
reaction time	0.132	interval	7 steps	1.04	1.00	1.01	0.99	1.01	1.01	1.01	1.04	1.05	1.57		3.05	3.01	3.10
		velocity		5.69	8.79	9.14	9.05	9.23	9.05	9.05	8.79	8.70	8.93	8.37	8.99	9.11	8.85
Baji, Balázs (HUN) (1989)	time	2.39	3.47	4.49	5.52	6.54	7.56	8.58	9.61	10.66	11.71		13.28	5 / 3			
reaction time	0.122	interval	7 steps	1.08	1.02	1.03	1.02	1.02	1.02	1.03	1.05	1.05	1.57		3.13	3.06	3.13
		velocity		5.74	8.46	8.96	8.87	8.96	8.96	8.87	8.70	8.70	8.93	8.28	8.76	8.96	8.76
Darien, Garfield (FRA) (1987)	time	2.38	3.44	4.46	5.48	6.52	7.54	8.56	9.58	10.62	11.72		13.30	6 / 4			
reaction time	0.115	interval	7 steps	1.06	1.02	1.02	1.04	1.02	1.02	1.02	1.04	1.10	1.58		3.10	3.08	3.16
		velocity		5.76	8.62	8.96	8.96	8.79	8.96	8.96	8.79	8.31	8.87	8.27	8.85	8.90	8.68
Merritt, Aries (USA) (1985)	time	2.43	3.48	4.50	5.51	6.53	7.55	8.57	9.60	10.66	11.72		13.31	9 / 5			
reaction time	0.163	interval	7 steps	1.05	1.02	1.01	1.02	1.02	1.03	1.06	1.06	1.59			3.08	3.06	3.15
		velocity		5.65	8.70	8.96	9.05	8.96	8.96	8.87	8.62	8.62	8.82	8.26	8.90	8.96	8.70
Brathwaite, Shane (BAR) (1990)	time	2.43	3.47	4.52	5.58	6.61	7.62	8.64	9.67	10.72	11.77		13.32	7 / 6			
reaction time	0.169	interval	7 steps	1.04	1.05	1.06	1.03	1.01	1.02	1.03	1.05	1.05	1.55		3.15	3.06	3.13
		velocity		5.65	8.79	8.70	8.62	8.87	9.05	8.96	8.87	8.70	9.05	8.26	8.70	8.96	8.76
Ortega, Orlando (ESP) (1991)	time	2.45	3.51	4.55	5.60	6.62	7.65	8.68	9.71	10.73	11.78		13.37	3 / 7			
reaction time	0.140	interval		1.06	1.04	1.05	1.02	1.03	1.03	1.03	1.02	1.05	1.59		3.15	3.08	3.10
		velocity		5.60	8.62	8.79	8.70	8.96	8.87	8.87	8.96	8.70	8.82	8.23	8.70	8.90	8.85
Parchment, Hansle (JAM) (199)	time	2.51	3.58	4.62	5.65	6.68	7.67	8.69	9.69	10.76	11.82		13.37	8 / 8			
reaction time	0.145	interval	7 steps	1.07	1.04	1.03	1.03	0.99	1.02	1.00	1.07	1.06	1.55		3.14	3.04	3.13
		velocity		5.47	8.54	8.79	8.87	8.87	9.23	8.96	9.14	8.54	8.62	8.23	8.73	9.02	8.76

Semi-Final 3

date 06-Aug-17 wind 0.3 m/s

Pollitt (2018) - biomechanical report for the IAAF World Champs - London 2017 - Men's 110m Hurdles

		H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
Baji, Balázs (HUN) (1989)	time	2.34	3.40	4.44	5.46	6.48	7.50	8.51	9.54	10.57	11.63		13.23	5 / 1			
reaction time	0.120	interval	7 steps	1.06	1.04	1.02	1.02	1.01	1.03	1.03	1.06	1.60			3.12	3.05	3.12
		velocity		5.86	8.62	8.79	8.96	8.96	8.87	8.87	8.62	8.76	8.31		8.79	8.99	8.79
Merritt, Aries (USA) (1985)	time	2.36	3.41	4.44	5.46	6.48	7.50	8.52	9.55	10.59	11.65		13.25	7 / 2			
reaction time	0.117	interval	7 steps	1.05	1.03	1.02	1.02	1.02	1.03	1.04	1.06	1.60			3.10	3.06	3.13
		velocity		5.81	8.70	8.87	8.96	8.96	8.87	8.79	8.62	8.76	8.30		8.85	8.96	8.76
Xie Wenjun (CHN) (1990)	time	2.50	3.60	4.63	5.65	6.67	7.70	8.71	9.74	10.78	11.85		13.36	4 / 3			
reaction time	0.171	interval		1.10	1.03	1.02	1.02	1.03	1.01	1.03	1.04	1.07	1.51		3.15	3.06	3.14
		velocity		5.49	8.31	8.87	8.96	8.96	8.87	8.79	8.54	9.28	8.23		8.70	8.96	8.73
Alkana, Antonio (RSA) (1990)	time	2.66	3.74	4.78	5.81	6.82	7.85	8.88	9.95	11.02	12.13		13.59	9 / 4			
reaction time	0.230	interval		1.08	1.04	1.03	1.01	1.03	1.03	1.07	1.07	1.11	1.46		3.15	3.07	3.25
		velocity		5.16	8.46	8.79	8.87	9.05	8.87	8.87	8.54	8.54	8.23	8.09	8.70	8.93	8.44
Lovett, Eddie (ISV) (1992)	time	2.49	3.56	4.62	5.67	6.71	7.77	8.81	9.90	10.99	12.10		13.67	6 / 5			
reaction time	0.180	interval	7 steps	1.07	1.06	1.05	1.04	1.06	1.04	1.09	1.09	1.11	1.57		3.18	3.14	3.29
		velocity		5.51	8.54	8.62	8.70	8.79	8.62	8.79	8.39	8.39	8.23	8.05	8.62	8.73	8.33
Souza, Éder Antonio (BRA) (199)	time	2.44	3.53	4.57	5.63	6.66	7.71	8.75	9.82	10.91	12.03		13.70	3 / 6			
reaction time	0.133	interval	7 steps	1.09	1.04	1.06	1.03	1.05	1.04	1.07	1.09	1.12	1.67		3.19	3.12	3.28
		velocity		5.62	8.39	8.79	8.62	8.87	8.70	8.79	8.54	8.39	8.16	8.03	8.60	8.79	8.36
Cabral, Johnathan (CAN) (199)	time	2.51	3.59	4.63	5.67	6.70	7.76	8.86	10.03	11.33	12.77		14.98	8 / 7			
reaction time	0.174	interval	7 steps	1.08	1.04	1.04	1.03	1.06	1.10	1.17	1.30	1.44	2.21		3.16	3.19	3.91
		velocity		5.47	8.46	8.79	8.79	8.87	8.62	8.31	7.81	7.03	6.35	7.34	8.68	8.60	7.01
O'Farrill, Yordan (CUB) (1993)	time	2.51	3.58	4.62	5.65	6.68							dnf	2 / --			
reaction time	0.147	interval	7 steps	1.07	1.04	1.03	1.03								3.14		
		velocity		5.47	8.54	8.79	8.87	8.87							8.73		

Semi-Final 2

date 06-Aug-17 wind 0.6 m/s

Pollitt (2018) - biomechanical report for the IAAF World Champs - London 2017 - Men's 110m Hurdles

		H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
Brathwaite, Shane (BAR) (1990)	time	2.40	3.44	4.46	5.49	6.52	7.53	8.56	9.61	10.65	11.72		13.26	5 / 1			

	reaction time	0.179	interval	7 steps	1.04	1.02	1.03	1.03	1.01	1.03	1.05	1.04	1.07	1.54		3.09	3.07	3.16	
			velocity	5.72	8.79	8.96	8.87	8.87	9.05	8.87	8.70	8.79	8.54	9.10	8.30	8.87	8.93	8.68	
Parchment, Hansle (JAM) (1996)	time			2.48	3.56	4.58	5.61	6.63	7.61	8.63	9.66	10.68	11.73		13.27	7 / 2			
	reaction time	0.181	interval	7 steps	1.08	1.02	1.03	1.02	0.98	1.02	1.03	1.02	1.05	1.54			3.13	3.02	3.10
			velocity	5.53	8.46	8.96	8.87	8.96	9.33	8.96	8.87	8.96	8.70	9.10	8.29		8.76	9.08	8.85
Allen, Devon (USA) (1994)	time			2.29	3.35	4.37	5.40	6.42	7.47	8.50	9.54	10.58	11.65		13.27	4 / 3			
	reaction time	0.126	interval		1.06	1.02	1.03	1.02	1.05	1.03	1.04	1.04	1.07	1.62			3.11	3.10	3.15
			velocity	5.99	8.62	8.96	8.87	8.96	8.70	8.87	8.79	8.79	8.54	8.65	8.29		8.82	8.85	8.70
Pozzi, Andy (GBR) (1992)	time			2.35	3.39	4.40	5.43	6.44	7.43	8.45	9.48	10.51	11.62		13.28	6 / 4			
	reaction time	0.137	interval	7 steps	1.04	1.01	1.03	1.01	0.99	1.02	1.03	1.03	1.11	1.66			3.08	3.02	3.17
			velocity	5.84	8.79	9.05	8.87	9.05	9.23	8.96	8.87	8.87	8.23	8.45	8.28		8.90	9.08	8.65
Czykier, Damian (POL) (1992)	time			2.45	3.52	4.57	5.62	6.65	7.64	8.65	9.71	10.77	11.84		13.42	9 / 5			
	reaction time	0.183	interval	7 steps	1.07	1.05	1.05	1.03	0.99	1.01	1.06	1.06	1.07	1.58			3.17	3.03	3.19
			velocity	5.60	8.54	8.70	8.70	8.87	9.23	9.05	8.62	8.62	8.54	8.87	8.20		8.65	9.05	8.60
Iribarne, Roger (CUB) (1996)	time			2.39	3.44	4.46	5.49	6.51	7.58	8.62	9.67	10.73	11.83		13.43	3 / 6			
	reaction time	0.175	interval	7 steps	1.05	1.02	1.03	1.02	1.07	1.04	1.05	1.06	1.10	1.60			3.10	3.13	3.21
			velocity	5.74	8.70	8.96	8.87	8.96	8.54	8.79	8.70	8.62	8.31	8.76	8.19		8.85	8.76	8.54
Contreras, Yidiel (ESP) (1992)	time			2.40	3.46	4.52	5.57	6.65	7.67	8.73	9.80	10.88	12.00		13.65	8 / 7			
	reaction time	0.157	interval	7 steps	1.06	1.06	1.05	1.08	1.02	1.06	1.07	1.08	1.12	1.65			3.17	3.16	3.27
			velocity	5.72	8.62	8.62	8.70	8.46	8.96	8.62	8.54	8.46	8.16	8.50	8.06		8.65	8.68	8.39
Bühler, Matthias (GER) (1986)	time			2.48	3.57	4.64	5.71	6.77	7.82	8.87	9.93	11.00	12.11		13.79	2 / 8			
	reaction time	0.158	interval		1.09	1.07	1.07	1.06	1.05	1.05	1.06	1.07	1.11	1.68			3.23	3.16	3.24
			velocity	5.53	8.39	8.54	8.54	8.62	8.70	8.70	8.62	8.54	8.23	8.35	7.98		8.49	8.68	8.46

Semi-Final 1

Semi-Final 1		date	06-Aug-17	wind	Pollitt (2018) - biomechanical report for the IAAF World Champs - London 2017 - Men's 110m Hurdles														
				H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
McLeod, Omar (JAM) (1994)	time			2.37	3.43	4.45	5.46	6.48	7.47	8.47	9.49	10.53	11.57		13.10	7 / 1			
	reaction time	0.135	interval		1.06	1.02	1.01	1.02	0.99	1.00	1.02	1.04	1.04	1.53			3.09	3.01	3.10
			velocity		5.79	8.62	8.96	9.05	8.96	9.23	9.14	8.96	8.79	8.79	9.16	8.40		8.87	9.11
Darien, Garfield (FRA) (1987)	time			2.34	3.39	4.41	5.43	6.45	7.45	8.45	9.48	10.52	11.57		13.17	6 / 2			
	reaction time	0.125	interval	7 steps	1.05	1.02	1.02	1.02	1.00	1.00	1.03	1.04	1.05	1.60			3.09	3.02	3.12
			velocity		5.86	8.70	8.96	8.96	8.96	9.14	9.14	8.87	8.79	8.70	8.76	8.35		8.87	9.08
Shubenkov, Sergey (ANA) (1999)	time			2.46	3.52	4.54	5.54	6.55	7.55	8.56	9.58	10.62	11.70		13.22	9 / 3			
	reaction time	0.160	interval	7 steps	1.06	1.02	1.00	1.01	1.00	1.01	1.02	1.04	1.08	1.52			3.08	3.02	3.14
			velocity		5.58	8.62	8.96	9.14	9.05	9.14	9.05	8.96	8.79	8.46	9.22	8.32		8.90	9.08
Ortega, Orlando (ESP) (1991)	time			2.45	3.51	4.54	5.56	6.60	7.61	8.60	9.62	10.66	11.71		13.23	4 / 4			
	reaction time	0.148	interval		1.06	1.03	1.02	1.04	1.01	0.99	1.02	1.04	1.05	1.52			3.11	3.04	3.11
			velocity		5.60	8.62	8.87	8.96	8.79	9.05	9.23	8.96	8.79	8.70	9.22	8.31		8.82	9.02
Trajkovic, Milan (CYP) (1992)	time			2.42	3.45	4.50	5.53	6.58	7.61	8.61	9.64	10.68	11.75		13.32	5 / 5			
	reaction time	0.146	interval	7 steps	1.03	1.05	1.03	1.05	1.03	1.00	1.03	1.04	1.07	1.57			3.11	3.08	3.14
			velocity		5.67	8.87	8.70	8.87	8.70	8.87	9.14	8.87	8.79	8.54	8.93	8.26		8.82	8.90
Harris, Aleec (USA) (1990)	time			2.41	3.49	4.53	5.56	6.60	7.62	8.66	9.70	10.75	11.82		13.40	8 / 6			
	reaction time	0.140	interval		1.08	1.04	1.03	1.04	1.02	1.04	1.04	1.05	1.07	1.58			3.15	3.10	3.16
			velocity		5.69	8.46	8.79	8.87	8.79	8.96	8.79	8.79	8.70	8.54	8.87	8.21		8.70	8.85
Masuno, Genta (JPN) (1993)	time			2.46	3.51	4.57	5.63	6.71	7.78	8.87	9.94	11.05	12.17		13.79	2 / 7			
	reaction time	0.160	interval		1.05	1.06	1.06	1.08	1.07	1.09	1.07	1.11	1.12	1.62			3.17	3.24	3.30
			velocity		5.58	8.70	8.62	8.62	8.46	8.54	8.39	8.54	8.23	8.16	8.65	7.98		8.65	8.46
Al-Youha, Yaqoub (KUW) (1999)	time			2.43	3.47	4.52									dnf	3 / --			
	reaction time	0.152	interval		1.04	1.05													
			velocity		5.65	8.79	8.70												

2017 Japanese National High School Championships (Yamagata, JPN) (99cm)

FINAL

FINAL	date 02-Aug-17		wind -0.7 m/s		Kota (2017) - 70th high school championships: JAF scientific committee - biomechanics data collection														
			H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10	
Katsuta, Kizuku (JPN) (1999)	time		2.68	3.78	4.88	5.98	7.08	8.20	9.33	10.44	11.59	12.73		14.19	7 / 1				
	reaction time	0.153		1.10	1.10	1.10	1.10	1.12	1.13	1.11	1.15	1.14	1.46			3.30	3.35	3.40	
	interval velocity		5.12	8.31	8.31	8.31	8.31	8.16	8.09	8.23	7.95	8.02	9.60	7.75		8.31	8.19	8.06	
Morito, Nobuaki (JPN) (1999)	time		2.74	3.87	4.97	6.09	7.20	8.31	9.41	10.54	11.64	12.81		14.30	4 / 2				
	reaction time	0.179		1.13	1.10	1.12	1.11	1.11	1.10	1.13	1.10	1.17	1.49			3.35	3.32	3.40	
	interval velocity		5.01	8.09	8.31	8.16	8.23	8.23	8.31	8.09	8.31	7.81	9.41	7.69		8.19	8.26	8.06	

Higuchi, Rikoto (JPN) (1999)	time	2.70	3.82	4.95	6.08	7.22	8.35	9.48	10.61	11.76	12.92		14.34	6 / 3				
	reaction time	0.169		1.12	1.13	1.13	1.14	1.13	1.13	1.13	1.15	1.16	1.42			3.38	3.40	3.44
	velocity		5.08	8.16	8.09	8.09	8.02	8.09	8.09	8.09	7.95	7.88	9.87		7.67	8.11	8.06	7.97
Nakata, Hidetoshi (JPN) (1999)	time	2.69	3.81	4.94	6.05	7.17	8.27	9.41	10.56	11.73	12.91		14.44	5 / 4				
	reaction time	0.130		1.12	1.13	1.11	1.12	1.10	1.14	1.15	1.17	1.18	1.53			3.36	3.36	3.50
	velocity		5.10	8.16	8.09	8.23	8.16	8.31	8.02	7.95	7.81	7.75	9.16		7.62	8.16	8.16	7.83
Abe, Ryoto (JPN) (2000)	time	2.65	3.76	4.92	6.03	7.20	8.34	9.48	10.63	11.82	12.99		14.48	8 / 5				
	reaction time	0.144		1.11	1.16	1.11	1.17	1.14	1.14	1.15	1.19	1.17	1.49			3.38	3.45	3.51
	velocity		5.18	8.23	7.88	8.23	7.81	8.02	8.02	7.95	7.68	7.81	9.41		7.60	8.11	7.95	7.81
Togochi, Hiromasa (JPN)	time	2.77	3.90	5.05	6.18	7.32	8.43	9.56	10.71	11.84	13.03		14.52	9 / 6				
	reaction time	0.149		1.13	1.15	1.13	1.14	1.11	1.13	1.15	1.13	1.19	1.49			3.41	3.38	3.47
	velocity		4.95	8.09	7.95	8.09	8.02	8.23	8.09	7.95	8.09	7.68	9.41		7.58	8.04	8.11	7.90
Hiraga, Kentaro (JPN) (1999)	time	2.70	3.83	4.96	6.09	7.25	8.35	9.49	10.67	11.83	13.03		14.54	2 / 7				
	reaction time	0.156		1.13	1.13	1.13	1.16	1.10	1.14	1.18	1.16	1.20	1.51			3.39	3.40	3.54
	velocity		5.08	8.09	8.09	8.09	7.88	8.31	8.02	7.75	7.88	7.62	9.28		7.57	8.09	8.06	7.75
Yamada, Nash (JPN) (1999)	time	2.71	3.86	5.11	6.31	7.46	8.58	9.72	10.87	12.03	13.22		14.79	3 / 8				
	reaction time	0.153		1.15	1.25	1.20	1.15	1.12	1.14	1.15	1.16	1.19	1.57			3.60	3.41	3.50
	velocity		5.06	7.95	7.31	7.62	7.95	8.16	8.02	7.95	7.88	7.68	8.93		7.44	7.62	8.04	7.83

2017 Windsprint (Sundsvall, SWE)

FINAL 2

		date	30-Jul-17	wind	1.0 m/s	H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
Levin, Anton (SWE) (1992)	time					2.68	3.78	4.87	5.95	7.03	8.14	9.23	10.34	11.47	12.61		14.09	/ 1			
	reaction time							1.10	1.09	1.08	1.11	1.09	1.11	1.13	1.14	1.48			3.27	3.28	3.38
	velocity					5.12	8.31	8.39	8.46	8.46	8.23	8.39	8.23	8.09	8.02	9.47	7.81		8.39	8.36	8.11
Holm, Jonatan (SWE) (1994)	time					2.69	3.79	4.89	5.98	7.07	8.19	9.28	10.38	11.64	12.91		14.49	/ 2			
	reaction time							1.10	1.10	1.09	1.12	1.09	1.10	1.26	1.27	1.58			3.29	3.30	3.63
	velocity					5.10	8.31	8.31	8.39	8.39	8.16	8.39	8.31	7.25	7.20	8.87	7.59		8.33	8.31	7.55
Gustafsson, Petter (SWE) (1999)	time					2.65	3.77	4.90	6.05	7.18	8.37	9.52	10.66	11.85	13.05		14.56	/ 3			
	reaction time							1.12	1.13	1.15	1.13	1.19	1.15	1.14	1.19	1.20	1.51		3.40	3.47	3.53
	velocity					5.18	8.16	8.09	7.95	8.09	7.68	7.95	8.02	7.68	7.62	9.28	7.55		8.06	7.90	7.77
Sjodin, Ludwig (SWE) (1995)	time					2.68	3.86	5.00	6.18	7.40	8.62	9.83	11.12	12.34	13.62		15.29	/ 4			
	reaction time							1.18	1.14	1.18	1.22	1.22	1.21	1.29	1.22	1.28	1.67		3.50	3.65	3.79
	velocity					5.12	7.75	8.02	7.75	7.49	7.49	7.55	7.09	7.49	7.14	8.40	7.19		7.83	7.51	7.23
Tay, Marwin (SWE) (1990)	time					2.69	3.82	4.92									dnf	/ --			
	reaction time							1.13	1.10												
	velocity					5.10	8.09	8.31													

FINAL

		date	30-Jul-17	wind	1.0 m/s	H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
Levin, Anton (SWE) (1992)	time					2.66	3.78	4.87	5.93	7.02	8.10	9.20	10.33	11.43	12.56		14.06	/ 1			
	reaction time							1.12	1.09	1.06	1.09	1.08	1.10	1.13	1.10	1.13	1.50		3.27	3.27	3.36
	velocity					5.16	8.16	8.39	8.62	8.39	8.46	8.31	8.09	8.31	8.09	9.35	7.82		8.39	8.39	8.16
Holm, Jonatan (SWE) (1994)	time					2.69	3.80	4.88	5.99	7.11	8.23	9.35	10.50	11.63	12.81		14.34	/ 2			
	reaction time							1.11	1.08	1.11	1.12	1.12	1.15	1.13	1.18	1.53			3.30	3.36	3.46
	velocity					5.10	8.23	8.46	8.23	8.16	8.16	8.16	7.95	8.09	7.75	9.16	7.67		8.31	8.16	7.92
Gustafsson, Petter (SWE) (1999)	time					2.68	3.81	4.94	6.07	7.22	8.36	9.51	10.69	11.86	13.00		14.54	/ 3			
	reaction time							1.13	1.13	1.13	1.15	1.14	1.15	1.18	1.17	1.14	1.54		3.39	3.44	3.49
	velocity					5.12	8.09	8.09	8.09	7.95	8.02	7.95	7.75	7.81	8.02	9.10	7.57		8.09	7.97	7.86
Tay, Marwin (SWE) (1990)	time					2.68	3.79	4.89	6.01	7.12	8.25	9.41	10.56	11.72	12.93		14.59	/ 4			
	reaction time							1.11	1.10	1.12	1.11	1.13	1.16	1.15	1.16	1.21	1.66		3.33	3.40	3.52
	velocity					5.12	8.23	8.31	8.16	8.23	8.09	7.88	7.95	7.88	7.55	8.45	7.54		8.23	8.06	7.79
Sjodin, Ludwig (SWE) (1995)	time					2.76	3.96	5.11	6.32	7.56	8.78						15.68	/ 5			
	reaction time							1.20	1.15	1.21	1.24	1.22				6.90			3.56		
	velocity					4.97	7.62	7.95	7.55	7.37	7.49					7.33	7.02		7.70		

2017 All Star Night (Hiratsuka, JPN)

FINAL

		date	22-Jul-17	wind	1.8 m/s	H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
Yazawa, Wataru (JPN) (1991)	time					2.59	3.66	4.71	5.74	6.78	7.82	8.87	9.93	11.00	12.07		13.49	6 / 1			
	reaction time	0.146					7 steps	1.07	1.05	1.03	1.04	1.05	1.06	1.07	1.07	1.42			3.15	3.13	3.20
	velocity					5.30	8.54	8.70	8.87	8.79	8.79	8.70	8.62	8.54	8.54	9.87	8.15		8.70	8.76	8.57

Kanai, Taio (JPN) (1995)	time	2.55	3.62	4.67	5.73	6.79	7.83	8.90	9.97	11.04	12.12		13.53	7 / 2				
	reaction time	0.127	interval	7 steps	1.07	1.05	1.06	1.06	1.04	1.07	1.07	1.07	1.08	1.41		3.18	3.17	3.22
	velocity				5.38	8.54	8.70	8.62	8.62	8.79	8.54	8.54	8.46	9.94	8.13	8.62	8.65	8.52
Kuriki, Anthony Tyrell (JPN) (1	time	2.62	3.72	4.77	5.82	6.87	7.92	8.98	10.06	11.12	12.19		13.60	3 / 3				
	reaction time	0.128	interval	7 steps	1.10	1.05	1.05	1.05	1.06	1.08	1.06	1.07	1.41			3.20	3.16	3.21
	velocity				5.24	8.31	8.70	8.70	8.70	8.62	8.46	8.62	8.54	9.94	8.09	8.57	8.68	8.54

2017 Galà dei Castelli (Bellinzona, SUI)**FINAL**

FINAL	date	18-Jul-17	wind	-0.6 m/s	(2017.07.18) - https://twitter.com/touchdown_time/media?lang=en&lang=en&lang=en&lang=en														
			H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10	
Merritt, Aries (USA) (1985)	time		2.57	3.61	4.64	5.64	6.64	7.66	8.67	9.70	10.73	11.77		13.20	4 / 1				
	reaction time		interval	7 steps	1.04	1.03	1.00	1.00	1.02	1.01	1.03	1.03	1.04	1.43			3.07	3.03	3.10
		velocity		5.34	8.79	8.87	9.14	9.14	8.96	9.05	8.87	8.87	8.79	9.80	8.33		8.93	9.05	8.85
Eaton, Jarret (USA) (1989)	time		2.53	3.58	4.59	5.59	6.62	7.67	8.71	9.78	10.87	12.01		13.52	3 / 2				
	reaction time		interval	7 steps	1.05	1.01	1.00	1.03	1.05	1.04	1.07	1.09	1.14	1.51			3.06	3.12	3.30
		velocity		5.42	8.70	9.05	9.14	8.87	8.70	8.79	8.54	8.39	8.02	9.28	8.14		8.96	8.79	8.31

2017 Spitzen Leichtathletik (Luzern, SUI)**FINAL**

INAL	date	11-Jul-17	wind -0.1 m/s			(2017.07.11) - https://twitter.com/touchdown_time/media?lang=en&lang=en&lang=en&lang=en													
			H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10	
Allen, Devon (USA) (1994)	time		2.60	3.64	4.68	5.71	6.73	7.75	8.77	9.82	10.88	11.94		13.31	5 / 1				
	reaction time			interval		1.04	1.04	1.03	1.02	1.02	1.02	1.05	1.06	1.06	1.37		3.11	3.06	3.17
	velocity		5.28	8.79	8.79	8.87	8.96	8.96	8.96	8.70	8.62	8.62	10.23	8.26		8.82	8.96	8.65	

2017 Müller's Anniversary Games (London, GBR) (TV Analysis)**FINAL**

		date	09-Jul-17	wind 0.0 m/s			Henson (2020) - Athlete First: 2017 year end hurdle report												
				H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
Merritt, Aries (USA) (1985)	time			2.52	3.56	4.58	5.56	6.58	7.58	8.58	9.58	10.62	11.62		13.09	4 / 1			
	reaction time	0.134	interval	7 steps	1.04	1.02	0.98	1.02	1.00	1.00	1.00	1.04	1.00	1.47			3.04	3.02	3.04
			velocity		5.44	8.79	8.96	9.33	8.96	9.14	9.14	9.14	8.79	9.14	9.54	8.40		9.02	9.08
Trajkovic, Milan (CYP) (1992)	time			2.52	3.58	4.60	5.60	6.62	7.64	8.66	9.70	10.74	11.80		13.25	5 / 2			
	reaction time	0.157	interval	7 steps	1.06	1.02	1.00	1.02	1.02	1.02	1.04	1.04	1.06	1.45	NR PB		3.08	3.06	3.14
			velocity		5.44	8.62	8.96	9.14	8.96	8.96	8.96	8.79	8.79	8.62	9.67	8.30		8.90	8.96
Brathwaite, Shane (BAR) (1990)	time			2.52	3.60	4.60	5.62	6.64	7.66	8.70	9.74	10.80	11.84		13.27	8 / 3			
	reaction time	0.135	interval	7 steps	1.08	1.00	1.02	1.02	1.02	1.04	1.04	1.06	1.04	1.43			3.10	3.08	3.14
			velocity		5.44	8.46	9.14	8.96	8.96	8.96	8.79	8.79	8.62	8.79	9.80	8.29		8.85	8.90
Allen, Devon (USA) (1994)	time			2.52	3.56	4.56	5.64	6.70	7.70	8.74	9.80	10.82	11.88		13.30	6 / 4			
	reaction time	0.104	interval		1.04	1.00	1.08	1.06	1.00	1.04	1.06	1.02	1.06	1.42			3.12	3.10	3.14
			velocity		5.44	8.79	9.14	8.46	8.62	9.14	8.79	8.62	8.96	8.62	9.87	8.27		8.79	8.85
Eaton, Jarret (USA) (1989)	time			2.44	3.48	4.48	5.48	6.48	7.52	8.56	9.62	10.72	11.80		13.34	9 / 5			
	reaction time	0.125	interval	7 steps	1.04	1.00	1.00	1.00	1.04	1.04	1.06	1.10	1.08	1.54			3.04	3.08	3.24
			velocity		5.62	8.79	9.14	9.14	9.14	8.79	8.79	8.62	8.31	8.46	9.10	8.25		9.02	8.90
Mallett, Aaron (USA) (1994)	time			2.58	3.66	4.74	5.76	6.78	7.82	8.84	9.88	10.92	11.98		13.47	1 / 6			
	reaction time	0.198	interval	7 steps	1.08	1.08	1.02	1.02	1.04	1.02	1.04	1.04	1.06	1.49			3.18	3.08	3.14
			velocity		5.32	8.46	8.46	8.96	8.96	8.79	8.96	8.79	8.79	8.62	9.41	8.17		8.62	8.90
Lovett, Eddie (ISV) (1992)	time			2.52	3.56	4.60	5.64	6.68	7.70	8.74	9.80	10.90	12.00		13.48	3 / 7			
	reaction time	0.144	interval	7 steps	1.04	1.04	1.04	1.04	1.02	1.04	1.06	1.10	1.10	1.48			3.12	3.10	3.26
			velocity		5.44	8.79	8.79	8.79	8.79	8.96	8.79	8.62	8.31	8.31	9.47	8.16		8.79	8.85
Omoregie, David (GBR) (1995)	time			2.52	3.64	4.74	5.80	6.84	7.88	8.92	9.98	11.06	12.16		13.70	7 / 8			
	reaction time	0.127	interval		1.12	1.10	1.06	1.04	1.04	1.04	1.06	1.08	1.10	1.54			3.28	3.12	3.24
			velocity		5.44	8.16	8.31	8.62	8.79	8.79	8.79	8.62	8.46	8.31	9.10	8.03		8.36	8.79
Xie Wenjun (CHN) (1990)	time			2.60	3.66	4.74	5.78	6.82	7.90	8.98	10.08	11.26	12.46		14.34	2 / 9			
	reaction time	0.157	interval		1.06	1.08	1.04	1.04	1.08	1.08	1.10	1.18	1.20	1.88			3.18	3.20	3.48
			velocity		5.28	8.62	8.46	8.79	8.79	8.46	8.46	8.31	7.75	7.62	7.46	7.67		8.62	8.57

2017 Nambu Memorial (Sapporo, JPN)**FINAL**

date 09-Jul-17		wind 2.7 m/s										(2017) - tfdata-store.com/2017/07/12/post-1025/						
		H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10	
Nomoto, Syuusei (JPN) (1995)	time	2.61	3.70	4.75	5.79	6.84	7.91	8.96	10.04	11.11	12.22		13.59	7 / 1				
	reaction time	interval	7 steps	1.09	1.05	1.04	1.05	1.07	1.05	1.08	1.07	1.11	1.37		3.18	3.17	3.26	
	velocity	5.26	8.39	8.70	8.79	8.70	8.54	8.70	8.46	8.54	8.23	10.23	8.09		8.62	8.65	8.41	
Furuya, Takumu (JPN) (1997)	time	2.64	3.73	4.78	5.83	6.87	7.92	8.98	10.06	11.14	12.26		13.70	4 / 2				
	reaction time	interval		1.09	1.05	1.05	1.04	1.05	1.06	1.08	1.08	1.12	1.44		3.19	3.15	3.28	
	velocity	5.20	8.39	8.70	8.70	8.79	8.70	8.62	8.46	8.46	8.16	9.74	8.03		8.60	8.70	8.36	

Iwafune, Yoichi (JPN) (1985)	time	2.68	3.78	4.85	5.93	7.00	8.06	9.15	10.26	11.37	12.52		14.01	5 / 3			
	reaction time	interval	1.10	1.07	1.08	1.07	1.06	1.09	1.11	1.11	1.15	1.49			3.25	3.22	3.37
	velocity	5.12	8.31	8.54	8.46	8.54	8.62	8.39	8.23	8.23	7.95	9.41	7.85		8.44	8.52	8.14
Abe, Hirofumi (JPN) (1985)	time	2.68	3.79	4.86	5.95	7.04	8.13	9.20	10.34	11.47	12.62		14.07	6 / 4			
	reaction time	interval	1.11	1.07	1.09	1.09	1.09	1.07	1.14	1.13	1.15	1.45			3.27	3.25	3.42
	velocity	5.12	8.23	8.54	8.39	8.39	8.39	8.54	8.02	8.09	7.95	9.67	7.82		8.39	8.44	8.02
Nishizawa, Masanori (JPN) (1985)	time	2.71	3.81	4.91	6.02	7.11	8.20	9.30	10.41	11.56	12.72		14.25	9 / 5			
	reaction time	interval	1.10	1.10	1.11	1.09	1.09	1.10	1.11	1.15	1.16	1.53			3.31	3.28	3.42
	velocity	5.06	8.31	8.31	8.23	8.39	8.39	8.31	8.23	7.95	7.88	9.16	7.72		8.28	8.36	8.02
Kubota, Kohei (JPN) (1989)	time	2.65	3.77	4.90	6.01	7.14	8.26	9.38	10.51	11.65	12.82		14.34	8 / 6			
	reaction time	interval	1.12	1.13	1.11	1.13	1.12	1.12	1.13	1.14	1.17	1.52			3.36	3.37	3.44
	velocity	5.18	8.16	8.09	8.23	8.09	8.16	8.16	8.09	8.02	7.81	9.22	7.67		8.16	8.14	7.97
Abe, Masaya (JPN) (1996)	time	2.69	3.83	4.95	6.08	7.23	8.35	9.47	10.61	11.76	12.93		14.45	2 / 7			
	reaction time	interval	1.14	1.12	1.13	1.15	1.12	1.12	1.14	1.15	1.17	1.52			3.39	3.39	3.46
	velocity	5.10	8.02	8.16	8.09	7.95	8.16	8.16	8.02	7.95	7.81	9.22	7.61		8.09	8.09	7.92
Nakao, Kengo (JPN)	time	2.82	4.01	5.19	6.34	7.49	8.62	9.80	10.96	12.13	13.31		14.84	3 / 8			
	reaction time	interval	1.19	1.18	1.15	1.15	1.13	1.18	1.16	1.17	1.18	1.53			3.52	3.46	3.51
	velocity	4.87	7.68	7.75	7.95	7.95	8.09	7.75	7.88	7.81	7.75	9.16	7.41		7.79	7.92	7.81
Degughi, Masaya (JPN)	time	2.76	3.87	5.04	6.18	7.37	8.53	9.71	10.90	12.09	13.33		14.85	1 / 9			
	reaction time	interval	1.11	1.17	1.14	1.19	1.16	1.18	1.19	1.19	1.24	1.52			3.42	3.53	3.62
	velocity	4.97	8.23	7.81	8.02	7.68	7.88	7.75	7.68	7.68	7.37	9.22	7.41		8.02	7.77	7.57

2017 István Gyulai Memorial (Székesfehérvár, HUN) (TV Analysis)

FINAL

INAL	date 04-Jul-17		wind 0.3 m/s										Henson (2020) - Athlete First: 2017 year end hurdle report						
			H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10	
McLeod, Omar (JAM) (1994)	time		2.48	3.50	4.50	5.50	6.50	7.50	8.48	9.52	10.54	11.56		12.96	4 / 1				
	reaction time	0.138	interval	1.02	1.00	1.00	1.00	1.00	0.98	1.04	1.02	1.02	1.40			3.02	2.98	3.08	
		velocity	5.53	8.96	9.14	9.14	9.14	9.14	9.33	8.79	8.96	8.96	10.01	8.49		9.08	9.20	8.90	
Shubenkov, Sergey (ANA) (1995)	time		2.56	3.58	4.60	5.56	6.54	7.54	8.54	9.56	10.58	11.60		13.01	6 / 2				
	reaction time	0.177	interval	7 steps	1.02	1.02	0.96	0.98	1.00	1.00	1.02	1.02	1.02	1.41		3.00	2.98	3.06	
		velocity	5.36	8.96	8.96	9.52	9.33	9.14	9.14	8.96	8.96	8.96	9.94	8.46		9.14	9.20	8.96	
Merritt, Aries (USA) (1985)	time		2.48	3.52	4.52	5.52	6.52	7.52	8.54	9.58	10.60	11.66		13.11	3 / 3				
	reaction time	0.126	interval	7 steps	1.04	1.00	1.00	1.00	1.00	1.02	1.04	1.02	1.06	1.45		3.04	3.02	3.12	
		velocity	5.53	8.79	9.14	9.14	9.14	9.14	8.96	8.79	8.96	8.62	9.67	8.39		9.02	9.08	8.79	
Baji, Balázs (HUN) (1989)	time		2.50	3.56	4.56	5.56	6.56	7.56	8.58	9.58	10.62	11.66		13.15	2 / 4				
	reaction time	0.139	interval	7 steps	1.06	1.00	1.00	1.00	1.00	1.02	1.00	1.04	1.04	1.49	NR PB	3.06	3.02	3.08	
		velocity	5.49	8.62	9.14	9.14	9.14	9.14	8.96	9.14	8.79	8.79	9.41	8.37		8.96	9.08	8.90	
Alkana, Antonio (RSA) (1990)	time		2.54	3.58	4.62	5.64	6.64	7.68	8.72	9.76	10.80	11.86		13.31	5 / 5				
	reaction time		interval	1.04	1.04	1.02	1.00	1.04	1.04	1.04	1.04	1.06	1.45			3.10	3.08	3.14	
		velocity	5.40	8.79	8.79	8.96	9.14	8.79	8.79	8.79	8.79	8.62	9.67	8.26		8.85	8.90	8.73	
Riley, Andrew (JAM) (1988)	time		2.54	3.62	4.64	5.66	6.70	7.74	8.78	9.84	10.88	11.96		13.39	1 / 6				
	reaction time	0.150	interval	7 steps	1.08	1.02	1.02	1.04	1.04	1.06	1.04	1.08	1.43			3.12	3.12	3.18	
		velocity	5.40	8.46	8.96	8.96	8.79	8.79	8.79	8.62	8.79	8.46	9.80	8.22		8.79	8.79	8.62	
Brathwaite, Shane (BAR) (1990)	time		2.52	3.56	4.62	5.64	6.66	7.70	8.76	9.80	10.86	11.94		13.39	7 / 7				
	reaction time	0.146	interval	7 steps	1.04	1.06	1.02	1.04	1.06	1.04	1.06	1.08	1.45			3.12	3.12	3.18	
		velocity	5.44	8.79	8.62	8.96	8.96	8.79	8.62	8.79	8.62	8.46	9.67	8.22		8.79	8.79	8.62	

2017 Meeting de Paris (Paris, FRA) (TV Analysis)

FINAL

Henson (2020) - Athlete First: 2017 year end hurdle report																						
		date	01-Jul-17	wind	-0.1 m/s	H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10	
Levy, Ronald (JAM) (1992)	time					2.44	3.46	4.50	5.50	6.48	7.50	8.54	9.54	10.58	11.66		13.05	8 / 1				
	reaction time	0.133	interval	7 steps		1.02	1.04	1.00	0.98	1.02	1.04	1.00	1.04	1.08	1.39		PB		3.06	3.04	3.12	
			velocity			5.62	8.96	8.79	9.14	9.33	8.96	8.79	9.14	8.79	8.46	10.09	8.43		8.96	9.02	8.79	
Pozzi, Andy (GBR) (1992)	time					2.48	3.50	4.50	5.48	6.48	7.48	8.50	9.54	10.58	11.66		13.14	3 / 2				
	reaction time	0.107	interval	7 steps		1.02	1.00	0.98	1.00	1.00	1.02	1.04	1.04	1.04	1.08	1.48		PB		3.00	3.02	3.16
			velocity			5.53	8.96	9.14	9.33	9.14	9.14	8.96	8.79	8.79	8.46	9.47	8.37		9.14	9.08	8.68	
Darien, Garfield (FRA) (1987)	time					2.50	3.54	4.56	5.56	6.54	7.58	8.58	9.60	10.62	11.68		13.15	7 / 3				
	reaction time	0.121	interval	7 steps		1.04	1.02	1.00	0.98	1.04	1.00	1.02	1.02	1.02	1.06	1.47			3.06	3.02	3.10	
			velocity			5.49	8.79	8.96	9.14	9.33	8.79	9.14	8.96	8.96	8.62	9.54	8.37		8.96	9.08	8.85	
Shubenkov, Sergey (ANA) (1995)	time					2.56	3.60	4.58	5.58	6.56	7.58	8.58	9.60	10.66	11.76		13.18	4 / 4				
	reaction time	0.125	interval	7 steps		1.04	0.98	1.00	0.98	1.02	1.00	1.02	1.02	1.06	1.10	1.42			3.02	3.00	3.18	
			velocity																			

	velocity	5.36	8.79	9.33	9.14	9.33	8.96	9.14	8.96	8.62	8.31	9.87	8.35		9.08	9.14	8.62
Alkana, Antonio (RSA) (1990)	time	2.52	3.56	4.56	5.58	6.58	7.60	8.62	9.66	10.70	11.78		13.24	1 / 5			
	reaction time	0.139		1.04	1.00	1.02	1.00	1.02	1.04	1.04	1.08	1.46			3.06	3.04	3.16
	interval																
	velocity	5.44	8.79	9.14	8.96	9.14	8.96	8.96	8.79	8.79	8.46	9.60	8.31		8.96	9.02	8.68
Trajkovic, Milan (CYP) (1992)	time	2.50	3.56	4.58	5.60	6.62	7.66	8.70	9.74	10.80	11.86		13.30	2 / 6			
	reaction time	0.142		1.06	1.02	1.02	1.04	1.04	1.04	1.06	1.06	1.44			3.10	3.10	3.16
	interval																
	velocity	5.49	8.62	8.96	8.96	8.96	8.79	8.79	8.79	8.62	8.62	9.74	8.27		8.85	8.85	8.68
McLeod, Omar (JAM) (1994)	time	2.48	3.52	4.54	5.56	6.56	7.56	8.58	9.62	10.70	11.82		13.41	5 / 7			
	reaction time	0.122		1.04	1.02	1.02	1.00	1.00	1.02	1.04	1.08	1.12	1.59		3.08	3.02	3.24
	interval																
	velocity	5.53	8.79	8.96	8.96	9.14	9.14	8.96	8.79	8.46	8.16	8.82	8.20		8.90	9.08	8.46

2017 Golden Spike (Ostrava, CZE) (TV Analysis)

FINAL

date 28-Jun-17 wind 0.6 m/s Henson (2020) - Athlete First: 2017 year end hurdle report

		H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
Darien, Garfield (FRA) (1987)	time	2.48	3.50	4.52	5.52	6.52	7.54	8.54	9.56	10.58	11.62		13.09	5 / 1			
	reaction time	0.151		1.02	1.00	1.00	1.02	1.00	1.02	1.02	1.04	1.47	PB		3.04	3.02	3.08
	interval																
	velocity	5.53	8.96	8.96	9.14	9.14	8.96	9.14	8.96	8.96	8.79	9.54	8.40		9.02	9.08	8.90
Baji, Balázs (HUN) (1989)	time	2.52	3.56	4.58	5.60	6.60	7.60	8.62	9.64	10.68	11.74		13.23	6 / 2			
	reaction time	0.152		1.04	1.02	1.00	1.00	1.02	1.02	1.04	1.06	1.49	NR PB		3.08	3.02	3.12
	interval																
	velocity	5.44	8.79	8.96	8.96	9.14	9.14	8.96	8.96	8.79	8.62	9.41	8.31		8.90	9.08	8.79
Trajkovic, Milan (CYP) (1992)	time	2.48	3.52	4.56	5.58	6.62	7.66	8.66	9.70	10.78	11.82		13.34	8 / 3			
	reaction time	0.164		1.04	1.04	1.02	1.04	1.00	1.04	1.08	1.04	1.52			3.10	3.08	3.16
	interval																
	velocity	5.53	8.79	8.79	8.96	8.79	8.79	9.14	8.79	8.46	8.79	9.22	8.25		8.85	8.90	8.68
Alkana, Antonio (RSA) (1990)	time	2.52	3.56	4.58	5.60	6.64	7.68	8.74	9.80	10.86	11.92		13.37	4 / 4			
	reaction time	0.155		1.04	1.02	1.02	1.04	1.06	1.06	1.06	1.06	1.45			3.08	3.14	3.18
	interval																
	velocity	5.44	8.79	8.96	8.96	8.79	8.79	8.62	8.62	8.62	8.62	9.67	8.23		8.90	8.73	8.62
Douvalidis, Konstadinos (GRE)	time	2.56	3.62	4.66	5.70	6.72	7.76	8.80	9.88	10.96	12.04		13.51	2 / 5			
	reaction time	0.154		1.06	1.04	1.04	1.02	1.04	1.04	1.08	1.08	1.47			3.14	3.10	3.24
	interval																
	velocity	5.36	8.62	8.79	8.79	8.96	8.79	8.79	8.46	8.46	8.46	9.54	8.14		8.73	8.85	8.46
Martinsen, Andreas (DEN) (1991)	time	2.54	3.62	4.66	5.70	6.74	7.78	8.82	9.90	10.96	12.06		13.55	3 / 6			
	reaction time	0.149		1.08	1.04	1.04	1.04	1.04	1.08	1.06	1.10	1.49			3.16	3.12	3.24
	interval																
	velocity	5.40	8.46	8.79	8.79	8.79	8.79	8.79	8.46	8.62	8.31	9.41	8.12		8.68	8.79	8.46
Czykier, Damian (POL) (1992)	time	2.56	3.62	4.68	5.70	6.72	7.78	8.82	9.88	10.96	12.06		13.56	7 / 7			
	reaction time	0.177		1.06	1.02	1.02	1.06	1.04	1.06	1.08	1.10	1.50			3.14	3.12	3.24
	interval																
	velocity	5.36	8.62	8.62	8.96	8.96	8.62	8.79	8.62	8.46	8.31	9.35	8.11		8.73	8.79	8.46
Noga, Artur (POL) (1988)	time	2.56	3.64	4.70	5.72	6.76	7.80	8.84	9.90	10.98	12.10		13.62	1 / 8			
	reaction time	0.164		1.08	1.06	1.02	1.04	1.04	1.06	1.08	1.12	1.52			3.16	3.12	3.26
	interval																
	velocity	5.36	8.46	8.62	8.96	8.79	8.79	8.79	8.62	8.46	8.16	9.22	8.08		8.68	8.79	8.41

2017 Japanese National Championships (Osaka, JPN)

FINAL

date 25-Jun-17 wind -0.2 m/s (2017) - tfdata-store.com/2017/07/13/post-1040/

		H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
Takayama, Shunya (JPN) (1994)	time	2.57	3.60	4.65	5.68	6.70	7.75	8.80	9.87	10.94	12.03		13.45	4 / 1			
	reaction time	0.138		1.03	1.05	1.03	1.02	1.05	1.07	1.07	1.09	1.42			3.11	3.12	3.23
	interval																
	velocity	5.34	8.87	8.70	8.87	8.96	8.70	8.70	8.54	8.54	8.39	9.87	8.18		8.82	8.79	8.49
Yazawa, Wataru (JPN) (1991)	time	2.61	3.69	4.74	5.78	6.83	7.88	8.93	9.99	11.07	12.17		13.61	8 / 2			
	reaction time	0.144		1.08	1.05	1.04	1.05	1.05	1.06	1.08	1.10	1.44			3.17	3.15	3.24
	interval																
	velocity	5.26	8.46	8.70	8.79	8.70	8.70	8.70	8.62	8.46	8.31	9.74	8.08		8.65	8.70	8.46
Masuno, Genta (JPN) (1993)	time	2.61	3.67	4.71	5.74	6.81	7.85	8.92	9.99	11.05	12.15		13.61	5 / 3			
	reaction time	0.156		1.06	1.04	1.03	1.07	1.04	1.07	1.06	1.10	1.46			3.13	3.18	3.23
	interval																
	velocity	5.26	8.62	8.79	8.87	8.54	8.79	8.54	8.54	8.62	8.31	9.60	8.08		8.76	8.62	8.49
Sato, Hiroyuki (JPN) (1990)	time	2.58	3.68	4.72	5.77	6.82	7.88	8.94	10.02	11.11	12.23		13.69	7 / 4			
	reaction time	0.159		1.10	1.04	1.05	1.05	1.06	1.06	1.08	1.09	1.12	1.46		3.19	3.17	3.29
	interval																
	velocity	5.32	8.31	8.79	8.70	8.70	8.62	8.62	8.46	8.39	8.16	9.60	8.04		8.60	8.65	8.33
Kanai, Taio (JPN) (1995)	time	2.58	3.65	4.71	5.78	6.85	7.92	9.00	10.09	11.20	12.31		13.77	6 / 5			
	reaction time	0.128		1.07	1.06	1.07	1.07	1.07	1.08	1.09	1.11	1.11	1.46		3.20	3.22	3.31
	interval																
	velocity	5.32	8.54	8.62	8.54	8.54	8.54	8.46	8.39	8.23	8.23	9.60	7.99		8.57	8.52	8.28
Kuriki, Anthony Tyrell (JPN) (1991)	time	2.67	3.74	4.81	5.86	6.93	8.00	9.07	10.16	11.25	12.36		13.80	2 / 6			
	reaction time	0.126		1.07	1.07	1.05	1.07	1.07	1.07	1.09	1.09	1.11	1.44		3.19	3.21	3.29
	interval																
	velocity	5.14	8.54	8.54	8.70	8.54	8.54	8.54	8.39	8.39	8.23	9.74	7.97		8.60	8.54	8.33

Ishikawa, Shuhei (JPN) (1995)	time	2.59	3.66	4.72	5.78	6.85	7.93	9.01	10.10	11.19	12.32		13.80	3 / 7				
	reaction time	0.136	interval	7 steps	1.07	1.06	1.06	1.07	1.08	1.08	1.09	1.09	1.13	1.48		3.19	3.23	3.31
	velocity	5.30	8.54	8.62	8.62	8.54	8.46	8.46	8.39	8.39	8.09	9.47	7.97		8.60	8.49	8.28	
Tanaka, Shinya (JPN) (1993)	time	2.61	3.69	4.75	5.80	6.86	7.93	9.00	10.11	11.22	12.37		13.92	9 / 8				
	reaction time	0.141	interval	7 steps	1.08	1.06	1.05	1.06	1.07	1.07	1.11	1.11	1.15	1.55		3.19	3.20	3.37
	velocity	5.26	8.46	8.62	8.70	8.62	8.54	8.54	8.23	8.23	7.95	9.05	7.90		8.60	8.57	8.14	

Semi-Final 2

date 24-Jun-17 wind 0.3 m/s

(2017) - tfdata-store.com/2017/07/13/post-1040/

		H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10	
Masuno, Genta (JPN) (1993)	time	2.60	3.68	4.71	5.73	6.76	7.81	8.87	9.93	11.01	12.09		13.54	7 / 1				
	reaction time	0.148	interval		1.08	1.03	1.02	1.03	1.05	1.06	1.06	1.08	1.08	1.45		3.13	3.14	3.22
	velocity	5.28	8.46	8.87	8.96	8.87	8.70	8.62	8.62	8.46	8.46	9.67	8.12		8.76	8.73	8.52	
Kanai, Taio (JPN) (1995)	time	2.59	3.67	4.71	5.77	6.83	7.90	8.97	10.06	11.16	12.27		13.71	6 / 2				
	reaction time	0.115	interval	7 steps	1.08	1.04	1.06	1.06	1.07	1.07	1.09	1.10	1.11	1.44		3.18	3.20	3.30
	velocity	5.30	8.46	8.79	8.62	8.62	8.54	8.54	8.39	8.31	8.23	9.74	8.02		8.62	8.57	8.31	
Ishikawa, Shuhei (JPN) (1995)	time	2.58	3.67	4.74	5.81	6.87	7.94	9.02	10.11	11.22	12.32		13.78	5 / 4				
	reaction time	0.124	interval	7 steps	1.09	1.07	1.07	1.06	1.07	1.08	1.09	1.11	1.10	1.46		3.23	3.21	3.30
	velocity	5.32	8.39	8.54	8.54	8.62	8.54	8.46	8.39	8.23	8.31	9.60	7.98		8.49	8.54	8.31	
Kagimoto, Masahiro (JPN) (1995)	time	2.64	3.72	4.79	5.88	6.95	8.03	9.11	10.21	11.33	12.45		13.93	8 / 6				
	reaction time	0.144	interval		1.08	1.07	1.09	1.07	1.08	1.10	1.12	1.12	1.48		3.24	3.23	3.34	
	velocity	5.20	8.46	8.54	8.39	8.54	8.46	8.46	8.31	8.16	8.16	9.47	7.90		8.46	8.49	8.21	
Furuya, Kyohei (JPN) (1996)	time	2.61	3.69	4.79	5.86	6.94	8.03	9.14	10.26	11.36	12.51		13.94	3 / 7				
	reaction time	0.102	interval		1.08	1.10	1.07	1.08	1.09	1.11	1.12	1.10	1.15	1.43		3.25	3.28	3.37
	velocity	5.26	8.46	8.31	8.54	8.46	8.39	8.23	8.16	8.31	7.95	9.80	7.89		8.44	8.36	8.14	
Fudaba, Hiroki (JPN) (1994)	time	2.68	3.77	4.85	5.91	6.98	8.05	9.15	10.27	11.40	12.54		14.06	9 / 8				
	reaction time	0.171	interval	7 steps	1.09	1.08	1.06	1.07	1.10	1.12	1.13	1.14	1.52		3.23	3.24	3.39	
	velocity	5.12	8.39	8.46	8.62	8.54	8.54	8.31	8.16	8.09	8.02	9.22	7.82		8.49	8.46	8.09	

Semi-Final 1

date 24-Jun-17 wind 0.6 m/s

(2017) - tfdata-store.com/2017/07/13/post-1040/

		H1-H10										Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10	
		H1	H2	H3	H4	H5	H6	H7	H8	H9	H10							
Takayama, Shunya (JPN) (1994)	time	2.58	3.62	4.65	5.68	6.71	7.75	8.80	9.85	10.89	12.00		13.44	4 / 1				
	reaction time	0.150	interval	1.04	1.03	1.03	1.03	1.04	1.05	1.05	1.04	1.11	1.44	PB	3.10	3.12	3.20	
	velocity	5.32	8.79	8.87	8.87	8.87	8.79	8.70	8.70	8.79	8.23	9.74	8.18		8.85	8.79	8.57	
Sato, Hiroyuki (JPN) (1990)	time	2.59	3.63	4.67	5.73	6.77	7.83	8.89	9.98	11.08	12.18		13.59	5 / 2				
	reaction time	0.169	interval	1.04	1.04	1.06	1.04	1.06	1.06	1.09	1.10	1.10	1.41	PB	3.14	3.16	3.29	
	velocity	5.30	8.79	8.79	8.62	8.79	8.62	8.62	8.39	8.31	8.31	9.94	8.09		8.73	8.68	8.33	
Yazawa, Wataru (JPN) (1991)	time	2.61	3.68	4.73	5.78	6.82	7.88	8.95	10.03	11.12	12.23		13.66	6 / 3				
	reaction time	0.146	interval	7 steps	1.07	1.05	1.05	1.04	1.06	1.07	1.08	1.09	1.11	1.43		3.17	3.17	3.28
	velocity	5.26	8.54	8.70	8.70	8.79	8.62	8.54	8.46	8.39	8.23	9.80	8.05		8.65	8.65	8.36	
Kuriki, Anthony Tyrell (JPN) (1991)	time	2.64	3.71	4.79	5.85	6.90	7.94	9.00	10.07	11.16	12.27		13.69	8 / 4				
	reaction time	0.145	interval	7 steps	1.07	1.08	1.06	1.05	1.04	1.06	1.07	1.09	1.11	1.42		3.21	3.15	3.27
	velocity	5.20	8.54	8.46	8.62	8.70	8.79	8.62	8.54	8.39	8.23	9.87	8.04		8.54	8.70	8.39	
Nomoto, Syuusei (JPN) (1995)	time	2.64	3.72	4.81	5.86	6.91	7.98	9.04	10.12	11.21	12.32		13.70	9 / 5				
	reaction time	0.156	interval	7 steps	1.08	1.09	1.05	1.05	1.07	1.06	1.08	1.09	1.11	1.38		3.22	3.18	3.28
	velocity	5.20	8.46	8.39	8.70	8.70	8.54	8.62	8.46	8.39	8.23	10.16	8.03		8.52	8.62	8.36	
Omoro, Hideki (JPN) (1990)	time	2.64	3.71	4.78	5.86	6.94	8.00	9.08	10.17	11.26	12.35		13.75	7 / 6				
	reaction time	0.147	interval	7 steps	1.07	1.07	1.08	1.08	1.06	1.08	1.09	1.09	1.40		3.22	3.22	3.27	
	velocity	5.20	8.54	8.54	8.46	8.46	8.62	8.46	8.39	8.39	8.39	10.01	8.00		8.52	8.52	8.39	
Nishizawa, Masanori (JPN) (1995)	time	2.66	3.75	4.82	5.91	6.97	8.03	9.11	10.20	11.32	12.41		13.87	3 / 7				
	reaction time	0.133	interval	1.09	1.07	1.09	1.06	1.06	1.08	1.09	1.12	1.09	1.46		3.25	3.20	3.30	
	velocity	5.16	8.39	8.54	8.39	8.62	8.62	8.46	8.39	8.16	8.39	9.60	7.93		8.44	8.57	8.31	

Heat 4

date 24-Jun-17 wind 0.0 m/s

(2017) - tfdata-store.com/2017/07/13/post-1040/

		H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10	
Tanaka, Shinya (JPN) (1993)	time	2.62	3.69	4.74	5.78	6.84	7.88	8.95	10.02	11.11	12.19		13.64	6 / 1				
	reaction time	0.148	interval	7 steps	1.07	1.05	1.04	1.06	1.04	1.07	1.07	1.09	1.08	1.45	PB	3.16	3.17	3.24
			velocity		5.24	8.54	8.70	8.79	8.62	8.79	8.54	8.54	8.39	8.46	9.67	8.06	8.68	8.65
Sato, Hiroyuki (JPN) (1990)	time	2.58	3.66	4.71	5.76	6.80	7.86	8.92	10.02	11.11	12.22		13.65	2 / 2				
	reaction time	0.165	interval		1.08	1.05	1.05	1.04	1.06	1.06	1.10	1.09	1.11	1.43		3.18	3.16	3.30
			velocity		5.32	8.46	8.70	8.70	8.79	8.62	8.62	8.31	8.39	8.23	9.80	8.06	8.62	8.68
Yazawa, Wataru (JPN) (1991)	time	2.61	3.69	4.76	5.80	6.85	7.90	8.96	10.03	11.12	12.23		13.65	8 / 3				
	reaction time	0.143	interval	7 steps	1.08	1.07	1.04	1.05	1.05	1.06	1.07	1.09	1.11	1.42		3.19	3.16	3.27

	velocity	5.26	8.46	8.54	8.79	8.70	8.70	8.62	8.54	8.39	8.23	9.87	8.06		8.60	8.68	8.39
Fujii, Ryota (JPN) (1996)	time	2.64	3.73	4.80	5.88	6.95	8.01	9.13	10.24	11.34	12.47		13.92	1 / 4			
reaction time	0.138	interval	1.09	1.07	1.08	1.07	1.06	1.12	1.11	1.10	1.13	1.45	=PB		3.24	3.25	3.34
	velocity	5.20	8.39	8.54	8.46	8.54	8.62	8.16	8.23	8.31	8.09	9.67	7.90		8.46	8.44	8.21
Baba, Kazuki (JPN) (1992)	time	2.67	3.77	4.83	5.88	6.97	8.06	9.15	10.27	11.39	12.53		13.93	4 / 5			
reaction time	0.136	interval	1.10	1.06	1.05	1.09	1.09	1.09	1.12	1.12	1.14	1.40	PB		3.21	3.27	3.38
	velocity	5.14	8.31	8.62	8.70	8.39	8.39	8.39	8.16	8.16	8.02	10.01	7.90		8.54	8.39	8.11
Iwafune, Yoichi (JPN) (1985)	time	2.71	3.81	4.88	5.97	7.05	8.15	9.24	10.37	11.50	12.63		14.14	7 / 6			
reaction time	0.151	interval	1.10	1.07	1.09	1.08	1.10	1.09	1.13	1.13	1.13	1.51			3.26	3.27	3.39
	velocity	5.06	8.31	8.54	8.39	8.46	8.31	8.39	8.09	8.09	8.09	9.28	7.78		8.41	8.39	8.09
Furuta, Kazuaki (JPN) (1992)	time	2.69	3.79	4.86	5.95	7.04	8.13	9.26	10.37	11.50	12.65		14.17	3 / 7			
reaction time	0.130	interval	7 steps	1.10	1.07	1.09	1.09	1.09	1.13	1.11	1.13	1.15	1.52		3.26	3.31	3.39
	velocity	5.10	8.31	8.54	8.39	8.39	8.39	8.09	8.23	8.09	7.95	9.22	7.76		8.41	8.28	8.09

Heat 3

Heat 3

		date	24-Jun-17	wind	0.5 m/s			(2017) - tfdata-store.com/2017/07/13/post-1040/												
				H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10	
Takayama, Shunya (JPN) (1994)	time			2.59	3.64	4.69	5.74	6.78	7.83	8.87	9.93	11.00	12.08		13.50	7 / 1				
	reaction time	0.159	interval		1.05	1.05	1.05	1.04	1.05	1.04	1.06	1.07	1.08	1.42	PB		3.15	3.13	3.21	
		velocity		5.30	8.70	8.70	8.70	8.79	8.70	8.79	8.62	8.54	8.46	9.87	8.15		8.70	8.76	8.54	
Kagimoto, Masahiro (JPN) (1995)	time			2.62	3.72	4.78	5.86	6.92	8.00	9.09	10.18	11.30	12.44		13.88	2 / 2				
	reaction time	0.140	interval		1.10	1.06	1.08	1.06	1.08	1.09	1.09	1.12	1.14	1.44			3.24	3.23	3.35	
		velocity		5.24	8.31	8.62	8.46	8.62	8.46	8.39	8.39	8.16	8.02	9.74	7.93		8.46	8.49	8.19	
Fudaba, Hiroki (JPN) (1994)	time			2.69	3.78	4.85	5.93	6.99	8.08	9.15	10.25	11.36	12.49		13.94	5 / 3				
	reaction time	0.165	interval	7 steps	1.09	1.07	1.08	1.06	1.09	1.07	1.10	1.11	1.13	1.45			3.24	3.22	3.34	
		velocity		5.10	8.39	8.54	8.46	8.62	8.39	8.54	8.31	8.23	8.09	9.67	7.89		8.46	8.52	8.21	
Ogata, Akihiro (JPN) (1993)	time			2.68	3.77	4.83	5.90	6.95	8.04	9.15	10.27	11.39	12.51		14.03	6 / 4				
	reaction time	0.155	interval	7 steps	1.09	1.06	1.07	1.05	1.09	1.11	1.12	1.12	1.12	1.52			3.22	3.25	3.36	
		velocity		5.12	8.39	8.62	8.54	8.70	8.39	8.23	8.16	8.16	8.16	9.22	7.84		8.52	8.44	8.16	
Maeda, Hirotake (JPN) (1996)	time			2.60	3.73	4.82	5.92	7.01	8.16	9.30	10.41	11.51	12.64		14.05	8 / 5				
	reaction time	0.137	interval		1.13	1.09	1.10	1.09	1.15	1.14	1.11	1.10	1.13	1.41			3.32	3.38	3.34	
		velocity		5.28	8.09	8.39	8.31	8.39	7.95	8.02	8.23	8.31	8.09	9.94	7.83		8.26	8.11	8.21	
Notoya, Yuta (JPN) (1989)	time			2.61	3.72	4.82	5.91	6.99	8.08	9.21	10.32	11.45	12.60		14.09	4 / 6				
	reaction time	0.119	interval	7 steps	1.11	1.10	1.09	1.08	1.09	1.13	1.11	1.13	1.15	1.49			3.30	3.30	3.39	
		velocity		5.26	8.23	8.31	8.39	8.46	8.39	8.09	8.23	8.09	7.95	9.41	7.81		8.31	8.31	8.09	
Furumoto, Tsubasa (JPN) (1999)	time			2.65	3.77	4.88	5.98	7.07	8.18	9.28	10.40	11.52	12.65		14.11	3 / 7				
	reaction time	0.120	interval		1.12	1.11	1.10	1.09	1.11	1.10	1.12	1.12	1.13	1.46			3.33	3.30	3.37	
		velocity		5.18	8.16	8.23	8.31	8.39	8.23	8.31	8.16	8.16	8.09	9.60	7.80		8.23	8.31	8.14	
Kobayashi, Fumihito (JPN) (1991)	time			2.76	3.88	5.05	6.26	7.45	8.65	9.90	11.14	12.38	13.63		15.27	9 / 8				
	reaction time	0.146	interval		1.12	1.17	1.21	1.19	1.20	1.25	1.24	1.24	1.25	1.64			3.50	3.64	3.73	
		velocity		4.97	8.16	7.81	7.55	7.68	7.62	7.31	7.37	7.37	7.31	8.55	7.20		7.83	7.53	7.35	

Heat 2

Heat 2

		date	24-Jun-17	wind 0.0 m/s			(2017) - tfdata-store.com/2017/07/13/post-1040/													
				H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10	
Masuno, Genta (JPN) (1993)	time			2.60	3.65	4.68	5.70	6.72	7.75	8.80	9.85	10.92	11.99		13.40	8 / 1				
	reaction time	0.156	interval		1.05	1.03	1.02	1.02	1.03	1.05	1.05	1.07	1.07	1.41	PB			3.10	3.10	3.19
		velocity		5.28	8.70	8.87	8.96	8.96	8.87	8.70	8.70	8.54	8.54	9.94	8.21			8.85	8.85	8.60
Kanai, Taio (JPN) (1995)	time			2.57	3.63	4.67	5.71	6.77	7.84	8.91	9.99	11.08	12.18		13.62	6 / 2				
	reaction time	0.135	interval	7 steps	1.06	1.04	1.04	1.06	1.07	1.07	1.08	1.09	1.10	1.44	PB			3.14	3.20	3.27
		velocity		5.34	8.62	8.79	8.79	8.62	8.54	8.54	8.46	8.39	8.31	9.74	8.08			8.73	8.57	8.39
Nomoto, Syuusei (JPN) (1995)	time			2.59	3.71	4.77	5.85	6.93	8.02	9.10	10.18	11.30	12.40		13.81	2 / 3				
	reaction time	0.154	interval	7 steps	1.12	1.06	1.08	1.08	1.09	1.08	1.08	1.12	1.10	1.41				3.26	3.25	3.30
		velocity		5.30	8.16	8.62	8.46	8.46	8.39	8.46	8.46	8.16	8.31	9.94	7.97			8.41	8.44	8.31
Nishizawa, Masanori (JPN) (1991)	time			2.63	3.71	4.79	5.86	6.93	8.01	9.08	10.18	11.28	12.39		13.83	4 / 4				
	reaction time	0.138	interval		1.08	1.08	1.07	1.07	1.08	1.07	1.10	1.10	1.11	1.44				3.23	3.22	3.31
		velocity		5.22	8.46	8.46	8.54	8.54	8.46	8.54	8.31	8.31	8.23	9.74	7.95			8.49	8.52	8.28
Yamamoto, Kenta (JPN) (1995)	time			2.68	3.75	4.83	5.91	6.97	8.04	9.13	10.23	11.32	12.43		13.84	3 / 5				
	reaction time	0.161	interval		1.07	1.08	1.08	1.06	1.07	1.09	1.10	1.09	1.11	1.41	PB			3.23	3.22	3.30
		velocity		5.12	8.54	8.46	8.46	8.62	8.54	8.39	8.31	8.39	8.23	9.94	7.95			8.49	8.52	8.31
Kishi, Kazuki (JPN) (1993)	time			2.65	3.75	4.85	5.95	7.06	8.14	9.27	10.39	11.53	12.67		14.15	9 / 6				
	reaction time	0.158	interval		1.10	1.10	1.10	1.11	1.08	1.13	1.12	1.14	1.14	1.48				3.30	3.32	3.40
		velocity		5.18	8.31	8.31	8.31	8.23	8.46	8.09	8.16	8.02	8.02	9.47	7.77			8.31	8.26	8.06

Kiriyama, Norihiro (JPN) (1992)	time	2.69	3.78	4.86	5.93	7.04	8.16	9.26	10.38	11.53	12.68		14.20	5 / 7			
	reaction time	0.176		1.09	1.08	1.07	1.11	1.12	1.10	1.12	1.15	1.15	1.52		3.24	3.33	3.42
	velocity	5.10	8.39	8.46	8.54	8.23	8.16	8.31	8.16	7.95	7.95	9.22	7.75		8.46	8.23	8.02
Okazaki, Masaki (JPN) (1994)	time	2.65	3.79	4.90	6.00	7.11	8.23	9.35	10.48	11.62	12.76		14.21	7 / 8			
	reaction time	0.142		1.14	1.11	1.10	1.11	1.12	1.12	1.13	1.14	1.14	1.45		3.35	3.35	3.41
	velocity	5.18	8.02	8.23	8.31	8.23	8.16	8.16	8.09	8.02	8.02	9.67	7.74		8.19	8.19	8.04

Heat 1

date 24-Jun-17		wind 1.3 m/s		(2017) - tfdata-store.com/2017/07/13/post-1040/													
Omoro, Hideki (JPN) (1990)	time	2.63	3.70	4.76	5.83	6.88	7.92	8.98	10.05	11.13	12.22		13.63	7 / 1			
	reaction time	0.161		1.07	1.06	1.07	1.05	1.04	1.06	1.07	1.08	1.09	1.41		3.20	3.15	3.24
	velocity	5.22	8.54	8.62	8.54	8.70	8.79	8.62	8.54	8.46	8.39	9.94	8.07		8.57	8.70	8.46
Ishikawa, Shuhei (JPN) (1995)	time	2.57	3.65	4.73	5.80	6.86	7.91	8.96	10.03	11.10	12.20		13.67	4 / 2			
	reaction time	0.156		1.08	1.08	1.07	1.06	1.05	1.05	1.07	1.07	1.10	1.47	PB	3.23	3.16	3.24
	velocity	5.34	8.46	8.46	8.54	8.62	8.70	8.70	8.54	8.54	8.31	9.54	8.05		8.49	8.68	8.46
Kuriki, Anthony Tyrell (JPN) (1991)	time	2.66	3.73	4.79	5.85	6.90	7.95	9.01	10.08	11.16	12.26		13.68	5 / 3			
	reaction time	0.140		1.07	1.06	1.06	1.05	1.05	1.06	1.07	1.08	1.10	1.42		3.19	3.16	3.25
	velocity	5.16	8.54	8.62	8.62	8.70	8.70	8.62	8.54	8.46	8.31	9.87	8.04		8.60	8.68	8.44
Furuya, Takumu (JPN) (1997)	time	2.66	3.75	4.80	5.87	6.90	7.95	9.02	10.10	11.18	12.30		13.74	6 / 4			
	reaction time	0.146		1.09	1.05	1.07	1.03	1.05	1.07	1.08	1.08	1.12	1.44		3.21	3.15	3.28
	velocity	5.16	8.39	8.70	8.54	8.87	8.70	8.54	8.46	8.46	8.16	9.74	8.01		8.54	8.70	8.36
Furuya, Kyohei (JPN) (1996)	time	2.63	3.69	4.76	5.83	6.88	7.95	9.06	10.16	11.28	12.43		13.89	9 / 5			
	reaction time	0.124		1.06	1.07	1.07	1.05	1.07	1.11	1.10	1.12	1.15	1.46		3.20	3.23	3.37
	velocity	5.22	8.62	8.54	8.54	8.70	8.54	8.23	8.31	8.16	7.95	9.60	7.92		8.57	8.49	8.14
Wado, Tatsuya (JPN) (1990)	time	2.63	3.74	4.80	5.88	6.95	8.01	9.09	10.18	11.28	12.40		13.91	8 / 6			
	reaction time	0.158		1.11	1.06	1.08	1.07	1.06	1.08	1.09	1.10	1.12	1.51		3.25	3.21	3.31
	velocity	5.22	8.23	8.62	8.46	8.54	8.62	8.46	8.39	8.31	8.16	9.28	7.91		8.44	8.54	8.28
Hiramatsu, Papdemba (JPN) (1991)	time	2.63	3.72	4.81	5.86	6.93	8.01	9.10	10.19	11.30	12.44		13.93	2 / 7			
	reaction time	0.168		1.09	1.09	1.05	1.07	1.08	1.09	1.09	1.11	1.14	1.49		3.23	3.24	3.34
	velocity	5.22	8.39	8.39	8.70	8.54	8.46	8.39	8.39	8.23	8.02	9.41	7.90		8.49	8.46	8.21
Murakami, Masao (JPN) (1991)	time	2.64	3.74	4.83	5.92	7.00	8.09	9.19	10.32	11.46	12.64		14.14	3 / 8			
	reaction time	0.167		1.10	1.09	1.09	1.08	1.09	1.10	1.13	1.14	1.18	1.50		3.28	3.27	3.45
	velocity	5.20	8.31	8.39	8.39	8.46	8.39	8.31	8.09	8.02	7.75	9.35	7.78		8.36	8.39	7.95

2017 Jamaican National Championships (Kingston, JAM) (TV Analysis)

FINAL

date 24-Jun-17		wind 0.7 m/s		Henson (2020) - Athlete First: 2017 year end hurdle report													
McLeod, Omar (JAM) (1994)	time	2.43	3.46	4.46	5.46	6.43	7.43	8.43	9.43	10.43	11.45		12.90	4 / 1			
	reaction time			1.03	1.00	1.00	0.97	1.00	1.00	1.00	1.02	1.45	NR PB		3.03	2.97	3.02
	velocity	5.65	8.87	9.14	9.14	9.42	9.14	9.14	9.14	9.14	8.96	9.67	8.53		9.05	9.23	9.08
Levy, Ronald (JAM) (1992)	time	2.43	3.46	4.48	5.50	6.50	7.50	8.52	9.53	10.56	11.62		13.13	5 / 2			
	reaction time			1.03	1.02	1.02	1.00	1.00	1.02	1.01	1.03	1.06	1.51		3.07	3.02	3.10
	velocity	5.65	8.87	8.96	8.96	9.14	9.14	8.96	9.05	8.87	8.62	9.28	8.38		8.93	9.08	8.85
Parchment, Hansle (JAM) (1990)	time	2.50	3.55	4.57	5.57	6.57	7.57	8.57	9.57	10.60	11.63		13.19	8 / 3			
	reaction time			1.05	1.02	1.00	1.00	1.00	1.00	1.03	1.03	1.56			3.07	3.00	3.06
	velocity	5.49	8.70	8.96	9.14	9.14	9.14	9.14	9.14	8.87	8.87	8.99	8.34		8.93	9.14	8.96
Russell, Dejour (JAM) (2000)	time	2.43	3.47	4.50	5.53	6.58	7.63	8.67	9.70	10.75	11.80		13.32	6 / 4			
	reaction time			1.04	1.03	1.03	1.05	1.05	1.04	1.03	1.05	1.05	1.52	AJR PB	3.10	3.14	3.13
	velocity	5.65	8.79	8.87	8.87	8.70	8.70	8.79	8.87	8.70	8.70	9.22	8.26		8.85	8.73	8.76

2017 Bauhaus Athletics (Stockholm, SWE) (TV Analysis)

FINAL

date 18-Jun-17		wind 3.5 m/s		Henson (2020) - Athlete First: 2017 year end hurdle report													
Ortega, Orlando (ESP) (1991)	time	2.56	3.56	4.58	5.62	6.62	7.62	8.62	9.60	10.62	11.64		13.09	4 / 1			
	reaction time	0.146		1.00	1.02	1.04	1.00	1.00	0.98	1.02	1.02	1.45			3.06	3.00	3.02
	velocity	5.36	9.14	8.96	8.79	9.14	9.14	9.14	9.33	8.96	8.96	9.67	8.40		8.96	9.14	9.08
Shubenkov, Sergey (ANA) (1991)	time	2.52	3.58	4.56	5.58	6.58	7.56	8.56	9.54	10.60	11.66		13.10	5 / 2			
	reaction time	0.141		1.06	0.98	1.02	1.00	0.98	1.00	0.98	1.06	1.06	1.44		3.06	2.98	3.10
	velocity	5.44	8.62	9.33	8.96	9.14	9.33	9.14	9.33	8.62	8.62	9.74	8.40		8.96	9.20	8.85
Brathwaite, Shane (BAR) (1990)	time	2.48	3.56	4.56	5.58	6.60	7.64	8.66	9.68	10.72	11.78		13.25	1 / 3			
	reaction time	0.134		1.08	1.00	1.02	1.02	1.04	1.02	1.02	1.04	1.06	1.47		3.10	3.08	3.12
	velocity	5.53	8.46	9.14	8.96	8.96	8.79	8.96	8.96	8.79	8.62	9.54	8.30		8.85	8.90	8.79

Manga, Aurel (FRA) (1992)	time	2.46	3.48	4.50	5.54	6.54	7.56	8.58	9.62	10.70	11.78		13.30		6 / 4				
	reaction time	0.147	interval	7 steps	1.02	1.02	1.04	1.00	1.02	1.02	1.04	1.08	1.08	1.52			3.08	3.04	3.20
		velocity	5.58	8.96	8.96	8.79	9.14	8.96	8.96	8.79	8.46	8.46	9.22	8.27			8.90	9.02	8.57
Czykier, Damian (POL) (1992)	time	2.54	3.62	4.64	5.64	6.66	7.70	8.72	9.74	10.78	11.86		13.36		7 / 5				
	reaction time	0.164	interval	7 steps	1.08	1.02	1.00	1.02	1.04	1.02	1.04	1.08	1.50				3.10	3.08	3.14
		velocity	5.40	8.46	8.96	9.14	8.96	8.79	8.96	8.96	8.79	8.46	9.35	8.23			8.85	8.90	8.73

2017 Golden Gala Pietro Mennea (Rome, ITA) (TV Analysis)**FINAL**

INAL	date		08-Jun-17		wind		-0.3 m/s		Henson (2020) - Athlete First: 2017 year end hurdle report									
			H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
Merritt, Aries (USA) (1985)	time		2.44	3.48	4.48	5.48	6.50	7.50	8.54	9.56	10.60	11.66		13.13	7 / 1			
	reaction time	0.132	interval	7 steps	1.04	1.00	1.00	1.02	1.00	1.04	1.02	1.04	1.06	1.47		3.04	3.06	3.12
			velocity	5.62	8.79	9.14	9.14	8.96	9.14	8.79	8.96	8.79	8.62	9.54	8.38		9.02	8.96
Ortega, Orlando (ESP) (1991)	time		2.58	3.62	4.64	5.66	6.68	7.66	8.68	9.70	10.72	11.76		13.17	6 / 2			
	reaction time	0.152	interval	7 steps	1.04	1.02	1.02	1.02	0.98	1.02	1.02	1.02	1.04	1.41		3.08	3.02	3.08
			velocity	5.32	8.79	8.96	8.96	8.96	9.33	8.96	8.96	8.96	8.79	9.94	8.35		8.90	9.08
Shubenkov, Sergey (ANA) (1991)	time		2.60	3.66	4.68	5.68	6.68	7.68	8.70	9.70	10.72	11.78		13.21	4 / 3			
	reaction time	0.154	interval	7 steps	1.06	1.02	1.00	1.00	1.00	1.02	1.00	1.02	1.06	1.43		3.08	3.02	3.08
			velocity	5.28	8.62	8.96	9.14	9.14	9.14	8.96	9.14	8.96	8.62	9.80	8.33		8.90	9.08
Pozzi, Andy (GBR) (1992)	time		2.48	3.52	4.52	5.52	6.52	7.50	8.52	9.58	10.62	11.74		13.24	5 / 4			
	reaction time	0.119	interval	7 steps	1.04	1.00	1.00	1.00	0.98	1.02	1.06	1.04	1.12	1.50		3.04	3.00	3.22
			velocity	5.53	8.79	9.14	9.14	9.14	9.33	8.96	8.62	8.79	8.16	9.35	8.31		9.02	9.14
Trajkovic, Milan (CYP) (1992)	time		2.52	3.54	4.56	5.58	6.62	7.66	8.70	9.74	10.82	11.90		13.39	9 / 5			
	reaction time	0.157	interval	7 steps	1.02	1.02	1.02	1.04	1.04	1.04	1.04	1.08	1.08	1.49		3.06	3.12	3.20
			velocity	5.44	8.96	8.96	8.96	8.79	8.79	8.79	8.79	8.46	8.46	9.41	8.22		8.96	8.79
Oliver, David (USA) (1982)	time		2.50	3.54	4.58	5.62	6.62	7.62	8.66	9.74	10.82	11.92		13.46	3 / 6			
	reaction time	0.141	interval	7 steps	1.04	1.04	1.04	1.00	1.00	1.04	1.08	1.08	1.10	1.54		3.12	3.04	3.26
			velocity	5.49	8.79	8.79	8.79	9.14	9.14	8.79	8.46	8.46	8.31	9.10	8.17		8.79	9.02
Cabral, Jonathan (CAN) (1992)	time		2.52	3.58	4.64	5.66	6.70	7.74	8.78	9.86	10.94	12.08		13.60	2 / 7			
	reaction time	0.130	interval	7 steps	1.06	1.06	1.02	1.04	1.04	1.04	1.08	1.08	1.14	1.52		3.14	3.12	3.30
			velocity	5.44	8.62	8.62	8.96	8.79	8.79	8.79	8.46	8.46	8.02	9.22	8.09		8.73	8.79
Perini, Lorenzo (ITA) (1994)	time		2.56	3.62	4.68	5.74	6.80	7.84	8.90	9.96	11.06	12.14		13.65	1 / 8			
	reaction time	0.141	interval		1.06	1.06	1.06	1.06	1.04	1.06	1.06	1.10	1.08	1.51		3.18	3.16	3.24
			velocity	5.36	8.62	8.62	8.62	8.62	8.79	8.62	8.62	8.31	8.46	9.28	8.06		8.62	8.68
Darien, Garfield (FRA) (1987)	time		2.52	3.56	4.58	5.60	6.72							15.64	8 / 9			
	reaction time	0.148	interval	7 steps	1.04	1.02	1.02	1.12								3.08		
			velocity	5.44	8.79	8.96	8.96	8.16						7.03		8.90		

2017 Fuse Sprint (Tottori, JPN)**FINAL**

		date	04-Jun-17		wind		3.2 m/s												(2017) - tfdata-store.com/2017/06/04/post-848/				
			H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10					
Masuno, Genta (JPN) (1993)	time	2.59	3.65	4.70	5.74	6.77	7.80	8.84	9.89	10.96	12.04			13.43	6 / 1								
	reaction time		interval	1.06	1.05	1.04	1.03	1.03	1.04	1.05	1.07	1.08	1.39			3.15	3.10	3.20					
	velocity	5.30	8.62	8.70	8.79	8.87	8.87	8.79	8.70	8.54	8.46	10.09	8.19			8.70	8.85	8.57					
Takayama, Shunya (JPN) (1994)	time	2.62	3.67	4.72	5.77	6.82	7.87	8.93	10.00	11.08	12.18			13.62	5 / 2								
	reaction time		interval	7 steps	1.05	1.05	1.05	1.05	1.06	1.07	1.08	1.10	1.44			3.15	3.16	3.25					
	velocity	5.24	8.70	8.70	8.70	8.70	8.70	8.62	8.54	8.46	8.31	9.74	8.08			8.70	8.68	8.44					
Tanaka, Shinya (JPN) (1993)	time	2.61	3.68	4.73	5.79	6.85	7.92	8.99	10.07	11.16	12.25			13.68	9 / 3								
	reaction time		interval	1.07	1.05	1.06	1.06	1.07	1.07	1.08	1.09	1.09	1.43			3.18	3.20	3.26					
	velocity	5.26	8.54	8.70	8.62	8.62	8.54	8.54	8.46	8.39	8.39	9.80	8.04			8.62	8.57	8.41					
Wado, Tatsuya (JPN) (1990)	time	2.58	3.66	4.70	5.75	6.81	7.87	8.92	9.99	11.08	12.18			13.68	8 / 4								
	reaction time		interval	1.08	1.04	1.05	1.06	1.06	1.05	1.07	1.09	1.10	1.50			3.17	3.17	3.26					
	velocity	5.32	8.46	8.79	8.70	8.62	8.62	8.70	8.54	8.39	8.31	9.35	8.04			8.65	8.65	8.41					
Sato, Hiroyuki (JPN) (1990)	time	2.59	3.66	4.75	5.83	6.89	7.97	9.04	10.14	11.24	12.35			13.79	7 / 5								
	reaction time		interval	1.07	1.09	1.08	1.06	1.08	1.07	1.10	1.10	1.11	1.44			3.24	3.21	3.31					
	velocity	5.30	8.54	8.39	8.46	8.62	8.46	8.54	8.31	8.31	8.23	9.74	7.98			8.46	8.54	8.28					
Katsuta, Kizuku (JPN) (1999)	time	2.69	3.78	4.90	5.99	7.09	8.20	9.32	10.45	11.57	12.74			14.20	3 / 6								
	reaction time		interval	1.09	1.12	1.09	1.10	1.11	1.12	1.13	1.12	1.17	1.46			3.30	3.33	3.42					
	velocity	5.10	8.39	8.16	8.39	8.31	8.23	8.16	8.09	8.16	7.81	9.60	7.75			8.31	8.23	8.02					
Morita, Shunichi (JPN) (1989)	time	2.76	3.83	4.95	6.05	7.14	8.24	9.37	10.50	11.63	12.78			14.24	2 / 7								

	reaction time	interval	1.07	1.12	1.10	1.09	1.10	1.13	1.13	1.13	1.15	1.46		3.29	3.32	3.41		
		velocity	4.97	8.54	8.16	8.31	8.39	8.31	8.09	8.09	8.09	7.95	9.60	7.72	8.33	8.26	8.04	
Heat 5																		
	date	04-Jun-17	wind	1.5 m/s		(2017) - tfdata-store.com/2017/06/04/post-848/												
			H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
Omoro, Hideki (JPN) (1990)	time		2.58	3.65	4.69	5.74	6.79	7.84	8.88	9.94	11.03	12.12		13.54	4 / 1			
	reaction time	interval		1.07	1.04	1.05	1.05	1.05	1.04	1.06	1.09	1.09	1.42			3.16	3.14	3.24
		velocity	5.32	8.54	8.79	8.70	8.70	8.70	8.79	8.62	8.39	8.39	9.87	8.12		8.68	8.73	8.46
Yazawa, Wataru (JPN) (1991)	time		2.60	3.68	4.74	5.78	6.82	7.87	8.93	9.99	11.07	12.16		13.60	6 / 2			
	reaction time	interval	7 steps	1.08	1.06	1.04	1.04	1.05	1.06	1.06	1.08	1.09	1.44			3.18	3.15	3.23
		velocity	5.28	8.46	8.62	8.79	8.79	8.70	8.62	8.62	8.46	8.39	9.74	8.09		8.62	8.70	8.49
Takayama, Shunya (JPN) (1994)	time		2.57	3.61	4.68	5.73	6.77	7.82	8.89	9.96	11.04	12.16		13.63	5 / 3			
	reaction time	interval		1.04	1.07	1.05	1.04	1.05	1.07	1.07	1.08	1.12	1.47			3.16	3.16	3.27
		velocity	5.34	8.79	8.54	8.70	8.79	8.70	8.54	8.54	8.46	8.16	9.54	8.07		8.68	8.68	8.39
Masuno, Genta (JPN) (1993)	time		2.63	3.69	4.78	5.83	6.89	7.94	9.02	10.09	11.18	12.27		13.69	2 / 4			
	reaction time	interval		1.06	1.09	1.05	1.06	1.05	1.08	1.07	1.09	1.09	1.42			3.20	3.19	3.25
		velocity	5.22	8.62	8.39	8.70	8.62	8.70	8.46	8.54	8.39	8.39	9.87	8.04		8.57	8.60	8.44
Sato, Hiroyuki (JPN) (1990)	time		2.61	3.68	4.74	5.78	6.84	7.90	8.97	10.07	11.18	12.30		13.75	7 / 5			
	reaction time	interval		1.07	1.06	1.04	1.06	1.06	1.07	1.10	1.11	1.12	1.45			3.17	3.19	3.33
		velocity	5.26	8.54	8.62	8.79	8.62	8.62	8.54	8.31	8.23	8.16	9.67	8.00		8.65	8.60	8.23
Wado, Tatsuya (JPN) (1990)	time		2.59	3.68	4.73	5.79	6.85	7.91	8.98	10.05	11.15	12.26		13.78	8 / 6			
	reaction time	interval		1.09	1.05	1.06	1.06	1.06	1.07	1.07	1.10	1.11	1.52			3.20	3.19	3.28
		velocity	5.30	8.39	8.70	8.62	8.62	8.62	8.54	8.54	8.31	8.23	9.22	7.98		8.57	8.60	8.36
Tanaka, Shinya (JPN) (1993)	time		2.63	3.70	4.79	5.85	6.93	8.02	9.10	10.18	11.31	12.42		13.89	1 / 7			
	reaction time	interval	7 steps	1.07	1.09	1.06	1.08	1.09	1.08	1.08	1.13	1.11	1.47			3.22	3.25	3.32
		velocity	5.22	8.54	8.39	8.62	8.46	8.39	8.46	8.46	8.09	8.23	9.54	7.92		8.52	8.44	8.26

2017 Korean National Championships (Kimchun, KOR)**FINAL**

FINAL	date	02-Jun-17	wind 2.6 m/s										Hong (2018) - performance of men's 110m hurdles using rhythmic units						
			H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10	
Kim Byung-Jun (KOR) (1991)	time		2.63	3.76	4.83	5.90	6.90	8.03	9.06	10.13	11.20	12.27		13.70	/ 1				
	reaction time			1.13	1.07	1.07	1.00	1.13	1.03	1.07	1.07	1.07	1.43			3.27	3.16	3.21	
	interval		5.22	8.09	8.54	8.54	9.14	8.09	8.87	8.54	8.54	8.54	9.80	8.03		8.39	8.68	8.54	
Lee Jung-Joon (KOR) (1984)	time		2.62	3.72	4.82	5.92	6.95	8.05	9.12	10.19	11.29	12.39		13.86	/ 2				
	reaction time			1.10	1.10	1.10	1.03	1.10	1.07	1.07	1.10	1.10	1.47			3.30	3.20	3.27	
	interval		5.24	8.31	8.31	8.31	8.87	8.31	8.54	8.54	8.31	8.31	9.54	7.94		8.31	8.57	8.39	
Kim Gyeong-Tae (KOR) (1997)	time		2.65	3.78	4.88	6.01	7.14	8.21	9.31	10.41	11.54	12.67		14.20	/ 3				
	reaction time			1.13	1.10	1.13	1.13	1.07	1.10	1.10	1.13	1.13	1.53			3.36	3.30	3.36	
	interval		5.18	8.09	8.31	8.09	8.09	8.54	8.31	8.31	8.09	8.09	9.16	7.75		8.16	8.31	8.16	

2017 Shimane High School Championships (Honshu, JPN)**FINAL**

FINAL

	date	28-May-17	wind	0.3 m/s	Shimizu (2018) - shimane high school championships 2018 - biomechanics data analysis														
			H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10	
勝田 築	time		2.73	3.87	5.00	6.10	7.27	8.37	9.67	10.93	12.17	13.40		14.91	4 / 1				
	reaction time			1.14	1.13	1.10	1.17	1.10	1.30	1.26	1.24	1.23	1.51				3.37	3.57	3.73
	velocity		5.03	8.02	8.09	8.31	7.81	8.31	7.03	7.25	7.37	7.43	9.28	7.38			8.14	7.68	7.35
京極 史弥	time		2.87	4.07	5.23	6.40	7.57	8.73	9.90	11.13	12.37	13.60		15.20	8 / 2				
	reaction time			1.20	1.16	1.17	1.17	1.16	1.17	1.23	1.24	1.23	1.60				3.53	3.50	3.70
	velocity		4.78	7.62	7.88	7.81	7.81	7.88	7.81	7.43	7.37	7.43	8.76	7.24			7.77	7.83	7.41
阿川 公希	time		2.80	4.00	5.20	6.37	7.57	8.77	9.97	11.20	12.47	13.70		15.29	7 / 3				
	reaction time			1.20	1.20	1.17	1.20	1.20	1.20	1.23	1.27	1.23	1.59				3.57	3.60	3.73
	velocity		4.90	7.62	7.62	7.81	7.62	7.62	7.62	7.43	7.20	7.43	8.82	7.19			7.68	7.62	7.35
遠所 勇歩	time		2.90	4.10	5.30	6.50	7.70	8.93	10.17	11.43	12.67	13.97		15.55	5 / 4				
	reaction time			1.20	1.20	1.20	1.20	1.23	1.24	1.26	1.24	1.30	1.58				3.60	3.67	3.80
	velocity		4.73	7.62	7.62	7.62	7.62	7.43	7.37	7.25	7.37	7.03	8.87	7.07			7.62	7.47	7.22

2017 Prefontaine Classic (Eugene, OR) (TV Analysis)**FINAL**

FINAL		date	27-May-17	wind	Henson (2020) - Athlete First: 2017 year end hurdle report														
				H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
McLeod, Omar (JAM) (1994)	time			2.48	3.52	4.52	5.52	6.52	7.54	8.52	9.52	10.54	11.58		13.01	4 / 1			
	reaction time	0.150	interval		1.04	1.00	1.00	1.00	1.02	0.98	1.00	1.02	1.04	1.43			3.04	3.00	3.06
			velocity	5.53	8.79	9.14	9.14	9.14	8.96	9.33	9.14	8.96	8.79	9.80	8.46		9.02	9.14	8.96
Levy, Ronald (JAM) (1992)	time			2.44	3.48	4.50	5.52	6.52	7.54	8.54	9.58	10.60	11.64		13.10	1 / 2			
	reaction time	0.151	interval	7 steps	1.04	1.02	1.02	1.00	1.02	1.00	1.04	1.02	1.04	1.46	PB		3.08	3.02	3.10

	velocity	5.62	8.79	8.96	8.96	9.14	8.96	9.14	8.79	8.96	8.79	9.60	8.40		8.90	9.08	8.85
Allen, Devon (USA) (1994)	time	2.48	3.52	4.54	5.54	6.56	7.56	8.58	9.60	10.60	11.64		13.11	5 / 3			
reaction time	0.131	interval	1.04	1.02	1.00	1.02	1.00	1.02	1.02	1.00	1.04	1.47			3.06	3.04	3.06
	velocity	5.53	8.79	8.96	9.14	8.96	9.14	8.96	8.96	9.14	8.79	9.54	8.39		8.96	9.02	8.96
Merritt, Aries (USA) (1985)	time	2.48	3.50	4.52	5.52	6.54	7.54	8.56	9.58	10.62	11.66		13.13	6 / 4			
reaction time	0.132	interval	7 steps	1.02	1.02	1.00	1.02	1.00	1.02	1.02	1.04	1.04	1.47		3.04	3.04	3.10
	velocity	5.53	8.96	8.96	9.14	8.96	9.14	8.96	8.96	8.79	8.79	9.54	8.38		9.02	9.02	8.85
Pozzi, Andy (GBR) (1992)	time	2.46	3.48	4.48	5.50	6.50	7.50	8.50	9.52	10.54	11.62		13.19	2 / 5			
reaction time	0.145	interval	7 steps	1.02	1.00	1.02	1.00	1.00	1.02	1.02	1.08	1.57	=PB		3.04	3.00	3.12
	velocity	5.58	8.96	9.14	8.96	9.14	9.14	8.96	8.96	8.46	8.93	8.34			9.02	9.14	8.79
Harris, Aleec (USA) (1990)	time	2.52	3.58	4.62	5.62	6.62	7.64	8.68	9.74	10.78	11.82		13.30	8 / 6			
reaction time	0.165	interval		1.06	1.04	1.00	1.00	1.02	1.04	1.06	1.04	1.04	1.48		3.10	3.06	3.14
	velocity	5.44	8.62	8.79	9.14	9.14	8.96	8.79	8.62	8.79	8.79	9.47	8.27		8.85	8.96	8.73
Oliver, David (USA) (1982)	time	2.48	3.56	4.58	5.60	6.62	7.64	8.68	9.72	10.78	11.86		13.40	7 / 7			
reaction time	0.172	interval	7 steps	1.08	1.02	1.02	1.02	1.04	1.04	1.06	1.08	1.54			3.12	3.08	3.18
	velocity	5.53	8.46	8.96	8.96	8.96	8.96	8.79	8.79	8.62	8.46	9.10	8.21		8.79	8.90	8.62
Bascou, Dimitri (FRA) (1987)	time	2.48	3.50	4.54	5.56	6.60	7.66	8.74	9.82	10.90	12.00		13.55	3 / 8			
reaction time	0.158	interval	7 steps	1.02	1.04	1.02	1.04	1.06	1.08	1.08	1.08	1.10	1.55		3.08	3.18	3.26
	velocity	5.53	8.96	8.79	8.96	8.79	8.62	8.46	8.46	8.46	8.31	9.05	8.12		8.90	8.62	8.41

2017 Kanto Collegiate Championships (Yokohama, JPN)

Division 1 FINAL

date 26-May-17 wind 0.4 m/s

(2017) - tfdata-store.com/2017/06/08/914/

		H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10	
Kanai, Taio (JPN) (1995)	time	2.58	3.63	4.68	5.74	6.80	7.87	8.93	10.01	11.09	12.21		13.63	5 / 1				
	reaction time	0.149	interval	7 steps	1.05	1.05	1.06	1.06	1.07	1.06	1.08	1.08	1.12	1.42	PB	3.16	3.19	3.28
		velocity	5.32	8.70	8.70	8.62	8.62	8.54	8.62	8.46	8.46	8.16	9.87	8.07		8.68	8.60	8.36

2017 Seiko Golden Grand Prix (Kawasaki, JPN)

FINAL

date 21-May-17 wind -1.8 m/s

(2017) - tfdata-store.com/2017/05/22/post-599/

		H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10	
Xie Wenjun (CHN) (1990)	time	2.61	3.66	4.71	5.75	6.79	7.82	8.86	9.92	10.98	12.06		13.51	3 / 1				
	reaction time	0.153	interval	1.05	1.05	1.04	1.04	1.03	1.04	1.06	1.06	1.08	1.45		3.14	3.11	3.20	
	velocity	5.26	8.70	8.70	8.79	8.79	8.87	8.79	8.62	8.62	8.46	9.67	8.14		8.73	8.82	8.57	
Trajkovic, Milan (CYP) (1992)	time	2.56	3.61	4.64	5.70	6.75	7.76	8.81	9.87	10.96	12.05		13.52	5 / 2				
	reaction time	0.155	interval	7 steps	1.05	1.03	1.06	1.05	1.01	1.05	1.06	1.09	1.09	1.47		3.14	3.11	3.24
	velocity	5.36	8.70	8.87	8.62	8.70	9.05	8.70	8.62	8.39	8.39	9.54	8.14		8.73	8.82	8.46	
Omoro, Hideki (JPN) (1990)	time	2.59	3.65	4.69	5.74	6.81	7.87	8.92	9.99	11.06	12.14		13.59	7 / 3				
	reaction time	0.141	interval	1.06	1.04	1.05	1.07	1.06	1.05	1.07	1.07	1.08	1.45		3.15	3.18	3.22	
	velocity	5.30	8.62	8.79	8.70	8.54	8.62	8.70	8.54	8.54	8.46	9.67	8.09		8.70	8.62	8.52	
Yazawa, Wataru (JPN) (1991)	time	2.64	3.72	4.78	5.84	6.89	7.95	9.02	10.07	11.15	12.24		13.69	6 / 4				
	reaction time	0.151	interval	7 steps	1.08	1.06	1.06	1.05	1.06	1.07	1.05	1.08	1.09	1.45		3.20	3.18	3.22
	velocity	5.20	8.46	8.62	8.62	8.70	8.62	8.54	8.70	8.46	8.39	9.67	8.04		8.57	8.62	8.52	
Swift, Greggmar (BAR) (1991)	time	2.55	3.61	4.68	5.73	6.77	7.84	8.92	10.00	11.10	12.22		13.72	1 / 5				
	reaction time	0.141	interval	1.06	1.07	1.05	1.04	1.07	1.08	1.08	1.10	1.12	1.50		3.18	3.19	3.30	
	velocity	5.38	8.62	8.54	8.70	8.79	8.54	8.46	8.46	8.31	8.16	9.35	8.02		8.62	8.60	8.31	
Masuno, Genta (JPN) (1993)	time	2.59	3.66	4.73	5.77	6.83	7.89	8.94	10.03	11.11	12.25		13.72	2 / 6				
	reaction time	0.150	interval	1.07	1.07	1.04	1.06	1.06	1.05	1.09	1.08	1.14	1.47		3.18	3.17	3.31	
	velocity	5.30	8.54	8.54	8.79	8.62	8.62	8.70	8.39	8.46	8.02	9.54	8.02		8.62	8.65	8.28	
Porter, Jeff (USA) (1985)	time	2.59	3.68	4.74	5.82	6.87	7.94	9.00	10.08	11.16	12.26		13.73	4 / 7				
	reaction time	0.125	interval	1.09	1.06	1.08	1.05	1.07	1.06	1.08	1.08	1.10	1.47		3.23	3.18	3.26	
	velocity	5.30	8.39	8.62	8.46	8.70	8.54	8.62	8.46	8.46	8.31	9.54	8.01		8.49	8.62	8.41	
Yang Wei-Ting (TPE) (1994)	time	2.67	3.75	4.83	5.89	6.94	7.99	9.06	10.13	11.20	12.31		13.79	8 / 8				
	reaction time	0.145	interval	1.08	1.08	1.06	1.05	1.05	1.07	1.07	1.07	1.11	1.48		3.22	3.17	3.25	
	velocity	5.14	8.46	8.46	8.62	8.70	8.70	8.54	8.54	8.54	8.54	8.23	9.47	7.98	8.52	8.65	8.44	

2017 East Japan Business Athletics Championships (Yuma, JPN)

FINAL

date 20-May-17 wind 1.1 m/s

(2017) - tfdata-store.com/2017/05/20/post-579/

		H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10	
Takayama, Shunya (JPN) (1994)	time	2.58	3.62	4.64	5.68	6.72	7.77	8.83	9.90	10.98	12.08		13.53	4 / 1				
	reaction time	interval		1.04	1.02	1.04	1.04	1.05	1.06	1.07	1.08	1.10	1.45	PB		3.10	3.15	3.25
	velocity	5.32	8.79	8.96	8.79	8.79	8.70	8.62	8.54	8.46	8.31	9.67	8.13		8.85	8.70	8.44	
Sato, Hiroyuki (JPN) (1990)	time	2.57	3.63	4.69	5.74	6.79	7.87	8.95	10.04	11.16	12.28		13.75	7 / 2				

reaction time	interval	1.06	1.06	1.05	1.05	1.08	1.08	1.09	1.12	1.12	1.47			3.17	3.21	3.33
	velocity	5.34	8.62	8.62	8.70	8.70	8.46	8.46	8.39	8.16	8.16	9.54	8.00	8.65	8.54	8.23
Tanaka, Shinya (JPN) (1993)	time	2.58	3.65	4.70	5.78	6.84	7.93	9.02	10.10	11.20	12.32		13.76	6 / 3		
reaction time	interval	7 steps	1.07	1.05	1.08	1.06	1.09	1.09	1.08	1.10	1.12	1.44	PB		3.20	3.24
	velocity	5.32	8.54	8.70	8.46	8.62	8.39	8.39	8.46	8.31	8.16	9.74	7.99		8.57	8.46
Ogata, Akihiro (JPN) (1993)	time	2.59	3.66	4.73	5.83	6.88	7.96	9.05	10.16	11.30	12.43		13.95	5 / 4		
reaction time	interval	7 steps	1.07	1.07	1.10	1.05	1.08	1.09	1.11	1.14	1.13	1.52			3.24	3.22
	velocity	5.30	8.54	8.54	8.31	8.70	8.46	8.39	8.23	8.02	8.09	9.22	7.89		8.46	8.52
Fudaba, Hiroki (JPN) (1994)	time	2.71	3.81	4.86	5.94	7.01	8.10	9.18	10.28	11.37	12.50		14.00	2 / 5		
reaction time	interval	7 steps	1.10	1.05	1.08	1.07	1.09	1.08	1.10	1.09	1.13	1.50			3.23	3.24
	velocity	5.06	8.31	8.70	8.46	8.54	8.39	8.46	8.31	8.39	8.09	9.35	7.86		8.49	8.46
Abe, Hirofumi (JPN) (1985)	time	2.61	3.71	4.80	5.88	6.95	8.06	9.14	10.26	11.40	12.53		14.02	3 / 6		
reaction time	interval		1.10	1.09	1.08	1.07	1.11	1.08	1.12	1.14	1.13	1.49	PB		3.27	3.26
	velocity	5.26	8.31	8.39	8.46	8.54	8.23	8.46	8.16	8.02	8.09	9.41	7.85		8.39	8.41
Okazaki, Masaki (JPN) (1994)	time	2.64	3.74	4.83	5.94	7.02	8.16	9.28	10.40	11.55	12.70		14.20	8 / 7		
reaction time	interval		1.10	1.09	1.11	1.08	1.14	1.12	1.12	1.15	1.15	1.50			3.30	3.34
	velocity	5.20	8.31	8.39	8.23	8.46	8.02	8.16	8.16	7.95	7.95	9.35	7.75		8.31	8.21
Morita, Shunichi (JPN) (1989)	time	2.77	3.90	5.01	6.11	7.23	8.35	9.48	10.63	11.80	12.98		14.62	9 / 8		
reaction time	interval		1.13	1.11	1.10	1.12	1.12	1.13	1.15	1.17	1.18	1.64			3.34	3.37
	velocity	4.95	8.09	8.23	8.31	8.16	8.16	8.09	7.95	7.81	7.75	8.55	7.52		8.21	8.14

2017 Shanghai Diamond League (Shanghai, CHN) (TV Analysis)

FINAL

date 13-May-17 wind 0.5 m/s

Henson (2020) - Athlete First: 2017 year end hurdle report

			H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
McLeod, Omar (JAM) (1994)	time		2.48	3.50	4.54	5.58	6.56	7.58	8.58	9.62	10.64	11.70		13.09	5 / 1			
reaction time	0.141	interval		1.02	1.04	1.04	0.98	1.02	1.00	1.04	1.02	1.06	1.39			3.10	3.00	3.12
		velocity	5.53	8.96	8.79	8.79	9.33	8.96	9.14	8.79	8.96	8.62	10.09	8.40		8.85	9.14	8.79
Ortega, Orlando (ESP) (1991)	time		2.50	3.56	4.58	5.58	6.58	7.58	8.60	9.60	10.64	11.70		13.15	6 / 2			
reaction time	0.136	interval	7 steps	1.06	1.02	1.00	1.00	1.00	1.02	1.00	1.04	1.06	1.45			3.08	3.02	3.10
		velocity	5.49	8.62	8.96	9.14	9.14	9.14	8.96	9.14	8.79	8.62	9.67	8.37		8.90	9.08	8.85
Xie Wenjun (CHN) (1990)	time		2.56	3.62	4.66	5.66	6.68	7.70	8.72	9.74	10.78	11.86		13.31	2 / 3			
reaction time	0.156	interval		1.06	1.04	1.00	1.02	1.02	1.02	1.02	1.04	1.08	1.45			3.10	3.06	3.14
		velocity	5.36	8.62	8.79	9.14	8.96	8.96	8.96	8.96	8.79	8.46	9.67	8.26		8.85	8.96	8.73
Parchment, Hansle (JAM) (1991)	time		2.54	3.58	4.58	5.60	6.62	7.62	8.66	9.70	10.76	11.82		13.35	3 / 4			
reaction time	0.143	interval	7 steps	1.04	1.00	1.02	1.02	1.00	1.04	1.04	1.06	1.06	1.53			3.06	3.06	3.16
		velocity	5.40	8.79	9.14	8.96	8.96	9.14	8.79	8.79	8.62	8.62	9.16	8.24		8.96	8.96	8.68
Shubenkov, Sergey (ANA) (1991)	time		2.56	3.64	4.66	5.70	6.70	7.70	8.76	9.80	10.86	11.92		13.35	4 / 5			
reaction time	0.153	interval	7 steps	1.08	1.02	1.04	1.00	1.00	1.06	1.04	1.06	1.06	1.43			3.14	3.06	3.16
		velocity	5.36	8.46	8.96	8.79	9.14	9.14	8.62	8.79	8.62	8.62	9.80	8.24		8.73	8.96	8.68
Merritt, Aries (USA) (1985)	time		2.54	3.58	4.58	5.58	6.62	7.66	8.66	9.74	10.78	11.86		13.36	7 / 6			
reaction time	0.136	interval	7 steps	1.04	1.00	1.00	1.04	1.04	1.00	1.08	1.04	1.08	1.50			3.04	3.08	3.20
		velocity	5.40	8.79	9.14	9.14	8.79	8.79	9.14	8.46	8.79	8.46	9.35	8.23		9.02	8.90	8.57
Traber, Gregor (GRE) (1992)	time		2.56	3.60	4.62	5.62	6.66	7.68	8.70	9.76	10.82	11.94		13.41	9 / 7			
reaction time	0.150	interval	7 steps	1.04	1.02	1.00	1.04	1.02	1.02	1.06	1.06	1.12	1.47			3.06	3.08	3.24
		velocity	5.36	8.79	8.96	9.14	8.79	8.96	8.96	8.62	8.62	8.16	9.54	8.20		8.96	8.90	8.46
Oliver, David (USA) (1982)	time		2.54	3.62	4.64	5.68	6.72	7.76	8.84	9.90	10.98	12.06		13.62	8 / 8			
reaction time	0.156	interval	7 steps	1.08	1.02	1.04	1.04	1.04	1.08	1.06	1.08	1.08	1.56			3.14	3.16	3.22
		velocity	5.40	8.46	8.96	8.79	8.79	8.79	8.46	8.62	8.46	8.46	8.99	8.08		8.73	8.68	8.52
Porter, Jeff (USA) (1985)	time		2.58	3.66	4.70	5.72	6.76	7.82	8.88	9.96	11.04	12.16		13.70	1 / 9			
reaction time	0.141	interval		1.08	1.04	1.02	1.04	1.06	1.06	1.08	1.08	1.12	1.54			3.14	3.16	3.28
		velocity	5.32	8.46	8.79	8.96	8.79	8.62	8.62	8.46	8.46	8.16	9.10	8.03		8.73	8.68	8.36

2017 Kansai Regional University Championships (Osaka, JPN)

FINAL

date 11-May-17 wind 1.6 m/s

(2017) - tfdata-store.com/2017/05/13/post-441/

			H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
Kagimoto, Masahiro (JPN) (1991)	time		2.64	3.72	4.81	5.89	6.98	8.06	9.15	10.25	11.35	12.48		13.93	7 / 1			
reaction time		interval		1.08	1.09	1.08	1.09	1.08	1.09	1.10	1.10	1.13	1.45			3.25	3.26	3.33
		velocity	5.20	8.46	8.39	8.46	8.39	8.46	8.39	8.31	8.31	8.09	9.67	7.90		8.44	8.41	8.23
Hatanaka, Yushi (JPN) (1994)	time		2.64	3.77	4.85	5.93	7.03	8.12	9.22	10.31	11.43	12.58		14.03	4 / 2			
reaction time		interval		1.13	1.08	1.08	1.10	1.09	1.10	1.09	1.12	1.15	1.45	=PB		3.29	3.29	3.36
		velocity	5.20	8.09	8.46	8.46	8.31	8.39	8.31	8.39	8.16	7.95	9.67	7.84		8.33	8.33	8.16

Maeda, Hirotake (JPN) (1996)	time	2.65	3.74	4.85	5.94	7.03	8.12	9.22	10.35	11.46	12.59		14.03	5 / 3			
	reaction time	interval	1.09	1.11	1.09	1.09	1.09	1.10	1.13	1.11	1.13	1.44	PB		3.29	3.28	3.37
	velocity	5.18	8.39	8.23	8.39	8.39	8.39	8.31	8.09	8.23	8.09	9.74	7.84		8.33	8.36	8.14
Kamimura, Ryuji (JPN) (1996)	time	2.68	3.78	4.91	6.01	7.10	8.20	9.32	10.42	11.54	12.69		14.16	9 / 4			
	reaction time	interval	1.10	1.13	1.10	1.09	1.10	1.12	1.10	1.12	1.15	1.47	PB		3.33	3.31	3.37
	velocity	5.12	8.31	8.09	8.31	8.39	8.31	8.16	8.31	8.16	7.95	9.54	7.77		8.23	8.28	8.14
Kameyama, Yuji (JPN) (1997)	time	2.74	3.86	4.98	6.10	7.21	8.30	9.40	10.52	11.64	12.81		14.33	6 / 5			
	reaction time	interval	1.12	1.12	1.12	1.11	1.09	1.10	1.12	1.12	1.17	1.52			3.36	3.30	3.41
	velocity	5.01	8.16	8.16	8.16	8.23	8.39	8.31	8.16	8.16	7.81	9.22	7.68		8.16	8.31	8.04
Aiso, Ibuki (JPN) (1996)	time	2.69	3.82	4.95	6.06	7.15	8.27	9.42	10.54	11.68	12.85		14.34	3 / 6			
	reaction time	interval	1.13	1.13	1.11	1.09	1.12	1.15	1.12	1.14	1.17	1.49			3.37	3.36	3.43
	velocity	5.10	8.09	8.09	8.23	8.39	8.16	7.95	8.16	8.02	7.81	9.41	7.67		8.14	8.16	7.99
Watase, Yuto (JPN) (1997)	time	2.65	3.75	4.87	5.97	7.08	8.21	9.33	10.48	11.63	12.81		14.38	2 / 7			
	reaction time	interval	1.10	1.12	1.10	1.11	1.13	1.12	1.15	1.15	1.18	1.57			3.32	3.36	3.48
	velocity	5.18	8.31	8.16	8.31	8.23	8.09	8.16	7.95	7.95	7.75	8.93	7.65		8.26	8.16	7.88
Miura, Yoshiaki (JPN) (1995)	time	2.68	3.80	4.91	6.01	7.15	8.26	9.38	10.52	11.70	12.88		14.42	8 / 8			
	reaction time	interval	1.12	1.11	1.10	1.14	1.11	1.12	1.14	1.18	1.18	1.54			3.33	3.37	3.50
	velocity	5.12	8.16	8.23	8.31	8.02	8.23	8.16	8.02	7.75	7.75	9.10	7.63		8.23	8.14	7.83

2017 Michitaka Memorial (Osaka, JPN)**Open FINAL**

date 07-May-17 wind -1.5 m/s

(2017) - tfd-data-store.com/2017/05/08/post-413/

		H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
Kim Byung-Jun (KOR) (1991)	time	2.67	3.74	4.79	5.82	6.87	7.91	8.96	10.03	11.12	12.21		13.68	2 / 1			
	reaction time	interval	7 steps	1.07	1.05	1.03	1.05	1.04	1.05	1.07	1.09	1.47			3.15	3.14	3.25
	velocity	5.14	8.54	8.70	8.87	8.70	8.79	8.70	8.54	8.39	8.39	9.54	8.04		8.70	8.73	8.44
Masuno, Genta (JPN) (1993)	time	2.63	3.70	4.77	5.81	6.90	7.96	9.04	10.12	11.21	12.30		13.74	8 / 2			
	reaction time	interval	1.07	1.07	1.04	1.09	1.06	1.08	1.08	1.09	1.09	1.44			3.18	3.23	3.26
	velocity	5.22	8.54	8.54	8.79	8.39	8.62	8.46	8.46	8.39	8.39	9.74	8.01		8.62	8.49	8.41
Yazawa, Wataru (JPN) (1991)	time	2.62	3.70	4.77	5.83	6.89	7.97	9.04	10.11	11.21	12.33		13.77	6 / 3			
	reaction time	interval	7 steps	1.08	1.07	1.06	1.06	1.08	1.07	1.07	1.10	1.12	1.44		3.21	3.21	3.29
	velocity	5.24	8.46	8.54	8.62	8.62	8.46	8.54	8.54	8.31	8.16	9.74	7.99		8.54	8.54	8.33
Sato, Hiroyuki (JPN) (1990)	time	2.64	3.72	4.78	5.86	6.93	8.00	9.08	10.18	11.28	12.39		13.85	7 / 4			
	reaction time	interval	1.08	1.06	1.08	1.07	1.07	1.08	1.10	1.10	1.11	1.46			3.22	3.22	3.31
	velocity	5.20	8.46	8.62	8.46	8.54	8.54	8.46	8.31	8.31	8.23	9.60	7.94		8.52	8.52	8.28
Wado, Tatsuya (JPN) (1990)	time	2.61	3.69	4.75	5.82	6.91	7.98	9.07	10.19	11.32	12.45		13.93	3 / 5			
	reaction time	interval	1.08	1.06	1.07	1.09	1.07	1.09	1.12	1.13	1.13	1.48			3.21	3.25	3.38
	velocity	5.26	8.46	8.62	8.54	8.39	8.54	8.39	8.16	8.09	8.09	9.47	7.90		8.54	8.44	8.11
Fudaba, Hiroki (JPN) (1994)	time	2.77	3.85	4.96	6.06	7.16	8.24	9.37	10.51	11.61	12.75		14.27	5 / 6			
	reaction time	interval	7 steps	1.08	1.11	1.10	1.10	1.08	1.13	1.14	1.10	1.14	1.52		3.29	3.31	3.38
	velocity	4.95	8.46	8.23	8.31	8.31	8.46	8.09	8.02	8.31	8.02	9.22	7.71		8.33	8.28	8.11

FINAL

date 07-May-17 wind -0.8 m/s

(2017) - tfd-data-store.com/2017/05/08/post-413/

		H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
Omoro, Hideki (JPN) (1990)	time	2.63	3.69	4.75	5.80	6.86	7.90	8.96	10.01	11.08	12.17		13.57	7 / 1			
	reaction time	interval	1.06	1.06	1.05	1.06	1.04	1.06	1.05	1.07	1.09	1.40			3.17	3.16	3.21
	velocity	5.22	8.62	8.62	8.70	8.62	8.79	8.62	8.70	8.54	8.39	10.01	8.11		8.65	8.68	8.54
Kim Byung-Jun (KOR) (1991)	time	2.60	3.68	4.74	5.81	6.88	7.93	8.99	10.04	11.11	12.20		13.68	8 / 2			
	reaction time	interval	7 steps	1.08	1.06	1.07	1.05	1.06	1.05	1.07	1.09	1.48			3.21	3.18	3.21
	velocity	5.28	8.46	8.62	8.54	8.54	8.70	8.62	8.70	8.54	8.39	9.47	8.04		8.54	8.62	8.54
Masuno, Genta (JPN) (1993)	time	2.63	3.70	4.76	5.83	6.88	7.93	8.99	10.07	11.15	12.24		13.69	4 / 3			
	reaction time	interval	1.07	1.06	1.07	1.05	1.05	1.06	1.08	1.08	1.09	1.45			3.20	3.16	3.25
	velocity	5.22	8.54	8.62	8.54	8.70	8.70	8.62	8.46	8.46	8.39	9.67	8.04		8.57	8.68	8.44
Yazawa, Wataru (JPN) (1991)	time	2.64	3.72	4.79	5.85	6.91	7.98	9.04	10.11	11.20	12.30		13.71	5 / 4			
	reaction time	interval	7 steps	1.08	1.07	1.06	1.06	1.07	1.07	1.09	1.10	1.41			3.21	3.19	3.26
	velocity	5.20	8.46	8.54	8.62	8.62	8.54	8.62	8.54	8.39	8.31	9.94	8.02		8.54	8.60	8.41
Sato, Hiroyuki (JPN) (1990)	time	2.68	3.75	4.81	5.88	6.95	8.01	9.10	10.19	11.29	12.41		13.87	3 / 5			
	reaction time	interval	1.07	1.06	1.07	1.07	1.06	1.09	1.09	1.10	1.12	1.46			3.20	3.22	3.31
	velocity	5.12	8.54	8.62	8.54	8.54	8.62	8.39	8.39	8.31	8.16	9.60	7.93		8.57	8.52	8.28
Fudaba, Hiroki (JPN) (1994)	time	2.72	3.81	4.92	6.00	7.10	8.19	9.29	10.39	11.52	12.63		14.14	1 / 6			
	reaction time	interval	7 steps	1.09	1.11	1.08	1.10	1.09	1.10	1.10	1.13	1.51			3.28	3.29	3.34

velocity 5.04 8.39 8.23 8.46 8.31 8.39 8.31 8.31 8.09 8.23 9.28 7.78 8.36 8.33 8.21

2017 Mito Invitational (Mito, JPN)

FINAL

date 05-May-17 wind -2.3 m/s

(2017) - tfdata-store.com/2017/05/06/post-423/

		H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
Sato, Hiroyuki (JPN) (1990)	time	2.62	3.69	4.77	5.84	6.94	8.00	9.09	10.17	11.27	12.41		13.88	5 / 1			
	reaction time																
	interval		1.07	1.08	1.07	1.10	1.06	1.09	1.08	1.10	1.14	1.47			3.22	3.25	3.32
	velocity	5.24	8.54	8.46	8.54	8.31	8.62	8.39	8.46	8.31	8.02	9.54	7.93		8.52	8.44	8.26
Tanaka, Shinya (JPN) (1993)	time	2.66	3.74	4.83	5.91	7.00	8.09	9.18	10.28	11.35	12.50		13.94	7 / 2			
	reaction time																
	interval	7 steps	1.08	1.09	1.08	1.09	1.09	1.09	1.10	1.07	1.15	1.44			3.25	3.27	3.32
	velocity	5.16	8.46	8.39	8.46	8.39	8.39	8.39	8.31	8.54	7.95	9.74	7.89		8.44	8.39	8.26
Furuya, Kyohei (JPN) (1996)	time	2.63	3.72	4.82	5.93	7.05	8.15	9.23	10.37	11.48	12.59		14.02	4 / 3			
	reaction time																
	interval		1.09	1.10	1.11	1.12	1.10	1.08	1.14	1.11	1.11	1.43			3.30	3.30	3.36
	velocity	5.22	8.39	8.31	8.23	8.16	8.31	8.46	8.02	8.23	8.23	9.80	7.85		8.31	8.31	8.16
Furukawa, Yutaro (JPN) (1985)	time	2.64	3.72	4.81	5.90	6.97	8.06	9.16	10.27	11.39	12.53		14.05	2 / 4			
	reaction time																
	interval		1.08	1.09	1.09	1.07	1.09	1.10	1.11	1.12	1.14	1.52			3.26	3.26	3.37
	velocity	5.20	8.46	8.39	8.39	8.54	8.39	8.31	8.23	8.16	8.02	9.22	7.83		8.41	8.41	8.14
Hiramatsu, Papdemba (JPN) (1991)	time	2.63	3.76	4.87	5.96	7.07	8.19	9.30	10.41	11.55	12.69		14.21	9 / 5			
	reaction time																
	interval	7 steps	1.13	1.11	1.09	1.11	1.12	1.11	1.11	1.14	1.14	1.52			3.33	3.34	3.39
	velocity	5.22	8.09	8.23	8.39	8.23	8.16	8.23	8.23	8.02	8.02	9.22	7.74		8.23	8.21	8.09
Kato, Yoshihiko (JPN) (1994)	time	2.72	3.88	5.04	6.17	7.28	8.39	9.49	10.60	11.70	12.83		14.31	8 / 6			
	reaction time																
	interval		1.16	1.16	1.13	1.11	1.11	1.10	1.11	1.10	1.13	1.48			3.45	3.32	3.34
	velocity	5.04	7.88	7.88	8.09	8.23	8.23	8.31	8.23	8.31	8.09	9.47	7.69		7.95	8.26	8.21
Kagimoto, Masahiro (JPN) (1995)	time	2.68	3.76	4.90	6.01	7.12	8.24	9.37	10.49	11.64	12.81		14.33	3 / 7			
	reaction time																
	interval		1.08	1.14	1.11	1.11	1.12	1.13	1.12	1.15	1.17	1.52			3.33	3.36	3.44
	velocity	5.12	8.46	8.02	8.23	8.23	8.16	8.09	8.16	7.95	7.81	9.22	7.68		8.23	8.16	7.97

Heat 3

date 05-May-17 wind -2.5 m/s

(2017) - tfdata-store.com/2017/05/06/post-423/

		H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
Sato, Hiroyuki (JPN) (1990)	time	2.60	3.68	4.76	5.80	6.88	7.95	9.04	10.13	11.24	12.38		13.86	5 / 1			
	reaction time																
	interval		1.08	1.08	1.04	1.08	1.07	1.09	1.09	1.11	1.14	1.48			3.20	3.24	3.34
	velocity	5.28	8.46	8.46	8.79	8.46	8.54	8.39	8.39	8.23	8.02	9.47	7.94		8.57	8.46	8.21
Tanaka, Shinya (JPN) (1993)	time	2.63	3.70	4.79	5.87	6.95	8.02	9.10	10.19	11.27	12.41		13.88	6 / 2			
	reaction time																
	interval	7 steps	1.07	1.09	1.08	1.08	1.07	1.08	1.09	1.08	1.14	1.47	PB		3.24	3.23	3.31
	velocity	5.22	8.54	8.39	8.46	8.46	8.54	8.46	8.39	8.46	8.02	9.54	7.93		8.46	8.49	8.28
Furukawa, Yutaro (JPN) (1985)	time	2.65	3.73	4.81	5.90	6.99	8.09	9.21	10.35	11.49	12.66		14.25	4 / 3			
	reaction time																
	interval		1.08	1.08	1.09	1.09	1.10	1.12	1.14	1.14	1.17	1.59			3.25	3.31	3.45
	velocity	5.18	8.46	8.46	8.39	8.39	8.31	8.16	8.02	8.02	7.81	8.82	7.72		8.44	8.28	7.95
Kumada, Ryo (JPN) (1994)	time	2.77	3.93	5.05	6.16	7.30	8.42	9.55	10.67	11.83	12.99		14.42	9 / 4			
	reaction time																
	interval	7 steps	1.16	1.12	1.11	1.14	1.12	1.13	1.12	1.16	1.16	1.43			3.39	3.39	3.44
	velocity	4.95	7.88	8.16	8.23	8.02	8.16	8.09	8.16	7.88	7.88	9.80	7.63		8.09	8.09	7.97
Kanai, Tadashi (JPN)	time	2.70	3.82	4.93	6.05	7.15	8.28	9.41	10.56	11.71	12.89		14.43	2 / 5			
	reaction time																
	interval		1.12	1.11	1.12	1.10	1.13	1.13	1.15	1.15	1.18	1.54			3.35	3.36	3.48
	velocity	5.08	8.16	8.23	8.16	8.31	8.09	8.09	7.95	7.95	7.75	9.10	7.62		8.19	8.16	7.88
Takahasai, Yusuke (JPN) (1996)	time	2.65	3.83	4.96	6.10	7.25	8.37	9.49	10.63	11.80	12.97		14.43	8 / 6			
	reaction time																
	interval		1.18	1.13	1.14	1.15	1.12	1.12	1.14	1.17	1.17	1.46			3.45	3.39	3.48
	velocity	5.18	7.75	8.09	8.02	7.95	8.16	8.16	8.02	7.81	7.81	9.60	7.62		7.95	8.09	7.88
Tsuchiya, Toru (JPN) (1995)	time	2.74	3.90	5.05	6.19	7.36	8.53	9.68	10.83	12.07	13.30		14.83	3 / 7			
	reaction time																
	interval	7 steps	1.16	1.15	1.14	1.17	1.17	1.15	1.15	1.24	1.23	1.53			3.45	3.49	3.62
	velocity	5.01	7.88	7.95	8.02	7.81	7.81	7.95	7.95	7.37	7.43	9.16	7.42		7.95	7.86	7.57

Heat 2

date 05-May-17 wind 0.6 m/s

(2017) - tfdata-store.com/2017/05/06/post-423/

		H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
Furumoto, Tsubasa (JPN) (1999)	time	2.64	3.73	4.84	5.91	6.99	8.08	9.17	10.27	11.41	12.54		14.01	5 / 1			
	reaction time																
	interval		1.09	1.11	1.07	1.08	1.09	1.09	1.10	1.14	1.13	1.47			3.27	3.26	3.37
	velocity	5.20	8.39	8.23	8.54	8.46	8.39	8.39	8.31	8.02	8.09	9.54	7.85		8.39	8.41	8.14
Hiramatsu, Papdemba (JPN) (1991)	time	2.59	3.70	4.80	5.88	6.98	8.06	9.16	10.29	11.41	12.56		14.05	2 / 2			
	reaction time																
	interval	7 steps	1.11	1.10	1.08	1.10	1.08	1.10	1.13	1.12	1.15	1.49			3.29	3.28	3.40
	velocity	5.30	8.23	8.31	8.46	8.31	8.46	8.31	8.09	8.16	7.95	9.41	7.83		8.33	8.36	8.06
Kagimoto, Masahiro (JPN) (1995)	time	2.67	3.77	4.83	5.93	7.03	8.12	9.21	10.34	11.48	12.64		14.11	6 / 3			
	reaction time																
	interval		1.10	1.06	1.10	1.10	1.09	1.09	1.13	1.14	1.16	1.47			3.26	3.28	3.43
	velocity	5.14	8.31	8.62	8.31	8.31	8.39	8.39	8.09	8.02	7.88	9.54	7.80		8.41	8.36	7.99

Aiso, Ibuki (JPN) (1996)	time	2.70	3.84	4.98	6.11	7.23	8.35	9.47	10.59	11.73	12.91		14.40	9 / 4			
	reaction time	interval		1.14	1.14	1.13	1.12	1.12	1.12	1.12	1.14	1.18	1.49		3.41	3.36	3.44
	velocity		5.08	8.02	8.02	8.09	8.16	8.16	8.16	8.16	8.02	7.75	9.41	7.64	8.04	8.16	7.97
Morita, Shunichi (JPN) (1989)	time	2.75	3.89	5.02	6.15	7.28	8.40	9.53	10.69	11.86	13.04		14.58	3 / 5			
	reaction time	interval		1.14	1.13	1.13	1.13	1.12	1.13	1.16	1.17	1.18	1.54		3.40	3.38	3.51
	velocity		4.99	8.02	8.09	8.09	8.09	8.16	8.09	7.88	7.81	7.75	9.10	7.54	8.06	8.11	7.81
Kawamura, Naoya (JPN) (1996)	time	2.76	3.91	5.06	6.20	7.34	8.46	9.58	10.74	11.89	13.07		14.62	8 / 6			
	reaction time	interval		1.15	1.15	1.14	1.14	1.12	1.12	1.16	1.15	1.18	1.55		3.44	3.38	3.49
	velocity		4.97	7.95	7.95	8.02	8.02	8.16	8.16	7.88	7.95	7.75	9.05	7.52	7.97	8.11	7.86

Heat 1

date 05-May-17 wind 0.3 m/s

(2017) - tfdata-store.com/2017/05/06/post-423/

		H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
Furuya, Kyohei (JPN) (1996)	time	2.64	3.74	4.82	5.90	7.00	8.09	9.18	10.30	11.42	12.55		14.00	5 / 1			
	reaction time	interval		1.10	1.08	1.08	1.10	1.09	1.12	1.12	1.13	1.45			3.26	3.28	3.37
	velocity		5.20	8.31	8.46	8.46	8.31	8.39	8.39	8.16	8.16	8.09	9.67	7.86	8.41	8.36	8.14
Kato, Yoshihiko (JPN) (1994)	time	2.72	3.82	4.93	6.03	7.12	8.23	9.35	10.46	11.60	12.74		14.22	9 / 2			
	reaction time	interval		1.10	1.11	1.10	1.09	1.11	1.12	1.11	1.14	1.14	1.48		3.31	3.32	3.39
	velocity		5.04	8.31	8.23	8.31	8.39	8.23	8.16	8.23	8.02	8.02	9.47	7.74	8.28	8.26	8.09
Kato, Takuma (JPN) (1998)	time	2.71	3.83	4.96	6.06	7.19	8.31	9.43	10.56	11.70	12.89		14.44	6 / 3			
	reaction time	interval		1.12	1.13	1.10	1.13	1.12	1.12	1.13	1.14	1.19	1.55		3.35	3.37	3.46
	velocity		5.06	8.16	8.09	8.31	8.09	8.16	8.16	8.09	8.02	7.68	9.05	7.62	8.19	8.14	7.92
Sakimoto, Kazuma (JPN) (1996)	time	2.71	3.84	4.99	6.14	7.27	8.41	9.56	10.72	11.88	13.07		14.62	2 / 4			
	reaction time	interval		1.13	1.15	1.15	1.13	1.14	1.15	1.16	1.16	1.19	1.55		3.43	3.42	3.51
	velocity		5.06	8.09	7.95	7.95	8.09	8.02	7.95	7.88	7.88	7.68	9.05	7.52	7.99	8.02	7.81
Tetsuguchi, Ren (JPN) (1994)	time	2.79	3.94	5.08	6.21	7.36	8.51	9.65	10.82	12.01	13.21		14.73	8 / 5			
	reaction time	interval	7 steps	1.15	1.14	1.13	1.15	1.15	1.14	1.17	1.19	1.20	1.52		3.42	3.44	3.56
	velocity		4.92	7.95	8.02	8.09	7.95	7.95	8.02	7.81	7.68	7.62	9.22	7.47	8.02	7.97	7.70
Igarashi, Rai (JPN) (1997)	time	2.76	3.91	5.06	6.23	7.38	8.54	9.70	10.86	12.05	13.24		14.84	3 / 6			
	reaction time	interval		1.15	1.15	1.17	1.15	1.16	1.16	1.16	1.19	1.19	1.60		3.47	3.47	3.54
	velocity		4.97	7.95	7.95	7.81	7.95	7.88	7.88	7.88	7.68	7.68	8.76	7.41	7.90	7.90	7.75

2017 Drake Relays (Des Moines, IA) (TV Analysis)**FINAL**

date 29-Apr-17 wind 1.8 m/s

Henson (2020) - Athlete First: 2017 year end hurdle report

		H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
McLeod, Omar (JAM) (1994)	time	2.47	3.53	4.53	5.53	6.53	7.53	8.55	9.57	10.57	11.61		13.04	4 / 1			
	reaction time	interval		1.06	1.00	1.00	1.00	1.02	1.02	1.00	1.04	1.43			3.06	3.02	3.06
	velocity		5.55	8.62	9.14	9.14	9.14	8.96	8.96	9.14	8.79	9.80	8.44		8.96	9.08	8.96
Pozzi, Andy (GBR) (1992)	time	2.50	3.53	4.53	5.53	6.50	7.53	8.53	9.55	10.60	11.66		13.24	3 / 2			
	reaction time	interval	7 steps	1.03	1.00	1.00	0.97	1.03	1.00	1.02	1.05	1.06	1.58		3.03	3.00	3.13
	velocity		5.49	8.87	9.14	9.14	9.42	8.87	9.14	8.96	8.70	8.62	8.87	8.31	9.05	9.14	8.76
Harris, Aleec (USA) (1990)	time	2.56	3.60	4.64	5.68	6.70	7.73	8.76	9.80	10.86	11.90		13.37	1 / 3			
	reaction time	interval		1.04	1.04	1.04	1.02	1.03	1.03	1.04	1.06	1.04	1.47		3.12	3.08	3.14
	velocity		5.36	8.79	8.79	8.79	8.96	8.87	8.87	8.79	8.62	8.79	9.54	8.23	8.79	8.90	8.73
Merritt, Aries (USA) (1985)	time	2.53	3.60	4.63	5.63	6.63	7.66	8.70	9.76	10.83	11.93		13.44	5 / 4			
	reaction time	interval	7 steps	1.07	1.03	1.00	1.00	1.03	1.04	1.06	1.07	1.10	1.51		3.10	3.07	3.23
	velocity		5.42	8.54	8.87	9.14	9.14	8.87	8.79	8.62	8.54	8.31	9.28	8.18	8.85	8.93	8.49
Riley, Andrew (JAM) (1988)	time	2.60	3.70	4.76	5.76	6.80	7.83	8.88	9.93	11.00	12.06		13.53	7 / 5			
	reaction time	interval	7 steps	1.10	1.06	1.00	1.04	1.03	1.05	1.05	1.07	1.06	1.47		3.16	3.12	3.18
	velocity		5.28	8.31	8.62	9.14	8.79	8.87	8.70	8.70	8.54	8.62	9.54	8.13	8.68	8.79	8.62
Adams, Spencer (USA) (1989)	time	2.50	3.56	4.60	5.63	6.66	7.70	8.73	9.80	10.90	12.00		13.56	8 / 6			
	reaction time	interval	7 steps	1.06	1.04	1.03	1.03	1.04	1.03	1.07	1.10	1.10	1.56		3.13	3.10	3.27
	velocity		5.49	8.62	8.79	8.87	8.87	8.79	8.87	8.54	8.31	8.31	8.99	8.11	8.76	8.85	8.39
Cabral, Jonathan (CAN) (1992)	time	2.63	3.70	4.73	5.76	6.80	7.80	8.83	9.90	10.98	12.10		13.64	6 / 7			
	reaction time	interval	7 steps	1.07	1.03	1.03	1.04	1.00	1.03	1.07	1.08	1.12	1.54		3.13	3.07	3.27
	velocity		5.22	8.54	8.87	8.87	8.79	9.14	8.87	8.54	8.46	8.16	9.10	8.06	8.76	8.93	8.39
Eaton, Jarret (USA) (1989)	time	2.50	3.56	4.60	5.60	6.64	7.68	8.76	9.86	10.96	12.10		13.68	2 / 8			
	reaction time	interval	7 steps	1.06	1.04	1.00	1.04	1.08	1.10	1.10	1.14	1.58			3.10	3.16	3.34
	velocity		5.49	8.62	8.79	9.14	8.79	8.79	8.46	8.31	8.31	8.02	8.87	8.04	8.85	8.68	8.21

2017 Mikio Oda Memorial (Hiroshima, JPN)**FINAL**

date 29-Apr-17 wind 1.5 m/s

(2017) - tfdata-store.com/2017/05/01/458/

		H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
Omoro, Hideki (JPN) (1990)	time	2.61	3.67	4.73	5.77	6.82	7.87	8.92	9.98	11.07	12.13		13.52	5 / 1			
	reaction time																
	interval		1.06	1.06	1.04	1.05	1.05	1.05	1.06	1.09	1.06	1.39			3.16	3.15	3.21
	velocity	5.26	8.62	8.62	8.79	8.70	8.70	8.70	8.62	8.39	8.62	10.09	8.14		8.68	8.70	8.54
Kim Byung-Jun (KOR) (1991)	time	2.63	3.69	4.75	5.80	6.85	7.90	8.96	10.01	11.08	12.15		13.64	4 / 2			
	reaction time																
	interval	7 steps	1.06	1.06	1.05	1.05	1.05	1.06	1.05	1.07	1.07	1.49			3.17	3.16	3.19
	velocity	5.22	8.62	8.62	8.70	8.70	8.70	8.62	8.70	8.54	8.54	9.41	8.06		8.65	8.68	8.60
Takayama, Shunya (JPN) (1994)	time	2.62	3.68	4.74	5.80	6.85	7.92	8.97	10.05	11.13	12.21		13.67	6 / 3			
	reaction time																
	interval		1.06	1.06	1.06	1.05	1.07	1.05	1.08	1.08	1.08	1.46			3.18	3.17	3.24
	velocity	5.24	8.62	8.62	8.62	8.70	8.54	8.70	8.46	8.46	8.46	9.60	8.05		8.62	8.65	8.46
Kanai, Taio (JPN) (1995)	time	2.60	3.68	4.77	5.84	6.90	7.96	9.03	10.11	11.18	12.28		13.73	7 / 4			
	reaction time																
	interval	7 steps	1.08	1.09	1.07	1.06	1.06	1.07	1.08	1.07	1.10	1.45			3.24	3.19	3.25
	velocity	5.28	8.46	8.39	8.54	8.62	8.62	8.54	8.46	8.54	8.31	9.67	8.01		8.46	8.60	8.44
Yazawa, Wataru (JPN) (1991)	time	2.66	3.75	4.82	5.88	6.93	8.01	9.07	10.13	11.21	12.30		13.74	3 / 5			
	reaction time																
	interval	7 steps	1.09	1.07	1.06	1.05	1.08	1.06	1.06	1.08	1.09	1.44			3.22	3.19	3.23
	velocity	5.16	8.39	8.54	8.62	8.70	8.46	8.62	8.62	8.46	8.39	9.74	8.01		8.52	8.60	8.49
Wado, Tatsuya (JPN) (1990)	time	2.57	3.67	4.72	5.78	6.85	7.91	9.01	10.07	11.21	12.32		13.82	2 / 6			
	reaction time																
	interval		1.10	1.05	1.06	1.07	1.06	1.10	1.06	1.14	1.11	1.50			3.21	3.23	3.31
	velocity	5.34	8.31	8.70	8.62	8.54	8.62	8.31	8.62	8.02	8.23	9.35	7.96		8.54	8.49	8.28
Kuriki, Anthony Tyrell (JPN) (1991)	time	2.66	3.75	4.83	5.90	6.98	8.06	9.16	10.25	11.36	12.48		13.95	8 / 7			
	reaction time																
	interval	7 steps	1.09	1.08	1.07	1.08	1.08	1.10	1.09	1.11	1.12	1.47			3.24	3.26	3.32
	velocity	5.16	8.39	8.46	8.54	8.46	8.46	8.31	8.39	8.23	8.16	9.54	7.89		8.46	8.41	8.26
Nomoto, Syuusei (JPN) (1995)	time	2.61	3.67	4.75	5.80	6.85	7.93	8.99	10.07	11.13	12.20		22.95	1 / 8			
	reaction time																
	interval	7 steps	1.06	1.08	1.05	1.05	1.08	1.06	1.08	1.06	1.07	10.75			3.19	3.19	3.21
	velocity	5.26	8.62	8.46	8.70	8.70	8.46	8.62	8.46	8.62	8.54	1.30	4.79		8.60	8.60	8.54

B Final

date 29-Apr-17 wind -1.5 m/s

(2017) - tfdata-store.com/2017/05/01/458/

		H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
Ogata, Akihiro (JPN) (1993)	time	2.64	3.74	4.83	5.90	6.98	8.07	9.16	10.25	11.39	12.52		14.04	8 / 1			
	reaction time																
	interval	7 steps	1.10	1.09	1.07	1.08	1.09	1.09	1.09	1.14	1.13	1.52			3.26	3.26	3.36
	velocity	5.20	8.31	8.39	8.54	8.46	8.39	8.39	8.39	8.02	8.09	9.22	7.83		8.41	8.41	8.16
Furuya, Kyohei (JPN) (1996)	time	2.60	3.71	4.81	5.89	6.98	8.06	9.18	10.28	11.44	12.58		14.07	4 / 2			
	reaction time																
	interval		1.11	1.10	1.08	1.09	1.08	1.12	1.10	1.16	1.14	1.49			3.29	3.29	3.40
	velocity	5.28	8.23	8.31	8.46	8.39	8.46	8.16	8.31	7.88	8.02	9.41	7.82		8.33	8.33	8.06
Sato, Hiroyuki (JPN) (1990)	time	2.60	3.67	4.74	5.83	6.93	8.03	9.12	10.25	11.37	12.54		14.08	3 / 3			
	reaction time																
	interval		1.07	1.07	1.09	1.10	1.10	1.09	1.13	1.12	1.17	1.54			3.23	3.29	3.42
	velocity	5.28	8.54	8.54	8.39	8.31	8.31	8.39	8.09	8.16	7.81	9.10	7.81		8.49	8.33	8.02
Tanaka, Shinya (JPN) (1993)	time	2.62	3.74	4.82	5.89	6.98	8.09	9.21	10.34	11.49	12.64		14.13	7 / 4			
	reaction time																
	interval	7 steps	1.12	1.08	1.07	1.09	1.11	1.12	1.13	1.15	1.15	1.49			3.27	3.32	3.43
	velocity	5.24	8.16	8.46	8.54	8.39	8.23	8.16	8.09	7.95	7.95	9.41	7.78		8.39	8.26	7.99
Fudaba, Hiroki (JPN) (1994)	time	2.72	3.83	4.92	6.00	7.09	8.18	9.29	10.40	11.54	12.68		14.20	1 / 5			
	reaction time																
	interval	7 steps	1.11	1.09	1.08	1.09	1.09	1.11	1.11	1.14	1.14	1.52			3.28	3.29	3.39
	velocity	5.04	8.23	8.39	8.46	8.39	8.39	8.23	8.23	8.02	8.02	9.22	7.75		8.36	8.33	8.09
Kagimoto, Masahiro (JPN) (1995)	time	2.67	3.75	4.86	5.97	7.06	8.18	9.31	10.47	11.60	12.76		14.28	6 / 6			
	reaction time																
	interval		1.08	1.11	1.11	1.09	1.12	1.13	1.16	1.13	1.16	1.52			3.30	3.34	3.45
	velocity	5.14	8.46	8.23	8.23	8.39	8.16	8.09	7.88	8.09	7.88	9.22	7.70		8.31	8.21	7.95
Fujii, Ryota (JPN) (1996)	time	2.61	3.73	4.84	5.97	7.09	8.19	9.34	10.49	11.65	12.81		14.36	2 / 7			
	reaction time																
	interval		1.12	1.11	1.13	1.12	1.10	1.15	1.15	1.16	1.16	1.55			3.36	3.37	3.47
	velocity	5.26	8.16	8.23	8.09	8.16	8.31	7.95	7.95	7.88	7.88	9.05	7.66		8.16	8.14	7.90
Furukawa, Yutaro (JPN) (1985)	time	2.67	3.77	4.88	5.98	7.08	8.20	9.35	10.50	11.68	12.87		14.61	5 / 8			
	reaction time																
	interval		1.10	1.11	1.10	1.10	1.12	1.15	1.15	1.18	1.19	1.74			3.31	3.37	3.52
	velocity	5.14	8.31	8.23	8.31	8.31	8.16	7.95	7.95	7.75	7.68	8.06	7.53		8.28	8.14	7.79

2017 Kyoto University Meeting (Kyoto, JPN)**FINAL**

date 09-Apr-17 wind 1.5 m/s

(2017) - tfdata-store.com/2017/04/12/post-362/

		H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
Kagimoto, Masahiro (JPN) (1995)	time	2.68	3.77	4.85	5.95	7.02	8.11	9.21	10.31	11.41	12.51		13.97	6 / 1			
	reaction time																
	interval		1.09	1.08	1.10	1.07	1.09	1.10	1.10	1.10	1.10	1.46			3.27	3.26	3.30
	velocity	5.12	8.39	8.46	8.31	8.54	8.39	8.31	8.31	8.31	8.31	9.60	7.87		8.39	8.41	8.31
Maeda, Hirotake (JPN) (1996)	time	2.65	3.78	4.89	5.99	7.09	8.19	9.30	10.41	11.51	12.64		14.09	5 / 2			
	reaction time																
	interval		1.13	1.11	1.10	1.10	1.10	1.11	1.11	1.10	1.13	1.45	PB		3.34	3.31	3.34

	velocity	5.18	8.09	8.23	8.31	8.31	8.31	8.23	8.23	8.31	8.09	9.67	7.81		8.21	8.28	8.21
Aiso, Ibuki (JPN) (1996)	time	2.70	3.84	4.95	6.04	7.17	8.28	9.38	10.50	11.64	12.82		14.33	4 / 3			
reaction time	interval		1.14	1.11	1.09	1.13	1.11	1.10	1.12	1.14	1.18	1.51			3.34	3.34	3.44
	velocity	5.08	8.02	8.23	8.39	8.09	8.23	8.31	8.16	8.02	7.75	9.28	7.68		8.21	8.21	7.97

2017 Tokyo Six School Meeting (Tokyo, JPN)

FINAL

date 08-Apr-17 wind 0.5 m/s (2017) - tfdata-store.com/2017/04/10/post-342/

		H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
Kanai, Taio (JPN) (1995)	time	2.60	3.67	4.75	5.80	6.86	7.93	8.99	10.09	11.16	12.27		13.71	6 / 1			
reaction time	interval	7 steps	1.07	1.08	1.05	1.06	1.07	1.06	1.10	1.07	1.11	1.44	PB		3.20	3.19	3.28
	velocity	5.28	8.54	8.46	8.70	8.62	8.54	8.62	8.31	8.54	8.23	9.74	8.02		8.57	8.60	8.36

Furuya, Takumu (JPN) (1997)	time	2.62	3.72	4.79	5.85	6.94	7.99	9.06	10.15	11.22	12.31		13.75	5 / 2			
reaction time	interval	7 steps	1.10	1.07	1.06	1.09	1.05	1.07	1.09	1.07	1.09	1.44			3.23	3.21	3.25
	velocity	5.24	8.31	8.54	8.62	8.39	8.70	8.54	8.39	8.54	8.39	9.74	8.00		8.49	8.54	8.44

Nomoto, Syuusei (JPN) (1995)	time	2.60	3.70	4.76	5.86	6.94	8.02	9.10	10.18	11.28	12.39		13.80	4 / 3			
reaction time	interval	7 steps	1.10	1.06	1.10	1.08	1.08	1.08	1.08	1.10	1.11	1.41	PB		3.26	3.24	3.29
	velocity	5.28	8.31	8.62	8.31	8.46	8.46	8.46	8.46	8.31	8.23	9.94	7.97		8.41	8.46	8.33

Heat 2

date 08-Apr-17 wind 0.8 m/s (2017) - tfdata-store.com/2017/04/10/post-342/

		H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
Kanai, Taio (JPN) (1995)	time	2.60	3.68	4.74	5.81	6.88	7.97	9.07	10.17	11.29	12.40		13.86	4 / 1			
reaction time	interval	7 steps	1.08	1.06	1.07	1.07	1.09	1.10	1.10	1.12	1.11	1.46			3.21	3.26	3.33
	velocity	5.28	8.46	8.62	8.54	8.54	8.39	8.31	8.31	8.16	8.23	9.60	7.94		8.54	8.41	8.23

Nomoto, Syuusei (JPN) (1995)	time	2.63	3.72	4.80	5.89	6.97	8.05	9.14	10.22	11.35	12.47		13.91	5 / 2			
reaction time	interval	7 steps	1.09	1.08	1.09	1.08	1.08	1.09	1.08	1.13	1.12	1.44	PB		3.26	3.25	3.33
	velocity	5.22	8.39	8.46	8.39	8.46	8.46	8.39	8.46	8.09	8.16	9.74	7.91		8.41	8.44	8.23

Heat 1

date 08-Apr-17 wind 1.3 m/s (2017) - tfdata-store.com/2017/04/10/post-342/

		H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
Furuya, Takumu (JPN) (1997)	time	2.64	3.72	4.82	5.86	6.92	7.98	9.06	10.14	11.23	12.34		13.84	5 / 1			
reaction time	interval	7 steps	1.08	1.10	1.04	1.06	1.06	1.08	1.08	1.09	1.11	1.50			3.22	3.20	3.28
	velocity	5.20	8.46	8.31	8.79	8.62	8.62	8.46	8.46	8.39	8.23	9.35	7.95		8.52	8.57	8.36

2017 Tokushima Athletics Carnival (Naruto, JPN)

FINAL

date 08-Apr-17 wind 0.7 m/s (2017) - tfdata-store.com/2017/04/09/post-285/

		H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
Omoro, Hideki (JPN) (1990)	time	2.65	3.74	4.79	5.86	6.91	7.96	9.03	10.10	11.16	12.23		13.63	6 / 1			
reaction time	interval		1.09	1.05	1.07	1.05	1.05	1.07	1.07	1.06	1.07	1.40			3.21	3.17	3.20
	velocity	5.18	8.39	8.70	8.54	8.70	8.70	8.54	8.54	8.62	8.54	10.01	8.07		8.54	8.65	8.57

2017 Osaka University Athletics Championships (Osaka, JPN)

FINAL

date 06-Apr-17 wind -1.8 m/s (2017) - tfdata-store.com/2017/04/07/post-249/

		H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
Hatanaka, Yushi (JPN) (1994)	time	2.71	3.84	4.95	6.04	7.15	8.28	9.42	10.58	11.75	12.91		14.46	6 / 1			
reaction time	interval	7 steps	1.13	1.11	1.09	1.11	1.13	1.14	1.16	1.17	1.16	1.55			3.33	3.38	3.49
	velocity	5.06	8.09	8.23	8.39	8.23	8.09	8.02	7.88	7.81	7.88	9.05	7.61		8.23	8.11	7.86

Watase, Yuto (JPN)	time	2.67	3.82	4.95	6.08	7.22	8.38	9.56	10.75	11.92	13.15		14.67	8 / 2			
reaction time	interval		1.15	1.13	1.13	1.14	1.16	1.18	1.19	1.17	1.23	1.52			3.41	3.48	3.59
	velocity	5.14	7.95	8.09	8.09	8.02	7.88	7.75	7.68	7.81	7.43	9.22	7.50		8.04	7.88	7.64

Shohei, Miyamoto (JPN)	time	2.75	3.91	5.08	6.24	7.39	8.53	9.70	10.89	12.05	13.23		14.79	4 / 3			
reaction time	interval	7 steps	1.16	1.17	1.16	1.15	1.14	1.17	1.19	1.16	1.18	1.56			3.49	3.46	3.53
	velocity	4.99	7.88	7.81	7.88	7.95	8.02	7.81	7.68	7.88	7.75	8.99	7.44		7.86	7.92	7.77

Maeda, Takanori (JPN) (1997)	time	2.74	3.88	5.05	6.21	7.36	8.52	9.71	10.91	12.10	13.30		14.87	5 / 4			
reaction time	interval		1.14	1.17	1.16	1.15	1.16	1.19	1.20	1.19	1.20	1.57			3.47	3.50	3.59
	velocity	5.01	8.02	7.81	7.88	7.95	7.88	7.68	7.62	7.68	7.62	8.93	7.40		7.90	7.83	7.64

Takaya, Yasuda (JPN)	time	2.73	3.90	5.07	6.23				10.98	12.19	13.42		14.97	9 / 5			
reaction time	interval		1.17	1.17	1.16				4.75	1.21	1.23	1.55			3.50		
	velocity	5.03	7.81	7.81	7.88				7.70	7.55	7.43	9.05	7.35		7.83		

Takahashi, Masahiro (JPN)	time	2.76	3.92				8.58	9.79	10.98	12.18	13.41		14.98	2 / 6			
reaction time	interval		1.16				4.66	1.21	1.19	1.20	1.23	1.57					3.62
	velocity	4.97	7.88				7.85	7.55	7.68	7.62	7.43	8.93	7.34				7.57

Kumagai, Gen (JPN)	time	2.71	3.89	5.06	6.27	7.46	8.69	9.91	11.08	12.33	13.58		15.11	3 / 7			
reaction time	interval		1.18	1.17	1.21	1.19	1.23	1.22	1.17	1.25	1.25	1.53			3.56	3.64	3.67
	velocity	5.06	7.75	7.81	7.55	7.68	7.43	7.49	7.81	7.31	7.31	9.16	7.28		7.70	7.53	7.47

Ariga, Takashi (JPN)	time	2.71	3.87	5.07	6.32	7.68							19.33	7 / 8			
-----------------------------	------	------	------	------	------	------	--	--	--	--	--	--	-------	-------	--	--	--

reaction time	interval	1.16	1.20	1.25	1.36		3.61
	velocity	5.06	7.88	7.62	7.31	6.72	7.60
						5.69	

2017 European Indoor Championships (Beograd, SRB)**FINAL**

date 03-Mar-17

(2017.03.05) - https://twitter.com/touchdown_time/media?lang=en&lang=en&lang=en&lang=en

		H1	H2	H3	H4	H5	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
Pozzi, Andy (GBR) (1992)	time	2.62	3.63	4.63	5.61	6.62		7.51	5 / 1			
	reaction time	interval	7 steps	1.01	1.00	0.98	0.89			2.99		
		velocity	5.24	9.05	9.14	9.33	10.92	7.99		9.17		
Martinot-Lagarde, Pascal (FRA)	time	2.59	3.63	4.64	5.65	6.65		7.52	4 / 2			
	reaction time	interval	7 steps	1.04	1.01	1.01	0.87			3.06		
		velocity	5.30	8.79	9.05	9.05	11.17	7.98		8.96		
Ortega, Orlando (ESP) (1991)	time	2.63	3.67	4.71	5.69	6.75		7.64	7 / 7			
	reaction time	interval	7 steps	1.04	1.04	0.98	0.89			3.06		
		velocity	5.22	8.79	8.79	9.33	10.92	7.85		8.96		

2017 German National Indoor Junior Championships (Sindelfingen, GER) (99cm)**FINAL**

date 25-Feb-17

Poser (2017) - mit mehrkampftraining zum hürdentitel

		H1	H2	H3	H4	H5	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
Petzke, Jonathan (GER) (1998)	time	2.66	3.72	4.80	5.86	6.93		7.81	1 / 1			
	reaction time	interval	1.06	1.08	1.06	1.07	0.88	PB		3.20		
		velocity	5.16	8.62	8.46	8.62	11.05	7.68		8.57		

Semi-Final 1

date 25-Feb-17

Poser (2017) - mit mehrkampftraining zum hürdentitel

		H1	H2	H3	H4	H5	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
Petzke, Jonathan (GER) (1998)	time	2.70	3.77	4.88	5.95	7.04		7.97	1 / 1			
	reaction time	interval	1.07	1.11	1.07	1.09	0.93			3.25		
		velocity	5.08	8.54	8.23	8.54	10.45	7.53		8.44		

2017 Japanese National Indoor Junior Championships (Osaka, JPN) (99c)**FINAL**

date 04-Feb-17

(2017.02.04) - https://twitter.com/touchdown_time/media?lang=en&lang=en&lang=en&lang=en

		H1	H2	H3	H4	H5	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
Higuchi, Rikuto (JPN) (1999)	time	2.70	3.80	4.89	5.99	7.09		7.98	3 / 1			
	reaction time	interval	1.10	1.09	1.10	1.10	0.89	PB		3.29		
		velocity	5.08	8.31	8.39	8.31	10.92	7.52		8.33		

2016 West Japan Carnival (Kita-Kyushu, JPN)**FINAL**

date 03-Nov-16

wind 1.1 m/s

(2016.11.07) - https://twitter.com/touchdown_time/media?lang=en&lang=en&lang=en&lang=en

		H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
Omuro, Hideki (JPN) (1990)	time	2.62	3.69	4.74	5.79	6.84	7.91	8.98	10.05	11.13	12.24		13.68	5 / 1			
	reaction time	interval	1.07	1.05	1.05	1.05	1.07	1.07	1.07	1.08	1.11	1.44			3.17	3.19	3.26
		velocity	5.24	8.54	8.70	8.70	8.54	8.54	8.54	8.46	8.23	9.74	8.04		8.65	8.60	8.41
Ishikawa, Shuhei (JPN) (1995)	time	2.58	3.67	4.74	5.84	6.93	8.01	9.09	10.19	11.28	12.41		13.88	1 / 2			
	reaction time	interval	1.09	1.07	1.10	1.09	1.08	1.08	1.10	1.09	1.13	1.47			3.26	3.25	3.32
		velocity	5.32	8.39	8.54	8.31	8.39	8.46	8.31	8.39	8.09	9.54	7.93		8.41	8.44	8.26

Heat 5

date 03-Nov-16

wind 1.5 m/s

(2016.11.07) - https://twitter.com/touchdown_time/media?lang=en&lang=en&lang=en&lang=en

		H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
Omuro, Hideki (JPN) (1990)	time	2.63	3.69	4.75	5.81	6.86	7.91	8.98	10.06	11.16	12.25		13.70	3 / 1			
	reaction time	interval	1.06	1.06	1.06	1.05	1.05	1.07	1.08	1.10	1.09	1.45			3.18	3.17	3.27
		velocity	5.22	8.62	8.62	8.62	8.70	8.54	8.46	8.31	8.39	9.67	8.03		8.62	8.65	8.39

2016 Japanese Junior Olympics (Yokohama, JPN) (A - 99cm) (B - 91cm)**FINAL - A**

date 30-Oct-16

wind 0.1 m/s

(2016.10.30) - https://twitter.com/touchdown_time/media?lang=en&lang=en&lang=en&lang=en

		H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
Kubota, Kosuke (JPN) (2001)	time	2.72	3.84	4.96	6.10	7.24	8.37	9.52	10.67	11.82	12.99		14.60	7 / 1			
	reaction time	interval	1.12	1.12	1.14	1.14	1.13	1.15	1.15	1.15	1.17	1.61			3.38	3.42	3.47
		velocity	5.04	8.16	8.16	8.02	8.02	8.09	7.95	7.95	7.95	7.81	8.71	7.53	8.11	8.02	7.90

FINAL - B

date 28-Oct-16

wind -0.8 m/s

(2016.10.28) - https://twitter.com/touchdown_time/media?lang=en&lang=en&lang=en&lang=en

		H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
Issei, Osaka (JPN) (2002)	time	2.76	3.91	5.06	6.21	7.33	8.49	9.63	10.82	12.00	13.21		14.87	4 / 1			
	reaction time	interval	1.15	1.15	1.15	1.12	1.16	1.14	1.19	1.18	1.21	1.66			3.45	3.42	3.58
		velocity	4.97	7.95	7.95	7.95	8.16	7.88	8.02	7.68	7.75	7.55	8.45	7.40	7.95	8.02	7.66
Shizuki, Kondo (JPN)	time	2.72	3.88	4.99	6.12	7.24	8.38	9.54	10.73	11.97	13.20		14.89	5 / 2			
	reaction time	interval	1.16	1.11	1.13	1.12	1.14	1.16	1.19	1.24	1.23	1.69			3.40	3.42	3.66
		velocity	5.04	7.88	8.23	8.09	8.16	8.02	7.88	7.68	7.37	7.43	8.30	7.39	8.06	8.02	7.49

Semi-Final 3 - B

date 28-Oct-16

wind -0.8 m/s

(2016.10.27) - https://twitter.com/touchdown_time/media?lang=en&lang=en&lang=en&lang=en

		H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
--	--	----	----	----	----	----	----	----	----	----	-----	--------	---------------	--------------	-------	-------	--------

Shizuki, Kondo (JPN)	time	2.75	3.91	5.03	6.16	7.33	8.49	9.66	10.87	12.10	13.38		15.14	7 / 1			
	reaction time	interval	1.16	1.12	1.13	1.17	1.16	1.17	1.21	1.23	1.28	1.76			3.41	3.50	3.72
	velocity	4.99	7.88	8.16	8.09	7.81	7.88	7.81	7.55	7.43	7.14	7.97	7.27		8.04	7.83	7.37

Heat 1 - B		date	28-Oct-16	wind	-0.6 m/s (2016.10.27) - https://twitter.com/touchdown_time/media?lang=en&lang=en&lang=en&lang=en														
				H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
Shizuki, Kondo (JPN)	time	2.74	3.89	5.02	6.17	7.31	8.46	9.60	10.79	12.00	13.24				14.89	8 / 1			
	reaction time	interval	1.15	1.13	1.15	1.14	1.15	1.14	1.19	1.21	1.24	1.65					3.43	3.43	3.64
	velocity	5.01	7.95	8.09	7.95	8.02	7.95	8.02	7.68	7.55	7.37	8.50	7.39				7.99	7.99	7.53

2016 Tashima Memorial (Yamaguchi, JPN)

FINAL

	date	23-Oct-16	wind	0.2 m/s	(2016.10.26) - https://twitter.com/touchdown_time/media?lang=en&lang=en&lang=en&lang=en																
					H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10	
Omuro, Hideki (JPN) (1990)	time				2.62	3.68	4.73	5.78	6.83	7.89	8.96	10.05	11.16	12.26		13.69	4 / 1				
	reaction time					interval	1.06	1.05	1.05	1.05	1.06	1.07	1.09	1.11	1.10	1.43			3.16	3.18	3.30
	velocity					5.24	8.62	8.70	8.70	8.70	8.62	8.54	8.39	8.23	8.31	9.80	8.04		8.68	8.62	8.31

2016 Japanese National Junior Championships (Mizuho, JPN) (99cm)

FINAL		date	21-Oct-16		wind		-0.2 m/s		(2016.10.21) - https://twitter.com/touchdown_time/media?lang=en&lang=en&lang=en&lang=en											
					H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
Kato, Takuma (JPN) (1998)	time	2.62	3.70	4.75	5.80	6.84	7.91	8.98	10.05	11.18	12.28				13.69		6 / 1			
	reaction time	interval	1.08	1.05	1.05	1.04	1.07	1.07	1.07	1.13	1.10	1.41						3.18	3.18	3.30
	velocity	5.24	8.46	8.70	8.70	8.79	8.54	8.54	8.54	8.09	8.31	9.94	8.04					8.62	8.62	8.31

2016 Japanese National Youth Championships (Mizuho, JPN) (91cm)

FINAL		date	21-Oct-16		wind		-0.1 m/s		(2016.10.21) - https://twitter.com/touchdown_time/media?lang=en&lang=en&lang=en&lang=en												
					H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10	
Higuchi, Rikuto (JPN) (1999)	time	2.63	3.70	4.75	5.81	6.87	7.92	8.98	10.06	11.12	12.23					13.64	3 / 1				
	reaction time			interval	1.07	1.05	1.06	1.06	1.05	1.06	1.08	1.06	1.11	1.41		NYR			3.18	3.17	3.25
	velocity	5.22	8.54	8.70	8.62	8.62	8.70	8.62	8.46	8.62	8.23	9.94				8.06			8.62	8.65	8.44

Semi-Final 3		date	21-Oct-16		wind		-0.2 m/s		(2016.11.11) - https://twitter.com/touchdown_time/media?lang=en&lang=en&lang=en&lang=en										
			H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10	
Higuchi, Rikuto (JPN) (1999)	time		2.60	3.66	4.71	5.76	6.81	7.87	8.94	10.02	11.10	12.22		13.70	4 / 1				
	reaction time			1.06	1.05	1.05	1.05	1.06	1.07	1.08	1.08	1.12	1.48				3.16	3.18	3.28
	velocity		5.28	8.62	8.70	8.70	8.70	8.62	8.54	8.46	8.46	8.16	9.47	8.03			8.68	8.62	8.36
Kawakami, Asuma (JPN) (1999)	time		2.67	3.76	4.87	5.96	7.04	8.14	9.24	10.35	11.49	12.63		14.09	5 / 2				
	reaction time			1.09	1.11	1.09	1.08	1.10	1.10	1.11	1.14	1.14	1.46				3.29	3.28	3.39
	velocity		5.14	8.39	8.23	8.39	8.46	8.31	8.31	8.23	8.02	8.02	9.60	7.81			8.33	8.36	8.09
Tokuoka, Rika (JPN)	time		2.64	3.76	4.85	5.93	7.01	8.11	9.22	10.35	11.51	12.67		14.22	8 / 3				
	reaction time			1.12	1.09	1.08	1.08	1.10	1.11	1.13	1.16	1.16	1.55				3.29	3.29	3.45
	velocity		5.20	8.16	8.39	8.46	8.46	8.31	8.23	8.09	7.88	7.88	9.05	7.74			8.33	8.33	7.95

Semi-Final 2		date	21-Oct-16		wind	1.6 m/s		(2016.11.08) - https://twitter.com/touchdown_time/media?lang=en&lang=en&lang=en&lang=en												
					H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
Fukamoto, Riku (JPN)	time	2.64	3.73	4.83	5.90	6.98	8.06	9.15	10.26	11.38	12.51				13.95		3 / 1			
	reaction time	interval	1.09	1.10	1.07	1.08	1.08	1.09	1.11	1.12	1.13	1.44						3.26	3.25	3.36
	velocity	5.20	8.39	8.31	8.54	8.46	8.46	8.39	8.23	8.16	8.09	9.74	7.89					8.41	8.44	8.16
Nagamatsu, Hiroya (JPN) (1999)	time	2.65	3.76	4.89	5.99	7.07	8.20	9.31	10.42	11.53	12.65				14.06		4 / 2			
	reaction time	interval	1.11	1.13	1.10	1.08	1.13	1.11	1.11	1.11	1.12	1.41						3.34	3.32	3.34
	velocity	5.18	8.23	8.09	8.31	8.46	8.09	8.23	8.23	8.23	8.16	9.94	7.82					8.21	8.26	8.21
Kubota, Taichi (JPN) (2000)	time	2.78	3.91	5.01	6.12	7.21	8.31	9.41	10.49	11.61	12.72				14.19		5 / 3			
	reaction time	interval	1.13	1.10	1.11	1.09	1.10	1.10	1.08	1.12	1.11	1.47						3.34	3.29	3.31
	velocity	4.94	8.09	8.31	8.23	8.39	8.31	8.31	8.46	8.16	8.23	9.54	7.75					8.21	8.33	8.28

Semi-Final 1		date	21-Oct-16	wind	(2016.11.08) - https://twitter.com/touchdown_time/media?lang=en&lang=en&lang=en&lang=en															
				H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10	
Katsuta, Chiku (JPN)	time			2.63	3.70	4.79	5.84	6.91	7.99	9.07	10.17	11.28	12.41		13.85	4 / 1				
	reaction time	interval			1.07	1.09	1.05	1.07	1.08	1.08	1.10	1.11	1.13	1.44				3.21	3.23	3.34
	velocity			5.22	8.54	8.39	8.70	8.54	8.46	8.46	8.31	8.23	8.09	9.74	7.94			8.54	8.49	8.21
Niwa, Hayato (JPN) (1999)	time			2.71	3.81	4.89	5.97	7.04	8.11	9.17	10.24	11.33	12.50		14.01	3 / 2				
	reaction time	interval			1.10	1.08	1.08	1.07	1.07	1.06	1.07	1.09	1.17	1.51				3.26	3.20	3.33
	velocity			5.06	8.31	8.46	8.46	8.54	8.54	8.62	8.54	8.39	7.81	9.28	7.85			8.41	8.57	8.23
Takaoka, Yudai (JPN) (1999)	time			2.66	3.76	4.84	5.93	7.03	8.11	9.21	10.35	11.47	12.58		14.06	6 / 3				
	reaction time	interval			1.10	1.08	1.09	1.10	1.08	1.10	1.14	1.12	1.11	1.48				3.27	3.28	3.37
	velocity			5.16	8.31	8.46	8.39	8.31	8.46	8.31	8.02	8.16	8.23	9.47	7.82			8.39	8.36	8.14

Heat 6

date 21-Oct-16 wind 1.2 m/s (2016.11.08) - https://twitter.com/touchdown_time/media?lang=en&lang=en&lang=en&lang=en

H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
----	----	----	----	----	----	----	----	----	-----	--------	---------------	--------------	-------	-------	--------

Higuchi, Rikuto (JPN) (1999)	time	2.63	3.69	4.75	5.79	6.86	7.92	9.00	10.10	11.21	12.36		13.98	4 / 1			
	reaction time	interval	1.06	1.06	1.04	1.07	1.06	1.08	1.10	1.11	1.15	1.62			3.16	3.21	3.36
	velocity	5.22	8.62	8.62	8.79	8.54	8.62	8.46	8.31	8.23	7.95	8.65	7.87		8.68	8.54	8.16
Shibaya, Takumi (JPN) (2000)	time	2.75	3.85	4.97	6.08	7.21	8.32	9.44	10.59	11.75	12.90		14.43	1 / 2			
	reaction time	interval	1.10	1.12	1.11	1.13	1.11	1.12	1.15	1.16	1.15	1.53			3.33	3.36	3.46
	velocity	4.99	8.31	8.16	8.23	8.09	8.23	8.16	7.95	7.88	7.95	9.16	7.62		8.23	8.16	7.92
Kudo, Masayoshi (JPN)	time	2.70	3.80	4.89	6.00	7.09	8.28	9.41	10.57	11.70	12.91		14.44	2 / 3			
	reaction time	interval	1.10	1.09	1.11	1.09	1.19	1.13	1.16	1.13	1.21	1.53			3.30	3.41	3.50
	velocity	5.08	8.31	8.39	8.23	8.39	7.68	8.09	7.88	8.09	7.55	9.16	7.62		8.31	8.04	7.83

Heat 5

date 21-Oct-16		wind 1.3 m/s		(2016.11.08) - https://twitter.com/touchdown_time/media?lang=en&lang=en&lang=en&lang=en													
Kubota, Taichi (JPN) (2000)		H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
	time	2.74	3.85	4.97	6.08	7.20	8.31	9.45	10.59	11.73	12.86		14.38	3 / 1			
	reaction time	interval	1.11	1.12	1.11	1.12	1.11	1.14	1.14	1.14	1.13	1.52			3.34	3.37	3.41
	velocity	5.01	8.23	8.16	8.23	8.16	8.23	8.02	8.02	8.02	8.09	9.22	7.65		8.21	8.14	8.04
Namba, Niroshi (JPN) (1999)	time	2.77	3.92	5.05	6.18	7.30	8.41	9.53	10.68	11.81	12.95		14.42	4 / 2			
	reaction time	interval	1.15	1.13	1.13	1.12	1.11	1.12	1.15	1.13	1.14	1.47			3.41	3.35	3.42
	velocity	4.95	7.95	8.09	8.09	8.16	8.23	8.16	7.95	8.09	8.02	9.54	7.63		8.04	8.19	8.02
Imai, Rikkoshi (JPN)	time	2.66	3.79	4.91	6.02	7.16	8.28	9.42	10.56	11.76	12.93		14.43	7 / 3			
	reaction time	interval	1.13	1.12	1.11	1.14	1.12	1.14	1.14	1.20	1.17	1.50			3.36	3.40	3.51
	velocity	5.16	8.09	8.16	8.23	8.02	8.16	8.02	8.02	7.62	7.81	9.35	7.62		8.16	8.06	7.81

Heat 4

date 21-Oct-16		wind 1.5 m/s		(2016.11.08) - https://twitter.com/touchdown_time/media?lang=en&lang=en&lang=en&lang=en													
Katsuta, Chiku (JPN)		H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
	time	2.67	3.75	4.83	5.90	6.97	8.04	9.14	10.23	11.33	12.51		14.05	1 / 1			
	reaction time	interval	1.08	1.08	1.07	1.07	1.07	1.10	1.09	1.10	1.18	1.54			3.23	3.24	3.37
	velocity	5.14	8.46	8.46	8.54	8.54	8.54	8.31	8.39	8.31	7.75	9.10	7.83		8.49	8.46	8.14
Takaoka, Yudai (JPN) (1999)	time	2.72	3.82	4.93	6.02	7.11	8.19	9.31	10.43	11.62	12.77		14.28	8 / 2			
	reaction time	interval	1.10	1.11	1.09	1.09	1.08	1.12	1.12	1.19	1.15	1.51			3.30	3.29	3.46
	velocity	5.04	8.31	8.23	8.39	8.39	8.46	8.16	8.16	7.68	7.95	9.28	7.70		8.31	8.33	7.92
Abe, Ryoto (JPN) (2000)	time	2.66	3.76	4.85	5.96	7.06	8.18	9.32	10.45	11.63	12.83		14.30	2 / 3			
	reaction time	interval	1.10	1.09	1.11	1.10	1.12	1.14	1.13	1.18	1.20	1.47			3.30	3.36	3.51
	velocity	5.16	8.31	8.39	8.23	8.31	8.16	8.02	8.09	7.75	7.62	9.54	7.69		8.31	8.16	7.81

Heat 3

date 21-Oct-16		wind 1.4 m/s		(2016.11.07) - https://twitter.com/touchdown_time/media?lang=en&lang=en&lang=en&lang=en													
Fukamoto, Riku (JPN)		H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
	time	2.62	3.71	4.78	5.83	6.89	7.97	9.05	10.17	11.27	12.41		13.86	2 / 1			
	reaction time	interval	1.09	1.07	1.05	1.06	1.08	1.08	1.12	1.10	1.14	1.45			3.21	3.22	3.36
	velocity	5.24	8.39	8.54	8.70	8.62	8.46	8.46	8.16	8.31	8.02	9.67	7.94		8.54	8.52	8.16
Morita, Kanon (JPN) (2000)	time	2.66	3.80	4.92	6.04	7.15	8.28	9.42	10.54	11.70	12.86		14.39	7 / 2			
	reaction time	interval	1.14	1.12	1.12	1.11	1.13	1.14	1.12	1.16	1.16	1.53			3.38	3.38	3.44
	velocity	5.16	8.02	8.16	8.16	8.23	8.09	8.02	8.16	7.88	7.88	9.16	7.64		8.11	8.11	7.97
Suzuki, Yusei (JPN) (2000)	time	2.64	3.76	4.89	6.01	7.13	8.27	9.42	10.55	11.72	12.91		14.46	4 / 3			
	reaction time	interval	1.12	1.13	1.12	1.12	1.14	1.15	1.13	1.17	1.19	1.55			3.37	3.41	3.49
	velocity	5.20	8.16	8.09	8.16	8.16	8.02	7.95	8.09	7.81	7.68	9.05	7.61		8.14	8.04	7.86

Heat 2

date 21-Oct-16		wind 2.3 m/s		(2016.11.07) - https://twitter.com/touchdown_time/media?lang=en&lang=en&lang=en&lang=en													
Niwa, Hayato (JPN) (1999)		H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
	time	2.69	3.81	4.91	5.98	7.04	8.10	9.18	10.28	11.38	12.50		14.00	5 / 1			
	reaction time	interval	1.12	1.10	1.07	1.06	1.06	1.08	1.10	1.10	1.12	1.50			3.29	3.20	3.32
	velocity	5.10	8.16	8.31	8.54	8.62	8.62	8.46	8.31	8.31	8.16	9.35	7.86		8.33	8.57	8.26
Nagamatsu, Hiroya (JPN) (1999)	time	2.67	3.77	4.86	6.00	7.10	8.22	9.33	10.45	11.59	12.72		14.19	8 / 2			
	reaction time	interval	1.10	1.09	1.14	1.10	1.12	1.11	1.12	1.14	1.13	1.47			3.33	3.33	3.39
	velocity	5.14	8.31	8.39	8.02	8.31	8.16	8.23	8.16	8.02	8.09	9.54	7.75		8.23	8.23	8.09
Tokuoka, Rika (JPN)	time	2.68	3.79	4.91	6.02	7.12	8.22	9.34	10.49	11.61	12.77		14.30	7 / 3			
	reaction time	interval	1.11	1.12	1.11	1.10	1.10	1.12	1.15	1.12	1.16	1.53			3.34	3.32	3.43
	velocity	5.12	8.23	8.16	8.23	8.31	8.31	8.16	7.95	8.16	7.88	9.16	7.69		8.21	8.26	7.99

Heat 1

date 21-Oct-16		wind 1.7 m/s		(2016.11.07) - https://twitter.com/touchdown_time/media?lang=en&lang=en&lang=en&lang=en													
Kawakami, Asuma (JPN) (1999)		H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
	time	2.67	3.78	4.88	5.97	7.08	8.20	9.31	10.43	11.56	12.71		14.21	7 / 1			
	reaction time	interval	1.11	1.10	1.09	1.11	1.12	1.11	1.12	1.13	1.15	1.50			3.30	3.34	3.40
	velocity	5.14	8.23	8.31	8.39	8.23	8.16	8.23	8.16	8.09	7.95	9.35	7.74		8.31	8.21	8.06
Hiraga, Kentaro (JPN) (1999)	time	2.68	3.82	4.92	6.04	7.14	8.24	9.36	10.49	11.63	12.79		14.28	6 / 2			
	reaction time	interval	1.14	1.10	1.12	1.10	1.10	1.12	1.13	1.14	1.16	1.49			3.36	3.32	3.43

reaction time	interval		1.16	1.19	1.19	1.19	1.18	1.28	1.23	1.23	1.24	1.55		3.54	3.65	3.70
	velocity	4.88	7.88	7.68	7.68	7.68	7.75	7.14	7.43	7.43	7.37	9.05	7.21	7.75	7.51	7.41

2016 Memorial van Damme (Brussels, BEL) (TV Analysis)

FINAL

		date	09-Sep-16	wind 0.2 m/s			Henson (2020) - Athlete First: 2016 year end hurdle report												
				H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
Ortega, Orlando (ESP) (1991)	time			2.50	3.54	4.57	5.57	6.57	7.57	8.57	9.59	10.61	11.64		13.08	5 / 1			
	reaction time	0.145	interval	7 steps	1.04	1.03	1.00	1.00	1.00	1.00	1.02	1.02	1.03	1.44			3.07	3.00	3.07
			velocity	5.49	8.79	8.87	9.14	9.14	9.14	9.14	8.96	8.96	8.87	9.74	8.41			8.93	9.14
Martinot-Lagarde, Pascal (FRA)	time			2.48	3.50	4.50	5.50	6.50	7.50	8.50	9.54	10.57	11.64		13.12	6 / 2			
	reaction time	0.160	interval	7 steps	1.02	1.00	1.00	1.00	1.00	1.00	1.04	1.03	1.07	1.48			3.02	3.00	3.14
			velocity	5.53	8.96	9.14	9.14	9.14	9.14	9.14	8.79	8.87	8.54	9.47	8.38			9.08	9.14
Belocian, Wilhem (FRA) (1995)	time			2.50	3.53	4.57	5.57	6.57	7.60	8.62	9.64	10.71	11.81		13.32	8 / 3			
	reaction time	0.153	interval		1.03	1.04	1.00	1.00	1.03	1.02	1.02	1.07	1.10	1.51			3.07	3.05	3.19
			velocity	5.49	8.87	8.79	9.14	9.14	8.87	8.96	8.96	8.54	8.31	9.28	8.26			8.93	8.99
Bascou, Dimitri (FRA) (1987)	time			2.50	3.57	4.57	5.57	6.57	7.60	8.64	9.67	10.75	11.86		13.37	4 / 4			
	reaction time	0.160	interval	7 steps	1.07	1.00	1.00	1.00	1.03	1.04	1.03	1.08	1.11	1.51			3.07	3.07	3.22
			velocity	5.49	8.54	9.14	9.14	9.14	8.87	8.79	8.87	8.46	8.23	9.28	8.23			8.93	8.93
Omoregie, David (GBR) (1995)	time			2.53	3.57	4.60	5.62	6.64	7.67	8.72	9.77	10.84	11.94		13.43	3 / 5			
	reaction time	0.153	interval		1.04	1.03	1.02	1.02	1.03	1.05	1.05	1.07	1.10	1.49			3.09	3.10	3.22
			velocity	5.42	8.79	8.87	8.96	8.96	8.87	8.70	8.70	8.54	8.31	9.41	8.19			8.87	8.85
Trajkovic, Milan (CYP) (1992)	time			2.50	3.53	4.57	5.57	6.59	7.61	8.66	9.74	10.81	11.90		13.44	7 / 6			
	reaction time	0.143	interval	7 steps	1.03	1.04	1.00	1.02	1.02	1.05	1.08	1.07	1.09	1.54			3.07	3.09	3.24
			velocity	5.49	8.87	8.79	9.14	8.96	8.96	8.70	8.46	8.54	8.39	9.10	8.18			8.93	8.87
Baji, Balázs (HUN) (1989)	time			2.53	3.60	4.64	5.67	6.70	7.70	8.74	9.81	10.88	11.99		13.50	2 / 7			
	reaction time	0.183	interval	7 steps	1.07	1.04	1.03	1.03	1.00	1.04	1.07	1.07	1.11	1.51			3.14	3.07	3.25
			velocity	5.42	8.54	8.79	8.87	8.87	9.14	8.79	8.54	8.54	8.23	9.28	8.15			8.73	8.93
Eaton, Jarret (USA) (1989)	time			2.50	3.57	4.60	5.60	6.62	7.66	8.72	9.77	10.83	11.94		13.54	9 / 8			
	reaction time	0.181	interval	7 steps	1.07	1.03	1.00	1.02	1.04	1.06	1.05	1.06	1.11	1.60			3.10	3.12	3.22
			velocity	5.49	8.54	8.87	9.14	8.96	8.79	8.62	8.70	8.62	8.23	8.76	8.12			8.85	8.79
Manga, Aurel (FRA) (1992)	time			2.50	3.57	4.60	5.62	6.64	7.69	8.75	9.81	10.91	12.06		13.64	1 / 9			
	reaction time	0.173	interval	7 steps	1.07	1.03	1.02	1.02	1.05	1.06	1.06	1.10	1.15	1.58			3.12	3.13	3.31
			velocity	5.49	8.54	8.87	8.96	8.96	8.70	8.62	8.62	8.31	7.95	8.87	8.06			8.79	8.76

2016 Japanese University Championships (Kumagaya, JPN)

FINAL

		date 04-Sep-16		wind -1.9 m/s		(2016.09.06) - https://twitter.com/touchdown_time/media?lang=en&lang=en&lang=en&lang=en												
		H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10	
Takayama, Shunya (JPN) (1994)	time	2.67	3.74	4.84	5.92	7.01	8.09	9.19	10.31	11.43	12.56		14.02	7 / 1				
	reaction time	0.159	interval	1.07	1.10	1.08	1.09	1.08	1.10	1.12	1.12	1.13	1.46		3.25	3.27	3.37	
	velocity	5.14	8.54	8.31	8.46	8.39	8.46	8.31	8.16	8.16	8.09	9.60	7.85		8.44	8.39	8.14	
Furuya, Takumu (JPN) (1997)	time	2.66	3.76	4.85	5.94	7.02	8.12	9.23	10.34	11.48	12.64		14.09	3 / 2				
	reaction time	0.149	interval	1.10	1.09	1.09	1.08	1.10	1.11	1.11	1.14	1.16	1.45		3.28	3.29	3.41	
	velocity	5.16	8.31	8.39	8.39	8.46	8.31	8.23	8.23	8.02	7.88	9.67	7.81		8.36	8.33	8.04	
Kuriki, Anthony Tyrell (JPN) (1994)	time	2.69	3.80	4.89	5.98	7.08	8.17	9.27	10.39	11.52	12.68		14.11	4 / 3				
	reaction time	0.142	interval	7 steps	1.11	1.09	1.09	1.10	1.09	1.10	1.12	1.13	1.16	1.43		3.29	3.29	3.41
	velocity	5.10	8.23	8.39	8.39	8.31	8.39	8.31	8.16	8.09	7.88	9.80	7.80		8.33	8.33	8.04	
Kanai, Taio (JPN) (1995)	time	2.64	3.75	4.85	5.94	7.03	8.17	9.30	10.44	11.57	12.73		14.25	6 / 4				
	reaction time	0.146	interval	7 steps	1.11	1.10	1.09	1.09	1.14	1.13	1.14	1.13	1.16	1.52		3.30	3.36	3.43
	velocity	5.20	8.23	8.31	8.39	8.39	8.02	8.09	8.02	8.09	7.88	9.22	7.72		8.31	8.16	7.99	
Kagimoto, Masahiro (JPN) (1995)	time	2.71	3.83	4.94	6.06	7.17	8.27	9.39	10.51	11.64	12.80		14.32	9 / 5				
	reaction time	0.169	interval	1.12	1.11	1.12	1.11	1.10	1.12	1.12	1.13	1.16	1.52		3.35	3.33	3.41	
	velocity	5.06	8.16	8.23	8.16	8.23	8.31	8.16	8.16	8.09	7.88	9.22	7.68		8.19	8.23	8.04	

Semi-Final 3

Semi-Final 3		date 04-Sep-16		wind -0.1 m/s				(2016.09.05) - https://twitter.com/touchdown_time/media?lang=en&lang=en&lang=en&lang=en											
				H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
Takayama, Shunya (JPN) (1994)	time			2.66	3.73	4.78	5.83	6.88	7.95	9.03	10.12	11.23	12.35		13.83	4 / 1			
	reaction time	0.151	interval		1.07	1.05	1.05	1.05	1.07	1.08	1.09	1.11	1.12	1.48			3.17	3.20	3.32
			velocity	5.16	8.54	8.70	8.70	8.70	8.54	8.46	8.39	8.23	8.16	9.47	7.95		8.65	8.57	8.26
Takeyoshi, Daiki (JPN) (1994)	time			2.66	3.74	4.82	5.90	7.00	8.09	9.22	10.35	11.49	12.65		14.15	6 / 2			
	reaction time	0.134	interval		1.08	1.08	1.08	1.10	1.09	1.13	1.13	1.14	1.16	1.50			3.24	3.32	3.43
			velocity	5.16	8.46	8.46	8.46	8.31	8.39	8.09	8.09	8.02	7.88	9.35	7.77		8.46	8.26	7.99
Tanaka, Shinya (JPN) (1993)	time			2.68	3.77	4.86	5.96	7.07	8.17	9.29	10.44	11.58	12.75		14.22	7 / 3			

reaction time	0.141	interval	7 steps	1.09	1.09	1.10	1.11	1.10	1.12	1.15	1.14	1.17	1.47			3.28	3.33	3.46
velocity				5.12	8.39	8.39	8.31	8.23	8.31	8.16	7.95	8.02	7.81	9.54	7.74	8.36	8.23	7.92

Semi-Final 2

date 04-Sep-16 wind 0.7 m/s

(2016.09.05) - https://twitter.com/touchdown_time/media?lang=en&lang=en&lang=en&lang=en

			H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
Kuriki, Anthony Tyrell (JPN) (1995)	time		2.65	3.75	4.83	5.91	6.99	8.09	9.19	10.27	11.37	12.48		13.91	5 / 1			
reaction time	0.135	interval	7 steps	1.10	1.08	1.08	1.08	1.10	1.10	1.08	1.10	1.11	1.43			3.26	3.28	3.29
velocity			5.18	8.31	8.46	8.46	8.46	8.31	8.31	8.46	8.31	8.23	9.80	7.91		8.41	8.36	8.33
Kanai, Taio (JPN) (1995)	time		2.62	3.70	4.78	5.85	6.94	8.03	9.14	10.26	11.36	12.49		13.93	4 / 2			
reaction time	0.135	interval	7 steps	1.08	1.08	1.07	1.09	1.09	1.11	1.12	1.10	1.13	1.44			3.23	3.29	3.35
velocity			5.24	8.46	8.46	8.54	8.39	8.39	8.23	8.16	8.31	8.09	9.74	7.90		8.49	8.33	8.19
Furuya, Takumu (JPN) (1997)	time		2.65	3.75	4.83	5.92	7.00	8.09	9.17	10.27	11.37	12.48		13.93	6 / 3			
reaction time	0.143	interval		1.10	1.08	1.09	1.08	1.09	1.08	1.10	1.10	1.11	1.45			3.27	3.25	3.31
velocity			5.18	8.31	8.46	8.39	8.46	8.39	8.46	8.31	8.31	8.23	9.67	7.90		8.39	8.44	8.28
Ishikawa, Shuhei (JPN) (1995)	time		2.61	3.74	4.81	5.93	7.01	8.12	9.22	10.34	11.44	12.56		14.07	7 / 4			
reaction time	0.157	interval	7 steps	1.13	1.07	1.12	1.08	1.11	1.10	1.12	1.10	1.12	1.51			3.32	3.29	3.34
velocity			5.26	8.09	8.54	8.16	8.46	8.23	8.31	8.16	8.31	8.16	9.28	7.82		8.26	8.33	8.21
Okazaki, Masaki (JPN) (1994)	time		2.65	3.77	4.85	5.94	7.05	8.14	9.23	10.35	11.49	12.63		14.10	2 / 5			
reaction time	1.530	interval		1.12	1.08	1.09	1.11	1.09	1.09	1.12	1.14	1.14	1.47			3.29	3.29	3.40
velocity			5.18	8.16	8.46	8.39	8.23	8.39	8.39	8.16	8.02	8.02	9.54	7.80		8.33	8.33	8.06
Furuya, Kyohei (JPN) (1996)	time		2.68	3.77	4.89	5.97	7.09	8.18	9.34	10.48	11.61	12.78		14.21	3 / 7			
reaction time	0.133	interval		1.09	1.12	1.08	1.12	1.09	1.16	1.14	1.13	1.17	1.43			3.29	3.37	3.44
velocity			5.12	8.39	8.16	8.46	8.16	8.39	7.88	8.02	8.09	7.81	9.80	7.74		8.33	8.14	7.97
Furumoto, Tsubasa (JPN) (1999)	time		2.70	3.80	4.93	6.04	7.14	8.24	9.34	10.47	11.66	12.85		14.36	8 / 8			
reaction time	0.133	interval		1.10	1.13	1.11	1.10	1.10	1.10	1.13	1.19	1.19	1.51			3.34	3.30	3.51
velocity			5.08	8.31	8.09	8.23	8.31	8.31	8.31	8.09	7.68	7.68	9.28	7.66		8.21	8.31	7.81

Semi-Final 1

date 04-Sep-16 wind 0.6 m/s

(2016.09.05) - https://twitter.com/touchdown_time/media?lang=en&lang=en&lang=en&lang=en

			H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
Yada, Gen (JPN) (1994)	time		2.70	3.85	4.95	6.03	7.12	8.21	9.31	10.40	11.50	12.63		14.01	5 / 1			
reaction time	0.135	interval		1.15	1.10	1.08	1.09	1.09	1.10	1.09	1.10	1.13	1.38			3.33	3.28	3.32
velocity			5.08	7.95	8.31	8.46	8.39	8.39	8.31	8.39	8.31	8.09	10.16	7.85		8.23	8.36	8.26
Kagimoto, Masahiro (JPN) (1995)	time		2.71	3.80	4.88	5.98	7.08	8.18	9.29	10.37	11.47	12.59		14.01	7 / 2			
reaction time	0.159	interval		1.09	1.08	1.10	1.10	1.10	1.11	1.08	1.10	1.12	1.42			3.27	3.31	3.30
velocity			5.06	8.39	8.46	8.31	8.31	8.31	8.23	8.46	8.31	8.16	9.87	7.85		8.39	8.28	8.31
Taue, Shun (JPN) (1997)	time		2.73	3.85	4.94	6.02	7.12	8.22	9.32	10.41	11.54	12.69		14.16	2 / 4			
reaction time	0.144	interval		1.12	1.09	1.08	1.10	1.10	1.10	1.09	1.13	1.15	1.47			3.29	3.30	3.37
velocity			5.03	8.16	8.39	8.46	8.31	8.31	8.31	8.39	8.09	7.95	9.54	7.77		8.33	8.31	8.14
Tsuneyoshi, Keisuke (JPN) (1995)	time		2.76	3.88	4.97	6.07	7.17	8.27	9.36	10.52	11.63	12.78		14.21	4 / 5			
reaction time	0.173	interval		1.12	1.09	1.10	1.10	1.10	1.09	1.16	1.11	1.15	1.43			3.31	3.29	3.42
velocity			4.97	8.16	8.39	8.31	8.31	8.31	8.39	7.88	8.23	7.95	9.80	7.74		8.28	8.33	8.02
Kawamura, Naoya (JPN) (1996)	time		2.78	3.89	4.98	6.07	7.17	8.28	9.38	10.50	11.62	12.77		14.23	3 / 6			
reaction time	0.160	interval		1.11	1.09	1.09	1.10	1.11	1.10	1.12	1.12	1.15	1.46			3.29	3.31	3.39
velocity			4.94	8.23	8.39	8.39	8.31	8.23	8.31	8.16	8.16	7.95	9.60	7.73		8.33	8.28	8.09
Ueno, Yuta (JPN) (1995)	time		2.71	3.83	4.95	6.05	7.15	8.28	9.40	10.54	11.68	12.84		14.31	8 / 7			
reaction time	0.136	interval		1.12	1.12	1.10	1.10	1.13	1.12	1.14	1.14	1.16	1.47			3.34	3.35	3.44
velocity			5.06	8.16	8.16	8.31	8.31	8.09	8.16	8.02	8.02	7.88	9.54	7.69		8.21	8.19	7.97
Hatanaka, Yushi (JPN) (1994)	time		2.69	3.81	4.91	6.03	7.12	8.27	9.40	10.56	11.69	12.86		14.34	6 / 8			
reaction time	0.151	interval		1.12	1.10	1.12	1.09	1.15	1.13	1.16	1.13	1.17	1.48			3.34	3.37	3.46
velocity			5.10	8.16	8.31	8.16	8.39	7.95	8.09	7.88	8.09	7.81	9.47	7.67		8.21	8.14	7.92

2016 Kyushu Championships (Kita-Kyushu, JPN)**FINAL**

date 27-Aug-16 wind -2.1 m/s

(2016.08.30) - https://twitter.com/touchdown_time/media?lang=en&lang=en&lang=en&lang=en

			H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
Nishi, Kyohei (JPN) (1995)	time		2.70	3.81	4.95	6.05	7.15	8.28	9.39	10.51	11.65	12.81		14.32	4 / 1			
reaction time		interval		1.11	1.14	1.10	1.10	1.13	1.11	1.12	1.14	1.16	1.51			3.35	3.34	3.42
velocity			5.08	8.23	8.02	8.31	8.31	8.09	8.23	8.16	8.02	7.88	9.28	7.68		8.19	8.21	8.02

2016 Athletissima (Lausanne, SUI) (TV Analysis)**FINAL**

date 25-Aug-16 wind 0.5 m/s

Henson (2020) - Athlete First: 2016 year end hurdle report

			H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
Ortega, Orlando (ESP) (1991)	time		2.52	3.56	4.58	5.60	6.60	7.60	8.62	9.64	10.64	11.68		13.11	5 / 1			
reaction time	0.133	interval	7 steps	1.04	1.02	1.02	1.00	1.00	1.02	1.02	1.00	1.04	1.43			3.08	3.02	3.06
velocity			5.44	8.79	8.96	8.96	9.14	9.14	8.96	8.96	9.14	8.79	9.80	8.39		8.90	9.08	8.96

McLeod, Omar (JAM) (1994)	time	2.48	3.54	4.56	5.58	6.58	7.58	8.62	9.64	10.64	11.70		13.12	4 / 2					
	reaction time	0.144	interval	1.06	1.02	1.02	1.00	1.00	1.04	1.02	1.00	1.06	1.42		3.10	3.04	3.08		
	velocity	5.53	8.62	8.96	8.96	9.14	9.14	8.79	8.96	9.14	8.62	9.87	8.38		8.85	9.02	8.90		
Bascou, Dimitri (FRA) (1987)	time	2.48	3.52	4.54	5.56	6.56	7.56	8.58	9.62	10.68	11.76		13.25	6 / 3					
	reaction time	0.139	interval	7 steps	1.04	1.02	1.02	1.00	1.00	1.02	1.04	1.06	1.08	1.49		3.08	3.02	3.18	
	velocity	5.53	8.79	8.96	8.96	9.14	9.14	8.96	8.79	8.62	8.46	9.41	8.30		8.90	9.08	8.62		
Belocian, Wilhem (FRA) (1995)	time	2.48	3.50	4.50	5.52	6.52	7.54	8.56	9.60	10.66	11.74		13.25	7 / 4					
	reaction time	0.126	interval	1.02	1.00	1.02	1.00	1.02	1.02	1.04	1.06	1.08	1.51	PB		3.04	3.04	3.18	
	velocity	5.53	8.96	9.14	8.96	9.14	8.96	8.96	8.79	8.62	8.46	9.28	8.30		9.02	9.02	8.62		
Baji, Balázs (HUN) (1989)	time	2.52	3.56	4.60	5.60	6.60	7.62	8.64	9.66	10.72	11.78		13.34	1 / 5					
	reaction time	0.141	interval	7 steps	1.04	1.04	1.00	1.00	1.02	1.02	1.06	1.06	1.56			3.08	3.04	3.14	
	velocity	5.44	8.79	8.79	9.14	9.14	8.96	8.96	8.96	8.62	8.62	8.99	8.25		8.90	9.02	8.73		
Martinot-Lagarde, Pascal (FRA)	time	2.52	3.56	4.58	5.60	6.62	7.64	8.68	9.72	10.76	11.84		13.37	2 / 6					
	reaction time	0.146	interval	7 steps	1.04	1.02	1.02	1.02	1.04	1.04	1.04	1.08	1.53			3.08	3.08	3.16	
	velocity	5.44	8.79	8.96	8.96	8.96	8.96	8.79	8.79	8.79	8.79	8.46	9.16	8.23		8.90	8.90	8.68	
Oliver, David (USA) (1982)	time	2.52	3.56	4.58	5.60	6.60	7.64	8.68	9.72	10.78	11.84		13.40	3 / 7					
	reaction time	0.146	interval	7 steps	1.04	1.02	1.02	1.00	1.04	1.04	1.04	1.06	1.06	1.56			3.08	3.08	3.16
	velocity	5.44	8.79	8.96	8.96	9.14	8.79	8.79	8.79	8.79	8.62	8.62	8.99	8.21		8.90	8.90	8.68	
Douvalidis, Konstadinos (GRE)	time	2.58	3.66	4.70	5.74	6.76	7.80	8.86	9.92	11.00	12.10		13.69	8 / 8					
	reaction time	0.136	interval	1.08	1.04	1.04	1.02	1.04	1.06	1.06	1.08	1.10	1.59			3.16	3.12	3.24	
	velocity	5.32	8.46	8.79	8.79	8.96	8.79	8.62	8.62	8.46	8.31	8.82	8.04		8.68	8.79	8.46		

2016 Olympic Games (Rio de Janeiro, BRA) (TV Analysis)

FINAL

date 16-Aug-16 wind 0.2 m/s

Henson (2020) - Athlete First: 2016 year end hurdle report

		date	10-Aug-10	wind	0.2 m/s	H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
McLeod, Omar (JAM) (1994)	time					2.44	3.50	4.52	5.54	6.54	7.54	8.56	9.60	10.60	11.62		13.05	5 / 1			
	reaction time	0.142	interval				1.06	1.02	1.02	1.00	1.00	1.02	1.04	1.00	1.02	1.43			3.10	3.02	3.06
			velocity				5.62	8.62	8.96	8.96	9.14	9.14	8.96	8.79	9.14	8.96	9.80	8.43		8.85	9.08
Ortega, Orlando (ESP) (1991)	time					2.56	3.62	4.64	5.64	6.64	7.64	8.64	9.68	10.70	11.74		13.17	7 / 2			
	reaction time	0.127	interval	7 steps			1.06	1.02	1.00	1.00	1.00	1.00	1.04	1.02	1.04	1.43			3.08	3.00	3.10
			velocity				5.36	8.62	8.96	9.14	9.14	9.14	8.79	8.96	8.79	9.80	8.35		8.90	9.14	8.85
Bascou, Dimitri (FRA) (1987)	time					2.50	3.52	4.54	5.56	6.56	7.62	8.66	9.72	10.76	11.80		13.24	6 / 3			
	reaction time	0.131	interval	7 steps			1.02	1.02	1.02	1.00	1.06	1.04	1.06	1.04	1.04	1.44			3.06	3.10	3.14
			velocity				5.49	8.96	8.96	8.96	9.14	8.62	8.79	8.62	8.79	8.79	9.74	8.31		8.96	8.85
Martinot-Lagarde, Pascal (FRA)	time					2.46	3.50	4.52	5.54	6.56	7.56	8.60	9.66	10.72	11.80		13.29	4 / 4			
	reaction time	0.142	interval	7 steps			1.04	1.02	1.02	1.02	1.00	1.04	1.06	1.06	1.08	1.49			3.08	3.06	3.20
			velocity				5.58	8.79	8.96	8.96	8.96	9.14	8.79	8.62	8.62	8.46	9.41	8.28		8.90	8.96
Allen, Devon (USA) (1994)	time					2.50	3.58	4.60	5.60	6.62	7.66	8.70	9.74	10.76	11.82		13.31	3 / 5			
	reaction time	0.135	interval				1.08	1.02	1.00	1.02	1.04	1.04	1.04	1.02	1.06	1.49			3.10	3.10	3.12
			velocity				5.49	8.46	8.96	9.14	8.96	8.79	8.79	8.79	8.96	8.62	9.41	8.26		8.85	8.85
Cabral, Jonathan (CAN) (1992)	time					2.58	3.64	4.68	5.72	6.74	7.76	8.80	9.84	10.88	11.96		13.40	2 / 6			
	reaction time	0.146	interval	7 steps			1.06	1.04	1.04	1.02	1.02	1.04	1.04	1.04	1.08	1.44			3.14	3.08	3.16
			velocity				5.32	8.62	8.79	8.79	8.96	8.96	8.79	8.79	8.79	8.46	9.74	8.21		8.73	8.90
Trajkovic, Milan (CYP) (1992)	time					2.50	3.56	4.62	5.64	6.66	7.72	8.76	9.82	10.88	11.94		13.41	8 / 7			
	reaction time	0.136	interval	7 steps			1.06	1.06	1.02	1.02	1.06	1.04	1.06	1.06	1.06	1.47			3.14	3.12	3.18
			velocity				5.49	8.62	8.62	8.96	8.96	8.62	8.79	8.62	8.62	8.62	9.54	8.20		8.73	8.79
Ash, Ronnie (USA) (1988)	time					2.52	3.58	4.60	5.60	6.60	7.60	8.62	9.64	10.70	11.72			9 / --			
	reaction time	0.160	interval	7 steps			1.06	1.02	1.00	1.00	1.00	1.02	1.02	1.06	1.02		DQ		3.08	3.02	3.10
			velocity				5.44	8.62	8.96	9.14	9.14	9.14	8.96	8.96	8.62	8.96			8.90	9.08	8.85

2016 Japanese National High School Championships (Okayama, JPN) (99cm)

FINAL

date 02-Aug-16 wind 1.2 m/s

Kota (2016) - 69th high school championships: JAF scientific committee - biomechanics data collection

		date: 02 Aug 10			wind: 12 m/s			Kato (2016) - 5th high school championships - JAF - Selection Committee - Bromocyanides data collection										
		H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10	
Kato, Takuma (JPN) (1998)	time	2.63	3.73	4.83	5.92	7.01	8.11	9.20	10.32	11.41	12.55		13.98	6 / 1				
	reaction time	0.127	interval	1.10	1.10	1.09	1.09	1.10	1.09	1.12	1.09	1.14	1.43	PB		3.29	3.28	3.35
	velocity	5.22	8.31	8.31	8.39	8.39	8.31	8.39	8.16	8.39	8.02	9.80	7.87		8.33	8.36	8.19	
Katsuta, Kizuku (JPN) (1999)	time	2.65	3.74	4.88	5.99	7.11	8.24	9.36	10.46	11.59	12.74		14.18	4 / 2				
	reaction time	0.147	interval	1.09	1.14	1.11	1.12	1.13	1.12	1.10	1.13	1.15	1.44	PB		3.34	3.37	3.38
	velocity	5.18	8.39	8.02	8.23	8.16	8.09	8.16	8.31	8.09	7.95	9.74	7.76		8.21	8.14	8.11	
Kichima, Kaito (JPN) (1998)	time	2.68	3.78	4.87	5.95	7.05	8.15	9.27	10.38	11.53	12.70		14.20	7 / 3				

	reaction time	0.161	interval		1.10	1.09	1.08	1.10	1.10	1.12	1.11	1.15	1.17	1.50	PB		3.27	3.32	3.43
			velocity	5.12	8.31	8.39	8.46	8.31	8.31	8.16	8.23	7.95	7.81	9.35	7.75		8.39	8.26	7.99
Tomioka, Tatsuto (JPN) (1998)	time			2.71	3.82	4.94	6.02	7.12	8.21	9.30	10.41	11.54	12.67		14.20	5 / 4			
	reaction time	0.138	interval		1.11	1.12	1.08	1.10	1.09	1.09	1.11	1.13	1.13	1.53	PB		3.31	3.28	3.37
			velocity	5.06	8.23	8.16	8.46	8.31	8.39	8.39	8.23	8.09	8.09	9.16	7.75		8.28	8.36	8.14
Higuchi, Rikuto (JPN) (1999)	time			2.68	3.78	4.86	5.96	7.07	8.21	9.35	10.47	11.60	12.77		14.25	2 / 5			
	reaction time	0.168	interval		1.10	1.08	1.10	1.11	1.14	1.14	1.12	1.13	1.17	1.48	PB		3.28	3.39	3.42
			velocity	5.12	8.31	8.46	8.31	8.23	8.02	8.02	8.16	8.09	7.81	9.47	7.72		8.36	8.09	8.02
Kawashima, Ryota (JPN) (1999)	time			2.69	3.82	4.94	6.05	7.16	8.29	9.41	10.52	11.67	12.83		14.33	9 / 6			
	reaction time	0.169	interval		1.13	1.12	1.11	1.11	1.13	1.12	1.11	1.15	1.16	1.50	PB		3.36	3.36	3.42
			velocity	5.10	8.09	8.16	8.23	8.23	8.09	8.16	8.23	7.95	7.88	9.35	7.68		8.16	8.16	8.02
Iwakiri, Yuki (JPN) (1998)	time			2.77	3.88	5.00	6.13	7.24	8.39	9.53	10.70	11.87	13.06		14.67	8 / 7			
	reaction time	0.160	interval		1.11	1.12	1.13	1.11	1.15	1.14	1.17	1.17	1.19	1.61			3.36	3.40	3.53
			velocity	4.95	8.23	8.16	8.09	8.23	7.95	8.02	7.81	7.81	7.68	8.71	7.50		8.16	8.06	7.77

2016 Spanish National Championships (Gijón, ESP) (TV Analysis)

FINAL

date	24-Jul-16	wind	1.0 m/s	Henson (2021) - Athlete First: 2016 year end hurdle report													
		H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
Ortega, Orlando (ESP) (1991)	time	2.50	3.56	4.57	5.56	6.54	7.56	8.56	9.56	10.56	11.60		13.09	4 / 1			
reaction time	0.161	interval	7 steps	1.06	1.01	0.99	0.98	1.02	1.00	1.00	1.04	1.49			3.06	3.00	3.04
velocity				5.49	8.62	9.05	9.23	9.33	8.96	9.14	9.14	8.79	9.41	8.40	8.96	9.14	9.02
Contreras, Yidiel (ESP) (1992)	time	2.44	3.56	4.57	5.58	6.60	7.62	8.66	9.71	10.80	11.90		13.45	5 / 2			
reaction time		interval	7 steps	1.12	1.01	1.01	1.02	1.04	1.05	1.09	1.10	1.55			3.14	3.08	3.24
velocity				5.62	8.16	9.05	9.05	8.96	8.96	8.79	8.70	8.39	8.31	9.05	8.73	8.90	8.46

2016 London Anniversary Games (London, GBR) (TV Analysis)

FINAL

date	23-Jul-16	wind	-1.0 m/s	Henson (2021) - Athlete First: 2016 year end hurdle report													
		H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
Bascou, Dimitri (FRA) (1987)	time	2.50	3.53	4.56	5.56	6.58	7.60	8.62	9.64	10.68	11.74		13.20	5 / 1			
reaction time	0.103	interval	7 steps	1.03	1.03	1.00	1.02	1.02	1.02	1.04	1.06	1.46			3.06	3.06	3.12
velocity				5.49	8.87	8.87	9.14	8.96	8.96	8.96	8.79	8.62	9.60	8.33	8.96	8.96	8.79
Traber, Gregor (GER) (1992)	time	2.53	3.58	4.60	5.63	6.65	7.68	8.72	9.80	10.86	11.95		13.45	3 / 2			
reaction time	0.150	interval	7 steps	1.05	1.02	1.03	1.02	1.03	1.04	1.08	1.06	1.09	1.50		3.10	3.09	3.23
velocity				5.42	8.70	8.96	8.87	8.96	8.87	8.79	8.46	8.62	8.39	9.35	8.85	8.87	8.49
Douvalidis, Konstadinos (GRE)	time	2.60	3.63	4.66	5.73	6.76	7.80	8.85	9.90	10.98	12.06		13.54	7 / 3			
reaction time	0.146	interval		1.03	1.03	1.07	1.03	1.04	1.05	1.05	1.08	1.08	1.48		3.13	3.12	3.21
velocity				5.28	8.87	8.87	8.54	8.87	8.79	8.70	8.70	8.46	9.47	8.12	8.76	8.79	8.54
Manga, Aurel (FRA) (1992)	time	2.46	3.52	4.58	5.62	6.63	7.74	8.82	9.88	11.02	12.08		13.59	9 / 4			
reaction time	0.155	interval	7 steps	1.06	1.06	1.04	1.01	1.11	1.08	1.06	1.14	1.06	1.51		3.16	3.20	3.26
velocity				5.58	8.62	8.62	8.79	9.05	8.23	8.46	8.62	8.02	9.28	8.09	8.68	8.57	8.41
Omorieg, David (GBR) (1995)	time	2.50	3.60	4.63	5.70	6.78	7.83	8.90	10.02	11.08	12.16		13.64	4 / 5			
reaction time	0.153	interval		1.10	1.03	1.07	1.08	1.05	1.07	1.12	1.06	1.08	1.48		3.20	3.20	3.26
velocity				5.49	8.31	8.87	8.54	8.46	8.70	8.54	8.16	8.62	9.47	8.06	8.57	8.57	8.41

2016 All-Star Night Athletics (Hiratsuka, JPN)

FINAL

date	23-Jul-16	wind	0.5 m/s	(2016.07.23) - https://twitter.com/touchdown_time/media?lang=en&lang=en&lang=en&lang=en													
		H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
Omuro, Hideki (JPN) (1990)	time	2.58	3.65	4.70	5.73	6.78	7.84	8.88	9.95	11.02	12.10		13.52	5 / 3			
reaction time	0.144	interval		1.07	1.05	1.03	1.05	1.06	1.04	1.07	1.07	1.08	1.42	PB	3.15	3.15	3.22
velocity				5.32	8.54	8.70	8.87	8.70	8.62	8.79	8.54	8.54	9.87	8.14	8.70	8.70	8.52
Tagaku, Takashi (JPN)	time	2.69	3.78	4.85	5.90	6.96	8.01	9.10	10.16	11.25	12.35		13.81	3 / 2			
reaction time	0.160	interval	7 steps	1.09	1.07	1.05	1.06	1.05	1.09	1.06	1.09	1.10	1.46		3.21	3.20	3.25
velocity				5.10	8.39	8.54	8.70	8.62	8.70	8.39	8.62	8.39	9.60	7.97	8.54	8.57	8.44

2016 IAAF World Junior Championships (Bydgoszcz, POL) (0.99cm)

FINAL

date	21-Jul-16	wind	0.2 m/s	(2016.07.24) - https://twitter.com/touchdown_time/media?lang=en&lang=en&lang=en&lang=en													
		H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
Krah, Marcus (USA) (1997)	0.145	no information available											PB	13.25	6 / 1		
Lattin, Amere (USA) (1997)	0.154	no information available											PB	13.30	4 / 2		
Furuya, Takumu (JPN) (1997)	time	2.62	3.66	4.68	5.69	6.70	7.72	8.74	9.76	10.81	11.86		13.31	7 / 3			
reaction time	0.152	interval		1.04	1.02	1.01	1.01	1.02	1.02	1.05	1.05	1.45	NJR		3.07	3.05	3.12
velocity				5.24	8.79	8.96	9.05	9.05	8.96	8.96	8.70	8.70	9.67	8.26	8.93	8.99	8.79
Russell, De'Jour (JAM) (2000)	0.172	no information available											13.39	5 / 4			
Żebrowski, Dawid (POL) (1997)	0.138	no information available											PB	13.45	3 / 5		

Nicholls, Machael (BAH) (1997) 0.145 no information available
 Weaver, James (GBR) (1997) 0.155 no information available
 Treston, Matthew (GBR) (1998) 0.132 no information available

13.45 2 / 6
 13.51 9 / 7
 13.55 8 / 8

2016 Six College Meeting (Sagamihara, JPN)

FINAL

		date	17-Jul-16	wind	0.3 m/s	(2016.07.18) - https://twitter.com/touchdown_time/media?lang=en&lang=en&lang=en&lang=en													
				H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
Okazaki, Masaki (JPN) (1994)	time			2.64	3.72	4.81	5.91	6.98	8.06	9.16	10.26	11.37	12.48		13.90	6 / 1			
	reaction time				1.08	1.09	1.10	1.07	1.08	1.10	1.10	1.11	1.11	1.42	PB		3.27	3.25	3.32
	interval			5.20	8.46	8.39	8.31	8.54	8.46	8.31	8.31	8.23	8.23	9.87	7.91		8.39	8.44	8.26
Fujii, Ryota (1996)	time			2.63	3.73	4.82	5.90	6.98	8.07	9.18	10.28	11.38	12.51		13.95	5 / 2			
	reaction time				1.10	1.09	1.08	1.08	1.09	1.11	1.10	1.10	1.13	1.44	PB		3.27	3.28	3.33
	interval			5.22	8.31	8.39	8.46	8.46	8.39	8.23	8.31	8.31	8.09	9.74	7.89		8.39	8.36	8.23

2016 Herculis Meeting International d'Athlétisme (Monaco, MON) (TV Analysis)

FINAL

		date	15-Jul-16	wind	Henson (2021) - Athlete First: 2016 year end hurdle report															
					H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
Ortega, Orlando (ESP) (1991)	time				2.48	3.52	4.54	5.54	6.54	7.54	8.56	9.56	10.56	11.60		13.04	3 / 1			
	reaction time	0.136	interval	7 steps	1.04	1.02	1.00	1.00	1.00	1.02	1.00	1.00	1.00	1.04	1.44	NR		3.06	3.02	3.04
			velocity		5.53	8.79	8.96	9.14	9.14	9.14	8.96	9.14	9.14	8.79	9.74	8.44		8.96	9.08	9.02
Bascou, Dimitri (FRA) (1987)	time				2.44	3.48	4.48	5.46	6.44	7.46	8.48	9.52	10.56	11.64		13.12	6 / 2			
	reaction time	0.136	interval	7 steps	1.04	1.00	0.98	0.98	1.02	1.02	1.04	1.04	1.04	1.08	1.48	PB		3.02	3.02	3.16
			velocity		5.62	8.79	9.14	9.33	9.33	8.96	8.96	8.79	8.79	8.46	9.47	8.38		9.08	9.08	8.68
Martinot-Lagarde, Pascal (FRA)	time				2.44	3.50	4.52	5.54	6.56	7.56	8.56	9.58	10.60	11.68		13.17	4 / 3			
	reaction time	0.132	interval	7 steps	1.06	1.02	1.02	1.02	1.00	1.00	1.02	1.02	1.02	1.08	1.49			3.10	3.02	3.12
			velocity		5.62	8.62	8.96	8.96	8.96	9.14	9.14	8.96	8.96	8.46	9.41	8.35		8.85	9.08	8.79
Belocian, Wilhem (FRA) (1995)	time				2.48	3.52	4.54	5.56	6.56	7.60	8.66	9.72	10.78	11.88		13.42	2 / 4			
	reaction time	0.129	interval			1.04	1.02	1.02	1.00	1.04	1.06	1.06	1.06	1.10	1.54			3.08	3.10	3.22
			velocity		5.53	8.79	8.96	8.96	9.14	8.79	8.62	8.62	8.62	8.31	9.10	8.20		8.90	8.85	8.52
Baji, Balázs (HUN) (1989)	time				2.48	3.54	4.56	5.58	6.60	7.60	8.64	9.68	10.72	11.86		13.45	8 / 5			
	reaction time	0.131	interval	7 steps	1.06	1.02	1.02	1.02	1.00	1.04	1.04	1.04	1.04	1.14	1.59			3.10	3.06	3.22
			velocity		5.53	8.62	8.96	8.96	8.96	9.14	8.79	8.79	8.79	8.02	8.82	8.18		8.85	8.96	8.52
Czykier, Damian (POL) (1992)	time				2.56	3.58	4.62	5.66	6.70	7.72	8.80	9.88	10.96	12.08		13.64	1 / 6			
	reaction time	0.157	interval	7 steps	1.02	1.04	1.04	1.04	1.02	1.08	1.08	1.08	1.08	1.12	1.56			3.10	3.14	3.28
			velocity		5.36	8.96	8.79	8.79	8.79	8.96	8.46	8.46	8.46	8.16	8.99	8.06		8.85	8.73	8.36
Alkana, Antonio (RSA) (1990)	time				2.48	3.54	4.60	5.64	6.68	7.76	8.80	9.88	11.04	12.12		13.66	7 / 7			
	reaction time	0.130	interval			1.06	1.06	1.04	1.04	1.08	1.04	1.08	1.16	1.08	1.54			3.16	3.16	3.32
			velocity		5.53	8.62	8.62	8.79	8.79	8.46	8.79	8.46	7.88	8.46	9.10	8.05		8.68	8.68	8.26
McLeod, Omar (JAM) (1994)	time				2.42	3.48	4.48	5.48	6.48	7.50	8.52	9.60	10.68	11.72		13.82	5 / 8			
	reaction time	0.128	interval			1.06	1.00	1.00	1.00	1.02	1.02	1.08	1.08	1.04	2.10			3.06	3.04	3.20
			velocity		5.67	8.62	9.14	9.14	9.14	8.96	8.96	8.46	8.46	8.79	6.68	7.96		8.96	9.02	8.57

2016 USA Olympic Trials (Eugene, OR) (TV Analysis)

FINAL

				Henson (2020) - Athlete First: 2016 year end hurdle report																	
		date	09-Jul-16	wind	1.0 m/s	H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
Allen, Devon (USA) (1994)	time			2.53	3.55	4.57	5.55	6.54	7.54	8.52	9.54	10.56	11.57				13.03	3 / 1			
	reaction time	0.157	interval		1.02	1.02	0.98	0.99	1.00	0.98	1.02	1.02	1.01	1.46			PB		3.02	2.97	3.05
			velocity		5.42	8.96	8.96	9.33	9.23	9.14	9.33	8.96	8.96	9.05	9.60		8.44		9.08	9.23	8.99
Ash, Ronnie (USA) (1988)	time			2.53	3.57	4.57	5.60	6.57	7.60	8.60	9.62	10.65	11.71				13.21	4 / 2			
	reaction time	0.203	interval	7 steps	1.04	1.00	1.03	0.97	1.03	1.00	1.02	1.03	1.06	1.50					3.07	3.00	3.11
			velocity		5.42	8.79	9.14	8.87	9.42	8.87	9.14	8.96	8.87	8.62	9.35		8.33		8.93	9.14	8.82
Porter, Jeff (USA) (1985)	time			2.50	3.53	4.53	5.55	6.57	7.60	8.62	9.67	10.69	11.73				13.21	6 / 3			
	reaction time	0.163	interval		1.03	1.00	1.02	1.02	1.03	1.02	1.05	1.02	1.04	1.48					3.05	3.07	3.11
			velocity		5.49	8.87	9.14	8.96	8.96	8.87	8.96	8.70	8.96	8.79	9.47		8.33		8.99	8.93	8.82
Merritt, Aries (USA) (1985)	time			2.47	3.49	4.49	5.49	6.47	7.47	8.49	9.54	10.60	11.66				13.22	1 / 4			
	reaction time	0.155	interval	7 steps	1.02	1.00	1.00	0.98	1.00	1.02	1.05	1.06	1.06	1.56					3.02	3.00	3.17
			velocity		5.55	8.96	9.14	9.14	9.33	9.14	8.96	8.70	8.62	8.62	8.99		8.32		9.08	9.14	8.65
Richardson, Jason (USA) (1986)	time			2.56	3.60	4.60	5.60	6.60	7.64	8.64	9.67	10.71	11.76				13.28	7 / 5			
	reaction time	0.212	interval	7 steps	1.04	1.00	1.00	1.00	1.04	1.00	1.03	1.04	1.05	1.52					3.04	3.04	3.12
			velocity		5.36	8.79	9.14	9.14	9.14	8.79	9.14	8.87	8.79	8.70	9.22		8.28		9.02	9.02	8.79
Eaton, Jarret (USA) (1989)	time			2.47	3.47	4.47	5.47	6.45	7.47	8.49	9.51	10.61	11.71				13.30	2 / 6			
	reaction time	0.196	interval	7 steps	1.00	1.00	1.00	0.98	1.02	1.02	1.02	1.10	1.10	1.59					3.00	3.02	3.22
			velocity		5.55	9.14	9.14	9.14	9.33	8.96	8.96	8.96	8.31	8.31	8.82		8.27		9.14	9.08	8.52

Harris, Aleec (USA) (1990)	time	2.53	3.60	4.64	5.67	6.70	7.74	8.81	9.84	10.91	12.00		13.56	8 / 7			
	reaction time	0.186	interval	1.07	1.04	1.03	1.03	1.04	1.07	1.03	1.07	1.09	1.56		3.14	3.14	3.19
	velocity	5.42	8.54	8.79	8.87	8.87	8.79	8.54	8.87	8.54	8.39	8.99	8.11		8.73	8.73	8.60

2016 European Championships (Amsterdam, NED)

FINAL

date 09-Jul-16		wind 0.0 m/s		(2016.07.09) - https://twitter.com/touchdown_time/media?lang=en&lang=en&lang=en&lang=en													
Bascou, Dimitri (FRA) (1987)	time	2.53	3.56	4.58	5.59	6.60	7.62	8.65	9.71	10.78	11.86		13.25	7 / 1			
	reaction time	0.141	interval	7 steps	1.03	1.02	1.01	1.01	1.02	1.03	1.06	1.07	1.08		3.06	3.06	3.21
	velocity	5.42	8.87	8.96	9.05	9.05	8.96	8.87	8.62	8.54	8.46	10.09	8.30		8.96	8.96	8.54

2016 Jamaican National Championships (Kingston, JAM) (TV Analysis)

FINAL

date 02-Jul-16		wind 0.4 m/s		Henson (2020) - Athlete First: 2016 year end hurdle report													
McLeod, Omar (JAM) (1994)	time	2.43	3.50	4.51	5.51	6.49	7.47	8.49	9.51	10.51	11.53		13.01	4 / 1			
	reaction time		interval	1.07	1.01	1.00	0.98	0.98	1.02	1.02	1.00	1.02	1.48		3.08	2.98	3.04
	velocity	5.65	8.54	9.05	9.14	9.33	9.33	8.96	8.96	9.14	8.96	9.47	8.46		8.90	9.20	9.02
Carter, Deuce (JAM) (1990)	time	2.46	3.53	4.51	5.49	6.49	7.47	8.49	9.54	10.58	11.64		13.21	5 / 2			
	reaction time		interval	7 steps	1.07	0.98	0.98	1.00	0.98	1.02	1.05	1.04	1.06	1.57	3.03	3.00	3.15
	velocity	5.58	8.54	9.33	9.33	9.14	9.33	8.96	8.70	8.79	8.62	8.93	8.33		9.05	9.14	8.70

2016 West Japan University Championships (Kyoto, JPN)

FINAL

date 01-Jul-16		wind -2.1 m/s		(2016.07.02) - https://twitter.com/touchdown_time/media?lang=en&lang=en&lang=en&lang=en													
Masashi, Kimoto (JPN)	time	2.66	3.74	4.84	5.93	7.01	8.10	9.20	10.31	11.44	12.58		14.01	4 / 1			
	reaction time		interval	1.08	1.10	1.09	1.08	1.09	1.10	1.11	1.13	1.14	1.43		3.27	3.27	3.38
	velocity	5.16	8.46	8.31	8.39	8.46	8.39	8.31	8.23	8.09	8.02	9.80	7.85		8.39	8.39	8.11
Yamamoto, Kenta (JPN) (1995)	time	2.66	3.78	4.86	5.97	7.07	8.17	9.26	10.38	11.50	12.63		14.08	6 / 2			
	reaction time		interval	1.12	1.08	1.11	1.10	1.10	1.09	1.12	1.12	1.13	1.45		3.31	3.29	3.37
	velocity	5.16	8.16	8.46	8.23	8.31	8.31	8.39	8.16	8.16	8.09	9.67	7.81		8.28	8.33	8.14

2016 Japanese National Championships (Nagoya, JPN)

FINAL

date 26-Jun-16		wind 2.4 m/s		Kota (2016) - race analysis of japanese top-ranked men 110m and women 100m hurdlers													
Yazawa, Wataru (JPN) (1991)	time	2.62	3.69	4.73	5.77	6.81	7.85	8.88	9.93	10.98	12.06		13.48	7 / 1			
	reaction time	0.138	interval	1.07	1.04	1.04	1.04	1.03	1.05	1.05	1.08	1.42			3.15	3.11	3.18
	velocity	5.24	8.54	8.79	8.79	8.79	8.79	8.87	8.70	8.70	8.46	9.87	8.16		8.70	8.82	8.62
Masuno, Genta (JPN) (1993)	time	2.64	3.70	4.75	5.79	6.82	7.86	8.92	9.96	11.03	12.09		13.51	5 / 2			
	reaction time	0.161	interval	1.06	1.05	1.04	1.03	1.04	1.06	1.04	1.07	1.06	1.42		3.15	3.13	3.17
	velocity	5.20	8.62	8.70	8.79	8.87	8.79	8.62	8.79	8.54	8.62	9.87	8.14		8.70	8.76	8.65
Kanai, Taio (JPN) (1995)	time	2.58	3.66	4.73	5.78	6.84	7.90	8.96	10.03	11.10	12.19		13.61	3 / 3			
	reaction time	0.136	interval	1.08	1.07	1.05	1.06	1.06	1.07	1.07	1.09	1.42			3.20	3.18	3.23
	velocity	5.32	8.46	8.54	8.70	8.62	8.62	8.62	8.54	8.54	8.39	9.87	8.08		8.57	8.62	8.49
Omoro, Hideki (JPN) (1990)	time	2.60	3.68	4.74	5.80	6.86	7.92	8.98	10.06	11.14	12.23		13.66	6 / 4			
	reaction time	0.144	interval	1.08	1.06	1.06	1.06	1.06	1.08	1.08	1.09	1.43			3.20	3.18	3.25
	velocity	5.28	8.46	8.62	8.62	8.62	8.62	8.62	8.46	8.46	8.39	9.80	8.05		8.57	8.62	8.44
Fudaba, Hiroki (JPN) (1994)	time	2.64	3.74	4.80	5.87	6.93	7.98	9.05	10.11	11.19	12.27		13.71	4 / 5			
	reaction time	0.140	interval	1.10	1.06	1.07	1.06	1.05	1.07	1.06	1.08	1.08	1.44		3.23	3.18	3.22
	velocity	5.20	8.31	8.62	8.54	8.62	8.70	8.54	8.62	8.46	8.46	9.74	8.02		8.49	8.62	8.52
Kagimoto, Masahiro (JPN) (1995)	time	2.64	3.74	4.81	5.89	6.97	8.02	9.11	10.18	11.25	12.36		13.78	1 / 6			
	reaction time	0.146	interval	1.10	1.07	1.08	1.08	1.05	1.09	1.07	1.07	1.11	1.42		3.25	3.22	3.25
	velocity	5.20	8.31	8.54	8.46	8.46	8.70	8.39	8.54	8.54	8.23	9.87	7.98		8.44	8.52	8.44

2016 Hokusei Shinetsu High School Meeting (Okayama, JPN) (99cm)

FINAL

date 18-Jun-16		wind -2.5 m/s		(2016.06.18) - https://twitter.com/touchdown_time/media?lang=en&lang=en&lang=en&lang=en													
Kato, Takuma (JPN) (1998)	time	2.68	3.76	4.88	5.98	7.10	8.24	9.40	10.56	11.73	12.89		14.39	7 / 1			
	reaction time		interval	1.08	1.12	1.10	1.12	1.14	1.16	1.16	1.17	1.16	1.50		3.30	3.42	3.49
	velocity	5.12	8.46	8.16	8.31	8.16	8.02	7.88	7.88	7.81	7.88	9.35	7.64		8.31	8.02	7.86

2016 Racers Grand Prix (Kingston, JAM) (TV Analysis)

FINAL

date 11-Jun-16		wind 0.2 m/s		Henson (2020) - Athlete First: 2016 year end hurdle report													
Oliver, David (USA) (1982)	time	2.43	3.46	4.46	5.46	6.46	7.43	8.46	9.50	10.50	11.56		13.09	4 / 1			
	reaction time		interval	7 steps	1.03	1.00	1.00	0.97	1.03	1.04	1.00	1.06	1.53		3.03	3.00	3.10
	velocity	5.65	8.87	9.14	9.14	9.14	9.42	8.87	8.79	9.14	8.62	9.16	8.40		9.05	9.14	8.85

Carter, Deuce (JAM) (1990)	time	2.46	3.50	4.50	5.50	6.48	7.48	8.50	9.56	10.58	11.66		13.20	2 / 2				
	reaction time	interval	7 steps	1.04	1.00	1.00	0.98	1.00	1.02	1.06	1.02	1.08	1.54	PB		3.04	3.00	3.16
	velocity		5.58	8.79	9.14	9.14	9.33	9.14	8.96	8.62	8.96	8.46	9.10	8.33		9.02	9.14	8.68
Eaton, Jarret (USA) (1989)	time	2.43	3.46	4.46	5.48	6.50	7.50	8.53	9.56	10.63	11.70		13.28	5 / 3				
	reaction time	interval	7 steps	1.03	1.00	1.02	1.02	1.00	1.03	1.03	1.07	1.07	1.58			3.05	3.05	3.17
	velocity		5.65	8.87	9.14	8.96	8.96	9.14	8.87	8.87	8.54	8.54	8.87	8.28		8.99	8.99	8.65

2016 Fuse Sprint (Tottori, JPN)**FINAL**

INAL		date	05-Jun-16	wind 1.4 m/s																Kota (2016) - race analysis of japanese top-ranked men 110m and women 100m hurdlers															
				H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10																
Yazawa, Wataru (JPN) (1991)	time			2.61	3.67	4.72	5.76	6.79	7.84	8.88	9.93	11.00	12.07		13.47	5 / 1																			
	reaction time		interval	7 steps	1.06	1.05	1.04	1.03	1.05	1.04	1.05	1.07	1.07	1.40	PB			3.15	3.12	3.19															
		velocity		5.26	8.62	8.70	8.79	8.87	8.70	8.79	8.70	8.54	8.54	10.01	8.17			8.70	8.79	8.60															
Masuno, Genta (JPN) (1993)	time			2.62	3.68	4.73	5.78	6.82	7.89	8.95	10.01	11.10	12.17		13.59	6 / 2																			
	reaction time		interval		1.06	1.05	1.05	1.04	1.07	1.06	1.06	1.09	1.07	1.42				3.16	3.17	3.22															
		velocity		5.24	8.62	8.70	8.70	8.79	8.54	8.62	8.62	8.39	8.54	9.87	8.09			8.68	8.65	8.52															
Omoro, Hideki (JPN) (1990)	time			2.62	3.68	4.72	5.77	6.81	7.86	8.95	10.03	11.13	12.21		13.65	7 / 3																			
	reaction time		interval		1.06	1.04	1.05	1.04	1.05	1.09	1.08	1.10	1.08	1.44				3.15	3.18	3.26															
		velocity		5.24	8.62	8.79	8.70	8.79	8.70	8.39	8.46	8.31	8.46	9.74	8.06			8.70	8.62	8.41															
Iwafane, Yoichi (JPN) (1985)	time			2.64	3.72	4.77	5.84	6.91	7.99	9.07	10.16	11.29	12.41		13.87	3 / 5																			
	reaction time		interval		1.08	1.05	1.07	1.07	1.08	1.08	1.09	1.13	1.12	1.46				3.20	3.23	3.34															
		velocity		5.20	8.46	8.70	8.54	8.54	8.46	8.46	8.39	8.09	8.16	9.60	7.93			8.57	8.49	8.21															
Fudaba, Hiroki (JPN) (1994)	time			2.67	3.75	4.83	5.89	6.96	8.05	9.14	10.23	11.33	12.46		13.96	8 / 8																			
	reaction time		interval		1.08	1.08	1.06	1.07	1.09	1.09	1.09	1.10	1.13	1.50				3.22	3.25	3.32															
		velocity		5.14	8.46	8.46	8.62	8.54	8.39	8.39	8.39	8.31	8.09	9.35	7.88			8.52	8.44	8.26															

Heat 3

Heat 3

	date	05-Jun-16	wind	1.6 m/s	(2016.06.05) - https://twitter.com/touchdown_time/media?lang=en&lang=en&lang=en&lang=en														
			H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10	
Yazawa, Wataru (JPN) (1991)	time		2.64	3.71	4.75	5.80	6.85	7.89	8.94	10.00	11.06	12.14		13.54	6 / 1				
reaction time	interval	7 steps	1.07	1.04	1.05	1.05	1.04	1.05	1.06	1.06	1.08	1.40		PB			3.16	3.14	3.20
	velocity		5.20	8.54	8.79	8.70	8.70	8.79	8.70	8.62	8.62	8.46	10.01	8.12			8.68	8.73	8.57

2016 Golden Gala Pietro Mennea (Roma, ITA) (TV Analysis)**FINAL**

INAL	date			02-Jun-16		wind		0.6 m/s		Henson (2020) - Athlete First: 2016 year end hurdle report												
				H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10			
Ortega, Orlando (ESP) (1991)	time			2.56	3.60	4.64	5.66	6.66	7.66	8.66	9.68	10.70	11.74		13.22	7 / 1						
	reaction time	0.136	interval	7 steps	1.04	1.04	1.02	1.00	1.00	1.00	1.02	1.02	1.04	1.48			3.10	3.00	3.08			
			velocity		5.36	8.79	8.79	8.96	9.14	9.14	9.14	8.96	8.96	8.79	9.47	8.32		8.85	9.14	8.90		
Martinot-Lagarde, Pascal (FRA)	time			2.50	3.54	4.56	5.58	6.58	7.58	8.62	9.68	10.70	11.78		13.29	5 / 2						
	reaction time	0.155	interval	7 steps	1.04	1.02	1.02	1.00	1.00	1.04	1.06	1.02	1.08	1.51			3.08	3.04	3.16			
			velocity		5.49	8.79	8.96	8.96	9.14	9.14	8.79	8.62	8.96	8.46	9.28	8.28		8.90	9.02	8.68		
Pozzi, Andy (GBR) (1992)	time			2.50	3.54	4.54	5.58	6.58	7.60	8.66	9.70	10.74	11.86		13.37	2 / 3						
	reaction time	0.139	interval	7 steps	1.04	1.00	1.04	1.00	1.02	1.06	1.04	1.04	1.12	1.51			3.08	3.08	3.20			
			velocity		5.49	8.79	9.14	8.79	9.14	8.96	8.62	8.79	8.79	8.16	9.28	8.23		8.90	8.90	8.57		
Brathwaite, Shane (BAR) (1990)	time			2.52	3.58	4.62	5.66	6.70	7.74	8.80	9.86	10.96	12.10		13.64	3 / 4						
	reaction time	0.133	interval	7 steps	1.06	1.04	1.04	1.04	1.04	1.06	1.06	1.10	1.14	1.54			3.14	3.14	3.30			
			velocity		5.44	8.62	8.79	8.79	8.79	8.79	8.62	8.62	8.31	8.02	9.10	8.06		8.73	8.73	8.31		
Belocian, Wilhem (FRA) (1995)	time			2.52	3.56	4.60	5.64	6.70	7.76	8.82	9.94	11.02	12.14		13.73	8 / 5						
	reaction time	0.132	interval		1.04	1.04	1.04	1.06	1.06	1.06	1.12	1.08	1.12	1.59			3.12	3.18	3.32			
			velocity		5.44	8.79	8.79	8.79	8.62	8.62	8.62	8.16	8.46	8.16	8.82	8.01		8.79	8.62	8.26		
Abate, Emanuele (ITA) (1985)	time			2.60	3.72	4.78	5.82	6.86	7.94	8.98	10.08	11.16	12.26		13.83	9 / 6						
	reaction time	0.149	interval	7 steps	1.12	1.06	1.04	1.04	1.08	1.04	1.10	1.08	1.10	1.57			3.22	3.16	3.28			
			velocity		5.28	8.16	8.62	8.79	8.79	8.46	8.79	8.31	8.46	8.31	8.93	7.95		8.52	8.68	8.36		
Perini, Lorenzo (ITA) (1994)	time			2.62	3.70	4.76	5.80	6.88	7.98	9.02	10.18	11.28	12.40		13.90	1 / 7						
	reaction time	0.151	interval		1.08	1.06	1.04	1.08	1.10	1.04	1.16	1.10	1.12	1.50			3.18	3.22	3.38			
			velocity		5.24	8.46	8.62	8.79	8.46	8.31	8.79	7.88	8.31	8.16	9.35	7.91		8.62	8.52	8.11		
Bascou, Dimitri (FRA) (1987)	time			2.48	3.58	4.66	5.70	6.72	7.74	8.80					dnf	4 / --						
	reaction time	0.149	interval	7 steps	1.10	1.08	1.04	1.02	1.02	1.06							3.22	3.10				
			velocity		5.53	8.31	8.46	8.79	8.96	8.96	8.62						8.52	8.85				
O'Farril, Yordan (CUB) (1993)	time			2.54	3.58	4.64	5.66	6.70	7.74	8.76					dnf	6 / --						
	reaction time	0.135	interval	7 steps	1.04	1.06	1.02	1.04	1.04	1.02							3.12	3.10				
			velocity		5.40	8.79	8.62	8.96	8.79	8.79	8.96						8.79	8.85				

2016 Miyazaki Prefecture High School Meeting (Miyazaki, JPN)

FINAL

		date	31-May-16	wind	-1.0 m/s	(2016.06.01) - https://twitter.com/touchdown_time/media?lang=en&lang=en&lang=en&lang=en													
				H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
Makoto, Ino (JPN)	time		2.74	3.89	5.06	6.24	7.41	8.58	9.75	10.91	12.09	13.29		14.80	5 / 1				
	reaction time	interval		1.15	1.17	1.18	1.17	1.17	1.17	1.16	1.18	1.20	1.51				3.50	3.51	3.54
		velocity		5.01	7.95	7.81	7.75	7.81	7.81	7.81	7.88	7.75	7.62	9.28	7.43		7.83	7.81	7.75

2016 Hiroshima Prefecture High School Meeting (Hiroshima, JPN)

FINAL

FINAL		date	29-May-16	wind	0.1 m/s (2016.05.30) - https://twitter.com/touchdown_time/media?lang=en&lang=en&lang=en&lang=en															
				H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10	
Toyama, Hiroki (JPN) (1999)	time		2.69	3.83	4.95	6.11	7.25	8.38	9.51	10.65	11.80	12.97			14.42	6 / 1				
	reaction time	interval	7 steps	1.14	1.12	1.16	1.14	1.13	1.13	1.14	1.15	1.17	1.45				3.42	3.40	3.46	
	velocity		5.10	8.02	8.16	7.88	8.02	8.09	8.09	8.02	7.95	7.81	9.67	7.63			8.02	8.06	7.92	

2016 Niigata Prefecture High School Meeting (Niigata, JPN)

FINAL

		date	29-May-16	wind	-0.6 m/s	(2016.05.30) - https://twitter.com/touchdown_time/media?lang=en&lang=en&lang=en&lang=en													
				H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
Kato, Takuma (JPN) (1998)	time		2.68	3.78	4.88	5.97	7.08	8.20	9.31	10.43	11.56	12.70			14.16	/ 1			
	reaction time	interval		1.10	1.10	1.09	1.11	1.12	1.11	1.12	1.13	1.14	1.46				3.29	3.34	3.39
	velocity		5.12	8.31	8.31	8.39	8.23	8.16	8.23	8.16	8.09	8.02	9.60		7.77		8.33	8.21	8.09
Imai, Junpei (JPN) (1998)	time		2.69	3.82	4.94	6.07	7.19	8.32	9.46	10.60	11.76	12.90			14.43	/ 2			
	reaction time	interval		1.13	1.12	1.13	1.12	1.13	1.14	1.14	1.16	1.14	1.53				3.38	3.39	3.44
	velocity		5.10	8.09	8.16	8.09	8.16	8.09	8.02	8.02	7.88	8.02	9.16		7.62		8.11	8.09	7.97

2016 Prefontaine Classic (Eugene, OR) (TV Analysis)

FINAL

		date	28-May-16	wind		0.7 m/s		Henson (2021) - Athlete First: 2016 year end hurdle report											
				H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
McLeod, Omar (JAM) (1994)	time		2.44	3.50	4.52	5.54	6.56	7.56	8.56	9.58	10.60	11.62		13.06	4 / 1				
	reaction time	interval		1.06	1.02	1.02	1.02	1.00	1.00	1.02	1.02	1.02	1.44				3.10	3.02	3.06
	velocity		5.62	8.62	8.96	8.96	8.96	9.14	9.14	8.96	8.96	8.96	9.74	8.42			8.85	9.08	8.96
Oliver, David (USA) (1982)	time		2.44	3.52	4.56	5.58	6.60	7.62	8.64	9.70	10.78	11.82		13.38	6 / 2				
	reaction time	interval	7 steps	1.08	1.04	1.02	1.02	1.02	1.02	1.06	1.08	1.04	1.56				3.14	3.06	3.18
	velocity		5.62	8.46	8.79	8.96	8.96	8.96	8.96	8.62	8.46	8.79	8.99	8.22			8.73	8.96	8.62
Porter, Jeff (USA) (1985)	time		2.52	3.58	4.62	5.64	6.68	7.74	8.78	9.82	10.88	11.96		13.48	1 / 3				
	reaction time	interval		1.06	1.04	1.02	1.04	1.06	1.04	1.04	1.06	1.08	1.52				3.12	3.14	3.18
	velocity		5.44	8.62	8.79	8.96	8.79	8.62	8.79	8.79	8.62	8.46	9.22	8.16			8.79	8.73	8.62
Merritt, Aries (USA) (1985)	time		2.46	3.52	4.54	5.56	6.60	7.62	8.64	9.68	10.74	11.86		13.51	3 / 4				
	reaction time	interval	7 steps	1.06	1.02	1.02	1.04	1.02	1.02	1.04	1.06	1.12	1.65				3.10	3.08	3.22
	velocity		5.58	8.62	8.96	8.96	8.79	8.96	8.96	8.79	8.62	8.16	8.50	8.14			8.85	8.90	8.52
Eaton, Jarret (USA) (1989)	time		2.44	3.48	4.52	5.54	6.56	7.60	8.66	9.72	10.82	11.94		13.52	7 / 5				
	reaction time	interval	7 steps	1.04	1.04	1.02	1.02	1.04	1.06	1.06	1.10	1.12	1.58				3.10	3.12	3.28
	velocity		5.62	8.79	8.79	8.96	8.96	8.79	8.62	8.62	8.31	8.16	8.87	8.14			8.85	8.79	8.36
Martinot-Lagarde, Pascal (FRA)	time		2.52	3.56	4.58	5.62	6.66	7.70	8.74	9.78	10.86	11.96		13.55	5 / 6				
	reaction time	interval	7 steps	1.04	1.02	1.04	1.04	1.04	1.04	1.04	1.08	1.10	1.59				3.10	3.12	3.22
	velocity		5.44	8.79	8.96	8.79	8.79	8.79	8.79	8.79	8.46	8.31	8.82	8.12			8.85	8.79	8.52
Cabral, Jonathan (CAN) (1992)	time		2.56	3.68	4.70	5.74	6.78	7.82	8.86	9.90	10.98	12.06		13.57	8 / 7				
	reaction time	interval	7 steps	1.12	1.02	1.04	1.04	1.04	1.04	1.04	1.08	1.08	1.51				3.18	3.12	3.20
	velocity		5.36	8.16	8.96	8.79	8.79	8.79	8.79	8.79	8.46	8.46	9.28	8.11			8.62	8.79	8.57
Parchment, Hansle (JAM) (1999)	time		2.52	3.58	4.60	5.60	6.60	7.60	8.60	9.64	10.72	11.88		13.58	2 / 8				
	reaction time	interval	7 steps	1.06	1.02	1.00	1.00	1.00	1.00	1.04	1.08	1.16	1.70				3.08	3.00	3.28
	velocity		5.44	8.62	8.96	9.14	9.14	9.14	9.14	8.79	8.46	7.88	8.25	8.10			8.90	9.14	8.36

2016 Kagoshima Prefecture High School Meeting (Kagoshima, JPN)

FINAL

INAL	date	28-May-16	wind	(2016.06.02) - https://twitter.com/touchdown_time/media?lang=en&lang=en&lang=en&lang=en																
			H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10		
Iwakiri, Yuki (JPN) (1998)	time		2.78	3.94	5.07	6.19	7.32	8.45	9.59	10.75	11.91	13.10		14.60	3 / 1					
	reaction time	interval		1.16	1.13	1.12	1.13	1.13	1.14	1.16	1.16	1.19	1.50				3.41	3.40	3.51	
	velocity		4.94	7.88	8.09	8.16	8.09	8.09	8.02	7.88	7.88	7.68	9.35	7.53			8.04	8.06	7.81	

2016 Meeting International Mohammed VI D'Athletisme (Rabat, MAR) (TV Analysis)

FINAL

		date	22-May-16	wind	1.4 m/s	Henson (2020) - Athlete First: 2016 year end hurdle report														
				H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10	
Oliver, David (USA) (1982)	time			2.46	3.48	4.48	5.50	6.50	7.52	8.54	9.54	10.58	11.62		13.12	7 / 1				
	reaction time	0.152	interval	7 steps	1.02	1.00	1.02	1.00	1.02	1.02	1.00	1.04	1.04	1.50			3.04	3.04	3.08	
			velocity		5.58	8.96	9.14	8.96	9.14	8.96	8.96	9.14	8.79	8.79	9.35	8.38		9.02	9.02	8.90
Ortega, Orlando (ESP) (1991)	time			2.50	3.54	4.54	5.52	6.52	7.54	8.56	9.60	10.62	11.66		13.13	6 / 2				

	velocity	5.44	8.62	8.79	8.96	8.96	8.96	8.96	8.79	8.79	8.62	9.35	8.25		8.79	8.96	8.73	
Bascou, Dimitri (FRA) (1987)	time	2.48	3.52	4.56	5.58	6.62	7.64	8.68	9.72	10.76	11.86		13.34	2 / 4				
	reaction time	0.141	interval	7 steps	1.04	1.04	1.02	1.04	1.02	1.04	1.04	1.10	1.48		3.10	3.10	3.18	
	velocity	5.53	8.79	8.79	8.96	8.79	8.96	8.79	8.79	8.79	8.31	9.47	8.25		8.85	8.85	8.62	
Oliver, David (USA) (1982)	time	2.48	3.54	4.56	5.60	6.64	7.66	8.70	9.76	10.84	11.90		13.38	7 / 5				
	reaction time	0.141	interval	7 steps	1.06	1.02	1.04	1.04	1.02	1.04	1.06	1.08	1.06	1.48		3.12	3.10	3.20
	velocity	5.53	8.62	8.96	8.79	8.79	8.96	8.79	8.62	8.46	8.62	9.47	8.22		8.79	8.85	8.57	
Richardson, Jason (USA) (1986)	time	2.54	3.60	4.62	5.64	6.66	7.70	8.74	9.78	10.84	11.92		13.44	9 / 6				
	reaction time	0.151	interval	7 steps	1.06	1.02	1.02	1.04	1.04	1.04	1.06	1.08	1.52		3.10	3.10	3.18	
	velocity	5.40	8.62	8.96	8.96	8.96	8.79	8.79	8.79	8.62	8.46	9.22	8.18		8.85	8.85	8.62	
Zhang Honglin (CHN) (1994)	time	2.54	3.60	4.64	5.68	6.74	7.82	8.90	10.02	11.10	12.22		13.79	1 / 7				
	reaction time	0.142	interval		1.06	1.04	1.06	1.08	1.08	1.12	1.08	1.12	1.57		3.14	3.22	3.32	
	velocity	5.40	8.62	8.79	8.79	8.62	8.46	8.46	8.16	8.46	8.16	8.93	7.98		8.73	8.52	8.26	

2016 Kansai University Meeting (Osaka, JPN)

FINAL

date 13-May-16 wind 0.5 m/s

(2016.05.30) - https://twitter.com/touchdown_time/media?lang=en&lang=en&lang=en&lang=en

		H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
Kagimoto, Masahiro (JPN) (1996)	time	2.64	3.72	4.80	5.85	6.92	7.98	9.04	10.11	11.20	12.30		13.72	5 / 1			
	reaction time		interval	1.08	1.08	1.05	1.07	1.06	1.06	1.07	1.09	1.10	1.42		3.21	3.19	3.26
	velocity	5.20	8.46	8.46	8.70	8.54	8.62	8.62	8.54	8.39	8.31	9.87	8.02		8.54	8.60	8.41

2016 Seiko Golden Grand Prix (Kawasaki, JPN)

FINAL

date 08-May-16 wind -0.6 m/s

Kota (2016) - race analysis of japanese top-ranked men 110m and women 100m hurdlers

		H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
Xie Wenjun (CHN) (1990)	time	2.58	3.63	4.66	5.68	6.70	7.72	8.76	9.79	10.85	11.93		13.36	5 / 1			
	reaction time	0.160	interval	1.05	1.03	1.02	1.02	1.04	1.03	1.06	1.08	1.43			3.10	3.08	3.17
	velocity	5.32	8.70	8.87	8.96	8.96	8.96	8.79	8.87	8.62	8.46	9.80	8.23		8.85	8.90	8.65
Zhang Honglin (CHN) (1994)	time	2.69	3.72	4.76	5.78	6.82	7.84	8.90	9.94	11.02	12.12		13.59	8 / 2			
	reaction time	0.162	interval	1.03	1.04	1.02	1.04	1.02	1.06	1.04	1.08	1.10	1.47		3.09	3.12	3.22
	velocity	5.10	8.87	8.79	8.96	8.79	8.96	8.62	8.79	8.46	8.31	9.54	8.09		8.87	8.79	8.52
Omoro, Hideki (JPN) (1990)	time	2.58	3.64	4.67	5.72	6.78	7.84	8.92	10.00	11.09	12.19		13.61	4 / 3			
	reaction time	0.123	interval	1.06	1.03	1.05	1.06	1.06	1.08	1.08	1.09	1.10	1.42		3.14	3.20	3.27
	velocity	5.32	8.62	8.87	8.70	8.62	8.62	8.46	8.46	8.39	8.31	9.87	8.08		8.73	8.57	8.39
Masuno, Genta (JPN) (1993)	time	2.63	3.71	4.76	5.80	6.87	7.91	8.97	10.05	11.13	12.22		13.67	3 / 4			
	reaction time	0.148	interval	1.08	1.05	1.04	1.07	1.04	1.06	1.08	1.08	1.09	1.45		3.17	3.17	3.25
	velocity	5.22	8.46	8.70	8.79	8.54	8.79	8.62	8.46	8.46	8.39	9.67	8.05		8.65	8.65	8.44
de Oliveira, João Vítor (BRA) (1991)	time	2.64	3.72	4.78	5.81	6.87	7.92	8.99	10.07	11.17	12.30		13.81	7 / 5			
	reaction time	0.128	interval	1.08	1.06	1.03	1.06	1.05	1.07	1.08	1.10	1.13	1.51		3.17	3.18	3.31
	velocity	5.20	8.46	8.62	8.87	8.62	8.70	8.54	8.46	8.31	8.09	9.28	7.97		8.65	8.62	8.28
Furuya, Takumu (JPN) (1997)	time	2.64	3.75	4.82	5.89	6.94	8.02	9.10	10.21	11.29	12.40		13.85	2 / 6			
	reaction time	0.149	interval	1.11	1.07	1.07	1.05	1.08	1.08	1.11	1.08	1.11	1.45		3.25	3.21	3.30
	velocity	5.20	8.23	8.54	8.54	8.70	8.46	8.46	8.23	8.46	8.23	9.67	7.94		8.44	8.54	8.31
Swift, Greggmar (BAR) (1991)	time	2.67	3.82	4.95	6.11	7.24	8.41	9.56	10.75	11.94	13.14		14.79	6 / 7			
	reaction time	0.153	interval	1.15	1.13	1.16	1.13	1.17	1.15	1.19	1.19	1.20	1.65		3.44	3.45	3.58
	velocity	5.14	7.95	8.09	7.88	8.09	7.81	7.95	7.68	7.68	7.62	8.50	7.44		7.97	7.95	7.66

2016 Kinan Memorial (Osaka, JPN)

FINAL

date 08-May-16 wind -0.2 m/s

(2016.05.17) - https://twitter.com/touchdown_time/media?lang=en&lang=en&lang=en&lang=en

		H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
Yazawa, Wataru (JPN) (1991)	time	2.66	3.74	4.80	5.85	6.91	7.97	9.02	10.08	11.14	12.22		13.62	6 / 1			
	reaction time		interval	7 steps	1.08	1.06	1.05	1.06	1.05	1.06	1.06	1.08	1.40		3.19	3.17	3.20
	velocity	5.16	8.46	8.62	8.70	8.62	8.62	8.70	8.62	8.62	8.46	10.01	8.08		8.60	8.65	8.57

2016 Doha Diamond League (Doha, QAT) (TV Analysis)

FINAL

date 06-May-16 wind 1.4 m/s

Henson (2020) - Athlete First: 2016 year end hurdle report

				Henson (2020) - Athlete's first 2010 year and made report															
				H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
McLeod, Omar (JAM) (1994)	time			2.50	3.54	4.54	5.54	6.52	7.50	8.50	9.54	10.56	11.62		13.05	4 / 1			
	reaction time	0.131	interval		1.04	1.00	1.00	0.98	0.98	1.00	1.04	1.02	1.06	1.43			3.04	2.96	3.12
	velocity			5.49	8.79	9.14	9.14	9.33	9.33	9.14	8.79	8.96	8.62	9.80	8.43		9.02	9.26	8.79
Parchment, Hansle (JAM) (1996)	time			2.56	3.62	4.64	5.64	6.64	7.64	8.64	9.64	10.66	11.70		13.10	7 / 2			
	reaction time	0.127	interval	7 steps	1.06	1.02	1.00	1.00	1.00	1.00	1.00	1.02	1.04	1.40			3.08	3.00	3.06
	velocity			5.36	8.62	8.96	9.14	9.14	9.14	9.14	9.14	8.96	8.79	10.01	8.40		8.90	9.14	8.96
Ortega, Orlando (ESP) (1991)	time			2.56	3.58	4.60	5.60	6.60	7.60	8.60	9.64	10.68	11.72		13.12	6 / 3			
	reaction time	0.118	interval	7 steps	1.02	1.02	1.00	1.00	1.00	1.00	1.04	1.04	1.04	1.40	NR		3.04	3.00	3.12

	velocity	5.36	8.96	8.96	9.14	9.14	9.14	9.14	8.79	8.79	8.79	10.01	8.38		9.02	9.14	8.79
Oliver, David (USA) (1982)	time	2.52	3.56	4.56	5.56	6.58	7.60	8.62	9.64	10.68	11.76		13.16	3 / 4			
reaction time	0.140	interval	7 steps	1.04	1.00	1.00	1.02	1.02	1.02	1.04	1.08	1.40			3.04	3.06	3.14
	velocity	5.44	8.79	9.14	9.14	8.96	8.96	8.96	8.96	8.79	8.46	10.01	8.36		9.02	8.96	8.73
Bascou, Dimitri (FRA) (1987)	time	2.56	3.60	4.64	5.64	6.64	7.68	8.72	9.76	10.82	11.90		13.33	2 / 5			
reaction time	0.153	interval	7 steps	1.04	1.04	1.00	1.00	1.04	1.04	1.04	1.06	1.08	1.43		3.08	3.08	3.18
	velocity	5.36	8.79	8.79	9.14	9.14	8.79	8.79	8.79	8.62	8.46	9.80	8.25		8.90	8.90	8.62
Merritt, Aries (USA) (1985)	time	2.52	3.60	4.64	5.68	6.70	7.72	8.76	9.80	10.84	11.92		13.37	5 / 6			
reaction time	0.125	interval	7 steps	1.08	1.04	1.04	1.02	1.02	1.04	1.04	1.08	1.45			3.16	3.08	3.16
	velocity	5.44	8.46	8.79	8.79	8.96	8.96	8.79	8.79	8.79	8.46	9.67	8.23		8.68	8.90	8.68
Adams, Spencer (USA) (1989)	time	2.52	3.56	4.60	5.64	6.68	7.72	8.76	9.80	10.88	11.96		13.44	8 / 7			
reaction time	0.124	interval	7 steps	1.04	1.04	1.04	1.04	1.04	1.04	1.08	1.08	1.48			3.12	3.12	3.20
	velocity	5.44	8.79	8.79	8.79	8.79	8.79	8.79	8.79	8.46	8.46	9.47	8.18		8.79	8.79	8.57
Alkana, Antonio (RSA) (1990)	time	2.56	3.60	4.66	5.72	6.74	7.78	8.82	9.88	10.94	12.04		13.50	1 / 8			
reaction time	0.139	interval		1.04	1.06	1.06	1.02	1.04	1.04	1.06	1.06	1.10	1.46		3.16	3.10	3.22
	velocity	5.36	8.79	8.62	8.62	8.96	8.79	8.79	8.62	8.62	8.31	9.60	8.15		8.68	8.85	8.52

2016 Mito Invitational (Mito, JPN)

FINAL

INAL	date	05-May-16	(2016.05.27) - https://twitter.com/touchdown_time/media?lang=en&lang=en&lang=en&lang=en															
			H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
Sato, Hiroyuki (JPN) (1990)	time	2.58	3.67	4.73	5.79	6.87	7.94	9.02	10.12	11.23	12.35		13.78	5 / 1				
	reaction time	interval		1.09	1.06	1.06	1.08	1.07	1.08	1.10	1.11	1.12	1.43			3.21	3.23	3.33
	velocity	5.32	8.39	8.62	8.62	8.46	8.54	8.46	8.31	8.23	8.16	9.80	7.98			8.54	8.49	8.23
Kagimoto, Masahiro (JPN) (1995)	time	2.66	3.74	4.81	5.89	6.96	8.04	9.12	10.22	11.32	12.45		13.86	4 / 2				
	reaction time	interval		1.08	1.07	1.08	1.07	1.08	1.08	1.10	1.10	1.13	1.41			3.23	3.23	3.33
	velocity	5.16	8.46	8.54	8.46	8.54	8.46	8.46	8.31	8.31	8.09	9.94	7.94			8.49	8.49	8.23

2016 Mikio Oda Memorial (Hiroshima, JPN)

FINAL

		date	29-Apr-16	wind -0.5 m/s			Kota (2016) - race analysis of japanese top-ranked men 110m and women 100m hurdlers												
				H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
Yazawa, Wataru (JPN) (1991)	time			2.69	3.78	4.84	5.89	6.94	8.00	9.06	10.13	11.22	12.33		13.78	5 / 1			
	reaction time				1.09	1.06	1.05	1.05	1.06	1.06	1.07	1.09	1.11	1.45			3.20	3.17	3.27
	velocity			5.10	8.39	8.62	8.70	8.70	8.62	8.62	8.54	8.39	8.23	9.67	7.98		8.57	8.65	8.39
Masuno, Genta (JPN) (1993)	time			2.66	3.72	4.78	5.83	6.90	7.99	9.06	10.16	11.27	12.39		13.86	3 / 2			
	reaction time				1.06	1.06	1.05	1.07	1.09	1.07	1.10	1.11	1.12	1.47			3.17	3.23	3.33
	velocity			5.16	8.62	8.62	8.70	8.54	8.39	8.54	8.31	8.23	8.16	9.54	7.94		8.65	8.49	8.23
Takayama, Shunya (JPN) (1994)	time			2.61	3.68	4.76	5.82	6.91	7.98	9.06	10.16	11.26	12.39		13.89	2 / 3			
	reaction time				1.07	1.08	1.06	1.09	1.07	1.08	1.10	1.10	1.13	1.50			3.21	3.24	3.33
	velocity			5.26	8.54	8.46	8.62	8.39	8.54	8.46	8.31	8.31	8.09	9.35	7.92		8.54	8.46	8.23
Furuya, Takumu (JPN) (1997)	time			2.67	3.75	4.84	5.90	6.96	8.03	9.12	10.21	11.32	12.44		13.92	7 / 4			
	reaction time				1.08	1.09	1.06	1.06	1.07	1.09	1.09	1.11	1.12	1.48			3.23	3.22	3.32
	velocity			5.14	8.46	8.39	8.62	8.62	8.54	8.39	8.39	8.23	8.16	9.47	7.90		8.49	8.52	8.26
Omoro, Hideki (JPN) (1990)	time			2.63	3.69	4.73	5.80	6.85	7.95	9.02	10.11	11.25	12.39		13.98	4 / 6			
	reaction time				1.06	1.04	1.07	1.05	1.10	1.07	1.09	1.14	1.14	1.59			3.17	3.22	3.37
	velocity			5.22	8.62	8.79	8.54	8.70	8.31	8.54	8.39	8.02	8.02	8.82	7.87		8.65	8.52	8.14

2015 Japanese National Junior Championships (Nagoya, JPN) (99cr)

FINAL

INAL	date	16-Oct-15	wind 0.1 m/s			Takashima (2015) - race analysis of 110m & 100m hurdles in Japan- results of main 2015 competitions													
			H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10	
Furuya, Takumu (JPN) (1997)	time		2.67	3.76	4.83	5.90	6.97	8.05	9.13	10.21	11.31	12.40		13.86	3 / 1				
	reaction time	interval		1.09	1.07	1.07	1.07	1.08	1.08	1.08	1.10	1.09	1.46				3.23	3.23	3.27
	velocity		5.14	8.39	8.54	8.54	8.54	8.46	8.46	8.46	8.31	8.39	9.60	7.94			8.49	8.49	8.39
Kanai, Nao (JPN) (1997)	time		2.65	3.74	4.81	5.89	6.97	8.04	9.12	10.22	11.32	12.45		13.91	5 / 2				
	reaction time	interval		1.09	1.07	1.08	1.08	1.07	1.08	1.10	1.10	1.13	1.46				3.24	3.23	3.33
	velocity		5.18	8.39	8.54	8.46	8.46	8.54	8.46	8.31	8.31	8.09	9.60	7.91			8.46	8.49	8.23
Kuriki, Anthony Tyrell (JPN) (1991)	time		2.67	3.79	4.89	5.98	7.05	8.14	9.25	10.35	11.46	12.58		14.07	4 / 3				
	reaction time	interval		1.12	1.10	1.09	1.07	1.09	1.11	1.10	1.11	1.12	1.49	PB			3.31	3.27	3.33
	velocity		5.14	8.16	8.31	8.39	8.54	8.39	8.23	8.31	8.23	8.16	9.41	7.82			8.28	8.39	8.23

2015 IAAF World Championships (Beijing, CHN) (TV Analysis)

FINAL

INAL	date	28-Aug-15	wind	0.2 m/s	Henson (2021) - Athlete First: major championships report															
					H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
Shubenkov, Sergey (RUS) (1991)		time			2.52	3.54	4.56	5.54	6.52	7.52	8.52	9.52	10.54	11.58		12.98	7 / 1			
	reaction time	0.139	interval	7 steps	1.02	1.02	0.98	0.98	1.00	1.00	1.00	1.02	1.04	1.40	NR			3.02	2.98	3.06

	velocity	5.44	8.96	8.96	9.33	9.33	9.14	9.14	9.14	8.96	8.79	10.01	8.47		9.08	9.20	8.96
Parchment, Hansle (JAM) (1990)	time	2.54	3.58	4.60	5.58	6.58	7.56	8.56	9.56	10.56	11.62		13.03	8 / 2			
	reaction time	0.143	interval	7 steps	1.04	1.02	0.98	1.00	0.98	1.00	1.00	1.06	1.41		3.04	2.98	3.06
	velocity	5.40	8.79	8.96	9.33	9.14	9.33	9.14	9.14	9.14	8.62	9.94	8.44		9.02	9.20	8.96
Merritt, Aries (USA) (1985)	time	2.48	3.52	4.52	5.52	6.52	7.52	8.52	9.54	10.58	11.62		13.04	4 / 3			
	reaction time	0.141	interval	7 steps	1.04	1.00	1.00	1.00	1.00	1.02	1.04	1.04	1.42		3.04	3.00	3.10
	velocity	5.53	8.79	9.14	9.14	9.14	9.14	9.14	8.96	8.79	8.79	9.87	8.44		9.02	9.14	8.85
Martinot-Lagarde, Pascal (FRA)	time	2.48	3.52	4.56	5.56	6.56	7.56	8.56	9.60	10.64	11.72		13.17	2 / 4			
	reaction time	0.150	interval	7 steps	1.04	1.04	1.00	1.00	1.00	1.04	1.04	1.08	1.45		3.08	3.00	3.16
	velocity	5.53	8.79	8.79	9.14	9.14	9.14	9.14	8.79	8.79	8.46	9.67	8.35		8.90	9.14	8.68
Bascou, Dimitri (FRA) (1987)	time	2.48	3.50	4.52	5.56	6.56	7.58	8.60	9.64	10.68	11.72		13.17	5 / 5			
	reaction time	0.137	interval	7 steps	1.02	1.02	1.04	1.00	1.02	1.04	1.04	1.04	1.45		3.08	3.04	3.12
	velocity	5.53	8.96	8.96	8.79	9.14	8.96	8.96	8.79	8.79	8.79	9.67	8.35		8.90	9.02	8.79
McLeod, Omar (JAM) (1994)	time	2.44	3.46	4.48	5.52	6.54	7.58	8.62	9.66	10.72	11.78		13.18	6 / 6			
	reaction time	0.135	interval		1.02	1.02	1.04	1.02	1.04	1.04	1.06	1.06	1.40		3.08	3.10	3.16
	velocity	5.62	8.96	8.96	8.79	8.96	8.79	8.79	8.79	8.62	8.62	10.01	8.35		8.90	8.85	8.68
Oliver, David (USA) (1982)	time	2.44	3.54	4.58	5.62	6.64	7.66	8.70	9.74	10.78	11.86		13.33	9 / 7			
	reaction time	0.157	interval	7 steps	1.10	1.04	1.04	1.02	1.02	1.04	1.04	1.08	1.47		3.18	3.08	3.16
	velocity	5.62	8.31	8.79	8.79	8.96	8.96	8.79	8.79	8.79	8.46	9.54	8.25		8.62	8.90	8.68
Darien, Garfield (FRA) (1987)	time	2.52	3.58	4.64	5.66	6.68	7.68	8.72	9.76	10.80	11.86		13.34	3 / 8			
	reaction time	0.178	interval	7 steps	1.06	1.06	1.02	1.02	1.04	1.04	1.06	1.48			3.14	3.06	3.14
	velocity	5.44	8.62	8.62	8.96	8.96	9.14	8.79	8.79	8.79	8.62	9.47	8.25		8.73	8.96	8.73

Semi-Final 2

date		27-Aug-15		wind		-0.2 m/s		Xu (2015) - the 15th world track and field special report: Men's and women's hurdle project										
		H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10	
163	time			4.70				8.83			11.99		13.39	7 / 6				
	interval							4.13			3.16	1.40				8.83	3.16	
	velocity			6.81				8.85			8.68	10.01	8.22			3.11	8.68	

2015 Japanese National High School Championships (Wakayama, JPN) (99cm)

FINAL

date		02-Aug-15	wind		Enomoto (2015) - 68th high school championships: JAF scientific committee - biomechanics data collection													
			H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
Kanae, Nao (JPN) (1997)	time		2.65	3.72	4.80	5.86	6.91	7.99	9.05	10.14	11.22	12.37		13.85	4 / 1			
	interval			1.07	1.08	1.06	1.05	1.08	1.06	1.09	1.08	1.15	1.48	PB		3.21	3.19	3.32
	velocity		5.18	8.54	8.46	8.62	8.70	8.46	8.62	8.39	8.46	7.95	9.47	7.94		8.54	8.60	8.26
Taue, Shun (JPN) (1997)	time		2.72	3.83	4.94	6.01	7.11	8.21	9.30	10.40	11.52	12.65		14.10	6 / 2			
	interval			1.11	1.11	1.07	1.10	1.10	1.09	1.10	1.12	1.13	1.45	=PB		3.29	3.29	3.35
	velocity		5.04	8.23	8.23	8.54	8.31	8.31	8.39	8.31	8.16	8.09	9.67	7.80		8.33	8.33	8.19
Hiramatsu, Papdamba (JPN) (1997)	time		2.64	3.73	4.82	5.91	6.99	8.09	9.22	10.36	11.49	12.64		14.19	5 / 3			
	interval			1.09	1.09	1.09	1.08	1.10	1.13	1.14	1.13	1.15	1.55	PB		3.27	3.31	3.42
	velocity		5.20	8.39	8.39	8.39	8.46	8.31	8.09	8.02	8.09	7.95	9.05	7.75		8.39	8.28	8.02
Igarashi, Rai (JPN) (1997)	time		2.67	3.81	4.91	6.04	7.15	8.29	9.41	10.56	11.71	12.87		14.36	2 / 4			
	interval			1.14	1.10	1.13	1.11	1.14	1.12	1.15	1.15	1.16	1.49	PB		3.37	3.37	3.46
	velocity		5.14	8.02	8.31	8.09	8.23	8.02	8.16	7.95	7.95	7.88	9.41	7.66		8.14	8.14	7.92
Toyoda, Masaki (JPN) (1998)	time		2.72	3.87	5.01	6.13	7.24	8.35	9.49	10.64	11.78	12.93		14.38	8 / 5			
	interval			1.15	1.14	1.12	1.11	1.11	1.14	1.15	1.14	1.15	1.45			3.41	3.36	3.44
	velocity		5.04	7.95	8.02	8.16	8.23	8.23	8.02	7.95	8.02	7.95	9.67	7.65		8.04	8.16	7.97
Kato, Takuma (JPN) (1998)	time		2.69	3.83	4.96	6.08	7.22	8.35	9.48	10.64	11.80	12.97		14.50	3 / 6			
	interval			1.14	1.13	1.12	1.14	1.13	1.13	1.16	1.16	1.17	1.53			3.39	3.40	3.49
	velocity		5.10	8.02	8.09	8.16	8.02	8.09	8.09	7.88	7.88	7.81	9.16	7.59		8.09	8.06	7.86
Takahashi, Rui (JPN) (1997)	time		2.73	3.86	5.00	6.13	7.28	8.46	9.57	10.75	11.93	13.04		14.57	7 / 7			
	interval			1.13	1.14	1.13	1.15	1.18	1.11	1.18	1.18	1.11	1.53			3.40	3.44	3.47
	velocity		5.03	8.09	8.02	8.09	7.95	7.75	8.23	7.75	7.75	8.23	9.16	7.55		8.06	7.97	7.90
Tomiyama, Hiroki (JPN) (1998)	time		2.75	3.88	5.01	6.15	7.29	8.42	9.60	10.76	11.94	13.13		14.69	9 / 8			
	interval			1.13	1.13	1.14	1.14	1.13	1.18	1.16	1.18	1.19	1.56			3.40	3.45	3.53
	velocity		4.99	8.09	8.09	8.02	8.02	8.09	7.75	7.88	7.75	7.68	8.99	7.49		8.06	7.95	7.77

2015 Italian National Championships (Torino, ITA)

FINAL

date		25-Jul-15		wind		1.8 m/s		Incalza (FIDAL - Centro Studi & Ricerche) (2015) - https://www.youtube.com/watch?v=Ytpg1XHNpfg										
		H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10	
1992)	time	2.66	3.74	4.79	5.82	6.85	7.91	8.96	9.99	11.09	12.17		13.59	6 / 1				
	interval		1.08	1.05	1.03	1.03	1.06	1.05	1.03	1.10	1.08	1.42			3.16	3.14	3.21	

velocity	5.16	8.46	8.70	8.87	8.87	8.62	8.70	8.87	8.31	8.46	9.87	8.09		8.68	8.73	8.54
Perini, Lorenzo (ITA) (1994) time	2.69	3.75	4.81	5.88	6.94	7.98	9.05	10.13	11.23	12.35		13.77	4 / 2			
reaction time	interval	1.06	1.06	1.07	1.06	1.04	1.07	1.08	1.10	1.12	1.42			3.19	3.17	3.30
velocity	5.10	8.62	8.62	8.54	8.62	8.79	8.54	8.46	8.31	8.16	9.87	7.99		8.60	8.65	8.31

2015 Pan American Games (Toronto, CAN)

FINAL

date 24-Jul-15 wind 0.8 m/s

(2016.01.29) - https://twitter.com/touchdown_time/media?lang=en&lang=en&lang=en&lang=en

		H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
Thomas, Mikel (TTO) (1987) time		2.57	3.62	4.63	5.64	6.65	7.66	8.69	9.71	10.75	11.83		13.17	8 / 2			
reaction time	0.160	interval	7 steps	1.05	1.01	1.01	1.01	1.03	1.02	1.04	1.08	1.34	NR		3.07	3.05	3.14
velocity		5.34	8.70	9.05	9.05	9.05	9.05	8.87	8.96	8.79	8.46	10.46	8.35		8.93	8.99	8.73

2015 Japanese National Championships (Niigata, JPN)

FINAL

date 28-Jun-15 wind -1.4 m/s

Takashima (2015) - race analysis of 110m & 100m hurdles in Japan- results of main 2015 competitions

		H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
Takayama, Shunya (JPN) (1994) time		2.58	3.64	4.69	5.76	6.84	7.91	8.98	10.08	11.18	12.30		13.81	4 / 1			
reaction time	0.135	interval	1.06	1.05	1.07	1.08	1.07	1.07	1.10	1.10	1.12	1.51			3.18	3.22	3.32
velocity		5.32	8.62	8.70	8.54	8.46	8.54	8.54	8.31	8.31	8.16	9.28	7.97		8.62	8.52	8.26
Furuya, Takumu (JPN) (1997) time		2.62	3.72	4.79	5.86	6.92	7.98	9.07	10.17	11.27	12.36		13.81	5 / 2			
reaction time	0.132	interval	1.10	1.07	1.07	1.06	1.06	1.09	1.10	1.10	1.09	1.45			3.24	3.21	3.29
velocity		5.24	8.31	8.54	8.54	8.62	8.62	8.39	8.31	8.31	8.39	9.67	7.97		8.46	8.54	8.33
Masuno, Genta (JPN) (1993) time		2.63	3.70	4.77	5.82	6.88	7.97	9.05	10.13	11.22	12.35		13.85	7 / 3			
reaction time	0.142	interval	1.07	1.07	1.05	1.06	1.09	1.08	1.08	1.09	1.13	1.50			3.19	3.23	3.30
velocity		5.22	8.54	8.54	8.70	8.62	8.39	8.46	8.46	8.39	8.09	9.35	7.94		8.60	8.49	8.31
Notoya, Yuta (JPN) (1989) time		2.67	3.73	4.79	5.85	6.92	7.99	9.08	10.16	11.27	12.37		13.87	6 / 4			
reaction time	0.136	interval	7 steps	1.06	1.06	1.06	1.07	1.09	1.08	1.11	1.10	1.50			3.18	3.23	3.29
velocity		5.14	8.62	8.62	8.62	8.54	8.54	8.39	8.46	8.23	8.31	9.35	7.93		8.62	8.49	8.33
Wado, Tatsuya (JPN) (1990) time		2.57	3.65	4.71	5.78	6.85	7.93	9.03	10.11	11.23	12.37		13.90	3 / 5			
reaction time	0.121	interval	1.08	1.06	1.07	1.07	1.08	1.10	1.08	1.12	1.14	1.53			3.21	3.25	3.34
velocity		5.34	8.46	8.62	8.54	8.54	8.46	8.31	8.46	8.16	8.02	9.16	7.91		8.54	8.44	8.21
Omuro, Hideki (JPN) (1990) time		2.62	3.70	4.77	5.84	6.94	8.01	9.10	10.21	11.31	12.42		13.90	8 / 6			
reaction time	0.133	interval	1.08	1.07	1.07	1.10	1.07	1.09	1.11	1.10	1.11	1.48			3.22	3.26	3.32
velocity		5.24	8.46	8.54	8.54	8.31	8.54	8.39	8.23	8.31	8.23	9.47	7.91		8.52	8.41	8.26
Nishizawa, Masanori (JPN) (1991) time		2.66	3.76	4.85	5.94	7.02	8.10	9.19	10.29	11.40	12.51		13.93	2 / 7			
reaction time	0.121	interval	1.10	1.09	1.09	1.08	1.08	1.09	1.10	1.11	1.11	1.42			3.28	3.25	3.32
velocity		5.16	8.31	8.39	8.39	8.46	8.46	8.39	8.31	8.23	8.23	9.87	7.90		8.36	8.44	8.26
Furukawa, Yutaro (JPN) (1985) time		2.63	3.70	4.77	5.83	6.89	7.97	9.09	10.20	11.34	12.48		14.00	9 / 8			
reaction time	0.150	interval	1.07	1.07	1.06	1.06	1.08	1.12	1.11	1.14	1.14	1.52			3.20	3.26	3.39
velocity		5.22	8.54	8.54	8.62	8.62	8.46	8.16	8.23	8.02	8.02	9.22	7.86		8.57	8.41	8.09

2015 Golden Grand Prix (Shanghai, CHN)

FINAL

date 17-May-15 wind 0.4 m/s

Wang (2016) - analysis of technical characteristics and performance prediction of rising star Xie Wenjun in 110m Hurdles

		H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
Xie Wenjun (CHN) (1990) time		2.55	3.58	4.61	5.64	6.65	7.67	8.69	9.74	10.82	11.92		13.36	/ 6			
reaction time	0.165	interval	1.03	1.03	1.03	1.01	1.02	1.02	1.05	1.08	1.10	1.44			3.09	3.05	3.23
velocity		5.38	8.87	8.87	8.87	9.05	8.96	8.96	8.70	8.46	8.31	9.74	8.23		8.87	8.99	8.49

2015 Shizuoka International Athletics Meeting (Fukuroi, JPN)

FINAL

date 03-May-15 wind -0.8 m/s

Takashima (2015) - race analysis of 110m & 100m hurdles in Japan- results of main 2015 competitions

		H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
Masuno, Genta (JPN) (1993) time		2.63	3.71	4.78	5.87	6.93	8.04	9.12	10.21	11.30	12.39		13.80	/ 1			
reaction time		interval	1.08	1.07	1.09	1.06	1.11	1.08	1.09	1.09	1.09	1.41			3.24	3.25	3.27
velocity		5.22	8.46	8.54	8.39	8.62	8.23	8.46	8.39	8.39	8.39	9.94	7.97		8.46	8.44	8.39
Furuya, Takumu (JPN) (1997) time		2.63	3.74	4.82	5.91	6.99	8.07	9.16	10.26	11.36	12.48		13.93	/ 2			
reaction time		interval	1.11	1.08	1.09	1.08	1.08	1.09	1.10	1.10	1.12	1.45			3.28	3.25	3.32
velocity		5.22	8.23	8.46	8.39	8.46	8.46	8.39	8.31	8.31	8.16	9.67	7.90		8.36	8.44	8.26
Sato, Hiroyuki (JPN) (1990) time		2.62	3.77	4.85	5.94	7.00	8.09	9.16	10.28	11.38	12.52		13.97	/ 3			
reaction time		interval	1.15	1.08	1.09	1.06	1.09	1.07	1.12	1.10	1.14	1.45			3.32	3.22	3.36
velocity		5.24	7.95	8.46	8.39	8.62	8.39	8.54	8.16	8.31	8.02	9.67	7.87		8.26	8.52	8.16
Fudaba, Hiroki (JPN) (1994) time		2.74	3.86	4.93	6.01	7.08	8.14	9.22	10.30	11.40	12.51		13.99	/ 4			
reaction time		interval	1.12	1.07	1.08	1.07	1.06	1.08	1.08	1.10	1.11	1.48			3.27	3.21	3.29
velocity		5.01	8.16	8.54	8.46	8.54	8.62	8.46	8.46	8.31	8.23	9.47	7.86		8.39	8.54	8.33
Omuro, Hideki (JPN) (1990) time		2.66	3.76	4.85	5.95	7.03	8.12	9.23	10.33	11.46	12.61		14.09	/ 6			
reaction time		interval	1.10	1.09	1.10	1.08	1.09	1.11	1.10	1.13	1.15	1.48			3.29	3.28	3.38

velocity 5.16 8.31 8.39 8.31 8.46 8.39 8.23 8.31 8.09 7.95 9.47 7.81 8.33 8.36 8.11

2015 Czech Republic National U20 Indoor Championships (Praha, CZE) (99cm)

FINAL

date 14-Feb-15

Feher (2015) - Final report from the U18 & U20 Czech Republic indoor national championships

		H1	H2	H3	H4	H5	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
Puliček, Tomáš (CZE) (1997)	time	2.60	3.72	4.82	5.90	7.03		7.98	3 / 1			
	reaction time	0.132	interval	1.12	1.10	1.08	0.95	PB		3.30		
	velocity	5.28	8.16	8.31	8.46	8.09	10.23	7.52		8.31		
Sklénár, David (CZE) (1997)	time	2.71	3.81	4.89	5.95	7.05		8.01	6 / 2			
	reaction time	0.184	interval	1.10	1.08	1.06	0.96	PB		3.24		
	velocity	5.06	8.31	8.46	8.62	8.31	10.13	7.49		8.46		
Monzar, Daniel (CZE) (1997)	time	2.77	3.91	5.03	6.15	7.28		8.23	7 / 3			
	reaction time	0.182	interval	1.14	1.12	1.12	0.95	PB		3.38		
	velocity	4.95	8.02	8.16	8.16	8.09	10.23	7.29		8.11		
Pirochta, Tomáš (CZE) (1997)	time	2.75	3.91	5.06	6.24	7.42		8.42	2 / 4			
	reaction time	0.284	interval	1.16	1.15	1.18	1.00			3.49		
	velocity	4.99	7.88	7.95	7.75	7.75	9.72	7.13		7.86		
Odvárka, Jirí (CZE) (1996)	time	2.74	4.04	5.24	6.41	7.59		8.57	4 / 5			
	reaction time	0.148	interval	1.30	1.20	1.17	0.98			3.67		
	velocity	5.01	7.03	7.62	7.81	7.75	9.92	7.00		7.47		
Sekyra, Daniel (CZE) (1996)	time	2.84	4.00	5.19	6.37	7.57		8.61	1 / 6			
	reaction time	0.191	interval	1.16	1.19	1.18	1.04			3.53		
	velocity	4.83	7.88	7.68	7.75	7.62	9.35	6.97		7.77		
Špičák, Daniel (CZE) (1997)	time	2.78	3.95	5.12	6.33	7.65		8.79	5 / 7			
	reaction time	0.138	interval	1.17	1.17	1.21	1.14			3.55		
	velocity	4.94	7.81	7.81	7.55	6.92	8.53	6.83		7.72		
Sykora, Miroslav (CZE) (1997)	time	2.90	4.08	5.64	7.06	8.66		10.20	8 / 8			
	reaction time	0.162	interval	1.18	1.56	1.42	1.54			4.16		
	velocity	4.73	7.75	5.86	6.44	5.71	6.31	5.88		6.59		

2015 Czech Republic National U18 Indoor Championships (Praha, CZE) (91cm)

FINAL

date 14-Feb-15

Feher (2015) - Final report from the U18 & U20 Czech Republic indoor national championships

		H1	H2	H3	H4	H5	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
Kijanica, Michael (CZE) (1998)	time	2.68	3.81	4.90	6.00	7.09		8.08	5 / 1			
	reaction time	0.175	interval	1.13	1.09	1.10	0.99			3.32		
	velocity	5.12	8.09	8.39	8.31	8.39	9.82	7.43		8.26		
Desensky, Lukáš (CZE) (1998)	time	2.71	3.83	4.96	6.06	7.17		8.11	2 / 2			
	reaction time	0.195	interval	1.12	1.13	1.10	0.94			3.35		
	velocity	5.06	8.16	8.09	8.31	8.23	10.34	7.40		8.19		
Mališka, Rostislav (CZE) (1998)	time	2.73	3.85	4.95	6.04	7.15		8.11	4 / 3			
	reaction time	0.185	interval	1.12	1.10	1.09	0.96			3.31		
	velocity	5.03	8.16	8.31	8.39	8.23	10.13	7.40		8.28		
Krivák, Dominik (CZE) (1999)	time	2.65	3.77	4.90	6.03	7.16		8.14	8 / 4			
	reaction time	0.109	interval	1.12	1.13	1.13	0.98			3.38		
	velocity	5.18	8.16	8.09	8.09	8.09	9.92	7.37		8.11		
Bárta, Dan (CZE) (1998)	time	2.70	3.86	5.00	6.09	7.23		8.20	6 / 5			
	reaction time	0.158	interval	1.16	1.14	1.09	0.97			3.39		
	velocity	5.08	7.88	8.02	8.39	8.02	10.02	7.32		8.09		
Smolka, Adam (CZE) (1998)	time	2.74	3.89	5.00	6.14	7.28		8.29	3 / 6			
	reaction time	0.153	interval	1.15	1.11	1.14	1.01			3.40		
	velocity	5.01	7.95	8.23	8.02	8.02	9.62	7.24		8.06		
Svoboda, Lukáš (CZE) (1998)	time	2.73	3.87	5.02	6.13	7.26		8.30	7 / 7			
	reaction time	0.158	interval	1.14	1.15	1.11	1.04			3.40		
	velocity	5.03	8.02	7.95	8.23	8.09	9.35	7.23		8.06		
Jelínek, Erik (CZE) (1998)	time	2.78	3.94	5.06	6.19	7.31		8.33	1 / 8			
	reaction time	0.153	interval	1.16	1.12	1.13	1.02			3.41		
	velocity	4.94	7.88	8.16	8.09	8.16	9.53	7.20		8.04		

2014 Japanese National Junior Championships (Nagoya, JPN) (99cr)

Semi-Final 1

date 03-Oct-14

wind 0.7 m/s

Kota (2017) - 70th high school championships: JAF scientific committee - biomechanics data collection

		H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
Furuya, Takumu (JPN) (1997)	time	2.62	3.71	4.81	5.87	6.93	8.01	9.10	10.18	11.28	12.37		13.83	1			

reaction time	interval		1.09	1.10	1.06	1.06	1.08	1.09	1.08	1.10	1.09	1.46	PB	3.25	3.23	3.27
	velocity	5.24	8.39	8.31	8.62	8.62	8.46	8.39	8.46	8.31	8.39	9.60	7.95	8.44	8.49	8.39

2014 Asian Games (Incheon, KOR)**Decathlon - Heat 2**

date 01-Oct-14		wind 1.9 m/s		Takeo (2014) - decathletes sprint ability and the relationship of the individual event performance													
		H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
(1990 time		2.74	3.86	4.97	6.09	7.21	8.33	9.47	10.59	11.73	12.89		14.33	/ 1			
166	interval	1.12	1.11	1.12	1.12	1.12	1.12	1.14	1.12	1.14	1.16	1.44			3.35	3.38	3.42
	velocity	5.01	8.16	8.23	8.16	8.16	8.16	8.02	8.16	8.02	7.88	9.74	7.68		8.19	8.11	8.02

Decathlon - Heat 1

date		01-Oct-14		wind		1.7 m/s		Takeo (2014) - decathletes sprint ability and the relationship of the individual event performance									
		H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
1986	time	2.78	3.97	5.15	6.31	7.47	8.65	9.85	11.07	12.30	13.54		15.16	/ 4			
1996	interval		1.19	1.18	1.16	1.16	1.18	1.20	1.22	1.23	1.24	1.62			3.53	3.54	3.69
	velocity	4.94	7.68	7.75	7.88	7.88	7.75	7.62	7.49	7.43	7.37	8.65	7.26		7.77	7.75	7.43

FINAL

date		30-Sep-14		wind		0.4 m/s		Wang (2016) - comparative analysis of the grey relational grade of segment times and final results of the 110m hurdles									
		H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
time		2.60	3.66	4.70	5.74	6.76	7.80	8.82	9.85	10.90	11.96		13.36	/ 1			
176	interval		1.06	1.04	1.04	1.02	1.04	1.02	1.03	1.05	1.06	1.40			3.14	3.08	3.14
	velocity	5.28	8.62	8.79	8.79	8.96	8.79	8.96	8.87	8.70	8.62	10.01	8.23		8.73	8.90	8.73

2014 Kanto Championships (Sagamihara, JPN)**Semi-Final 1**

date		31-Aug-14		wind		-0.2 m/s		(2017.03.01) - https://twitter.com/touchdown_time/media?lang=en&lang=en&lang=en&lang=en									
		H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
997)	time	2.62	3.72	4.80	5.85	6.93	8.00	9.08	10.17	11.28	12.40		13.89	6 / 1			
	interval		1.10	1.08	1.05	1.08	1.07	1.08	1.09	1.11	1.12	1.49	PB		3.23	3.23	3.32
	velocity	5.24	8.31	8.46	8.70	8.46	8.54	8.46	8.39	8.23	8.16	9.41	7.92		8.49	8.49	8.26

2014 European Championships (Zurich, SUI)**Heat 4**

date		13-Aug-14		wind		0.7 m/s		Incalza (2014) - https://www.youtube.com/watch?v=wFOM-j9iZHE&list=UU7691fXeyNFIMROfkJ-WWUg									
		H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
Shubenkov, Sergey (RUS) (1999)	time	2.63	3.68	4.71	5.74	6.75	7.77	8.81	9.85	10.90	11.94		13.29	8 / 1			
	interval		1.05	1.03	1.03	1.01	1.02	1.04	1.04	1.05	1.04	1.35			3.11	3.07	3.13
	velocity	5.22	8.70	8.87	8.87	9.05	8.96	8.79	8.79	8.70	8.79	10.39	8.28		8.82	8.93	8.76
Fofana, Hassane (ITA) (1992)	time	2.64	3.71	4.74	5.79	6.83	7.86	8.89	9.93	10.99	12.08		13.55	2 / 5			
	interval		1.07	1.03	1.05	1.04	1.03	1.03	1.04	1.06	1.09	1.47	PB		3.15	3.10	3.19
	velocity	5.20	8.54	8.87	8.70	8.79	8.87	8.87	8.79	8.62	8.39	9.54	8.12		8.70	8.85	8.60

Heat 2

date		13-Aug-14		Incalza (2014) - https://www.youtube.com/watch?v=nliXgBWbl_k&list=UU7691fXeyNFIMROfkJ-WWUg													
		H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
987	time	2.58	3.61	4.64	5.68	6.70	7.73	8.77	9.81	10.86	11.92		13.35	8 / 1			
148	interval		1.03	1.03	1.04	1.02	1.03	1.04	1.04	1.05	1.06	1.43			3.10	3.09	3.15
	velocity	5.32	8.87	8.87	8.79	8.96	8.87	8.79	8.79	8.70	8.62	9.80	8.24		8.85	8.87	8.70
987	time	2.55	3.64	4.67	5.73	6.77	7.80	8.90	9.97	11.05	12.14		13.54	6 / 4			
138	interval		1.09	1.03	1.06	1.04	1.03	1.10	1.07	1.08	1.09	1.40			3.18	3.17	3.24
	velocity	5.38	8.39	8.87	8.62	8.79	8.87	8.31	8.54	8.46	8.39	10.01	8.12		8.62	8.65	8.46

Heat 1

date		13-Aug-14		wind		-0.7 m/s		Incalza (2014) - https://www.youtube.com/watch?v=4CScuS4aCYI&list=UU7691fXeyNFIMROfkJ-WWUg									
		H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
(FR)		time	2.56	3.59	4.60	5.56	6.54	7.58	8.60	9.64	10.68	11.76	13.29	3 / 1			
152		interval		1.03	1.01	0.96	0.98	1.04	1.02	1.04	1.04	1.08	1.53		3.00	3.04	3.16
		velocity	5.36	8.87	9.05	9.52	9.33	8.79	8.96	8.79	8.79	8.46	9.16	8.28	9.14	9.02	8.68
(ITA)		time	2.64	3.72	4.78	5.81	6.84	7.90	8.97	10.06	11.16	12.28	13.77	4 / 6			
141		interval		1.08	1.06	1.03	1.03	1.06	1.07	1.09	1.10	1.12	1.49		3.17	3.16	3.31
		velocity	5.20	8.46	8.62	8.87	8.87	8.62	8.54	8.39	8.31	8.16	9.41	7.99	8.65	8.68	8.28

2014 Japanese National High School Championships (Kofu, JPN) (99cm)**FINAL**

date		03-Aug-14				wind -2.4 m/s													Yanagiya (2014) - research on athlete performance and technique- 2014 data book												
		H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10														
1997)	time	2.64	3.74	4.82	5.89	6.97	8.05	9.16	10.27	11.39	12.55		14.05	4 / 1																	
	interval		1.10	1.08	1.07	1.08	1.08	1.11	1.11	1.12	1.16	1.50			3.25	3.27	3.39														
	velocity	5.20	8.31	8.46	8.54	8.46	8.46	8.23	8.23	8.16	7.88	9.35	7.83		8.44	8.39	8.09														
1996)	time	2.74	3.89	4.98	6.12	7.22	8.36	9.51	10.67	11.85	13.04		14.52	5 / 2																	
	interval		1.15	1.09	1.14	1.10	1.14	1.15	1.16	1.18	1.19	1.48			3.38	3.39	3.53														
	velocity	5.01	7.95	8.39	8.02	8.31	8.02	7.95	7.88	7.75	7.68	9.47	7.58		8.11	8.09	7.77														
	time	2.74	3.88	4.99	6.11	7.23	8.36	9.50	10.65	11.81	13.00		14.55	6 / 3																	
	interval		1.14	1.11	1.12	1.12	1.13	1.14	1.15	1.16	1.19	1.55			3.37	3.39	3.50														
	velocity	5.01	8.02	8.23	8.16	8.16	8.09	8.02	7.95	7.88	7.68	9.05	7.56		8.14	8.09	7.83														

2014 Czech Republic National Championships (Ostrava, CZE)

FINAL

FINAL

		date	02-Aug-14	wind -1.2 m/s				Feher (2014) - Final report from the Czech Republic national championships											
				H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
Svoboda, Petr (CZE) (1984)	time			2.57	3.63	4.67	5.68	6.71	7.76	8.80	9.85	10.91	12.01		13.50	4 / 1			
	reaction time	0.140	interval		1.06	1.04	1.01	1.03	1.05	1.04	1.05	1.06	1.10	1.49			3.11	3.12	3.21
			velocity	5.34	8.62	8.79	9.05	8.87	8.70	8.79	8.70	8.62	8.31	9.41	8.15		8.82	8.79	8.54
Polách, Josef (CZE) (1990)	time			2.70	3.82	4.93	6.05	7.15	8.26	9.37	10.52	11.63	12.83		14.39	3 / 2			
	reaction time	0.154	interval		1.12	1.11	1.12	1.10	1.11	1.11	1.15	1.11	1.20	1.56			3.35	3.32	3.46
			velocity	5.08	8.16	8.23	8.16	8.31	8.23	8.23	7.95	8.23	7.62	8.99	7.64		8.19	8.26	7.92
Říha, Michal (CZE) (1994)	time			2.66	3.85	5.02	6.15	7.28	8.45	9.61	10.75	11.91	13.09		14.66	6 / 3			
	reaction time	0.159	interval		1.19	1.17	1.13	1.13	1.17	1.16	1.14	1.16	1.18	1.57			3.49	3.46	3.48
			velocity	5.16	7.68	7.81	8.09	8.09	7.81	7.88	8.02	7.88	7.75	8.93	7.50		7.86	7.92	7.88
Penáz, Petr (CZE) (1991)	time			2.77	3.90	5.02	6.10	7.20	8.38	9.55	10.70	11.87	13.05		14.87	5 / 4			
	reaction time	0.172	interval		1.13	1.12	1.08	1.10	1.18	1.17	1.15	1.17	1.18	1.82			3.33	3.45	3.50
			velocity	4.95	8.09	8.16	8.46	8.31	7.75	7.81	7.95	7.81	7.75	7.70	7.40		8.23	7.95	7.83
Pokorný, Martin (CZE) (1988)	time			2.75	3.90	5.05	6.17	7.31	8.47	9.64	10.80	11.99	13.22		14.89	8 / 5			
	reaction time	0.135	interval		1.15	1.15	1.12	1.14	1.16	1.17	1.16	1.19	1.23	1.67			3.42	3.47	3.58
			velocity	4.99	7.95	7.95	8.16	8.02	7.88	7.81	7.88	7.68	7.43	8.40	7.39		8.02	7.90	7.66
Kavka, Tomáš (CZE) (1989)	time			2.77	3.93	5.03	6.21	7.35	8.53	9.69	10.85	12.08	13.27		14.95	7 / 6			
	reaction time	0.122	interval		1.16	1.10	1.18	1.14	1.18	1.16	1.16	1.23	1.19	1.68			3.44	3.48	3.58
			velocity	4.95	7.88	8.31	7.75	8.02	7.75	7.88	7.88	7.43	7.68	8.35	7.36		7.97	7.88	7.66
Sajdok, Stanislav (CZE) (1983)	time			2.68	3.85	4.99	6.15	7.34	8.50	9.70	10.93	12.15	13.43		15.24	2 / 7			
	reaction time	0.124	interval		1.17	1.14	1.16	1.19	1.16	1.20	1.23	1.22	1.28	1.81			3.47	3.55	3.73
			velocity	5.12	7.81	8.02	7.88	7.68	7.88	7.62	7.43	7.49	7.14	7.75	7.22		7.90	7.72	7.35
Krejčí, Michal (CZE) (1981)	time			2.74	3.93	5.11	6.29	7.48	8.72	9.94	11.17	12.42	13.72		15.45	1 / 8			
	reaction time	0.147	interval		1.19	1.18	1.18	1.19	1.24	1.22	1.23	1.25	1.30	1.73			3.55	3.65	3.78
			velocity	5.01	7.68	7.75	7.75	7.68	7.37	7.49	7.43	7.31	7.03	8.10	7.12		7.72	7.51	7.25

Heat 2

Heat 2

		date	02-Aug-14	wind	0.8 m/s	Feher (2014) - Final report from the Czech Republic national championships													
				H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
Penáz, Petr (CZE) (1991)	time			2.82	3.93	5.03	6.13	7.21	8.31	9.40	10.49	11.60	12.72		14.21	5 / 1			
	reaction time	0.190	interval		1.11	1.10	1.10	1.08	1.10	1.09	1.09	1.11	1.12	1.49			3.31	3.27	3.32
			velocity	4.87	8.23	8.31	8.31	8.46	8.31	8.39	8.39	8.23	8.16	9.41	7.74		8.28	8.39	8.26
Polách, Josef (CZE) (1990)	time			2.73	3.85	4.94	6.05	7.15	8.26	9.35	10.46	11.61	12.76		14.26	4 / 2			
	reaction time	0.143	interval		1.12	1.09	1.11	1.10	1.11	1.09	1.11	1.15	1.15	1.50			3.32	3.30	3.41
			velocity	5.03	8.16	8.39	8.23	8.31	8.23	8.39	8.23	7.95	7.95	9.35	7.71		8.26	8.31	8.04
Kavka, Tomáš (CZE) (1989)	time			2.84	4.00	5.15	6.25	7.40	8.55	9.70	10.85	12.05	13.28		14.83	6 / 3			
	reaction time	0.150	interval		1.16	1.15	1.10	1.15	1.15	1.15	1.15	1.20	1.23	1.55			3.41	3.45	3.58
			velocity	4.83	7.88	7.95	8.31	7.95	7.95	7.95	7.95	7.62	7.43	9.05	7.42		8.04	7.95	7.66
Sajdok, Stanislav (CZE) (1983)	time			2.70	3.87	4.98	6.14	7.28	8.45	9.63	10.81	12.03	13.29		14.96	3 / 4			
	reaction time	0.159	interval		1.17	1.11	1.16	1.14	1.17	1.18	1.18	1.22	1.26	1.67			3.44	3.49	3.66
			velocity	5.08	7.81	8.23	7.88	8.02	7.81	7.75	7.75	7.49	7.25	8.40	7.35		7.97	7.86	7.49
Vašátko, Stanislav (CZE) (1992)	time			2.87	4.08	5.27	6.48	7.70	8.94	10.18	11.44	12.70	13.98		15.64	1 / 5			
	reaction time	0.169	interval		1.21	1.19	1.21	1.22	1.24	1.24	1.26	1.26	1.28	1.66			3.61	3.70	3.80
			velocity	4.78	7.55	7.68	7.55	7.49	7.37	7.37	7.25	7.25	7.14	8.45	7.03		7.60	7.41	7.22

Heat 1

Heat 1

		date	02-Aug-14	wind -1.6 m/s				Feher (2014) - Final report from the Czech Republic national championships											
				H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
Svoboda, Petr (CZE) (1984)	time			2.57	3.65	4.70	5.75	6.81	7.87	8.92	9.98	11.05	12.15		13.64	4 / 1			
	reaction time	0.133	interval		1.08	1.05	1.05	1.06	1.06	1.05	1.06	1.07	1.10	1.49			3.18	3.17	3.23
			velocity	5.34	8.46	8.70	8.70	8.62	8.62	8.70	8.62	8.54	8.31	9.41	8.06		8.62	8.65	8.49
Říha, Michal (CZE) (1994)	time			2.70	3.86	5.02	6.15	7.31	8.47	9.65	10.82	12.00	13.20		14.74	3 / 2			
	reaction time	0.171	interval		1.16	1.16	1.13	1.16	1.16	1.18	1.17	1.18	1.20	1.54			3.45	3.50	3.55
			velocity	5.08	7.88	7.88	8.09	7.88	7.88	7.75	7.81	7.75	7.62	9.10	7.46		7.95	7.83	7.72
Pokorný, Martin (CZE) (1988)	time			2.75	3.93	5.11	6.26	7.41	8.56	9.75	10.91	12.12	13.34		15.02	6 / 3			
	reaction time	0.156	interval		1.18	1.18	1.15	1.15	1.15	1.19	1.16	1.21	1.22	1.68			3.51	3.49	3.59
			velocity	4.99	7.75	7.75	7.95	7.95	7.95	7.68	7.88	7.55	7.49	8.35	7.32		7.81	7.86	7.64
Krejčí, Michal (CZE) (1981)	time			2.75	3.97	5.16	6.35	7.54	8.72	9.90	11.14	12.38	13.64		15.27	2 / 4			
	reaction time	0.106	interval		1.22	1.19	1.19	1.19	1.18	1.18	1.24	1.24	1.26	1.63			3.60	3.55	3.74
			velocity	4.99	7.49	7.68	7.68	7.68	7.75	7.75	7.37	7.37	7.25	8.60	7.20		7.62	7.72	7.33
Frolík, Petr (CZE) (1991)	time			2.88	4.07	5.26	6.46	7.63	8.85	10.07	11.29	12.48	13.74		15.36	7 / 5			

reaction time	0.186	interval	1.19	1.19	1.20	1.17	1.22	1.22	1.22	1.19	1.26	1.62		3.58	3.61	3.67
velocity	4.76	7.68	7.68	7.62	7.81	7.49	7.49	7.49	7.68	7.25	8.65	7.16		7.66	7.60	7.47
Dinga, Libor (CZE) (1984)	time	2.83	4.08	5.30	6.51	7.72	8.95	10.19	11.47	12.73	13.99		15.66	1 / 6		
reaction time	0.152	interval	1.25	1.22	1.21	1.21	1.23	1.24	1.28	1.26	1.26	1.67		3.68	3.68	3.80
velocity	4.85	7.31	7.49	7.55	7.55	7.43	7.37	7.14	7.25	7.25	8.40	7.02		7.45	7.45	7.22
Sedlák, Václav (CZE) (1993)	time	2.73	3.88	5.00	6.13	7.24	8.35							5 / --		
reaction time	0.151	interval	1.15	1.12	1.13	1.11	1.11						DQ	3.40		
velocity	5.03	7.95	8.16	8.09	8.23	8.23								8.06		

2014 Czech Republic National U20 Championships (Trinec, CZE) (99cm)

FINAL

		date	28-Jun-14	wind -2.2 m/s					Feher (2014) - Final report from the U18 & U20 Czech Republic national championships											
				H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10	
Homola, Tomáš (CZE) (1995)	time			2.68	3.80	4.90	5.99	7.10	8.20	9.30	10.43	11.56	12.73		14.28	4 / 1				
	reaction time	0.147	interval		1.12	1.10	1.09	1.11	1.10	1.10	1.13	1.13	1.17	1.55			3.31	3.31	3.43	
			velocity	5.12	8.16	8.31	8.39	8.23	8.31	8.31	8.09	8.09	7.81	9.05	7.70		8.28	8.28	7.99	
Tuček, Martin (CZE) (1995)	time			2.68	3.82	4.94	6.07	7.20	8.31	9.42	10.56	11.70	12.91		14.51	6 / 2				
	reaction time	0.148	interval		1.14	1.12	1.13	1.13	1.11	1.11	1.14	1.14	1.21	1.60			3.39	3.35	3.49	
			velocity	5.12	8.02	8.16	8.09	8.09	8.23	8.23	8.02	8.02	7.55	8.76	7.58		8.09	8.19	7.86	
Doležal, Jan (CZE) (1996)	time			2.74	3.89	4.99	6.09	7.20	8.34	9.45	10.58	11.75	12.96		14.52	3 / 3				
	reaction time	0.157	interval		1.15	1.10	1.10	1.11	1.14	1.11	1.13	1.17	1.21	1.56			3.35	3.36	3.51	
			velocity	5.01	7.95	8.31	8.31	8.23	8.02	8.23	8.09	7.81	7.55	8.99	7.58		8.19	8.16	7.81	
Večera, Jindřich (CZE) (1995)	time			2.70	3.83	4.94	6.06	7.17	8.28	9.43	10.56	11.73	12.92		14.55	5 / 4				
	reaction time	0.171	interval		1.13	1.11	1.12	1.11	1.11	1.15	1.13	1.17	1.19	1.63			3.36	3.37	3.49	
			velocity	5.08	8.09	8.23	8.16	8.23	8.23	7.95	8.09	7.81	7.68	8.60	7.56		8.16	8.14	7.86	
Rykl, Radoš (CZE) (1995)	time			2.77	3.93	5.06	6.19	7.32	8.50	9.65	10.80	11.97	13.13		14.75	7 / 5				
	reaction time	0.177	interval		1.16	1.13	1.13	1.13	1.18	1.15	1.15	1.17	1.16	1.62			3.42	3.46	3.48	
			velocity	4.95	7.88	8.09	8.09	8.09	7.75	7.95	7.95	7.81	7.88	8.65	7.46		8.02	7.92	7.88	
Schmöger, Martin (CZE) (1995)	time			2.83	4.00	5.15	6.29	7.46	8.59	9.78	10.97	12.16	13.35		15.00	2 / 6				
	reaction time	0.174	interval		1.17	1.15	1.14	1.17	1.13	1.19	1.19	1.19	1.19	1.65			3.46	3.49	3.57	
			velocity	4.85	7.81	7.95	8.02	7.81	8.09	7.68	7.68	7.68	7.68	8.50	7.33		7.92	7.86	7.68	
Kubeš, Jirí (CZE) (1995)	time			2.79	3.93	5.10	6.25	7.40	8.56	9.75	10.93	12.14	13.40		15.29	1 / 7				
	reaction time	0.186	interval		1.14	1.17	1.15	1.15	1.16	1.19	1.18	1.21	1.26	1.89			3.46	3.50	3.65	
			velocity	4.92	8.02	7.81	7.95	7.95	7.88	7.68	7.75	7.55	7.25	7.42	7.19		7.92	7.83	7.51	
Decker, Jan (CZE) (1995)	time			2.86	4.07	5.28	6.46	7.66	8.87	10.07	11.26	12.50	13.73		15.40	8 / 8				
	reaction time	0.235	interval		1.21	1.21	1.18	1.20	1.21	1.20	1.19	1.24	1.23	1.67			3.60	3.61	3.66	
			velocity	4.80	7.55	7.55	7.75	7.62	7.55	7.62	7.68	7.37	7.43	8.40	7.14		7.62	7.60	7.49	

Heat 3

Heat 3

		date	28-Jun-14	wind -2.5 m/s					Feher (2014) - Final report from the U18 & U20 Czech Republic national championships											
				H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10	
Doležal, Jan (CZE) (1996)	time			2.75	3.87	4.97	6.07	7.19	8.33	9.45	10.60	11.77	12.92		14.52	3 / 1				
	reaction time	0.524	interval		1.12	1.10	1.10	1.12	1.14	1.12	1.15	1.17	1.15	1.60				3.32	3.38	3.47
	velocity		4.99	8.16	8.31	8.31	8.16	8.02	8.16	7.95	7.81	7.95	8.76	7.58				8.26	8.11	7.90
Rykl, Radoš (CZE) (1995)	time			2.85	4.00	5.17	6.34	7.48	8.68	9.83	11.00	12.22	13.39		15.04	6 / 2				
	reaction time	0.253	interval		1.15	1.17	1.17	1.14	1.20	1.15	1.17	1.22	1.17	1.65				3.49	3.49	3.56
	velocity		4.81	7.95	7.81	7.81	8.02	7.62	7.95	7.81	7.49	7.81	8.50	7.31				7.86	7.86	7.70
Kubeš, Jirí (CZE) (1995)	time			2.80	3.95	5.11	6.25	7.43	8.62	9.80	11.00	12.20	13.43		15.06	5 / 3				
	reaction time	0.203	interval		1.15	1.16	1.14	1.18	1.19	1.18	1.20	1.20	1.23	1.63				3.45	3.55	3.63
	velocity		4.90	7.95	7.88	8.02	7.75	7.68	7.75	7.62	7.62	7.43	8.60	7.30				7.95	7.72	7.55
Perutka, Martin (CZE) (1995)	time			2.77	3.96	5.17	6.38	7.60	8.84	10.10	11.36	12.66	14.15		15.80	2 / 4				
	reaction time	0.139	interval		1.19	1.21	1.21	1.22	1.24	1.26	1.26	1.30	1.49	1.65				3.61	3.72	4.05
	velocity		4.95	7.68	7.55	7.55	7.49	7.37	7.25	7.25	7.03	6.13	8.50	6.96				7.60	7.37	6.77
Konečný, Michael (CZE) (1995)	time			2.82	4.02	5.20	6.43	7.64	8.88	10.23	11.59	12.90	14.22		15.97	8 / 5				
	reaction time	0.144	interval		1.20	1.18	1.23	1.21	1.24	1.35	1.36	1.31	1.32	1.75				3.61	3.80	3.99
	velocity		4.87	7.62	7.75	7.43	7.55	7.37	6.77	6.72	6.98	6.92	8.01	6.89				7.60	7.22	6.87
Sekyra, Daniel (CZE) (1996)	time			2.95	4.16	5.40	6.62	7.87	9.10	10.37	11.62	12.91	14.24		16.00	4 / 6				
	reaction time	0.247	interval		1.21	1.24	1.22	1.25	1.23	1.27	1.25	1.29	1.33	1.76				3.67	3.75	3.87
	velocity		4.65	7.55	7.37	7.49	7.31	7.43	7.20	7.31	7.09	6.87	7.97	6.88				7.47	7.31	7.09
Rezníček, Tomáš (CZE) (1995)	time			2.98	4.38	5.75	7.19	8.60	10.11	11.63	13.15	14.62	16.10		18.00	7 / 7				
	reaction time	0.163	interval		1.40	1.37	1.44	1.41	1.51	1.52	1.52	1.47	1.48	1.90				4.21	4.44	4.47
	velocity		4.60	6.53	6.67	6.35	6.48	6.05	6.01	6.01	6.22	6.18	7.38	6.11				6.51	6.18	6.13

Heat 2

Heat 2

		date	28-Jun-14	wind -2.9 m/s			Feher (2014) - Final report from the U18 & U20 Czech Republic national championships												
				H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
Večera, Jindrich (CZE) (1995)	time			2.75	3.85	4.99	6.09	7.19	8.30	9.45	10.59	11.74	12.92		14.55	3 / 1			
	reaction time	0.184	interval		1.10	1.14	1.10	1.10	1.11	1.15	1.14	1.15	1.18	1.63			3.34	3.36	3.47
	velocity			4.99	8.31	8.02	8.31	8.31	8.23	7.95	8.02	7.95	7.75	8.60	7.56		8.21	8.16	7.90
Tuček, Martin (CZE) (1995)	time			2.73	3.86	4.99	6.13	7.29	8.43	9.60	10.77	11.99	13.20		15.00	4 / 2			
	reaction time	0.154	interval		1.13	1.13	1.14	1.16	1.14	1.17	1.17	1.22	1.21	1.80			3.40	3.47	3.60
	velocity			5.03	8.09	8.09	8.02	7.88	8.02	7.81	7.81	7.49	7.55	7.79	7.33		8.06	7.90	7.62
Sirota, Martin (CZE) (1995)	time			2.88	4.07	5.27	6.44	7.66	8.86	10.07	11.31	12.53	13.81		15.48	8 / 3			
	reaction time	0.249	interval		1.19	1.20	1.17	1.22	1.20	1.21	1.24	1.22	1.28	1.67			3.56	3.63	3.74
	velocity			4.76	7.68	7.62	7.81	7.49	7.62	7.55	7.37	7.49	7.14	8.40	7.11		7.70	7.55	7.33
Pilar, Lukáš (CZE) (1995)	time			2.88	4.05	5.23	6.44	7.64	8.86	10.05	11.29	12.53	13.78		15.52	7 / 4			
	reaction time	0.212	interval		1.17	1.18	1.21	1.20	1.22	1.19	1.24	1.24	1.25	1.74			3.56	3.61	3.73
	velocity			4.76	7.81	7.75	7.55	7.62	7.49	7.68	7.37	7.37	7.31	8.06	7.09		7.70	7.60	7.35
Šilhavy, Václav (CZE) (1995)	time			2.96	4.14	5.38	6.59	7.78	9.00	10.22	11.42	12.69	13.95		15.61	2 / 5			
	reaction time	0.217	interval		1.18	1.24	1.21	1.19	1.22	1.22	1.20	1.27	1.26	1.66			3.63	3.63	3.73
	velocity			4.64	7.75	7.37	7.55	7.68	7.49	7.49	7.62	7.20	7.25	8.45	7.05		7.55	7.55	7.35
Dostražil, Tomáš (CZE) (1996)	time			2.85	4.09	5.30	6.52	7.75	9.00	10.22	11.52	12.78	14.11		15.87	6 / 6			
	reaction time	0.184	interval		1.24	1.21	1.22	1.23	1.25	1.22	1.30	1.26	1.33	1.76			3.67	3.70	3.89
	velocity			4.81	7.37	7.55	7.49	7.43	7.31	7.49	7.03	7.25	6.87	7.97	6.93		7.47	7.41	7.05

Heat 1

Heat 1		date	28-Jun-14	wind -1.6 m/s			Feher (2014) - Final report from the U18 & U20 Czech Republic national championships												
				H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
Homola, Tomáš (CZE) (1995)	time			2.65	3.78	4.89	5.99	7.11	8.23	9.34	10.50	11.67	12.84		14.43	4 / 1			
	reaction time	0.176	interval		1.13	1.11	1.10	1.12	1.12	1.11	1.16	1.17	1.17	1.59			3.34	3.35	3.50
			velocity	5.18	8.09	8.23	8.31	8.16	8.16	8.23	7.88	7.83	7.79	8.82	7.62		8.21	8.19	7.83
Schmöger, Martin (CZE) (1995)	time			2.81	3.94	5.13	6.27	7.43	8.58	9.68	10.94	12.16	13.38		15.02	5 / 2			
	reaction time	0.215	interval		1.13	1.19	1.14	1.16	1.15	1.10	1.26	1.22	1.22	1.64			3.46	3.41	3.70
			velocity	4.88	8.09	7.68	8.02	7.88	7.95	8.31	7.25	7.49	7.49	8.55	7.32		7.92	8.04	7.41
Decker, Jan (CZE) (1995)	time			2.78	3.95	5.12	6.30	7.48	8.69	9.90	11.10	12.34	13.60		15.34	6 / 3			
	reaction time	0.206	interval		1.17	1.17	1.18	1.18	1.21	1.21	1.20	1.24	1.26	1.74			3.52	3.60	3.70
			velocity	4.94	7.81	7.81	7.75	7.75	7.55	7.55	7.62	7.37	7.25	8.06	7.17		7.79	7.62	7.41
Odvárka, Jirí (CZE) (1996)	time			2.80	4.03	5.23	6.43	7.64	8.88	10.11	11.38	12.68	14.02		15.83	7 / 4			
	reaction time	0.162	interval		1.23	1.20	1.20	1.21	1.24	1.23	1.27	1.30	1.34	1.81			3.63	3.68	3.91
			velocity	4.90	7.43	7.62	7.62	7.55	7.37	7.43	7.20	7.03	6.82	7.75	6.95		7.55	7.45	7.01
Holek, Jakub (CZE) (1995)	time			2.99	4.26	5.52	6.77	7.98	9.20	10.44	11.66	12.91	14.18		15.89	3 / 5			
	reaction time	0.287	interval		1.27	1.26	1.25	1.21	1.22	1.24	1.22	1.25	1.27	1.71			3.78	3.67	3.74
			velocity	4.59	7.20	7.25	7.31	7.55	7.49	7.37	7.49	7.31	7.20	8.20	6.92		7.25	7.47	7.33
Ružička, Martin (CZE) (1995)	time			2.95	4.18	5.41	6.62	7.85	9.15	10.39	11.70	12.97	14.26		15.99	2 / 6			
	reaction time	0.244	interval		1.23	1.23	1.21	1.23	1.30	1.24	1.31	1.27	1.29	1.73			3.67	3.77	3.87
			velocity	4.65	7.43	7.43	7.55	7.43	7.03	7.37	6.98	7.20	7.09	8.10	6.88		7.47	7.27	7.09
Král, Aleš (CZE) (1996)	time			2.81	4.10	5.40	6.66	7.95	9.27	10.58	11.95	13.23	14.60		16.34	8 / 7			
	reaction time	0.165	interval		1.29	1.30	1.26	1.29	1.32	1.31	1.37	1.28	1.37	1.74			3.85	3.92	4.02
			velocity	4.88	7.09	7.03	7.25	7.09	6.92	6.98	6.67	7.14	6.67	8.06	6.73		7.12	6.99	6.82

2014 Czech Republic National U18 Championships (Trinec, CZE) (91cm)

FINAL

FINAL	date		28-Jun-14		wind -3.7 m/s					Feher (2014) - Final report from the U18 & U20 Czech Republic national championships												
			H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10				
Pulíček, Tomáš (CZE) (1997)	time		2.66	3.74	4.80	5.87	6.92	7.99	9.06	10.13	11.22	12.35		13.89	4 / 1							
	reaction time	0.177	interval		1.08	1.06	1.07	1.05	1.07	1.07	1.07	1.09	1.13	1.54		3.21	3.19	3.29				
		velocity		5.16	8.46	8.62	8.54	8.70	8.54	8.54	8.54	8.39	8.09	9.10	7.92		8.54	8.60	8.33			
Sklenár, David (CZE) (1997)	time		2.67	3.77	4.86	5.94	7.02	8.11	9.22	10.35	11.49	12.66		14.25	5 / 2							
	reaction time	0.159	interval		1.10	1.09	1.08	1.08	1.09	1.11	1.13	1.14	1.17	1.59		3.27	3.28	3.44				
		velocity		5.14	8.31	8.39	8.46	8.46	8.39	8.23	8.09	8.02	7.81	8.82	7.72		8.39	8.36	7.97			
Kisiala, Jan (CZE) (1998)	time		2.88	4.04	5.18	6.32	7.45	8.57	9.77	10.96	12.17	13.39		14.93	2 / 3							
	reaction time	0.177	interval		1.16	1.14	1.14	1.13	1.12	1.20	1.19	1.21	1.22	1.54		3.44	3.45	3.62				
		velocity		4.76	7.88	8.02	8.02	8.09	8.16	7.62	7.68	7.55	7.49	9.10	7.37		7.97	7.95	7.57			
Nardelli, Alexander (CZE) (1997)	time		2.86	4.02	5.18	6.35	7.52	8.71	9.90	11.08	12.27	13.45		15.07	6 / 4							
	reaction time	0.229	interval		1.16	1.16	1.17	1.17	1.19	1.19	1.18	1.19	1.18	1.62		3.49	3.55	3.55				
		velocity		4.80	7.88	7.88	7.81	7.81	7.68	7.68	7.75	7.68	7.75	8.65	7.30		7.86	7.72	7.72			
Špičák, Daniel (CZE) (1997)	time		2.96	4.11	5.29	6.48	7.65	8.80	9.95	11.15	12.34	13.55		15.09	8 / 5							

reaction time	0.288	interval	1.15	1.18	1.19	1.17	1.15	1.15	1.20	1.19	1.21	1.54			3.52	3.47	3.60
velocity	4.64		7.95	7.75	7.68	7.81	7.95	7.95	7.62	7.68	7.55	9.10	7.29		7.79	7.90	7.62
Mališka, Rostislav (CZE) (1998)	time	2.91	4.07	5.25	6.43	7.59	8.78	9.95	11.15	12.36	13.58		15.24	1 / 6			
reaction time	0.193	interval	1.16	1.18	1.18	1.16	1.19	1.17	1.20	1.21	1.22	1.66			3.52	3.52	3.63
velocity	4.71		7.88	7.75	7.75	7.88	7.68	7.81	7.62	7.55	7.49	8.45	7.22		7.79	7.79	7.55
Nemec, Jan (CZE) (1997)	time	2.91	4.09	5.26	6.46	7.70	8.91	10.11	11.33	12.52	13.75		15.43	7 / 7			
reaction time	0.231	interval	1.18	1.17	1.20	1.24	1.21	1.20	1.22	1.19	1.23	1.68			3.55	3.65	3.64
velocity	4.71		7.75	7.81	7.62	7.37	7.55	7.62	7.49	7.68	7.43	8.35	7.13		7.72	7.51	7.53
Bárta, Dan (CZE) (1998)	time	2.80	3.97	5.13	6.28	7.44	8.59	9.83					dnf	3 / --			
reaction time	0.169	interval		1.17	1.16	1.15	1.16	1.15	1.24						3.48	3.55	
velocity	4.90		7.81	7.88	7.95	7.88	7.95	7.37							7.88	7.72	

Heat 3

date	28-Jun-14	wind	-1.6 m/s	Feher (2014) - Final report from the U18 & U20 Czech Republic national championships													
		H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
Bárta, Dan (CZE) (1998)	time	2.79	3.94	5.08	6.22	7.33	8.50	9.67	10.86	12.08	13.28		14.90	4 / 1			
reaction time	0.166	interval	1.15	1.14	1.14	1.11	1.17	1.17	1.19	1.22	1.20	1.62			3.43	3.45	3.61
velocity	4.92		7.95	8.02	8.02	8.23	7.81	7.81	7.68	7.49	7.62	8.65	7.38		7.99	7.95	7.60
Kislaša, Jan (CZE) (1998)	time	2.88	4.05	5.20	6.35	7.50	8.65	9.81	10.98	12.18	13.39		14.99	5 / 2			
reaction time	0.168	interval	1.17	1.15	1.15	1.15	1.16	1.17	1.20	1.21	1.60				3.47	3.46	3.58
velocity	4.76		7.81	7.95	7.95	7.95	7.95	7.88	7.81	7.62	7.55	8.76	7.34		7.90	7.92	7.66
Špičák, Daniel (CZE) (1997)	time	2.83	4.04	5.21	6.39	7.56	8.74	9.92	11.16	12.37	13.61		15.18	1 / 3			
reaction time	0.170	interval	1.21	1.17	1.18	1.17	1.18	1.18	1.24	1.21	1.24	1.57			3.56	3.53	3.69
velocity	4.85		7.55	7.81	7.75	7.81	7.75	7.75	7.37	7.55	7.37	8.93	7.25		7.70	7.77	7.43
Desensky, Lukáš (CZE) (1998)	time	2.77	3.95	5.12	6.32	7.49	8.70	9.88	11.08	12.28	13.53		15.20	2 / 4			
reaction time	0.201	interval	1.18	1.17	1.20	1.17	1.21	1.18	1.20	1.20	1.25	1.67			3.55	3.56	3.65
velocity	4.95		7.75	7.81	7.62	7.81	7.55	7.75	7.62	7.62	7.31	8.40	7.24		7.72	7.70	7.51
Slaviček, Lukáš (CZE) (1997)	time	2.82	3.99	5.20	6.36	7.57	8.80	10.01	11.20	12.42	13.60		15.44	8 / 5			
reaction time	0.180	interval	1.17	1.21	1.16	1.21	1.23	1.21	1.19	1.22	1.18	1.84			3.54	3.65	3.59
velocity	4.87		7.81	7.55	7.88	7.55	7.43	7.55	7.68	7.49	7.75	7.62	7.12		7.75	7.51	7.64
Guzik, Vojtech (CZE) (1997)	time	2.83	4.04	5.23	6.41	7.60	8.80	10.01	11.21	12.50	13.79		15.54	6 / 6			
reaction time	0.170	interval	1.21	1.19	1.18	1.19	1.20	1.21	1.20	1.29	1.29	1.75			3.58	3.60	3.78
velocity	4.85		7.55	7.68	7.75	7.68	7.62	7.55	7.62	7.09	7.09	8.01	7.08		7.66	7.62	7.25
Hromek, Vít (CZE) (1997)	time	3.01	4.25	5.55	6.84	8.09	9.37	10.65	11.90	13.15	14.48		16.31	3 / 7			
reaction time	0.356	interval	1.24	1.30	1.29	1.25	1.28	1.28	1.25	1.25	1.33	1.83			3.83	3.81	3.83
velocity	4.56		7.37	7.03	7.09	7.31	7.14	7.14	7.31	7.31	6.87	7.66	6.74		7.16	7.20	7.16

Heat 2

date	28-Jun-14	wind	-4.1 m/s	Feher (2014) - Final report from the U18 & U20 Czech Republic national championships													
		H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
Sklenár, David (CZE) (1997)	time	2.73	3.83	4.93	6.04	7.12	8.21	9.33	10.44	11.50	12.65		14.30	3 / 1			
reaction time	0.197	interval	1.10	1.10	1.11	1.08	1.09	1.12	1.11	1.06	1.15	1.65			3.31	3.29	3.32
velocity	5.03		8.31	8.31	8.23	8.46	8.39	8.16	8.23	8.62	7.95	8.50	7.69		8.28	8.33	8.26
Mališka, Rostislav (CZE) (1998)	time	2.86	4.05	5.19	6.38	7.53	8.70	9.87	11.03	12.25	13.50		15.15	4 / 2			
reaction time	0.228	interval	1.19	1.14	1.19	1.15	1.17	1.17	1.16	1.22	1.25	1.65			3.52	3.49	3.63
velocity	4.80		7.68	8.02	7.68	7.95	7.81	7.81	7.88	7.49	7.31	8.50	7.26		7.79	7.86	7.55
Kijanica, Michael (CZE) (1998)	time	2.87	4.03	5.20	6.38	7.55	8.74	9.93	11.15	12.35	13.62		15.36	7 / 3			
reaction time	0.249	interval	1.16	1.17	1.18	1.17	1.19	1.19	1.22	1.20	1.27	1.74			3.51	3.55	3.69
velocity	4.78		7.88	7.81	7.75	7.81	7.68	7.68	7.49	7.62	7.20	8.06	7.16		7.81	7.72	7.43
Pirochta, Tomáš (CZE) (1997)	time	2.85	4.05	5.24	6.46	7.67	8.87	10.10	11.28	12.53	13.79		15.54	6 / 4			
reaction time	0.190	interval	1.20	1.19	1.22	1.21	1.20	1.23	1.18	1.25	1.26	1.75			3.61	3.64	3.69
velocity	4.81		7.62	7.68	7.49	7.55	7.62	7.43	7.75	7.31	7.25	8.01	7.08		7.60	7.53	7.43
Capouch, Tomáš (CZE) (1997)	time	2.86	4.06	5.27	6.49	7.75	8.98	10.21	11.46	12.67	13.91		15.62	2 / 5			
reaction time	0.193	interval	1.20	1.21	1.22	1.26	1.23	1.23	1.25	1.21	1.24	1.71			3.63	3.72	3.70
velocity	4.80		7.62	7.55	7.49	7.25	7.43	7.43	7.31	7.55	7.37	8.20	7.04		7.55	7.37	7.41
Sykora, Miroslav (CZE) (1997)	time	2.95	4.08	5.23	6.38	7.55	8.72	9.94	11.25	12.53	13.87		15.72	5 / 6			
reaction time	0.213	interval	1.13	1.15	1.15	1.17	1.17	1.22	1.31	1.28	1.34	1.85			3.43	3.56	3.93
velocity	4.65		8.09	7.95	7.95	7.81	7.81	7.49	6.98	7.14	6.82	7.58	7.00		7.99	7.70	6.98
Hoffmann, Marek (CZE) (1998)	time	2.93	4.11	5.29	6.49	7.70	8.95	10.21	11.58	12.93	14.45		16.54	8 / 7			
reaction time	0.215	interval	1.18	1.18	1.20	1.21	1.25	1.26	1.37	1.35	1.52	2.09			3.56	3.72	4.24
velocity	4.68		7.75	7.75	7.62	7.55	7.31	7.25	6.67	6.77	6.01	6.71	6.65		7.70	7.37	6.47

Heat 1

date 28-Jun-14 wind -0.3 m/s Feher (2014) - Final report from the U18 & U20 Czech Republic national championships

		H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
Puliček, Tomáš (CZE) (1997)	time	2.74	3.82	4.93	6.00	7.05	8.10	9.17	10.24	11.33	12.44		14.05	3 / 1			
	reaction time	0.247															
	interval		1.08	1.11	1.07	1.05	1.05	1.07	1.07	1.09	1.11	1.61			3.26	3.17	3.27
	velocity	5.01	8.46	8.23	8.54	8.70	8.70	8.54	8.54	8.39	8.23	8.71	7.83		8.41	8.65	8.39
Nardelli, Alexander (CZE) (1997)	time	2.87	4.04	5.21	6.37	7.50	8.65	9.79	10.96	12.12	13.34		14.93	8 / 2			
	reaction time	0.372															
	interval		1.17	1.17	1.16	1.13	1.15	1.14	1.17	1.16	1.22	1.59			3.50	3.42	3.55
	velocity	4.78	7.81	7.81	7.88	8.09	7.95	8.02	7.81	7.88	7.49	8.82	7.37		7.83	8.02	7.72
Nemec, Jan (CZE) (1997)	time	2.89	4.09	5.30	6.44	7.60	8.76	9.94	11.11	12.28	13.47		15.05	4 / 3			
	reaction time	0.236															
	interval		1.20	1.21	1.14	1.16	1.16	1.18	1.17	1.17	1.19	1.58			3.55	3.50	3.53
	velocity	4.75	7.62	7.55	8.02	7.88	7.88	7.75	7.81	7.81	7.68	8.87	7.31		7.72	7.83	7.77
Kraus, Patrik (CZE) (1998)	time	2.84	4.08	5.30	6.53	7.70	8.90	10.08	11.31	12.57	13.81		15.52	2 / 4			
	reaction time	0.179															
	interval		1.24	1.22	1.23	1.17	1.20	1.18	1.23	1.26	1.24	1.71			3.69	3.55	3.73
	velocity	4.83	7.37	7.49	7.43	7.81	7.62	7.75	7.43	7.25	7.37	8.20	7.09		7.43	7.72	7.35
Babiš, Ivo (CZE) (1997)	time	2.99	4.16	5.33	6.49	7.72	8.95	10.15	11.38	12.60	13.96		15.74	6 / 5			
	reaction time	0.258															
	interval		1.17	1.17	1.16	1.23	1.23	1.20	1.23	1.22	1.36	1.78			3.50	3.66	3.81
	velocity	4.59	7.81	7.81	7.88	7.43	7.43	7.62	7.43	7.49	6.72	7.88	6.99		7.83	7.49	7.20
Záhora, Vojtěch (CZE) (1998)	time	2.92	4.15	5.36	6.62	7.82	9.04	10.30	11.53	12.79	14.10		15.84	5 / 6			
	reaction time	0.180															
	interval		1.23	1.21	1.26	1.20	1.22	1.26	1.23	1.26	1.31	1.74			3.70	3.68	3.80
	velocity	4.70	7.43	7.55	7.25	7.62	7.49	7.25	7.43	7.25	6.98	8.06	6.94		7.41	7.45	7.22
Večera, Vincent (CZE) (1998)	time	2.88	4.15	5.37	6.69	8.00	9.28	10.57	11.87	13.16	14.48		16.25	7 / 7			
	reaction time	0.219															
	interval		1.27	1.22	1.32	1.31	1.28	1.29	1.30	1.29	1.32	1.77			3.81	3.88	3.91
	velocity	4.76	7.20	7.49	6.92	6.98	7.14	7.09	7.03	7.09	6.92	7.92	6.77		7.20	7.07	7.01

2014 Japanese National Championships (Fukushima, JPN)

FINAL

FINAL	date	08-Jun-14	wind	(2017.02.22) - https://twitter.com/touchdown_time/media?lang=en&lang=en&lang=en&lang=en															
			H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10	
Masuno, Genta (JPN) (1993)	time		2.64	3.71	4.78	5.82	6.87	7.92	8.98	10.04	11.10	12.18		13.58	5 / 1				
	reaction time	0.144																	
	interval			1.07	1.07	1.04	1.05	1.05	1.06	1.06	1.06	1.08	1.40	PB			3.18	3.16	3.20
	velocity		5.20	8.54	8.54	8.79	8.70	8.70	8.62	8.62	8.62	8.46	10.01	8.10			8.62	8.68	8.57
Yazawa, Wataru (JPN) (1991)	time		2.60	3.69	4.74	5.78	6.81	7.87	8.93	10.00	11.07	12.15		13.59	4 / 2				
	reaction time	0.137																	
	interval			7 steps	1.09	1.05	1.04	1.03	1.06	1.07	1.07	1.08	1.44	=PB			3.18	3.15	3.22
	velocity		5.28	8.39	8.70	8.79	8.87	8.62	8.62	8.54	8.54	8.46	9.74	8.09			8.62	8.70	8.52

Semi-Final 2

	date	08-Jun-14	wind	-0.5 m/s	(2017.02.26) - https://twitter.com/touchdown_time/media?lang=en&lang=en&lang=en&lang=en												
Yazawa, Wataru (JPN) (1991)	time	2.63	3.71	4.75	5.79	6.84	7.89	8.94	10.01	11.08	12.17		13.60	6 / 1			
	reaction time	0.147															
	interval		7 steps	1.08	1.04	1.04	1.05	1.05	1.07	1.07	1.09	1.43			3.16	3.15	3.23
	velocity	5.22	8.46	8.79	8.79	8.70	8.70	8.70	8.54	8.54	8.39	9.80	8.09		8.68	8.70	8.49

Semi-Final 1

	date	08-Jun-14	wind	0.2 m/s	(2017.02.27) - https://twitter.com/touchdown_time/media?lang=en&lang=en&lang=en&lang=en												
Masuno, Genta (JPN) (1993)	time	2.65	3.71	4.75	5.80	6.85	7.90	8.95	10.02	11.09	12.19		13.61	5 / 1			
	reaction time	0.148															
	interval		1.06	1.04	1.05	1.05	1.05	1.05	1.07	1.07	1.10	1.42	=PB		3.15	3.15	3.24
	velocity	5.18	8.62	8.79	8.70	8.70	8.70	8.70	8.54	8.54	8.31	9.87	8.08		8.70	8.70	8.46

2014 Japanese National Multi-Event Championships (Nagano, JPN)

Decathlon - Heat 1

	date	01-Jun-14	wind	0.1 m/s	Takeo (2014) - decathletes sprint ability and the relationship of the individual event performance												
Nakamura, Akihiko (JPN) (1990)	time	2.68	3.79	4.93	6.04	7.14	8.27	9.40	10.52	11.64	12.77		14.21	5 / 2			
	reaction time																
	interval		1.11	1.14	1.11	1.10	1.13	1.13	1.12	1.12	1.13	1.44			3.36	3.36	3.37
	velocity	5.12	8.23	8.02	8.23	8.31	8.09	8.09	8.16	8.16	8.09	9.74	7.74		8.16	8.16	8.14
Ushiro, Keisuke (JPN) (1986)	time	2.76	3.92	5.06	6.21	7.38	8.54	9.71	10.91	12.08	13.30		14.90	1 / 6			
	reaction time																
	interval		1.16	1.14	1.15	1.17	1.16	1.17	1.20	1.17	1.22	1.60			3.45	3.50	3.59
	velocity	4.97	7.88	8.02	7.95	7.81	7.88	7.81	7.62	7.81	7.49	8.76	7.38		7.95	7.83	7.64

2014 Golden Grand Prix (Shanghai, CHN)

FINAL

	date	18-May-14	wind	-0.5 m/s	Wang (2016) - comparative analysis of the grey relational grade of segment times and final results of the 110m hurdles												
Xie Wenjun (CHN) (1990)	time	2.54	3.60	4.62	5.62	6.62	7.64	8.66	9.70	10.74	11.82		13.23	1 / 1			
	reaction time	0.136															
	interval		1.06	1.02	1.00	1.00	1.02	1.02	1.04	1.04	1.08	1.41	PB		3.08	3.04	3.16
	velocity	5.40	8.62	8.96	9.14	9.14	8.96	8.96	8.79	8.79	8.46	9.94	8.31		8.90	9.02	8.68

2014 Nittai University Distance Meeting (Wakayama, JPN)

Decathlon - Heat 2

	date	27-Apr-14	wind	0.0 m/s	Takeo (2014) - decathletes sprint ability and the relationship of the individual event performance												
Ushiro, Keisuke (JPN) (1986)	time	2.79	3.92	5.07	6.24	7.41	8.58	9.78	10.97	12.16	13.38		14.95	5 / 1			
	reaction time																
	interval		1.13	1.15	1.17	1.17	1.17	1.20	1.19	1.19	1.22	1.57			3.45	3.54	3.60

velocity 4.92 8.09 7.95 7.81 7.81 7.81 7.62 7.68 7.68 7.49 8.93 7.36 7.95 7.75 7.62
Matsuo (2014) - research on athlete performance and technique- 2014 data book

Decathlon - Heat 1

date 27-Apr-14

wind 1.0 m/s

Takeo (2014) - decathletes sprint ability and the relationship of the individual event performance

		H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
Nakamura, Akihiko (JPN) (1990)	time	2.71	3.84	4.97	6.09	7.21	8.32	9.43	10.54	11.67	12.80		14.23	4 / 1			<i>Takeo (2014)</i>
	reaction time	interval	1.13	1.13	1.12	1.12	1.11	1.11	1.11	1.13	1.13	1.43			3.38	3.34	3.37
	velocity	5.06	8.09	8.09	8.16	8.16	8.23	8.23	8.23	8.09	8.09	9.80	7.73		8.11	8.21	8.14
Otobe, Takumi (JPN) (1989)	time	2.70	3.90	5.06	6.21	7.36	8.52	9.66	10.84	12.00	13.19		14.74	6 / 3			<i>Matsuo (2014)</i>
	reaction time	interval	1.20	1.16	1.15	1.15	1.16	1.14	1.18	1.16	1.19	1.55	PB		3.51	3.45	3.53
	velocity	5.08	7.62	7.88	7.95	7.95	7.88	8.02	7.75	7.88	7.68	9.05	7.46		7.81	7.95	7.77

2014 Italian National Indoor Championships (Ancona, ITA)**FINAL**

date 22-Feb-14

FIDAL - Centro Studi & Ricerche (2014) - www.youtube.com/watch?v=bF4QEg7shU

		H1	H2	H3	H4	H5						Run-In	Official Time	Lane / Place	H1-H4
Dal Molin, Paolo (ITA) (1987)	time	2.56	3.63	4.68	5.69	6.72							7.60	3 / 1	
	reaction time	interval	1.07	1.05	1.01	1.03						0.88			3.13
	velocity	5.36	8.54	8.70	9.05	8.87						11.05	7.89		8.76

2013 Japanese Corporate Championships (Hiratsuka, JPN)**FINAL**

date 19-Oct-13

wind 6.7 m/s

(2016.11.18) - https://twitter.com/touchdown_time/media?lang=en&lang=en&lang=en&lang=en

		H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
Yazawa, Wataru (JPN) (1991)	time	2.72	3.80	4.85	5.89	6.93	7.95	9.00	10.06	11.12	12.21		13.60	4 / 1			
	reaction time	interval	1.08	1.05	1.04	1.04	1.02	1.05	1.06	1.06	1.09	1.39			3.17	3.11	3.21
	velocity	5.04	8.46	8.70	8.79	8.79	8.96	8.70	8.62	8.62	8.39	10.09	8.09		8.65	8.82	8.54

2013 IAAF World Championships (Moscow, RUS) (TV Analysis)**FINAL**

date 12-Aug-13

wind 0.3 m/s

Henson (2021) - Athlete First: major championships report

		H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10	
Oliver, David (USA) (1982)	time	2.44	3.48	4.48	5.46	6.46	7.44	8.44	9.46	10.48	11.52		13.00	4 / 1				
	reaction time	0.147	interval	7 steps	1.04	1.00	0.98	1.00	0.98	1.00	1.02	1.02	1.04	1.48		3.02	2.98	3.08
		velocity	5.62	8.79	9.14	9.33	9.14	9.33	9.14	8.96	8.96	8.79	9.47	8.46		9.08	9.20	8.90
Wilson, Ryan (USA) (1980)	time	2.52	3.56	4.56	5.54	6.52	7.50	8.50	9.52	10.54	11.62		13.13	9 / 2				
	reaction time	0.167	interval	7 steps	1.04	1.00	0.98	0.98	1.00	1.02	1.02	1.08	1.51		3.02	2.96	3.12	
		velocity	5.44	8.79	9.14	9.33	9.33	9.33	9.14	8.96	8.96	8.46	9.28	8.38		9.08	9.26	8.79
Shubenkov, Sergey (RUS) (199	time	2.56	3.58	4.58	5.56	6.58	7.62	8.66	9.68	10.72	11.80		13.24	5 / 3				
	reaction time	0.154	interval		1.02	1.00	0.98	1.02	1.04	1.02	1.04	1.08	1.44		3.00	3.10	3.14	
		velocity	5.36	8.96	9.14	9.33	8.96	8.79	8.79	8.96	8.79	8.46	9.74	8.31		9.14	8.85	8.73
Richardson, Jason (USA) (198	time	2.56	3.56	4.56	5.58	6.58	7.58	8.60	9.62	10.64	11.68		13.27	7 / 4				
	reaction time	0.158	interval	7 steps	1.00	1.00	1.02	1.00	1.00	1.02	1.02	1.02	1.04	1.59		3.02	3.02	3.08
		velocity	5.36	9.14	9.14	8.96	9.14	9.14	8.96	8.96	8.96	8.79	8.82	8.29		9.08	9.08	8.90
Sharman, William (GBR) (1984)	time	2.54	3.56	4.60	5.60	6.62	7.64	8.68	9.72	10.76	11.84		13.30	2 / 5				
	reaction time	0.151	interval	7 steps	1.02	1.04	1.00	1.02	1.04	1.04	1.04	1.08	1.46		3.06	3.08	3.16	
		velocity	5.40	8.96	8.79	9.14	8.96	8.96	8.79	8.79	8.79	8.46	9.60	8.27		8.96	8.90	8.68
Merritt, Aries (USA) (1985)	time	2.50	3.54	4.54	5.54	6.54	7.54	8.58	9.62	10.68	11.78		13.31	8 / 6				
	reaction time	0.147	interval	7 steps	1.04	1.00	1.00	1.00	1.04	1.04	1.06	1.10	1.53		3.04	3.04	3.20	
		velocity	5.49	8.79	9.14	9.14	9.14	9.14	8.79	8.79	8.62	8.31	9.16	8.26		9.02	9.02	8.57
Martinot-Lagarde, Thomas (FR	time	2.52	3.56	4.56	5.60	6.64	7.66	8.68	9.72	10.80	11.90		13.42	6 / 7				
	reaction time	0.162	interval	7 steps	1.04	1.00	1.04	1.04	1.02	1.04	1.08	1.10	1.52		3.08	3.08	3.22	
		velocity	5.44	8.79	9.14	8.79	8.79	8.96	8.96	8.79	8.46	8.31	9.22	8.20		8.90	8.90	8.52
Riley, Andrew (JAM) (1988)	time	2.48	3.52	4.54	5.54	6.56	7.56	8.58	9.56	10.74	11.92		13.51	3 / 8				
	reaction time	0.135	interval	7 steps	1.04	1.02	1.00	1.02	1.02	0.98	1.18	1.18	1.59		3.06	3.04	3.34	
		velocity	5.53	8.79	8.96	9.14	8.96	9.14	8.96	9.33	7.75	7.75	8.82	8.14		8.96	9.02	8.21

2013 Windsprint (Sundsvall, SWE)**FINAL**

date 28-Jul-13

wind 1.7 m/s

(2013) - www.ssg.se/windsprint/Splittider/

		H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
Nossmys, Philip (SWE) (1982)	time	2.62	3.68	4.73	5.78	6.81	7.86	8.91	9.96	11.02	12.08		13.49	3 / 1			
	reaction time	interval	1.06	1.05	1.05	1.03	1.05	1.05	1.05	1.06	1.06	1.41			3.16	3.13	3.17
	velocity	5.24	8.62	8.70	8.70	8.87	8.70	8.70	8.70	8.62	8.62	9.94	8.15		8.68	8.76	8.65
Townsend, Fred (USA) (1982)	time	2.62	3.69	4.73	5.78	6.81	7.86	8.93	9.99	11.04	12.13		13.53	5 / 2			
	reaction time	interval	7 steps	1.07	1.04	1.05	1.03	1.05	1.07	1.06	1.05	1.09	1.40		3.16	3.15	3.20
	velocity	5.24	8.54	8.79	8.70	8.87	8.70	8.54	8.62	8.70	8.39	10.01	8.13		8.68	8.70	8.57
Buhler, Mattias (GER) (1986)	time	2.72	3.78	4.84	5.87	6.92	7.95	8.99	10.03	11.07	12.14		13.64	7 / 3			
	reaction time	interval	1.06	1.06	1.03	1.05	1.03	1.04	1.04	1.04	1.07	1.50			3.15	3.12	3.15
	velocity	5.04	8.62	8.62	8.87	8.70	8.87	8.79	8.79	8.79	8.54	9.35	8.06		8.70	8.79	8.70

de Vries, Ruan (RSA) (1986)	time	2.68	3.77	4.83	5.89	6.95	8.00	9.04	10.10	11.16	12.24		13.70	6 / 4			
	reaction time	interval	1.09	1.06	1.06	1.06	1.05	1.04	1.06	1.06	1.08	1.46			3.21	3.15	3.20
	velocity	5.12	8.39	8.62	8.62	8.62	8.70	8.79	8.62	8.62	8.46	9.60	8.03		8.54	8.70	8.57
Vukicevic, Vladimir (NOR) (1991)	time	2.62	3.68	4.73	5.80	6.85	7.92	8.99	10.06	11.13	12.23		13.70	4 / 5			
	reaction time	interval	1.06	1.05	1.07	1.05	1.07	1.07	1.07	1.07	1.10	1.47			3.18	3.19	3.24
	velocity	5.24	8.62	8.70	8.54	8.70	8.54	8.54	8.54	8.54	8.31	9.54	8.03		8.62	8.60	8.46
Shaker, Amir (SWE) (1989)	time	2.70	3.79	4.84	5.91	6.97	8.05	9.14	10.21	11.29	12.40		13.87	2 / 6			
	reaction time	interval	1.09	1.05	1.07	1.06	1.08	1.09	1.07	1.08	1.11	1.47			3.21	3.23	3.26
	velocity	5.08	8.39	8.70	8.54	8.62	8.46	8.39	8.54	8.46	8.23	9.54	7.93		8.54	8.49	8.41
Widlund, Hampus (SWE) (1993)	time	2.76	3.89	5.00	6.11	7.23	8.38	9.53	10.65	11.78	12.93		14.46	1 / 7			
	reaction time	interval	7 steps 1.13	1.11	1.11	1.12	1.15	1.15	1.12	1.13	1.15	1.53			3.35	3.42	3.40
	velocity	4.97	8.09	8.23	8.23	8.16	7.95	7.95	8.16	8.09	7.95	9.16	7.61		8.19	8.02	8.06
Andersson, Andreas (SWE) (1981)	time	2.78	3.96	5.08	6.20	7.35	8.56	9.75	10.94	12.14	13.35		14.97	8 / 8			
	reaction time	interval	1.18	1.12	1.12	1.15	1.21	1.19	1.19	1.20	1.21	1.62	PB		3.42	3.55	3.60
	velocity	4.94	7.75	8.16	8.16	7.95	7.55	7.68	7.68	7.62	7.55	8.65	7.35		8.02	7.72	7.62

2013 Italian National Junior Championships (Rieti, ITA) (99cm)

FINAL

INAL	date	15-Jun-13	wind	1.6 m/s										FIDAL - Centro Studi & Ricerche (2013) - www.youtube.com/watch?v=IGr3fQGxe1c							
			H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10			
	Perini, Lorenzo (ITA) (1994)	time	2.70	3.76	4.80	5.82	6.83	7.84	8.86	9.89	10.92	11.95		13.39	6 / 1						
	reaction time	interval		1.06	1.04	1.02	1.01	1.01	1.02	1.03	1.03	1.03	1.44	NR		3.12	3.04	3.09			
		velocity	5.08	8.62	8.79	8.96	9.05	9.05	8.96	8.87	8.87	8.87	9.74	8.22		8.79	9.02	8.87			

2013 Czech Republic National Championships (Tábor, CZE)

FINAL

INAL		date	15-Jun-13	wind 0.4 m/s			Feher (2013) - Final report from the men's & women's Czech Republic national championships												
				H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
Mazáč, Martin (CZE) (1990)	time			2.66	3.72	4.76	5.78	6.79	7.81	8.88	9.95	11.00	12.04		13.48	4 / 1			
	reaction time	0.151	interval		1.06	1.04	1.02	1.01	1.02	1.07	1.07	1.05	1.04	1.44	PB		3.12	3.10	3.16
			velocity	5.16	8.62	8.79	8.96	9.05	8.96	8.54	8.54	8.70	8.79	9.74	8.16		8.79	8.85	8.68
Penáz, Petr (CZE) (1991)	time			2.78	3.85	4.93	5.96	7.00	8.04	9.10	10.18	11.25	12.29		13.77	5 / 2			
	reaction time	0.202	interval		1.07	1.08	1.03	1.04	1.04	1.06	1.08	1.07	1.04	1.48	PB		3.18	3.14	3.19
			velocity	4.94	8.54	8.46	8.87	8.79	8.79	8.62	8.46	8.54	8.79	9.47	7.99		8.62	8.73	8.60
Carda, Jan (CZE) (1989)	time			2.66	3.76	4.84	5.91	6.99	8.09	9.23	10.37	11.51	12.69		14.22	2 / 3			
	reaction time	0.135	interval		1.10	1.08	1.07	1.08	1.10	1.14	1.14	1.14	1.18	1.53	PB		3.25	3.32	3.46
			velocity	5.16	8.31	8.46	8.54	8.46	8.31	8.02	8.02	8.02	7.75	9.16	7.74		8.44	8.26	7.92
Sedlák, Václav (CZE) (1993)	time			2.73	3.85	4.96	6.05	7.13	8.25	9.40	10.53	11.68	12.87		14.36	3 / 4			
	reaction time	0.147	interval		1.12	1.11	1.09	1.08	1.12	1.15	1.13	1.15	1.19	1.49			3.32	3.35	3.47
			velocity	5.03	8.16	8.23	8.39	8.46	8.16	7.95	8.09	7.95	7.68	9.41	7.66		8.26	8.19	7.90
Kavka, Tomáš (CZE) (1989)	time			2.80	3.92	5.05	6.13	7.24	8.38	9.50	10.65	11.79	12.95		14.54	6 / 5			
	reaction time	0.161	interval		1.12	1.13	1.08	1.11	1.14	1.12	1.15	1.14	1.16	1.59			3.33	3.37	3.45
			velocity	4.90	8.16	8.09	8.46	8.23	8.02	8.16	7.95	8.02	7.88	8.82	7.57		8.23	8.14	7.95
Lukáš, Marek (CZE) (1991)	time			2.75	3.87	5.00	6.10	7.24	8.38	9.53	10.68	11.85	13.05		14.58	1 / 6			
	reaction time	0.177	interval		1.12	1.13	1.10	1.14	1.14	1.15	1.15	1.17	1.20	1.53	PB		3.35	3.43	3.52
			velocity	4.99	8.16	8.09	8.31	8.02	8.02	7.95	7.95	7.81	7.62	9.16	7.54		8.19	7.99	7.79
Krejčí, Michal (CZE) (1981)	time			2.75	3.88	5.00	6.13	7.24	8.38	9.50	10.65	11.84	13.03		14.62	8 / 7			
	reaction time	0.149	interval		1.13	1.12	1.13	1.11	1.14	1.12	1.15	1.19	1.19	1.59			3.38	3.37	3.53
			velocity	4.99	8.09	8.16	8.09	8.23	8.02	8.16	7.95	7.68	7.68	8.82	7.52		8.11	8.14	7.77
Nejedly, Adam (CZE) (1988)	time			2.78	3.95	5.07	6.18	7.32	8.45	9.62	10.78	11.95	13.14		14.68	7 / 8			
	reaction time	0.145	interval		1.17	1.12	1.11	1.14	1.13	1.17	1.16	1.17	1.19	1.54			3.40	3.44	3.52
			velocity	4.94	7.81	8.16	8.23	8.02	8.09	7.81	7.88	7.81	7.68	9.10	7.49		8.06	7.97	7.79

Heat 3

Heat 3	date	15-Jun-13	wind	-0.3 m/s	Feher (2013) - Final report from the men's & women's Czech Republic national championships														
			H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10	
Sedlák, Václav (CZE) (1993)	time		2.75	3.91	5.03	6.13	7.26	8.38	9.52	10.67	11.84	13.00		14.56	4 / 1				
	reaction time	0.131		1.16	1.12	1.10	1.13	1.12	1.14	1.15	1.17	1.16	1.56				3.38	3.39	3.48
	interval		4.99	7.88	8.16	8.31	8.09	8.16	8.02	7.95	7.81	7.88	8.99	7.55			8.11	8.09	7.88
Kavka, Tomáš (CZE) (1989)	time		2.81	3.98	5.08	6.19	7.32	8.45	9.58	10.70	11.87	13.03		14.59	5 / 2				
	reaction time	0.161		1.17	1.10	1.11	1.13	1.13	1.13	1.12	1.17	1.16	1.56				3.38	3.39	3.45
	interval		4.88	7.81	8.31	8.23	8.09	8.09	8.09	8.16	7.81	7.88	8.99	7.54			8.11	8.09	7.95
Lukáš, Marek (CZE) (1991)	time		2.78	3.91	5.03	6.17	7.32	8.46	9.62	10.79	12.00	13.16		14.75	3 / 3				
	reaction time	0.176		1.13	1.12	1.14	1.15	1.14	1.16	1.17	1.21	1.16	1.59				3.39	3.45	3.54
	interval		4.94	8.09	8.16	8.02	7.95	8.02	7.88	7.81	7.55	7.88	8.82	7.46			8.09	7.95	7.75

Uher, Jakub (CZE) (1983)	time	2.78	3.94	5.07	6.23	7.39	8.53	9.73	10.90	12.11	13.29		14.92	6 / 4			
	reaction time	0.170		1.16	1.13	1.16	1.16	1.14	1.20	1.17	1.21	1.18	1.63		3.45	3.50	3.56
	velocity	4.94	7.88	8.09	7.88	7.88	8.02	7.62	7.81	7.55	7.75	8.60	7.37		7.95	7.83	7.70
Frolík, Petr (CZE) (1991)	time	2.92	4.13	5.31	6.51	7.75	8.98	10.26	11.51	12.81	14.09		15.78	2 / 5			
	reaction time	0.204		1.21	1.18	1.20	1.24	1.23	1.28	1.25	1.30	1.28	1.69		3.59	3.75	3.83
	velocity	4.70	7.55	7.75	7.62	7.37	7.43	7.14	7.31	7.03	7.14	8.30	6.97		7.64	7.31	7.16
Pavelka, Lukáš (CZE) (1990)	time	2.92	4.12	5.33	6.55	7.78	9.05	10.31	11.55	12.89	14.20		15.95	7 / 6			
	reaction time	0.207		1.20	1.21	1.22	1.23	1.27	1.26	1.24	1.34	1.31	1.75		3.63	3.76	3.89
	velocity	4.70	7.62	7.55	7.49	7.43	7.20	7.25	7.37	6.82	6.98	8.01	6.90		7.55	7.29	7.05

Heat 2

Heat 2

	date	15-Jun-13	wind	-0.6 m/s	Feher (2013) - Final report from the men's & women's Czech Republic national championships														
			H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10	
Penáz, Petr (CZE) (1991)	time		2.79	3.90	5.00	6.07	7.13	8.19	9.26	10.35	11.45	12.58		14.08	4 / 1				
	reaction time	0.185		1.11	1.10	1.07	1.06	1.06	1.07	1.09	1.10	1.13	1.50				3.28	3.19	3.32
	velocity		4.92	8.23	8.31	8.54	8.62	8.62	8.54	8.39	8.31	8.09	9.35	7.81			8.36	8.60	8.26
Nejedlý, Adam (CZE) (1988)	time		2.81	3.95	5.06	6.18	7.31	8.44	9.59	10.75	11.91	13.08		14.63	3 / 2				
	reaction time	0.142		1.14	1.11	1.12	1.13	1.13	1.15	1.16	1.16	1.17	1.55				3.37	3.41	3.49
	velocity		4.88	8.02	8.23	8.16	8.09	8.09	7.95	7.88	7.88	7.81	9.05	7.52			8.14	8.04	7.86
Carda, Jan (CZE) (1989)	time		2.70	3.83	4.96	6.05	7.17	8.30	9.48	10.68	11.85	13.08		14.64	5 / 3				
	reaction time	0.141		1.13	1.13	1.09	1.12	1.13	1.18	1.20	1.17	1.23	1.56				3.35	3.43	3.60
	velocity		5.08	8.09	8.09	8.39	8.16	8.09	7.75	7.62	7.81	7.43	8.99	7.51			8.19	7.99	7.62
Pokorný, Martin (CZE) (1988)	time		2.77	3.94	5.10	6.26	7.43	8.58	9.75	10.98	12.20	13.46		15.21	6 / 4				
	reaction time	0.155		1.17	1.16	1.16	1.17	1.15	1.17	1.23	1.22	1.26	1.75				3.49	3.49	3.71
	velocity		4.95	7.81	7.88	7.88	7.81	7.95	7.81	7.43	7.49	7.25	8.01	7.23			7.86	7.86	7.39
Dinga, Libor (CZE) (1984)	time		2.95	4.18	5.37	6.48	7.80	9.00	10.25	11.46	12.73	14.02		15.61	2 / 5				
	reaction time	0.166		1.23	1.19	1.11	1.32	1.20	1.25	1.21	1.27	1.29	1.59				3.53	3.77	3.77
	velocity		4.65	7.43	7.68	8.23	6.92	7.62	7.31	7.55	7.20	7.09	8.82	7.05			7.77	7.27	7.27
Lindhart, David (CZE) (1989)	time		2.97	4.20	5.42	6.65	7.84	9.07	10.30	11.55	12.81		15.79	7 / 6					
	reaction time	0.262		1.23	1.22	1.23	1.19	1.23	1.23	1.25	1.26						3.68	3.65	
	velocity		4.62	7.43	7.49	7.43	7.68	7.43	7.43	7.31	7.25			6.97			7.45	7.51	

Heat 1

Heat 1		date	15-Jun-13	wind -0.6 m/s			Feher (2013) - Final report from the men's & women's Czech Republic national championships												
				H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
Mazáč, Martin (CZE) (1990)	time			2.66	3.75	4.80	5.85	6.89	7.95	8.98	10.06	11.12	12.20		13.69	4 / 1			
	reaction time	0.161	interval		1.09	1.05	1.05	1.04	1.06	1.03	1.08	1.06	1.08	1.49			3.19	3.13	3.22
			velocity	5.16	8.39	8.70	8.70	8.79	8.62	8.87	8.46	8.62	8.46	9.41	8.04			8.60	8.76
Krejčí, Michal (CZE) (1981)	time			2.71	3.86	4.99	6.13	7.27	8.40	9.54	10.70	11.88	13.08		14.66	5 / 2			
	reaction time	0.126	interval		1.15	1.13	1.14	1.14	1.13	1.14	1.16	1.18	1.20	1.58			3.42	3.41	3.54
			velocity	5.06	7.95	8.09	8.02	8.02	8.09	8.02	7.88	7.75	7.62	8.87	7.50			8.02	8.04
Polách, Josef (CZE) (1990)	time			2.74	3.88	5.00	6.17	7.31	8.48	9.59	10.77	11.95	13.12		14.76	3 / 3			
	reaction time	0.151	interval		1.14	1.12	1.17	1.14	1.17	1.11	1.18	1.18	1.17	1.64			3.43	3.42	3.53
			velocity	5.01	8.02	8.16	7.81	8.02	7.81	8.23	7.75	7.75	7.81	8.55	7.45			7.99	8.02
Novotny, Šimon (CZE) (1992)	time			2.86	4.06	5.27	6.47	7.67	8.88	10.11	11.35	12.62		15.58	6 / 4				
	reaction time	0.170	interval		1.20	1.21	1.20	1.20	1.21	1.23	1.24	1.27					3.61	3.64	
			velocity	4.80	7.62	7.55	7.62	7.62	7.55	7.43	7.37	7.20			7.06			7.60	7.53
Štastny, Michal (CZE) (1992)	time			2.88	4.15	5.43	6.75	8.07		10.73	12.09			16.54	7 / 5				
	reaction time	0.193	interval		1.27	1.28	1.32	1.32		2.66	1.36						3.87	3.98	
			velocity	4.76	7.20	7.14	6.92	6.92		6.87	6.72			6.65			7.09	6.89	

2013 Japanese National Championships (Chofu, JPN)

FINAL

INAL			date	09-Jun-13	wind 1.3 m/s				Kota (2013) - race analysis of leading Japanese men's 110m hurdlers - from 2011-13												
					H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10	
Yazawa, Wataru (JPN) (1991)	time				2.60	3.69	4.76	5.81	6.86	7.89	8.95	10.01	11.07	12.18		13.59	7 / 1				
	reaction time				7 steps	1.09	1.07	1.05	1.05	1.03	1.06	1.06	1.06	1.11	1.41	PB		3.21	3.14	3.23	
	velocity				5.28	8.39	8.54	8.70	8.70	8.87	8.62	8.62	8.62	8.23	9.94	8.09		8.54	8.73	8.49	
Sato, Hiroyuki (JPN) (1990)	time				2.60	3.67	4.72	5.77	6.83	7.88	8.93	10.01	11.09	12.20		13.61	8 / 2				
	reaction time					1.07	1.05	1.05	1.06	1.05	1.05	1.08	1.08	1.11	1.41	PB		3.17	3.16	3.27	
	velocity				5.28	8.54	8.70	8.70	8.62	8.70	8.70	8.46	8.46	8.23	9.94	8.08		8.65	8.68	8.39	
Omuro, Hideki (JPN) (1990)	time				2.58	3.64	4.70	5.76	6.82	7.87	8.97	10.05	11.13	12.22		13.67	5 / 4				
	reaction time					1.06	1.06	1.06	1.06	1.05	1.10	1.08	1.08	1.09	1.45			3.18	3.21	3.25	
	velocity				5.32	8.62	8.62	8.62	8.62	8.70	8.31	8.46	8.46	8.39	9.67	8.05		8.62	8.54	8.44	

Semi-Final 2

emi-Final 2		date	09-Jun-13	wind	Kota (2013) - race analysis of leading Japanese men's 110m hurdlers - from 2011-13															
				H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10	
Yazawa, Wataru (JPN) (1991)	time			2.62	3.71	4.75	5.82	6.86	7.91	8.97	10.03	11.11	12.21		13.63	6 / 1				
	reaction time			interval	7 steps	1.09	1.04	1.07	1.04	1.05	1.06	1.08	1.10	1.42			3.20	3.15	3.24	
	velocity			5.24	8.39	8.79	8.54	8.79	8.70	8.62	8.62	8.46	8.31	9.87	8.07		8.57	8.70	8.46	
Sato, Hiroyuki (JPN) (1990)	time			2.60	3.68	4.73	5.79	6.86	7.91	8.98	10.06	11.20	12.29		13.71	4 / 3				
	reaction time			interval		1.08	1.05	1.06	1.07	1.05	1.07	1.08	1.14	1.09	1.42		3.19	3.19	3.31	
	velocity			5.28	8.46	8.70	8.62	8.54	8.70	8.54	8.46	8.02	8.39	9.87	8.02		8.60	8.60	8.28	
Mita, Kyohei (JPN) (1991)	time			2.69	3.79	4.88	5.96	7.04	8.11	9.19	10.26	11.35	12.45		13.88	2 / 5				
	reaction time			interval		1.10	1.09	1.08	1.08	1.07	1.08	1.07	1.09	1.10	1.43		3.27	3.23	3.26	
	velocity			5.10	8.31	8.39	8.46	8.46	8.54	8.46	8.54	8.39	8.31	9.80	7.93		8.39	8.49	8.41	

Semi-Final 1

emi-Final 1		date	09-Jun-13		wind		1.5 m/s		Kota (2013) - race analysis of leading Japanese men's 110m hurdlers - from 2011-13												
					H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10	
Omuro, Hideki (JPN) (1990)	time				2.61	3.66	4.71	5.75	6.80	7.86	8.93	10.01	11.07	12.16		13.58	7 / 1				
	reaction time				interval		1.05	1.05	1.04	1.05	1.06	1.07	1.08	1.06	1.09	1.42		3.14	3.18	3.23	
	velocity				5.26	8.70	8.70	8.79	8.70	8.62	8.54	8.46	8.62	8.39	9.87	8.10		8.73	8.62	8.49	

Heat 4

Heat 4	date	08-Jun-13	wind	Kota (2013) - race analysis of leading Japanese men's 110m hurdlers - from 2011-13															
				H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
	Yazawa, Wataru (JPN) (1991)	time		2.61	3.70	4.75	5.79	6.83	7.88	8.95	10.03	11.12	12.21		13.62	6 / 1			
	reaction time	interval	7 steps	1.09	1.05	1.04	1.04	1.05	1.07	1.08	1.09	1.09	1.41		PB		3.18	3.16	3.26
	velocity		5.26	8.39	8.70	8.79	8.79	8.70	8.54	8.46	8.39	8.39	9.94		8.08		8.62	8.68	8.41

Heat 3

Heat 3	date	08-Jun-13	wind -0.4 m/s										Kota (2013) - race analysis of leading Japanese men's 110m hurdlers - from 2011-13							
			H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10		
	Wado, Tatsuya (JPN) (1990)	time	2.63	3.74	4.82	5.91	7.02	8.11	9.22	10.32	11.45	12.63		14.15	5 / 4					
	reaction time	interval		1.11	1.08	1.09	1.11	1.09	1.11	1.10	1.13	1.18	1.52			3.28	3.31	3.41		
	velocity		5.22	8.23	8.46	8.39	8.23	8.39	8.23	8.31	8.09	7.75	9.22	7.77		8.36	8.28	8.04		

Heat 2

Heat 2

	date	08-Jun-13	wind	0.1 m/s	Kota (2013) - race analysis of leading Japanese men's 110m hurdlers - from 2011-13														
			H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10	
Omuro, Hideki (JPN) (1990)	time		2.62	3.70	4.76	5.83	6.91	8.00	9.08	10.19	11.29	12.42		13.85	6 / 1				
	reaction time		interval		1.08	1.06	1.07	1.08	1.09	1.08	1.11	1.10	1.13	1.43			3.21	3.25	3.34
		velocity		5.24	8.46	8.62	8.54	8.46	8.39	8.46	8.23	8.31	8.09	9.80	7.94		8.54	8.44	8.21

Heat 1

Heat 1

	date	08-Jun-13	wind	-0.1 m/s	Kota (2013) - race analysis of leading Japanese men's 110m hurdlers - from 2011-13														
			H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10	
Sato, Hiroyuki (JPN) (1990)	time		2.65	3.72	4.79	5.86	6.93	8.00	9.08	10.19	11.29	12.41		13.87	4 / 1				
	reaction time			1.07	1.07	1.07	1.07	1.07	1.08	1.11	1.10	1.12	1.46				3.21	3.22	3.33
	velocity		5.18	8.54	8.54	8.54	8.54	8.54	8.46	8.23	8.31	8.16	9.60	7.93			8.54	8.52	8.23
Mita, Kyohei (JPN) (1991)	time		2.71	3.82	4.90	5.98	7.06	8.14	9.23	10.32	11.42	12.52		13.96	3 / 5				
	reaction time			1.11	1.08	1.08	1.08	1.08	1.09	1.09	1.10	1.10	1.44				3.27	3.25	3.29
	velocity		5.06	8.23	8.46	8.46	8.46	8.46	8.39	8.39	8.31	8.31	9.74	7.88			8.39	8.44	8.33

2013 Japanese National Multi-Event Championships (Nagano, JPN)

Decathlon - Heat 3

Decathlon - Heat 3		date	02-Jun-13	wind	0.0 m/s															Takeo (2014) - decathletes sprint ability and the relationship of the individual event performance														
				H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10															
Ushiro, Keisuke (JPN) (1986)	time			2.80	4.00	5.20	6.39	7.61	8.83	10.10	11.38	12.69	13.99		15.66	5 / 3																		
	reaction time				1.20	1.20	1.19	1.22	1.22	1.27	1.28	1.31	1.30	1.67			3.59	3.71	3.89															
	velocity			4.90	7.62	7.62	7.68	7.49	7.49	7.20	7.14	6.98	7.03	8.40	7.02		7.64	7.39	7.05															

Decathlon - Heat 1

Decathlon - Heat 1		date	02-Jun-13		wind	0.7 m/s		Takeo (2014) - decathletes sprint ability and the relationship of the individual event performance												
			H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10		
Nakamura, Akihiko (JPN) (1990)	time		2.68	3.81	4.94	6.06	7.17	8.29	9.38	10.53	11.68	12.81		14.24	6 / 2					
	reaction time	interval		1.13	1.13	1.12	1.11	1.12	1.09	1.15	1.15	1.13	1.43			3.38	3.32	3.43		
		velocity	5.12	8.09	8.09	8.16	8.23	8.16	8.39	7.95	7.95	8.09	9.80	7.72		8.11	8.26	7.99		

2013 Golden Grand Prix (Shanghai, CHN)

FINAL

FINAL			date	18-May-13	wind	Wang (2016) - comparative analysis of the grey relational grade of segment times and final results of the 110m hurdles														
					H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
Xie Wenjun (CHN) (1990)	time				2.54	3.60	4.63	5.64	6.68	7.70	8.74	9.77	10.81	11.87		13.28	/ 3			
	reaction time	0.137		interval		1.06	1.03	1.01	1.04	1.02	1.04	1.03	1.04	1.06	1.41	PB		3.10	3.10	3.13
	velocity			5.40	8.62	8.87	9.05	8.79	8.96	8.79	8.87	8.79	8.62	9.94	8.28		8.85	8.85	8.76	

2013 Seiko Golden Grand Prix (Tokyo, JPN)

FINAL

		date	05-May-13	wind	-1.4 m/s	Kota (2013) - race analysis of leading Japanese men's 110m hurdlers - from 2011-13													
				H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
Omuro, Hideki (JPN) (1990)	time			2.66	3.75	4.83	5.93	7.01	8.08	9.17	10.28	11.37	12.50		13.94	3 / 5			
	reaction time			interval		1.09	1.08	1.10	1.08	1.07	1.09	1.11	1.09	1.13	1.44		3.27	3.24	3.33
	velocity			5.16	8.39	8.46	8.31	8.46	8.54	8.39	8.23	8.39	8.09	9.74	7.89		8.39	8.46	8.23

Sato, Hiroyuki (JPN) (1990)	time	2.65	3.76	4.85	5.95	7.04	8.13	9.21	10.32	11.43	12.56		14.02	2 / 7			
reaction time	interval		1.11	1.09	1.10	1.09	1.09	1.08	1.11	1.11	1.13	1.46			3.30	3.26	3.35
	velocity	5.18	8.23	8.39	8.31	8.39	8.39	8.46	8.23	8.23	8.09	9.60	7.85		8.31	8.41	8.19

2013 Mito Invitational Athletics Meeting (Mito, JPN)**FINAL**

		date	04-May-13		wind -1.3 m/s			Kota (2013) - race analysis of leading Japanese men's 110m hurdlers - from 2011-13												
			H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10		
Yazawa, Wataru (JPN) (1991)	time		2.67	3.80	4.90	5.99	7.06	8.13	9.22	10.32	11.42	12.53		13.97	/ 1					
	reaction time	interval		1.13	1.10	1.09	1.07	1.07	1.09	1.10	1.10	1.11	1.44			3.32	3.23	3.31		
		velocity	5.14	8.09	8.31	8.39	8.54	8.54	8.39	8.31	8.31	8.23	9.74	7.87		8.26	8.49	8.28		
Wado, Tatsuya (JPN) (1990)	time		2.60	3.93	5.15	6.34	7.46	8.60	9.71	10.83	11.97	13.13		14.68	/ 8					
	reaction time	interval		1.33	1.22	1.19	1.12	1.14	1.11	1.12	1.14	1.16	1.55			3.74	3.37	3.42		
		velocity	5.28	6.87	7.49	7.68	8.16	8.02	8.23	8.16	8.02	7.88	9.05	7.49		7.33	8.14	8.02		

Heat 2

Heat 2

	date	04-May-13	wind	0.8 m/s	Kota (2013) - race analysis of leading Japanese men's 110m hurdlers - from 2011-13													
			H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
Wado, Tatsuya (JPN) (1990)	time		2.64	3.72	4.84	5.93	7.03	8.11	9.20	10.32	11.46	12.62		14.14	/ 2			
reaction time	interval			1.08	1.12	1.09	1.10	1.08	1.09	1.12	1.14	1.16	1.52			3.29	3.27	3.42
	velocity		5.20	8.46	8.16	8.39	8.31	8.46	8.39	8.16	8.02	7.88	9.22	7.78		8.33	8.39	8.02

Heat 1

Heat 1	date	04-May-13	wind	Kota (2013) - race analysis of leading Japanese men's 110m hurdlers - from 2011-13															
				H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
	Yazawa, Wataru (JPN) (1991)	time		2.63	3.73	4.79	5.85	6.91	7.97	9.05	10.12	11.22	12.32		13.74	/ 1			
	reaction time	interval			1.10	1.06	1.06	1.06	1.06	1.08	1.07	1.10	1.10	1.42			3.22	3.20	3.27
		velocity		5.22	8.31	8.62	8.62	8.62	8.62	8.46	8.54	8.31	8.31	9.87	8.01		8.52	8.57	8.39

2013 Chinese National Grand Prix (Zhaoqing, CHN)**FINAL**

FINAL	date	28-Apr-12	wind	0.1 m/s	Wang (2016) - comparative analysis of the grey relational grade of segment times and final results of the 110m hurdles													
					H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4
Xie Wenjun (CHN) (1990)	time		2.60	3.66	4.69	5.75	6.79	7.83	8.87	9.91	10.95	12.03		13.44	/ 1			
	reaction time			1.06	1.03	1.06	1.04	1.04	1.04	1.04	1.08	1.41				3.15	3.12	3.16
	velocity		5.28	8.62	8.87	8.62	8.79	8.79	8.79	8.79	8.46	9.94	8.18			8.70	8.79	8.68

2013 Mikio Oda Memorial (Hiroshima, JPN)**FINAL**

INAL	date	29-Apr-13	wind 0.7 m/s			Kota (2013) - race analysis of leading Japanese men's 110m hurdlers - from 2011-13												
			H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
Omuro, Hideki (JPN) (1990)	time		2.63	3.71	4.76	5.82	6.87	7.94	9.01	10.11	11.19	12.27		13.70	/ 1			
	reaction time			1.08	1.05	1.06	1.05	1.07	1.07	1.10	1.08	1.08	1.43			3.19	3.19	3.26
	velocity		5.22	8.46	8.70	8.62	8.70	8.54	8.54	8.31	8.46	8.46	9.80	8.03		8.60	8.60	8.41
Yazawa, Wataru (JPN) (1991)	time		2.64	3.74	4.82	5.90	6.98	8.03	9.11	10.18	11.27	12.40		13.86	/ 4			
	reaction time			1.10	1.08	1.08	1.08	1.05	1.08	1.07	1.09	1.13	1.46			3.26	3.21	3.29
	velocity		5.20	8.31	8.46	8.46	8.46	8.70	8.46	8.54	8.39	8.09	9.60	7.94		8.41	8.54	8.33
Mita, Kyohei (JPN) (1991)	time		2.70	3.80	4.90	5.97	7.03	8.11	9.19	10.28	11.39	12.48		13.92	/ 5			
	reaction time			1.10	1.10	1.07	1.06	1.08	1.08	1.09	1.11	1.09	1.44			3.27	3.22	3.29
	velocity		5.08	8.31	8.31	8.54	8.62	8.46	8.46	8.39	8.23	8.39	9.74	7.90		8.39	8.52	8.33
Sato, Hiroyuki (JPN) (1990)	time		2.61	3.69	4.75	5.84	6.91	8.00	9.08	10.19	11.33	12.50		13.99	/ 6			
	reaction time			1.08	1.06	1.09	1.07	1.09	1.08	1.11	1.14	1.17	1.49			3.23	3.24	3.42
	velocity		5.26	8.46	8.62	8.39	8.54	8.39	8.46	8.23	8.02	7.81	9.41	7.86		8.49	8.46	8.02

Heat 3

Heat 3

	date	29-Apr-13	wind	2.4 m/s	Kota (2013) - race analysis of leading Japanese men's 110m hurdlers - from 2011-13														
			H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10	
Mita, Kyohei (JPN) (1991)	time		2.66	3.76	4.83	5.90	6.96	8.03	9.10	10.17	11.26	12.34		13.77	/ 1				
	reaction time			1.10	1.07	1.07	1.06	1.07	1.07	1.07	1.09	1.08	1.43				3.24	3.20	3.24
	interval																		
	velocity		5.16	8.31	8.54	8.54	8.62	8.54	8.54	8.54	8.39	8.46	9.80	7.99			8.46	8.57	8.46
Wado, Tatsuya (JPN) (1990)	time		2.59	3.66	4.75	5.81	6.87	7.95	9.00	10.09	11.12	12.52		14.26	/ 6				
	reaction time			1.07	1.09	1.06	1.06	1.08	1.05	1.09	1.03	1.40	1.74				3.22	3.19	3.52
	interval																		
	velocity		5.30	8.54	8.39	8.62	8.62	8.46	8.70	8.39	8.87	6.53	8.06	7.71			8.52	8.60	7.79

Heat 2

Heat 2

	date	29-Apr-13	wind	0.4 m/s	Kota (2013) - race analysis of leading Japanese men's 110m hurdlers - from 2011-13													
			H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
Sato, Hiroyuki (JPN) (1990)	time		2.64	3.71	4.78	5.84	6.91	7.98	9.07	10.18	11.29	12.43		13.90	/ 2			
reaction time	interval			1.07	1.07	1.06	1.07	1.07	1.09	1.11	1.11	1.14	1.47			3.20	3.23	3.36
	velocity		5.20	8.54	8.54	8.62	8.54	8.54	8.39	8.23	8.23	8.02	9.54	7.91		8.57	8.49	8.16

Heat 1

Heat 1	date	29-Apr-13		wind	1.5 m/s		Kota (2013) - race analysis of leading Japanese men's 110m hurdlers - from 2011-13													
				H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10	
	Omuro, Hideki (JPN) (1990)	time	2.64	3.69	4.76	5.81	6.87	7.93	9.01	10.09	11.20	12.31			13.76	/ 1				
		reaction time			1.05	1.07	1.05	1.06	1.06	1.08	1.08	1.11	1.11	1.45				3.17	3.20	3.30
		velocity	5.20	8.70	8.54	8.70	8.62	8.62	8.46	8.46	8.23	8.23	9.67		7.99			8.65	8.57	8.31

Yazawa, Wataru (JPN) (1991)	time	2.63	3.73	4.79	5.86	6.91	8.00	9.08	10.18	11.28	12.42		13.85	/ 4			
reaction time	interval		1.10	1.06	1.07	1.05	1.09	1.08	1.10	1.10	1.14	1.43			3.23	3.22	3.34
	velocity	5.22	8.31	8.62	8.54	8.70	8.39	8.46	8.31	8.31	8.02	9.80	7.94		8.49	8.52	8.21

2013 Wakayama Athletics Meeting (Wakayama, JPN)**Decathlon - Heat 2**

date 28-Apr-13 wind 0.8 m/s

Takeo (2014) - decathletes sprint ability and the relationship of the individual event performance

		H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
Ushiro, Keisuke (JPN) (1986)	time	2.80	3.97	5.12	6.29	7.46	8.64	9.82	11.03	12.21	13.43		15.05	3 / 1			
reaction time	interval		1.17	1.15	1.17	1.17	1.18	1.18	1.21	1.18	1.22	1.62			3.49	3.53	3.61
	velocity	4.90	7.81	7.95	7.81	7.81	7.75	7.75	7.55	7.75	7.49	8.65	7.31		7.86	7.77	7.60

Decathlon - Heat 1

date 28-Apr-13 wind 1.5 m/s

Takeo (2014) - decathletes sprint ability and the relationship of the individual event performance

		H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
Nakamura, Akihiko (JPN) (1990)	time	2.69	3.81	4.91	6.03	7.12	8.23	9.34	10.46	11.58	12.72		14.16	6 / 2			
reaction time	interval		1.12	1.10	1.12	1.09	1.11	1.11	1.12	1.12	1.14	1.44			3.34	3.31	3.38
	velocity	5.10	8.16	8.31	8.16	8.39	8.23	8.23	8.16	8.16	8.02	9.74	7.77		8.21	8.28	8.11

2013 European Indoor Championships (Göteborg, SWE)**Heptathlon - Heat 1**

date 03-Mar-13

Blomkvist (2013) - www.elitlandslag.se/SprintHäck/KortHäck.aspx

		H1	H2	H3	H4	H5	Run-In	Official Time	Lane / Place	H1-H4
Olson, Petter (SWE) (1991)	time	2.70	3.85	4.95	6.05	7.16		8.11	4 / 2	
reaction time	0.150		1.15	1.10	1.10	1.11	0.95			3.35
	velocity	5.08	7.95	8.31	8.31	8.23	10.23	7.40		8.19
Rosenquist, Fabian (SWE) (1991)	time	2.72	3.85	4.97	6.07	7.21		8.30	7 / 3	
reaction time	0.138		1.13	1.12	1.10	1.14	1.09			3.35
	velocity	5.04	8.09	8.16	8.31	8.02	8.92	7.23		8.19

FINAL

date 01-Mar-13

Blomkvist (2013) - www.elitlandslag.se/SprintHäck/KortHäck.aspx

		H1	H2	H3	H4	H5	Run-In	Official Time	Lane / Place	H1-H4
Shubenkov, Sergey (RUS) (1991)	time	2.58	3.62	4.62	5.62	6.64		7.49	6 / 1	
reaction time	0.161		1.04	1.00	1.00	1.02	0.85	=PB		3.04
	velocity	5.32	8.79	9.14	9.14	8.96	11.44	8.01		9.02
Dal Molin, Paolo (ITA) (1987)	time	2.51	3.55	4.60	5.62	6.66		7.51	4 / 2	
reaction time	0.138		1.04	1.05	1.02	1.04	0.85	NR		3.11
	velocity	5.47	8.79	8.70	8.96	8.79	11.44	7.99		8.82
Martinot Lagarde, Pascal (FRA) (1989)	time	2.57	3.64	4.65	5.65	6.65		7.53	3 / 3	
reaction time	0.155		1.07	1.01	1.00	1.00	0.88	=PB		3.08
	velocity	5.34	8.54	9.05	9.14	9.14	11.05	7.97		8.90
Baji, Balázs (HUN) (1989)	time	2.55	3.61	4.65	5.66	6.67		7.56	8 / 4	
reaction time	0.159		7 steps	1.06	1.04	1.01	0.89	=NR		3.11
	velocity	5.38	8.62	8.79	9.05	9.05	10.92	7.94		8.82
Balnuweit, Erik (GER) (1988)	time	2.55	3.60	4.62	5.65	6.68		7.58	5 / 5	
reaction time	0.146		7 steps	1.05	1.02	1.03	0.90	PB		3.10
	velocity	5.38	8.70	8.96	8.87	8.87	10.80	7.92		8.85
Lynsha, Maksim (BLR) (1985)	time	2.56	3.60	4.64	5.66	6.68		7.58	2 / 1	
reaction time	0.162		1.04	1.04	1.02	1.02	0.90	PB		3.10
	velocity	5.36	8.79	8.79	8.96	8.96	10.80	7.92		8.85
Douvalidis, Konstadinos (GRE) (1985)	time	2.62	3.68	4.72	5.76	6.78		7.64	1 / 7	
reaction time	0.153		1.06	1.04	1.04	1.02	0.86			3.14
	velocity	5.24	8.62	8.79	8.79	8.96	11.30	7.85		8.73
Shabanov, Konstantin (RUS) (1985)	time	2.59	3.64	4.67	5.71	6.77		7.66	7 / 8	
reaction time	0.137		1.05	1.03	1.04	1.06	0.89			3.12
	velocity	5.30	8.70	8.87	8.79	8.62	10.92	7.83		8.79

Semi-Final 1

date 01-Mar-13

Blomkvist (2013) - www.elitlandslag.se/SprintHäck/KortHäck.aspx

		H1	H2	H3	H4	H5	Run-In	Official Time	Lane / Place	H1-H4
Nossmys, Philip (SWE) (1982)	time	2.62	3.71	4.77	5.82	6.86		7.78	2 / 6	
reaction time	0.153		1.09	1.06	1.05	1.04	0.92			3.20
	velocity	5.24	8.39	8.62	8.70	8.79	10.57	7.71		8.57

Heat 4

date 01-Mar-13

Blomkvist (2013) - www.elitlandslag.se/SprintHäck/KortHäck.aspx

		H1	H2	H3	H4	H5	Run-In	Official Time	Lane / Place	H1-H4
Lööv, Filip (SWE) (1992)	time	2.66	3.80	4.90	5.97	7.08		8.02	2 / 7	
reaction time	0.167		1.14	1.10	1.07	1.11	0.94			3.31
	velocity	5.16	8.02	8.31	8.54	8.23	10.34	7.48		8.28

Heat 2

date 01-Mar-13

Blomkvist (2013) - www.elitlandslag.se/SprintHäck/KortHäck.aspx

		H1	H2	H3	H4	H5	Run-In	Official Time	Lane / Place	H1-H4
Nossmy, Philip (SWE) (1982)	time	2.63	3.73	4.80	5.84	6.89		7.76	1 / 4	
	reaction time	0.162	interval	1.10	1.07	1.04	0.87			3.21
	velocity	5.22	8.31	8.54	8.79	8.70	11.17	7.73		8.54

Heat 1

date 01-Mar-13

Blomkvist (2013) - www.elitlandslag.se/SprintHäck/KortHäck.aspx

		H1	H2	H3	H4	H5	Run-In	Official Time	Lane / Place	H1-H4
Brorson, Alexander (SWE) (1995)	time	2.60	3.70	4.77	5.84	6.93		7.85	2 / 6	
	reaction time	0.156	interval	1.10	1.07	1.09	0.92			3.24
	velocity	5.28	8.31	8.54	8.54	8.39	10.57	7.64		8.46

2013 Italian National Indoor Junior Championships (Ancona, ITA) (99cm)

FINAL

date 23-Feb-13

FIDAL - Centro Studi & Ricerche (2013) - www.youtube.com/watch?v=Egf_xlSp-DI

		H1	H2	H3	H4	H5	Run-In	Official Time	Lane / Place	H1-H4
Perini, Lorenzo (ITA) (1994)	time	2.71	3.80	4.86	5.91	6.95		7.90	6 / 1	
	reaction time		interval	1.09	1.06	1.04	0.95			3.20
	velocity	5.06	8.39	8.62	8.70	8.79	10.23	7.59		8.57

2013 Italian National Indoor Youth Championships (Ancona, ITA) (91cm)

FINAL

date 23-Feb-13

FIDAL - Centro Studi & Ricerche (2013) - www.youtube.com/watch?v=Xy9UwrfVMj6g

		H1	H2	H3	H4	H5	Run-In	Official Time	Lane / Place	H1-H4
Biondo, Giuseppe (ITA) (1996)	time	2.66	3.78	4.85	5.90	6.95		7.94	6 / 1	
	reaction time		interval	1.12	1.07	1.05	0.99	PB		3.24
	velocity	5.16	8.16	8.54	8.70	8.70	9.82	7.56		8.46
Beria, Matteo (ITA) (1996)	time	2.66	3.78	4.87	5.95	7.00		7.99	7 / 2	
	reaction time		interval	1.12	1.09	1.05	0.99			3.29
	velocity	5.16	8.16	8.39	8.46	8.70	9.82	7.51		8.33

2012 Memorial Van Damme (Brussels, BEL) (TV Analysis)

FINAL

date 07-Sep-12

wind 0.3 m/s

Henson (2020) - Athlete First: 2012 Aries Merritt world record season

May (2012) - <http://www.fgs.uni-halle.de>

		Date of Competition		Wind direction and speed										May (2012)		http://www.iaaf.org/iaaf-henry-henson			
				H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
Merritt, Aries (USA) (1985)	time			2.52	3.56	4.56	5.53	6.50	7.48	8.46	9.44	10.44	11.46		12.80	6 / 1			May (2012)
	reaction time	0.145	interval	7 steps	1.04	1.00	0.97	0.97	0.98	0.98	0.98	1.00	1.02	1.34	WR		3.01	2.93	3.00
			velocity	5.44	8.79	9.14	9.42	9.42	9.33	9.33	9.33	9.14	8.96	10.46	8.59		9.11	9.36	9.14
Richardson, Jason (USA) (1986)	time			2.50	3.54	4.52	5.50	6.50	7.50	8.50	9.50	10.54	11.58		13.05	5 / 2			Henson (2020)
	reaction time	0.147	interval	7 steps	1.04	0.98	0.98	1.00	1.00	1.00	1.00	1.04	1.04	1.47			3.00	3.00	3.08
			velocity	5.49	8.79	9.33	9.33	9.14	9.14	9.14	9.14	8.79	8.79	9.54	8.43		9.14	9.14	8.90
Parchment, Hansle (JAM) (1990)	time			2.56	3.62	4.62	5.60	6.60	7.58	8.58	9.60	10.62	11.66		13.14	4 / 3			Henson (2020)
	reaction time	0.157	interval	7 steps	1.06	1.00	0.98	1.00	0.98	1.00	1.02	1.02	1.04	1.48			3.04	2.98	3.08
			velocity	5.36	8.62	9.14	9.33	9.14	9.33	9.14	8.96	8.96	8.79	9.47	8.37		9.02	9.20	8.90
Shubenkov, Sergey (RUS) (1988)	time			2.52	3.56	4.56	5.58	6.58	7.58	8.62	9.66	10.70	11.78		13.21	1 / 4			Henson (2020)
	reaction time	0.165	interval		1.04	1.00	1.02	1.00	1.00	1.04	1.04	1.04	1.08	1.43			3.06	3.04	3.16
			velocity	5.44	8.79	9.14	8.96	9.14	9.14	8.79	8.79	8.79	8.46	9.80	8.33		8.96	9.02	8.68
Oliver, David (USA) (1990)	time			2.52	3.56	4.58	5.56	6.56	7.58	8.58	9.60	10.62	11.72		13.21	2 / 5			Henson (2020)
	reaction time	0.171	interval	7 steps	1.04	1.02	0.98	1.00	1.02	1.00	1.02	1.02	1.10	1.49			3.04	3.02	3.14
			velocity	5.44	8.79	8.96	9.33	9.14	8.96	9.14	8.96	8.96	8.31	9.41	8.33		9.02	9.08	8.73
Fourie, Lehann (RSA) (1987)	time			2.48	3.52	4.52	5.52	6.52	7.54	8.58	9.60	10.62	11.70		13.24	3 / 6			Henson (2020)
	reaction time	0.133	interval	7 steps	1.04	1.00	1.00	1.00	1.02	1.04	1.02	1.02	1.08	1.54	AR		3.04	3.06	3.12
			velocity	5.53	8.79	9.14	9.14	9.14	8.96	8.79	8.96	8.96	8.46	9.10	8.31		9.02	8.96	8.79
Wilson, Ryan (USA) (1980)	time			2.52	3.56	4.56	5.56	6.58	7.60	8.64	9.66	10.70	11.76		13.26	8 / 7			Henson (2020)
	reaction time	0.144	interval	7 steps	1.04	1.00	1.00	1.02	1.02	1.04	1.02	1.04	1.06	1.50			3.04	3.08	3.12
			velocity	5.44	8.79	9.14	9.14	8.96	8.96	8.79	8.96	8.79	8.62	9.35	8.30		9.02	8.90	8.79
Brathwaite, Ryan (BAR) (1988)	time			2.48	3.52	4.56	5.56	6.56	7.56	8.58	9.62	10.70	11.80		13.39	9 / 8			Henson (2020)
	reaction time	0.156	interval		1.04	1.04	1.00	1.00	1.00	1.02	1.04	1.08	1.10	1.59			3.08	3.02	3.22
			velocity	5.53	8.79	8.79	9.14	9.14	9.14	8.96	8.79	8.46	8.31	8.82	8.22		8.90	9.08	8.52
Clarke, Lawrence (GBR) (1990)	time			2.50	3.58	4.62	5.68	6.74	7.80	8.84	9.92	11.00	12.14		13.71	7 / 9			Henson (2020)
	reaction time	0.148	interval		1.08	1.04	1.06	1.06	1.06	1.04	1.08	1.08	1.14	1.57			3.18	3.16	3.30
			velocity	5.49	8.46	8.79	8.62	8.62	8.62	8.79	8.46	8.46	8.02	8.93	8.02		8.62	8.68	8.31

2012 ISTAF (Berlin, GER) (TV Analysis)

FINAL

date 02-Sep-12

wind 1.2 m/s

Henson (2021) - Athlete First: 2012 Aries Merritt world record season

		H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
Merritt, Aries (USA) (1985)	time	2.43	3.45	4.47	5.47	6.47	7.47	8.47	9.47	10.49	11.51		12.97	4 / 1			
	reaction time	0.164	interval	7 steps	1.02	1.02	1.00	1.00	1.00	1.00	1.02	1.46			3.04	3.00	3.04
	velocity	5.65	8.96	8.96	9.14	9.14	9.14	9.14	9.14	8.96	8.96	9.60	8.48		9.02	9.14	9.02

Wilson, Ryan (USA) (1980)	time	2.57	3.63	4.67	5.70	6.70	7.74	8.74	9.77	10.84	11.91		13.45	7 / 2				
	reaction time	0.192	interval	7 steps	1.06	1.04	1.03	1.00	1.04	1.00	1.03	1.07	1.07	1.54		3.13	3.04	3.17
			velocity		5.34	8.62	8.79	8.87	9.14	8.79	9.14	8.87	8.54	8.54	9.10	8.18	8.76	9.02

2012 AVIVA Grand Prix (Birmingham, GBR) (TV Analysis)

FINAL

		date	26-Aug-12	wind -0.9 m/s					Henson (2020) - Athlete First: 2012 Aries Merritt world record season												
				H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10		
Merritt, Aries (USA) (1985)	time			2.44	3.46	4.47	5.48	6.46	7.46	8.46	9.46	10.46	11.50		12.95	3 / 1					
	reaction time	0.146	interval	7 steps	1.02	1.01	1.01	0.98	1.00	1.00	1.00	1.00	1.04	1.45			3.04	2.98	3.04		
			velocity		5.62	8.96	9.05	9.05	9.33	9.14	9.14	9.14	9.14	8.79	9.67	8.49		9.02	9.20	9.02	
Richardson, Jason (USA) (1986)	time			2.48	3.52	4.52	5.52	6.52	7.50	8.48	9.48	10.50	11.52		12.98	2 / 2					
	reaction time	0.165	interval	7 steps	1.04	1.00	1.00	1.00	0.98	0.98	1.00	1.02	1.02	1.46	=PB		3.04	2.96	3.04		
			velocity		5.53	8.79	9.14	9.14	9.14	9.33	9.33	9.14	8.96	8.96	9.60	8.47		9.02	9.26	9.02	
Oliver, David (USA) (1990)	time			2.50	3.53	4.56	5.59	6.59	7.62	8.64	9.68	10.74	11.78		13.28	4 / 3					
	reaction time	0.154	interval	7 steps	1.03	1.03	1.03	1.00	1.03	1.02	1.04	1.06	1.04	1.50			3.09	3.05	3.14		
			velocity		5.49	8.87	8.87	8.87	9.14	8.87	8.96	8.79	8.62	8.79	9.35	8.28		8.87	8.99	8.73	
Wilson, Ryan (USA) (1980)	time			2.54	3.60	4.63	5.64	6.64	7.66	8.70	9.73	10.77	11.83		13.34	1 / 4					
	reaction time	0.156	interval	7 steps	1.06	1.03	1.01	1.00	1.02	1.04	1.03	1.04	1.06	1.51			3.10	3.06	3.13		
			velocity		5.40	8.62	8.87	9.05	9.14	8.96	8.79	8.87	8.79	8.62	9.28	8.25		8.85	8.96	8.76	
Brathwaite, Ryan (BAR) (1988)	time			2.52	3.56	4.60	5.62	6.60	7.70	8.74	9.76	10.80	11.86		13.42	7 / 5					
	reaction time	0.160	interval		1.04	1.04	1.02	0.98	1.10	1.04	1.02	1.04	1.06	1.56			3.10	3.12	3.12		
			velocity		5.44	8.79	8.79	8.96	9.33	8.31	8.79	8.96	8.79	8.62	8.99	8.20		8.85	8.79	8.79	
Clarke, Lawrence (GBR) (1990)	time			2.45	3.50	4.57	5.60	6.62	7.66	8.72	9.78	10.84	11.94		13.52	6 / 6					
	reaction time	0.146	interval		1.05	1.07	1.03	1.02	1.04	1.06	1.06	1.06	1.10	1.58			3.15	3.12	3.22		
			velocity		5.60	8.70	8.54	8.87	8.96	8.79	8.62	8.62	8.62	8.31	8.87	8.14		8.70	8.79	8.52	
Fourie, Lehann (RSA) (1987)	time			2.48	3.52	4.54	5.56	6.58	7.60	8.70	9.81	10.87	11.99		13.59	8 / 7					
	reaction time	0.139	interval	7 steps	1.04	1.02	1.02	1.02	1.02	1.10	1.11	1.06	1.12	1.60			3.08	3.14	3.29		
			velocity		5.53	8.79	8.96	8.96	8.96	8.96	8.31	8.23	8.62	8.16	8.76	8.09		8.90	8.73	8.33	
Parchment, Hansle (JAM) (1990)	time			2.54	3.63	4.75	5.85	6.97	8.00	9.05	10.12	11.16	12.22		13.76	5 / 8					
	reaction time	0.155	interval	7 steps	1.09	1.12	1.10	1.12	1.03	1.05	1.07	1.04	1.06	1.54			3.31	3.20	3.17		
			velocity		5.40	8.39	8.16	8.31	8.16	8.87	8.70	8.54	8.79	8.62	9.10	7.99		8.28	8.57	8.65	

2012 Olympic Games (London, GBR)

Decathlon - Heat 1

Decathlon - Heat 1		date	09-Aug-12	wind	-0.9 m/s															Takeo (2014) - decathletes sprint ability and the relationship of the individual event performance														
				H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10															
Ushiro, Keisuke (JPN) (1986)	time			2.80	3.99	5.20	6.42	7.62	8.85	10.06	11.30	12.56	13.84		15.47	6 / 5																		
	reaction time	0.186	interval		1.19	1.21	1.22	1.20	1.23	1.21	1.24	1.26	1.28	1.63			3.62	3.64	3.78															
			velocity	4.90	7.68	7.55	7.49	7.62	7.43	7.55	7.37	7.25	7.14	8.60	7.11		7.57	7.53	7.25															

Henson (2020) - Athlete First: 2012 Aries Merritt world record season (TV Analysis)

FINAL

		date	08-Aug-12	wind -0.3 m/s			Graubner (2012) - http://www.fgs.uni-halle.de (TV Analysis)												
				H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
Merritt, Aries (USA) (1985)	time			2.54	3.55	4.55	5.54	6.53	7.52	8.53	9.53	10.55	11.57		12.92	6 / 1			Graubner (2012)
	reaction time	0.143	interval	7 steps	1.01	1.00	0.99	0.99	0.99	1.01	1.00	1.02	1.02	1.35	PB		3.00	2.99	3.04
			velocity		5.40	9.05	9.14	9.23	9.23	9.23	9.05	9.14	8.96	8.96	10.39	8.51		9.14	9.17
Richardson, Jason (USA) (1986)	time			2.54	3.54	4.54	5.54	6.56	7.58	8.58	9.58	10.58	11.62		13.04	4 / 2			Henson (2020)
	reaction time	0.194	interval	7 steps	1.00	1.00	1.00	1.02	1.02	1.00	1.00	1.00	1.04	1.42			3.00	3.04	3.04
			velocity		5.40	9.14	9.14	9.14	8.96	8.96	9.14	9.14	9.14	8.79	9.87	8.44		9.14	9.02
Parchment, Hansle (JAM) (1990)	time			2.56	3.62	4.62	5.62	6.62	7.62	8.62	9.62	10.64	11.68		13.12	7 / 3			Henson (2020)
	reaction time	0.172	interval	7 steps	1.06	1.00	1.00	1.00	1.00	1.00	1.00	1.02	1.04	1.44	NR		3.06	3.00	3.06
			velocity		5.36	8.62	9.14	9.14	9.14	9.14	9.14	9.14	8.96	8.79	9.74	8.38		8.96	9.14
Clarke, Lawrence (GBR) (1990)	time			2.50	3.54	4.56	5.58	6.62	7.66	8.74	9.78	10.82	11.90		13.39	2 / 4			Henson (2020)
	reaction time	0.169	interval		1.04	1.02	1.02	1.04	1.04	1.08	1.04	1.04	1.08	1.49			3.08	3.16	3.16
			velocity		5.49	8.79	8.96	8.96	8.79	8.79	8.46	8.79	8.79	8.46	9.41	8.22		8.90	8.68
Brathwaite, Ryan (BAR) (1988)	time			2.48	3.56	4.60	5.62	6.62	7.66	8.70	9.78	10.82	11.90		13.40	8 / 5			Henson (2020)
	reaction time	0.163	interval		1.08	1.04	1.02	1.00	1.04	1.04	1.08	1.04	1.08	1.50			3.14	3.08	3.20
			velocity		5.53	8.46	8.79	8.96	9.14	8.79	8.79	8.46	8.79	8.46	9.35	8.21		8.73	8.90
Ortega, Orlando (CUB) (1991)	time			2.54	3.62	4.64	5.64	6.66	7.68	8.70	9.74	10.80	11.90		13.43	9 / 6			Henson (2020)
	reaction time	0.135	interval	7 steps	1.08	1.02	1.00	1.02	1.02	1.02	1.04	1.06	1.10	1.53			3.10	3.06	3.20
			velocity		5.40	8.46	8.96	9.14	8.96	8.96	8.96	8.79	8.62	8.31	9.16	8.19		8.85	8.96
Fourie, Lehann (RSA) (1987)	time			2.48	3.56	4.58	5.58	6.62	7.66	8.74	9.78	10.90	12.02		13.53	3 / 7			Henson (2020)
	reaction time	0.136	interval	7 steps	1.08	1.02	1.00	1.04	1.04	1.08	1.04	1.12	1.12	1.51			3.10	3.16	3.28
			velocity		5.53	8.46	8.96	9.14	8.79	8.79	8.46	8.79	8.16	8.16	9.28	8.13		8.85	8.68

Robles, Dayron (CUB) (1986)	time	2.46	3.48	4.50	5.50	6.50	7.72						42.86	5 / --		Henson (2020)
	reaction time	0.159	interval	7 steps	1.02	1.02	1.00	1.00	1.22				DQ		3.04	
	velocity	5.58	8.96	8.96	9.14	9.14	7.49						2.57		9.02	

Semi-Final 3 (TV Analysis)

date 08-Aug-12 wind 0.1 m/s

Henson (2020) - Athlete First: 2012 Aries Merritt world record season

		H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
Robles, Dayron (CUB) (1986)	time	2.48	3.48	4.48	5.46	6.46	7.46	8.50	9.54	10.58	11.62		13.10	7 / 1			
	reaction time	0.162	interval	7 steps	1.00	1.00	0.98	1.00	1.00	1.04	1.04	1.04	1.48		2.98	3.04	3.12
	velocity	5.53	9.14	9.14	9.33	9.14	9.14	8.79	8.79	8.79	8.79	9.47	8.40		9.20	9.02	8.79
Parchment, Hansle (JAM) (1990)	time	2.54	3.58	4.62	5.62	6.62	7.62	8.62	9.62	10.64	11.66		13.14	5 / 2			
	reaction time	0.172	interval	7 steps	1.04	1.04	1.00	1.00	1.00	1.00	1.02	1.02	1.48	NR	3.08	3.00	3.04
	velocity	5.40	8.79	8.79	9.14	9.14	9.14	9.14	9.14	8.96	8.96	9.47	8.37		8.90	9.14	9.02
Fourie, Lehann (RSA) (1987)	time	2.44	3.52	4.54	5.56	6.56	7.58	8.58	9.64	10.70	11.78		13.28	6 / 3			
	reaction time	0.139	interval	7 steps	1.08	1.02	1.02	1.00	1.02	1.00	1.06	1.06	1.08	1.50	3.12	3.02	3.20
	velocity	5.62	8.46	8.96	8.96	9.14	8.96	9.14	8.62	8.62	8.46	9.35	8.28		8.79	9.08	8.57

Henson (2020) - Athlete First: 2012 Aries Merritt world record season (TV Analysis)

Semi-Final 2

date 08-Aug-12 wind 0.1 m/s

Wang (2016) - comparative analysis of the grey relational grade of segment times and final results of the 110m hurdles

		Sample data analysis of the grey relational grade of segment times and final results of the four horses																
		H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10	
Merritt, Aries (USA) (1985)	time	2.48	3.50	4.52	5.50	6.50	7.48	8.46	9.46	10.48	11.50		12.94	7 / 1			Henson (2020)	
	reaction time	0.147	interval	7 steps	1.02	1.02	0.98	1.00	0.98	0.98	1.00	1.02	1.44		3.02	2.96	3.04	
		velocity	5.53	8.96	8.96	9.33	9.14	9.33	9.33	9.14	8.96	8.96	9.74	8.50		9.08	9.26	9.02
Braithwaite, Ryan (BAR) (1988)	time	2.48	3.50	4.52	5.52	6.52	7.54	8.56	9.58	10.60	11.70		13.23	4 / 2			Henson (2020)	
	reaction time	0.159	interval		1.02	1.02	1.00	1.00	1.02	1.02	1.02	1.10	1.53		3.04	3.04	3.14	
		velocity	5.53	8.96	8.96	9.14	9.14	8.96	8.96	8.96	8.96	8.31	9.16	8.31		9.02	9.02	8.73
Xie Wenjun (CHN) (1990)	time	2.60	3.68	4.72	5.73	6.74	7.78	8.80	9.84	10.89	11.96		13.34	9 / 3			Wang (2016)	
	reaction time	0.158	interval		1.08	1.04	1.01	1.01	1.04	1.02	1.04	1.05	1.07	1.38	PB	3.13	3.07	3.16
		velocity	5.28	8.46	8.79	9.05	9.05	8.79	8.96	8.79	8.70	8.54	10.16	8.25		8.76	8.93	8.68

Semi-Final 1 (TV Analysis)

date 08-Aug-12 wind -0.5 m/s

Henson (2020) - Athlete First: 2012 Aries Merritt world record season

		H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
Richardson, Jason (USA) (1986)	time	2.46	3.50	4.50	5.48	6.46	7.44	8.44	9.46	10.50	11.54		13.13	5 / 1			
	reaction time	0.160	interval	7 steps	1.04	1.00	0.98	0.98	0.98	1.00	1.02	1.04	1.59		3.02	2.96	3.10
	velocity	5.58	8.79	9.14	9.33	9.33	9.33	9.14	8.96	8.79	8.79	8.82	8.38		9.08	9.26	8.85
Ortega, Orlando (CUB) (1991)	time	2.52	3.56	4.60	5.60	6.60	7.62	8.62	9.62	10.66	11.74		13.26	6 / 2			
	reaction time	0.151	interval	7 steps	1.04	1.04	1.00	1.00	1.02	1.00	1.04	1.08	1.52		3.08	3.02	3.12
	velocity	5.44	8.79	8.79	9.14	9.14	8.96	9.14	9.14	8.79	8.46	9.22	8.30		8.90	9.08	8.79
Clarke, Lawrence (GBR) (1990)	time	2.48	3.52	4.54	5.54	6.58	7.58	8.62	9.66	10.72	11.80		13.31	4 / 3			
	reaction time	0.159	interval		1.04	1.02	1.00	1.04	1.00	1.04	1.06	1.08	1.51	PB	3.06	3.08	3.18
	velocity	5.53	8.79	8.96	9.14	8.79	9.14	8.79	8.79	8.62	8.46	9.28	8.26		8.96	8.90	8.62

Henson (2020) - Athlete First: 2012 Aries Merritt world record season (TV Analysis)

Heat 5

date 07-Aug-12 wind 0.7 m/s

Wang (2016) - comparative analysis of the grey relational grade of segment times and final results of the 110m hurdles

		H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
Merritt, Aries (USA) (1985)	time	2.46	3.50	4.48	5.50	6.50	7.50	8.50	9.50	10.50	11.58		13.07	7 / 1			Henson (2020)
	reaction time	0.142	interval	7 steps	1.04	0.98	1.02	1.00	1.00	1.00	1.00	1.08	1.49		3.04	3.00	3.08
	velocity	5.58	8.79	9.33	8.96	9.14	9.14	9.14	9.14	9.14	8.46	9.41	8.42		9.02	9.14	8.90
Xie Wenjun (CHN) (1990)	time	2.55	3.61	4.65	5.70	6.75	7.79	8.85	9.89	10.95	12.03		13.43	2 / 3			Wang (2016)
	reaction time	0.143	interval		1.06	1.04	1.05	1.05	1.04	1.06	1.04	1.08	1.40		3.15	3.15	3.18
	velocity	5.38	8.62	8.79	8.70	8.70	8.79	8.62	8.79	8.62	8.46	10.01	8.19		8.70	8.70	8.62

2012 Herculis Meeting International D'Athlétisme (Monaco, MON) (TV Analysis)

FINAL

date 20-Jul-12 wind 0.0 m/s

Henson (2020) - Athlete First: 2012 Aries Merritt world record season

		H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
Merritt, Aries (USA) (1985)	time	2.38	3.40	4.44	5.42	6.40	7.38	8.38	9.38	10.42	11.46		12.93	5 / 1			
	reaction time	0.112	interval	7 steps	1.02	1.04	0.98	0.98	0.98	1.00	1.00	1.04	1.47	=PB	3.04	2.96	3.08
	velocity	5.76	8.96	8.79	9.33	9.33	9.33	9.14	9.14	8.79	8.79	9.54	8.51		9.02	9.26	8.90
Richardson, Jason (USA) (1986)	time	2.50	3.52	4.52	5.52	6.50	7.50	8.48	9.48	10.52	11.56		13.07	4 / 2			
	reaction time	0.162	interval	7 steps	1.02	1.00	1.00	0.98	1.00	0.98	1.00	1.04	1.51		3.02	2.96	3.08
	velocity	5.49	8.96	9.14	9.14	9.33	9.14	9.33	9.14	8.79	8.79	9.28	8.42		9.08	9.26	8.90
Shubenkov, Sergey (RUS) (1988)	time	2.46	3.50	4.52	5.54	6.54	7.52	8.52	9.54	10.56	11.60		13.09	7 / 3			
	reaction time	0.143	interval		1.04	1.02	1.02	1.00	0.98	1.00	1.02	1.04	1.49	NR	3.08	2.98	3.08
	velocity	5.58	8.79	8.96	8.96	9.14	9.33	9.14	8.96	8.96	8.79	9.41	8.40		8.90	9.20	8.90
Oliver, David (USA) (1990)	time	2.46	3.50	4.52	5.54	6.52	7.52	8.54	9.54	10.58	11.64		13.14	6 / 4			
	reaction time	0.154	interval	7 steps	1.04	1.02	1.02	0.98	1.00	1.02	1.00	1.04	1.06	1.50	3.08	3.00	3.10
	velocity	5.58	8.79	8.96	8.96	9.33	9.14	8.96	9.14	8.79	8.62	9.35	8.37		8.90	9.14	8.85

Porter, Jeff (USA) (1985)	time	2.46	3.50	4.53	5.53	6.55	7.55	8.55	9.58	10.62	11.71		13.19	3 / 5			
	reaction time	0.153		1.04	1.03	1.00	1.02	1.00	1.00	1.03	1.04	1.09	1.48		3.07	3.02	3.16
	velocity		5.58	8.79	8.87	9.14	8.96	9.14	9.14	8.87	8.79	8.39	9.47	8.34	8.93	9.08	8.68
Darien, Garfield (FRA) (1987)	time	2.48	3.54	4.54	5.56	6.56	7.58	8.60	9.64	10.71	11.81		13.27	2 / 6			
	reaction time	0.161		1.06	1.00	1.02	1.00	1.02	1.02	1.04	1.07	1.10	1.46		3.08	3.04	3.21
	velocity		5.53	8.62	9.14	8.96	9.14	8.96	8.79	8.54	8.31	9.60	8.29		8.90	9.02	8.54
Riley, Andrew (JAM) (1988)	time	2.46	3.50	4.53	5.55	6.57	7.64	8.67	9.71	10.72	11.81		13.28	8 / 7			
	reaction time	0.138		1.04	1.03	1.02	1.02	1.07	1.03	1.04	1.01	1.09	1.47		3.09	3.12	3.14
	velocity		5.58	8.79	8.87	8.96	8.96	8.54	8.87	8.79	9.05	8.39	9.54	8.28	8.87	8.79	8.73
Bascou, Dimitri (FRA) (1987)	time	2.50	3.55	4.60	5.64	6.67	7.71	8.77	9.84	10.91	12.01		13.54	1 / 8			
	reaction time	0.159		7 steps	1.05	1.05	1.04	1.03	1.04	1.07	1.07	1.10	1.53		3.14	3.13	3.24
	velocity		5.49	8.70	8.70	8.79	8.87	8.79	8.62	8.54	8.54	8.31	9.16	8.12	8.73	8.76	8.46

2012 AVIVA London Grand Prix (London, GBR) (TV Analysis)**FINAL**

INAL	date		13-Jul-12		wind		0.6 m/s		Henson (2020) - Athlete First: 2012 Aries Merritt world record season												
			H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10			
Merritt, Aries (USA) (1985)	time		2.46	3.46	4.48	5.46	6.44	7.42	8.42	9.42	10.46	11.46		12.93	4 / 1						
	reaction time	0.137	interval	7 steps	1.00	1.02	0.98	0.98	0.98	1.00	1.00	1.04	1.00	1.47	=PB		3.00	2.96	3.04		
			velocity	5.58	9.14	8.96	9.33	9.33	9.33	9.14	9.14	8.79	9.14	9.54	8.51		9.14	9.26	9.02		
Richardson, Jason (USA) (1986)	time		2.48	3.50	4.50	5.50	6.50	7.50	8.48	9.48	10.54	11.58		13.06	3 / 2						
	reaction time	0.166	interval	7 steps	1.02	1.00	1.00	1.00	0.98	1.00	1.06	1.04	1.48			3.02	2.98	3.10			
			velocity	5.53	8.96	9.14	9.14	9.14	9.33	9.14	8.62	8.79	9.47	8.42		9.08	9.20	8.85			
Wilson, Ryan (USA) (1980)	time		2.52	3.56	4.60	5.58	6.56	7.56	8.58	9.60	10.64	11.70		13.18	7 / 3						
	reaction time	0.153	interval	7 steps	1.04	1.04	0.98	0.98	1.00	1.02	1.02	1.04	1.06	1.48		3.06	3.00	3.12			
			velocity	5.44	8.79	8.79	9.33	9.33	9.14	8.96	8.96	8.79	8.62	9.47	8.35		8.96	9.14	8.79		
Porter, Jeff (USA) (1985)	time		2.48	3.56	4.56	5.56	6.60	7.64	8.68	9.74	10.80	11.86		13.32	5 / 4						
	reaction time	0.154	interval		1.08	1.00	1.00	1.04	1.04	1.04	1.06	1.06	1.06	1.46		3.08	3.12	3.18			
			velocity	5.53	8.46	9.14	9.14	8.79	8.79	8.79	8.62	8.62	8.62	9.60	8.26		8.90	8.79	8.62		
Brown, Joel (USA) (1980)	time		2.52	3.60	4.64	5.64	6.68	7.72	8.76	9.80	10.86	11.92		13.39	6 / 5						
	reaction time	0.149	interval		1.08	1.04	1.00	1.04	1.04	1.04	1.06	1.06	1.47			3.12	3.12	3.16			
			velocity	5.44	8.46	8.79	9.14	8.79	8.79	8.79	8.79	8.62	8.62	9.54	8.22		8.79	8.79	8.68		
Hicks, Antwon (USA) (1983)	time		2.48	3.54	4.60	5.60	6.62	7.66	8.72	9.78	10.84	11.92		13.46	1 / 6						
	reaction time	0.165	interval	7 steps	1.06	1.06	1.00	1.02	1.04	1.06	1.06	1.06	1.08	1.54		3.12	3.12	3.20			
			velocity	5.53	8.62	8.62	9.14	8.96	8.79	8.62	8.62	8.62	8.46	9.10	8.17		8.79	8.79	8.57		
Pozzi, Andy (GBR) (1992)	time		2.52	3.56										dnf	2 / --						
	reaction time	0.160	interval		1.04																
			velocity	5.44	8.79																

Heat 2

Heat 2		date		13-Jul-12		wind		-0.9 m/s		Henson (2021) - Athlete First: 2012 Aries Merritt world record season												
				H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10			
Merritt, Aries (USA) (1985)				time	2.48	3.52	4.52	5.52	6.52	7.54	8.56	9.58	10.60	11.66		13.14	4 / 1					
reaction time				0.139	7 steps	1.04	1.00	1.00	1.00	1.02	1.02	1.02	1.02	1.06	1.48			3.04	3.04	3.10		
				interval																		
				velocity	5.53	8.79	9.14	9.14	9.14	8.96	8.96	8.96	8.96	8.62	9.47	8.37		9.02	9.02	8.85		
Richardson, Jason (USA) (1986)				time	2.48	3.54	4.56	5.58	6.60	7.62	8.64	9.66	10.68	11.76		13.23	3 / 2					
reaction time				0.161	7 steps	1.06	1.02	1.02	1.02	1.02	1.02	1.02	1.02	1.08	1.47			3.10	3.06	3.12		
				interval																		
				velocity	5.53	8.62	8.96	8.96	8.96	8.96	8.96	8.96	8.96	8.46	9.54	8.31		8.85	8.96	8.79		
Wilson, Ryan (USA) (1980)				time	2.56	3.60	4.60	5.60	6.60	7.62	8.64	9.68	10.72	11.78		13.28	5 / 3					
reaction time				0.157	7 steps	1.04	1.00	1.00	1.00	1.02	1.02	1.04	1.04	1.06	1.50			3.04	3.04	3.14		
				interval																		
				velocity	5.36	8.79	9.14	9.14	9.14	8.96	8.96	8.79	8.79	8.62	9.35	8.28		9.02	9.02	8.73		

2012 USA Olympic Trials (Eugene, OR)**FINAL**

NAME	date	30-Jun-12	wind	1.2 m/s	Henson (2020) - Athlete First: 2012 Aries Merritt world record season															
					H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
Merritt, Aries (USA) (1985)	time				2.37	3.40	4.40	5.39	6.37	7.36	8.34	9.34	10.36	11.37		12.93	5 / 1			
	reaction time	0.138	interval	7 steps	1.03	1.00	0.99	0.98	0.99	0.98	1.00	1.02	1.01	1.56	PB			3.02	2.95	3.03
			velocity	5.79	8.87	9.14	9.23	9.33	9.23	9.33	9.14	8.96	9.05	8.99	8.51			9.08	9.29	9.05
Richardson, Jason (USA) (1986)	time				2.42	3.44	4.44	5.44	6.45	7.44	8.44	9.44	10.44	11.44		12.98	6 / 2			
	reaction time	0.185	interval	7 steps	1.02	1.00	1.00	1.01	0.99	1.00	1.00	1.00	1.00	1.54	=PB			3.02	3.00	3.00
			velocity	5.67	8.96	9.14	9.14	9.05	9.23	9.14	9.14	9.14	9.14	9.10	8.47			9.08	9.14	9.14
Porter, Jeff (USA) (1985)	time				2.40	3.42	4.43	5.42	6.44	7.44	8.44	9.47	10.51	11.53		13.08	4 / 3			
	reaction time	0.157	interval			1.02	1.01	0.99	1.02	1.00	1.00	1.03	1.04	1.02	1.55	PB		3.02	3.02	3.09
			velocity	5.72	8.96	9.05	9.23	8.96	9.14	9.14	8.87	8.79	8.96	9.05	8.41			9.08	9.08	8.87

Hicks, Antwon (USA) (1983)	time	2.37	3.44	4.44	5.44	6.44	7.44	8.46	9.47	10.51	11.56		13.14	2 / 4				
	reaction time	0.158	interval	7 steps	1.07	1.00	1.00	1.00	1.00	1.02	1.01	1.04	1.05	1.58		3.07	3.02	3.10
		velocity	5.79	8.54	9.14	9.14	9.14	9.14	8.96	9.05	8.79	8.70	8.87	8.37		8.93	9.08	8.85
Oliver, David (USA) (1982)	time	2.43	3.45	4.47	5.47	6.45	7.44	8.44	9.45	10.51	11.58		13.17	3 / 5				
	reaction time	0.165	interval	7 steps	1.02	1.02	1.00	0.98	0.99	1.00	1.01	1.06	1.07	1.59		3.04	2.97	3.14
		velocity	5.65	8.96	8.96	9.14	9.33	9.23	9.14	9.05	8.62	8.54	8.82	8.35		9.02	9.23	8.73
Faulk, Dexter (USA) (1984)	time	2.42	3.46	4.48	5.50	6.56	7.60	8.62	9.64	10.69	11.74		13.23	8 / 6				
	reaction time	0.142	interval		1.04	1.02	1.06	1.04	1.02	1.02	1.05	1.05	1.49		3.08	3.12	3.12	
		velocity	5.67	8.79	8.96	8.96	8.62	8.79	8.96	8.96	8.70	8.70	9.41	8.31		8.90	8.79	8.79
Wilson, Ryan (USA) (1980)	time	2.43	3.45	4.45	5.42	6.40	7.39	8.37	9.47	10.54	11.61		13.24	7 / 7				
	reaction time	0.179	interval	7 steps	1.02	1.00	0.97	0.98	0.99	1.10	1.07	1.07	1.63		2.99	2.95	3.24	
		velocity	5.65	8.96	9.14	9.42	9.33	9.23	9.33	8.31	8.54	8.54	8.60	8.31		9.17	9.29	8.46
Akins, Tyron (USA) (1986)	time	2.40	3.47	4.50	5.54	6.59	7.61	8.61					dnf	1 / --				
	reaction time	0.179	interval		1.07	1.03	1.04	1.05	1.02	1.00					3.14	3.07		
		velocity	5.72	8.54	8.87	8.79	8.70	8.96	9.14						8.73	8.93		

2012 Chinese National Grand Prix Final (Fuzhou, CHN)

FINAL

date 24-Jun-12		wind 0.5 m/s		Wang (2016) - comparative analysis of the grey relational grade of segment times and final results of the 110m hurdles													
Xie Wenjun (CHN) (1990)		H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
	time	2.64	3.70	4.76	5.78	6.80	7.84	8.88	9.92	10.98	12.06		13.42	1 / 1			
	reaction time		interval	1.06	1.06	1.02	1.02	1.04	1.04	1.06	1.08	1.36	PB		3.14	3.10	3.18
	velocity	5.20	8.62	8.62	8.96	8.96	8.79	8.79	8.79	8.62	8.46	10.31	8.20		8.73	8.85	8.62

2012 Windsprint (Sundsvall, SWE)

FINAL

date 17-Jun-12		wind 1.1 m/s		Blomkvist (2012) - www.sub13.se													
Townsend, Fred (USA) (1982)		H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
	time	2.64	3.68	4.72	5.75	6.77	7.78	8.83	9.86	10.92	11.98		13.39	4 / 1			
	reaction time		interval	1.04	1.04	1.03	1.02	1.01	1.05	1.03	1.06	1.06	1.41	PB	3.11	3.08	3.15
	velocity	5.20	8.79	8.79	8.87	8.96	9.05	8.70	8.87	8.62	8.62	9.94	8.22		8.82	8.90	8.70
Nossmy, Philip (SWE) (1982)	time	2.65	3.74	4.76	5.81	6.84	7.88	8.94	9.99	11.04	12.13		13.56	5 / 2			
	reaction time		interval	1.09	1.02	1.05	1.03	1.04	1.06	1.05	1.05	1.09	1.43		3.16	3.13	3.19
	velocity	5.18	8.39	8.96	8.70	8.87	8.79	8.62	8.70	8.70	8.39	9.80	8.11		8.68	8.76	8.60
Vukicevic, Vladimir (NOR) (1991)	time	2.67	3.76	4.83	5.89	6.94	8.00	9.06	10.12	11.19	12.30		13.77	3 / 3			
	reaction time		interval	1.09	1.07	1.06	1.05	1.06	1.06	1.07	1.11	1.47			3.22	3.17	3.24
	velocity	5.14	8.39	8.54	8.62	8.70	8.62	8.62	8.62	8.54	8.23	9.54	7.99		8.52	8.65	8.46
de Vries, Ruan (RSA) (1986)	time	2.71	3.81	4.89	5.97	7.04	8.10	9.17	10.24	11.32	12.41		13.90	6 / 4			
	reaction time		interval	1.10	1.08	1.07	1.06	1.07	1.07	1.08	1.09	1.49			3.26	3.20	3.24
	velocity	5.06	8.31	8.46	8.46	8.54	8.62	8.54	8.54	8.46	8.39	9.41	7.91		8.41	8.57	8.46
McDonald, Ian (IRL) (1984)	time	2.74	3.84	4.94	6.03	7.12	8.21	9.35	10.48	11.62	12.77		14.36	1 / 5			
	reaction time		interval	1.10	1.10	1.09	1.09	1.14	1.13	1.14	1.15	1.59			3.29	3.32	3.42
	velocity	5.01	8.31	8.31	8.39	8.39	8.39	8.02	8.09	8.02	7.95	8.82	7.66		8.33	8.26	8.02
Stavang, Eivind (NOR) (1982)	time	2.72	3.85	4.99	6.12	7.23	8.36	9.49	10.64	11.79	12.95		14.47	2 / 6			
	reaction time		interval	1.13	1.14	1.13	1.11	1.13	1.13	1.15	1.15	1.16	1.52		3.40	3.37	3.46
	velocity	5.04	8.09	8.02	8.09	8.23	8.09	8.09	7.95	7.95	7.88	9.22	7.60		8.06	8.14	7.92
Shaker, Amir (SWE) (1989)	time	2.76	3.87	4.97	6.06	7.15	8.28	9.42	10.58	11.74	12.91		14.52	8 / 7			
	reaction time		interval	1.11	1.10	1.09	1.09	1.13	1.14	1.16	1.16	1.17	1.61		3.30	3.36	3.49
	velocity	4.97	8.23	8.31	8.39	8.39	8.09	8.02	7.88	7.88	7.81	8.71	7.58		8.31	8.16	7.86
Brorsson, Alexander (SWE) (1985)	time	2.67	3.76	4.81	5.89	7.03	8.16	9.26					dnf	7 / --			
	reaction time		interval	1.09	1.05	1.08	1.14	1.13	1.10						3.22	3.37	
	velocity	5.14	8.39	8.70	8.46	8.02	8.09	8.31							8.52	8.14	

Heat 2

date 17-Jun-12		wind 0.0 m/s		Blomkvist (2012) - www.sub13.se													
Brorsson, Alexander (SWE) (1985)		H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
	time	2.70	3.80	4.87	5.93	7.00	8.08	9.17	10.28	11.38	12.51		13.99	4 / 1			
	reaction time		interval	1.10	1.07	1.06	1.07	1.08	1.09	1.11	1.10	1.13	1.48		3.23	3.24	3.34
	velocity	5.08	8.31	8.54	8.62	8.54	8.46	8.39	8.23	8.31	8.09	9.47	7.86		8.49	8.46	8.21

2012 Japanese National Championships (Osaka, JPN)

FINAL

date 10-Jun-12		wind -0.6 m/s		Kota (2013) - race analysis of leading Japanese men's 110m hurdlers - from 2011-13													
Mita, Kyohei (JPN) (1991)		H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
	time	2.67	3.78	4.86	5.94	7.02	8.10	9.18	10.27	11.38	12.47		13.92	2 / 4			
	reaction time		interval	1.11	1.08	1.08	1.08	1.08	1.09	1.11	1.09	1.45			3.27	3.24	3.29
	velocity	5.14	8.23	8.46	8.46	8.46	8.46	8.46	8.39	8.23	8.39	9.67	7.90		8.39	8.46	8.33
Sato, Hiroyuki (JPN) (1990)	time	2.62	3.74	4.84	5.92	6.99	8.07	9.18	10.28	11.39	12.53		13.99	4 / 6			

reaction time	interval	1.12	1.10	1.08	1.07	1.08	1.11	1.10	1.11	1.14	1.46			3.30	3.26	3.35
	velocity	5.24	8.16	8.31	8.46	8.54	8.46	8.23	8.31	8.23	8.02	9.60	7.86	8.31	8.41	8.19

Semi-Final 2

Semi-Final 2		date	10-Jun-12	wind	Kota (2013) - race analysis of leading Japanese men's 110m hurdlers - from 2011-13															
				H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10	
Yazawa, Wataru (JPN) (1991)	time		2.61	3.67	4.72	5.78	6.85	7.90	8.99	10.08	11.18	12.32			13.75	5 / 1				
	reaction time			1.06	1.05	1.06	1.07	1.05	1.09	1.09	1.10	1.14	1.43					3.17	3.21	3.33
	velocity		5.26	8.62	8.70	8.62	8.54	8.70	8.39	8.39	8.31	8.02	9.80		8.00			8.65	8.54	8.23
Sato, Hiroyuki (JPN) (1990)	time		2.60	3.68	4.74	5.80	6.85	7.91	9.00	10.08	11.20	12.32			13.78	6 / 2				
	reaction time			1.08	1.06	1.06	1.05	1.06	1.09	1.08	1.12	1.12	1.46					3.20	3.20	3.32
	velocity		5.28	8.46	8.62	8.62	8.70	8.62	8.39	8.46	8.16	8.16	9.60		7.98			8.57	8.57	8.26

Semi-Final 1

Semi-Final 1		date	10-Jun-12	wind	Kota (2013) - race analysis of leading Japanese men's 110m hurdlers - from 2011-13														
				H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
Mita, Kyohei (JPN) (1991)	time		2.68	3.78	4.87	5.95	7.01	8.09	9.16	10.23	11.31	12.38			13.79	3 / 4			
	reaction time			1.10	1.09	1.08	1.06	1.08	1.07	1.07	1.08	1.07	1.41				3.27	3.21	3.22
	velocity		5.12	8.31	8.39	8.46	8.62	8.46	8.54	8.54	8.46	8.54	9.94	7.98			8.39	8.54	8.52
Wado, Tatsuya (JPN) (1990)	time		2.61	3.70	4.78	5.86	6.93	8.02	9.11	10.23	11.35	12.48			13.98	4 / 8			
	reaction time			1.09	1.08	1.08	1.07	1.09	1.09	1.12	1.12	1.13	1.50				3.25	3.25	3.37
	velocity		5.26	8.39	8.46	8.46	8.54	8.39	8.39	8.16	8.16	8.09	9.35	7.87			8.44	8.44	8.14

Heat 4

Heat 4

	date	09-Jun-12	wind		-0.2 m/s		Kota (2013) - race analysis of leading Japanese men's 110m hurdlers - from 2011-13												
			H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10	
Sato, Hiroyuki (JPN) (1990)	time		2.61	3.68	4.73	5.80	6.84	7.92	8.99	10.08	11.19	12.31		13.78	7 / 1				
	reaction time	interval		1.07	1.05	1.07	1.04	1.08	1.07	1.09	1.11	1.12	1.47				3.19	3.19	3.32
	velocity		5.26	8.54	8.70	8.54	8.79	8.46	8.54	8.39	8.23	8.16	9.54	7.98			8.60	8.60	8.26

Heat 3

Heat 3

	date	09-Jun-12	wind	0.0 m/s	Kota (2013) - race analysis of leading Japanese men's 110m hurdlers - from 2011-13														
			H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10	
Mita, Kyohei (JPN) (1991)	time		2.68	3.78	4.89	5.97	7.04	8.11	9.20	10.28	11.38	12.49		13.92	6 / 4				
reaction time	interval			1.10	1.11	1.08	1.07	1.07	1.09	1.08	1.10	1.11	1.43				3.29	3.23	3.29
	velocity		5.12	8.31	8.23	8.46	8.54	8.54	8.39	8.46	8.31	8.23	9.80	7.90			8.33	8.49	8.33

Heat 2

Heat 2

	date	09-Jun-12	wind	-1.1 m/s	Kota (2013) - race analysis of leading Japanese men's 110m hurdlers - from 2011-13														
			H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10	
Yazawa, Wataru (JPN) (1991)	time		2.62	3.71	4.77	5.84	6.91	7.99	9.10	10.21	11.32	12.46		13.94	5 / 2				
	reaction time			1.09	1.06	1.07	1.07	1.08	1.11	1.11	1.11	1.14	1.48				3.22	3.26	3.36
	velocity		5.24	8.39	8.62	8.54	8.54	8.46	8.23	8.23	8.23	8.02	9.47	7.89			8.52	8.41	8.16

Heat 1

Heat 1

	date	09-Jun-12	wind	0.8 m/s	Kota (2013) - race analysis of leading Japanese men's 110m hurdlers - from 2011-13														
			H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10	
Wado, Tatsuya (JPN) (1990)	time		2.59	3.68	4.76	5.82	6.89	7.98	9.07	10.19	11.30	12.45		13.98	5 / 2				
	reaction time			1.09	1.08	1.06	1.07	1.09	1.09	1.12	1.11	1.15	1.53				3.23	3.25	3.38
	velocity		5.30	8.39	8.46	8.62	8.54	8.39	8.39	8.16	8.23	7.95	9.16	7.87			8.49	8.44	8.11

2012 Japanese National Multi-Event Championships (Nagano, JPN)**Decathlon - Heat 2**

Decathlon - Heat 2																			
	date	03-Jun-12	wind	-1.7 m/s															
			H1	H2	H3	H4	H5	H6	Takeo (2014) - decathletes sprint ability and the relationship of the individual event performance										
									H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10	
Ushiro, Keisuke (JPN) (1986)	time		2.76	3.93	5.10	6.25	7.41	8.58	9.78	10.98	12.19	13.42		15.01	5 / 2				
	reaction time			1.17	1.17	1.15	1.16	1.17	1.20	1.20	1.21	1.23	1.59				3.49	3.53	3.64
	velocity		4.97	7.81	7.81	7.95	7.88	7.81	7.62	7.62	7.55	7.43	8.82	7.33			7.86	7.77	7.53

Decathlon - Heat 1

Decathlon - Heat 1	date	03-Jun-12	wind	1.7 m/s																Takeo (2014) - decathletes sprint ability and the relationship of the individual event performance															
				H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10																
	Nakamura, Akihiko (JPN) (1990)	time		2.67	3.76	4.84	5.94	7.03	8.12	9.21	10.34	11.47	12.63		14.11	5 / 2																			
		reaction time	interval		1.09	1.08	1.10	1.09	1.09	1.09	1.13	1.13	1.16	1.48				3.27	3.27	3.42															
			velocity	5.14	8.39	8.46	8.31	8.39	8.39	8.39	8.09	8.09	7.88	9.47	7.80			8.39	8.39	8.02															

2012 Prefontaine Classic (Eugene, OR) (TV Analysis)**FINAL**

		date	02-Jun-12	wind	Henson (2020) - Athlete First: 2012 Aries Merritt world record season														
				H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
Liu Xiang (CHN) (1983)	time			2.50	3.52	4.52	5.50	6.48	7.50	8.48	9.48	10.48	11.50		12.87	5 / 1			
	reaction time	0.131	interval	7 steps	1.02	1.00	0.98	0.98	1.02	0.98	1.00	1.00	1.02	1.37			3.00	2.98	3.02
			velocity		5.49	8.96	9.14	9.33	9.33	8.96	9.33	9.14	9.14	8.96	10.23	8.55		9.14	9.20
Merritt, Aries (USA) (1985)	time			2.52	3.54	4.56	5.58	6.56	7.56	8.56	9.54	10.54	11.58		12.96	3 / 2			
	reaction time	0.151	interval	7 steps	1.02	1.02	1.02	0.98	1.00	1.00	0.98	1.00	1.04	1.38			3.06	2.98	3.02
			velocity		5.44	8.96	8.96	8.96	9.33	9.14	9.14	9.33	9.14	8.79	10.16	8.49		8.96	9.20
Richardson, Jason (USA) (1986)	time			2.52	3.56	4.56	5.56	6.56	7.56	8.56	9.58	10.62	11.66		13.11	4 / 3			
	reaction time	0.159	interval	7 steps	1.04	1.00	1.00	1.00	1.00	1.00	1.02	1.04	1.04	1.45			3.04	3.00	3.10
			velocity		5.44	8.79	9.14	9.14	9.14	9.14	9.14	8.96	8.79	8.79	9.67	8.39		9.02	9.14

Faulk, Dexter (USA) (1984)	time	2.52	3.56	4.58	5.60	6.60	7.60	8.62	9.64	10.66	11.74		13.12	7 / 4			
	reaction time	0.143	interval	1.04	1.02	1.02	1.00	1.00	1.02	1.02	1.02	1.08	1.38		3.08	3.02	3.12
	velocity	5.44	8.79	8.96	8.96	9.14	9.14	8.96	8.96	8.96	8.46	10.16	8.38		8.90	9.08	8.79
Oliver, David (USA) (1982)	time	2.52	3.56	4.56	5.56	6.54	7.56	8.60	9.60	10.64	11.72		13.13	6 / 5			
	reaction time	0.171	interval	7 steps 1.04	1.00	1.00	0.98	1.02	1.04	1.00	1.04	1.08	1.41		3.04	3.04	3.12
	velocity	5.44	8.79	9.14	9.14	9.33	8.96	8.79	9.14	8.79	8.46	9.94	8.38		9.02	9.02	8.79
Wilson, Ryan (USA) (1980)	time	2.62	3.66	4.68	5.68	6.68	7.70	8.74	9.76	10.80	11.88		13.29	8 / 6			
	reaction time	0.194	interval	7 steps 1.04	1.02	1.00	1.00	1.02	1.04	1.02	1.04	1.08	1.41		3.06	3.06	3.14
	velocity	5.24	8.79	8.96	9.14	9.14	8.96	8.79	8.96	8.79	8.46	9.94	8.28		8.96	8.96	8.73
Eaton, Ashton (USA) (1988)	time	2.56	3.60	4.64	5.66	6.68	7.72	8.76	9.78	10.82	11.90		13.34	1 / 7			
	reaction time	0.168	interval	7 steps 1.04	1.04	1.02	1.02	1.04	1.02	1.04	1.08	1.44			3.10	3.10	3.14
	velocity	5.36	8.79	8.79	8.96	8.96	8.79	8.79	8.96	8.79	8.46	9.74	8.25		8.85	8.85	8.73
Turner, Andy (GBR) (1980)	time	2.60	3.64	4.68	5.72	6.74	7.76	8.82	9.86	10.92	12.00		13.46	2 / 8			
	reaction time	0.149	interval	1.04	1.04	1.04	1.02	1.06	1.04	1.06	1.08	1.46			3.12	3.10	3.18
	velocity	5.28	8.79	8.79	8.79	8.96	8.96	8.62	8.79	8.62	8.46	9.60	8.17		8.79	8.85	8.62

2012 Chinese National Grand Prix (Zibo, CHN)**FINAL**

date 27-May-12		wind 0.1 m/s	Wang (2016) - comparative analysis of the grey relational grade of segment times and final results of the 110m hurdles														
Xie Wenjun (CHN) (1990)	time	H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
	reaction time	2.64	3.72	4.76	5.82	6.86	7.88	8.92	9.95	11.00	12.08		13.48	/ 1			
	velocity	5.20	8.46	8.79	8.62	8.79	8.96	8.79	8.87	8.70	8.46	10.01	8.16		8.62	8.85	8.68

2012 Golden Grand Prix (Shanghai, CHN)**FINAL**

Henson (2020) - Athlete First: 2012 Aries Merritt world record season (TV Analysis)

date 19-May-12		wind 0.4 m/s	Feng (2016) - Liu Xiang's seven-step starting: discussion on the key issues of 110m hurdles technical training														
Liu Xiang (CHN) (1983)	time	H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
	reaction time	2.38	3.38	4.45	5.42	6.42	7.45	8.42	9.46	10.43	11.47		12.97	5 / 1			
	velocity	5.76	9.14	8.54	9.42	9.14	8.87	9.42	8.79	9.42	8.79	9.35	8.48		9.02	9.14	8.99
Oliver, David (USA) (1982)	time	2.50	3.52	4.50	5.50	6.50	7.48	8.50	9.52	10.56	11.62		13.13	7 / 2			Henson (2020)
	reaction time	0.192	interval	7 steps 1.02	0.98	1.00	1.00	0.98	1.02	1.02	1.04	1.06	1.51		3.00	3.00	3.12
	velocity	5.49	8.96	9.33	9.14	9.14	9.33	8.96	8.96	8.79	8.62	9.28	8.38		9.14	9.14	8.79
Richardson, Jason (USA) (1986)	time	2.48	3.52	4.54	5.54	6.52	7.54	8.56	9.56	10.60	11.66		13.16	4 / 3			Henson (2020)
	reaction time	0.168	interval	7 steps 1.04	1.02	1.00	0.98	1.02	1.00	1.04	1.06	1.50			3.06	3.02	3.10
	velocity	5.53	8.79	8.96	9.14	9.33	8.96	8.96	9.14	8.79	8.62	9.35	8.36		8.96	9.08	8.85
Merritt, Aries (USA) (1985)	time	2.34	3.38	4.42	5.46	6.50	7.54	8.58	9.62	10.66	11.70		13.26	6 / 4			Feng (2016)
	reaction time	0.142	interval	7 steps 1.04	1.04	1.04	1.04	1.04	1.04	1.04	1.04	1.56			3.12	3.12	3.12
	velocity	5.86	8.79	8.79	8.79	8.79	8.79	8.79	8.79	8.79	8.79	8.99	8.30		8.79	8.79	8.79
Faulk, Dexter (USA) (1984)	time	2.52	3.58	4.62	5.64	6.66	7.70	8.74	9.80	10.86	11.94		13.43	2 / 5			Henson (2020)
	reaction time	0.177	interval	1.06	1.04	1.02	1.02	1.04	1.06	1.06	1.08	1.49			3.12	3.10	3.20
	velocity	5.44	8.62	8.79	8.96	8.96	8.79	8.79	8.62	8.62	8.46	9.41	8.19		8.79	8.85	8.57
Craddock, Kevin (USA) (1987)	time	2.44	3.52	4.54	5.56	6.58	7.62	8.68	9.76	10.84	11.94		13.51	9 / 6			Henson (2020)
	reaction time	0.167	interval	7 steps 1.08	1.02	1.02	1.02	1.04	1.06	1.08	1.08	1.10	1.57		3.12	3.12	3.26
	velocity	5.62	8.46	8.96	8.96	8.96	8.79	8.62	8.46	8.46	8.31	8.93	8.14		8.79	8.79	8.41
Shi Dongpeng (CHN) (1984)	time	2.52	3.60	4.66	5.70	6.74	7.78	8.82	9.86	10.96	12.06		13.60	1 / 7			Henson (2020)
	reaction time	0.155	interval	1.08	1.06	1.04	1.04	1.04	1.04	1.10	1.10	1.54			3.18	3.12	3.24
	velocity	5.44	8.46	8.62	8.79	8.79	8.79	8.79	8.79	8.31	8.31	9.10	8.09		8.62	8.79	8.46
Xie Wenjun (CHN) (1990)	time	2.54	3.64	4.68	5.74	6.78	7.82	8.88	9.96	11.06	12.14		13.64	8 / 8			Henson (2020)
	reaction time	0.179	interval	1.10	1.04	1.06	1.04	1.04	1.06	1.08	1.10	1.08	1.50		3.20	3.14	3.26
	velocity	5.40	8.31	8.79	8.62	8.79	8.79	8.62	8.46	8.31	8.46	9.35	8.06		8.57	8.73	8.41
Thomas, Dwight (JAM) (1980)	time	2.58	3.68	4.78	5.86	6.90	7.94	8.98	10.04	11.12	12.20		13.72	3 / 9			Henson (2020)
	reaction time	0.174	interval	7 steps 1.10	1.10	1.08	1.04	1.04	1.06	1.08	1.08	1.52			3.28	3.12	3.22
	velocity	5.32	8.31	8.31	8.46	8.79	8.79	8.79	8.62	8.46	8.46	9.22	8.02		8.36	8.79	8.52

2012 Colorful Daegu Meeting (Daegu, KOR) (TV Analysis)**FINAL**

date 16-May-12		wind -0.4 m/s	Henson (2021) - Athlete First: 2012 Aries Merritt world record season														
Merritt, Aries (USA) (1985)	time	H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
	reaction time	2.53	3.60	4.60	5.60	6.60	7.60	8.64	9.67				13.21	3 / 1			
	velocity	5.42	8.54	9.14	9.14	9.14	9.14	8.79	8.87				8.33		8.93	9.02	
Oliver, David (USA) (1982)	time	2.56	3.60	4.60	5.60	6.60	7.60	8.64	9.67	10.71			13.22	5 / 2			
	reaction time	0.180	interval	7 steps 1.04	1.00	1.00	1.00	1.04	1.03	1.04					3.04	3.04	
	velocity	5.36	8.79	9.14	9.14	9.14	9.14	8.79	8.87	8.79			8.32		9.02	9.02	

Richardson, Jason (USA) (1986)	time	2.65	3.70	4.70	5.72	6.74	7.77	8.77	9.79			13.34	4 / 3				
	reaction time	0.234	interval	7 steps	1.05	1.00	1.02	1.02	1.03	1.00	1.02				3.07	3.05	
	velocity	5.18	8.70	9.14	8.96	8.96	8.87	9.14	8.96			8.25			8.93	8.99	

2012 Kanto (Tokyo, JPN)

FINAL

INAL	date	13-May-12	wind	Karube (2013) - the analysis of hurdle clearance and interval sprint in men's 110m sprint hurdle															
			H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10	
Omuro, Hideki (JPN) (1990)	time		2.67	3.78	4.86	5.94	7.05	8.11	9.20	10.28	11.37	12.47		13.88	/ 2				
	reaction time			1.11	1.08	1.08	1.11	1.06	1.09	1.08	1.09	1.10	1.41				3.27	3.26	3.27
	interval																		
	velocity		5.14	8.23	8.46	8.46	8.23	8.62	8.39	8.46	8.39	8.31	9.94	7.93			8.39	8.41	8.39

2012 Seiko Golden Grand Prix (Kawasaki, JPN)

FINAL

INAL	date	06-May-12	wind	0.2 m/s	Kota (2013) - race analysis of leading Japanese men's 110m hurdlers - from 2011-13														
			H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10	
Liu Xiang (CHN) (1983)	time		2.51	3.57	4.57	5.59	6.58	7.59	8.61	9.62	10.63	11.67		13.09	4 / 1			Karube (2013)	
	reaction time		interval	7 steps	1.06	1.00	1.02	0.99	1.01	1.02	1.01	1.01	1.04	1.42			3.08	3.02	3.06
		velocity		5.47	8.62	9.14	8.96	9.23	9.05	8.96	9.05	9.05	8.79	9.87	8.40		8.90	9.08	8.96
Wado, Tatsuya (JPN) (1990)	time		2.55	3.64	4.71	5.75	6.80	7.86	8.93	10.00	11.09	12.22		13.71	1 / 4			Kota (2013)	
	reaction time			1.09	1.07	1.04	1.05	1.06	1.07	1.07	1.09	1.13	1.49	PB			3.20	3.18	3.29
		velocity		5.38	8.39	8.54	8.79	8.70	8.62	8.54	8.54	8.39	8.09	9.41	8.02		8.57	8.62	8.33
Sato, Hiroyuki (JPN) (1990)	time		2.62	3.71	4.76	5.84	6.90	7.97	9.06	10.14	11.25	12.38		13.83	6 / 6			Kota (2013)	
	reaction time			1.09	1.05	1.08	1.06	1.07	1.09	1.08	1.11	1.13	1.45				3.22	3.22	3.32
		velocity		5.24	8.39	8.70	8.46	8.62	8.54	8.39	8.46	8.23	8.09	9.67	7.95		8.52	8.52	8.26

2012 Mito Invitational Athletics Meeting (Mito, JPN)

FINAL

		date	04-May-12	wind -0.3 m/s			Kota (2013) - race analysis of leading Japanese men's 110m hurdlers - from 2011-13												
				H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
Sato, Hiroyuki (JPN) (1990)	time		2.60	3.67	4.74	5.81	6.87	7.95	9.02	10.09	11.19	12.31			13.77	/ 1			
	reaction time			1.07	1.07	1.07	1.06	1.08	1.07	1.07	1.10	1.12	1.46				3.21	3.21	3.29
	interval																8.54	8.54	8.33
	velocity		5.28	8.54	8.54	8.54	8.62	8.46	8.54	8.54	8.31	8.16	9.60	7.99					
Omuro, Hideki (JPN) (1990)	time		2.65	3.72	4.80	5.87	6.91	7.99	9.07	10.16	11.25	12.35			13.78	/ 3			
	reaction time			1.07	1.08	1.07	1.04	1.08	1.08	1.09	1.09	1.10	1.43				3.22	3.20	3.28
	interval																8.52	8.57	8.36
	velocity		5.18	8.54	8.46	8.54	8.79	8.46	8.46	8.39	8.39	8.31	9.80	7.98					
Wado, Tatsuya (JPN) (1990)	time		2.57	3.68	4.75	5.81	6.90	7.96	9.04	10.13	11.26	12.38			13.88	/ 5			
	reaction time			1.11	1.07	1.06	1.09	1.06	1.08	1.09	1.13	1.12	1.50				3.24	3.23	3.34
	interval																8.46	8.49	8.21
	velocity		5.34	8.23	8.54	8.62	8.39	8.62	8.46	8.39	8.09	8.16	9.35	7.93					

2012 Chinese National Grand Prix (Wuhan, CHN)

FINAL

FINAL	date	29-Apr-12	wind	1.1 m/s	Wang (2016) - comparative analysis of the grey relational grade of segment times and final results of the 110m hurdles													
					H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4
Xie Wenjun (CHN) (1990)	time		2.64	3.72	4.76	5.82	6.86	7.88	8.92	9.95	11.00	12.08		13.48	/ 1			
	reaction time			1.08	1.04	1.06	1.04	1.02	1.04	1.03	1.05	1.08	1.40			3.18	3.10	3.16
	interval																	
	velocity		5.20	8.46	8.79	8.62	8.79	8.96	8.79	8.87	8.70	8.46	10.01	8.16		8.62	8.85	8.68

2012 Mikio Oda Memorial (Hiroshima, JPN)

A FINAL

FINAL		date	29-Apr-12		wind		2.1 m/s		Kota (2013) - race analysis of leading Japanese men's 110m hurdlers - from 2011-13									
			H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
Sato, Hiroyuki (JPN) (1990)	time		2.61	3.66	4.71	5.74	6.79	7.84	8.91	9.99	11.07	12.17		13.59	3 / 2			
	reaction time			1.05	1.05	1.03	1.05	1.05	1.07	1.08	1.08	1.10	1.42			3.13	3.17	3.26
	interval															8.76	8.65	8.41
	velocity		5.26	8.70	8.70	8.87	8.70	8.70	8.54	8.46	8.46	8.31	9.87	8.09				
Yazawa, Wataru (JPN) (1991)	time		2.59	3.65	4.71	5.78	6.85	7.93	8.99	10.06	11.15	12.27		13.69	6 / 3			
	reaction time			1.06	1.06	1.07	1.07	1.08	1.06	1.07	1.09	1.12	1.42			3.19	3.21	3.28
	interval															8.60	8.54	8.36
	velocity		5.30	8.62	8.62	8.54	8.54	8.46	8.62	8.54	8.39	8.16	9.87	8.04				
Wado, Tatsuya (JPN) (1990)	time		2.58	3.65	4.73	5.78	6.84	7.92	8.98	10.04	11.15	12.26		13.70	5 / 4			
	reaction time			1.07	1.08	1.05	1.06	1.08	1.06	1.06	1.11	1.11	1.44			3.20	3.20	3.28
	interval															8.57	8.57	8.36
	velocity		5.32	8.54	8.46	8.70	8.62	8.46	8.62	8.62	8.23	8.23	9.74	8.03				

B FINAL

FINAL

	date	29-Apr-12	wind	2.1 m/s	Kota (2013) - race analysis of leading Japanese men's 110m hurdlers - from 2011-13														
			H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10	
Omuro, Hideki (JPN) (1990)	time		2.60	3.70	4.76	5.84	6.92	8.00	9.08	10.17	11.27	12.39		13.83	7 / 2				
	reaction time			1.10	1.06	1.08	1.08	1.08	1.08	1.09	1.10	1.12	1.44				3.24	3.24	3.31
	interval																8.46	8.46	8.28
	velocity		5.28	8.31	8.62	8.46	8.46	8.46	8.46	8.39	8.31	8.16	9.74	7.95					
Mita, Kyohei (JPN) (1991)	time		2.69	3.78	4.86	5.94	7.03	8.10	9.19	10.28	11.37	12.46		13.87	6 / 3				
	reaction time			1.09	1.08	1.08	1.09	1.07	1.09	1.09	1.09	1.09	1.41	PB			3.25	3.25	3.27
	interval																8.44	8.44	8.39
	velocity		5.10	8.39	8.46	8.46	8.39	8.54	8.39	8.39	8.39	8.39	9.94	7.93					

2012 Tokyo Athletics Championships (Chofu, JPN)

Decathlon - Heat ?

date 22-Apr-12 wind -0.1 m/s

Takeo (2014) - decathletes sprint ability and the relationship of the individual event performance

		H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
Nakamura, Akihiko (JPN) (1990)	time	2.71	3.83	4.94	6.05	7.15	8.26	9.35	10.46	11.59	12.72		14.15	???			
	reaction time			1.12	1.11	1.11	1.10	1.11	1.09	1.11	1.13	1.43			3.34	3.30	3.37
	interval																
	velocity	5.06	8.16	8.23	8.23	8.31	8.23	8.39	8.23	8.09	8.09	9.80	7.77		8.21	8.31	8.14
Ushiro, Keisuke (JPN) (1986)	time	2.79	3.95	5.15	6.31	7.50	8.73	9.95	11.17	12.41	13.64		15.25	???			
	reaction time			1.16	1.20	1.16	1.19	1.23	1.22	1.22	1.24	1.61			3.52	3.64	3.69
	interval																
	velocity	4.92	7.88	7.62	7.88	7.68	7.43	7.49	7.49	7.37	7.43	8.71	7.21		7.79	7.53	7.43

2012 Arkansas Twilight (Fayetteville, AR)**FINAL**

date 04-May-12 wind 1.0 m/s

Henson (2020) - Athlete First: 2012 Aries Merritt world record season

		H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
Merritt, Aries (USA) (1985)	time	2.50	3.53	4.55	5.57	6.57	7.54	8.54	9.54	10.54			13.03	5 / 1			
	reaction time			7 steps	1.03	1.02	1.00	0.97	1.00	1.00					3.07	2.97	
	interval																
	velocity	5.49	8.87	8.96	8.96	9.14	9.42	9.14	9.14	9.14			8.44		8.93	9.23	

Heat 3

date 04-May-12 wind 2.9 m/s

Henson (2020) - Athlete First: 2012 Aries Merritt world record season

		H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
Merritt, Aries (USA) (1985)	time	2.50	3.53	4.53	5.53	6.52	7.51	8.50	9.51	10.52	11.54		12.99	5 / 1			
	reaction time			7 steps	1.03	1.00	0.99	0.99	1.01	1.01	1.02	1.45			3.03	2.97	3.04
	interval																
	velocity	5.49	8.87	9.14	9.14	9.23	9.23	9.23	9.05	9.05	8.96	9.67	8.47		9.05	9.23	9.02

2012 IAAF World Indoor Championships (Istanbul, TUR) (TV Analysis)**FINAL**

date 11-Mar-12

Henson (2020) - Athlete First: 2012 Aries Merritt world record season

		H1	H2	H3	H4	H5						Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
Merritt, Aries (USA) (1985)	time	2.48	3.48	4.50	5.50	6.50							7.44	4 / 1			
	reaction time	0.135		7 steps	1.00	1.02	1.00	1.00				0.94			3.02		
	interval											10.34	8.06		9.08		
	velocity	5.53	9.14	8.96	9.14	9.14											
Liu Xiang (CHN) (1983)	time	2.50	3.52	4.54	5.56	6.58							7.49	3 / 2			
	reaction time	0.167		7 steps	1.02	1.02	1.02	1.02				0.91			3.06		
	interval											10.68	8.01		8.96		
	velocity	5.49	8.96	8.96	8.96	8.96											
Martinot Lagarde, Pascal (FRA)	time	2.50	3.54	4.56	5.58	6.58							7.53	6 / 3			
	reaction time	0.141		1.04	1.02	1.00						0.95			3.08		
	interval											10.23	7.97		8.90		
	velocity	5.49	8.79	8.96	8.96	9.14											
Pozzi, Andy (GBR) (1992)	time	2.54	3.56	4.58	5.60	6.60							7.58	5 / 4			
	reaction time	0.155		1.02	1.02	1.00						0.98			3.06		
	interval											9.92	7.92		8.96		
	velocity	5.40	8.96	8.96	8.96	9.14											
Shabanov, Konstantin (RUS) (1985)	time	2.58	3.60	4.62	5.64	6.66							7.60	2 / 5			
	reaction time	0.202		1.02	1.02	1.02						0.94			3.06		
	interval											10.34	7.89		8.96		
	velocity	5.32	8.96	8.96	8.96	8.96											
Abate, Emanuele (ITA) (1985)	time	2.60	3.62	4.64	5.66	6.68							7.63	7 / 6			
	reaction time	0.155		7 steps	1.02	1.02	1.02	1.02				0.95			3.06		
	interval											10.23	7.86		8.96		
	velocity	5.28	8.96	8.96	8.96	8.96											
Fourie, Lehann (RSA) (1987)	time	2.52	3.58	4.62	5.66	6.70							7.69	8 / 7			
	reaction time	0.147		7 steps	1.06	1.04	1.04	1.04				0.99			3.14		
	interval											9.82	7.80		8.73		
	velocity	5.44	8.62	8.79	8.79	8.79											
Noga, Artur (POL) (1988)	time	2.60	3.68	4.68	5.70	6.76							7.74	1 / 8			
	reaction time	0.157		1.08	1.00	1.02	1.06					0.98			3.10		
	interval											9.92	7.75		8.85		
	velocity	5.28	8.46	9.14	8.96	8.62											

2012 USATF National Indoor Championships (Albuquerque, NM) (Altitude) (TV Analysis)**FINAL**

date 26-Feb-12

Henson (2020) - Athlete First: 2012 Aries Merritt world record season

		H1	H2	H3	H4	H5						Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
Merritt, Aries (USA) (1985)	time	2.50	3.52	4.56	5.54	6.54							7.43	6 / 1			
	reaction time	0.138		7 steps	1.02	1.04	0.98	1.00				0.89	PB		3.04		
	interval											10.92	8.08		9.02		
	velocity	5.49	8.96	8.79	9.33	9.14											

2012 Czech Republic National Indoor Championships (Praha, CZE)**FINAL**

date 25-Feb-12

Feher (2012) - Czech Republic indoor national championships - men's 60m hurdles

		H1	H2	H3	H4	H5						Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
Mazáč, Martin (CZE) (1990)	time	2.68	3.76	4.77	5.82	6.84							7.81	4 / 1			
	reaction time	0.138		1.08	1.01	1.05	1.02					0.97			3.14		
	interval											10.02	7.68		8.73		
	velocity	5.12	8.46	9.05	8.70	8.96											
Sedlák, Václav (CZE) (1993)	time	2.67	3.78	4.85	5.93	7.00							7.89	5 / 2			
	reaction time	0.141		1.11	1.07	1.08	1.07					0.89			3.26		
	interval											10.92	7.60		8.41		
	velocity	5.14	8.23	8.54	8.46	8.54											

Penáz, Petr (CZE) (1991)	time	2.72	3.83	4.90	5.93	7.02		7.99	3 / 3	
	reaction time	0.156		1.11	1.07	1.03	1.09	0.97		3.21
	velocity		5.04	8.23	8.54	8.87	8.39	10.02	7.51	8.54
Kavka, Tomáš (CZE) (1989)	time	2.74	3.88	4.94	6.01	7.10		8.05	6 / 4	
	reaction time	0.153		1.14	1.06	1.07	1.09	0.95		3.27
	velocity		5.01	8.02	8.62	8.54	8.39	10.23	7.45	8.39
Helcelet, Adam Sebastian (CZE)	time	2.70	3.83	4.88	5.97	7.09		8.06	7 / 5	
	reaction time	0.140		1.13	1.05	1.09	1.12	0.97		3.27
	velocity		5.08	8.09	8.70	8.39	8.16	10.02	7.44	8.39
Pokorny, Martin (CZE) (1988)	time	2.72	3.84	4.94	6.03	7.14		8.17	1 / 6	
	reaction time	0.156		1.12	1.10	1.09	1.11	1.03		3.31
	velocity		5.04	8.16	8.31	8.39	8.23	9.44	7.34	8.28
Sedlák, Pavel (CZE) (1983)	time	2.67	3.80	4.89	6.02	7.16		8.23	8 / 7	
	reaction time	0.149		1.13	1.09	1.13	1.14	1.07		3.35
	velocity		5.14	8.09	8.39	8.09	8.02	9.08	7.29	8.19
Nejedly, Adam (CZE) (1988)	time	2.78	3.95	5.06	6.20	7.29		8.27	2 / 8	
	reaction time	0.187		1.17	1.11	1.14	1.09	0.98		3.42
	velocity		4.94	7.81	8.23	8.02	8.39	9.92	7.26	8.02

Heat 3

date 25-Feb-12

Feher (2012) - Czech Republic indoor national championships - men's 60m hurdles

			H1	H2	H3	H4	H5	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
Sedlák, Václav (CZE) (1993)	time	2.70	3.82	4.88	5.99	7.05		7.98	4 / 1				
	reaction time	0.141		1.12	1.06	1.11	1.06	0.93			3.29		
	velocity		5.08	8.16	8.62	8.23	8.62	10.45	7.52		8.33		
Helcelet, Adam Sebastian (CZE)	time	2.69	3.82	4.90	5.96	7.13		8.13	5 / 2				
	reaction time	0.135		1.13	1.08	1.06	1.17	1.00			3.27		
	velocity		5.10	8.09	8.46	8.62	7.81	9.72	7.38		8.39		
Nejedly, Adam (CZE) (1988)	time	2.75	3.91	5.01	6.12	7.23		8.18	3 / 3				
	reaction time	0.137		1.16	1.10	1.11	1.11	0.95			3.37		
	velocity		4.99	7.88	8.31	8.23	8.23	10.23	7.33		8.14		
Pokorny, Martin (CZE) (1988)	time	2.71	3.85	4.95	6.06	7.17		8.20	6 / 4				
	reaction time	0.154		1.14	1.10	1.11	1.11	1.03			3.35		
	velocity		5.06	8.02	8.31	8.23	8.23	9.44	7.32		8.19		
Ján, Jan (CZE) (1993)	time	2.75	3.92	5.04	6.21	7.34		8.39	2 / 5				
	reaction time	0.141		1.17	1.12	1.17	1.13	1.05			3.46		
	velocity		4.99	7.81	8.16	7.81	8.09	9.26	7.15		7.92		
Frolík, Petr (CZE) (1991)	time	2.82	4.01	5.19	6.37	7.54		8.56	7 / 6				
	reaction time	0.172		1.19	1.18	1.18	1.17	1.02			3.55		
	velocity		4.87	7.68	7.75	7.75	7.81	9.53	7.01		7.72		
Král, Lukáš (CZE) (1992)	time	2.87	4.09	5.32	6.54	7.77		8.84	1 / 7				
	reaction time	0.202		1.22	1.23	1.22	1.23	1.07			3.67		
	velocity		4.78	7.49	7.43	7.49	7.43	9.08	6.79		7.47		
Dinga, Libor (CZE) (1984)	time	2.92	4.15	5.38	6.60	7.85		8.92	8 / 8				
	reaction time	0.174		1.23	1.23	1.22	1.25	1.07			3.68		
	velocity		4.70	7.43	7.43	7.49	7.31	9.08	6.73		7.45		

Heat 2

date 25-Feb-12

Feher (2012) - Czech Republic indoor national championships - men's 60m hurdles

			H1	H2	H3	H4	H5	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
Penáz, Petr (CZE) (1991)	time	2.75	3.85	4.92	5.97	7.03		8.01	4 / 1				
	reaction time	0.168		1.10	1.07	1.05	1.06	0.98			3.22		
	velocity		4.99	8.31	8.54	8.70	8.62	9.92	7.49		8.52		
Kavka, Tomáš (CZE) (1989)	time	2.75	3.87	4.95	6.02	7.11		8.09	3 / 2				
	reaction time	0.143		1.12	1.08	1.07	1.09	0.98			3.27		
	velocity		4.99	8.16	8.46	8.54	8.39	9.92	7.42		8.39		
Krejčí, Michal (CZE) (1981)	time	2.76	3.90	5.04	6.16	7.28		8.27	5 / 3				
	reaction time	0.108		1.14	1.14	1.12	1.12	0.99			3.40		
	velocity		4.97	8.02	8.02	8.16	8.16	9.82	7.26		8.06		
Lukáš, Marek (CZE) (1991)	time	2.66	3.83	4.99	6.14	7.29		8.27	2 / 4				
	reaction time	0.111		1.17	1.16	1.15	1.15	0.98			3.48		
	velocity		5.16	7.81	7.88	7.95	7.95	9.92	7.26		7.88		

Hunčovsky, Ondrej (CZE) (1990) time 2.80 4.01 5.22 6.42 7.62
 reaction time 0.141 interval 1.21 1.21 1.20 1.20
 velocity 4.90 7.55 7.55 7.62 7.62

8.67 7 / 5
 1.05 3.62
 9.26 6.92 7.57

Vondráček, Jan (CZE) (1991) time 2.88 4.06 5.27 6.48 7.66
 reaction time 0.157 interval 1.18 1.21 1.21 1.18
 velocity 4.76 7.75 7.55 7.55 7.75

8.68 1 / 6
 1.02 3.60
 9.53 6.91 7.62

Šebrle, Roman (CZE) (1974) time 2.85 4.07 5.30 6.54 7.83
 reaction time 0.140 interval 1.22 1.23 1.24 1.29
 velocity 4.81 7.49 7.43 7.37 7.09

9.05 8 / 7
 1.22 3.69
 7.97 6.63 7.43

Heat 1

date 25-Feb-12

Feher (2012) - Czech Republic indoor national championships - men's 60m hurdles

		H1	H2	H3	H4	H5	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
Mazáč, Martin (CZE) (1990)	time	2.70	3.80	4.83	5.89	6.91		7.84	4 / 1			
	reaction time	0.161					0.93				3.19	
	interval		1.10	1.03	1.06	1.02						
	velocity	5.08	8.31	8.87	8.62	8.96	10.45	7.65			8.60	

Sedlák, Pavel (CZE) (1983) time 2.66 3.80 4.90 6.04 7.18
 reaction time 0.135 interval 1.14 1.10 1.14 1.14
 velocity 5.16 8.02 8.31 8.02 8.02

8.20 5 / 2
 1.02 3.38
 9.53 7.32 8.11

Haloun, Lukáš (CZE) (1991) time 2.70 3.86 5.00 6.13 7.30
 reaction time 0.122 interval 1.16 1.14 1.13 1.17
 velocity 5.08 7.88 8.02 8.09 7.81

8.27 2 / 3
 0.97 3.43
 10.02 7.26 7.99

Hedvičák, Jaroslav (CZE) (1990) time 2.74 3.90 5.05 6.18 7.33
 reaction time 0.149 interval 1.16 1.15 1.13 1.15
 velocity 5.01 7.88 7.95 8.09 7.95

8.32 6 / 4
 0.99 3.44
 9.82 7.21 7.97

Carda, Jan (CZE) (1989) time 2.76 3.90 5.04 6.16 7.40
 reaction time 0.157 interval 1.14 1.14 1.12 1.24
 velocity 4.97 8.02 8.02 8.16 7.37

8.51 3 / 5
 1.11 3.40
 8.76 7.05 8.06

Šimánek, Jindřich (CZE) (1979) time 2.85 4.01 5.19 6.35 7.52
 reaction time 0.189 interval 1.16 1.18 1.16 1.17
 velocity 4.81 7.88 7.75 7.88 7.81

8.55 7 / 6
 1.03 3.50
 9.44 7.02 7.83

Juránek, Martin (CZE) (1991) time 2.85 4.06 5.29 6.54 7.76
 reaction time 0.174 interval 1.21 1.23 1.25 1.22
 velocity 4.81 7.55 7.43 7.31 7.49

8.78 1 / 7
 1.02 3.69
 9.53 6.83 7.43

Linhart, David (CZE) (1989) time 2.92 4.15 5.37 6.60 7.83
 reaction time 0.175 interval 1.23 1.22 1.23 1.23
 velocity 4.70 7.43 7.49 7.43 7.43

8.93 8 / 8
 1.10 3.68
 8.84 6.72 7.45

2012 Italian National Indoor Championships (Ancona, ITA)

FINAL

date 25-Feb-12

FIDAL - Centro Studi & Ricerche (2012) - www.youtube.com/watch?v=SIDALf6RdBY

		H1	H2	H3	H4	H5	Run-In	Official Time	Lane / Place	H1-H4
Dal Molin, Paolo (ITA) (1987)	time	2.60	3.66	4.71	5.75	6.82		7.75	6 / 1	
	reaction time						0.93			3.15
	interval		1.06	1.05	1.04	1.07				
	velocity	5.28	8.62	8.70	8.79	8.54	10.45	7.74		8.70

2012 Italian National Indoor Junior Championships (Ancona, ITA) (99cm)

FINAL

date 18-Feb-12

FIDAL - Centro Studi & Ricerche (2012) - www.youtube.com/watch?v=SU_c9jUKwsw

		H1	H2	H3	H4	H5	Run-In	Official Time	Lane / Place	H1-H4
Perini, Lorenzo (ITA) (1994)	time	2.68	3.77	4.85	5.90	6.95		7.91	3 / 1	
	reaction time						0.96	PB		3.22
	interval		1.09	1.08	1.05	1.05				
	velocity	5.12	8.39	8.46	8.70	8.70	10.13	7.59		8.52

2012 Italian National Indoor Youth Championships (Ancona, ITA) (91cm)

FINAL

date 18-Feb-12

FIDAL - Centro Studi & Ricerche (2012) - www.youtube.com/watch?v=SIDALf6RdBY

		H1	H2	H3	H4	H5	Run-In	Official Time	Lane / Place	H1-H4
Faragona, Alessandro (ITA) (1994)	time	2.68	3.78	4.83	5.87	6.93		7.93	4 / 1	
	reaction time						1.00	PB		3.19
	interval	7 steps	1.10	1.05	1.04	1.06				
	velocity	5.12	8.31	8.70	8.79	8.62	9.72	7.57		8.60

2012 Aviva Grand Prix (Birmingham, GBR)

FINAL

date 18-Feb-12

Wang (2012) - analysis of the time characteristics of Liu Xiang in his first 110m hurdles competition of 2012

		H1	H2	H3	H4	H5	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
Liu Xiang (CHN) (1983)	time	2.54	3.54	4.54	5.54	6.51		7.41	1 / 1			
	reaction time	0.154					0.90	AR PB			3.00	
	interval		1.00	1.00	1.00	0.97						
	velocity	5.40	9.14	9.14	9.14	9.42	10.80	8.10			9.14	

2012 Leonard Hilton Memorial Invitational (Houston, TX)

FINAL

date 13-Jan-12

Henson (2020) - Athlete First: 2012 Aries Merritt world record season

		H1	H2	H3	H4	H5												
Merritt, Aries (USA) (1985)	time	2.53	3.53	4.53	5.52	6.52								7.46	5 / 1			
	reaction time	interval	7 steps	1.00	1.00	0.99	1.00						0.94			2.99		
		velocity	5.42	9.14	9.14	9.23	9.14						10.34	8.04		9.17		

2011 IAAF World Championships (Daegu, KOR) (TV Analysis)

FINAL

date 29-Aug-11 wind -1.1 m/s

Henson (2021) - Athlete First: major championships report

		H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
Robles, Dayron (CUB) (1986)	time	2.40	3.45	4.47	5.47	6.47	7.47	8.47	9.51	10.54	11.64		13.14	5 / 1			
	reaction time	0.150	7 steps	1.05	1.02	1.00	1.00	1.00	1.04	1.03	1.10	1.50	DQ		3.07	3.00	3.17
		velocity	5.72	8.70	8.96	9.14	9.14	9.14	8.79	8.87	8.31	9.35	8.37		8.93	9.14	8.65
Richardson, Jason (USA) (1986)	time	2.50	3.53	4.55	5.54	6.54	7.54	8.55	9.57	10.61	11.66		13.16	3 / 2			
	reaction time	0.178	7 steps	1.03	1.02	0.99	1.00	1.01	1.02	1.04	1.05	1.50			3.04	3.01	3.11
		velocity	5.49	8.87	8.96	9.23	9.14	9.05	8.96	8.79	8.70	9.35	8.36		9.02	9.11	8.82
Liu Xiang (CHN) (1983)	time	2.48	3.52	4.52	5.52	6.52	7.51	8.51	9.52	10.54	11.62		13.27	6 / 3			
	reaction time	0.164	7 steps	1.04	1.00	1.00	0.99	1.00	1.01	1.02	1.08	1.65			3.04	2.99	3.11
		velocity	5.53	8.79	9.14	9.14	9.23	9.14	9.05	8.96	8.46	8.50	8.29		9.02	9.17	8.82
Turner, Andy (GBR) (1980)	time	2.53	3.58	4.64	5.67	6.71	7.74	8.77	9.82	10.89	11.92		13.44	1 / 4			
	reaction time	0.146		1.05	1.06	1.03	1.04	1.03	1.03	1.05	1.07	1.03	1.52		3.14	3.10	3.15
		velocity	5.42	8.70	8.62	8.87	8.79	8.87	8.70	8.54	8.87	9.22	8.18		8.73	8.85	8.70
Oliver, David (USA) (1982)	time	2.48	3.50	4.58	5.62	6.67	7.71	8.77	9.80	10.84	11.91		13.44	4 / 5			
	reaction time	0.171	7 steps	1.02	1.08	1.04	1.05	1.04	1.06	1.03	1.04	1.07	1.53		3.14	3.15	3.14
		velocity	5.53	8.96	8.46	8.79	8.70	8.79	8.62	8.87	8.79	8.54	9.16	8.18	8.73	8.70	8.73
Merritt, Aries (USA) (1985)	time	2.50	3.54	4.57	5.57	6.61	7.71	8.79	9.88	10.94	12.01		13.67	7 / 6			
	reaction time	0.140		1.04	1.03	1.00	1.04	1.10	1.08	1.09	1.06	1.07	1.66		3.07	3.22	3.22
		velocity	5.49	8.79	8.87	9.14	8.79	8.31	8.46	8.39	8.62	8.54	8.45	8.05	8.93	8.52	8.52
Sharman, William (GBR) (1984)	time	2.52	3.62	4.68	5.74	6.77	7.81	8.86	9.92	11.00	12.11		13.67	8 / 6			
	reaction time	0.178	7 steps	1.10	1.06	1.06	1.03	1.04	1.05	1.06	1.08	1.11	1.56		3.22	3.12	3.25
		velocity	5.44	8.31	8.62	8.62	8.87	8.79	8.70	8.62	8.46	8.23	8.99	8.05	8.52	8.79	8.44
Thomas, Dwight (JAM) (1980)	time	2.57	3.68										dnf	2 / --			
	reaction time	0.153	7 steps	1.11													
		velocity	5.34	8.23													

Decathlon - Heat 1

date 28-Aug-11 wind -0.1 m/s

Hisashi (2012) - performance analysis of Japanese record holder Keisuke Ushiro

		H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
Ushiro, Keisuke (JPN) (1986)	time	2.77	3.92	5.09	6.28	7.45	8.66	9.88	11.10	12.34	13.57		15.20	5 / 6			
	reaction time	0.167		1.15	1.17	1.19	1.17	1.22	1.22	1.24	1.23	1.63			3.51	3.60	3.69
		velocity	4.95	7.95	7.81	7.68	7.81	7.55	7.49	7.49	7.37	7.43	8.60	7.24	7.81	7.62	7.43

2011 Windsprint (Sundsvall, SWE)

FINAL

date 31-Jul-11 wind 2.8 m/s

Blomkvist (2011) - www.sub13.se

		H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
Townsend, Fred (USA) (1982)	time	2.62	3.68	4.71	5.75	6.76	7.78	8.83	9.87	10.93	11.99		13.41	1 / 1			
	reaction time		interval	1.06	1.03	1.04	1.01	1.02	1.05	1.04	1.06	1.42			3.13	3.08	3.16
		velocity	5.24	8.62	8.87	8.79	9.05	8.96	8.70	8.79	8.62	8.62	9.87	8.20	8.76	8.90	8.68
Vukicevic, Vladimir (NOR) (1991)	time	2.67	3.74	4.81	5.89	6.98	8.07	9.17	10.28	11.39	12.52		14.04	1 / 2			
	reaction time		interval	1.07	1.07	1.08	1.09	1.10	1.11	1.11	1.13	1.52			3.22	3.28	3.35
		velocity	5.14	8.54	8.54	8.46	8.39	8.39	8.31	8.23	8.23	8.09	9.22	7.83	8.52	8.36	8.19
Martinsen, Andreas (DEN) (1991)	time	2.71	3.81	4.87	5.94	7.01	8.08	9.17	10.31	11.44	12.59		14.19	1 / 3			
	reaction time		interval	1.10	1.06	1.07	1.07	1.09	1.14	1.13	1.15	1.60			3.23	3.23	3.42
		velocity	5.06	8.31	8.62	8.54	8.54	8.39	8.02	8.09	7.95	8.76	7.75		8.49	8.49	8.02
Bentsen, Jon Yde (DEN) (1985)	time	2.72	3.85	4.95	6.04	7.14	8.25	9.37	10.49	11.64	12.81		14.42	1 / 4			
	reaction time		interval	1.13	1.10	1.09	1.10	1.11	1.12	1.15	1.17	1.61			3.32	3.33	3.44
		velocity	5.04	8.09	8.31	8.39	8.31	8.23	8.16	8.16	7.95	8.71	7.63		8.26	8.23	7.97
Acson, Dennis (SWE) (1981)	time	2.74	3.85	4.97	6.09	7.24							14.99	1 / 5			
	reaction time		interval	1.11	1.12	1.12	1.15								3.35		
		velocity	5.01	8.23	8.16	8.16	7.95						7.34		8.19		

2011 Asian Championships (Kobe, JPN)

FINAL

date 10-Jul-11 wind -2.1 m/s

Hisashi (2012) - performance analysis of Japanese record holder Keisuke Ushiro

		H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
Yazawa, Wataru (JPN) (1991)	time	2.62	3.72	4.80	5.91	7.01	8.11	9.23	10.37	11.51	12.69		14.21	1 / 4			
	reaction time		interval	1.10	1.08	1.11	1.10	1.12	1.14	1.14	1.18	1.52			3.29	3.32	3.46
		velocity	5.24	8.31	8.46	8.23	8.31	8.16	8.02	8.02	7.75	9.22	7.74		8.33	8.26	7.92

Decathlon - Heat 1

date 08-Jul-11 wind 0.3 m/s

Takeo (2014) - decathletes sprint ability and the relationship of the individual event performance

			H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
Nakamura, Akihiko (JPN) (1990)	time		2.70	3.83	4.95	6.07	7.21	8.34	9.48	10.66	11.82	13.00		14.50	/ 1			
	reaction time		1.13		1.12	1.12	1.14	1.13	1.14	1.18	1.16	1.18	1.50			3.37	3.41	3.52
	interval																	
	velocity		5.08	8.09	8.16	8.16	8.02	8.09	8.02	7.75	7.88	7.75	9.35	7.59		8.14	8.04	7.79

2011 European Team Championships (Stockholm, SWE)

Race 1

date 19-Jun-11 wind -0.8 m/s

Blomkvist (2011) - www.sub13.se

			H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
Quiñónez, Jackson (ESP) (1980)	time		2.56	3.64	4.68	5.72	6.76	7.80	8.88	9.96	11.08	12.20		13.71	3 / 1			
	reaction time	0.171														3.16	3.16	3.32
	interval		1.08	1.04	1.04	1.04	1.04	1.04	1.08	1.08	1.12	1.12	1.51			8.68	8.68	8.26
	velocity		5.36	8.46	8.79	8.79	8.79	8.79	8.46	8.46	8.16	8.16	9.28	8.02				
Nossmy, Phillip (SWE) (1982)	time		2.60	3.68	4.72	5.80	6.84	7.92	8.98	10.06	11.16	12.24		13.73	5 / 2			
	reaction time	0.174														3.20	3.18	3.26
	interval		1.08	1.04	1.08	1.04	1.08	1.06	1.08	1.10	1.10	1.08	1.49			8.57	8.62	8.41
	velocity		5.28	8.46	8.79	8.46	8.79	8.46	8.62	8.46	8.31	8.46	9.41	8.01				
Lynsha, Maksim (BLR) (1985)	time		2.60	3.66	4.72	5.80	6.84	7.92	8.98	10.06	11.16	12.24		13.76	6 / 3			
	reaction time	0.173														3.20	3.18	3.26
	interval		1.06	1.06	1.08	1.04	1.08	1.06	1.08	1.10	1.10	1.08	1.52			8.57	8.62	8.41
	velocity		5.28	8.62	8.62	8.46	8.79	8.46	8.62	8.46	8.31	8.46	9.22	7.99				
Mazáč, Martin (CZE) (1990)	time		2.66	3.72	4.80	5.84	6.90	7.96	9.02	10.12	11.28	12.40		14.00	1 / 4			
	reaction time	0.179														3.18	3.18	3.38
	interval		1.06	1.08	1.04	1.06	1.06	1.06	1.10	1.16	1.16	1.12	1.60			8.62	8.62	8.11
	velocity		5.16	8.62	8.46	8.79	8.62	8.62	8.62	8.31	7.88	8.16	8.76	7.86				
Dabó, Rasul (POR) (1989)	time		2.60	3.72	4.80	5.86	6.96	8.03	9.12	10.28	11.40	12.56		14.14	2 / 5			
	reaction time	0.148														3.26	3.26	3.44
	interval		1.12	1.08	1.06	1.10	1.07	1.09	1.16	1.12	1.16	1.16	1.58			8.41	8.41	7.97
	velocity		5.28	8.16	8.46	8.62	8.31	8.54	8.39	7.88	8.16	7.88	8.87	7.78				
Kopanayko, Serhiy (UKR) (1981)	time		2.68	3.76	4.88	5.96	7.04	8.11	9.26	10.40	11.62	12.84		14.57	4 / 6			
	reaction time	0.166														3.28	3.30	3.58
	interval		1.08	1.12	1.08	1.08	1.07	1.15	1.14	1.22	1.22	1.22	1.73			8.36	8.31	7.66
	velocity		5.12	8.46	8.16	8.46	8.46	8.54	7.95	8.02	7.49	7.49	8.10	7.55				

2011 Folksam GP (Göteborg, SWE)

FINAL

date 11-Jun-11 wind 0.6 m/s

Blomkvist (2011) - www.sub13.se

			H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
Akins, Ty (USA) (1986)	time		2.44	3.48	4.52	5.56	6.60	7.64	8.68	9.72	10.76	11.84		13.41	/ 1			
	reaction time															3.12	3.12	3.16
	interval		1.04	1.04	1.04	1.04	1.04	1.04	1.04	1.04	1.04	1.08	1.57			8.79	8.79	8.68
	velocity		5.62	8.79	8.79	8.79	8.79	8.79	8.79	8.79	8.79	8.46	8.93	8.20				
Berger, Dominic (USA) (1986)	time		2.48	3.56	4.60	5.64	6.68	7.72	8.76	9.80	10.86	11.92		13.49	/ 2			
	reaction time															3.16	3.12	3.16
	interval		1.08	1.04	1.04	1.04	1.04	1.04	1.04	1.04	1.06	1.06	1.57			8.68	8.79	8.68
	velocity		5.53	8.46	8.79	8.79	8.79	8.79	8.79	8.79	8.62	8.62	8.93	8.15				
Townsend, Fred (USA) (1982)	time		2.52	3.56	4.60	5.64	6.68	7.72	8.78	9.84	10.90	11.98		13.57	/ 3			
	reaction time															3.12	3.14	3.20
	interval		1.04	1.04	1.04	1.04	1.04	1.06	1.06	1.06	1.06	1.08	1.59			8.79	8.73	8.57
	velocity		5.44	8.79	8.79	8.79	8.79	8.79	8.62	8.62	8.62	8.46	8.82	8.11				
Nossmy, Phillip (SWE) (1982)	time		2.52	3.60	4.68	5.72	6.78	7.82	8.88	9.94	11.00	12.10		13.71	/ 4			
	reaction time															3.20	3.16	3.22
	interval		1.08	1.08	1.04	1.06	1.04	1.06	1.06	1.06	1.06	1.10	1.61			8.57	8.68	8.52
	velocity		5.44	8.46	8.46	8.79	8.62	8.79	8.62	8.62	8.62	8.31	8.71	8.02				
Baji, Balázs (HUN) (1989)	time		2.53	3.64	4.72	5.78	6.84	7.88	8.96	10.04	11.12	12.20		13.80	/ 5			
	reaction time															3.25	3.18	3.24
	interval		1.11	1.08	1.06	1.06	1.04	1.08	1.08	1.08	1.08	1.08	1.60			8.44	8.62	8.46
	velocity		5.42	8.23	8.46	8.62	8.62	8.79	8.46	8.46	8.46	8.46	8.76	7.97				
Shaker, Amir (IRQ) (1989)	time		2.64	3.72	4.84	5.90	6.96	8.05	9.16	10.26	11.38	12.56		14.27	/ 6			
	reaction time															3.26	3.26	3.40
	interval		1.08	1.12	1.06	1.06	1.09	1.11	1.10	1.12	1.18	1.18	1.71			8.41	8.41	8.06
	velocity		5.20	8.46	8.16	8.62	8.62	8.39	8.23	8.31	8.16	7.75	8.20	7.71				
Blaschke, Joakim (SWE) (1977)	time		2.60	3.72	4.84	5.96	7.12	8.28	9.42	10.58	11.76	12.94		14.72	/ 7			
	reaction time															3.36	3.46	3.52
	interval		1.12	1.12	1.12	1.16	1.16	1.14	1.16	1.18	1.18	1.18	1.78			8.16	7.92	7.79
	velocity		5.28	8.16	8.16	8.16	7.88	7.88	8.02	7.88	7.75	7.75	7.88	7.47				

2011 Japanese National Championships (Kumagaya, JPN)

FINAL

date 11-Jun-11 wind -0.1 m/s

Hisashi (2012) - performance analysis of Japanese record holder Keisuke Ushiro

			H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
Yazawa, Wataru (JPN) (1991)	time		2.64	3.72	4.80	5.87	6.94	8.01	9.11	10.20	11.30	12.42		13.86	/ 1			
	reaction time															3.23	3.24	3.31
	interval		1.08	1.08	1.07	1.07	1.07	1.07	1.10	1.09	1.10	1.12	1.44	PB		8.49	8.46	8.28
	velocity		5.20	8.46	8.46	8.54	8.54	8.54	8.31	8.39	8.31	8.16	9.74	7.94				
Wado, Tatsuya (JPN) (1990)	time		2.59	3.68	4.77	5.85	6.91	7.99	9.09	10.19	11.29	12.44		13.95	/ 4			
	reaction time															3.26	3.24	3.35
	interval		1.09	1.09	1.08	1.06	1.08	1.10	1.10	1.10	1.10	1.15	1.51			8.41	8.46	8.19
	velocity		5.30	8.39	8.39	8.46	8.62	8.46	8.31	8.31	8.31	7.95	9.28	7.89				

Omuro, Hideki (JPN) (1990)	time	2.63	3.74	4.84	5.92	7.02	8.10	9.21	10.31	11.41	12.53		13.98	/ 6			
	reaction time	interval	1.11	1.10	1.08	1.10	1.08	1.11	1.10	1.10	1.12	1.45			3.29	3.29	3.32
	velocity	5.22	8.23	8.31	8.46	8.31	8.46	8.23	8.31	8.31	8.16	9.67	7.87		8.33	8.33	8.26

Semi-Final 2

date 11-Jun-11 wind -0.1 m/s Hisashi (2012) - performance analysis of Japanese record holder Keisuke Ushiro

		H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
Yazawa, Wataru (JPN) (1991)	time	2.59	3.70	4.79	5.87	6.95	8.03	9.12	10.23	11.34	12.48		13.95	/ 1			
	reaction time	interval	1.11	1.09	1.08	1.08	1.08	1.09	1.11	1.11	1.14	1.47			3.28	3.25	3.36
	velocity	5.30	8.23	8.39	8.46	8.46	8.46	8.39	8.23	8.23	8.02	9.54	7.89		8.36	8.44	8.16

Wado, Tatsuya (JPN) (1990)	time	2.59	3.68	4.75	5.86	6.93	8.03	9.14	10.24	11.36	12.51		14.04	/ 3			
	reaction time	interval	1.09	1.07	1.11	1.07	1.10	1.11	1.10	1.12	1.15	1.53			3.27	3.28	3.37
	velocity	5.30	8.39	8.54	8.23	8.54	8.31	8.23	8.31	8.16	7.95	9.16	7.83		8.39	8.36	8.14

Semi-Final 1

date 11-Jun-11 wind -2.0 m/s Hisashi (2012) - performance analysis of Japanese record holder Keisuke Ushiro

		H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
Omuro, Hideki (JPN) (1990)	time	2.67	3.76	4.85	5.94	7.03	8.14	9.26	10.39	11.52	12.66		14.09	/ 2			
	reaction time	interval	1.09	1.09	1.09	1.09	1.11	1.12	1.13	1.13	1.14	1.43			3.27	3.32	3.40
	velocity	5.14	8.39	8.39	8.39	8.39	8.23	8.16	8.09	8.09	8.02	9.80	7.81		8.39	8.26	8.06

Sato, Hiroyuki (JPN) (1990)	time	2.65	3.75	4.84	5.94	7.03	8.12	9.23	10.35	11.49	12.65		14.17	/ 7			
	reaction time	interval	1.10	1.09	1.10	1.09	1.09	1.11	1.12	1.14	1.16	1.52			3.29	3.29	3.42
	velocity	5.18	8.31	8.39	8.31	8.39	8.39	8.23	8.16	8.02	7.88	9.22	7.76		8.33	8.33	8.02

Heat 2

date 10-Jun-11 wind 0.1 m/s Hisashi (2012) - performance analysis of Japanese record holder Keisuke Ushiro

		H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
Yazawa, Wataru (JPN) (1991)	time	2.61	3.68	4.74	5.81	6.87	7.95	9.05	10.16	11.29	12.43		13.89	/ 1			
	reaction time	interval	1.07	1.06	1.07	1.06	1.08	1.10	1.11	1.13	1.14	1.46	PB		3.20	3.24	3.38
	velocity	5.26	8.54	8.62	8.54	8.62	8.46	8.31	8.23	8.09	8.02	9.60	7.92		8.57	8.46	8.11

2011 Japanese National Multi-Event Championships (Kawasaki, JPN)

Decathlon - Heat ?

date 05-Jun-11 wind 0.5 m/s Hisashi (2012) - performance analysis of Japanese record holder Keisuke Ushiro

		H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
Ushiro, Keisuke (JPN) (1986)	time	2.75	3.92	5.08	6.25	7.41	8.58	9.75	10.92	12.14	13.33		14.93	???			
	reaction time	interval	1.17	1.16	1.17	1.16	1.17	1.17	1.17	1.22	1.19	1.60	PB		3.50	3.50	3.58
	velocity	4.99	7.81	7.88	7.81	7.88	7.81	7.81	7.81	7.49	7.68	8.76	7.37		7.83	7.83	7.66

Decathlon - Heat ?

date 05-Jun-11 wind 1.4 m/s Takeo (2014) - decathletes sprint ability and the relationship of the individual event performance

		H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
Nakamura, Akihiko (JPN) (1990)	time	2.70	3.83	4.95	6.06	7.17	8.28	9.40	10.52	11.65	12.79		14.25	???			
	reaction time	interval	1.13	1.12	1.11	1.11	1.11	1.12	1.12	1.13	1.14	1.46			3.36	3.34	3.39
	velocity	5.08	8.09	8.16	8.23	8.23	8.23	8.16	8.16	8.09	8.02	9.60	7.72		8.16	8.21	8.09

2011 Mikio Oda Memorial (Hiroshima, JPN)

FINAL

date 29-Apr-11 wind 2.4 m/s Kota (2013) - race analysis of leading Japanese men's 110m hurdlers - from 2011-13

		H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
Omuro, Hideki (JPN) (1990)	time	2.59	3.67	4.72	5.80	6.87	7.90	8.98	10.04	11.14	12.23		13.66	/ 2			
	reaction time	interval	1.08	1.05	1.08	1.07	1.03	1.08	1.06	1.10	1.09	1.43			3.21	3.18	3.25
	velocity	5.30	8.46	8.70	8.46	8.54	8.87	8.46	8.62	8.31	8.39	9.80	8.05		8.54	8.62	8.44

Yazawa, Wataru (JPN) (1991)	time	2.62	3.71	4.78	5.84	6.90	7.96	9.04	10.14	11.25	12.36		13.79	/ 6			
	reaction time	interval	1.09	1.07	1.06	1.06	1.06	1.08	1.10	1.11	1.11	1.43			3.22	3.20	3.32
	velocity	5.24	8.39	8.54	8.62	8.62	8.62	8.46	8.31	8.23	8.23	9.80	7.98		8.52	8.57	8.26

Heat 3

date 29-Apr-11 wind 2.4 m/s Kota (2013) - race analysis of leading Japanese men's 110m hurdlers - from 2011-13

		H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
Wado, Tatsuya (JPN) (1990)	time	2.62	3.70	4.77	5.83	6.95	8.03	9.13	10.22	11.31	12.48		13.97	/ 5			
	reaction time	interval	1.08	1.07	1.06	1.12	1.08	1.10	1.09	1.09	1.17	1.49			3.21	3.30	3.35
	velocity	5.24	8.46	8.54	8.62	8.16	8.46	8.31	8.39	8.39	7.81	9.41	7.87		8.54	8.31	8.19

Heat 1

date 29-Apr-11 wind 0.7 m/s Kota (2013) - race analysis of leading Japanese men's 110m hurdlers - from 2011-13

		H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
Sato, Hiroyuki (JPN) (1990)	time	2.59	3.66	4.74	5.82	6.92	8.03	9.13	10.25	11.35	12.51		14.00	/ 5			
	reaction time	interval	1.07	1.08	1.08	1.10	1.11	1.10	1.12	1.10	1.16	1.49			3.23	3.31	3.38
	velocity	5.30	8.54	8.46	8.46	8.31	8.23	8.31	8.16	8.31	7.88	9.41	7.86		8.49	8.28	8.11

2011 Wakayama Athletics Meeting (Wakayama, JPN)

Decathlon - Heat 1

date 24-Apr-11 wind -0.1 m/s Hisashi (2012) - performance analysis of Japanese record holder Keisuke Ushiro

		H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
Ushiro, Keisuke (JPN) (1986)	time	2.82	3.98	5.17	6.35	7.56	8.77	10.00	11.21	12.43	13.64		15.28	8 / 2			
	reaction time	interval	1.16	1.19	1.18	1.21	1.21	1.23	1.21	1.22	1.21	1.64			3.53	3.65	3.64
	velocity	4.87	7.88	7.68	7.75	7.55	7.55	7.43	7.55	7.49	7.55	8.55	7.20		7.77	7.51	7.53

2011 Swedish National Indoor Championships (Sätra, SWE)

FINAL

date 28-Feb-11

Blomkvist (2011) - www.sub13.se

		H1	H2	H3	H4	H5	Run-In	Official Time	Lane / Place	H1-H4
Nossmys, Philipp (SWE) (1982)	time	2.66	3.71	4.77	5.80	6.83		7.73	/ 1	
	reaction time		interval	1.05	1.06	1.03	0.90			3.14
	velocity	5.16	8.70	8.62	8.87	8.87	10.80	7.76		8.73

2011 German National Indoor Junior Championships (Leverkusen , GER) (99cm)

FINAL

date 19-Feb-11

May (2011) - zwei jugendliche hürdensprinter im vergleich

		H1	H2	H3	H4	H5	Run-In	Official Time	Lane / Place	H1-H4
Traber, Gregor (GER) (1992)	time	2.60	3.66	4.70	5.73	6.76		7.66	4 / 1	
	reaction time		interval	1.06	1.04	1.03	0.90			3.13
	velocity	5.28	8.62	8.79	8.87	8.87	10.80	7.83		8.76
Mählmann, René (GER) (1993)	time	2.66	3.74	4.80	5.87	6.96		7.91	6 / 2	
	reaction time		interval	1.08	1.06	1.07	0.95			3.21
	velocity	5.16	8.46	8.62	8.54	8.39	10.23	7.59		8.54

2011 Landskamp FIN-SWE-NOR (Tampere, FIN)

FINAL

date 05-Feb-11

Blomkvist (2011) - www.sub13.se

		H1	H2	H3	H4	H5	Run-In	Official Time	Lane / Place	H1-H4
Heinä, Joonas-Ville (FIN) (1989)	time	2.56	3.64	4.68	5.76	6.84		7.87	7 / 1	
	reaction time		interval	1.08	1.04	1.08	1.03			3.20
	velocity	5.36	8.46	8.79	8.46	8.46	9.44	7.62		8.57
Vukicevic, Vladimir (NOR) (1991)	time	2.52	3.68	4.76	5.82	6.92		7.95	3 / 2	
	reaction time		interval	1.16	1.08	1.10	1.03			3.30
	velocity	5.44	7.88	8.46	8.62	8.31	9.44	7.55		8.31
Korkealaakso, Antti (FIN) (1984)	time	2.60	3.72	4.80	5.88	6.96		7.99	4 / 3	
	reaction time		interval	1.12	1.08	1.08	1.03			3.28
	velocity	5.28	8.16	8.46	8.46	8.46	9.44	7.51		8.36
Blaschke, Joakim (SWE) (1977)	time	2.60	3.72	4.80	5.90	7.00		8.06	5 / 4	
	reaction time		interval	1.12	1.08	1.10	1.06			3.30
	velocity	5.28	8.16	8.46	8.31	8.31	9.17	7.44		8.31
Stavang, Eivind (NOR) (1982)	time	2.60	3.76	4.86	5.96	7.08		8.13	6 / 5	
	reaction time		interval	1.16	1.10	1.12	1.05			3.36
	velocity	5.28	7.88	8.31	8.31	8.16	9.26	7.38		8.16
Askros, Rafael (SWE) (1981)	time	2.68	3.86	4.94	6.04	7.12		8.15	2 / 6	
	reaction time		interval	1.18	1.08	1.10	1.03			3.36
	velocity	5.12	7.75	8.46	8.31	8.46	9.44	7.36		8.16

2010 Swedish National Championships (Falun, SWE)

FINAL

date 22-Aug-10

wind -0.7 m/s

Blomkvist (2010) - www.sub13.se

		H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
Nossmys, Philip (SWE) (1982)	time	2.70	3.79	4.85	5.92	6.98	8.06	9.15	10.29	11.42	12.55		14.14	/ 1			
	reaction time		interval	1.09	1.06	1.07	1.06	1.08	1.09	1.14	1.13	1.59			3.22	3.23	3.40
	velocity	5.08	8.39	8.62	8.54	8.62	8.46	8.39	8.02	8.09	8.09	8.82	7.78		8.52	8.49	8.06
Askros, Rafael (SWE) (1981)	time	2.79	3.91	5.02	6.14	7.25	8.36	9.50	10.64	11.80	12.97		14.45	/ 2			
	reaction time		interval	1.12	1.11	1.12	1.11	1.14	1.14	1.16	1.17	1.48			3.35	3.36	3.47
	velocity	4.92	8.16	8.23	8.16	8.23	8.23	8.02	8.02	7.88	7.81	9.47	7.61		8.19	8.16	7.90
Blaschke, Joakim (SWE) (1977)	time	2.70	3.84	4.97	6.09	7.21	8.35	9.48	10.64	11.82	13.00		14.57	/ 3			
	reaction time		interval	1.14	1.13	1.12	1.14	1.13	1.16	1.18	1.18	1.57			3.39	3.39	3.52
	velocity	5.08	8.02	8.09	8.16	8.16	8.02	8.09	7.88	7.75	7.75	8.93	7.55		8.09	8.09	7.79
Shaker, Amir (SWE) (1989)	time	2.80	3.94	5.07	6.20	7.32	8.45	9.59	10.77	11.94	13.11		14.68	/ 4			
	reaction time		interval	1.14	1.13	1.13	1.12	1.13	1.14	1.18	1.17	1.57	PB		3.40	3.39	3.52
	velocity	4.90	8.02	8.09	8.09	8.16	8.09	8.02	7.75	7.81	7.81	8.93	7.49		8.06	8.09	7.79
Tay, Marvin (SWE) (1990)	time	2.71	3.82	4.95	6.07	7.20	8.37	9.53	10.70	11.85	13.02		14.77	/ 5			
	reaction time		interval	1.11	1.13	1.12	1.13	1.17	1.16	1.17	1.15	1.75			3.36	3.46	3.49
	velocity	5.06	8.23	8.09	8.16	8.09	7.81	7.88	7.81	7.95	7.81	8.01	7.45		8.16	7.92	7.86
Forsberg, Anton (SWE) (1991)	time	2.82	4.00	5.15	6.30	7.43	8.59	9.78	10.97	12.16	13.39		15.02	/ 6			
	reaction time		interval	1.18	1.15	1.15	1.16	1.19	1.19	1.19	1.23	1.63			3.48	3.48	3.61
	velocity	4.87	7.75	7.95	7.95	8.09	7.88	7.68	7.68	7.68	7.43	8.60	7.32		7.88	7.88	7.60
Gustafsson, Josua (SWE) (1981)	time	2.80	3.97	5.17	6.33	7.49	8.69	9.89	11.10	12.31	13.54		15.17	/ 7			
	reaction time		interval	1.17	1.20	1.16	1.16	1.20	1.20	1.21	1.21	1.63			3.53	3.56	3.65
	velocity	4.90	7.81	7.62	7.88	7.88	7.62	7.62	7.55	7.55	7.43	8.60	7.25		7.77	7.70	7.51
Chiguer, Adam (SWE) (1991)	time	2.83	4.04	5.27	6.53	7.77	9.05	10.31	11.60	12.89	14.20		15.94	/ 8			

reaction time	interval	1.21	1.23	1.26	1.24	1.28	1.26	1.29	1.29	1.31	1.74		3.70	3.78	3.89	
	velocity	4.85	7.55	7.43	7.25	7.37	7.14	7.25	7.09	7.09	6.98	8.06	6.90	7.41	7.25	7.05

2010 Windsprint (Sundsvall, SWE)

FINAL

		date 08-Aug-10		wind 1.1 m/s										Blomkvist (2010) - www.sub13.se				
		H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10	
Townsend, Fred (USA) (1982)	time	2.61	3.68	4.74	5.79	6.82	7.88	8.94	9.99	11.08	12.16		13.64	/ 1				
	reaction time		1.07	1.06	1.05	1.03	1.06	1.06	1.05	1.09	1.08	1.48			3.18	3.15	3.22	
	velocity	5.26	8.54	8.62	8.70	8.87	8.62	8.62	8.70	8.39	8.46	9.47	8.06		8.62	8.70	8.52	
Nossmy, Philip (SWE) (1982)	time	2.67	3.74	4.82	5.86	6.91	7.97	9.04	10.09	11.17	12.26		13.67	/ 2				
	reaction time		1.07	1.08	1.04	1.05	1.06	1.07	1.05	1.08	1.09	1.41			3.19	3.18	3.22	
	velocity	5.14	8.54	8.46	8.79	8.70	8.62	8.54	8.70	8.46	8.39	9.94	8.05		8.60	8.62	8.52	
Porter, Jeff (USA) (1985)	time	2.63	3.71	4.76	5.85	6.91	8.00	9.08	10.14	11.23	12.32		13.79	/ 3				
	reaction time		1.08	1.05	1.09	1.06	1.09	1.08	1.06	1.09	1.09	1.47			3.22	3.23	3.24	
	velocity	5.22	8.46	8.70	8.39	8.62	8.39	8.46	8.62	8.39	8.39	9.54	7.98		8.52	8.49	8.46	
Hudson, Matt (GBR) (1988)	time	2.63	3.72	4.79	5.84	6.90	7.96	9.05	10.15	11.26	12.40		13.91	/ 4				
	reaction time		1.09	1.07	1.05	1.06	1.06	1.09	1.10	1.11	1.14	1.51			3.21	3.21	3.35	
	velocity	5.22	8.39	8.54	8.70	8.62	8.62	8.39	8.31	8.23	8.02	9.28	7.91		8.54	8.54	8.19	
Herring, Aubrey (USA) (1978)	time	2.68	3.75	4.84	5.90	6.96	8.04	9.13	10.23	11.36	12.49		14.07	/ 5				
	reaction time		1.07	1.09	1.06	1.06	1.08	1.09	1.10	1.13	1.13	1.58			3.22	3.23	3.36	
	velocity	5.12	8.54	8.39	8.62	8.62	8.46	8.39	8.31	8.09	8.09	8.87	7.82		8.52	8.49	8.16	
Al-Ameen, Alex (GBR) (1989)	time	2.75	3.86	4.95	6.03	7.11	8.21	9.31	10.42	11.53	12.67		14.20	/ 6				
	reaction time		1.11	1.09	1.08	1.08	1.10	1.10	1.11	1.11	1.14	1.53			3.28	3.28	3.36	
	velocity	4.99	8.23	8.39	8.46	8.46	8.31	8.31	8.23	8.23	8.02	9.16	7.75		8.36	8.36	8.16	
Askros, Rafael (SWE) (1981)	time	2.79	3.91	5.01	6.11	7.20	8.33	9.44	10.58	11.74	12.90		14.43	/ 7				
	reaction time		1.12	1.10	1.10	1.09	1.13	1.11	1.14	1.16	1.16	1.53			3.32	3.33	3.46	
	velocity	4.92	8.16	8.31	8.31	8.39	8.09	8.23	8.02	7.88	7.88	9.16	7.62		8.26	8.23	7.92	

2010 Areva Meeting (Saint-Denis, FRA)

FINAL

FINAL		date	16-Jul-10	wind 0.5 m/s										Wang (2013) - research upon time characters of Liu Xiang's 110m hurdles							
				H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10		
Oliver, David (USA) (1982)	time			2.53	3.53	4.56	5.56	6.53	7.50	8.50	9.50	10.50	11.50		12.87	/ 1					
	reaction time 0.161	interval	7 steps	1.00	1.03	1.00	0.97	0.97	1.00	1.00	1.00	1.00	1.00	1.37	AR PB		3.03	2.94	3.00		
		velocity		5.42	9.14	8.87	9.14	9.42	9.42	9.14	9.14	9.14	9.14	10.23	8.55		9.05	9.33	9.14		

2010 Folsam GP (Sollentuna, SWE)

FINAL

		date	wind 0.4 m/s										Blomkvist (2010) - www.sub13.se					
		01-Jul-10	H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
Phillips, Richard (JAM) (1983)	time		2.48	3.54	4.57	5.60	6.63	7.67	8.70	9.75	10.81	11.88		13.52	5 / 1			
	reaction time		7 steps	1.06	1.03	1.03	1.03	1.04	1.03	1.05	1.06	1.07	1.64			3.12	3.10	3.18
	velocity		5.53	8.62	8.87	8.87	8.87	8.79	8.87	8.70	8.62	8.54	8.55	8.14		8.79	8.85	8.62
Nossmy, Philip (SWE) (1982)	time		2.49	3.56	4.62	5.66	6.70	7.75	8.80	9.86	10.91	11.99		13.56	4 / 2			
	reaction time			1.07	1.06	1.04	1.04	1.05	1.05	1.06	1.05	1.08	1.57			3.17	3.14	3.19
	velocity		5.51	8.54	8.62	8.79	8.79	8.70	8.70	8.62	8.70	8.46	8.93	8.11		8.65	8.73	8.60
Clarke, Lawrence (GBR) (1990)	time		2.51	3.59	4.67	5.71	6.76	7.90	8.85	9.89	10.99	12.09		13.71	6 / 3			
	reaction time			1.08	1.08	1.04	1.05	1.14	0.95	1.04	1.10	1.10	1.62			3.20	3.14	3.24
	velocity		5.47	8.46	8.46	8.79	8.70	8.02	9.62	8.79	8.31	8.31	8.65	8.02		8.57	8.73	8.46
Thornton, Kirk (USA) (1986)	time		2.51	3.57	4.64	5.69	6.73	7.79	8.85	9.93	11.01	12.12		13.77	3 / 4			
	reaction time		7 steps	1.06	1.07	1.05	1.04	1.06	1.06	1.08	1.08	1.11	1.65			3.18	3.16	3.27
	velocity		5.47	8.62	8.54	8.70	8.79	8.62	8.62	8.46	8.46	8.23	8.50	7.99		8.62	8.68	8.39
Inocêncio, Mateus (BRA) (1981)	time		2.53	3.62	4.69	5.73	6.78	7.85	8.93	10.03	11.14	12.28		13.94	2 / 5			
	reaction time			1.09	1.07	1.04	1.05	1.07	1.08	1.10	1.11	1.14	1.66			3.20	3.20	3.35
	velocity		5.42	8.39	8.54	8.79	8.70	8.54	8.46	8.31	8.23	8.02	8.45	7.89		8.57	8.57	8.19
Smit, Louw (RSA) (1988)	time		2.53	3.65	4.73									14.12	8 / 6			
	reaction time			1.12	1.08													
	velocity		5.42	8.16	8.46									7.79				
Askros, Rafael (SWE) (1981)	time		2.60	3.73	4.83	5.93	7.03	8.15	9.27	10.40	11.56	12.76		14.48	1 / 7			
	reaction time			1.13	1.10	1.10	1.10	1.12	1.12	1.13	1.16	1.20	1.72			3.33	3.34	3.49
	velocity		5.28	8.09	8.31	8.31	8.31	8.16	8.16	8.09	7.88	7.62	8.15	7.60		8.23	8.21	7.86
Blaschke, Joakim (SWE) (1977)	time		2.53	3.65	4.79									14.55	7 / 8			
	reaction time			1.12	1.14													
	velocity		5.42	8.16	8.02									7.56				

B FINAL

		date	01-Jul-10	wind	0.8 m/s											Blomkvist (2010) - www.sub13.se			
				H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
Askros, Rafael (SWE) (1981)	time			2.56	3.67	4.77	5.88	6.99	8.10	9.23	10.38	11.56	12.74		14.55	/ 1			
	reaction time				1.11	1.10	1.11	1.11	1.11	1.13	1.15	1.18	1.18	1.81			3.32	3.35	3.51
	velocity			5.36	8.23	8.31	8.23	8.23	8.23	8.09	7.95	7.75	7.75	7.75	7.56		8.26	8.19	7.81
Stavang, Eivind (NOR) (1982)	time			2.45	3.64	4.80	5.95	7.09	8.22	9.42	10.58	11.79	12.99		14.81	/ 2			
	reaction time				1.19	1.16	1.15	1.14	1.13	1.20	1.16	1.21	1.20	1.82			3.50	3.47	3.57
	velocity			5.60	7.68	7.88	7.95	8.02	8.09	7.62	7.88	7.55	7.62	7.70	7.43		7.83	7.90	7.68
Forsberg, Anton (SWE) (1991)	time			2.60	3.77	4.90	6.05	7.24	8.33	9.55	10.72	11.97	13.28		15.16	/ 3			
	reaction time				1.17	1.13	1.15	1.19	1.09	1.22	1.17	1.25	1.31	1.88	PB		3.45	3.50	3.73
	velocity			5.28	7.81	8.09	7.95	7.68	8.39	7.49	7.81	7.31	6.98	7.46	7.26		7.95	7.83	7.35
Nilsson, Nicklas (SWE) (1987)	time			2.66	3.86	5.04	6.23	7.43	8.66		11.07				15.42	/ 4			
	reaction time				1.20	1.18	1.19	1.20	1.23		2.41				PB		3.57		
	velocity			5.16	7.62	7.75	7.68	7.62	7.43		7.59				7.13		7.68		
Almgren, Daniel (SWE) (1979)	time			2.66	3.87	5.04	6.25	7.43	8.66		11.09				15.44	/ 5			
	reaction time				1.21	1.17	1.21	1.18	1.23		2.43						3.59		
	velocity			5.16	7.55	7.81	7.55	7.75	7.43		7.52				7.12		7.64		

2010 European Team Championships, 1st League (Budapest, HUN)**FINAL**

		date	20-Jun-10	wind	1.1 m/s											Blomkvist (2010) - www.sub13.se			
				H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
Kiss, Dániel (HUN) (1982)	time			2.49	3.52	4.54	5.53	6.54	7.54	8.58	9.58	10.67	11.75		13.32	5 / 1			
	reaction time	0.160			1.03	1.02	0.99	1.01	1.00	1.04	1.00	1.09	1.08	1.57	NR		3.04	3.05	3.17
	velocity			5.51	8.87	8.96	9.23	9.05	9.14	8.79	9.14	8.39	8.46	8.93	8.26		9.02	8.99	8.65
van der Westen, Marcel (NED)	time			2.46	3.53	4.57	5.59	6.64	7.68	8.75	9.81	10.90	11.98		13.61	3 / 2			
	reaction time	0.130			1.07	1.04	1.02	1.05	1.04	1.07	1.06	1.09	1.08	1.63			3.13	3.16	3.23
	velocity			5.58	8.54	8.79	8.96	8.70	8.79	8.54	8.62	8.39	8.46	8.60	8.08		8.76	8.68	8.49
Nossmy, Philip (SWE) (1982)	time			2.51	3.58	4.67	5.71	6.76	7.84	8.89	9.98	11.04	12.16		13.79	4 / 3			
	reaction time	0.190			1.07	1.09	1.04	1.05	1.08	1.05	1.09	1.06	1.12	1.63			3.20	3.18	3.27
	velocity			5.47	8.54	8.39	8.79	8.70	8.46	8.70	8.39	8.62	8.16	8.60	7.98		8.57	8.62	8.39
Mihailescu, Alexandru (ROU)	time			2.51	3.64	4.69	5.75	6.80	7.86	8.93	10.00	11.11	12.22		13.87	2 / 4			
	reaction time	0.150			1.13	1.05	1.06	1.05	1.06	1.07	1.07	1.11	1.11	1.65			3.24	3.18	3.29
	velocity			5.47	8.09	8.70	8.62	8.70	8.62	8.54	8.54	8.23	8.23	8.50	7.93		8.46	8.62	8.33
Mazáč, Martin (CZE) (1990)	time			2.57	3.67	4.76	5.80	6.87	7.93	9.01	10.12	11.20	12.33		13.95	6 / 5			
	reaction time	0.176			1.10	1.09	1.04	1.07	1.06	1.08	1.11	1.08	1.13	1.62			3.23	3.21	3.32
	velocity			5.34	8.31	8.39	8.79	8.54	8.62	8.46	8.23	8.46	8.09	8.65	7.89		8.49	8.54	8.26
Ferreira, João (POR) (1986)	time			2.62	3.67	4.75	5.80	6.87	7.96	9.04	10.13	11.24	12.36		13.97	7 / 6			
	reaction time	0.168			1.05	1.08	1.05	1.07	1.09	1.08	1.09	1.11	1.12	1.61			3.18	3.24	3.32
	velocity			5.24	8.70	8.46	8.70	8.54	8.39	8.46	8.39	8.23	8.16	8.71	7.87		8.62	8.46	8.26

2010 ExxonMobil Bislett Games (Oslo, NOR) (99cm)**Junior Final**

		date	04-Jun-10	wind	0.3 m/s											Blomkvist (2010) - www.sub13.se			
				H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
Meredith, Jack (GBR) (1992)	time			2.44	3.52	4.52	5.52	6.56	7.56	8.58	9.62	10.68	11.76		13.32	5 / 1			
	reaction time	0.147			1.08	1.00	1.00	1.04	1.00	1.02	1.04	1.06	1.08	1.56	NJR		3.08	3.06	3.18
	velocity			5.62	8.46	9.14	9.14	8.79	9.14	8.96	8.79	8.62	8.46	8.99	8.26		8.90	8.96	8.62
Vukicevic, Vladimir (NOR) (1991)	time			2.48	3.52	4.52	5.54	6.56	7.56	8.60	9.64	10.72	11.80		13.42	4 / 2			
	reaction time	0.158			1.04	1.00	1.02	1.02	1.00	1.04	1.04	1.08	1.08	1.62	NJR		3.06	3.06	3.20
	velocity			5.53	8.79	9.14	8.96	8.96	9.14	8.79	8.79	8.46	8.46	8.65	8.20		8.96	8.96	8.57
Lööv, Filip (SWE) (1992)	time			2.56	3.62	4.68	5.76	6.84	7.89	9.00	10.08	11.16	12.30		13.95	2 / 3			
	reaction time	0.214			1.06	1.06	1.08	1.08	1.05	1.11	1.08	1.08	1.14	1.65			3.20	3.24	3.30
	velocity			5.36	8.62	8.62	8.46	8.46	8.70	8.23	8.46	8.46	8.02	8.50	7.89		8.57	8.46	8.31
McLean, James (GBR) (1991)	time			2.46	3.52	4.56	5.60	6.68	7.80	8.88	10.00	11.14	12.32		13.97	3 / 4			
	reaction time	0.151			1.06	1.04	1.04	1.08	1.12	1.08	1.12	1.14	1.18	1.65	PB		3.14	3.28	3.44
	velocity			5.58	8.62	8.79	8.79	8.46	8.16	8.46	8.16	8.02	7.75	8.50	7.87		8.73	8.36	7.97

2010 NCAA National Championships (Eugene, OR)**Decathlon - Heat 2**

		date	01-Jun-10	wind	1.6 m/s											Deca Newsletter (2010) - eaton on ncaa three-peat			
				H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
Eaton, Ashton (USA) (1988)	time			2.4	3.5	4.7	5.7	6.7	7.7	8.8	9.9	11.1	12.1		13.68	6 / 1			
	reaction time				1.1	1.2	1.0	1.0	1.0	1.1	1.1	1.2	1.0	1.58			3.30	3.10	3.30
	velocity			5.72	8.31	7.62	9.14	9.14	9.14	8.31	8.31	7.62	9.14	8.87	8.04		8.31	8.85	8.31

2010 Shanghai Diamond League (Shanghai, CHN)

FINAL

date 23-May-10 wind -0.4 m/s

He (2014) - kinematic analysis of Xie Wenjun's hurdle technique

		H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
Xie Wenjun (CHN) (1990)	time	2.64	3.54	4.54	5.44	6.64	7.74	8.84	9.84	10.84	11.84		13.58	/ 4			
	reaction time	0.165	interval	0.90	1.00	0.90	1.20	1.10	1.10	1.00	1.00	1.74			2.80	3.40	3.00
	velocity	5.20	10.16	9.14	10.16	7.62	8.31	8.31	9.14	9.14	9.14	8.06	8.10		9.79	8.06	9.14

2010 Colorful Daegu Meeting (Daegu, KOR)

FINAL

date 19-May-10 wind -0.2 m/s

Hong (2018) - performance of men's 110m hurdles using rhythmic units

		H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
Oliver, David (USA) (1982)	time	2.61	3.66	4.68	5.66	6.66	7.64	8.64	9.66	10.68	11.75		13.11	/ 1			
	reaction time	0.170	7 steps	1.05	1.02	0.98	1.00	0.98	1.00	1.02	1.02	1.36			3.05	2.98	3.11
	interval														8.99	9.20	8.82
	velocity	5.26	8.70	8.96	9.33	9.14	9.33	9.14	8.96	8.96	8.54	10.31	8.39				
Robles, Dayron (CUB) (1986)	time	2.69	3.76	4.76	5.74	6.76	7.76	8.76	9.78	10.83	11.88		13.26	/ 2			
	reaction time	0.216	7 steps	1.07	1.00	0.98	1.02	1.00	1.02	1.05	1.05	1.38			3.05	3.02	3.12
	interval														8.99	9.08	8.79
	velocity	5.10	8.54	9.14	9.33	8.96	9.14	9.14	8.96	8.70	8.70	10.16	8.30				
Thomas, Dwight (JAM) (1980)	time	2.65	3.73	4.73	5.73	6.75	7.78	8.83	9.86	10.93	11.99		13.40	/ 3			
	reaction time	0.161	interval	1.08	1.00	1.00	1.02	1.03	1.05	1.03	1.07	1.06	1.41		3.08	3.10	3.16
	interval														8.90	8.85	8.68
	velocity	5.18	8.46	9.14	9.14	8.96	8.87	8.70	8.87	8.54	8.62	9.94	8.21				

2010 IAAF World Indoor Championships (Doha, QAT)

FINAL

date 14-Mar-10

Kuitunen (2010) - biomechanical analysis of the 14th IAAF world indoor championships

		H1	H2	H3	H4	H5		Run-In	Official Time	Lane / Place	H1-H4
Robles, Dayron (CUB) (1986)	time	2.54	3.54	4.52	5.49	6.47			7.34	6 / 1	
	reaction time	0.196	7 steps	1.00	0.98	0.97		0.87	CR		2.95
	interval										9.29
	velocity	5.40	9.14	9.33	9.42	9.33		11.17	8.17		
Trammell, Terrence (USA) (1971)	time	2.48	3.50	4.50	5.49	6.48			7.36	4 / 2	
	reaction time	0.149	interval	1.02	1.00	0.99		0.88	AR		3.01
	interval										9.11
	velocity	5.53	8.96	9.14	9.23	9.23		11.05	8.15		
Oliver, David (USA) (1982)	time	2.53	3.56	4.56	5.56	6.56			7.44	8 / 3	
	reaction time	0.149	7 steps	1.03	1.00	1.00		0.88	PB		3.03
	interval										9.05
	velocity	5.42	8.87	9.14	9.14	9.14		11.05	8.06		
Borisov, Yevgeniy (RUS) (1984)	time	2.53	3.57	4.58	5.59	6.63			7.51	3 / 4	
	reaction time	0.167	7 steps	1.04	1.01	1.01		0.88			3.06
	interval										8.96
	velocity	5.42	8.79	9.05	9.05	8.79		11.05	7.99		
Svoboda, Petr (CZE) (1984)	time	2.57	3.63	4.62	5.65	6.70			7.58	5 / 5	
	reaction time	0.134	1.06	0.99	1.03	1.05		0.88			3.08
	interval										8.90
	velocity	5.34	8.62	9.23	8.87	8.70		11.05	7.92		
Wignall, Maurice (JAM) (1976)	time	2.60	3.66	4.68	5.70	6.75			7.60	1 / 6	
	reaction time	0.135	1.06	1.02	1.02	1.05		0.85			3.10
	interval										8.85
	velocity	5.28	8.62	8.96	8.96	8.70		11.44	7.89		
Liu Xiang (CHN) (1983)	time	2.61	3.67	4.70	5.73	6.77			7.65	7 / 7	
	reaction time	0.151	1.06	1.03	1.03	1.04		0.88			3.12
	interval										8.79
	velocity	5.26	8.62	8.87	8.87	8.79		11.05	7.84		
Kiss, Dániel (HUN) (1982)	time	2.73	3.75	4.82	5.89	6.93			7.81	2 / 8	
	reaction time	0.182	1.02	1.07	1.07	1.04		0.88			3.16
	interval										8.68
	velocity	5.03	8.96	8.54	8.54	8.79		11.05	7.68		

Semi-Final 3

date 14-Mar-10

Blomkvist (2010) - 100/110m Hurdles: Development

		H1	H2	H3	H4	H5		Run-In	Official Time	Lane / Place	H1-H4
Nossmy, Philip (SWE) (1982)	time	2.63	3.68	4.75	5.77	6.81			7.69	1 / 5	
	reaction time	0.157	1.05	1.07	1.02	1.04		0.88			3.14
	interval										8.73
	velocity	5.22	8.70	8.54	8.96	8.79		11.05	7.80		

2010 Swedish Indoor National Championships (Sätra, SWE)

FINAL

date 28-Feb-10

Blomkvist (2010) - www.sub13.se

		H1	H2	H3	H4	H5		Run-In	Official Time	Lane / Place	H1-H4
Nossmy, Philip (SWE) (1982)	time	2.66	3.71	4.77	5.80	6.83			7.73	/ 1	
	reaction time		1.05	1.06	1.03	1.03		0.90			3.14
	interval										8.73
	velocity	5.16	8.70	8.62	8.87	8.87		10.80	7.76		

2010 GE Galan (Stockholm, SWE)

FINAL

date 10-Feb-10

Blomkvist (2010) - www.sub13.se

		H1	H2	H3	H4	H5		Run-In	Official Time	Lane / Place	H1-H4
Robles, Dayron (CUB) (1986)	time	2.35	3.39	4.39	5.37	6.38			7.49	/ 1	
	reaction time		7 steps	1.04	1.00	0.98		1.11			3.02
	interval										9.08
	velocity	5.84	8.79	9.14	9.33	9.05		8.76	8.01		

Kiss, Dániel (HUN) (1982)	time	2.50	3.53	4.56	5.58	6.60			7.66	/ 2	
	reaction time	interval		1.03	1.03	1.02	1.02				3.08
	velocity	5.49	8.87	8.87	8.96	8.96		1.06			8.90
Thomas, Dwight (JAM) (1980)	time	2.42	3.48	4.51	5.55	6.58			7.66	/ 3	
	reaction time	interval		1.06	1.03	1.04	1.03		1.08		3.13
	velocity	5.67	8.62	8.87	8.79	8.87		9.00	7.83		8.76
Nossmy, Philip (SWE) (1982)	time	2.45	3.51	4.56	5.58	6.60			7.68	/ 4	
	reaction time	interval		1.06	1.05	1.02	1.02		1.08		3.13
	velocity	5.60	8.62	8.70	8.96	8.96		9.00	7.81		8.76
Wignall, Maurice (JAM) (1976)	time	2.45	3.52	4.57	5.61	6.67			7.78	/ 5	
	reaction time	interval		1.07	1.05	1.04	1.06		1.11		3.16
	velocity	5.60	8.54	8.70	8.79	8.62		8.76	7.71		8.68

2010 Samsung Galan (Göteborg, SWE)**Heat 2**

date 02-Feb-10

Blomkvist (2010) - www.sub13.se

		H1	H2	H3	H4	H5	Run-In	Official Time	Lane / Place	H1-H4
Oliver, David (USA) (1982)	time	2.45	3.52	4.53	5.54	6.58		7.57	/ 1	
	reaction time	interval	7 steps	1.07	1.01	1.01	1.04	0.99		3.09
	velocity	5.60	8.54	9.05	9.05	8.79		9.82	7.93	8.87
Nossmy, Philip (SWE) (1982)	time	2.56	3.65	4.72	5.76	6.80		7.78	/ 2	
	reaction time	interval		1.09	1.07	1.04	1.04	0.98		3.20
	velocity	5.36	8.39	8.54	8.79	8.79		9.92	7.71	8.57
Baillie, Chris (GBR) (1981)	time	2.52	3.60	4.66	5.72	6.78		7.80	/ 3	
	reaction time	interval		1.08	1.06	1.06	1.06	1.02		3.20
	velocity	5.44	8.46	8.62	8.62	8.62		9.53	7.69	8.57
Porter, Jeff (USA) (1985)	time	2.54	3.64	4.68	5.76	6.88		7.87	/ 4	
	reaction time	interval		1.10	1.04	1.08	1.12	0.99		3.22
	velocity	5.40	8.31	8.79	8.46	8.16		9.82	7.62	8.52
Sparrman, Markus (SWE) (1988)	time	2.56	3.72	4.86	5.96	7.10		8.14	/ 5	
	reaction time	interval		1.16	1.14	1.10	1.14	1.04		3.40
	velocity	5.36	7.88	8.02	8.31	8.02		9.35	7.37	8.06
Blaschke, Joakim (SWE) (1977)	time	2.60	3.76	4.90	6.04	7.18		8.29	/ 6	
	reaction time	interval		1.16	1.14	1.14	1.14	1.11		3.44
	velocity	5.28	7.88	8.02	8.02	8.02		8.76	7.24	7.97

2010 Five Nations (Glasgow, GBR)**FINAL**

date 30-Jan-10

Blomkvist (2010) - www.sub13.se

		H1	H2	H3	H4	H5	Run-In	Official Time	Lane / Place	H1-H4
Nossmy, Philip (SWE) (1977)	time	2.46	3.50	4.55	5.58	6.61		7.71	/ 3	
	reaction time	0.186	interval		1.04	1.05	1.03	1.10		3.12
	velocity		5.58	8.79	8.70	8.87	8.87	8.84	7.78	8.79

2009 Seiko Super Grand Prix (Kawasaki, JPN)**FINAL**

date 23-Sep-09 wind -1.3 m/s

Ryutaro (2010) - an attempt or a creation of a race pattern standard model for the 110m hurdles

		H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
Thomas, Dwight (JAM) (1980)	time	2.60	3.66	4.70	5.73	6.74	7.78	8.83	9.85	10.95	11.98		13.37	/ 1			
	reaction time	interval		1.06	1.04	1.03	1.01	1.04	1.05	1.02	1.10	1.03	1.39		3.13	3.10	3.15
	velocity	5.28	8.62	8.79	8.87	9.05	8.79	8.70	8.96	8.31	8.87	10.09	8.23		8.76	8.85	8.70
Moses, Yume (JPN) (1988)	time	2.60	3.68	4.74	5.80	6.87	7.94	9.02	10.10	11.18	12.28		13.77	/ 2			
	reaction time	interval		1.08	1.06	1.06	1.07	1.08	1.08	1.08	1.10	1.49			3.20	3.22	3.26
	velocity	5.28	8.46	8.62	8.62	8.54	8.54	8.46	8.46	8.46	8.31	9.41	7.99		8.57	8.52	8.41
Shabanov, Konstantin (RUS) (1977)	time	2.61	3.69	4.75	5.79	6.85	7.91	9.02	10.14	11.26	12.41		13.96	/ 3			
	reaction time	interval		1.08	1.06	1.04	1.06	1.11	1.12	1.12	1.15	1.55			3.18	3.23	3.39
	velocity	5.26	8.46	8.62	8.79	8.62	8.62	8.23	8.16	8.16	7.95	9.05	7.88		8.62	8.49	8.09

2009 IAAF World Championships (Berlin, GER)**FINAL**

date 20-Aug-09 wind 0.1 m/s

Graubner (2009) - biomechanical analyses at the 12th IAAF world championships

		H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
Brathwaite, Ryan (BAR) (1988)	time	2.54	3.60	4.61	5.60	6.59	7.60	8.62	9.65	10.70	11.74		13.14	4 / 1			
	reaction time	0.157	interval		1.06	1.01	0.99	0.99	1.01	1.02	1.03	1.05	1.04		3.06	3.02	3.12
	velocity		5.40	8.62	9.05	9.23	9.23	9.05	8.96	8.87	8.70	8.79	10.01		8.96	9.08	8.79
Trammell, Terrence (USA) (1977)	time	2.56	3.60	4.61	5.60	6.60	7.62	8.63	9.66	10.71	11.77		13.15	5 / 2			
	reaction time	0.141	interval		1.04	1.01	0.99	1.00	1.02	1.01	1.03	1.05	1.06		3.04	3.03	3.14
	velocity		5.36	8.79	9.05	9.23	9.14	8.96	9.05	8.87	8.70	8.62	10.16		9.02	9.05	8.73

Payne, David (USA) (1982)	time	2.54	3.58	4.61	5.61	6.62	7.64	8.66	9.69	10.74	11.78		13.15	3 / 3			
	reaction time	0.122		1.04	1.03	1.00	1.01	1.02	1.02	1.03	1.05	1.04	1.37		3.07	3.05	3.12
	velocity		5.40	8.79	8.87	9.14	9.05	8.96	8.96	8.87	8.70	8.79	10.23	8.37	8.93	8.99	8.79
Sharman, William (GBR) (1984)	time	2.54	3.60	4.62	5.64	6.65	7.68	8.73	9.78	10.84	11.91		13.30	6 / 4			
	reaction time	0.125		1.06	1.02	1.02	1.01	1.03	1.05	1.05	1.06	1.07	1.39	PB	3.10	3.09	3.18
	velocity		5.40	8.62	8.96	8.96	9.05	8.87	8.70	8.70	8.62	8.54	10.09	8.27	8.85	8.87	8.62
Wignall, Maurice (JAM) (1976)	time	2.61	3.66	4.70	5.72	6.75	7.78	8.81	9.85	10.90	11.98		13.31	8 / 5			
	reaction time	0.151		1.05	1.04	1.02	1.03	1.03	1.04	1.05	1.08	1.33			3.11	3.09	3.17
	velocity		5.26	8.70	8.79	8.96	8.87	8.87	8.79	8.70	8.46	10.54	8.26		8.82	8.87	8.65
Svoboda, Petr (CZE) (1984)	time	2.58	3.62	4.64	5.66	6.69	7.74	8.77	9.82	10.87	11.94		13.38	7 / 6			
	reaction time	0.144		1.04	1.02	1.02	1.03	1.05	1.03	1.05	1.05	1.07	1.44		3.08	3.11	3.17
	velocity		5.32	8.79	8.96	8.96	8.87	8.70	8.87	8.70	8.70	8.54	9.74	8.22	8.90	8.82	8.65
Thomas, Dwight (JAM) (1990)	time	2.66	3.73	4.77	5.80	6.84	7.88	8.90	9.95	11.00	12.14		13.56	2 / 7			
	reaction time	0.145		1.07	1.04	1.03	1.04	1.02	1.05	1.05	1.14	1.42			3.14	3.10	3.24
	velocity		5.16	8.54	8.79	8.87	8.79	8.79	8.96	8.70	8.70	8.02	9.87	8.11	8.73	8.85	8.46
Ji Wei (CHN) (1984)	time	2.68	3.77	4.80	5.84	6.86	7.92	8.96	10.02	11.09	12.17		13.57	1 / 8			
	reaction time	0.144		1.09	1.03	1.04	1.02	1.06	1.06	1.07	1.08	1.40			3.16	3.12	3.21
	velocity		5.12	8.39	8.87	8.79	8.96	8.62	8.79	8.62	8.54	8.46	10.01	8.11	8.68	8.79	8.54

Semi-Final 3

date 20-Aug-09 wind 0.1 m/s Graubner (2009) - biomechanical analyses at the 12th IAAF world championships

		H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
Sharman, William (GBR) (1984)	time	2.56	3.58	4.63	5.65	6.66	7.69	8.72	9.77	10.82	11.90		13.38	8 / 1			
	reaction time	0.110		1.02	1.05	1.02	1.01	1.03	1.05	1.05	1.08	1.48	PB		3.09	3.07	3.18
	velocity		5.36	8.96	8.70	8.96	9.05	8.87	8.87	8.70	8.70	8.46	9.47	8.22	8.87	8.93	8.62
Wignall, Maurice (JAM) (1976)	time	2.60	3.67	4.70	5.73	6.77	7.80	8.84	9.90	10.96	12.03		13.43	4 / 2			
	reaction time	0.162		1.07	1.03	1.03	1.04	1.03	1.06	1.06	1.07	1.40			3.13	3.11	3.19
	velocity		5.28	8.54	8.87	8.87	8.79	8.87	8.79	8.62	8.62	8.54	10.01	8.19	8.76	8.82	8.60
Kiss, Dániel (HUN) (1982)	time	2.67	3.72	4.76	5.76	6.80	7.83	8.86	9.90	10.96	12.04		13.45	3 / 3			
	reaction time	0.156		1.05	1.04	1.00	1.04	1.03	1.04	1.06	1.08	1.41			3.09	3.10	3.18
	velocity		5.14	8.70	8.79	9.14	8.79	8.87	8.79	8.62	8.46	9.94	8.18		8.87	8.85	8.62
Sands, Shamar (BAH) (1985)	time	2.51	3.58	4.62	5.64	6.68	7.73	8.78	9.86	10.94	12.04		13.47	1 / 4			
	reaction time	0.144		1.07	1.04	1.02	1.04	1.05	1.05	1.08	1.08	1.10	1.43		3.13	3.14	3.26
	velocity		5.47	8.54	8.79	8.96	8.79	8.70	8.46	8.46	8.31	9.80	8.17		8.76	8.73	8.41
Schwarzer, Helge (GER) (1985)	time	2.59	3.69	4.77	5.82	6.88	7.92	8.98	10.04	11.12	12.24		13.72	6 / 7			
	reaction time	0.132		1.10	1.08	1.05	1.06	1.04	1.06	1.06	1.08	1.12	1.48		3.23	3.16	3.26
	velocity		5.30	8.31	8.46	8.70	8.62	8.79	8.62	8.62	8.46	8.16	9.47	8.02	8.49	8.68	8.41

Semi-Final 2

date 20-Aug-09 wind 0.6 m/s Graubner (2009) - biomechanical analyses at the 12th IAAF world championships

		H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
Brathwaite, Ryan (BAR) (1988)	time	2.57	3.59	4.62	5.63	6.63	7.63	8.65	9.69	10.73	11.79		13.18	4 / 1			
	reaction time	0.169		1.02	1.03	1.01	1.00	1.02	1.04	1.04	1.06	1.39	NR		3.06	3.02	3.14
	velocity		5.34	8.96	8.87	9.05	9.14	8.96	8.79	8.79	8.62	10.09	8.35		8.96	9.08	8.73
Payne, David (USA) (1982)	time	2.60	3.65	4.68	5.69	6.71	7.75	8.78	9.80	10.83	11.89		13.24	7 / 2			
	reaction time	0.151		1.05	1.03	1.01	1.02	1.04	1.03	1.02	1.03	1.06	1.35		3.09	3.09	3.11
	velocity		5.28	8.70	8.87	9.05	8.96	8.79	8.87	8.96	8.87	8.62	10.39	8.31	8.87	8.87	8.82
Thomas, Dwight (JAM) (1980)	time	2.65	3.73	4.77	5.80	6.81	7.84	8.86	9.89	10.91	11.97		13.37	5 / 3			
	reaction time	0.151		1.08	1.04	1.03	1.01	1.03	1.02	1.03	1.02	1.06	1.40		3.15	3.06	3.11
	velocity		5.18	8.46	8.79	8.87	9.05	8.87	8.96	8.87	8.96	8.62	10.01	8.23	8.70	8.96	8.82
Shi Dongpeng (CHN) (1984)	time	2.63	3.73	4.77	5.79	6.81	7.83	8.86	9.90	10.95	12.03		13.42	3 / 4			
	reaction time	0.144		1.10	1.04	1.02	1.02	1.03	1.04	1.05	1.08	1.39			3.16	3.07	3.17
	velocity		5.22	8.31	8.79	8.96	8.96	8.96	8.87	8.79	8.70	8.46	10.09	8.20	8.68	8.93	8.65
Sedoc, Gregory (NED) (1981)	time	2.63	3.66	4.70	5.73	6.77	7.80	8.84	9.90	10.95	12.03		13.45	8 / 5			
	reaction time	0.162		1.03	1.04	1.03	1.04	1.03	1.04	1.06	1.05	1.08	1.42		3.10	3.11	3.19
	velocity		5.22	8.87	8.79	8.87	8.79	8.87	8.79	8.62	8.70	8.46	9.87	8.18	8.85	8.82	8.60
Lynsha, Maksim (BLR) (1985)	time	2.59	3.62	4.67	5.70	6.73	7.79	8.85	9.89	10.97	12.05		13.46	1 / 6			
	reaction time	0.154		1.03	1.05	1.03	1.06	1.06	1.04	1.08	1.08	1.41	PB		3.11	3.15	3.20
	velocity		5.30	8.87	8.70	8.87	8.87	8.62	8.62	8.79	8.46	8.46	9.94	8.17	8.82	8.70	8.57
John, Alexander (GER) (1986)	time	2.65	3.71	4.77	5.84	6.89	7.95	9.03	10.09	11.17	12.27		13.64	6 / 8			
	reaction time	0.185		1.06	1.06	1.07	1.05	1.06	1.08	1.06	1.08	1.10	1.37		3.19	3.19	3.24
	velocity		5.18	8.62	8.62	8.54	8.70	8.62	8.46	8.62	8.46	8.31	10.23	8.06	8.60	8.60	8.46

Semi-Final 1

date 20-Aug-09 wind 0.9 m/s

Graubner (2009) - biomechanical analyses at the 12th IAAF world championships

			H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
Trammell, Terrence (USA) (1971)	time		2.54	3.58	4.59	5.61	6.62	7.65	8.68	9.71	10.76	11.83		13.24	6 / 1			
	reaction time	0.160		1.04	1.01	1.02	1.01	1.03	1.03	1.03	1.05	1.07	1.41			3.07	3.07	3.15
	interval															8.93	8.93	8.70
Svoboda, Petr (CZE) (1984)	time		2.61	3.65	4.66	5.68	6.69	7.71	8.74	9.80	10.87	11.95		13.33	3 / 2			
	reaction time	0.149		1.04	1.01	1.02	1.01	1.02	1.03	1.06	1.07	1.08	1.38			3.07	3.06	3.21
	interval															8.93	8.96	8.54
Ji Wei (CHN) (1984)	time		2.64	3.71	4.74	5.77	6.79	7.82	8.84	9.89	10.95	12.01		13.41	4 / 3			
	reaction time	0.147		1.07	1.03	1.03	1.02	1.03	1.02	1.05	1.06	1.06	1.40			3.13	3.07	3.17
	interval															8.76	8.93	8.65
Noga, Artur (POL) (1988)	time		2.67	3.73	4.77	5.78	6.79	7.91	8.84	9.89	10.93	12.01		13.43	7 / 4			
	reaction time	0.174		1.06	1.04	1.01	1.01	1.12	0.93	1.05	1.04	1.08	1.42			3.11	3.06	3.17
	interval															8.82	8.96	8.65
Villar, Paulo (COL) (1978)	time		2.55	3.60	4.63	5.65	6.68	7.71	8.75	9.81	10.89	11.97		13.44	5 / 5			
	reaction time	0.171		1.05	1.03	1.02	1.03	1.03	1.04	1.06	1.08	1.08	1.47			3.10	3.10	3.22
	interval															8.85	8.85	8.52

Decathlon - Heat 5

date 20-Aug-09 wind 0.3 m/s

Graubner (2009) - biomechanical analyses at the 12th IAAF world championships

			H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
Vos, Ingmar (NED) (1986)	time		2.76	3.91	5.04	6.16	7.28	8.40	9.52	10.66	11.80	12.99		14.51	4 / 1			
	reaction time	0.178		1.15	1.13	1.12	1.12	1.12	1.12	1.14	1.14	1.19	1.52	PB		3.40	3.36	3.47
	interval															8.06	8.16	7.90
Bouraada, Larbi (ALG) (1988)	time		2.74	3.88	5.04	6.18	7.31	8.49	9.64	10.78	11.94	13.12		14.57	5 / 2			
	reaction time	0.140		1.14	1.16	1.14	1.13	1.18	1.15	1.14	1.16	1.18	1.45			3.44	3.46	3.48
	interval															7.97	7.92	7.88
Wiberg, Nicklas (SWE) (1985)	time		2.84	4.00	5.11	6.20	7.33	8.48	9.64	10.86	12.02	13.22		14.75	7 / 3			
	reaction time	0.208		1.16	1.11	1.09	1.13	1.15	1.16	1.22	1.16	1.20	1.53			3.36	3.44	3.58
	interval															8.16	7.97	7.66

Decathlon - Heat 4

date 20-Aug-09 wind 0.0 m/s

Graubner (2009) - biomechanical analyses at the 12th IAAF world championships

			H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
Karpov, Dmitriy (KAZ) (1981)	time		2.73	3.85	4.94	6.03	7.17	8.31	9.45	10.61	11.73	12.91		14.46	4 / 1			
	reaction time	0.141		1.12	1.09	1.09	1.14	1.14	1.14	1.16	1.12	1.18	1.55			3.30	3.42	3.46
	interval															8.31	8.02	7.92
Sysoyev, Aleksey (RUS) (1985)	time		2.87	3.95	5.10	6.29	7.49	8.65	9.83	11.02	12.23	13.42		14.97	6 / 5			
	reaction time	0.150		1.08	1.15	1.19	1.20	1.16	1.18	1.19	1.21	1.19	1.55			3.42	3.54	3.59
	interval															8.02	7.75	7.64
Almgren, Daniel (SWE) (1979)	time		2.89	4.08	5.25	6.39	7.53	8.75	9.93	11.11	12.31	13.52		15.14	3 / 6			
	reaction time	0.193		1.19	1.17	1.14	1.14	1.22	1.18	1.18	1.20	1.21	1.62			3.50	3.54	3.59
	interval															7.83	7.75	7.64

Decathlon - Heat 3

date 20-Aug-09 wind -0.1 m/s

Graubner (2009) - biomechanical analyses at the 12th IAAF world championships

			H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
Krauchanka, Andrei (BLR) (1981)	time		2.77	3.89	4.99	6.07	7.17	8.25	9.35	10.45	11.56	12.67		14.13	4 / 1			
	reaction time	0.175		1.12	1.10	1.08	1.10	1.08	1.10	1.10	1.11	1.11	1.46			3.30	3.28	3.32
	interval															8.31	8.36	8.26
Coertzen, Willem (RSA) (1982)	time		2.75	3.86	4.97	6.08	7.17	8.27	9.39	10.49	11.63	12.79		14.26	6 / 2			
	reaction time	0.152		1.11	1.11	1.11	1.09	1.10	1.12	1.10	1.14	1.16	1.47	PB		3.33	3.31	3.40
	interval															8.23	8.28	8.06
Barras, Romain (FRA) (1980)	time		2.72	3.85	4.96	6.07	7.15	8.27	9.37	10.49	11.61	12.75		14.27	8 / 3			
	reaction time	0.135		1.13	1.11	1.11	1.08	1.12	1.10	1.12	1.12	1.14	1.52			3.35	3.30	3.38
	interval															8.19	8.31	8.11
Kasyanov, Oleksiy (UKR) (1985)	time		2.73	3.85	4.95	6.07	7.18	8.31	9.45	10.59	11.77	12.96		14.44	7 / 4			
	reaction time	0.163		1.12	1.10	1.12	1.11	1.13	1.14	1.14	1.18	1.19	1.48			3.34	3.38	3.51
	interval															8.21	8.11	7.81
Šebrle, Roman (CZE) (1974)	time		2.77	3.91	5.01	6.11	7.21	8.35	9.44	10.57	11.73	12.90		14.44	2 / 5			
	reaction time	0.165		1.14	1.10	1.10	1.10	1.14	1.09	1.13	1.16	1.17	1.54			3.34	3.33	3.46
	interval															8.21	8.23	7.92
Cleve, Moritz (GER) (1987)	time		2.77	3.87	5.05	6.17	7.28	8.37	9.49	10.67	11.81	12.99		14.54	5 / 6			
	reaction time	0.162		1.10	1.18	1.12	1.11	1.09	1.12	1.18	1.14	1.18	1.55			3.40	3.32	3.50
	interval															8.06	8.26	7.83

Decathlon - Heat 2

date 20-Aug-09

wind 0.3 m/s

Graubner (2009) - biomechanical analyses at the 12th IAAF world championships

			H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
Hardee, Trey (USA) (1984)	time		2.65	3.75	4.82	5.87	6.93	8.03	9.09	10.21	11.29	12.41		13.86	1 / 1			
	reaction time	0.134		1.10	1.07	1.05	1.06	1.10	1.06	1.12	1.08	1.12	1.45			3.22	3.22	3.32
	velocity		5.18	8.31	8.54	8.70	8.62	8.31	8.62	8.16	8.46	8.16	9.67	7.94		8.52	8.52	8.26
García, Yordanis (CUB) (1988)	time		2.61	3.69	4.78	5.89	6.99	8.11	9.21	10.30	11.45	12.59		14.08	4 / 2			
	reaction time	0.156		1.08	1.09	1.11	1.10	1.12	1.10	1.09	1.15	1.14	1.49			3.28	3.32	3.38
	velocity		5.26	8.46	8.39	8.23	8.31	8.16	8.31	8.39	7.95	8.02	9.41	7.81		8.36	8.26	8.11
Raja, Andres (EST) (1982)	time		2.67	3.79	4.91	6.00	7.10	8.19	9.29	10.43	11.55	12.71		14.22	7 / 3			
	reaction time	0.151		1.12	1.12	1.09	1.10	1.09	1.10	1.14	1.12	1.16	1.51			3.33	3.29	3.42
	velocity		5.14	8.16	8.16	8.39	8.31	8.39	8.31	8.02	8.16	7.88	9.28	7.74		8.23	8.33	8.02
Behrenbruch, Pascal (GER) (1982)	time		2.75	3.87	4.97	6.07	7.17	8.28	9.39	10.50	11.63	12.77		14.24	2 / 4			
	reaction time	0.156		1.12	1.10	1.10	1.10	1.11	1.11	1.11	1.13	1.14	1.47			3.32	3.32	3.38
	velocity		4.99	8.16	8.31	8.31	8.31	8.23	8.23	8.23	8.09	8.02	9.54	7.72		8.26	8.26	8.11
Eaton, Ashton (USA) (1988)	time		2.70	3.79	4.87	5.97	7.07	8.17	9.27	10.45	11.61	12.75		14.28	5 / 5			
	reaction time	0.178		1.09	1.08	1.10	1.10	1.10	1.10	1.18	1.16	1.14	1.53			3.27	3.30	3.48
	velocity		5.08	8.39	8.46	8.31	8.31	8.31	8.31	7.75	7.88	8.02	9.16	7.70		8.39	8.31	7.88
Suárez, Leonel (CUB) (1987)	time		2.65	3.78	4.89	6.01	7.13	8.25	9.39	10.51	11.68	12.85		14.45	8 / 7			
	reaction time	0.139		1.13	1.11	1.12	1.12	1.12	1.14	1.12	1.17	1.17	1.60			3.36	3.38	3.46
	velocity		5.18	8.09	8.23	8.16	8.16	8.16	8.02	8.16	7.81	7.81	8.76	7.61		8.16	8.11	7.92

Decathlon - Heat 1

date 20-Aug-09

wind 0.3 m/s

Graubner (2009) - biomechanical analyses at the 12th IAAF world championships

			H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
Pogorelov, Aleksandr (RUS) (1984)	time		2.82	3.94	5.04	6.15	7.25	8.34	9.41	10.52	11.64	12.74		14.19	8 / 1			
	reaction time	0.146		1.12	1.10	1.11	1.10	1.09	1.07	1.11	1.12	1.10	1.45			3.33	3.26	3.33
	velocity		4.87	8.16	8.31	8.23	8.31	8.39	8.54	8.23	8.16	8.31	9.67	7.75		8.23	8.41	8.23
Díaz, Yúnior (CUB) (1987)	time		2.69	3.86	5.00	6.13	7.26	8.38	9.51	10.86	11.86	13.08		14.56	6 / 2			
	reaction time	0.155		1.17	1.14	1.13	1.13	1.12	1.13	1.35	1.00	1.22	1.48			3.44	3.38	3.57
	velocity		5.10	7.81	8.02	8.09	8.09	8.16	8.09	6.77	9.14	7.49	9.47	7.55		7.97	8.11	7.68
Müller, Norman (GER) (1985)	time		2.79	3.93	5.05	6.17	7.30	8.42	9.55	10.70	11.86	13.40		14.59	5 / 3			
	reaction time	0.172		1.14	1.12	1.12	1.13	1.12	1.13	1.15	1.16	1.54	1.19			3.38	3.38	3.85
	velocity		4.92	8.02	8.16	8.16	8.09	8.16	8.09	7.95	7.88	5.94	11.78	7.54		8.11	8.11	7.12

Heat 6

date 19-Aug-09

wind 0.1 m/s

Graubner (2009) - biomechanical analyses at the 12th IAAF world championships

			H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
Kiss, Dániel (HUN) (1982)	time		2.70	3.77	4.78	5.79	6.79	7.79	8.81	9.83	10.87	11.97		13.34	7 / 1			
	reaction time	0.158		1.07	1.01	1.01	1.00	1.00	1.02	1.02	1.04	1.10	1.37	NR		3.09	3.02	3.16
	velocity		5.08	8.54	9.05	9.05	9.14	9.14	8.96	8.96	8.79	8.31	10.23	8.25		8.87	9.08	8.68
Villar, Paulo (COL) (1978)	time		2.58	3.66	4.71	5.75	6.77	7.81	8.86	9.91	10.97	12.07		13.52	2 / 2			
	reaction time	0.162		1.08	1.05	1.04	1.02	1.04	1.05	1.05	1.06	1.10	1.45			3.17	3.11	3.21
	velocity		5.32	8.46	8.70	8.79	8.96	8.79	8.70	8.70	8.62	8.31	9.67	8.14		8.65	8.82	8.54
Payne, David (USA) (1982)	time		2.61	3.69	4.75	5.77	6.80	7.85	8.89	9.93	10.99	12.05		13.54	6 / 3			
	reaction time	0.150		1.08	1.06	1.02	1.03	1.05	1.04	1.04	1.06	1.06	1.49			3.16	3.12	3.16
	velocity		5.26	8.46	8.62	8.96	8.87	8.70	8.79	8.79	8.62	8.62	9.41	8.12		8.68	8.79	8.68
Capetillo, Dayron (CUB) (1987)	time		2.63	3.67	4.74	5.77	6.79	7.83	8.87	9.93	11.05	12.15		13.61	4 / 4			
	reaction time	0.168		7 steps	1.04	1.07	1.03	1.02	1.04	1.06	1.12	1.10	1.46			3.14	3.10	3.28
	velocity		5.22	8.79	8.54	8.87	8.96	8.79	8.79	8.62	8.16	8.31	9.60	8.08		8.73	8.85	8.36
Bühler, Matthias (GER) (1986)	time		2.71	3.79	4.87	5.95	7.00	8.05	9.11	10.19	11.27	12.35		13.75	8 / 6			
	reaction time	0.131		1.08	1.08	1.08	1.05	1.05	1.06	1.08	1.08	1.08	1.40			3.24	3.16	3.24
	velocity		5.06	8.46	8.46	8.46	8.70	8.70	8.62	8.46	8.46	8.46	10.01	8.00		8.46	8.68	8.46

Heat 5

date 19-Aug-09

wind 0.0 m/s

Graubner (2009) - biomechanical analyses at the 12th IAAF world championships

			H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
Ji Wei (CHN) (1984)	time		2.64	3.71	4.75	5.79	6.81	7.86	8.91	9.95	11.03	12.11		13.51	2 / 1			
	reaction time	0.124		1.07	1.04	1.04	1.02	1.05	1.05	1.04	1.08	1.08	1.40			3.15	3.12	3.20
	velocity		5.20	8.54	8.79	8.79	8.96	8.70	8.70	8.79	8.46	8.46	10.01	8.14		8.70	8.79	8.57
Trammell, Terrence (USA) (1977)	time		2.61	3.67	4.71	5.72	6.73	7.80	8.87	9.93	11.01	12.11		13.51	8 / 2			
	reaction time	0.202		1.06	1.04	1.01	1.01	1.07	1.07	1.06	1.08	1.10	1.40			3.11	3.15	3.24
	velocity		5.26	8.62	8.79	9.05	9.05	8.54	8.54	8.62	8.46	8.31	10.01	8.14		8.82	8.70	8.46
Sedoc, Gregory (NED) (1981)	time		2.63	3.69	4.74	5.77	6.79	7.83	8.90	9.95	11.03	12.14		13.54	7 / 3			
	reaction time	0.145		1.06	1.05	1.03	1.02	1.04	1.07	1.05	1.08	1.11	1.40			3.14	3.13	3.24
	velocity		5.22	8.62	8.70	8.87	8.96	8.79	8.54	8.70	8.46	8.23	10.01	8.12		8.73	8.76	8.46

Bascou, Dimitri (FRA) (1987)	time	2.65	3.72	4.75	5.77	6.80	7.85	8.91	9.97	11.05	12.13		13.55	6 / 4			
	reaction time	0.158		1.07	1.03	1.02	1.03	1.05	1.06	1.06	1.08	1.08	1.42		3.12	3.14	3.22
	velocity	5.18	8.54	8.87	8.96	8.87	8.70	8.62	8.62	8.46	8.46	9.87	8.12		8.79	8.73	8.52
Sands, Shamar (BAH) (1985)	time	2.63	3.69	4.75	5.79	6.81	7.86	8.90	9.95	11.03	12.13		13.57	4 / 5			
	reaction time	0.243		1.06	1.06	1.04	1.02	1.05	1.04	1.05	1.08	1.10	1.44		3.16	3.11	3.23
	velocity	5.22	8.62	8.62	8.79	8.96	8.70	8.79	8.70	8.46	8.31	9.74	8.11		8.68	8.82	8.49

Heat 4date 19-Aug-09 wind 0.5 m/s *Graubner (2009) - biomechanical analyses at the 12th IAAF world championships*

		H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
Brathwaite, Ryan (BAR) (1988)	time	2.57	3.62	4.63	5.64	6.65	7.67	8.71	9.80	10.85	11.91		13.35	2 / 1			
	reaction time	0.149		1.05	1.01	1.01	1.01	1.02	1.04	1.09	1.05	1.06	1.44		3.07	3.07	3.20
	velocity	5.34	8.70	9.05	9.05	9.05	8.96	8.79	8.39	8.70	8.62	9.74	8.24		8.93	8.93	8.57
John, Alexander (GER) (1986)	time	2.63	3.70	4.75	5.81	6.84	7.88	8.93	9.96	11.01	12.08		13.41	5 / 2			
	reaction time	0.114		1.07	1.05	1.06	1.03	1.04	1.05	1.03	1.05	1.07	1.33		3.18	3.12	3.15
	velocity	5.22	8.54	8.70	8.62	8.87	8.79	8.70	8.87	8.70	8.54	10.54	8.20		8.62	8.79	8.70
Sharman, William (GBR) (1984)	time	2.61	3.67	4.71	5.75	6.81	7.85	8.90	9.97	11.03	12.11		13.52	1 / 3			
	reaction time	0.143		7 steps 1.06	1.04	1.04	1.06	1.04	1.05	1.07	1.06	1.08	1.41		3.14	3.15	3.21
	velocity	5.26	8.62	8.79	8.79	8.62	8.79	8.70	8.54	8.62	8.46	9.94	8.14		8.73	8.70	8.54
Olijar, Stanislavs (LAT) (1979)	time	2.72	3.79	4.83	5.85	6.87	7.91	8.96	10.00	11.07	12.15		13.59	8 / 4			
	reaction time	0.160		1.07	1.04	1.02	1.02	1.04	1.05	1.04	1.07	1.08	1.44		3.13	3.11	3.19
	velocity	5.04	8.54	8.79	8.96	8.96	8.79	8.70	8.79	8.54	8.46	9.74	8.09		8.76	8.82	8.60

Heat 3date 19-Aug-09 wind 0.2 m/s *Graubner (2009) - biomechanical analyses at the 12th IAAF world championships*

		H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
Svoboda, Petr (CZE) (1984)	time	2.63	3.69	4.72	5.74	6.75	7.78	8.85	9.91	10.99	12.09		13.56	6 / =1			
	reaction time	0.152		1.06	1.03	1.02	1.01	1.03	1.07	1.06	1.08	1.10	1.47		3.11	3.11	3.24
	velocity	5.22	8.62	8.87	8.96	9.05	8.87	8.54	8.62	8.46	8.31	9.54	8.11		8.82	8.82	8.46
Darien, Garfield (FRA) (1987)	time	2.67	3.71	4.75	5.80	6.83	7.87	8.93	9.99	11.07	12.15		13.56	2 / =1			
	reaction time	0.149		1.04	1.04	1.05	1.03	1.04	1.06	1.06	1.08	1.08	1.41		3.13	3.13	3.22
	velocity	5.14	8.79	8.79	8.70	8.87	8.79	8.62	8.62	8.46	8.46	9.94	8.11		8.76	8.76	8.52
Noga, Artur (POL) (1988)	time	2.70	3.77	4.83	5.87	6.90	7.93	8.97	10.02	11.07	12.14		13.56	3 / 3			
	reaction time	0.176		1.07	1.06	1.04	1.03	1.04	1.05	1.05	1.07	1.42			3.17	3.10	3.17
	velocity	5.08	8.54	8.62	8.79	8.87	8.87	8.79	8.70	8.70	8.54	9.87	8.11		8.65	8.85	8.65

Heat 2date 19-Aug-09 wind -0.2 m/s *Graubner (2009) - biomechanical analyses at the 12th IAAF world championships*

		H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
Wignall, Maurice (JAM) (1976)	time	2.63	3.71	4.75	5.79	6.89	7.97	9.01	10.08	11.15	12.23		13.62	3 / 1			
	reaction time	0.168		1.08	1.04	1.04	1.10	1.08	1.07	1.07	1.08	1.39			3.16	3.22	3.22
	velocity	5.22	8.46	8.79	8.79	8.31	8.46	8.79	8.54	8.54	8.46	10.09	8.08		8.68	8.52	8.52
Schwarzer, Helge (GER) (1985)	time	2.59	3.63	4.71	5.74	6.78	7.84	8.91	9.99	11.07	12.17		13.66	7 / 2			
	reaction time	0.136		1.04	1.08	1.03	1.04	1.06	1.07	1.08	1.08	1.10	1.49		3.15	3.17	3.26
	velocity	5.30	8.79	8.46	8.87	8.79	8.62	8.54	8.46	8.46	8.31	9.41	8.05		8.70	8.65	8.41
Robles, Dayron (CUB) (1986)	time	2.71	3.81	4.91	5.99	7.03	8.08	9.11	10.14	11.17	12.21		13.67	2 / 3			
	reaction time	0.178		7 steps 1.10	1.10	1.08	1.04	1.05	1.03	1.03	1.03	1.04	1.46		3.28	3.12	3.10
	velocity	5.06	8.31	8.31	8.46	8.79	8.70	8.87	8.87	8.87	8.79	9.60	8.05		8.36	8.79	8.85

Heat 1date 19-Aug-09 wind -0.4 m/s *Graubner (2009) - biomechanical analyses at the 12th IAAF world championships*

		H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
Shi Dongpeng (CHN) (1984)	time	2.68	3.75	4.79	5.82	6.84	7.90	8.95	10.00	11.06	12.12		13.56	2 / 1			
	reaction time	0.155		1.07	1.04	1.03	1.02	1.06	1.05	1.05	1.06	1.06	1.44		3.14	3.13	3.17
	velocity	5.12	8.54	8.79	8.87	8.96	8.62	8.70	8.70	8.62	8.62	9.74	8.11		8.73	8.76	8.65
Thomas, Dwight (JAM) (1980)	time	2.64	3.70	4.78	5.82	6.86	7.89	8.94	10.00	11.08	12.16		13.57	7 / 2			
	reaction time	0.138		1.06	1.08	1.04	1.04	1.03	1.05	1.06	1.08	1.08	1.41		3.18	3.12	3.22
	velocity	5.20	8.62	8.46	8.79	8.79	8.87	8.70	8.62	8.46	8.46	9.94	8.11		8.62	8.79	8.52
Quiñónez, Jackson (ESP) (1980)	time	2.70	3.76	4.79	5.82	6.84	7.88	8.92	9.98	11.06	12.16		13.63	8 / 3			
	reaction time	0.221		1.06	1.03	1.03	1.02	1.04	1.04	1.06	1.08	1.10	1.47		3.12	3.10	3.24
	velocity	5.08	8.62	8.87	8.87	8.96	8.79	8.79	8.62	8.46	8.31	9.54	8.07		8.79	8.85	8.46
Fourie, Lehann (RSA) (1987)	time	2.62	3.66	4.72	5.78	6.84	7.88	8.96	10.02	11.10	12.21		13.67	3 / 4			
	reaction time	0.139		1.04	1.06	1.06	1.06	1.04	1.08	1.06	1.08	1.11	1.46		3.16	3.18	3.25
	velocity	5.24	8.79	8.62	8.62	8.62	8.79	8.46	8.62	8.46	8.23	9.60	8.05		8.68	8.62	8.44

2009 Windsprint (Sundsvall, SWE)**FINAL**date 25-Jul-09 wind 1.6 m/s *(2009) - www.ssg.se/en/Windsprint/Split-times/*

		H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
Townsend, Fredrick (USA) (1986)	time		3.67			6.79			9.98				13.56	1 / 1			

reaction time	interval		3.12	3.19	3.58	
velocity		6.23	8.79	8.60	9.02	8.11

2009 Nambu Chuhei Memorial (Sapporo, JPN)**FINAL**

FINAL

	date	09-Jul-09	wind	1.2 m/s	Ryutaro (2010) - an attempt or a creation of a race pattern stadard model for the 110m hurdles													
			H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
Tanonaka, Tasuku (JPN) (1978)	time		2.63	3.69	4.75	5.81	6.87	7.94	9.02	10.10	11.18	12.26		13.72	/ 1			
	reaction time	interval		1.06	1.06	1.06	1.06	1.07	1.08	1.08	1.08	1.08	1.46			3.18	3.21	3.24
		velocity	5.22	8.62	8.62	8.62	8.62	8.54	8.46	8.46	8.46	8.46	9.60	8.02		8.62	8.54	8.46
Ohashi, Yuji (JPN) (1983)	time		2.63	3.70	4.76	5.81	6.85	7.91	8.97	10.04	11.14	12.25		13.72	/ 2			
	reaction time	interval		1.07	1.06	1.05	1.04	1.06	1.06	1.07	1.10	1.11	1.47			3.18	3.16	3.28
		velocity	5.22	8.54	8.62	8.70	8.79	8.62	8.62	8.54	8.31	8.23	9.54	8.02		8.62	8.68	8.36
Iwafune, Yoichi (JPN) (1985)	time		2.65	3.74	4.82	5.87	6.93	8.00	9.06	10.14	11.25	12.38		13.87	/ 3			
	reaction time	interval		1.09	1.08	1.05	1.06	1.07	1.06	1.08	1.11	1.13	1.49			3.22	3.19	3.32
		velocity	5.18	8.39	8.46	8.70	8.62	8.54	8.62	8.46	8.23	8.09	9.41	7.93		8.52	8.60	8.26
Naito, Masato (JPN) (1980)	time		2.67	3.76	4.81	5.87	6.91	8.00	9.08	10.18	11.31	12.42		13.88	/ 4			
	reaction time	interval		1.09	1.05	1.06	1.04	1.09	1.08	1.10	1.13	1.11	1.46			3.20	3.21	3.34
		velocity	5.14	8.39	8.70	8.62	8.79	8.39	8.46	8.31	8.09	8.23	9.60	7.93		8.57	8.54	8.21
Shuto, Takayuki (JPN) (1986)	time		2.73	3.83	4.91	5.99	7.08	8.16	9.25	10.36	11.45	12.57		13.94	/ 5			
	reaction time	interval		1.10	1.08	1.08	1.09	1.08	1.09	1.11	1.09	1.12	1.37			3.26	3.26	3.32
		velocity	5.03	8.31	8.46	8.46	8.39	8.46	8.39	8.23	8.39	8.16	10.23	7.89		8.41	8.41	8.26
Irie, Yukito (JPN) (1983)	time		2.44	3.54	4.65	5.74	6.82	7.91	9.01	10.11	11.22	12.33		13.98	/ 6			
	reaction time	interval		1.10	1.11	1.09	1.08	1.09	1.10	1.10	1.11	1.11	1.65			3.30	3.27	3.32
		velocity	5.62	8.31	8.23	8.39	8.46	8.39	8.31	8.31	8.23	8.23	8.50	7.87		8.31	8.39	8.26

2009 Japanese National Championships (Hiroshima, JPN)**FINAL**

FINAL

		date	27-Jun-09	wind 1.6 m/s			Ryutaro (2010) - an attempt or a creation of a race pattern stadard model for the 110m hurdles												
				H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
Tanonaka, Tasuku (JPN) (1978)	time			2.59	3.67	4.72	5.76	6.80	7.87	8.91	9.97	11.05	12.13		13.57	3 / 1			
	reaction time		interval		1.08	1.05	1.04	1.04	1.07	1.04	1.06	1.08	1.08	1.44			3.17	3.15	3.22
		velocity		5.30	8.46	8.70	8.79	8.79	8.54	8.79	8.62	8.46	8.46	9.74	8.11		8.65	8.70	8.52
Shuto, Takayuki (JPN) (1986)	time			2.65	3.75	4.81	5.90	6.94	8.01	9.08	10.15	11.27	12.32		13.77	4 / 2			
	reaction time		interval		1.10	1.06	1.09	1.04	1.07	1.07	1.07	1.12	1.05	1.45	PB		3.25	3.18	3.24
		velocity		5.18	8.31	8.62	8.39	8.79	8.54	8.54	8.54	8.16	8.70	9.67	7.99		8.44	8.62	8.46
Naito, Masato (JPN) (1980)	time			2.61	3.71	4.76	5.81	6.86	7.92	8.99	10.08	11.17	12.27		13.81	5 / 3			
	reaction time		interval		1.10	1.05	1.05	1.05	1.06	1.07	1.09	1.09	1.10	1.54			3.20	3.18	3.28
		velocity		5.26	8.31	8.70	8.70	8.70	8.62	8.54	8.39	8.39	8.31	9.10	7.97		8.57	8.62	8.36
Ohashi, Yuji (JPN) (1983)	time			2.66	3.73	4.76	5.81	6.86	7.90	8.99	10.07	11.17	12.28		13.81	7 / 3			
	reaction time		interval		1.07	1.03	1.05	1.05	1.04	1.09	1.08	1.10	1.11	1.53			3.15	3.18	3.29
		velocity		5.16	8.54	8.87	8.70	8.70	8.79	8.39	8.46	8.31	8.23	9.16	7.97		8.70	8.62	8.33
Yoshinaga, Kazuyuki (JPN) (1985)	time			2.63	3.71	4.78	5.85	6.91	8.01	9.10	10.19	11.27	12.38		13.86	1 / 5			
	reaction time		interval		1.08	1.07	1.07	1.06	1.10	1.09	1.09	1.08	1.11	1.48	PB		3.22	3.25	3.28
		velocity		5.22	8.46	8.54	8.54	8.62	8.31	8.39	8.39	8.46	8.23	9.47	7.94		8.52	8.44	8.36
Ida, Masayuki (JPN) (1987)	time			2.68	3.77	4.87	5.95	7.03	8.09	9.17	10.25	11.33	12.44		13.86	2 / 5			
	reaction time		interval		1.09	1.10	1.08	1.08	1.06	1.08	1.08	1.08	1.11	1.42			3.27	3.22	3.27
		velocity		5.12	8.39	8.31	8.46	8.46	8.62	8.46	8.46	8.46	8.23	9.87	7.94		8.39	8.52	8.39
Moses, Yume (JPN) (1988)	time			2.64	3.75	4.81	5.87	6.92	7.99	9.07	10.17	11.29	12.41		13.89	6 / 7			
	reaction time		interval		1.11	1.06	1.06	1.05	1.07	1.08	1.10	1.12	1.12	1.48			3.23	3.20	3.34
		velocity		5.20	8.23	8.62	8.62	8.70	8.54	8.46	8.31	8.16	8.16	9.47	7.92		8.49	8.57	8.21
Furukawa, Yutaro (JPN) (1985)	time			2.63	3.72	4.80	5.86	6.93	7.99	9.09	10.17	11.27	12.43		13.96	8 / 8			
	reaction time		interval		1.09	1.08	1.06	1.07	1.06	1.10	1.08	1.10	1.16	1.53			3.23	3.23	3.34
		velocity		5.22	8.39	8.46	8.62	8.54	8.62	8.31	8.46	8.31	7.88	9.16	7.88		8.49	8.49	8.21

Semi-Final 1

Semi-Final 1	date	27-Jun-09	wind		0.5 m/s		Akirabun (2010) - analysis of speed and stride on straight track events in the 2009 season											
			H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
	Naito, Masato (JPN) (1980)	time	2.50	3.59	4.65	5.69	6.74	7.80	8.86	9.93	11.02	12.14		13.79	6 / 1			
	reaction time	interval		1.09	1.06	1.04	1.05	1.06	1.06	1.07	1.09	1.12	1.65			3.19	3.17	3.28
		velocity	5.49	8.39	8.62	8.79	8.70	8.62	8.62	8.54	8.39	8.16	8.50	7.98		8.60	8.65	8.36

Heat 5

Heat 5	date	26-Jun-09		wind		0.6 m/s		Akirabun (2010) - analysis of speed and stride on straight track events in the 2009 season												
				H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10	
	Furukawa, Yutaro (JPN) (1985)	time		2.41	3.51	4.57	5.63	6.70	7.77	8.85	9.93	11.04	12.16		13.82	2 / 1				
		reaction time	interval			1.10	1.06	1.06	1.07	1.07	1.08	1.08	1.11	1.12	1.66			3.22	3.22	3.31
			velocity		5.69	8.31	8.62	8.62	8.54	8.54	8.46	8.46	8.23	8.16	8.45	7.96		8.52	8.52	8.28

Heat 4

date 26-Jun-09 wind 0.3 m/s

Akirabun (2010) - analysis of speed and stride on straight track events in the 2009 season

		H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
Yahata, Kenji (JPN) (1980)	time	2.44	3.52	4.60	5.66	6.73	7.81	8.91	10.01	11.11	12.23		13.91	3 / 1			
	reaction time		1.08	1.08	1.06	1.07	1.08	1.10	1.10	1.10	1.12	1.68			3.22	3.25	3.32
	velocity	5.62	8.46	8.46	8.62	8.54	8.46	8.31	8.31	8.31	8.16	8.35	7.91		8.52	8.44	8.26

Heat 3

date 26-Jun-09 wind 0.6 m/s

Akirabun (2010) - analysis of speed and stride on straight track events in the 2009 season

		H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
Ohashi, Yuji (JPN) (1983)	time	2.46	3.55	4.61	5.65	6.70	7.76	8.83	9.90	11.00	12.12		13.78	3 / 1			
	reaction time		1.09	1.06	1.04	1.05	1.06	1.07	1.07	1.10	1.12	1.66			3.19	3.18	3.29
	velocity	5.58	8.39	8.62	8.79	8.70	8.62	8.54	8.54	8.31	8.16	8.45	7.98		8.60	8.62	8.33

Decathlon - Heat 2

date 26-Jun-09 wind 1.6 m/s

Hisashi (2010) - the run performance analysis in the decathlon sprint events

		H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
Ikeda, Daisuke (JPN) (1986)	time	2.50	3.64	4.79	5.91	7.04	8.18	9.32	10.48	11.64	12.82		14.53	4 / 1			
	reaction time		1.14	1.15	1.12	1.13	1.14	1.14	1.16	1.16	1.18	1.71			3.41	3.41	3.50
	velocity	5.49	8.02	7.95	8.16	8.09	8.02	8.02	7.88	7.88	7.75	8.20	7.57		8.04	8.04	7.83
Tanaka, Hiromasa (JPN) (1981)	time	2.53	3.71	4.89	6.08	7.27	8.48	9.70	10.92	12.17	13.44		15.23	1 / 4			
	reaction time		1.18	1.18	1.19	1.19	1.21	1.22	1.22	1.25	1.27	1.79			3.55	3.62	3.74
	velocity	5.42	7.75	7.75	7.68	7.68	7.55	7.49	7.49	7.31	7.20	7.83	7.22		7.72	7.57	7.33
Ushiro, Keisuke (JPN) (1986)	time	2.59	3.78	4.97	6.16	7.36	8.57	9.79	11.06	12.33	13.52		15.34	5 / 7			
	reaction time		1.19	1.19	1.19	1.20	1.21	1.22	1.27	1.27	1.19	1.82	PB		3.57	3.63	3.73
	velocity	5.30	7.68	7.68	7.68	7.62	7.55	7.49	7.20	7.20	7.68	7.70	7.17		7.68	7.55	7.35

2009 ISTAF (Berlin, GER)

FINAL

date 14-Jun-09 wind 0.7 m/s

Graubner (2009) - <http://www.fgs.uni-halle.de>

		H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
Faulk, Dexter (USA) (1984)	time	2.55	3.61	4.63	5.66	6.69	7.70	8.75	9.77	10.81	11.86		13.18	4 / 1			
	reaction time 0.147		1.06	1.02	1.03	1.03	1.01	1.05	1.02	1.04	1.05	1.32	PB		3.11	3.09	3.11
	velocity	5.38	8.62	8.96	8.87	8.87	9.05	8.70	8.96	8.79	8.70	10.62	8.35		8.82	8.87	8.82
Wilson, Ryan (USA) (1980)	time	2.59	3.64	4.65	5.66	6.65	7.69	8.71	9.73	10.77	11.83		13.21	6 / 2			
	reaction time 0.171		1.05	1.01	1.01	0.99	1.04	1.02	1.02	1.04	1.06	1.38			3.07	3.05	3.12
	velocity	5.30	8.70	9.05	9.05	9.23	8.79	8.96	8.96	8.79	8.62	10.16	8.33		8.93	8.99	8.79
Payne, David (USA) (1982)	time	2.59	3.63	4.66	5.68	6.69	7.71	8.74	9.78	10.83	11.87		13.22	3 / 3			
	reaction time 0.181		1.04	1.03	1.02	1.01	1.02	1.03	1.04	1.05	1.04	1.35			3.09	3.06	3.13
	velocity	5.30	8.79	8.87	8.96	9.05	8.96	8.87	8.79	8.70	8.79	10.39	8.32		8.87	8.96	8.76
Sands, Shamar (BAH) (1985)	time	2.59	3.65	4.69	5.71	6.73	7.77	8.81	9.87	10.93	12.09		13.56	8 / 4			
	reaction time 0.234		1.06	1.04	1.02	1.02	1.04	1.04	1.06	1.06	1.16	1.47			3.12	3.10	3.28
	velocity	5.30	8.62	8.79	8.96	8.96	8.79	8.79	8.62	8.62	7.88	9.54	8.11		8.79	8.85	8.36
Wignall, Maurice (JAM) (1976)	time	2.61	3.69	4.71	5.78	6.80	7.85	8.93	9.99	11.08	12.19		13.62	5 / 5			
	reaction time 0.189		1.08	1.02	1.07	1.02	1.05	1.08	1.06	1.09	1.11	1.43			3.17	3.15	3.26
	velocity	5.26	8.46	8.96	8.54	8.96	8.70	8.46	8.62	8.39	8.23	9.80	8.08		8.65	8.70	8.41
Olijar, Stanislav (LAT) (1979)	time	2.69	3.77	4.81	5.82	6.86	7.91	8.97	10.03	11.13	12.23		13.66	2 / 6			
	reaction time 0.172		1.08	1.04	1.01	1.04	1.05	1.06	1.06	1.10	1.10	1.43			3.13	3.15	3.26
	velocity	5.10	8.46	8.79	9.05	8.79	8.70	8.62	8.62	8.31	8.31	9.80	8.05		8.76	8.70	8.41
Doucouré, Ladji (FRA) (1983)	time	2.65	3.73	4.76	5.80	6.83	7.90	8.97	10.05	11.13	12.24		13.73	7 / 7			
	reaction time 0.206		1.08	1.03	1.04	1.03	1.07	1.07	1.08	1.08	1.11	1.49			3.15	3.17	3.27
	velocity	5.18	8.46	8.87	8.79	8.87	8.54	8.54	8.46	8.46	8.23	9.41	8.01		8.70	8.65	8.39
Mathiszik, Willi (GER) (1984)	time	2.68	3.79	4.84	5.89	6.95	8.03	9.13	10.19	11.31	12.43		13.86	9 / 8			
	reaction time 0.204		1.11	1.05	1.05	1.06	1.08	1.10	1.06	1.12	1.12	1.43			3.21	3.24	3.30
	velocity	5.12	8.23	8.70	8.70	8.62	8.46	8.31	8.62	8.16	8.16	9.80	7.94		8.54	8.46	8.31
Bühler, Matthias (GER) (1986)	time	2.68	3.77	4.87	5.93	6.98	8.05	9.13	10.21	11.31	12.43		13.87	1 / 9			
	reaction time 0.134		1.09	1.10	1.06	1.05	1.07	1.08	1.08	1.10	1.12	1.44			3.25	3.20	3.30
	velocity	5.12	8.39	8.31	8.62	8.70	8.54	8.46	8.46	8.31	8.16	9.74	7.93		8.44	8.57	8.31

2009 University of Barcelona Hurdle Analysis (Barcelona, ESP)

Time Trials

date Jun-09 wind -0.4 m/s

Padulles (2010) - valoración funcional / análisis técnico biomecánico del dayron robes

		H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
Robles, Dayron (CUB) (1986)	time	2.34	3.39	4.42	5.44	6.49	7.54	8.60	9.64	10.70	11.80		13.51	5 /			
	reaction time 0.184		7 steps	1.05	1.03	1.02	1.05	1.05	1.04	1.06	1.10	1.71			3.10	3.16	3.20
	velocity	5.86	8.70	8.87	8.96	8.70	8.70	8.62	8.79	8.62	8.31	8.20	8.14		8.85	8.68	8.57

Time Trials

date Jun-09 wind -0.5 m/s

Padulles (2010) - valoración funcional / análisis técnico biomecánico del dayron robes

		H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
Quiñónez, Jackson (ESP) (1980)	time	2.37	3.48	4.56	5.61	6.70	7.79	8.91	10.01	11.15	12.32		14.11	5 /			

reaction time	0.148	interval	1.11	1.08	1.05	1.09	1.09	1.12	1.10	1.14	1.17	1.79			3.24	3.30	3.41
velocity			5.79	8.23	8.46	8.70	8.39	8.39	8.16	8.31	8.02	7.81	7.83	7.80	8.46	8.31	8.04

2009 Osaka Grand Prix (Osaka, JPN)**FINAL**

		date	09-May-09	wind	0.2 m/s	Ryutaro (2010) - an attempt or a creation of a race pattern stadard model for the 110m hurdles													
				H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
Sands, Shamar (BAH) (1985)	time			2.55	3.59	4.60	5.63	6.62	7.65	8.71	9.76	10.84	11.94		13.40	/ 1			
	reaction time				1.04	1.01	1.03	0.99	1.03	1.06	1.05	1.08	1.10	1.46	NR		3.08	3.08	3.23
	velocity			5.38	8.79	9.05	8.87	9.23	8.87	8.62	8.70	8.46	8.31	9.60	8.21		8.90	8.90	8.49
Shi Dongpeng (CHN) (1984)	time			2.64	3.70	4.74	5.79	6.83	7.84	8.88	9.95	11.00	12.09		13.48	/ 2			
	reaction time				1.06	1.04	1.05	1.04	1.01	1.04	1.07	1.05	1.09	1.39			3.15	3.09	3.21
	velocity			5.20	8.62	8.79	8.70	8.79	9.05	8.79	8.54	8.70	8.39	10.09	8.16		8.70	8.87	8.54
Ji Wei (CHN) (1984)	time			2.63	3.69	4.73	5.77	6.78	7.80	8.81	9.84	10.91	11.99		13.51	/ 3			
	reaction time				1.06	1.04	1.04	1.01	1.02	1.01	1.03	1.07	1.08	1.52			3.14	3.04	3.18
	velocity			5.22	8.62	8.79	8.79	9.05	8.96	9.05	8.87	8.54	8.46	9.22	8.14		8.73	9.02	8.62
Tanonaka, Tasuku (JPN) (1978)	time			2.60	3.69	4.73	5.79	6.84	7.89	8.96	10.02	11.10	12.19		13.67	/ 4			
	reaction time				1.09	1.04	1.06	1.05	1.05	1.07	1.06	1.08	1.09	1.48			3.19	3.17	3.23
	velocity			5.28	8.39	8.79	8.62	8.70	8.70	8.54	8.62	8.46	8.39	9.47	8.05		8.60	8.65	8.49
Nomoto, Hideki (JPN) (1984)	time			2.61	3.69	4.77	5.83	6.90	7.97	9.05	10.15	11.25	12.37		13.94	/ 5			
	reaction time				1.08	1.08	1.06	1.07	1.07	1.08	1.10	1.10	1.12	1.57			3.22	3.22	3.32
	velocity			5.26	8.46	8.46	8.62	8.54	8.54	8.46	8.31	8.31	8.16	8.93	7.89		8.52	8.52	8.26

2009 Mikio Oda Memorial (Hiroshima, JPN)**FINAL**

		date	29-Apr-09	wind	Ryutaro (2010) - an attempt or a creation of a race pattern stadard model for the 110m hurdles														
				H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
Yin Jing (CHN) (1988)	time			2.60	3.67	4.70	5.73	6.75	7.77	8.81	9.86	10.90	11.96		13.40	/ 1			
	reaction time				1.07	1.03	1.03	1.02	1.02	1.04	1.05	1.04	1.06	1.44	PB		3.13	3.08	3.15
	velocity			5.28	8.54	8.87	8.87	8.96	8.96	8.79	8.70	8.79	8.62	9.74	8.21		8.76	8.90	8.70
Tanonaka, Tasuku (JPN) (1978)	time			2.60	3.68	4.74	5.80	6.85	7.90	8.96	10.04	11.12	12.20		13.65	/ 2			
	reaction time				1.08	1.06	1.06	1.05	1.05	1.06	1.08	1.08	1.08	1.45			3.20	3.16	3.24
	velocity			5.28	8.46	8.62	8.62	8.70	8.70	8.62	8.46	8.46	8.46	9.67	8.06		8.57	8.68	8.46
Ohashi, Yuji (JPN) (1983)	time			2.62	3.71	4.75	5.83	6.87	7.95	9.01	10.08	11.18	12.29		13.76	/ 3			
	reaction time				1.09	1.04	1.08	1.04	1.08	1.06	1.07	1.10	1.11	1.47			3.21	3.18	3.28
	velocity			5.24	8.39	8.79	8.46	8.79	8.46	8.62	8.54	8.31	8.23	9.54	7.99		8.54	8.62	8.36
Park Tae-Kyong (KOR) (1980)	time			2.64	3.71	4.77	5.84	6.89	7.96	9.08	10.13	11.23	12.33		13.77	/ 4			
	reaction time				1.07	1.06	1.07	1.05	1.07	1.12	1.05	1.10	1.10	1.44			3.20	3.24	3.25
	velocity			5.20	8.54	8.62	8.54	8.70	8.54	8.16	8.70	8.31	8.31	9.74	7.99		8.57	8.46	8.44
Yahata, Kenji (JPN) (1983)	time			2.63	3.71	4.78	5.83	6.88	7.94	8.99	10.08	11.17	12.29		13.77	/ 4			
	reaction time				1.08	1.07	1.05	1.05	1.06	1.05	1.09	1.09	1.12	1.48			3.20	3.16	3.30
	velocity			5.22	8.46	8.54	8.70	8.70	8.62	8.70	8.39	8.39	8.16	9.47	7.99		8.57	8.68	8.31
Irie, Yukito (JPN) (1983)	time			2.63	3.72	4.75	5.83	6.91	7.96	9.05	10.14	11.25	12.37		13.79	/ 6			
	reaction time				1.09	1.03	1.08	1.08	1.05	1.09	1.09	1.11	1.12	1.42	PB		3.20	3.22	3.32
	velocity			5.22	8.39	8.87	8.46	8.46	8.70	8.39	8.39	8.23	8.16	9.87	7.98		8.57	8.52	8.26
Aoki, Yuto (JPN) (1984)	time			2.65	3.74	4.81	5.84	6.92	7.98	9.07	10.16	11.28	12.37		13.80	/ 7			
	reaction time				1.09	1.07	1.03	1.08	1.06	1.09	1.09	1.12	1.09	1.43			3.19	3.23	3.30
	velocity			5.18	8.39	8.54	8.87	8.46	8.62	8.39	8.39	8.16	8.39	9.80	7.97		8.60	8.49	8.31
Nomoto, Hideki (JPN) (1984)	time			2.66	3.74	4.82	5.88	6.93	8.00	9.08	10.16	11.27	12.40		13.99	/ 8			
	reaction time				1.08	1.08	1.06	1.05	1.07	1.08	1.08	1.11	1.13	1.59			3.22	3.20	3.32
	velocity			5.16	8.46	8.46	8.62	8.70	8.54	8.46	8.46	8.23	8.09	8.82	7.86		8.52	8.57	8.26

Heat 1

Heat 1		date	29-Apr-09		wind	0.6 m/s		Akirabun (2010) - analysis of speed and stride on straight track events in the 2009 season												
			H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10		
Park Tae-Kyong (KOR) (1980)	time		2.45	3.54	4.61	5.68	6.74	7.82	8.93	10.03	11.13	12.23		13.87	/ 1					
	reaction time	interval		1.09	1.07	1.07	1.06	1.08	1.11	1.10	1.10	1.10	1.64			3.23	3.25	3.30		
		velocity		5.60	8.39	8.54	8.54	8.62	8.46	8.23	8.31	8.31	8.31	8.55	7.93		8.49	8.44	8.31	

2008 Seiko Super Grand Prix (Kawasaki, JPN)**FINAL**

100m Sprint Super Grand Prix (Kawasaki, JPN)

FINAL

	date	23-Sep-08	wind	-0.7 m/s	Akirabun (2009) - race analysis of 2008 men's and women's 100m and 110m & 100m hurdles														
			H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10	
Lee Jung-Joon (KOR) (1984)	time		2.48	3.57	4.65	5.71	6.78	7.83	8.87	9.95	11.03	12.10		13.71	/ 1				
	reaction time	interval		1.09	1.08	1.06	1.07	1.05	1.04	1.08	1.08	1.07	1.61			3.23	3.16	3.23	
	velocity		5.53	8.39	8.46	8.62	8.54	8.70	8.79	8.46	8.46	8.54	8.71	8.02		8.49	8.68	8.49	
Herring, Aubrey (USA) (1978)	time		2.45	3.54	4.60	5.64	6.66	7.74	8.81	9.89	10.98	12.12		13.77	/ 2				
	reaction time	interval		1.09	1.06	1.04	1.02	1.08	1.07	1.08	1.09	1.14	1.65			3.19	3.17	3.31	

velocity	5.60	8.39	8.62	8.79	8.96	8.46	8.54	8.46	8.39	8.02	8.50	7.99		8.60	8.65	8.28
Tanonaka, Tasuku (JPN) (1978) time	2.44	3.52	4.59	5.66	6.73	7.80	8.89	9.99	11.10	12.20		13.88	/ 4			
reaction time	interval	1.08	1.07	1.07	1.07	1.07	1.09	1.10	1.11	1.10	1.68			3.22	3.23	3.31
velocity	5.62	8.46	8.54	8.54	8.54	8.39	8.31	8.23	8.31	8.35	7.93			8.52	8.49	8.28
Aoki, Yuto (JPN) (1984) time	2.59	3.69	4.75	5.83	6.89	8.00	9.07	10.22	11.31	12.43		14.16	/ 6			
reaction time	interval	1.10	1.06	1.08	1.06	1.11	1.07	1.15	1.09	1.12	1.73			3.24	3.24	3.36
velocity	5.30	8.31	8.62	8.46	8.62	8.23	8.54	7.95	8.39	8.16	8.10	7.77		8.46	8.46	8.16

2008 Golden Grand Prix (Shanghai, CHN)

FINAL

INAL	date 20-Sep-08		wind 0.2 m/s		Wang (2016) - comparative analysis of the grey relational grade of segment times and final results of the 110m hurdles													
			H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
Xie Wenjun (CHN) (1990)	time		2.66	3.70	4.74	5.78	6.80	7.84	8.90	9.94	11.04	12.08		13.47	/ 2			
	reaction time	0.223		1.04	1.04	1.04	1.02	1.04	1.06	1.04	1.10	1.04	1.39	PB		3.12	3.12	3.18
	velocity		5.16	8.79	8.79	8.79	8.96	8.79	8.62	8.79	8.31	8.79	10.09	8.17		8.79	8.79	8.62

2008 IAAF World Athletic Final (Stuttgart, GER)

FINAL

		date	14-Sep-08	wind -1.3 m/s										Graubner (2009) - http://www.fgs.uni-halle.de						
				H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10	
Oliver, David (USA) (1982)	time			2.65	3.66	4.67	5.68	6.69	7.69	8.71	9.74	10.79	11.84		13.22	6 / 1				
	reaction time	0.181	interval		1.01	1.01	1.01	1.01	1.00	1.02	1.03	1.05	1.05	1.38			3.03	3.03	3.13	
			velocity		5.18	9.05	9.05	9.05	9.05	9.14	8.96	8.87	8.70	8.70	10.16	8.32		9.05	9.05	8.76
Svoboda, Petr (CZE) (1984)	time			2.65	3.69	4.70	5.70	6.72	7.74	8.78	9.82	10.87	11.93		13.33	8 / 2				
	reaction time	0.191	interval		1.04	1.01	1.00	1.02	1.02	1.04	1.04	1.05	1.06	1.40			3.05	3.08	3.15	
			velocity		5.18	8.79	9.05	9.14	8.96	8.96	8.79	8.79	8.70	8.62	10.01	8.25		8.99	8.90	8.70
Wilson, Ryan (USA) (1980)	time			2.71	3.78	4.80	5.83	6.85	7.88	8.92	9.98	11.05	12.13		13.54	3 / 3				
	reaction time	0.163	interval		1.07	1.02	1.03	1.02	1.03	1.04	1.06	1.07	1.08	1.41			3.12	3.09	3.21	
			velocity		5.06	8.54	8.96	8.87	8.96	8.87	8.79	8.62	8.54	8.46	9.94	8.12		8.79	8.87	8.54
Merritt, Aries (USA) (1985)	time			2.66	3.72	4.77	5.80	6.83	7.87	8.91	9.96	11.02	12.10		13.56	4 / 4				
	reaction time	0.156	interval		1.06	1.05	1.03	1.03	1.04	1.04	1.05	1.06	1.08	1.46			3.14	3.11	3.19	
			velocity		5.16	8.62	8.70	8.87	8.87	8.79	8.79	8.70	8.62	8.46	9.60	8.11		8.73	8.82	8.60
van der Westen, Marcel (NED)	time			2.64	3.69	4.72	5.76	6.79	7.83	8.87	9.93	11.00	12.09		13.57	1 / 5				
	reaction time	0.136	interval		1.05	1.03	1.04	1.03	1.04	1.04	1.06	1.07	1.09	1.48			3.12	3.11	3.22	
			velocity		5.20	8.70	8.87	8.79	8.87	8.79	8.79	8.62	8.54	8.39	9.47	8.11		8.79	8.82	8.52
Akins, Ty (USA) (1986)	time			2.61	3.69	4.76	5.80	6.86	7.93	9.00	10.06	11.13	12.22		13.62	5 / 6				
	reaction time	0.150	interval		1.08	1.07	1.04	1.06	1.07	1.07	1.06	1.07	1.09	1.40			3.19	3.20	3.22	
			velocity		5.26	8.46	8.54	8.79	8.62	8.54	8.54	8.62	8.54	8.39	10.01	8.08		8.60	8.57	8.52
Turner, Andy (GBR) (1980)	time			2.65	3.74	4.79	5.83	6.88	7.93	8.98	10.06	11.14	12.24		13.68	2 / 7				
	reaction time	0.125	interval		1.09	1.05	1.04	1.05	1.05	1.05	1.08	1.08	1.10	1.44			3.18	3.15	3.26	
			velocity		5.18	8.39	8.70	8.79	8.70	8.70	8.70	8.46	8.46	8.31	9.74	8.04		8.62	8.70	8.41
Brown, Joel (USA) (1980)	time			2.66	3.75	4.83	5.88	6.94	8.02	9.11	10.18	11.29	12.38		13.81	7 / 8				
	reaction time	0.139	interval		1.09	1.08	1.05	9.00	1.08	1.09	1.07	1.11	1.09	1.43			3.22	3.23	3.27	
			velocity		5.16	8.39	8.46	8.70	1.02	8.46	8.39	8.54	8.23	8.39	9.80	7.97		8.52	8.49	8.39

2008 Olympic Games (Beijing, CHN)

FINAL

		date	21-Aug-08	wind 0.1 m/s			Tsiokanos (2018) - race pattern of men's 110m hurdles- time analysis of olympic hurdle performance												
				H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
Robles, Dayron (CUB) (1986)	time			2.50	3.53	4.50	5.49	6.48	7.48	8.48	9.50	10.53	11.56		12.93	6 / 1			
	reaction time	0.183	interval	7 steps	1.03	0.97	0.99	0.99	1.00	1.00	1.02	1.03	1.03	1.37			2.99	2.99	3.08
			velocity		5.49	8.87	9.42	9.23	9.23	9.14	9.14	8.96	8.87	8.87	10.23	8.51		9.17	9.17
Payne, David (USA) (1982)	time			2.58	3.62	4.64	5.65	6.68	7.67	8.69	9.73	10.76	11.80		13.17	5 / 2			
	reaction time	0.175	interval		1.04	1.02	1.01	1.03	0.99	1.02	1.04	1.03	1.04	1.37			3.07	3.04	3.11
			velocity		5.32	8.79	8.96	9.05	8.87	9.23	8.96	8.79	8.87	8.79	10.23	8.35		8.93	9.02
Oliver, David (USA) (1982)	time			2.58	3.64	4.67	5.66	6.68	7.68	8.67	9.73	10.76	11.80		13.18	7 / 3			
	reaction time	0.158	interval		1.06	1.03	0.99	1.02	1.00	0.99	1.06	1.03	1.04	1.38			3.08	3.01	3.13
			velocity		5.32	8.62	8.87	9.23	8.96	9.14	9.23	8.62	8.87	8.79	10.16	8.35		8.90	9.11
Doucouré, Ladj (FRA) (1983)	time			2.58	3.64	4.66	5.68	6.69	7.68	8.69	9.73	10.78	11.84		13.24	8 / 4			
	reaction time	0.170	interval		1.06	1.02	1.02	1.01	0.99	1.01	1.04	1.05	1.06	1.40			3.10	3.01	3.15
			velocity		5.32	8.62	8.96	8.96	9.05	9.23	9.05	8.79	8.70	8.62	10.01	8.31		8.85	9.11
Noga, Artur (POL) (1988)	time			2.66	3.70	4.74	5.77	6.78	7.81	8.84	9.87	10.94	12.00		13.36	4 / 5			
	reaction time	0.169	interval		1.04	1.04	1.03	1.01	1.03	1.03	1.03	1.07	1.06	1.36			3.11	3.07	3.16
			velocity		5.16	8.79	8.79	8.87	9.05	8.87	8.87	8.87	8.54	8.62	10.31	8.23		8.82	8.93
Wignall, Maurice (JAM) (1976)		time		2.58	3.64	4.68	5.72	6.77	7.80	8.86	9.90	10.97	12.09		13.46	2 / 6			

reaction time	0.163	interval		1.06	1.04	1.04	1.05	1.03	1.06	1.04	1.07	1.12	1.37			3.14	3.14	3.23
		velocity	5.32	8.62	8.79	8.79	8.70	8.87	8.62	8.79	8.54	8.16	10.23	8.17		8.73	8.73	8.49
Phillips, Richard (JAM) (1983)	time	2.62	3.70	4.73	5.77	6.80	7.82	8.90	9.96	11.02	12.15			13.60	3 / 7			
reaction time	0.154	interval		1.08	1.03	1.04	1.03	1.02	1.08	1.06	1.06	1.13	1.45			3.15	3.13	3.25
		velocity	5.24	8.46	8.87	8.79	8.87	8.96	8.46	8.62	8.62	8.09	9.67	8.09		8.70	8.76	8.44
Quiñónez, Jackson (ESP) (1980)	time	2.64	3.72	4.76	5.79	6.82	7.87	8.92	9.99	11.10	12.20			13.69	9 / 8			
reaction time	0.187	interval		1.08	1.04	1.03	1.03	1.05	1.05	1.07	1.11	1.10	1.49			3.15	3.13	3.28
		velocity	5.20	8.46	8.79	8.87	8.87	8.70	8.70	8.54	8.23	8.31	9.41	8.04		8.70	8.76	8.36

2008 Swiss National Championships (Fribourg, SUI)

Heat 1

	date	03-Aug-08	wind	1.0 m/s	Weber (2008) - wettkampfauswertung 110m hürden													
			H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
Kundert, Andreas (SUI) (1984)	time	2.67	3.74	4.78	5.83	6.86	7.92	8.97	10.03	11.09	12.15			13.53	1 / 1			
reaction time	interval		1.07	1.04	1.05	1.03	1.06	1.05	1.06	1.06	1.06	1.38				3.16	3.14	3.18
	velocity	5.14	8.54	8.79	8.70	8.87	8.62	8.70	8.62	8.62	8.62	10.16	8.13			8.68	8.73	8.62

2008 USA Olympic Trials (Eugene, OR)

FINAL

	date	06-Jul-08	wind	3.5 m/s	USATF Hurdle Development (2008)													
			H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
Oliver, David (USA) (1985)	time	2.57	3.59	4.60	5.59	6.57	7.56	8.55	9.54	10.53	11.57			12.95	4 / 1			
reaction time	0.191	interval		1.02	1.01	0.99	0.98	0.99	0.99	0.99	0.99	1.04	1.38			3.02	2.96	3.02
	velocity	5.34	8.96	9.05	9.23	9.33	9.23	9.23	9.23	9.23	9.23	8.79	10.16	8.49		9.08	9.26	9.08
Trammell, Terrence (USA) (1971)	time	2.52	3.54	4.53	5.54	6.52	7.52	8.53	9.54	10.56	11.59			13.00	6 / 2			
reaction time	0.157	interval		1.02	0.99	1.01	0.98	1.00	1.01	1.01	1.02	1.03	1.41			3.02	2.99	3.06
	velocity	5.44	8.96	9.23	9.05	9.33	9.14	9.05	9.05	8.96	8.87	9.94	8.46			9.08	9.17	8.96
Payne, David (USA) (1982)	time	2.55	3.64	4.65	5.69	6.71	7.72	8.74	9.75	10.79	11.86			13.25	5 / 3			
reaction time	0.161	interval		1.09	1.01	1.04	1.02	1.01	1.02	1.01	1.04	1.07	1.39			3.14	3.05	3.12
	velocity	5.38	8.39	9.05	8.79	8.96	9.05	8.96	9.05	8.79	8.54	10.09	8.30			8.73	8.99	8.79
Merritt, Aries (USA) (1985)	time	2.59	3.65	4.70	5.74	6.74	7.76	8.79	9.83	10.84	11.89			13.27	1 / 4			
reaction time	0.136	interval		1.06	1.05	1.04	1.00	1.02	1.03	1.04	1.01	1.05	1.38			3.15	3.05	3.10
	velocity	5.30	8.62	8.70	8.79	9.14	8.96	8.87	8.79	9.05	8.70	10.16	8.29			8.70	8.99	8.85
Hicks, Antwon (USA) (1983)	time	2.57	3.64	4.69	5.74	6.76	7.76	8.78	9.81	10.85	11.91			13.33	3 / 5			
reaction time	0.170	interval	7 steps	1.07	1.05	1.05	1.02	1.00	1.02	1.03	1.04	1.06	1.42			3.17	3.04	3.13
	velocity	5.34	8.54	8.70	8.70	8.96	9.14	8.96	8.87	8.79	8.62	9.87	8.25			8.65	9.02	8.76
Brunson, Andrew (USA) (1986)	time	2.57	3.64	4.67	5.72	6.74	7.77	8.81	9.86	10.91	12.00			13.43	7 / 6			
reaction time	0.150	interval		1.07	1.03	1.05	1.02	1.03	1.04	1.05	1.05	1.09	1.43			3.15	3.09	3.19
	velocity	5.34	8.54	8.87	8.70	8.96	8.87	8.79	8.70	8.70	8.39	9.80	8.19			8.70	8.87	8.60
Akins, Ty (USA) (1986)	time	2.57	3.67	4.74	5.79	6.84	7.89	8.94	9.98	11.03	12.19			13.82	8 / 7			
reaction time	0.175	interval		1.10	1.07	1.05	1.05	1.05	1.04	1.05	1.16	1.63				3.22	3.15	3.25
	velocity	5.34	8.31	8.54	8.70	8.70	8.70	8.70	8.79	8.70	7.88	8.60	7.96			8.52	8.70	8.44
Moore, Anwar (USA) (1979)	time	2.52	3.58	4.59	5.59	6.57	7.57	8.57	9.59	10.60	11.73			16.64	2 / 8			
reaction time	0.171	interval		1.06	1.01	1.00	0.98	1.00	1.00	1.02	1.01	1.13	4.91			3.07	2.98	3.16
	velocity	5.44	8.62	9.05	9.14	9.33	9.14	9.14	8.96	9.05	8.09	2.86	6.61			8.93	9.20	8.68

2008 Windsprint (Sundsvall, SWE)

FINAL

	date	06-Jul-08	wind	0.5 m/s	(2008) - www.ssg.se/en/Windsprint/Split-times/													
			H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
Townsend, Fredrick (USA) (1981)	time		3.72				6.78			10.04				13.63	4 / 1			
reaction time	interval						3.06			3.26			3.59					
	velocity		6.15				8.96			8.41			9.00	8.07				
Kronberg, Robert (SWE) (1976)	time		3.76				6.85			10.06				13.69	5 / 2			
reaction time	interval						3.09			3.21			3.63					
	velocity		6.08				8.87			8.54			8.90	8.04				
Blaschke, Joakim (SWE) (1977)	time		3.84				7.22			10.73				14.64	3 / 3			
reaction time	interval						3.38			3.51			3.91					
	velocity		5.95				8.11			7.81			8.26	7.51				
Lindén, Samuel (SWE) (1986)	time		3.97				7.42			10.93				14.97	2 / 4			
reaction time	interval						3.45			3.51			4.04					
	velocity		5.76				7.95			7.81			8.00	7.35				

2008 Nambu Chuhei Memorial (Harodate, JPN)

FINAL

	date	06-Jul-08	wind	0.5 m/s	Akirabun (2009) - race analysis of 2008 men's and women's 100m and 110m & 100m hurdles													
			H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
Tanonaka, Tasuku (JPN) (1978)	time	2.44	3.52	4.57	5.62	6.67	7.74	8.81	9.88	10.95	12.04			13.64	1 / 1			

Ohashi, Yuji (JPN) (1983)	time	2.48	3.57	4.62	5.67	6.72	7.76	8.85	9.95	11.04	12.14		13.81	/ 2			
	reaction time	interval	1.09	1.05	1.05	1.05	1.04	1.09	1.10	1.09	1.10	1.67			3.19	3.18	3.29
	velocity	5.53	8.39	8.70	8.70	8.70	8.79	8.39	8.31	8.39	8.31	8.40	7.97		8.60	8.62	8.33
Nomoto, Hideki (JPN) (1984)	time	2.49	3.60	4.67	5.73	6.81	7.90	8.99	10.10	11.22	12.35		14.04	/ 4			
	reaction time	interval	1.11	1.07	1.06	1.08	1.09	1.09	1.11	1.12	1.13	1.69			3.24	3.26	3.36
	velocity	5.51	8.23	8.54	8.62	8.46	8.39	8.39	8.23	8.16	8.09	8.30	7.83		8.46	8.41	8.16

2008 Spanish National Indoor Championships (Valencia, ESP)**FINAL**

date 24-Feb-08

Frutos (2008) - 7.52 record de españa de 60mv masculino de jackson quiñóñez: estudio biomecánico

		H1	H2	H3	H4	H5						Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
Quiñóñez, Jackson (ESP) (1980)	time	2.56	3.62	4.62	5.62	6.64							7.52	/ 1			
	reaction time	interval	1.06	1.00	1.00	1.02						0.88	NR		3.06		
	velocity	5.36	8.62	9.14	9.14	8.96						11.05	7.98		8.96		

2007 Seiko Super Grand Prix (Yokohama, JPN)**FINAL**

date 30-Sep-07

wind -1.6 m/s

Akirabun (2008) - 2007 speed analysis of the men's & women's 100m, 100m hurdles, & 110m hurdles

		H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
Naito, Masato (JPN) (1980)	time	2.48	3.53	4.58	5.63	6.68	7.76	8.86	9.95	11.06	12.14		13.83	/ 3			
	reaction time	interval	1.05	1.05	1.05	1.05	1.08	1.10	1.09	1.11	1.08	1.69			3.15	3.23	3.28
	velocity	5.53	8.70	8.70	8.70	8.70	8.46	8.31	8.39	8.23	8.46	8.30	7.95		8.70	8.49	8.36
Moore, Anwar (USA) (1979)	time	2.31	3.37	4.42	5.48	6.56	7.65	8.79	9.93	11.09	12.23		14.08	/ 6			
	reaction time	interval	1.06	1.05	1.06	1.08	1.09	1.14	1.14	1.16	1.14	1.85			3.17	3.31	3.44
	velocity	5.94	8.62	8.70	8.62	8.46	8.39	8.02	8.02	7.88	8.02	7.58	7.81		8.65	8.28	7.97

2007 Golden Grand Prix (Shanghai, CHN)**FINAL**

date 28-Sep-07

wind -0.4 m/s

Ma (2008) - analysis of Liu Xiang's time characteristic in men's 110m hurdles during golden grand prix in Shanghai

		H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10	
Robles, Dayron (CUB) (1986)	time	2.27	3.24	4.26	5.33	6.31	7.30	8.24	9.17	10.28	11.35		13.01	/ 1				
	reaction time	0.186	interval	7 steps	0.97	1.02	1.07	0.98	0.99	0.94	0.93	1.11	1.07	1.66		3.06	2.91	3.11
		velocity	6.04	9.42	8.96	8.54	9.33	9.23	9.72	9.83	8.23	8.54	8.45	8.46		8.96	9.42	8.82
Liu Xiang (CHN) (1983)	time	2.26	3.22	4.36	5.44	6.40	7.42	8.29	9.32	10.42	11.46		13.21	/ 3				
	reaction time	0.164	interval		0.96	1.14	1.08	0.96	1.02	0.87	1.03	1.10	1.04	1.75		3.18	2.85	3.17
		velocity	6.07	9.52	8.02	8.46	9.52	8.96	10.51	8.87	8.31	8.79	8.01	8.33		8.62	9.62	8.65

2007 IAAF World Athletic Final (Stuttgart, GER)**FINAL**

date 23-Sep-07

wind 0.0 m/s

Graubner (2007) - <http://www.fgs.uni-halle.de>

		H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10	
Robles, Dayron (CUB) (1986)	time	2.51	3.55	4.54	5.53	6.53	7.51	8.51	9.51	10.53	11.57		12.92	4 / 1				
	reaction time	0.158	interval	7 steps	1.04	0.99	0.99	1.00	0.98	1.00	1.00	1.02	1.04	1.35	CR / AR	3.02	2.98	3.06
		velocity	5.47	8.79	9.23	9.23	9.14	9.33	9.14	9.14	8.96	8.79	10.39	8.51		9.08	9.20	8.96
Payne, David (USA) (1982)	time	2.57	3.61	4.64	5.65	6.67	7.68	8.69	9.69	10.71	11.74		13.08	5 / 2				
	reaction time	0.146	interval		1.04	1.03	1.01	1.02	1.01	1.00	1.02	1.03	1.34		3.08	3.04	3.05	
		velocity	5.34	8.79	8.87	9.05	8.96	9.05	9.05	9.14	8.96	8.87	10.46	8.41		8.90	9.02	8.99
Trammell, Terrence (USA) (1971)	time	2.53	3.57	4.59	5.58	6.59	7.61	8.63	9.67	10.71	11.75		13.15	3 / 3				
	reaction time	0.140	interval		1.04	1.02	0.99	1.01	1.02	1.02	1.04	1.04	1.40		3.05	3.05	3.12	
		velocity	5.42	8.79	8.96	9.23	9.05	8.96	8.96	8.79	8.79	8.79	10.01	8.37		8.99	8.99	8.79
Moore, Anwar (USA) (1979)	time	2.59	3.65	4.66	5.67	6.67	7.67	8.68	9.70	10.73	11.80		13.18	7 / 4				
	reaction time	0.161	interval		1.06	1.01	1.01	1.00	1.00	1.01	1.02	1.03	1.07	1.38		3.08	3.01	3.12
		velocity	5.30	8.62	9.05	9.05	9.14	9.14	9.05	8.96	8.87	8.54	10.16	8.35		8.90	9.11	8.79
Doucouré, Ladji (FRA) (1983)	time	2.61	3.65	4.67	5.69	6.70	7.72	8.74	9.77	10.81	11.87		13.27	1 / 5				
	reaction time	0.160	interval		1.04	1.02	1.02	1.01	1.02	1.02	1.03	1.04	1.06	1.40		3.08	3.05	3.13
		velocity	5.26	8.79	8.96	8.96	9.05	8.96	8.96	8.87	8.79	8.62	10.01	8.29		8.90	8.99	8.76
Johnson, Allen (USA) (1971)	time	2.59	3.63	4.66	5.68	6.71	7.74	8.77	9.81	10.87	11.95		13.36	2 / 6				
	reaction time	0.158	interval		1.04	1.03	1.02	1.03	1.03	1.04	1.06	1.08	1.41		3.09	3.09	3.18	
		velocity	5.30	8.79	8.87	8.96	8.87	8.87	8.87	8.79	8.62	8.46	9.94	8.23		8.87	8.87	8.62
Demidyuk, Sergiy (UKR) (1982)	time	2.68	3.72	4.76	5.79	6.81	7.83	8.86	9.90	10.95	12.01		13.37	8 / 7				
	reaction time	0.214	interval		1.04	1.04	1.03	1.02	1.03	1.04	1.05	1.06	1.36		3.11	3.07	3.15	
		velocity	5.12	8.79	8.79	8.87	8.96	8.96	8.87	8.79	8.70	8.62	10.31	8.23		8.82	8.93	8.70
Wilson, Ryan (USA) (1980)	time	2.67	3.71	4.73	5.74	6.75	7.75	8.75	9.77				dnf	6 / --				
	reaction time	0.142	interval		1.04	1.02	1.01	1.00	1.00	1.02					3.07	3.01		
		velocity	5.14	8.79	8.96	9.05	9.05	9.14	9.14	8.96						8.93	9.11	

2007 IAAF World Championships (Osaka, JPN)**FINAL**

date 31-Aug-07

wind 1.7 m/s

Xiao (2014) - development process of the characterisits about 110m hurdle training in China

Ae (2008) - IAAF biomechanics team - a booklet of biomechanics data

H1 H2 H3 H4 H5 H6 H7 H8 H9 H10 Run-In Official Time Lane / Place H1-H4 H4-H7 H7-H10

Liu Xiang (CHN) (1983)	time	2.57	3.61	4.58	5.58	6.57	7.55	8.55	9.55	10.53	11.57		12.95	9 / 1			Ae (2008)
	reaction time	0.161		1.04	0.97	1.00	0.99	0.98	1.00	1.00	0.98	1.04	1.38		3.01	2.97	3.02
	velocity		5.34	8.79	9.42	9.14	9.23	9.33	9.14	9.14	9.33	8.79	10.16	8.49	9.11	9.23	9.08
Trammell, Terrence (USA) (1971)	time	2.53	3.53	4.52	5.53	6.53	7.52	8.53	9.55	10.57	11.60		12.99	5 / 2			Ae (2008)
	reaction time	0.205		1.00	0.99	1.01	1.00	0.99	1.01	1.02	1.02	1.03	1.39		3.00	3.00	3.07
	velocity		5.42	9.14	9.23	9.05	9.14	9.23	9.05	8.96	8.96	8.87	10.09	8.47	9.14	9.14	8.93
Payne, David (USA) (1982)	time	2.52	3.55	4.57	5.58	6.60	7.60	8.62	9.62	10.63	11.67		13.02	7 / 3			Ae (2008)
	reaction time	0.141		1.03	1.02	1.01	1.02	1.00	1.02	1.00	1.01	1.04	1.35	PB	3.06	3.04	3.05
	velocity		5.44	8.87	8.96	9.05	8.96	9.14	8.96	9.14	9.05	8.79	10.39	8.45	8.96	9.02	8.99
Robles, Dayron (CUB) (1986)	0.164	no information available 7 steps											13.15	4 / 4			
Shi Dongpeng (CHN) (1984)	time	2.59	3.62	4.63	5.62	6.60	7.57	8.57	9.57	10.59	11.62		13.19	6 / 5			Xiao (2014)
	reaction time	0.150		1.03	1.01	0.99	0.98	0.97	1.00	1.00	1.02	1.03	1.57	PB	3.03	2.95	3.05
	velocity		5.30	8.87	9.05	9.23	9.33	9.42	9.14	9.14	8.96	8.87	8.93	8.34	9.05	9.29	8.99
Demydyuk, Serhly (UKR) (1982)	0.136	no information available											NR	13.22	3 / 6		
Quiñónez, Jackson (ESP) (1980)	0.172	no information available											NR	13.33	2 / 7		
Wignall, Maurice (JAM) (1976)	0.144	no information available												13.39	8 / 8		

Semi-Final 3

date 30-Aug-07		wind -1.0 m/s		Shibayama (2008) - analysis of racing patterns in the men's 110m & women's 100m sprint hurdles													
		H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
Liu Xiang (CHN) (1983)	time	2.63	3.65	4.67	5.67	6.69	7.69	8.71	9.74	10.77	11.82		13.25	6 / 2			
	reaction time	0.158		1.02	1.02	1.00	1.02	1.00	1.02	1.03	1.03	1.05	1.43		3.04	3.04	3.11
	velocity		5.22	8.96	8.96	9.14	8.96	9.14	8.96	8.87	8.87	8.70	9.80	8.30	9.02	9.02	8.82

Semi-Final 2

date 30-Aug-07		wind 0.4 m/s		Shibayama (2008) - analysis of racing patterns in the men's 110m & women's 100m sprint hurdles													
		H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
Trammell, Terrence (USA) (1971)	time	2.55	3.58	4.60	5.62	6.62	7.64	8.64	9.67	10.72	11.79		13.23	7 / 1			
	reaction time	0.223		1.03	1.02	1.02	1.00	1.02	1.00	1.03	1.05	1.07	1.44		3.07	3.02	3.15
	velocity		5.38	8.87	8.96	8.96	9.14	8.96	9.14	8.87	8.70	8.54	9.74	8.31	8.93	9.08	8.70

Semi-Final 1

date 30-Aug-07		wind -0.4 m/s		Shibayama (2008) - analysis of racing patterns in the men's 110m & women's 100m sprint hurdles													
		H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
Payne, David (USA) (1982)	time	2.61	3.66	4.68	5.68	6.68	7.68	8.71	9.73	10.76	11.83		13.19	7 / 1			
	reaction time	0.166		1.05	1.02	1.00	1.00	1.03	1.02	1.03	1.07	1.36			3.07	3.03	3.12
	velocity		5.26	8.70	8.96	9.14	9.14	8.87	8.96	8.87	8.54	10.31	8.34		8.93	9.05	8.79

Heat 5

date 29-Aug-07		wind 0.6 m/s		Shibayama (2008) - analysis of racing patterns in the men's 110m & women's 100m sprint hurdles													
		H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
Payne, David (USA) (1982)	time	2.62	3.69	4.71	5.71	6.73	7.75	8.77	9.80	10.83	11.86		13.27	3 / 2			
	reaction time	0.163		1.07	1.02	1.00	1.02	1.02	1.03	1.03	1.03	1.41			3.09	3.06	3.09
	velocity		5.24	8.54	8.96	9.14	8.96	8.96	8.87	8.87	8.87	9.94	8.29		8.87	8.96	8.87

Heat 4

date 29-Aug-07		wind 0.6 m/s		Shibayama (2008) - analysis of racing patterns in the men's 110m & women's 100m sprint hurdles													
		H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
Tanonaka, Tasuku (JPN) (1978)	time	2.64	3.69	4.76	5.81	6.84	7.89	8.94	10.03	11.11	12.19		13.61	5 / 3			
	reaction time	0.140		1.05	1.07	1.03	1.05	1.05	1.09	1.08	1.08	1.42			3.17	3.13	3.25
	velocity		5.20	8.70	8.54	8.70	8.87	8.70	8.39	8.46	8.46	9.87	8.08		8.65	8.76	8.44

Heat 3

date 29-Aug-07		wind 0.2 m/s		Shibayama (2008) - analysis of racing patterns in the men's 110m & women's 100m sprint hurdles													
		H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
Naito, Masato (JPN) (1980)	time	2.62	3.68	4.70	5.73	6.73	7.77	8.83	9.90	10.98	12.08		13.54	9 / 3			
	reaction time	0.164		1.06	1.02	1.03	1.04	1.06	1.07	1.08	1.10	1.46			3.11	3.10	3.25
	velocity		5.24	8.62	8.96	8.87	9.14	8.79	8.62	8.54	8.46	8.31	9.60	8.12	8.82	8.85	8.44

Heat 2

date 29-Aug-07		wind -0.6 m/s		Shibayama (2008) - analysis of racing patterns in the men's 110m & women's 100m sprint hurdles													
		H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
Liu Xiang (CHN) (1983)	time	2.63	3.68	4.71	5.74	6.77	7.79	8.82	9.84	10.89	11.96		13.36	4 / 1			
	reaction time	0.141		1.05	1.03	1.03	1.02	1.03	1.02	1.05	1.07	1.40			3.11	3.08	3.14
	velocity		5.22	8.70	8.87	8.87	8.87	8.96	8.87	8.96	8.70	8.54	10.01	8.23	8.82	8.90	8.73

Heat 1

date 29-Aug-07		wind 0.4 m/s		Shibayama (2008) - analysis of racing patterns in the men's 110m & women's 100m sprint hurdles													
		H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
Trammell, Terrence (USA) (1971)	time	2.63	3.66	4.73	5.75	6.77	7.79	8.84	9.91	10.96	12.01		13.40	5 / 1			
	reaction time	0.223		1.03	1.07	1.02	1.02	1.05	1.07	1.05	1.05	1.39			3.12	3.09	3.17
	velocity		5.22	8.87	8.54	8.96	8.96	8.70	8.54	8.70	8.70	10.09	8.21		8.79	8.87	8.65

2007 Windsprint (Sundsvall, SWE)

FINAL

date 24-Jul-07		wind 1.3 m/s		(2007) - www.ssg.se/en/Windsprint/Split-times/													
		H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
Kronberg, Robert (SWE) (1976)	time		3.68			6.82			10.01				13.65	1 / 1			
	reaction time					3.14			3.19			3.64					

	velocity	6.21	8.73	8.60	8.87	8.06	
Thomas, Chris (USA) (1981)	time	3.73	6.92	10.19		13.95	/ 2
reaction time	interval		3.19	3.27	3.76		
	velocity	6.13	8.60	8.39	8.59	7.89	
Andrews, Ron (USA) (1979)	time	3.74	6.98	10.22		13.97	/ 3
reaction time	interval		3.24	3.24	3.75		
	velocity	6.11	8.46	8.46	8.61	7.87	
Askros, Rafael (SWE) (1981)	time	3.76	7.19	10.59		14.46	/ 4
reaction time	interval		3.43	3.40	3.87		
	velocity	6.08	7.99	8.06	8.35	7.61	
Acson, Dennis (SWE) (1981)	time	3.80	7.21	10.63		14.62	/ 5
reaction time	interval		3.41	3.42	3.99		
	velocity	6.02	8.04	8.02	8.10	7.52	
Blaschke, Joakim (SWE) (1977)	time	3.87	7.23	10.68		14.79	/ 6
reaction time	interval		3.36	3.45	4.11		
	velocity	5.91	8.16	7.95	7.86	7.44	
Sjöberg, Christian (SWE) (1984)	time	3.90	7.29	10.72		14.84	/ 7
reaction time	interval		3.39	3.43	4.12		
	velocity	5.86	8.09	7.99	7.84	7.41	
Olsson, Carl-Johan (SWE) (1981)	time	3.95	7.48	11.07		15.33	/ 8
reaction time	interval		3.53	3.59	4.26		
	velocity	5.79	7.77	7.64	7.58	7.18	

2007 European Cup (Munich, GER)

FINAL

		date	wind 0.4 m/s										Graubner (2007) - http://www.fgs.uni-halle.de					
			H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
Doucouré, Ladji (FRA) (1983)	time		2.57	3.61	4.63	5.63	6.64	7.65	8.67	9.70	10.77	11.87		13.35	/ 1			
	reaction time	0.152		1.04	1.02	1.00	1.01	1.01	1.02	1.03	1.07	1.10	1.48			3.06	3.04	3.20
	velocity		5.34	8.79	8.96	9.14	9.05	9.05	8.96	8.87	8.54	8.31	9.47	8.24		8.96	9.02	8.57
Turner, Andy (GBR) (1980)	time		2.62	3.66	4.69	5.73	6.76	7.80	8.85	9.91	10.98	12.07		13.48	/ 2			
	reaction time	0.156		1.04	1.03	1.04	1.03	1.04	1.05	1.06	1.07	1.09	1.41			3.11	3.12	3.22
	velocity		5.24	8.79	8.87	8.79	8.87	8.79	8.70	8.62	8.54	8.39	9.94	8.16		8.82	8.79	8.52
Blaschek, Thomas (GER) (1981)	time		2.63	3.67	4.71	5.73	6.75	7.78	8.83	9.89	10.97	12.07		13.51	/ 3			
	reaction time	0.133		1.04	1.04	1.02	1.02	1.03	1.05	1.06	1.08	1.10	1.44			3.10	3.10	3.24
	velocity		5.22	8.79	8.79	8.96	8.96	8.87	8.70	8.62	8.46	8.31	9.74	8.14		8.85	8.85	8.46
Peremota, Igor (RUS) (1981)	time		2.61	3.63	4.67	5.69	6.73	7.77	8.83	9.93	11.03	12.15		13.61	/ 4			
	reaction time	0.132		1.02	1.04	1.02	1.04	1.04	1.06	1.10	1.10	1.12	1.46			3.08	3.14	3.32
	velocity		5.26	8.96	8.79	8.96	8.79	8.79	8.62	8.31	8.31	8.16	9.60	8.08		8.90	8.73	8.26
Douvalidis, Konstadinos (GRE)	time		2.63	3.70	4.74	5.79	6.83	7.91	8.97	10.03	11.10	12.20		13.63	/ 5			
	reaction time	0.141		1.07	1.04	1.05	1.04	1.08	1.06	1.06	1.07	1.10	1.43			3.16	3.18	3.23
	velocity		5.22	8.54	8.79	8.70	8.79	8.46	8.62	8.62	8.54	8.31	9.80	8.07		8.68	8.62	8.49
Deghelt, Adrien (BEL) (1985)	time		2.65	3.71	4.75	5.80	6.85	7.91	8.96	10.03	11.13	12.23		13.67	/ 6			
	reaction time	0.149		1.06	1.04	1.05	1.05	1.06	1.05	1.07	1.10	1.10	1.44	PB		3.15	3.16	3.27
	velocity		5.18	8.62	8.79	8.70	8.70	8.62	8.70	8.54	8.31	8.31	9.74	8.05		8.70	8.68	8.39
Kubaszewski, Mariusz (POL) (1981)	time		2.64	3.73	4.79	5.83	6.91	7.95	9.03	10.08	11.17	12.29		13.74	/ 7			
	reaction time	0.127		1.09	1.06	1.04	1.08	1.04	1.08	1.05	1.09	1.12	1.45			3.19	3.20	3.26
	velocity		5.20	8.39	8.62	8.79	8.46	8.79	8.46	8.70	8.39	8.16	9.67	8.01		8.60	8.57	8.41
Vahnyakov, Oleksandr (UKR) (1981)	time		2.80	3.93	5.02	6.08	7.15	8.24	9.35	10.47	11.57	12.69		14.21	/ 8			
	reaction time	0.243		1.13	1.09	1.06	1.07	1.09	1.11	1.12	1.10	1.12	1.52			3.28	3.27	3.34
	velocity		4.90	8.09	8.39	8.62	8.54	8.39	8.23	8.16	8.31	8.16	9.22	7.74		8.36	8.39	8.21

2007 Athletissima Meeting (Lausanne, SUI)

Race 2

date 10-Jul-07		wind 0.0 m/s										Lu (2008) - 110m hurdle technical characteristics of Shi Dongpeng							
		H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10		
Shi Dongpeng (CHN) (1984)	time	2.63	3.71	4.76	5.79	6.80	7.80	8.82	9.85	10.90	11.97		13.42	/ 8					
	reaction time		interval	1.08	1.05	1.03	1.01	1.00	1.02	1.03	1.05	1.07	1.45			3.16	3.03	3.15	
		velocity	5.22	8.46	8.70	8.87	9.05	9.14	8.96	8.87	8.70	8.54	9.67	8.20		8.68	9.05	8.70	

2007 USATF National Championship (Indianapolis, IN)

FINAL

USATF Men's Hurdle Development (2007)																	
date	24-Jun-07	wind 0.5 m/s										Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
		H1	H2	H3	H4	H5	H6	H7	H8	H9	H10						
Trammell, Terrence (USA) (1971)	time	2.55	3.59	4.59	5.59	6.59	7.59	8.59	9.59	10.61	11.70		13.08	3 / 1			

reaction time	interval	1.04	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.02	1.09	1.38		3.04	3.00	3.11
	velocity	5.38	8.79	9.14	9.14	9.14	9.14	9.14	9.14	8.96	8.39	10.16	8.41	9.02	9.14	8.82
Arnold, Dominique (USA) (197	time	2.62	3.67	4.70	5.69	6.69	7.69	8.69	9.69	10.71	11.73		13.17	5 / 2		
reaction time	interval		1.05	1.03	0.99	1.00	1.00	1.00	1.00	1.02	1.02	1.44		3.07	3.00	3.04
	velocity	5.24	8.70	8.87	9.23	9.14	9.14	9.14	9.14	8.96	8.96	9.74	8.35	8.93	9.14	9.02
Oliver, David (USA) (1982)	time	2.65	3.67	4.72	5.72	6.72	7.72	8.74	9.73	10.76	11.78		13.18	6 / 3		
reaction time	interval		1.02	1.05	1.00	1.00	1.00	1.02	0.99	1.03	1.02	1.40		3.07	3.02	3.04
	velocity	5.18	8.96	8.70	9.14	9.14	9.14	8.96	9.23	8.87	8.96	10.01	8.35	8.93	9.08	9.02
Payne, David (USA) (1982)	time	2.59	3.65	4.65	5.69	6.71	7.72	8.74	9.76	10.81	11.86		13.22	4 / 4		
reaction time	interval		1.06	1.00	1.04	1.02	1.01	1.02	1.02	1.05	1.05	1.36		3.10	3.05	3.12
	velocity	5.30	8.62	9.14	8.79	8.96	9.05	8.96	8.96	8.70	8.70	10.31	8.32	8.85	8.99	8.79
Moore, Anwar (USA) (1979)	time	2.62	3.67	4.70	5.72	6.72	7.71	8.70	9.74	10.78	11.85		13.23	8 / 5		
reaction time	interval		1.05	1.03	1.02	1.00	0.99	0.99	1.04	1.04	1.07	1.38		3.10	2.98	3.15
	velocity	5.24	8.70	8.87	8.96	9.14	9.23	9.23	8.79	8.79	8.54	10.16	8.31	8.85	9.20	8.70
Merritt, Aries (USA) (1985)	time	2.65	3.72	4.74	5.76	6.76	7.77	8.76	9.79	10.83	11.85		13.24	7 / 6		
reaction time	interval		1.07	1.02	1.02	1.00	1.01	0.99	1.03	1.04	1.02	1.39		3.11	3.00	3.09
	velocity	5.18	8.54	8.96	8.96	9.14	9.05	9.23	8.87	8.79	8.96	10.09	8.31	8.82	9.14	8.87
Johnson, Allen (USA) (1971)	time	2.59	3.62	4.64	5.64	6.67	7.69	8.76	9.78	10.93	12.03		13.60	2 / 7		
reaction time	interval		1.03	1.02	1.00	1.03	1.02	1.07	1.02	1.15	1.10	1.57		3.05	3.12	3.27
	velocity	5.30	8.87	8.96	9.14	8.87	8.96	8.54	8.96	7.95	8.31	8.93	8.09	8.99	8.79	8.39
Wilson, Ryan (USA) (1980)	time	2.62	3.79	4.80	5.91	7.02	8.41						dnf	1 / --		
reaction time	interval		1.17	1.01	1.11	1.11	1.39							3.29		
	velocity	5.24	7.81	9.05	8.23	8.23	6.58							8.33		

Semi-Final 2

Semi-Final 2		date	wind -0.9 m/s										USATF Men's Hurdle Development (2007)						
			H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10	
Trammell, Terrence (USA) (1971)	time	2.59	3.64	4.65	5.69	6.71	7.72	8.73	9.78	10.82	11.86			13.26	2 / 1				
	reaction time			1.05	1.01	1.04	1.02	1.01	1.01	1.05	1.04	1.04	1.40			3.10	3.04	3.13	
	interval velocity	5.30	8.70	9.05	8.79	8.96	9.05	9.05	8.70	8.79	8.79	10.01	8.30			8.85	9.02	8.76	
Oliver, David (USA) (1982)	time	2.65	3.70	4.74	5.74	6.76	7.76	8.79	9.81	10.88	11.93			13.29	4 / 2				
	reaction time			1.05	1.04	1.00	1.02	1.00	1.03	1.02	1.07	1.05	1.36			3.09	3.05	3.14	
	interval velocity	5.18	8.70	8.79	9.14	8.96	9.14	8.87	8.96	8.54	8.70	10.31	8.28			8.87	8.99	8.73	
Wilson, Ryan (USA) (1980)	time	2.65	3.70	4.74	5.74	6.72	7.74	8.78	9.81	10.86	11.91			13.33	3 / 3				
	reaction time			1.05	1.04	1.00	0.98	1.02	1.04	1.03	1.05	1.05	1.42			3.09	3.04	3.13	
	interval velocity	5.18	8.70	8.79	9.14	9.33	8.96	8.79	8.87	8.70	8.70	9.87	8.25			8.87	9.02	8.76	
Moore, Anwar (USA) (1979)	time	2.62	3.67	4.69	5.72	6.77	7.81	8.86	9.88	10.92	11.98			13.35	5 / 4				
	reaction time			1.05	1.02	1.03	1.05	1.04	1.05	1.02	1.04	1.06	1.37			3.10	3.14	3.12	
	interval velocity	5.24	8.70	8.96	8.87	8.70	8.79	8.70	8.96	8.79	8.62	10.23	8.24			8.85	8.73	8.79	
Faulk, Dexter (USA) (1984)	time	2.65	3.69	4.72	5.76	6.81	7.86	8.90	9.99	11.04	12.12			13.52	1 / 5				
	reaction time			1.04	1.03	1.04	1.05	1.05	1.04	1.09	1.05	1.08	1.40			3.11	3.14	3.22	
	interval velocity	5.18	8.79	8.87	8.79	8.70	8.70	8.79	8.39	8.70	8.46	10.01	8.14			8.82	8.73	8.52	
Yarbrough, Linnie (USA) (1982)	time	2.70	3.75	4.79	5.82	6.86	7.89	8.94	9.99	11.06	12.11			13.55	6 / 6				
	reaction time			1.05	1.04	1.03	1.04	1.03	1.05	1.05	1.07	1.05	1.44			3.12	3.12	3.17	
	interval velocity	5.08	8.70	8.79	8.87	8.79	8.87	8.70	8.70	8.54	8.70	9.74	8.12			8.79	8.79	8.65	
Yarbrough, John (USA) (1985)	time	2.69	3.75	4.80	5.84	6.87	7.92	8.98	10.02	11.08	12.16			13.61	7 / 7				
	reaction time			1.06	1.05	1.04	1.03	1.05	1.06	1.04	1.06	1.08	1.45			3.15	3.14	3.18	
	interval velocity	5.10	8.62	8.70	8.79	8.87	8.70	8.62	8.79	8.62	8.46	9.67	8.08			8.70	8.73	8.62	
Odom, Marlon (USA) (1982)	time	2.70	3.75	4.82	5.89	6.92	7.96	9.03	10.11	11.21	12.31			13.79	8 / 8				
	reaction time			1.05	1.07	1.07	1.03	1.04	1.07	1.08	1.10	1.10	1.48			3.19	3.14	3.28	
	interval velocity	5.08	8.70	8.54	8.54	8.87	8.79	8.54	8.46	8.31	8.31	9.47	7.98			8.60	8.73	8.36	

Semi-Final 1

emi-Final 1		date	wind -0.6 m/s										USATF Men's Hurdle Development (2007)						
			H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10	
Payne, David (USA) (1982)	time	2.57	3.64	4.65	5.69	6.69	7.72	8.76	9.78	10.82	11.88			13.28	4 / 1				
	reaction time	interval		1.07	1.01	1.04	1.00	1.03	1.04	1.02	1.04	1.06	1.40			3.12	3.07	3.12	
		velocity	5.34	8.54	9.05	8.79	9.14	8.87	8.79	8.96	8.79	8.62	10.01	8.28		8.79	8.93	8.79	
Arnold, Dominique (USA) (197	time	2.62	3.67	4.69	5.71	6.72	7.72	8.74	9.79	10.81	11.88			13.35	3 / 2				
	reaction time	interval		1.05	1.02	1.02	1.01	1.00	1.02	1.05	1.02	1.07	1.47			3.09	3.03	3.14	
		velocity	5.24	8.70	8.96	8.96	9.05	9.14	8.96	8.70	8.96	8.54	9.54	8.24		8.87	9.05	8.73	
Merritt, Aries (USA) (1985)	time	2.62	3.69	4.70	5.76	6.79	7.79	8.83	9.88	10.89	11.95			13.37	5 / 3				
	reaction time	interval		1.07	1.01	1.06	1.03	1.00	1.04	1.05	1.01	1.06	1.42			3.14	3.07	3.12	

	velocity	5.24	8.54	9.05	8.62	8.87	9.14	8.79	8.70	9.05	8.62	9.87	8.23		8.73	8.93	8.79
Johnson, Allen (USA) (1971)	time	2.57	3.64	4.64	5.67	6.69	7.71	8.76	9.81	10.86	11.95		13.42	8 / 4			
	reaction time		1.07	1.00	1.03	1.02	1.02	1.05	1.05	1.05	1.09	1.47			3.10	3.09	3.19
	interval																
	velocity	5.34	8.54	9.14	8.87	8.96	8.96	8.70	8.70	8.70	8.39	9.54	8.20		8.85	8.87	8.60
Brown, Joel (USA) (1980)	time	2.65	3.72	4.77	5.82	6.84	7.91	8.94	9.98	11.03	12.10		13.52	1 / 5			
	reaction time		1.07	1.05	1.05	1.02	1.07	1.03	1.04	1.05	1.07	1.42			3.17	3.12	3.16
	interval																
	velocity	5.18	8.54	8.70	8.70	8.96	8.54	8.87	8.79	8.70	8.54	9.87	8.14		8.65	8.79	8.68
Cooper, Jermaine (USA) (1980)	time	2.60	3.67	4.74	5.79	6.84	7.89	8.98	10.06	11.16	12.26		13.75	2 / 6			
	reaction time		1.07	1.07	1.05	1.05	1.05	1.09	1.08	1.10	1.10	1.49			3.19	3.19	3.28
	interval																
	velocity	5.28	8.54	8.54	8.70	8.70	8.70	8.39	8.46	8.31	8.31	9.41	8.00		8.60	8.60	8.36
Hicks, Antwon (USA) (1983)	time	2.59	3.69	4.79	5.86	6.89	7.96	8.99	10.08	11.14	12.25		13.75	6 / 7			
	reaction time		1.10	1.10	1.07	1.03	1.07	1.03	1.09	1.06	1.11	1.50			3.27	3.13	3.26
	interval																
	velocity	5.30	8.31	8.31	8.54	8.87	8.54	8.87	8.39	8.62	8.23	9.35	8.00		8.39	8.76	8.41
Porter, Jeff (USA) (1985)	time	2.67	3.74	4.79	5.86	6.92	7.99	9.06	10.14	11.24	12.36		13.83	7 / 8			
	reaction time		1.07	1.05	1.07	1.06	1.07	1.07	1.08	1.10	1.12	1.47			3.19	3.20	3.30
	interval																
	velocity	5.14	8.54	8.70	8.54	8.62	8.54	8.54	8.46	8.31	8.16	9.54	7.95		8.60	8.57	8.31

Heat 4

heat 4		date	23-Jun-07	wind	-2.8 m/s										USATF Men's Hurdle Development (2007)						
				H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10		
Oliver, David (USA) (1982)	time		2.64	3.69	4.77	5.81	6.84	7.91	8.96	10.01	11.08	12.15		13.56		7 / 1					
	reaction time			1.05	1.08	1.04	1.03	1.07	1.05	1.05	1.07	1.07	1.41				3.17	3.15	3.19		
	interval																8.65	8.70	8.60		
	velocity		5.20	8.70	8.46	8.79	8.87	8.54	8.70	8.70	8.54	8.54	9.94	8.11							
Yarbrough, Linnie (USA) (1982)	time		2.64	3.70	4.74	5.79	6.87	7.92	8.98	10.04	11.11	12.18		13.61		5 / 2					
	reaction time			1.06	1.04	1.05	1.08	1.05	1.06	1.06	1.07	1.07	1.43				3.15	3.19	3.20		
	interval																8.70	8.60	8.57		
	velocity		5.20	8.62	8.79	8.70	8.46	8.70	8.62	8.62	8.54	8.54	9.80	8.08							
Trammell, Terrence (USA) (1971)	time		2.62	3.72	4.77	5.82	6.89	7.96	9.06	10.13	11.23	12.30		13.78		2 / 3					
	reaction time			1.10	1.05	1.05	1.07	1.07	1.10	1.07	1.10	1.07	1.48				3.20	3.24	3.24		
	interval																8.57	8.46	8.46		
	velocity		5.24	8.31	8.70	8.70	8.54	8.54	8.31	8.54	8.31	8.54	9.47	7.98							
Kelley, Kai (USA) (1986)	time		2.64	3.77	4.84	5.92	7.01	8.10	9.19	10.28	11.39	12.52		14.00		1 / 4					
	reaction time			1.13	1.07	1.08	1.09	1.09	1.09	1.09	1.11	1.13	1.48				3.28	3.27	3.33		
	interval																8.36	8.39	8.23		
	velocity		5.20	8.09	8.54	8.46	8.39	8.39	8.39	8.39	8.23	8.09	9.47	7.86							
Mack, Thomas (USA) (1985)	time		2.64	3.74	4.82	5.91	7.01	8.11	9.20	10.31	11.43	12.56		14.03		4 / 5					
	reaction time			1.10	1.08	1.09	1.10	1.10	1.09	1.11	1.12	1.13	1.47				3.27	3.29	3.36		
	interval																8.39	8.33	8.16		
	velocity		5.20	8.31	8.46	8.39	8.31	8.31	8.39	8.23	8.16	8.09	9.54	7.84							
Akins, Ty (USA) (1986)	time		2.69	3.82	4.90	6.02	7.12	8.19	9.29	10.37	11.49	12.61		14.09		3 / 6					
	reaction time			1.13	1.08	1.12	1.10	1.07	1.10	1.08	1.12	1.12	1.48				3.33	3.27	3.32		
	interval																8.23	8.39	8.26		
	velocity		5.10	8.09	8.46	8.16	8.31	8.54	8.31	8.46	8.16	8.16	9.47	7.81							
Jiles-Tindall, Julius (USA) (1981)	time		2.69	3.79	4.90	5.99	7.08	8.16	9.28	10.38	11.47	12.61		14.10		6 / 7					
	reaction time			1.10	1.11	1.09	1.09	1.08	1.12	1.10	1.09	1.14	1.49				3.30	3.29	3.33		
	interval																8.31	8.33	8.23		
	velocity		5.10	8.31	8.23	8.39	8.39	8.46	8.16	8.31	8.39	8.02	9.41	7.80							
Herring, Aubrey (USA) (1978)	time		2.64	3.79	4.89	5.99	7.07	8.17	9.29	10.41	11.51	12.66		14.27		8 / 8					
	reaction time			1.15	1.10	1.10	1.08	1.10	1.12	1.12	1.10	1.15	1.61				3.35	3.30	3.37		
	interval																8.19	8.31	8.14		
	velocity		5.20	7.95	8.31	8.31	8.46	8.31	8.16	8.16	8.31	7.95	8.71	7.71							

Heat 3

Heat 3

	date	23-Jun-07	wind -3.2 m/s										USATF Men's Hurdle Development (2007)						
			H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10	
Moore, Anwar (USA) (1979)	time		2.60	3.67	4.69	5.76	6.79	7.87	8.93	10.00	11.06	12.15		13.62	3 / 1				
	reaction time			1.07	1.02	1.07	1.03	1.08	1.06	1.07	1.06	1.09	1.47				3.16	3.17	3.22
	velocity		5.28	8.54	8.96	8.54	8.87	8.46	8.62	8.54	8.62	8.39	9.54	8.08			8.68	8.65	8.52
Wilson, Ryan (USA) (1980)	time		2.65	3.70	4.77	5.86	6.89	7.93	8.98	10.04	11.11	12.18		13.64	5 / 2				
	reaction time			1.05	1.07	1.09	1.03	1.04	1.05	1.06	1.07	1.07	1.46				3.21	3.12	3.20
	velocity		5.18	8.70	8.54	8.39	8.87	8.79	8.70	8.62	8.54	8.54	9.60	8.06			8.54	8.79	8.57
Bramlett, Ron (USA) (1979)	time		2.62	3.67	4.75	5.84	6.89	7.97	9.05	10.13	11.21	12.31		13.81	2 / 3				
	reaction time			1.05	1.08	1.09	1.05	1.08	1.08	1.08	1.08	1.10	1.50				3.22	3.21	3.26
	velocity		5.24	8.70	8.46	8.39	8.70	8.46	8.46	8.46	8.46	8.31	9.35	7.97			8.52	8.54	8.41
Hughes, Robby (USA) (1978)	time		2.60	3.75	4.85	5.94	7.02	8.09	9.19	10.28	11.38	12.50		13.96	6 / 4				
	reaction time			1.15	1.10	1.09	1.08	1.07	1.10	1.09	1.10	1.12	1.46				3.34	3.25	3.31
	velocity		5.28	7.95	8.31	8.39	8.46	8.54	8.31	8.39	8.31	8.16	9.60	7.88			8.21	8.44	8.28
Walker, Josh (USA) (1982)	time		2.67	3.79	4.85	5.94	6.99	8.09	9.18	10.26	11.36	12.46		13.97	1 / 5				
	reaction time			1.12	1.06	1.09	1.05	1.10	1.09	1.08	1.10	1.10	1.51				3.27	3.24	3.28
	velocity		5.14	8.16	8.62	8.39	8.70	8.31	8.39	8.46	8.31	8.31	9.28	7.87			8.39	8.46	8.36

Taylor, Logan (USA) (1986)	time	2.64	3.75	4.84	5.94	7.06	8.12	9.22	10.36	11.46	12.63		14.12	7 / 6			
	reaction time	interval	1.11	1.09	1.10	1.12	1.06	1.10	1.14	1.10	1.17	1.49			3.30	3.28	3.41
	velocity	5.20	8.23	8.39	8.31	8.16	8.62	8.31	8.02	8.31	7.81	9.41	7.79		8.31	8.36	8.04
Berger, Dominic (USA) (1986)	time	2.70	3.82	4.90	5.99	7.08	8.16	9.26	10.38	11.48	12.61		14.35	8 / 7			
	reaction time	interval	1.12	1.08	1.09	1.09	1.08	1.10	1.12	1.10	1.13	1.74			3.29	3.27	3.35
	velocity	5.08	8.16	8.46	8.39	8.39	8.46	8.31	8.16	8.31	8.09	8.06	7.67		8.33	8.39	8.19
DuBose, Jacoby (USA) (1982)	time	2.79	3.90	5.01	6.10	7.22	8.34	9.44	10.58	11.71	12.85		14.37	4 / 8			
	reaction time	interval	1.11	1.11	1.09	1.12	1.12	1.10	1.14	1.13	1.14	1.52			3.31	3.34	3.41
	velocity	4.92	8.23	8.23	8.39	8.16	8.16	8.31	8.02	8.09	8.02	9.22	7.65		8.28	8.21	8.04

Heat 2

date 23-Jun-07 wind -1.9 m/s USATF Men's Hurdle Development (2007)

		H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
Arnold, Dominique (USA) (1977)	time	2.74	3.80	4.87	5.91	6.89	7.92	8.96	10.01	11.09	12.24		13.66	6 / 1			
	reaction time	interval	1.06	1.07	1.04	0.98	1.03	1.04	1.05	1.08	1.15	1.42			3.17	3.05	3.28
	velocity	5.01	8.62	8.54	8.79	9.33	8.87	8.79	8.70	8.46	7.95	9.87	8.05		8.65	8.99	8.36
Merritt, Aries (USA) (1985)	time	2.75	3.89	4.92	5.99	7.06	8.11	9.16	10.21	11.26	12.35		13.68	2 / 2			
	reaction time	interval	1.14	1.03	1.07	1.07	1.05	1.05	1.05	1.05	1.09	1.33			3.24	3.17	3.19
	velocity	4.99	8.02	8.87	8.54	8.54	8.70	8.70	8.70	8.70	8.39	10.54	8.04		8.46	8.65	8.60
Yarbrough, John (USA) (1985)	time	2.70	3.77	4.84	5.86	6.89	7.94	9.03	10.09	11.18	12.26		13.69	4 / 3			
	reaction time	interval	1.07	1.07	1.02	1.03	1.05	1.09	1.06	1.09	1.08	1.43			3.16	3.17	3.23
	velocity	5.08	8.54	8.54	8.96	8.87	8.70	8.39	8.62	8.39	8.46	9.80	8.04		8.68	8.65	8.49
Brown, Joel (USA) (1980)	time	2.75	3.80	4.85	5.92	6.96	8.02	9.08	10.14	11.21	12.33		13.70	3 / 4			
	reaction time	interval	1.05	1.05	1.07	1.04	1.06	1.06	1.06	1.07	1.12	1.37			3.17	3.16	3.25
	velocity	4.99	8.70	8.70	8.54	8.79	8.62	8.62	8.62	8.54	8.16	10.23	8.03		8.65	8.68	8.44
Mitchum, Eric (USA) (1984)	time	2.74	3.79	4.87	5.91	6.95	8.02	9.09	10.18	11.28	12.40		13.82	5 / 5			
	reaction time	interval	1.05	1.08	1.04	1.04	1.07	1.07	1.09	1.10	1.12	1.42			3.17	3.18	3.31
	velocity	5.01	8.70	8.46	8.79	8.79	8.54	8.54	8.39	8.31	8.16	9.87	7.96		8.65	8.62	8.28
Thomas, Chris (USA) (1981)	time	2.82	3.90	4.99	6.06	7.12	8.19	9.27	10.34	11.43	12.55		13.90	8 / 6			
	reaction time	interval	1.08	1.09	1.07	1.06	1.07	1.08	1.07	1.09	1.12	1.35			3.24	3.21	3.28
	velocity	4.87	8.46	8.39	8.54	8.62	8.54	8.46	8.54	8.39	8.16	10.39	7.91		8.46	8.54	8.36
McDowell, Jon (USA) (1984)	time	2.84	3.94	5.00	6.11	7.18	8.26	9.36	10.46	11.58	12.71		14.05	1 / 7			
	reaction time	interval	1.10	1.06	1.11	1.07	1.08	1.10	1.10	1.12	1.13	1.34			3.27	3.25	3.35
	velocity	4.83	8.31	8.62	8.23	8.54	8.46	8.31	8.31	8.16	8.09	10.46	7.83		8.39	8.44	8.19
Hon, Brandon (USA) (1979)	time	2.74	3.85	4.95	6.06	7.14	8.22	9.32	10.46	11.64	12.90		14.49	7 / 8			
	reaction time	interval	1.11	1.10	1.11	1.08	1.08	1.10	1.14	1.18	1.26	1.59			3.32	3.26	3.58
	velocity	5.01	8.23	8.31	8.23	8.46	8.46	8.31	8.02	7.75	7.25	8.82	7.59		8.26	8.41	7.66

Heat 1

date 23-Jun-07 wind 0.1 m/s USATF Men's Hurdle Development (2007)

		H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
Payne, David (USA) (1982)	time	2.59	3.65	4.70	5.72	6.73	7.76	8.78	9.83	10.88	11.95		13.43	8 / 1			
	reaction time	interval	1.06	1.05	1.02	1.01	1.03	1.02	1.05	1.05	1.07	1.48			3.13	3.06	3.17
	velocity	5.30	8.62	8.70	8.96	9.05	8.87	8.96	8.70	8.70	8.54	9.47	8.19		8.76	8.96	8.65
Hicks, Antwon (USA) (1983)	time	2.57	3.64	4.70	5.77	6.81	7.82	8.88	9.91	10.96	12.05		13.49	7 / 2			
	reaction time	interval	1.07	1.06	1.07	1.04	1.01	1.06	1.03	1.05	1.09	1.44			3.20	3.11	3.17
	velocity	5.34	8.54	8.62	8.54	8.79	9.05	8.62	8.87	8.70	8.39	9.74	8.15		8.57	8.82	8.65
Johnson, Allen (USA) (1971)	time	2.57	3.62	4.64	5.67	6.71	7.73	8.77	9.86	10.93	12.02		13.52	3 / 3			
	reaction time	interval	1.05	1.02	1.03	1.04	1.02	1.04	1.09	1.07	1.09	1.50			3.10	3.10	3.25
	velocity	5.34	8.70	8.96	8.87	8.79	8.96	8.79	8.39	8.54	8.39	9.35	8.14		8.85	8.85	8.44
Faulk, Dexter (USA) (1984)	time	2.56	3.65	4.67	5.72	6.87	7.93	9.01	10.08	11.18	12.23		13.67	1 / 4			
	reaction time	interval	1.09	1.02	1.05	1.15	1.06	1.08	1.07	1.10	1.05	1.44			3.16	3.29	3.22
	velocity	5.36	8.39	8.96	8.70	7.95	8.62	8.46	8.54	8.31	8.70	9.74	8.05		8.68	8.33	8.52
Cooper, Jermaine (USA) (1980)	time	2.58	3.67	4.74	5.82	6.87	7.92	8.99	10.04	11.18	12.27		13.75	2 / 5			
	reaction time	interval	1.09	1.07	1.08	1.05	1.05	1.07	1.05	1.14	1.09	1.48			3.24	3.17	3.28
	velocity	5.32	8.39	8.54	8.46	8.70	8.70	8.54	8.70	8.02	8.39	9.47	8.00		8.46	8.65	8.36
Odom, Marlon (USA) (1982)	time	2.70	3.77	4.82	5.89	6.94	7.99	9.03	10.07	11.16	12.28		13.76	4 / 6			
	reaction time	interval	1.07	1.05	1.07	1.05	1.04	1.04	1.09	1.12	1.48				3.19	3.14	3.25
	velocity	5.08	8.54	8.70	8.54	8.70	8.70	8.79	8.79	8.39	8.16	9.47	7.99		8.60	8.73	8.44
Porter, Jeff (USA) (1985)	time	2.62	3.67	4.77	5.84	6.91	7.95	9.04	10.11	11.21	12.27		13.80	5 / 7			
	reaction time	interval	1.05	1.10	1.07	1.07	1.04	1.09	1.07	1.10	1.06	1.53			3.22	3.20	3.23
	velocity	5.24	8.70	8.31	8.54	8.54	8.79	8.39	8.54	8.31	8.62	9.16	7.97		8.52	8.57	8.49

Lindsey, Justin (USA) (1979)	time	2.65	3.70	4.80	5.84	6.92	7.99	9.09	10.19	11.29	12.39		13.88	6 / 8			
	reaction time	interval	1.05	1.10	1.04	1.08	1.07	1.10	1.10	1.10	1.10	1.49			3.19	3.25	3.30
	velocity		5.18	8.70	8.31	8.79	8.46	8.54	8.31	8.31	8.31	8.31	9.41	7.93	8.60	8.44	8.31

2007 USATF National Junior Championship (Indianapolis, IN) (99cm)

FINAL

date 21-Jun-07 wind 2.7 m/s

USATF Men's Hurdle Development (2007)

		H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
Ames, Brendan (USA) (1988)	time	2.59	3.67	4.70	5.69	6.67	7.66	8.64	9.63	10.64	11.68		13.15	/ 1			
	reaction time	interval	1.08	1.03	0.99	0.98	0.99	0.98	0.99	1.01	1.04	1.47			3.10	2.95	3.04
	velocity		5.30	8.46	8.87	9.23	9.33	9.23	9.23	9.05	8.79	9.54	8.37		8.85	9.29	9.02
Dutch, Johnny (USA) (1989)	time	2.64	3.67	4.70	5.69	6.67	7.67	8.68	9.74	10.81	11.88		13.36	/ 2			
	reaction time	interval	1.03	1.03	0.99	0.98	1.00	1.01	1.06	1.07	1.07	1.48			3.05	2.99	3.20
	velocity		5.20	8.87	8.87	9.23	9.33	9.14	9.05	8.62	8.54	9.47	8.23		8.99	9.17	8.57
Brown, Bryce (USA) (1988)	time	2.67	3.77	4.80	5.82	6.86	7.91	8.94	9.96	11.01	12.08		13.42	/ 3			
	reaction time	interval	1.10	1.03	1.02	1.04	1.05	1.03	1.02	1.05	1.07	1.34			3.15	3.12	3.14
	velocity		5.14	8.31	8.87	8.96	8.79	8.70	8.87	8.96	8.70	8.54	10.46	8.20	8.70	8.79	8.73
Prentice, Terry (USA) (1989)	time	2.70	3.77	4.89	5.91	6.94	7.96	8.98	10.01	11.06	12.10		13.52	/ 4			
	reaction time	interval	1.07	1.12	1.02	1.03	1.02	1.02	1.03	1.05	1.04	1.42			3.21	3.07	3.12
	velocity		5.08	8.54	8.16	8.96	8.87	8.96	8.87	8.70	8.79	9.87	8.14		8.54	8.93	8.79
Osaghae, Omo (USA) (1988)	time	2.72	3.80	4.87	5.92	6.96	7.99	9.06	10.13	11.18	12.28		13.71	/ 5			
	reaction time	interval	1.08	1.07	1.05	1.04	1.03	1.07	1.07	1.05	1.10	1.43			3.20	3.14	3.22
	velocity		5.04	8.46	8.54	8.70	8.79	8.87	8.54	8.54	8.70	8.31	9.80	8.02	8.57	8.73	8.52
Reed, Darius (USA) (1988)	time	2.65	3.72	4.75	5.77	6.81	7.86	8.93	9.98	11.09	12.21		13.79	/ 6			
	reaction time	interval	1.07	1.03	1.02	1.04	1.05	1.07	1.05	1.11	1.12	1.58			3.12	3.16	3.28
	velocity		5.18	8.54	8.87	8.96	8.79	8.70	8.54	8.70	8.23	8.16	8.87	7.98	8.79	8.68	8.36
McDowell, Andrew (USA) (1988)	time	2.69	3.74	4.77	5.82	6.89	7.94	8.98	10.09	11.18	12.30		13.84	/ 7			
	reaction time	interval	1.05	1.03	1.05	1.07	1.05	1.04	1.11	1.09	1.12	1.54			3.13	3.16	3.32
	velocity		5.10	8.70	8.87	8.70	8.54	8.70	8.79	8.23	8.39	8.16	9.10	7.95	8.76	8.68	8.26
Nunley, Booker (USA) (1990)	time	2.69	3.77	4.85	5.95	7.01	8.14	9.26	10.39	11.51	12.68		14.28	/ 8			
	reaction time	interval	1.08	1.08	1.10	1.06	1.13	1.12	1.13	1.12	1.17	1.60			3.26	3.31	3.42
	velocity		5.10	8.46	8.46	8.31	8.62	8.09	8.16	8.09	8.16	7.81	8.76	7.70	8.41	8.28	8.02

2007 National Grand Prix (Zhaoqing, CHN)

FINAL

date 19-May-07 wind 0.4 m/s

Zhao (2009) - kinematic research on the 110m hurdle technique of Shi Dongpeng

		H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
Shi Dongpeng (CHN) (1984)	time	2.63	3.70	4.73	5.75	6.77	7.79	8.82	9.86	10.92	11.99		13.47	/ 1			
	reaction time	interval	1.07	1.03	1.02	1.02	1.02	1.03	1.04	1.06	1.07	1.48			3.12	3.07	3.17
	velocity		5.22	8.54	8.87	8.96	8.96	8.87	8.79	8.62	8.54	9.47	8.17		8.79	8.93	8.65

2007 Osaka Grand Prix (Osaka, JPN)

FINAL

date 05-May-07 wind 0.8 m/s

Akirabun (2008) - 2007 speed analysis of the men's & women's 100m, 100m hurdles, & 110m hurdles

		H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
Liu Xiang (CHN) (1983)	time	2.43	3.49	4.51	5.50	6.50	7.50	8.51	9.53	10.55	11.58		13.14	/ 1			
	reaction time	interval	1.06	1.02	0.99	1.00	1.00	1.01	1.02	1.02	1.03	1.56			3.07	3.01	3.07
	velocity		5.65	8.62	8.96	9.23	9.14	9.14	9.05	8.96	8.96	8.87	8.99	8.37	8.93	9.11	8.93
Tanonaka, Tasuku (JPN) (1978)	time	2.40	3.48	4.52	5.56	6.62	7.68	8.74	9.82	10.90	11.99		13.59	/ 3			
	reaction time	interval	1.08	1.04	1.04	1.06	1.06	1.06	1.08	1.08	1.09	1.60			3.16	3.18	3.25
	velocity		5.72	8.46	8.79	8.79	8.62	8.62	8.46	8.46	8.39	8.76	8.09		8.68	8.62	8.44
Naito, Masato (JPN) (1980)	time	2.43	3.51	4.55	5.58	6.62	7.66	8.71	9.78	10.87	11.96		13.60	/ 4			
	reaction time	interval	1.08	1.04	1.03	1.04	1.04	1.05	1.07	1.09	1.09	1.64			3.15	3.13	3.25
	velocity		5.65	8.46	8.79	8.87	8.79	8.70	8.54	8.39	8.39	8.55	8.09		8.70	8.76	8.44

2007 Mikio Oda Memorial (Hiroshima, JPN)

FINAL

date 29-Apr-07 wind 0.0 m/s

Akirabun (2008) - 2007 speed analysis of the men's & women's 100m, 100m hurdles, & 110m hurdles

		H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
Naito, Masato (JPN) (1980)	time	2.42	3.50	4.55	5.59	6.64	7.69	8.75	9.82	10.90	12.01		13.64	/ 1			
	reaction time	interval	1.08	1.05	1.04	1.05	1.05	1.06	1.07	1.08	1.11	1.63			3.17	3.16	3.26
	velocity		5.67	8.46	8.70	8.79	8.70	8.62	8.54	8.46	8.23	8.60	8.06		8.65	8.68	8.41
Tanonaka, Tasuku (JPN) (1978)	time	2.42	3.49	4.54	5.59	6.65	7.71	8.78	9.87	10.98	12.10		13.73	/ 1			
	reaction time	interval	1.07	1.05	1.05	1.06	1.06	1.07	1.09	1.11	1.12	1.63			3.17	3.19	3.32
	velocity		5.67	8.54	8.70	8.70	8.62	8.54	8.39	8.23	8.16	8.60	8.01		8.65	8.60	8.26

Heat 2

date 29-Apr-07 wind 2.6 m/s

Akirabun (2008) - 2007 speed analysis of the men's & women's 100m, 100m hurdles, & 110m hurdles

		H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
Tanonaka, Tasuku (JPN) (1978)	time	2.40	3.49	4.55	5.60	6.65	7.70	8.76	9.83	10.90	11.99		13.60	/ 1			

reaction time	interval	1.09	1.06	1.05	1.05	1.05	1.06	1.07	1.07	1.09	1.61		3.20	3.16	3.23	
	velocity	5.72	8.39	8.62	8.70	8.70	8.70	8.62	8.54	8.54	8.39	8.71	8.09	8.57	8.68	8.49

2006 IAAF World Cup (Athens, GRE) (TV Analysis)

FINAL

date 17-Sep-06		wind 0.4 m/s										Graubner (2009) - http://www.fgs.uni-halle.de						
		H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10	
Johnson, Allen (USA) (1971)	time	2.55	3.54	4.54	5.53	6.51	7.50	8.49	9.49	10.51	11.55		12.96	3 / 1				
reaction time	0.164	interval	0.99	1.00	0.99	0.98	0.99	0.99	1.00	1.02	1.04	1.41	CR		2.98	2.96	3.06	
		velocity	5.38	9.23	9.14	9.23	9.33	9.23	9.23	9.14	8.96	8.79	9.94	8.49	9.20	9.26	8.96	
Liu, Xiang (CHN) (1983)	0.185	no information available											13.03	2 / 2				
Robles, Dayron (CUB) (1986)	0.174	no information available											13.06	8 / 3				
Olijars, Stanislavs (LAT) (1979)	0.147	no information available											13.15	6 / 4				
Lavanne, Cedric (FRA) (1980)	0.148	no information available											13.56	7 / 5				
Peremota, Igor (RUS) (1981)	0.190	no information available											13.57	1 / 6				
Mortimer, James (NZL) (1983)	0.171	no information available											13.94	5 / 7				
Marridis, Theopistos (GRE) (1981)	0.241	no information available											13.96	9 / 8				
Ben Ahmed, Aymen (TUN) (1981)	0.231	no information available											14.08	4 / 9				

2006 IAAF World Athletic Final (Stuttgart, GER) (TV Analysis)

FINAL

INAL	date 09-Sep-06		wind -0.6 m/s										Graubner (2009) - http://www.fgs.uni-halle.de						
			H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10	
Liu Xiang (CHN) (1983)	time		2.57	3.59	4.58	5.57	6.55	7.53	8.54	9.55	10.56	11.57		12.93	5 / 1				
reaction time	0.108	interval		1.02	0.99	0.99	0.98	0.98	1.01	1.01	1.01	1.01	1.36	CR		3.00	2.97	3.03	
		velocity	5.34	8.96	9.23	9.23	9.33	9.33	9.05	9.05	9.05	9.05	10.31	8.51		9.14	9.23	9.05	
Robles, Dayron (CUB) (1986)	0.163	no information available													13.00	4 / 2			
Johnson, Allen (USA) (1971)	0.159	no information available													13.01	8 / 3			
Trammell, Terrence (USA) (1971)	0.140	no information available													13.22	3 / 4			
Oliver, David (USA) (1982)	0.175	no information available													13.24	2 / 5			
Merritt, Aries (USA) (1985)	0.135	no information available													13.25	6 / 6			
Olijars, Stanislavs (LAT) (1979)	0.167	no information available													13.27	1 / 7			
Blaschek, Thomas (GER) (1981)	0.169	no information available													13.28	7 / 8			

2006 European Championships (Göteborg, SWE) (TV Analysis)

FINAL

INAL	date 12-Aug-06		wind -1.0 m/s										Graubner (2009) - http://www.fgs.uni-halle.de					
			H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
Olijar, Stanislav (LAT) (1979)	time		2.64	3.68	4.70	5.71	6.69	7.68	8.70	9.72	10.76	11.84		13.24	6 / 1			
	reaction time	0.151		1.04	1.02	1.01	0.98	0.99	1.02	1.02	1.04	1.08	1.40			3.07	2.99	3.14
	interval		5.20	8.79	8.96	9.05	9.33	9.23	8.96	8.96	8.79	8.46	10.01	8.31		8.93	9.17	8.73
	velocity																	
Blaschek, Thomas (GER) (1981)			no information available											13.46	3 / 2			
Turner, Andy (GBR) (1980)			no information available											13.52	2 / 3			
Peremota, Igor (RUS) (1981)		7 steps	no information available											13.55	4 / 4			
Kronber, Robert (SWE) (1976)			no information available											13.57	7 / 5			
Wermann, Jens (GER) (1985)			no information available											13.73	1 / 6			
Kiss, Daniel (HUN) (1982)			no information available											13.77	8 / 7			
Demvdvuk, Serhiy (UKR) (1982)			no information available											13.96	5 / 8			

Heat 4

Heat 4

	date	11-Aug-06	wind -1.1 m/s										Weber (2006) - wettkampfauswertung 110m hürden						
			H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10	
Bitzi, Ivan (SUI) (1975)	time		2.71	3.77	4.82	5.87	6.91	7.96	9.02	10.12	11.21	12.30		13.68	4 / 4				
	reaction time	interval		1.06	1.05	1.05	1.04	1.05	1.06	1.10	1.09	1.09	1.38				3.16	3.15	3.28
		velocity	5.06	8.62	8.70	8.70	8.79	8.70	8.62	8.31	8.39	8.39	10.16	8.04		8.68	8.70	8.36	

2006 German National Junior Championships (Wattenscheid, GER) (99cm)

FINAL

INAL	date	22-Jul-06	wind 0.8 m/s										Killing (2006) - Werrmann und Dittmer im vergleich						
			H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10	
Dittmer, Paul (GER) (1987)	time	2.64	3.73	4.79	5.85	6.91	7.97	9.05	10.13	11.23	12.34		13.83	/ 1					
	reaction time	interval	1.09	1.06	1.06	1.06	1.06	1.08	1.08	1.10	1.11	1.49	PB		3.21	3.20	3.29		
	velocity	5.20	8.39	8.62	8.62	8.62	8.62	8.46	8.46	8.31	8.23	9.41	7.95		8.54	8.57	8.33		

2006 German National Championships (Ulm, GER)

FINAL

INAL	date 15-Jul-06		wind 1.0 m/s										Killing (2006) - Werrmann und Dittmer im vergleich						
			H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10	
Werrmann, Jens (GER) (1985)	time		2.65	3.75	4.80	5.85	6.91	7.98	9.02	10.08	11.15	12.21		13.62	/ 2				
	reaction time	0.137	interval	1.10	1.05	1.05	1.06	1.07	1.04	1.06	1.07	1.06	1.41	PB		3.20	3.17	3.19	
			velocity	5.18	8.31	8.70	8.70	8.62	8.54	8.79	8.62	8.54	8.62	9.94	8.08	8.57	8.65	8.60	

Heat 1

Heat 1	date 15-Jul-06		wind -0.3 m/s										Killing (2006) - Werrmann und Dittmer im vergleich						
			H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10	
	Werrmann, Jens (GER) (1985)	time	2.68	3.78	4.84	5.90	6.95	8.03	9.09	10.16	11.24	12.37		13.78	/ 1				
	reaction time	0.146	interval		1.10	1.06	1.06	1.05	1.08	1.06	1.07	1.08	1.13	1.41	PB		3.22	3.19	3.28
			velocity	5.12	8.31	8.62	8.62	8.70	8.46	8.62	8.54	8.46	8.09	9.94	7.98		8.52	8.60	8.36

2006 Athletissima Meeting (Lausanne, SUI)**Race 2**

date 11-Jul-06 wind 1.1 m/s

*Li (2013) - technical characteristics of Merritt, the 110m hurdle champion at London Olympics**Xiao (2014) - development process of the characteristics about 110m hurdle training in China*

		H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
Liu Xiang (CHN) (1983)	time	2.52	3.54	4.55	5.56	6.54	7.51	8.49	9.47	10.50	11.53		12.88	2 / 1			<i>Xiao (2014)</i>
	reaction time		1.02	1.01	1.01	0.98	0.97	0.98	0.98	1.03	1.03	1.35	WR		3.04	2.93	3.04
	velocity	5.44	8.96	9.05	9.05	9.33	9.42	9.33	9.33	8.87	8.87	10.39	8.54		9.02	9.36	9.02
Arnold, Dominique (USA) (197	time	2.59	3.61	4.59	5.55	6.51	7.47	8.45	9.45	10.47	11.48		12.90	4 / 2			<i>Li (2013)</i>
	reaction time		1.02	0.98	0.96	0.96	0.96	0.98	1.00	1.02	1.01	1.42	AR		2.96	2.90	3.03
	velocity	5.30	8.96	9.33	9.52	9.52	9.52	9.33	9.14	8.96	9.05	9.87	8.53		9.26	9.46	9.05
Trammell, Terrence (USA) (197	time	2.54	3.55	4.55	5.51	6.48	7.47	8.49	9.51	10.53	11.59		13.02	5 / 3			<i>Li (2013)</i>
	reaction time		1.01	1.00	0.96	0.97	0.99	1.02	1.02	1.02	1.06	1.43	=PB		2.97	2.98	3.10
	velocity	5.40	9.05	9.14	9.52	9.42	9.23	8.96	8.96	8.96	8.62	9.80	8.45		9.23	9.20	8.85

2006 USATF National Championship (Indianapolis, IN)**FINAL**

date 25-Jun-06 wind -0.4 m/s

USATF Men's Hurdle Development (2006)

		H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
Arnold, Dominique (USA) (197	time	2.55	3.59	4.57	5.56	6.56	7.57	8.56	9.59	10.61	11.68		13.10	/ 1			
	reaction time		1.04	0.98	0.99	1.00	1.01	0.99	1.03	1.02	1.07	1.42			3.01	3.00	3.12
	velocity	5.38	8.79	9.33	9.23	9.14	9.05	9.23	8.87	8.96	8.54	9.87	8.40		9.11	9.14	8.79
Trammell, Terrence (USA) (197	time	2.57	3.62	4.62	5.62	6.64	7.64	8.64	9.68	10.71	11.75		13.14	/ 2			
	reaction time		1.05	1.00	1.00	1.02	1.00	1.00	1.04	1.03	1.04	1.39			3.05	3.02	3.11
	velocity	5.34	8.70	9.14	9.14	8.96	9.14	9.14	8.79	8.87	8.79	10.09	8.37		8.99	9.08	8.82
Wilson, Ryan (USA) (1980)	time	2.62	3.65	4.66	5.66	6.66	7.64	8.64	9.68	10.71	11.76		13.22	/ 3			
	reaction time		1.03	1.01	1.00	1.00	0.98	1.00	1.04	1.03	1.05	1.46	PB		3.04	2.98	3.12
	velocity	5.24	8.87	9.05	9.14	9.14	9.33	9.14	8.79	8.87	8.70	9.60	8.32		9.02	9.20	8.79
Hughes, Robby (USA) (1978)	time	2.62	3.64	4.66	5.71	6.69	7.74	8.78	9.81	10.86	11.93		13.30	/ 4			
	reaction time		1.02	1.02	1.05	0.98	1.05	1.04	1.03	1.05	1.07	1.37			3.09	3.07	3.15
	velocity	5.24	8.96	8.96	8.70	9.33	8.70	8.79	8.87	8.70	8.54	10.23	8.27		8.87	8.93	8.70
Payne, David (USA) (1982)	time	2.59	3.67	4.69	5.72	6.72	7.76	8.78	9.84	10.93	12.01		13.42	/ 5			
	reaction time		1.08	1.02	1.03	1.00	1.04	1.02	1.06	1.09	1.08	1.41			3.13	3.06	3.23
	velocity	5.30	8.46	8.96	8.87	9.14	8.79	8.96	8.62	8.39	8.46	9.94	8.20		8.76	8.96	8.49
Merritt, Aries (USA) (1985)	time	2.67	3.75	4.81	5.84	6.86	7.86	8.89	9.93	10.98	12.03		13.43	/ 6			
	reaction time		1.08	1.06	1.03	1.02	1.00	1.03	1.04	1.05	1.05	1.40			3.17	3.05	3.14
	velocity	5.14	8.46	8.62	8.87	8.96	9.14	8.87	8.79	8.70	8.70	10.01	8.19		8.65	8.99	8.73
Herring, Aubrey (USA) (1978)	time	2.60	3.65	4.69	5.74	6.77	7.81	8.83	9.86	10.93	12.00		13.45	/ 7			
	reaction time		1.05	1.04	1.05	1.03	1.04	1.02	1.03	1.07	1.07	1.45			3.14	3.09	3.17
	velocity	5.28	8.70	8.79	8.70	8.87	8.79	8.96	8.87	8.54	8.54	9.67	8.18		8.73	8.87	8.65
Hicks, Antwon (USA) (1983)	time	2.65	3.74	4.79	5.84	6.89	7.93	8.98	10.08	11.13	12.23		13.71	/ 8			
	reaction time		1.09	1.05	1.05	1.05	1.04	1.05	1.10	1.05	1.10	1.48			3.19	3.14	3.25
	velocity	5.18	8.39	8.70	8.70	8.70	8.79	8.70	8.31	8.70	8.31	9.47	8.02		8.60	8.73	8.44

2006 Osaka Grand Prix (Osaka, JPN)**FINAL**

date 06-May-06 wind 1.2 m/s

Akirabun (2007) - speed analysis of the 100m and hurdles with a laser system

		H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
Liu Xiang (CHN) (1983)	time	2.42	3.48	4.49	5.50	6.51	7.53	8.55	9.58	10.60	11.67		13.22	/ 1			
	reaction time		1.06	1.01	1.01	1.01	1.02	1.02	1.03	1.02	1.07	1.55			3.08	3.05	3.12
	velocity	5.67	8.62	9.05	9.05	9.05	8.96	8.96	8.87	8.96	8.54	9.05	8.32		8.90	8.99	8.79
Naito, Masato (JPN) (1980)	time	2.46	3.53	4.58	5.60	6.64	7.69	8.75	9.82	10.91	12.02		13.64	/ 3			
	reaction time		1.07	1.05	1.02	1.04	1.05	1.06	1.07	1.09	1.11	1.62			3.14	3.15	3.27
	velocity	5.58	8.54	8.70	8.96	8.79	8.70	8.62	8.54	8.39	8.23	8.65	8.06		8.73	8.70	8.39
Tanonaka, Tasuku (JPN) (1978)	time	2.45	3.53	4.60	5.65	6.70	7.77	8.85	9.93	11.03	12.13		13.72	/ 4			
	reaction time		1.08	1.07	1.05	1.05	1.07	1.08	1.08	1.10	1.10	1.59			3.20	3.20	3.28
	velocity	5.60	8.46	8.54	8.70	8.70	8.54	8.46	8.46	8.31	8.31	8.82	8.02		8.57	8.57	8.36

2005 Chinese National Games (Nanjing, CHN)**FINAL**

date 20-Oct-05 wind 0.4 m/s

Wang (2013) - analysis on the technical characteristics of Liu Xiang's and Shi Dongpeng's 110m Hurdles

		H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
Liu Xiang (CHN) (1983)	time	2.60	3.63	4.65	5.64	6.65	7.65	8.65	9.67	10.69	11.71		13.10	/ 1			
	reaction time		1.03	1.02	0.99	1.01	1.00	1.00	1.02	1.02	1.02	1.39			3.04	3.01	3.06
	velocity	5.28	8.87	8.96	9.23	9.05	9.14	9.14	8.96	8.96	8.96	10.09	8.40		9.02	9.11	8.96
Shi Dongpeng (CHN) (1984)	time	2.70	3.75	4.77	5.78	6.78	7.81	8.84	9.89	10.93	11.98		13.44	/ 2			
	reaction time		1.05	1.02	1.01	1.00	1.03	1.03	1.05	1.04	1.05	1.46			3.08	3.06	3.14
	velocity	5.08	8.70	8.96	9.05	9.14	8.87	8.87	8.70	8.79	8.70	9.60	8.18		8.90	8.96	8.73

2005 IAAF World Championships (Helsinki, FIN)**FINAL**

FRACTION	date		12-Aug-05		wind		-0.2 m/s		Longuevre (2005) - Helsinki 2005 - Les haies: Cinq qualifiées et un titre mondial												
					H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10	
Doucouré, Ladji (FRA) (1983)	time				2.30	3.30	4.30	5.28	6.26	7.26	8.26	9.26	10.32	11.36		13.07	5 / 1				
	reaction time	0.164	interval			1.00	1.00	0.98	0.98	1.00	1.00	1.00	1.06	1.04	1.71			2.98	2.98	3.10	
	velocity				5.97	9.14	9.14	9.33	9.33	9.14	9.14	9.14	8.62	8.79	8.20	8.42		9.20	9.20	8.85	
Liu Xiang (CHN) (1983)	0.142	no information available														13.08	2 / 2				
Johnson, Allen (USA) (1971)	0.147	no information available														13.10	7 / 3				
Arnold, Dominique (USA) (1971)	0.173	no information available														13.13	8 / 4				
Trammell, Terrence (USA) (1971)	0.222	no information available														13.20	6 / 5				
Brown, Joel (USA) (1980)	0.160	no information available														13.47	1 / 6				
Wignall, Maurice (JAM) (1976)	0.167	no information available														13.47	3 / 7				
Inocêncio, Mateus Facho (BRA)	0.157	no information available														13.48	7 / 8				

2005 Chinese National Grand Prix (Tianjin, CHN)**FINAL**

		date	01-Aug-05	wind -0.8 m/s			Zhang (2005) - technical characteristics analysis of 110m hurdles race of Liu Xiang												
				H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
Liu Xiang (CHN) (1983)	time			2.56	3.59	4.59	5.57	6.58	7.58	8.60	9.62	10.65	11.69		13.06	/ 1			
	reaction time		interval		1.03	1.00	0.98	1.01	1.00	1.02	1.02	1.03	1.04	1.37	=PB		3.01	3.03	3.09
		velocity			5.36	8.87	9.14	9.33	9.05	9.14	8.96	8.96	8.87	8.79	10.23	8.42		9.11	9.05

2005 USATF National Championship (Carson, CA)**FINAL**

2005 USAATF National Championship (Carlson, CA)										USATF Women's Sprint Development (2005)											
		date	24-Jun-05	wind	0.2 m/s	H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
Johnson, Allen (USA) (1971)	time					2.56	3.57	4.58	5.58	6.56	7.56	8.58	9.56	10.58	11.64		12.99	2 / 1			
	reaction time		interval				1.01	1.01	1.00	0.98	1.00	1.02	0.98	1.02	1.06	1.35			3.02	3.00	3.06
		velocity				5.36	9.05	9.05	9.14	9.33	9.14	8.96	9.33	8.96	8.62	10.39	8.47		9.08	9.14	8.96
Arnold, Dominique (USA) (1971)	time					2.61	3.66	4.66	5.66	6.61	7.60	8.56	9.61	10.61	11.62		13.01	5 / 2			
	reaction time		interval				1.05	1.00	1.00	0.95	0.99	0.96	1.05	1.00	1.01	1.39	PB		3.05	2.90	3.06
		velocity				5.26	8.70	9.14	9.14	9.62	9.23	9.52	8.70	9.14	9.05	10.09	8.46		8.99	9.46	8.96
Trammell, Terrence (USA) (1971)	time					2.54	3.57	4.61	5.61	6.61	7.61	8.63	9.64	10.64	11.69		13.02	6 / 3			
	reaction time		interval				1.03	1.04	1.00	1.00	1.00	1.02	1.01	1.00	1.05	1.33	PB		3.07	3.02	3.06
		velocity				5.40	8.87	8.79	9.14	9.14	9.14	8.96	9.05	9.14	8.70	10.54	8.45		8.93	9.08	8.96
Brown, Joel (USA) (1980)	time					2.61	3.67	4.71	5.73	6.74	7.78	8.80	9.81	10.86	11.91		13.28	4 / 4			
	reaction time		interval				1.06	1.04	1.02	1.01	1.04	1.02	1.01	1.05	1.05	1.37	PB		3.12	3.07	3.11
		velocity				5.26	8.62	8.79	8.96	9.05	8.79	8.96	9.05	8.70	8.70	10.23	8.28		8.79	8.93	8.82
Payne, David (USA) (1982)	time					2.62	3.67	4.71	5.73	6.76	7.81	8.83	9.84	10.90	11.91		13.33	7 / 5			
	reaction time		interval				1.05	1.04	1.02	1.03	1.05	1.02	1.01	1.06	1.01	1.42	PB		3.11	3.10	3.08
		velocity				5.24	8.70	8.79	8.96	8.87	8.70	8.96	9.05	8.62	9.05	9.87	8.25		8.82	8.85	8.90
Hicks, Antwon (USA) (1983)	time					2.61	3.67	4.71	5.71	6.76	7.78	8.83	9.84	10.89	11.95		13.35	1 / 6			
	reaction time		interval	7 steps			1.06	1.04	1.00	1.05	1.02	1.05	1.01	1.05	1.06	1.40	PB		3.10	3.12	3.12
		velocity				5.26	8.62	8.79	9.14	8.70	8.96	8.70	9.05	8.70	8.62	10.01	8.24		8.85	8.79	8.79
Merritt, Aries (USA) (1985)	time					2.66	3.74	4.81	5.81	6.84	7.88	8.93	9.91	10.97	12.03		13.39	8 / 7			
	reaction time		interval				1.08	1.07	1.00	1.03	1.04	1.05	0.98	1.06	1.06	1.36			3.15	3.12	3.10
		velocity				5.16	8.46	8.54	9.14	8.87	8.79	8.70	9.33	8.62	8.62	10.31	8.22		8.70	8.79	8.85
Oliver, David (USA) (1982)	time					2.66	3.71	4.76	5.76	6.76	7.81	8.83	9.86	10.92	11.97		13.46	3 / 8			
	reaction time		interval				1.05	1.05	1.00	1.00	1.05	1.02	1.03	1.06	1.05	1.49			3.10	3.07	3.14
		velocity				5.16	8.70	8.70	9.14	9.14	8.70	8.96	8.87	8.62	8.70	9.41	8.17		8.85	8.93	8.73
Bramlett, Ron (USA) (1979)	time					2.59	3.62	4.71	5.73	6.74	7.78	8.83	9.87	10.92	12.01		13.58	9 / 9			
	reaction time		interval				1.03	1.09	1.02	1.01	1.04	1.05	1.04	1.05	1.09	1.57			3.14	3.10	3.18
		velocity				5.30	8.87	8.39	8.96	9.05	8.79	8.70	8.79	8.70	8.39	8.93	8.10		8.73	8.85	8.62

2004 Olympic Games (Athens, GRE)**FINAL**

		date	27-Aug-04	wind 0.3 m/s										Ruiz (2004) - las carreras con vallas de juegos olímpicos de atenas '04							
				H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10		
Liu Xiang (CHN) (1983)	time			2.41	3.43	4.43	5.42	6.41	7.39	8.39	9.39	10.41	11.44		12.91	4 / 1					
	reaction time	0.139	interval		1.02	1.00	0.99	0.99	0.98	1.00	1.00	1.02	1.03	1.47	=WR / OR		3.01	2.97	3.05		
		velocity		5.69	8.96	9.14	9.23	9.23	9.33	9.14	9.14	8.96	8.87	9.54	8.52		9.11	9.23	8.99		
Trammell, Terrence (USA) (1971)	time			2.40	3.42	4.43	5.45	6.45	7.45	8.48	9.50	10.54	11.60		13.18	6 / 2					
	reaction time	0.154	interval		1.02	1.01	1.02	1.00	1.00	1.03	1.02	1.04	1.06	1.58			3.05	3.03	3.12		
		velocity		5.72	8.96	9.05	8.96	9.14	9.14	8.87	8.96	8.79	8.62	8.87	8.35		8.99	9.05	8.79		
García, Anier (CUB) (1976)	time			2.47	3.51	4.53	5.55	6.55	7.55	8.57	9.61	10.65	11.69		13.2	7 / 3					
	reaction time	0.169	interval		1.04	1.02	1.02	1.00	1.00	1.02	1.04	1.04	1.04	1.51			3.08	3.02	3.12		
		velocity		5.55	8.79	8.96	8.96	9.14	9.14	8.96	8.79	8.79	8.79	9.28	8.33		8.90	9.08	8.79		

Wignall, Maurice (JAM) (1976)	time	2.47	3.50	4.53	5.55	6.55	7.56	8.60	9.64	10.67	11.73		13.21	5 / 4				
	reaction time	0.164		1.03	1.03	1.02	1.00	1.01	1.04	1.04	1.03	1.06	1.48			3.08	3.05	3.13
	velocity		5.55	8.87	8.87	8.96	9.14	9.05	8.79	8.79	8.87	8.62	9.47	8.33		8.90	8.99	8.76
Olijars, Stanislavs (LAT) (1979)	time	2.49	3.52	4.54	5.55	6.56	7.56	8.57	9.61	10.67	11.74		13.21	2 / 5				
	reaction time	0.174		1.03	1.02	1.01	1.01	1.00	1.01	1.04	1.06	1.07	1.47			3.06	3.02	3.17
	velocity		5.51	8.87	8.96	9.05	9.05	9.14	9.05	8.79	8.62	8.54	9.54	8.33		8.96	9.08	8.65
Allen, Charles (CAN) (1977)	time	2.46	3.50	4.53	5.57	6.63	7.66	8.71	9.75	10.82	11.88		13.48	1 / 6				
	reaction time	0.139		1.04	1.03	1.04	1.06	1.03	1.05	1.04	1.07	1.06	1.60			3.11	3.14	3.17
	velocity		5.58	8.79	8.87	8.79	8.62	8.87	8.70	8.79	8.54	8.62	8.76	8.16		8.82	8.73	8.65
Inocêncio, Mateus Facho (BRA)	time	2.50	3.58	4.63	5.67	6.69	7.72	8.75	9.81	10.89	11.95		13.49	7 / 7				
	reaction time	0.169		1.08	1.05	1.04	1.02	1.03	1.03	1.06	1.08	1.06	1.54			3.17	3.08	3.20
	velocity		5.49	8.46	8.70	8.79	8.96	8.87	8.87	8.62	8.46	8.62	9.10	8.15		8.65	8.90	8.57
Doucouré, Ladj (FRA) (1983)	time	2.52	3.54	4.55	5.56	6.55	7.54	8.54	9.55	10.58	11.63		13.76	3 / 8				
	reaction time	0.204		1.02	1.01	1.01	0.99	0.99	1.00	1.01	1.03	1.05	2.13			3.04	2.98	3.09
	velocity		5.44	8.96	9.05	9.05	9.23	9.23	9.14	9.05	8.87	8.70	6.58	7.99		9.02	9.20	8.87

Semi-Final 2

emi-Final 2		date 26-Aug-04	wind 0.0 m/s										Liagonne (2007) - Historique: Le 110m haies en chiffres						
Doucouré, Ladj (FRA) (1983)			H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10	
	time		2.48	3.50	4.49	5.48	6.48	7.46	8.48	9.48	10.50	11.56		13.06	5 / 1				
	reaction time	0.200	interval		1.02	0.99	0.99	1.00	0.98	1.02	1.00	1.02	1.06	1.50	NR / PB		3.00	3.00	3.08
			velocity		5.53	8.96	9.23	9.23	9.14	9.33	8.96	9.14	8.96	8.62	9.35	8.42	9.14	9.14	8.90

Heat 1

Heat 1	date 24-Aug-04		wind 1.5 m/s										<i>Liagonne (2007) - Historique: Le 110m haies en chiffres</i>						
			H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10	
	Doucouré, Ladji (FRA) (1983)	time	2.52	3.56	4.55	5.54	6.54	7.54	8.55	9.56	10.60	11.64		13.18	8 / 1				
	reaction time	0.205	interval		1.04	0.99	0.99	1.00	1.00	1.01	1.01	1.04	1.04	1.54	NR / PB		3.02	3.01	3.09
		velocity	5.44	8.79	9.23	9.23	9.14	9.14	9.05	9.05	8.79	8.79	9.10	8.35		9.08	9.11	8.87	
<i>(2016.05.12) - https://twitter.com/touchdown_time/media?lang=en&lang=en&lang=en&lang=en</i>																			
Tanigawa, Satoru (JPN) (1972)	time	2.55	3.60	4.63	5.65	6.68	7.71	8.78	9.83	10.90	12.00		13.39	5 / 3					
	reaction time	0.128	interval		1.05	1.03	1.02	1.03	1.07	1.05	1.07	1.10	1.39	NR / PB		3.10	3.13	3.22	
		velocity	5.38	8.70	8.87	8.96	8.87	8.87	8.54	8.70	8.54	8.31	10.09	8.22		8.85	8.76	8.52	

2004 Chinese National Grand Prix (Tianjin, CHN)

FINAL

		date 01-Aug-04		wind -0.8 m/s		Wang (2016) - comparative analysis of the grey relational grade of segment times and final results of the 110m hurdles												
Liu Xiang (CHN) (1983)		H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10	
	time	2.57	3.61	4.62	5.62	6.62	7.63	8.63	9.65	10.67	11.72		13.06	/ 1				
	reaction time		1.04	1.01	1.00	1.00	1.01	1.00	1.02	1.02	1.05	1.34	=PB		3.05	3.01	3.09	
	velocity	5.34	8.79	9.05	9.14	9.14	9.05	9.14	8.96	8.96	8.70	10.46	8.42		8.99	9.11	8.87	

2004 Atletica (Genève, SUI)

FINAL

date 08-May-04		wind -1.2 m/s										Weber (2004) - wettkampfauswertung 110m hürden						
Monachon, Raphael (SUI) (1973)		H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10	
	time	2.70	3.78	4.80	5.84	6.86	7.96	9.02	10.09	11.16	12.26		13.64	/ 3				
	reaction time			1.08	1.02	1.04	1.02	1.10	1.06	1.07	1.07	1.10	1.38			3.14	3.18	3.24
	velocity		5.08	8.46	8.96	8.79	8.96	8.31	8.62	8.54	8.54	8.31	10.16	8.06		8.73	8.62	8.46

2004 Osaka Grand Prix (Osaka, JPN)

FINAL

date 08-May-04		wind -1.2 m/s				Zhang (2005) - technical characteristics analysis of 110m hurdles race of Liu Xiang												
Liu Xiang (CHN) (1983)		H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10	
	time	2.56	3.59	4.59	5.57	6.58	7.58	8.60	9.62	10.65	11.69		13.06	/ 1				
	reaction time			1.03	1.00	0.98	1.01	1.00	1.02	1.02	1.03	1.04	1.37	AR		3.01	3.03	3.09
	velocity	5.36	8.87	9.14	9.33	9.05	9.14	8.96	8.96	8.87	8.79	10.23		8.42		9.11	9.05	8.87

2004 Mt. SAC Relays (Walnut, CA)

Decathlon - Heat 7

date 15-Apr-04		wind 2.8 m/s										Henson (2004) - USATF OTC sport science							
Smith, Chad (USA) (1974)		H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10		
	time	2.55	3.67	4.75	5.83	6.92	8.02	9.12	10.22	11.37	12.52		14.19	/ 1					
	reaction time			1.12	1.08	1.08	1.09	1.10	1.10	1.15	1.15	1.67				3.28	3.29	3.40	
	velocity		5.38	8.16	8.46	8.46	8.39	8.31	8.31	8.31	7.95	7.95	8.40	7.75		8.36	8.33	8.06	

2003 IAAF World Championships (Paris, FRA)

FINAL

date		30-Aug-03		wind		0.3 m/s		Leynier (2003) - Paris 2003 - Haies: Le 110m haies									
		H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
Johnson, Allen (USA) (1971)	time	2.56	3.60	4.60	5.60	6.60	7.62	8.62	9.64	10.68	11.72		13.12	4 / 1			Leynier (2003)
	reaction time	0.159	interval	1.04	1.00	1.00	1.02	1.00	1.02	1.04	1.04	1.40			3.04	3.02	3.10
	velocity	5.36	8.79	9.14	9.14	9.14	8.96	9.14	8.96	8.79	8.79	10.01	8.38		9.02	9.08	8.85
Trammell, Terrence (USA) (1971)	time	2.57	3.61	4.63	5.63	6.63	7.63	8.65	9.67	10.73	11.79		13.20	6 / 2			Xu (2004)
	reaction time	0.154	interval	1.04	1.02	1.00	1.00	1.02	1.02	1.06	1.06	1.41			3.06	3.02	3.14
	velocity	5.34	8.79	8.96	9.14	9.14	9.14	8.96	8.96	8.62	8.62	9.94	8.33		8.96	9.08	8.73

Liu Xiang (CHN) (1983)	time	2.58	3.63	4.65	5.65	6.67	7.68	8.70	9.74	10.80	11.85		13.23	5 / 3				<i>Xu (2004)</i>
reaction time	0.140	interval		1.05	1.02	1.00	1.02	1.01	1.02	1.04	1.06	1.05	1.38			3.07	3.05	3.15
		velocity	5.32	8.70	8.96	9.14	8.96	9.05	8.96	8.79	8.62	8.70	10.16	8.31		8.93	8.99	8.70
Wade, Larry (USA) (1974)	0.157	no information available											13.34	1 / 4				
Phillips, Chris (USA) (1972)	0.150	no information available											DV 13.36	8 / 5				
de Souza, Márcio (BRA) (1975)	0.155	no information available											13.48	7 / 6 5				
Shi Dongpeng (CHN) (1984)	0.145	no information available											13.55	3 / 7 6				
Hernández, Yoel (CUB) (1977)	0.152	no information available											13.57	2 / 8 7				

2003 Athletissima Meeting (Lausanne, SUI)**FINAL**

	date	01-Jul-03	wind	0.0 m/s	<i>Zhang (2005) - technical characteristics analysis of 110m hurdles race of Liu Xiang</i> <i>Xu (2004) - an analysis of Liu Xiang's technique in 100m hurdle final at the Athens Olympic Games</i>													
			H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
Johnson, Allen (USA) (1971)	time		2.56	3.58	4.60	5.60	6.58	7.58	8.60	9.60	10.62	11.66		13.06	1 / 1			<i>Xu (2004)</i>
reaction time		interval		1.02	1.02	1.00	0.98	1.00	1.02	1.00	1.02	1.04	1.40			3.04	3.00	3.06
		velocity	5.36	8.96	8.96	9.14	9.33	9.14	8.96	9.14	8.96	8.79	10.01	8.42		9.02	9.14	8.96
Olijar, Stanislav (LAT) (1979)	time		2.58	3.60	4.60	5.60	6.58	7.58	8.60	9.60	10.64	11.70		13.08	2 / 2			<i>Zhang (2005)</i>
reaction time		interval		1.02	1.00	1.00	0.98	1.00	1.02	1.00	1.04	1.06	1.38	NR PB		3.02	3.00	3.10
		velocity	5.32	8.96	9.14	9.14	9.33	9.14	8.96	9.14	8.79	8.62	10.16	8.41		9.08	9.14	8.85
Liu Xiang (CHN) (1983)	time		2.57	3.61	4.63	5.63	6.65	7.65	8.67	9.69	10.73	11.79		13.17	3 / 3			<i>Xu (2004)</i>
reaction time		interval		1.04	1.02	1.00	1.02	1.00	1.02	1.02	1.04	1.06	1.38			3.06	3.04	3.12
		velocity	5.34	8.79	8.96	9.14	8.96	9.14	8.96	8.96	8.79	8.62	10.16	8.35		8.96	9.02	8.79

2003 USATF National Junior Championships (Palo Alto, CA) (99cm)**FINAL**

	date	22-Jun-03	wind	-0.4 m/s	<i>USATF Women's Sprint Development (2003)</i>													
			H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
Ferguson, Kenneth (USA) (1981)	time		2.55	3.62	4.67	5.69	6.72	7.75	8.79	9.86	10.94	12.04		13.60	4 / 1			
reaction time		interval		1.07	1.05	1.02	1.03	1.03	1.04	1.07	1.08	1.10	1.56			3.14	3.10	3.25
		velocity	5.38	8.54	8.70	8.96	8.87	8.87	8.79	8.54	8.46	8.31	8.99	8.09		8.73	8.85	8.44
Faulk, Dexter (USA) (1984)	time		2.55	3.64	4.72	5.76	6.82	7.89	8.94	10.03	11.11	12.19		13.75	5 / 2			
reaction time		interval		1.09	1.08	1.04	1.06	1.07	1.05	1.09	1.08	1.08	1.56			3.21	3.18	3.25
		velocity	5.38	8.39	8.46	8.79	8.62	8.54	8.70	8.39	8.46	8.46	8.99	8.00		8.54	8.62	8.44
Mitchum, Eric (USA) (1984)	time		2.58	3.67	4.76	5.82	6.87	7.92	8.99	10.06	11.16	12.28		13.81	7 / 3			
reaction time		interval		1.09	1.09	1.06	1.05	1.05	1.07	1.07	1.10	1.12	1.53			3.24	3.17	3.29
		velocity	5.32	8.39	8.39	8.62	8.70	8.70	8.54	8.54	8.31	8.16	9.16	7.97		8.46	8.65	8.33
Loyd, Michael (USA) (1984)	time		2.58	3.73	4.82	5.87	6.97	8.09	9.21	10.34	11.48	12.63		14.28	8 / 4			
reaction time		interval		1.15	1.09	1.05	1.10	1.12	1.12	1.13	1.14	1.15	1.65			3.29	3.34	3.42
		velocity	5.32	7.95	8.39	8.70	8.31	8.16	8.16	8.09	8.02	7.95	8.50	7.70		8.33	8.21	8.02
Craddock, Kevin (USA) (1987)	time		2.60	3.72	4.86	5.94	7.07	8.19	9.31	10.44	11.56	12.73		14.36	6 / 5			
reaction time		interval		1.12	1.14	1.08	1.13	1.12	1.12	1.13	1.12	1.17	1.63			3.34	3.37	3.42
		velocity	5.28	8.16	8.02	8.46	8.09	8.16	8.16	8.09	8.16	7.81	8.60	7.66		8.21	8.14	8.02
Whitely, Aaron (USA)	time		2.68	3.77	4.90	5.96	7.07	8.19	9.33	10.46	11.64	12.82		14.48	3 / 6			
reaction time		interval		1.09	1.13	1.06	1.11	1.12	1.14	1.13	1.18	1.18	1.66			3.28	3.37	3.49
		velocity	5.12	8.39	8.09	8.62	8.23	8.16	8.02	8.09	7.75	7.75	8.45	7.60		8.36	8.14	7.86
Jiles-Tindall, Julius (USA) (1981)	time		2.70	3.84	4.94	6.04	7.14	8.29	9.41	10.54	11.71	12.89		14.51	2 / 7			
reaction time		interval		1.14	1.10	1.10	1.10	1.15	1.12	1.13	1.17	1.18	1.62			3.34	3.37	3.48
		velocity	5.08	8.02	8.31	8.31	8.31	7.95	8.16	8.09	7.81	7.75	8.65	7.58		8.21	8.14	7.88
English, Andre (USA) (1984)	time		2.63	3.75	4.89	6.01	7.12	8.24	9.39	10.54	11.69	12.85		14.52	9 / 8			
reaction time		interval		1.12	1.14	1.12	1.11	1.12	1.15	1.15	1.15	1.16	1.67			3.38	3.38	3.46
		velocity	5.22	8.16	8.02	8.16	8.23	8.16	7.95	7.95	7.95	7.88	8.40	7.58		8.11	8.11	7.92

2003 USATF National Championship (Palo Alto, CA)**FINAL**

	date	21-Jun-03	wind	-2.4 m/s	<i>USATF Women's Sprint Development (2003)</i>													
			H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
Johnson, Allen (USA) (1971)	time		2.35	3.44	4.44	5.49	6.51	7.52	8.56	9.59	10.64	11.70		13.37	6 / 1			
reaction time		interval		1.09	1.00	1.05	1.02	1.01	1.04	1.03	1.05	1.06	1.67			3.14	3.07	3.14
		velocity	5.84	8.39	9.14	8.70	8.96	9.05	8.79	8.87	8.70	8.62	8.40	8.23		8.73	8.93	8.73
Trammell, Terrence (USA) (1971)	time		2.34	3.44	4.44	5.50	6.52	7.52	8.54	9.58	10.64	11.70		13.38	3 / 2			
reaction time		interval		1.10	1.00	1.06	1.02	1.00	1.02	1.04	1.06	1.06	1.68			3.16	3.04	3.16
		velocity	5.86	8.31	9.14	8.62	8.96	9.14	8.96	8.79	8.62	8.62	8.35	8.22		8.68	9.02	8.68
Wade, Larry (USA) (1974)	time		2.40	3.47	4.52	5.59	6.61	7.61	8.62	9.64	10.69	11.75		13.43	4 / 3			
reaction time		interval		1.07	1.05	1.07	1.02	1.00	1.01	1.02	1.05	1.06	1.68			3.19	3.03	3.13
		velocity	5.72	8.54	8.70	8.54	8.96	9.14	9.05	8.96	8.70	8.62	8.35	8.19		8.60	9.05	8.76
Phillips, Christopher (USA) (1971)	time		2.42	3.50	4.55	5.56	6.62	7.64	8.68	9.71	10.78	11.88		13.55	7 / 4			

reaction time	interval	1.08	1.05	1.01	1.06	1.02	1.04	1.03	1.07	1.10	1.67			3.14	3.12	3.20
	velocity	5.67	8.46	8.70	9.05	8.62	8.96	8.79	8.87	8.54	8.31	8.40	8.12	8.73	8.79	8.57
Bramlett, Ron (USA) (1979)	time	2.39	3.47	4.51	5.57	6.61	7.66	8.69	9.73	10.83	11.95		13.66	2 / 5		
reaction time	interval		1.08	1.04	1.06	1.04	1.05	1.03	1.04	1.10	1.12	1.71		3.18	3.12	3.26
	velocity	5.74	8.46	8.79	8.62	8.79	8.70	8.87	8.79	8.31	8.16	8.20	8.05	8.62	8.79	8.41
Crear, Mark (USA) (1968)	time	2.44	3.54	4.59	5.66	6.69	7.70	8.73	9.78	10.83	11.93		13.68	8 / 6		
reaction time	interval		1.10	1.05	1.07	1.03	1.01	1.03	1.05	1.05	1.10	1.75		3.22	3.07	3.20
	velocity	5.62	8.31	8.70	8.54	8.87	9.05	8.87	8.70	8.70	8.31	8.01	8.04	8.52	8.93	8.57
Walker, Josh (USA) (1982)	time	2.37	3.52	4.57	5.64	6.71	7.76	8.83	9.89	10.96	12.08		13.84	1 / 7		
reaction time	interval		1.15	1.05	1.07	1.07	1.05	1.07	1.06	1.07	1.12	1.76		3.27	3.19	3.25
	velocity	5.79	7.95	8.70	8.54	8.54	8.70	8.54	8.62	8.54	8.16	7.97	7.95	8.39	8.60	8.44
Ross, Duane (USA) (1972)	no information available												dnf	5 / --		

2003 Chinese National Grand Prix (Nanning, CHN)**Race 1**

Table 1	date	05-Apr-03	wind	0.4 m/s	Wang (2016) - comparative analysis of the grey relational grade of segment times and final results of the 110m hurdles													
			H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
	Liu Xiang (CHN) (1983)	time	2.63	3.73	4.73	5.73	6.75	7.77	8.82	9.87	10.92	11.98		13.32	/ 1			
	reaction time	interval		1.10	1.00	1.00	1.02	1.02	1.05	1.05	1.05	1.06	1.34			3.10	3.09	3.16
		velocity	5.22	8.31	9.14	9.14	8.96	8.96	8.70	8.70	8.70	8.62	10.46	8.26		8.85	8.87	8.68

2002 Chinese National Grand Prix (Shanghai, CHN)**FINAL**

FINAL	date	19-May-02	wind	Huang (2002) - technical research on the men's 110m hurdles at the 9th national games														
				H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7
Liu Xiang (CHN) (1983)	time		2.59	3.63	4.64	5.69	6.71	7.73	8.76	9.80	10.84	11.91		13.36	/ 1			
	reaction time	interval		1.04	1.01	1.05	1.02	1.02	1.03	1.04	1.04	1.07	1.45			3.10	3.07	3.15
		velocity	5.30	8.79	9.05	8.70	8.96	8.96	8.87	8.79	8.79	8.54	9.67	8.23		8.85	8.93	8.70

2002 IAAF World Cup (Madrid, ESP)**FINAL**

		date	21-Sep-02	wind -2.2 m/s				Sanchez (2002) - copa del mundo de Madrid '02: análisis de la carreras con villas											
				H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
Garcia, Anier (CUB) (1976)	time			2.47	3.49	4.50	5.50	6.50	7.51	8.52	9.52	10.56	11.61		13.10	9 / 1			
	reaction time	0.136	interval		1.02	1.01	1.00	1.00	1.01	1.01	1.00	1.04	1.05	1.49			3.03	3.02	3.09
		velocity		5.55	8.96	9.05	9.14	9.14	9.05	9.05	9.14	8.79	8.70	9.41	8.40		9.05	9.08	8.87
Johnson, Allen (USA) (1971)	time			2.48	3.49	4.51	5.51	6.52	7.55	8.59	9.65	10.71	11.78		13.45	5 / 2			
	reaction time	0.135	interval		1.01	1.02	1.00	1.01	1.03	1.04	1.06	1.06	1.07	1.67			3.03	3.08	3.19
		velocity		5.53	9.05	8.96	9.14	9.05	8.87	8.79	8.62	8.62	8.54	8.40	8.18		9.05	8.90	8.60
Olijars, Stanislav (LAT) (1979)	time			2.52	3.54	4.57	5.60	6.64	7.70	8.76	9.83	10.90	11.97		13.58	3 / 3			
	reaction time	0.151	interval		1.02	1.03	1.03	1.04	1.06	1.06	1.07	1.07	1.07	1.61			3.08	3.16	3.21
		velocity		5.44	8.96	8.87	8.87	8.79	8.62	8.62	8.54	8.54	8.54	8.71	8.10		8.90	8.68	8.54
Bownes, Shaun (RSA) (1970)	time			2.58	3.62	4.65	5.66	6.69	7.75	8.83	9.89	10.97	12.07		13.67	1 / 4			
	reaction time	0.146	interval	7 steps	1.04	1.03	1.01	1.03	1.06	1.08	1.06	1.08	1.10	1.60			3.08	3.17	3.24
		velocity		5.32	8.79	8.87	9.05	8.87	8.62	8.46	8.62	8.46	8.31	8.76	8.05		8.90	8.65	8.46
Vivancos, Felipe (ESP) (1980)	time			2.53	3.61	4.65	5.69	6.74	7.81	8.90	9.99	11.08	12.18		13.79	7 / 5			
	reaction time	0.124	interval		1.08	1.04	1.04	1.05	1.07	1.09	1.09	1.09	1.10	1.61			3.16	3.21	3.28
		velocity		5.42	8.46	8.79	8.79	8.70	8.54	8.39	8.39	8.39	8.31	8.71	7.98		8.68	8.54	8.36
Schwarthoff, Florian (GER) (1979)	time			2.59	3.65	4.70	5.72	6.77	7.83	8.93	10.02	11.10	12.20		13.79	4 / 6			
	reaction time	0.176	interval	7 steps	1.06	1.05	1.02	1.05	1.06	1.10	1.09	1.08	1.10	1.59			3.13	3.21	3.27
		velocity		5.30	8.62	8.70	8.96	8.70	8.62	8.31	8.39	8.46	8.31	8.82	7.98		8.76	8.54	8.39
Ewen, Tim (AUS) (1977)	time			2.69	3.78	4.85	5.93	6.97	8.03	9.13	10.22	11.31	12.42		14.10	8 / 7			
	reaction time	0.140	interval		1.09	1.07	1.08	1.04	1.06	1.10	1.09	1.09	1.11	1.68			3.24	3.20	3.29
		velocity		5.10	8.39	8.54	8.46	8.79	8.62	8.31	8.39	8.39	8.39	8.23	8.35	7.80		8.46	8.57
Liu Xiang (CHN) (1983)		0.156	no information available												dnf	6 / --			
Jackson, Colin (GBR) (1967)		0.138	no information available												DQ	2 / --			

2002 European Championships (Munich, GER)**FINAL**

INAL	date	10-Aug-02	wind	0.4 m/s	Graubner (2009) - http://www.fgs.uni-halle.de														
			H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10	
Jackson, Colin (GBR) (1967)	time		2.57	3.59	4.59	5.59	6.59	7.61	8.63	9.66	10.71	11.77		13.11	/ 1				
	reaction time	0.117		1.02	1.00	1.00	1.00	1.02	1.02	1.03	1.05	1.06	1.34				3.02	3.04	3.14
	interval																		
	velocity		5.34	8.96	9.14	9.14	9.14	8.96	8.96	8.87	8.70	8.62	10.46	8.39		9.08	9.02	8.73	
Olijar, Stanislavs (LAT) (1979)	time		2.59	3.61	4.63	5.62	6.63	7.63	8.65	9.67	10.73	11.78		13.22	/ 2				
	reaction time	0.157		1.02	1.02	0.99	1.01	1.00	1.02	1.02	1.06	1.05	1.44				3.03	3.03	3.13
	interval																		
	velocity		5.30	8.96	8.96	9.23	9.05	9.14	8.96	8.96	8.62	8.70	9.74	8.32		9.05	9.05	8.76	

Kohutek, Artur (POL) (1971)	time	2.62	3.67	4.69	5.71	6.71	7.73	8.77	9.79	10.85	11.91		13.32	/ 3			
	reaction time	0.157		1.05	1.02	1.02	1.00	1.02	1.04	1.02	1.06	1.06	1.41		3.09	3.06	3.14
	velocity		5.24	8.70	8.96	8.96	9.14	8.96	8.79	8.96	8.62	8.62	9.94	8.26	8.87	8.96	8.73
Schwarthoff, Florian (GER) (1971)	time	2.61	3.67	4.69	5.71	6.71	7.75	8.77	9.81	10.85	11.93		13.37	/ 4			
	reaction time	0.164		1.06	1.02	1.02	1.00	1.04	1.02	1.04	1.04	1.08	1.44		3.10	3.06	3.16
	velocity		5.26	8.62	8.96	8.96	9.14	8.79	8.96	8.79	8.79	8.46	9.74	8.23	8.85	8.96	8.68
Fenner, Mike (GER) (1971)	time	2.63	3.67	4.71	5.73	6.75	7.77	8.83	9.85	10.91	11.97		13.39	/ 5			
	reaction time	0.161		1.04	1.04	1.02	1.02	1.02	1.06	1.02	1.06	1.06	1.42		3.10	3.10	3.14
	velocity		5.22	8.79	8.79	8.96	8.96	8.96	8.62	8.96	8.62	8.62	9.87	8.22	8.85	8.85	8.73
Favaro, Denis (ITA) (1972)	time	2.73	3.79	4.85	5.89	6.93	7.95	8.99	10.03	11.09	12.18		13.59	/ 6			
	reaction time	0.202		1.06	1.06	1.04	1.04	1.02	1.04	1.04	1.06	1.09	1.41		3.16	3.10	3.19
	velocity		5.03	8.62	8.62	8.79	8.79	8.96	8.79	8.79	8.62	8.39	9.94	8.09	8.68	8.85	8.60
Kronberg, Robert (SWE) (1976)	time	2.63	3.67	4.71	5.76	6.77	7.85	8.89	9.95	11.06	12.15		13.63	/ 7			
	reaction time	0.134		1.04	1.04	1.05	1.01	1.08	1.04	1.06	1.11	1.09	1.48		3.13	3.13	3.26
	velocity		5.22	8.79	8.79	8.70	9.05	8.46	8.79	8.62	8.23	8.39	9.47	8.07	8.76	8.76	8.41
Videnov, Zhivko (BUL) (1977)	time	2.63	3.69	4.73	5.75	6.79	7.85	8.90	10.01	11.09	12.25		13.67	/ 8			
	reaction time	0.146		1.06	1.04	1.02	1.04	1.06	1.05	1.11	1.08	1.16	1.42		3.12	3.15	3.35
	velocity		5.22	8.62	8.79	8.96	8.79	8.62	8.70	8.23	8.46	7.88	9.87	8.05	8.79	8.70	8.19

Semi-Final 2

date 10-Aug-02 wind 0.8 m/s Graubner (2009) - <http://www.fgs.uni-halle.de>

		H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
Jackson, Colin (GBR) (1967)	time	2.58	3.62	4.64	5.66	6.66	7.68	8.70	9.73	10.76	11.82		13.21	/ 1			
	reaction time	0.112		1.04	1.02	1.02	1.00	1.02	1.03	1.03	1.06	1.39			3.08	3.04	3.12
	velocity		5.32	8.79	8.96	8.96	9.14	8.96	8.87	8.87	8.62	10.09	8.33		8.90	9.02	8.79
Fenner, Mike (GER) (1971)	time	2.62	3.66	4.70	5.72	6.76	7.78	8.80	9.84	10.92	11.99		13.44	/ 2			
	reaction time	0.138		1.04	1.04	1.02	1.04	1.02	1.04	1.08	1.07	1.45			3.10	3.08	3.19
	velocity		5.24	8.79	8.79	8.96	8.79	8.96	8.96	8.79	8.46	8.54	9.67	8.18	8.85	8.90	8.60
Videnov, Zhivko (BUL) (1977)	time	2.64	3.68	4.70	5.74	6.76	7.80	8.82	9.88	10.94	12.04		13.45	/ 3			
	reaction time	0.128		1.04	1.02	1.04	1.02	1.04	1.02	1.06	1.06	1.10	1.41		3.10	3.08	3.22
	velocity		5.20	8.79	8.96	8.79	8.96	8.79	8.96	8.62	8.62	8.31	9.94	8.18	8.85	8.90	8.52
Kronberg, Robert (SWE) (1976)	time	2.58	3.64	4.66	5.72	6.74	7.80	8.86	9.92	10.99	12.06		13.53	/ 4			
	reaction time	0.103		1.06	1.02	1.06	1.02	1.06	1.06	1.06	1.07	1.07	1.47		3.14	3.14	3.20
	velocity		5.32	8.62	8.96	8.62	8.96	8.62	8.62	8.62	8.54	8.54	9.54	8.13	8.73	8.73	8.57

Semi-Final 1

date 10-Aug-02 wind -0.5 m/s Graubner (2009) - <http://www.fgs.uni-halle.de>

		H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
Olijar, Stanislavs (LAT) (1979)	time	2.60	3.63	4.65	5.67	6.68	7.71	8.75	9.79	10.85	11.94		13.34	/ 1			
	reaction time	0.143		1.03	1.02	1.02	1.01	1.03	1.04	1.04	1.06	1.09	1.40		3.07	3.08	3.19
	velocity		5.28	8.87	8.96	8.96	9.05	8.87	8.79	8.79	8.62	8.39	10.01	8.25	8.93	8.90	8.60
Kohutek, Artur (POL) (1971)	time	2.65	3.70	4.73	5.74	6.75	7.77	8.82	9.85	10.93	12.01		13.44	/ 2			
	reaction time	0.135		1.05	1.03	1.01	1.01	1.02	1.05	1.03	1.08	1.08	1.43		3.09	3.08	3.19
	velocity		5.18	8.70	8.87	9.05	9.05	8.96	8.70	8.87	8.46	8.46	9.80	8.18	8.87	8.90	8.60
Schwarthoff, Florian (GER) (1971)	time	2.63	3.71	4.73	5.75	6.79	7.83	8.86	9.89	10.97	12.03		13.52	/ 3			
	reaction time	0.149		1.08	1.02	1.02	1.04	1.04	1.03	1.03	1.08	1.06	1.49		3.12	3.11	3.17
	velocity		5.22	8.46	8.96	8.96	8.79	8.79	8.87	8.87	8.46	8.62	9.41	8.14	8.79	8.82	8.65

Heat 5

date 09-Aug-02 wind 0.0 m/s Graubner (2009) - <http://www.fgs.uni-halle.de>

		H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
Videnov, Zhivko (BUL) (1977)	time	2.69	3.72	4.73	5.77	6.79	7.84	8.88	9.93	11.01	12.11		13.52	/ 1			
	reaction time	0.139		1.03	1.01	1.04	1.02	1.05	1.04	1.05	1.08	1.10	1.41		3.08	3.11	3.23
	velocity		5.10	8.87	9.05	8.79	8.96	8.70	8.79	8.70	8.46	8.31	9.94	8.14	8.90	8.82	8.49
Lichtenegger, Elmar (AUT) (1971)	time	2.57	3.62	4.65	5.67	6.70	7.75	8.82	9.88	10.97	12.07		13.53	/ 2			
	reaction time	0.150		1.05	1.03	1.02	1.03	1.05	1.07	1.06	1.09	1.10	1.46		3.10	3.15	3.25
	velocity		5.34	8.70	8.87	8.96	8.87	8.70	8.54	8.62	8.39	8.31	9.60	8.13	8.85	8.70	8.44

Heat 4

date 09-Aug-02 wind -0.1 m/s Graubner (2009) - <http://www.fgs.uni-halle.de>

		H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
Olijar, Stanislavs (LAT) (1979)	time	2.69	3.71	4.73	5.76	6.79	7.81	8.87	9.93	10.99	12.07		13.50	/ 1			
	reaction time	0.157		1.02	1.02	1.03	1.03	1.02	1.06	1.06	1.06	1.08	1.43		3.07	3.11	3.20
	velocity		5.10	8.96	8.96	8.87	8.87	8.96	8.62	8.62	8.62	8.46	9.80	8.15	8.93	8.82	8.57
Jarrett, Tony (GBR) (1968)	time	2.66	3.72	4.78	5.80	6.86	7.90	8.96	10.01	11.09	12.19		13.63	/ 2			
	reaction time	0.153		1.06	1.06	1.02	1.06	1.04	1.06	1.05	1.08	1.10	1.44		3.14	3.16	3.23
	velocity		5.16	8.62	8.62	8.96	8.62	8.79	8.62	8.70	8.46	8.31	9.74	8.07	8.73	8.68	8.49

Heat 3

date 09-Aug-02 wind 2.0 m/s

Graubner (2009) - <http://www.fgs.uni-halle.de>

		H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
Giacconi, Andrea (ITA) (1974)	time	2.61	3.69	4.75	5.79	6.81	7.87	8.91	9.97	11.02	12.11		13.55	/ 1			
	reaction time	0.116													3.18	3.12	3.20
	interval		1.08	1.06	1.04	1.02	1.06	1.04	1.06	1.05	1.09	1.44			8.62	8.79	8.57
	velocity	5.26	8.46	8.62	8.79	8.96	8.62	8.79	8.62	8.70	8.39	9.74	8.12				
Lavanne, Cédric (FRA) (1980)	time	2.65	3.73	4.76	5.81	6.84	7.87	8.91	9.98	11.03	12.13		13.58	/ 2			
	reaction time	0.142													3.16	3.10	3.22
	interval		1.08	1.03	1.05	1.03	1.03	1.04	1.07	1.05	1.10	1.45	PB		8.68	8.85	8.52
	velocity	5.18	8.46	8.87	8.70	8.87	8.87	8.79	8.54	8.70	8.31	9.67	8.10				

Heat 2

date 09-Aug-02 wind 0.5 m/s

Graubner (2009) - <http://www.fgs.uni-halle.de>

		H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
Kohutek, Artur (POL) (1971)	time	2.68	3.74	4.78	5.78	6.80	7.83	8.85	9.92	10.98	12.05		13.49	/ 1			
	reaction time	0.152													3.10	3.07	3.20
	interval		1.06	1.04	1.00	1.02	1.03	1.02	1.07	1.06	1.07	1.44			8.85	8.93	8.57
	velocity	5.12	8.62	8.79	9.14	8.96	8.87	8.96	8.54	8.62	8.54	9.74	8.15				
Fenner, Mike (GER) (1971)	time	2.64	3.70	4.76	5.81	6.84	7.88	8.94	9.99	11.05	12.14		13.57	/ 2			
	reaction time	0.134													3.17	3.13	3.20
	interval		1.06	1.06	1.05	1.03	1.04	1.06	1.05	1.06	1.09	1.43			8.65	8.76	8.57
	velocity	5.20	8.62	8.62	8.70	8.87	8.79	8.62	8.70	8.62	8.39	9.80	8.11				

Heat 1

date 09-Aug-02 wind -0.9 m/s

Graubner (2009) - <http://www.fgs.uni-halle.de>

		H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
Jackson, Colin (GBR) (1967)	time	2.59	3.63	4.66	5.67	6.69	7.72	8.75	9.79	10.85	11.95		13.41	/ 1			
	reaction time	0.107													3.08	3.08	3.20
	interval		1.04	1.03	1.01	1.02	1.03	1.03	1.04	1.06	1.10	1.46			8.90	8.90	8.57
	velocity	5.30	8.79	8.87	9.05	8.96	8.87	8.87	8.79	8.62	8.31	9.60	8.20				
Schwarthoff, Florian (GER) (19)	time	2.62	3.69	4.74	5.75	6.75	7.80	8.85	9.89	10.97	12.05		13.53	/ 2			
	reaction time	0.135													3.13	3.10	3.20
	interval		1.07	1.05	1.01	1.00	1.05	1.05	1.04	1.08	1.08	1.48			8.76	8.85	8.57
	velocity	5.24	8.54	8.70	9.05	9.14	8.70	8.70	8.79	8.46	8.46	9.47	8.13				

2001 Chinese National Games (Guangzhou, CHN)

FINAL

date 19-Nov-01 wind 0.4 m/s

Huang (2002) - technical research on the men's 110m hurdles at the 9th national games

		H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
Chen Yanhao (CHN) (1972)	time	2.61	3.67	4.69	5.72	6.73	7.76	8.80	9.85	10.89	11.95		13.42	/ 2			
	reaction time														3.11	3.08	3.15
	interval		1.06	1.02	1.03	1.01	1.03	1.04	1.05	1.04	1.06	1.47			8.82	8.90	8.70
	velocity	5.26	8.62	8.96	8.87	9.05	8.87	8.79	8.70	8.79	8.62	9.54	8.20				

2001 IAAF World Championships (Edmonton, CAN)

FINAL

date 09-Aug-01 wind -0.3 m/s

		H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
Johnson, Allen (USA) (1971)	0.150	no information available											13.04	3 / 1			
Garcia, Anier (CUB) (1976)	0.150	no information available											13.07	4 / 2			
Dorival, Dudley (HAI) (1975)	0.134	no information available											13.25	8 / 3			
Hernández, Yoel (CUB) (1977)	0.151	no information available											13.30	6 / 4			
Kronberg, Robert (SWE) (1976)	0.136	no information available											13.51	1 / 5			
Pechonkin, Yevgeniy (RUS) (19)	0.131	no information available											13.52	5 / 6			
Wallace, Dawane (USA) (1976)	0.159	no information available											13.76	7 / 7			
Bownes, Shaun (RSA) (1970)	0.133	no information available											13.84	2 / 8			

2001 ISTAF (Berlin, GER)

FINAL

date 31-Aug-09 wind 1.0 m/s

Graubner (2009) - <http://www.fgs.uni-halle.de>

		H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
Johnson, Allen (USA) (1971)	time	2.58	3.59	4.60	5.58	6.58	7.58	8.58	9.60	10.62	11.66		13.04	4 / 1			
	reaction time	0.170													3.00	3.00	3.08
	interval		1.01	1.01	0.98	1.00	1.00	1.00	1.02	1.02	1.04	1.38			9.14	9.14	8.90
	velocity	5.32	9.05	9.05	9.33	9.14	9.14	9.14	8.96	8.96	8.79	10.16	8.44				
García, Anier (CUB) (1976)	time	2.57	3.60	4.61	5.60	6.61	7.64	8.66	9.68	10.69	11.76		13.15	5 / 2			
	reaction time	0.168													3.03	3.06	3.10
	interval		1.03	1.01	0.99	1.01	1.03	1.02	1.02	1.01	1.07	1.39			9.05	8.96	8.85
	velocity	5.34	8.87	9.05	9.23	9.05	8.87	8.96	8.96	9.05	8.54	10.09	8.37				
Arnold, Dominique (USA) (197)	time	2.58	3.60	4.63	5.67	6.68	7.67	8.69	9.70	10.74	11.80		13.21	3 / 3			
	reaction time	0.167													3.09	3.02	3.11
	interval		1.02	1.03	1.04	1.01	0.99	1.02	1.01	1.04	1.06	1.41			8.87	9.08	8.82
	velocity	5.32	8.96	8.87	8.79	9.05	9.23	8.96	9.05	8.79	8.62	9.94	8.33				
Crear, Mark (USA) (1968)		no information available											13.42	2 / 4			
Dorival, Dudley (HAI) (1975)		no information available											13.58	6 / 5			
Crews, Jerome (GER) (1977)	time	2.66	3.74	4.80	5.85	6.91	7.98	9.04	10.14	11.24	12.35		13.70	7 / 6			
	reaction time	0.180													3.19	3.19	3.31
	interval		1.08	1.06	1.05	1.06	1.07	1.06	1.10	1.10	1.11	1.35			8.60	8.60	8.28
	velocity	5.16	8.46	8.62	8.70	8.62	8.54	8.62	8.31	8.31	8.23	10.39	8.03				
Fenner, Mike (GER) (1971)	time	2.62	3.70	4.79	5.86	6.92	8.01	9.09	10.18	11.27	12.38		13.83	1 / 7			
	reaction time	0.140													3.24	3.23	3.29
	interval		1.08	1.09	1.07	1.06	1.09	1.08	1.09	1.09	1.11	1.45			8.46	8.49	8.33
	velocity	5.24	8.46	8.39	8.54	8.62	8.39	8.46	8.39	8.39	8.23	9.67	7.95				

Schwarthoff, Florian (GER) (19	time	2.68	3.76	4.80	5.82	6.84	7.90	8.94	10.00	10.10	12.20		13.85	7 / 8				
	reaction time	0.161	interval	7 steps	1.08	1.04	1.02	1.02	1.06	1.04	1.06	0.10	2.10	1.65		3.14	3.12	3.26
			velocity		5.12	8.46	8.79	8.96	8.96	8.62	8.79	8.62	91.40	4.35	8.50	7.94	8.73	8.79

2001 European Cup (Bremen, GER)

FINAL

date 24-Jun-01 wind 1.3 m/s

Graubner (2007) - <http://www.fgs.uni-halle.de>

				H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
Pechonkin, Yevgeniy (RUS) (19	time			2.61	3.66	4.68	5.69	6.71	7.76	8.79	9.83	10.90	11.98		13.38	8 / 1			
	reaction time	0.128	interval		1.05	1.02	1.01	1.02	1.05	1.03	1.04	1.07	1.08	1.40			3.08	3.10	3.19
	velocity			5.26	8.70	8.96	9.05	8.96	8.70	8.87	8.79	8.54	8.46	10.01	8.22		8.90	8.85	8.60
Schwarthoff, Florian (GER) (19	time			2.60	3.65	4.74	5.81	6.85	7.87	8.91	9.96	11.06	12.14		13.57	3 / 2			
	reaction time	0.129	interval	7 steps	1.05	1.09	1.07	1.04	1.02	1.04	1.05	1.10	1.08	1.43			3.21	3.10	3.23
	velocity			5.28	8.70	8.39	8.54	8.79	8.96	8.79	8.70	8.31	8.46	9.80	8.11		8.54	8.85	8.49
Jarrett, Tony (GBR) (1968)	time			2.64	3.70	4.76	5.80	6.84	7.91	8.95	10.01	11.10	12.17		13.58	4 / 3			
	reaction time	0.150	interval		1.06	1.06	1.04	1.04	1.07	1.04	1.06	1.09	1.07	1.41			3.16	3.15	3.22
	velocity			5.20	8.62	8.62	8.79	8.79	8.54	8.79	8.62	8.39	8.54	9.94	8.10		8.68	8.70	8.52
Pizzoli, Emiliano (ITA) (1974)	time			2.63	3.73	4.79	5.83	6.88	7.93	8.98	10.04	11.14	12.25		13.71	6 / 4			
	reaction time	0.138	interval		1.10	1.06	1.04	1.05	1.05	1.05	1.06	1.10	1.11	1.46			3.20	3.15	3.27
	velocity			5.22	8.31	8.62	8.79	8.70	8.70	8.70	8.62	8.31	8.23	9.60	8.02		8.57	8.70	8.39
Mehlich, Krzysztof (POL) (1974	time			2.63	3.72	4.78	5.85	6.92	8.00	9.08	10.16	11.26	12.39		13.86	5 / 5			
	reaction time	0.157	interval	7 steps	1.09	1.06	1.07	1.07	1.08	1.08	1.08	1.10	1.13	1.47			3.22	3.23	3.31
	velocity			5.22	8.39	8.62	8.54	8.54	8.46	8.46	8.46	8.31	8.09	9.54	7.94		8.52	8.49	8.28
Montesinos, Hipólito (ESP) (19	time			2.66	3.72	4.78	5.84	6.91	8.00	9.08	10.18	11.28	12.42		13.88	2 / 6			
	reaction time	0.144	interval		1.06	1.06	1.06	1.07	1.09	1.08	1.10	1.10	1.14	1.46			3.18	3.24	3.34
	velocity			5.16	8.62	8.62	8.62	8.54	8.39	8.46	8.31	8.31	8.02	9.60	7.93		8.62	8.46	8.21
Doucouré, Ladji (FRA) (1983)	time			2.70	3.78	4.82	5.84	6.86	7.91	8.98	10.03	11.17	12.44		14.02	7 / 7			
	reaction time	0.164	interval		1.08	1.04	1.02	1.02	1.05	1.07	1.05	1.14	1.27	1.58			3.14	3.14	3.46
	velocity			5.08	8.46	8.79	8.96	8.96	8.70	8.54	8.70	8.02	7.20	8.87	7.85		8.73	8.73	7.92
Piétris, Dimitrios (GRE) (1979)	time			2.78	3.87	4.93	6.02	7.09	8.17	9.23	10.31	11.45	12.56		14.04	1 / 8			
	reaction time	0.155	interval		1.09	1.06	1.09	1.07	1.08	1.06	1.08	1.14	1.11	1.48			3.24	3.21	3.33
	velocity			4.94	8.39	8.62	8.39	8.54	8.46	8.62	8.46	8.02	8.23	9.47	7.83		8.46	8.54	8.23

2001 Chinese National Grand Prix (Shanghai, CHN)

FINAL

date 06-May-01 wind 1.9 m/s

Fang (2001) - analysis on technical characteristics in Liu Xiang's 110m hurdles

		H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10	
Liu Xiang (CHN) (1983)	time	2.63	3.73	4.73	5.73	6.75	7.77	8.82	9.87	10.92	11.98		13.32	/ 1				
	reaction time			1.10	1.00	1.00	1.02	1.05	1.05	1.05	1.06	1.34	WYB 17 / PB		3.10	3.09	3.16	
	interval	5.22	8.31	9.14	9.14	8.96	8.96	8.70	8.70	8.70	8.62	10.46	8.26		8.85	8.87	8.68	
Chen Yanhao (CHN) (1972)	time	2.64	3.69	4.75	5.78	6.81	7.83	8.86	9.90	10.97	12.02		13.37	/ 2				
	reaction time			1.05	1.06	1.03	1.03	1.02	1.03	1.04	1.07	1.05	1.35	PB		3.14	3.08	3.16
	interval	5.20	8.70	8.62	8.87	8.87	8.96	8.87	8.79	8.54	8.70	10.39	8.23		8.73	8.90	8.68	
Shi Dongpeng (CHN) (1984)	time	2.71	3.78	4.79	5.81	6.83	7.86	8.87	9.91	10.99	12.07		13.43	/ 3				
	reaction time			1.07	1.01	1.02	1.02	1.03	1.01	1.04	1.08	1.08	1.36	WYR NR PB		3.10	3.06	3.20
	interval	5.06	8.54	9.05	8.96	8.96	8.87	9.05	8.79	8.46	8.46	10.31	8.19		8.85	8.96	8.57	
Qi Zhen (CHN) (1976)	time	2.66	3.77	4.86	5.90	6.93	8.00	9.07	10.13	11.29	12.45		13.95	/ 5				
	reaction time			1.11	1.09	1.04	1.03	1.07	1.07	1.06	1.16	1.16	1.50		3.24	3.17	3.38	
	interval	5.16	8.23	8.39	8.79	8.87	8.54	8.54	8.62	7.88	7.88	9.35	7.89		8.46	8.65	8.11	

2001 Chinese National Grand Prix (Ningbo, CHN)

FINAL

date 28-Apr-01 wind 2.5 m/s

Fang (2001) - analysis on technical characteristics in Liu Xiang's 110m hurdles

		H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
Liu Xiang (CHN) (1983)	time	2.66	3.74	4.78	5.82	6.85	7.88	8.90	9.92	10.97	12.04		13.45	/ 1			
	reaction time	interval		1.08	1.04	1.04	1.03	1.03	1.02	1.02	1.05	1.07	1.41		3.16	3.08	3.14
	velocity			5.16	8.46	8.79	8.79	8.87	8.87	8.96	8.96	8.70	8.54	9.94	8.18	8.68	8.90
Chen Yanhao (CHN) (1972)	time	2.66	3.74	4.78	5.83	6.87	7.89	8.93	9.97	11.02	12.06		13.47	/ 2			
	reaction time	interval		1.08	1.04	1.05	1.04	1.02	1.04	1.04	1.05	1.04	1.41	PB	3.17	3.10	3.13
	velocity			5.16	8.46	8.79	8.70	8.79	8.96	8.79	8.79	8.70	8.79	9.94	8.17	8.65	8.85

2000 Olympic Games (Sydney, AUS)

FINAL

date 25-Sep-00 wind 0.6 m/s

		H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
García, Anier (CUB) (1976)	0.172	no information available										NR	13.00	3 / 1			
Trammel, Terrence (USA) (1978	0.221	no information available										PB	13.16	4 / 2			
Crear, Mark (USA) (1968)	0.214	no information available											13.22	6 / 3			
Johnson, Allen (USA) (1971)	0.182	no information available											13.23	5 / 4			

Jackson, Colin (GBR) (1967)	0.139	no information available
Schwarthoff, Florian (GER) (19	0.184	no information available
Dorival, Dudley (HAI) (1975)	0.161	no information available
Kronberg, Robert (SWE) (1976	0.244	no information available

13.28	1 / 5
13.42	7 / 6
13.49	2 / 7
13.61	8 / 8

2000 ISTAF (Berlin, GER)

FINAL

date 01-Sep-00

wind -0.4 m/s

Graubner (2009) - <http://www.fgs.uni-halle.de>

		H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
Trammel, Terrence (USA) (1978	time	2.54	3.59	4.62	5.63	6.66	7.68	8.72	9.74	10.78	11.85		13.28	/ 1			
	reaction time	0.154													3.09	3.09	3.13
	interval		1.05	1.03	1.01	1.03	1.02	1.04	1.02	1.04	1.07	1.43			8.87	8.87	8.76
	velocity	5.40	8.70	8.87	9.05	8.87	8.96	8.79	8.96	8.79	8.54	9.80	8.28				
Arnold, Dominique (USA) (197	time	2.55	3.63	4.66	5.68	6.70	7.73	8.76	9.78	10.82	11.88		13.31	/ 2			
	reaction time	0.157													3.13	3.08	3.12
	interval		1.08	1.03	1.02	1.02	1.03	1.03	1.02	1.04	1.06	1.43			8.76	8.90	8.79
	velocity	5.38	8.46	8.87	8.96	8.96	8.87	8.87	8.96	8.79	8.62	9.80	8.26				
Schwarthoff, Florian (GER) (19	time	2.56	3.62	4.68	5.70	6.70	7.75	8.78	9.82	10.86	11.96		13.40	/ 3			
	reaction time	0.148													3.14	3.08	3.18
	interval		1.06	1.06	1.02	1.00	1.05	1.03	1.04	1.04	1.10	1.44			8.73	8.90	8.62
	velocity	5.36	8.62	8.62	8.96	9.14	8.70	8.87	8.79	8.79	8.31	9.74	8.21				
Balzer, Falk (GER) (1973)	time	2.60	3.67	4.70	5.72	6.72	7.73	8.78	9.80	10.88	12.00		13.47	/ 4			
	reaction time	0.158													3.12	3.06	3.22
	interval		1.07	1.03	1.02	1.00	1.01	1.05	1.02	1.08	1.12	1.47			8.79	8.96	8.52
	velocity	5.28	8.54	8.87	8.96	9.14	9.05	8.70	8.96	8.46	8.16	9.54	8.17				

1999 IAAF World Championships (Sevilla, ESP)

FINAL

date 25-Aug-99

wind 1.0 m/s

Sanchez (1999) - Sevilla '99: análisis de la carreras con villas

		H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
Jackson, Colin (GBR) (1967)	time	2.51	3.52	4.52	5.50	6.51	7.52	8.52	9.55	10.58	11.64		13.04	5 / 1			
	reaction time	0.116													2.99	3.02	3.12
	interval		1.01	1.00	0.98	1.01	1.01	1.00	1.03	1.03	1.06	1.40			9.17	9.08	8.79
	velocity	5.47	9.05	9.14	9.33	9.05	9.05	9.14	8.87	8.87	8.62	10.01	8.44				
Garcia, Anier (CUB) (1976)	time	2.52	3.53	4.54	5.53	6.53	7.52	8.51	9.53	10.59	11.65		13.07	3 / 2			
	reaction time	0.158											AR		3.01	2.98	3.14
	interval		1.01	1.01	0.99	1.00	0.99	0.99	1.02	1.06	1.06	1.42			9.11	9.20	8.73
	velocity	5.44	9.05	9.05	9.23	9.14	9.23	9.23	8.96	8.62	8.62	9.87	8.42				
Ross, Duane (USA) (1972)	time	2.47	3.49	4.51	5.53	6.52	7.54	8.56	9.58	10.61	11.68		13.12	4 / 3			
	reaction time	0.141											PB		3.06	3.03	3.12
	interval		1.02	1.02	1.02	0.99	1.02	1.02	1.02	1.03	1.07	1.44			8.96	9.05	8.79
	velocity	5.55	8.96	8.96	8.96	9.23	8.96	8.96	8.96	8.87	8.54	9.74	8.38				
Dees, Tony (USA) (1963)	time	2.47	3.51	4.53	5.52	6.53	7.53	8.56	9.58	10.64	11.72		13.22	8 / 4			
	reaction time	0.148													3.05	3.04	3.16
	interval		1.04	1.02	0.99	1.01	1.00	1.03	1.02	1.06	1.08	1.50			8.99	9.02	8.68
	velocity	5.55	8.79	8.96	9.23	9.05	9.14	8.87	8.96	8.62	8.46	9.35	8.32				
Balzer, Falk (GER) (1973)	time	2.60	3.64	4.65	5.67	6.69	7.71	8.71	9.72	10.76	11.84		13.26	2 / 5			
	reaction time	0.153													3.07	3.04	3.13
	interval		1.04	1.01	1.02	1.02	1.02	1.00	1.01	1.04	1.08	1.42			8.93	9.02	8.76
	velocity	5.28	8.79	9.05	8.96	8.96	8.96	9.14	9.05	8.79	8.46	9.87	8.30				
Hernández, Yoel (CUB) (1977)	time	2.57	3.61	4.63	5.66	6.65	7.65	8.68	9.71	10.73	11.83		13.30	7 / 6			
	reaction time	0.129													3.09	3.02	3.15
	interval		1.04	1.02	1.03	0.99	1.00	1.03	1.03	1.02	1.10	1.47			8.87	9.08	8.70
	velocity	5.34	8.79	8.96	8.87	9.23	9.14	8.87	8.87	8.96	8.31	9.54	8.27				
Schwarthoff, Florian (GER) (19	time	2.56	3.59	4.64	5.64	6.67	7.67	8.69	9.73	10.80	11.89		13.39	6 / 7			
	reaction time	0.132													3.08	3.05	3.20
	interval		7 steps	1.03	1.05	1.00	1.03	1.00	1.02	1.04	1.07	1.09	1.50		8.90	8.99	8.57
	velocity	5.36	8.87	8.70	9.14	8.87	9.14	8.96	8.79	8.54	8.39	9.35	8.22				
Nsenga, Jonathan (BEL) (1973)	time	2.61	3.65	4.72	5.77	6.80	7.86	8.90	9.93	11.01	12.08		13.54	1 / 8			
	reaction time	0.142													3.16	3.13	3.18
	interval		1.04	1.07	1.05	1.03	1.06	1.04	1.03	1.08	1.07	1.46			8.68	8.76	8.62
	velocity	5.26	8.79	8.54	8.70	8.87	8.62	8.79	8.87	8.46	8.54	9.60	8.12				

Semi-Final 2

date 24-Aug-99

wind 0.0 m/s

Sanchez (1999) - Sevilla '99: análisis de la carreras con villas

		H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
Garcia, Anier (CUB) (1976)	time	2.51	3.55	4.55	5.55	6.55	7.54	8.56	9.60	10.64	11.72		13.18	5 / 1			
	reaction time	0.139													3.04	3.01	3.16
	interval		1.04	1.00	1.00	1.00	0.99	1.02	1.04	1.04	1.08	1.46			9.02	9.11	8.68
	velocity	5.47	8.79	9.14	9.14	9.14	9.23	8.96	8.79	8.79	8.46	9.60	8.35				
Jackson, Colin (GBR) (1967)	time	2.50	3.52	4.53	5.54	6.54	7.56	8.57	9.60	10.66	11.72		13.19	6 / 2			
	reaction time	0.110													3.04	3.03	3.15
	interval		1.02	1.01	1.01	1.00	1.02	1.01	1.03	1.06	1.06	1.47			9.02	9.05	8.70
	velocity	5.49	8.96	9.05	9.05	9.14	8.96	9.05	8.87	8.62	8.62	9.54	8.34				
Dees, Tony (USA) (1963)	time	2.43	3.45	4.49	5.49	6.50	7.51	8.54	9.60	10.70	11.81		13.30	4 / 3			
	reaction time	0.123													3.06	3.05	3.27
	interval		1.02	1.04	1.00	1.01	1.01	1.03	1.06	1.10	1.11	1.49			8.96	8.99	8.39
	velocity	5.65	8.96	8.79	9.14	9.05	9.05	8.87	8.62	8.31	8.23	9.41	8.27				
Nsenga, Jonathan (BEL) (1973)	time	2.56	3.61	4.66	5.70	6.71	7.74	8.77	9.86	10.91	12.00		13.47	7 / 4			
	reaction time	0.134													3.14	3.07	3.23
	interval		1.05	1.05	1.04	1.01	1.03	1.03	1.09	1.05	1.09	1.47			8.73	8.93	8.49
	velocity	5.36	8.70	8.70	8.79	9.05	8.87	8.87	8.39	8.70	8.39	9.54	8.17				

Philibert, Dan (FRA) (1970)	time	2.58	3.59	4.62	5.64	6.65	7.66	8.70	9.77	10.84	11.92		13.49	8 / 5			
	reaction time	0.113		1.01	1.03	1.02	1.01	1.01	1.04	1.07	1.07	1.08	1.57		3.06	3.06	3.22
	velocity		5.32	9.05	8.87	8.96	9.05	9.05	8.79	8.54	8.54	8.46	8.93	8.15	8.96	8.96	8.52
Olijar, Stanislav (CZE) (1979)	time	2.59	3.64	4.67	5.67	6.68	7.70	8.73	9.81	10.95	12.09		13.70	3 / 6			
	reaction time	0.137		1.05	1.03	1.00	1.01	1.02	1.03	1.08	1.14	1.14	1.61		3.08	3.06	3.36
	velocity		5.30	8.70	8.87	9.14	9.05	8.96	8.87	8.46	8.02	8.02	8.71	8.03	8.90	8.96	8.16
Dorival, Dudley (HAI) (1975)	time	2.60	3.62	4.65	5.70	6.74	7.82	8.87	9.96	11.05	12.20		13.86	2 / 7			
	reaction time	0.151		1.02	1.03	1.05	1.04	1.08	1.05	1.09	1.09	1.15	1.66		3.10	3.17	3.33
	velocity		5.28	8.96	8.87	8.70	8.79	8.46	8.70	8.39	8.39	7.95	8.45	7.94	8.85	8.65	8.23

Semi-Final 1

date 24-Aug-99		wind 0.0 m/s		Sanchez (1999) - Sevilla '99: análisis de la carreras con villas													
Ross, Duane (USA) (1972)	time	2.53	3.56	4.56	5.58	6.57	7.56	8.59	9.60	10.63	11.68		13.14	3 / 1			
	reaction time	0.151		1.03	1.00	1.02	0.99	0.99	1.03	1.01	1.03	1.05	1.46		3.05	3.01	3.09
	velocity		5.42	8.87	9.14	8.96	9.23	9.23	8.87	9.05	8.87	8.70	9.60	8.37	8.99	9.11	8.87
Schwarthoff, Florian (GER) (1979)	time	2.52	3.56	4.59	5.59	6.60	7.61	8.61	9.64	10.67	11.75		13.30	8 / 2			
	reaction time	0.135		1.04	1.03	1.00	1.01	1.01	1.00	1.03	1.03	1.08	1.55		3.07	3.02	3.14
	velocity		5.44	8.79	8.87	9.14	9.05	9.05	9.14	8.87	8.87	8.46	9.05	8.27	8.93	9.08	8.73
Hernández, Yoel (CUB) (1977)	time	2.60	3.61	4.64	5.64	6.66	7.65	8.68	9.71	10.77	11.83		13.32	4 / 3			
	reaction time	0.143		1.01	1.03	1.00	1.02	0.99	1.03	1.03	1.06	1.06	1.49		3.04	3.04	3.15
	velocity		5.28	9.05	8.87	9.14	8.96	9.23	8.87	8.87	8.62	8.62	9.41	8.26	9.02	9.02	8.70
Balzer, Falk (GER) (1973)	time	2.58	3.61	4.62	5.63	6.65	7.66	8.70	9.74	10.80	11.88		13.32	6 / 4			
	reaction time	0.136		1.03	1.01	1.01	1.02	1.01	1.04	1.04	1.06	1.08	1.44		3.05	3.07	3.18
	velocity		5.32	8.87	9.05	9.05	8.96	9.05	8.79	8.79	8.62	8.46	9.74	8.26	8.99	8.93	8.62
Coghlan, Peter (IRL) (1975)	time	2.59	3.62	4.67	5.67	6.70	7.72	8.72	9.78	10.86	11.97		13.35	1 / 5			
	reaction time	0.161		1.03	1.05	1.00	1.03	1.02	1.00	1.06	1.08	1.11	1.38		3.08	3.05	3.25
	velocity		5.30	8.87	8.70	9.14	8.87	8.96	9.14	8.62	8.46	8.23	10.16	8.24	8.90	8.99	8.44
Batte, Erick (CUB) (1974)	time	2.54	3.58	4.60	5.64	6.64	7.66	8.67	9.72	10.76	11.86		13.40	7 / 6			
	reaction time	0.136		1.04	1.02	1.04	1.00	1.02	1.01	1.05	1.04	1.10	1.54		3.10	3.03	3.19
	velocity		5.40	8.79	8.96	8.79	9.14	8.96	9.05	8.70	8.79	8.31	9.10	8.21	8.85	9.05	8.60
Korving, Robin (NED) (1974)	time	2.59	3.62	4.66	5.67	6.72	7.74	8.78	9.84	10.91	11.96		13.45	2 / 7			
	reaction time	0.133		1.03	1.04	1.01	1.05	1.02	1.04	1.06	1.07	1.05	1.49		3.08	3.11	3.18
	velocity		5.30	8.87	8.79	9.05	8.70	8.96	8.79	8.62	8.54	8.70	9.41	8.18	8.90	8.82	8.62
Lichteneggar, Elmar (AUT) (1976)	time	2.55	3.57	4.62	5.66	6.69	7.73	8.78	9.85	10.90	12.00		13.52	5 / 8			
	reaction time	0.133		1.02	1.05	1.04	1.03	1.04	1.05	1.07	1.05	1.10	1.52		3.11	3.12	3.22
	velocity		5.38	8.96	8.70	8.79	8.87	8.79	8.70	8.54	8.70	8.31	9.22	8.14	8.82	8.79	8.52

Quarter-Final 2

date 23-Aug-99		wind -0.2 m/s		Sanchez (1999) - Sevilla '99: análisis de la carreras con villas													
Jackson, Colin (GBR) (1967)	time	2.50	3.52	4.53	5.53	6.52	7.53	8.54	9.57	10.63	11.71		13.21	6 / 1			
	reaction time	0.117		1.02	1.01	1.00	0.99	1.01	1.01	1.03	1.06	1.08	1.50		3.03	3.01	3.17
	velocity		5.49	8.96	9.05	9.14	9.23	9.05	9.05	8.87	8.62	8.46	9.35	8.33	9.05	9.11	8.65

Quarter-Final 1

date 23-Aug-99		wind -0.4 m/s		Sanchez (1999) - Sevilla '99: análisis de la carreras con villas													
Ross, Duane (USA) (1972)	time	2.52	3.55	4.56	5.57	6.56	7.58	8.58	9.59	10.63	11.68		13.13	5 / 1			
	reaction time	0.139		1.03	1.01	1.01	0.99	1.02	1.00	1.01	1.04	1.05	1.45	PB	3.05	3.01	3.10
	velocity		5.44	8.87	9.05	9.05	9.23	8.96	9.14	9.05	8.79	8.70	9.67	8.38	8.99	9.11	8.85
Garcia, Anier (CUB) (1976)	time	2.54	3.57	4.58	5.58	6.58	7.58	8.59	9.62	10.68	11.74		13.20	3 / 2			
	reaction time	0.169		1.03	1.01	1.00	1.00	1.01	1.03	1.06	1.06	1.46			3.04	3.01	3.15
	velocity		5.40	8.87	9.05	9.14	9.14	9.05	8.87	8.62	8.62	9.60	8.33		9.02	9.11	8.70

Heat 6

date 23-Aug-99		wind 0.0 m/s		Sanchez (1999) - Sevilla '99: análisis de la carreras con villas													
Garcia, Anier (CUB) (1976)	time	2.53	3.54	4.54	5.55	6.55	7.56	8.59	9.63	10.71	11.80		13.40	1 / 1			
	reaction time	0.155		1.01	1.00	1.01	1.00	1.01	1.03	1.04	1.08	1.09	1.60		3.02	3.04	3.21
	velocity		5.42	9.05	9.14	9.05	9.14	9.05	8.87	8.79	8.46	8.39	8.76	8.21	9.08	9.02	8.54

Heat 5

date 23-Aug-99		wind -0.5 m/s		Sanchez (1999) - Sevilla '99: análisis de la carreras con villas													
Jackson, Colin (GBR) (1967)	time	2.49	3.50	4.50	5.50	6.51	7.52	8.55	9.57	10.61	11.67		13.19	1 / 1			
	reaction time	0.115		1.01	1.00	1.00	1.01	1.01	1.03	1.02	1.04	1.06	1.52		3.01	3.05	3.12
	velocity		5.51	9.05	9.14	9.14	9.05	9.05	8.87	8.96	8.79	8.62	9.22	8.34	9.11	8.99	8.79

Heat 1

date 23-Aug-99		wind 0.6 m/s		Sanchez (1999) - Sevilla '99: análisis de la carreras con villas													
----------------	--	--------------	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

		H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
Ross, Duane (USA) (1972)	time	2.53	3.55	4.57	5.58	6.59	7.64	8.69	9.77	10.82	11.89		13.39	1 / 1			
	reaction time	0.162	interval	1.02	1.02	1.01	1.01	1.05	1.05	1.08	1.05	1.07	1.50		3.05	3.11	3.20
	velocity	5.42	8.96	8.96	9.05	9.05	8.70	8.70	8.46	8.70	8.54	9.35	8.22		8.99	8.82	8.57

1999 Sydney Grand Prix (Sydney, AUS)

FINAL

date 20-Feb-99 wind 0.6 m/s

Arnold (2009) - UK Athletics - Hurdling Masterclass

		H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
Jackson, Colin (GBR) (1967)	time	2.57	3.62	4.66	5.68	6.70	7.72	8.76	9.79	10.85	11.92		13.30	/ 1			
	reaction time		interval	1.05	1.04	1.02	1.02	1.04	1.03	1.06	1.07	1.38			3.11	3.08	3.16
	velocity	5.34	8.70	8.79	8.96	8.96	8.96	8.79	8.87	8.62	8.54	10.16	8.27		8.82	8.90	8.68

1998 European Championships (Budapest, HUN)

FINAL

date 22-Aug-98 wind 1.5 m/s

Sanchez (1998) - Budapest '98: análisis de la carreras con villas

		H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
Jackson, Colin (GBR) (1967)	time	2.56	3.58	4.60	5.58	6.59	7.59	8.57	9.59	10.63	11.69		13.02	3 / 1			
	reaction time		interval	1.02	1.02	0.98	1.01	1.00	0.98	1.02	1.04	1.06	1.33	CR	3.02	2.99	3.12
	velocity	5.36	8.96	8.96	9.33	9.05	9.14	9.33	8.96	8.79	8.62	10.54	8.45		9.08	9.17	8.79
Balzer, Falk (GER) (1973)	time	2.60	3.62	4.63	5.64	6.67	7.67	8.68	9.68	10.71	11.77		13.12	6 / 2			
	reaction time		interval	1.02	1.01	1.01	1.03	1.00	1.01	1.00	1.03	1.06	1.35	PB	3.04	3.04	3.09
	velocity	5.28	8.96	9.05	9.05	8.87	9.14	9.05	9.14	8.87	8.62	10.39	8.38		9.02	9.02	8.87
Korving, Robin (NED) (1974)	time	2.60	3.62	4.66	5.67	6.68	7.70	8.70	9.72	10.77	11.81		13.20	5 / 3			
	reaction time		interval	1.02	1.04	1.01	1.01	1.02	1.00	1.02	1.05	1.04	1.39	NR	3.07	3.03	3.11
	velocity	5.28	8.96	8.79	9.05	9.05	8.96	9.14	8.96	8.70	8.79	10.09	8.33		8.93	9.05	8.82
Schwarthoff, Florian (GER) (1971)	time	2.59	3.64	4.68	5.68	6.70	7.70	8.71	9.75	10.78	11.83		13.23	4 / 4			
	reaction time		interval	1.05	1.04	1.00	1.02	1.00	1.01	1.04	1.03	1.05	1.40		3.09	3.03	3.12
	velocity	5.30	8.70	8.79	9.14	8.96	9.14	9.05	8.79	8.87	8.70	10.01	8.31		8.87	9.05	8.79
Kohutek, Artur (POL) (1971)	time	2.60	3.62	4.65	5.67	6.67	7.70	8.72	9.73	10.75	11.83		13.29	7 / 5			
	reaction time		interval	1.02	1.03	1.02	1.00	1.03	1.02	1.01	1.02	1.08	1.46		3.07	3.05	3.11
	velocity	5.28	8.96	8.87	8.96	9.14	8.87	8.96	9.05	8.96	8.46	9.60	8.28		8.93	8.99	8.82
Jarrett, Tony (GBR) (1968)	time	2.58	3.61	4.66	5.69	6.74	7.74	8.74	9.76	10.80	11.89		13.32	8 / 6			
	reaction time		interval	1.03	1.05	1.03	1.05	1.00	1.00	1.02	1.04	1.09	1.43		3.11	3.05	3.15
	velocity	5.32	8.87	8.70	8.87	8.70	9.14	9.14	8.96	8.79	8.39	9.80	8.26		8.82	8.99	8.70
Fenner, Mike (GER) (1971)	time	2.61	3.63	4.68	5.72	6.74	7.77	8.82	9.85	10.88	11.97		13.38	2 / 7			
	reaction time		interval	1.02	1.05	1.04	1.02	1.03	1.05	1.03	1.03	1.09	1.41		3.11	3.10	3.15
	velocity	5.26	8.96	8.70	8.79	8.96	8.87	8.70	8.87	8.87	8.39	9.94	8.22		8.82	8.85	8.70
Nsenga, Jonathan (BEL) (1973)	time	2.62	3.66	4.71	5.76	6.81	7.83	8.87	9.94	11.02	12.12		13.54	1 / 8			
	reaction time		interval	1.04	1.05	1.05	1.02	1.04	1.07	1.08	1.10	1.42			3.14	3.11	3.25
	velocity	5.24	8.79	8.70	8.70	8.70	8.96	8.79	8.54	8.46	8.31	9.87	8.12		8.73	8.82	8.44

1998 Spanish National Championships (San Sebastián, ESP)

FINAL

date 02-Aug-98 wind 0.1 m/s

Ruiz (1998) - campeonato des españa absoluto: análisis de la carreras con villas

		H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
López, Francisco (ESP) (1973)	time	2.64	3.72	4.82	5.94	6.98	8.05	9.12	10.25	11.37	12.52		14.12	5 / 1			
	reaction time		interval	1.08	1.10	1.12	1.04	1.07	1.07	1.13	1.12	1.15	1.60		3.30	3.18	3.40
	velocity	5.20	8.46	8.31	8.16	8.79	8.54	8.54	8.09	8.16	7.95	8.76	7.79		8.31	8.62	8.06
Santos, Miguel (ESP) (1973)	time	2.66	3.75	4.85	5.98	7.04	8.16	9.28	10.38	11.50	12.63		14.17	6 / 2			
	reaction time		interval	1.09	1.10	1.13	1.06	1.12	1.12	1.10	1.12	1.13	1.54		3.32	3.30	3.35
	velocity	5.16	8.39	8.31	8.09	8.62	8.16	8.16	8.31	8.16	8.09	9.10	7.76		8.26	8.31	8.19
Vega, Javier (ESP) (1975)	time	2.66	3.75	4.84	5.95	7.02	8.14	9.24	10.37	11.50	12.64		14.21	7 / 3			
	reaction time		interval	1.09	1.09	1.11	1.07	1.12	1.10	1.13	1.13	1.14	1.57		3.29	3.29	3.40
	velocity	5.16	8.39	8.39	8.23	8.54	8.16	8.31	8.09	8.09	8.02	8.93	7.74		8.33	8.33	8.06
Pérez, Eduardo (ESP) (1970)	time	2.69	3.76	4.89	6.03	7.08	8.20	9.32	10.45	11.58	12.78		14.33	1 / 4			
	reaction time		interval	1.07	1.13	1.14	1.05	1.12	1.13	1.13	1.20	1.55			3.34	3.29	3.46
	velocity	5.10	8.54	8.09	8.02	8.70	8.16	8.16	8.09	8.09	7.62	9.05	7.68		8.21	8.33	7.92
Belmar, Félix (ESP) (1978)	time	2.70	3.82	4.96	6.06	7.20	8.36	9.51	10.70	11.86	13.08		14.73	8 / 5			
	reaction time		interval	1.12	1.14	1.10	1.14	1.16	1.15	1.19	1.16	1.22	1.65		3.36	3.45	3.57
	velocity	5.08	8.16	8.02	8.31	8.02	7.88	7.95	7.68	7.88	7.49	8.50	7.47		8.16	7.95	7.68

1998 ??? (Chengdu, CHN)

FINAL

date 12-May-98 wind ???

Yuan (1998) - Chen Yanhao analysis of 110m results

		H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
Chen Yanhao (CHN) (1972)	time	2.56	3.60	4.64	5.67	6.70	7.72	8.84	9.87	10.92	11.99		13.50	/ 1			
	reaction time		interval	1.04	1.04	1.03	1.02	1.12	1.03	1.05	1.07	1.51			3.11	3.17	3.15
	velocity	5.36	8.79	8.79	8.87	8.87	8.96	8.16	8.87	8.70	8.54	9.28	8.15		8.82	8.65	8.70

1997 Chinese National Games (Shanghai, CHN)

FINAL

date 19-Oct-97 wind 0.0 m/s

Zhang (1999) - analysis of 110m hurdles of Chen Yanhao

		H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
Chen Yanhao (CHN) (1972)	time	2.52	3.57	4.63	5.66	6.67	7.70	8.82	9.84	10.99	11.96		13.41	/ 1			
	reaction time		1.05	1.06	1.03	1.01	1.03	1.12	1.02	1.15	0.97	1.45			3.14	3.16	3.14
	velocity	5.44	8.70	8.62	8.87	9.05	8.87	8.16	8.96	7.95	9.42	9.67	8.20		8.73	8.68	8.73

1997 IAAF World Championships (Athens, GRE)

FINAL

date 07-Aug-97 wind 0.0 m/s

Hommel (1997) - biomechanical research project at the VIth world championships in athletics, Athens 1997

		H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
Johnson, Allen (USA) (1971)	time	2.56	3.56	4.54	5.54	6.52	7.54	8.52	9.57	10.59	11.62		12.93	4 / 1			
	reaction time	0.128	1.00	0.98	1.00	0.98	1.02	0.98	1.05	1.02	1.03	1.31			2.98	2.98	3.10
	interval																
	velocity	5.36	9.14	9.33	9.14	9.33	8.96	9.33	8.70	8.96	8.87	10.70	8.51		9.20	9.20	8.85
Jackson, Colin (GBR) (1967)	time	2.58	3.62	4.62	5.62	6.62	7.62	8.62	9.64	10.68	11.72		13.05	5 / 2			
	reaction time	0.118	1.04	1.00	1.00	1.00	1.00	1.00	1.02	1.04	1.04	1.33			3.04	3.00	3.10
	interval																
	velocity	5.32	8.79	9.14	9.14	9.14	9.14	9.14	8.96	8.79	8.79	10.54	8.43		9.02	9.14	8.85
Kováč, Igor (SVK) (1969)	time	2.62	3.68	4.70	5.72	6.72	7.74	8.78	9.78	10.82	11.86		13.18	1 / 3			
	reaction time	0.161	1.06	1.02	1.02	1.00	1.02	1.04	1.00	1.04	1.04	1.32			3.10	3.06	3.08
	interval																
	velocity	5.24	8.62	8.96	8.96	9.14	8.96	8.79	9.14	8.79	8.79	10.62	8.35		8.85	8.96	8.90
Schwarthoff, Florian (GER) (1971)	time	2.58	3.64	4.64	5.66	6.68	7.70	8.72	9.74	10.78	11.84		13.20	3 / 4			
	reaction time	0.141	7 steps	1.06	1.00	1.02	1.02	1.02	1.02	1.04	1.06	1.36			3.08	3.06	3.12
	interval																
	velocity	5.32	8.62	9.14	8.96	8.96	8.96	8.96	8.96	8.79	8.62	10.31	8.33		8.90	8.96	8.79
Philibert, Dan (FRA) (1970)	time	2.62	3.70	4.70	5.70	6.70	7.72	8.74	9.76	10.82	11.86		13.26	6 / 5			
	reaction time	0.118	1.08	1.00	1.00	1.00	1.02	1.02	1.02	1.06	1.04	1.40	PB		3.08	3.04	3.12
	interval																
	velocity	5.24	8.46	9.14	9.14	9.14	8.96	8.96	8.96	8.62	8.79	10.01	8.30		8.90	9.02	8.79
Reese, Terry (USA) (1967)	time	2.66	3.72	4.74	5.74	6.76	7.78	8.80	9.82	10.90	12.00		13.30	7 / 6			
	reaction time	0.117	1.06	1.02	1.00	1.02	1.02	1.02	1.02	1.08	1.10	1.30	=PB		3.08	3.06	3.20
	interval																
	velocity	5.16	8.62	8.96	9.14	8.96	8.96	8.96	8.96	8.46	8.31	10.78	8.27		8.90	8.96	8.57
Crear, Mark (USA) (1968)	time	2.60	3.64	4.65	5.66	6.66	7.76	8.78	9.86	10.92	12.06		13.55	8 / 7			
	reaction time	0.134	1.04	1.01	1.01	1.00	1.10	1.02	1.08	1.06	1.14	1.49			3.06	3.12	3.28
	interval																
	velocity	5.28	8.79	9.05	9.05	9.14	8.31	8.96	8.46	8.62	8.02	9.41	8.12		8.96	8.79	8.36
Kohutek, Artur (POL) (1971)													DNS	2 / --			

1997 European Cup (Munich, GER)

FINAL

date 22-Jun-97 wind 0.2 m/s

Jung (2003) - <http://www.fgs.uni-halle.de>

		H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
Schwarthoff, Florian (GER) (1971)	time	2.56	3.62	4.63	5.63	6.63	7.66	8.69	9.70	10.74	11.80		13.20	/ 1			
	reaction time		7 steps	1.01	1.00	1.00	1.03	1.03	1.01	1.04	1.06	1.40			3.07	3.06	3.11
	interval																
	velocity	5.36	8.62	9.05	9.14	9.14	8.87	8.87	9.05	8.79	8.62	10.01	8.33		8.93	8.96	8.82
Jackson, Colin (GBR) (1967)	time	2.57	3.60	4.62	5.62	6.62	7.64	8.68	9.72	10.77	11.85		13.28	/ 2			
	reaction time		1.03	1.02	1.00	1.00	1.02	1.04	1.04	1.05	1.08	1.43			3.05	3.06	3.17
	interval																
	velocity	5.34	8.87	8.96	9.14	9.14	8.96	8.79	8.79	8.70	8.46	9.80	8.28		8.99	8.96	8.65
Kislykh, Andrey (RUS) (1976)	time	2.65	3.70	4.71	5.73	6.74	7.78	8.84	9.90	10.97	12.05		13.53	/ 3			
	reaction time		1.05	1.01	1.02	1.01	1.04	1.06	1.06	1.07	1.08	1.48			3.08	3.11	3.21
	interval																
	velocity	5.18	8.70	9.05	8.96	9.05	8.79	8.62	8.62	8.54	8.46	9.47	8.13		8.90	8.82	8.54
Clarico, Vincent (FRA) (1966)	time	2.66	3.72	4.78	5.82	6.88	7.92	9.00	10.06	11.16	12.28		13.67	/ 4			
	reaction time		1.06	1.06	1.04	1.06	1.04	1.08	1.06	1.10	1.12	1.39			3.16	3.18	3.28
	interval																
	velocity	5.16	8.62	8.62	8.79	8.62	8.79	8.46	8.62	8.31	8.16	10.09	8.05		8.68	8.62	8.36
Gundersen, Gaute (NOR) (1972)	time	2.66	3.74	4.78	5.82	6.86	7.91	8.98	10.04	11.14	12.26		13.70	/ 5			
	reaction time		1.08	1.04	1.04	1.04	1.05	1.07	1.06	1.10	1.12	1.44			3.16	3.16	3.28
	interval																
	velocity	5.16	8.46	8.79	8.79	8.79	8.70	8.54	8.62	8.31	8.16	9.74	8.03		8.68	8.68	8.36
Re, Mauro (ITA) (1967)	time	2.70	3.76	4.81	5.84	6.89	7.94	9.00	10.06	11.18	12.30		13.76	/ 6			
	reaction time		1.06	1.05	1.03	1.05	1.05	1.06	1.06	1.12	1.12	1.46			3.14	3.16	3.30
	interval																
	velocity	5.08	8.62	8.70	8.87	8.70	8.70	8.62	8.62	8.16	8.16	9.60	7.99		8.73	8.68	8.31
Mágos, Stamatios (GRE) (1972)	time	2.66	3.74	4.82	5.88	6.94	8.02	9.12	10.24	11.36	12.52		13.97	/ 7			
	reaction time		1.08	1.08	1.06	1.06	1.08	1.10	1.12	1.12	1.16	1.45			3.22	3.24	3.40
	interval																
	velocity	5.16	8.46	8.46	8.62	8.62	8.46	8.31	8.16	8.16	7.88	9.67	7.87		8.52	8.46	8.06
de los Santos, Miguel (ESP) (1971)	time	2.77	3.87	4.94	6.02	7.08	8.17	9.29	10.40	11.52	12.70		14.16	/ 8			
	reaction time		1.10	1.07	1.08	1.06	1.09	1.12	1.11	1.12	1.18	1.46			3.25	3.27	3.41
	interval																
	velocity	4.95	8.31	8.54	8.46	8.62	8.39	8.16	8.23	8.16	7.75	9.60	7.77		8.44	8.39	8.04

1996 Memorial Van Damme (Brussels, BEL)

FINAL

INAL	date	23-Aug-96	wind	0.2 m/s	Shi (2013) - comprehensive research on the "three characteristics" of the men's 110m hurdles														
			H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10	
Johnson, Allen (USA) (1971)	time		2.56	3.56	4.54	5.54	6.52	7.53	8.51	9.56	10.58	11.61		12.92	/ 1				
	reaction time	interval		1.00	0.98	1.00	0.98	1.01	0.98	1.05	1.02	1.03	1.31	=AR =PB		2.98	2.97	3.10	
		velocity	5.36	9.14	9.33	9.14	9.33	9.05	9.33	8.70	8.96	8.87	10.70	8.51		9.20	9.23	8.85	

1996 Olympic Games (Atlanta, GA)

FINAL

NAME	date	29-Jul-96		Butler (2016) - IAAF statistics handbook- special edition: Games of the XXXI Olympiad- Rio 2016															
				wind 0.6 m/s	H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7
Johnson, Allen (USA) (1971)	time			2.34	3.38	4.39	5.36	6.32	7.30	8.30	9.33	10.36	11.38		12.95	6 / 1			
	reaction time	0.170	interval		1.04	1.01	0.97	0.96	0.98	1.00	1.03	1.03	1.02	1.57	OR		3.02	2.94	3.08
			velocity	5.86	8.79	9.05	9.42	9.52	9.33	9.14	8.87	8.87	8.96	8.93	8.49		9.08	9.33	8.90
Crear, Mark (USA) (1968)	time			2.35	3.38	4.37	5.39	6.39	7.41	8.46	9.47	10.51	11.56		13.09	8 / 2			
	reaction time	0.124	interval		1.03	0.99	1.02	1.00	1.02	1.05	1.01	1.04	1.05	1.53			3.04	3.07	3.10
			velocity	5.84	8.87	9.23	8.96	9.14	8.96	8.70	9.05	8.79	8.70	9.16	8.40		9.02	8.93	8.85
Schwarthoff, Florian (GER) (1967)	time			2.38	3.46	4.44	5.43	6.44	7.44	8.44	9.49	10.54	11.61		13.17	3 / 3			
	reaction time	0.164	interval	7 steps	1.08	0.98	0.99	1.01	1.00	1.00	1.05	1.05	1.07	1.56			3.05	3.01	3.17
			velocity	5.76	8.46	9.33	9.23	9.05	9.14	9.14	8.70	8.70	8.54	8.99	8.35		8.99	9.11	8.65
Jackson, Colin (GBR) (1967)	time			2.36	3.42	4.42	5.48	6.43	7.40	8.48	9.54	10.56	11.62		13.19	5 / 4			
	reaction time	0.133	interval		1.06	1.00	1.06	0.95	0.97	1.08	1.06	1.02	1.06	1.57			3.12	3.00	3.14
			velocity	5.81	8.62	9.14	8.62	9.62	9.42	8.46	8.62	8.96	8.62	8.93	8.34		8.79	9.14	8.73
Valle, Emilio (CUB) (1967)	time			2.36	3.44	4.46	5.46	6.48	7.49	8.50	9.54	10.58	11.61		13.20	7 / 5			
	reaction time	0.179	interval		1.08	1.02	1.00	1.02	1.01	1.01	1.04	1.04	1.03	1.59			3.10	3.04	3.11
			velocity	5.81	8.46	8.96	9.14	8.96	9.05	9.05	8.79	8.79	8.87	8.82	8.33		8.85	9.02	8.82
Swift, Eugene (USA) (1964)	time			2.36	3.42	4.44	5.47	6.48	7.47	8.46	9.50	10.55	11.62		13.23	4 / 6			
	reaction time	0.151	interval		1.06	1.02	1.03	1.01	0.99	0.99	1.04	1.05	1.07	1.61			3.11	2.99	3.16
			velocity	5.81	8.62	8.96	8.87	9.05	9.23	9.23	8.79	8.70	8.54	8.71	8.31		8.82	9.17	8.68
Vander-Kuyp, Kyle (AUS) (1971)	time			2.40	3.46	4.54	5.58	6.64	7.61	8.68	9.64	10.78	11.83		13.40	2 / 7			
	reaction time	0.160	interval		1.06	1.08	1.04	1.06	0.97	1.07	0.96	1.14	1.05	1.57			3.18	3.10	3.15
			velocity	5.72	8.62	8.46	8.79	8.62	9.42	8.54	9.52	8.02	8.70	8.93	8.21		8.62	8.85	8.70
Batte, Erik (CUB) (1974)	time			2.40	3.48	4.52	5.58	6.63	7.62	8.68	9.64	10.78	11.83		13.43	1 / 8			
	reaction time	0.160	interval		1.08	1.04	1.06	1.05	0.99	1.06	0.96	1.14	1.05	1.60			3.18	3.10	3.15
			velocity	5.72	8.46	8.79	8.62	8.70	9.23	8.62	9.52	8.02	8.70	8.76	8.19		8.62	8.85	8.70

1996 USA Olympic Trials (Atlanta, GA)

FINAL

		date	23-Jun-96	wind 0.9 m/s			McNichols (2000) - Hurdle technique study through video analysis												
				H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
Johnson, Allen (USA) (1971)	time			2.36	3.40	4.40	5.36	6.36	7.33	8.32	9.32	10.36	11.42		12.92	6 / 1			
	reaction time		interval		1.04	1.00	0.96	1.00	0.97	0.99	1.00	1.04	1.06	1.50	AR PB		3.00	2.96	3.10
			velocity		5.81	8.79	9.14	9.52	9.14	9.42	9.23	9.14	8.79	8.62	9.35	8.51		9.14	9.26
Crear, Mark (USA) (1968)	time			2.38	3.40	4.73	5.40	6.42	7.39	8.42	9.38	10.46	11.52		13.05	4 / 2			
	reaction time		interval		1.02	1.33	0.67	1.02	0.97	1.03	0.96	1.08	1.06	1.53			3.02	3.02	3.10
			velocity		5.76	8.96	6.87	13.64	8.96	9.42	8.87	9.52	8.46	8.62	9.16	8.43		9.08	9.08
Swift, Eugene (USA) (1964)	time			2.38	3.43	4.46	5.44	6.43	7.52	8.54	9.55	10.62	11.66		13.21	7 / 3			
	reaction time		interval		1.05	1.03	0.98	0.99	1.09	1.02	1.01	1.07	1.04	1.55	PB		3.06	3.10	3.12
			velocity		5.76	8.70	8.87	9.33	9.23	8.39	8.96	9.05	8.54	8.79	9.05	8.33		8.96	8.85
Brown, Steven (USA) (1969)	time			2.40	3.46	4.46	5.44	6.44	7.52	8.58	9.63	10.66	11.72		13.28	8 / 4			
	reaction time		interval		1.06	1.00	0.98	1.00	1.08	1.06	1.05	1.03	1.06	1.56			3.04	3.14	3.14
			velocity		5.72	8.62	9.14	9.33	9.14	8.46	8.62	8.70	8.87	8.62	8.99	8.28		9.02	8.73
Kingdom, Roger (USA) (1962)	time			2.45	3.50	4.53	5.52	6.55	7.54	8.55	9.62	10.65	11.77		13.34	1 / 5			
	reaction time		interval		1.05	1.03	0.99	1.03	0.99	1.01	1.07	1.03	1.12	1.57			3.07	3.03	3.22
			velocity		5.60	8.70	8.87	9.23	8.87	9.23	9.05	8.54	8.87	8.16	8.93	8.25		8.93	9.05
Dees, Tony (USA) (1963)	time			2.45	3.47	4.52	5.51	6.57	7.62	8.68	9.80	10.88	12.01		13.67	2 / 6			
	reaction time		interval		1.02	1.05	0.99	1.06	1.05	1.06	1.12	1.08	1.13	1.66			3.06	3.17	3.33
			velocity		5.60	8.96	8.70	9.23	8.62	8.70	8.62	8.16	8.46	8.09	8.45	8.05		8.96	8.65
Harrington, Larry (USA) (1970)	time			2.42											dnf	3 / --			
	reaction time		interval																
			velocity		5.67														
Pierce, Jack (USA) (1962)	time			2.38											dnf	5 / --			
	reaction time		interval																
			velocity		5.76														

1996 LIVE (Nurnberg, GER)**FINAL**

date 07-Jun-96 wind 0.0 m/s

Graubner (2009) - <http://www.fgs.uni-halle.de>

		H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
Schwarthoff, Florian (GER) (19	time	2.57	3.62	4.62	5.62	6.62	7.64	8.65	9.67	10.69	11.73		13.11	/ 1			
reaction time	interval		1.05	1.00	1.00	1.00	1.02	1.01	1.02	1.04	1.38				3.05	3.03	3.08
	velocity	5.34	8.70	9.14	9.14	9.14	8.96	9.05	8.96	8.96	8.79	10.16	8.39		8.99	9.05	8.90

1995 IAAF World Championships (Göteborg, SWE)**FINAL**

date 12-Aug-95 wind -0.1 m/s

		H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
Johnson, Allen (USA) (1971)	0.143	no information available										PB	13.00	6 / 1			
Jarrett, Tony (GBR) (1968)	0.129	no information available											13.04	3 / 2			
Kingdom, Roger (USA) (1962)	0.153	no information available											13.19	4 / 3			
Pierce, Jack (USA) (1962)	0.136	no information available											13.27	5 / 4			
Vander-Kuyp, Kyle (AUS) (1971)	0.129	no information available											13.30	1 / 5			
Philibert, Dan (FRA) (1970)	0.152	no information available											13.34	8 / 6			
Batte, Erik (CUB) (1974)	0.143	no information available										PB	13.38	2 / 7			
Valle, Emilo (CUB) (1967)	0.188	no information available											13.43	7 / 8			

1995 German National Championships (Bremen, GER)**FINAL**

date 02-Jul-95 wind -0.8 m/s

Graubner (2009) - <http://www.fgs.uni-halle.de>

		H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
Schwarthoff, Florian (GER) (19	time	2.54	3.59	4.61	5.60	6.59	7.60	8.58	9.59	10.61	11.65		13.05	/ 1			
reaction time	interval		1.05	1.02	0.99	0.99	1.01	0.98	1.01	1.02	1.04	1.40	NR		3.06	2.98	3.07
	velocity	5.40	8.70	8.96	9.23	9.23	9.05	9.33	9.05	8.96	8.79	10.01	8.43		8.96	9.20	8.93

1994 Asian Games (Hiroshima, JPN)**FINAL**

date 15-Oct-94 wind 1.6 m/s

Yuan (1998) - Chen Yanhao analysis of 110m results

		H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
Chen Yanhao (CHN) (1972)	time	2.58	3.62	4.66	5.68	6.63	7.70	8.74	9.78	10.84	11.92		13.39	/ 2			
reaction time	interval		1.04	1.04	1.02	0.95	1.07	1.04	1.04	1.06	1.08	1.47	PB		3.10	3.06	3.18
	velocity	5.32	8.79	8.79	8.96	9.62	8.54	8.79	8.79	8.62	8.46	9.54	8.22		8.85	8.96	8.62

1994 Hallensportfest (Sindelfingen, GER)**FINAL**

date 6-Mar-94

Vazel (2020) - coaching observations

		H1	H2	H3	H4	H5						Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
Jackson, Colin (GBR) (1967)	time	2.51	3.53	4.52	5.51	6.49							7.30	2 / 1			
reaction time	interval		1.02	0.99	0.99	0.98						0.81	WR		3.00		
	velocity	5.47	8.96	9.23	9.23	9.33						12.00	8.22		9.14		

1993 Chinese National Games (Beijing, CHN)**FINAL**

date 13-Sep-93 wind -0.5 m/s

Yuan (1998) - Chen Yanhao analysis of 110m results

		H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
Chen Yanhao (CHN) (1972)	time	2.59	3.69	4.77	5.83	6.89	7.93	8.98	10.03	11.11	12.22		13.59	/ 1			
reaction time	interval		1.10	1.08	1.06	1.06	1.04	1.05	1.05	1.08	1.11	1.37	PB		3.24	3.15	3.24
	velocity	5.30	8.31	8.46	8.62	8.62	8.79	8.70	8.70	8.46	8.23	10.23	8.09		8.46	8.70	8.46

1993 IAAF World Championships (Stuttgart, GER)**FINAL**

date 20-Aug-93 wind 0.5 m/s

Graubner (2009) - <http://www.fgs.uni-halle.de>

		H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
Jackson, Colin (GBR) (1967)	time	2.53	3.54	4.53	5.51	6.49	7.47	8.47	9.48	10.51	11.55		12.91	3 / 1			
reaction time	0.122 interval		1.01	0.99	0.98	0.98	0.98	1.00	1.01	1.03	1.04	1.36	WR		2.98	2.96	3.08
	velocity	5.42	9.05	9.23	9.33	9.33	9.33	9.14	9.05	8.87	8.79	10.31	8.52		9.20	9.26	8.90
Jarrett, Tony (GBR) (1968)	0.159	no information available											13.00	5 / 2			
Pierce, Jack (USA) (1962)	0.170	no information available											13.06	6 / 3			
Valle, Emilo (CUB) (1967)	0.171	no information available											13.20	2 / 4			
Schwarthoff, Florian (GER) (19	0.128	no information available											13.27	1 / 5			
Kazanovs, Igors (LAT) (1963)	0.166	no information available											13.38	7 / 6			
Koszewski, Dietmar (GER) (196	0.153	no information available											13.60	8 / 7			
Dees, Tony (USA) (1963)	0.160	no information available											14.13	4 / 8			

7 steps

1992 Olympic Games (Barcelona, ESP)**FINAL**

date 03-Aug-92 wind 0.8 m/s

Knight (1992) - 1992 Olympic report: the hurdling events

		H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
McKoy, Mark (CAN) (1961)	time	2.46	3.51	4.52	5.53	6.53	7.56	8.57	9.61	10.66	11.73		13.12	4 / 1			
reaction time	0.155 interval		1.05	1.01	1.01	1.00	1.03	1.01	1.04	1.05	1.07	1.39			3.07	3.04	3.16
	velocity	5.58	8.70	9.05	9.05	9.14	8.87	9.05	8.79	8.70	8.54	10.09	8.38		8.93	9.02	8.68
Dees, Tony (USA) (1963)	time	2.53	3.56	4.56	5.57	6.59	7.60	8.64	9.69	10.78	11.85		13.24	7 / 2			
reaction time	0.159 interval		1.03	1.00	1.01	1.02	1.01	1.04	1.05	1.09	1.07	1.39			3.04	3.07	3.21
	velocity	5.42	8.87	9.14	9.05	8.96	9.05	8.79	8.70	8.39	8.54	10.09	8.31		9.02	8.93	8.54

Pierce, Jack (USA) (1962)	time	2.60	3.66	4.69	5.70	6.75	7.78	8.81	9.84	10.89	11.92		13.26	6 / 3			
	reaction time	0.173	interval	1.06	1.03	1.01	1.05	1.03	1.03	1.03	1.05	1.03	1.34		3.10	3.11	3.11
	velocity	5.28	8.62	8.87	9.05	8.70	8.87	8.87	8.87	8.70	8.87	10.46	8.30		8.85	8.82	8.82
Jarrett, Tony (GBR) (1968)	time	2.63	3.69	4.71	5.72	6.75	7.78	8.79	9.82	10.88	11.93		13.26	8 / 4			
	reaction time	0.163	interval	1.06	1.02	1.01	1.03	1.03	1.01	1.03	1.06	1.05	1.33		3.09	3.07	3.14
	velocity	5.22	8.62	8.96	9.05	8.87	8.87	9.05	8.87	8.62	8.70	10.54	8.30		8.87	8.93	8.73
Schwarthoff, Florian (GER) (1968)	time	2.56	3.57	4.61	5.61	6.61	7.64	8.69	9.74	10.81	11.87		13.29	5 / 5			
	reaction time	0.175	interval	7 steps 1.01	1.04	1.00	1.00	1.03	1.05	1.05	1.07	1.06	1.42		3.05	3.08	3.18
	velocity	5.36	9.05	8.79	9.14	9.14	8.87	8.70	8.70	8.54	8.62	9.87	8.28		8.99	8.90	8.62
Valle, Emilio (CUB) (1967)	time	2.60	3.68	4.71	5.74	6.75	7.78	8.82	9.86	10.93	12.02		13.41	2 / 6			
	reaction time	0.171	interval	1.08	1.03	1.03	1.01	1.03	1.04	1.04	1.07	1.09	1.39		3.14	3.08	3.20
	velocity	5.28	8.46	8.87	8.87	9.05	8.87	8.79	8.79	8.54	8.39	10.09	8.20		8.73	8.90	8.57
Jackson, Colin (GBR) (1967)	time	2.57	3.59	4.58	5.66	6.77	7.80	8.83	9.87	10.88	11.90		13.46	3 / 7			
	reaction time	0.133	interval	1.02	0.99	1.08	1.11	1.03	1.03	1.04	1.01	1.02	1.56		3.09	3.17	3.07
	velocity	5.34	8.96	9.23	8.46	8.23	8.87	8.87	8.79	9.05	8.96	8.99	8.17		8.87	8.65	8.93
Teape, Hugh (GBR) (1963)	time	2.64	3.78	4.84	5.97	6.99	8.03	9.10	10.19	11.34	12.57		14.00	1 / 8			
	reaction time	0.156	interval	1.14	1.06	1.13	1.02	1.04	1.07	1.09	1.15	1.23	1.43		3.33	3.13	3.47
	velocity	5.20	8.02	8.62	8.09	8.96	8.79	8.54	8.39	7.95	7.43	9.80	7.86		8.23	8.76	7.90

1992 Fresno Relays (Fresno, CA)

Heat 1

Heat 1	date	04-Apr-92	wind	0.0 m/s	Wang (2009) - research of training characteristic during different stages in the development of 110m hurdles														
			H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10	
	Li Tong (CHN) (1967)	time	2.54	3.64	4.67	5.71	6.76	7.83	8.90	9.99	11.09	12.21		13.59	/ 1				
	reaction time	interval		1.10	1.03	1.04	1.05	1.07	1.07	1.09	1.10	1.12	1.38				3.17	3.19	3.31
		velocity	5.40	8.31	8.87	8.79	8.70	8.54	8.54	8.39	8.31	8.16	10.16	8.09			8.65	8.60	8.28

1991 IAAF World Championships (Tokyo, JPN)

FINAL

FEDERLE (2003) - http://www.fgs.uni-halle.de	date	29-Aug-91	wind	0.7 m/s											Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
					H1	H2	H3	H4	H5	H6	H7	H8	H9	H10						
Foster, Greg (USA) (1958)	0.148	no information available											13.06	4 / 1						
Pierce, Jack (USA) (1962)	0.114	no information available											13.06	6 / 2						
Jarrett, Tony (GBR) (1968)	0.160	no information available											13.25	5 / 3						
McKoy, Mark (CAN) (1961)	0.128	no information available											13.30	3 / 4						
Philibert, Dan (FRA) (1970)	0.134	no information available											13.33	2 / 5						
Shishkin, Vladimir (URS) (1964)	0.161	no information available											13.39	1 / 6						
Schwarthoff, Florian (GER) (1968)	0.154	no information available											13.41	7 / 7						
Li Tong (CHN) (1967)	0.161	no information available											13.46	8 / 8						

1991 European Cup (Frankfurt, GER)

FINAL

NAME	date	30-Jun-91	wind -0.9 m/s										Federle (2003) - http://www.fgs.uni-halle.de						
		H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10		
Jackson, Colin (GBR) (1967)	time	2.55	3.61	4.64	5.66	6.69	7.72	8.75	9.79	10.83	11.89		13.31	/ 1					
	reaction time	interval	1.06	1.03	1.02	1.03	1.03	1.03	1.04	1.04	1.06	1.42			3.11	3.09	3.14		
	velocity	5.38	8.62	8.87	8.96	8.87	8.87	8.87	8.79	8.79	8.62	9.87	8.26		8.82	8.87	8.73		
Schwarthoff, Florian (GER) (1968)	time	2.54	3.61	4.66	5.69	6.74	7.79	8.83	9.86	10.90	11.98		13.43	/ 2					
	reaction time	interval	1.07	1.05	1.03	1.05	1.05	1.04	1.03	1.04	1.08	1.45			3.15	3.14	3.15		
	velocity	5.40	8.54	8.70	8.87	8.70	8.70	8.79	8.87	8.79	8.46	9.67	8.19		8.70	8.73	8.70		
Touret, Philippe (FRA) (1967)	time	2.63	3.72	4.77	5.82	6.87	7.93	8.97	10.07	11.09	12.20		13.63	/ 3					
	reaction time	interval	1.09	1.05	1.05	1.05	1.06	1.04	1.10	1.02	1.11	1.43			3.19	3.15	3.23		
	velocity	5.22	8.39	8.70	8.70	8.70	8.62	8.79	8.31	8.96	8.23	9.80	8.07		8.60	8.70	8.49		
Kazanov, Igor (URS) (1963)	time	2.54	3.64	4.69	5.75	6.79	7.86	8.93	9.99	11.09	12.18		13.64	/ 4					
	reaction time	interval	1.10	1.05	1.06	1.04	1.07	1.07	1.06	1.10	1.09	1.46			3.21	3.18	3.25		
	velocity	5.40	8.31	8.70	8.62	8.79	8.54	8.54	8.62	8.31	8.39	9.60	8.06		8.54	8.62	8.44		
Kováč, Igor (TCH) (1969)	time	2.63	3.72	4.78	5.86	6.92	7.99	9.05	10.14	11.25	12.45		13.94	/ 5					
	reaction time	interval	1.09	1.06	1.08	1.06	1.07	1.06	1.09	1.11	1.20	1.49			3.23	3.19	3.40		
	velocity	5.22	8.39	8.62	8.46	8.62	8.54	8.62	8.39	8.23	7.62	9.41	7.89		8.49	8.60	8.06		
Frigerio, Fausto (ITA) (1966)		no information available											14.03	/ 6					
Bakos, György (HUN) (1960)		no information available											14.20	/ 7					
Georgiev, Georgi (BUL) (1968)		no information available											14.21	/ 8					

1990 Asian Games (Beijing, CHN)

FINAL

INAL	date	28-Sep-90	wind	Feng (1993) - an analysis of the racing technique in the entire course of the men's 110m hurdles															
			H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10	
Yu Zhicheng (CHN) (1963)	time		2.70	3.79	4.86	5.94	6.99	8.06	9.14	10.22	11.32	12.40		13.82	3 / 1				
	reaction time	interval		1.09	1.07	1.08	1.05	1.07	1.08	1.08	1.10	1.08	1.42	=AR			3.24	3.20	3.26
		velocity	5.08	8.39	8.54	8.46	8.70	8.54	8.46	8.46	8.31	8.46	9.87	7.96			8.46	8.57	8.41

1989 IAAF World Cup (Barcelona, ESP)

FINAL

date 10-Sep-89 wind 2.6 m/s

Gill (1990) - atletismo (I) carreras y marcha

			H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
Kingdom, Roger (USA) (1962)	time		2.55	3.60	4.61	5.61	6.58	7.56	8.54	9.54	10.54	11.57		12.87	9 / 1			
	reaction time			1.05	1.01	1.00	0.97	0.98	0.98	1.00	1.00	1.03	1.30			3.06	2.93	3.03
	velocity		5.38	8.70	9.05	9.14	9.42	9.33	9.33	9.14	9.14	8.87	10.78	8.55		8.96	9.36	9.05
Jackson, Colin (GBR) (1967)	time		2.55	3.59	4.60	5.61	6.61	7.60	8.61	9.62	10.63	11.67		12.95	4 / 2			
	reaction time			1.04	1.01	1.01	1.00	0.99	1.01	1.01	1.01	1.04	1.28			3.06	3.00	3.06
	velocity		5.38	8.79	9.05	9.05	9.14	9.23	9.05	9.05	9.05	8.79	10.95	8.49		8.96	9.14	8.96
Valle, Emilio (CUB) (1967)	time		2.59	3.65	4.67	5.71	6.70	7.70	8.72	9.72	10.77	11.83		13.21	5 / 3			
	reaction time			1.06	1.02	1.04	0.99	1.00	1.02	1.00	1.05	1.06	1.38			3.12	3.01	3.11
	velocity		5.30	8.62	8.96	8.79	9.23	9.14	8.96	9.14	8.70	8.62	10.16	8.33		8.79	9.11	8.82
Pohland, Holger (GDR) (1963)			no information available											13.50	2 / 4			
Moracho, Javier (ESP) (1957)	time		2.60	3.67	4.73	5.78	6.86	7.92	9.02	10.12	11.23	12.39		13.83	6 / 5			
	reaction time			7 steps	1.07	1.06	1.05	1.08	1.06	1.10	1.10	1.11	1.16	1.44		3.18	3.24	3.37
	velocity		5.28	8.54	8.62	8.70	8.46	8.62	8.31	8.31	8.23	7.88	9.74	7.95		8.62	8.46	8.14
Iwasaki, Toshikiko (JPN) (1967)			no information available											14.00	3 / 6			
Mbadugha, Ikechukwu (NGR) (1966)			no information available											14.05	8 / 7			
Caliguri, John (AUS) (1962)			no information available											14.18	1 / 8			

1988 Olympic Games (Seoul, KOR)

FINAL

date 26-Sep-88 wind 1.5 m/s

Brüggemann (1990) - scientific research project at the games of the XXXIV Olympiad, Seoul 1988

			H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
Kingdom, Roger (USA) (1962)	time		2.55	3.60	4.61	5.61	6.61	7.59	8.59	9.59	10.62	11.64		12.98	4 / 1			
	reaction time	0.142		1.05	1.01	1.00	1.00	0.98	1.00	1.00	1.03	1.02	1.34	OR		3.06	2.98	3.05
	velocity		5.38	8.70	9.05	9.14	9.14	9.33	9.14	9.14	8.87	8.96	10.46	8.47		8.96	9.20	8.99
Jackson, Colin (GBR) (1967)	time		2.57	3.61	4.65	5.67	6.70	7.72	8.75	9.79	10.82	11.87		13.28	1 / 2			
	reaction time	0.131		1.04	1.04	1.02	1.03	1.02	1.03	1.04	1.03	1.05	1.41			3.10	3.08	3.12
	velocity		5.34	8.79	8.79	8.96	8.87	8.96	8.87	8.79	8.87	8.70	9.94	8.28		8.85	8.90	8.79
Campbell, Tonie (USA) (1960)	time		2.60	3.66	4.69	5.69	6.69	7.71	8.73	9.78	10.83	11.95		13.38	6 / 3			
	reaction time	0.161		7 steps	1.06	1.03	1.00	1.02	1.02	1.05	1.05	1.12	1.43			3.09	3.04	3.22
	velocity		5.28	8.62	8.87	9.14	9.14	8.96	8.96	8.70	8.70	8.16	9.80	8.22		8.87	9.02	8.52
Shishkin, Vladimir (URS) (1964)	time		2.57	3.63	4.67	5.71	6.74	7.79	8.83	9.87	10.96	12.06		13.51	3 / 4			
	reaction time	0.139		7 steps	1.06	1.04	1.04	1.03	1.05	1.04	1.04	1.09	1.10	1.45		3.14	3.12	3.23
	velocity		5.34	8.62	8.79	8.79	8.87	8.70	8.79	8.79	8.39	8.31	9.67	8.14		8.73	8.79	8.49
Ridgeon, Jon (GBR) (1967)	time		2.58	3.66	4.70	5.73	6.80	7.86	8.91	9.99	11.06	12.13		13.52	7 / 5			
	reaction time	0.150		1.08	1.04	1.03	1.07	1.06	1.05	1.08	1.07	1.07	1.39			3.15	3.18	3.22
	velocity		5.32	8.46	8.79	8.87	8.54	8.62	8.70	8.46	8.54	8.54	10.09	8.14		8.70	8.62	8.52
Jarrett, Tony (GBR) (1968)	time		2.58	3.65	4.73	5.77	6.84	7.89	8.93	9.99	11.05	12.15		13.54	2 / 6			
	reaction time	0.164		1.07	1.08	1.04	1.07	1.05	1.04	1.06	1.06	1.10	1.39			3.19	3.16	3.22
	velocity		5.32	8.54	8.46	8.79	8.54	8.70	8.79	8.62	8.62	8.31	10.09	8.12		8.60	8.68	8.52
McCoy, Mark (CAN) (1961)	time		2.49	3.55	4.57	5.62	6.65	7.77	8.86	9.97	11.08	12.17		13.61	8 / 7			
	reaction time	0.142		1.06	1.02	1.05	1.03	1.12	1.09	1.11	1.11	1.09	1.44			3.13	3.24	3.31
	velocity		5.51	8.62	8.96	8.70	8.87	8.16	8.39	8.23	8.23	8.39	9.74	8.08		8.76	8.46	8.28
Blake, Arthur (USA) (1966)	time		2.48	3.57	4.61	5.65	6.67	7.68	8.85	10.01	11.21	12.40		13.96	5 / 8			
	reaction time	0.160		1.09	1.04	1.04	1.02	1.01	1.17	1.16	1.20	1.19	1.56			3.17	3.20	3.55
	velocity		5.53	8.39	8.79	8.79	8.96	9.05	7.81	7.88	7.62	7.68	8.99	7.88		8.65	8.57	7.72

Semi-Final 2

date 26-Sep-88 wind -0.7 m/s

Brüggemann (1990) - scientific research project at the games of the XXXIV Olympiad, Seoul 1988

			H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
Kingdom, Roger (USA) (1962)	time		2.60	3.66	4.68	5.67	6.66	7.68	8.71	9.76	10.83	11.91		13.37	3 / 1			
	reaction time	0.192		1.06	1.02	0.99	0.99	1.02	1.03	1.05	1.07	1.08	1.46			3.07	3.04	3.20
	velocity		5.28	8.62	8.96	9.23	9.23	8.96	8.87	8.70	8.54	8.46	9.60	8.23		8.93	9.02	8.57
Blake, Arthur (USA) (1966)	time		2.61	3.65	4.70	5.74	6.78	7.82	8.86	9.95	11.02	12.10		13.52	4 / 2			
	reaction time	0.154		1.04	1.05	1.04	1.04	1.04	1.04	1.09	1.07	1.08	1.42			3.13	3.12	3.24
	velocity		5.26	8.79	8.70	8.79	8.79	8.79	8.79	8.39	8.54	8.46	9.87	8.14		8.76	8.79	8.46
McCoy, Mark (CAN) (1961)	time		2.49	3.54	4.61	5.66	6.72	7.77	8.82	9.91	10.99	12.08		13.54	5 / 3			
	reaction time	0.144		1.05	1.07	1.05	1.06	1.05	1.05	1.09	1.08	1.09	1.46			3.17	3.16	3.26
	velocity		5.51	8.70	8.54	8.70	8.62	8.70	8.70	8.39	8.46	8.39	9.60	8.12		8.65	8.68	8.41
Jarrett, Tony (GBR) (1968)	time		2.54	3.62	4.69	5.74	6.78	7.82	8.90	9.96	11.04	12.11		13.56	6 / 4			

reaction time	0.152	interval		1.08	1.07	1.05	1.04	1.04	1.08	1.06	1.08	1.07	1.45		3.20	3.16	3.21
		velocity	5.40	8.46	8.54	8.70	8.79	8.79	8.46	8.62	8.46	8.54	9.67	8.11	8.57	8.68	8.54
Hudec, Jiri (TCH) (1964)	time	2.56	3.64	4.68	5.74	6.78	7.82	8.90	9.99	11.11	12.23			13.73	1 / 5		
reaction time	0.158	interval		1.08	1.04	1.06	1.04	1.08	1.09	1.12	1.12	1.50			3.18	3.16	3.33
		velocity	5.36	8.46	8.79	8.62	8.79	8.79	8.46	8.39	8.16	8.16	9.35	8.01	8.62	8.68	8.23
Yu Zhicheng (CHN) (1963)	time	2.62	3.74	4.82	5.88	6.94	8.03	9.11	10.23	11.35	12.47			13.94	7 / 6		
reaction time	0.143	interval		1.12	1.08	1.06	1.06	1.09	1.08	1.12	1.12	1.47			3.26	3.23	3.36
		velocity	5.24	8.16	8.46	8.62	8.62	8.39	8.46	8.16	8.16	8.16	9.54	7.89	8.41	8.49	8.16
Touret, Philippe (FRA) (1967)	time	2.55	3.62	4.70	5.78	6.86	7.97	9.07	10.19	11.31	12.43			13.96	8 / 7		
reaction time	0.155	interval		1.07	1.08	1.08	1.08	1.11	1.10	1.12	1.12	1.53			3.23	3.29	3.36
		velocity	5.38	8.54	8.46	8.46	8.46	8.23	8.31	8.16	8.16	8.16	9.16	7.88	8.49	8.33	8.16
Ylöstalo, Mikael (FIN) (1963)	time	2.61	3.70	4.78	5.86	6.98	8.06	9.18	10.28	11.39	12.51			14.09	2 / 8		
reaction time	0.146	interval		1.09	1.08	1.08	1.12	1.08	1.12	1.10	1.11	1.12	1.58		3.25	3.32	3.33
		velocity	5.26	8.39	8.46	8.46	8.16	8.46	8.16	8.31	8.23	8.16	8.87	7.81	8.44	8.26	8.23

Semi-Final 1

date 26-Sep-88		wind 0.0 m/s		Brüggemann (1990) - scientific research project at the games of the XXXIV Olympiad, Seoul 1988													
		H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
Shishkin, Vladimir (URS) (1964)	time	2.60	3.67	4.70	5.73	6.75	7.78	8.81	9.87	10.93	12.01		13.46	5 / 1			
reaction time	0.160	interval	7 steps	1.07	1.03	1.03	1.02	1.03	1.03	1.06	1.06	1.08	1.45		3.13	3.08	3.20
		velocity	5.28	8.54	8.87	8.87	8.96	8.87	8.87	8.62	8.62	8.46	9.67	8.17	8.76	8.90	8.57
Campbell, Tonie (USA) (1960)	time	2.60	3.65	4.68	5.68	6.70	7.73	8.77	9.81	10.89	11.97		13.47	6 / 2			
reaction time	0.163	interval	7 steps	1.05	1.03	1.00	1.02	1.03	1.04	1.04	1.08	1.08	1.50		3.08	3.09	3.20
		velocity	5.28	8.70	8.87	9.14	8.96	8.87	8.79	8.79	8.46	8.46	9.35	8.17	8.90	8.87	8.57
Jackson, Colin (GBR) (1967)	time	2.58	3.69	4.73	5.74	6.78	7.81	8.85	9.89	10.97	12.05		13.55	4 / 3			
reaction time	0.137	interval		1.11	1.04	1.01	1.04	1.03	1.04	1.04	1.08	1.08	1.50		3.16	3.11	3.20
		velocity	5.32	8.23	8.79	9.05	8.79	8.87	8.79	8.79	8.46	8.46	9.35	8.12	8.68	8.82	8.57
Ridgeon, Jon (GBR) (1967)	time	2.60	3.73	4.77	5.84	6.89	7.97	9.05	10.13	11.20	12.31		13.68	1 / 4			
reaction time	0.170	interval		1.13	1.04	1.07	1.05	1.08	1.08	1.08	1.07	1.11	1.37		3.24	3.21	3.26
		velocity	5.28	8.09	8.79	8.54	8.70	8.46	8.46	8.46	8.54	8.23	10.23	8.04	8.46	8.54	8.41
Caristan, Stéphane (FRA) (1966)	time	2.60	3.72	4.76	5.80	6.85	7.89	8.96	10.05	11.13	12.25		13.71	3 / 5			
reaction time	0.150	interval		1.12	1.04	1.04	1.05	1.04	1.07	1.09	1.08	1.12	1.46		3.20	3.16	3.29
		velocity	5.28	8.16	8.79	8.79	8.70	8.79	8.54	8.39	8.46	8.16	9.60	8.02	8.57	8.68	8.33
Sala, Carles (ESP) (1960)	time	2.60	3.73	4.81	5.85	6.90	7.97	9.05	10.15	11.25	12.37		13.85	7 / 6			
reaction time	0.127	interval		1.13	1.08	1.04	1.05	1.07	1.08	1.10	1.10	1.12	1.48		3.25	3.20	3.32
		velocity	5.28	8.09	8.46	8.79	8.70	8.54	8.46	8.31	8.31	8.16	9.47	7.94	8.44	8.57	8.26
Cuypers, Alain (BEL) (1967)	time	2.69	3.80	4.86	5.95	7.01	8.09	9.16	10.29	11.38	12.50		13.92	8 / 7			
reaction time	0.169	interval		1.11	1.06	1.09	1.06	1.08	1.07	1.13	1.09	1.12	1.42		3.26	3.21	3.34
		velocity	5.10	8.23	8.62	8.39	8.62	8.46	8.54	8.09	8.39	8.16	9.87	7.90	8.41	8.54	8.21
Bucknor, Richard (JAM) (1966)	time	2.62	3.73	4.82	5.93	7.00	8.09	9.17	10.29	11.41	12.53		13.98	2 / 8			
reaction time	0.166	interval		1.11	1.09	1.11	1.07	1.09	1.08	1.12	1.12	1.12	1.45		3.31	3.24	3.36
		velocity	5.24	8.23	8.39	8.23	8.54	8.39	8.46	8.16	8.16	8.16	9.67	7.87	8.28	8.46	8.16

1987 Chinese National Games (Guangzhou, CHN)

Heat 1

date 28-Nov-87		wind 1.6 m/s		Wang (2009) - research of training characteristic during different stages in the development of 110m hurdles													
		H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
Yu Zhicheng (CHN) (1963)	time	2.62	3.73	4.85	5.93	6.95	7.98	9.02	10.04	11.12	12.24		13.72	1 / 2			
reaction time		interval		1.11	1.12	1.08	1.02	1.03	1.04	1.02	1.08	1.12	1.48		3.31	3.09	3.22
		velocity	5.24	8.23	8.16	8.46	8.96	8.87	8.79	8.96	8.46	8.16	9.47	8.02	8.28	8.87	8.52

1987 IAAF World Championships (Rome, ITA)

FINAL

date 03-Sep-87		wind 0.5 m/s		Susanka (1988) - time analysis of the sprint hurdle events at the II World Championships in Athletics													
		H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
Foster, Greg (USA) (1958)	time	2.60	3.64	4.64	5.64	6.64	7.65	8.67	9.72	10.76	11.81		13.21	3 / 1			
reaction time		interval		1.04	1.00	1.00	1.01	1.02	1.05	1.04	1.05	1.40			3.04	3.03	3.14
		velocity	5.28	8.79	9.14	9.14	9.05	8.96	8.70	8.79	8.70	10.01	8.33		9.02	9.05	8.73
Ridgeon, Jon (GBR) (1967)	time	2.60	3.64	4.70	5.72	6.76	7.79	8.82	9.88	10.93	11.96		13.29	6 / 2			
reaction time		interval		1.04	1.06	1.02	1.04	1.03	1.06	1.05	1.03	1.33			3.12	3.10	3.14
		velocity	5.28	8.79	8.62	8.96	8.79	8.87	8.87	8.62	8.70	8.87	10.54	8.28	8.79	8.85	8.73
Jackson, Colin (GBR) (1967)	time	2.60	3.65	4.71	5.75	6.76	7.80	8.83	9.88	10.94	11.99		13.38	2 / 3			
reaction time		interval		1.05	1.06	1.04	1.01	1.04	1.03	1.05	1.06	1.05	1.39		3.15	3.08	3.16
		velocity	5.28	8.70	8.62	8.79	9.05	8.79	8.87	8.70	8.62	8.70	10.09	8.22	8.70	8.90	8.68
Pierce, Jack (USA) (1962)	no information available												13.41	5 / 4			

Kazanovs, Igors (URS) (1963)	no information available
Sala, Carles (ESP) (1960)	no information available
McKoy, Mark (CAN) (1961)	no information available
Bryggare, Arto (FIN) (1958)	

13.48	1 / 5
13.55	7 / 6
13.71	4 / 7
DNS	8 / --

Semi-Final 2

emi-Final 2		date	01-Sep-87	wind	-1.2 m/s															Susanka (1988) - time analysis of the sprint hurdle events at the II World Championships in Athletics														
				H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10															
Jackson, Colin (GBR) (1967)	time			2.64	3.68	4.77	5.80	6.84	7.88	8.92	10.00	11.08	12.17		13.58	6 / 4																		
	reaction time				1.04	1.09	1.03	1.04	1.04	1.04	1.08	1.08	1.09	1.41			3.16	3.12	3.25															
	velocity			5.20	8.79	8.39	8.87	8.79	8.79	8.79	8.46	8.46	8.39	9.94	8.10		8.68	8.79	8.44															

Semi-Final 1

Semi-Final 1		date	01-Sep-87	wind	Susanka (1988) - time analysis of the sprint hurdle events at the II World Championships in Athletics														
				H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
Ridgeon, Jon (GBR) (1967)	time			2.60	3.65	4.72	5.73	6.75	7.78	8.85	9.89	10.93	11.98		13.34	3 / 1			
	reaction time				1.05	1.07	1.01	1.02	1.03	1.07	1.04	1.04	1.05	1.36			3.13	3.12	3.13
	velocity			5.28	8.70	8.54	9.05	8.96	8.87	8.54	8.79	8.79	8.70	10.31	8.25		8.76	8.79	8.76
Foster, Greg (USA) (1958)	time			2.60	3.65	4.68	5.70	6.72	7.76	8.79	9.85	10.93	11.99		13.41	6 / 2			
	reaction time				1.05	1.03	1.02	1.02	1.04	1.03	1.06	1.08	1.06	1.42			3.10	3.09	3.20
	velocity			5.28	8.70	8.87	8.96	8.96	8.79	8.87	8.62	8.46	8.62	9.87	8.20		8.85	8.87	8.57

Heat 3

Heat 3

	date	01-Sep-87	wind	0.2 m/s	Susanka (1988) - time analysis of the sprint hurdle events at the II World Championships in Athletics														
				H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
Ridgeon, Jon (GBR) (1967)	time			2.72	3.77	4.81	5.85	6.88	7.94	8.98	10.02	11.06	12.15		13.46	/ 1			
	reaction time				1.05	1.04	1.04	1.03	1.06	1.04	1.04	1.04	1.09	1.31			3.13	3.13	3.17
	velocity			5.04	8.70	8.79	8.79	8.87	8.62	8.79	8.79	8.79	8.39	10.70	8.17		8.76	8.76	8.65

Heat 1

Heat 1		date	01-Sep-87	wind	1.4 m/s	Susanka (1988) - time analysis of the sprint hurdle events at the II World Championships in Athletics													
				H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
Foster, Greg (USA) (1958)	time			2.60	3.63	4.64	5.64	6.64	7.62	8.63	9.67	10.72	11.79		13.20	8 / 1			
	reaction time				1.03	1.01	1.00	1.00	0.98	1.01	1.04	1.05	1.07	1.41	CR		3.04	2.99	3.16
	velocity			5.28	8.87	9.05	9.14	9.14	9.33	9.05	8.79	8.70	8.54	9.94	8.33		9.02	9.17	8.68
Jackson, Colin (GBR) (1967)	time			2.61	3.67	4.70	5.71	6.73	7.73	8.75	9.79	10.87	11.95		13.37	7 / 2			
	reaction time				1.06	1.03	1.01	1.02	1.00	1.02	1.04	1.08	1.08	1.42	PB		3.10	3.04	3.20
	velocity			5.26	8.62	8.87	9.05	8.96	9.14	8.96	8.79	8.46	8.46	9.87	8.23		8.85	9.02	8.57

1986 European Championships (Stuttgart, GER)**FINAL**

INAL	date	30-Aug-86	wind	2.0 m/s	Hommel (1988) - nsa photosequence 4 - 110m hurdles - stephane caristan (FRA)														
			H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10	
Caristan, Stéphane (FRA) (1966)	time		2.59	3.62	4.63	5.62	6.63	7.66	8.66	9.72	10.76	11.86		13.20	3 / 1				
	reaction time	interval		1.03	1.01	0.99	1.01	1.03	1.00	1.06	1.04	1.10	1.34	CR / NR		3.03	3.04	3.20	
		velocity	5.30	8.87	9.05	9.23	9.05	8.87	9.14	8.62	8.79	8.31	10.46	8.33		9.05	9.02	8.57	
Bryggare, Arto (FIN) (1958)	7 steps		no information available											13.42	6 / 2				
Sala, Carlos (ESP) (1960)			no information available											13.50	4 / 3				
Walker, Nigel (GBR) (1963)			no information available											13.52	5 / 4				
Oschkenat, Andreas (GDR) (1962)			no information available											13.55	2 / 5				
Ridgeon, Jonathan (GBR) (1967)			no information available											13.70	7 / 6				
Giurgian, Liviu (ROU) (1962)	7 steps		no information available											13.71	1 / 7				
Bakos, György (HUN) (1960)	7 steps		no information available											13.84	8 / 8				

Semi-Final 1

emi-Final 1	date	29-Aug-86	wind	0.9 m/s																	Hommel (1988) - nsa photosequence 4 - 110m hurdles - stephane caristan (FRA)																
				H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10																		
	Caristan, Stéphane (FRA) (1966)	time	2.59	3.59	4.61	5.65	6.67	7.69	8.71	9.75	10.81	11.85		13.28	/ 1																						
		reaction time			1.00	1.02	1.04	1.02	1.02	1.04	1.06	1.04	1.43	CR / NR			3.06	3.06	3.14																		
		velocity	5.30	9.14	8.96	8.79	8.96	8.96	8.96	8.79	8.62	8.79	9.80	8.28			8.96	8.96	8.73																		

1986 IAAF World Junior Championships (Athens, GRE) (1.07m)**FINAL**

INAL	date	19-Jul-86		Miskos (1986) - IAAF biomechanical research - Athens 1986 - hurdle races																
				H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10	
Jackson, Colin (GBR) (1967)	time			2.58	3.63	4.68	5.70	6.71	7.79	8.83	9.90	10.98	12.07		13.44	/ 1				
	reaction time				1.05	1.05	1.02	1.01	1.08	1.04	1.07	1.08	1.09	1.37	CR / AJR		3.12	3.13	3.24	
	velocity			5.32	8.70	8.70	8.96	9.05	8.46	8.79	8.54	8.46	8.39	10.23	8.18		8.79	8.76	8.46	
Ridgeon, Jon (GBR) (1967)	time			2.62	3.73	4.79	5.89	6.93	8.04	9.14	10.24	11.33	12.49		13.91	/ 2				
	reaction time				1.11	1.06	1.10	1.04	1.11	1.10	1.10	1.09	1.16	1.42			3.27	3.25	3.35	
	velocity			5.24	8.23	8.62	8.31	8.79	8.23	8.31	8.31	8.39	7.88	9.87	7.91		8.39	8.44	8.19	
Valle, Emilio (CUB) (1967)	time			2.62	3.50	4.83	5.91	6.98	8.07	9.19	10.30	11.42	12.54		14.00	/ 3				
	reaction time				0.88	1.33	1.08	1.07	1.09	1.12	1.11	1.12	1.12	1.46			3.29	3.28	3.35	
	velocity			5.24	10.39	6.87	8.46	8.54	8.39	8.16	8.23	8.16	8.16	9.60	7.86		8.33	8.36	8.19	
Touret, Philippe (FRA) (1967)				no information available												14.11	/ 4			
Tomaschek, Norbert (AUT) (1967)				no information available												14.43	/ 5			

Weimann, Thomas (AUT) (1967) no information available
 Ivashchenko, Vyacheslav (URS) (1967) no information available
 Ryabukhin, Mikhail (URS) (1967) no information available

14.46 / 6
 14.48 / 7
 14.49 / 8

1986 East German National Championships (Jena, GDR)

FINAL

date 27-Jun-86 wind 1.7 m/s

Aubert (1987) - mesures et observations biomecaniques a propos des courses de haies hautes

		H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
Pohland, Holger (GDR) (1963)	time	2.61	3.66	4.68	5.71	6.76	7.81	8.88	9.94	11.03	12.12		13.56	/ 1			
	reaction time														3.10	3.17	3.24
	interval		1.05	1.02	1.03	1.05	1.05	1.07	1.06	1.09	1.09	1.44			8.85	8.65	8.46
	velocity	5.26	8.70	8.96	8.87	8.70	8.70	8.54	8.62	8.39	8.39	9.74	8.11				
Oschkenat, Andreas (GDR) (19	time	2.62	3.68	4.73	5.76	6.82	7.88	8.95	10.01	11.10	12.17		13.63	/ 2			
	reaction time														3.14	3.19	3.22
	interval		1.06	1.05	1.03	1.06	1.06	1.07	1.06	1.09	1.07	1.46			8.73	8.60	8.52
	velocity	5.24	8.62	8.70	8.87	8.62	8.62	8.54	8.62	8.39	8.54	9.60	8.07				
Kitz, Silvio (GDR) (1966)	time	2.62	3.74	4.84	5.95	7.06	8.17	9.29	10.40	11.51	12.66		14.12	/ 3			
	reaction time														3.33	3.34	3.37
	interval		1.12	1.10	1.11	1.11	1.11	1.12	1.11	1.11	1.15	1.46			8.23	8.21	8.14
	velocity	5.24	8.16	8.31	8.23	8.23	8.23	8.16	8.23	8.23	7.95	9.60	7.79				

1986 French National Indoor Championship (Lievin, FRA)

FINAL

date 26-Jan-86

Aubert (1986) - indoor - video - haies, etude des performances

		H1	H2	H3	H4	H5	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
Caristan, Stéphane (FRA) (196	time	2.54	3.61	4.64	5.65	6.66		7.55	/ 1			
	reaction time						0.89	NR		3.11		
	interval		1.07	1.03	1.01	1.01				8.82		
	velocity	5.40	8.54	8.87	9.05	9.05	10.92	7.95				
Aubert, Philippe (FRA) (1957)	time	2.61	3.69	4.78	5.83	6.91		7.82	/ 2			
	reaction time						0.91			3.22		
	interval		1.08	1.09	1.05	1.08				8.52		
	velocity	5.26	8.46	8.39	8.70	8.46	10.68	7.67				
Chaucoy, Eric (FRA) (1963)	time	2.61	3.71	4.78	5.86	6.93		7.89	/ 3			
	reaction time						0.96			3.25		
	interval		1.10	1.07	1.08	1.07				8.44		
	velocity	5.26	8.31	8.54	8.46	8.54	10.13	7.60				
Darien, Daniel (FRA) (1961)	time	2.60	3.70	4.79	5.90	7.00		7.91	/ 4			
	reaction time						0.91			3.30		
	interval		1.10	1.09	1.11	1.10				8.31		
	velocity	5.28	8.31	8.39	8.23	8.31	10.68	7.59				
Richard, Thierry (FRA) (1961)	time	2.59	3.66	4.75	5.84	7.00		8.09	/ 5			
	reaction time						1.09			3.25		
	interval		1.07	1.09	1.09	1.16				8.44		
	velocity	5.30	8.54	8.39	8.39	7.88	8.92	7.42				
Boussebart, Pascal (FRA) (195	time	2.63	3.69	4.83	6.00	7.14		8.11	/ 6			
	reaction time						0.97			3.37		
	interval		1.06	1.14	1.17	1.14				8.14		
	velocity	5.22	8.62	8.02	7.81	8.02	10.02	7.40				

Semi-Final

date 26-Jan-86

Aubert (1986) - indoor - video - haies, etude des performances

		H1	H2	H3	H4	H5	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
Caristan, Stéphane (FRA) (196	time	2.63	3.68	4.72	5.73	6.76		7.65	/ 1			
	reaction time						0.89			3.10		
	interval		1.05	1.04	1.01	1.03				8.85		
	velocity	5.22	8.70	8.79	9.05	8.87	10.92	7.84				
Aubert, Philippe (FRA) (1957)	time	2.66	3.75	4.79	5.86	6.95		7.86	/ 2			
	reaction time						0.91			3.20		
	interval		1.09	1.04	1.07	1.09				8.57		
	velocity	5.16	8.39	8.79	8.54	8.39	10.68	7.63				
Chaucoy, Eric (FRA) (1963)	time	2.61	3.72	4.78	5.84	6.93		7.87	/ 3			
	reaction time						0.94			3.23		
	interval		1.11	1.06	1.06	1.09				8.49		
	velocity	5.26	8.23	8.62	8.62	8.39	10.34	7.62				
Darien, Daniel (FRA) (1961)	time	2.67	3.76	4.85	5.95	7.04		7.94	/ 4			
	reaction time						0.90			3.28		
	interval		1.09	1.09	1.10	1.09				8.36		
	velocity	5.14	8.39	8.39	8.31	8.39	10.80	7.56				
Richard, Thierry (FRA) (1961)	time	2.68	3.78	4.85	5.93	7.07		7.82	/ 5			
	reaction time						0.75			3.25		
	interval		1.10	1.07	1.08	1.14				8.44		
	velocity	5.12	8.31	8.54	8.46	8.02	12.96	7.67				
Boussebart, Pascal (FRA) (195	time	2.62	3.73	4.83	5.93	7.02		7.94	/ 6			
	reaction time						0.92			3.31		
	interval		1.11	1.10	1.10	1.09				8.28		
	velocity	5.24	8.23	8.31	8.31	8.39	10.57	7.56				
Pawowicz, Piotr (POL)	time	2.68	3.79	4.89	5.97	7.10		8.00	/ 1			
	reaction time						0.90			3.29		
	interval		1.11	1.10	1.08	1.13				8.33		
	velocity	5.12	8.23	8.31	8.46	8.09	10.80	7.50				

Chevallier, Franck (FRA) (1964)	time	2.72	3.82	4.95	6.05	7.14		8.08	/ 2	
	reaction time	interval	1.10	1.13	1.10	1.09	0.94			3.33
	velocity	5.04	8.31	8.09	8.31	8.39	10.34	7.43		8.23
Clarico, Vincent (FRA) (1966)	time	2.64	3.76	4.86	5.99	7.09		8.03	/ 3	
	reaction time	interval	1.12	1.10	1.13	1.10	0.94			3.35
	velocity	5.20	8.16	8.31	8.09	8.31	10.34	7.47		8.19
St. Gilles, Olivier (FRA)	time	2.68	3.81	4.90	5.98	7.07		8.03	/ 4	
	reaction time	interval	1.13	1.09	1.08	1.09	0.96			3.30
	velocity	5.12	8.09	8.39	8.46	8.39	10.13	7.47		8.31
Gabriel, G. (christian?)	time	2.66	3.76	4.91	6.01	7.10		8.08	/ 5	
	reaction time	interval	1.10	1.15	1.10	1.09	0.98			3.35
	velocity	5.16	8.31	7.95	8.31	8.39	9.92	7.43		8.19
Vesin, Yannick (FRA)	no information available									

Heats

date 25-Jan-86

Aubert (1986) - indoor - video - haies, etude des performances

		H1	H2	H3	H4	H5	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
Caristan, Stéphane (FRA) (1966)	time	2.58	3.68	4.72	5.78	6.81		7.69	/ 1			
	reaction time	interval	1.10	1.04	1.06	1.03	0.88			3.20		
	velocity	5.32	8.31	8.79	8.62	8.87	11.05	7.80		8.57		
Boussemart, Pascal (FRA) (1954)	time	2.64	3.75	4.83	5.91	6.98		7.94	/ 6			
	reaction time	interval	1.11	1.08	1.08	1.07	0.96			3.27		
	velocity	5.20	8.23	8.46	8.46	8.54	10.13	7.56		8.39		
Pawowicz, Piotr (POL)	time	2.75	3.88	5.00	6.08	7.20		8.06	/ 1			
	reaction time	interval	1.13	1.12	1.08	1.12	0.86			3.33		
	velocity	4.99	8.09	8.16	8.46	8.16	11.30	7.44		8.23		
Chevallier, Franck (FRA) (1964)	time	2.67	3.77	4.88	5.98	7.08		8.05	/ 2			
	reaction time	interval	1.10	1.11	1.10	1.10	0.97			3.31		
	velocity	5.14	8.31	8.23	8.31	8.31	10.02	7.45		8.28		
Clarico, Vincent (FRA) (1966)	time	2.62	3.76	4.88	6.01	7.12		8.08	/ 3			
	reaction time	interval	1.14	1.12	1.13	1.11	0.96			3.39		
	velocity	5.24	8.02	8.16	8.09	8.23	10.13	7.43		8.09		
St. Gilles, Olivier (FRA) (1963)	time	2.78	3.90	5.02	6.10	7.20		8.07	/ 4			
	reaction time	interval	1.12	1.12	1.08	1.10	0.87			3.32		
	velocity	4.94	8.16	8.16	8.46	8.31	11.17	7.43		8.26		
Gabriel, G.	time	2.70	3.85	4.98	6.08	7.24		8.16	/ 5			
	reaction time	interval	1.15	1.13	1.10	1.16	0.92			3.38		
	velocity	5.08	7.95	8.09	8.31	7.88	10.57	7.35		8.11		
Vésin, Yannick (FRA) (1954)	time	2.64	3.78	4.88	5.95	7.07		8.06	/ 6			
	reaction time	interval	1.14	1.10	1.07	1.12	0.99			3.31		
	velocity	5.20	8.02	8.31	8.54	8.16	9.82	7.44		8.28		
Gonigam, Philippe (FRA) (1963)	time	2.72	3.84	4.96	6.07	7.20		8.18	/ 5			
	reaction time	interval	1.12	1.12	1.11	1.13	0.98			3.35		
	velocity	5.04	8.16	8.16	8.23	8.09	9.92	7.33		8.19		
Viala, Didier (FRA)	time	2.69	3.82	4.94	6.08	7.22		8.19	/ 6			
	reaction time	interval	1.13	1.12	1.14	1.14	0.97			3.39		
	velocity	5.10	8.09	8.16	8.02	8.02	10.02	7.33		8.09		

1985 European Junior Championships (Cottbus, GDR) (99cm)

FINAL

date 23-Aug-85 wind 1.8 m/s

Warburton (1985) - 1985 european junior championships - hurdles

		H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
Ridgeon, Johathan (GBR) (1966)	time	2.32	3.39	4.40	5.39	6.42	7.46	8.53	9.60	10.65	11.72		13.46	/ 1			
	reaction time	interval	1.07	1.01	0.99	1.03	1.04	1.07	1.07	1.05	1.07	1.74	PB		3.07	3.14	3.19
	velocity	5.91	8.54	9.05	9.23	8.87	8.79	8.54	8.54	8.70	8.54	8.06	8.17		8.93	8.73	8.60
Jackson, Colin (GBR) (1967)	time	2.32	3.41	4.44	5.51	6.57	7.63	8.69	9.76	10.87	11.96		13.69	/ 2			
	reaction time	interval	1.09	1.03	1.07	1.06	1.06	1.06	1.07	1.11	1.09	1.73			3.19	3.18	3.27
	velocity	5.91	8.39	8.87	8.54	8.62	8.62	8.62	8.54	8.23	8.39	8.10	8.04		8.60	8.62	8.39

Semi-Final 2

date 23-Aug-85 wind -0.8 m/s

Warburton (1985) - 1985 european junior championships - hurdles

		H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
Jackson, Colin (GBR) (1967)	time	2.42	3.51	4.57	5.64	6.70	7.73	8.86	9.91	11.00	12.13		14.05	/ 1			
	reaction time	interval	1.09	1.06	1.07	1.06	1.03	1.13	1.05	1.09	1.13	1.92			3.22	3.22	3.27
	velocity	5.67	8.39	8.62	8.54	8.62	8.87	8.09	8.70	8.39	8.09	7.30	7.83		8.52	8.52	8.39

Semi-Final 1

date 23-Aug-85 wind 1.1 m/s

Warburton (1985) - 1985 european junior championships - hurdles

		H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
Ridgeon, Johathan (GBR) (1961)	time	2.41	3.54	4.55	5.65	6.65	7.76	8.83	9.89	10.89	12.02		13.75	/ 1			
	reaction time		1.13	1.01	1.10	1.00	1.11	1.07	1.06	1.00	1.13	1.73			3.24	3.18	3.19
	velocity	5.69	8.09	9.05	8.31	9.14	8.23	8.54	8.62	9.14	8.09	8.10	8.00		8.46	8.62	8.60

Heat 3

date 23-Aug-85 wind 1.9 m/s

Warburton (1985) - 1985 european junior championships - hurdles

		H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
Ridgeon, Johathan (GBR) (1961)	time	2.47	3.52	4.58	5.70	6.75	7.82	8.89	10.05	11.09	12.23		14.09	/ 1			
	reaction time		1.05	1.06	1.12	1.05	1.07	1.07	1.16	1.04	1.14	1.86			3.23	3.19	3.34
	velocity	5.55	8.70	8.62	8.16	8.70	8.54	8.54	7.88	8.79	8.02	7.54	7.81		8.49	8.60	8.21

Heat 2

date 23-Aug-85 wind 1.7 m/s

Warburton (1985) - 1985 european junior championships - hurdles

		H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
Jackson, Colin (GBR) (1967)	time	2.32	3.40	4.47	5.51	6.62	7.74	8.82	9.88	10.94	12.08		13.97	/ 1			
	reaction time		1.08	1.07	1.04	1.11	1.12	1.08	1.06	1.06	1.14	1.89			3.19	3.31	3.26
	velocity	5.91	8.46	8.54	8.79	8.23	8.16	8.46	8.62	8.62	8.02	7.42	7.87		8.60	8.28	8.41

1984 Olympic Games (Los Angeles, CA)

FINAL

date 06-Aug-84 wind -0.4 m/s

		H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
Kingdom, Roger (USA) (1962)	0.137	no information available										OR	13.20	8 / 1			
Foster, Greg (USA) (1958)	0.150	no information available											13.23	1 / 2			
Bryggare, Arto (FIN) (1958)	0.136	no information available											13.40	2 / 3			
McKoy, Mark (CAN) (1961)	0.149	no information available											13.45	3 / 4			
Campbell, Tonie (USA) (1960)	0.144	no information available											13.55	7 / 5			
Caristan, Stéphane (FRA) (1961)	0.151	no information available											13.71	6 / 6			
Sala, Carlos (ESP) (1960)	0.159	no information available											13.80	4 / 7			
Glass, Jeff (CAN) (1962)	0.170	no information available											14.15	5 / 8			

1983 IAAF World Championships (Helsinki, FIN)

FINAL

date 12-Aug-83 wind 1.3 m/s

Knoke (1984) - the hurdles

		H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
Foster, Greg (USA) (1958)	time	2.58	3.68	4.62	5.62	6.61	7.61	8.60	9.61	10.68	11.81		13.42	5 / 1			
	reaction time	0.191		1.10	0.94	1.00	0.99	1.00	0.99	1.01	1.07	1.13	1.61		3.04	2.98	3.21
	velocity	5.32	8.31	9.72	9.14	9.23	9.14	9.23	9.05	8.54	8.09	8.71	8.20		9.02	9.20	8.54
Bryggare, Arto (FIN) (1958)	time	2.53	3.67	4.63	5.62	6.67	7.67	8.81	9.82	10.87	11.95		13.46	7 / 2			
	reaction time	0.159		7 steps	1.14	0.96	0.99	1.05	1.00	1.14	1.01	1.05	1.08	1.51	3.09	3.19	3.14
	velocity	5.42	8.02	9.52	9.23	8.70	9.14	8.02	9.05	8.70	8.46	9.28	8.17		8.87	8.60	8.73
Gault, Willie (USA) (1960)	time	2.54	3.57	4.64	5.69	6.72	7.74	8.78	9.83	10.97	12.05		13.48	8 / 3			
	reaction time	0.161		1.03	1.07	1.05	1.03	1.02	1.04	1.05	1.14	1.08	1.43		3.15	3.09	3.27
	velocity	5.40	8.87	8.54	8.70	8.87	8.96	8.79	8.70	8.02	8.46	9.80	8.16		8.70	8.87	8.39
McKoy, Mark (CAN) (1961)	0.132	no information available											13.56	2 / 4			
Munkelt, Thomas (GDR) (1952)		no information available											13.66	4 / 5			
Bakos, György (HUN) (1960)		no information available											13.68	1 / 6			
Radev, Ventsislav (BUL) (1961)		no information available											13.73	6 / 7			
Turner, Sam (USA) (1957)		no information available											13.82	3 / 8			

Semi-Final 2

date 12-Aug-83 wind -0.7 m/s

Knoke (1984) - the hurdles

		H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
Wright, Don (AUS) (1959)	time	2.66	3.82	4.96	5.98	7.04	8.42	9.57	10.58	11.72	12.88		14.28	3 / 7			
	reaction time		1.16	1.14	1.02	1.06	1.38	1.15	1.01	1.14	1.16	1.40			3.32	3.59	3.31
	velocity	5.16	7.88	8.02	8.96	8.62	6.62	7.95	9.05	8.02	7.88	10.01	7.70		8.26	7.64	8.28

Heat 4

date 11-Aug-83 wind 2.1 m/s

Knoke (1984) - the hurdles

		H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
Wright, Don (AUS) (1959)	time	2.55	3.68	4.77	5.75	6.81	7.82	9.00	10.04	11.10	12.19		13.70	/ 2			
	reaction time		1.13	1.09	0.98	1.06	1.01	1.18	1.04	1.06	1.09	1.51			3.20	3.25	3.19
	velocity	5.38	8.09	8.39	9.33	8.62	9.05	7.75	8.79	8.62	8.39	9.28	8.03		8.57	8.44	8.60

1982 European Championships (Athens, GRE)

FINAL

date 11-Sep-82 wind -0.9 m/s

Heß (1985) - entwicklungsaspekte die leichtathletischen sprint-/hürdendisziplinen im olympiazklus 1981/84

		H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
Munkelt, Thomas (GDR) (1952)	time	2.58		4.70		6.77		8.84			11.98		13.41	/ 1			
	reaction time			2.12		2.07		2.07			3.14	1.43					3.14
	velocity	5.32		8.62		8.83		8.83			8.73	9.80	8.20				8.73
Prokofyev, Andrey (URS) (1959)	time	2.56		4.64		6.72		8.81			12.01		13.46	/ 2			
	reaction time			2.08		2.08		2.09			3.20	1.45	PB				3.20
	velocity	5.36		8.79		8.79		8.75			8.57	9.67	8.17				8.57

1981 International Abendsportfest (Koblenz, FRG)**FINAL**

INAL	date	26-Aug-81	wind	0.0 m/s	Heß (1985) - entwicklungsaspekte die leichtathletischen sprint-/hürden disziplinen im olympiazklus 1981/84													
			H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
Nehemiah, Renaldo (USA) (195)	time		2.50		4.53		6.54		8.54			11.63		13.04	/ 1			
	reaction time	interval			2.03		2.01		2.00			3.09	1.41					3.09
	velocity		5.49		9.00		9.09		9.14			8.87	9.94	8.44				8.87

1981 USSR vs. USA (Leningrad, URS)**FINAL**

INAL		date	10-Jul-81	wind	Chanon (1987) - analyse comparee des caracteristiques des courses de 100m haies et 110m haies														
				H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
Foster, Greg (USA) (1958)	time			2.48	3.58	4.60	5.61	6.62	7.67	8.72	9.82	10.92	12.02		13.30	/ 1			
	reaction time				1.10	1.02	1.01	1.01	1.05	1.05	1.10	1.10	1.10	1.28			3.13	3.11	3.30
	velocity			5.53	8.31	8.96	9.05	9.05	8.70	8.70	8.31	8.31	8.31	10.95	8.27		8.76	8.82	8.31
Cowling, Larry (USA) (1960)	time			2.48	3.59	4.62	5.65	6.68	7.75	8.82	9.94	11.06	12.21		13.53	/ 2			
	reaction time				1.11	1.03	1.03	1.03	1.07	1.07	1.12	1.12	1.15	1.32			3.17	3.17	3.39
	velocity			5.53	8.23	8.87	8.87	8.87	8.54	8.54	8.16	8.16	7.95	10.62	8.13		8.65	8.65	8.09
Chervanyev, Yuriy (URS) (1958)	time			2.50	3.61	4.65	5.68	6.72	7.78	8.87	10.00	11.17	12.42		13.74	/ 3			
	reaction time				1.11	1.04	1.03	1.04	1.06	1.09	1.13	1.17	1.25	1.32			3.18	3.19	3.55
	velocity			5.49	8.23	8.79	8.87	8.79	8.62	8.39	8.09	7.81	7.31	10.62	8.01		8.62	8.60	7.72
Shabanov, Georgiy (URS) (1961)	time			2.50	3.61	4.66	5.69	6.74	7.80	8.90	10.06	11.24	12.49		13.81	/ 4			
	reaction time				1.11	1.05	1.03	1.05	1.06	1.10	1.16	1.18	1.25	1.32			3.19	3.21	3.59
	velocity			5.49	8.23	8.70	8.87	8.70	8.62	8.31	7.88	7.75	7.31	10.62	7.97		8.60	8.54	7.64

1981 USSR vs. GDR (Tbilisi, URS)**FINAL**

NAME	date	26-Jun-81	wind	???	Chanon (1987) - analyse comparee des caracteristiques des courses de 100m haies et 110m haies												
		H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
Munkelt, Thomas (GDR) (1952)	time	2.56	3.59	4.75	5.78	6.81	7.86	8.94	10.00	11.14	12.32		13.69	/ 1			
	reaction time		1.03	1.16	1.03	1.03	1.05	1.08	1.06	1.14	1.18	1.37			3.22	3.16	3.38
	velocity	5.36	8.87	7.88	8.87	8.87	8.70	8.46	8.62	8.02	7.75	10.23	8.04		8.52	8.68	8.11
Shabanov, Georgiy (URS) (1961)	time	2.48	3.61	4.69	5.72	6.74	7.81	8.90	10.00	11.20	12.44		13.82	/ 2			
	reaction time		1.13	1.08	1.03	1.02	1.07	1.09	1.10	1.20	1.24	1.38			3.24	3.18	3.54
	velocity	5.53	8.09	8.46	8.87	8.96	8.54	8.39	8.31	7.62	7.37	10.16	7.96		8.46	8.62	7.75
Chervanyev, Yuriy (URS) (1958)	time	2.49	3.62	4.71	5.73	6.75	7.82	8.96	10.12	11.30	12.56		13.93	/ 3			
	reaction time		1.13	1.09	1.02	1.02	1.07	1.14	1.16	1.18	1.26	1.37			3.24	3.23	3.60
	velocity	5.51	8.09	8.39	8.96	8.96	8.54	8.02	7.88	7.75	7.25	10.23	7.90		8.46	8.49	7.62
Dittrich, Thomas (GDR) (1954)	time	2.50	3.65	4.78	5.80	6.82	7.89	9.04	10.19	11.39	12.64		13.94	/ 4			
	reaction time		1.15	1.13	1.02	1.02	1.07	1.15	1.15	1.20	1.25	1.30			3.30	3.24	3.60
	velocity	5.49	7.95	8.09	8.96	8.96	8.54	7.95	7.95	7.62	7.31	10.78	7.89		8.31	8.46	7.62

1980 Olympic Games (Moscow, URS)**FINAL**

NAME	date	27-Jul-80		wind 0.9 m/s										Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
		H1	H2	H3	H4	H5	H6	H7	H8	H9	H10								
Munkelt, Thomas (GDR) (1952)	0.129	no information available											13.39	3 / 1					
Casañas, Alejandro (CUB) (195)	0.144	no information available											13.40	6 / 2					
Puchkov, Aleksandr (URS) (195)	0.145	no information available											13.44	4 / 3					
Prokofyev, Andrey (URS) (1959)	0.145	no information available											13.49	5 / 4					
Pusty, Jan (POL) (1952)	0.149	no information available											13.68	1 / 5					
Bryggare, Arto (FIN) (1958)	1.750	no information available											13.76	2 / 6					
Moracho, Javier (ESP) (1957)	0.151	no information available											13.78	8 / 7					
Chervanyev, Yuriy (URS) (1958)	0.166	no information available											15.80	7 / 8					

1976 Olympic Games (Montreal, CAN)**FINAL**

		date	28-Jul-76		wind 0.0 m/s										Ewen (1977) - montreal 1976				
			H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10	
Drut, Guy (FRA) (1950)	time		2.5	3.5	4.6	5.6	6.6	7.6	8.7	9.7	10.7	11.8		13.30	5 / 1			Ewen (1977)	
	reaction time	interval		1.0	1.1	1.0	1.0	1.0	1.1	1.0	1.0	1.1	1.5			3.10	3.10	3.10	
		velocity	5.49	9.14	8.31	9.14	9.14	9.14	8.31	9.14	9.14	8.31	9.35	8.27		8.85	8.85	8.85	
Casañas, Alejandro (CUB) (195	time		2.6	3.6	4.7	5.7	6.8	7.8	8.8	9.8	10.9	11.9		13.33	7 / 2			Ewen (1977)	
	reaction time	interval		1.0	1.1	1.0	1.1	1.0	1.0	1.0	1.1	1.0	1.4	PB		3.10	3.10	3.10	
		velocity	5.28	9.14	8.31	9.14	8.31	9.14	9.14	9.14	8.31	9.14	9.80	8.25		8.85	8.85	8.85	
Davenport, Willie (USA) (1943)	time		2.6	3.6	4.6	5.7	6.7	7.7	8.8	9.8	10.9	11.9		13.38	3 / 3			Ewen (1977)	
	reaction time	interval		1.0	1.0	1.1	1.0	1.0	1.1	1.0	1.1	1.0	1.5			3.10	3.10	3.10	
		velocity	5.28	9.14	9.14	8.31	9.14	9.14	8.31	9.14	8.31	9.14	9.47	8.22		8.85	8.85	8.85	
Foster, Charles (USA) (1953)			no information available											13.41	6 / 4				
Munkelt, Thomas (GDR) (1952)		time	2.59	3.67	4.73	5.75	6.77	7.83	8.87	9.91	10.97	12.04		13.44	1 / 5			Tepper (1978)	

reaction time	interval	7 steps	1.08	1.06	1.02	1.02	1.06	1.04	1.04	1.06	1.07	1.40	PB		3.16	3.12	3.17
	velocity		5.30	8.46	8.62	8.96	8.96	8.62	8.79	8.79	8.62	8.54	10.01	8.18	8.68	8.79	8.65
Owens, James (USA) (1955)		no information available												13.73	4 / 6		
Kulebyakin, Vyacheslav (URS) (1950)		no information available												13.93	8 / 7		
Myasnikov, Victor (URS) (1948)		no information available												13.94	2 / 8		

Heat 3

eat 3

date	26-Jul-76	wind	0.0 m/s	Lamare (1976) - les jeux de la XXI olympia, Montreal 1976													
		H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
Munkelt, Thomas (GDR) (1952)	time	2.3			5.5	6.6							13.69	1 / 1			
	reaction time	interval	7 steps		3.22	1.12									3.22		
		velocity	6.04		8.52	8.16							8.04		8.52		

Heat 2

heat 2		date	26-Jul-76		wind	0.0 m/s												Lamare (1976) - les jeux de la XXI olympia, Montreal 1976				
			H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10				
Casañas, Alejandro (CUB) (1955)		time	2.3			5.5	6.6							13.73	1 / 2							
reaction time		interval				3.18	1.08									3.18						
		velocity	5.97			8.62	8.46							8.01		8.62						

Heat 1

heat 1		date	26-Jul-76		wind		0.0 m/s		Lamare (1976) - les jeux de la XXI olympia, Montreal 1976									
			H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
Foster, Charles (USA) (1953)		time	2.4			5.5	6.6							13.68	6 / 1			
reaction time		interval				3.15	1.07										3.15	
		velocity	5.84			8.70	8.54							8.04			8.70	

1975 International "Pravda" Sportfest (Sotschi, URS)

FINAL

date 30-May-75		wind -0.2 m/s										Artyschenko (1977) - zwischen den hürden geht es um die sekunde, teil 1						
		H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10	
Myasnikov, Viktor (URS) (1948)	time	2.4	3.5	4.6	5.7	6.8	7.8	8.9	9.9	11.0	12.1		13.7	/ 1				
	reaction time		interval	1.06	1.10	1.16	1.04	1.04	1.10	1.04	1.10	1.62			3.32	3.18	3.18	
		velocity	5.72	8.62	8.31	7.88	8.79	8.79	8.31	8.79	8.79	8.31	8.65	8.03		8.26	8.62	8.62
Kulebyakin, Vyacheslav (URS)	time	2.2	3.3	4.5	5.6	6.6	7.7	8.8	9.9	11.0	12.1		13.8	/ 2				
	reaction time		interval	1.10	1.16	1.09	1.05	1.07	1.09	1.13	1.06	1.09	1.73		3.35	3.21	3.28	
		velocity	6.15	8.31	7.88	8.39	8.70	8.54	8.39	8.09	8.62	8.39	8.10	7.97		8.19	8.54	8.36
Moshiashvili, Anatoliy (URS) (1951)	time	2.2	3.2	4.4	5.5	6.7	7.8	8.9	10.1	11.2			dnf	/ --				
	reaction time		interval	1.07	1.13	1.17	1.13	1.13	1.15	1.15					3.37	3.39		
		velocity	6.35	8.54	8.09	7.81	8.09	8.09	8.09	7.95	7.95					8.14	8.09	

Heat ??

eat ??	date	30-May-75	wind	???	Artyschenko (1977) - zwischen den hürden geht es um die sekunde, teil 1													
			H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
Kulebyakin, Vyacheslav (URS)	time	2.4	3.33	4.49	5.58	6.63	7.70	8.79	9.92	10.98	12.07			13.6	???			
	reaction time	interval		1.14	1.12	1.12	1.10	1.08	1.12	1.14	1.16	1.22	1.52			3.18	3.21	3.28
		velocity	5.72	8.02	8.16	8.16	8.31	8.46	8.16	8.02	7.88	7.49	9.22	8.09		8.62	8.54	8.36

1972 Olympic Games (Munich, GER)

FINAL

		date	07-Sep-72	wind	0.3 m/s													Kay (1972) - BAAB Munich Technical Report			
				H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10		
Milburn, Rodney (USA) (1950)	time			2.5	3.5	4.5	5.6	6.6	7.6	8.6	9.6	10.6	11.7		13.24	5 / 1			Kay (1972)		
	reaction time		interval		1.0	1.0	1.1	1.0	1.0	1.0	1.0	1.1	1.54	WR			3.10	3.00	3.10		
		velocity		5.49	9.14	9.14	8.31	9.14	9.14	9.14	9.14	9.14	8.31	9.10	8.31		8.85	9.14	8.85		
Drut, Guy (FRA) (1950)	time			2.5	3.6	4.6	5.6	6.6	7.7	8.7	9.7	10.7	11.8		13.34	8 / 2			Kay (1972)		
	reaction time		interval		1.1	1.0	1.0	1.0	1.1	1.0	1.0	1.1	1.54				3.10	3.10	3.10		
		velocity		5.49	8.31	9.14	9.14	9.14	8.31	9.14	9.14	9.14	8.31	9.10	8.25		8.85	8.85	8.85		
Hill, Thomas (USA) (1949)	time			2.5	3.5	4.6	5.6	6.6	7.6	8.7	9.8	10.9	12.0		13.48	7 / 3			Kay (1972)		
	reaction time		interval		1.0	1.1	1.0	1.0	1.0	1.1	1.1	1.1	1.48				3.10	3.10	3.30		
		velocity		5.49	9.14	8.31	9.14	9.14	9.14	8.31	8.31	8.31	8.31	9.47	8.16		8.85	8.85	8.31		
Davenport, Willie (USA) (1943)	time			2.5	3.5	4.6	5.6	6.6	7.6	8.7	9.8	10.9	11.9		13.50	1 / 4			Kay (1972)		
	reaction time		interval		1.0	1.1	1.0	1.0	1.0	1.1	1.1	1.1	1.0	1.60			3.10	3.10	3.20		
		velocity		5.49	9.14	8.31	9.14	9.14	9.14	8.31	8.31	8.31	9.14	8.76	8.15		8.85	8.85	8.57		
Siebeck, Frank (GDR) (1949)	time			2.5	3.6	4.6	5.6	6.7	7.8	8.9	9.9	11.1	12.2		13.71	6 / 5			Kay (1972)		
	reaction time		interval	7 steps	1.1	1.0	1.0	1.1	1.1	1.1	1.0	1.2	1.1	1.51			3.10	3.30	3.30		
		velocity		5.49	8.31	9.14	9.14	8.31	8.31	8.31	9.14	7.62	8.31	9.28	8.02		8.85	8.31	8.31		
Wodzynski, Leszek (POL) (1946)			no information available													13.72	3 / 6				
Nádenicek, Lubomir (TCH) (1950)	time			2.5			5.7								13.76	2 / 7			Letzelter (1977)		
	reaction time		interval				3.2										3.20				
		velocity		5.60			8.57								7.99		8.57				
Cech, Petr (TCH) (1944)			no information available													13.86	4 / 8				

Heat 4

date 03-Sep-72 wind 0.9 m/s

Letzelter (1977) - Hurdensprint: geschwindigkeitsverlauf und spezielle eigenschaften

		H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
McDonald, Rich (CAN) (1949)	time	2.5			5.8								14.36	8 / 3			
	reaction time				3.3										3.32		
	interval				3.3												
	velocity	5.44			8.26								7.66		8.26		
Tørring, Jesper (DEN) (1947)	time	2.6			6.0								14.50	1 / 5			
	reaction time				3.4										3.38		
	interval				3.4												
	velocity	5.34			8.11								7.59		8.11		

Heat 3

date 03-Sep-72 wind -1.6 m/s

Letzelter (1977) - Hurdensprint: geschwindigkeitsverlauf und spezielle eigenschaften

		H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
Pascoe, Alan (GBR) (1947)	time	2.5			5.8								14.08	3 / 3			
	reaction time				3.3										3.26		
	interval				3.3												
	velocity	5.51			8.41								7.81		8.41		

1970 Commonwealth Games (Edinburgh, GBR)**FINAL**

date 18-Jul-70 wind 2.9 m/s

LeMasurier (1972) - track speed- hurdles, sprints, and relays

		H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
Hemery, David (GBR) (1944)	time	2.55	3.65	4.65	5.70	6.75	7.80	8.85	9.90	11.00	12.10	(13.66)	13.6	1 / 1			
	reaction time		1.10	1.00	1.05	1.05	1.05	1.05	1.05	1.10	1.10	1.50			3.15	3.15	3.25
	interval		1.10	1.00	1.05	1.05	1.05	1.05	1.05	1.10	1.10	1.50					
	velocity	5.38	8.31	9.14	8.70	8.70	8.70	8.70	8.70	8.31	8.31	9.35	8.09		8.70	8.70	8.44
Baird, Malcolm (AUS) (1948)		no information available											13.8	4 / 2			
Murray, Godfrey (JAM) (1950)		no information available											14.0	3 / 3			
Neeland, George (CAN)		no information available											14.2	5 / 4			
Donnelly, Brian (CAN)		no information available											14.2	7 / 5			
Macdonald, Richard (CAN) (1949)		no information available											14.4	2 / 6			
Ohene-Frempong, Kwaku (GHA)		no information available											14.7	6 / 7			
Pascoe, Alan (GBR) (1947)		no information available											dnf	8 / 8			

1968 Olympic Games (Mexico City, MEX)**FINAL**

date 17-Oct-68 wind 0.0 m/s

		H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
Davenport, Willie (USA) (1943)		no information available											13.3	4 / 1	(13.33)		
Hall, Ervin (USA) (1947)		no information available											13.4	6 / 2	(13.42)		
Ottoz, Eddy (ITA) (1944)		no information available											13.4	3 / 3	(13.46)		
Coleman, Leon (USA) (1944)		no information available											13.6	7 / 4	(13.67)		
Trzmiel, Werner (FRG) (1942)		no information available											13.6	1 / 5	(13.68)		
Forssander, Bo (SWE) (1942)		no information available											13.7	2 / 6	(13.73)		
Duriez, Marcel (FRA) (1940)		no information available											13.7	8 / 7	(13.77)		
Schoebel, Pierre (FRA) (1942)		no information available											14.0	7 / 8	(14.02)		

1964 Olympic Games (Tokyo, JPN)**FINAL**

date 18-Oct-64 wind 2.0 m/s

		H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
Jones, Hayes (USA) (1938)		no information available											13.6	6 / 1	(13.67)		
Lindgren, Blaine (USA) (1939)		no information available											13.7	1 / 2	(13.74)		
Mikhailov, Anatoliy (URS) (1936)		no information available											13.7	8 / 3	(13.78)		
Ottoz, Eddy (ITA) (1944)		no information available											13.8	3 / 4	(13.84)		
Singh, Gurbachan (IND) (1939)		no information available											14.0	4 / 5	(14.09)		
Duriez, Marcel (FRA) (1940)		no information available											14.0	5 / 6	(14.09)		
Cornacchia, Giovanni (ITA) (1939)		no information available											14.1	7 / 7	(14.12)		
Mazza, Giorgio (ITA) (1939)		no information available											14.1	2 / 8	(14.17)		

1960 Olympic Games (Rome, ITA)**FINAL**

date 05-Sep-60 wind 0.0 m/s

		H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
Calhoun, Lee (USA) (1933)		no information available											13.8	2 / 1	(13.98)		
May, Willie (USA) (1936)		no information available											13.8	1 / 2	(13.99)		
Jones, Hayes (USA) (1938)		no information available											14.0	6 / 3	(14.17)		
Lauer, Martin (EUA/FRG) (1937)		no information available											14.0	3 / 4	(14.20)		
Gardner, Keith (BWI) (1929)		no information available											14.4	4 / 5	(14.55)		
Chistyakov, Valentin (URS) (1939)		no information available											14.6	6 / 5	(14.71)		

1956 Olympic Games (Melbourne, AUS)**FINAL**

date 26-Nov-56 wind -1.9 m/s

		H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
Calhoun, Lee (USA) (1933)		no information available											13.5	4 / 1	(13.70)		
Davis, Jack (USA) (1930)		no information available											13.5	7 / 2	(13.73)		
Shankle, Joel (USA) (1933)		no information available											14.1	3 / 3	(14.25)		
Lauer, Martin (EUA/FRG) (1937)		no information available											14.5	8 / 4	(14.67)		
Lorger, Stanko (YUG) (1931)		no information available											14.5	5 / 5	(14.68)		

Stolyarov, Boris (URS) (1932) no information available

14.6

6 / 6 (14.71)

1952 Olympic Games (Helsinki, FIN)

FINAL

date 24-Jul-52 wind *legal*

	H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
Dillard, Harrison (USA) (1923)			no information available								OR	13.7	4 / 1	(13.91)		
Davis, Jack (USA) (1930)			no information available								OR	13.7	1 / 2	(14.00)		
Barnard, Arthur (USA) (1929)			no information available									14.1	5 / 3	(14.40)		
Bulanchik, Yevgeniy (URS) (1922)			no information available									14.5	2 / 4	(14.73)		
Doubleday, Ken (AUS) (1926)			no information available									14.7	3 / 5	(14.82)		
Weinberg, Ray (AUS) (1929)			no information available									14.8	6 / 6	(15.15)		

1948 Olympic Games (London, GBR)

FINAL

date 04-Aug-48 wind *against*

	H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
Porter, William (USA) (1926)			no information available								OR	13.9	3 / 1			
Scott, Clyde (USA) (1924)			no information available									14.1	7 / 2			
Dixon, Craig (USA) (1926)			no information available									14.1	2 / 3			
Triulzi, Alberto (ARG) (1928)			no information available										5 / 4			
Gardner, Peter (AUS) (1925)			no information available										6 / 5			
Linman, Håkan (SWE) (1915)			no information available										4 / 6			

1936 Olympic Games (Berlin, GER)

FINAL

date 06-Aug-36 wind

	H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
Towns, Forrest (USA) (1914)			no information available									14.2	3 / 1			
Finlay, Don (GBR) (1909)			no information available									14.4	2 / 2			
Pollard, Frederick (USA) (1915)			no information available									14.4	7 / 3			
Linman, Håkan (SWE) (1915)			no information available									14.4	4 / 4			
Thornton, John (GBR) (1911)			no information available									14.7	5 / 5			
O'Connor, Larry (CAN) (1916)			no information available									15.0	6 / 6			

1932 Olympic Games (Los Angeles, CA)

FINAL

date 03-Aug-32 wind

	H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
Saling, George (USA) (1909)			no information available									14.6	5 / 1	(14.57)		
Beard, Percy (USA) (1908)			no information available									14.7	2 / 2	(14.72)		
Finlay, Don (GBR) (1909)			no information available									14.8	3 / 3	(14.79)		
Keller, Jack (USA) (1911)			no information available									14.8	7 / 4	(14.83)		
Burghley, Lord David (GBR) (1905)			no information available									14.8	6 / 5	(14.84)		
Welscher, Willi (GER) (1906)			no information available								(knocked down 4 hurdles)	DQ	15.3e	4 / 6		

1928 Olympic Games (Amsterdam, NED)

FINAL

date 01-Aug-28 wind

	H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
Atkinson, Sydney (RSA) (1901)			no information available									14.8	3 / 1			
Anderson, Stephen (USA) (1906)			no information available									14.8e	4 / 2			
Collier, John (USA) (1907)			no information available									14.9e	2 / 3			
Dye, Leighton (USA) (1901)			no information available									15.0e	6 / 4			
Weightman-Smith, George (RSA) (1905)			no information available									15.0e	1 / 5			
Gaby, Fred (GBR) (1895)			no information available									15.2e	5 / 6			

1924 Olympic Games (Paris, FRA)

FINAL

date 09-Jul-24 wind

	H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
Kinsey, Daniel (USA) (1902)			no information available									15.0	5 / 1			
Atkinson, Sydney (RSA) (1901)			no information available									15.0	4 / 2			
Guthrie, George (USA) (1904)			no information available								(knocked down 3 hurdles)	DQ	15.1	1 / 3		
Pettersson, Sten-Pelle (SWE) (1902)			no information available									15.4	6 / 4 3			
Christiernsson, Carl-Axel (SWE) (1898)			no information available									15.5	3 / 5 4			
Anderson, Karl (USA) (1900)			no information available										2 / 6 5			

1920 Olympic Games (Antwerp, BEL)

FINAL

date 18-Aug-20 wind

	H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
Thomson, Earl (CAN) (1895)			no information available								WR	14.8	6 / 1			
Barron, Harold (USA) (1894)			no information available									15.0e	1 / 2			
Murray, Frederick (USA) (1894)			no information available									15.2e	4 / 3			
Wilson, Harry (NZL) (1896)			no information available									15.2e	2 / 4			
Smith, Walker (USA) (1896)			no information available									15.3e	5 / 5			
Christiernsson, Carl-Axel (SWE) (1898)			no information available									15.5e	3 / 6			

1912 Olympic Games (Stockholm, SWE)

FINAL

date 12-Jul-12 wind

	H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
Kelly, Fred (USA) (1891)	no information available											15.1	5 / 1			
Wendell, James (USA) (1890)	no information available											15.2	2 / 2			
Hawkins, Martin (USA) (1888)	no information available											15.3	6 / 3			
Case, John (USA) (1889)	no information available											15.3	1 / 4			
Powell, Kenneth (GBR) (1885)	no information available											15.5	3 / 5			
Nicholson, John (USA) (1889)	no information available											dnf	4 / --			

1908 Olympic Games (London, GBR) (587yd Track)**FINAL**

date 25-Jul-08

wind

	H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
Smithson, Forrest (USA) (1884)	no information available										WR	15.0	2 / 1			
Garrels, John (USA) (1885)	no information available											15.7e	1 / 2			
Shaw, Arthur (USA) (1886)	no information available											15.8e	4 / 3			
Rand, William (USA) (1886)	no information available											16.0e	3 / 4			

1906 Intercalated Games (Athens, GRE)**FINAL**

date 01-May-06

wind

	H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
Leavitt, Robert (USA) (1883)	no information available											16.2	2 / 1			
Healey, Alfred (GBR) (1879)	no information available											16.2e	4 / 2			
Dunker, Vincent (RSA) (1884)	no information available											16.3e	3 / 3			
Friend, Hugo (USA) (1882)	no information available											16.4e	1 / 4			
Molinie, Henri (FRA) (1874)	no information available												5 / 5			

1904 Olympic Games (St. Louis, MO)**FINAL**

date 03-Sep-04

wind

	H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
Schule, Fred (USA) (1879)	no information available											16.0	4 / 1			
Shideler, Thaddeus (USA) (1883)	no information available											16.2e	1 / 2			
Ashburner, Lesley (USA) (1883)	no information available											16.4e	2 / 3			
Castleman, Frank (USA) (1877)	no information available												3 / 4			

1900 Olympic Games (Paris, FRA) (500m Grass Track)**FINAL**

date 07-Jul-00

wind

	H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
Kraenzlein, Alvin (USA) (1876)	no information available										OR	15.4				
McLean, John (USA) (1878)	no information available											15.8e				
Moloney, Frederick (USA) (1882)	no information available											15.9e				
Lécuyer, Jean (FRA) (1876)	no information available															
Pritchard, Norman (GBR) (1875)	no information available											dnf				

1896 Olympic Games (Athens, GRE) (333.33m Trac)**FINAL**

date 07-Apr-96

wind

	H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Run-In	Official Time	Lane / Place	H1-H4	H4-H7	H7-H10
Curtis, Thomas (USA) (1873)	no information available										OR	17.6				
Goulding, Grantley (GBR) (1874)	no information available											17.7e				